



DesignWare® Cores PCI Express Controller

Reference Manual

DWC PCIe Endpoint

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Revision History

Revision history for PCI Express Controller EP Reference Manual:

Date	Version	Description
April 2021	5.95a	A version of this chapter with change tracking is available at https://www.synopsys.com/dw/doc.php/iip/DWC_pcie/5.95a/doc/DWC_pcie_ctl_ep_reference_change.pdf

1

Parameter Descriptions

This chapter details all the configuration parameters. **You can use the coreConsultant GUI configuration reports to determine the complete configuration state of the controller.** Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

The parameter descriptions in this chapter include the **Enabled:** attribute which indicates the values required to be set on other parameters before you can change the value of this parameter.

These tables define all of the configuration options for this component.

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- [“AMBAConfig Parameters” on page 287](#)

1.1 Main Features Config Parameters

Table 1-1 Main Features Config Parameters

Label	Description
Core Type	
Device Type	Select the Device Type of the PCIe controller. You can choose among: EP: Endpoint, RP: Root Port, DM: Dual Mode (pin-selectable EP or RP), and SW: Switch Port (pin-selectable Upstream/Downstream Switch Port). This selection must be in line with your current license key. Values: ■ EP (0) ■ RP (1) ■ DM (2) ■ SW (3) ■ NOT_SET (99) Default Value: 99 (NOT_SET) Enabled: Always Parameter Type: Feature Setting Parameter Name: CC_DEVICE_TYPE
PCIe EP Device Type	Select the EP Device Type. You can choose between EP and EP_LEGACY. Values: ■ PCIE_EP (0x0) ■ PCIE_EP_LEGACY (0x1) Default Value: PCIE_EP Enabled: CC_DEVICE_TYPE==CC_EP Parameter Name: CX_EP_DEVICE_TYPE
PCIe Modes Supported	You can configure the controller to support Conventional PCIe or PCIe over CIO. Values: ■ Single Conventional PCIe (0) ■ PCIe over CIO (3) Default Value: Single Conventional PCIe Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_PCIE_MODE

Label	Description
AXI Bridge	
AMBA Enable	Use the AXI bridge to interface to your application. For more information, see the <i>Databook</i>. Additional configuration panes becomes visible when you set the AMBA_INTERFACE configuration parameter. Values: ■ None (0) ■ Rsvd (1) ■ AXI3 (2) ■ AXI4 (3) ■ AXI5 (4) Default Value: None Enabled: $!(CC_DEVICE_TYPE == CC_SW)$ Parameter Type: Feature Setting Parameter Name: AMBA_INTERFACE
ACE Lite Interface	Enables ACE Lite. Values: 0, 1 Default Value: 0 Enabled: $AMBA_INTERFACE \geq 3$ Parameter Type: Feature Setting Parameter Name: CC_ACELITE_ENABLE
Embedded DMA	
DMA Enable	Enable the embedded DMA controller. For more information, see the 'DMA' chapter of the <i>Databook</i>. An additional configuration pane becomes visible when you set the CC_DMA_ENABLE configuration parameter. Values: ■ false (0) ■ true (1) Default Value: false Enabled: $!(CC_DEVICE_TYPE == CC_SW)$ Parameter Type: Feature Setting Parameter Name: CC_DMA_ENABLE

Label	Description
Multifunction	
Number of Functions	Number of (physical) functions to support (in EP mode). There is a complete set of all of the 'Function/BAR Configuration Parameters' parameters for each of the CX_NFUNC functions. The additional configuration panes for each additional function becomes visible when you set the CX_NFUNC configuration parameter to a value greater than 1. CX_ARI_ENABLE must be 1 when you have more than 8 functions. Values: 1, ..., 64 Default Value: 1 Enabled: CC_DEVICE_TYPE!=CC_RC && CC_DEVICE_TYPE!=CC_SW Parameter Type: Feature Setting Parameter Name: CX_NFUNC
Support ARI	Alternate Routing ID (ARI) Capability enable. For more information, see the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: CX_SRIOV_ENABLE Enabled: ((CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM) && !CX_SRIOV_ENABLE) Parameter Type: Feature Setting Parameter Name: CX_ARI_ENABLE
Individual Bus Number Per Function	Determines if all functions use the same bus number (of function #0) or if they use the individual bus numbers assigned to them by host software. Determines the size of the cfg_pbus_num and cfg_pbus_dev_num buses: one set of bits for all functions or separate bit-fields for each configured function. Not applicable when CX_ARI_ENABLE=1. ■ True: Use individual bus numbers ■ False: Use same bus number Values: ■ false (0) ■ true (1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_RC && CC_DEVICE_TYPE!=CC_SW && CX_ARI_ENABLE!=1 && CX_NW<8 Parameter Type: Feature Setting Parameter Name: MULTI_DEVICE_AND_BUS_PER_FUNC_EN

Label	Description
CCIX Features	
Support CCIX Transport Protocol	Configure the controller to support CCIX Transport Protocol. Values: ■ false (0) ■ true (1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_CCIX_ENABLE
Support ESM Mode	Specifies if the PHY supports ESM (20.0 GT/s and 25 GT/s) Mode. Values: ■ false (0) ■ true (1) Default Value: CX_CCIX_ENABLE==1 Enabled: CX_CCIX_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CCIX_ESM_SUPPORT
Support CCIX TLP Interface	Specifies if CCIX Interface and optimized TLP is supported. Values: ■ false (0) ■ true (1) Default Value: CX_CCIX_ENABLE==1 Enabled: CX_CCIX_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CCIX_INTERFACE_ENABLE
Support CXS Interface	Configure the controller to support CXS Interface. Values: ■ false (0) ■ true (1) Default Value: CX_CXL_CXS_ENABLE ? 1 : 0 Enabled: CX_CCIX_INTERFACE_ENABLE CX_CXL_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_ENABLE

Label	Description
CXL Features	
Support Compute Express Link CXL	Configure the controller to support Compute Express Link CXL Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_MAX_PCIE_SPEED>=3 Parameter Type: Feature Setting Parameter Name: CX_CXL_ENABLE
Support Desired Version of Compute Express Link CXL	Configure the controller to support the desired version of Compute Express Link CXL Values: ■ CXL1.1 (1) ■ CXL2.0 (2) Default Value: CXL2.0 Enabled: CX_CXL_ENABLE && !CX_IS_SW Parameter Type: Feature Setting Parameter Name: CX_CXL_VERSION
Enable CXS+ Interface for CXL Controller	Enables CXS+ Interface for CXL Controller Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_CXL_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_CXS_ENABLE

1.2 Basic Features Config / PCIe Basic Features Config Parameters

Table 1-2 Basic Features Config / PCIe Basic Features Config Parameters

Label	Description
Phy Type	
Phy Selection	Select the type of PHY you are using. PHYs are instantiated outside the controller, interfacing through the standard PIPE I/F. You must select the "Custom PHY" option and follow the configuration guidelines presented in the section "Integrating the controller with the PHY" in the <i>User Guide</i>. A PHY simulation-only model is also supplied by Synopsys. This block cannot be implemented and it is provided for simulation purposes only. You might temporarily use this for simulation purposes until you receive a simulation model from your PHY vendor. Note that different PHY models may have different configuration requirements. Review your configuration settings when switching between the example PHY and the PHY model provided by your PHY vendor. Values: ■ CIO PHY (9) ■ Example PIPE PHY (8) ■ Custom PHY (7) Default Value: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 9 : 8 Enabled: Always Parameter Type: Feature Setting Parameter Name: PHY_TYPE
Max PCIe Speed	Maximum supported link speed that the controller supports. Values: ■ Gen1 (2.5GT/s) (1) ■ Gen2 (5.0GT/s) (2) ■ Gen3 (8.0GT/s) (3) ■ Gen4 (16.0GT/s) (4) ■ Gen5 (32.0GT/s) (5) Default Value: Gen1 (2.5GT/s) Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_MAX_PCIE_SPEED

Label	Description
Core-Phy Combination	Read Only parameter to indicate PHY-MAC combination is supported or not Values: ■ Not Supported (0) ■ Supported (1) Default Value: function_of (CX_MAX_PCIE_SPEED CX_MAC_SMODE_GEN1 CX_MAC_SMODE_GEN2 CX_MAC_SMODE_GEN3 CX_MAC_SMODE_GEN4 CX_MAC_SMODE_GEN5 CX_PHY_SMODE_GEN1 CX_PHY_SMODE_GEN2 CX_PHY_SMODE_GEN3 CX_PHY_SMODE_GEN4 CX_PHY_SMODE_GEN5) Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_MAC_PHY_FREQ_VALID
Core Gen1 Freq/Width	MAC Gen1 Frequency/Width Mode Values: ■ g1_1s (250Mhz) (1) ■ g1_2s (125Mhz) (2) ■ g1_4s (62.5Mhz) (4) Default Value: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 1 : 2 Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : 1 Parameter Type: Feature Setting Parameter Name: CX_MAC_SMODE_GEN1
Phy Gen1 Freq/Width	PHY Gen1 Frequency/Width Mode Values: ■ g1_1s (250Mhz) (1) ■ g1_2s (125Mhz) (2) ■ g1_4s (62.5Mhz) (4) ■ g1_1s_dt2 (500Mhz) (102) ■ g1_1s_dt4 (1000Mhz) (104) ■ g1_2s_dt2 (250Mhz) (202) ■ g1_2s_dt4 (500Mhz) (204) ■ g1_2s_dt8 (1000Mhz) (208) ■ g1_4s_dt2 (125Mhz) (402) ■ g1_4s_dt4 (250Mhz) (404) ■ g1_4s_dt8 (500Mhz) (408) ■ g1_4s_dt16 (1000Mhz) (4016) Default Value: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 1 : 2 Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : 1 Parameter Type: Feature Setting Parameter Name: CX_PHY_SMODE_GEN1

Label	Description
Core Gen2 Freq/Width	MAC Gen2 Frequency/Width Mode Values: ■ Disabled (0) ■ g2_1s (500Mhz) (1) ■ g2_2s (250Mhz) (2) ■ g2_4s (125Mhz) (4) Default Value: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 1 : (CX_MAX_PCIE_SPEED>=2) ? 2 : 0 Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : (CX_MAX_PCIE_SPEED>=2) Parameter Type: Feature Setting Parameter Name: CX_MAC_SMODE_GEN2
Phy Gen2 Freq/Width	PHY Gen2 Frequency/Width Mode Values: ■ Disabled (0) ■ g2_1s (500Mhz) (1) ■ g2_2s (250Mhz) (2) ■ g2_4s (125Mhz) (4) ■ g2_1s_dt2 (1000Mhz) (102) ■ g2_2s_dt2 (500Mhz) (202) ■ g2_2s_dt4 (1000Mhz) (204) ■ g2_4s_dt2 (250Mhz) (402) ■ g2_4s_dt4 (500Mhz) (404) ■ g2_4s_dt8 (1000Mhz) (408) Default Value: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 1 : (CX_MAX_PCIE_SPEED>=2) ? 2 : 0 Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : (CX_MAX_PCIE_SPEED>=2) Parameter Type: Feature Setting Parameter Name: CX_PHY_SMODE_GEN2

Label	Description
Core Gen3 Freq/Width	MAC Gen3 Frequency/Width Mode Values: ■ Disabled (0) ■ g3_1s (1000Mhz) (1) ■ g3_2s (500Mhz) (2) ■ g3_4s (250Mhz) (4) ■ g3_8s (125Mhz) (8) ■ g3_16s (62.5Mhz) (16) Default Value: (CX_MAX_PCIE_SPEED<3) ? 0 : (CX_MAC_SMODE_GEN1==1) && (CX_MAC_SMODE_GEN2==1) ? 1 : (CX_MAC_SMODE_GEN1==2) && (CX_MAC_SMODE_GEN2==2) ? 2 : (CX_MAC_SMODE_GEN1==4) && (CX_MAC_SMODE_GEN2==4) ? 4 : (CX_MAC_SMODE_GEN1==1) && (CX_MAC_SMODE_GEN2==2) ? 4 : (CX_MAC_SMODE_GEN1==2) && (CX_MAC_SMODE_GEN2==4) ? 8 : 2 Enabled: CX_MAX_PCIE_SPEED>=3 Parameter Type: Feature Setting Parameter Name: CX_MAC_SMODE_GEN3
Phy Gen3 Freq/Width	PHY Gen3 Frequency/Width Mode Values: ■ Disabled (0) ■ g3_1s (1000Mhz) (1) ■ g3_2s (500Mhz) (2) ■ g3_4s (250Mhz) (4) ■ g3_8s (125Mhz) (8) ■ g3_16s (62.5Mhz) (16) ■ g3_4s_dt2 (500Mhz) (402) ■ g3_4s_dt4 (1000Mhz) (404) ■ g3_8s_dt2 (250Mhz) (802) ■ g3_8s_dt4 (500Mhz) (804) ■ g3_8s_dt8 (1000Mhz) (808) Default Value: (CX_MAX_PCIE_SPEED>=3) ? 2 : 0 Enabled: CX_MAX_PCIE_SPEED>=3 Parameter Type: Feature Setting Parameter Name: CX_PHY_SMODE_GEN3

Label	Description
Core Gen4 Freq/Width	MAC Gen4 Frequency/Width Mode Values: ■ Disabled (0) ■ g4_2s (1000Mhz) (2) ■ g4_4s (500Mhz) (4) ■ g4_8s (250Mhz) (8) ■ g4_16s (125Mhz) (16) Default Value: (CX_MAX_PCIE_SPEED<4) ? 0 : (CX_MAC_SMODE_GEN1==2) && (CX_MAC_SMODE_GEN2==2) ? 2 : (CX_MAC_SMODE_GEN1==4) && (CX_MAC_SMODE_GEN2==4) ? 4 : (CX_MAC_SMODE_GEN1==1) && (CX_MAC_SMODE_GEN2==2) ? 8 : (CX_MAC_SMODE_GEN1==2) && (CX_MAC_SMODE_GEN2==4) ? 16 : 2 Enabled: CX_MAX_PCIE_SPEED>=4 Parameter Type: Feature Setting Parameter Name: CX_MAC_SMODE_GEN4
Phy Gen4 Freq/Width	PHY Gen4 Frequency/Width Mode Values: ■ Disabled (0) ■ g4_2s (1000Mhz) (2) ■ g4_4s (500Mhz) (4) ■ g4_8s (250Mhz) (8) ■ g4_16s (125Mhz) (16) ■ g4_4s_dt2 (1000Mhz) (402) ■ g4_8s_dt2 (500Mhz) (802) ■ g4_8s_dt4 (1000Mhz) (804) Default Value: (CX_MAX_PCIE_SPEED>=4) ? 2 : 0 Enabled: CX_MAX_PCIE_SPEED>=4 Parameter Type: Feature Setting Parameter Name: CX_PHY_SMODE_GEN4

Label	Description
Core Gen5 Freq/Width	MAC Gen5 Frequency/Width Mode Values: ■ Disabled (0) ■ g5_2s (2000Mhz) (2) ■ g5_4s (1000Mhz) (4) ■ g5_8s (500Mhz) (8) ■ g5_16s (250Mhz) (16) Default Value: (CX_MAX_PCIE_SPEED<5) ? 0 : (CX_MAC_SMODE_GEN1==2) && (CX_MAC_SMODE_GEN2==2) ? 2 : (CX_MAC_SMODE_GEN1==4) && (CX_MAC_SMODE_GEN2==4) ? 4 : (CX_MAC_SMODE_GEN1==1) && (CX_MAC_SMODE_GEN2==2) ? 8 : (CX_MAC_SMODE_GEN1==2) && (CX_MAC_SMODE_GEN2==4) ? 16 : 2 Enabled: CX_MAX_PCIE_SPEED>=5 Parameter Type: Feature Setting Parameter Name: CX_MAC_SMODE_GEN5
Phy Gen5 Freq/Width	PHY Gen5 Frequency/Width Mode Values: ■ Disabled (0) ■ g5_2s (2000Mhz) (2) ■ g5_4s (1000Mhz) (4) ■ g5_8s (500Mhz) (8) ■ g5_16s (250Mhz) (16) Default Value: (CX_MAX_PCIE_SPEED>=5) ? 2 : 0 Enabled: CX_MAX_PCIE_SPEED>=5 Parameter Type: Feature Setting Parameter Name: CX_PHY_SMODE_GEN5
CCIX ESM0 Speed	CCIX ESM0 Speed Values: ■ Disabled (0) ■ 8.0GT/s, 16.0GT/s (1) Default Value: (CX_CCIX_ESM_SUPPORT) ? 1 : 0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_CCIX_ESM0_SPEED
CCIX ESM1 Speed	CCIX ESM1 Speed Values: ■ Disabled (0) ■ 20.0GT/s, 25.0GT/s (1) Default Value: (CX_CCIX_ESM_SUPPORT) ? 1 : 0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_CCIX_ESM1_SPEED

Label	Description
Core CCIX ESM0 Freq/Width	MAC CCIX ESM0 Frequency/Width Mode Values: ■ Disabled (0) ■ ccix3_2s (500Mhz/1000Mhz) (2) ■ ccix3_4s (250Mhz/500Mhz) (4) ■ ccix3_8s (125Mhz/250Mhz) (8) ■ ccix3_16s (62.5Mhz/125Mhz) (16) Default Value: (CX_CCIX_ESM_SUPPORT) ? CX_MAC_SMODE_GEN3 : 0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_MAC_SMODE_ESM0
Phy CCIX ESM0 Freq/Width	PHY CCIX ESM0 Frequency/Width Mode Values: ■ Disabled (0) ■ ccix3_2s (500Mhz/1000Mhz) (2) ■ ccix3_4s (250Mhz/500Mhz) (4) ■ ccix3_8s (125Mhz/250Mhz) (8) ■ ccix3_16s (62.5Mhz/125Mhz) (16) Default Value: (CX_CCIX_ESM_SUPPORT) ? CX_PHY_SMODE_GEN3 : 0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_PHY_SMODE_ESM0
Core CCIX ESM1 Freq/Width	MAC CCIX ESM1 Frequency/Width Mode Values: ■ Disabled (0) ■ ccix4_2s (1250Mhz/1562.5Mhz) (2) ■ ccix4_4s (625Mhz/781.25Mhz) (4) ■ ccix4_8s (312.5Mhz/390.625Mhz) (8) ■ ccix4_16s (156.25Mhz/195.3125Mhz) (16) Default Value: (CX_CCIX_ESM_SUPPORT) ? CX_MAC_SMODE_GEN4 : 0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_MAC_SMODE_ESM1

Label	Description
Phy CCIX ESM1 Freq/Width	PHY CCIX ESM1 Frequency/Width Mode Values: ■ Disabled (0) ■ ccix4_2s (1250Mhz/1562.5Mhz) (2) ■ ccix4_4s (625Mhz/781.25Mhz) (4) ■ ccix4_8s (312.5Mhz/390.625Mhz) (8) ■ ccix4_16s (156.25Mhz/195.3125Mhz) (16) Default Value: (CX_CCIX_ESM_SUPPORT) ? CX_PHY_SMODE_GEN4 : 0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_PHY_SMODE_ESM1
Maximum Link Width	Maximum number of lanes that the controller supports. For more information on what combinations of modes, frequencies and link widths are supported for each controller, see Table 1-1 in the "Frequency, Speed, and Width Support" section in the Product Overview chapter of the <i>Databook</i>. Values: ■ x1 (1) ■ x2 (2) ■ x4 (4) ■ x8 (8) ■ x16 (16) Default Value: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 16 : 4 Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : CX_S_CPCIE_MODE Parameter Type: Feature Setting Parameter Name: CX_NL
Phy Rx multi-bit startblock	Enables PIPE Rx multi-bit startblock Mode for 8s/16s. When disabled and PIPE width is 8s/16s, the Controller only accepts 64-bit-aligned SKP-OS in Gen3/4/5 speed. When enabled and PIPE width is 8s/16s, the Controller can accept 32-bit-aligned SKP-OS in Gen3/4/5 Speed. Possible SKP-OS length is 32/64/96/128/160/192/224-bit considering two retimers on the link. Values: ■ Disabled (0) ■ Enabled (1) Default Value: ((SNPS_RSVDPARAM_71==1) (CX_PIPE_SERDES_ARCH && CX_MAX_PCIE_SPEED>=3 && (CX_PHY_NB_GEN3>=8 CX_PHY_NB_GEN4>=8 CX_PHY_NB_GEN5>=8)))? 1 : 0 Enabled: ((CX_MAX_PCIE_SPEED>=3 && (CX_PHY_SMODE_GEN3>=8 CX_PHY_SMODE_GEN4>=8 CX_PHY_SMODE_GEN5>=8)) && !CX_PIPE_SERDES_ARCH) Parameter Type: Feature Setting Parameter Name: CX_PIPERX_MULTI_BLOCK

Label	Description
Freq Step (b/w PHY and Core)	If this automatically-derived parameter is defined, then a module called <code>freq_step</code> (<i>workspace/src/common/freq_step.v</i>) is placed in between the PIPE I/O and the controller's internal pipe interface. This module steps up/down the signals to/from the pipe interface. For example, the controller can run at 62.5 MHz (4 symbols per clock) and the pipe can run at 250 MHz (1 symbol per clock). This read-only parameter is derived automatically. For more information, see 'Frequency, Speed and Width Support' in the Product Overview chapter of <i>Databook</i>. Values: ■ No (0) ■ Yes (1) Default Value: (CX_MAC_SMODE_GEN1 != CX_PHY_SMODE_GEN1) (CX_MAC_SMODE_GEN2 != CX_PHY_SMODE_GEN2) (CX_MAC_SMODE_GEN3 != CX_PHY_SMODE_GEN3) (CX_MAC_SMODE_GEN4 != CX_PHY_SMODE_GEN4) (CX_MAC_SMODE_GEN5 != CX_PHY_SMODE_GEN5) (CX_PIPERX_MULTI_BLOCK==1) Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_FREQ_STEP_EN
Freq Step (b/w PL and DL)	Specifies if the controller supports for frequency step module between the Physical Layer and the DataLink Layer. Values: ■ No (0) ■ Yes(1:1) (1) ■ Yes(1:2) (2) ■ Yes(1:4) (4) Default Value: No Enabled: SNPS_RSVDPARAM_49 Parameter Name: CX_FREQ_STEP_DL
Datapath Width (PL)	Specifies the width of the datapath in dwords. Displayed in bits. This read-only parameter is derived automatically as $CX_NL * CX_NB$, rounded up to 1, 2, 4, or 8. For more information, see the "Frequency, Speed, and Width Support" section in the Product Overview chapter of the <i>Databook</i>. For the definition of CX_NB, see the description of "Controller PIPE Lane Width" ($CX_PIPE_WIDTH_$). Values: ■ 32-bit (1) ■ 64-bit (2) ■ 128-bit (4) ■ 256-bit (8) ■ 512-bit (16) Default Value: [<functionof> CX_NB CX_NL] Enabled: Always Parameter Name: CX_PL_NW

Label	Description
Datapath Width (DL/TL/ADM)	Specifies the width of the datapath in dwords. Displayed in bits. This read-only parameter is derived automatically as $CX_NL * CX_DL_NB$, rounded up to 1, 2, 4, or 8. For more information, see the "Frequency, Speed, and Width Support" section in the Product Overview chapter of the <i>Databook</i>. For the definition of CX_NB, see the description of "Controller PIPE Lane Width" ($CX_PIPE_WIDTH_$). Values: ■ 32-bit (1) ■ 64-bit (2) ■ 128-bit (4) ■ 256-bit (8) ■ 512-bit (16) Default Value: [$\langle functionof \rangle CX_DL_NB CX_NL$] Enabled: Always Parameter Name: CX_NW
PIPE Spec	
PIPE Version Options	Select PIPE Version. ■ PIPE 4.2 ■ PIPE 4.3 □ Use $mac_phy_powerdown$ signal for L1 substate/L1.CPM transition □ Use Synopsys specific Message Bus Interface for Lane Margining ■ PIPE 4.4.1 □ Use $mac_phy_powerdown$ signal for L1 substate/L1.CPM transition □ Use 5-bit LocalPresetIndex encoding for Gen4 data rate □ Use Message Bus Interface for Lane Margining for Gen4 data rate ■ PIPE 5.1.1 □ Use $mac_phy_powerdown$ signal for L1 substate/L1.CPM transition □ Use PIPE Low Pin Count Interface □ Use SerDes Architecture (Optional) □ Use PCLK as PHY input optionally Contact your PHY vendor for guidelines on PIPE Spec Version. For more information on how to configure this parameter for Synopsys PHYs, see the <i>User Guide</i>. Values: ■ PIPE4.2 and below (0) ■ PIPE4.3 (1) ■ PIPE4.4.1 (2) ■ PIPE5.1.1 (3) Default Value: $CX_PCIE_MODE == PCIE_OVER_CIO) ? 2 : ((CX_GEN5_SPEED CX_CCIX_ESM_SUPPORT) ? 2 : [\langle functionof \rangle]$ Enabled: $(CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : CX_CPCIE_ENABLE$ Parameter Type: Feature Setting Parameter Name: CX_PIPE_VER

Label	Description
SerDes Architecture	Enable PIPE SerDes Architecture. Values: ■ Original PIPE architecture (0) ■ SerDes architecture (1) Default Value: Original PIPE architecture Enabled: CX_CPCIE_ENABLE && CX_PIPE5_SUPPORT Parameter Type: Feature Setting Parameter Name: CX_PIPE_SERDES_ARCH
Lanes Turn-off for SerDes Architecture	Enable functionality to turn off a lane using serdes_pipe_turnoff_lanes signal, for SerDes architecture. When enabled, lanes between the controller and the PHY can be disconnected or dynamically turned off. For more information, see the "PIPE Interface" section in the "Signal Interfaces" chapter of the Databook. Values: ■ Disabled (0) ■ Enabled (1) Default Value: Disabled Enabled: CX_CPCIE_ENABLE && CX_PIPE_SERDES_ARCH Parameter Type: Feature Setting Parameter Name: CX_SERDES_ARCH_LANES_TURNOFF
PIPE PCLK Mode	Select the PIPE PCLK mode. <i>Note:</i> Any PCLK as PHY input mode requires PIPE4.4.1 and above. ■ 0: PCLK as PHY output. ■ 1: PCLK as PHY input with Fixed MAX PCLK frequency. ■ 2: PCLK as PHY input with Dynamic MAX PCLK frequency. ■ 3: PCLK as PHY input with MAX PCLK to PCLK frequency ratio 2:1. Values: ■ PCLK as PHY output (0) ■ PCLK as PHY input Fixed MAXPCLK (1) ■ PCLK as PHY input Dynamic MAXPCLK (2) ■ PCLK as PHY input fixed ratio MAXPCLK (3) Default Value: (CX_CXL_ENABLE==1 && DWC_PCIE_FPGA==0) ? 3 : 0 Enabled: CX_PIPE_VER>=2 Parameter Type: Feature Setting Parameter Name: CX_PIPE_PCLK_MODE

Label	Description
Drift Buffer	When enabled, the controller implements the drift buffer. Values: ■ Disabled (0) ■ Enabled (1) Default Value: CX_PIPE_SERDES_ARCH && CX_CXL_ENABLE && CX_IS_EP Enabled: CX_CPCIE_ENABLE && CX_PIPE_SERDES_ARCH && CX_CXL_ENABLE Parameter Type: Feature Setting Parameter Name: CX_DRIFT_BUFFER_SUPPORT
PCIe Basespec 4.0 Feature	
Max number of Retimer	Specifies maximum number of Retimers that can be present in your system. This setting impacts default buffer size of Retry, SOT buffer and Receive queue. Values: 0x0, 0x1, 0x2 Default Value: ((CX_GEN4_MODE != GEN4_DISABLED)==1) ? 2 : 0 Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_MAX_RETIMER
Total Retimer Latency (Symbol times)	Specifies total round trip Latency of Retimer. The default value is calculated from CX_MAX_RETIMER setting and reference Retimer latency table in PCIe Base Spec. This setting impacts retry buffer auto-sizing and default credits of Receive queue. If you know Latency value of Retimer in your system, you should adjust the value. Otherwise, use the default setting. Values: 0, ..., 2048 Default Value: [<functionof> CX_MAX_RETIMER CX_SRIS_SUPPORT CX_MAX_MTU CX_GEN5_SPEED CX_GEN4_SPEED CX_GEN3_SPEED CX_GEN2_SPEED] Enabled: Always Parameter Name: CX_RETIMER_LATENCY
Data Link Feature Enabled	The Data Link Feature Capability is an optional Extended Capability that is required for Downstream Ports that support 16.0 GTs. It is optional in other Downstream Ports. It is optional in Functions associated with an Upstream Port. It is not applicable in Functions that are not associated with a Port (For example, Root Complex Integrated Endpoints, Root Complex Event Collectors). Ports that implement this protocol contain the Data Link Feature Extended Capability as described in Section 7.37 of the <i>PCIe Specification 4.0</i>. Values: ■ false (0) ■ true (1) Default Value: CX_GEN4_SPEED FC_SCALE_EN Enabled: 0 Parameter Type: Feature Setting Parameter Name: DL_FEATURE_EN

Label	Description
Scaled Flow Control Enabled	Link performance can be affected when there are insufficient flow control credits available to account for the Link round trip time. This effect becomes more noticeable at higher Link speeds and the limitation of 127 header credits and 2047 data credits can limit performance. The Scaled Flow Control mechanism is designed to address this limitation. Scaled Flow control is described in Section 3.4.2 of the <i>PCIe Specification 4.0</i>. Values: ■ false (0) ■ true (1) Default Value: CX_GEN4_MODE != GEN4_DISABLED Enabled: !(CX_GEN4_MODE != GEN4_DISABLED)) Parameter Type: Feature Setting Parameter Name: FC_SCALE_EN
Data Link Feature Exchange Enable	Default value for the Data Link Feature Exchange Enable field in the 'Data Link Feature Capabilities Register' Values: 0x0, 0x1 Default Value: DL_FEATURE_EN Enabled: DL_FEATURE_EN Parameter Type: Register Default Setting Parameter Name: DL_FEATURE_EXCHANGE_ENABLE
10-Bit Tag	Configure the controller to support the extended tag range using a 10-bit tag field. Values: 0, 1 Default Value: (((CX_GEN4_SPEED_VALUE == 1) (CX_MAX_TAG > 255) (CX_REMOTE_MAX_TAG > 255)) && (AMBA_INTERFACE != 1))? 1: 0 Enabled: ((AMBA_INTERFACE != 1) && (CX_GEN4_SPEED_VALUE == 0) && (CX_MAX_TAG <= 255) && (CX_REMOTE_MAX_TAG <= 255)) Parameter Type: Feature Setting Parameter Name: CX_10BITS_TAG
PCIe Basespec 5.0 Feature	
Support Equalization bypass to highest rate	When enabled, the controller supports Equalization bypass to highest rate. Values: 0x0, 0x1 Default Value: CX_MAX_PCIE_SPEED >= 5 Enabled: ((CX_MAX_PCIE_SPEED >= 5)) Parameter Type: Feature Setting Parameter Name: CX_SKIP_LOWER_RATE_EQ
Support No Equalization Needed	When enabled, the controller supports No Equalization Needed. Contact your PHY vendor for the value that you should use. Values: 0x0, 0x1 Default Value: 0x0 Enabled: ((CX_MAX_PCIE_SPEED >= 5)) Parameter Type: Feature Setting Parameter Name: CX_NO_EQ_NEEDED

Label	Description
Request Transmitter Precode	When enabled, the controller requests the remote transmitter to use Precoding by setting the Precoding Request bit in the EQ TS it transmits prior to entry to Recovery.Speed. Values: ■ false (0x0) ■ true (0x1) Default Value: CX_MAX_PCIE_SPEED >= 5 Enabled: CX_MAX_PCIE_SPEED >= 5 Parameter Type: Register Default Setting Parameter Name: CX_DEFAULT_TX_PRECODE_REQ
Default Value for Gen5 Loopback EQ	Default value for Loopback to do Equalization at Gen5 rate. Values: 0x0, 0x1 Default Value: CX_MAX_PCIE_SPEED >= 5 Enabled: CX_MAX_PCIE_SPEED >= 5 Parameter Type: Feature Setting Parameter Name: CX_DEFAULT_GEN5_EQ_FOR_LOOPBACK
Modified TS Format	Enable Modified TS format support. Values: ■ false (0) ■ true (1) Default Value: CX_GEN5_SPEED CX_AP_SUPPORT Enabled: CX_GEN3_MODE != GEN3_DISABLED Parameter Type: Feature Setting Parameter Name: CX_MODIFIED_TS_FORMAT_SUPPORT
Alternate Protocols (AP)	Enable Alternate Protocols (AP) support. Values: ■ false (0) ■ true (1) Default Value: CX_CXL_ENABLE Enabled: CX_MAX_PCIE_SPEED >= 3 Parameter Type: Feature Setting Parameter Name: CX_AP_SUPPORT

1.3 Basic Features Config / Common Basic Features Config Parameters

Table 1-3 Basic Features Config / Common Basic Features Config Parameters

Label	Description
PCIe Automotive Support	
Automotive safety package	Automotive safety package Values: 0, 1 Default Value: 0 Enabled: Always Parameter Type: Register Default Setting Parameter Name: CX_AUTOMOTIVE_ENABLE
PCIe Max Payload	
PCIe Max Payload Supported	The largest packet payload (Maximum Transfer Unit) that the device supports. Specified in bytes and not dwords. This is distinct from the maximum operating payload (Max_Payload_Size) which may be set by software. This value is used to set memory sizes and places other restrictions on the size of structures. The device supports any valid payload size equal to or smaller than the value specified here. Values: 128, 256, 512, 1024, 2048, 4096 Default Value: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 128 : 256 Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : 1 Parameter Type: Performance Setting Parameter Name: CX_MAX_MTU

Label	Description
Simultaneous Outstanding Requests	
Max Outbound NP Requests	Specifies the maximum number of simultaneous outbound PCIe non-posted requests in total for all functions. The default value is sufficient for typical endpoint applications. This is used to size: - the Receive 'Completion Timeout Look-Up Table' - the completion Header Queue RAM when you configure completions in store-and-forward or cut-through modes - the tag Manager memories in the AXI bridge slave The controller can always accept 8-bit tags (IDs 0 to 255) from the PCIe wire or your application, regardless of the value of this parameter. Note: The value range in the GUI is displayed as CX_MAX_TAG+1. The actual parameter value in the RTL represents the maximum tag number because the tags are numbered 0 to CX_MAX_TAG If you enable the DMA controller then: - CC_NUM_DMA_RD_TAG tags are reserved for use by the DMA controller. - The minimum value of CX_MAX_TAG allowed is 3 Values: 2 (1) 4 (3) 8 (7) 16 (15) 32 (31) 64 (63) 128 (127) 256 (255) 512 (511) 768 (767) Default Value: RECOMMENDED_TAGS Enabled: ((CC_DEVICE_TYPE!=CC_SW)) Parameter Type: Performance Setting Parameter Name: CX_MAX_TAG
Address Translation	
Internal Address Translation Unit (iATU)	Enable the instantiation of the internal address translation unit (iATU). For more information, see the 'Internal Address Translation (iATU)' section of the <i>Databook</i>. Values: - false (0x0) - true (0x1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_INTERNAL_ATU_ENABLE

Label	Description
Number of Outbound Address Translation Regions	Specify the number of address regions to be mapped by the internal Address Translation Unit in the outbound direction. For more information, see the 'Internal Address Translation (iATU)' section of the <i>Databook</i>. Values: 0, ..., 256 Default Value: 2 Enabled: CX_INTERNAL_ATU_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ATU_NUM_OUTBOUND_REGIONS
Number of Inbound Address Translation Regions	Specifies the number of address regions to be mapped by the internal Address Translation Unit in the inbound direction. For more information, see the 'Internal Address Translation (iATU)' section of the <i>Databook</i>. Values: 0, ..., 256 Default Value: 2 Enabled: CX_INTERNAL_ATU_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ATU_NUM_INBOUND_REGIONS
Minimum size of an Address Translation Region	Specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. Smaller Regions require more levels of decode logic. For more information, see the 'Internal Address Translation (iATU)' section of the <i>Databook</i>. Values: ■ 4k (4096) ■ 8k (8192) ■ 16k (16384) ■ 32k (32768) ■ 64k (65536) Default Value: 64k Enabled: CX_INTERNAL_ATU_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ATU_MIN_REGION_SIZE

Label	Description
Maximum size of an Address Translation Region	Specifies the maximum allowable size of an Address Translation Region in iATU for both inbound and outbound TLPs. This parameter determines the number of programmable bits in the iATU Upper Limit Address Register for both inbound and outbound. For more information, see the 'Internal Address Translation (iATU)' section of the <i>Databook</i>. Values: ■ 4G (0) ■ 8G (1) ■ 16G (2) ■ 32G (3) ■ 64G (4) ■ 128GB (5) ■ 256GB (6) ■ 512GB (7) ■ 1TB (8) ■ 2TB (9) ■ 4TB (10) ■ 8TB (11) ■ 16TB (12) ■ 32TB (13) ■ 64TB (14) ■ 128TB (15) ■ 256TB (16) ■ 512TB (17) ■ 1PB (18) ■ 2PB (19) ■ 4PB (20) ■ 8PB (21) ■ 16PB (22) ■ 32PB (23) ■ 64PB (24) ■ 128PB (25) ■ 256PB (26) ■ 512PB (27) ■ 1EB (28) ■ 2EB (29) ■ 4EB (30) ■ 8EB (31) ■ 16EB (32) Default Value: 4G Enabled: CX_INTERNAL_ATU_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ATU_MAX_REGION_SIZE

Label	Description
Receive Queue (VC0)	
Receive Posted Queue Mode (VC0)	The queue mode for the posted TLP receive queue for VC0. ■ Bypass: There is no receive queue in this mode, your application must be able to accept all traffic as back-pressure is disabled in the mode. ■ Store-and-forward: TLPs are stored into queue; TLP is advertised only after the full TLP is stored into the queue. ■ Cut-through: TLPs are stored into queue and presented to your application at the same time it is being stored into the queue. For more information, see "Receive Queue Buffers" in the Architecture chapter of the <i>Databook</i>. Values: ■ Store/Fwd (0x1) ■ Cut Thru (0x2) ■ Bypass (0x4) Default Value: Store/Fwd Enabled: $!(CX_NW \geq 8 \ \&\& \ CX_RADMQ_MODE \neq 2)$ Parameter Type: Feature Setting Parameter Name: RADM_P_QMODE_VC0
Receive Non-Posted Queue Mode (VC0)	The queue mode for the non-posted TLP receive queue for VC0. ■ Store-and-forward: TLPs are stored into queue; TLP is advertised only after the full TLP is stored into the queue. For more information, see "Receive Queue Buffers" in the Architecture chapter of the <i>Databook</i>. Values: ■ Store/Fwd (0x1) Default Value: $CX_RADMQ_MODE == 0 ? RADM_P_QMODE_VC0 : 1$ Enabled: $(CX_RADMQ_MODE \neq 0 \ \&\& \ !(CX_NW \geq 8 \ \&\& \ CX_RADMQ_MODE \neq 2))$ Parameter Type: Feature Setting Parameter Name: RADM_NP_QMODE_VC0

Label	Description
Receive Completions Queue Mode (VC0)	The queue mode for the completion TLP receive queue for VC0. ■ Bypass: There is no receive queue in this mode, your application must be able to accept all traffic as back-pressure is disabled in the mode. ■ Store-and-forward: TLPs are stored into queue; TLP is advertised only after the full TLP is stored into the queue. ■ Cut-through: TLPs are stored into queue and presented to your application at the same time it is being stored into the queue. For more information, see "Receive Queue Buffers" in the Architecture chapter of the <i>Databook</i>. Values: ■ Store/Fwd (0x1) ■ Cut Thru (0x2) ■ Bypass (0x4) Default Value: (CC_DEVICE_TYPE==CC_SW CX_NW>=8) ? 1 : 4 Enabled: (!(CX_NW>=8 && CX_RADMQ_MODE!=2)) Parameter Type: Feature Setting Parameter Name: RADM_CPL_QMODE_VC0
Completion Queue Management Feature	Enable the NP throttle that halts the client interface if there is not enough space for the CPL in the RADM_CPLQ. Values: 0, 1 Default Value: (CC_DEVICE_TYPE!=CC_SW) && !(CX_P2P_ENABLE) && TRGT1_POPULATE && (RADM_CPL_QMODE_VC0<4) Enabled: ((CC_DEVICE_TYPE!=CC_SW) && TRGT1_POPULATE && (RADM_CPL_QMODE_VC0<4)) Parameter Type: Feature Setting Parameter Name: CX_CPLQ_MANAGEMENT_ENABLE
Technology	
FPGA	Select this option if you are implementing the controller in an FPGA. This adds the required pipelining (and corresponding logic to control pipelined logic) to enable the synthesis tool to meet FPGA timing. For more information FPGA implementation, see 'Synthesizing the Controller for an FPGA' section of the <i>User Guide</i>. Values: 0, 1 Default Value: 0 Enabled: Always Parameter Type: Feature Setting Parameter Name: DWC_PCIE_FPGA

Label	Description
Individual Pipeline Control	Enable independent control of inter-module pipelines to trade-off latency and gates for ease of timing closure. Additional configuration panes becomes visible when you set this parameter. Values: ■ Disable (0) ■ Enable (1) Default Value: Disable Enabled: Always Parameter Type: Performance Setting Parameter Name: CX_CUSTOM_PIPELINING
Technology Speed	The speed of the target technology relative to the clock frequency and architecture. Possible values are SLOW or FAST. When you have selected "Individual Pipeline Control" (CX_CUSTOM_PIPELINING set to "1"), then this parameter sets the default for many of the individual inter-module pipeline enabling parameters to "1" when CX_TECHNOLOGY is SLOW. This parameter is forced to SLOW when you are using an FPGA (DWC_PCIE_FPGA parameter set to "1"). An ASIC technology might be FAST if running at 125 MHz, but the SLOW setting might be needed to meet timing at 250 MHz in an x8/x16 architecture. Values: ■ SLOW (0x0) ■ FAST (0x2) Default Value: DWC_PCIE_FPGA ? 0 : 2 Enabled: !DWC_PCIE_FPGA (CX_CUSTOM_PIPELINING==0) Parameter Type: Performance Setting Parameter Name: CX_TECHNOLOGY
MultiStream Client Interface	
Enable MultiStream Client Interface	Enable the MultiStream Client Interface. The MultiStream Client Interface improves support for applications requiring multiple Virtual Channels, providing three dedicated client interfaces (Cpl, NP, P) per Virtual Channel and enhancing VC Arbitration capabilities. Values: 0, 1 Default Value: 0 Enabled: CC_DMA_ENABLE==0 && AMBA_INTERFACE==0 Parameter Name: CX_MULTISTREAM

1.4 DMA Configuration Parameters

Table 1-4 DMA Configuration Parameters

Label	Description
Basic Features	
Number of DMA Write Channels	Number of write channels implemented in the DMA logic. A DMA write copies data from your local application to the remote link partner. For more information, see the 'DMA' chapter of the <i>Databook</i>. Values: 1, ..., 64 Default Value: 2 Enabled: CC_DMA_ENABLE Parameter Type: Feature Setting Parameter Name: CC_NUM_DMA_WR_CHAN
Number of DMA Read Channels	Number of read channels implemented in the DMA logic. A DMA read copies data from the remote link partner to your local application. For more information, see the 'DMA' chapter of the <i>Databook</i>. Values: 1, ..., 64 Default Value: 2 Enabled: CC_DMA_ENABLE Parameter Type: Feature Setting Parameter Name: CC_NUM_DMA_RD_CHAN
Advanced Features	
Number of DMA Read PCIe Tags	Specifies how many of the CX_MAX_TAG+1 PCIe tags are reserved for DMA MRd generation. ■ CC_NUM_DMA_RD_TAG of the CX_MAX_TAG+1 tags are reserved for use by the DMA controller (and are no longer available to your application) for generation of non-posted DMA traffic. These are in the number range CX_MAX_TAG + 1 - CC_NUM_DMA_RD_TAG to CX_MAX_TAG. ■ The remaining CX_MAX_TAG + 1 - CC_NUM_DMA_RD_TAG tags are assigned to non-DMA transfers. These are in the number range 0 to CX_MAX_TAG - CC_NUM_DMA_RD_TAG. This parameter is also used to size the DMA Read buffer. The minimum value for CX_MAX_TAG is "3" when DMA is enabled. Values: 2, 4, 8, 16, 32, 64, 128, 256, 512 Default Value: (CX_MAX_TAG==3) ? 2 : (CX_MAX_TAG==7) ? 4 : (CX_MAX_TAG==15) ? 8 : (CX_MAX_TAG==31) ? 16 : (CX_MAX_TAG==63) ? 32 : (CX_MAX_TAG==127) ? 64 : (CX_MAX_TAG==255) ? 128 : (CX_MAX_TAG==511) ? 256 : (CX_MAX_TAG>=767) ? 256 : 2 Enabled: CC_DMA_ENABLE Parameter Type: Performance Setting Parameter Name: CC_NUM_DMA_RD_TAG

Label	Description
DMA Handshake Enable	Enable DMA Handshake feature. The Handshake feature is available for each DMA Write/Read channel. For more information, see the 'DMA' chapter of the <i>Databook</i> Values: ■ false (0) ■ true (1) Default Value: false Enabled: CC_DMA_ENABLE && !CX_DMA_PF_ENABLE Parameter Name: CC_DMA_HSHAKE
Hyper-DMA	Enable Hyper-DMA code, that brings true descriptor pre-fetch. This parameter must be set together with CC_DMA_ENABLE. Values: 0, 1 Default Value: 0 Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_DMA_PF_ENABLE
Write Engine Descriptors Queue Depth	Hyper-DMA LL queue (pre-fetch) depth for the DMA write engine Values: 1, ..., 1024 Default Value: 1 Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CX_DMA_WR_LLQ
Read Engine Descriptors Queue Depth	Hyper-DMA LL queue (pre-fetch) depth for the DMA read engine Values: 1, ..., 1024 Default Value: 1 Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CX_DMA_RD_LLQ
Hyper-DMA Legacy Map Mode	Hyper DMA LegacyMap Mode Values: 0, 1 Default Value: !CX_DMA_PF_ENABLE Enabled: CX_DMA_PF_ENABLE Parameter Name: CC_LEGACY_DMA_MAP

Label	Description
Hyper-DMA Channel Separation	Indicates the channel separation Hyper-DMA configuration registers into memory space. Values: 0 (CHSEP_256): 256 separated 1 (CHSEP_512): 512 separated 2 (CHSEP_1K): 1k separated 3 (CHSEP_2K): 2k separated 4 (CHSEP_4K): 4k separated 5 (CHSEP_8K): 8k separated 6 (CHSEP_16K): 16k separated 7 (CHSEP_32K): 32k separated Values: ■ CHSEP_256 (0) ■ CHSEP_512 (1) ■ CHSEP_1K (2) ■ CHSEP_2K (3) ■ CHSEP_4K (4) ■ CHSEP_8K (5) ■ CHSEP_16K (6) ■ CHSEP_32K (7) Default Value: CHSEP_256 Enabled: !CC_LEGACY_DMA_MAP Parameter Type: Feature Setting Parameter Name: CX_DMAREG_CHADDR_SPACE

1.5 Basic AXI Config Parameters

Table 1-5 Basic AXI Config Parameters

Label	Description
AXI General and Data/Address	
Support Slave Wrap Bursts	Enable support of wrapping bursts at the AXI bridge slave. For more information, see the "Supported AXI Burst Operations" section in the "AXI Bridge Module" chapter of the <i>Databook</i>. Values: 0, 1 Default Value: 0 Enabled: AMBA_INTERFACE>=2 Parameter Type: Feature Setting Parameter Name: CC_SLV_WRAP_ENABLE
Max Number of Outstanding AXI Slave NP Write Requests	Specifies the maximum number of outstanding AXI Slave NP Write Requests. To enable P writes to pass NP writes (and avoid deadlocks by allowing peer-to-peer posted traffic to flow freely through the RC); the controller uses a "Slave Non-Posted Write Set-Aside Buffer" to offload the NP writes from the AXI Write Channel when NP reads are stalled. Offloading NP writes thereby avoids blocking of P on the AXI Write Channel. You configure the maximum number of offloaded NP writes using this parameter. Values: 2, 4, 8, 16, 32, 64 Default Value: 32 Enabled: ((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4)) Parameter Type: Feature Setting Parameter Name: CC_SLV_NUM_OUTSTND_CPU_WR_REQ
Master Address Width	Specifies the width of the AXI master address bus (excluding parity bits [AXI only] if any) Values: ■ 32 (32) ■ 64 (64) Default Value: CC_DTIM_ENABLE ? 64 : (AHB_POPULATED ? SLAVE_BUS_ADDR_WIDTH : 32) Enabled: (MASTER_POPULATED && (AXI_POPULATED AHB_POPULATED)) Parameter Type: Feature Setting Parameter Name: MASTER_BUS_ADDR_WIDTH
Master Data Width	Specifies the width of the AXI master databus (excluding parity bits [AXI only] if any) Values: 32, 64, 128, 256, 512 Default Value: (CX_AXI_RCHUNK_ENABLE && (PCIE_CORE_DATA_BUS_WD % 128 != 0)) ? 128 : PCIE_CORE_DATA_BUS_WD Enabled: MASTER_POPULATED Parameter Type: Feature Setting Parameter Name: MASTER_BUS_DATA_WIDTH

Label	Description
Slave Address Width	Specifies the width of the AXI slave address bus (excluding parity bits [AXI only] if any). Values: ■ 32 (32) ■ 64 (64) Default Value: CC_DTIM_ENABLE ? 64 : 32 Enabled: (SLAVE_POPULATED && (AXI_POPULATED AHB_POPULATED)) Parameter Type: Feature Setting Parameter Name: SLAVE_BUS_ADDR_WIDTH
Slave Data Width	Specifies the width of the AXI slave data bus (excluding parity bits [AXI only] if any). Values: 32, 64, 128, 256, 512 Default Value: CX_NW * 32 Enabled: SLAVE_POPULATED Parameter Type: Feature Setting Parameter Name: SLAVE_BUS_DATA_WIDTH
Integrated MSI Reception Module Enable	Instantiate the iMSI-RX: Integrated MSI Receiver in the AXI bridge to detect and terminate inbound MSI TLPs. For more information, see the Interrupt section in the "Controller Operations" chapter of the <i>Databook</i>. Values: 0, 1 Default Value: 0 Enabled: ((CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_DM) && AMBA_INTERFACE!=0) Parameter Type: Feature Setting Parameter Name: CX_MSI_CTRL_ENABLE
AXI MSI Controller FIFO Depth	AXI MSI interrupt controller FIFO depth. Number of maximum interrupt vectors AXI MSI controller can hold before halting the target interface. Values: 2, 3, 4, 5 Default Value: 2 Enabled: ((CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_DM) && CX_MSI_CTRL_EN) Parameter Type: Feature Setting Parameter Name: CX_MSI_CTRL_FIFO_DEPTH

Label	Description
AXI Clocking	
Asynchronous AXI Slave Clock	Include slave asynchronous clock crossing circuitry in AXI bridge. The AXI bridge assumes that <code>slv_aclk</code> and <code>dbislv_aclk</code> are asynchronous to <code>core_clk</code> and it implements the appropriate clock crossings using pre-existing FIFOs and inserting extra FIFOs when necessary. - When your AXI clocks are synchronous to <code>core_clk</code> then you can eliminate all unnecessary clock crossing circuitry by unsetting the <code>SLV_CLK_DIFF_ENABLE</code> configuration parameter. - When you are using a dynamic frequency Gen/Gear 2/3 core, the frequency of <code>core_clk</code> changes across speed modes. In this case, your AXI clock cannot be synchronous to <code>core_clk</code> and you cannot set <code>SLV_CLK_DIFF_ENABLE</code> to "0". Values: 0, 1 Default Value: (AXI_POPULATED && SLAVE_POPULATED) AHB_CLK_DIFF_ENABLE Enabled: AXI_POPULATED && SLAVE_POPULATED Parameter Type: Feature Setting Parameter Name: SLV_CLK_DIFF_ENABLE
Asynchronous AXI Master Clock	Include master asynchronous clock crossing circuitry in AXI bridge. The AXI bridge assumes that <code>mstr_aclk</code> is asynchronous to <code>core_clk</code> and it implements the appropriate clock crossings using pre-existing FIFOs and inserting extra FIFOs when necessary. - When your AXI clock is synchronous to <code>core_clk</code> then you can eliminate all unnecessary clock crossing circuitry by unsetting the <code>MSTR_CLK_DIFF_ENABLE</code> configuration parameter. - When you are using a dynamic frequency Gen/Gear 2/3 core, the frequency of <code>core_clk</code> changes across speed modes. In this case, your AXI clock cannot be synchronous to <code>core_clk</code> and you cannot set <code>MSTR_CLK_DIFF_ENABLE</code> to "0". Values: 0, 1 Default Value: (AXI_POPULATED && MASTER_POPULATED) AHB_CLK_DIFF_ENABLE Enabled: AXI_POPULATED && MASTER_POPULATED Parameter Type: Feature Setting Parameter Name: MSTR_CLK_DIFF_ENABLE
AXI Slave Max Payload	
Slave Interface Burst Length	Specify the maximum burst length (in beats) of an AXI transfer at the slave interface. An application master might execute any burst up to a maximum byte size given by <code>CC_SLV_BURST_LEN * (CC_SLV_BUS_DATA_WIDTH / 8)</code> bytes. This parameter controls the default size of the slave request decomposer data RAM inside the AXI bridge based on the maximum MemWr transfer size. This parameter uses the unit of burst beat. Values: 4, 8, 16, 32, 64, 128, 256 Default Value: (AHB_POPULATED && (SLAVE_BUS_DATA_WIDTH==32))? 32 : 16 Enabled: SLAVE_POPULATED Parameter Type: Performance Setting Parameter Name: CC_SLV_BURST_LEN

Label	Description
Slave Interface Burst Length (5-Bit Mode)	Maximum burst length (in beats) of an AXI transfer at the slave interface, when Extended Tag Support is disabled. An application master may execute any burst up to a maximum byte size given by $CC_SLV_BURST_LEN_5BIT * (CC_SLV_BUS_DATA_WIDTH / 8)$ bytes. If your system actively disables Extended Tag Support and $CC_SLV_BURST_LEN_5BIT$ is lower than $CC_SLV_BURST_LEN$, your application master must ensure this limit is followed. Note: "Extended Tag Support" refers to the register: $PfX_PCIE_CAP/DEVICE_CONTROL_DEVICE_STATUS/PCIE_CAP_EXT_TAG_EN$. Values: 1, ..., 256 Default Value: $(CC_SLV_BURST_LEN_5BIT_INT > CC_SLV_BURST_LEN) ? CC_SLV_BURST_LEN : CC_SLV_BURST_LEN_5BIT_INT$ Enabled: SLAVE_POPULATED Parameter Name: CC_SLV_BURST_LEN_5BIT
AXI Slave MTU	Slave Maximum Transfer Unit. The application master (at the AXI Slave Interface) must not request or write more than CC_SLV_MTU bytes in a single AXI burst. This value is used to set memory sizes in the bridge (slave request decomposer data FIFO and slave response composer). This parameter is derived automatically as $CC_SLV_BURST_LEN * (SLAVE_BUS_DATA_WIDTH / 8)$ and is read-only. Values: -2147483648, ..., 2147483647 Default Value: $((CC_SLV_BURST_LEN * SLAVE_BUS_DATA_WIDTH / 8) > 4096) ? 4096 : (CC_SLV_BURST_LEN * SLAVE_BUS_DATA_WIDTH / 8)$ Enabled: 0 Parameter Type: Feature Setting Parameter Name: CC_SLV_MTU
AXI Master Max Payload	
Master Interface Burst Length	Specifies the maximum burst length (in beats) of an AXI transfer at the master interface. ■ A value $< MASTER_BUS_DATA_WIDTH/8$ forces $MSTR_FIXED_SIZE_ENABLE=1$ because the master cannot produce a sufficiently long burst of bytes. Therefore the master uses dword access and not byte access for narrow or NCBE reads; which means that such accesses to non-prefetchable memory might result in corrupting not-to-be-read data. ■ A value of 8 also forces $MSTR_FIXED_SIZE_ENABLE=1$. AXI Note: The value of 8 is only possible if $MASTER_BUS_DATA_WIDTH > 32$ because the minimum allowable master burst transfer size (CC_MSTR_MTU) is 64 bytes. Values: 4, 8, 16, 32, 64, 128, 256 Default Value: 16 Enabled: $(MASTER_POPULATED \&\& !(AHB_POPULATED \&\& (CC_MSTR_BURST_TYPE == 0)))$ Parameter Type: Performance Setting Parameter Name: CC_MSTR_BURST_LEN

Label	Description
AXI Master MTU	Master Maximum Transfer Unit. The bridge master do not request or write more than CC_MSTR_MTU bytes in a single AXI burst. This parameter is derived automatically as $\max\{64, \text{CC_MSTR_BURST_LEN} * (\text{MASTER_BUS_DATA_WIDTH} / 8)\}$ and is read-only. Values: 64, ..., 4096 Default Value: $((\text{CC_MSTR_BURST_LEN} * \text{MASTER_BUS_DATA_WIDTH} / 8) \geq 64) ? (\text{CC_MSTR_BURST_LEN} * \text{MASTER_BUS_DATA_WIDTH} / 8) : 64$ Enabled: 0 Parameter Type: Feature Setting Parameter Name: CC_MSTR_MTU
Master Page Boundary Bit	Specifies an address page boundary of 128, 256, 512, 1K, 2K or 4K (bytes). ■ The AXI bridge ensures that the inbound request do not cross the selected page boundary when driven onto the AXI master interface. ■ The Master Page Boundary (CC_MSTR_PAGE_BOUNDARY_BYTES) is calculated as $2^{(\text{CC_MSTR_PAGE_BOUNDARY_PW}-1)}$. ■ For more information, see "Decomposition" in the <i>Databook</i>. <i>Note:</i> The Master Page Boundary $\{2^{(\text{CC_MSTR_PAGE_BOUNDARY_PW}-1)}\}$ must be $\geq \max\{\text{CC_MSTR_MTU}, \text{CC_MSTR_RD_REQ_SIZE}\}$ Values: 8, ..., 13 Default Value: $((\text{AMBA_INTERFACE}==2) \parallel (\text{AMBA_INTERFACE}==3) \parallel (\text{AMBA_INTERFACE}==4)) ? 13 : 11$ Enabled: MASTER_POPULATED Parameter Type: Performance Setting Parameter Name: CC_MSTR_PAGE_BOUNDARY_PW
Master Page Boundary	For more information, see description of Master Page Boundary Bit (CC_MSTR_PAGE_BOUNDARY_PW). <i>Note:</i> This parameter is derived automatically as $2^{(\text{CC_MSTR_PAGE_BOUNDARY_PW}-1)}$ and is read-only. Values: -2147483648, ..., 2147483647 Default Value: $\text{CC_MSTR_PAGE_BOUNDARY_PW} - 1$ Enabled: Always Parameter Type: Performance Setting Parameter Name: CC_MSTR_PAGE_BOUNDARY_BYTES
PCIe Max Payload	
PCIe Max Payload Supported	This parameter is a read-only reflection of the CX_MAX_MTU parameter. Values: 0, ..., 65535 Default Value: CX_MAX_MTU Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_MAX_MTU_

Label	Description
AXI DBI	
Dedicated DBI Slave Enable	Enables the AXI Slave DBI interface. For more information, see the "Local Bus Controller (LBC)" section of the <i>Databook</i>. Values: 0, 1 Default Value: AMBA_POPULATED && !SHARED_DBI_ENABLED Enabled: AMBA_POPULATED && !SHARED_DBI_ENABLED Parameter Type: Feature Setting Parameter Name: DBI_4SLAVE_POPULATED
Shared DBI Slave Enable	Indicates that the slave interface for PCIe outbound traffic is also used for DBI access. For more information, see the "Local Bus Controller (LBC)" section in the "Controller Operations" chapter of the <i>Databook</i>. >Note: Valid for AXI only but cannot be enabled in EP/DM when (CX_ARI_ENABLE=1 or CX_SRIOV_ENABLE=1) Values: 0, 1 Default Value: 0 Enabled: (AXI_POPULATED && (!CX_AMBA_ARI_OR_IOV SLAVE_BUS_ADDR_WIDTH_IS_64)) Parameter Type: Feature Setting Parameter Name: SHARED_DBI_ENABLED
DBI Slave Address Width	Specifies the width of the dedicated DBI slave address bus. This is a read-only parameter. Values: ■ 32 (32) ■ 33 (33) Default Value: CX_ARI_ENABLE ? 33 : 32 Enabled: Always Parameter Type: Feature Setting Parameter Name: CC_DBI_SLV_BUS_ADDR_WIDTH
DBI Slave Data Width	Specifies the width of the dedicated DBI slave databus. Values: 32, 64, 128 Default Value: 32 Enabled: ((AMBA_INTERFACE==1)) Parameter Type: Feature Setting Parameter Name: CC_DBI_SLV_BUS_DATA_WIDTH

1.6 Basic AXI Config / PCIe TAGs and AXI IDs Parameters

Table 1-6 Basic AXI Config / PCIe TAGs and AXI IDs Parameters

Label	Description
Slave	
Max Outbound NP Requests	This parameter is a read-only reflection of the CX_MAX_TAG parameter. Values: ■ 2 (1) ■ 4 (3) ■ 8 (7) ■ 16 (15) ■ 32 (31) ■ 64 (63) ■ 128 (127) ■ 256 (255) ■ 512 (511) ■ 768 (767) Default Value: CX_MAX_TAG Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_MAX_TAG_
AXI Slave Tags	The maximum number of non-posted AXI requests issued to the bridge slave that are outstanding at one time over the PCIe link. For more information, see "Outbound Bridge Tag Management" in the "AXI Bridge Module" chapter of the <i>Databook</i>. ■ For AXI, the default value is CX_MAX_TAG+1. Note that CX_MAX_TAG represents the number of PCIe TAGs minus 1. ■ When DMA is enabled, the default value for AXI is (CX_MAX_TAG + 1)/2. Values: 2, 4, 8, 16, 32, 64, 128, 256, 512, 768 Default Value: ((AHB_ENABLED==1) && (CC_AHB_SNUM_MASTERS<=2)) ? 2 : ((AHB_ENABLED==1) && (CC_AHB_SNUM_MASTERS<=4)) ? 4 : ((AHB_ENABLED==1) && (CC_AHB_SNUM_MASTERS<=8)) ? 8 : ((AHB_ENABLED==1) && (CC_AHB_SNUM_MASTERS<=16)) ? 16 : (CC_DMA_ENABLE_VALUE==1) ? ((CX_MAX_TAG==3) ? 2 : (CX_MAX_TAG==7) ? 4 : (CX_MAX_TAG==15) ? 8 : (CX_MAX_TAG==31) ? 16 : (CX_MAX_TAG==63) ? 32 : (CX_MAX_TAG==127) ? 64 : (CX_MAX_TAG==255) ? 128 : (CX_MAX_TAG==511) ? 256 : (CX_MAX_TAG>=767) ? 512 : 2) : ((CX_MAX_TAG<=767) ? CX_MAX_TAG + 1 : 768) Enabled: SLAVE_POPULATED && (AHB_ENABLED==0) Parameter Type: Performance Setting Parameter Name: CC_MAX_SLV_TAG

Label	Description
High Priority Master	
AXI High Priority Master IDs	Represents the maximum number of AXI master IDs and the maximum number of outstanding NP transactions the AXI High Priority master can handle. This value must be smaller than CC_MAX_MSTR_TAGS_AXI. The Low Priority master handles (CC_MAX_MSTR_TAGS_AXI-CC_MAX_MSTRH_TAGS_AXI). <i>Note:</i> This parameter is only visible when High Priority Master is enabled. Values: 1, ..., (CC_MAX_MSTR_TAGS_AXI-1) Default Value: CC_MAX_MSTR_TAGS_AXI/2 Enabled: CC_HP_MASTER Parameter Type: Performance Setting Parameter Name: CC_MAX_MSTRH_TAGS_AXI
Master	
AXI Master IDs	Represents the maximum number of AXI master IDs and the maximum number of outstanding NP transactions the AXI bridge can handle. It is used to size the master completion RAM which can be large if CC_MAX_MSTR_TAGS_AXI is large. You should set CC_MAX_MSTR_TAGS_AXI to the minimum number of outstanding AXI requests that are required to saturate the AXI bus and/or supply the throughput required by the controller. If decomposition occurs, the number of simultaneously outstanding AXI requests is larger than this value. For more information, see the 'Inbound Bridge Tag Management' section in the AXI Bridge chapter of the <i>Databook</i>. This parameter is used to size the master ID busses with the value $\log_2\{\text{CC_MAX_MSTR_TAGS_AXI}\}$. Values: 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024 Default Value: 32 Enabled: ((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4)) Parameter Type: Performance Setting Parameter Name: CC_MAX_MSTR_TAGS_AXI

1.7 Device-Wide Optional Non-PCIe Config Parameters

Table 1-7 Device-Wide Optional Non-PCIe Config Parameters

Label	Description
Application Interface Options	
Target Completion LUT Enable	Enable the Target Completion Lookup Table to be connected internally between the RTRGT1 and XALI0/1/2 interfaces. This feature provides two functions: ■ The application does not need to provide certain TLP header information when transmitting completions on XALI0/1/2 because the controller stores and tracks the information in the internal Target Completion Lookup Table. ■ When the application has not generated a completion for an incoming request within the required time interval, then the Target Completion Timeout indicates that a Timeout has occurred. The Target Completion Lookup Table (and Target Completion Timeout event) should not be confused with the Completion Lookup table (and Completion Timeout event) described in 'Received Completion TLP Processing' in the <i>Databook</i>. For more information, see the 'Target Completion Lookup Table' section of the "Signal Interfaces" chapter of the <i>Databook</i>. The size of this LUT is controlled using the 'Max Forwarded NP Requests' (CX_REMOTE_MAX_TAG) parameter. <i>Note:</i> This feature is not available if Peer-to-Peer support is enabled. <i>Note:</i> When the AXI bridge is used (AMBA_INTERFACE=1, 2 or 3) or when DMA is enabled, the Target Completion Lookup Table is automatically activated. Values: ■ false (0) ■ true (1) Default Value: (CC_DMA_ENABLE) ? 1 : (AMBA_POPULATED) ? 1 : 0 Enabled: ((TRGT1_POPULATE==1 && !(CC_DMA_ENABLE) && CC_DEVICE_TYPE!=CC_SW && CX_P2P_ENABLE==0 && AMBA_POPULATED==0)) Parameter Type: Feature Setting Parameter Name: TRGT_CPL_LUT_EN

Label	Description
Max Forwarded NP Requests	The function depends on the nature of the application interface. ■ AXI: You should leave this parameter at its default value. ■ Native controller: Specifies the maximum number of received non-posted requests that the controller forwards to your application at one time, when the Target Completion LUT (TRGT_CPL_LUT_EN) is enabled. The default value is sufficient for typical applications. It does not control the number of non-posted requests that the controller can receive and buffer from the PCIe wire, which is controlled by flow control credits. Values: ■ 2_Tags (1) ■ 4_Tags (3) ■ 8_Tags (7) ■ 16_Tags (15) ■ 32_Tags (31) ■ 64_Tags (63) ■ 128_Tags (127) ■ 256_Tags (255) ■ 512_Tags (511) ■ 1024_Tags (1023) Default Value: CC_HP_MASTER ? CC_MAX_MSTR_TAGS_AXI-1 : AHB_POPULATED ? 1 : (AXI_POPULATED ? CC_MAX_MSTR_TAGS - 1 : ((CX_MAX_TAG == 767) ? 1023 : CX_MAX_TAG)) Enabled: ((AMBA_POPULATED) ? 0 : (TRGT_CPL_LUT_EN)) Parameter Type: Performance Setting Parameter Name: CX_REMOTE_MAX_TAG
Enable Address Alignment	When enabled (and input client*_addr_align_en input is asserted), the controller supports address alignment and generates the first and last byte enables (FBE, LBE) based on the address and number of bytes of the TLP requested from the client interface. When CX_ECRC_EN is not set, GLOB_ADDR_ALIGN_EN should normally be disabled. However, if your application requires this to be enabled, then the address alignment pin (client0/1/2_addr_align_en) should only be high for those TLPs without ECRC. For TLPs with ECRC that are being transmitted by the Application, the address alignment pin (client0/1/2_addr_align_en) should be de-asserted for that TLP. <i>Note:</i> This parameter cannot be changed to True if the AXI Bridge is present or if Peer2Peer (P2P) support is enabled (CX_P2P_ENABLE=1). For more information, see 'Transmit Address Alignment' section in the "Signal Descriptions" chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: false Enabled: !AMBA_POPULATED && CX_P2P_ENABLE==0 Parameter Type: Feature Setting Parameter Name: GLOB_ADDR_ALIGN_EN

Label	Description
Byte Count Storage	Determines whether or not to store the byte count in the receive completion LUT. If the controller does not receive a completion for each request, then the completion timeout mechanism terminates the request. It also indicates on the <code>radm_timeout_cpl_len[11:0]</code> output port, the number of bytes remaining to be delivered when the completion timed out. When the byte count is not stored in the LUT, the controller is unable to detect when it is receiving a completion with an incorrect byte count field. Therefore, completions with incorrect byte count is treated as valid completions instead of being treated as unexpected completions. The controller does not detect missing completions and the receive completion LUT does not timeout. Values: ■ false (0) ■ true (1) Default Value: true Enabled: Always Parameter Type: Feature Setting Parameter Name: RADM_CPL_LUT_STORE_BYTE_CNT
Application Return CRD Enable	Include optional top-level ports for the application to update the Flow Control credits counters. The optional ports are the <code>app_*_ca</code> ports on the SII (For more information, see 'System Information Interface (SII)' section in the "Signal Descriptions" chapter of the <i>Databook</i>). <i>Note:</i> The application is not expected to return completion credits since completion credits are infinite according to the <i>PCI Express 3.1 Specification</i>. <i>Note:</i> Not applicable when AXI Bridge is enabled. Values: ■ false (0) ■ true (1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: APP_RETURN_CRD_EN
Lane/Link Options	
Enable Manual Lane Flip (Conventional PCIe)	Include support for manual lane flip. For more information, see the 'Lane Reversal' chapter in the <i>Databook</i>. Include the <code>tx_lane_flip_en</code> and <code>rx_lane_flip_en</code> inputs to manually control TX/RX Lane Reversal. Values: ■ false (0) ■ true (1) Default Value: $((CX_LANE_REVERSE \parallel CX_GEN5_SPEED) \&\& CX_NL > 1) ? 1 : 0$ Enabled: $(CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : 1$ Parameter Type: Feature Setting Parameter Name: CX_LANE_FLIP_CTRL_EN

Label	Description
Enable Auto Lane Flip and Reversal	Include support for auto lane flip and reversal. When you set this parameter, the controller implements logic to: ■ flip the lanes autonomously in Detect LTSSM state when lane0 is not detected, and ■ reverse the lanes autonomously in Configuration LTSSM state when logical lane0 receives TS ordered sets with lane number different from '0'. For more information, see the description of the AUTO_LANE_FLIP_CTRL_EN register in the <i>Register Descriptions</i> document and 'Lane Reversal' chapter in the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: (CX_LANE_FLIP_CTRL_EN) ? 1 : 0 Enabled: CX_LANE_FLIP_CTRL_EN Parameter Type: Feature Setting Parameter Name: CX_AUTO_LANE_FLIP_CTRL_EN
Enable Partial Auto Lane Flip and Reversal	Enables support for partial auto flipping or reversal of the lanes. ■ Full flip or reversal: Lane0 is connected to Lane NL-1. ■ Partial flip or reversal: Lane0 is connected to either Lane NL/2-1 or Lane NL/4-1 or Lane NL/8-1 and so on. For more information, see the 'Lane Reversal' chapter in the <i>Databook</i>. Values: ■ Full (0) ■ Full or Partial (1) ■ Full and Partial (2) Default Value: (CX_LANE_REVERSE) ? 2 : 1 Enabled: CX_AUTO_LANE_FLIP_CTRL_EN Parameter Type: Feature Setting Parameter Name: CX_AUTO_LANE_FLIP_MUX_ARCH
Enable Forced Lane Flip	When this parameter is set to true, and FORCE_LANE_FLIP field of GEN2_CTRL_OFF register is set to 1, you can forcefully connect the physical lane specified in LANE_UNDER_TEST field of GEN2_CTRL_OFF register to logical Lane0 of the controller after reset. Only the physical lane specified in LANE_UNDER_TEST can be flipped to logical Lane0, starting from Detect LTSSM state. That is, only x1 can be formed, all the other physical lanes are turned off. Values: ■ false (0) ■ true (1) Default Value: CX_LANE_FLIP_CTRL_EN Enabled: CX_LANE_FLIP_CTRL_EN Parameter Type: Feature Setting Parameter Name: CX_FORCE_LANE_FLIP_EN

Label	Description
Default Link Number	Default Link Number value that the Upstream Device advertises to the Downstream Link partner over the TS sequence. It is also the default value for the Link Number field in the 'Port Force Link Register'. This register is used to force the link to the state specified by the Link State field through a link re-negotiation. For more information, see the Registers chapter of the <i>Databook</i> . Values: 0x0, ..., 0xff Default Value: 0x4 Enabled: CX_CPCIE_ENABLE && CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: DEFAULT_LINK_NUM
Upconfigure	Default value for the UPCONFIGURE_SUPPORT field in the MULTI_LANE_CONTROL_OFF register. You must set the UPCONFIGURE_SUPPORT field in MULTI_LANE_CONTROL_OFF so that the controller advertises '1' in the Upconfigure Capability bit of the TS2 OS. This allows you to upsize or downsize the link width through Configuration state without bringing the link down. For more information, see the "Link Establishment" section in the "Controller Operations" chapter of the Databook. Values: 0x0, 0x1 Default Value: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : (CX_CPCIE_ENABLE && (CX_NL > 1)) ? 1 : 0 Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : (CX_NL > 1) Parameter Type: Feature Setting Parameter Name: DEFAULT_UPCONFIGURE_SUPPORT
Lane Deskew Options	
Disable Lane Deskew	Disable the deskewing mechanism. Disable the deskewing mechanism only if connecting to a multi-Lane PHY that implements Lane-to-Lane deskewing. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: ■ false (0x0) ■ true (0x1) Default Value: CX_PCIE_MODE == PCIE_OVER_CIO) ? 1 : (CX_NL == 1 Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : (CX_NL > 1) Parameter Type: Feature Setting Parameter Name: CX_DESKEW_DISABLE

Label	Description
Deskew Fifo Depth for C-PCIe	Specifies the depth of fifos used in the deskew logic for C-PCIe. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: 2, ..., 25 Default Value: $(CX_CCIX_ESM_SUPPORT == 1) \&\& (CX_MAX_PCIE_SPEED == 5) \&\& (CX_NB == 4) ? 14 : (CX_CCIX_ESM_SUPPORT == 1) \&\& (CX_NB == 2) ? 22 : (CX_CCIX_ESM_SUPPORT == 1) \&\& (CX_NB == 4) ? 13 :$ $(CX_CCIX_ESM_SUPPORT == 1) \&\& (CX_NL == 1) \&\& (CX_NB == 16) ? 4 :$ $(CX_CCIX_ESM_SUPPORT == 1) \&\& (CX_NL != 1) \&\& (CX_NB == 16) ? 12 :$ $(CX_MAX_PCIE_SPEED == 5) \&\& (CX_NB == 4) ? 14 : (CX_MAX_PCIE_SPEED == 5) \&\& (CX_NB == 8) ? 12 : (CX_MAX_PCIE_SPEED == 5) \&\& (CX_NL == 1) \&\&$ $(CX_NB == 16) ? 3 : (CX_MAX_PCIE_SPEED == 4) \&\&$ $(CX_GEN4_DYNAMIC_WIDTH == 1) \&\& (CX_NB == 16) ? 12 :$ $(CX_MAX_PCIE_SPEED == 4) \&\& (CX_GEN4_DYNAMIC_WIDTH == 1) \&\& (CX_NB == 8) ? 14 : (CX_MAX_PCIE_SPEED == 4) \&\& (CX_GEN4_DYNAMIC_FREQ == 1)$ $\&\& (CX_NB == 4) ? 12 : (CX_MAX_PCIE_SPEED == 4) \&\&$ $(CX_GEN4_DYNAMIC_FREQ == 1) \&\& (CX_NB == 2) ? 20 :$ $(CX_MAX_PCIE_SPEED == 3) \&\& (CX_GEN3_DYNAMIC_WIDTH == 1) \&\& (CX_NB == 8) ? 12 : (CX_MAX_PCIE_SPEED == 3) \&\& (CX_GEN3_DYNAMIC_WIDTH == 1)$ $\&\& (CX_NB == 4) ? 14 : (CX_MAX_PCIE_SPEED == 3) \&\&$ $(CX_GEN3_DYNAMIC_FREQ == 1) \&\& (CX_NB == 4) ? 11 :$ $(CX_MAX_PCIE_SPEED == 3) \&\& (CX_GEN3_DYNAMIC_FREQ == 1) \&\& (CX_NB == 2) ? 12 : (CX_MAX_PCIE_SPEED == 3) \&\& (CX_GEN3_DYNAMIC_FREQ == 1)$ $\&\& (CX_NB == 1) ? 19 : (CX_MAX_PCIE_SPEED == 2) \&\&$ $(CX_GEN2_DYNAMIC_WIDTH == 1) \&\& (CX_NB == 4) ? 12 :$ $(CX_MAX_PCIE_SPEED == 2) \&\& (CX_GEN2_DYNAMIC_WIDTH == 1) \&\& (CX_NB == 2) ? 14 : (CX_MAX_PCIE_SPEED == 2) \&\& (CX_GEN2_DYNAMIC_FREQ == 1)$ $\&\& (CX_NB == 4) ? 11 : (CX_MAX_PCIE_SPEED == 2) \&\&$ $(CX_GEN2_DYNAMIC_FREQ == 1) \&\& (CX_NB == 2) ? 12 :$ $(CX_MAX_PCIE_SPEED == 2) \&\& (CX_GEN2_DYNAMIC_FREQ == 1) \&\& (CX_NB == 1) ? 14 : (CX_MAX_PCIE_SPEED == 1) \&\& (CX_GEN2_MODE ==$ $GEN2_DISABLED) \&\& (CX_NB == 4) ? 11 : (CX_MAX_PCIE_SPEED == 1) \&\&$ $(CX_GEN2_MODE == GEN2_DISABLED) \&\& (CX_NB == 2) ? 12 :$ $(CX_MAX_PCIE_SPEED == 1) \&\& (CX_GEN2_MODE == GEN2_DISABLED) \&\&$ $(CX_NB == 1) ? 14 : 14$ Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_DESKEW_DEPTH_CPCIE
+ Gen1 Wire Skew (Lrx-skew ns)	This is read-only parameter to indicate how much wire skew for Gen1 speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: 0, ..., 60 Default Value: 20 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_WIRE_SKEW_GEN1_NS

Label	Description
+ Gen2 Wire Skew (Lrx-skew ns)	This is read-only parameter to indicate how much wire skew for Gen2 speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: 0, ..., 60 Default Value: ((CX_GEN2_MODE != GEN2_DISABLED)==1) ? 8 : 0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_WIRE_SKEW_GEN2_NS
+ Gen3 Wire Skew (Lrx-skew ns)	This is read-only parameter to indicate how much wire skew for Gen3 speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: 0, ..., 60 Default Value: ((CX_GEN3_MODE != GEN3_DISABLED)==1) ? 6 : 0 Enabled: CX_GEN3_SPEED==1 && SNPS_RSVDPARAM_65==1 Parameter Type: Feature Setting Parameter Name: CX_WIRE_SKEW_GEN3_NS
+ Gen4 Wire Skew (Lrx-skew ns)	This is read-only parameter to indicate how much wire skew for Gen4 speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: 0, ..., 60 Default Value: ((CX_GEN4_MODE != GEN4_DISABLED)==1) ? 5 : 0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_WIRE_SKEW_GEN4_NS
+ Gen5 Wire Skew (Lrx-skew ns)	This is read-only parameter to indicate how much wire skew for Gen5 speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: 0, ..., 60 Default Value: ((CX_MAX_PCIE_SPEED >= 5)==1) ? 5 : 0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_WIRE_SKEW_GEN5_NS
+ CCIX ESM 20G Wire Skew (Lrx-skew ns)	This is read-only parameter to indicate how much wire skew for CCIX ESM 20G speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: 0, ..., 60 Default Value: (CX_CCIX_ESM_SUPPORT==1) ? 4 : 0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_WIRE_SKEW_CCIX20G_NS

Label	Description
+ CCIX ESM 25G Wire Skew (Lrx-skew ns)	This is read-only parameter to indicate how much wire skew for CCIX ESM 25G speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: 0, ..., 60 Default Value: (CX_CCIX_ESM_SUPPORT==1) ? 4 : 0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_WIRE_SKEW_CCIX25G_NS
+ Gen1 PHY Skew (pipe_clk cycles)	This is read-only parameter to indicate how much PHY skew for Gen1 speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: -2147483648, ..., 2147483647 Default Value: CX_DESKEW_DISABLE==1 CX_NL==1 ? 0 : (CX_SKEW_PIPEIF_GEN1_PCLK - CX_WIRE_SKEW_GEN1_PCLK) Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_PHY_SKEW_GEN1_PCLK
+ Gen2 PHY Skew (pipe_clk cycles)	This is read-only parameter to indicate how much PHY skew for Gen2 speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: -2147483648, ..., 2147483647 Default Value: CX_DESKEW_DISABLE==1 CX_NL==1 CX_GEN2_SPEED==0 ? 0 : (CX_SKEW_PIPEIF_GEN2_PCLK - CX_WIRE_SKEW_GEN2_PCLK) Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_PHY_SKEW_GEN2_PCLK
+ Gen3 PHY Skew (pipe_clk cycles)	This is read-only parameter to indicate how much PHY skew for Gen3 speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: -2147483648, ..., 2147483647 Default Value: CX_DESKEW_DISABLE==1 CX_NL==1 CX_GEN3_SPEED==0 ? 0 : (CX_SKEW_PIPEIF_GEN3_PCLK - CX_WIRE_SKEW_GEN3_PCLK) Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_PHY_SKEW_GEN3_PCLK

Label	Description
+ Gen4 PHY Skew (pipe_clk cycles)	This is read-only parameter to indicate how much PHY skew for Gen4 speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: -2147483648, ..., 2147483647 Default Value: CX_DESKEW_DISABLE==1 CX_NL==1 CX_GEN4_SPEED==0) ? 0 : (CX_SKEW_PIPEIF_GEN4_PCLK - CX_WIRE_SKEW_GEN4_PCLK Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_PHY_SKEW_GEN4_PCLK
+ Gen5 PHY Skew (pipe_clk cycles)	This is read-only parameter to indicate how much PHY skew for Gen5 speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: -2147483648, ..., 2147483647 Default Value: CX_DESKEW_DISABLE==1 CX_NL==1 CX_GEN5_SPEED==0) ? 0 : (CX_SKEW_PIPEIF_GEN5_PCLK - CX_WIRE_SKEW_GEN5_PCLK Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_PHY_SKEW_GEN5_PCLK
+ CCIX ESM 20G PHY Skew (pipe_clk cycles)	This is read-only parameter to indicate how much PHY skew for CCIX ESM 20G speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: -2147483648, ..., 2147483647 Default Value: CX_DESKEW_DISABLE==1 CX_NL==1 CX_CCIX_ESM_SUPPORT==0) ? 0 : (CX_SKEW_PIPEIF_CCIX20G_PCLK - CX_WIRE_SKEW_CCIX20G_PCLK Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_PHY_SKEW_CCIX20G_PCLK
+ CCIX ESM 25G PHY Skew (pipe_clk cycles)	This is read-only parameter to indicate how much PHY skew for CCIX ESM 25G speed your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) ensures. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: -2147483648, ..., 2147483647 Default Value: CX_DESKEW_DISABLE==1 CX_NL==1 CX_CCIX_ESM_SUPPORT==0) ? 0 : (CX_SKEW_PIPEIF_CCIX25G_PCLK - CX_WIRE_SKEW_CCIX25G_PCLK Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_PHY_SKEW_CCIX25G_PCLK

Label	Description
+ Margin for Deskew Buffer Setting	This is read-only parameter to indicate how much margin your deskew buffer setting (CX_DESKEW_DEPTH_CPCIE) has. For more information, see the "Lane Deskew" chapter of the <i>Databook</i>. Values: 0, ..., 10 Default Value: 1 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_SKEW_MAC_MARGIN_CORECLK
Layer 1/2 Options	
Default Gen1/2 Replay Timer Adjustment	Default value for the 'Timer Modifier for Replay Timer' field in the 'Timer Control and Max Function Number' register. Each increment in this value increases the replay timer by 64. For more information, see the 'Transmit Replay' section of the Controller Operations chapter in the <i>Databook</i>. This parameter is for conventional PCIe mode. Values: 0x0, ..., 0x1f Default Value: $((CX_MAX_NFTS*4)/CX_NB)+(CX_CPCIE_INTERNAL_DELAY*3))/64 + (((((CX_MAX_NFTS*4)/CX_NB)+(CX_CPCIE_INTERNAL_DELAY*3))\%64) ? 1 : 0$ Enabled: ((CX_S_CPCIE_MODE)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_REPLAY_ADJ
Default Gen3/4/5 Replay Timer Adjustment	Default value (in Gen3/4/5 mode) for the 'Timer Modifier for Replay Timer' field in the 'Timer Control and Max Function Number' register at Gen3/4/5 speed. Must consider an EIEOS is transmitted after every 32 FTS. Each increment in this value increases the replay timer by 256. For more information, see the 'Transmit Replay' section of the Controller Operations chapter in the <i>Databook</i>. Values: 0x0, ..., 0x1f Default Value: $((CX_MAX_NFTS*65)/(CX_NB*1024)) + ((CX_MAX_NFTS*65)/(CX_NB*32768)) + ((CX_CPCIE_INTERNAL_DELAY*3)/256) + 3$ Enabled: CX_GEN3_SPEED == 1 && CX_CPCIE_ENABLE Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_REPLAY_ADJ
Default FC Watch Dog Disable	Default value of the Disable FC Watchdog Timer bit of the 'Symbol Timer Register and Filter Mask 1' Register, in the 'Registers' section of the <i>Databook</i>. Values: 0, 1 Default Value: 0 Enabled: Always Parameter Type: Feature Setting Parameter Name: DEFAULT_FC_WATCH_DOG_DISABLE

Label	Description
Cfg Directed Speed Change	Enables the controller to initiate a Gen1 -> Gen2 and Gen2 -> Gen3 speed change after the link is initialized. Only applicable for Gen2 or Gen3 configured controllers. This is the default value of the "Directed Speed Change" field of the "Link width and Speed Change Control" register. Values: 0, 1 Default Value: 0 Enabled: CX_GEN2_MODE != GEN2_DISABLED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN2_SPEED_CHANGE
Power Gating	
Power Gating/UPF	Enables support for power gating using the UPF flow. Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CX_CPCIE_ENABLE && !(AMBA_INTERFACE==1)) Parameter Type: Feature Setting Parameter Name: CX_ENHANCED_PM_EN
Power Gating/PHY	Enable power gating support in the PHY. When setting this parameter the PHY UPF is read during PHY interoperability testing. If using a third party PHY, you need to setup the PHY UPF following the guidelines provided in the <i>User Guide</i>, "Integrating the controller with the PHY". Values: ■ false (0) ■ true (1) Default Value: CX_ENHANCED_PM_EN Enabled: ((CX_S_CPCIE_MODE)) Parameter Type: Feature Setting Parameter Name: CX_PHY_ENHANCED_PM_EN
VAUX Available	Indicates that VAUX power domain is available in L2 when VMAIN is removed. Values: 0, 1 Default Value: 0 Enabled: CX_ENHANCED_PM_EN CX_PHY_ENHANCED_PM_EN Parameter Type: Feature Setting Parameter Name: CX_EXISTS_VAUX
PHY L2 Power Gating	Indicates that power gating is supported in the PHY in L2 Values: 0, 1 Default Value: CX_L2_PG_EN Enabled: CX_PHY_ENHANCED_PM_EN && CX_EXISTS_VAUX Parameter Type: Feature Setting Parameter Name: CX_PHY_L2_PG_EN

Label	Description
PHY L1 Power Gating	Enable support for power-gating L1 sub-states (L1.2) in C-PCIe PHY. Values: 0, 1 Default Value: CX_L12_PG_EN Enabled: ((CX_PHY_ENHANCED_PM_EN) && (CX_L1_SUBSTATES_ENABLE)) Parameter Type: Feature Setting Parameter Name: CX_PHY_L1_PG_EN
L2 Power Gating	Indicates that power gating in L2 is supported Values: 0, 1 Default Value: (CX_EXISTS_VAUX) ? 1 : 0 Enabled: CX_EXISTS_VAUX Parameter Type: Feature Setting Parameter Name: CX_L2_PG_EN
L1 Power Gating	Enable this parameter for power-gating L1 substates (L1.2) in C-PCIe. If you want to disable L1 power gating, set app_l1_pwr_off_en =0. app_l1_pwr_off_en has to be set or clear before entering L1. It cannot be used to toggle power gating. Values: 0, 1 Default Value: 0 Enabled: ((CX_ENHANCED_PM_EN) && (CX_L1_SUBSTATES_ENABLE)) Parameter Type: Feature Setting Parameter Name: CX_L12_PG_EN
L1 Power Gating Mode	Select always-on power islands or retention registers for L1 power gating. If you want to disable L1 power gating, set app_l1_pwr_off_en =0. app_l1_pwr_off_en has to be set or clear before entering L1. It cannot be used to toggle power gating. Values: ■ Power Islands (0) ■ Retention Registers (1) Default Value: Power Islands Enabled: ((CX_L12_PG_EN) && (CX_L1_SUBSTATES_ENABLE)) Parameter Type: Feature Setting Parameter Name: CX_L1_RETENTION
L1 Retention Register Type	This parameter determines the type of retention register inferred in the UPF file for L1 power gating. Values: ■ Single Pin Retention (1) ■ Balloon Retention (2) Default Value: Balloon Retention Enabled: CX_L1_RETENTION Parameter Type: Feature Setting Parameter Name: CX_RETENTION_TYPE

Label	Description
L2 Level Shifter	Enable level shifting on the boundary between VAUX and VMAIN power domains Values: 0, 1 Default Value: 0 Enabled: CX_ENHANCED_PM_EN && CX_L2_PG_EN Parameter Type: Feature Setting Parameter Name: CX_LEVEL_SHIFT_EN
Power Switch Enable Polarity	Determine the polarity and naming of the power switch enable output. ■ 0: pm_en_vmain_n ■ 1: pm_en_vmain Values: ■ LOW (0) ■ HIGH (1) Default Value: LOW Enabled: CX_L12_PG_EN Parameter Type: Feature Setting Parameter Name: CX_PSW_EN_ACTIVE_LOW
Power Switch Acknowledge Handshake after PERST#	If this bit is set to 0 the power management logic waits for the power switch acknowledge to be deasserted after PERST# is asserted. If this bit is set to 1 the power management logic does not check the power switch acknowledge after PERST# is asserted. Values: 0, 1 Default Value: 0 Enabled: CX_L2_PG_EN Parameter Type: Feature Setting Parameter Name: CX_VMAIN_ACK_CTRL_RST_VALUE
ASPM Options	
L1 Entry Latency	Default value for the L1 Entrance Latency of the 'Ack Frequency and L0-L1 ASPM Control' register. For more information, see the Registers section of the <i>Databook</i>. The default value is acceptable unless a Custom or Generic PHY is used. Values: ■ 1 us (0x0) ■ 2 us (0x1) ■ 4 us (0x2) ■ 8 us (0x3) ■ 16 us (0x4) Default Value: 8 us Enabled: Always Parameter Type: Register Default Setting Parameter Name: DEFAULT_L1_ENTR_LATENCY

Label	Description
L0S Entry Latency	Default value for the L0s Entrance Latency field of the 'Ack Frequency and L0-L1 ASPM Control' register. For more information, see the Registers section of the <i>Databook</i>. The default value is acceptable unless a Custom or Generic PHY is used. Values: ■ 1 us (0x0) ■ 2 us (0x1) ■ 3 us (0x2) ■ 4 us (0x3) ■ 5 us (0x4) ■ 6 us (0x5) ■ 7 us (0x6) ■ also 7us (0x7) Default Value: 4 us Enabled: Always Parameter Type: Register Default Setting Parameter Name: DEFAULT_L0S_ENTR_LATENCY
Enable ASPM L1 Timeout	Enable the ASPM L1 timer so that the controller automatically goes to L1 when the timer expires and the conditions in the <i>PCI Express 3.1 Specification</i> are met. If you disable the APSM L1 timer, the application can request L1 entry by asserting <code>app_req_entr_l1</code>. The ASPM L1 timer should be larger than the L0s timer. Otherwise, ASPM L0s and L1 timeout may not function properly. Values: ■ false (0x0) ■ true (0x1) Default Value: CX_PL_MODE != 1 Enabled: CX_PL_MODE != 1 Parameter Type: Feature Setting Parameter Name: CX_ASPM_TIMEOUT_ENTR_L1_EN

Label	Description
Error Detection	
ECRC Generation and Checking	Include the ECRC generation and checking hardware. This is distinct from the software control that may be used to enable or disable ECRC. When disabled, ECRC checking and insertion logic is not included in the controller in order to reduce gates. This is especially important for large architectures, in which these blocks can consume timing margin as well as area. May be disabled for smaller gate size if the controller is placed in a system where it's guaranteed that received TLPs do not contain ECRC and your application does not transmit ECRC from the application (client) interfaces. <i>Note:</i> This parameter is always '1' when the DMA controller is selected or when the AXI Bridge is used. Values: ■ false (0) ■ true (1) Default Value: CX_ADM_ADAPTOR_ENABLE==1 ? 0 : 1 Enabled: (TRGT1_POPULATE && !(AMBA_POPULATED) && !(CC_DMA_ENABLE)) Parameter Type: Feature Setting Parameter Name: CX_ECRC_ENABLE
ECRC Stripping	Include ECRC stripping hardware. If enabled, the controller strips ECRC from all incoming packets. The controller always strips the ECRC when you have selected the AXI bridge module, or when DMA controller is selected. The controller never strips the ECRC when you enable CX_P2P_ENABLE. Values: ■ false (0) ■ true (1) Default Value: !(TRGT1_POPULATE) (CX_RAM_PROTECTION_MODE==2 && CX_RADMQ_MODE!=2) (CX_ECRC_ENABLE && CC_DEVICE_TYPE!=CC_SW && !(CX_P2P_ENABLE)) (((RADM_NP_QMODE_VC0==2) (RADM_NP_QMODE_VC1==2) (RADM_NP_QMODE_VC2==2) (RADM_NP_QMODE_VC3==2) (RADM_NP_QMODE_VC4==2) (RADM_NP_QMODE_VC5==2) (RADM_NP_QMODE_VC6==2) (RADM_NP_QMODE_VC7==2)) && (ARC_WIDTH==32)) CX_CPLQ_MANAGEMENT_ENABLE Enabled: (TRGT1_POPULATE && !(AMBA_POPULATED) && !(CC_DMA_ENABLE) && !(CC_DEVICE_TYPE==CC_SW) && !(CX_P2P_ENABLE) && !(CX_RAM_PROTECTION_MODE==2 && CX_RADMQ_MODE!=2) && !(((RADM_NP_QMODE_VC0==2) (RADM_NP_QMODE_VC1==2) (RADM_NP_QMODE_VC2==2) (RADM_NP_QMODE_VC3==2) (RADM_NP_QMODE_VC4==2) (RADM_NP_QMODE_VC5==2) (RADM_NP_QMODE_VC6==2) (RADM_NP_QMODE_VC7==2)) && (ARC_WIDTH==32)) CX_CPLQ_MANAGEMENT_ENABLE) Parameter Type: Feature Setting Parameter Name: CX_ECRC_STRIP_ENABLE

Label	Description
Extra Interface Options	
Include 3rd Client Interface	Determines whether to include top-level ports for the optional third application transmit client interface (XALI2). Enabled for AXI configurations. Values: ■ false (0) ■ true (1) Default Value: (AXI_POPULATED CX_DMA_NATIVE CX_XADM_CLI_INTF_DEDICATED)? 1 : 0 Enabled: !AMBA_POPULATED && !CX_XADM_CLI_INTF_DEDICATED Parameter Type: Feature Setting Parameter Name: CLIENT2_POPULATED
Enable Diagnostic Bus	Enables routing of diag_status_bus and diag_ctrl_bus signals to the SII interface. For more information, see 'SII: Diagnostic Control Signals' in the "Signal Descriptions" chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: CX_RAS_DES_ENABLE Enabled: !CX_RAS_DES_ENABLE Parameter Type: Feature Setting Parameter Name: DIAGNOSTIC_ENABLE
Application Error Reporting	Determines whether to include input ports for application-detected error reporting. For more information, see the "Application Error Reporting Interface" section on the "Advanced Error Handling For Received TLPs" chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: CX_DPC_ENABLE Enabled: Always Parameter Type: Feature Setting Parameter Name: APP_RETURN_ERR_EN
Poisoned CfgWr Monitoring	Enable the SII: Detect Poisoned CfgWr interface. Your application can use this interface to monitor received poisoned CfgWr. A received poisoned CfgWr is advertised on this interface irrespectively of whether the RADM filter is set to discard it or not. Values: ■ false (0) ■ true (1) Default Value: !CX_IS_RC && CX_CXL_IDE_ENABLE Enabled: !(CC_DEVICE_TYPE==CC_RC) ? 1 : 0 Parameter Type: Feature Setting Parameter Name: CX_DETECT_POISONED_CFGWR

Label	Description
LBC-ELBI-DBI Advanced Options	
DBI ReadOnly Write Enable	Default value of the DBI Read-Only Write Enable port logic register. For more information, see the "Register Configuration Space" chapter of the <i>Databook</i>. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: Always Parameter Type: Register Default Setting Parameter Name: CX_DBI_RO_WR_EN
Disable Port Logic wire side write	When set, it disables the write access to port logic space from the wire. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: Always Parameter Type: Register Default Setting Parameter Name: CX_PL_WIRE_WR_DISABLE
ELBI Address Bus Width	Defines the width of the ELBI address bus for the Local Bus Controller (LBC). Only 12-bits are needed to access the 4KB (1K DWORDS) of the PCI Express Configuration space per function. However, in order to access more than 4 KB of ELBI register space, the default value of CX_LBC_EXT_AW can be increased up to a maximum of 32. For more information, see the 'Local Bus Controller (LBC)' section in the Architecture chapter of the <i>Databook</i>. Values: 12, ..., (FLT_Q_ADDR_WIDTH<=32) ? FLT_Q_ADDR_WIDTH : 32 Default Value: 32 Enabled: ((CC_DEVICE_TYPE!=CC_RC)) Parameter Type: Feature Setting Parameter Name: CX_LBC_EXT_AW
ELBI Data Bus Width	Defines the width of the ELBI databus for the Local Bus Controller (LBC) in dwords. There are 32 or 64 bits of ext_lbc_din for each function in your controller configuration. (128 bits is planned for future release). Values: ■ 32-bit (1) ■ 64-bit (2) Default Value: 32-bit Enabled: ((CC_DEVICE_TYPE!=CC_RC)) Parameter Type: Feature Setting Parameter Name: CX_LBC_NW

Label	Description
Uninverted ELBI Databus	Removes requirement for data swapping logic at application layer for accesses to unaligned addresses for a 64 bit ELBI databus. Values: ■ false (0) ■ true (1) Default Value: (CX_LBC_NW == 2) && CX_CXL_ENABLE Enabled: CX_LBC_NW == 2 Parameter Type: Feature Setting Parameter Name: CX_UNINVERTED_ELBI_DATA
DBI Full Access with BAR and Function Number	Provide support to allow DBI full access with BAR number and function number. Setting the DBI_MULTI_FUNC_BAR_EN configuration parameter allows the DBI interface to direct an access to a particular BAR#. This helps the ELBI application logic to fully decode an access to the ELBI space when multiple BARs have been mapped to RTRGT0 and need to be accessed through the DBI. By default, DBI_MULTI_FUNC_BAR_EN is set to 0, indicating a single BAR at the ELBI. For more information, see the 'Local Bus Controller (LBC)' section in the Architecture chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: MSIX_TABLE_EN CX_CXL_ENABLE Enabled: ((CC_DEVICE_TYPE!=CC_RC) && (CC_DEVICE_TYPE!=CC_SW)) Parameter Type: Feature Setting Parameter Name: DBI_MULTI_FUNC_BAR_EN

Label	Description
Application Configuration Register Address	This configuration parameter is deprecated and should no longer be used. Specifies the default value of register CONFIG_LIMIT_REG. It specifies a limit above which incoming configuration requests are routed to the destination interface defined by TARGET_ABOVE_CONFIG_LIMIT. The CONFIG_LIMIT parameter is normally set to a limit that divides the controller's configuration space registers from the application's configuration space registers. The controller LBC module uses this limit to direct a configuration request to the CDM or ELBI/RTRGT1. The application must set a proper value based on its extended configuration registers. For more information, see the 'Local Bus Controller (LBC)' section in the Controller Operations chapter of the <i>Databook</i>. ■ This value indicates a DWORD address and not a byte address. ■ The default setting of 0x3FF corresponds to the 4K upper limit of configuration space and so TRGT0 consumes all CFG transactions by default. CONFIG_LIMIT must be set to a value lower than this to have an effect. Normally, you would never set this to less than 0xD00 which is the top of the Synopsys Port Logic register space. Note: Superseded by the Configuration Intercept Controller(CIC) interface, enabled when CX_CONFIG_INTERCEPT_ENABLE=1. The CIC provides a much more flexible way for the application to control routing of configuration transactions. Values: 0x0, ..., 0x3ff Default Value: 0x3ff Enabled: CC_DEVICE_TYPE!=CC_RC Parameter Type: Feature Setting Parameter Name: CONFIG_LIMIT
Application Configuration Register Location	Specifies the default value of register TARGET_ABOVE_CONFIG_LIMIT. Target interface destination for configuration requests with an address greater than CONFIG_LIMIT. Normally you would set this to '1' (ELBI). Values: ■ ELBI (1) ■ Target_1 (2) Default Value: TRGT1_POPULATE==1 ? 2 : 1 Enabled: TRGT1_POPULATE==1 && CC_DEVICE_TYPE!=CC_RC Parameter Type: Feature Setting Parameter Name: TARGET_ABOVE_CONFIG_LIMIT

Label	Description
Memory Map Port Logic Registers	Enables routing of memory (non-CFG) transactions to the port logic (PL) configuration registers. This is memory mapping of the PL register space. PL registers (which by default are accessed by CFG requests) can also (at the same time) be accessed by MEM requests through the use of the ENABLE_MEM_MAP_PL_REG, PL_FUNC_NUM and PL_BAR_NUM configuration parameters. These can be used to map the PL registers to any BAR of any function. All MEM requests that match PL_BAR_NUM (when ENABLE_MEM_MAP_PL_REG=1) and whose address offset is in the range 0x700-0x8FF is routed to the PL registers. The BAR corresponding to PL_BAR_NUM must also be mapped/assigned to RTRGT0. For more information, see the "Local Bus Controller (LBC)" section in the "Controller Operations" chapter of the <i>Databook</i>. Values: 0, 1 Default Value: 0 Enabled: CC_DEVICE_TYPE!=CC_RC Parameter Type: Feature Setting Parameter Name: ENABLE_MEM_MAP_PL_REG
Port Logic Function	Indicates which function is used to map the Port Logic configuration registers into memory space. For more information, see the ENABLE_MEM_MAP_PL_REG parameter. Values: 0, ..., 7 Default Value: 0 Enabled: ENABLE_MEM_MAP_PL_REG==1 Parameter Type: Feature Setting Parameter Name: PL_FUNC_NUM
Port Logic BAR	Indicates which BAR is used to map the Port Logic configuration registers into memory space. For more information, see the ENABLE_MEM_MAP_PL_REG parameter. Values: ■ BAR0 (0) ■ BAR1 (1) ■ BAR2 (2) ■ BAR3 (3) ■ BAR4 (4) ■ BAR5 (5) Default Value: BAR0 Enabled: ENABLE_MEM_MAP_PL_REG==1 Parameter Type: Feature Setting Parameter Name: PL_BAR_NUM
Port Logic Register Address Limit	Indicates the limit address for BAR matched memory mapped Port Logic configuration registers. Values: 2304, ..., 3208 Default Value: 3208 Enabled: ENABLE_MEM_MAP_PL_REG==1 Parameter Type: Feature Setting Parameter Name: PL_LIMIT

Label	Description
Configuration Intercept Enable	Enables Configuration Intercept feature. Allows your application logic to modify the behavior of Rx CFG requests that are accessing the controller's internal registers. For more information, see "Configuration Intercept Controller (CIC) for USP" section in the "Controller Operations" chapter of the <i>Databook</i>. This is an advanced feature that is only required in exceptional applications. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_RC Parameter Type: Feature Setting Parameter Name: CX_CONFIG_INTERCEPT_ENABLE
Memory Map ATU UNROLL Registers	Enables routing of memory (non-CFG) transactions to the Unroll ATU configuration registers. Unroll registers can be accessed by MEM requests through the use of the ENABLE_MEM_MAP_UNROLL_ATU_REG, UNROLL_FUNC_NUM and UNROLL_BAR_NUM configuration parameters. These can be used to map the UNROLL registers to any BAR of any function. All MEM requests that match UNROLL_BAR_NUM (when ENABLE_MEM_MAP_UNROLL_ATU_REG=1) and whose address offset is in the range UNROLL_ATU_OFFSET_BAR-UNROLL_ATU_OFFSET_BAR+UNROLL_ATU_SIZE-1 is routed to the UNROLL registers. The BAR corresponding to UNROLL_BAR_NUM must also be mapped/assigned to RTRGT0. For more information, see the "Local Bus Controller (LBC)" section in the "Controller Operations" chapter of the <i>Databook</i>. Values: 0, 1 Default Value: 0 Enabled: CC_DEVICE_TYPE!=CC_RC && CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE Parameter Type: Feature Setting Parameter Name: ENABLE_MEM_MAP_UNROLL_ATU_REG
Offset BAR address for ATU UNROLL Registers	Indicates the offset BAR address to allocate the UNROLL ATU configuration registers into memory space. Values: 0x0, ..., 0xffffffffffffff Default Value: UNROLL_DMA_OFFSET_BAR + UNROLL_DMA_SIZE Enabled: ENABLE_MEM_MAP_UNROLL_ATU_REG==1 Parameter Type: Feature Setting Parameter Name: UNROLL_ATU_OFFSET_BAR

Label	Description
Memory Map DMA UNROLL Registers	Enables routing of memory (non-CFG) transactions to the Unroll DMA configuration registers. Unroll DMA registers can be accessed by MEM requests through the use of the ENABLE_MEM_MAP_UNROLL_DMA_REG, UNROLL_FUNC_NUM and UNROLL_BAR_NUM configuration parameters. These can be used to map the UNROLL registers to any BAR of any function. All MEM requests that match UNROLL_BAR_NUM (when ENABLE_MEM_MAP_UNROLL_DMA_REG=1) and whose address offset is in the range UNROLL_DMA_OFFSET_BAR to UNROLL_DMA_OFFSET_BAR+UNROLL_DMA_SIZE-1 is routed to the UNROLL registers. The BAR corresponding to UNROLL_BAR_NUM must also be mapped/assigned to RTRGT0. For more information, see the "Local Bus Controller (LBC)" section in the "Controller Operations" chapter of the <i>Databook</i>. Values: 0, 1 Default Value: 0 Enabled: CC_DEVICE_TYPE!=CC_RC && CC_DMA_ENABLE && CC_UNROLL_ENABLE Parameter Type: Feature Setting Parameter Name: ENABLE_MEM_MAP_UNROLL_DMA_REG
Offset BAR address for DMA UNROLL Registers	Indicates the offset BAR address to allocate the UNROLL DMA configuration registers into memory space. Values: 0x0, ..., 0xffffffffffff Default Value: 0x0 Enabled: ENABLE_MEM_MAP_UNROLL_DMA_REG Parameter Type: Feature Setting Parameter Name: UNROLL_DMA_OFFSET_BAR
UNROLL Function	Indicates which function is used to map the Unroll configuration registers into memory space. For more information, see the ENABLE_MEM_MAP_UNROLL_ATU_REG or ENABLE_MEM_MAP_UNROLL_DMA_REG parameter. Values: 0, ..., 7 Default Value: 0 Enabled: ENABLE_MEM_MAP_UNROLL_ATU_REG==1 ENABLE_MEM_MAP_UNROLL_DMA_REG==1 Parameter Type: Feature Setting Parameter Name: UNROLL_FUNC_NUM

Label	Description
UNROLL BAR	Indicates which BAR is used to map the UNROLL configuration registers into memory space. For more information, see the ENABLE_MEM_MAP_UNROLL_ATU_REG or ENABLE_MEM_MAP_UNROLL_DMA_REG parameter. Values: ■ BAR0 (0) ■ BAR1 (1) ■ BAR2 (2) ■ BAR3 (3) ■ BAR4 (4) ■ BAR5 (5) Default Value: BAR0 Enabled: ENABLE_MEM_MAP_UNROLL_ATU_REG==1 ENABLE_MEM_MAP_UNROLL_DMA_REG==1 Parameter Type: Feature Setting Parameter Name: UNROLL_BAR_NUM
Receive Filter Options	
Number of Address Bits Passed to the Application	Indicates the number of address bits that are passed through the receive queues and are sent to the application on the RTRGT1 interface (radm_trgt1_addr). ■ DM/RP: The default is 64. the minimum number of bits allowed is normally 32. ■ SW: The default is 64 and the parameter is read-only. ■ EP: The default is 64 when <code>VENDOR_MESSAGE_SUPPORT DEFAULT_TARGET ((CX_INTERNAL_ATU_ENABLE ADDR_TRANSLATION_SUPPORT_EN) & CX_CLIENT_PAR_MODE != 0)</code>, otherwise 32. The minimum number of bits allowed is normally 32. Values: 26, ..., 64 Default Value: <code>(VENDOR_MESSAGE_SUPPORT DEFAULT_TARGET CX_CXL_ENABLE (CC_DEVICE_TYPE!=CC_EP) ADDR_TRANSLATION_SUPPORT_EN && (CX_CLIENT_PAR_MODE!=0 ECC_PROTECTION_EN!=0)) ? 64 : 32</code> Enabled: <code>!(VENDOR_MESSAGE_SUPPORT DEFAULT_TARGET (CC_DEVICE_TYPE==CC_SW) ADDR_TRANSLATION_SUPPORT_EN && (CX_CLIENT_PAR_MODE!=0 ECC_PROTECTION_EN!=0))</code> Parameter Type: Feature Setting Parameter Name: FLT_Q_ADDR_WIDTH

Label	Description
Application Receives Full Message Header	When you want to route the third header dword of received messages to your application interface (RRTRGT1 or AXI bridge master interface), then you must manually set FLT_Q_ADDR_WIDTH to 64 bits, or set this parameter to "1". Setting this parameter to "1" forces FLT_Q_ADDR_WIDTH to 64 bits. This adds gates and makes some RAMs wider. The complete message header is always routed to the SII interface regardless of this parameter setting. Values: ■ false (0) ■ true (1) Default Value: CX_CCIX_INTERFACE_ENABLE Enabled: CC_DEVICE_TYPE==CC_EP Parameter Type: Feature Setting Parameter Name: VENDOR_MESSAGE_SUPPORT
Terminate Received Messages	Drop message TLPs and do not pass them to the application on RTRGT1. The controller processes received messages and decodes the header before sending it to the application logic on the System Information Interface (SII). By default, received messages are dropped silently and not passed to the application (or AXI bridge master if present) on RTRGT1. To have all decoded messages also sent to the application then do not set this parameter. ■ True: Discard message after decoding ■ False: Pass message TLP to application on RTRGT1 (or AXI bridge master if present) <i>Note:</i> When The "Receive Posted Queue Mode" is Bypass (RADM_P_QMODE_VCO = 0x4), received messages are not dropped and are always passed to the application irrespective of the setting of this parameter. For more information, see "Routing of Received Messages to SII and optionally to Application" in the "Message Reception" section in the Controller Operations chapter of the <i>Databook</i>. Your application can override the value of this option at runtime by writing to the 'Symbol Timer Register and Filter Mask Register 1'. These registers allow you to override any decisions (regarding MSG routing) made at configuration time by the FLT_DROP_MSG, DEFAULT_FILTER_MASK_1 and DEFAULT_FILTER_MASK_2 configuration parameters. Values: ■ false (0x0) ■ true (0x1) Default Value: true Enabled: Always Parameter Type: Register Default Setting Parameter Name: FLT_DROP_MSG

Label	Description
Receive Filter Rule Mask Register 1	Default value for Filter Mask Register 1 ('Symbol Timer Register and Filter Mask Register 1'). This register is used to set the number of symbol times to wait between transmitting SKP ordered sets, and also to mask the RADM Filtering and Error Handling Rules. There are several mask bits used to turn off the filtering and error handling rules. For more information, see the 'Receive Filtering' section in Controller Operations chapter of the <i>Databook</i>. Values: 0x0, ..., 0xffff Default Value: (FLT_DROP_MSG ? 0 : 0x2000) (CX_SRIOV_ENABLE ? 0x0008 : 0) Enabled: Always Parameter Type: Register Default Setting Parameter Name: DEFAULT_FILTER_MASK_1
Receive Filter Rule Mask Register 2	Default value for 'Filter Mask Register 2'. This register is used to mask the RADM Filtering and Error Handling Rules. There are several mask bits used to turn off the filtering and error handling rules. For more information, see the 'Receive Filtering' section in Controller Operations chapter of the <i>Databook</i>. Values: 0x0, ..., 0xffffffff Default Value: CX_CXL_ENABLE ? 0x00000001 : 0 Enabled: Always Parameter Type: Register Default Setting Parameter Name: DEFAULT_FILTER_MASK_2
Mask Completion Timeout Errors	Mask detection of completion timeout errors. When selected, the controller does not automatically report completion timeout errors. Your application must check for completion timeouts and report them using the app_err_bus input signal on the application error return interface. The 'Application Error Reporting' (APP_RETURN_ERR_EN) parameter must be enabled to activate the application error return interface. Values: ■ false (0) ■ true (1) Default Value: false Enabled: APP_RETURN_ERR_EN==1 Parameter Type: Feature Setting Parameter Name: CPL_TIMEOUT_ERR_MASK

Label	Description
Enable Optional Checks	Adds optional protocol checks including byte enable and flow control. Violations of the byte enable check (Section 2.2.5 of the <i>PCIe Specification</i>) and the address/length check (Section 2.2.7 of the <i>PCIe Specification</i>) is treated as malformed TLPs. Also, violations of the various flow control checks results in a Flow Control Protocol Error (FCPE). Flow control checks are described in Section 2.6.1 of the <i>PCIe Specification</i>. Values: ■ false (0) ■ true (1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: ENABLE_OPTIONAL_CHECKS
Receive Errored-TLP Forwarding Options	
Forward Errored-TLPs To Application	Forward all incoming I/O or MEM requests with UR/CA/CRS status to your application. By default, all such requests are dropped in store-and-forward mode (after corresponding error reporting). A completion with UR status is generated for non-posted requests. For more information, see the description of the DEFAULT_TARGET port logic register. See also the "ECRC Handling" and "Request TLP Routing Rules" in the "Receive Routing" sections in the "Controller Operations" chapter of the <i>Databook</i>. ■ When you set this parameter, you should also set CX_MASK_UR_CA_4_TRGT1 ■ This parameter sets the default value of the DEFAULT_TARGET field in the MISC_CONTROL_1_OFF port logic register. ■ This parameter only applies to EP, DM in EP mode, or SW in USP mode. It does not apply to a DSP. Values: ■ Drop (0x0) ■ Forward (0x1) Default Value: Drop Enabled: TRGT1_POPULATE==1 && CC_DEVICE_TYPE!=CC_RC Parameter Type: Register Default Setting Parameter Name: DEFAULT_TARGET

Label	Description
Suppress Errors for Forwarded Unsupported Requests	Suppress error logging, Error Message generation, and CPL generation (for non-posted requests) for requests TLPs with UR filtering status that you have chosen to forward to the application (when you set the DEFAULT_TARGET field). For more information, see the description of the UR_CA_MASK_4_TRGT1 field, "ECRC Handling" and "Request TLP Routing Rules" in the "Receive Routing" section in the "Controller Operations" chapter of the <i>Databook</i>. ■ This parameter sets the default value of the UR_CA_MASK_4_TRGT1 field in the MISC_CONTROL_1_OFF port logic register. ■ You should set this if you have set the DEFAULT_TARGET parameter to '1'. ■ This parameter only applies to EP, or DM in EP mode. It does not apply to a DSP. Values: ■ Report_UR_ERR (0x0) ■ Suppress_UR_ERR (0x1) Default Value: Report_UR_ERR Enabled: CC_DEVICE_TYPE!=CC_SW && CC_DEVICE_TYPE!=CC_RC Parameter Type: Feature Setting Parameter Name: CX_MASK_UR_CA_4_TRGT1
Indicate ECRC Error Using Application I/O	Passes the ECRC error notification to your application on the corresponding radm_<trgt1 cpl byp>_ecrc_err output. Otherwise the controller do not assert the output when it receives a TLP with an ECRC error. ■ This parameter is only used for queues configured in cut-through or store-and-forward mode. It does not apply to bypass mode. ■ The default setting of this parameter is sufficient for most applications. For more information, see the 'Error Detection For Received TLPs' section in the 'Controller Operations' chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_RADMQ_MODE==2 && AMBA_INTERFACE==0 Parameter Type: Feature Setting Parameter Name: ECRC_ERR_PASS_THROUGH
Addressing ECAM	
SBSA ECAM Addressing	Configure ECAM addressing for Downstream Ports. The Downstream Port controller handles Memory Transactions from the Host CPU by either targeting local CDM registers, or converting to Type 0 Configuration Requests for Endpoint devices, or converting to Type 1 Configuration Requests for Downstream Ports on the PCIe fabric. Values: 0, 1 Default Value: 0 Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Feature Setting Parameter Name: CX_SBSA_ECAM

1.8 Advanced AXI Config Parameters

Table 1-8 Advanced AXI Config Parameters

Label	Description
Distributed Translation Interface Master (DTIM)	
DTIM Enable	Enables the Root Ports Distributed Translation Interface Master (DTIM). The DTI Master interfaces to an external Memory Management Unit (MMU). The MMU provides address translation functions for the memory accessing PCIe AXI masters. Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_DM) && MASTER_POPULATED Parameter Type: Feature Setting Parameter Name: CC_DTIM_ENABLE
Root Port ID	The Root Port ID determines the upper 16-bits of the PCIe DTIM Stream ID (SID). Values: 0x0, ..., 0xffff Default Value: 0xffff Enabled: CC_DTIM_ENABLE Parameter Type: Feature Setting Parameter Name: CC_ROOT_PORT_ID
DTIM Translation Request Tokens Requested	This parameter indicates the maximum number of translation request tokens requested by the DTI Master, i.e. the maximum number of DTI-ATS translation requests outstanding at any point in time. It is used to flow control the number of outstanding translation requests sent by the DTI Master AXI4-Stream Master. Values: 1, ..., 256 Default Value: 32 Enabled: CC_DTIM_ENABLE Parameter Type: Feature Setting Parameter Name: CC_DTIM_NUM_TRANS_TOKENS_REQUESTED
DTIM Invalidate Request Tokens Granted	This parameter indicates the maximum number of invalidate request tokens granted by the DTI Master, i.e. the maximum number of DTI-ATS invalidate requests outstanding at any point in time. It is used to determine the depth of the DTI Master AXI4-Stream Slave Request queue. Values: 4, 8, 16 Default Value: 16 Enabled: CC_DTIM_ENABLE Parameter Type: Feature Setting Parameter Name: CC_DTIM_NUM_INV_TOKENS_GRANTED

Label	Description
DTIM PCIe Max Num Outstanding Invalidate Requests	The maximum number of outstanding PCIe invalidate requests that can be sent on the wire. The theoretical maximum is 32 requests per BDF. Values: 16, 32, 64, 128, 256 Default Value: 32 Enabled: CC_DTIM_ENABLE Parameter Type: Feature Setting Parameter Name: CC_DTIM_PCIE_MAX_NUM_INV_REQ
Advanced Slave Options	
AXI Slave ID Width	Specify the width of the AXI slave interface ID bus. You can increase or decrease CC_SLV_BUS_ID_WIDTH but your application master must never issue more than CC_MAX_SLV_TAG different AXI ID's (for non-posted requests) because the bridge is still only configured to handle CC_MAX_SLV_TAG different AXI ID's. CC_SLV_BUS_ID_WIDTH sets the width of each entry in the Slave Response Application-Tag Look Up Table, but it does not determine the number of available TAGs. Default value is ceil [log2{CC_MAX_SLV_TAG}]. For more information, see "Increasing Size of AXI Slave ID Bus" in the AXI chapter of the <i>Databook</i>.) Values: 1, ..., 32 Default Value: [<functionof> CC_MAX_SLV_TAG] Enabled: SLAVE_POPULATED && AXI_POPULATED Parameter Type: Performance Setting Parameter Name: CC_SLV_BUS_ID_WIDTH
AXI Slave Multiple Requests per ID Enable	Enable support for multiple request per AXI ID on the AXI Slave Values: 0, 1 Default Value: 1 Enabled: ((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4)) Parameter Type: Feature Setting Parameter Name: CC_AXI_SLV_MULTIPLE_REQ_PER_ID_EN
Slave Composition Segment Byte Length	Specify the buffer size for each outbound completion, in bytes. When the AXI slave receives the read request whose default value is equal to the max slave read request size, ensures that there is no performance overhead caused by the bridge. Decreasing this value decreases the size of the RAM OB_CCMP_DATA_RAM proportionately, but when receiving read requests whose completion exceeds this byte length, there will be throttling of request interface for one clock cycle per additional segment allocation. Values: 32, 64, 128, 256, 512, 1024, 2048, 4096 Default Value: CC_COMPOSER_SIZE_DEFAULT Enabled: SLAVE_POPULATED Parameter Type: Performance Setting Parameter Name: CC_RADMX_COMPOSER_MAX_SEG_BYTE_LEN

Label	Description
Advanced Master Options	
AXI Master Fixed Size Transfer Enable	Specifies that the AXI master interface (under certain conditions) does a single bus-wide access when doing narrow read transfers rather than multiple byte-wide accesses. When you set CC_MSTR_BURST_LEN to < MASTER_BUS_DATA_WIDTH/8, then MSTR_FIXED_SIZE_ENABLE=1 because the master cannot produce a sufficiently long burst of bytes. Therefore the master uses dword access and not byte access for narrow or NCBE reads; which means that such accesses to non-prefetchable memory might result in corrupting not-to-be-read data. For more information on this and other restrictions, see the description of the CC_MSTR_BURST_LEN parameter and the 'Supported AXI Transfer Sizes' section in the AXI chapter of the <i>Databook</i>. Values: 0, 1 Default Value: (((CC_MSTR_BURST_LEN==8) && (MASTER_BUS_DATA_WIDTH==64)) (CC_MSTR_BURST_LEN < MASTER_BUS_DATA_WIDTH/8)) ? 1 : 0 Enabled: (AXI_POPULATED && MASTER_POPULATED && !((CC_MSTR_BURST_LEN==8) && (MASTER_BUS_DATA_WIDTH==64))) Parameter Type: Performance Setting Parameter Name: MSTR_FIXED_SIZE_ENABLE
AXI Master Sub Bus Size Read Enable	Specifies that the master performs a narrow read transfer in one beat when the PCIe request size is a power of two and the AXI address is AXI transfer-size aligned. For more information, see "Supported AXI Transfer Sizes" section of the <i>Databook</i>. <i>Note:</i> AXI only. Values: 0, 1 Default Value: 0 Enabled: AXI_POPULATED && MASTER_POPULATED Parameter Type: Performance Setting Parameter Name: MSTR_SUB_BUS_SIZE_READ_ENABLE
AXI Master Sub Bus Size Write Enable	Specifies that the master performs a narrow write transfer in one beat when the PCIe request size is a power of two and the AXI address is AXI transfer-size aligned. For more information, see "Supported AXI Transfer Sizes" section of the <i>Databook</i>. <i>Note:</i> AXI only. Values: 0, 1 Default Value: 0 Enabled: AXI_POPULATED && MASTER_POPULATED Parameter Type: Performance Setting Parameter Name: MSTR_SUB_BUS_SIZE_WRITE_ENABLE

Label	Description
AXI Master NCBE Read Decompose Disable	Specifies that the master does not decompose NCBE read requests into a stream of single-byte requests. For more information, see 'Non Contiguous Byte Enable (NCBE) Support' section of the <i>Databook</i>. 0: Enables read NCBE decomposition into single-byte requests. 1: Disables read NCBE decomposition into single-byte requests. Values: 0, 1 Default Value: !MSTR_SUB_BUS_SIZE_READ_ENABLE && MSTR_FIXED_SIZE_ENABLE (CX_AXI_RCHUNK_ENABLE Enabled: MASTER_POPULATED && AXI_POPULATED Parameter Type: Performance Setting Parameter Name: CC_AXI_NCBE_NODECOMP
AXI Master Read Response Interleaving Enable	Enable support for interleaved read responses on the AXI Master Values: 0, 1 Default Value: 1 Enabled: ((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4)) Parameter Type: Feature Setting Parameter Name: CX_AXI_MSTR_RD_RSP_INTERLEAVING_EN
Inbound Posted Tracker	
Inbound Posted Tracker Enable	Enables Inbound Posted Tracker. Allows P to pass P on multi-VC configuration. Allows RO CPL and NP ordering disabling features. Values: 0, 1 Default Value: CC_HP_MASTER Enabled: AMBA_INTERFACE > 1 Parameter Type: Feature Setting Parameter Name: CC_IB_WREQ_PTRK_EN
AXI5 Optional Features	
Unique ID Indicator	Enable AXI5 Unique ID Indicator. Values: 0, 1 Default Value: 0 Enabled: AMBA_INTERFACE == 4 Parameter Type: Feature Setting Parameter Name: CX_AXI_UIDI_ENABLE
Read Data Chunking	Enable AXI5 Read Data Chunking. Values: 0, 1 Default Value: 0 Enabled: AMBA_INTERFACE == 4 Parameter Type: Feature Setting Parameter Name: CX_AXI_RCHUNK_ENABLE

Label	Description
Read Data Chunking Prefetch Disable	Enable support for non-prefetchable devices by disabling realignment of mstr_araddr to 16. <i>Warning:</i> ■ This disables chunking for AR channel requests where the address is not 16-byte aligned ■ This parameter only refers to the logic on the Read Data Chunking feature. You still need to ensure all other configuration option allows support for non-prefetchable devices Values: 0, 1 Default Value: 0 Enabled: (AMBA_INTERFACE == 4) && (CX_AXI_RCHUNK_ENABLE) Parameter Type: Feature Setting Parameter Name: CX_AXI_RCHUNK_NO_PREFETCH

1.9 Device-Wide PCIe Features and Capabilities Config / MSI/MSI-X Capability Parameters

Table 1-9 Device-Wide PCIe Features and Capabilities Config / MSI/MSI-X Capability Parameters

Label	Description
MSI/MSI-X	
MSI Capability	Include the MSI capability structure. Values: ■ false (0x0) ■ true (0x1) Default Value: true Enabled: Always Parameter Type: Feature Setting Parameter Name: MSI_CAP_ENABLE
MSI PVM Support	Support MSI Capability Per Vector Masking Values: ■ false (0) ■ true (1) Default Value: MSI_CAP_ENABLE && CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=CC_RC Enabled: !CX_SRIOV_ENABLE && MSI_CAP_ENABLE && CC_DEVICE_TYPE!=CC_RC Parameter Type: Feature Setting Parameter Name: MSI_PVM_EN
MSI IO Enable	Include optional top-level ports for MSI, as listed in 'Message Signaled Interrupt (MSI) Interface' section of the <i>Databook</i>. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: MSI_CAP_ENABLE==1 Parameter Type: Feature Setting Parameter Name: MSI_IO
Default Extended Message Data for MSI Capability	Default value for the Extended Message Data Capable field in the 'MSI Control Register'. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: MSI_CAP_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: DEFAULT_EXT_MSI_DATA_CAPABLE

Label	Description
MSI-X Capability	Include the MSI-X capability structure. Values: ■ false (0x0) ■ true (0x1) Default Value: CX_SRIOV_ENABLE ? 1 : 0 Enabled: Always Parameter Type: Feature Setting Parameter Name: MSIX_CAP_ENABLE
Enable Integrated MSI-X Generation Module	Implements the MSI-X generation logic with the data/address table and PBA in the controller. For more information, see the Interrupts section in the "Controller Operations" chapter of the Databook. Not available when Extensible Virtual Function (CX_EXTENSIBLE_VFUNC) is enabled. Values: 0, 1 Default Value: 0 Enabled: (((CC_DEVICE_TYPE==CC_EP) (CC_DEVICE_TYPE==CC_DM)) && !CX_EXTENSIBLE_VFUNC) Parameter Name: MSIX_TABLE_EN
MSI-X IO Enable	Include optional top-level ports for MSI-X, as listed in the 'MSI-X Interface' section of the <i>Databook</i>. If included, the MSI-X ports are used only when the controller is operating as an upstream port. Values: ■ false (0x0) ■ true (0x1) Default Value: MSIX_CAP_ENABLE Enabled: MSIX_CAP_ENABLE==1 Parameter Type: Feature Setting Parameter Name: MSIX_IO

1.10 Device-Wide PCIe Features and Capabilities Config / PCIe Capability Parameters

Table 1-10 Device-Wide PCIe Features and Capabilities Config / PCIe Capability Parameters

Label	Description
PCIe Capability	
CRS Software Visibility Default	Default value for the CRS Software Visibility bit in the Root Capabilities Register. Values: 0x0, 0x1 Default Value: 0x1 Enabled: Always Parameter Type: Register Default Setting Parameter Name: DEFAULT_CRIS_SW_VISIBILITY_CAP
ASPM Latencies	
L0S Exit Latency	Default value for the L0s Exit Latency field in the 'Link Capabilities Register'. Values: ■ Less than 64ns (0x0) ■ 64ns to less than 128ns (0x1) ■ 128ns to less than 256ns (0x2) ■ 256ns to less than 512ns (0x3) ■ 512ns to less than 1us (0x4) ■ 1us to less than 2us (0x5) ■ 2us to 4us (0x6) ■ More than 4us (0x7) Default Value: [<functionof> ((CX_GEN2_SPEED==1) ? ((DEFAULT_GEN2_N_FTS > CX_MAX_NFTS_TMP) ? DEFAULT_GEN2_N_FTS : CX_MAX_NFTS_TMP) : CX_MAX_NFTS_TMP)] Enabled: ((CX_S_CPCIE_MODE)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_L0S_EXIT_LATENCY

Label	Description
L0S Exit Latency (Common Clock)	Default value for the L0s Exit Latency field in the 'Link Capabilities Register' when common clock is used. Common Clock operation cannot be fully enabled (through the Common Clock Configuration field of the Link Control register) unless you observe the following configuration parameter relationships: ■ CX_NFTS != CX_COMM_NFTS ■ DEFAULT_L0S_EXIT_LATENCY != DEFAULT_COMM_L0S_EXIT_LATENCY ■ DEFAULT_L1_EXIT_LATENCY != DEFAULT_COMM_L1_EXIT_LATENCY Values: ■ Less than 64ns (0x0) ■ 64ns to less than 128ns (0x1) ■ 128ns to less than 256ns (0x2) ■ 256ns to less than 512ns (0x3) ■ 512ns to less than 1us (0x4) ■ 1us to less than 2us (0x5) ■ 2us to 4us (0x6) ■ More than 4us (0x7) Default Value: [<functionof> CX_COMM_NFTS] Enabled: SLOT_CLK_CONFIG && CX_CPCIE_ENABLE Parameter Type: Register Default Setting Parameter Name: DEFAULT_COMM_L0S_EXIT_LATENCY
L1 Exit Latency	Default value for the L1 Exit Latency field in the 'Link Capabilities Register'. This parameter represents a characteristic of the PHY being used. It measures the total time from when the controller initiates a transition from P1 to P0 until the PHY begins providing valid receive data to the controller. This parameter is for conventional PCIe mode only. Values: ■ Less than 1us (0x0) ■ 1us to less than 2us (0x1) ■ 2us to less than 4us (0x2) ■ 4us to less than 8us (0x3) ■ 8us to less than 16us (0x4) ■ 16us to less than 32us (0x5) ■ 32us to 64us (0x6) ■ More than 64us (0x7) Default Value: 32us to 64us Enabled: CX_S_CPCIE_MODE Parameter Type: Register Default Setting Parameter Name: DEFAULT_L1_EXIT_LATENCY

Label	Description
L1 Exit Latency (common clk)	Default value for the L1 Exit Latency field in the 'Link Capabilities Register', when common clock is used. It measures the total time from when the controller initiates a transition from P1 to P0 until the PHY begins providing valid receive data to the controller. Common Clock operation cannot be fully enabled (through the Common Clock Configuration field of the Link Control register) unless you observe the following configuration parameter relationships: ■ CX_NFTS != CX_COMM_NFTS ■ DEFAULT_L0S_EXIT_LATENCY != DEFAULT_COMM_L0S_EXIT_LATENCY ■ DEFAULT_L1_EXIT_LATENCY != DEFAULT_COMM_L1_EXIT_LATENCY This parameter is for Conventional PCIe mode only. Values: ■ Less than 1us (0x0) ■ 1us to less than 2us (0x1) ■ 2us to less than 4us (0x2) ■ 4us to less than 8us (0x3) ■ 8us to less than 16us (0x4) ■ 16us to less than 32us (0x5) ■ 32us to 64us (0x6) ■ More than 64us (0x7) Default Value: 32us to 64us Enabled: CX_CPCIE_ENABLE && SLOT_CLK_CONFIG Parameter Type: Register Default Setting Parameter Name: DEFAULT_COMM_L1_EXIT_LATENCY
Use Platform Reference Clock	Default value for the Slot Clock Configuration bit in the Link Status register. Values: ■ false (0x0) ■ true (0x1) Default Value: true Enabled: Always Parameter Type: Register Default Setting Parameter Name: SLOT_CLK_CONFIG

Label	Description
Active State Link PM Support	Default value for the Active State Link PM Support field in the Link Capabilities register. When SRIS is supported, L0s is not supported and must not be advertised in the Capability register. Values: ■ No ASPM Support (0) ■ L0s Supported (1) ■ L1 Supported (2) ■ L0s and L1 Supported (3) Default Value: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 2 : (CX_SRIS_SUPPORT && !CX_ADM_ADAPTOR_ENABLE) ? 2 : (CX_ADM_ADAPTOR_ENABLE) ? 0 : CX_CXL_ENABLE ? 2 : 3 Enabled: CX_PCIE_MODE != PCIE_OVER_CIO Parameter Type: Register Default Setting Parameter Name: AS_LINK_PM_SUPT
Endpoint L0s Acceptable Latency	Default value for the Endpoint L0s Acceptable Latency field in the 'Device Capabilities Register'. This should be >= DEFAULT_L0S_EXIT_LATENCY (or DEFAULT_COMM_L0S_EXIT_LATENCY), and you must ensure that there is sufficient buffering in your EP for this latency setting. Values: ■ Less than 64ns (0x0) ■ 64ns to less than 128ns (0x1) ■ 128ns to less than 256ns (0x2) ■ 256ns to less than 512ns (0x3) ■ 512ns to less than 1us (0x4) ■ 1us to less than 2us (0x5) ■ 2us to 4us (0x6) ■ More than 4us (0x7) Default Value: More than 4us Enabled: CX_CPCIE_ENABLE && CC_DEVICE_TYPE!=CC_RC && CC_DEVICE_TYPE!=CC_SW Parameter Type: Register Default Setting Parameter Name: DEFAULT_EP_L0S_ACCPT_LATENCY

Label	Description
Endpoint L1 Acceptable Latency	Default value for the Endpoint L1 Acceptable Latency field in the 'Device Capabilities Register.' This should be $\geq$ DEFAULT_L1_EXIT_LATENCY (or DEFAULT_COMM_L1_EXIT_LATENCY), and you must ensure that there is sufficient buffering in your EP for this latency setting. This parameter is for conventional PCIe mode only. Values: ■ Less than 1us (0x0) ■ 1us to less than 2us (0x1) ■ 2us to less than 4us (0x2) ■ 4us to less than 8us (0x3) ■ 8us to less than 16us (0x4) ■ 16us to less than 32us (0x5) ■ 32us to 64us (0x6) ■ More than 64us (0x7) Default Value: More than 64us Enabled: CX_CPCIE_ENABLE && CC_DEVICE_TYPE!=CC_RC && CC_DEVICE_TYPE!=CC_SW Parameter Type: Register Default Setting Parameter Name: DEFAULT_EP_L1_ACCPT_LATENCY
ASPM Optionality Compliance	Software uses this bit to determine whether to enable ASPM or whether to run ASPM compliance tests Values: ■ false (0x0) ■ true (0x1) Default Value: true Enabled: Always Parameter Type: Register Default Setting Parameter Name: ASPM_OPTIONALITY_COMPLIANCE
Misc	
Port Number	Default value for the Port Number field in the Link Capabilities Register. Values: 0x0, ..., 0xff Default Value: 0x0 Enabled: Always Parameter Type: Register Default Setting Parameter Name: PORT_NUM

Label	Description
RCB Support	Support the ability of configuration software to indicate the Read Completion Boundary value. Values: ■ false (0x0) ■ true (0x1) Default Value: true Enabled: CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM Parameter Type: Register Default Setting Parameter Name: CX_RCB_SUPPORT
Slot Control	
Physical Slot Number	Default value for the Physical Slot Number field in the Slot Capabilities Register. Values: 0x0, ..., 0x1fff Default Value: 0x0 Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SLOT_PHY_SLOT_NUM
Slot Power Limit Scale	Default value for the Slot Power Limit Scale field in the Slot Capabilities Register. Values: ■ "1.0x" (0x0) ■ "0.1x" (0x1) ■ "0.01x" (0x2) ■ "0.001x" (0x3) Default Value: "1.0x" Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SET_SLOT_PWR_LIMIT_SCALE
Slot Power Limit Value	Default value for the Slot Power Limit Value field in the Slot Capabilities Register. Values: 0x0, ..., 0xff Default Value: 0x0 Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SET_SLOT_PWR_LIMIT_VAL
Slot is Hot-Plug Capable	Default value for the Hot-Plug Capable bit in the Slot Capabilities Register. When set indicates that this slot is capable of supporting Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SLOT_HP_CAPABLE

Label	Description
Slot Support Hot-Plug Surprise	Default value for the Hot-Plug Surprise bit in the Slot Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SLOT_HP_SURPRISE
Disable Hot-Plug Software Notification	Default value for the No Command Complete Support bit in the Slot Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SLOT_NO_CC_SUPPORT
Electromechanical Interlock Implemented	Default value for the Electromechanical Interlock Present bit in the Slot Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SLOT_EML_PRESENT
Slot Power Indicator Present	Default value for the Power Indicator Present bit in the Slot Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SLOT_PWR_IND_PRESENT

Label	Description
Slot Attention Indicator Present	Default value for the Attention Indicator Present bit in the Slot Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SLOT_ATTEN_IND_PRESENT
Slot MRL Sensor Present	Default value for the MRL Sensor Present bit in the Slot Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SLOT_MRL_SENSOR_PRESENT
Slot Power Controller Present	Default value for the Power Controller Present bit in the Slot Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SLOT_PWR_CTRL_PRESENT
Slot Attention Button Present	Default value for the Attention Button Present bit in the Slot Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SLOT_ATTEN_BUTTON_PRESENT

Label	Description
Surprise Down	
Surprise Down Error Reporting Enabled	Adds optional protocol check capability for Surprise Down Error Reporting. Downstream Ports that are Surprise Down Error Reporting Capable (Section 7.8.6 of the <i>PCIe Specification</i>) must treat this transition from DL_Active to DL_Inactive as a Surprise Down error, except in the cases presented in (Section 3.2.1 of the <i>PCIe Specification</i>) where this error detection is blocked. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Feature Setting Parameter Name: SURPRISE_LINK_DOWN_SUPPORTED
Clock PM	
Clock PM Support	Default value for the Clock Power Management bit in the Link Capabilities Register. Values: 0x0, 0x1 Default Value: 0x0 Enabled: (CX_PIPE_VER==0 && CX_L1_SUBSTATES_ENABLE==0) ? 0 : 1 Parameter Type: Register Default Setting Parameter Name: DEFAULT_CLK_PM_CAP

1.11 Device-Wide PCIe Features and Capabilities Config / PF Extended Capabilities Parameters

Table 1-11 Device-Wide PCIe Features and Capabilities Config / PF Extended Capabilities Parameters

Label	Description
PF Extended Capabilities	
Advanced Error Reporting (AER) Enable	Support PCI Express Advanced Error Reporting. Required if ECRC or ARI are enabled. If you want to disable this parameter, you must first disable CX_ECRC_ENABLE and CX_ARI_ENABLE. Values: ■ false (0x0) ■ true (0x1) Default Value: true Enabled: CX_ECRC_ENABLE==0 && CX_ARI_ENABLE==0 Parameter Type: Feature Setting Parameter Name: AER_ENABLE
Serial Number Capability	Include Device Serial Number capability structure. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: SERIAL_CAP_ENABLE
+ Device Serial Number (1st DW)	Specifies the first 32 bits of the device serial number Values: 0x0, ..., 0xffffffff Default Value: 0x0 Enabled: SERIAL_CAP_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: DEFAULT_SN_DW1
+ Device Serial Number (2nd DW)	Specifies the second 32 bits of the device serial number Values: 0x0, ..., 0xffffffff Default Value: 0x0 Enabled: SERIAL_CAP_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: DEFAULT_SN_DW2

Label	Description
Power Budget Capability	Include Power Budgeting capability structure Values: ■ false (0x0) ■ true (0x1) Default Value: EMERGENCY_POWER_REDUCTION_ENABLE ? 1 : 0 Enabled: !EMERGENCY_POWER_REDUCTION_ENABLE Parameter Type: Feature Setting Parameter Name: PWR_BUDGET_CAP_ENABLE
+ Power Budget System Allocated	Default value for the System Allocated bit in the Power Budget Capability register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: PWR_BUDGET_CAP_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: DEFAULT_PWR_BUDGET_SYS_ALLOC
Address Translation Services Support	Support PCI-SIG ATS (address translation services) and allow your application to use/set the Address Type (AT) field and No Write (NW) fields of the TLP header. For more information, see the 'PCI-SIG Address Translation Services (ATS)' section of the <i>Databook</i>. For generic application<->system address translation, see the 'Address Translation Services (ATS)' section of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: CC_DTIM_ENABLE Enabled: (CC_DEVICE_TYPE!=CC_SW) && !CC_DTIM_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ATS_ENABLE
+ Page Request Services Support	Support PCI-SIG PRS (page request services) and allow your EP application to use/set the memory page request mechanism. For more information, see the Controller Operations chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: CC_DTIM_ENABLE Enabled: (CX_ATS_ENABLE==1) && !CC_DTIM_ENABLE Parameter Type: Feature Setting Parameter Name: CX_PRS_ENABLE

Label	Description
Access Control Services Support	Support PCI-SIG ACS (Access Control Services) and allow your application to use/set the access control mechanism. For more information, see the Access Control Services chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: false Enabled: ((CC_DEVICE_TYPE!=CC_EP) (CX_SRIOV_ENABLE (CX_NFUNC > 1))) Parameter Type: Feature Setting Parameter Name: CX_ACS_ENABLE
+ ACS Function Groups Support	Support PCI-SIG ACS Function Groups and allow your EP application to assign PFs to Function Groups . For more information, see the Access Control Services chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: CX_ACS_ENABLE && CX_SRIOV_ENABLE Enabled: CX_ACS_ENABLE && CX_ARI_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ACS_FUNC_GRP
Latency Tolerance Reporting (LTR)	Support the Latency Tolerance Reporting Mechanism. For more information see the "Messages" section in the "Controller Operations" chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: false Enabled: Always Parameter Type: Feature Setting. Parameter Name: CX_LTR_M_ENABLE
TLP Processing Hints	Support TLP Processing Hints. For more information, see TLP Processing Hints section in the "Controller Operations" chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_SW Parameter Type: Feature Setting. Parameter Name: CX_TPH_ENABLE

Label	Description
Dynamic Power Allocation (DPA)	Function: Support the Dynamic Power Allocation (DPA) capability. Values: ■ false (0) ■ true (1) Default Value: false Enabled: ((CC_DEVICE_TYPE!=CC_RC) && (CC_DEVICE_TYPE!=CC_SW)) Parameter Type: Feature Setting Parameter Name: CX_DPA_ENABLE
+ DPA Maximum Substate	Function: The default value of the maximum substate number which is the total number of supported substates minus one. A value of 0 indicates support for one substate. This value represents the maximum configurable value of the Substate_Max field in the DPA Capability register. A write to this register field through the DBI cannot alter this to a larger value. Values: 0x0, ..., 0x1f Default Value: 0x1f Enabled: ((CC_DEVICE_TYPE!=CC_RC) && (CC_DEVICE_TYPE!=CC_SW) && CX_DPA_ENABLE) Parameter Type: Feature Setting Parameter Name: DEFAULT_DPA_SUBSTATE_MAX
ARI Forwarding	When set, Alternate Routing ID (ARI) Forwarding is supported and the ARI Forwarding Supported field in the PCIe Device Capabilities 2 Register is set. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Feature Setting Parameter Name: CX_ARI_FWD_CAP

1.12 Device-Wide PCIe Features and Capabilities Config / VC Capability Parameters

Table 1-12 Device-Wide PCIe Features and Capabilities Config / VC Capability Parameters

Label	Description
Virtual Channel Capability	
Number of Virtual Channels	The number of Virtual Channels (VCs) to be supported. The controller supports up to eight VCs. Values: 1, 2, 3, 4, 5, 6, 7, 8 Default Value: 1 Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_NVC
Virtual Channel Support	Support PCIe virtual channel Capability (required if there are multiple VCs). Values: ■ false (0x0) ■ true (0x1) Default Value: (CX_NVC==1 && CX_MULTISTREAM==0) ? 0 : 1 Enabled: ((CX_NVC==1) && (CX_MULTISTREAM==0)) Parameter Type: Feature Setting Parameter Name: VC_ENABLE
VC Arbitration Capability	Default value for the VC Arbitration Capability field in the Port VC Capability Register 2. Values: 0x0, ..., 0xf Default Value: ((CX_MULTISTREAM)) ? 0xF : 0x1 Enabled: VC_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: DEFAULT_VC_ARB_32
Low Priority Extended VC Count	Default value for the Low Priority Extended VC Count field in the Port VC Capability Register 1. Values: 0x0, ..., CX_NVC Default Value: 0x0 Enabled: ((VC_ENABLE==1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_LOW_PRI_EXT_VC_CNT

1.13 Device-Wide PCIe Features and Capabilities Config / Slot ID Capability Parameters

Table 1-13 Device-Wide PCIe Features and Capabilities Config / Slot ID Capability Parameters

Label	Description
Slot ID Capability	
Slot ID Capability	Slot ID Capability structure enable Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CC_DEVICE_TYPE==CC_SW Parameter Type: Feature Setting Parameter Name: SLOT_CAP_ENABLE
Slot First In Chassis	Default value for the "First In Chassis" field in the Slot Numbering Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: ((SLOT_CAP_ENABLE==1)) Parameter Type: Register Default Setting Parameter Name: FIRST_IN_CHASSIS
Slot Number	Default value for the "Add-In Card Slots Provided" field in the Slot Numbering Capabilities Register. Values: 0x0, ..., 0x1f Default Value: 0x0 Enabled: SLOT_CAP_ENABLE==1 Parameter Name: SLOT_NUM

1.14 Device-Wide PCIe Features and Capabilities Config / AtomicOp Support Options Parameters

Table 1-14 Device-Wide PCIe Features and Capabilities Config / AtomicOp Support Options Parameters

Label	Description
AtomicOp Support Options	
AtomicOps	Enable AtomicOps support. Values: ■ false (0) ■ true (1) Default Value: false Enabled: (!AMBA_POPULATED (AMBA_INTERFACE==4)) Parameter Type: Feature Setting. Parameter Name: CX_ATOMIC_ENABLE
AtomicOp Routing	Enable AtomicOp routing support. Values: ■ false (0x0) ■ true (0x1) Default Value: CX_ATOMIC_ENABLE && (CC_DEVICE_TYPE!=CC_EP) Enabled: (CC_DEVICE_TYPE!=CC_EP) && CX_ATOMIC_ENABLE Parameter Type: Feature Setting. Parameter Name: CX_ATOMIC_ROUTING_EN
32-bit AtomicOp Completer	Enable 32-bit AtomicOp Completer support. Values: ■ false (0x0) ■ true (0x1) Default Value: CX_ATOMIC_ENABLE Enabled: CX_ATOMIC_ENABLE Parameter Type: Feature Setting. Parameter Name: CX_ATOMIC_32_CPL_EN
64-bit AtomicOp Completer	Enable 64-bit AtomicOp Completer support. Values: ■ false (0x0) ■ true (0x1) Default Value: CX_ATOMIC_ENABLE Enabled: CX_ATOMIC_ENABLE Parameter Type: Feature Setting. Parameter Name: CX_ATOMIC_64_CPL_EN

Label	Description
128-bit CAS Completer	Enable 128-bit operand Compare and Swap (CAS) AtomicOp completion support. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CX_ATOMIC_ENABLE Parameter Type: Feature Setting. Parameter Name: CX_ATOMIC_128_CAS_EN

1.15 Device-Wide PCIe Features and Capabilities Config / Readiness Support Options Parameters

Table 1-15 Device-Wide PCIe Features and Capabilities Config / Readiness Support Options Parameters

Label	Description
General Options	
Support Immediate Readiness	Enable Immediate Readiness support in the controller. When you set the Immediate Readiness bit in the PCI header Status Register, the function is configuration-ready. Software is exempt from all requirements to delay configuration accesses following any type of reset or exit from low-power states. The function always responds to a valid configuration request targeting the function with a completion indicating successful completion status, and not CRS. For more information, see Readiness Notifications in the Controller Operations chapter of the <i>Databook</i>. Values: 0, 1 Default Value: 0 Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_RN_IMM_EN
RTR Capability Supported	Enable Readiness Time Reporting support in the controller. RTR provides an optional mechanism for describing the time required for a device or function to become configuration-ready. Software is permitted to issue requests to the device or function (following any type of reset or exit from low-power states) after waiting for the time advertised in this capability and need not wait for the (longer) times required elsewhere. The function then responds to a valid configuration request targeting the function with a completion indicating successful completion status, and not CRS. For more information, see Readiness Notifications in the Controller Operations chapter of the <i>Databook</i>. Values: 0, 1 Default Value: 0 Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_RN_RTR_EN

Label	Description
FRS/DRS Options	
Function Readiness Status (FRS) Support	Enable FRS support in the controller. FRS provides an optional mechanism for messaging the host software when a function has become configuration-ready. The controller autonomously sends a Vendor-Defined Type 1 Message (VDM) with no payload following reset or exit from low-power states. Software is permitted to issue requests to the function (following any type of reset or exit from low-power states) after receiving an FRS message from this function and need not wait for the (longer) times required elsewhere. The function then responds to a valid configuration request targeting the function with a completion indicating successful completion status, and not CRS. For more information, see Readiness Notifications in the Controller Operations chapter of the <i>Databook</i>. Values: 0, 1 Default Value: 0 Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_RN_FRS_SUPPORTED
Device Readiness Status (DRS) Support	Enable DRS support in the controller. DRS provides an optional mechanism for messaging the host software when a device has become configuration-ready. The controller autonomously sends a Vendor-Defined Type 1 Message (VDM) with no payload following reset or exit from low-power states. Software is permitted to issue requests to the device (following any type of reset or exit from low-power states) after receiving an DRS message from this device and need not wait for the (longer) times required elsewhere. The device then responds to a valid configuration request targeting the device with a completion indicating successful completion status, and not CRS. For more information, see Readiness Notifications in the Controller Operations chapter of the <i>Databook</i>. Values: 0, 1 Default Value: CX_RN_FRS_SUPPORTED==1 Enabled: CX_RN_FRS_SUPPORTED==0 Parameter Type: Feature Setting Parameter Name: CX_RN_DRS_SUPPORTED
FRS Queue Max Depth	Sets the maximum FRS queue depth in root ports (Must be at least 4 for greater than 128 bit Datapath Width). Values: 0x2, 0x4, 0x8, 0x10, 0x20, 0x40, 0x80, 0x100, 0x200, 0x400, 0x800 Default Value: 0x8 Enabled: (CX_RN_FRS_SUPPORTED==1 && (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_DM)) Parameter Type: Performance Setting Parameter Name: CX_RN_FRS_QUEUE_MAX_DEPTH

Label	Description
FRS Queue Interrupt Message Number	Default for the FRS_INT_MESSAGE_NUMBER field in the FRSQ_CAP_OFF register. Values: 0, ..., 31 Default Value: 0 Enabled: (CX_RN_FRS_SUPPORTED==1 && (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_DM)) Parameter Type: Register Default Setting Parameter Name: CX_RN_FRS_INT_MSG_NUM
FRS Queue Interrupt Enable	Default for the FRS_INTERRUPT_ENABLE field in the FRSQ_CONTROL_FRSQ_STATUS_OFF register. Values: 0, 1 Default Value: 0 Enabled: (CX_RN_FRS_SUPPORTED==1 && (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_DM)) Parameter Type: Register Default Setting Parameter Name: CX_RN_FRS_INT_ENABLE

1.16 Device-Wide PCIe Features and Capabilities Config / Lightweight Notification Support Options Parameters

Table 1-16 Device-Wide PCIe Features and Capabilities Config / Lightweight Notification Support Options Parameters

Label	Description
Lightweight Notification Support Options	
Lightweight Notification Enable	When enabled, the controller supports the Lightweight Notification capability. Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CC_DEVICE_TYPE!=CC_SW && (AMBA_INTERFACE!=1) && CX_RADMQ_MODE==2) Parameter Name: CX_LN_ENABLE
Lightweight Notification Completer Cacheline Size	Lightweight Notification System Cache Line Size (RC). Values: 0, ..., 2 Default Value: 0 Enabled: CX_LN_ENABLE && (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE == CC_DM) Parameter Name: CX_LN_COMPLETER_CACHELINE_SIZE

1.17 Device-Wide PCIe Features and Capabilities Config / Other Optional PCIe Features Parameters

Table 1-17 Device-Wide PCIe Features and Capabilities Config / Other Optional PCIe Features Parameters

Label	Description
Other Optional PCIe Features	
ID Based Ordering	Support ID Based Ordering (IDO). The controller does not perform any additional ordering when ID-based ordering (IDO) is enabled. Your application is expected to do any IDO. Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CC_DEVICE_TYPE!=CC_SW) && !AHB_POPULATED Parameter Type: Feature Setting. Parameter Name: CX_IDO_ENABLE
Completion Timeout Ranges Enable	Support Completion Timeout Ranges. For more information, see 'Completion Timeout Range' section in the 'Controller Operations' chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 1 : 0 Enabled: ((CC_DEVICE_TYPE!=CC_SW)) Parameter Type: Feature Setting Parameter Name: CX_CPL_TO_RANGES_ENABLE
Vital Product Data (VPD)	Include Vital Product Data (VPD) capability structure Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: VPD_CAP_ENABLE

Label	Description
Optimized Buffer Flush/Fill Type Supported	Indicates if the device supports Optimized Buffer Flush/Fill (OBFF) using the WAKE# signal, messages or both. Values: ■ Not Supported (0) ■ OBFF Messages Only (1) ■ WAKE# Signalling Only (2) ■ WAKE# Signalling and OBFF Messages (3) Default Value: Not Supported Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_OBFF_SUPPORT
Enable Crosslink Support	Enables support for the controller to negotiate a crosslink, with a switch from EP to RP and RP to EP. When a port negotiates a crosslink connection, the port changes its behavior from a Downstream Port to an Upstream Port and vice versa. Crosslink is supported only in DM and SW controllers because these products support both Upstream and Downstream Ports. For more information, see <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CX_CPCIE_ENABLE && (CC_DEVICE_TYPE==CC_DM CC_DEVICE_TYPE==CC_SW)) Parameter Type: Feature Setting Parameter Name: CX_CROSSLINK_ENABLE
Peer-to-peer Support	Support peer-to-peer transactions in RC. For more information, see 'Peer-to-Peer Support (P2P)' in the <i>Databook</i>. Values: 0, 1 Default Value: 0 Enabled: ((CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_DM) && (AMBA_INTERFACE==0) && (CC_DMA_ENABLE==0)) Parameter Type: Feature Setting Parameter Name: CX_P2P_ENABLE
SRIS Support	When enabled, the controller implements logic for Separate Reference Clocks with Independent SSC (SRIS). Gen5 port must support SRIS/non-SRIS mode selection mechanism. Values: ■ false (0) ■ true (1) Default Value: CX_MAX_PCIE_SPEED >= 5 Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_SRIS_SUPPORT

Label	Description
Downstream Port Containment (DPC)	Enable Downstream Port Containment (DPC) capability support. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Feature Setting Parameter Name: CX_DPC_ENABLE
RP PIO Error Controls Support (eDPC)	Enable RP PIO Error Controls (eDPC) support for DSP. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_DPC_ENABLE && CC_DEVICE_TYPE!=CC_EP && CC_DEVICE_TYPE!=CC_SW Parameter Type: Feature Setting Parameter Name: CX_DPC_RP_PIO_EXTNS

1.18 Device-Wide PCIe Features and Capabilities Config / SR-IOV Related Features Parameters

Table 1-18 Device-Wide PCIe Features and Capabilities Config / SR-IOV Related Features Parameters

Label	Description
SR-IOV Related Features	
Support SR-IOV	SR-IOV Capability enable. For more information, see the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: false Enabled: ((CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM)) Parameter Type: Feature Setting Parameter Name: CX_SRIOV_ENABLE
Function Level Reset Support	Enables controller support for Function Level Reset (FLR). For more information, see the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: CX_SRIOV_ENABLE Enabled: ((CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM) && !CX_SRIOV_ENABLE) Parameter Type: Feature Setting Parameter Name: CX_FLR_ENABLE
Number of Virtual Functions.	Read-only parameter that specifies the total number of virtual functions supported by the controller. When CX_EXTENSIBLE_VFUNC=1, it is the total of both internal VFs and external VFs. Values: 2, ..., SNPS_RSVDPARAM_61 Default Value: Sum(CX_MAX_VF_i, i=0..CX_NFUNC-1) Enabled: false Parameter Name: CX_NVFUNC
External Virtual Function Registers	Enables the Extensible Virtual function which allows you to implement VFs capability registers in your application logic. For more information, see the SRIOV section in the Controller Operations chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: CX_EXTENSIBLE_VFUNC

Label	Description
Number of (Internal) Virtual Functions.	The total number of VF registers implemented internally in the controller. ■ Normally, this is a read-only parameter that specifies the total number of virtual functions supported by the controller. ■ When CX_EXTENSIBLE_VFUNC=1, it is a writable parameter that specifies the total number of virtual functions implemented internally in the controller. Internal VFs are mapped from the lowest function number; that is, PF0_VF1. For more information, see the SR-IOV section in the Controller Operations chapter of the <i>Databook</i>. Values: 0, ..., SNPS_RSVDPARAM_19 Default Value: CX_EXTENSIBLE_VFUNC ? 0 : CX_NVFUNC Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_INTERNAL_NVFUNC

1.19 Device-Wide PCIe Features and Capabilities Config / VF Extended Capabilities Parameters

Table 1-19 Device-Wide PCIe Features and Capabilities Config / VF Extended Capabilities Parameters

Label	Description
VF Capabilities/Features	
Header Log Shared amongst VFs of a PF	Header Log Sharing amongst VF of a PF Values: 0x0, 0x1 Default Value: 0x0 Enabled: VF_AER_EN==1 && !CX_EXTENSIBLE_VFUNC Parameter Type: Feature Setting. Parameter Name: VF_HDR_LOG_SHARED
VF Shared Header Log Depth per each PF	VF Shared Header Log Depth Values: 0x2, 0x4, 0x8, 0x10, 0x20 Default Value: 0x2 Enabled: VF_HDR_LOG_SHARED Parameter Type: Feature Setting. Parameter Name: VF_HDR_LOG_SHARED_DEPTH
VF Individual Header Log Depth	VF Header Log Depth Values: 0x1, 0x2, 0x4 Default Value: 0x1 Enabled: VF_HDR_LOG_SHARED==0 && VF_AER_EN Parameter Type: Feature Setting. Parameter Name: VF_HDR_LOG_DEPTH
Virtual Function Dependency Link Support	Enables support for VF dependency link. Valid only if CX_NFUNC > 1. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CX_SRIOV_ENABLE && (CX_NFUNC > 1) Parameter Type: Feature Setting Parameter Name: CX_VF_DEPENDENCY_LINK_SUPP
VF Power Management Capability	Enable Power Management Capability for virtual functions. Not available if Extensible Virtual Function (CX_EXTENSIBLE_VFUNC) is enabled. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CX_SRIOV_ENABLE && !CX_EXTENSIBLE_VFUNC Parameter Type: Feature Setting Parameter Name: VF_PM_CAP_ENABLE

Label	Description
VF AER Capability	Enable AER Capability for virtual functions. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CX_SRIOV_ENABLE && AER_ENABLE && INTERNAL_VF_ENABLE Parameter Type: Feature Setting Parameter Name: VF_AER_ENABLE
VF TPH Capability	Include VF TPH Capability structure Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CX_SRIOV_ENABLE && CX_TPH_ENABLE && INTERNAL_VF_ENABLE Parameter Type: Feature Setting Parameter Name: VF_TPH_ENABLE_VALUE
VF ATS Capability	Include VF ATS Capability structure Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_SRIOV_ENABLE && CX_ATS_ENABLE && INTERNAL_VF_ENABLE Parameter Type: Feature Setting Parameter Name: VF_ATS_ENABLE
VF TPH Steering Tag Table Size	TLP Processing Hints Steering Tag Table Size for virtual functions. ■ All Virtual functions have the same table size set by this parameter. ■ Software reads this field to determine the ST Table Size N, which is encoded as N-1. For example, a returned value of 00000000011 indicates a table size of 4. ■ There is an upper limit of 64 entries when the ST Table is located in the TPH Requester Capability structure. Values: 1, ..., 2048 Default Value: 1 Enabled: (VF_TPH_ENABLE && (TPH_ST_TABLE_LOC_0 != 0)) Parameter Type: Feature Setting Parameter Name: VF_TPH_ST_TABLE_SIZE

Label	Description
VF TPH Steering Tag Table Size	TLP Processing Hints Steering Tag Table Size for external virtual functions. ■ All Virtual functions have the same table size set by this parameter. ■ Software reads this field to determine the ST Table Size N, which is encoded as N-1. For example, a returned value of 0000000011 indicates a table size of 4. ■ There is an upper limit of 64 entries when the ST Table is located in the TPH Requester Capability structure. Values: 1, ..., 2048 Default Value: 1 Enabled: (EXT_VF_TPH_ENABLE && (TPH_ST_TABLE_LOC_0 != 0)) Parameter Type: Feature Setting Parameter Name: EXT_VF_TPH_ST_TABLE_SIZE
VF ACS Capability (Device-Wide)	Enable Access Control Services for VFs. Values: 0, 1 Default Value: CX_ACS_ENABLE==1 && CX_SRIOV_ENABLE==1 && INTERNAL_VF_ENABLE==1 Enabled: CX_ACS_ENABLE==1 && CX_SRIOV_ENABLE==1 && INTERNAL_VF_ENABLE==1 Parameter Type: Feature Setting Parameter Name: VF_ACS_ENABLE
VF ACS Function Groups (Device-Wide)	ACS Function Group Support for VFs Values: 0, 1 Default Value: VF_ACS_ENABLE==1 && CX_SRIOV_ENABLE==1 && INTERNAL_VF_ENABLE==1 && CX_ACS_FUNC_GRP==1 Enabled: VF_ACS_ENABLE==1 && CX_SRIOV_ENABLE==1 && INTERNAL_VF_ENABLE==1 && CX_ACS_FUNC_GRP==1 Parameter Type: Feature Setting Parameter Name: VF_ACS_FUNC_GRP
VF MSI/MSI-X Capabilities	
VF MSI Capability	Enable MSI Capability for virtual functions. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CX_SRIOV_ENABLE && INTERNAL_VF_ENABLE Parameter Type: Feature Setting Parameter Name: VF_MSI_CAP_ENABLE

Label	Description
VF MSI-X Capability	Enable MSI-X Capability for virtual functions. Values: ■ false (0x0) ■ true (0x1) Default Value: CX_SRIOV_ENABLE && (CX_INTERNAL_NVFUNC != 0) Enabled: CX_SRIOV_ENABLE && INTERNAL_VF_ENABLE Parameter Type: Feature Setting Parameter Name: VF_MSIX_CAP_ENABLE
VF MSI-X Table BIR (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Table BAR Indicator Register (BIR) field in the VF MSI-X Table Offset and BIR Register. Indicates which BAR is used to map the MSI-X Table into memory space. Values: ■ BAR0 (0x0) ■ BAR1 (0x1) ■ BAR2 (0x2) ■ BAR3 (0x3) ■ BAR4 (0x4) ■ BAR5 (0x5) Default Value: BAR0 Enabled: VF_MSIX_CAP_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: VF_MSIX_TABLE_BIR_n
VF MSI-X Table Offset (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Table Offset field in the VF MSI-X Table Offset and BIR Register. Values: 0x0, ..., 0x1ffffff Default Value: 0x0 Enabled: ((VF_MSIX_CAP_ENABLE==1)) Parameter Type: Register Default Setting Parameter Name: VF_MSIX_TABLE_OFFSET_n
VF MSI-X PBA BIR (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Pending Bit Array (PBA) BAR Indicator Register (BIR) field in the VF MSI-X PBA Offset and BIR Register. Indicates which BAR is used to map the MSI-X PBA into memory space. Values: ■ BAR0 (0x0) ■ BAR1 (0x1) ■ BAR2 (0x2) ■ BAR3 (0x3) ■ BAR4 (0x4) ■ BAR5 (0x5) Default Value: BAR0 Enabled: VF_MSIX_CAP_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: VF_MSIX_PBA_BIR_n

Label	Description
VF MSI-X PBA Offset (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the PBA Offset field in the VF MSI-X PBA Offset and BIR Register. Values: 0x0, ..., 0x1ffffff Default Value: VF_MSIX_PBA_OFFSET_0_BASELINE >> 3) + ((VF_MSIX_PBA_OFFSET_0_BASELINE & 0b111) != 0 Enabled: ((VF_MSIX_CAP_ENABLE==1)) Parameter Type: Register Default Setting Parameter Name: VF_MSIX_PBA_OFFSET_n
VF Readiness Capabilities	
VF RTR Capability (Device-Wide)	Enable Readiness Time Reporting for VF's Values: 0, 1 Default Value: CX_RN_RTR_EN==1 && CX_SRIOV_ENABLE==1 && INTERNAL_VF_ENABLE==1 Enabled: CX_RN_RTR_EN==1 && CX_SRIOV_ENABLE==1 && INTERNAL_VF_ENABLE==1 Parameter Type: Feature Setting Parameter Name: VF_RTR_ENABLE
VF Immediate Readiness Support	Enable Immediate Readiness support for VFs in the controller. When you set the Immediate Readiness bit in the PCI header Status Register, the function is configuration-ready. Software is exempt from all requirements to delay configuration accesses following any type of reset or exit from low-power states. The function always responds to a valid configuration request targeting the function with a completion indicating successful completion status, and not CRS. For more information, see Readiness Notifications in the Controller Operations chapter of the <i>Databook</i>. Values: 0, 1 Default Value: CX_RN_IMM_EN==1 && CX_SRIOV_ENABLE==1 && INTERNAL_VF_ENABLE==1 Enabled: CX_RN_IMM_EN==1 && CX_SRIOV_ENABLE==1 && INTERNAL_VF_ENABLE==1 Parameter Type: Feature Setting Parameter Name: VF_IMM_ENABLE
VF Lightweight Notification Capabilities	
VF Lightweight Notification Enable	When enabled, controller supports Lightweight Notification Capability for VFs. Values: 0, 1 Default Value: CX_LN_ENABLE==1 && CX_SRIOV_ENABLE==1 && INTERNAL_VF_ENABLE==1 Enabled: CX_LN_ENABLE==1 && CX_SRIOV_ENABLE==1 && INTERNAL_VF_ENABLE==1 Parameter Name: VF_LN_ENABLE

Label	Description
VF Features	
VF Stride Always 1	Enable logic optimization when you never use a VF stride greater than one. For more information, see the Single Root I/O Virtualization (SR-IOV) section in the <i>Databook</i>. You should enable this parameter when possible to save gates and ease timing closure. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_SRIOV_ENABLE Parameter Type: Performance Setting Parameter Name: CX_VF_STRIDE_ALWAYS_ONE
Programmable VF allocation	Enable programmable allocation of VFs to PFs. For more information, see the Programmable Virtual Function Allocation section in the <i>Databook</i>. You should disable this parameter when you do not need it, to save gates and ease timing closure. Values: ■ false (0) ■ true (1) Default Value: true Enabled: CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: DYNAMIC_VF_ENABLE
External VF Capabilities (Extensible IOV only)	
External VF RTR Capability	Readiness Time Reporting Capability external for virtual functions. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_SRIOV_ENABLE && CX_EXTENSIBLE_VFUNC && CX_RN_RTR_EN Parameter Type: Feature Setting Parameter Name: EXT_VF_RTR_ENABLE
External VF ACS Capability	Enable Access Control Services for external virtual functions. This parameter cannot be enabled in this version. Values: ■ false (0) ■ true (1) Default Value: false Enabled: 0 Parameter Type: Feature Setting Parameter Name: EXT_VF_ACS_ENABLE

Label	Description
External VF Lightweight Notification Capability	Lightweight Notification Capability for external virtual functions. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_SRIOV_ENABLE && CX_EXTENSIBLE_VFUNC && CX_LN_ENABLE Parameter Type: Feature Setting Parameter Name: EXT_VF_LN_ENABLE
External VF Immediate Readiness Capability	Readiness Time Reporting Capability for external virtual functions. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_SRIOV_ENABLE && CX_EXTENSIBLE_VFUNC && CX_RN_IMM_EN Parameter Type: Feature Setting Parameter Name: EXT_VF_IMM_ENABLE
External VF MSI Capability	MSI Capability for external virtual functions. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CX_SRIOV_ENABLE && CX_EXTENSIBLE_VFUNC Parameter Type: Feature Setting Parameter Name: EXT_VF_MSI_CAP_ENABLE
External VF MSI-X Capability	MSIX Capability for external virtual functions. Values: ■ false (0x0) ■ true (0x1) Default Value: CX_EXTENSIBLE_VFUNC Enabled: CX_SRIOV_ENABLE && CX_EXTENSIBLE_VFUNC Parameter Type: Feature Setting Parameter Name: EXT_VF_MSIX_CAP_ENABLE
External VF AER Capability	AER Capability for external virtual functions. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CX_SRIOV_ENABLE && CX_EXTENSIBLE_VFUNC && AER_ENABLE Parameter Type: Feature Setting Parameter Name: EXT_VF_AER_ENABLE

Label	Description
External VF TPH Capability	TPH Capability for external virtual functions. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CX_SRIOV_ENABLE && CX_EXTENSIBLE_VFUNC && CX_TPH_ENABLE Parameter Type: Feature Setting Parameter Name: EXT_VF_TPH_ENABLE_VALUE
External VF ATS Capability	ATS Capability for external virtual functions. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_SRIOV_ENABLE && CX_EXTENSIBLE_VFUNC && CX_ATS_ENABLE Parameter Type: Feature Setting Parameter Name: EXT_VF_ATS_ENABLE

1.20 Device-Wide PCIe Features and Capabilities Config / L1 Substates/L1.CPM Capability Parameters

Table 1-20 Device-Wide PCIe Features and Capabilities Config / L1 Substates/L1.CPM Capability Parameters

Label	Description
L1 Substates Capability	
L1 Substates Support	Enable support for L1 substates. For more information, see the Power Management section of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : (CX_CPCIE_ENABLE) Parameter Type: Feature Setting Parameter Name: CX_L1_SUBSTATES_ENABLE
Configure Other Register Defaults	Select here if you want to configure the defaults for the L1 substates capability register. An extra window appears under this page. To access it, click the plus/minus symbol in the hierarchy view on the left panel. This is optional but not usually recommended. You can also change these register defaults using the DBI before (or after) the link comes up. Values: 0, 1 Default Value: 0 Enabled: Always Parameter Type: Register Default Setting Parameter Name: CX_L1_SUBSTATES_DEFAULTS_VISIBLE
L1 Substates/L1.CPM Transition Options	
L1 Substate signaling	This is read-only parameter to indicate whether the controller uses mac_phy_powerdown signal for L1-Substate transition or uses legacy side-band interface signals. You can change the setting by CX_PIPE_VER parameter located in "Basic Feature Config". Values: ■ Side band (0) ■ PowerDown (1) Default Value: CX_PIPE_VER >= 1 Enabled: SNPS_RSVDPARAM_35 Parameter Type: Feature Setting Parameter Name: CX_PIPE43_SUPPORT

Label	Description
Bypass Asynchronous Power Change Handshake	Bypass the handshake between phystatus and AsyncPowerChangeAck in P1.1 or P1.2 powerdown state when your PHY vendor does not support it. Values: 0, 1 Default Value: 0 Enabled: CX_PIPE43_SUPPORT Parameter Type: Feature Setting Parameter Name: CX_PIPE43_ASYNC_HS_BYPASS
P1.CPM Encoding	Powerdown state encoding on the mac_phy_powerdown[3:0] output for P1.CPM. Contact your PHY vendor for the value that you should use. Values: 0x0, ..., 0xf Default Value: 0x4 Enabled: CX_PIPE43_SUPPORT Parameter Type: Feature Setting Parameter Name: CX_PIPE43_P1CPM_ENCODING
P1.1 Encoding	Powerdown state encoding on the mac_phy_powerdown[3:0] output for P1.1. Contact your PHY vendor for the value that you should use. Values: 0x4, ..., 0xf Default Value: 0x5 Enabled: CX_PIPE43_SUPPORT Parameter Type: Feature Setting Parameter Name: CX_PIPE43_P1_1_ENCODING
P1.2 Encoding	Powerdown state encoding on the mac_phy_powerdown[3:0] output for P1.2. Contact your PHY vendor for the value that you should use. Values: 0x4, ..., 0xf Default Value: 0x6 Enabled: CX_PIPE43_SUPPORT Parameter Type: Feature Setting Parameter Name: CX_PIPE43_P1_2_ENCODING
P1.CPM Entry Sequence	Specifies P1.CPM Entry sequence. Contact your PHY vendor for the sequence that you should use. ■ 0: P0 -> P1 -> P1.CPM ■ 1: P0 -> P1.CPM Values: 0, 1 Default Value: 0 Enabled: CX_PIPE43_SUPPORT Parameter Type: Feature Setting Parameter Name: CX_PIPE43_P0_P1CPM

Label	Description
P1.1 and P1.2 Exit Sequence	Specifies P1.1 and P1.2 Exit sequence. Contact your PHY vendor for the sequence that you should use. ■ 0: P1.1(P1.2) -> P1 ■ 1: P1.1(P1.2) -> P1.CPM -> P1 <i>Note:</i> P1.1(P1.2) -> P1.CPM -> P1 is not supported in PCLK as PHY input. Values: 0, 1 Default Value: 0 Enabled: CX_PIPE43_SUPPORT && CX_L1_SUBSTATES_ENABLE && !CX_PIPE_PCLK_AS_PHY_INPUT Parameter Type: Feature Setting Parameter Name: CX_PIPE43_P1CPM_P1

1.21 Device-Wide PCIe Features and Capabilities Config / L1 Substates/L1.CPM Capability / L1 Substates Capability Register Defaults Parameters

Table 1-21 Device-Wide PCIe Features and Capabilities Config / L1 Substates/L1.CPM Capability / L1 Substates Capability Register Defaults Parameters

Label	Description
L1 Substates Capability Register Defaults	
Default Aux Clock Frequency [MHz]	The frequency (in MHz) of the aux_clk that is supplied by the application during low power states. This parameter is used to set the default value of the Auxiliary Clock Frequency Control Port Logic Register Values: 1, ..., 1000 Default Value: 10 Enabled: Always Parameter Type: Register Default Setting Parameter Name: DEFAULT_AUX_CLK_FREQ
Default Port Common Mode Restore Time [us]	Default value of the Port Common_Mode_Restore_Time field in the L1 Substates Capability register. For downstream ports, it is also the default value of the Common Mode Restore Time field in the L1 Substates Control 1 register. Values: 0, ..., 255 Default Value: 10 Enabled: CX_L1_SUBSTATES_ENABLE Parameter Type: Register Default Setting Parameter Name: DEFAULT_L1SUB_PORT_T_COMM_MODE
Default Port Power ON Time Scale	Default value of the Port T_POWER_ON Scale field in the L1 Substates Capability register. Values: ■ 2 us (0x0) ■ 10 us (0x1) ■ 100 us (0x2) Default Value: 2 us Enabled: CX_L1_SUBSTATES_ENABLE Parameter Type: Register Default Setting Parameter Name: DEFAULT_L1SUB_PORT_T_POWER_ON_SCALE
Default Port Power ON Time Value	Default value of the Port T_POWER_ON Value field in the L1 Substates Capability register. Values: 0, ..., 31 Default Value: 5 Enabled: CX_L1_SUBSTATES_ENABLE Parameter Type: Register Default Setting Parameter Name: DEFAULT_L1SUB_PORT_T_POWER_ON_VALUE

1.22 Device-Wide PCIe Features and Capabilities Config / TLP Prefix Support Parameters

Table 1-22 Device-Wide PCIe Features and Capabilities Config / TLP Prefix Support Parameters

Label	Description
TLP Prefix Support	
Number of Prefixes	The number of TLP Prefixes that the controller supports. When CX_NPRFX is greater than zero, then each function in the PCIe controller supports the extended FMT field in TLP prefixes and headers. When CX_NPRFX is zero, then the extended FMT field is not supported. The controller does not support TLP prefixes for the DMA module. Range: 0-8 (excluding 1,2,3 as a switch support that supports prefixes must support up to four End-End TLP Prefixes). Values: 0, 1, 2, 3, 4, 5, 6, 7 Default Value: 0 Enabled: Always Parameter Type: Feature Setting. Parameter Name: CX_NPRFX
Support PASID	Support the PASID capability for untranslated addresses. Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CX_TLP_PREFIX_ENABLE && (CC_DEVICE_TYPE!=CC_RC) && (CC_DEVICE_TYPE!=CC_SW)) Parameter Type: Feature Setting. Parameter Name: CX_PASID_ENABLE
Support PRG PASID Prefix	Any PRG Response Message with PASID capability for untranslated addresses, is required to have a PASID TLP Prefix. Values: 0, 1 Default Value: 0 Enabled: CX_PASID_ENABLE Parameter Type: Feature Setting. Parameter Name: CX_PRG_PASID_REQUIRED
Configure Other Register Defaults	Select here if you want to configure the defaults for the PASID capability register. An extra window will appear under this page. To access it, click the plus/minus symbol in the hierarchy view on the left panel. This is optional but not usually recommended. You can also change these register defaults using the DBI before (or after) the link comes up. Values: 0, 1 Default Value: 0 Enabled: (CX_PASID_ENABLE && (CC_DEVICE_TYPE!=CC_RC) && (CC_DEVICE_TYPE!=CC_SW)) Parameter Name: CX_PASID_DEFAULTS_VISIBLE

1.23 Device-Wide PCIe Features and Capabilities Config / TLP Prefix Support / PASID Capability Register Defaults Parameters

Table 1-23 Device-Wide PCIe Features and Capabilities Config / TLP Prefix Support / PASID Capability Register Defaults Parameters

Label	Description
PASID Capability Register Defaults	
Default Max PASID Width	The default value for the width of the PASID field supported by the endpoint. ■ n indicates support for PASID values 0 through $(2^n)-1$. ■ 0 indicates support for a single PASID. ■ 20 indicates support for all PASID values (20 bits). Values: 0, ..., 20 Default Value: 0 Enabled: (CX_PASID_ENABLE && (CC_DEVICE_TYPE!=CC_RC) && (CC_DEVICE_TYPE!=CC_SW)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_MAX_PASID_WIDTH
Default Privileged Mode Support	The default value for Privileged Mode Supported. ■ 1: Endpoint supports operating in Privileged and Non-Privileged modes, and supports sending requests that have the Privileged Mode Requested bit Set. ■ 0: Endpoint never Sets the Privileged Mode Requested bit. Values: 0, 1 Default Value: 1 Enabled: (CX_PASID_ENABLE && (CC_DEVICE_TYPE!=CC_RC) && (CC_DEVICE_TYPE!=CC_SW)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_PRVLGD_MODE_SPPRT
Default Execute Permission Support	The default value for Execute Permission Supported. ■ 1: Endpoint supports sending TLPs that have the Execute Requested bit Set. ■ 0: Endpoint never Sets the Execute Requested bit. Values: 0, 1 Default Value: 1 Enabled: (CX_PASID_ENABLE && (CC_DEVICE_TYPE!=CC_RC) && (CC_DEVICE_TYPE!=CC_SW)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_EXCT_PRMSN_SPPRT

1.24 Device-Wide PCIe Features and Capabilities Config / Precision Time Management Support Options Parameters

Table 1-24 Device-Wide PCIe Features and Capabilities Config / Precision Time Management Support Options Parameters

Label	Description
Precision Time Management Support Options	
Precision Time Measurement Enable	When enabled, the controller supports the Precision Time Measurement capability. Values: ■ false (0) ■ true (1) Default Value: false Enabled: (((CX_PCIE_MODE == SINGLE_CPCIE) (CX_PCIE_MODE == PCIE_OVER_CIO))) Parameter Type: Feature Setting Parameter Name: CX_PTM_ENABLE
PTM External Master Time	When enabled, the PTM local clock may be updated through an External Master Time input. Values: ■ false (0) ■ true (1) Default Value: (CC_DEVICE_TYPE==CC_EP) ? 0 : 1 Enabled: CX_PTM_ENABLE Parameter Type: Feature Setting Parameter Name: CX_PTM_EXTERNAL_MASTER_TIME
PTM Local Clock Granularity	Set Local Clock Granularity for PTM. Values: 0x0, ..., 0xff Default Value: (CX_PTM_ROOT_CAPABLE==0) ? 0 : (CX_FREQ == FREQ_500) ? 2 : (CX_FREQ == FREQ_250) ? 4 : (CX_FREQ == FREQ_125) ? 8 : (CX_FREQ == FREQ_62_5) ? 16 : 32 Enabled: CX_PTM_ENABLE Parameter Type: Feature Setting Parameter Name: CX_PTM_LOCAL_CLK_GRAN

1.25 Device-Wide PCIe Features and Capabilities Config / Secondary PCIe Extended Capability Parameters

Table 1-25 Device-Wide PCIe Features and Capabilities Config / Secondary PCIe Extended Capability Parameters

Label	Description
Gen3 EQ Control Register	
USP 8G Rx Preset Hint	Upstream Port 8.0 GT/s Receiver Preset Hint Values: 0x0, ..., 0x7 Default Value: 0x7 Enabled: ((CX_GEN3_MODE != GEN3_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_USP_RX_PRESET_HINT0
USP 8G Tx Preset	Upstream Port 8.0 GT/s Transmitter Preset Values: 0x0, ..., 0xf Default Value: 0xf Enabled: ((CX_GEN3_MODE != GEN3_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_USP_TX_PRESET0
DSP 8G Rx Preset Hint	Downstream Port 8.0 GT/s Receiver Preset Hint Values: 0x0, ..., 0x7 Default Value: 0x7 Enabled: ((CX_GEN3_MODE != GEN3_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_DSP_RX_PRESET_HINT0
DSP 8G Tx Preset	Downstream Port 8.0 GT/s Transmitter Preset Values: 0x0, ..., 0xf Default Value: 0xf Enabled: ((CX_GEN3_MODE != GEN3_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_DSP_TX_PRESET0
Gen4 EQ Control Register	
USP 16G Rx Preset Hint	Upstream Port 16.0 GT/s Receiver Preset Hint Values: 0x0, ..., 0x7 Default Value: 0x7 Enabled: ((CX_GEN4_MODE != GEN4_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_USP_16G_RX_PRESET_HINT0

Label	Description
USP 16G Tx Preset	Upstream Port 16.0 GT/s Transmitter Preset Values: 0x0, ..., 0xf Default Value: 0xf Enabled: ((CX_GEN4_MODE != GEN4_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_USP_16G_TX_PRESET0
DSP 16G Rx Preset Hint	Downstream Port 16.0 GT/s Receiver Preset Hint Values: 0x0, ..., 0x7 Default Value: 0x7 Enabled: ((CX_GEN4_MODE != GEN4_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_DSP_16G_RX_PRESET_HINT0
DSP 16G Tx Preset	Downstream Port 16.0 GT/s Transmitter Preset Values: 0x0, ..., 0xf Default Value: 0xf Enabled: ((CX_GEN4_MODE != GEN4_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_DSP_16G_TX_PRESET0
Gen5 EQ Control Register	
USP 32G Tx Preset	Upstream Port 32.0 GT/s Transmitter Preset Values: 0x0, ..., 0xf Default Value: 0xf Enabled: ((CX_MAX_PCIE_SPEED >= 5)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_USP_32G_TX_PRESET0
DSP 32G Tx Preset	Downstream Port 32.0 GT/s Transmitter Preset Values: 0x0, ..., 0xf Default Value: 0xf Enabled: ((CX_MAX_PCIE_SPEED >= 5)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_DSP_32G_TX_PRESET0

1.26 Device-Wide PCIe Features and Capabilities Config / CCIX Transport DVSEC Parameters

Table 1-26 Device-Wide PCIe Features and Capabilities Config / CCIX Transport DVSEC Parameters

Label	Description
CCIX Transport DVSEC	
USP 20GT/s Transmitter Preset	Upstream Port 20GT/s Transmitter Preset Values: 0x0, ..., 0xf Default Value: 0xf Enabled: CX_CCIX_ESM_SUPPORT Parameter Type: Feature Setting Parameter Name: DEFAULT_ESM_USP_TX_PRESET_20G0
USP 25GT/s Transmitter Preset	Upstream Port 25GT/s Transmitter Preset Values: 0x0, ..., 0xf Default Value: 0xf Enabled: CX_CCIX_ESM_SUPPORT Parameter Type: Feature Setting Parameter Name: DEFAULT_ESM_USP_TX_PRESET_25G0
DSP 20GT/s Transmitter Preset	Downstream Port 20GT/s Transmitter Preset Values: 0x0, ..., 0xf Default Value: 0xf Enabled: CX_CCIX_ESM_SUPPORT Parameter Type: Feature Setting Parameter Name: DEFAULT_ESM_DSP_TX_PRESET_20G0
DSP 25GT/s Transmitter Preset	Downstream Port 25GT/s Transmitter Preset Values: 0x0, ..., 0xf Default Value: 0xf Enabled: CX_CCIX_ESM_SUPPORT Parameter Type: Feature Setting Parameter Name: DEFAULT_ESM_DSP_TX_PRESET_25G0
ESM Quick EQ Timeout	Specifies the quick EQ time Values: ■ Quick EQ is not Supported (0) ■ 8ms/16ms (1) ■ 24ms/32ms (2) ■ 50ms/58ms (3) ■ 100ms/108ms (4) ■ 200ms/208ms (5) Default Value: Quick EQ is not Supported Enabled: CX_CCIX_ESM_SUPPORT Parameter Type: Feature Setting Parameter Name: DEFAULT_CX_CCIX_ESM_QUICK_EQ_TIME

Label	Description
ESM re-calibration is needed after ESM Data Rate is updated	Specifies if ESM re-calibration is needed after ESM Data Rate is updated Values: ■ Not needed (0) ■ Needed (1) Default Value: Not needed Enabled: CX_CCIX_ESM_SUPPORT Parameter Type: Feature Setting Parameter Name: DEFAULT_CX_CCIX_ESM_RECAL_NEEDED
ESM Calibration Time	Specifies the maximum calibration time Values: ■ 10us (0) ■ 50us (1) ■ 100us (2) ■ 500us (3) ■ 1ms (4) ■ 5ms (5) ■ 10ms (6) ■ 50ms (7) Default Value: 10us Enabled: CX_CCIX_ESM_SUPPORT Parameter Type: Feature Setting Parameter Name: DEFAULT_CX_CCIX_ESM_CAL_TIME
ESM Reach Length for ESM Data Rate1	Specifies the reach length for ESM Data Rate1 Values: ■ Short Reach(SR) (0) ■ Long Reach(LR) (1) ■ SR and LR (2) Default Value: Short Reach(SR) Enabled: CX_CCIX_ESM_SUPPORT Parameter Type: Feature Setting Parameter Name: DEFAULT_CX_CCIX_ESM_REACH_LENGTH

Label	Description
ESM Extended EQ2 timeout for USP	Specifies the ESM Extended EQ Phase2 timeout value for USP ESM Data Rate1 Values: ■ 24ms/32ms (0) ■ 50ms/58ms (1) ■ 100ms/108ms (2) ■ 200ms/208ms (3) ■ 400ms/408ms (4) ■ 600ms/608ms (5) Default Value: 24ms/32ms Enabled: CX_CCIX_ESM_SUPPORT Parameter Type: Feature Setting Parameter Name: DEFAULT_CX_CCIX_ESM_EXT_EQ2_USP_TIMEOUT
ESM Extended EQ3 timeout for DSP	Specifies the ESM Extended EQ Phase3 timeout value for DSP ESM Data Rate1 Values: ■ 24ms/32ms (0) ■ 50ms/58ms (1) ■ 100ms/108ms (2) ■ 200ms/208ms (3) ■ 400ms/408ms (4) ■ 600ms/608ms (5) Default Value: 24ms/32ms Enabled: CX_CCIX_ESM_SUPPORT Parameter Type: Feature Setting Parameter Name: DEFAULT_CX_CCIX_ESM_EXT_EQ3_DSP_TIMEOUT

1.27 Device-Wide PCIe Features and Capabilities Config / Native PCIe Enclosure Management Support Options Parameters

Table 1-27 Device-Wide PCIe Features and Capabilities Config / Native PCIe Enclosure Management Support Options Parameters

Label	Description
Native PCIe Enclosure Management Support Options	
Native PCIe Enclosure Management Enable	When enabled, the controller supports the Native PCIe Enclosure management capability. Values: ■ false (0) ■ true (1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: NPEM_ENABLE
NPEM Enclosure-Specific Capabilities	Default value for the NPEM Enclosure Specific Capabilities in the NPEM Capability register. Values: 0x0, ..., 0xfe Default Value: 0x0 Enabled: NPEM_ENABLE Parameter Type: Register Default Setting Parameter Name: NPEM_ENCLOSURE_SPECIFIC_CAPABILITIES

1.28 Device-Wide PCIe Features and Capabilities Config / Emergency Power Reduction Support Options Parameters

Table 1-28 Device-Wide PCIe Features and Capabilities Config / Emergency Power Reduction Support Options Parameters

Label	Description
Emergency Power Reduction Support Options	
Emergency Power Reduction Enabled	Configure the controller to support Emergency Power Reduction mechanism Values: ■ false (0) ■ true (1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: EMERGENCY_POWER_REDUCTION_ENABLE
Emergency Power Reduction Supported	Indicates support level of the optional Emergency Power Reduction State feature Values: 0x0, ..., 0x3 Default Value: 0x0 Enabled: EMERGENCY_POWER_REDUCTION_ENABLE Parameter Type: Register Default Setting Parameter Name: EMERGENCY_POWER_REDUCTION_SUPPORTED
Emergency Power Reduction Initialization	Indicates whether a function requires complete or partial initialization upon exit from the Power Reduction State. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: EMERGENCY_POWER_REDUCTION_ENABLE Parameter Type: Register Default Setting Parameter Name: EMERGENCY_POWER_REDUCTION_INIT_REQUIRED

1.29 Device-Wide PCIe Features and Capabilities Config / Deferrable Memory Write Support Options Parameters

Table 1-29 Device-Wide PCIe Features and Capabilities Config / Deferrable Memory Write Support Options Parameters

Label	Description
Deferrable Memory Write Support Options	
Deferrable Memory Write (DMWr)	Enable Deferrable Memory Write (DMWr) support. Values: ■ false (0) ■ true (1) Default Value: CX_CXL_ENABLE Enabled: CX_GEN3_MODE != GEN3_DISABLED Parameter Type: Feature Setting. Parameter Name: CX_DEF_MEM_WR_ENABLE
Deferrable Memory Write (DMWr) Routing	Enable Deferrable Memory Write (DMWr) routing support. Applicable only to Switch USPs, Switch DSPs, and Root Ports (AXI Bridge not supported). Values: 0x0, 0x1 Default Value: CX_DEF_MEM_WR_ENABLE && (CC_DEVICE_TYPE==CC_SW) Enabled: (CC_DEVICE_TYPE!=CC_EP) && CX_DEF_MEM_WR_ENABLE Parameter Type: Feature Setting. Parameter Name: CX_DEF_MEM_WR_ROUTING_EN
Deferrable Memory Write (DMWr) Completer	Enable Deferrable Memory Write (DMWr) completer support. Values: ■ false (0x0) ■ true (0x1) Default Value: CX_DEF_MEM_WR_ENABLE && (CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM) Enabled: CX_DEF_MEM_WR_ENABLE Parameter Type: Feature Setting. Parameter Name: CX_DEF_MEM_WR_CPL_EN
Deferrable Memory Write (DMWr) Requester	Enable Deferrable Memory Write (DMWr) requester support. Values: ■ false (0x0) ■ true (0x1) Default Value: CX_DEF_MEM_WR_ENABLE && (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_DM) Enabled: CX_DEF_MEM_WR_ENABLE && (CC_DEVICE_TYPE!=CC_SW) Parameter Type: Feature Setting. Parameter Name: CX_DEF_MEM_WR_REQ_EN

Label	Description
Deferrable Memory Write (DMWr) Lengths	Enable Deferrable Memory Write (DMWr) Lengths support. Values: ■ up to 64 bytes (0x0) ■ up to 128 bytes (0x1) Default Value: up to 64 bytes Enabled: CX_DEF_MEM_WR_ENABLE Parameter Type: Feature Setting. Parameter Name: CX_DEF_MEM_WR_LEN_SUPP

1.30 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / PCI Express Capability Parameters

Table 1-30 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / PCI Express Capability Parameters

Label	Description
PCI Express Capability	
PCIe Capabilities Interrupt Message Number (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Interrupt Message Number field in the PCI Express Capabilities Register. Values: 0x0, ..., 0x1f Default Value: 0x0 Enabled: Always Parameter Type: Register Default Setting Parameter Name: PCIE_CAP_INT_MSG_NUM_n
Is Port Connected to Slot (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates that the PCI Express link associated with this Port is connected to a slot. Values: 0x0, 0x1 Default Value: 0x0 Enabled: CC_DEVICE_TYPE!=CC_EP Parameter Type: Register Default Setting Parameter Name: SLOT_IMPLEMENTED_n
Support No-Snoop (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Enable No Snoop bit in the Device Control Register. Values: 0x0, 0x1 Default Value: 0x0 Enabled: Always Parameter Type: Register Default Setting Parameter Name: DEFAULT_NO_SNOOP_SUPPORTED_n
Enable Root RCB (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Read Completion Boundary (RCB) bit in the Link Control Register. Values: ■ 64-bytes (0x0) ■ 128-bytes (0x1) Default Value: 64-bytes Enabled: CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_DM Parameter Type: Register Default Setting Parameter Name: ROOT_RCB_n

1.31 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / MSI Register Configuration Parameters

Table 1-31 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / MSI Register Configuration Parameters

Label	Description
MSI Register Configuration	
MSI Multiple Message Capable (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Multiple Message Capable field in the 'MSI Control Register'. Values: 0x0, ..., 0x5 Default Value: 0x0 Enabled: MSI_CAP_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: DEFAULT_MULTI_MSI_CAPABLE_n

1.32 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / MSI-X Register Configuration (PF0) Parameters

Table 1-32 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / MSI-X Register Configuration (PF0) Parameters

Label	Description
MSI-X Register Configuration (PF0)	
MSI-X Table Size (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the MSI-X Table Size field in the MSI-X Control Register. Values: 0x0, ..., 0x7ff Default Value: 0x0 Enabled: MSIX_CAP_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: MSIX_TABLE_SIZE_n
MSI-X Table BIR (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Table BAR Indicator Register (BIR) field in the MSI-X Table Offset and BIR Register. Values: ■ BAR0 (0x0) ■ BAR1 (0x1) ■ BAR2 (0x2) ■ BAR3 (0x3) ■ BAR4 (0x4) ■ BAR5 (0x5) Default Value: BAR0 Enabled: MSIX_CAP_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: MSIX_TABLE_BIR_n
MSI-X Table Offset (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Table Offset field in the MSI-X Table Offset and BIR Register. Values: 0x0, ..., 0x1ffffff Default Value: MSIX_TABLE_OFFSET_0_BASELINE >> 3) + ((MSIX_TABLE_OFFSET_0_BASELINE & 0b111) != 0 Enabled: ((MSIX_CAP_ENABLE==1)) Parameter Type: Register Default Setting Parameter Name: MSIX_TABLE_OFFSET_n

Label	Description
MSI-X PBA BIR (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Pending Bit Array (PBA) BIR field in the MSI-X PBA Offset and BIR Register. Values: ■ BAR0 (0x0) ■ BAR1 (0x1) ■ BAR2 (0x2) ■ BAR3 (0x3) ■ BAR4 (0x4) ■ BAR5 (0x5) Default Value: BAR0 Enabled: ((MSIX_CAP_ENABLE==1)) Parameter Type: Register Default Setting Parameter Name: MSIX_PBA_BIR_n
MSI-X PBA Offset (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the PBA Offset field in the MSI-X PBA Offset and BIR Register. Values: 0x0, ..., 0x1ffffff Default Value: MSIX_PBA_OFFSET_0_BASELINE >> 3) + ((MSIX_PBA_OFFSET_0_BASELINE & 0b111) != 0 Enabled: ((MSIX_CAP_ENABLE==1)) Parameter Type: Register Default Setting Parameter Name: MSIX_PBA_OFFSET_n

1.33 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / Advanced Error Register Configuration Parameters

Table 1-33 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / Advanced Error Register Configuration Parameters

Label	Description
Advanced Error Register Configuration	
Default ECRC Check Capability (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the ECRC Check Capable bit in the Advanced Capabilities and Control Register. Values: ■ false (0x0) ■ true (0x1) Default Value: (CX_ECRC_ENABLE==1) ? 1 : 0 Enabled: CX_ECRC_ENABLE==1 && AER_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: DEFAULT_ECRC_CHK_CAP_n
Default ECRC Generation Capability (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the ECRC Generation Capability bit in the Advanced Capabilities and Control Register. Values: ■ false (0x0) ■ true (0x1) Default Value: (CX_ECRC_ENABLE==1) ? 1 : 0 Enabled: CX_ECRC_ENABLE==1 && AER_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: DEFAULT_ECRC_GEN_CAP_n
Advanced Error Interrupt Message Number (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Advanced Error Interrupt Message Number field of the Root Error Status Register. Values: 0x0, ..., 0x1f Default Value: 0x0 Enabled: AER_ENABLE==1 && CC_DEVICE_TYPE!=CC_EP && CC_DEVICE_TYPE!=CC_SW Parameter Type: Register Default Setting Parameter Name: AER_INT_MSG_NUM_n
AER Header Log Depth (PF#i) (for n = 0; n <= CX_NFUNC-1)	AER Header log depth. Values: 0x1, 0x2, 0x4 Default Value: 0x1 Enabled: AER_ENABLE==1 Parameter Type: Feature Setting. Parameter Name: CX_HDR_LOG_DEPTH_n

1.34 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / TLP Processing Hints Register Configuration (PF0) Parameters

Table 1-34 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / TLP Processing Hints Register Configuration (PF0) Parameters

Label	Description
TLP Processing Hints Register Configuration (PF0)	
Interrupt Vector Mode Support (PF#i) (for n = 0; n <= CX_NFUNC-1)	TLP Processing Hints supports Interrupt Vector Mode. If set indicates that the function supports the Interrupt Vector Mode of operation. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CX_TPH_ENABLE==1 Parameter Type: Feature Setting Parameter Name: TPH_IVEC_n
Device Specific Mode Support (PF#i) (for n = 0; n <= CX_NFUNC-1)	TLP Processing Hints supports Device Specific Mode. If set indicates that the function supports the Device Specific Mode of operation. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: CX_TPH_ENABLE==1 Parameter Type: Feature Setting Parameter Name: TPH_DS_n
Steering Tag Table Location (PF#i) (for n = 0; n <= CX_NFUNC-1)	TLP Processing Hints Steering Tag Table Location. Value indicates if and where the ST Table is located. Values: ■ Not Present (0) ■ TPH Requester Capability Structure (1) ■ MSI-X Table Structure (2) Default Value: Not Present Enabled: (CX_TPH_ENABLE==1 && (TPH_DS_0 TPH_IVEC_0)) Parameter Type: Feature Setting Parameter Name: TPH_ST_TABLE_LOC_n
Steering Tag Table Size (PF#i) (for n = 0; n <= CX_NFUNC-1)	TLP Processing Hints Steering Tag Table Size. Software reads this field to determine the ST Table Size N, which is encoded as N-1. For example, a returned value of 0000000011 indicates a table size of 4. Values: 1, ..., 2048 Default Value: 1 Enabled: (CX_TPH_ENABLE==1 && (TPH_ST_TABLE_LOC_0 != 0)) Parameter Type: Feature Setting Parameter Name: TPH_ST_TABLE_SIZE_n

1.35 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / ATS Register Configuration (PF0) Parameters

Table 1-35 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / ATS Register Configuration (PF0) Parameters

Label	Description
ATS Register Configuration (PF0)	
ATS Invalidate Queue Depth (PF#i) (for n = 0; n <= CX_NFUNC-1)	The number of Invalidate Requests that the Function can accept before putting backpressure on the upstream connection. A value of 0 indicates the function can accept 32 Invalidate requests. Values: 0, ..., 31 Default Value: 0 Enabled: CX_ATS_ENABLE==1 Parameter Type: Feature Setting Parameter Name: ATS_INV_Q_DPTH_n
ATS Smallest Translation Unit (PF#i) (for n = 0; n <= CX_NFUNC-1)	This value indicates to the Function the minimum number of 4096-byte blocks that is indicated in a Translation Completion or Invalidate Request. This is a power of 2 multiplier and the number of blocks is 2^{STU}. A value of 0 indicates 1 block and a value of 31 indicates 2³¹ blocks (or 8 TB total). Values: 0, ..., 31 Default Value: 0 Enabled: CX_ATS_ENABLE==1 Parameter Type: Feature Setting Parameter Name: ATS_STU_n
ATS Global Invalidate Supported	This is the default value (after reset) of the Global Invalidate Supported bit field in the ATS Capability register. If set, the function supports Invalidation Requests that have the Global Invalidate bit set. If clear, the function ignores the Global Invalidate bit in all Invalidate Requests. This bit is 0b if the function does not support the PASID TLP Prefix. Values: 0x0, 0x1 Default Value: 0x0 Enabled: CX_ATS_ENABLE && CX_PASID_ENABLE Parameter Type: Feature Setting Parameter Name: DEFAULT_ATS_GLOBAL_INVAL_SPPRT
PRS Outstanding Capacity (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for PRS Outstanding Capacity in PF0. For more information, see the PRS_OUTSTANDING_CAPACITY field in the PRS_REQ_CAPACITY_REG register. Values: 0x0, ..., 0xffffffff Default Value: 0x1 Enabled: (CX_PRS_ENABLE==1) && (CC_DEVICE_TYPE != CC_SW) && (CC_DEVICE_TYPE != CC_RC) Parameter Type: Register Default Setting Parameter Name: CX_PRS_OUTSTANDING_CAPACITY_VALUE_n

1.36 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / ACS Register Configuration (PF0) Parameters

Table 1-36 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / ACS Register Configuration (PF0) Parameters

Label	Description
ACS Source Validation Supported (PF0)	This is the default value (after reset) of the ACS Source Validation bit field in the ACS Capability register in PF0. A value of 1, indicates that the component implements ACS Source Validation. Values: 0x0, 0x1 Default Value: CC_DEVICE_TYPE!=CC_EP Enabled: CX_ACS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ACS_SRC_VALID
ACS Translation Blocking Supported (PF0)	This is the default value (after reset) of the ACS Translation Blocking bit field in the ACS Capability register in PF0. A value of 1, indicates that the component implements ACS Translation Blocking. Values: 0x0, 0x1 Default Value: CC_DEVICE_TYPE!=CC_EP Enabled: CX_ACS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ACS_AT_BLOCK
ACS P2P Request Redirect (PF#i) (for n = 0; n <= CX_NFUNC-1)	This is the default value (after reset) of the ACS P2P Request Redirect bit field in the ACS Capability register in PF0. A value of 1, indicates that the component implements ACS P2P Request Redirect. Values: 0x0, 0x1 Default Value: 0x1 Enabled: CX_ACS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ACS_P2P_REQ_REDIRECT_n
ACS P2P Completion Redirect (PF#i) (for n = 0; n <= CX_NFUNC-1)	This is the default value (after reset) of the ACS P2P Completion Redirect bit field in the ACS Capability register in PF0. A value of 1, indicates that the component implements ACS P2P Completion Redirect. Values: 0x0, 0x1 Default Value: 0x1 Enabled: CX_ACS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ACS_P2P_COMPL_REDIRECT_n

Label	Description
ACS Upstream Forwarding Supported (PF0)	This is the default value (after reset) of the ACS Upstream Forwarding bit field in the ACS Capability register in PF0. A value of 1, indicates that the component implements ACS Upstream Forwarding. Values: 0x0, 0x1 Default Value: CC_DEVICE_TYPE!=CC_EP Enabled: CX_ACS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ACS_UP_FORWARD
ACS P2P Egress Control (PF#i) (for n = 0; n <= CX_NFUNC-1)	This is the default value (after reset) of the ACS P2P Egress Control bit field in the ACS Capability register in PF0. A value of 1, indicates that the component implements ACS P2P Egress Control. Values: 0x0, 0x1 Default Value: 0x1 Enabled: CX_ACS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ACS_P2P_EGRESS_CTRL_n
ACS Direct Translated P2P (PF#i) (for n = 0; n <= CX_NFUNC-1)	This is the default value (after reset) of the ACS Direct Translated P2P bit field in the ACS Capability register in PF0. A value of 1, indicates that the component implements ACS Direct Translated P2P. Values: 0x0, 0x1 Default Value: CC_DEVICE_TYPE==CC_SW ((CC_DEVICE_TYPE==CC_DM CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_EP) && (CX_ATS_ENABLE==1)) Enabled: CX_ACS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ACS_P2P_DIRECT_TRANSL_n
ACS Egress Control Vector Size (PF#i) (for n = 0; n <= CX_NFUNC-1)	This is the default value (after reset) of the Egress Control Vector Size field in the ACS Capability register in PF0. Encodes 01h-FFh directly indicate the number of applicable bits in the Egress Control Vector; the encoding 00h indicates 256 bits. Values: 0x0, ..., 0xff Default Value: CC_DEVICE_TYPE==CC_SW CC_DEVICE_TYPE==CC_RC) ? 0x4 : (CX_ACS_FUNC_GRP ? 0x8 : ((CX_NFUNC > 1 CX_SRIOV_ENABLE) ? CX_NFUNC : 0x4) Enabled: CX_ACS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_ACS_EGRESS_CTRL_SIZE_n

Label	Description
Upstream Port ACS Egress Control Vector Mask (PF#i) (for n = 0; n <= CX_NFUNC-1)	This determines the default value of the read only bits in the Upstream Port Egress Control Vector (after reset) in PF0. Set bits to 1 to indicate Read Only. Values: 0x0, ..., 0xffffffffffffffffffffffffffffffff Default Value: [<functionof> CX_ACS_FUNC_GRP 0] Enabled: (CX_ACS_ENABLE && (CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM)) Parameter Type: Feature Setting Parameter Name: CX_ACS_UP_EGRESS_CTRL_MASK_n
Downstream Port ACS Egress Control Vector Mask (PF#i) (for n = 0; n <= CX_NFUNC-1)	This determines the default value of the read only bits in the Downstream Port Egress Control Vector (after reset) in PF0. Set bits to 1 to indicate Read Only. Values: 0x0, ..., 0xffffffffffffffffffffffffffffffff Default Value: 0x1 Enabled: CX_ACS_ENABLE && CC_DEVICE_TYPE!=CC_EP Parameter Type: Feature Setting Parameter Name: CX_ACS_DW_EGRESS_CTRL_MASK_n

1.37 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / Lightweight Notification Configuration (PF0) Parameters

Table 1-37 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / Lightweight Notification Configuration (PF0) Parameters

Label	Description
Lightweight Notification Configuration (PF0)	
LN Registration Max Default (PF#i) (for n = 0; n <= CX_NFUNC-1)	This parameter sets the default of the LNR_REGISTRATION_MAX field in the LNR_CAP_OFF register for PF0. Values: 0x0, ..., 0x1f Default Value: 0x0 Enabled: CX_LN_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: CX_LN_REG_MAX_VALUE_n
LNR-128 Supported Default (PF#i) (for n = 0; n <= CX_NFUNC-1)	This parameter sets the default of the LNR_128_SUPPORTED field in the LNR_CAP_OFF register for PF0. Values: 0x0, 0x1 Default Value: 0x0 Enabled: CX_LN_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: CX_LN_128_SUPPORTED_VALUE_n
LNR-64 Supported Default (PF#i) (for n = 0; n <= CX_NFUNC-1)	This parameter sets the default of the LNR_64_SUPPORTED field in the LNR_CAP_OFF register for PF0. Values: 0x0, 0x1 Default Value: 0x0 Enabled: CX_LN_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: CX_LN_64_SUPPORTED_VALUE_n

1.38 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / Readiness Configuration (PF0) Parameters

Table 1-38 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / Readiness Configuration (PF0) Parameters

Label	Description
Immediate Readiness Configuration (PF0)	
Immediate Readiness (PF#i) (for n = 0; n <= CX_NFUNC-1)	Enable Immediate Readiness support in the controller for PF0. Values: 0x0, 0x1 Default Value: CX_RN_IMM_EN==1 Enabled: CX_RN_IMM_EN==1 Parameter Type: Feature Setting Parameter Name: CX_RN_IMM_VALUE_n
Immediate Readiness D0 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Enable Immediate Readiness on Return to D0 support in the controller for PF0. Values: 0x0, 0x1 Default Value: CX_RN_IMM_EN==1 Enabled: CX_RN_IMM_EN==1 Parameter Type: Feature Setting Parameter Name: CX_RN_IMM_D0_VALUE_n
Readiness Time Reporting Configuration (PF0)	
RTR Valid (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR Valid field in PF0. Values: 0x0, 0x1 Default Value: CX_RN_RTR_EN==1 Enabled: CX_RN_RTR_EN==1 Parameter Type: Register Default Setting Parameter Name: CX_RN_RTR_VALID_n
RTR DL Up Time (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR DL Up Time field in PF0. Values: 0x0, ..., 0xa1e Default Value: 0x0 Enabled: CX_RN_RTR_EN==1 Parameter Type: Register Default Setting Parameter Name: CX_RN_RTR_DL_UP_VALUE_n
RTR Reset Time (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR Reset Time field in PF0. Values: 0x0, ..., 0xa1e Default Value: 0x0 Enabled: CX_RN_RTR_EN==1 Parameter Type: Register Default Setting Parameter Name: CX_RN_RTR_RESET_VALUE_n

Label	Description
RTR D3hot to D0 Time (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR D3hot to D0 Time field in PF0. Values: 0x0, ..., 0x80a Default Value: 0x0 Enabled: CX_RN_RTR_EN==1 Parameter Type: Register Default Setting Parameter Name: CX_RN_RTR_D3D0_VALUE_n
RTR FLR Time (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR FLR Time field in PF0. Values: 0x0, ..., 0xa1e Default Value: 0x0 Enabled: CX_RN_RTR_EN==1 && CX_FLR_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: CX_RN_RTR_FLR_VALUE_n
VF RTR Valid (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR Valid field in PF0 VFs. Values: 0x0, 0x1 Default Value: CX_RN_RTR_VALID_0 && VF_RTR_ENABLE Enabled: VF_RTR_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: VF_RN_RTR_VALID_n
VF Reset Time (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR Reset Time field in PF0 VFs. Values: 0x0, ..., 0xa1e Default Value: CX_RN_RTR_RESET_VALUE_0 Enabled: VF_RTR_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: VF_RN_RTR_RESET_VALUE_n
VF D3hot to D0 Time (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR D3hot to D0 Time field in PF0 VFs. Values: 0x0, ..., 0x80a Default Value: CX_RN_RTR_D3D0_VALUE_0 Enabled: VF_RTR_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: VF_RN_RTR_D3D0_VALUE_n
VF FLR Time (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR FLR Time field in PF0 VFs. Values: 0x0, ..., 0xa1e Default Value: CX_RN_RTR_FLR_VALUE_0 Enabled: VF_RTR_ENABLE==1 && CX_FLR_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: VF_RN_RTR_FLR_VALUE_n

Label	Description
External VF RTR Valid (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR Valid field in PF0 VFs. Values: 0x0, 0x1 Default Value: CX_RN_RTR_VALID_0 && EXT_VF_RTR_ENABLE Enabled: EXT_VF_RTR_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: EXT_VF_RN_RTR_VALID_n
External VF Reset Time (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR Reset Time field in PF0 VFs. Values: 0x0, ..., 0xa1e Default Value: CX_RN_RTR_RESET_VALUE_0 Enabled: EXT_VF_RTR_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: EXT_VF_RN_RTR_RESET_VALUE_n
External VF D3hot to D0 Time (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR D3hot to D0 Time field in PF0 VFs. Values: 0x0, ..., 0x80a Default Value: CX_RN_RTR_D3D0_VALUE_0 Enabled: EXT_VF_RTR_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: EXT_VF_RN_RTR_D3D0_VALUE_n
External VF FLR Time (PF#i) (for n = 0; n <= CX_NFUNC-1)	Register default for RTR FLR Time field in PF0 VFs. Values: 0x0, ..., 0xa1e Default Value: CX_RN_RTR_FLR_VALUE_0 Enabled: EXT_VF_RTR_ENABLE==1 && CX_FLR_ENABLE==1 Parameter Type: Register Default Setting Parameter Name: EXT_VF_RN_RTR_FLR_VALUE_n
FRS Messaging Configuration (PF0)	
FRS Supported (PF#i) (for n = 0; n <= CX_NFUNC-1)	Function Readiness Status messaging supported for function PF0. Values: 0x0, 0x1 Default Value: CX_RN_FRS_SUPPORTED==1 Enabled: CX_RN_FRS_SUPPORTED==1 Parameter Type: Feature Setting Parameter Name: CX_RN_FRS_VALUE_n

1.39 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / Power Management Register Configuration Parameters

Table 1-39 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / Power Management Register Configuration Parameters

Label	Description
Power Management Register Configuration	
PME Support (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the PME_Support field in the Power Management Capabilities Register. Values: 0x0, ..., 0x1f Default Value: 0x1b Enabled: Always Parameter Type: Register Default Setting Parameter Name: PME_SUPPORT_n
D1 Support (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the D1 Support bit in the Power Management Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: true Enabled: Always Parameter Type: Register Default Setting Parameter Name: D1_SUPPORT_n
D2 Support (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the D2 Support bit in the Power Management Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: Always Parameter Type: Register Default Setting Parameter Name: D2_SUPPORT_n
Device Specific Initialization (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Device Specific Initialization (DSI) bit in the Power Management Capabilities Register. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: Always Parameter Type: Register Default Setting Parameter Name: DEV_SPEC_INIT_n

Label	Description
Auxiliary Current (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Aux Current field in the Power Management Capabilities Register. Values: ■ 0mA (0x0) ■ 55mA (0x1) ■ 100mA (0x2) ■ 160mA (0x3) ■ 220mA (0x4) ■ 270mA (0x5) ■ 320mA (0x6) ■ 375mA (0x7) Default Value: 375mA Enabled: Always Parameter Type: Register Default Setting Parameter Name: AUX_CURRENT_n
No Reset on D3hot->D0 Transition (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the No Soft Reset bit in the Power Management Control and Status Register. When set, you should not reset any controller registers when transitioning from D3hot to D0. Therefore, you should not assert the non_sticky_rst_n and sticky_rst_n inputs. Values: ■ false (0x0) ■ true (0x1) Default Value: true Enabled: Always Parameter Type: Register Default Setting Parameter Name: DEFAULT_NO_SOFT_RESET_n

1.40 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / PCI Register Configuration Parameters

Table 1-40 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / PCI Register Configuration Parameters

Label	Description
PCI Register Configuration	
Device Identification Number (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Device ID field in the Device ID and Vendor ID Register. Values: 0x0, ..., 0xffff Default Value: 0xabcd Enabled: Always Parameter Type: Register Default Setting Parameter Name: CX_DEVICE_ID_n
Vendor Identification Number (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Vendor ID field in the Device ID and Vendor ID Register. Values: 0x0, ..., 0xffff Default Value: 0x16c3 Enabled: Always Parameter Type: Register Default Setting Parameter Name: CX_VENDOR_ID_n
Device Revision Number (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Revision ID field in the Revision ID Register Values: 0x0, ..., 0xff Default Value: 0x1 Enabled: Always Parameter Type: Register Default Setting Parameter Name: CX_REVISION_ID_n
Subsystem Device ID (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Subsystem ID field in the Subsystem ID and Subsystem Vendor ID Register. Values: 0x0, ..., 0xffff Default Value: 0x0 Enabled: CC_DEVICE_TYPE!=CC_RC && CC_DEVICE_TYPE!=CC_SW Parameter Type: Register Default Setting Parameter Name: SUBSYS_DEV_ID_n
Subsystem Vendor ID (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Subsystem Vendor ID field in the Subsystem ID and Subsystem Vendor ID Register. Values: 0x0, ..., 0xffff Default Value: 0x0 Enabled: CC_DEVICE_TYPE!=CC_RC && CC_DEVICE_TYPE!=CC_SW Parameter Type: Register Default Setting Parameter Name: SUBSYS_VENDOR_ID_n

Label	Description
Include ROM BAR (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the expansion ROM BAR registers. Values: 0x0, 0x1 Default Value: 0x1 Enabled: Always Parameter Type: Feature Setting Parameter Name: ROM_BAR_ENABLED_n
ROM BAR Mask (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the default of the Expansion ROM BAR Mask register. ■ The BAR Mask register specifies which bits of the Expansion ROM BAR are non-writable by host software, which determines the size of the BAR. ■ The maximum value for ROM_MASK_N is 0xFFFFFFFF because the maximum space that can be claimed by an Expansion ROM BAR is 16 MB. ■ For example: 32'hFFFF =BAR size of 2¹⁶. Values: 0x7ff, ..., 0xffffffff Default Value: 0xffff Enabled: ROM_BAR_ENABLED_n==1 Parameter Type: Register Default Setting Parameter Name: ROM_MASK_n
Programmable ROM BAR Mask (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines if the Expansion ROM BAR Mask register is writable by application software. If writing to a Expansion ROM BAR Mask register is enabled, your application can write to the Expansion ROM BAR Mask register through the DBI by asserting dbi_cs2 in addition to dbi_cs. Values: 0x0, 0x1 Default Value: 0x0 Enabled: ROM_BAR_ENABLED_n==1 Parameter Type: Feature Setting Parameter Name: ROM_MASK_WRITABLE_n
Specify ROM BAR Target Interface (PF#i) (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match the Expansion ROM BAR to either RTRGT0 or RTRGT1. ■ For example, setting ROM_FUNC0_BAR0_TARGET_MAP to 1 maps all incoming requests for the Expansion ROM BAR of function 0 to RTRGT1. ■ If TRGT1_POPULATE =0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: ROM_BAR_ENABLED_n==1 && TRGT1_POPULATE==1 Parameter Type: Feature Setting Parameter Name: ROM_FUNCn_TARGET_MAP

Label	Description
Configure Other PCI Register Defaults (PF#i) (for n = 0; n <= CX_NFUNC-1)	Select here if you want to configure the defaults for the other PCI Standard Header registers. An extra window appears under this page. To access it, click the plus/minus symbol in the hierarchy view on the left panel. This is optional but not usually recommended. You can also change these register defaults using the DBI before (or after) the link comes up. Values: 0, 1 Default Value: 0 Enabled: Always Parameter Type: Register Default Setting Parameter Name: CX_PCI_HEADER_DEFAULTS_VISIBLE_n

1.41 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / PCI Register Configuration / PF0 PCI Register Defaults Parameters

Table 1-41 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / PCI Register Configuration / PF0 PCI Register Defaults Parameters

Label	Description
PF0 PCI Register Defaults	
Base Class Code (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Base Class Code field in the Class Code Register. Values: 0x0, ..., 0xff Default Value: 0x0 Enabled: Always Parameter Type: Register Default Setting Parameter Name: BASE_CLASS_CODE_n
Sub Class Code (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Subclass Code field in the Class Code Register. Values: 0x0, ..., 0xff Default Value: 0x0 Enabled: Always Parameter Type: Register Default Setting Parameter Name: SUB_CLASS_CODE_n
Programming Interface Code (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Programming Interface field in the Class Code Register. Values: 0x0, ..., 0xff Default Value: 0x0 Enabled: Always Parameter Type: Register Default Setting Parameter Name: IF_CODE_n
CardBus CIS Address Pointer (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the CardBus CIS Pointer Register. Values: 0x0, ..., 0xffff Default Value: 0x0 Enabled: CC_DEVICE_TYPE!=CC_RC && CC_DEVICE_TYPE!=CC_SW Parameter Type: Register Default Setting Parameter Name: CARDBUS_CIS_PTR_n
Interrupt Pin (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the Interrupt Pin Register. Values: ■ None (0x0) ■ INTA (0x1) ■ INTB (0x2) ■ INTC (0x3) ■ INTD (0x4) Default Value: INTA Enabled: CC_DEVICE_TYPE!=CC_RC && CC_DEVICE_TYPE!=CC_SW Parameter Type: Register Default Setting Parameter Name: INT_PIN_MAPPING_n

Label	Description
IO Address Decode (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the 32-Bit I/O Space bit in the I/O Base and I/O Limit Register. Values: ■ 16-bit (0x0) ■ 32-bit (0x1) Default Value: CC_DEVICE_TYPE==CC_SW Enabled: !(CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_SW)) Parameter Type: Register Default Setting Parameter Name: IO_DECODE_32_n
Memory Address Decode (PF#i) (for n = 0; n <= CX_NFUNC-1)	Default value for the 64-Bit Memory Addressing bit in the Prefetchable Memory Base and Limit Register. Values: ■ 32-bit (0x0) ■ 64-bit (0x1) Default Value: CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_SW CC_DEVICE_TYPE==CC_DM Enabled: CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_SW CC_DEVICE_TYPE==CC_DM Parameter Type: Register Default Setting Parameter Name: MEM_DECODE_64_n

1.42 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / BAR Setup For Physical Function 0 (PF0) Parameters

Table 1-42 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / BAR Setup For Physical Function 0 (PF0) Parameters

Label	Description
BAR_0 / BAR_1	
Include BAR_0 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the BAR0 registers. Values: 0x0, 0x1 Default Value: (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_SW) ? 0 : 1 Enabled: !(CC_DEVICE_TYPE==CC_RC) Parameter Type: Feature Setting Parameter Name: BAR0_ENABLED_n
BAR_0 is Memory or I/O (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines whether the BAR is for memory or I/O (bit 0 of the BAR). Values: ■ Memory (0x0) ■ I/O (0x1) Default Value: Memory Enabled: BAR0_ENABLED_n==1 Parameter Type: Register Default Setting Parameter Name: MEM0_SPACE_DECODER_n
BAR_0 is Prefetchable (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines if a memory BAR is for prefetchable memory (bit 3 of a memory BAR). Values: 0x0, 0x1 Default Value: 0x0 Enabled: BAR0_ENABLED_n==1 && MEM0_SPACE_DECODER_n==0 Parameter Type: Register Default Setting Parameter Name: PREFETCHABLE0_n
BAR_0 Bit Size (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the type as 32-bit or 64-bit; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: (MEM0_SPACE_DECODER_0==1 BAR0_ENABLED_0==0) ? 0 : 2 Enabled: BAR0_ENABLED_n==1 && MEM0_SPACE_DECODER_n==0 Parameter Type: Register Default Setting Parameter Name: BAR0_TYPE_n

Label	Description
BAR_0 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	BAR0 Sizing Scheme ■ Fixed Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ Resizable BAR: Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ Resizable BAR (0x2) Default Value: Fixed Mask Enabled: BAR0_ENABLED_n==1 Parameter Type: Feature Setting Parameter Name: BAR0_SIZING_SCHEME_n
RBAR0 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for BAR0. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that BAR0 operates correctly with the BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xfffffffff0 Default Value: 0x10 Enabled: CX_BAR0_RESIZABLE_n==1 Parameter Type: Feature Setting Parameter Name: CX_BAR0_RESOURCE_AVAIL_n

Label	Description
BAR_0 Mask (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the default value of the BAR Mask register. The BAR Mask register specifies which bits of the BAR are non-writable by host software, which determines the size of the BAR. When the BAR is Resizable (BARn_SIZING_SCHEME_N =2), then the default value of this mask is determined from the CX_BARn_RESOURCE_AVAIL_N parameter. Values: 0xff, ..., 0xffffffffffff Default Value: CX_BAR0_RESIZABLE_n==1 ? [::DWC_pcie_ctl::pcie_cc_bar_mask CX_BAR0_RESOURCE_AVAIL_n] : MEM0_SPACE_DECODER_n==1 ? 0xFF : 0xFFFF Enabled: BAR0_ENABLED_n==1 && MEM0_SPACE_DECODER_n==0 && CX_BAR0_RESIZABLE_n==0 Parameter Type: Register Default Setting Parameter Name: BAR0_MASK_n
Specify Target Interface for BAR_0 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Direct incoming Requests that pass filtering and match BAR_0 to either RTRGT0 or RTRGT1. For example, setting MEM_FUNC0_BAR0_TARGET_MAP to 1 maps all incoming requests for BAR0 of function 0 to RTRGT1. If TRGT1_POPULATE = 0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: BAR0_ENABLED_n==1 && TRGT1_POPULATE==1 Parameter Type: Feature Setting Parameter Name: MEM_FUNCn_BAR0_TARGET_MAP
Include BAR_1 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the BAR1 registers. Values: 0x0, 0x1 Default Value: 0x0 Enabled: BAR0_TYPE_0==0 && CC_DEVICE_TYPE!=CC_RC Parameter Type: Feature Setting Parameter Name: BAR1_ENABLED_n
BAR_1 is Memory or I/O (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines whether the BAR is for memory or I/O (bit 0 of the BAR). Values: ■ Memory (0x0) ■ I/O (0x1) Default Value: Memory Enabled: BAR1_ENABLED_0==1 Parameter Type: Register Default Setting Parameter Name: MEM1_SPACE_DECODER_n

Label	Description
BAR_1 is Prefetchable (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines if a memory BAR is for prefetchable memory (bit 3 of a memory BAR). Values: 0x0, 0x1 Default Value: 0x0 Enabled: BAR1_ENABLED_0==1 && MEM1_SPACE_DECODER_0==0 Parameter Type: Register Default Setting Parameter Name: PREFETCHABLE1_n
BAR_1 Bit Size (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the type as 32-bit or 64-bit; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: 32-bit Enabled: 0 Parameter Type: Register Default Setting Parameter Name: BAR1_TYPE_n
BAR_1 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	BAR1 Sizing Scheme ■ Fixed Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ Resizable BAR: Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ Resizable BAR (0x2) Default Value: Fixed Mask Enabled: BAR1_ENABLED_0==1 Parameter Type: Feature Setting Parameter Name: BAR1_SIZING_SCHEME_n

Label	Description
RBAR1 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for BAR1. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that BAR1 operates correctly with the BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xffff0 Default Value: 0x10 Enabled: CX_BAR1_RESIZABLE_0==1 Parameter Type: Register Default Setting Parameter Name: CX_BAR1_RESOURCE_AVAIL_n
BAR_1 Mask (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the default value of the BAR Mask register. The BAR Mask register specifies which bits of the BAR are non-writable by host software, which determines the size of the BAR. When the BAR is Resizable (BARN_SIZING_SCHEME_N =2), then the default value of this mask is determined from the CX_BARn_RESOURCE_AVAIL_N parameter. Values: 0xff, ..., 0xffffffff Default Value: CX_BAR1_RESIZABLE_0==1 ? [<functionof> CX_BAR1_RESOURCE_AVAIL_0] : MEM1_SPACE_DECODER_0==1 ? 0xFF : 0xFFFF Enabled: BAR1_ENABLED_0==1 && MEM1_SPACE_DECODER_0==0 && CX_BAR1_RESIZABLE_0==0 Parameter Type: Register Default Setting Parameter Name: BAR1_MASK_n
Specify Target Interface for BAR_1 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match BAR_1 to either RTRGT0 or RTRGT1. For example, setting MEM_FUNC0_BAR1_TARGET_MAP to 1 maps all incoming requests for BAR1 of function 0 to RTRGT1. If TRGT1_POPULATE = 0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: ((BAR1_ENABLED_0==1 && TRGT1_POPULATE==1) && CC_DEVICE_TYPE!=CC_RC) Parameter Type: Register Default Setting Parameter Name: MEM_FUNCn_BAR1_TARGET_MAP

Label	Description
VF BAR_0 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	VF BAR0 Sizing Scheme. ■ Fixed Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ VF Resizable BAR: VF Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ VF Resizable BAR (0x2) Default Value: Fixed Mask Enabled: VF_BAR0_ENABLED_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR0_SIZING_SCHEME_n
VF RBAR0 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for VF BAR0. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that VF BAR0 operates correctly with the VF BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xfffffffff0 Default Value: 0x10 Enabled: VF_BAR0_RESIZABLE_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR0_RESOURCE_AVAIL_n

Label	Description
VF BAR_1 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	VF BAR1 Sizing Scheme. ■ Fixed Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ VF Resizable BAR: VF Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ VF Resizable BAR (0x2) Default Value: Fixed Mask Enabled: VF_BAR1_ENABLED_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR1_SIZING_SCHEME_n
VF BAR_2 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	VF BAR2 Sizing Scheme. ■ Fixed Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ VF Resizable BAR: VF Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ VF Resizable BAR (0x2) Default Value: Fixed Mask Enabled: VF_BAR2_ENABLED_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR2_SIZING_SCHEME_n

Label	Description
VF BAR_3 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	VF BAR3 Sizing Scheme. ■ Fixed Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ VF Resizable BAR: VF Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ VF Resizable BAR (0x2) Default Value: Fixed Mask Enabled: VF_BAR3_ENABLED_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR3_SIZING_SCHEME_n
VF BAR_4 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	VF BAR4 Sizing Scheme. ■ Fixed Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ VF Resizable BAR: VF Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ VF Resizable BAR (0x2) Default Value: Fixed Mask Enabled: VF_BAR4_ENABLED_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR4_SIZING_SCHEME_n

Label	Description
VF BAR_5 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	VF BAR5 Sizing Scheme. ■ Fixed Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the VF BAR mask value using the VF_BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ VF Resizable BAR: VF Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ VF Resizable BAR (0x2) Default Value: Fixed Mask Enabled: VF_BAR5_ENABLED_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR5_SIZING_SCHEME_n
BAR_2 / BAR_3	
Include BAR_2 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the BAR2 registers. Values: 0x0, 0x1 Default Value: (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_SW) ? 0 : 1 Enabled: CC_DEVICE_TYPE!=CC_RC && CC_DEVICE_TYPE!=CC_SW Parameter Type: Feature Setting Parameter Name: BAR2_ENABLED_n
BAR_2 is Memory or I/O (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines whether the BAR is for memory or I/O (bit 0 of the BAR). Values: ■ Memory (0x0) ■ I/O (0x1) Default Value: Memory Enabled: BAR2_ENABLED_0==1 Parameter Type: Register Default Setting Parameter Name: MEM2_SPACE_DECODER_n
BAR_2 is Prefetchable (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines if a memory BAR is for prefetchable memory (bit 3 of a memory BAR). Values: 0x0, 0x1 Default Value: 0x0 Enabled: BAR2_ENABLED_0==1 && MEM2_SPACE_DECODER_0==0 Parameter Type: Register Default Setting Parameter Name: PREFETCHABLE2_n

Label	Description
BAR_2 Bit Size (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the type as 32-bit or 64-bit; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: 32-bit Enabled: BAR2_ENABLED_0==1 && MEM2_SPACE_DECODER_0==0 Parameter Type: Register Default Setting Parameter Name: BAR2_TYPE_n
BAR_2 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	BAR2 Sizing Scheme ■ Fixed Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ Resizable BAR: Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ Resizable BAR (0x2) Default Value: Fixed Mask Enabled: BAR2_ENABLED_0==1 Parameter Type: Feature Setting Parameter Name: BAR2_SIZING_SCHEME_n
RBAR2 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for BAR2. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that BAR2 operates correctly with the BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xfffffffff0 Default Value: 0x10 Enabled: CX_BAR2_RESIZABLE_0==1 Parameter Type: Register Default Setting Parameter Name: CX_BAR2_RESOURCE_AVAIL_n

Label	Description
BAR_2 Mask (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the default value of the BAR Mask register. The BAR Mask register specifies which bits of the BAR are non-writable by host software, which determines the size of the BAR. When the BAR is Resizable (BARn_SIZING_SCHEME_N =2), then the default value of this mask is determined from the CX_BARn_RESOURCE_AVAIL_N parameter. Values: 0xff, ..., 0xfffffffffffff Default Value: CX_BAR2_RESIZABLE_0==1 ? [<functionof> CX_BAR2_RESOURCE_AVAIL_0] : MEM2_SPACE_DECODER_0==1 ? 0xFF : 0xFFFFF Enabled: BAR2_ENABLED_0==1 && MEM2_SPACE_DECODER_0==0 && CX_BAR2_RESIZABLE_0==0 Parameter Type: Register Default Setting Parameter Name: BAR2_MASK_n
Specify Target Interface for BAR_2 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match BAR_2 to either RTRGT0 or RTRGT1. For example, setting MEM_FUNC0_BAR2_TARGET_MAP to 1 maps all incoming requests for BAR2 of function 0 to RTRGT1. If TRGT1_POPULATE = 0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: BAR2_ENABLED_0==1 && TRGT1_POPULATE==1 Parameter Type: Feature Setting Parameter Name: MEM_FUNCn_BAR2_TARGET_MAP
Include BAR_3 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the BAR3 registers. Values: 0x0, 0x1 Default Value: 0x0 Enabled: BAR2_TYPE_0==0 Parameter Type: Feature Setting Parameter Name: BAR3_ENABLED_n
BAR_3 is Memory or I/O (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines whether the BAR is for memory or I/O (bit 0 of the BAR). Values: ■ Memory (0x0) ■ I/O (0x1) Default Value: Memory Enabled: BAR3_ENABLED_0==1 Parameter Type: Register Default Setting Parameter Name: MEM3_SPACE_DECODER_n

Label	Description
BAR_3 is Prefetchable (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines if a memory BAR is for prefetchable memory (bit 3 of a memory BAR). Values: 0x0, 0x1 Default Value: 0x0 Enabled: BAR3_ENABLED_0==1 && MEM3_SPACE_DECODER_0==0 Parameter Type: Register Default Setting Parameter Name: PREFETCHABLE3_n
BAR_3 Bit Size (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the type as 32-bit or 64-bit; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: 32-bit Enabled: 0 Parameter Type: Register Default Setting Parameter Name: BAR3_TYPE_n
BAR_3 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	BAR3 Sizing Scheme ■ Fixed Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ Resizable BAR: Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ Resizable BAR (0x2) Default Value: Fixed Mask Enabled: BAR3_ENABLED_0==1 Parameter Type: Feature Setting Parameter Name: BAR3_SIZING_SCHEME_n

Label	Description
RBAR3 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for BAR3. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that BAR3 operates correctly with the BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xffff0 Default Value: 0x10 Enabled: CX_BAR3_RESIZABLE_0==1 Parameter Type: Register Default Setting Parameter Name: CX_BAR3_RESOURCE_AVAIL_n
BAR_3 Mask (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the default value of the BAR Mask register. The BAR Mask register specifies which bits of the BAR are non-writable by host software, which determines the size of the BAR. When the BAR is Resizable (BARn_SIZING_SCHEME_N =2), then the default value of this mask is determined from the CX_BARn_RESOURCE_AVAIL_N parameter. Values: 0xff, ..., 0xffffffff Default Value: CX_BAR3_RESIZABLE_0==1 ? [<functionof> CX_BAR3_RESOURCE_AVAIL_0] : MEM3_SPACE_DECODER_0==1 ? 0xFF : 0xFFFF Enabled: BAR3_ENABLED_0==1 && MEM3_SPACE_DECODER_0==0 && CX_BAR3_RESIZABLE_0==0 Parameter Type: Register Default Setting Parameter Name: BAR3_MASK_n
Specify Target Interface for BAR_3 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match BAR_3 to either RTRGT0 or RTRGT1. For example, setting MEM_FUNC0_BAR3_TARGET_MAP to 1 maps all incoming requests for BAR3 of function 0 to RTRGT1. If TRGT1_POPULATE = 0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: BAR3_ENABLED_0==1 && TRGT1_POPULATE==1 Parameter Type: Register Default Setting Parameter Name: MEM_FUNCn_BAR3_TARGET_MAP

Label	Description
VF RBAR2 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for VF BAR2. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that VF BAR2 operates correctly with the VF BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xffffffff0 Default Value: 0x10 Enabled: VF_BAR2_RESIZABLE_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR2_RESOURCE_AVAIL_n
BAR_4 / BAR_5	
Include BAR_4 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the BAR4 registers. Values: 0x0, 0x1 Default Value: (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_SW) ? 0 : 1 Enabled: Always Parameter Type: Feature Setting Parameter Name: BAR4_ENABLED_n
BAR_4 is Memory or I/O (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines whether the BAR is for memory or I/O (bit 0 of the BAR). Values: ■ Memory (0x0) ■ I/O (0x1) Default Value: CC_DEVICE_TYPE!=CC_RC && BAR4_ENABLED_0==1 Enabled: BAR4_ENABLED_0==1 Parameter Type: Register Default Setting Parameter Name: MEM4_SPACE_DECODER_n
BAR_4 is Prefetchable (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines if a memory BAR is for prefetchable memory (bit 3 of a memory BAR). Values: 0x0, 0x1 Default Value: 0x0 Enabled: BAR4_ENABLED_0==1 && MEM4_SPACE_DECODER_0==0 Parameter Type: Register Default Setting Parameter Name: PREFETCHABLE4_n

Label	Description
BAR_4 Bit Size (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the type as 32-bit or 64-bit; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: 32-bit Enabled: BAR4_ENABLED_0==1 && MEM4_SPACE_DECODER_0==0 Parameter Type: Register Default Setting Parameter Name: BAR4_TYPE_n
BAR_4 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	BAR4 Sizing Scheme ■ Fixed Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ Resizable BAR: Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ Resizable BAR (0x2) Default Value: Fixed Mask Enabled: BAR4_ENABLED_0==1 Parameter Type: Feature Setting Parameter Name: BAR4_SIZING_SCHEME_n
RBAR4 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for BAR4. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that BAR4 operates correctly with the BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xfffffffff0 Default Value: 0x10 Enabled: CX_BAR4_RESIZABLE_0==1 Parameter Type: Register Default Setting Parameter Name: CX_BAR4_RESOURCE_AVAIL_n

Label	Description
BAR_4 Mask (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the default value of the BAR Mask register. The BAR Mask register specifies which bits of the BAR are non-writable by host software, which determines the size of the BAR. When the BAR is Resizable (BARn_SIZING_SCHEME_N =2), then the default value of this mask is determined from the CX_BARn_RESOURCE_AVAIL_N parameter. Values: 0xff, ..., 0xfffffffffffff Default Value: CX_BAR4_RESIZABLE_0==1 ? [<functionof> CX_BAR4_RESOURCE_AVAIL_0] : MEM4_SPACE_DECODER_0==1 ? 0xFF : 0xFFF Enabled: BAR4_ENABLED_0==1 && MEM4_SPACE_DECODER_0==0 && CX_BAR4_RESIZABLE_0==0 Parameter Type: Register Default Setting Parameter Name: BAR4_MASK_n
Specify Target Interface for BAR_4 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match BAR_4 to either RTRGT0 or RTRGT1. For example, setting MEM_FUNC0_BAR4_TARGET_MAP to 1 maps all incoming requests for BAR4 of function 0 to RTRGT1. If TRGT1_POPULATE = 0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: BAR4_ENABLED_0==1 && TRGT1_POPULATE==1 Parameter Type: Feature Setting Parameter Name: MEM_FUNCn_BAR4_TARGET_MAP
Include BAR_5 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the BAR5 registers. Values: 0x0, 0x1 Default Value: 0x0 Enabled: BAR4_TYPE_0==0 Parameter Type: Feature Setting Parameter Name: BAR5_ENABLED_n
BAR_5 is Memory or I/O (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines whether the BAR is for memory or I/O (bit 0 of the BAR). Values: ■ Memory (0x0) ■ I/O (0x1) Default Value: Memory Enabled: BAR5_ENABLED_0==1 Parameter Type: Register Default Setting Parameter Name: MEM5_SPACE_DECODER_n

Label	Description
BAR_5 is Prefetchable (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines if a memory BAR is for prefetchable memory (bit 3 of a memory BAR). Values: 0x0, 0x1 Default Value: 0x0 Enabled: BAR5_ENABLED_0==1 && MEM5_SPACE_DECODER_0==0 Parameter Type: Register Default Setting Parameter Name: PREFETCHABLE5_n
BAR_5 Bit Size (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the type as 32-bit or 64-bit; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: 32-bit Enabled: 0 Parameter Type: Register Default Setting Parameter Name: BAR5_TYPE_n
BAR_5 Sizing Scheme (PF#i) (for n = 0; n <= CX_NFUNC-1)	BAR5 Sizing Scheme ■ Fixed Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You cannot change it in hardware. ■ Programmable Mask: You set the BAR mask value using the BARn_MASK_N configuration parameter. You can change it in hardware by writing to the Mask Register through the local DBI (only). Increases the register count. ■ Resizable BAR: Resizable BAR feature as defined in the PCIe Specification. You can change the 'Supported Resource Sizes' field in the Control Register through the local DBI (only). Values: ■ Fixed Mask (0x0) ■ Programmable Mask (0x1) ■ Resizable BAR (0x2) Default Value: Fixed Mask Enabled: BAR5_ENABLED_0==1 Parameter Type: Feature Setting Parameter Name: BAR5_SIZING_SCHEME_n

Label	Description
RBAR5 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for BAR5. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that BAR5 operates correctly with the BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xffff0 Default Value: 0x10 Enabled: CX_BAR5_RESIZABLE_0==1 Parameter Type: Register Default Setting Parameter Name: CX_BAR5_RESOURCE_AVAIL_n
BAR_5 Mask (PF#i) (for n = 0; n <= CX_NFUNC-1)	Determines the default value of the BAR Mask register. The BAR Mask register specifies which bits of the BAR are non-writable by host software, which determines the size of the BAR. When the BAR is Resizable (BARn_SIZING_SCHEME_N =2), then the default value of this mask is determined from the CX_BARn_RESOURCE_AVAIL_N parameter. Values: 0xff, ..., 0xffffffff Default Value: CX_BAR5_RESIZABLE_0==1 ? [<functionof> CX_BAR5_RESOURCE_AVAIL_0] : MEM5_SPACE_DECODER_0==1 ? 0xFF : 0xFFFF Enabled: BAR5_ENABLED_0==1 && MEM5_SPACE_DECODER_0==0 && CX_BAR5_RESIZABLE_0==0 Parameter Type: Register Default Setting Parameter Name: BAR5_MASK_n
Specify Target Interface for BAR_5 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match BAR_5 to either RTRGT0 or RTRGT1. For example, setting MEM_FUNC0_BAR5_TARGET_MAP to 1 maps all incoming requests for BAR5 of function 0 to RTRGT1. If TRGT1_POPULATE = 0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: BAR5_ENABLED_0==1 && TRGT1_POPULATE==1 Parameter Type: Feature Setting Parameter Name: MEM_FUNCn_BAR5_TARGET_MAP

Label	Description
VF RBAR4 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for VF BAR4. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that VF BAR4 operates correctly with the VF BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xfffffffff0 Default Value: 0x10 Enabled: VF_BAR4_RESIZABLE_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR4_RESOURCE_AVAIL_n
VF RBAR1 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for VF BAR1. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that VF BAR1 operates correctly with the VF BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xfffffffff0 Default Value: 0x10 Enabled: VF_BAR1_RESIZABLE_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR1_RESOURCE_AVAIL_n

Label	Description
VF RBAR3 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for VF BAR3. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that VF BAR3 operates correctly with the VF BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xfffffffff0 Default Value: 0x10 Enabled: VF_BAR3_RESIZABLE_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR3_RESOURCE_AVAIL_n
VF RBAR5 Usable Resource Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Indicates usable resource sizes for VF BAR5. ■ 3-0: RsvdP ■ 4: 1MB ■ 5: 2MB ■ . ■ . ■ . ■ 22: 256GB ■ 23: 512GB ■ 31-24: RsvdP Example: 0x00000030 indicates that VF BAR5 operates correctly with the VF BAR sized to either 1MB or 2MB. Values: 0x10, ..., 0xfffffffff0 Default Value: 0x10 Enabled: VF_BAR5_RESIZABLE_n==1 Parameter Type: Feature Setting Parameter Name: VF_BAR5_RESOURCE_AVAIL_n

1.43 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / SR-IOV Register Configuration PF0 Parameters

Table 1-43 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / SR-IOV Register Configuration PF0 Parameters

Label	Description
SR-IOV Register Configuration PF0	
Number of Virtual Functions (PF#i) (for n = 0; n <= CX_NFUNC)	Number of virtual functions supported for this physical function. Default value of the InitialVFs field in the SR-IOV Capability registers. When Extensible Virtual Function (CX_EXTENSIBLE_VFUNC) is enabled, this parameter indicates the total of both internal VFs and external VFs. Values: 0x0, ..., SNPS_RSVDPARAM_61 Default Value: 0x2 Enabled: CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: CX_MAX_VF_n
Virtual Function Dependency Link (PF#i) (for n = 0; n <= CX_NFUNC-1)	Specifies the 8-bit VF dependency link for the PF. Valid only if CX_NFUNC>1 and CX_VF_DEPENDENCY_LINK_SUPP=1. Values: 0x0, ..., 0xff Default Value: 0x0 Enabled: CX_VF_DEPENDENCY_LINK_SUPP && CX_SRIOV_ENABLE && (CX_NFUNC > 1) Parameter Type: Feature Setting Parameter Name: CX_VF_DEPENDENCY_LINK_n
First VF Offset with ARI (PF#i) (for n = 0; n <= CX_NFUNC-1)	Specifies the Routing ID offset of the first VF in this physical function in an ARI capable hierarchy. When CX_VF_STRIDE_ALWAYS_ONE=1 this parameter is disabled because it is not used. Values: 0x0, ..., 0xffff Default Value: (CX_NFUNC>=33) ? 64 : (CX_NFUNC>=17) ? 32 : (CX_NFUNC>=9) ? 16 : (CX_NFUNC>=5) ? 8 : (CX_NFUNC>=3) ? 4 : CX_NFUNC Enabled: CX_SRIOV_ENABLE && !CX_VF_STRIDE_ALWAYS_ONE Parameter Type: Register Default Setting Parameter Name: FIRST_VF_OFFSET_ARI_CAP_HIER1_n
First VF Offset without ARI (PF#i) (for n = 0; n <= CX_NFUNC-1)	Specifies the Routing ID offset of the first VF in this physical function in a non-ARI capable hierarchy. When CX_VF_STRIDE_ALWAYS_ONE=1 this parameter is disabled because it is not used. Values: 0x0, ..., 0xffff Default Value: 0x100 Enabled: CX_SRIOV_ENABLE && !CX_VF_STRIDE_ALWAYS_ONE Parameter Type: Register Default Setting Parameter Name: FIRST_VF_OFFSET_ARI_CAP_HIER0_n

Label	Description
VF Stride with ARI (PF#i) (for n = 0; n <= CX_NFUNC-1)	Specifies the Routing ID stride from one VF to the next one in this physical function in an ARI capable hierarchy Values: 0x0, ..., 0xffff Default Value: CX_VF_STRIDE_ALWAYS_ONE) ? 1 : ((CX_NFUNC>=33) ? 64 : (CX_NFUNC>=17) ? 32 : (CX_NFUNC>=9) ? 16 : (CX_NFUNC>=5) ? 8 : (CX_NFUNC>=3) ? 4 : CX_NFUNC Enabled: CX_SRIOV_ENABLE && !CX_VF_STRIDE_ALWAYS_ONE Parameter Type: Register Default Setting Parameter Name: VF_STRIDE_ARI_CAP_HIER1_n
VF Stride without ARI (PF#i) (for n = 0; n <= CX_NFUNC-1)	Specifies the Routing ID offset from one VF to the next one in this physical function in a non-ARI capable hierarchy Values: 0x0, ..., 0xffff Default Value: (CX_VF_STRIDE_ALWAYS_ONE) ? 0x0001 : 0x0100 Enabled: CX_SRIOV_ENABLE && !CX_VF_STRIDE_ALWAYS_ONE Parameter Type: Register Default Setting Parameter Name: VF_STRIDE_ARI_CAP_HIER0_n
Virtual Function Device Identification Number (PF#i) (for n = 0; n <= CX_NFUNC-1)	Specifies the 16-bit virtual function Device ID for this physical function. It may be different from the physical function's. Values: 0x0, ..., 0xffff Default Value: CX_DEVICE_ID_0 Enabled: CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: CX_VF_DEVICE_ID_n
Virtual Function Supported Page Sizes (PF#i) (for n = 0; n <= CX_NFUNC-1)	Specifies the 32-bit supported page sizes for virtual functions for the physical function. Values: 0x553, ..., 0xffff Default Value: 0x553 Enabled: CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: CX_VF_SUPP_PAGE_SIZE_n
ARI Capable Hierarchy Preserved (PF#i) (for n = 0; n <= CX_NFUNC-1)	Only present in PF0. Read Only Zero in other PFs. Specifies if ARI Capable Hierarchy bit is preserved across certain power state transitions Values: 0x0, 0x1 Default Value: CX_SRIOV_ENABLE Enabled: CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: CX_ARI_CAP_HIER_PRSD_n

1.44 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / Virtual Function BARs for PF0 Parameters

Table 1-44 Per-Function PCIe Capabilities and BAR Config / Physical Function 0 (PF0) / Virtual Function BARs for PF0 Parameters

Label	Description
Virtual Function BAR_0 / BAR_1 for PF0	
Include VF BAR_0 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the VF BAR0 registers . Values: 0x0, 0x1 Default Value: (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_SW) ? 0 : 1 Enabled: (CC_DEVICE_TYPE!=CC_RC) && CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: VF_BAR0_ENABLED_n
PF#i VF BAR_0 is Prefetchable (for n = 0; n <= CX_NFUNC-1)	Virtual Function BAR0 Memory Prefetchable. When set indicates VF BAR0 Memory BAR is a prefetchable BAR Values: 0x0, 0x1 Default Value: 0x0 Enabled: VF_BAR0_ENABLED_0==1 && CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: VF_PREFETCHABLE0_n
PF#i VF BAR_0 Bit Size (for n = 0; n <= CX_NFUNC-1)	Determines the BAR type as 32-bit or 64-bit in function 0; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: (VF_BAR0_ENABLED_0==0) ? 0 : 2 Enabled: VF_BAR0_ENABLED_0==1 && CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: VF_BAR0_TYPE_n
PF#i VF BAR_0 Mask (for n = 0; n <= CX_NFUNC-1)	Virtual Function BAR0 Mask. Determines the default of the VF BAR Mask register. The VF BAR Mask register specifies which bits of the VF BAR are non-writable by host software, which determines the size of the BAR. Values: 0xff, ..., 0xffffffffffff Default Value: VF_BAR0_RESIZABLE_0==1?[<functionof> VF_BAR0_RESOURCE_AVAIL_0]:(VF_BAR0_TYPE_0==0 ? 0xFFFF : 0xFFFFF) Enabled: VF_BAR0_ENABLED_0==1 && CX_SRIOV_ENABLE && VF_BAR0_RESIZABLE_0==0 Parameter Type: Register Default Setting Parameter Name: VF_BAR0_MASK_n

Label	Description
Specify Target Interface for PF#i VF BAR_0 (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match this VF BAR to either RTRGT0 or RTRGT1. For example, setting VF_MEM_FUNC0_BAR0_TARGET_MAP to 1 maps all incoming requests for VF BAR0 of function 0 to RTRGT1. If TRGT1_POPULATE = 0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. <i>Note:</i> This feature is not applicable for RC. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: ((VF_BAR0_ENABLED_0==1 && TRGT1_POPULATE==1) && CC_DEVICE_TYPE!=CC_RC) && CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: VF_MEM_FUNCn_BAR0_TARGET_MAP
Include VF BAR_1 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the VF BAR1 registers . Values: 0x0, 0x1 Default Value: 0x0 Enabled: (VF_BAR0_TYPE_0==0) && (CC_DEVICE_TYPE!=CC_RC) && CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: VF_BAR1_ENABLED_n
PF#i VF BAR_1 is Prefetchable (for n = 0; n <= CX_NFUNC-1)	Virtual Function BAR1 Memory Prefetchable. When set indicates VF BAR1 Memory BAR is a prefetchable BAR Values: 0x0, 0x1 Default Value: 0x0 Enabled: VF_BAR1_ENABLED_0==1 && CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: VF_PREFETCHABLE1_n
PF#i VF BAR_1 Bit Size (for n = 0; n <= CX_NFUNC-1)	Determines the BAR type as 32-bit or 64-bit in function 0; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: 32-bit Enabled: 0 Parameter Type: Register Default Setting Parameter Name: VF_BAR1_TYPE_n

Label	Description
PF#i VF BAR_1 Mask (for n = 0; n <= CX_NFUNC-1)	Virtual Function BAR1 Mask. Determines the default of the VF BAR Mask register. The VF BAR Mask register specifies which bits of the VF BAR are non-writable by host software, which determines the size of the BAR Values: 0xff, ..., 0xffffffff Default Value: VF_BAR1_RESIZABLE_0==1?[<functionof> VF_BAR1_RESOURCE_AVAIL_0]: 0xFFFF Enabled: VF_BAR1_ENABLED_0==1 && CX_SRIOV_ENABLE && VF_BAR1_RESIZABLE_0==0 Parameter Type: Register Default Setting Parameter Name: VF_BAR1_MASK_n
Specify Target Interface for PF#i VF BAR_1 (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match this VF BAR to either RTRGT0 or RTRGT1. For example, setting VF_MEM_FUNC0_BAR0_TARGET_MAP to 1 maps all incoming requests for VF BAR0 of function 0 to RTRGT1. If TRGT1_POPULATE=0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. <i>Note:</i> This feature is not applicable for RC. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: ((VF_BAR1_ENABLED_0==1 && TRGT1_POPULATE==1) && CC_DEVICE_TYPE!=CC_RC) && CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: VF_MEM_FUNCn_BAR1_TARGET_MAP
Virtual Function BAR_2 / BAR_3 for PF0	
Include VF BAR_2 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the VF BAR2 registers . Values: 0x0, 0x1 Default Value: (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_SW) ? 0 : 1 Enabled: CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: VF_BAR2_ENABLED_n
PF#i VF BAR_2 is Prefetchable (for n = 0; n <= CX_NFUNC-1)	BAR2 Memory Prefetchable. When set indicates VF BAR2 Memory BAR is a prefetchable BAR Values: 0x0, 0x1 Default Value: 0x0 Enabled: VF_BAR2_ENABLED_0==1 && CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: VF_PREFETCHABLE2_n

Label	Description
PF#i VF BAR_2 Bit Size (for n = 0; n <= CX_NFUNC-1)	Determines the BAR type as 32-bit or 64-bit in function 0; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: 32-bit Enabled: VF_BAR2_ENABLED_0==1 && CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: VF_BAR2_TYPE_n
PF#i VF BAR_2 Mask (for n = 0; n <= CX_NFUNC-1)	Virtual Function BAR2 Mask. Determines the default of the VF BAR Mask register. The VF BAR Mask register specifies which bits of the VF BAR are non-writable by host software, which determines the size of the BAR Values: 0xff, ..., 0xffffffffffff Default Value: VF_BAR2_RESIZABLE_0==1? [<functionof> VF_BAR2_RESOURCE_AVAIL_0]: (VF_BAR2_TYPE_0==0 ? 0xFFFF : 0xFFFFF) Enabled: VF_BAR2_ENABLED_0==1 && CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: VF_BAR2_MASK_n
Specify Target Interface for PF#i VF BAR_2 (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match this VF BAR to either RTRGT0 or RTRGT1. For example, setting VF_MEM_FUNC0_BAR0_TARGET_MAP to 1 maps all incoming requests for VF BAR0 of function 0 to RTRGT1. If TRGT1_POPULATE =0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: VF_BAR2_ENABLED_0==1 && TRGT1_POPULATE==1 && CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: VF_MEM_FUNCn_BAR2_TARGET_MAP
Include VF BAR_3 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the VF BAR3 registers . Values: 0x0, 0x1 Default Value: 0x0 Enabled: VF_BAR2_TYPE_0==0 && CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: VF_BAR3_ENABLED_n

Label	Description
PF#i VF BAR_3 is Prefetchable (for n = 0; n <= CX_NFUNC-1)	Virtual Function BAR3 Memory Prefetchable. When set indicates VF BAR3 Memory BAR is a prefetchable BAR Values: 0x0, 0x1 Default Value: 0x0 Enabled: VF_BAR3_ENABLED_0==1 && CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: VF_PREFETCHABLE3_n
PF#i VF BAR_3 Bit Size (for n = 0; n <= CX_NFUNC-1)	Determines the BAR type as 32-bit or 64-bit in function 0; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: 32-bit Enabled: 0 Parameter Type: Register Default Setting Parameter Name: VF_BAR3_TYPE_n
PF#i VF BAR_3 Mask (for n = 0; n <= CX_NFUNC-1)	Virtual Function BAR3 Mask. Determines the default of the VF BAR Mask register. The VF BAR Mask register specifies which bits of the VF BAR are non-writable by host software, which determines the size of the BAR Values: 0xff, ..., 0xffffffff Default Value: VF_BAR3_RESIZABLE_0==1?[<functionof> VF_BAR3_RESOURCE_AVAIL_0]: 0xFFFF Enabled: VF_BAR3_ENABLED_0==1 && CX_SRIOV_ENABLE && VF_BAR3_RESIZABLE_0==0 Parameter Type: Register Default Setting Parameter Name: VF_BAR3_MASK_n
Specify Target Interface for PF#i VF BAR_3 (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match this VF BAR to either RTRGT0 or RTRGT1. For example, setting VF_MEM_FUNC0_BAR0_TARGET_MAP to 1 maps all incoming requests for VF BAR0 of function 0 to RTRGT1. If TRGT1_POPULATE =0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: VF_BAR3_ENABLED_0==1 && TRGT1_POPULATE==1 && CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: VF_MEM_FUNCn_BAR3_TARGET_MAP

Label	Description
Virtual Function BAR_4 / BAR_5 for PF0	
Include VF BAR_4 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the VF BAR4 registers . Values: 0x0, 0x1 Default Value: (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_SW) ? 0 : 1 Enabled: CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: VF_BAR4_ENABLED_n
PF#i VF BAR_4 is Prefetchable (for n = 0; n <= CX_NFUNC-1)	Virtual Function BAR4 Memory Prefetchable. When set indicates VF BAR4 Memory BAR is a prefetchable BAR Values: 0x0, 0x1 Default Value: 0x0 Enabled: VF_BAR4_ENABLED_0==1 && CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: VF_PREFETCHABLE4_n
PF#i VF BAR_4 Bit Size (for n = 0; n <= CX_NFUNC-1)	Determines the BAR type as 32-bit or 64-bit in function 0; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: 32-bit Enabled: VF_BAR4_ENABLED_0==1 && CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: VF_BAR4_TYPE_n
PF#i VF BAR_4 Mask (for n = 0; n <= CX_NFUNC-1)	Virtual Function BAR4 Mask. Determines the default of the VF BAR Mask register. The VF BAR Mask register specifies which bits of the VF BAR are non-writable by host software, which determines the size of the BAR Values: 0xff, ..., 0xffffffffffff Default Value: VF_BAR4_RESIZABLE_0==1?[<functionof> VF_BAR4_RESOURCE_AVAIL_0]: 0xFFF Enabled: VF_BAR4_ENABLED_0==1 && CX_SRIOV_ENABLE && VF_BAR4_RESIZABLE_0==0 Parameter Type: Register Default Setting Parameter Name: VF_BAR4_MASK_n

Label	Description
Specify Target Interface for PF#i VF BAR_4 (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match this VF BAR to either RTRGT0 or RTRGT1. For example, setting VF_MEM_FUNC0_BAR0_TARGET_MAP to 1 maps all incoming requests for VF BAR0 of function 0 to RTRGT1. If TRGT1_POPULATE = 0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: VF_BAR4_ENABLED_0==1 && TRGT1_POPULATE==1 && CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: VF_MEM_FUNCn_BAR4_TARGET_MAP
Include VF BAR_5 (PF#i) (for n = 0; n <= CX_NFUNC-1)	Include the VF BAR5 registers . Values: 0x0, 0x1 Default Value: 0x0 Enabled: VF_BAR4_TYPE_0==0 && CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: VF_BAR5_ENABLED_n
PF#i VF BAR_5 is Prefetchable (for n = 0; n <= CX_NFUNC-1)	Virtual Function BAR5 Memory Prefetchable. When set indicates VF BAR5 Memory BAR is a prefetchable BAR Values: 0x0, 0x1 Default Value: 0x0 Enabled: VF_BAR5_ENABLED_0==1 && CX_SRIOV_ENABLE Parameter Type: Register Default Setting Parameter Name: VF_PREFETCHABLE5_n
PF#i VF BAR_5 Bit Size (for n = 0; n <= CX_NFUNC-1)	Determines the BAR type as 32-bit or 64-bit in function 0; (bits [2:1] for a memory BAR). Values: ■ 32-bit (0x0) ■ 64-bit (0x2) Default Value: 32-bit Enabled: 0 Parameter Type: Register Default Setting Parameter Name: VF_BAR5_TYPE_n

Label	Description
PF#i VF BAR_5 Mask (for n = 0; n <= CX_NFUNC-1)	Virtual Function BAR5 Mask. Determines the default of the VF BAR Mask register. The VF BAR Mask register specifies which bits of the VF BAR are non-writable by host software, which determines the size of the BAR Values: 0xff, ..., 0xffffffff Default Value: VF_BAR5_RESIZABLE_0==1?[<functionof> VF_BAR5_RESOURCE_AVAIL_0]: 0xFFFF Enabled: VF_BAR5_ENABLED_0==1 && CX_SRIOV_ENABLE && VF_BAR5_RESIZABLE_0==0 Parameter Type: Register Default Setting Parameter Name: VF_BAR5_MASK_n
Specify Target Interface for PF#i VF BAR_5 (for n = 0; n <= CX_NFUNC-1)	Direct incoming requests that pass filtering and match this VF BAR to either RTRGT0 or RTRGT1. For example, setting VF_MEM_FUNC0_BAR0_TARGET_MAP to 1 maps all incoming requests for VF BAR0 of function 0 to RTRGT1. If TRGT1_POPULATE =0 (no RTRGT1 interface), then the map-by-BAR parameters have no effect; all requests that pass filtering are routed to RTRGT0. Values: ■ Target_0 (0x0) ■ Target_1 (0x1) Default Value: (TRGT1_POPULATE==1) ? 1 : 0 Enabled: VF_BAR5_ENABLED_0==1 && TRGT1_POPULATE==1 && CX_SRIOV_ENABLE Parameter Type: Feature Setting Parameter Name: VF_MEM_FUNCn_BAR5_TARGET_MAP

1.45 Advanced RAM Config Parameters

Table 1-45 Advanced RAM Config Parameters

Label	Description
RAM Timing	
Single Port RAM Read Access Time [ps]	Specifies the single port RAM read access time. Used by the synthesis. Values: -2147483648, ..., 2147483647 Default Value: 800 Enabled: Always Parameter Type: Performance Setting Parameter Name: RAM1P_RD_ACCESS
Single Port RAM Address/Data Setup Time [ps]	Specifies the single port RAM data setup. Used by the synthesis". Values: -2147483648, ..., 2147483647 Default Value: 350 Enabled: Always Parameter Type: Performance Setting Parameter Name: RAM1P_ADDR_SU
Dual Port RAM Read Access Time [ps]	Specifies the dual port RAM read access time. Used by the synthesis. Values: -2147483648, ..., 2147483647 Default Value: 600 Enabled: Always Parameter Type: Performance Setting Parameter Name: RAM2P_RD_ACCESS
Dual Port RAM Address/Data Setup Time [ps]	Specifies the dual port RAM data setup. Used by the synthesis. Values: -2147483648, ..., 2147483647 Default Value: 350 Enabled: Always Parameter Type: Performance Setting Parameter Name: RAM2P_ADDR_SU

Label	Description
RAM Architecture	
Single-port RAM model architecture	There are two possible architectures for the single-port RAMs used in the PCIe controller: a) with a registered read port data output and b) with a registered read port address input. coreConsultant uses this parameter to select the architecture used in the single-port RAM models. It has no effect on the RTL implementation. Note: when DesignWare RAMs are selected only registered read port data architecture is supported for single-port RAMs 0: Registered read port data output RAM model 1: Registered read port address input RAM model Values: - Registered Read Port Data Output (0) - Registered Read Port Address Input (1) Default Value: Registered Read Port Data Output Enabled: Always Parameter Type: Feature Setting Parameter Name: RAM1P_ARCH
Dual-port RAM model architecture	There are two possible architectures for the dual-port RAMs used in the PCIe controller: a) with a registered read port data output and b) with a registered read port address input. coreConsultant uses this parameter to select the architecture used in the dual-port RAM models. It has no effect on the RTL implementation. 0: Registered read port data output RAM model 1: Registered read port address input RAM model Values: - Registered Read Port Data Output (0) - Registered Read Port Address Input (1) Default Value: Registered Read Port Data Output Enabled: Always Parameter Type: Feature Setting Parameter Name: RAM2P_ARCH
RAM Output Registering/Retiming	
Receive Queue RAM Output Pipeline	Specifies if a pipeline is inserted for the Receive Data/Header Queue RAM output signals. This is a read-only signal for informational purposes. A non-optional read data retiming register is always inserted on the RAM outputs. Values: - No (0) - Yes (1) Default Value: Yes Enabled: 0 Parameter Type: Performance Setting Parameter Name: CX_RAMQGRAM_REGOUT_

Label	Description
SOT RAM Output Pipeline	Specifies if a pipeline is inserted for the SOT RAM output signals. Values: ■ No (0) ■ Yes (1) Default Value: ((CX_MAX_CORECLK_FREQ >= 500) && CX_RASDP > 0) (CX_MAX_CORECLK_FREQ > 500) Enabled: Always Parameter Type: Performance Setting Parameter Name: CX_RETRYSTOTRAM_REGOUT
Retry RAM Output Pipeline	Specifies if a pipeline is inserted for the Retry RAM output signals. Values: ■ No (0) ■ Yes (1) Default Value: ((CX_MAX_CORECLK_FREQ >= 500) && CX_RASDP > 0) (CX_MAX_CORECLK_FREQ > 500) Enabled: Always Parameter Type: Performance Setting Parameter Name: CX_RETRYRAM_REGOUT
RAM Read Data Latency	
RADM Receive Queue RAM Read Cycle Latency	Specifies the Receive Queue Header and Data RAM read cycle latency. The controller samples the read data CX_RADM_RAM_RD_LATENCY clock cycles after the read address is valid at the Header/Data RADM RAM interface. <i>Note:</i> The RAM read data is always retimed/registered. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_RADMQ_MODE==2 Parameter Type: Performance Setting Parameter Name: CX_RADM_RAM_RD_LATENCY

Label	Description
Receive Serialization Queue RAM Read Cycle Latency	Specifies the Receive Serialization Queue RAM read cycle latency. The controller samples the read data CX_RADM_FORMQ_RAM_RD_LATENCY clock cycles after the read address is valid at the RADM FORMQ RAM interface. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_NW==16 Parameter Type: Performance Setting Parameter Name: CX_RADM_FORMQ_RAM_RD_LATENCY
Retry RAM Read Cycle Latency	Specifies the Retry RAM read cycle latency. The controller samples the read data CX_RETRY_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) Default Value: 1 Cycle Enabled: Always Parameter Type: Performance Setting Parameter Name: CX_RETRY_RAM_RD_LATENCY
SOT RAM Read Cycle Latency	Specifies the Retry SOT RAM read cycle latency. The controller samples the read data CX_RETRY_SOT_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) Default Value: 1 Cycle Enabled: Always Parameter Type: Performance Setting Parameter Name: CX_RETRY_SOT_RAM_RD_LATENCY

Label	Description
AXI Master Completion Buffer RAM Read Latency	Specifies the MCB RAM read cycle latency. The controller samples the read data CC_MSTR_CPL_SEG_BUF_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: MASTER_POPULATED && AXI_POPULATED Parameter Type: Performance Setting Parameter Name: CC_MSTR_CPL_SEG_BUF_RAM_RD_LATENCY
AXI Slave Completion Composer RAM Read Latency	Specifies the AXI Slave Completion Composer RAM read cycle latency. The controller samples the read data CC_OB_CCMP_DATA_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: ((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4)) Parameter Type: Performance Setting Parameter Name: CC_OB_CCMP_DATA_RAM_RD_LATENCY
AXI Slave Posted Decomposer RAM Read Latency	Specifies the AXI Slave Posted Decomposer RAM read cycle latency. The controller samples the read data CC_OB_PDCMP_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: SLAVE_POPULATED && AXI_POPULATED Parameter Type: Performance Setting Parameter Name: CC_OB_PDCMP_RAM_RD_LATENCY

Label	Description
AXI Slave Non-Posted decomposer RAM Read Latency	Specifies the AXI Slave Non-Posted decomposer RAM read cycle latency. The controller samples the read data CC_OB_NPDCMP_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: SLAVE_POPULATED && AXI_POPULATED Parameter Type: Performance Setting Parameter Name: CC_OB_NPDCMP_RAM_RD_LATENCY
AXI Slave Non-Posted Write Set-Aside Buffer Read Latency	Specifies the AXI Slave Non-Posted Write Set-Aside Buffer read cycle latency. The controller samples the read data CC_SLV_NPW_SAB_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: SLAVE_POPULATED && AXI_POPULATED Parameter Type: Performance Setting Parameter Name: CC_SLV_NPW_SAB_RAM_RD_LATENCY
AXI Master Completion Buffer CDC RAM Read Latency	Specifies the AXI Master Completion Buffer CDC RAM read cycle latency. The controller samples the read data CC_IB_MCPL_CDC_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: MASTER_POPULATED && AXI_POPULATED Parameter Type: Performance Setting Parameter Name: CC_IB_MCPL_CDC_RAM_RD_LATENCY

Label	Description
AXI Inbound Read Request CDC RAM Read Latency	Specifies the AXI Inbound Read Request CDC RAM read cycle latency. The controller samples the read data CC_IB_RD_REQ_CDC_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: AXI_POPULATED && MASTER_POPULATED && MSTR_CLK_DIFF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IB_RD_REQ_CDC_RAM_RD_LATENCY
AXI Master Non-posted Ordering Queue RAM Read Latency	Specifies the AXI Master Non-posted Ordering Queue RAM read cycle latency. The controller samples the read data CC_IB_RD_REQ_ORDR_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: MASTER_POPULATED && AXI_POPULATED Parameter Type: Performance Setting Parameter Name: CC_IB_RD_REQ_ORDR_RAM_RD_LATENCY
AXI Inbound Write Request CDC RAM Read Latency	Specifies the AXI Inbound Write Request CDC RAM read cycle latency. The controller samples the read data CC_IB_WR_REQ_CDC_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: AXI_POPULATED && MASTER_POPULATED && MSTR_CLK_DIFF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IB_WR_REQ_CDC_RAM_RD_LATENCY

Label	Description
AXI Inbound Write Posted Request Tracker RAM Read Latency	Specifies the AXI Inbound Write Posted Request Tracker RAM read cycle latency. The controller samples the read data CC_IB_WR_REQ_PTRK_RAM_RD_LATENCY clock cycles after the read address is valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: AXI_POPULATED && MASTER_POPULATED && CC_IB_WREQ_PTRK_EN Parameter Type: Performance Setting Parameter Name: CC_IB_WR_REQ_PTRK_RAM_RD_LATENCY
DTIM AXI Stream Slave Request Queue RAM Read Cycle Latency	Specifies the DTIM AXI Stream Slave Request Queue RAM read cycle latency. The controller samples the read data CX_DTIM_REQQ_RAM_RD_LATENCY clock cycles after the read address is valid at the DTIM AXI Stream Slave Request Queue RAM interface. <i>Note:</i> The RAM read data is always retimed/registered. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CC_DTIM_ENABLE Parameter Type: Performance Setting Parameter Name: CC_DTIM_REQQ_RAM_RD_LATENCY
MSIX Table RAM Read Cycle Latency	Specifies the internal MSIX table RAM read cycle latency. The controller samples the read data CX_MSIX_TABLE_RAM_RD_LATENCY clock cycles after the read address is valid at the MSIX table RAM interface. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: MSIX_TABLE_EN==1 Parameter Type: Performance Setting Parameter Name: CX_MSIX_TABLE_RAM_RD_LATENCY

Label	Description
HDMA Read engine Linked List Controller RAM Read Cycle Latency	Specifies the RAM latency of the HDMA Linked List Controller of the Read engine. The controller samples the read data CC_IF_RDCTRL_LL_RAM_LATENCY clock cycles after the read address been valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IF_RDCTRL_LL_RAM_LATENCY
HDMA Write engine Linked List Controller RAM Read Cycle Latency	Specifies the RAM latency of the HDMA Linked List Controller of the Write engine. The controller samples the read data CC_IF_WRCTRL_LL_RAM_LATENCY clock cycles after the read address been valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IF_WRCTRL_LL_RAM_LATENCY
HDMA Write engine Pre-Fetched Descriptor Queue RAM Read Cycle Latency	Specifies the RAM latency of the HDMA Pre-Fetched Descriptor Queue of the Write engine. The controller samples the read data CC_IF_WRCTX_LLQ_OVERLAY_RAM_LATENCY clock cycles after the read address been valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IF_WRCTX_LLQ_OVERLAY_RAM_LATENCY

Label	Description
HDMA Read engine Pre-Fetched Descriptor Queue RAM Read Cycle Latency	Specifies the RAM latency of the HDMA Pre-Fetched Descriptor Queue of the Read engine. The controller samples the read data CC_IF_RDCTX_LLQ_OVERLAY_RAM_LATENCY clock cycles after the read address been valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IF_RDCTX_LLQ_OVERLAY_RAM_LATENCY
HDMA Read engine Memory Read Context LUT RAM Read Cycle Latency	Specifies the RAM latency of the HDMA Memory Read Context LUT of the Read engine. The controller samples the read data CC_IF_RD_CTXC2W_RAM_LATENCY clock cycles after the read address been valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IF_RD_CTXC2W_RAM_LATENCY
HDMA Write engine Memory Read Context LUT RAM Read Cycle Latency	Specifies the RAM latency of the HDMA Memory Read Context LUT of the Write engine. The controller samples the read data CC_IF_WR_CTXC2W_RAM_LATENCY clock cycles after the read address been valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IF_WR_CTXC2W_RAM_LATENCY

Label	Description
HDMA Read engine Memory Write Context LUT RAM Read Cycle Latency	Specifies the RAM latency of the HDMA Memory Write Context LUT of the Read engine. The controller samples the read data CC_IF_RD_CTXSTSH_RAM_LATENCY clock cycles after the read address been valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IF_RD_CTXSTSH_RAM_LATENCY
HDMA Write engine Memory Write Context LUT RAM Read Cycle Latency	Specifies the RAM latency of the HDMA Memory Write Context LUT of the Write engine. The controller samples the read data CC_IF_WR_CTXSTSH_RAM_LATENCY clock cycles after the read address been valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IF_WR_CTXSTSH_RAM_LATENCY
HDMA Read engine MSI setup RAM Read Cycle Latency	Specifies the RAM latency of the MSI setup of the Read engine. The controller samples the read data CC_IF_RD_MSI_RAM_LATENCY clock cycles after the read address been valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IF_RD_MSI_RAM_LATENCY

Label	Description
HDMA Write engine MSI setup RAM Read Cycle Latency	Specifies the RAM latency of the MSI setup of the Write engine. The controller samples the read data CC_IF_WR_MSI_RAM_LATENCY clock cycles after the read address been valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CX_DMA_PF_ENABLE Parameter Type: Performance Setting Parameter Name: CC_IF_WR_MSI_RAM_LATENCY
DMA Read Buffer Controller RAM Read Cycle Latency	Specifies the RAM latency of the DMA Read Buffer RAM. The controller samples the read data CC_DMA_SEG_BUF_RAM_LATENCY clock cycles after the read address been valid. The default of '1' corresponds to the typical read cycle access time for synchronous memories. Values: ■ 1 Cycle (1) ■ 2 Cycles (2) ■ 3 Cycles (3) ■ 4 Cycles (4) Default Value: 1 Cycle Enabled: CC_DMA_ENABLE Parameter Type: Performance Setting Parameter Name: CC_DMA_SEG_BUF_RAM_LATENCY

1.46 Advanced PHY Config / General Options Parameters

Table 1-46 Advanced PHY Config / General Options Parameters

Label	Description
General Options	
Elasticity Buffer Mode	Determines Elasticity Buffer Mode. For Original PIPE architecture, contact your PHY vendor for the value that you should use. Only applicable to configurations not using a Frequency Step. Values: ■ Nominal Half Full Buffer Mode (0) ■ Nominal Empty Buffer Mode (1) Default Value: CX_PCIE_MODE == PCIE_OVER_CIO) ? 1 : ([<functionof>] CX_PIPE_SERDES_ARCH) && (CX_FREQ_STEP_EN==0 CX_PIPERX_MULTI_BLOCK==1) && CX_GEN3_SPEED Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : (CX_CPCIE_ENABLE==1 && (CX_FREQ_STEP_EN==0 CX_PIPERX_MULTI_BLOCK==1) && CX_GEN3_SPEED && CX_PIPE_SERDES_ARCH==0) Parameter Type: Feature Setting Parameter Name: CX_PIPE_HYBRID_MODE
Selectable De-emphasis	Selectable De-emphasis. Only for Gen2 downstream ports operating at 5 GT/s. This bit selects the level of de-emphasis the link operates at. Values: ■ -6 dB (0x0) ■ -3.5 dB (0x1) Default Value: -6 dB Enabled: CX_GEN2_SPEED==1 && CX_CPCIE_ENABLE && CC_DEVICE_TYPE!=CC_EP Parameter Type: Feature Setting Parameter Name: SEL_DE_EMPHASIS
PIPE PCLK Change Optional Behavior	Enable the optional PIPE5 behavior for the PclkChangeOk/PclkChangeAck handshake. When enabled, the controller always performs a PclkChangeOk/PclkChangeAck handshake with the PHY whenever a rate change occurs, even if the new and old rate use the same Pclk frequency. PHY_INTEROP_CTRL_OFF.PIPE_OPTIONAL_PCLKCHANGE_HS register uses this parameter as a reset value. Values: ■ PCLK Change Optional Behavior Disabled (0) ■ PCLK Change Optional Behavior Enabled (1) Default Value: PCLK Change Optional Behavior Enabled Enabled: CX_PIPE_PCLK_MODE==1 Parameter Name: CX_PIPE_OPTIONAL_PCLKCHANGE_HS

Label	Description
PHY Perst on Warm Reset	Select the PHY Perst on Warm Reset value for the PMC to drive pm_req_phy_perst during Warm Resets (PERST# without cycling of power). PHY_INTEROP_CTRL_OFF.PHY_PERST_ON_WARM_RESET register uses this parameter as the reset value. Values: ■ DO NOT drive pm_req_phy_perst (0) ■ Drive pm_req_phy_perst during a Warm Reset (1) Default Value: CX_PHY_ENHANCED_PM_EN==1 ? 0 : 1 Enabled: Always Parameter Name: DEFAULT_PHY_PERST_ON_WARM_RESET
Max PCLK frequency (MHz)	Frequency (in MHz) of the max_pclk output clock from the PHY. Contact your PHY vendor for the value that you should use. Values: ■ MaxPclk not available (0) ■ 125 MHz (125) ■ 250 MHz (250) ■ 500 MHz (500) ■ 1 GHz (1000) ■ 2 GHz (2000) ■ 4 GHz (4000) Default Value: (CX_PIPE_PCLK_MODE==0 CX_PIPE_PCLK_MODE==2) ? 0 : (CX_PIPE_PCLK_MODE==3) ? 2000 : 1000 Enabled: Always Parameter Name: CX_PHY_MAX_PCLK_FREQ_MHZ
RxStandby Control	
RxStandby Signal Default	Default value of RxStandby signal (mac_phy_rxstandby) on all lanes after reset. <i>Note:</i> Contact your PHY vendor for guidelines on how to properly configure the RxStandby behavior. Refer to the <i>User Guide</i> for guidelines on how to configure this parameter for Synopsys PHYs. Values: ■ 0 (0x0) ■ 1 (0x1) Default Value: CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : (PHY_TYPE==CC_GENERIC_PHY Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : (CX_CPCIE_ENABLE && PHY_TYPE!=CC_GENERIC_PHY) Parameter Type: Feature Setting Parameter Name: CX_RXSTANDBY_DEFAULT

Label	Description
RxStandby Signal Operation Control	Bits 0..5 determine if the controller asserts the RxStandby signal (mac_phy_rxstandby) in the indicated condition. Bit 6 enables the controller to perform the RxStandby/RxStandbyStatus handshake. ■ [0]: Rx EIOS and subsequent T TX-IDLE-MIN ■ [1]: Rate Change ■ [2]: Inactive lane for upconfigure/downconfigure ■ [3]: PowerDown=P1,P1.CPM,P1.1,P1.2 or P2 (When CX_PIPE43_SUPPORT==1, this bit must be set to 0) ■ [4]: RxL0s.Idle ■ [5]: EI Infer in L0 ■ [6]: Execute RxStandby/RxStandbyStatus Handshake <i>Note:</i> Contact your PHY vendor for guidelines on how to properly configure the RxStandby behavior. Refer to the <i>User Guide</i> for guidelines on how to configure this parameter for Synopsys PHYs. Values: 0x0, ..., 0x7f Default Value: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : (PHY_TYPE==CC_GENERIC_PHY) ? 0x7F : (CX_PIPE_SERDES_ARCH) ? 0x7F : 0x44 Enabled: (CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : (CX_CPCIE_ENABLE && PHY_TYPE!=CC_GENERIC_PHY) Parameter Type: Feature Setting Parameter Name: CX_RXSTANDBY_CONTROL
Synopsys PHY Features	
PHY Register Viewport and PRBI/CRPI.	Enables PHY register viewport and PRBI/CRPI for Synopsys PHY. The controller implements the PHY register viewport in its Port Logic register space, and the Register Bus Interface (PRBI) which communicates with the Phy's Control Register Parallel Interface (CRPI), to enable access to registers in the PHY. The PHY registers can be indirectly written to and read from through a configuration wire access or local DBI access. This feature can be used with Synopsys PHYs that have the Control Register Parallel Interface and not for PIPE Message Bus Interface. For more information, see "PHY Port Logic Registers" in the "Register Module, LBC, and DBI" section in the Controller Operations chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: false Enabled: !(CX_PL_REG_DISABLE) && CX_CPCIE_ENABLE Parameter Type: Feature Setting Parameter Name: CX_PHY_VIEWPORT_ENABLE

Label	Description
P2.NoBeacon Enable	Enable support for P2.NoBeacon. When this parameter is set, mac_phy_powerdown drives P2.NoBeacon encoding, instead of P2 encoding, when the link goes to L2. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_PIPE_VER>=1 Parameter Type: Feature Setting Parameter Name: CX_P2NOBEACON_ENABLE
P2.NoBeacon Encoding	Powerdown state encoding on the mac_phy_powerdown[3:0] output for P2.NoBeacon. Contact your PHY vendor for the value that you should use. Values: 0x0, ..., 0xf Default Value: 0xf Enabled: CX_P2NOBEACON_ENABLE==1 Parameter Type: Feature Setting Parameter Name: CX_PIPE43_P2NOBEACON_ENCODING

1.47 Advanced PHY Config / PHY Timing Parameters

Table 1-47 Advanced PHY Config / PHY Timing Parameters

Label	Description
PHY Timing	
Number of Fast Training (NFTS) Sequences	Specifies the number of Fast Training Sequences the controller advertises during link training. This is used to inform the link partner of the controllers ability to recover synchronization after a low power state. This number should come from your PHY vendor. This parameter sets the default value of the ACK_N_FTS field of the 'Ack Frequency and L0-L1 ASPM Control Register'. For more information, see <i>Register Descriptions</i>. For SerDes architecture, this number should be the following: ■ CX_NFTS = 32 + the number of FTS that your PHY vendor requires to detect bit lock. Values: 1, ..., 255 Default Value: ((CX_PHY_NB>8) CX_PIPE_SERDES_ARCH) ? 32 : 15 Enabled: CX_S_CPCIE_MODE Parameter Type: Register Default Setting Parameter Name: CX_NFTS
NFTS when using common clock	Specifies the number of Fast Training Sequences (NFTS) the controller advertises during link training, when common clock configuration is set. This parameter sets the default value of the Common Clock ACK_N_FTS field of the 'Ack Frequency and L0-L1 ASPM Control Register'. For more information, see <i>Register Descriptions</i>. Common Clock operation cannot be fully enabled (through the Common Clock Configuration field of the Link Control register) unless you observe the following configuration parameter relationships: ■ CX_NFTS != CX_COMM_NFTS ■ DEFAULT_L0S_EXIT_LATENCY != DEFAULT_COMM_L0S_EXIT_LATENCY ■ DEFAULT_L1_EXIT_LATENCY != DEFAULT_COMM_L1_EXIT_LATENCY Default value is set automatically unless Custom PHY or Sample PHY is used. For SerDes architecture, this number should be the following: ■ CX_NFTS = 32 + the number of FTS that your PHY vendor requires to detect bit lock. Values: 1, ..., 255 Default Value: CX_NFTS Enabled: CX_CPCIE_ENABLE && SLOT_CLK_CONFIG Parameter Type: Register Default Setting Parameter Name: CX_COMM_NFTS

Label	Description
NFTS Sequences at 5 Gb/s	Specifies the number of Fast Training Sequences (NFTS) the controller advertises during link training when running at Gen 2 or higher speed. This is used to inform the link partner the controllers ability to recover synchronization after a low power state. This number should come from your PHY vendor. This parameter is for Conventional PCIe mode only. For SerDes architecture, this number should be the following: ■ $CX_NFTS = 32 + \text{the number of FTS that your PHY vendor requires to detect bit lock.}$ Values: 0, ..., 255 Default Value: CX_NFTS Enabled: (CX_CPCIE_ENABLE && (CX_GEN2_SPEED == 1)) Parameter Type: Feature Setting Parameter Name: DEFAULT_GEN2_N_FTS
PHY Tx Delay	PHY Transmitter delay in clock cycles. This must be specified by your PHY provider and is the (worst case) delay in PIPE clock cycles from the PIPE interface to the TX Phy serial pins. This parameter in conjunction with 'PHY Rx Delay' is used to calculate the delay in getting an ACK DLLP back from the link partner, for a TLP sent by your controller. It is then used to calculate the size of the Retry Buffer. This parameter is for Conventional PCIe mode only. Values: 0, ..., 500 Default Value: 6 Enabled: CX_S_CPCIE_MODE Parameter Type: Feature Setting Parameter Name: CX_PHY_TX_DELAY_PHY
PHY Rx Delay	PHY Receiver delay in clock cycles. This must be specified by your PHY provider and is the (worst case) delay in PIPE clock cycles from the RX Phy serial pins to the PIPE interface. This parameter in conjunction with 'PHY Tx Delay' is used to calculate the delay in getting an ACK DLLP back from the link partner, for a TLP sent by your controller. It is then used to calculate the size of the Retry Buffer. This parameter is for Conventional PCIe mode only. Values: 0, ..., 500 Default Value: 14 Enabled: CX_S_CPCIE_MODE Parameter Type: Feature Setting Parameter Name: CX_PHY_RX_DELAY_PHY

1.48 Advanced PHY Config / Gen3 PHY Equalization Config Parameters

Table 1-48 Advanced PHY Config / Gen3 PHY Equalization Config Parameters

Label	Description
General Config	
Gen3 controller as EQ Master Asserts RxEqEval for PHY to do Rx adaptation and Evaluation Regardless of Rx TS1s Detection	When set to '1', the controller as Gen3 EQ master asserts RxEqEval to instruct the PHY to do Rx adaptation and evaluation after a 500ns timeout from a new preset request. This is the default of the RXEQ_RGRDLESS_RXTS field in GEN3_RELATED_OFF when in Gen3 mode. Values: ■ No (0) ■ Yes (1) Default Value: Yes Enabled: ((CX_GEN3_MODE != GEN3_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_RXEQ_RGRDLESS_RXTS
Gen3 EQ Include Initial FOM	Gen3 EQ Include Initial FOM. Include or not the FOM feedback from the initial preset evaluation performed in the EQ Master, when finding the highest FOM among all preset evaluations. ■ 0: Do not include ■ 1: Include This is the default of the GEN3_EQ_FOM_INC_INITIAL_EVAL field in GEN3_EQ_CONTROL_OFF when in Gen3 mode. Values: 0, 1 Default Value: 0 Enabled: ((CX_GEN3_MODE != GEN3_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_EQ_FOM_INC_INITIAL_EVAL
Gen3 EQ Preset Request Vector	Gen3 EQ Preset Request Vector. Requesting of Presets during the initial part of the EQ Master Phase. Encoding scheme is as follows: ■ Bit [15:0] = 0x0: No preset is requested and evaluated in EQ Master Phase. ■ Bit [i] = 1: "Preset=i" is requested and evaluated in EQ Master Phase. This is the default of the GEN3_EQ_PSET_REQ_VEC field in GEN3_EQ_CONTROL_OFF when in Gen3 mode. You must contact your PHY vendor to ensure 24 ms timeout does not occur in presets (FOM) requests in EQ master phase. Values: 0x0, ..., 0x7ff Default Value: 0x59f Enabled: ((CX_GEN3_MODE != GEN3_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_EQ_PSET_REQ_VEC

Label	Description
Gen3 EQ Feedback Mode	Gen3 EQ Feedback Mode. ■ 0000b: Direction Change. Figure Of Merit method is also used by setting "Preset Request Vector". ■ 0001b: Figure Of Merit. Only Figure Of Merit method is used. ■ 0010b: Reserved ■ Reserved ■ 1111b: Reserved This is the default of the GEN3_EQ_FB_MODE field in GEN3_EQ_CONTROL_OFF when in Gen3 mode. Values: ■ Direction Change (0) ■ Figure Of Merit (1) Default Value: CX_GEN3_MODE != GEN3_DISABLED Enabled: ((CX_GEN3_MODE != GEN3_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_EQ_FB_MODE
Gen3 EQ Master Phase Exit Mode	Gen3 EQ Master Phase Exit Mode After 24 ms Timeout ■ 0b: Exit to Recovery.Speed state ■ 1b: Exit to EQ Phase 3 for USP controller or Recovery.RcvrLock state for DSP controller This is the default of the GEN3_EQ_PHASE23_EXIT_MODE field in GEN3_EQ_CONTROL_OFF when in Gen3 mode. Setting to '1b' affects Direction Change EQ Feedback Mode. EQ requests for Figure Of Merit mode complete before 24 ms timeout. See DEFAULT_GEN3_EQ_PSET_REQ_VEC for more. Values: ■ To Recovery.Speed (0) ■ To EQ 3 for USP or to Recovery.RcvrLock for DSP (1) Default Value: To Recovery.Speed Enabled: ((CX_GEN3_MODE != GEN3_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_EQ_PHASE23_EXIT_MODE

Label	Description
Programmable Table Mode	
Gen3/Gen4/Gen5 EQ Presets to Coefficients Mapping Modes	For Gen3/Gen4/Gen5 equalization, there are three possible implementations for the mapping of presets to coefficients: ■ Dynamic PHY: The coefficients are dynamically mapped in the PHY. ■ Dynamic MAC: The coefficients are dynamically mapped in the MAC. ■ Programmable Table: The coefficients are programmed in a table by your application. The coefficients obtained from a preset are then used to drive the <code>mac_phy_txdeemph[17:0]</code> output in the transmit path of the relevant port. Note: Gen3 and Gen4 cannot use different mapping modes. Values: ■ Dynamic PHY (0) ■ Dynamic MAC (1) ■ Programmable Table (2) Default Value: Dynamic PHY Enabled: ((CX_GEN3_MODE != GEN3_DISABLED)) Parameter Type: Feature Setting Parameter Name: CX_GEN3_EQ_PSET_COEF_MAP_MODE
Gen3 EQ Programmable Default Local FS Value	When the Gen3 Equalization Presets to Coefficients Mapping Mode is Programmable Table, this parameter is used to set the default local Full Swing (FS) value for Gen3 data rate. The FS value is specific to the PHY implementation. It is needed by the controller because it is advertised during Gen3 equalization Phase 1, and it is used to determine if the coefficients meet the rules, as described in Rules for Transmitter Coefficients, section 4.2.3.1 of the <i>PCI Express Base Specification, Rev 3.0</i>. This is the default of the GEN3_EQ_LOCAL_FS field in GEN3_EQ_LOCAL_FS_LF_OFF when in Gen3 mode. Values: 12, ..., 63 Default Value: 48 Enabled: CX_GEN3_SPEED && CX_GEN3_EQ_PSET_COEF_MAP_MODE==2 Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_EQ_LOCAL_FS
Gen3 EQ Programmable Default Local LF Value	When the Gen3 Equalization Presets to Coefficients Mapping Mode is Programmable Table, this parameter is used to set the default local Low Frequency (LF) value for Gen3 data rate. The LF value is specific to the PHY implementation. It is needed by the controller because it is advertised during Gen3 equalization Phase1, and it is used to determine if the coefficients meet the rules, as described in Rules for Transmitter Coefficients, section 4.2.3.1 of the <i>PCI Express Base Specification, Rev 3.0</i>. This is the default of the GEN3_EQ_LOCAL_LF field in GEN3_EQ_LOCAL_FS_LF_OFF when in Gen3 mode. Values: 0, ..., 63 Default Value: 24 Enabled: CX_GEN3_SPEED && CX_GEN3_EQ_PSET_COEF_MAP_MODE==2 Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_EQ_LOCAL_LF

Label	Description
Coefficient Convergence Mode	
Gen3 EQ Coefficient Convergence Support	When enabled the controller implements additional logic for checking that the fine tuning process of the remote transmitter that is executed during Phase 2 (when controller is USP) or Phase 3 (controller is DSP) is converging toward a stable point. This setting is both for Gen3 and Gen4. Gen3 and Gen4 cannot have different settings. Values: ■ false (0) ■ true (1) Default Value: CX_GEN3_MODE != GEN3_DISABLED Enabled: ((CX_GEN3_MODE != GEN3_DISABLED)) Parameter Type: Feature Setting Parameter Name: CX_GEN3_EQ_COEF_CONV_SUPPORTED
Gen3 EQ Coefficient Convergence Queue Depth	When Equalization Convergence Support (CX_GEN3_EQ_COEF_CONV_SUPPORTED) is enabled, this parameter provides the depth of the queue used to store the coefficients in use by the remote transmitter in successive evaluation cycles. The queue is used to check if the successive coefficient values are converging toward a stable value. This setting is both for Gen3 and Gen4. Gen3 and Gen4 cannot have different setting. Values: 2, ..., 16 Default Value: 2 Enabled: (CX_GEN3_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Feature Setting Parameter Name: CX_GEN3_EQ_COEFQ_DEPTH
Gen3 EQ Convergence Window Aperture for C+1.	Gen3 EQ Convergence Window Aperture for C+1. Post-cursor coefficients maximum delta within the convergence window depth. This is the default of the GEN3_EQ_FMDC_MAX_POST_CUSROR_DELTA field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen3 mode. Values: 0, ..., 15 Default Value: 0 Enabled: (CX_GEN3_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_EQ_FMDC_MAX_POST_CUSROR_DELTA

Label	Description
Gen3 EQ Convergence Window Aperture for C-1.	Gen3 EQ Convergence Window Aperture for C-1. Pre-cursor coefficients maximum delta within the convergence window depth. This is the default of the GEN3_EQ_FMDC_MAX_PRE_CURSOR_DELTA field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen3 mode. Values: 0, ..., 15 Default Value: 0 Enabled: (CX_GEN3_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_EQ_FMDC_MAX_PRE_CURSOR_DELTA
Gen3 EQ Convergence Window Depth	Gen3 EQ Convergence Window Depth. Number of consecutive evaluations considered in Phase 2/3 when determining if optimal coefficients have been found. This is the default of the GEN3_EQ_FMDC_N_EVALS field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen3 mode. Values: 0, ..., 16 Default Value: 2 Enabled: (CX_GEN3_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_EQ_FMDC_N_EVALS
Gen3 EQ Minimum Time (in ms) To Remain in EQ Master Phase	Gen3 EQ Minimum Time (in ms) To Remain in EQ Master Phase. The LTSSM stays in EQ Master phase for at least this amount of time, before starting to check for convergence of the coefficients. This is the default of the GEN3_EQ_FMDC_T_MIN_PHASE23 field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen3 mode. Values: 0, ..., 24 Default Value: 0 Enabled: (CX_GEN3_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN3_EQ_FMDC_T_MIN_PHASE23

1.49 Advanced PHY Config / Gen4 PHY Equalization Config Parameters

Table 1-49 Advanced PHY Config / Gen4 PHY Equalization Config Parameters

Label	Description
General Config	
Gen4 controller as EQ Master Asserts RxEqEval for PHY to do Rx Adaptation and Evaluation Regardless of Rx TS1s Detection	When set to '1', the controller as Gen4 EQ master asserts RxEqEval to instruct the PHY to do Rx adaptation and evaluation after a 500ns timeout from a new preset request. This is the default of the RXEQ_RGRDLESS_RXTS field in GEN3_RELATED_OFF when in Gen4 mode. Values: ■ No (0) ■ Yes (1) Default Value: Yes Enabled: ((CX_GEN4_MODE != GEN4_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXEQ_RGRDLESS_RXTS
Gen4 EQ Include Initial FOM	Gen4 EQ Include Initial FOM. Include or not the FOM feedback from the initial preset evaluation performed in the EQ Master, when finding the highest FOM among all preset evaluations. ■ 0: Do not include ■ 1: Include This is the default of the GEN3_EQ_FOM_INC_INITIAL_EVAL field in GEN3_EQ_CONTROL_OFF when in Gen4 mode. Values: 0, 1 Default Value: 0 Enabled: ((CX_GEN4_MODE != GEN4_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_EQ_FOM_INC_INITIAL_EVAL
Gen4 EQ Preset Request Vector	Gen4 EQ Preset Request Vector. Requesting of Presets during the initial part of the EQ Master Phase. Encoding scheme is as follows: ■ Bit [15:0] =0x0: No preset is requested and evaluated in EQ Master Phase. ■ Bit [i] =1: "Preset=i" is requested and evaluated in EQ Master Phase. This is the default of the GEN3_EQ_PSET_REQ_VEC field in GEN3_EQ_CONTROL_OFF when in Gen4 mode. You must contact your PHY vendor to ensure 24 ms timeout does not occur in presets (FOM) requests in EQ master phase. Values: 0x0, ..., 0x7ff Default Value: 0x59f Enabled: ((CX_GEN4_MODE != GEN4_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_EQ_PSET_REQ_VEC

Label	Description
Gen4 EQ Feedback Mode	Gen4 EQ Feedback Mode. ■ 0000b: Direction Change. Figure Of Merit method is also used by setting "Preset Request Vector". ■ 0001b: Figure Of Merit. Only Figure Of Merit method is used. ■ 0010b: Reserved ■ Reserved ■ 1111b: Reserved This is the default of the GEN3_EQ_FB_MODE field in GEN3_EQ_CONTROL_OFF when in Gen4 mode. Values: ■ Direction Change (0) ■ Figure Of Merit (1) Default Value: CX_GEN4_MODE != GEN4_DISABLED Enabled: ((CX_GEN4_MODE != GEN4_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_EQ_FB_MODE
Gen4 EQ Master Phase Exit Mode	Gen4 EQ Master Phase Exit Mode After 24 ms Timeout ■ 0b: Exit to Recovery.Speed state ■ 1b: Exit to EQ Phase 3 for USP controller or Recovery.RcvrLock state for DSP controller This is the default of the GEN3_EQ_PHASE23_EXIT_MODE field in GEN3_EQ_CONTROL_OFF when in Gen4 mode. Setting to '1b' affects Direction Change EQ Feedback Mode. EQ requests for Figure Of Merit mode complete before 24 ms timeout. See DEFAULT_GEN4_EQ_PSET_REQ_VEC for more. Values: ■ To Recovery.Speed (0) ■ To EQ 3 for USP or to Recovery.RcvrLock for DSP (1) Default Value: To Recovery.Speed Enabled: ((CX_GEN4_MODE != GEN4_DISABLED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_EQ_PHASE23_EXIT_MODE

Label	Description
Programmable Table Mode	
Gen4 EQ Programmable Default Local FS Value	When the Gen3 Equalization Presets to Coefficients Mapping Mode is Programmable Table, this parameter is used to set the default local Full Swing (FS) value for Gen4 data rate. The FS value is specific to the PHY implementation. It is needed by the controller because it is advertised during Gen4 equalization Phase 1, and it is used to determine if the coefficients meet the rules, as described in Rules for Transmitter Coefficients, section TBD of the <i>PCI Express Base Specification, Rev 4.0</i>. This is the default of the GEN3_EQ_LOCAL_FS field in GEN3_EQ_LOCAL_FS_LF_OFF when in Gen4 mode. Values: 12, ..., 63 Default Value: 48 Enabled: CX_GEN4_SPEED && CX_GEN3_EQ_PSET_COEF_MAP_MODE==2 Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_EQ_LOCAL_FS
Gen4 EQ Programmable Default Local LF Value	When the Gen3 Equalization Presets to Coefficients Mapping Mode is Programmable Table, this parameter is used to set the default local Low Frequency (LF) value for Gen4 data rate. The LF value is specific to the PHY implementation. It is needed by the controller because it is advertised during Gen4 equalization Phase1, and it is used to determine if the coefficients meet the rules, as described in Rules for Transmitter Coefficients, section TBD of the <i>PCI Express Base Specification, Rev 4.0</i>. This is the default of the GEN3_EQ_LOCAL_LF field in GEN3_EQ_LOCAL_FS_LF_OFF when in Gen4 mode. Values: 0, ..., 63 Default Value: 24 Enabled: CX_GEN4_SPEED && CX_GEN3_EQ_PSET_COEF_MAP_MODE==2 Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_EQ_LOCAL_LF
Coefficient Convergence Mode	
Gen4 EQ Convergence Window Aperture for C+1.	Gen4 EQ Convergence Window Aperture for C+1. Post-cursor coefficients maximum delta within the convergence window depth. This is the default of the GEN3_EQ_FMDC_MAX_POST_CUSROR_DELTA field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen4 mode. Values: 0, ..., 15 Default Value: 0 Enabled: (CX_GEN4_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_EQ_FMDC_MAX_POST_CUSROR_DELTA

Label	Description
Gen4 EQ Convergence Window Aperture for C-1.	Gen4 EQ Convergence Window Aperture for C-1. Pre-cursor coefficients maximum delta within the convergence window depth. This is the default of the GEN3_EQ_FMDC_MAX_PRE_CURSOR_DELTA field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen4 mode. Values: 0, ..., 15 Default Value: 0 Enabled: (CX_GEN4_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_EQ_FMDC_MAX_PRE_CURSOR_DELTA
Gen4 EQ Convergence Window Depth	Gen4 EQ Convergence Window Depth. Number of consecutive evaluations considered in Phase 2/3 when determining if optimal coefficients have been found. This is the default of the GEN3_EQ_FMDC_N_EVALS field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen4 mode. Values: 0, ..., 16 Default Value: 2 Enabled: (CX_GEN4_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_EQ_FMDC_N_EVALS
Gen4 EQ Minimum Time (in ms) To Remain in EQ Master Phase	Gen4 EQ Minimum Time (in ms) To Remain in EQ Master Phase. The LTSSM stays in EQ Master phase for at least this amount of time, before starting to check for convergence of the coefficients. This is the default of the GEN3_EQ_FMDC_T_MIN_PHASE23 field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen4 mode. Values: 0, ..., 24 Default Value: 0 Enabled: (CX_GEN4_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_EQ_FMDC_T_MIN_PHASE23

1.50 Advanced PHY Config / Gen5 PHY Equalization Config Parameters

Table 1-50 Advanced PHY Config / Gen5 PHY Equalization Config Parameters

Label	Description
General Config	
Gen5 controller as EQ Master Asserts RxEqEval for PHY to do Rx Adaptation and Evaluation Regardless of Rx TS1s Detection	When set to '1', the controller as Gen5 EQ master asserts RxEqEval to instruct the PHY to do Rx adaptation and evaluation after a 500ns timeout from a new preset request. This is the default of the RXEQ_RGRDLESS_RXTS field in GEN3_RELATED_OFF when in Gen5 mode. Values: ■ No (0) ■ Yes (1) Default Value: Yes Enabled: ((CX_MAX_PCIE_SPEED >= 5)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXEQ_RGRDLESS_RXTS
Gen5 EQ Include Initial FOM	Gen5 EQ Include Initial FOM. Include or not the FOM feedback from the initial preset evaluation performed in the EQ Master, when finding the highest FOM among all preset evaluations. ■ 0: Do not include ■ 1: Include This is the default of the GEN3_EQ_FOM_INC_INITIAL_EVAL field in GEN3_EQ_CONTROL_OFF when in Gen5 mode. Values: 0, 1 Default Value: 0 Enabled: ((CX_MAX_PCIE_SPEED >= 5)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_EQ_FOM_INC_INITIAL_EVAL
Gen5 EQ Preset Request Vector	Gen5 EQ Preset Request Vector. Requesting of Presets during the initial part of the EQ Master Phase. Encoding scheme is as follows: ■ Bit [15:0] = 0x0: No preset is requested and evaluated in EQ Master Phase. ■ Bit [i] = 1: "Preset=i" is requested and evaluated in EQ Master Phase. This is the default of the GEN3_EQ_PSET_REQ_VEC field in GEN3_EQ_CONTROL_OFF when in Gen5 mode. You must contact your PHY vendor to ensure 24 ms timeout does not occur in presets (FOM) requests in EQ master phase. Values: 0x0, ..., 0x7ff Default Value: 0x59f Enabled: ((CX_MAX_PCIE_SPEED >= 5)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_EQ_PSET_REQ_VEC

Label	Description
Gen5 EQ Feedback Mode	Gen5 EQ Feedback Mode. ■ 0000b: Direction Change. Figure Of Merit method is also used by setting "Preset Request Vector". ■ 0001b: Figure Of Merit. Only Figure Of Merit method is used. ■ 0010b: Reserved ■ Reserved ■ 1111b: Reserved This is the default of the GEN3_EQ_FB_MODE field in GEN3_EQ_CONTROL_OFF when in Gen5 mode. Values: ■ Direction Change (0) ■ Figure Of Merit (1) Default Value: CX_MAX_PCIE_SPEED >= 5 Enabled: ((CX_MAX_PCIE_SPEED >= 5)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_EQ_FB_MODE
Gen5 EQ Master Phase Exit Mode	Gen5 EQ Master Phase Exit Mode After 24 ms Timeout ■ 0b: Exit to Recovery.Speed state ■ 1b: Exit to EQ Phase 3 for USP controller or Recovery.RcvrLock state for DSP controller This is the default of the GEN3_EQ_PHASE23_EXIT_MODE field in GEN3_EQ_CONTROL_OFF when in Gen5 mode. Setting to '1b' affects Direction Change EQ Feedback Mode. EQ requests for Figure Of Merit mode complete before 24 ms timeout. See DEFAULT_GEN5_EQ_PSET_REQ_VEC for more. Values: ■ To Recovery.Speed (0) ■ To EQ 3 for USP or to Recovery.RcvrLock for DSP (1) Default Value: To Recovery.Speed Enabled: ((CX_MAX_PCIE_SPEED >= 5)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_EQ_PHASE23_EXIT_MODE

Label	Description
Programmable Table Mode	
Gen5 EQ Programmable Default Local FS Value	When the Gen3 Equalization Presets to Coefficients Mapping Mode is Programmable Table, this parameter is used to set the default local Full Swing (FS) value for Gen5 data rate. The FS value is specific to the PHY implementation. It is needed by the controller because it is advertised during Gen5 equalization Phase 1, and it is used to determine if the coefficients meet the rules, as described in Rules for Transmitter Coefficients, section TBD of the <i>PCI Express Base Specification, Rev 5.0</i>. This is the default of the GEN3_EQ_LOCAL_FS field in GEN3_EQ_LOCAL_FS_LF_OFF when in Gen5 mode. Values: 12, ..., 63 Default Value: 48 Enabled: CX_GEN5_SPEED && CX_GEN3_EQ_PSET_COEF_MAP_MODE==2 Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_EQ_LOCAL_FS
Gen5 EQ Programmable Default Local LF Value	When the Gen3 Equalization Presets to Coefficients Mapping Mode is Programmable Table, this parameter is used to set the default local Low Frequency (LF) value for Gen5 data rate. The LF value is specific to the PHY implementation. It is needed by the controller because it is advertised during Gen5 equalization Phase1, and it is used to determine if the coefficients meet the rules, as described in Rules for Transmitter Coefficients, section TBD of the <i>PCI Express Base Specification, Rev 5.0</i>. This is the default of the GEN3_EQ_LOCAL_LF field in GEN3_EQ_LOCAL_FS_LF_OFF when in Gen5 mode. Values: 0, ..., 63 Default Value: 24 Enabled: CX_GEN5_SPEED && CX_GEN3_EQ_PSET_COEF_MAP_MODE==2 Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_EQ_LOCAL_LF
Coefficient Convergence Mode	
Gen5 EQ Convergence Window Aperture for C+1.	Gen5 EQ Convergence Window Aperture for C+1. Post-cursor coefficients maximum delta within the convergence window depth. This is the default of the GEN3_EQ_FMDC_MAX_POST_CUSROR_DELTA field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen5 mode. Values: 0, ..., 15 Default Value: 0 Enabled: (CX_GEN5_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_EQ_FMDC_MAX_POST_CUSROR_DELTA

Label	Description
Gen5 EQ Convergence Window Aperture for C-1.	Gen5 EQ Convergence Window Aperture for C-1. Pre-cursor coefficients maximum delta within the convergence window depth. This is the default of the GEN3_EQ_FMDC_MAX_PRE_CURSOR_DELTA field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen5 mode. Values: 0, ..., 15 Default Value: 0 Enabled: (CX_GEN5_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_EQ_FMDC_MAX_PRE_CURSOR_DELTA
Gen5 EQ Convergence Window Depth	Gen5 EQ Convergence Window Depth. Number of consecutive evaluations considered in Phase 2/3 when determining if optimal coefficients have been found. This is the default of the GEN3_EQ_FMDC_N_EVALS field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen5 mode. Values: 0, ..., 16 Default Value: 2 Enabled: (CX_GEN5_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_EQ_FMDC_N_EVALS
Gen5 EQ Minimum Time (in ms) To Remain in EQ Master Phase	Gen5 EQ Minimum Time (in ms) To Remain in EQ Master Phase. The LTSSM stays in EQ Master phase for at least this amount of time, before starting to check for convergence of the coefficients. This is the default of the GEN3_EQ_FMDC_T_MIN_PHASE23 field in GEN3_EQ_FB_MODE_DIR_CHANGE_OFF when in Gen5 mode. Values: 0, ..., 24 Default Value: 0 Enabled: (CX_GEN5_SPEED && (CX_GEN3_EQ_COEF_CONV_SUPPORTED == 1)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_EQ_FMDC_T_MIN_PHASE23

1.51 Advanced PHY Config / Gen4 PHY Lane Margining Config Parameters

Table 1-51 Advanced PHY Config / Gen4 PHY Lane Margining Config Parameters

Label	Description
Gen4 Lane Margining	
MNumTimingSteps	Specify the number of time steps from default for 16.0 GT/s (MNumTimingSteps). Contact your PHY vendor for the value that you should use. Values: 6, ..., 63 Default Value: 6 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_NUM_TIMING_STEPS
MMaxTimingOffset	Specify the offset from default at maximum step value for 16.0 GT/s (MMaxTimingOffset). Contact your PHY vendor for the value that you should use. Values: 0, ..., 50 Default Value: 20 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_MAX_TIMING_OFFSET
MNumVoltageSteps	Specify the number of voltage steps from default for 16.0 GT/s (MNumVoltageSteps). Contact your PHY vendor for the value that you should use. Values: 32, ..., 127 Default Value: 32 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_NUM_VOLTAGE_STEPS
MMaxVoltageOffset	Specify the offset from default at maximum step value as percentage of one voltage for 16.0 GT/s (MMaxVoltageOffset). Contact your PHY vendor for the value that you should use. Values: 0, ..., 50 Default Value: 5 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_MAX_VOLTAGE_OFFSET

Label	Description
MSamplingRateVoltage	Specify the ratio of bits tested to bits received during voltage margining for 16.0 GT/s (MSamplingRateVoltage). Contact your PHY vendor for the value that you should use. <i>Note:</i> If MIndErrorSampler = 0, the controller always reports MSamplingRateVoltage = 0x3F regardless of this parameter value when the controller receives Report MSamplingRateVoltage Command from System Software. Values: 0, ..., 63 Default Value: 0 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_SAMPLE_RATE_VOLTAGE
MSamplingRateTiming	Specify the ratio of bits tested to bits received during timing margining for 16.0 GT/s (MSamplingRateTiming). Contact your PHY vendor for the value that you should use. <i>Note :</i> that if MIndErrorSampler = 0, the controller always reports MSamplingRateTiming=0x3F regardless of this parameter value when the controller receives Report MSamplingRateTiming Command from System Software. Values: 0, ..., 63 Default Value: 0 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_SAMPLE_RATE_TIMING
MVoltageSupported	Enable voltage margining support for 16.0 GT/s (MVoltageSupported). Contact your PHY vendor for the value that you should use. Values: 0, 1 Default Value: 1 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_VOLTAGE_SUPPORTED
MIndLeftRightTiming	Enable the independent left/right timing margin support for 16.0 GT/s (MIndLeftRightTiming). Contact your PHY vendor for the value that you should use. Values: 0, 1 Default Value: 1 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_IND_LEFT_RIGHT_TIMING
MIndUpDownVoltage	Enable the independent up and down voltage margining support for 16.0 GT/s (MIndUpDownVoltage). Contact your PHY vendor for the value that you should use. Values: 0, 1 Default Value: 1 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_IND_UP_DOWN_VOLTAGE

Label	Description
MIndErrorSampler	Indicates Error sampler is independent for 16.0 GT/s (MIndErrorSampler). Contact your PHY vendor for the value that you should use. Values: 0, 1 Default Value: 1 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_IND_ERROR_SAMPLER
MSampleReportingMethod	Indicates the sample reporting method for 16.0 GT/s (MSampleReportingMethod). Contact your PHY vendor for the value that you should use. ■ Enabled(1): Sample frequency is supported ■ Disabled(2): Sample count is supported <i>Note:</i> If MIndErrorSampler =0, the controller always reports MSampleReportingMethod =1 regardless of this parameter value when the controller receives Report Margin Control Capabilities Command from System Software. Values: 0, 1 Default Value: 0 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_SAMPLE_REPORTING_METHOD
MMaxLanes	Specify the maximum number of lanes minus 1 that can be margined at the same time for 16.0 GT/s (MMaxLanes). Contact your PHY vendor for the value that you should use. Values: 0, ..., 31 Default Value: 1 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN4_RXMARGIN_MAXLANES
Common Lane Margining	
Margining uses Driver Software	Default value of Margining uses Driver Software. Contact your PHY vendor for the value that you should use. Values: 0, 1 Default Value: 0 Enabled: ((CX_GEN4_SPEED)) Parameter Type: Register Default Setting Parameter Name: DEFAULT_MARGINING_USES_DRIVER_SOFTWARE

1.52 Advanced PHY Config / Gen5 PHY Lane Margining Config Parameters

Table 1-52 Advanced PHY Config / Gen5 PHY Lane Margining Config Parameters

Label	Description
Gen5 Lane Margining	
MNumTimingSteps	Specify the number of time steps from default for 32.0 GT/s (MNumTimingSteps). Contact your PHY vendor for the value that you should use. Values: 6, ..., 63 Default Value: DEFAULT_GEN4_RXMARGIN_NUM_TIMING_STEPS Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_NUM_TIMING_STEPS
MMaxTimingOffset	Specify the offset from default at maximum step value for 32.0 GT/s (MMaxTimingOffset). Contact your PHY vendor for the value that you should use. Values: 0, ..., 50 Default Value: DEFAULT_GEN4_RXMARGIN_MAX_TIMING_OFFSET Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_MAX_TIMING_OFFSET
MNumVoltageSteps	Specify the number of voltage steps from default for 32.0 GT/s (MNumVoltageSteps). Contact your PHY vendor for the value that you should use. Values: 32, ..., 127 Default Value: DEFAULT_GEN4_RXMARGIN_NUM_VOLTAGE_STEPS Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_NUM_VOLTAGE_STEPS
MMaxVoltageOffset	Specify the offset from default at maximum step value as percentage of one voltage for 32.0 GT/s (MMaxVoltageOffset). Contact your PHY vendor for the value that you should use. Values: 0, ..., 50 Default Value: DEFAULT_GEN4_RXMARGIN_MAX_VOLTAGE_OFFSET Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_MAX_VOLTAGE_OFFSET

Label	Description
MSamplingRateVoltage	Specify the ratio of bits tested to bits received during voltage margining for 32.0 GT/s (MSamplingRateVoltage). Contact your PHY vendor for the value that you should use. <i>Note:</i> If MIndErrorSampler = 0, the controller always reports MSamplingRateVoltage = 0x3F regardless of this parameter value when the controller receives Report MSamplingRateVoltage Command from System Software. Values: 0, ..., 63 Default Value: DEFAULT_GEN4_RXMARGIN_SAMPLE_RATE_VOLTAGE Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_SAMPLE_RATE_VOLTAGE
MSamplingRateTiming	Specify the ratio of bits tested to bits received during timing margining for 32.0 GT/s (MSamplingRateTiming). Contact your PHY vendor for the value that you should use. <i>Note:</i> If MIndErrorSampler = 0, the controller always reports MSamplingRateTiming = 0x3F regardless of this parameter value when the controller receives Report MSamplingRateTiming Command from System Software. Values: 0, ..., 63 Default Value: DEFAULT_GEN4_RXMARGIN_SAMPLE_RATE_TIMING Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_SAMPLE_RATE_TIMING
MVoltageSupported	Enable voltage margining support for 32.0 GT/s (MVoltageSupported). Contact your PHY vendor for the value that you should use. Values: 0, 1 Default Value: DEFAULT_GEN4_RXMARGIN_VOLTAGE_SUPPORTED Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_VOLTAGE_SUPPORTED
MIndLeftRightTiming	Enable the independent left/right timing margin supported for 32.0 GT/s (MIndLeftRightTiming). Contact your PHY vendor for the value that you should use. Values: 0, 1 Default Value: DEFAULT_GEN4_RXMARGIN_IND_LEFT_RIGHT_TIMING Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_IND_LEFT_RIGHT_TIMING
MIndUpDownVoltage	Enable the independent up and down voltage margining supported for 32.0 GT/s (MIndUpDownVoltage). Contact your PHY vendor for the value that you should use. Values: 0, 1 Default Value: DEFAULT_GEN4_RXMARGIN_IND_UP_DOWN_VOLTAGE Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_IND_UP_DOWN_VOLTAGE

Label	Description
MIndErrorSampler	Indicates Error sampler is independent for 32.0 GT/s (MIndErrorSampler). Contact your PHY vendor for the value that you should use. Values: 0, 1 Default Value: DEFAULT_GEN4_RXMARGIN_IND_ERROR_SAMPLER Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_IND_ERROR_SAMPLER
MSampleReportingMethod	Indicates the sample reporting method for 32.0 GT/s (MSampleReportingMethod). Contact your PHY vendor for the value that you should use. ■ Enabled(1): Sample frequency is supported ■ Disabled(2): Sample count is supported <i>Note:</i> If MIndErrorSampler =0, the controller always reports MSampleReportingMethod =1 regardless of this parameter value when the controller receives Report Margin Control Capabilities Command from System Software. Values: 0, 1 Default Value: DEFAULT_GEN4_RXMARGIN_SAMPLE_REPORTING_METHOD Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_SAMPLE_REPORTING_METHOD
MMaxLanes	Specify the maximum number of lanes minus 1 that can be margined at the same time for 32.0 GT/s (MMaxLanes). Contact your PHY vendor for the value that you should use. Values: 0, ..., 31 Default Value: DEFAULT_GEN4_RXMARGIN_MAXLANES Enabled: CX_MARGIN_ENABLE && CX_GEN5_SPEED Parameter Type: Register Default Setting Parameter Name: DEFAULT_GEN5_RXMARGIN_MAXLANES

1.53 Advanced PHY Config / PHY Message Bus Config Parameters

Table 1-53 Advanced PHY Config / PHY Message Bus Config Parameters

Label	Description
Message Bus	
RX Message Bus Write Buffer Depth	Default value of RX Message Bus Write Buffer Depth. This is an implementation specific parameter specified in PIPE Specification. Contact your PHY vendor for the value that you should use. Values: 0, ..., 15 Default Value: (CX_PIPE5_SUPPORT) ? 5 : (CX_PIPE44_SUPPORT) ? 2 : 0 Enabled: (CX_GEN4_SPEED && CX_PIPE44_SUPPORT) CX_PIPE5_SUPPORT Parameter Type: Register Default Setting Parameter Name: DEFAULT_RX_MESSAGE_BUS_WRITE_BUFFER_DEPTH
TX Message Bus Min Write Buffer Depth	Default value of TX Message Bus Minimum Write Buffer Depth. This is an implementation specific parameter specified in PIPE Specification. Contact your PHY vendor for the value that you should use. Values: 0, ..., 8 Default Value: (CX_PIPE5_SUPPORT) ? 5 : (CX_PIPE44_SUPPORT) ? 2 : 0 Enabled: (CX_GEN4_SPEED && CX_PIPE44_SUPPORT) CX_PIPE5_SUPPORT Parameter Type: Register Default Setting Parameter Name: DEFAULT_TX_MESSAGE_BUS_MIN_WRITE_BUFFER_DEPTH

1.54 Advanced Transmit Config Parameters

Table 1-54 Advanced Transmit Config Parameters

Label	Description
Advanced Transmit Config	
Block VDMs in Non-D0 State	Block transmit of Vendor Defined Messages when in a non-D0 state. For more information, see the 'Power Management' section of the Architecture chapter of the <i>Databook</i>. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_BLOCK_VDM_TLP
Block CCIX Transmit Interface	Block outbound transactions on CCIX interface when in low power modes. For more information, see the 'Power Management' section of the Architecture chapter of the <i>Databook</i>. Values: ■ false (0x0) ■ true (0x1) Default Value: true Enabled: AMBA_INTERFACE==0 && CC_DEVICE_TYPE!=CC_RC && CX_CCIX_INTERFACE_ENABLE==1 Parameter Type: Feature Setting Parameter Name: CX_CCIX_BLOCK_NEW_TLP
General Config	
Block Client 0 Interface	Block outbound transactions on XALI0 when in low power modes. For more information, see the 'Power Management' section of the Architecture chapter of the <i>Databook</i>. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_CLIENT0_BLOCK_NEW_TLP

Label	Description
Block Client 1 Interface	Block outbound transactions on XALI1 when in low power modes. For more information, see the 'Power Management' section of the Architecture chapter of the <i>Databook</i>. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_CLIENT1_BLOCK_NEW_TLP
Block Client 2 Interface	Block outbound transactions on XALI2 when in low power modes. For more information, see the 'Power Management' section of the Architecture chapter of the <i>Databook</i>. Values: ■ false (0x0) ■ true (0x1) Default Value: false Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_CLIENT2_BLOCK_NEW_TLP
Populate ports for available credit buses	Include top-level ports to provide available credit information to the application. For more information, see 'SII: Transmit Control Signals' section of the <i>Databook</i>. This parameter enables the population of output ports for application monitoring of run-time available credit information for VCn buses: ■ xadm_ph_cdts [NVC*8-1:0] : available VC0-VCn header posted credits ■ xadm_nph_cdts [NVC*8-1:0] : available VC0-VCn header non-posted credits ■ xadm_cplh_cdts [NVC*8-1:0] : available VC0-VCn header completion credits ■ xadm_pd_cdts [NVC*12-1:0] : available VC0-VCn data posted credits ■ xadm_npd_cdts [NVC*12-1:0] : available VC0-VCn data non-posted credits ■ xadm_cpld_cdts [NVC*12-1:0] : available VC0-VCn data completion credits Information for lower order VCs is presented on the lower-order bits. Values: 0, 1 Default Value: 0 Enabled: ((AMBA_INTERFACE==0)) Parameter Type: Feature Setting Parameter Name: XADM_CRD_EN

Label	Description
Client Interface TLP pullback feature	Enable the transmit clients to cancel a TLP that is currently submitted for transmission. Values: ■ false (0) ■ true (1) Default Value: (CC_DMA_ENABLE && (AMBA_INTERFACE!=0) && (CX_MULTISTREAM==0)) ? 1 : 0 Enabled: ((AMBA_INTERFACE==0) && (CX_MULTISTREAM==0)) Parameter Type: Feature Setting Parameter Name: CLIENT_PULLBACK
VC-Based Transmit Arbitration	
Transmit Client Arbitration Scheme	Selects the arbitration method for transmitted TLPs, as described in the "Transmit TLP Arbitration" section of the <i>Databook</i>. ■ 0: VC Based. Provides a VC based programmable weighted round robin arbitration (WRR) using two different arbitration methods for the two groups of VCs. ■ 1: Round Robin (RR). Provides round robin arbitration between the three transmit clients. This is the default method. ■ 2: Strict Priority. Provides strict priority between the three transmit clients. XALI0 is lowest, XALI1 is higher, XALI2 (if implemented) is highest. You can only change this parameter when CX_NVC > 1. Values: ■ VC Based (0) ■ Round Robin (1) ■ Strict Priority (2) Default Value: Round Robin Enabled: CX_NVC > 1 && CX_MULTISTREAM==0 Parameter Type: Feature Setting Parameter Name: CX_XADM_ARB_MODE
Enable LPVC WRR Weights Writable	Determines whether the WRR arbitration VC weight values in VC Transmit Arbitration Register 1 and VC Transmit Arbitration Register 2 (in the register section of the <i>Databook</i>) are writable through the DBI. Values: ■ false (0) ■ true (1) Default Value: ((CX_MULTISTREAM)==1) ? 1 : 0 Enabled: VC_ENABLE==1 && CX_XADM_ARB_MODE==0 && CX_XADM_VC_ARB_WRR==0 Parameter Type: Feature Setting Parameter Name: CX_LPVC_WRR_WEIGHT_WRITABLE

Label	Description
VC ID #0 Weight	WRR Weighting for VC ID #0. If LPVC WRR Weights Writable is true, your application can change the value by writing to VC Transmit Arbitration Register 1 (in the register section of the Databook) through the DBI. Values: 0x0, ..., 0x7f Default Value: ((CX_MULTISTREAM)) ? 0x0 : 0xF Enabled: VC_ENABLE==1 && CX_XADM_ARB_MODE==0 && CX_XADM_VC_ARB_WRR==0 Parameter Type: Feature Setting Parameter Name: LPVC_WRR_WEIGHT_VC0
VC ID #1 Weight	WRR Weighting for VC ID #1. If LPVC WRR Weights Writable is true, your application can change the value by writing to VC Transmit Arbitration Register 1 (in the register section of the Databook) through the DBI. Values: 0x0, ..., 0x7f Default Value: 0x0 Enabled: VC_ENABLE==1 && CX_XADM_ARB_MODE==0 && CX_XADM_VC_ARB_WRR==0 Parameter Type: Feature Setting Parameter Name: LPVC_WRR_WEIGHT_VC1
VC ID #2 Weight	WRR Weighting for VC ID #2. If LPVC WRR Weights Writable is true, your application can change the value by writing to VC Transmit Arbitration Register 1 (in the register section of the Databook) through the DBI. Values: 0x0, ..., 0x7f Default Value: 0x0 Enabled: VC_ENABLE==1 && CX_XADM_ARB_MODE==0 && CX_XADM_VC_ARB_WRR==0 Parameter Type: Feature Setting Parameter Name: LPVC_WRR_WEIGHT_VC2
VC ID #3 Weight	WRR Weighting for VC ID #3. If LPVC WRR Weights Writable is true, your application can change the value by writing to VC Transmit Arbitration Register 1 (in the register section of the Databook) through the DBI. Values: 0x0, ..., 0x7f Default Value: 0x0 Enabled: VC_ENABLE==1 && CX_XADM_ARB_MODE==0 && CX_XADM_VC_ARB_WRR==0 Parameter Type: Feature Setting Parameter Name: LPVC_WRR_WEIGHT_VC3

Label	Description
VC ID #4 Weight	WRR Weighting for VC ID #4. If LPVC WRR Weights Writable is true, your application can change the value by writing to VC Transmit Arbitration Register 1 (in the register section of the Databook) through the DBI. Values: 0x0, ..., 0x7f Default Value: 0x0 Enabled: VC_ENABLE==1 && CX_XADM_ARB_MODE==0 && CX_XADM_VC_ARB_WRR==0 Parameter Type: Feature Setting Parameter Name: LPVC_WRR_WEIGHT_VC4
VC ID #5 Weight	WRR Weighting for VC ID #5. If LPVC WRR Weights Writable is true, your application can change the value by writing to VC Transmit Arbitration Register 1 (in the register section of the Databook) through the DBI. Values: 0x0, ..., 0x7f Default Value: 0x0 Enabled: VC_ENABLE==1 && CX_XADM_ARB_MODE==0 && CX_XADM_VC_ARB_WRR==0 Parameter Type: Feature Setting Parameter Name: LPVC_WRR_WEIGHT_VC5
VC ID #6 Weight	WRR Weighting for VC ID #6. If LPVC WRR Weights Writable is true, your application can change the value by writing to VC Transmit Arbitration Register 1 (in the register section of the Databook) through the DBI. Values: 0x0, ..., 0x7f Default Value: 0x0 Enabled: VC_ENABLE==1 && CX_XADM_ARB_MODE==0 && CX_XADM_VC_ARB_WRR==0 Parameter Type: Feature Setting Parameter Name: LPVC_WRR_WEIGHT_VC6
VC ID #7 Weight	WRR Weighting for VC ID #7. If LPVC WRR Weights Writable is true, your application can change the value by writing to VC Transmit Arbitration Register 1 (in the register section of the Databook) through the DBI. Values: 0x0, ..., 0x7f Default Value: 0x0 Enabled: VC_ENABLE==1 && CX_XADM_ARB_MODE==0 && CX_XADM_VC_ARB_WRR==0 Parameter Type: Feature Setting Parameter Name: LPVC_WRR_WEIGHT_VC7

Label	Description
Transmit Completion	
Compare Completion Credit	Enable the controller to compare the actual size of a requested Completion (as opposed to a maximum size Completion) against available Completion credits before transmitting the Completion. This parameter saves gates and improves timing when set to false. - Typically, an EP must have enough buffer space to receive all CPL data before it sends a MRd transaction. Therefore, an RC may assume infinite CPL credits. - There is no need to compare the CPL length with available CPL credits. Therefore, setting this parameter to false can save gate count by eliminating the compare logic. - If a PCIe device has specific or more strict requirement for checking a peer components CplID buffer capability, it sets a threshold that at least 8 CplID credits are available before a PCIe device can transmit a CPLD. Values: 0, 1 Default Value: 1 Enabled: Always Parameter Type: Feature Setting Parameter Name: CPL_LEN_CMP_ENABLE
Transmit Posted	
Compare Posted Credit	Enable the controller to compare the size of a requested posted payload length against available posted credits before transmitting the posted TLP. Values: 0, 1 Default Value: 1 Enabled: Always Parameter Type: Feature Setting Parameter Name: P_LEN_CMP_ENABLE
Application Credit Control	
Populate ports to allow the application to control credit usage from internal MSG/CPL	Application credit control. For more information, see 'SII: Transmit Control Signals' section of the <i>Databook</i>. This parameter enables the application to control credits used internally by the controller for internally generated MSG/CPL. Values: 0, 1 Default Value: 0 Enabled: ((AMBA_INTERFACE==0) && (CX_CCIX_INTERFACE_ENABLE==0) && (CX_MULTISTREAM==0)) Parameter Type: Feature Setting Parameter Name: APP_CREDIT_CTRL

1.55 Advanced Pipeline Config Parameters

Table 1-55 Advanced Pipeline Config Parameters

Label	Description
Pipeline Latencies	
RDLH TLP Extract Input	Input pipeline stage to RDLH TLP extraction module Values: ■ 0 (0) ■ 1 (1) Default Value: 1 Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_RDLH_REGIN
Miscellaneous Pipeline Enables	
CRC Pipeline Depth - XTLH	CRC Pipeline Latency value for xtlh. This value represents the number of pipeline stages needed to calculate and compare ECRC. controller supported latencies: 1,2 Values: ■ 1 (1) ■ 2 (2) Default Value: 1 + ((CX_NW > 1) ? ((CX_TECHNOLOGY == 0) ((CX_GEN4_MODE == GEN4_DF) && CX_NW==16) (CX_MAX_CORECLK_FREQ > 500)) : 0) Enabled: CX_CUSTOM_PIPELINING && (CX_NW > 1) Parameter Name: CX_CRC_LATENCY_XTLH
CRC Pipeline Depth - RTLH	CRC Pipeline Latency value for rtlh. This value represents the number of pipeline stages needed to calculate and compare ECRC. controller supported latencies: 1,2 Values: ■ 1 (1) ■ 2 (2) Default Value: 1 + ((CX_NW > 1) ? ((CX_TECHNOLOGY == 0) ((CX_GEN4_MODE == GEN4_DF) && CX_NW==16) (CX_MAX_CORECLK_FREQ > 500)) : 0) Enabled: CX_CUSTOM_PIPELINING && (CX_NW > 1) Parameter Name: CX_CRC_LATENCY_RTLH
CRC Pipeline Depth - XDLH	CRC Pipeline Latency value for xdlh. This value represents the number of pipeline stages needed to calculate and compare LCRC. controller supported latencies: 1,2 Values: ■ 1 (1) ■ 2 (2) Default Value: 1 + ((CX_NW > 1) ? ((CX_TECHNOLOGY == 0) ((CX_GEN4_MODE == GEN4_DF) && CX_NW==16) ((CX_MAX_CORECLK_FREQ >= 500) && CX_RASDP > 0) (CX_MAX_CORECLK_FREQ > 500)) : 0) Enabled: CX_CUSTOM_PIPELINING && (CX_NW > 1) Parameter Name: CX_CRC_LATENCY_XDLH

Label	Description
CRC Pipeline Depth - RDLH	CRC Pipeline Latency value for rdlh. This value represents the number of pipeline stages needed to calculate and compare LCRC. controller supported latencies: 1,2 Values: ■ 1 (1) ■ 2 (2) Default Value: 1 + ((CX_NW > 1) ? ((CX_TECHNOLOGY == 0) ((CX_GEN4_MODE == GEN4_DF) && CX_NW==16) (CX_MAX_CORECLK_FREQ > 500)) : 0) Enabled: CX_CUSTOM_PIPELINING && (CX_NW > 1) Parameter Name: CX_CRC_LATENCY_RDLH
LTSSM	This parameter adds selected pipelines in the LTSSM, providing trade-off of latency and gates for ease of timing closure. This parameter is for Conventional PCIe mode only. Values: 0, 1 Default Value: CX_TECHNOLOGY == 0 Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_SMLH_PIPELINE_LTSSM
Retry Buffer	Retry Buffer Pipeline Latency value. This value represents the number of pipeline stages used in retry buffer, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: (CX_TECHNOLOGY == 0) ((CX_GEN2_MODE == 0) && (CX_FREQ == 1) && (ARC_WIDTH < 128)) Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_XDLH_PIPELINE_RBUF
Intermodule Pipeline Enables	
SerDes Module Input for Sync Header Bypass	Inserts a pipeline at the following SerDes architecture related input signals for the path used when sync header bypass is enabled, providing trade-off of latency and gates for ease of timing closure. ■ phy_mac_rxdata ■ phy_mac_rxvalid Values: 0, 1 Default Value: 0 Enabled: (CX_CUSTOM_PIPELINING) && (CX_CPCIE_ENABLE) && (CX_PIPE_SERDES_ARCH) && (CX_CXL_ENABLE) Parameter Type: Performance Setting Parameter Name: CX_SERDES_ARCH_RX_SH_BYPASS_REGIN

Label	Description
Receive Serialization Queue Write Pipeline	Inserts a pipeline at the Receive Serialization Queue RAM write interface. Values: 0, 1 Default Value: CX_TECHNOLOGY==0 (CX_MAX_CORECLK_FREQ >= 1000) (CX_RASDP_EN && ((CX_GEN2_MODE==GEN2_DF && CX_FREQ==FREQ_250) (CX_GEN3_MODE==GEN3_DF && CX_FREQ==FREQ_125) (CX_GEN4_MODE==GEN4_DF && CX_FREQ==FREQ_62_5))) Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_RADM_FORMQ_WR_REGOUT
Receive MAC Layer (RML) Byte Order Input	Specifies insertion of register pipeline at the inputs of RMLH Byte Order, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: ([<functionof> CX_NB CX_NL] >= 128) ? (CX_GEN3_MODE == 0) ? 1 : ((CX_TECHNOLOGY == 0) ? 1 : 0) : 0 Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_RMLH_PKT_FINDER_REGIN
Receive MAC Layer (RML) PIPE Output	Specifies insertion of register pipeline at the outputs of RMLH PIPE, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: (CX_TECHNOLOGY == 2) ? ((CX_CXL_ENABLE) ? 0 : (((CX_NB == 1) (CX_GEN2_SPEED)) & (CX_NL > 1)) ? 1:0) : 1 Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_RMLH_PIPE_REGOUT
Receive MAC Layer (RML) Deskew Input	Specifies insertion of register pipeline at the Input of Deskew, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: 0 Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_RMLH_DESKEW_REGIN
Receive Data Link Layer (RDL) TLP Extract Output	Specifies insertion of register pipeline at the outputs of RDLH TLP Extract, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: ((CX_TECHNOLOGY == 0)) ? 1 : (((CX_NB == 1) (CX_2ND_SPEED)) & (CX_NL > 4)) ? 1: 0 Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_RDLH_TLP_EXTRACT_REGOUT

Label	Description
Receive Transaction Layer (RTL) TLP Extraction	Insert a pipeline at the output of the Receive Transaction Layer TLP extraction module. Values: 0, 1 Default Value: 1 Enabled: CX_RTLH_SIMPLE_EXTRACT && CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_RTLH_TLP_EXTRACT_REGOUT
Receive Transaction Layer (RTL) TLP FC credit check	Specifies insertion of a local register pipeline for Flow Control credit check calculation in the Receive Transaction Layer TLP check module for ease of timing closure. Values: 0, 1 Default Value: (CX_TECHNOLOGY == 0) (CX_MAX_CORECLK_FREQ > 781) && (CX_NW==16) && (CX_RASDP>2 Enabled: (CX_CUSTOM_PIPELINING && (CX_NW==8 CX_NW==16)) Parameter Type: Performance Setting Parameter Name: CX_RTLH_FC_CHECK_REGOUT
Receive Header and Data Queue Write RAM Outputs	Insert a pipeline at the Receive Header and Data Queue RAM Write interfaces. Values: 0, 1 Default Value: CX_TECHNOLOGY==0 (CX_MAX_CORECLK_FREQ > 500) Enabled: CX_RADMQ_MODE==2 && CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_RADM_RAM_WR_REGOUT
Receive Input Queue Manager Output	Specifies insertion of register pipeline at the output of the receive queue input queue manager before the receive queue RAM inputs, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: 1 Enabled: CX_RADMQ_MODE==2 Parameter Type: Performance Setting Parameter Name: CX_RADM_INQ_MGR_REGOUT
Transmit Transaction Layer (XTL) Output	Specifies insertion of register pipeline at the outputs of XTLH, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: ((CX_ECRC_ENABLE == 1) (((CX_RASDP > 0) && (CX_NW <= 2))) ? 1 : (((CX_NB == 1) (CX_2ND_SPEED)) & (CX_NL > 1)) ? 1: 0) Enabled: (CX_CUSTOM_PIPELINING && !CX_ECRC_ENABLE && !((CX_RASDP>0) && (CX_NW <= 2))) Parameter Type: Performance Setting Parameter Name: CX_XTLH_CTRL_REGOUT

Label	Description
Transmit Data Link Layer (XDL) Output	Specifies insertion of register pipeline at the outputs of XDLH, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: ((CX_TECHNOLOGY == 0)) ? 1 : 1 Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_XDLH_TLP_REGOUT
Transmit MAC Layer (XML) PIPE Output	Specifies insertion of register pipeline at the outputs of XMLH PIPE, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: ((CX_TECHNOLOGY == 0)) ? 1 : (CX_CXL_ENABLE) ? 0 : 1 Enabled: CX_CUSTOM_PIPELINING && !CX_MPCIE_ENABLE Parameter Type: Performance Setting Parameter Name: CX_XMLH_PIPE_REGOUT
Transmit MAC Layer (XML) Precoding Output	Specifies insertion of register pipeline at the outputs of XMLH Precoding, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: (SNPS_RSVDPARAM_37 == 1) ? 1 : (CX_TECHNOLOGY == 0) ? 0 : (CX_CXL_ENABLE && CX_GEN5_SPEED) ? 1 : 0 Enabled: CX_CUSTOM_PIPELINING && CX_GEN5_SPEED Parameter Type: Performance Setting Parameter Name: CX_XMLH_PRECODING_REGOUT
Error Log Output	Specifies insertion of register pipeline at the Outputs Error Log, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: ((CX_TECHNOLOGY == 0) (CX_SRIOV_ENABLE && CX_NVFUNC>32) (NFUNC_GTR_THAN_32==1)) ? 1 : 0 Enabled: CX_CUSTOM_PIPELINING==1 Parameter Type: Performance Setting Parameter Name: CX_ERROR_LOG_REGOUT
Receive Filter To Receive Queue	Specifies insertion of register pipeline between receive filter and receive queue input manager, providing trade-off of latency and gates for ease of timing closure. Values: 0, ..., 2 Default Value: (CX_INTERNAL_ATU_ENABLE && ((CX_ATU_NUM_INBOUND_REGIONS > 32) (CX_RASDP!=0 && CX_MAX_CORECLK_FREQ > 500))) ? 2 : ADDR_TRANSLATION_SUPPORT_EN SATA_CAP_ENABLE CX_GEN3_DYNAMIC_FREQ Enabled: (CX_CUSTOM_PIPELINING==1 && (ADDR_TRANSLATION_SUPPORT_EN==0 && SATA_CAP_ENABLE==0 CX_INTERNAL_ATU_ENABLE)) Parameter Type: Performance Setting Parameter Name: CX_FLT_Q_REGOUT

Label	Description
Radm_bypass pipeline	Specifies insertion of register pipeline in the radm_bypass interface, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: CX_RASDP > 0) (CX_MAX_CORECLK_FREQ > 500) (CX_INTERNAL_ATU_ENABLE && AMBA_INTERFACE==0 Enabled: Always Parameter Type: Performance Setting Parameter Name: CX_RADM_BYPASS_REGOUT
XADM Hdr/Data Align To Output Formation	Specifies insertion of register pipeline before XADM output formation providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: (CX_INTERNAL_ATU_ENABLE && CX_ATU_NUM_OUTBOUND_REGIONS > 32) (CX_RAS_DES_EINJ_ENABLE) Enabled: CX_CUSTOM_PIPELINING==1 Parameter Type: Performance Setting Parameter Name: CX_XADM_FORMATION_REGIN
XADM CplQ Mng pipeline from RADM	Specifies insertion of register pipeline in XADM CplQ Management for RADM Timeout/TRGT1 interface. Values: 0, 1 Default Value: CX_CPLQ_MANAGEMENT_ENABLE==1 && CX_MAX_TAG_GTR_255==1 Enabled: CX_CUSTOM_PIPELINING==1 && CX_CPLQ_MANAGEMENT_ENABLE==1 Parameter Type: Performance Setting Parameter Name: CX_XADM_CPLQ_MNG_RADM_PIPE
Pfvf_to_vfindx To RADM Filter Output	Specifies insertion of a local register pipeline for the pfvf_to_vfindx output in the RADM_FILTER providing area trade off for ease of timing closure. Values: 0, 1 Default Value: CX_SRIOV_ENABLE && (CX_NVFUNC>32 CX_NFUNC>32) Enabled: CX_SRIOV_ENABLE==1 Parameter Type: Performance Setting Parameter Name: VFINDEX_CALC_REGOUT
Rid_to_pfvf Inputs	Specifies insertion of register pipeline for the rid_to_pfvf module inputs in the RADM_FILTER and RADM_CPL_LUT blocks, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: CX_SRIOV_ENABLE && DWC_PCIE_FPGA Enabled: CX_CUSTOM_PIPELINING==1 && CX_SRIOV_ENABLE Parameter Type: Performance Setting Parameter Name: CX_RID_REGIN

Label	Description
Rid_to_pfvf Outputs	Specifies insertion of register pipeline for the rid_to_pfvf module output in the RADM_FILTER and RADM_CPL_LUT blocks, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: CX_SRIOV_ENABLE RADM_VFINDEX_REGOUT Enabled: CX_CUSTOM_PIPELINING==1 && CX_ARI_ENABLE && !RADM_VFINDEX_REGOUT Parameter Type: Performance Setting Parameter Name: CX_RID_REGOUT
Vendor MSI Interface Inputs	Specifies insertion of an input register pipeline for the Vendor MSI interface Values: 0, 1 Default Value: ((CX_TECHNOLOGY == 0) (CX_SRIOV_ENABLE ==1)) ? 1 : 0 Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_VEN_MSI_REGIN
MSG GEN output pipeline	Specifies insertion of register pipeline at the output of msg_gen module. This eases timing closure when many VFs are used together with RASDP. Values: 0, 1 Default Value: CX_SRIOV_ENABLE==1 && CX_NVFUNC>32 && CX_MULTISTREAM==0 && CX_RASDP_EN==1 Enabled: CX_CUSTOM_PIPELINING Parameter Name: CX_MSG_GEN_REGOUT
Application Error Return Interface Inputs	Specifies insertion of an input register pipeline for the Application Error Return interface Values: 0, 1 Default Value: ((CX_TECHNOLOGY == 0) (CX_SRIOV_ENABLE ==1) (CX_MAX_CORECLK_FREQ > 500)) ? 1 : 0 Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_APP_ERR_REGIN
Lane Flip RX Input	Specifies insertion of register pipeline at the inputs of Lane Flip block signals from PHY, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: (CX_DRIFT_BUFFER_SUPPORT ? 1 : (CX_MAX_CORECLK_FREQ >= 500)) Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_LANEFLIP_RX_REGIN

Label	Description
Lane Flip TX Output	Specifies insertion of register pipeline at the outputs of Lane Flip block signals to PHY, providing trade-off of latency and gates for ease of timing closure. Values: 0, 1 Default Value: ((CX_CXL_ENABLE && CX_PIPE_SERDES_ARCH) ? 0 : (CX_MAX_CORECLK_FREQ >= 500)) Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_LANEFLIP_TX_REGOUT
RADM Completion LUT Fetch Pipeline	Specifies insertion of register pipeline at the fetch stage of the RADM Completion Timeout mechanism Values: 0, 1 Default Value: 1 Enabled: CX_CUSTOM_PIPELINING Parameter Type: Performance Setting Parameter Name: CX_RADM_LUT_TO_FETCH_PIPE
RADM Filter to LUT Pipeline	Specifies insertion of register pipeline between the CPL filter and the CPL LUT Values: 0, 1 Default Value: CX_RADM_CPL_LUT_PIPE_EN && CX_SRIOV_ENABLE && CX_NW==8 && CPL_LUT_VFINDEX_REGOUT Enabled: CX_CUSTOM_PIPELINING && CX_RADM_CPL_LUT_PIPE_EN && CX_SRIOV_ENABLE && CX_NW==8 && CPL_LUT_VFINDEX_REGOUT Parameter Type: Performance Setting Parameter Name: CX_RADM_FILTER_TO_LUT_PIPE
Application FLR Interface Inputs Pipeline	Specifies insertion of an input register pipeline for the Application FLR interface to ease timing closure. Values: 0, 1 Default Value: CX_FLR_ENABLE && (CX_MAX_CORECLK_FREQ >=500) Enabled: CX_CUSTOM_PIPELINING && CX_FLR_ENABLE Parameter Type: Performance Setting Parameter Name: CX_APP_FLR_PIPE_EN
Application DRS Interface Inputs Pipeline	Specifies insertion of an input register pipeline for the Application DRS interface to ease timing closure. Values: 0, 1 Default Value: 0 Enabled: (CX_CUSTOM_PIPELINING && CX_RN_DRS_SUPPORTED && (CC_DEVICE_TYPE!=CC_RC)) Parameter Type: Performance Setting Parameter Name: CX_APP_DRS_PIPE_EN

Label	Description
External VFs Access Interface Inputs	Specifies insertion of an input register pipeline for the External VFs Access interface to ease timing closure. Values: 0, 1 Default Value: CX_EXTENSIBLE_VFUNC && CX_NVFUNC>256 Enabled: CX_CUSTOM_PIPELINING && CX_EXTENSIBLE_VFUNC Parameter Type: Performance Setting Parameter Name: CX_EXTENSIBLE_VF_REGIN

1.56 Advanced Buffer Config / Retry and SOT Buffer Worksheet Parameters

Table 1-56 Advanced Buffer Config / Retry and SOT Buffer Worksheet Parameters

Label	Description
General Configuration	
Enable Auto Size of Retry Buffer	Enable or disable automatic size calculation for the retry buffer.: ■ True: Enable Auto Size: The sizes of the retry buffer and SOT buffer are automatically calculated based on the "Maximum Payload Size Supported value", the link width, and device latencies. ■ False: Disable Auto Size: Automatic buffer sizing is disabled. The value that you enter for Retry Buffer Depth (CX_RBUF_DEPTH) directly sets the depth of the retry buffer. For more information, see the "Transmit Replay" section of the "Architecture" chapter of the <i>Databook</i>. Values: ■ false (0x0) ■ true (0x1) Default Value: true Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_RBUF_AUTOSIZE
Total Retimer Latency (Symbol times)	Read-only parameter that indicate total round trip Latency of Retimer calculated from CX_MAX_RETIMER setting and reference Retimer latency table in PCIe Base Spec, used for retry buffer auto-sizing. Values: -2147483648, ..., 2147483647 Default Value: CX_RETIMER_LATENCY Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_RETIMER_LATENCY2
Internal Delay / Link Partner Delay	Read-only parameter that indicates the sum of the MAC/PHY delays, used for retry buffer auto-sizing, AckNak Timer adjustment and replay timer adjustment. Values: 0, ..., 2048 Default Value: CX_PHY_TX_DELAY_MAC + CX_PHY_TX_DELAY_PHY + CX_PHY_TX_DELAY_SERDES_PIPE + CX_PHY_RX_DELAY_MAC + CX_PHY_RX_DELAY_PHY + CX_PHY_RX_DELAY_SERDES_PIPE Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_INTERNAL_DELAY

Label	Description
Retry Buffer Configuration	
Retry Buffer Depth	The depth of the retry buffer. When retry buffer auto-sizing is enabled (CX_RBUF_AUTOSIZE), "Retry Buffer Depth" is read-only and indicates the automatically calculated depth of the retry buffer. When retry buffer auto-sizing is disabled, "Retry Buffer Depth" is the user-specified depth of the retry buffer. Note that if the autosize feature is disabled, then the minimum value specified must be large enough to allow normal operation. The default value is the automatically calculated value. For more information, see the 'Transmit Replay' section of the Architecture chapter of the <i>Databook</i>. Values: SNPS_RSVDPARAM_54, ..., 4294967295 Default Value: [<functionof> CX_DL_NB CX_NL CX_MAX_MTU CX_ECRC_ENABLE CX_TLP_PREFIX_ENABLE_VALUE CX_NPRFX CX_INTERNAL_DELAY CX_MAX_L0S_LTIME CX_GEN5_SPEED CX_GEN4_SPEED CX_GEN3_SPEED CX_5GTS_SPEED CX_RETIMER_LATENCY] Enabled: ((!CX_RBUF_AUTOSIZE)) Parameter Type: Feature Setting Parameter Name: CX_RBUF_DEPTH
Retry Buffer Width	Read-only parameter that indicates the width of the retry buffer. Values: 34, ..., 686 Default Value: RBUF_PROT_WD + [<functionof> CX_DL_NB CX_NL CX_TLP_PREFIX_ENABLE CX_GEN3_SPEED] Enabled: Always Parameter Type: Feature Setting Parameter Name: RBUF_WIDTH

Label	Description
SOT Buffer Configuration	
Minimum SOT Depth	When retry buffer auto-sizing is enabled, the value displayed here corresponds to the number of minimum-sized TLPs (3 DWORDs) that can reside in the retry buffer. ■ The 'Start Of TLP (SOT)' buffer stores the starting address of each unacknowledged TLP stored in the retry buffer. ■ The SOT buffer requires one entry for each TLP that can be stored in the retry buffer. ■ The selected retry buffer size determines the size of the SOT buffer. ■ The actual SOT buffer depth is a combination of the value displayed here and the additional requirements that the actual SOT buffer depth must be at least 32 and must be a power of 2. When retry buffer auto-sizing is disabled, 'Minimum SOT Depth' is the user-specified depth of the SOT buffer. ■ The selected size must allow the retry buffer to store the maximum number of shortest TLPs (3 DWORDs). ■ The actual SOT buffer depth (SOTBUF_DEPTH) is calculated by adjusting CX_SOTBUF_DEPTH to be at least 32, and rounding up to the next power-of-2. For more information, see the 'Transmit Replay' section of the 'Architecture' chapter of the <i>Databook</i>. Values: -2147483648, ..., 2147483647 Default Value: [<functionof> CX_DL_NB CX_NL CX_MAX_MTU CX_ECRC_ENABLE CX_TLP_PREFIX_ENABLE_VALUE CX_NPRFX CX_INTERNAL_DELAY CX_MAX_L0S_LTIME CX_GEN5_SPEED CX_GEN4_SPEED CX_GEN3_SPEED CX_5GTS_SPEED CX_RETIMER_LATENCY] Enabled: ((CX_RBUF_AUTOSIZE==0)) Parameter Type: Feature Setting Parameter Name: CX_SOTBUF_DEPTH
SOT Buffer Depth	Read-only parameter that indicates the depth of the SOT buffer. Values: 32, ..., 4294967295 Default Value: [<functionof> CX_DL_NB CX_NL CX_MAX_MTU CX_ECRC_ENABLE CX_TLP_PREFIX_ENABLE_VALUE CX_NPRFX CX_INTERNAL_DELAY CX_MAX_L0S_LTIME CX_RBUF_AUTOSIZE CX_SOTBUF_DEPTH CX_GEN5_SPEED CX_GEN4_SPEED CX_GEN3_SPEED CX_5GTS_SPEED CX_RETIMER_LATENCY] Enabled: Always Parameter Type: Feature Setting Parameter Name: SOTBUF_DEPTH
SOT Buffer Width	Read-only parameter that indicates the width of the SOT buffer. Values: 0, ..., 4294967295 Default Value: CX_RAS_EN ? (CX_RAS_SOTBUF_PROT_WD + [<functionof> CX_RBUF_DEPTH]) : [<functionof> CX_RBUF_DEPTH] Enabled: Always Parameter Type: Feature Setting Parameter Name: SOTBUF_WIDTH

1.57 Advanced Buffer Config / Segmented-Buffer Options Parameters

Table 1-57 Advanced Buffer Config / Segmented-Buffer Options Parameters

Label	Description
Segmented-Buffer Options	
Receive VC Arbitration	The VC arbitration scheme for sending received TLPs to your application. ■ Strict Priority: Higher number VC IDs have higher priority ■ Round Robin: Round robin arbitration between VCs Notes: ■ This parameter is only applicable when CX_NVC > 1. ■ It is possible to change the VC Arbitration Scheme (during device setup by software) by writing to the VC_ORDERING_RX_Q field in the VC0_P_RX_Q_C-TRL_OFF register. Values: ■ Round Robin (0x0) ■ Strict Priority (0x1) Default Value: Round Robin Enabled: CX_RADMQ_MODE==2 Parameter Type: Register Default Setting Parameter Name: CX_RADM_STRICT_VC_PRIORITY
Enable Dynamic FC Credit Adjustment	Enable your application to update the advertised FC credit values by writing to the VC0 Posted Receive Queue Control port logic register. Values: ■ false (0) ■ true (1) Default Value: FC_SCALE_EN Enabled: CX_RADMQ_MODE==2 Parameter Type: Feature Setting Parameter Name: CX_DYNAMIC_FC_CREDIT

1.58 Advanced Buffer Config / Ordering Rules Configuration (Segmented-Buffer) Parameters

Table 1-58 Advanced Buffer Config / Ordering Rules Configuration (Segmented-Buffer) Parameters

Label	Description
Ordering Rules Configuration (Segmented-Buffer)	
Support Relaxed Ordering	Allows completions to be sent to your application out of order. For more information, see Receive Queues in the Architecture chapter of the <i>Databook</i>. Values: ■ false (0) ■ true (1) Default Value: true Enabled: CX_RADMQ_MODE==2 Parameter Type: Feature Setting Parameter Name: RELAXED_ORDER_SUPPORT
Ordering Rules	Specifies the ordering scheme for sending received TLPs to the application: ■ 0: Strict Priority: The priority order is Posted, then Completion, then Non-Posted ■ 1: PCI Ordering Rules: The arbitration is according to the ordering rules in the <i>PCI Express 3.1 Specification</i>. For more information, see the "Native Controller Receive Ordering Schemes" section in the "Advanced Ordering Information" appendix of the <i>Databook</i>. It is possible to change the ordering rules used (by software) by writing to the appropriate queue control register 'TLP Type Ordering'. Values: ■ Strict Priority (0x0) ■ PCIe Ordering Rules (0x1) Default Value: PCIe Ordering Rules Enabled: CX_RADMQ_MODE==2 Parameter Type: Feature Setting Parameter Name: CX_RADM_ORDERING_RULES

1.59 Advanced Buffer Config / Receive Serialization Queue Parameters

Table 1-59 Advanced Buffer Config / Receive Serialization Queue Parameters

Label	Description
Receive Serialization Queue	
Enable Receive Serialization Queue Size Configuration	The Receive Serialization Queue Size Configuration feature applies to 512-bit data width configurations and allows the size of the serialization queue to be adjusted while maintaining throughput at the application interface. Values: 0, 1 Default Value: 0 Enabled: CX_NW==16 Parameter Type: Feature Setting Parameter Name: RX_SERIALIZATION_Q_CTRL
Queue Size (% of the available TLP buffer space)	This is the size of the serialization queue relative to the storage capacity of the credit queues expressed as a percentage of credit queue size. The serialisation queue margin is excluded. Values: 0, ..., 100 Default Value: CX_NW==16 ? [<functionof> CX_RADM_RXQ_NUM_QDWS > CX_RADM_SERQ_OPT_AF_NUM_QDWS] ? [<functionof> [<functionof> CX_RADM_SERQ_OPT_AF_NUM_QDWS_PER_RAM*100] CX_RADM_RXQ_NUM_QDWS_PER_SERQ_RAM] : 100 : 1 Enabled: RX_SERIALIZATION_Q_CTRL Parameter Type: Feature Setting Parameter Name: CX_RADM_SERQ_TO_RXQ_SIZE_RATIO
Queue Size (Bytes)	Receive Serialization Queue size expressed in Bytes (additional control bits, parity bits, and so forth are not considered) Values: -2147483648, ..., 2147483647 Default Value: (CX_RADM_SERQ_AF_NUM_QDWS_PER_RAM*CX_RADM_SERQ_NUM_RAM)*16 Enabled: Always Parameter Name: CX_RADM_SERQ_AF_NUM_BYTES

1.60 Advanced RX Queue Credit and Size Config / Cplq_Mng Calculator Parameters

Table 1-60 Advanced RX Queue Credit and Size Config / Cplq_Mng Calculator Parameters

Label	Description
Completion Queue Size dependant parameters	
Round Trip Latency[ns]	Specifies your expected Round Trip Latency in ns. Includes: ■ Local Controller and PHY latencies ■ Remote Controller and PHY latencies ■ Retimers ■ Response time of the remote side Values: 0, ..., 65535 Default Value: 500 Enabled: Always Parameter Type: Feature Setting Parameter Name: ROUND_TRIP_LATENCY
TLP Efficiency[%]	Specifies the TLP Efficiency in %. Smaller payload has less data efficiency (Data/(Hdr+Data)) and therefore requires less data buffer. Values: 0, ..., 65535 Default Value: 100*CX_MAX_MTU/(CX_MAX_MTU+20) Enabled: 0 Parameter Type: Feature Setting Parameter Name: TLP_EFFICIENCY
Recommended Number of TAGs for the Critical Request Size	
Critical Request size[bytes]	Specifies the smallest request size for which you want to maintain throughput. Used to calculate the recommended number of TAGs (CPLQ_MNG_DDP/MIN_RD_REQ_SIZE). Recommended size equal to or greater than 64 bytes. Values: 32, 64, 128, 256, 512, 1024, 2048, 4096 Default Value: 64 Enabled: Always Parameter Name: MIN_RD_REQ_SIZE
Recommended number of TAGs	Recommended number of TAGs based on Latency and Critical Request Size specified. Values: 0, ..., 65535 Default Value: [<functionof> CPLQ_MNG_DDP MIN_RD_REQ_SIZE] Enabled: 0 Parameter Type: Feature Setting Parameter Name: RECOMMENDED_TAGS

Label	Description
Completion Queue Size	
Header Completion Queue size[Hdr]	Specifies the size of the Completion Data Queue/RAM in hdr units without the overhead needed by the Queue logic. Derived from the number of TAGs. Values: 0, ..., 65535 Default Value: MIN_RD_REQ_SIZE >= 2*CX_RCB) ? (1+MIN_RD_REQ_SIZE/CX_RCB)*(CX_MAX_TAG+1) : 2*(CX_MAX_TAG+1) Enabled: CX_CPLQ_MANAGEMENT_ENABLE Parameter Type: Feature Setting Parameter Name: CPLQ_MNG_HDP
Data Completion Queue size[bytes]	Specifies the size of the Completion Data Queue/RAM in bytes without the overhead needed by the Queue logic. This is a recommendation based on your bandwidth, max TLP size and expected latency. The Completion Queue Management feature can result in throttling of outbound read request. It is your responsibility to guarantee sufficient completion queue buffer to meet your performance requirements. Values: 0, ..., 65535 Default Value: [<functionof> CX_MAX_PCIE_SPEED CX_NL ROUND_TRIP_LATENCY TLP_EFFICIENCY CX_APP_RD_REQ_SIZE CX_NW RADM_CPL_QMODE_VC0] Enabled: Always Parameter Type: Feature Setting Parameter Name: CPLQ_MNG_DDP
Completion Queue Depth	
Header Completion Queue Depth[Hdr/CX_NHQ]	Specifies the depth of the completion header queue including overhead for queue logic. Values: 0, ..., 65535 Default Value: RADM_CPLQ_HDP_VC0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CPLQ_MNG_TOTAL_HDP
Data Completion Queue Depth[NW]	Specifies the depth of the completion data queue including overhead for queue logic. Values: 0, ..., 65535 Default Value: RADM_CPLQ_DDP_VC0 Enabled: 0 Parameter Type: Feature Setting Parameter Name: CPLQ_MNG_TOTAL_DDP

1.61 Advanced RX Queue Credit and Size Config / VC 0 Parameters

Table 1-61 Advanced RX Queue Credit and Size Config / VC 0 Parameters

Label	Description
Posted Queue Advertised Credits	
Hdr Scale Factor	Initial VC0 Posted header scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: (CX_HSCALE == 16) ? 3 : (CX_HSCALE == 4) ? 2 : 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_PQ_HSCALE_VC0
Hdr Credits	The number of posted TLP header credits to advertise for VC0. Values: 0, ..., 127 Default Value: RADM_P_QMODE_VC0==4 ? 0 : CX_CALC_HDEPTH / CX_HSCALE Enabled: Always Parameter Type: Feature Setting Parameter Name: RADM_PQ_HCRD_VC0
Data Scale Factor	Initial VC0 Posted data scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: (CX_DSCALE == 16) ? 3 : (CX_DSCALE == 4) ? 2 : 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_PQ_DSCALE_VC0
Data Credits	The number posted TLP data credits to advertise for VC0. Values: 0, ..., 2047 Default Value: RADM_P_QMODE_VC0 ==4 ? 0 : CX_CALC_DDEPTH / CX_DSCALE Enabled: Always Parameter Type: Feature Setting Parameter Name: RADM_PQ_DCRD_VC0

Label	Description
Non-Posted Queue Advertised Credits	
Hdr Scale Factor	Initial VC0 Non-Posted header scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: (CX_HSCALE == 16) ? 3 : (CX_HSCALE == 4) ? 2 : 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_NPQ_HSCALE_VC0
Hdr Credits	The number non-posted TLP header credits to advertise for VC0. Values: 1, ..., 127 Default Value: (CX_CALC_HDEPTH==0) ? 1 : CX_CALC_HDEPTH / CX_HSCALE Enabled: Always Parameter Type: Feature Setting Parameter Name: RADM_NPQ_HCRD_VC0
Data Scale Factor	Initial VC0 Non-Posted data scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_NPQ_DSCALE_VC0
Data Credits	The number non-posted TLP data credits to advertise for VC0. Values: 1, ..., 2047 Default Value: (CX_NP_CALC_DDEPTH==0) ? 1 : CX_NP_CALC_DDEPTH / CX_NP_DSCALE Enabled: Always Parameter Type: Feature Setting Parameter Name: RADM_NPQ_DCRD_VC0

Label	Description
Completion Queue Advertised Credits	
Hdr Scale Factor	Initial VC0 Completion header scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: (CX_HSCALE == 16) ? 3 : (CX_HSCALE == 4) ? 2 : 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_HSCALE_VC0
Hdr Credits	The number completion TLP header credits to advertise for VC0. Values: 0, ..., 127 Default Value: (RADM_CPL_QMODE_VC0==4 CC_DEVICE_TYPE!=CC_SW) ? 0 : CX_CALC_HDEPTH / CX_HSCALE Enabled: Always Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_HCRD_VC0
Data Scale Factor	Initial VC0 Completion data scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: (CX_DSCALE == 16) ? 3 : (CX_DSCALE == 4) ? 2 : 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_DSCALE_VC0
Data Credits	The number of completion TLP data credits to advertise for VC0. Values: 0, ..., 2047 Default Value: (RADM_CPL_QMODE_VC0==4 CC_DEVICE_TYPE!=CC_SW) ? 0 : CX_CALC_DDEPTH / CX_DSCALE Enabled: Always Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_DCRD_VC0

Label	Description
Additional VC 0 Options	
Max Outbound Read Request For Buffered CPLs	This parameter is used to set the depth of the receive completion data queue (CX_CPLQ_DDP_VC*) when completions are in store-and-forward or cut-through modes, and completion credits are infinite. ■ It is the maximum <i>individual</i> MRd size that your application (or AXI bridge slave) makes. ■ For more information, see the Receive Queues section in the Architecture chapter of the <i>Databook</i>. ■ When an AXI bridge is enabled, CX_APP_RD_REQ_SIZE is calculated automatically and you cannot change it. Values: 32, 64, 128, 256, 512, 1024, 2048, 4096 Default Value: !AMBA_POPULATED ? CX_MAX_MTU : (CC_DMA_ENABLE ? CX_MAX_APP_RD_REQ_SIZE_AMBA_DMA : CC_SLV_MTU) Enabled: !AMBA_POPULATED && (RADM_CPL_QMODE_VC0!=4) Parameter Name: CX_APP_RD_REQ_SIZE
Posted Buffer Depth	
Hdr	Specifies the depth of the Posted Header Queue/RAM. The number of entries in the Posted Header buffer for VC0. This option is read-only if the queue is bypassed. Note: for 256-bit configurations this depth corresponds to one half of the total header storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: RADM_P_QMODE_VC0==4) ? (CX_RADMQ_MODE==2) ? RADM_SEG_BUF_MIN_DPT : 0 : ([<functionof> CX_RADMQ_MODE RADM_PQ_HCRD_VC0 RADM_NPQ_HCRD_VC0 RADM_CPLQ_HCRD_VC0 RADM_CPL_QMODE_VC0 CX_NHQ CUT_THROUGH_INVOLVED CX_MAX_TAG CX_APP_RD_REQ_SIZE CX_RCB RADM_PQ_HSCALE_VC0] Enabled: RADM_P_QMODE_VC0!=4 && RADM_DEPTH_DECOUPLE_VC0==1 Parameter Type: Feature Setting Parameter Name: RADM_PQ_HDP_VC0

Label	Description
Data	Specifies the depth of the Posted Data Queue/RAM. The number of entries in the Posted Data buffer for VC0. This option is read-only if the queue is bypassed. Note: for 256-bit configurations this depth corresponds to one half of the total data storage capacity for this particular queue type; for 512-bit configurations this depth corresponds to one quarter of the total data storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: $\text{RADM_P_QMODE_VC0}==4 ? (\text{CX_RADMQ_MODE}==2) ? \text{RADM_SEG_BUF_MIN_DPT} : 0 : ([\langle \text{function of} \rangle \text{CX_RADMQ_MODE} \text{RADM_PQ_HCRD_VC0 RADM_PQ_DCRD_VC0 RADM_NPQ_HCRD_VC0 RADM_NPQ_DCRD_VC0 RADM_CPLQ_HDP_VC0 RADM_CPLQ_DCRD_VC0 RADM_CPL_QMODE_VC0 CX_NW CX_NHQ CC_DEVICE_TYPE CC_SW CX_ECRC_STRIP_ENABLE RADM_SEG_BUF_CT_DPT_ADJ CX_MAX_TAG CX_APP_RD_REQ_SIZE RADM_PQ_DS_SCALE_VC0 RADM_PQ_HS_SCALE_VC0])$ Enabled: $\text{RADM_P_QMODE_VC0}!=4 \ \&\& \ \text{RADM_DEPTH_DECOUPLE_VC0}==1$ Parameter Type: Feature Setting Parameter Name: RADM_PQ_DDP_VC0
Non-Posted Buffer Depth	
Hdr	Specifies the depth of the Non-Posted Header Queue/RAM. The number of entries in the Non-Posted Header buffer for VC0. Note: for 256-bit configurations this depth corresponds to one half of the total header storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: $\text{CX_RADMQ_MODE}==0 \ \ \text{RADM_NP_QMODE_VC0}==4) ? (\text{CX_RADMQ_MODE}==2) ? \text{RADM_SEG_BUF_MIN_DPT} : 0 : ([\langle \text{function of} \rangle \text{CX_RADMQ_MODE RADM_NPQ_HCRD_VC0 CUT_THROUGH_INVOLVED CX_NHQ RADM_NPQ_HS_SCALE_VC0}]$ Enabled: $\text{CX_RADMQ_MODE}!=0 \ \&\& \ \text{RADM_NP_QMODE_VC0}!=4 \ \&\& \ \text{RADM_DEPTH_DECOUPLE_VC0}==1$ Parameter Type: Feature Setting Parameter Name: RADM_NPQ_HDP_VC0

Label	Description
Data	Specifies the depth of the Non-Posted Data Queue/RAM. The number of entries in the Non-Posted Data buffer for VC0. Note: for 256-bit configurations this depth corresponds to one half of the total data storage capacity for this particular queue type; for 512-bit configurations this depth corresponds to one quarter of the total data storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: CX_RADMQ_MODE==0 RADM_NP_QMODE_VC0==4) ? (CX_RADMQ_MODE==2) ? RADM_SEG_BUF_MIN_DPT : 0 : ([<functionof> 0 CX_RADMQ_MODE RADM_NPQ_HCRD_VC0 RADM_NPQ_DCRD_VC0 CUT_THROUGH_INVOLVED CX_NW CX_NHQ CC_DEVICE_TYPE CC_SW CX_ECRC_STRIP_ENABLE RADM_NPQ_DSCALE_VC0 RADM_NPQ_HSCALE_VC0] Enabled: CX_RADMQ_MODE!=0 && RADM_NP_QMODE_VC0!=4 && RADM_DEPTH_DECOUPLE_VC0==1 Parameter Type: Feature Setting Parameter Name: RADM_NPQ_DDP_VC0
Completion Buffer Depth	
Hdr	Specifies the depth of the Completion Header Queue/RAM. The number of entries in the Completion Header buffer for VC0. This option is read-only for all DMA and/or AMBA configs and also if the queue is bypassed. Note: for 256-bit configurations this depth corresponds to one half of the total header storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: CX_RADMQ_MODE==0 RADM_CPL_QMODE_VC0==4) ? (CX_RADMQ_MODE==2) ? RADM_SEG_BUF_MIN_DPT : 0 : ([<functionof> CX_CPLQ_MANAGEMENT_ENABLE CPLQ_MNG_HDP CX_RADMQ_MODE RADM_CPLQ_HCRD_VC0 CUT_THROUGH_INVOLVED CX_NHQ CX_MAX_TAG CX_APP_RD_REQ_SIZE CX_RCB RADM_CPLQ_HSCALE_VC0] Enabled: ((AMBA_INTERFACE == 0 && CC_DMA_ENABLE == 0) CX_CPLQ_MANAGEMENT_ENABLE) && CX_RADMQ_MODE!=0 && RADM_CPL_QMODE_VC0!=4 && (RADM_DEPTH_DECOUPLE_VC0==1 RADM_CPLQ_HCRD_VC0 == 0) Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_HDP_VC0

Label	Description
Data	Specifies the depth of the Completion Data Queue/RAM. The number of entries in the Completion Data buffer for VC0. This option is read-only for all DMA and/or AMBA configs and also if the queue is bypassed. Note: for 256-bit configurations this depth corresponds to one half of the total data storage capacity for this particular queue type; for 512-bit configurations this depth corresponds to one quarter of the total data storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: $CX_RADMQ_MODE == 0 \parallel RADM_CPL_QMODE_VC0 == 4$? $(CX_RADMQ_MODE == 2) ? RADM_SEG_BUF_MIN_DPT : 0$: ([<functionof> $CX_CPLQ_MANAGEMENT_ENABLE$ $CPLQ_MNG_DDP$ CX_RADMQ_MODE $RADM_CPLQ_HDP_VC0$ $RADM_CPLQ_DCRD_VC0$ $CUT_THROUGH_INVOLVED$ CX_NHQ CX_NW CX_MAX_TAG $CX_APP_RD_REQ_SIZE$ CC_DEVICE_TYPE CC_SW $CX_ECRC_STRIP_ENABLE$ $RADM_SEG_BUF_CT_DPT_ADJ$ $RADM_CPLQ_DSCALE_VC0$ $RADM_CPLQ_HSCALE_VC0$] Enabled: $((AMBA_INTERFACE == 0 \&\& CC_DMA_ENABLE == 0) \parallel CX_CPLQ_MANAGEMENT_ENABLE) \&\& CX_RADMQ_MODE != 0 \&\& RADM_CPL_QMODE_VC0 != 4 \&\& (RADM_DEPTH_DECOUPLE_VC0 == 1 \parallel RADM_CPLQ_DCRD_VC0 == 0)$ Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_DDP_VC0

1.62 Advanced RX Queue Credit and Size Config / VC 1 Parameters

Table 1-62 Advanced RX Queue Credit and Size Config / VC 1 Parameters

Label	Description
Posted Queue Mode and Advertised Credits	
P Q Mode (VC#i) (for n = 1; n <= CX_NVC)	The queue mode for the posted tlp receive queue for VC1. ■ Bypass: There is no receive queue in this mode, your application must be able to accept all traffic as back-pressure is disabled in the mode. ■ Store-and-forward: TLPs are stored into queue; TLP is advertised only after the full TLP is stored into the queue. ■ Cut-through: TLPs are stored into queue and presented to your application at the same time it is being stored into the queue. For more information, see "Receive Queue Buffers" in the Architecture chapter of the <i>Databook</i>. Values: ■ Store/Fwd (0x1) ■ Cut Thru (0x2) ■ Bypass (0x4) Default Value: RADM_P_QMODE_VC0 Enabled: CX_NVC>1 && CX_RADMQ_MODE==2 Parameter Type: Feature Setting Parameter Name: RADM_P_QMODE_VCn
P Header Scaling Factor (VC#i) (for n = 1; n <= CX_NVC)	Initial VC1 Posted header scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: (CX_HSCALE == 16) ? 3 : (CX_HSCALE == 4) ? 2 : 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_PQ_HSCALE_VCn
P Data Scaling Factor (VC#i) (for n = 1; n <= CX_NVC)	Initial VC1 Posted data scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: (CX_DSCALE == 16) ? 3 : (CX_DSCALE == 4) ? 2 : 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_PQ_DSCALE_VCn

Label	Description
P Header Credits (VC#i) (for n = 1; n <= CX_NVC)	Specifies the # of Posted Hdr Credits to Advertise. Values: 0, ..., 127 Default Value: CX_NVC<=1 RADM_P_QMODE_VC1==4 ? 0 : RADM_PQ_HCRD_VC0 Enabled: CX_NVC>1 && CX_RADMQ_MODE==2 Parameter Type: Feature Setting Parameter Name: RADM_PQ_HCRD_VCn
P Data Credits (VC#i) (for n = 1; n <= CX_NVC)	Specifies the # of Posted Data Credits to Advertise. One data credit = 128 bits of data Values: 0, ..., 2047 Default Value: CX_NVC<=1 RADM_P_QMODE_VC1==4 ? 0 : RADM_PQ_DCRD_VC0 Enabled: CX_NVC>1 && CX_RADMQ_MODE==2 Parameter Type: Feature Setting Parameter Name: RADM_PQ_DCRD_VCn
Non-Posted Queue Mode and Advertised Credits	
NP Q Mode (VC#i) (for n = 1; n <= CX_NVC)	The queue mode for the non-posted TLP receive queue for VC1. ■ Store-and-forward: TLPs are stored into queue; TLP is advertised only after the full TLP is stored into the queue. For more information, see "Receive Queue Buffers" in the Architecture chapter of the <i>Databook</i> . Values: ■ Store/Fwd (0x1) Default Value: RADM_NP_QMODE_VC0 Enabled: CX_NVC>1 && CX_RADMQ_MODE==2 Parameter Type: Feature Setting Parameter Name: RADM_NP_QMODE_VCn
NP Header Scaling Factor (VC#i) (for n = 1; n <= CX_NVC)	Initial VC1 Non-Posted header scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: (CX_HSCALE == 16) ? 3 : (CX_HSCALE == 4) ? 2 : 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_NPQ_HSCALE_VCn

Label	Description
NP Data Scaling Factor (VC#i) (for n = 1; n <= CX_NVC)	Initial VC1 Non-Posted data scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_NPQ_DSCALE_VCn
NP Header Credits (VC#i) (for n = 1; n <= CX_NVC)	Specifies the # of Non-Posted Hdr Credits to Advertise. Values: 1, ..., 127 Default Value: CX_NVC<=1 ? 1 : RADM_NPQ_HCRD_VC0 Enabled: CX_NVC>1 && CX_RADMQ_MODE==2 Parameter Type: Feature Setting Parameter Name: RADM_NPQ_HCRD_VCn
NP Data Credits (VC#i) (for n = 1; n <= CX_NVC)	Specifies the # of Non-Posted Data Credits to Advertise. One data credit = 128 bits of data Values: 1, ..., 2047 Default Value: CX_ATOMIC_ENABLE ? (CX_NVC<=1) ? 1 : RADM_NPQ_DCRD_VC0 : (RADM_NPQ_DSCALE_VC0 > 1) ? 512 : 1 Enabled: CX_NVC>1 && CX_RADMQ_MODE==2 Parameter Type: Feature Setting Parameter Name: RADM_NPQ_DCRD_VCn
Completion Queue Mode and Advertised Credits	
CPL Q Mode (VC#i) (for n = 1; n <= CX_NVC)	The queue mode for the completion tlp receive queue for VC1. ■ Bypass: There is no receive queue in this mode, your application must be able to accept all traffic as back-pressure is disabled in the mode. ■ Store-and-forward: TLPs are stored into queue; TLP is advertised only after the full TLP is stored into the queue. ■ Cut-through: TLPs are stored into queue and presented to your application at the same time it is being stored into the queue. For more information, see "Receive Queue Buffers" in the Architecture chapter of the <i>Databook</i> . Values: ■ Store/Fwd (0x1) ■ Cut Thru (0x2) ■ Bypass (0x4) Default Value: RADM_CPL_QMODE_VC0 Enabled: CX_NVC>1 && CX_RADMQ_MODE==2 Parameter Type: Feature Setting Parameter Name: RADM_CPL_QMODE_VCn

Label	Description
CPL Header Scaling Factor (VC#i) (for n = 1; n <= CX_NVC)	Initial VC1 Completion header scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: (CX_HSCALE == 16) ? 3 : (CX_HSCALE == 4) ? 2 : 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_HSCALE_VCn
CPL Data Scaling Factor (VC#i) (for n = 1; n <= CX_NVC)	Initial VC1 Completion data scaling factor to advertise Flow Control credit width. Values: ■ 1 (0x1) ■ 4 (0x2) ■ 16 (0x3) Default Value: (CX_DSCALE == 16) ? 3 : (CX_DSCALE == 4) ? 2 : 1 Enabled: FC_SCALE_EN Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_DSCALE_VCn
CPL Header Credits (VC#i) (for n = 1; n <= CX_NVC)	Specifies the # of Completion Hdr Credits to Advertise. Values: 0, ..., 127 Default Value: (CX_NVC <= 1 RADM_CPL_QMODE_VC1 == 4) ? 0 : RADM_CPLQ_HCRD_VC0 Enabled: CX_NVC > 1 && CX_RADMQ_MODE == 2 Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_HCRD_VCn
CPL Data Credits (VC#i) (for n = 1; n <= CX_NVC)	Specifies the # of Completion Data Credits to Advertise. One data credit = 128 bits of data Values: 0, ..., 2047 Default Value: (CX_NVC <= 1 RADM_CPL_QMODE_VC1 == 4) ? 0 : RADM_CPLQ_DCRD_VC0 Enabled: CX_NVC > 1 && CX_RADMQ_MODE == 2 Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_DCRD_VCn

Label	Description
Posted Buffer Depth	
P Header Queue Depth (VC#i) (for n = 1; n <= CX_NVC)	Specifies the depth of the Posted Header Queue/RAM. The number of entries in the Posted Header buffer for VC1. This option is read-only if the queue is bypassed. Note: for 256-bit configurations this depth corresponds to one half of the total header storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: CX_NVC<=1 ? 0 : RADM_P_QMODE_VC1==4 ? (CX_RADMQ_MODE==2) ? RADM_SEG_BUF_MIN_DPT : 0 : ([<functionof> CX_RADMQ_MODE RADM_PQ_HCRD_VC1 CX_NHQ CUT_THROUGH_INVOLVED RADM_PQ_HSCALE_VC1]) Enabled: CX_NVC>1 && CX_RADMQ_MODE==2 && RADM_P_QMODE_VC1!=4 && RADM_DEPTH_DECOUPLE_VC1==1 Parameter Type: Feature Setting Parameter Name: RADM_PQ_HDP_VCn
P Data Queue Depth (VC#i) (for n = 1; n <= CX_NVC)	Specifies the depth of the Posted Data Queue/RAM. The number of entries in the Posted Data buffer for VC1. This option is read-only if the queue is bypassed. Note: for 256-bit configurations this depth corresponds to one half of the total data storage capacity for this particular queue type; for 512-bit configurations this depth corresponds to one quarter of the total data storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: CX_NVC<=1 ? 0 : RADM_P_QMODE_VC1==4 ? (CX_RADMQ_MODE==2) ? RADM_SEG_BUF_MIN_DPT : 0 : ([<functionof> RADM_PQ_HCRD_VC1 RADM_PQ_DCRD_VC1 CX_NW CX_NHQ CC_DEVICE_TYPE CC_SW CX_ECRC_STRIP_ENABLE RADM_SEG_BUF_CT_DPT_ADJ RADM_PQ_DSCALE_VC1 RADM_PQ_HSCALE_VC1]) Enabled: CX_NVC>1 && CX_RADMQ_MODE==2 && RADM_P_QMODE_VC1!=4 && RADM_DEPTH_DECOUPLE_VC1==1 Parameter Type: Feature Setting Parameter Name: RADM_PQ_DDP_VCn
Non-Posted Buffer Depth	
NP Header Queue Depth (VC#i) (for n = 1; n <= CX_NVC)	Specifies the depth of the Non-Posted Header Queue/RAM. The number of entries in the Non-Posted Header buffer for VC1. Note: for 256-bit configurations this depth corresponds to one half of the total header storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: CX_NVC<=1 ? 0 : RADM_NP_QMODE_VC1==4 ? (CX_RADMQ_MODE==2) ? RADM_SEG_BUF_MIN_DPT : 0 : ([<functionof> CX_RADMQ_MODE RADM_NPQ_HCRD_VC1 CUT_THROUGH_INVOLVED CX_NHQ RADM_NPQ_HSCALE_VC1]) Enabled: CX_NVC>1 && CX_RADMQ_MODE==2 && RADM_NP_QMODE_VC1!=4 && RADM_DEPTH_DECOUPLE_VC1==1 Parameter Type: Feature Setting Parameter Name: RADM_NPQ_HDP_VCn

Label	Description
NP Data Queue Depth (VC#i) (for n = 1; n <= CX_NVC)	Specifies the depth of the Non-Posted Data Queue/RAM. The number of entries in the Non-Posted Data buffer for VC1. Note: for 256-bit configurations this depth corresponds to one half of the total data storage capacity for this particular queue type; for 512-bit configurations this depth corresponds to one quarter of the total data storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: CX_NVC<=1 ? 0 : RADM_NP_QMODE_VC1==4 ? ((CX_RADMQ_MODE==2) ? RADM_SEG_BUF_MIN_DPT : 0) : ([<functionof> 1 CX_RADMQ_MODE RADM_NPQ_HCRD_VC1 RADM_NPQ_DCRD_VC1 CUT_THROUGH_INVOLVED CX_NW CX_NHQ CC_DEVICE_TYPE CC_SW CX_ECRC_STRIP_ENABLE RADM_NPQ_DSCALE_VC1 RADM_NPQ_HSCALE_VC1]) Enabled: CX_NVC>1 && CX_RADMQ_MODE==2 && RADM_NP_QMODE_VC1!=4 && RADM_DEPTH_DECOUPLE_VC1==1 Parameter Type: Feature Setting Parameter Name: RADM_NPQ_DDP_VCn
Completion Buffer Depth	
CPL Header Queue Depth (VC#i) (for n = 1; n <= CX_NVC)	Specifies the depth of the Completion Header Queue/RAM. The number of entries in the Completion Header buffer for VC1. This option is read-only for all DMA and/or AMBA configs and also if the queue is bypassed. Note: for 256-bit configurations this depth corresponds to one half of the total header storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: CX_NVC<=1 ? 0 : RADM_CPL_QMODE_VC1==4 ? ((CX_RADMQ_MODE==2) ? RADM_SEG_BUF_MIN_DPT : 0) : ([<functionof> CX_CPLQ_MANAGEMENT_ENABLE CPLQ_MNG_HDP CX_RADMQ_MODE RADM_CPLQ_HCRD_VC1 CUT_THROUGH_INVOLVED CX_NHQ CX_MAX_TAG CX_APP_RD_REQ_SIZE CX_RCB RADM_CPLQ_HSCALE_VC1]) Enabled: ((AMBA_INTERFACE == 0 && CC_DMA_ENABLE == 0) CX_CPLQ_MANAGEMENT_ENABLE) && CX_NVC>1 && CX_RADMQ_MODE==2 && RADM_CPL_QMODE_VC1!=4 && (RADM_DEPTH_DECOUPLE_VC1==1 RADM_CPLQ_HCRD_VC1 == 0) Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_HDP_VCn

Label	Description
CPL Data Queue Depth (VC#i) (for n = 1; n <= CX_NVC)	Specifies the depth of the Completion Data Queue/RAM. The number of entries in the Completion Data buffer for VC1. This option is read-only for all DMA and/or AMBA configs and also if the queue is bypassed. Note: for 256-bit configurations this depth corresponds to one half of the total data storage capacity for this particular queue type; for 512-bit configurations this depth corresponds to one quarter of the total data storage capacity for this particular queue type. Values: 0, ..., 65535 Default Value: CX_NVC<=1 ? 0 : RADM_CPL_QMODE_VC1==4 ? ((CX_RADMQ_MODE==2) ? RADM_SEG_BUF_MIN_DPT : 0) : ([<functionof> CX_CPLQ_MANAGEMENT_ENABLE CPLQ_MNG_DDP CX_RADMQ_MODE RADM_CPLQ_HDP_VC1 RADM_CPLQ_DCRD_VC1 CUT_THROUGH_INVOLVED CX_NHQ CX_NW CX_MAX_TAG CX_APP_RD_REQ_SIZE CC_DEVICE_TYPE CC_SW CX_ECRC_STRIP_ENABLE RADM_SEG_BUF_CT_DPT_ADJ RADM_CPLQ_DSCALE_VC1 RADM_CPLQ_HSCALE_VC1]) Enabled: ((AMBA_INTERFACE == 0 && CC_DMA_ENABLE == 0) CX_CPLQ_MANAGEMENT_ENABLE) && CX_NVC>1 && CX_RADMQ_MODE==2 && RADM_CPL_QMODE_VC1!=4 && (RADM_DEPTH_DECOUPLE_VC1==1 RADM_CPLQ_DCRD_VC1 == 0) Parameter Type: Feature Setting Parameter Name: RADM_CPLQ_DDP_VCn

1.63 Advanced RAS Config Parameters

Table 1-63 Advanced RAS Config Parameters

Label	Description
RAS DES (Debug/Statistics/Error Injection)	
RAS D.E.S feature enable	Enables RAS D.E.S functions (debug functions, error injection and statistical analysis). Values: ■ false (0) ■ true (1) Default Value: (CX_AUTOMOTIVE_ENABLE==1)? 1: 0 Enabled: Always Parameter Type: Feature Setting Parameter Name: CX_RAS_DES_ENABLE
- Event Counter	Enables RAS D.E.S event counters. Values: ■ false (0) ■ true (1) Default Value: CX_RAS_DES_ENABLE Enabled: CX_RAS_DES_ENABLE Parameter Type: Feature Setting Parameter Name: CX_RAS_DES_EC_ENABLE
- Time Based Analysis	Enables RAS D.E.S time based analysis function. Values: ■ false (0) ■ true (1) Default Value: CX_RAS_DES_ENABLE Enabled: CX_RAS_DES_ENABLE Parameter Type: Feature Setting Parameter Name: CX_RAS_DES_TBA_ENABLE
- Error Injection	Enables RAS D.E.S error injection function. Values: ■ false (0) ■ true (1) Default Value: CX_RAS_DES_ENABLE Enabled: CX_RAS_DES_ENABLE Parameter Type: Feature Setting Parameter Name: CX_RAS_DES_EINJ_ENABLE

Label	Description
- Silicon Debug	Enables RAS D.E.S silicon debug function. Values: ■ false (0) ■ true (1) Default Value: CX_RAS_DES_ENABLE Enabled: CX_RAS_DES_ENABLE Parameter Type: Feature Setting Parameter Name: CX_RAS_DES_SD_ENABLE
- Debug Signals Enable	Determines whether to include the debug signals that provide RAS D.E.S. feature information in top-level ports. Values: ■ false (0) ■ true (1) Default Value: CX_RAS_DES_ENABLE Enabled: CX_RAS_DES_ENABLE Parameter Type: Feature Setting Parameter Name: CX_RAS_DES_DBGIO_ENABLE
RAS DP (Data Protection)	
Datapath Protection Enable (ECC/Parity)	Select datapath protection features. Values: ■ None (0) ■ Parity (8-bit Even) (1) ■ Parity (8-bit Odd) (2) ■ ECC: With Error Correction (3) ■ ECC: No Error Correction (4) Default Value: (CX_AUTOMOTIVE_ENABLE==1)? 2: 0 Enabled: CX_RADMQ_MODE==2 Parameter Name: CX_RASDP
RAM Protection Enable (ECC)	Select RAM ECC protection. RAM ECC protection is always selected when you have enabled datapath protection (CX_RASDP >0). Single-bit RAM ECC correction is always enabled unless you disable single-bit datapath ECC correction (CX_RASDP =4). Values: 0, 1 Default Value: (CX_RASDP==0)? 0: 1 Enabled: (CX_RADMQ_MODE==2)? ((CX_RASDP==0 && AMBA_INTERFACE==0 && CC_DMA_ENABLE==0)? 1: 0): 0 Parameter Name: CX_RASDP_RAM_PROT

Label	Description
XTLH Merge FIFO ECC Disable	Disable error correction for xtlh_merge FIFO data and header outputs, for datapath widths greater than 64 bits. This parameter is enabled when RAS DP is enabled. Values: ■ Enable (0) ■ Disable (1) Default Value: Enable Enabled: CX_RASDP != 0 && ARC_WIDTH > 64 Parameter Name: CX_DISABLE_ECC_XTLH_MERGE
AXI Interface Protection	
AXI Interface Protection Enable (ECC/Parity)	Select AXI Interface protection features. Values: ■ Legacy Mode (0) ■ Parity in data, address and control (1) ■ ECC in data, Parity in address and control (2) Default Value: Legacy Mode Enabled: ((CX_RASDP==0)? 0: 1) Parameter Name: CX_AXI_IF_TYPE_OF_PROT
ECC granularity for AXI Interface	Select the ECC Granularity for the AXI Interface Values: ■ Full Width (0) ■ 32-bits (1) ■ 64-bits (2) Default Value: Full Width Enabled: (((CX_AXI_IF_TYPE_OF_PROT == 2) && CX_RASDP_EN)==1) Parameter Name: CX_AXI_IF_ECC_GRAN
ECC 32-bit Encoding	Select the AXI Interface ECC 32-bit Encoding Values: ■ Standard (0) ■ ARC Mode (1) Default Value: Standard Enabled: (((CX_AXI_IF_TYPE_OF_PROT == 2) && CX_RASDP_EN)==1) Parameter Name: CX_AXI_IF_ECC_MODE

1.64 Memory Map Parameters

Table 1-64 Memory Map Parameters

Label	Description
Memory Map	
Memory Map Position	The register map of an upstream port (as determined by device_type[3:0]) is different to that of a downstream port. When generating DocBook XML or HTML register reports, coreConsultant uses this parameter to determine which memory map to generate. This parameter has no effect on the RTL implementation. ■ 0: Upstream Port ■ 1: Downstream Port Values: ■ Upstream Port (0) ■ Downstream Port (1) Default Value: (CC_DEVICE_TYPE==CC_RC) (CC_DEVICE_TYPE==CC_DM) (CC_DEVICE_TYPE==CC_SW) Enabled: ((CC_DEVICE_TYPE==CC_SW CC_DEVICE_TYPE==CC_DM)) Parameter Name: CX_MEMORY_MAP_POSITION
Memory Map View	Specifies the current memory map view. There are two buses that can access the controller's register space: DBI and WIRE (over PCIe protocol). The register map as perceived from the remote link partner (wire view) is different to that of the local application (DBI view). For example, many registers that are RO over the wire are also RW over the DBI. When generating DocBook XML or HTML register reports, coreConsultant uses this parameter to determine which memory map view to generate. This parameter has no effect on the RTL implementation. ■ 0: DBI ■ 1: WIRE ■ 2: DBI2 Values: ■ DBI (0) ■ WIRE (1) ■ DBI2 (2) Default Value: DBI Enabled: Always Parameter Name: CX_MEMORY_MAP_VIEW

Label	Description
Memory Map Unroll View	Specifies whether to generate unroll (that is, iATU and DMA registers only) DocBook XML or HTML register reports or not. coreConsultant uses this parameter to determine whether to generate the unroll (that is, iATU and DMA registers only) register report or not. This parameter has no effect on the RTL implementation. ■ 0: NO_UNROLL ■ 1: UNROLL Values: ■ NO_UNROLL (0) ■ UNROLL (1) Default Value: NO_UNROLL Enabled: (CC_UNROLL_ENABLE && (CX_MEMORY_MAP_VIEW <2)) Parameter Name: CX_UNROLL_VIEW
Memory Map Membar0 View	Specifies whether to generate RCRB MEMBAR0 WIRE View DocBook XML or HTML register reports or not. coreConsultant uses this parameter to determine whether to generate the Membar0(that is, MEMBAR0 wire view registers only) register report or not. This parameter has no effect on the RTL implementation. Values: ■ NO_MEMBAR0_WIRE (0) ■ MEMBAR0_WIRE (1) Default Value: NO_MEMBAR0_WIRE Enabled: CX_CXL_ENABLE Parameter Name: CX_MEMBAR0_VIEW

1.65 Automotive Features Selection Parameters

Table 1-65 Automotive Features Selection Parameters

Label	Description
Enable CDM Register Checking	
CDM Register Checking Enable	When enabled, the controller contains CDM Register Checking feature. Values: ■ false (0) ■ true (1) Default Value: (CX_AUTOMOTIVE_ENABLE==1)? 1: 0 Enabled: (((CX_PCIE_MODE == SINGLE_CPCIE) (CX_PCIE_MODE == PCIE_OVER_CIO))) Parameter Type: Feature Setting Parameter Name: CX_CDM_REG_CHK_EN

1.66 CXS Configuration Parameters

Table 1-66 CXS Configuration Parameters

Label	Description
CXS Generic Config	
CXS Synchronous Design	If set to True, this parameter results in a CXS implementation using a single clock domain. The PCIe IP clock (core_clk) is used to drive CXS interface. Values: ■ false (0) ■ true (1) Default Value: CX_CXS_DATAFLITWIDTH==CX_CXS_CCIXDATAWIDTH Enabled: (CX_CXS_ENABLE && (CX_CXS_DATAFLITWIDTH!=CX_CXS_CCIXDATAWIDTH)) Parameter Type: Feature Setting Parameter Name: CX_CXS_SYNCHRONOUS
CXS Synchronous Design	Specifies the maximum operating frequency of CXS Controller (MHz). The default value is the Core and CXS Data Flit width ratio of the core_clk frequency. Values: -2147483648, ..., 2147483647 Default Value: $\text{int}((\text{CX_CXS_CCIXDATAWIDTH} * 1.0 / \text{CX_CXS_DATAFLITWIDTH}) * \text{CX_MAX_CORECLK_FREQ})$ Enabled: CX_CXS_ENABLE && !CX_CXS_SYNCHRONOUS Parameter Type: Feature Setting Parameter Name: CX_CXS_FREQUENCY
CXS Data Flit Width	Configures CXS Data Flit Width (maps to CXSDATAFLITWIDTH in CXS specification). Values: 256, 512, 1024 Default Value: (((CX_NW == 16) ? 512 : (CX_NW == 8) ? 256 : 128) == 512) ? 512 : 256 Enabled: CX_CXS_ENABLE && !CX_CXL_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_DATAFLITWIDTH
CXS Maximum Packets per Flit	Maximum number of packets that can be present in a single flit of data (maps to CXSMAXPKTPERFLIT in CXS specification). Values: 2, 3, 4 Default Value: ((CX_CXS_DATAFLITWIDTH==256) CX_CXL_CXS_ENABLE) ? 2 : 4 Enabled: (CX_CXS_ENABLE && (CX_CXS_DATAFLITWIDTH>256)) Parameter Type: Feature Setting Parameter Name: CX_CXS_MAXPKTPERFLIT

Label	Description
CXS Receiver Continuous Data	If set to True, this receiver requires that after a packet is started it is completed in consecutive cycles if enough credits are available (maps to CXSCONTINUOUSDATA in CXS specification). <i>Note:</i> Once this parameter is set to True, this attribute may not be overwritten through CXS Receiver configuration register. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_RX_CONTINUOUSDATA
CXS Transmitter Continuous Data	If set to True, this transmitter do not begin a packet until it can deliver the complete packet in consecutive cycles as long as credits are available (maps to CXSCONTINUOUSDATA in CXS specification). <i>Note:</i> This parameter only sets the Store-Forward mode default value in CXS Transmitter configuration register which may be overwritten later. Values: ■ false (0) ■ true (1) Default Value: true Enabled: CX_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_TX_CONTINUOUSDATA
CXS Receiver Error Full Packet	If set to True, this receiver requires that the length of every packet (including packets ending with EndError) match the packet length specified in the packet header (maps to CXSErrorFULLPKT in CXS specification). Values: ■ false (0) ■ true (1) Default Value: true Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_CXS_RX_ERRORFULLPKT

Label	Description
CXS Transmitter Error Full Packet	If set to True, if this transmitter is unable to complete a packet and must end the packet with an EndError indication, the transmitter sends the number of bytes specified in the packet header before ending the packet (maps to CXSErrorFullPkt in CXS specification). Values: ■ false (0) ■ true (1) Default Value: false Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_CXS_TX_ERRORFULLPKT
CXS Datapath Protection Config	
CXS Receiver Data Check	Configures data check support on CXSRXDATA, CXSRXCNTL (maps to CXSDATACHECK in CXS specification): 0 - None 1 - Odd Parity; 1 bit per byte on CXSDATA and 1 bit for CXSCNTL 2 - SECDED: ECC on a 64-bit granularity; 8 bits per 64 bits of data and 8 bits for CXSCNTL (zero extended to 64 bits) The default value of this parameter is derived from CX_RASDP setting. Values: ■ None (0) ■ Odd Parity (1) ■ SECDED (2) Default Value: (CX_CXS_ENABLE && (CX_RASDP > 0)) ? (CX_RASDP-1) : 0 Enabled: CX_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_RX_DATACHECK
CXS Transmitter Data Check	Configures data check support on CXSTXDATA, CXSTXCNTL (maps to CXSDATACHECK in CXS specification): 0 - None 1 - Odd Parity; 1 bit per byte on CXSDATA and 1 bit for CXSCNTL 2 - SECDED: ECC on a 64-bit granularity; 8 bits per 64 bits of data and 8 bits for CXSCNTL (zero extended to 64 bits) The default value of this parameter is derived from CX_RASDP setting. Values: ■ None (0) ■ Odd Parity (1) ■ SECDED (2) Default Value: (CX_CXS_ENABLE && (CX_RASDP > 0)) ? (CX_RASDP-1) : 0 Enabled: CX_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_TX_DATACHECK

Label	Description
CXS Receiver Signal Replication	Signal replication on CXSRXVALID, CXSRXCRDGNT, CXSRXCRDRTN (maps to CXSREPLICATION in CXS specification). Duplicate: CXSRXVALIDCHK, CXSRXCRDGNTCHK, and CXSRXCRDRTNCHK are each single bit signals duplicating the value of the corresponding control signal. Triplicate: CXSRXVALIDCHK, CXSRXCRDGNTCHK, and CXSRXCRDRTNCHK are each two-bit signals with each of the 2 bits having the same value as the corresponding control signal. Values: ■ None (0) ■ Duplicate (1) ■ Triplicate (2) Default Value: None Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_CXS_RX_REPLICATION
CXS Transmitter Signal Replication	Signal replication on CXSTXVALID, CXSTXCRDGNT, CXSTXCRDRTN (maps to CXSREPLICATION in CXS specification). Duplicate: CXSTXVALIDCHK, CXSTXCRDGNTCHK, and CXSTXCRDRTNCHK are each single bit signals duplicating the value of the corresponding control signal. Triplicate: CXSTXVALIDCHK, CXSTXCRDGNTCHK, and CXSTXCRDRTNCHK are each two-bit signals with each of the 2 bits having the same value as the corresponding control signal. Values: ■ None (0) ■ Duplicate (1) ■ Triplicate (2) Default Value: None Enabled: 0 Parameter Type: Feature Setting Parameter Name: CX_CXS_TX_REPLICATION
CXS Controller Config	
CXS Receiver Credits	Configures the number of CXS Receiver credits. Values: 1, ..., 15 Default Value: 15 Enabled: CX_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_RX_CREDITS
CXS Transmitter Deactivation Timer	CXS Transmitter Deactivation Timer Values: 0, ..., 65535 Default Value: 0 Enabled: CX_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_TX_DEACT_TIMER_VALUE

Label	Description
CXS Transmitter Output Register Enable	Enable output register on CXS Transmitter Data/Control bus outputs. Values: ■ false (0) ■ true (1) Default Value: true Enabled: CX_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_TX_OUTPUT_REG
CXS Receiver Input Register Enable	Enable input register on CXS Receiver Data/Control bus inputs. Values: ■ false (0) ■ true (1) Default Value: true Enabled: CX_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_RX_INPUT_REG
CXS RX/TX FIFO Bypass Feature for CXL DL	Enable CXS RX/TX FIFO RAM bypass feature. Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_CXL_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_CXS_FIFO_BYPASS_ENABLE
CXS Controller RAM Config	
CXS RX/TX FIFO RAM Read Latency	CXS RX/TX FIFO RAM Read Latency. Values: 1, 2 Default Value: 1 Enabled: CX_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_RAM_READ_LATENCY
CXS RX/TX FIFO RAM Pipeline Enable	Enable pipeline register on CXS RX/TX FIFO RAM output. Values: ■ false (0) ■ true (1) Default Value: (CX_CXS_RAM_READ_LATENCY > 1) ? 0 : 1 Enabled: CX_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXS_RAM_PIPELINE_ENABLE

1.67 CXL Configuration Parameters

Table 1-67 CXL Configuration Parameters

Label	Description
CXL Generic Config	
CXL.cache Enable	Enables CXL.cache protocol. Values: 0, 1 Default Value: CX_CXL_ENABLE Enabled: CX_CXL_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_CACHE_EN
CXL.mem Enable	Enables CXL.mem protocol. Values: 0, 1 Default Value: CX_CXL_ENABLE Enabled: CX_CXL_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_MEM_EN
Enable CCIX 2.0 Support	Enables CCIX Protocol over 64B CXL flits. Values: 0, 1 Default Value: 0 Enabled: CX_CXL_ENABLE && CX_CXL_CXS_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_CCIX_ENABLE
CXL Vendor IDLink CXL MLD	Define the CXL Vendor ID Values: 0x0, ..., 0xffff Default Value: 0x1e98 Enabled: CX_CXL_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_VENDOR_ID
Support Compute Express Link CXL MLD	Configure the controller to support Compute Express Link Multi Logic Device For Multic Logical Devices, Number of Physical Functions configured should be greater than Number of Logical devices Values: ■ false (0) ■ true (1) Default Value: false Enabled: ((CX_CXL_2_0_ENABLE) && (CX_CXL_MEM_EN == 1) && (CC_DEVICE_TYPE!=CC_RC)) Parameter Type: Feature Setting Parameter Name: CX_CXL_MLD_EN

Label	Description
CXL support for DPC trigger by FMLD	If set, indicates this CXL support DPC by FM owned LD in MLD supported switch controller Values: ■ false (0) ■ true (1) Default Value: (CX_IS_SW) && (CX_CXL_MLD_EN) && (CX_DPC_ENABLE) Enabled: ((CX_IS_SW) && (CX_CXL_MLD_EN) && (CX_DPC_ENABLE)) Parameter Type: Feature Setting Parameter Name: CX_CXL_IS_SW_MLD_DPC_ENABLE
CXL capability to do CXL_Reset	If set, indicates this CXL capability to do CXL_Reset Values: ■ false (0) ■ true (1) Default Value: false Enabled: ((CX_CXL_2_0_ENABLE) && (CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RESET_EN
CXL_Reset Timeout value	If set, indicates maximum time device may take to complete CXL_Reset Values: ■ 10ms (0) ■ 100ms (1) ■ 1s (2) ■ 10s (3) ■ 100s (4) Default Value: 10ms Enabled: CX_CXL_RESET_EN Parameter Type: Feature Setting Parameter Name: CX_CXL_RESET_TIMEOUT
CXL_Reset Mem Clear Capability	If set, indicates device is capable of clearing or randomizing of Volatile HDM ranges during CXL_Reset Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_CXL_RESET_EN Parameter Type: Feature Setting Parameter Name: CX_CXL_RESET_MEM_CLR_EN

Label	Description
CXL Device is a Logical Device	If set, indicates device is a Logical Device Values: ■ false (0) ■ true (1) Default Value: false Enabled: CX_CXL_2_0_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_LOGICAL_DEVICE_EN
CXL capability to do Cache writeback and Invalidate	If set, indicates this CXL capability to do Cache writeback and Invalidate Values: ■ false (0) ■ true (1) Default Value: CX_CXL_2_0_ENABLE && (CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM) && CX_CXL_CACHE_EN Enabled: ((CX_CXL_2_0_ENABLE) && (CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_CACHE_WR_INVLD_EN
CXL Device Cache Size Unit	CXL Device cache size unit 0 - Cache Size is not reported 1 - 64KB 2 - 1MB Values: ■ NA (0) ■ 64K (1) ■ 1MB (2) Default Value: NA Enabled: ((CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_DEVICE_CACHE_SIZE_UNIT
CXL Device Cache Size	CXL Device cache size in units of CXL Device Cache size unit Range 0 to 255 Values: 0, ..., 255 Default Value: 0 Enabled: ((CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_DEVICE_CACHE_SIZE
CXL Retry Buffer Depth	Specifies CXL Retry Buffer Depth Values: 32, 64, 128, 256 Default Value: 256 Enabled: CX_CXL_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_RBUF_DEPTH

Label	Description
CXL.Cache Enable bit default value	Default value for CXL.Cache Enable bit in CXL Port DVSEC for Upstream port Values: 0, 1 Default Value: 0 Enabled: ((CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM) && (CX_CXL_CACHE_EN)) Parameter Type: Register Default Value Parameter Name: DEFAULT_CXL_PORT_USP_CACHE_ENABLE
CXL.Mem Enable bit default value	Default value for CXL.Mem Enable bit in CXL Port DVSEC for Upstream port Values: 0, 1 Default Value: 0 Enabled: ((CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM) && (CX_CXL_MEM_EN)) Parameter Type: Register Default Value Parameter Name: DEFAULT_CXL_PORT_USP_MEM_ENABLE
CXL 2.0 Enable bit default value	Default value for CXL 2.0 Enable bit in CXL Port DVSEC for Upstream port Values: 0, 1 Default Value: CX_CXL_2_0_ENABLE Enabled: ((CX_CXL_2_0_ENABLE) && (CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM)) Parameter Type: Register Default Value Parameter Name: DEFAULT_CXL_PORT_USP_CXL2P0_ENABLE
CXL MLD Enable bit default value	Default value for CXL MLD Enable bit in CXL Port DVSEC for Upstream port Values: 0, 1 Default Value: 0 Enabled: CX_CXL_MLD_EN Parameter Type: Register Default Value Parameter Name: DEFAULT_CXL_PORT_USP_MLD_ENABLE
CXL APN Swith Mode bit default value	Default value for CXL Alternate Protocol Negotiation Switch Mode bit in CX_VLSM_CR_REG Values: ■ Based on CXL 2.0 specification (0) ■ Special negotiation (1) Default Value: Based on CXL 2.0 specification Enabled: ((CX_CXL_2_0_ENABLE) && (CC_DEVICE_TYPE==CC_SW)) Parameter Type: Feature Setting Parameter Name: DEFAULT_CXL_APN_SW_MODE

Label	Description
Array_size field of the CXL Capability Header Register in the MEMBAR0	Defines the number of elements in the CXL_Capability array, not including the CXL_Capability_Header element. Values: 0x0, ..., 0xff Default Value: 0x4 Enabled: CX_CXL_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_MBAR0_CAP_ARRAY_SIZE
CXL Memory Config	
CXL Memory Initialization Mode	If set, indicates this CXL.mem capable device initializes memory with assistance from hardware and firmware located on the device. If clear, indicates memory is initialized by host software such as device driver Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CX_CXL_MEM_EN && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_MEM_HWINIT_MODE
Number of HDM ranges implemented	Number of HDM ranges implemented by the CXL device and reported through this function Values: ■ Zero Range (0) ■ One Range (1) ■ Two Ranges (2) Default Value: CX_CXL_MEM_EN Enabled: (CX_CXL_MEM_EN && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_HDM_COUNT
Flex Bus Range 1 Memory Size High	Corresponds to bits 63:32 of Flex Bus Range 1 memory size Values: 0x0, ..., 0xffffffff Default Value: 0x0 Enabled: (CX_CXL_MEM_EN && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_1_MEM_SIZE_HIGH
Flex Bus Range 1 Memory Size Low	Corresponds to bits 31:28 of Flex Bus Range 1 memory size Values: 0x0, ..., 0xf Default Value: 0x0 Enabled: (CX_CXL_MEM_EN && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_1_MEM_SIZE_LOW

Label	Description
Flex Bus Range 1 Memory Info Valid	If set, indicates that the Flex Bus Range 1 Size high and Size Low registers are valid Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CX_CXL_MEM_EN && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_1_MEM_INFO_VALID
Flex Bus Range 1 Memory Active	If set, indicates that the Flex Bus Range 1 memory is fully initialized and available for software use Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CX_CXL_MEM_EN && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_1_MEM_ACTIVE
Flex Bus Range 1 Media Type	Indicates the memory media characteristics for Flex Bus Range 1 Values: ■ Volatile Memory (0) ■ Non-volatile Memory (1) ■ Use CDAT (2) Default Value: CX_CXL_2_0_ENABLE ? 2 : 0 Enabled: (CX_CXL_MEM_EN && !CX_CXL_2_0_ENABLE && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_1_MEDIA_TYPE
Flex Bus Range 1 Memory Class	Indicates the class of memory for Flex Bus Range 1 Values: ■ Memory Class (0) ■ Storage Class (1) ■ Use CDAT (2) Default Value: CX_CXL_2_0_ENABLE ? 2 : 0 Enabled: (CX_CXL_MEM_EN && !CX_CXL_2_0_ENABLE && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_1_MEM_CLASS

Label	Description
Flex Bus Range 1 Desired Interleave	Represents the memory interleaving desired by the device for Flex Bus Range 1 Values: ■ No Interleave (0) ■ 256 Byte Granularity (1) ■ 4K Interleave (2) ■ 512 Bytes (3) ■ 1024 Bytes (4) ■ 2048 Bytes (5) ■ 8192 Bytes (6) ■ 16384 Bytes (7) Default Value: No Interleave Enabled: (CX_CXL_MEM_EN && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_1_DESIRED_INTERLEAVE
Flex Bus Range 1 Memory Active Timeout	Indicates the value of Memory_Active_Timeout for Flex Bus Range 1 Size Low Register Values: ■ 1s (0) ■ 4s (1) ■ 16s (2) ■ 64s (3) ■ 256s (4) Default Value: 1s Enabled: (CX_CXL_MEM_EN && (CX_IS_EP CX_IS_DM)) Parameter Type: Register Default Value Parameter Name: CX_CXL_RANGE_1_MEM_ACTIVE_TIMEOUT
Flex Bus Range 2 Memory Size High	Corresponds to bits 63:32 of Flex Bus Range 2 memory size Values: 0x0, ..., 0xffffffff Default Value: 0x0 Enabled: (CX_CXL_MEM_EN && (CX_CXL_HDM_COUNT == 2)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_2_MEM_SIZE_HIGH
Flex Bus Range 2 Memory Size Low	Corresponds to bits 31:28 of Flex Bus Range 2 memory size Values: 0x0, ..., 0xf Default Value: 0x0 Enabled: (CX_CXL_MEM_EN && (CX_CXL_HDM_COUNT == 2)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_2_MEM_SIZE_LOW

Label	Description
Flex Bus Range 2 Memory Info Valid	If set, indicates that the Flex Bus Range 2 Size high and Size Low registers are valid Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CX_CXL_MEM_EN && (CX_CXL_HDM_COUNT == 2)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_2_MEM_INFO_VALID
Flex Bus Range 2 Memory Active	If set, indicates that the Flex Bus Range 2 memory is fully initialized and available for software use Values: ■ false (0) ■ true (1) Default Value: false Enabled: (CX_CXL_MEM_EN && (CX_CXL_HDM_COUNT == 2)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_2_MEM_ACTIVE
Flex Bus Range 2 Media Type	Indicates the memory media characteristics for Flex Bus Range 2 Values: ■ Volatile Memory (0) ■ Non-volatile Memory (1) ■ Use CDAT (2) Default Value: CX_CXL_2_0_ENABLE ? 2 : 0 Enabled: (CX_CXL_MEM_EN && !CX_CXL_2_0_ENABLE && (CX_CXL_HDM_COUNT == 2)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_2_MEDIA_TYPE
Flex Bus Range 2 Memory Class	Indicates the class of memory for Flex Bus Range 2 Values: ■ Memory Class (0) ■ Storage Class (1) ■ Use CDAT (2) Default Value: CX_CXL_2_0_ENABLE ? 2 : 0 Enabled: (CX_CXL_MEM_EN && !CX_CXL_2_0_ENABLE && (CX_CXL_HDM_COUNT == 2)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_2_MEM_CLASS

Label	Description
Flex Bus Range 2 Desired Interleave	Represents the memory interleaving desired by the device for Flex Bus Range 2 Values: ■ No Interleave (0) ■ 256 Byte Granularity (1) ■ 4K Interleave (2) ■ 512 Bytes (3) ■ 1024 Bytes (4) ■ 2048 Bytes (5) ■ 8192 Bytes (6) ■ 16384 Bytes (7) Default Value: No Interleave Enabled: (CX_CXL_MEM_EN && (CX_CXL_HDM_COUNT == 2)) Parameter Type: Feature Setting Parameter Name: CX_CXL_RANGE_2_DESIRED_INTERLEAVE
Flex Bus Range 2 Memory Active Timeout	Indicates the value of Memory_Active_Timeout for Flex Bus Range 2 Size Low Register Values: ■ 1s (0) ■ 4s (1) ■ 16s (2) ■ 64s (3) ■ 256s (4) Default Value: 1s Enabled: (CX_CXL_MEM_EN && (CX_CXL_HDM_COUNT == 2)) Parameter Type: Register Default Value Parameter Name: CX_CXL_RANGE_2_MEM_ACTIVE_TIMEOUT
CXL Device CTS Config	
CXL Device Test Cap DVSEC Enable	Enables CXL Device Test Capability DVSEC in the RCiEP Values: 0, 1 Default Value: 0 Enabled: (CX_CXL_ENABLE && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_DEVICE_CTS_EN

Label	Description
CXL Device CTS Capabilities	Defines the CXL Device CTS Capabilities . The field values and their meaning are described below : Bit 0 - SelfChecking : Set to 1 if Device supports Self Checking Bit 1 - Algorithm1a : Set to 1 if Device supports hardware for test Algorithm 1a Bit 2 - Algorithm1b : Set to 1 if Device supports hardware for test Algorithm 1b Bit 3 - Algorithm2 : Set to 1 if Device supports hardware for test Algorithm 2 Bit 4 - RdCurr : Set to 1 if Device supports CXL.cache and RdCurr opcodes as requester Bit 5 - RdOwn : Set to 1 if Device supports CXL.cache and RdOwn opcodes as requester Bit 6 - RdShared : Set to 1 if Device supports CXL.cache and RdShared opcodes as requester Bit 7 - RdAny : Set to 1 if Device supports CXL.cache and RdAny opcodes as requester. Bit 8 - RdOwnNoData : Set to 1 if Device supports CXL.cache and RdOwnNoData opcodes as requester Bit 9 - ItoMWr : Set to 1 if Device supports CXL.cache and ItoMWr opcodes as requester Bit 10 - MemWr : Set to 1 if Device supports CXL.cache and MemWr opcodes as requester Bit 11 - CLFlush : Set to 1 if Device supports CXL.cache and CLFlush opcodes as requester Bit 12 - CleanEvict : Set to 1 if Device supports CXL.cache and CleanEvict opcodes as requester Bit 13 - DirtyEvict : Set to 1 if Device supports CXL.cache and DirtyEvict opcodes as requester Bit 14 - CleanEvictNoData : Set to 1 if Device supports CXL.cache and CleanEvictNoData opcodes as requester Bit 15 - WOWrInv : Set to 1 if Device supports CXL.cache and WOWrInv opcodes as requester Bit 16 - WOWrInvF : Set to 1 if Device supports CXL.cache and WOWrInvF opcodes as requester Bit 17 - WrInv : Set to 1 if Device supports CXL.cache and WrInv opcodes as requester Bit 18 - CacheFlushed : Set to 1 if Device supports CXL.cache and CacheFlushed opcodes as requester Bit 19 - UnexpectedCompletion : Set to 1 if Device supports sending an unexpected completion on CXL.io Bit 20 - CompletionTimeoutInjection : Set to 1 if Device supports dropping a read in the completion timeout scenario Bits 28:21 - ConfigurationSize : Size in Bytes of Test configuration control registers Values: 0x0, ..., 0x1ffffff Default Value: 0x0 Enabled: CX_CXL_DEVICE_CTS_EN Parameter Type: Feature Setting Parameter Name: CX_CXL_DEVICE_CTS_CAP

Label	Description
Cache size supported by the device	Cache size supported by the device Values: 0x0, ..., 0x3fff Default Value: 0x0 Enabled: (CX_CXL_DEVICE_CTS_EN && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_CACHE_SIZE
Cache Size Unit	Units of advertised cache size in CacheSize field Values: ■ Bytes (0) ■ KiloBytes (1) ■ MegaBytes (2) Default Value: Bytes Enabled: (CX_CXL_DEVICE_CTS_EN && (CX_IS_EP CX_IS_DM)) Parameter Type: Feature Setting Parameter Name: CX_CXL_CACHE_SIZE_UNITS
CXL Device CTS Base Reg Type	CXL Device CTS Base register type Values: ■ 32 bit wide (0) ■ 64 bit wide (2) Default Value: 32 bit wide Enabled: CX_CXL_DEVICE_CTS_EN Parameter Type: Feature Setting Parameter Name: CX_CXL_CTS_BASE_REG_TYPE
CXL Channel Interface Config	
CXL Channel Interface Credits	Configures the number of CXL Transmitter Channel Interface credits. Values: 2, ..., 8 Default Value: 2 Enabled: CX_CXL_ENABLE && CX_CXL_CHI_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_CHI_MAXCREDITS

Label	Description
CXL Channel Interface Data Check	CXL Channel Interface data protection: 0 - None; 1 - Odd Parity; 1 bit per byte on Channel Interface Message; 2 - SECDED: ECC on a 64-bit granularity. The default value of this parameter is derived from CX_RASDP setting. Values: ■ None (0) ■ Odd Parity (1) ■ SECDED (2) Default Value: (CX_CXL_ENABLE && CX_CXL_CHI_ENABLE && (CX_RASDP > 0)) ? 2 : 0 Enabled: CX_CXL_ENABLE && CX_CXL_CHI_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_CHI_DATACHK
CXL Receive Queue Config	
CXL.cache Request Receive Queue Depth	Select CXL.cache Request Receive Queue Depth: 0: 64; 1: 128; 2: 256; 3: 512 (default); 4: 1024. Values: ■ 64 (0) ■ 128 (1) ■ 256 (2) ■ 512 (3) ■ 1024 (4) Default Value: 512 Enabled: CX_CXL_CACHE_EN && CX_CXL_CHI_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_CACHE_REQ_DP
CXL.cache Response Receive Queue Depth	Select CXL.cache Response Receive Queue Depth: 0: 64; 1: 128; 2: 256; 3: 512 (default); 4: 1024. Values: ■ 64 (0) ■ 128 (1) ■ 256 (2) ■ 512 (3) ■ 1024 (4) Default Value: 512 Enabled: CX_CXL_CACHE_EN && CX_CXL_CHI_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_CACHE_RSP_DP

Label	Description
CXL.cache Data Receive Queue Depth	Select CXL.cache Data Receive Queue Depth: 0: 64; 1: 128; 2: 256; 3: 512 (default); 4: 1024. Values: ■ 64 (0) ■ 128 (1) ■ 256 (2) ■ 512 (3) ■ 1024 (4) Default Value: 512 Enabled: CX_CXL_CACHE_EN && CX_CXL_CHI_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_CACHE_DQ_DP
CXL.mem NDR Receive Queue Depth	Select CXL.mem No Data Request or Response message, Receive Queue Depth 0: 64; 1: 128; 2: 256; 3: 512 (default); 4: 1024. Values: ■ 64 (0) ■ 128 (1) ■ 256 (2) ■ 512 (3) ■ 1024 (4) Default Value: 512 Enabled: CX_CXL_MEM_EN && CX_CXL_CHI_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_MEM_NDR_DP
CXL.mem Data Receive Queue Depth	Select CXL.mem Data Receive Queue Depth: 0: 64; 1: 128; 2: 256; 3: 512 (default); 4: 1024. Values: ■ 64 (0) ■ 128 (1) ■ 256 (2) ■ 512 (3) ■ 1024 (4) Default Value: 512 Enabled: CX_CXL_MEM_EN && CX_CXL_CHI_ENABLE Parameter Type: Feature Setting Parameter Name: CX_CXL_MEM_DQ_DP

1.68 AMBAConfig Parameters

Table 1-68 AMBAConfig Parameters

Label	Description
AHB Slave and DBI Immediate ERROR Response	Enables an immediate error response from the slave and DBI slave for an illegal AHB access. For more information, see the "slv_hresp" signal description in the <i>Databook</i>. If the parameter RETURN_ERR_RESP is set then the bridge slave interface performs immediate error checking on the slave interface protocol signals (slv_hsize, slv_haddr, slv_req_misc_info[25:22]) and return an ERROR status on slv_hresp immediately. It does not transmit the outbound request TLP. The following two checks are done: ■ HSIZE check. For example, for a data width of 32-bits an immediate ERROR response is got if slv_hsize is not 0,1 or 2. ■ CFG & I/O transfers are checked for byte enable, hsize and address alignment. For more information, see the "Zero-Byte Transfers Over the AHB Bridge (Flush Semantics)2 section of the <i>Databook</i>. The recommended setting for this parameter is "True". Values: ■ False (0) ■ True (1) Default Value: True Enabled: AHB_POPULATED && SLAVE_POPULATED Parameter Type: Feature Setting Parameter Name: RETURN_ERR_RESP

Signal Descriptions

This chapter details all possible I/O signals in the controller. For configurable IP titles, your actual configuration might not contain all of these signals.

Inputs are on the left of the signal diagrams; outputs are on the right.

Attention: For configurable IP titles, do not use this document to determine the exact I/O footprint of the controller. It is for reference purposes only.

When you configure the controller in coreConsultant, you must access the I/O signals for your actual configuration at `workspace/report/IO.html` or `workspace/report/IO.xml` after you have completed the report creation activity. That report comes from the exact same source as this chapter but removes all the I/O signals that are not in your actual configuration. This does not apply to non-configurable IP titles. In addition, all parameter expressions are evaluated to actual values. Therefore, the widths might change depending on your actual configuration.

Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

In addition to describing the function of each signal, the signal descriptions in this chapter include the following information:

- **Active State:** Indicates whether the signal is active high or active low. When a signal is not intended to be used in a particular application, then this signal needs to be tied or driven to the inactive state (opposite of the active state).
- **Registered:** Indicates whether or not the signal is registered directly inside the IP boundary without intervening logic (excluding simple buffers). A value of *No* does not imply that the signal is not synchronous, only that there is some combinatorial logic between the signal's origin or destination register and the boundary of the controller. A value of *N/A* indicates that this information is not provided for this IP title.
- **Synchronous to:** Indicates which clocks in the IP sample this input (drive for an output) when considering all possible configurations. A particular configuration might not have all of the clocks listed. This clock might not be the same as the clock that your application logic should use to clock (sample/ drive) this pin. For more details, consult the clock section in the databook.
- **Exists:** Name of configuration parameter that populates this signal in your configuration.

- **Validated by:** Assertion or de-assertion of signals that validates the signal being described.

Synchronous To

The Synchronous To: attribute indicates which controller clocks sample an input (or drive an output) when considering all possible configurations. In many cases an output has components driven (or an input sampled) by multiple clocks. The attribute lists all clocks which drive or sample for all possible configurations. It is an automatically generated list accumulated over all configurations. Your particular configuration might not have all of the clocks listed.

- When there is only one clock in the list, this clock is normally but not always the same as the clock that your application logic should use to clock (sample/drive) this pin.
- When there is more than one clock in the list:
 - A {} is placed around the clock which is used in generating the synthesis in/out timing delay constraints and which is also used in Spyglass boundary CDC checking.
 - In most cases, this is also the clock that you should use to drive/sample the pin in normal operational (non low-power) mode.
 - If more than one clock has a {}, that clock is configuration-dependent and you should look in your SDC file (as explained in the *Synthesizing to a Device Outside of coreConsultant* section of the User Guide) to see which one is relevant. Normally only applies to the PIPE outputs which are clocked on pipe_clk or core_clk depending on CX_FREQ_STEP_EN.
- When you see *perbitclock* in the clock list, it indicates that there is a different clock for some of the bits in the multi-bit port, and/or that some of the bits in a multi-bit port falls into the various categories as defined next by *None*.
- When you see *None_as* in the clock list, it indicates the following and is equivalent/mapped to core_clk for CDC synthesis.
 - Asynchronous reset with synchronous de-assertion.
- When you see *None* in the clock list, it indicates any of the following and is equivalent/mapped to core_clk for CDC and synthesis.
 - Direct or combinatorial feed-through.
 - Gated-off (that is, unlocked) inputs.
 - Unused inputs.
 - Hard-coded outputs.
 - Asynchronous outputs.
 - Asynchronous resets.
- For asynchronous inputs (excluding asynchronous resets), CDC synchronizers are added in the RTL, and are detailed in *Port Clocking and Input Synchronizers* in the *Databook*. Asynchronous inputs are indicated with *Synchronous To: myclock*, where myclock is the synchronizer clock.
- Asynchronous outputs are indicated with *Synchronous To: None*.

For more information on Generalized interface clocking guidelines qualifying the synchronous to: attribute, see **2.3.2 General Clock Relationships** in *Clock Requirements* section of the *Databook*.

The I/O signals are grouped as follows:

- “AXI Clock and Reset Signals” on page 295

- [“AXI Master Interface Signals” on page 300](#)
- [“AXI Master Interface Signals \(2nd Master\)” on page 338](#)
- [“AXI Optional DBI Slave Interface Signals” on page 374](#)
- [“AXI Slave Interface Signals” on page 393](#)
- [“CXL Controller RAS Interface Signals” on page 427](#)
- [“CXL Controller RAS Statistics Interface Signals” on page 428](#)
- [“CXL Interface Signals” on page 429](#)
- [“CXS Rx Interface Signals” on page 439](#)
- [“CXS Tx Interface Signals” on page 443](#)
- [“XALI0 Interface Signals” on page 448](#)
- [“XALI1 Interface Signals” on page 462](#)
- [“XALI2 Interface Signals” on page 475](#)
- [“XALI_CCIX Interface Signals” on page 487](#)
- [“ADM adaptor Interface Signals” on page 489](#)
- [“Bypass Interface Signals” on page 491](#)
- [“RTRGT1 Interface Signals” on page 503](#)
- [“TRGT1_CCIX Interface Signals” on page 520](#)
- [“Clock and Reset \(APM\) Signals” on page 523](#)
- [“Clock and Reset Signals” on page 529](#)
- [“DBI Interface Signals” on page 539](#)
- [“ELBI Interface Signals” on page 544](#)
- [“Master Completion Asynchronous Clock Crossing FIFO RAM Signals \(AXI Master #2\)” on page 549](#)
- [“Master Completion Asynchronous Clock Crossing FIFO RAM Signals \(AXI\)” on page 552](#)
- [“Master Completion Buffer RAM Signals \(AXI Master #2\)” on page 555](#)
- [“Master Completion Buffer RAM Signals \(AXI\)” on page 558](#)
- [“Master Non-posted Request Buffer RAM Signals \(AXI\)” on page 561](#)
- [“Master Posted Tracker Data Buffer RAM Signals \(AXI Master #2\)” on page 564](#)
- [“Master Posted Tracker Data Buffer RAM Signals \(AXI\)” on page 567](#)
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2.1 AXI Clock and Reset Signals

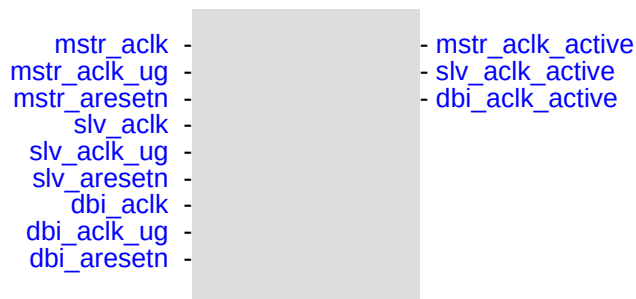


Table 2-1 AXI Clock and Reset Signals

Port Name	I/O	Description
mstr_aclk	I	Gated AXI Master Clock. AXI clock for AXI master interface after gating with mstr_aclk_active. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
mstr_aclk_ug	I	Ungated AXI Master Clock. Always-on AXI clock for AXI master interface. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
mstr_aclk_active	O	AXI Master Clock Active (Not a part of AXI Low power signaling interface). This signal can be used to gate mstr_aclk. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk_ug Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mstr_aresetn	I	AXI Master Reset. AXI reset for master interface. Synopsys recommends that your application resets the AXI bridge master when it is asserting a reset to the controller. Asynchronously asserted with a synchronous de-assertion, relative to its corresponding clock. Only one reset cycle is required to reset the logic associated with the corresponding clock. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
slv_aclk	I	Gated Slave Clock. AXI clock for AXI slave interface gated with slv_aclk_active. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
slv_aclk_ug	I	Ungated Slave Clock. Always-on AXI clock for AXI slave interface. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
slv_aclk_active	O	AXI Slave clock active (Not a part of AXI Low Power Signaling interface). This signal can be used to gate slv_aclk. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
slv_aresetn	I	Slave Reset. AXI reset for slave interface. Synopsys recommends that your application resets the AXI bridge slave when it is asserting a reset to the controller. Asynchronously asserted with a synchronous de-assertion, relative to its corresponding clock. Only one reset cycle is required to reset the logic associated with the corresponding clock. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
dbi_aclk	I	Gated DBI Clock. AXI clock for AXI DBI slave interface gated with dbi_aclk_active. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
dbi_aclk_ug	I	Ungated DBI Clock. Always-on AXI clock for AXI DBI slave interface. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
dbi_aclk_active	O	AXI DBI Slave clock active (Not a part of AXI Low Power Signaling interface). This signal can be used to gate dbi_aclk. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk_ug,{dbi_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
dbi_aresetn	I	DBI Reset. AXI reset for DBI slave interface. Synopsys recommends that your application resets the AXI bridge DBI slave when it is asserting a reset to the controller. Asynchronously asserted with a synchronous de-assertion, relative to its corresponding clock. Only one reset cycle is required to reset the logic associated with the corresponding clock. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

2.2 AXI Master Interface Signals

mstr_awready	-	mstr_awid
mstr_awready_p	-	mstr_awvalid
mstr_wready	-	mstr_awvalid_p
mstr_wready_p	-	mstr_awaddr
mstr_bid	-	mstr_awuser
mstr_bvalid	-	mstr_awlen
mstr_bvalid_p	-	mstr_awsiz
mstr_bresp	-	mstr_awburst
mstr_buser	-	mstr_awlock
mstr_bmisc_info_tlprrfx	-	mstr_awcache
mstr_bmisc_info_tlpIn	-	mstr_awprot
mstr_bmisc_info_cid	-	mstr_awmisc_info
mstr_bmisc_info_cpl_stat	-	mstr_awmisc_info_last_dcmp_tlp
mstr_bidunq	-	mstr_awmisc_info_hdr_34dw
mstr_arready	-	mstr_awmisc_info_tlprrfx
mstr_arready_p	-	mstr_awmisc_info_tlpIn
mstr_rid	-	mstr_awmisc_info_tph
mstr_rvalid	-	mstr_awmisc_info_dma
mstr_rvalid_p	-	mstr_awmisc_info_nw
mstr_rlast	-	mstr_awmisc_info_ats
mstr_rdata	-	mstr_awmisc_info_func_num
mstr_rchunkv	-	mstr_awmisc_info_vfunc_num
mstr_rchunknum	-	mstr_awmisc_info_vfunc_active
mstr_rchunkstrb	-	mstr_awmisc_info_ep
mstr_ruser	-	mstr_awmisc_info_id0
mstr_rresp	-	mstr_awqos
mstr_rmisc_info	-	mstr_awbar
mstr_rmisc_info_tlprrfx	-	mstr_awdomain
mstr_rmisc_info_tlpIn	-	mstr_awsnoop
mstr_rmisc_info_cid	-	mstr_awatop
mstr_rmisc_info_cpl_stat	-	mstr_awidunq
mstr_ridunq	-	mstr_wdata
mstr_csysreq	-	mstr_wuser
mstrh_rchunkv	-	mstr_wstrb
mstrh_rchunknum	-	mstr_wlast
mstrh_rchunkstrb	-	mstr_wvalid
mstrh_rmisc_info_cid	-	mstr_wvalid_p
		mstr_wid
		mstr_bready
		mstr_bready_p
		mstr_arid
		mstr_arvalid
		mstr_arvalid_p
		mstr_archunken
		mstr_araddr
		mstr_aruser
		mstr_arlen
		mstr_arsize
		mstr_arburst
		mstr_arlock
		mstr_arcache
		mstr_arprot
		mstr_rready
		mstr_rready_p
		mstr_armisc_info
		mstr_armisc_info_last_dcmp_tlp
		mstr_armisc_info_tlprrfx
		mstr_armisc_info_tlpIn
		mstr_armisc_info_tph

- mstr_armisc_info_dma
- mstr_armisc_info_nw
- mstr_armisc_info_ats
- mstr_armisc_info_func_num
- mstr_armisc_info_vfunc_num
- mstr_armisc_info_vfunc_active
- mstr_armisc_info_id0
- mstr_armisc_info_zeroread
- mstr_argos
- mstr_arbar
- mstr_ardomain
- mstr_arsnoop
- mstr_aridunq
- mstr_csysack
- mstr_cactive
- mstrh_awuser
- mstrh_armisc_info_zeroread

Table 2-2 AXI Master Interface Signals

Port Name	I/O	Description
mstr_awid[(MSTR_ID_WD-1):0]	O	AXI Master Write ID. Indicates ID of the burst. Memory writes and Messages are always assigned ID 0; Non-Posted writes are assigned unique IDs, from CC_MAX_MSTR_TAGS_AXI to 2 * CC_MAX_MSTR_TAGS_AXI-1. Posted write responses are accepted immediately by the AXI Bridge but are not forwarded to the wire (they are dropped inside the AXI master); Non-Posted write responses are returned to the requester. For more information, see 'Inbound Bridge Tag Management'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awvalid	O	AXI Master Write Address Valid. Indicates valid write address and control information are available. Address and control information remain stable until awready signal is high. ■ 0: Address and control information not available ■ 1: Address and control information available Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mstr_awvalid_p	O	AXI Master Write Address Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awaddr[(MSTR_ADDR_WD-1):0]	O	AXI Master Write Address. Specifies the address of the first transfer in a write burst transaction. Associated control signals are used to determine addresses of remaining transfers in the burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awuser[(MSTR_AW_USER_WIDTH-1):0]	O	AXI Master Write User. The controller sends all the AXI Master AW channel parity protection bits through this signal. In legacy mode, only mstr_awaddr is protected. ■ mstr_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH-1:0] contains the protection bits of mstr_awaddr ■ mstr_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH] contains the protection bits of mstr_awid ■ mstr_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH+1] contains the protection bits of mstr_awlen ■ mstr_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH+2] contains the protection bits of {mstr_awburst,mstr_awlock,mstr_awsz} ■ mstr_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH+3] contains the protection bits of {mstr_awcache,mstr_awprot} ■ mstr_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH+4] contains the protection bits of mstr_awqos ■ mstr_awuser[CC_MSTR_AW_USER_WIDTH-1:CC_MSTR_BUS_ADDR_PROT_WIDTH+5] contains the protection bits of mstr_awatop, all mstr_awmisc_info signals, and acelite signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: None,perbitclk,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid

Port Name	I/O	Description
mstr_awlen[(MSTR_BURST_LEN_PW-1):0]	O	AXI Master Write Burst Length. Specifies the exact number of transfers in a burst; determines the number of data transfers associated with the address. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None,perbitclk,{mstr_ack} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awsz[2:0]	O	AXI Master Write Burst Size. Indicates size of each transfer in burst. Byte lane strobes indicate exactly which byte lanes to update. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None,perbitclk,{mstr_ack} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awburst[1:0]	O	AXI Master Write Burst. Determines the type of burst used. AXI master interface of the controller generates only INCR Bursts. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None,perbitclk,{mstr_ack} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awlock[(MSTR_LOCK_WD-1):0]	O	AXI Master Write Lock. Provides additional information about the characteristics of the transfer. The controller does not use this signal and sets all bits inactive. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_ack,{None} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid

Port Name	I/O	Description
mstr_awcache[3:0]	O	AXI Master Write Cache. Indicates bufferable, cacheable, write-through, and write-back attributes of transaction. Set automatically by the controller when the CFG_MSTR_AWCACHE_MODE field of COHERENCY_CONTROL_REG_3 is '0'. Otherwise, set by the CFG_MSTR_AWCACHE_VALUE field of COHERENCY_CONTROL_REG_3 when CFG_MSTR_AWCACHE_MODE is '1'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awprot[2:0]	O	AXI Master Write Protection. Indicates unprivileged, non-secure, or data protection access level. The controller assigns this signal a value of 010. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awready	I	AXI Master Write Address Ready. Indicates that the slave is ready to accept address and associated control signals. ■ 0: Slave not ready ■ 1: Slave ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk_ug,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awready_p	I	AXI Master Write Address Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid

Port Name	I/O	Description
mstr_awmisc_info[(MSTR_MISC_INFO_PW-1):0]	O	AXI Master Write Misc Information. AXI master write transaction associated miscellaneous information from the TLP received by the native PCIe controller. This is an additional signal. It is not a part of the standard AXI interface. ■ Bus number, device number, function number, and register numbers are overlaid to the address bus when type is configuration transactions. ■ Message contents are also overlaid to the address bus for MSG TLPs. Bits [7:0] indicate the message code when the type is message. For more information, see 'AXI Master Interface Sideband Signals'. The bit fields are mapped as follows: ■ [7:0]: Reserved except for MSG TLP. Bits [7:0] indicate the message code when the type is message. ■ [8:6]: BAR number of TLP ■ [9]: TLP is an I/O ■ [10]: TLP is in ROM range ■ [13:11]: TLP's Function number. Function numbering starts at '0'. This field is not used when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. For more information, see mstr_awmisc_info_func_num. ■ [14]: NS bit of the TLP ■ [15]: RO bit of the TLP ■ [18:16]: TC bits of the TLP ■ [23:19]: TYPE bits of the TLP ■ [39:24]: Requester ID bits of the TLP ■ [CX_TAG_SIZE-1:40]: TLP's TAG Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None,perbitclk,{mstr_ack} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awmisc_info_last_dcmp_tlp	O	AXI Master Write Request last TLP bit. The signal indicates that the TLP in transaction is the last TLP of the write request. The master interface asserts this output on the last beat of a decomposed request. It is continuously asserted for a TLP that is not decomposed. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid

Port Name	I/O	Description
mstr_awmisc_info_hdr_34dw[63:0]	O	AXI Master 3rd and 4th header DW. The bus conveys the 3rd and 4th header DWs in a PCIe Msg/MsgD to the application. <i>Note:</i> The controller receives the data bytes in little endian format (+0, +1, and so on as defined in Figure 2-24 Message Request Header of the <i>PCI Express Base Specification, Revision 4.0, Version 0.7</i>), but it provides these bytes to your application in big endian format. For example, byte +3 is provided to the application in bits [7:0]. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None,perbitclk,{mstr_ack} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awmisc_info_tlpprfx[(PRFX_WIDTH-1):0]	O	AXI Master Write Request TLP Prefixes. The bits [31:0] represent the first prefix received. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that mstr_awmisc_info_tlpprfx[31:0] has prefix byte #0 in the lower byte position of the dword mstr_awmisc_info_tlpprfx[31:0]. That is, FMT=bits 7:5, and Type =bits 4:0. If no prefix is present for a given TLP, then that DW, including the FMT field is all zeros. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_TLP_PREFIX_ENABLE) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awmisc_info_tlpLn	O	AXI Master Write Request TLP Lightweight Notification (LN) bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_LN_ENABLE) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid

Port Name	I/O	Description
mstr_awmisc_info_tph[10:0]	O	AXI Master Write Request TLP Processing Hints. The bits are mapped as follows: ■ [0]: TH (TLP Processing Hint present) ■ [2:1] PH (TLP Processing Hint) ■ [10:3] ST (Steering Tag) Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_TPH_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awmisc_info_dma[((CC_NUM_DMA_RD_CHAN_WIDTH+DC_DMA_TYPE_WIDTH)-1):0]	O	DMA Bits of the AXI Write Master Transaction. This signal can be used for debugging. For more information, see 'Programming Examples'. This is a signal that the application can optionally use. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Bits [MSB:3] the indicate DMA channel number and bits [2:0] indicate the DMA request type as follows: ■ 000: Non-DMA request ■ 001: Reserved ■ 010: Reserved ■ 011: DMA Read channel MWr request ■ 100: Reserved ■ 101: Reserved ■ 110: Reserved ■ 111: Reserved Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_DMA_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid

Port Name	I/O	Description
mstr_awmisc_info_nw	O	ATS No Write (NW) Bit of the AXI Master Write Transaction Request. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (ATS_RX_ENABLE) && (CX_ATS_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awmisc_info_ats[1:0]	O	AT Bits of the AXI Master Write Transaction Request. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (ATS_RX_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awmisc_info_func_num[(PF_WD-1):0]	O	Function Number of the AXI Master Write Transaction. This signal is used instead of bits [13:11] of mstr_awmisc_info when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Function numbering starts at '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && ((AMBA_INTERFACE != 0 && (CX_SRIOV_ENABLE CX_ARI_ENABLE))) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid

Port Name	I/O	Description
mstr_awmisc_info_vfunc_num[(VF_WD-2):0]	O	Virtual Function Number of the AXI Master Write Transaction. Indicates which VF has been targeted by the received request. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, mstr_awmisc_info_vfunc_num = 0 refers to the first VF of the PF that is identified by mstr_awmisc_info_func_num. For more information of VF numbering, see 'Programming View'. This is a signal that the application can optionally use. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_SRIOV_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awmisc_info_vfunc_active	O	VF Active Indication of the AXI Master Write Transaction. Indicates that the received request matched a VF BAR. 0: No VF is active and mstr_awmisc_info_vfunc_num is invalid. A PF is valid and identified by mstr_awmisc_info_func_num. 1: A VF is active and is identified by mstr_awmisc_info_vfunc_num. This is a signal that the application can optionally use. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_SRIOV_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awmisc_info_ep	O	AXI Master Write mapping to the EP bit in the PCIe protocol domain. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid

Port Name	I/O	Description
mstr_awmisc_info_ido	O	IDO Bit of the AXI Master Write Transaction. Enables ID-base ordering on inbound requests. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_IDO_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_awqos[3:0]	O	AXI Master Write QoS. The controller always drives this output to '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awbar[1:0]	O	AXI Master Write Barrier Type. You can use this signal to distinguish between normal and transaction barriers, and between different barrier types. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: mstr_aclk,{None} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awdomain[1:0]	O	AXI Master Write Shareability Domain. Indicates the shareability domain to which the transaction applies. Set by the controller or by the CFG_MSTR_AWDOMAIN_VALUE field of the COHERENCY_CONTROL_REG_3 register. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid

Port Name	I/O	Description
mstr_awsnoop[2:0]	O	AXI Master Write Shareable Transaction Type. You can use this signal to distinguish between the different types of shareable write transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: mstr_aclk,{None} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awatop[5:0]	O	AXI Master Write Atomic Operation. Indicates the Atomic Operation encoding. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_ATOMIC_ENABLE) Synchronous To: None,perbitclk,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_awidunq	O	AXI Master Write Unique ID Indicator. Indicates whether mstr_awid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_AXI_UIDI_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_awvalid
mstr_wready	I	AXI Master Write Ready. Indicates that slave can accept write data. ■ 0: Slave not ready ■ 1: Slave ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk_ug,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_wvalid

Port Name	I/O	Description
mstr_wready_p	I	AXI Master Write Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_wvalid
mstr_wdata[(MSTR_DATA_WD-1):0]	O	AXI Master Write Data. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_wvalid
mstr_wuser[(MSTR_W_USER_WIDTH-1):0]	O	AXI Master Write User. The controller sends all of the Master W channel protection bits through this signal. You can configure the CX_AXI_IF_TYPE_OF_PROT parameter to specify the type of protection. In legacy mode, only mstr_wdata is protected. ■ mstr_wuser[CC_MSTR_BUS_DATA_PROT_WIDTH-1:0] contains the protection bits of mstr_wdata ■ mstr_wuser[CC_MSTR_BUS_DATA_PROT_WIDTH] contains the protection bits of mstr_wid ■ mstr_wuser[CC_MSTR_BUS_DATA_PROT_WIDTH+1] contains the protection bits of mstr_wstrb ■ mstr_wuser[CC_MSTR_BUS_DATA_PROT_WIDTH+2] contains the protection bits of mstr_wlast Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_wvalid

Port Name	I/O	Description
mstr_wstrb[(MSTR_WSTRB_WD-1):0]	O	AXI Master Write Strobes. Indicates which byte lanes to update in memory. There is one write strobe for each eight bits of write data bus. mstr_wstrb[n] is associated with mstr_wdata[8*n + 7 : 8*n] Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_wvalid
mstr_wlast	O	AXI Master Write Last. Indicates last transfer in write burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_wvalid
mstr_wvalid	O	AXI Master Write Valid. Indicates valid write data and strobes are available. ■ 0: Write data and strobes not available ■ 1: Write data and strobes available Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mstr_wvalid_p	O	AXI Master Write Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_wvalid

Port Name	I/O	Description
mstr_wid[(MSTR_ID_WD-1):0]	O	AXI Master Write ID. AXI master write identification ID. This signal is the ID tag of the write data transfer. Memory writes and Messages are always assigned ID 0; Non-Posted writes are assigned unique IDs, from CC_MAX_MSTR_TAGS_AXI to 2 * CC_MAX_MSTR_TAGS_AXI-1. For more information, see 'Inbound Bridge Tag Management'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (!((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_wvalid
mstr_bready	O	AXI Master Response Ready. Indicates master can accept response information. ■ 0: Master not ready ■ 1: Master ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_bvalid
mstr_bready_p	O	AXI Master Response Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_bvalid

Port Name	I/O	Description
mstr_bid[(MSTR_ID_WD-1):0]	I	AXI Master Write Response ID. The identification tag of the write response. The BID value must match the AWID value of the write transaction to which the slave is responding. This signal is designed for non-posted writes such as a PCIe configuration write or an I/O write, if they are applicable in a particular application. When I/O or configuration writes are not supported in a particular application, this signal can be tied to '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_bvalid
mstr_bvalid	I	AXI Master Write Response Valid. Indicates that valid write response is available. ■ 0: Write response not available ■ 1: Write response available Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Ug,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mstr_bvalid_p	I	AXI Master Write Response Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_bvalid

Port Name	I/O	Description
mstr_bresp[1:0]	I	AXI Master Write Response. Indicates status of write transaction. AXI error responses (SLVERR or DECERR) are mapped to PCIe Completion Abort (for non-posted requests). For more information, see the 'Advanced Error Handling For Received TLPs' advanced information chapter. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_bvalid
mstr_buser[(MSTR_B_USER_WIDTH-1):0]	I	AXI Master Write Response User. You can send the Master B channel protection bits through this signal. In legacy mode, this signal does not exist. ■ mstr_buser[0] contains the protection bits of mstr_bid ■ mstr_buser[1] contains the protection bits of mstr_bresp ■ mstr_buser[CC_MSTR_B_USER_WIDTH-1:2] contains the protection bits of all mstr_bmisc_info signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk_ug,perbitclk,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_bvalid

Port Name	I/O	Description
mstr_bmisc_info_tlpprfx[(PRFX_WIDTH-1):0]	I	AXI Master Write Response TLP Prefixes. The field [31:0] represents the first prefix to be transmitted. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that mstr_bmisc_info_tlpprfx[31:0] has prefix byte #0 in the lower byte position of the dword mstr_bmisc_info_tlpprfx[31:0]. That is, FMT =bits 7:5, and Type =bits 4:0. Your application is responsible for passing Local and End-End prefixes in the correct order for transmission. Local prefixes must precede End-End prefixes. Any parity bits that your application appends (when RAS data protection is enabled) to the most significant bits are routed but not checked by the controller. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_MSTR_RESP_TLP_PREFIX_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_bvalid
mstr_bmisc_info_tlpIn	I	AXI Master Write Response TLP Lightweight Notification (LN) bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_LN_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_bvalid
mstr_bmisc_info_cid[15:0]	I	AXI Master Write Response Completer ID. Sets the Completer ID value for AXI master write responses. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_AXI_MSTR_MISC_INFO_CID_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: mstr_bvalid

Port Name	I/O	Description
mstr_bmisc_info_cpl_stat[2:0]	I	AXI Master Write Response selection bus. This controls the response to be sent on the wire in the case of successful write requests. The controller always sends a CA (Completer Abort) response when it receives SLVERR/DECERR. For more information, see the "Master AXI-to-PCIe Error Mapping" section in the AXI chapter of the <i>Databook</i>. ■ 000: SC (Successful Completion) ■ 001: UR (Unsupported Request) ■ 010: RRS (Request Retry Status) ■ 100: CA (Completer Abort) ■ All others: Reserved Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk_ug,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_bvalid
mstr_bidung	I	AXI Master Write Response Unique ID Indicator. Indicates whether mstr_bid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_AXI_UIDI_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_bvalid
mstr_arid[(MSTR_ID_WD-1):0]	O	AXI Master Read Address ID. AXI master read address identification tag for read address signals. This signal provides a unique ID from 0 to CC_MAX_MSTR_TAGS_AXI-1 for each read transfer, unless decomposition has occurred in which case all AXI transfers related to the same PCIe request have the same ID. For more information, see 'Inbound Bridge Tag Management'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None,perbitclk,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid

Port Name	I/O	Description
mstr_arvalid	O	AXI Master Read Address Valid. When high, indicates read address and control information is valid and remains stable until already is high. 0: Address and control information not valid 1: Address and control information valid Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mstr_arvalid_p	O	AXI Master Read Address Valid Parity bit. In legacy mode, this signal does not exist. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_archunken	O	AXI Master Read Address Chunk Enable. Indicates whether the data for this transaction can be chunked or not. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_AXI_RCHUNK_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_araddr[(MSTR_ADDR_WD-1):0]	O	AXI Master Read Address. Gives initial address of read burst transaction. Only start address of burst is provided, and control signals issued alongside address show how address is calculated for remaining transfers in burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid

Port Name	I/O	Description
mstr_aruser[(MSTR_AR_USER_WIDTH-1):0]	O	AXI Master Read User. The controller sends all the Master AR channel parity protection bits through this signal. In legacy mode, only mstr_araddr is protected. ■ mstr_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH-1:0] contains the protection bits of mstr_araddr ■ mstr_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH] contains the protection bits of mstr_arid ■ mstr_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH+1] contains the protection bits of mstr_arlen ■ mstr_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH+2] contains the protection bits of {mstr_arburst,mstr_arlock,mstr_arsize} ■ mstr_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH+3] contains the protection bits of {mstr_arcache,mstr_arprot} ■ mstr_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH+4] contains the protection bits of mstr_arqos ■ mstr_aruser[CC_MSTR_AR_USER_WIDTH-1:CC_MSTR_BUS_ADDR_PROT_WIDTH+5] contains the protection bits of all mstr_ar-misc_info signals and acelite signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: None,perbitclk,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_arlen[(MSTR_BURST_LEN_PW-1):0]	O	AXI Master Read Burst Length. Gives exact number of transfers in burst; determines number of data transfers associated with address. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None,perbitclk,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_arsize[2:0]	O	AXI Master Read Burst Size. Indicates size of each transfer in burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None,perbitclk,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid

Port Name	I/O	Description
mstr_arburst[1:0]	O	AXI Master Read Burst. Determines the type of burst used. AXI master interface of the controller generates only INCR Bursts. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None,perbitclk,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_arlock[(MSTR_LOCK_WD-1):0]	O	AXI Master Read Lock. Encoded value indicates whether the transfer is a normal, exclusive, or locked access. The controller does not use this signal and sets all bits inactive. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk,{None} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_arcache[3:0]	O	AXI Master Read Cache. Indicates bufferable, cacheable, read-through, and read-back attributes of transaction. Set automatically by the controller when the CFG_MSTR_ARCACHE_MODE field of COHERENCY_CONTROL_REG_3 is '0'. Otherwise, set by the CFG_MSTR_ARCACHE_VALUE field of COHERENCY_CONTROL_REG_3 when CFG_MSTR_ARCACHE_MODE is '1'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_aprot[2:0]	O	AXI Master Read Protection. Indicates unprivileged, non-secure, or data protection access level. The controller assigns this signal a value of 010. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid

Port Name	I/O	Description
mstr_rready	O	AXI Master Read Ready. Indicates master can accept read data and response information. ■ 0: Master not ready ■ 1: Master ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid
mstr_rready_p	O	AXI Master Read Ready Parity bit. In legacy mode, this signal does not exist. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid

Port Name	I/O	Description
mstr_armisc_info[(MSTR_MISC_INFO_PW-1):0]	O	AXI Master Read Transaction Associated Misc Information (from the TLP received by the native PCIe controller). This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. ■ [5:0]: Reserved ■ [8:6]: BAR number of TLP. Bits [7:0] also indicate the message code when the type is message. ■ [9]: TLP is an I/O ■ [10]: TLP is in ROM range ■ [13:11]: TLP's Function number. Function numbering starts at '0'. This field is not used when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. For more information, see mstr_armisc_info_func_num. ■ [14]: TLP's NS bit ■ [15]: TLP's RO bit ■ [18:16]: TLP's TC bits ■ [23:19]: TLP's TYPE. For more information, see 'I/O and CFG Transaction Handling' ■ [39:24]: TLP's requester ID ■ [CX_TAG_SIZE-1:40]: TLP's TAG Bus number, device number, function number, and register numbers are overlaid as follows to the address bus for CFG TLPs. ■ bus_number =addr[31:24] ■ dev_number =addr[23:19] ■ func_number =addr[18:16] ■ ext_reg_number =addr[11:8] ■ reg_number =addr[7:2] Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None,perbitclk,{mstr_ack} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_armisc_info_last_dcmp_tlp	O	AXI Master Read Request last TLP bit. The signal indicates that the TLP in transaction is the last TLP of the read request. The master interface asserts this output on the last beat of a decomposed request. It is continuously asserted for a TLP that is not decomposed. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid

Port Name	I/O	Description
mstr_armisc_info_tlpprfx[(PRFX_WIDTH-1):0]	O	AXI Master Read Request TLP Prefixes. The field [31:0] represents the first prefix received. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that mstr_armisc_info_tlpprfx[31:0] has prefix byte #0 in the lower byte position of the dword mstr_armisc_info_tlpprfx[31:0]. That is, FMT=bits 7:5, and Type =bits 4:0. If no prefix is present for a given TLP, then that DW, including the FMT field is all zeros. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_TLP_PREFIX_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_armisc_info_tlpIn	O	AXI Master Read Request TLP Lightweight Notification (LN) bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_LN_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_armisc_info_tph[10:0]	O	AXI Master Read Request TLP Processing Hints. The bits are mapped as follows: ■ [0]: TH (TLP Processing Hint present) ■ [2:1] PH (TLP Processing Hint) ■ [10:3] ST (Steering Tag) Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_TPH_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid

Port Name	I/O	Description
mstr_armisc_info_dma[(((CC_NUM_DMA_MAX_CHAN_WIDTH+DC_DMA_TYPE_WIDTH)-1):0]	O	DMA bits of the AXI Read Master Transaction. This signal can be used for debugging. For more information, see 'Programming Examples'. This is a signal that the application can optionally use. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Bits [MSB:3] the indicate DMA channel number and bits [2:0] indicate the DMA request type as follows: ■ 000: Non-DMA request ■ 001: DMA Write channel MRd request ■ 010: Reserved ■ 011: Reserved ■ 100: Reserved ■ 101: DMA Write channel LL Element MRd request ■ 110: Reserved ■ 111: DMA Read channel LL Element MRd request Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_DMA_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_armisc_info_nw	O	ATS No Write (NW) Bit of the AXI Master Read Transaction Request. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (ATS_RX_ENABLE) && (CX_ATS_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid

Port Name	I/O	Description
mstr_armisc_info_ats[1:0]	O	ATS Bits of the AXI Master Read Transaction Request. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is a signal that the application can optionally use. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (ATS_RX_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_armisc_info_func_num[(PF_WD-1):0]	O	Function Number of the AXI Master Read Transaction. It is used instead of bits [13:11] of mstr_armisc_info when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Function numbering starts at '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && ((AMBA_INTERFACE != 0 && (CX_SRIOV_ENABLE CX_ARI_ENABLE))) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_armisc_info_vfunc_num[(VF_WD-2):0]	O	Virtual Function Number of the AXI Master Read Transaction. Indicates which VF has been targeted by the received request. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, mstr_armisc_info_vfunc_num =0 refers to the first VF of the PF that is identified by mstr_armisc_info_func_num. For more information of VF numbering, see 'Programming View'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_SRIOV_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid

Port Name	I/O	Description
mstr_armisc_info_vfunc_active	O	VF Active Indication of the AXI Master Read Transaction. Indicates that the received request matched a VF BAR. 0: No VF is active and mstr_armisc_info_vfunc_num is invalid. A PF is valid and identified by mstr_armisc_info_func_num. 1: A VF is active and is identified by mstr_armisc_info_vfunc_num. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_SRIOV_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_armisc_info_ido	O	IDO bit of the AXI Master Read Transaction. Enables ID-base ordering on inbound requests. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_IDO_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_armisc_info_zeroread	O	AXI Master Read mapping to a zero length read in the PCIe protocol domain. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid

Port Name	I/O	Description
mstr_arready	I	AXI Master Read Address Ready. Indicates that slave is ready to accept address and associated control signals. 0: Slave not ready 1: Slave ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Ug,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_arready_p	I	AXI Master Read Address Ready Parity bit. In legacy mode, this signal does not exist. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_arqos[3:0]	O	AXI Master Read QoS. The controller always drives this output to '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: mstr_aclk,{None} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_arbar[1:0]	O	AXI Master Read Barrier Type. You can use this signal to distinguish between normal and transaction barriers, and between different barrier types. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid

Port Name	I/O	Description
mstr_ardomain[1:0]	O	AXI Master Read Shareability Domain. Indicates the shareability domain to which the transaction applies. Set by the controller or by the CFG_MSTR_ARDOMAIN_VALUE field of the COHERENCY_CONTROL_REG_3 register. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_arsnoop[3:0]	O	AXI Master Read Shareable Transaction Type. You can use this signal to distinguish between the different types of shareable read transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_aridunq	O	AXI Master Read Unique ID Indicator. Indicates whether mstr_arid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_AXI_UIDI_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_arvalid
mstr_rid[(MSTR_ID_WD-1):0]	I	AXI Master Read ID. This signal is required by the AXI bridge to identify a response for a particular master request. For more information, see 'Inbound Bridge Tag Management'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None,perbitclk,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid

Port Name	I/O	Description
mstr_rvalid	I	AXI Master Read Valid. Indicates that required read data is available and read transfer can complete. 0: Read data not available 1: Read data available Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Ug,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mstr_rvalid_p	I	AXI Master Read Valid Parity bit. In legacy mode, this signal does not exist. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid
mstr_rlast	I	AXI Master Read Last. Indicates last transfer in read burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid
mstr_rdata[(MSTR_DATA_WD-1):0]	I	AXI Master Read Data. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Ug,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid

Port Name	I/O	Description
mstr_rchunkv	I	AXI Master Read Chunk Valid. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_AXI_RCHUNK_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid
mstr_rchunknum[(CC_AXI_RCHUNKNUM_WD-1):0]	I	AXI Master Read Chunk Number. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_AXI_RCHUNK_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rchunkv
mstr_rchunkstrb[(CC_AXI_RCHUNKSTRB_WD-1):0]	I	AXI Master Read Chunk Strobe. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_AXI_RCHUNK_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rchunkv

Port Name	I/O	Description
mstr_ruser[(MSTR_R_USER_WIDTH-1):0]	I	AXI Master Read Response User. You can use this signal to send all the Master R channel protection bits to the controller. You can configure the CX_AXI_IF_TYPE_OF_PROT parameter to specify the type of protection. In legacy mode, only mstr_rdata is protected. ■ mstr_ruser[CC_MSTR_BUS_DATA_PROT_WIDTH-1:0] contains the protection bits of mstr_rdata ■ mstr_ruser[CC_MSTR_BUS_DATA_PROT_WIDTH] contains the protection bits of mstr_rid ■ mstr_ruser[CC_MSTR_BUS_DATA_PROT_WIDTH+1] contains the protection bits of {mstr_rresp,mstr_rlast} ■ mstr_ruser[CC_MSTR_R_USER_WIDTH-1:CC_MSTR_BUS_DATA_PROT_WIDTH+2] contains the protection bits of all mstr_awmisc_info signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: mstr_aclk Ug,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid
mstr_rresp[1:0]	I	AXI Master Read Response. Indicates status of read transaction. AXI error responses (SLVERR or DECERR) are mapped to PCIe Completion Abort (for non-posted requests). For more information, see the 'Advanced Error Handling For Received TLPs' advanced information chapter. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid

Port Name	I/O	Description
mstr_misc_info[(MSTR_RESP_MISC_IN FO_PW-1):0]	I	AXI Master Read Response Transaction Associated Misc Information. This is a signal that the application can optionally use. It is not a part of standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. All reserved bits must be connected to logic '0'. ■ [1:0]: Reserved ■ [2]: Reserved. ■ [6:3]: Reserved ■ [7]: TLP's EP bit ■ [12:8]: Reserved. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk_ug,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid
mstr_misc_info_tlprrfx[(PRFX_WIDTH-1):0]	I	AXI Master Read Response TLP Prefixes. The field [31:0] represents the first prefix to be transmitted. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that mstr_misc_info_tlprrfx[31:0] has prefix byte #0 in the lower byte position of the dword mstr_misc_info_tlprrfx[31:0]. That is, FMT =bits 7:5, and Type =bits 4:0. Your application is responsible for passing Local and End-End prefixes in the correct order for transmission. Local prefixes must precede End-End prefixes. Any parity bits that your application appends (when RAS data protection is enabled) to the most significant bits are routed but not checked by the controller. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_MSTR_RESP_TLP_PREFIX_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid

Port Name	I/O	Description
mstr_rmisc_info_tlpIn	I	AXI Master Read Response TLP Lightweight Notification (LN) bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_LN_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid
mstr_rmisc_info_cid[15:0]	I	Master Read Response Completer ID. Sets the Completer ID value for AXI master read responses. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_AXI_MSTR_MISC_INFO_CID_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: mstr_rvalid
mstr_rmisc_info_cpl_stat[2:0]	I	AXI Master Read Response selection bus. This bus controls the response sent on the PCIe wire in the case of successful read requests. The controller always sends a CA (Completer Abort) response when it receives SLVERR/DECERR. For more information, see the "Master AXI-to-PCIe Error Mapping" section in the AXI chapter of the <i>Databook</i>. ■ 000: SC (Successful Completion) ■ 001: UR (Unsupported Request) ■ 010: RRS (Request Retry Status) ■ 100: CA (Completer Abort) ■ All others: Reserved Your application must drive the same value on mstr_rmisc_info_cpl_stat throughout the complete response. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: mstr_aclk_ug,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid

Port Name	I/O	Description
mstr_ridung	I	AXI Master Read Response Unique ID Indicator. Indicates whether mstr_rid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CX_AXI_UIDI_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_rvalid
mstr_csysack	O	AXI Master Low-Power Request Acknowledgment. Indicates acknowledgment from AXI bridge master to a system low-power request. The controller does not support AXI power modes and always denies any request the mstr_csysreq input. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: mstr_csysreq
mstr_cactive	O	AXI Master Clock Active. Indicates that AXI bridge master requires a clock signal. The controller does not support AXI power modes and always drives this output high. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstr_csysreq
mstr_csysreq	I	AXI Master System Low-Power Request. System low-power request from system clock controller for AXI bridge master to enter low-power state. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

Port Name	I/O	Description
mstrh_awuser[(MSTR_AW_USER_WIDTH-1):0]	O	AXI High Priority Master Write User. The controller sends all the High Priority Master AW channel parity protection bits through this signal. In legacy mode, only mstrh_awaddr is protected. ■ mstrh_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH-1:0] contains the protection bits of mstrh_awaddr ■ mstrh_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH] contains the protection bits of mstrh_awid ■ mstrh_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH+1] contains the protection bits of mstrh_awlen ■ mstrh_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH+2] contains the protection bits of {mstrh_awburst,mstrh_awlock,mstrh_awsz} ■ mstrh_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH+3] contains the protection bits of {mstrh_awcache,mstrh_awprot} ■ mstrh_awuser[CC_MSTR_BUS_ADDR_PROT_WIDTH+4] contains the protection bits of mstrh_awqos ■ mstrh_awuser[CC_MSTR_AW_USER_WIDTH-1:CC_MSTR_BUS_ADDR_PROT_WIDTH+5] contains the protection bits of mstrh_awatop, all mstrh_awmisc_info signals, and accel signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_armisc_info_zeroread	O	AXI High Priority Master Read mapping to a zero length read in the PCIe protocol domain. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_rchunkv	I	AXI High Priority Master Read Chunk Valid. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_AXI_RCHUNK_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid
mstrh_rchunknum[(CC_AXI_RCHUNKNUM_WD-1):0]	I	AXI High Priority Master Read Chunk Number. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_AXI_RCHUNK_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rchunkv
mstrh_rchunkstrb[(CC_AXI_RCHUNKSTRB_WD-1):0]	I	AXI High Priority Master Read Chunk Strobe. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_AXI_RCHUNK_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rchunkv
mstrh_rmisc_info_cid[15:0]	I	AXI High Priority Master Read Response Completer ID. Sets the Completer ID value for AXI master read responses. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CC_AXI_MSTR_MISC_INFO_CID_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: mstrh_rvalid

2.3 AXI Master Interface Signals (2nd Master)

mstrh_awready	-	mstrh_awid
mstrh_awready_p	-	mstrh_awvalid
mstrh_wready	-	mstrh_awvalid_p
mstrh_wready_p	-	mstrh_awaddr
mstrh_bid	-	mstrh_awlen
mstrh_bvalid	-	mstrh_awsize
mstrh_bvalid_p	-	mstrh_awburst
mstrh_bresp	-	mstrh_awlock
mstrh_buser	-	mstrh_awcache
mstrh_bmisc_info_tlpn	-	mstrh_awprot
mstrh_bmisc_info_cpl_stat	-	mstrh_awmisc_info
mstrh_bmisc_info_cpl_stat	-	mstrh_awmisc_info_last_dcmp_tlp
mstrh_bidunq	-	mstrh_awmisc_info_hdr_34dw
mstrh_arready	-	mstrh_awmisc_info_tlpprfx
mstrh_arready_p	-	mstrh_awmisc_info_tlpn
mstrh_rid	-	mstrh_awmisc_info_tph
mstrh_rvalid	-	mstrh_awmisc_info_dma
mstrh_rvalid_p	-	mstrh_awmisc_info_nw
mstrh_rlast	-	mstrh_awmisc_info_ats
mstrh_rdata	-	mstrh_awmisc_info_func_num
mstrh_ruser	-	mstrh_awmisc_info_vfunc_num
mstrh_rresp	-	mstrh_awmisc_info_vfunc_active
mstrh_rmisc_info	-	mstrh_awmisc_info_ep
mstrh_rmisc_info_tlpn	-	mstrh_awmisc_info_id0
mstrh_rmisc_info_cpl_stat	-	mstrh_awqos
mstrh_ridunq	-	mstrh_awbar
mstrh_csysreq	-	mstrh_awdomain
		mstrh_awsnoop
		mstrh_awatop
		mstrh_awidunq
		mstrh_wdata
		mstrh_wuser
		mstrh_wstrb
		mstrh_wlast
		mstrh_wvalid
		mstrh_wvalid_p
		mstrh_wid
		mstrh_bready
		mstrh_bready_p
		mstrh_arid
		mstrh_arvalid
		mstrh_arvalid_p
		mstrh_archunken
		mstrh_araddr
		mstrh_aruser
		mstrh_arsize
		mstrh_arburst
		mstrh_arlock
		mstrh_arcache
		mstrh_arprot
		mstrh_rready
		mstrh_rready_p
		mstrh_armisc_info
		mstrh_armisc_info_last_dcmp_tlp
		mstrh_armisc_info_tlpprfx
		mstrh_armisc_info_tlpn
		mstrh_armisc_info_tph
		mstrh_armisc_info_dma

- mstrh_armisc_info_nw
- mstrh_armisc_info_at
- mstrh_armisc_info_func_num
- mstrh_armisc_info_vfunc_num
- mstrh_armisc_info_vfunc_active
- mstrh_armisc_info_id
- mstrh_arqos
- mstrh_arbar
- mstrh_ardomain
- mstrh_arsnoop
- mstrh_aridunq
- mstrh_csysack
- mstrh_cactive

Table 2-3 AXI Master Interface Signals (2nd Master)

Port Name	I/O	Description
mstrh_awid[(MSTR_ID_WD-1):0]	O	AXI High Priority Master Write ID. Indicates ID of the burst. Memory writes and Messages are always assigned ID 0; Non-Posted writes are assigned unique IDs, from CC_MAX_MSTR_TAGS_AXI to 2 * CC_MAX_MSTR_TAGS_AXI-1. Posted write responses are accepted immediately by the AXI Bridge but are not forwarded to the wire (they are dropped inside the AXI master); Non-Posted write responses are returned to the requester. For more information, see 'Inbound Bridge Tag Management'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awvalid	O	AXI High Priority Master Write Address Valid. Indicates valid write address and control information are available. Address and control information remain stable until awready signal is high. ■ 0: Address and control information not available ■ 1: Address and control information available Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mstrh_awvalid_p	O	AXI High Priority Master Write Address Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awaddr[(MSTR_ADDR_WD-1):0]	O	AXI High Priority Master Write Address. Specifies the address of the first transfer in a write burst transaction. Associated control signals are used to determine addresses of remaining transfers in the burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awlen[(MSTR_BURST_LEN_PW-1):0]	O	AXI High Priority Master Write Burst Length. Specifies the exact number of transfers in a burst; determines the number of data transfers associated with the address. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awsz[2:0]	O	AXI High Priority Master Write Burst Size. Indicates size of each transfer in burst. Byte lane strobes indicate exactly which byte lanes to update. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid

Port Name	I/O	Description
mstrh_awburst[1:0]	O	AXI High Priority Master Write Burst. Determines the type of burst used. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awlock[(MSTR_LOCK_WD-1):0]	O	AXI High Priority Master Write Lock. Provides additional information about the characteristics of the transfer. The controller does not use this signal and sets all bits inactive. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awcache[3:0]	O	AXI High Priority Master Write Cache. Indicates bufferable, cacheable, write-through, and write-back attributes of transaction. Set automatically by the controller when the CFG_MSTR_AWCACHE_MODE field of COHERENCY_CONTROL_REG_3 is '0'. Otherwise, set by the CFG_MSTR_AWCACHE_VALUE field of COHERENCY_CONTROL_REG_3 when CFG_MSTR_AWCACHE_MODE is '1'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid

Port Name	I/O	Description
mstrh_awprot[2:0]	O	AXI High Priority Master Write Protection. Indicates unprivileged, non-secure, or data protection access level. The controller assigns this signal a value of 010. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awready	I	AXI High Priority Master Write Address Ready. Indicates that the slave is ready to accept address and associated control signals. ■ 0: Slave not ready ■ 1: Slave ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awready_p	I	AXI High Priority Master Write Address Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid

Port Name	I/O	Description
mstrh_awmisc_info[(MSTR_MISC_INFO_PW-1):0]	O	AXI High Priority Master Write Misc Information. AXI master write transaction associated miscellaneous information from the TLP received by the native PCIe controller. This is a signal that the application can optionally use. It is not a part of the standard AXI interface. ■ Bus number, device number, function number, and register numbers are overlaid to the address bus when type is configuration transactions. ■ Message contents are also overlaid to the address bus for MSG TLPs. Bits [7:0] indicate the message code when the type is message. For more information, see 'AXI Master Interface Sideband Signals'. The bit fields are mapped as follows: ■ [5:0]: Reserved except for MSG TLP. Bits [7:0] indicate the message code when the type is message. ■ [8:6]: BAR number of TLP ■ [9]: TLP is an I/O ■ [10]: TLP is in ROM range ■ [13:11]: Function number of the TLP. Function numbering starts at '0'. This field is not used when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. For more information, see mstrh_awmisc_info_func_num. ■ [14]: NS bit of the TLP ■ [15]: RO bit of the TLP ■ [18:16]: TC bits of the TLP ■ [23:19]: TLP's TYPE ■ [39:24]: Requester ID of the TLP ■ [CX_TAG_SIZE-1:40]: TLP's TAG Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstrh_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid

Port Name	I/O	Description
mstrh_awmisc_info_last_dcmp_tlp	O	AXI High Priority Master Write Request last TLP bit. The signal indicates that the TLP in transaction is the last TLP of the write request. The master interface asserts this output on the last beat of a decomposed request. It is continuously asserted for a TLP that is not decomposed. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awmisc_info_hdr_34dw[63:0]	O	AXI High Priority Master 3rd and 4th header DW. The bus conveys the 3rd and 4th header DWs in a PCIe Msg/MsgD to the application Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awmisc_info_tlpprfx[(PRFX_WIDTH-1):0]	O	AXI High Priority Master Write Request TLP Prefixes. The field [31:0] represents the first prefix received. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that mstrh_awmisc_info_tlpprfx[31:0] has prefix byte #0 in the lower byte position of the dword mstrh_awmisc_info_tlpprfx[31:0]. That is, FMT=bits 7:5, and Type =bits 4:0. If no prefix is present for a given TLP, then that DW, including the FMT field is all zeros. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_TLP_PREFIX_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid

Port Name	I/O	Description
mstrh_awmisc_info_tlpIn	O	AXI High Priority Master Write Request TLP Lightweight Notification (LN) bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_LN_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awmisc_info_tph[10:0]	O	AXI High Priority Master Write Request TLP Processing Hints. The bits are mapped as follows: ■ [0]: TH (TLP Processing Hint present) ■ [2:1] PH (TLP Processing Hint) ■ [10:3] ST (Steering Tag) Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_TPH_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid

Port Name	I/O	Description
mstrh_awmisc_info_dma[((CC_NUM_DMA_RD_CHAN_WIDTH+DC_DMA_TYPE_WIDTH)-1):0]	O	DMA bits of the AXI Write High Priority Master Transaction. This signal can be used for debugging. For more information, see 'Programming Examples'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Bits [5:3] indicate DMA channel number and bits [2:0] indicate the DMA request type as follows: ■ 000: Non-DMA request ■ 001: DMA Write channel MRd request ■ 010: Non-DMA completion ■ 011: DMA Read channel MWr request ■ 100: Reserved ■ 101: DMA Write channel LL Element MRd request ■ 110: Reserved ■ 111: DMA Read channel LL Element MRd request Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CC_DMA_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awmisc_info_nw	O	ATS No Write (NW) Bit of the AXI High Priority Master Write Transaction Request. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (ATS_RX_ENABLE) && (CX_ATS_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid

Port Name	I/O	Description
mstrh_awmisc_info_ats[1:0]	O	AT Bits of the AXI High Priority Master Write Transactions Request. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (ATS_RX_ENABLE) Synchronous To: mstrh_aclck Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awmisc_info_func_num[(PF_WD-1):0]	O	Function Number of the AXI High Priority Master Write Transaction. It is used instead of bits [13:11] of mstrh_awmisc_info when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Function numbering starts at '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && ((AMBA_INTERFACE != 0 && (CX_SRIOV_ENABLE CX_ARI_ENABLE))) Synchronous To: mstrh_aclck Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awmisc_info_vfunc_num[(VF_WD-2):0]	O	Virtual Function Number of the AXI High Priority Master Write Transaction. Indicates which VF has been targeted by the received request. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, mstrh_awmisc_info_vfunc_num =0 refers to the first VF of the PF that is identified by mstrh_awmisc_info_func_num. For more information of VF numbering, see 'Programming View'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_SRIOV_ENABLE) Synchronous To: mstrh_aclck Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid

Port Name	I/O	Description
mstrh_awmisc_info_vfunc_active	O	VF Active Indication of the AXI High Priority Master Write Transaction. Indicates that the received request matched a VF BAR. 0: No VF is active and mstrh_awmisc_info_vfunc_num is invalid. A PF is valid and identified by mstrh_awmisc_info_func_num. 1: A VF is active and is identified by mstrh_awmisc_info_vfunc_num. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_SRIOV_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awmisc_info_ep	O	AXI High Priority Master Write mapping to the EP bit in the PCIe protocol domain. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awmisc_info_ido	O	IDO Bit of the AXI High Priority Master Write Transaction. Enables ID-base ordering on inbound requests. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_IDO_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_awqos[3:0]	O	AXI High Priority Master Write QoS. The controller always drives this output to '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awbar[1:0]	O	AXI High Priority Master Write Barrier Type. You can use this signal to distinguish between normal and transaction barriers, and between different barrier types. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awdomain[1:0]	O	AXI High Priority Master Write Shareability Domain. Indicates the shareability domain to which the transaction applies. Set by the controller or by the CFG_MSTR_AWDOMAIN_VALUE field of the COHERENCY_CONTROL_REG_3 register. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CC_ACELITE_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid

Port Name	I/O	Description
mstrh_awsnoop[2:0]	O	AXI High Priority Master Write Shareable Transaction Type. You can use this information to distinguish between the different types of shareable write transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awatop[5:0]	O	AXI High Priority Master Write Atomic Operation. Indicates the Atomic Operation encoding. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_ATOMIC_ENABLE) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_awidunq	O	AXI High Priority Master Write Unique ID Indicator. Indicates whether mstrh_awid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_AXI_UIDI_ENABLE) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_awvalid
mstrh_wready	I	AXI High Priority Master Write Ready. Indicates that slave can accept write data. ■ 0: Slave not ready ■ 1: Slave ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_wvalid

Port Name	I/O	Description
mstrh_wready_p	I	AXI High Priority Master Write Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_wvalid
mstrh_wdata[(MSTR_DATA_WD-1):0]	O	AXI High Priority Master Write Data. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_wvalid
mstrh_wuser[(MSTR_W_USER_WIDTH-1):0]	O	AXI High Priority Master Write User. The controller sends all the High Priority Master W channel protection bits through this signal. You can configure CX_AXI_IF_TYPE_OF_PROT parameter to specify the type of protection. In legacy mode, only mstrh_wdata is protected. ■ mstrh_wuser[CC_MSTR_BUS_DATA_PROT_WIDTH-1:0] contains the protection bits of mstrh_wdata ■ mstrh_wuser[CC_MSTR_BUS_DATA_PROT_WIDTH] contains the protection bits of strh_wid ■ mstrh_wuser[CC_MSTR_BUS_DATA_PROT_WIDTH+1] contains the protection bits of mstrh_wstrb ■ mstrh_wuser[CC_MSTR_BUS_DATA_PROT_WIDTH+2] contains the protection bits of mstrh_wlast Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_wvalid

Port Name	I/O	Description
mstrh_wstrb[(MSTR_WSTRB_WD-1):0]	O	AXI High Priority Master Write Strobes. Indicates which byte lanes to update in memory. There is one write strobe for each eight bits of write data bus. mstrh_wstrb[n] is associated with mstrh_wdata[8*n + 7 : 8*n] Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_wvalid
mstrh_wlast	O	AXI High Priority Master Write Last. Indicates last transfer in write burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_wvalid
mstrh_wvalid	O	AXI High Priority Master Write Valid. Indicates valid write data and strobes are available. ■ 0: Write data and strobes not available ■ 1: Write data and strobes available Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mstrh_wvalid_p	O	AXI High Priority Master Write Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_wvalid

Port Name	I/O	Description
mstrh_wid[(MSTR_ID_WD-1):0]	O	AXI High Priority Master Write ID. AXI master write identification ID. This signal is the ID tag of the write data transfer. Memory writes and Messages are always assigned ID 0; Non-Posted writes are assigned unique IDs, from CC_MAX_MSTR_TAGS_AXI to 2 * CC_MAX_MSTR_TAGS_AXI-1. For more information, see 'Inbound Bridge Tag Management'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && !((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_wvalid
mstrh_bready	O	AXI High Priority Master Response Ready. Indicates master can accept response information. ■ 0: High Priority Master not ready ■ 1: High Priority Master ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_bvalid
mstrh_bready_p	O	AXI High Priority Master Response Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_bvalid

Port Name	I/O	Description
mstrh_bid[(MSTR_ID_WD-1):0]	I	AXI High Priority Master Write Response ID. The identification tag of the write response. The BID value must match the AWID value of the write transaction to which the slave is responding. This signal is designed for non-posted writes such as a PCIe configuration write or an I/O write, if they are applicable in a particular application. When I/O or configuration writes are not supported in a particular application, this signal can be tied to '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_bvalid
mstrh_bvalid	I	AXI High Priority Master Write Response Valid. Indicates that valid write response is available. ■ 0: Write response not available ■ 1: Write response available Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mstrh_bvalid_p	I	AXI High Priority Master Write Response Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_bvalid

Port Name	I/O	Description
mstrh_bresp[1:0]	I	AXI High Priority Master Write Response. Indicates status of write transaction. AXI error responses (SLVERR or DECERR) are mapped to PCIe Completion Abort (for non-posted requests). For more information, see the 'Advanced Error Handling For Received TLPs' advanced information chapter. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_bvalid
mstrh_buser[(MSTR_B_USER_WIDTH-1):0]	I	AXI High Priority Master Write Response User. You can send all the High Priority Master B channel protection bits through this signal. In legacy mode, this signal does not exist. ■ mstrh_buser[0] contains the protection bits of mstrh_bid ■ mstrh_buser[1] contains the protection bits of mstrh_bresp ■ mstrh_buser[CC_MSTR_B_USER_WIDTH-1:2] contains the protection bits of all mstrh_bmisc_info signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_bvalid
mstrh_bmisc_info_tlpln	I	AXI High Priority Master Write Response TLP Lightweight Notification (LN) bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_LN_ENABLE) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_bvalid

Port Name	I/O	Description
mstrh_bmisc_info_cid[15:0]	I	AXI High Priority Master Write Response Completer ID. Sets the Completer ID value for AXI master write responses. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CC_AXI_MSTR_MISC_INFO_CID_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: mstrh_bvalid
mstrh_bmisc_info_cpl_stat[2:0]	I	AXI High Priority Master Write Response selection bus. This controls the response to be sent on the wire in the case of successful write requests. The controller always sends a CA (Completer Abort) response when it receives SLVERR/DECERR. For more information, see the "Master AXI-to-PCIe Error Mapping" section in the AXI chapter of the <i>Databook</i>. ■ 000: SC (Successful Completion) ■ 001: UR (Unsupported Request) ■ 010: RRS (Request Retry Status) ■ 100: CA (Completer Abort) ■ All others: Reserved Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_bvalid
mstrh_bidunq	I	AXI High Priority Master Write Response Unique ID Indicator. Indicates whether mstrh_bid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_AXI_UIDI_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_bvalid

Port Name	I/O	Description
mstrh_arid[(MSTR_ID_WD-1):0]	O	AXI High Priority Master Read Address ID. AXI High Priority master read address identification tag for read address signals. This signal provides a unique ID from 0 to CC_MAX_MSTR_TAGS_AXI-1 for each read transfer, unless decomposition has occurred in which case all AXI transfers related to the same PCIe request have the same ID. For more information, see 'Inbound Bridge Tag Management'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_arvalid	O	AXI High Priority Master Read Address Valid. When high, indicates read address and control information is valid and remains stable until arready is high. ■ 0: Address and control information not valid ■ 1: Address and control information valid Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mstrh_arvalid_p	O	AXI High Priority Master Read Address Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_archunken	O	AXI High Priority Master Read Address Chunk Enable. Indicates whether the data for this transaction can be chunked or not. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_AXI_RCHUNK_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_araddr[(MSTR_ADDR_WD-1):0]	O	AXI High Priority Master Read Address. Gives initial address of read burst transaction. Only start address of burst is provided, and control signals issued alongside address show how address is calculated for remaining transfers in burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_aruser[(MSTR_AR_USER_WIDTH-1):0]	O	AXI High Priority Master Read User. The controller receives all the High Priority Master AR channel parity protection bits sent through this signal. In legacy mode, only mstrh_araddr is protected. ■ mstrh_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH-1:0] contains the protection bits of mstrh_araddr ■ mstrh_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH] contains the protection bits of mstrh_arid ■ mstrh_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH+1] contains the protection bits of mstrh_arlen ■ mstrh_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH+2] contains the protection bits of {mstrh_arburst,mstrh_arlock,mstrh_arsize} ■ mstrh_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH+3] contains the protection bits of {mstrh_arcache,mstrh_arprot} ■ mstrh_aruser[CC_MSTR_BUS_ADDR_PROT_WIDTH+4] contains the protection bits of mstrh_arqos ■ mstrh_aruser[CC_MSTR_AR_USER_WIDTH-1:CC_MSTR_BUS_ADDR_PROT_WIDTH+5] contains the protection bits of all mstrh_ar-misc_info signals and acelite signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_arlen[(MSTR_BURST_LEN_PW-1):0]	O	AXI High Priority Master Read Burst Length. Gives exact number of transfers in burst; determines number of data transfers associated with address. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_arsize[2:0]	O	AXI High Priority Master Read Burst Size. Indicates size of each transfer in burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_arburst[1:0]	O	AXI High Priority Master Read Burst. Determines the type of burst used. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_arlock[(MSTR_LOCK_WD-1):0]	O	AXI High Priority Master Read Lock. Encoded value indicates whether the transfer is a normal, exclusive, or locked access. The controller does not use this signal and sets all bits inactive. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_arcache[3:0]	O	AXI High Priority Master Read Cache. Indicates bufferable, cacheable, read-through, and read-back attributes of transaction. Set automatically by the controller when the CFG_MSTR_ARCACHE_MODE field of COHERENCY_CONTROL_REG_3 is '0'. Otherwise, set by the CFG_MSTR_ARCACHE_VALUE field of COHERENCY_CONTROL_REG_3 when CFG_MSTR_ARCACHE_MODE is '1'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_arprot[2:0]	O	AXI High Priority Master Read Protection. Indicates unprivileged, non-secure, or data protection access level. The controller assigns this signal a value of 010. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_rready	O	AXI High Priority Master Read Ready. Indicates master can accept read data and response information. ■ 0: Master not ready ■ 1: Master ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid

Port Name	I/O	Description
mstrh_rready_p	O	AXI High Priority Master Read Response Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid
mstrh_armisc_info[(MSTR_MISC_INFO_PW-1):0]	O	AXI High Priority Master Read Transaction Associated Misc Information (from the TLP received by the native PCIe controller). This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. ■ [5:0]: Reserved ■ [8:6]: BAR number of TLP. Bits [7:0] also indicate the message code when the type is message. ■ [9]: TLP is an I/O ■ [10]: TLP is in ROM range ■ [13:11]: TLP's Function number. Function numbering starts at '0'. This field is not used when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. For more information, see mstrh_armisc_info_func_num. ■ [14]: TLP's NS bit ■ [15]: TLP's RO bit ■ [18:16]: TLP's TC bits ■ [23:19]: TLP's TYPE. For more information, see 'I/O and CFG Transaction Handling' ■ [39:24]: TLP's requester ID ■ [CX_TAG_SIZE-1:40]: TLP's TAG Bus number, device number, function number, and register numbers are overlaid as follows to the address bus for CFG TLPs. ■ bus_number =addr[31:24] ■ dev_number =addr[23:19] ■ func_number =addr[18:16] ■ ext_reg_number =addr[11:8] ■ reg_number =addr[7:2] Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_armisc_info_last_dcmp_tlp	O	AXI High Priority Master Read Request last TLP bit. The signal indicates that the TLP in transaction is the last TLP of the read request. The master interface asserts this output on the last beat of a decomposed request. It is continuously asserted for a TLP that is not decomposed. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_armisc_info_tlpprfx[(PRFX_WIDTH-1):0]	O	AXI High Priority Master Read Request TLP Prefixes. The field [31:0] represents the first prefix received. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that mstrh_armisc_info_tlpprfx[31:0] has prefix byte #0 in the lower byte position of the dword mstrh_armisc_info_tlpprfx[31:0]. That is, FMT=bits 7:5, and Type =bits 4:0. If no prefix is present for a given TLP, then that DW, including the FMT field is all zeros. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_TLP_PREFIX_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_armisc_info_tlpLn	O	AXI High Priority Master Read Request TLP Lightweight Notification (LN) bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_LN_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_armisc_info_tph[10:0]	O	AXI High Priority Master Read Request TLP Processing Hints. The bits are mapped as follows: ■ [0]: TH (TLP Processing Hint present) ■ [2:1] PH (TLP Processing Hint) ■ [10:3] ST (Steering Tag) Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_TPH_ENABLE) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_armisc_info_dma[((CC_NUM_DMA_MAX_CHAN_WIDTH+DC_DMA_TYPE_WIDTH)-1):0]	O	DMA bits of the AXI Read High Priority Master Transaction. The signal can be used for debugging. For more information, see 'Programming Examples'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Bits [5:3] indicate DMA channel number and bits [2:0] indicate the DMA request type as follows: ■ 000: Non-DMA request ■ 001: DMA Write channel MRd request ■ 010: Non-DMA completion ■ 011: DMA Read channel MWr request ■ 100: Reserved ■ 101: DMA Write channel LL Element MRd request ■ 110: Reserved ■ 111: DMA Read channel LL Element MRd request Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CC_DMA_ENABLE) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_armisc_info_nw	O	ATS No Write (NW) Bit of the AXI High Priority Master Read Transaction. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (ATS_RX_ENABLE) && (CX_ATS_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_armisc_info_ats[1:0]	O	ATS Bits of the AXI High Priority Master Read Transaction. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (ATS_RX_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_armisc_info_func_num[(PF_WD-1):0]	O	Function Number of the AXI High Priority Master Read Transaction. It is used instead of bits [13:11] of mstrh_armisc_info when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Function numbering starts at '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && ((AMBA_INTERFACE != 0 && (CX_SRIOV_ENABLE CX_ARI_ENABLE))) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_armisc_info_vfunc_num[(VF_WD-2):0]	O	Virtual Function Number of the AXI High Priority Master Read Transaction. Indicates which VF has been targeted by the received request. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, mstrh_armisc_info_vfunc_num = 0 refers to the first VF of the PF that is identified by mstrh_armisc_info_func_num. For more information of VF numbering, see 'Programming View'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_SRIOV_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_armisc_info_vfunc_active	O	VF Active Indication of the AXI High Priority Master Read Transaction. Indicates that the received request matched a VF BAR. 0: No VF is active and mstrh_armisc_info_vfunc_num is invalid. A PF is valid and identified by mstrh_armisc_info_func_num. 1: A VF is active and is identified by mstrh_armisc_info_vfunc_num. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_SRIOV_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_armisc_info_ido	O	IDO Bit of the AXI High Priority Master Read Transaction. Enables ID-base ordering on inbound requests. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_IDO_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_arready	I	AXI High Priority Master Read Address Ready. Indicates that slave is ready to accept address and associated control signals. 0: Slave not ready 1: Slave ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_arready_p	I	AXI High Priority Master Read Address Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && ((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstrh_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_arqos[3:0]	O	AXI High Priority Master Read QoS. The controller always drives this output to '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_arbar[1:0]	O	AXI High Priority Master Read Barrier Type. You can use this information to distinguish between normal and transaction barriers, and between different barrier types. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid

Port Name	I/O	Description
mstrh_ardomain[1:0]	O	AXI High Priority Master Read Shareability Domain. Indicates the shareability domain to which the transaction applies. Set by the controller or by the CFG_MSTR_ARDOMAIN_VALUE field of the COHERENCY_CONTROL_REG_3 register. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CC_ACELITE_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_arsnoop[3:0]	O	AXI High Priority Master Read Shareable Transaction Type. You can use this information to distinguish between the different types of shareable read transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_aridunq	O	AXI High Priority Master Read Unique ID Indicator. Indicates whether mstrh_arid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_AXI_UIDI_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_arvalid
mstrh_rid[(MSTR_ID_WD-1):0]	I	AXI High Priority Master Read ID. This signal is required by the AXI bridge to identify a response for a particular master request. For more information, see 'Inbound Bridge Tag Management'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid

Port Name	I/O	Description
mstrh_rvalid	I	AXI High Priority Master Read Valid. Indicates that required read data is available and read transfer can complete. 0: Read data not available 1: Read data available Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mstrh_rvalid_p	I	AXI High Priority Master Read Response Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid
mstrh_rlast	I	AXI High Priority Master Read Last. Indicates last transfer in read burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid
mstrh_rdata[(MSTR_DATA_WD-1):0]	I	AXI High Priority Master Read Data. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid

Port Name	I/O	Description
mstrh_ruser[(MSTR_R_USER_WIDTH-1):0]	I	AXI High Priority Master Read Response User. You can send all the High Priority Master R channel protection bits through this signal. You can configure the CX_AXI_IF_TYPE_OF_PROT parameter to specify the type of protection. In legacy mode, only mstrh_rdata is protected. ■ mstrh_ruser[CC_MSTR_BUS_DATA_PROT_WIDTH-1:0] contains the protection bits of mstrh_rdata ■ mstrh_ruser[CC_MSTR_BUS_DATA_PROT_WIDTH] contains the protection bits of mstrh_rid ■ mstrh_ruser[CC_MSTR_BUS_DATA_PROT_WIDTH+1] contains the protection bits of {mstrh_rresp,mstrh_rlast} ■ mstrh_ruser[CC_MSTR_R_USER_WIDTH-1:CC_MSTR_BUS_DATA_PROT_WIDTH+2] contains the protection bits of all mstrh_awmisc_info signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: mstr_ahclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid
mstrh_rresp[1:0]	I	AXI High Priority Master Read Response. Indicates status of read transaction. AXI error responses (SLVERR or DECERR) are mapped to PCIe Completion Abort (for non-posted requests). For more information, see the 'Advanced Error Handling For Received TLPs' advanced information chapter. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_ahclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid

Port Name	I/O	Description
mstrh_misc_info[(MSTR_RESP_MISC_INFO_PW-1):0]	I	AXI High Priority Master Read Response Transaction Associated Misc Information. This is an additional signal. It is not part of standard AXI interface. For more information, see 'AXI Master Interface Sideband Signals'. All reserved bits must be connected to logic '0'. ■ [1:0]: Reserved ■ [2]: Reserved. ■ [6:3]: Reserved ■ [7]: EP bit of the TLP ■ [12:8]: Reserved. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid
mstrh_misc_info_tlpn	I	AXI High Priority Master Read Response TLP Lightweight Notification (LN) bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_LN_ENABLE) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid

Port Name	I/O	Description
mstrh_rmisc_info_cpl_stat[2:0]	I	AXI High Priority Master Read Response selection bus. This bus controls the response sent on the PCIe wire in the case of successful read requests. The controller always sends a CA (Completer Abort) response when it receives SLVERR/DECERR. For more information, see the "Master AXI-to-PCIe Error Mapping" section in the AXI chapter of the <i>Databook</i>. ■ 000: SC (Successful Completion) ■ 001: UR (Unsupported Request) ■ 010: RRS (Request Retry Status) ■ 100: CA (Completer Abort) ■ All others: Reserved Your application must drive the same value on mstrh_rmisc_info_cpl_stat throughout the complete response. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid
mstrh_ridunq	I	AXI High Priority Master Read Response Unique ID Indicator. Indicates whether mstrh_rid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) && (CX_AXI_UIDI_ENABLE) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_rvalid
mstrh_csysack	O	AXI High Priority Master Low-Power Request Acknowledgment. Indicates acknowledgment from AXI bridge master to a system low-power request. The controller does not support AXI power modes and always denies any request the mstrh_csysreq input. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: mstrh_csysreq

Port Name	I/O	Description
mstrh_cactive	O	AXI High Priority Master Clock Active. Indicates that AXI bridge master requires a clock signal. The controller does not support AXI power modes and always drives this output high. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mstrh_csysreq
mstrh_csysreq	I	AXI High Priority Master System Low-Power Request. System low-power request from system clock controller for AXI bridge master to enter low-power state. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (CC_HP_MASTER) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

2.4 AXI Optional DBI Slave Interface Signals

dbi_awid	-	dbi_awready
dbi_awidunq	-	dbi_awready_p
dbi_awaddr	-	dbi_wready
dbi_awuser	-	dbi_wready_p
dbi_awlen	-	dbi_bid
dbi_awsiz	-	dbi_bidunq
dbi_awburst	-	dbi_bresp
dbi_awlock	-	dbi_bvalid
dbi_awcache	-	dbi_bvalid_p
dbi_awprot	-	dbi_rid
dbi_awvalid	-	dbi_ridunq
dbi_awvalid_p	-	dbi_rdata
dbi_awqos	-	dbi_ruser
dbi_awbar	-	dbi_rresp
dbi_awdomain	-	dbi_rlast
dbi_awsnoop	-	dbi_rvalid
dbi_wid	-	dbi_rvalid_p
dbi_wdata	-	dbi_arready
dbi_wuser	-	dbi_arready_p
dbi_wstrb	-	dbi_csysack
dbi_wlast	-	dbi_cactive
dbi_wvalid	-	
dbi_wvalid_p	-	
dbi_bready	-	
dbi_bready_p	-	
dbi_arid	-	
dbi_aridunq	-	
dbi_araddr	-	
dbi_aruser	-	
dbi_arlen	-	
dbi_arsiz	-	
dbi_arburst	-	
dbi_arlock	-	
dbi_arcache	-	
dbi_arprot	-	
dbi_arvalid	-	
dbi_arvalid_p	-	
dbi_rready	-	
dbi_rready_p	-	
dbi_arqos	-	
dbi_arbar	-	
dbi_ardomain	-	
dbi_arsnoop	-	
dbi_csysreq	-	

Table 2-4 AXI Optional DBI Slave Interface Signals

Port Name	I/O	Description
dbi_awid[(DBI_SLV_ID_WD-1):0]	I	AXI DBI Slave Write ID. Indicates ID of the burst. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awidunq	I	AXI DBI Slave Write Address Unique ID Indicator. Indicates whether dbi_awid is unique in in-flight transactions. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CX_AXI_UIDI_ENABLE) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awaddr[(DBI_SLV_ADDR_WD-1):0]	I	AXI DBI Slave Write Address. Provides the initial address of a write burst transaction. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid

Port Name	I/O	Description
dbi_awuser[(DBI_SLV_AW_USER_WIDTH-1):0]	I	AXI DBI Slave Write User. You can send all the DBI Slave AW channel parity protection bits through this signal. In legacy mode this signal does not exist. - dbi_awuser[CC_DBI_BUS_ADDR_PROT_WIDTH-1:0] contains the protection bits of slv_awaddr - dbi_awuser[CC_DBI_BUS_ADDR_PROT_WIDTH] contains the protection bits of dbi_awid - dbi_awuser[CC_DBI_BUS_ADDR_PROT_WIDTH+1] contains the protection bits of dbi_awlen - dbi_awuser[CC_DBI_BUS_ADDR_PROT_WIDTH+2] contains the protection bits of {dbi_awburst,dbi_awlock,dbi_awszsize} - dbi_awuser[CC_DBI_BUS_ADDR_PROT_WIDTH+3] contains the protection bits of {dbi_awcache,dbi_awprot} - dbi_awuser[CC_DBI_BUS_ADDR_PROT_WIDTH+4] contains the protection bits of dbi_awqos - dbi_awuser[CC_DBI_SLV_AW_USER_WIDTH-1:CC_DBI_BUS_ADDR_PROT_WIDTH+5] contains the protection bits of all acelite signals. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awlen[(DBISLV_BURST_LEN_PW-1):0]	I	AXI DBI Slave Write Burst Length. Specifies the exact number of transfers in a burst; determines the number of data transfers associated with the address. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awszsize[2:0]	I	AXI DBI Slave Write Burst Size. Indicates size of each transfer in burst. Byte lane strobes indicate exactly which byte lanes to update. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid

Port Name	I/O	Description
dbi_awburst[1:0]	I	AXI DBI Slave Write Burst. Determines the type of burst used. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awlock[(DBI_SLV_LOCK_WD-1):0]	I	AXI DBI Slave Write Lock. Provides additional information about characteristics of transfer. This signal is not used in the controller. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awcache[3:0]	I	AXI DBI Slave Write Cache. Indicates bufferable, cacheable, write-through, and write-back attributes of transaction. This signal is not used in the controller. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awprot[2:0]	I	AXI DBI Slave Write Protection. Indicates normal, privileged, or secure protection level of transaction and whether transaction is data access or instruction access. This signal is not used in the controller. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid

Port Name	I/O	Description
dbi_awvalid	I	AXI DBI Slave Write Address Valid. Indicates that valid write address and control information are available. Address and control information remain stable until awready signal is high. 0: Address and control information not available 1: Address and control information available This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: aux_clk,dbi_aclk_ug,{dbi_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
dbi_awvalid_p	I	AXI DBI Slave Write Address Valid Parity bit. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awready	O	AXI DBI Slave Write Address Ready. Indicates that the slave is ready to accept address and associated control signals. 0: DBI Slave not ready 1: DBI Slave ready Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk_ug,{dbi_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awready_p	O	AXI DBI Slave Write Address Ready Parity bit. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid

Port Name	I/O	Description
dbi_awqos[3:0]	I	AXI DBI Slave Write QoS. Indicates relative priority of the request. This signal is not used internally. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awbar[1:0]	I	AXI DBI Slave Write Barrier Type. You can use this signal to distinguish between normal and transaction barriers, and between different barrier types. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awdomain[1:0]	I	AXI DBI Slave Write Shareability Domain. Indicates the shareability domain to which the transaction applies. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid
dbi_awsnoop[2:0]	I	AXI DBI Slave Write Shareable Transaction Type. You can use this signal to distinguish between the different types of shareable write transactions. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_awvalid

Port Name	I/O	Description
dbi_wready	O	AXI DBI Slave Write Ready. Indicates that slave can accept write data. 0: Slave not ready 1: Slave ready Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Ug,{dbi_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_wvalid
dbi_wready_p	O	AXI DBI Slave Write Ready Parity bit. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_wvalid
dbi_wid[(DBI_SLV_ID_WD-1):0]	I	AXI DBI Slave Write ID. AXI slave write identification tag of write data transfer. The controller internally always sets this value to '0' all bits. This signal is not used in the controller. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (!((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_wvalid
dbi_wdata[(DBI_SLV_DATA_WD-1):0]	I	AXI DBI Slave Write Data. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_wvalid

Port Name	I/O	Description
dbi_wuser[(DBI_SLV_W_USER_WIDTH-1):0]	I	AXI DBI Slave Write User. You can send all the DBI Slave W channel protection bits through this signal. You can configure the CX_AXI_IF_TYPE_OF_PROT parameter to specify the type of protection. In legacy mode this signal does not exist. - dbi_wuser[CC_DBI_BUS_DATA_PROT_WIDTH-1:0] contains the protection bits of dbi_wdata - dbi_wuser[CC_DBI_BUS_DATA_PROT_WIDTH] contains the protection bits of dbi_wid - dbi_wuser[CC_DBI_BUS_DATA_PROT_WIDTH+1] contains the protection bits of dbi_wstrb - dbi_wuser[CC_DBI_BUS_DATA_PROT_WIDTH+2] contains the protection bits of dbi_wlast Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_wvalid
dbi_wstrb[((DBI_SLV_DATA_WD/8)-1):0]	I	AXI DBI Slave Write Strobes. Indicates which byte lanes to update in memory. There is one write strobe for each eight bits of write data bus. dbi_wstrb[n] is associated with the following byte of dbi_wdata[8*n + 7 : 8*n] Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_wvalid
dbi_wlast	I	AXI DBI Slave Write Last. Indicates last transfer in write burst. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_ack Registered: N/A Power Domain: PD_VAUX Active State: High

Port Name	I/O	Description
dbi_wvalid	I	AXI DBI Slave Write Valid. Indicates valid write data and strobes are available. 0: Write data and strobes not available 1: Write data and strobes available This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: aux_clk,dbi_aclk_ug,{dbi_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
dbi_wvalid_p	I	AXI DBI Slave Write Valid Parity bit. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_wvalid
dbi_bid[(DBI_SLV_ID_WD-1):0]	O	AXI DBI Slave Write Response ID. This signal is not used by the controller. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_bvalid
dbi_bidunq	O	AXI DBI Slave Write Response Unique ID Indicator. Indicates whether dbi_bid is unique in in-flight transactions. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CX_AXI_UIDI_ENABLE) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_bvalid

Port Name	I/O	Description
dbi_bresp[1:0]	O	AXI DBI Slave Write Response. Indicates status of write transaction. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_bvalid
dbi_bvalid	O	AXI DBI Slave Write Response Valid. Indicates that valid write response is available. 0: write response not available 1: write response available Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
dbi_bvalid_p	O	AXI DBI Slave Write Response Valid Parity bit. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_bvalid
dbi_bready	I	AXI DBI Slave Response Ready. Indicates SLAVE can accept response information. 0: SLAVE not ready 1: SLAVE ready Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_bvalid

Port Name	I/O	Description
dbi_bready_p	I	AXI DBI Slave Write Response Ready Parity bit. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_bvalid
dbi_arid[(DBI_SLV_ID_WD-1):0]	I	AXI DBI Slave Read Address. AXI slave read address identification tag for read address signals. The controller internally always sets this value to '0' all bits. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_aridunq	I	AXI DBI Slave Read Address Unique ID Indicator. Indicates whether dbi_arid is unique in in-flight transactions. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CX_AXI_UIDI_ENABLE) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_araddr[(DBI_SLV_ADDR_WD-1):0]	I	AXI DBI Slave Read Address. Provides the initial address of a read burst transaction. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid

Port Name	I/O	Description
dbi_aruser[(DBI_SLV_AR_USER_WIDTH-1):0]	I	AXI DBI Slave Read User. You can send all the DBI Slave AR channel parity protection bits through this signal. In legacy mode, this signal does not exist. - dbi_aruser[CC_DBI_BUS_ADDR_PROT_WIDTH-1:0] contains the protection bits of dbi_araddr - dbi_aruser[CC_DBI_BUS_ADDR_PROT_WIDTH] contains the protection bits of dbi_arid - dbi_aruser[CC_DBI_BUS_ADDR_PROT_WIDTH+1] contains the protection bits of dbi_arlen - dbi_aruser[CC_DBI_BUS_ADDR_PROT_WIDTH+2] contains the protection bits of {dbi_arburst,dbi_arlock,dbi_arsize} - dbi_aruser[CC_DBI_BUS_ADDR_PROT_WIDTH+3] contains the protection bits of {dbi_arcache,dbi_arprot} - dbi_aruser[CC_DBI_BUS_ADDR_PROT_WIDTH+4] contains the protection bits of dbi_arqos - dbi_aruser[CC_DBI_SLV_AR_USER_WIDTH-1:CC_DBI_BUS_ADDR_PROT_WIDTH+5] contains the protection bits of all acelite signals Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_arlen[(DBISLV_BURST_LEN_PW-1):0]	I	AXI DBI Slave Read Burst Length. Gives the exact number of transfers in a burst; determines the number of data transfers associated with the address. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_arsize[2:0]	I	AXI DBI Slave Read Burst Size. Indicates size of each transfer in burst. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid

Port Name	I/O	Description
dbi_arburst[1:0]	I	AXI DBI Slave Read Burst. Determines the address for each transfer within a burst. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_arlock[(DBI_SLV_LOCK_WD-1):0]	I	AXI DBI Slave Read Lock. Encoded value indicates whether the transfer is a normal, exclusive, or locked access. This signal is not used in the controller. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_arcache[3:0]	I	AXI DBI Slave Read Cache. Indicates bufferable, cacheable, read-through, and read-back attributes of transaction. This signal is not used in the controller. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_arprot[2:0]	I	AXI DBI Slave Read Protection. Indicates normal, privileged, or secure protection level of transaction and whether transaction is data access or instruction access. This signal is not used in the controller. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid

Port Name	I/O	Description
dbi_arvalid	I	AXI DBI Slave Read Address Valid. When high, indicates read address and control information is valid and remains stable until arready is high. 0: address and control information not valid 1: address and control information valid This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: aux_clk,dbi_aclk_ug,{dbi_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
dbi_arvalid_p	I	AXI DBI Slave Read Address Valid Parity bit. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_rready	I	AXI DBI Slave Read Ready. Indicates slave can accept read data and response information. 0: SLAVE not ready 1: SLAVE ready Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_rvalid
dbi_rready_p	I	AXI DBI Slave Read Ready Parity bit. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_rvalid

Port Name	I/O	Description
dbi_rid[(DBI_SLV_ID_WD-1):0]	O	AXI DBI Slave Read ID. This signal is not actually required to be connected; it is not used and therefore is not connected internal to the component. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_rvalid
dbi_ridunq	O	AXI DBI Read Data Unique ID Indicator. Indicates whether dbi_rid is unique in in-flight transactions. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CX_AXI_UIDI_ENABLE) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_rvalid
dbi_rdata[(DBI_SLV_DATA_WD-1):0]	O	AXI DBI Slave Read Data. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_rvalid

Port Name	I/O	Description
dbi_ruser[(DBI_SLV_R_USER_WIDTH-1):0]	O	AXI DBI Slave Read User. The controller sends all the DBI Slave R channel protection bits through this signal. You can configure the CX_AXI_IF_TYPE_OF_PROT parameter to specify the type of protection. In legacy mode, this signal does not exist. - dbi_ruser[CC_DBI_BUS_DATA_PROT_WIDTH-1:0] contains the protection bits of dbi_rdata - dbi_ruser[CC_DBI_BUS_DATA_PROT_WIDTH] contains the protection bits of dbi_rid - dbi_ruser[CC_DBI_BUS_DATA_PROT_WIDTH+1] contains the protection bits of {dbi_rresp,dbi_rlast} Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_rvalid
dbi_rresp[1:0]	O	AXI DBI Slave Read Response. Indicates status of read transaction. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_rvalid
dbi_rlast	O	AXI DBI Slave Read Last. Indicates last transfer in read burst. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_rvalid

Port Name	I/O	Description
dbi_rvalid	O	AXI DBI Slave Read Valid. Indicates that required read data is available and read transfer can complete. 0: read data not available 1: read data available Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
dbi_rvalid_p	O	AXI DBI Slave Read Valid Parity bit. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_rvalid
dbi_arready	O	AXI DBI Slave Read Address Ready. Indicates that the slave is ready to accept the address and associated control signals. 0: Slave not ready 1: Slave ready Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: dbi_aclk_ug,{dbi_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_arready_p	O	AXI DBI Slave Read Address Ready Parity bit. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid

Port Name	I/O	Description
dbi_arqos[3:0]	I	AXI DBI Slave Read QoS. Indicates relative priority of the request. This signal is not used internally. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_arbar[1:0]	I	AXI DBI Slave Read Barrier Type. You can use this signal to distinguish between normal and transaction barriers, and between different barrier types. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_ardomain[1:0]	I	AXI DBI Slave Read Shareability Domain. Indicates the shareability domain to which the transaction applies. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid
dbi_arsnoop[3:0]	I	AXI DBI Slave Read Shareable Transaction Type. You can use this signal to distinguish between the different types of shareable read transactions. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_ACELITE_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_arvalid

Port Name	I/O	Description
dbi_csysreq	I	AXI DBI Slave System Low-Power Request. System low-power request from system clock controller for AXI bridge DBI slave to enter low-power state. AXI power modes are not currently supported. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
dbi_csysack	O	AXI DBI Slave Low-Power Request Acknowledgment. Indicates acknowledgment from AXI bridge DBI slave to a system low-power request. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: dbi_csysreq
dbi_cactive	O	AXI DBI Slave Clock Active. Indicates that AXI bridge DBI slave requires a clock signal: Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_csysreq

2.5 AXI Slave Interface Signals

slv_awid	-	slv_awready
slv_awaddr	-	slv_awready_p
slv_awuser	-	slv_wready
slv_awlen	-	slv_wready_p
slv_awsiz	-	slv_bid
slv_awburst	-	slv_bresp
slv_awlock	-	slv_buser
slv_awcache	-	slv_bidunq
slv_awprot	-	slv_bvalid
slv_awvalid	-	slv_bvalid_p
slv_awvalid_p	-	slv_bmisc_info
slv_awqos	-	slv_bmisc_info_tlpprfx
slv_awbar	-	slv_bmisc_info_tlpIn
slv_awdomain	-	slv_rid
slv_awsnoop	-	slv_rdata
slv_awatop	-	slv_ruser
slv_awidunq	-	slv_rresp
slv_awmisc_info	-	slv_rlast
slv_awmisc_info_hdr_34dw	-	slv_rvalid
slv_awmisc_info_tlpprfx	-	slv_rvalid_p
slv_awmisc_info_tlpIn	-	slv_rmisc_info
slv_awmisc_info_rid	-	slv_rmisc_info_tlpprfx
slv_awmisc_info_p_tag	-	slv_rmisc_info_tlpIn
slv_awmisc_info_tph	-	slv_rmisc_info_id0
slv_awmisc_info_atu_bypass	-	slv_ridunq
slv_awmisc_info_nw	-	slv_arready
slv_awmisc_info_ats	-	slv_arready_p
slv_awmisc_info_func_num	-	slv_csysack
slv_awmisc_info_vfunc_active	-	slv_cactive
slv_awmisc_info_vfunc_num	-	dbi_buser
slv_awmisc_info_id0	-	
slv_wid	-	
slv_wdata	-	
slv_wuser	-	
slv_wstrb	-	
slv_wlast	-	
slv_wvalid	-	
slv_wvalid_p	-	
slv_wmisc_info_ep	-	
slv_wmisc_info_silentDrop	-	
slv_bready	-	
slv_bready_p	-	
slv_arid	-	
slv_araddr	-	
slv_aruser	-	
slv_arlen	-	
slv_arsize	-	
slv_arburst	-	
slv_arlock	-	
slv_arcache	-	
slv_arprot	-	
slv_arvalid	-	
slv_arvalid_p	-	
slv_armisc_info	-	
slv_armisc_info_tlpprfx	-	
slv_armisc_info_tlpIn	-	
slv_armisc_info_rid	-	
slv_armisc_info_tph	-	
slv_armisc_info_atu_bypass	-	

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slv_armisc_info_nw -
slv_armisc_info_ats -
slv_armisc_info_func_num -
slv_armisc_info_vfunc_active -
slv_armisc_info_vfunc_num -
slv_armisc_info_id0 -
slv_aridung -
slv_rready -
slv_rready_p -
slv_arqos -
slv_arbar -
slv_ardomain -
slv_arsnoop -
slv_csysreq -

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Table 2-5 AXI Slave Interface Signals

Port Name	I/O	Description
slv_awid[(SLV_ID_WD-1):0]	I	AXI Slave Write ID. Indicates ID of the write transfer. This signal is an input to the AXI bridge from an AXI master. This signal can contain unique or identical IDs for each AXI write transfer. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awaddr[(SLV_ADDR_WD-1):0]	I	AXI Slave Write Address. Specifies the address of the first transfer in a write burst transaction. Associated control signals are used to determine the addresses of remaining transfers in the burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid

Port Name	I/O	Description
slv_awuser[(SLV_AW_USER_WIDTH-1):0]	I	AXI Slave Write User. You can send all the Slave AW channel parity protection bits through this signal. In legacy mode, only slv_awaddr is protected. ■ slv_awuser[CC_SLV_BUS_ADDR_PROT_WIDTH-1:0] contains the protection bits of slv_awaddr ■ slv_awuser[CC_SLV_BUS_ADDR_PROT_WIDTH] contains the protection bits of slv_awid ■ slv_awuser[CC_SLV_BUS_ADDR_PROT_WIDTH+1] contains the protection bits of slv_awlen ■ slv_awuser[CC_SLV_BUS_ADDR_PROT_WIDTH+2] contains the protection bits of {slv_awburst,slv_awlock,slv_awsz} ■ slv_awuser[CC_SLV_BUS_ADDR_PROT_WIDTH+3] contains the protection bits of {slv_awcache,slv_awprot} ■ slv_awuser[CC_SLV_BUS_ADDR_PROT_WIDTH+4] contains the protection bits of slv_awqos ■ slv_awuser[CC_SLV_AW_USER_WIDTH-1:CC_SLV_BUS_ADDR_PROT_WIDTH+5] contains the protection bits of all slv_awatop, slv_awmisc_info signals, and accel signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0)) ? 0: 1) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awlen[(SLV_BURST_LEN_PW-1):0]	I	AXI Slave Write Burst Length. Specifies the exact number of transfers in a burst; determines number of data transfers associated with the address. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awsz[2:0]	I	AXI Slave Write Burst Size. Indicates size of each transfer in burst. Byte lane strobes indicate exactly which byte lanes to update. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid

Port Name	I/O	Description
slv_awburst[1:0]	I	AXI Slave Write Burst. Determines the type of burst used. Only INCR-BURST is supported when CC_SLV_WRAP_ENABLE =0. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awlock[(SLV_LOCK_WD-1):0]	I	AXI Slave Write Lock. Provides additional information about characteristics of transfer. The controller slave does not support Locked/Exclusive transfers Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: {None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awcache[3:0]	I	AXI Slave Write Cache. Indicates bufferable, cacheable, write-through, and write-back attributes of transaction. This signal is not used in the controller. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: {None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awprot[2:0]	I	AXI Slave Write Protection. Indicates normal, privileged, or secure protection level of transaction and whether transaction is data access or instruction access. This signal is not used in the controller. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: {None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid

Port Name	I/O	Description
slv_awvalid	I	AXI Slave Write Address Valid. Indicates valid write address and control information are available. Address and control information remain stable until awready signal is high. 0: Address and control information not available 1: Address and control information available This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: aux_clk,slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
slv_awvalid_p	I	AXI Slave Write Address Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid_p
slv_awready	O	AXI Slave Write Address Ready. Indicates that the slave is ready to accept the address and associated control signals. 0: Slave not ready 1: Slave ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid

Port Name	I/O	Description
slv_awready_p	O	AXI Slave Write Address Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awqos[3:0]	I	AXI Slave Write QoS. This signal is not used internally. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: {None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awbar[1:0]	I	AXI Slave Write Barrier Type. You can use this signal to distinguish between normal and transaction barriers, and between different barrier types. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: slv_aclk_ug,{None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awdomain[1:0]	I	AXI Slave Write Shareability Domain. Indicates the shareability domain to which the transaction applies. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: slv_aclk_ug,{None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid

Port Name	I/O	Description
slv_awsnoop[2:0]	I	AXI Slave Write Shareable Transaction Type. You can use this signal to distinguish between the different types of shareable write transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: slv_aclk_ug,{None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awatop[5:0]	I	AXI Slave Write Atomic Operation. Indicates the Atomic Operation encoding. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_ATOMIC_ENABLE) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awidunq	I	AXI Slave Write Unique ID Indicator. Indicates whether slv_awid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_AXI_UIDI_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid

Port Name	I/O	Description
slv_awmisc_info[(SLV_MISC_INFO_PW-1):0]	I	AXI Slave Write Transaction Associated Misc Information. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. You must set all reserved bits to '0'. ■ [4:0]: TYPE bits of the TLP. For more information, see 'I/O and CFG Transaction Handling' ■ [5]: Serialize NP Requests ■ [6]: EP bit of the TLP ■ [7]: Reserved ■ [8]: NS bit of the TLP ■ [9]: RO bit of the TLP ■ [12:10]: TC bits of the TLP ■ [20:13]: MSG code of the TLP ■ [21]: AXI transaction is a DBI access. This is for SHARED DBI mode only. ■ [24:22]: Function number of the TLP. Function numbering starts at '0'. Not used when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. For more information, see slv_awmisc_info_func_num. This field is only present when multifunction support (CX_NFUNC > 1) is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: None,slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awmisc_info_hdr_34dw[63:0]	I	AXI Slave 3rd and 4th header DWs. The application drives this bus with the 3rd and 4th Header DWs it intends to send on a PCIe Msg/MsgD. Note: The data is in big endian format, that is, slv_awmisc_info_hdr_34dw[7:0] contains byte 15 of header DW. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: None,slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid

Port Name	I/O	Description
slv_awmisc_info_tlpprfx[(PRFX_WIDTH-1):0]	I	AXI Slave Write Request TLP Prefixes. The field [31:0] represents the first prefix to be transmitted. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that slv_awmisc_info_tlpprfx[31:0] has prefix byte #0 in the lower byte position of the dword slv_awmisc_info_tlpprfx[31:0]. That is, FMT =bits 7:5, and Type =bits 4:0. Your application is responsible for passing Local and End-End prefixes in the correct order for transmission. Local prefixes must precede End-End prefixes. Any parity bits that your application appends (when RAS data protection is enabled) to the most significant bits are routed but not checked by the controller. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_TLP_PREFIX_ENABLE) Synchronous To: slv_aclk Ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awmisc_info_tlpIn	I	AXI Slave Write Request TLP Lightweight Notification (LN) bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_LN_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awmisc_info_rid[15:0]	I	AXI Slave Write Request Requester ID. Sets the Requester ID value for AXI slave write requests Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CC_AXI_SLV_MISC_INFO_RID_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: NA Validated by: slv_awvalid

Port Name	I/O	Description
slv_awmisc_info_p_tag[(TAG_SIZE-1):0]	I	AXI Slave Write Request Tag. Sets the TAG number for output posted requests. It is expected that your application normally sets this to '0' except when generating ATS invalidate requests. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awmisc_info_tph[10:0]	I	AXI Slave Write Request TLP Processing Hints. The bits are mapped as follows: ■ [0]: TH (TLP Processing Hint present) ■ [2:1] PH (TLP Processing Hint) ■ [10:3] ST (Steering Tag) Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_TPH_ENABLE) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awmisc_info_atu_bypass	I	AXI Slave Write Request Internal ATU Bypass. When set it indicates that this request should not be processed by the internal address translation unit Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_INTERNAL_ATU_ENABLE) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid

Port Name	I/O	Description
slv_awmisc_info_nw	I	ATS No Write (NW) Bit of the AXI Slave Write Transaction Request. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (ATS_TX_ENABLE) Synchronous To: slv_aclk Ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awmisc_info_ats[1:0]	I	AT Bits of the AXI Slave Write Transaction Request. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (ATS_TX_ENABLE) Synchronous To: slv_aclk Ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awmisc_info_func_num[(PF_WD-1):0]	I	Function Number of the AXI Slave Write Transaction. It is used instead of bits [24:22] of slv_awmisc_info when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Function numbering starts at '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && ((AMBA_INTERFACE != 0 && (CX_SRIOV_ENABLE CX_ARI_ENABLE))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid

Port Name	I/O	Description
slv_awmisc_info_vfunc_active	I	VF Active Indication of the AXI Slave Write Transaction. Indicates the VF that is sending the request. 0: No VF is active and slv_awmisc_info_vfunc_num is invalid. A PF is valid and identified by slv_awmisc_info_func_num. 1: A VF is active and is identified by slv_awmisc_info_vfunc_num. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_SRIOV_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awmisc_info_vfunc_num[(VF_WD-2):0]	I	Virtual Function Number of the AXI Slave Write Transaction. Identifies the VF that is sending the request. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, slv_awmisc_info_vfunc_num = 0 refers to the first VF of the PF that is identified by slv_awmisc_info_func_num. For more information of VF numbering, see 'Programming View'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_SRIOV_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid
slv_awmisc_info_ido	I	IDO bit of the AXI Slave Write Transaction. Enables ID-base ordering on outbound requests. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_IDO_ENABLE) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_awvalid

Port Name	I/O	Description
slv_wready	O	AXI Slave Write Ready. Indicates that slave can accept write data. 0: Slave not ready 1: Slave ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_wvalid
slv_wready_p	O	AXI Slave Write Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_wvalid
slv_wid[(SLV_ID_WD-1):0]	I	AXI Slave Write ID. AXI slave write identification tag of a write data transfer. It should be the same as slv_awid for each write transfer. The bridge slave only supports sequential access where it matches each address channel transfer to write data channel transfer in order. This signal is not used internally. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (!((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_wvalid
slv_wdata[(SLV_DATA_WD-1):0]	I	AXI Slave Write Data. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_wvalid

Port Name	I/O	Description
slv_wuser[(SLV_W_USER_WIDTH-1):0]	I	AXI Slave Write User. You can send all the Slave W channel protection bits through this signal. You can configure the CX_AXI_IF_TYPE_OF_PROT parameter to specify the type of protection. In legacy mode, only slv_wdata is protected. ■ slv_wuser[CC_SLV_BUS_DATA_PROT_WIDTH-1:0] contains the protection bits of slv_wdata ■ slv_wuser[CC_SLV_BUS_DATA_PROT_WIDTH] contains the protection bits of slv_wid ■ slv_wuser[CC_SLV_BUS_DATA_PROT_WIDTH+1] contains the protection bits of slv_wstrb ■ slv_wuser[CC_SLV_BUS_DATA_PROT_WIDTH+2] contains the protection bits of slv_wlast ■ slv_wuser[CC_SLV_W_USER_WIDTH-1:CC_SLV_BUS_DATA_PROT_WIDTH+3] contains the protection bits of all slv_w-misc_info signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_wvalid
slv_wstrb[(SLV_WSTRB_WD-1):0]	I	AXI Slave Write Strokes. Indicates which byte lanes to update in memory. There is one write stroke for each eight bits of write data bus. slv_wstrb[n] is associated with slv_wdata[8*n +7 : 8*n] Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_wvalid
slv_wlast	I	AXI Slave Write Last. Indicates last transfer in write burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_wvalid

Port Name	I/O	Description
slv_wvalid	I	AXI Slave Write Valid. Indicates valid write data and strobes are available. 0: Write data and strobes not available 1: Write data and strobes available This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: aux_clk,slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
slv_wvalid_p	I	AXI Slave Write Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_wvalid
slv_wmisc_info_ep	I	AXI Slave Write Data transaction related misc information. This is an optional signal that your application can use to poison write requests. When asserted, the controller sets the Poisoned TLP (EP) bit in the TLP header of the current and subsequent Write Data transactions. It is not a part of the standard AXI interface. For more information, see "AXI Slave Interface Sideband Signals" section in the <i>Databook</i>. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_wvalid

Port Name	I/O	Description
slv_wmisc_info_silentDrop	I	AXI Slave Write Data transaction related misc information. This is an optional signal which can be used by the application to drop the current transaction delivered into the Bridge. It is not a part of the standard AXI interface. For more information, see "AXI Slave Interface Sideband Signals" section in the <i>Databook</i>. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_wvalid
slv_bid[(SLV_ID_WD-1):0]	O	AXI Slave Write Response ID. This signal indicates the write response ID which correlates to a transfer that has an identical ID Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_bvalid
slv_bresp[1:0]	O	AXI Slave Write Response. Indicates status of write transaction as follows: ■ 00: OKAY ■ 01: EXOKAY (not supported) ■ 10: SLVERR ■ 11: DECERR The mapping of PCIe completion errors (for non-posted requests) to AXI error responses (SLVERR or DECERR) is detailed in the "AXI Bridge Module" chapter of the <i>Databook</i>. EXOKAY response is not supported. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_bvalid

Port Name	I/O	Description
slv_buser[(SLV_B_USER_WIDTH-1):0]	O	AXI Slave Write Response User. The controller sends all the Slave B channel protection bits through this signal. In legacy mode, this signal does not exist. ■ slv_buser[0] contains the protection bits of slv_bid ■ slv_buser[1] contains the protection bits of slv_bresp ■ slv_buser[CC_SLV_B_USER_WIDTH-1:2] contains the protection bits of all slv_bmisc_info signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_bvalid
slv_bidunq	O	AXI Slave Write Response Unique ID Indicator. Indicates whether slv_bid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_AXI_UIDI_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_bvalid
slv_bvalid	O	AXI Slave Write Response Valid. Indicates that valid write response is available. ■ 0: write response not available ■ 1: write response available Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
slv_bvalid_p	O	AXI Slave Write Response Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_bvalid
slv_bready	I	AXI Slave Response Ready. Indicates SLAVE can accept response information. ■ 0: Slave not ready ■ 1: Slave ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_bvalid
slv_bready_p	I	AXI Slave Write Response Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_bvalid

Port Name	I/O	Description
slv_bmisc_info[(SLV_RESP_MISC_INFO_PW-1):0]	O	Associated Misc Information of the AXI Slave Write Transaction Response. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. ■ [0]: Completion timeout bit ■ [1]: EP bit of the TLP ■ [4:2]: TC bits of the TLP ■ [5]: NS bit of the TLP ■ [6]: RO bit of the TLP ■ [9:7]: Completion status field of the TLP ■ [10]: Non-posted request ■ [13:11]: Function number of the TLP. This field [13:11] is only present when multifunction support (CX_NFUNC > 1) is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_bvalid
slv_bmisc_info_tlpprfx[(PRFX_WIDTH-1):0]	O	AXI Slave Write Response TLP Prefixes. The field [31:0] represents the first prefix received. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX=1), this means that slv_bmisc_info_tlpprfx[31:0] has prefix byte #0 in the lower byte position of the dword slv_bmisc_info_tlpprfx[31:0]. That is, FMT=bits 7:5, and Type =bits 4:0. If no prefix is present for a given TLP, then that DW, including the FMT field is all zeros. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CC_SLV_RESP_TLP_PREFIX_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_bvalid

Port Name	I/O	Description
slv_bmisc_info_tlpIn	O	AXI Slave Write Response TLP Lightweight Notification (LN) bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_LN_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_bvalid
slv_arid[(SLV_ID_WD-1):0]	I	AXI Slave Read Address. The ID of a slave read transfer. This signal is an input to the AXI bridge from an AXI master. This signal can contain unique or identical IDs for each AXI write transfer. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_araddr[(SLV_ADDR_WD-1):0]	I	AXI Slave Read Address. Gives initial address of read burst transaction. Only the start address of the burst is provided, and control signals issued alongside the address show how the address is calculated for the remaining transfers in the burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid

Port Name	I/O	Description
slv_aruser[(SLV_AR_USER_WIDTH-1):0]	I	AXI Slave Read User. You can send all the Slave AR channel parity protection bits sent through this signal. In legacy mode, only slv_araddr is protected. ■ slv_aruser[CC_SLV_BUS_ADDR_PROT_WIDTH-1:0] contains the protection bits of slv_araddr ■ slv_aruser[CC_SLV_BUS_ADDR_PROT_WIDTH] contains the protection bits of slv_arid ■ slv_aruser[CC_SLV_BUS_ADDR_PROT_WIDTH+1] contains the protection bits of slv_arlen ■ slv_aruser[CC_SLV_BUS_ADDR_PROT_WIDTH+2] contains the protection bits of {slv_arburst,slv_arlock,slv_arsize} ■ slv_aruser[CC_SLV_BUS_ADDR_PROT_WIDTH+3] contains the protection bits of {slv_arcache,slv_arprot} ■ slv_aruser[CC_SLV_BUS_ADDR_PROT_WIDTH+4] contains the protection bits of slv_arqos ■ slv_aruser[CC_SLV_AR_USER_WIDTH-1:CC_SLV_BUS_ADDR_PROT_WIDTH+5] contains the protection bits of all slv_ar-misc_info signals and acelite signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_arlen[(SLV_BURST_LEN_PW-1):0]	I	AXI Slave Read Burst Length. Gives the exact number of transfers in the burst; determines the number of data transfers associated with the address. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_arsize[2:0]	I	AXI Slave Read Burst Size. Indicates size of each transfer in burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid

Port Name	I/O	Description
slv_arburst[1:0]	I	AXI Slave Read Burst. Determines the type of burst used. Only INCR-BURST is supported when CC_SLV_WRAP_ENABLE =0. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_arlock[(SLV_LOCK_WD-1):0]	I	AXI Slave Read Lock. Encoded value indicates whether the transfer is a normal, exclusive, or locked access. This signal is not used in the controller. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: {None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_arcache[3:0]	I	AXI Slave Read Cache. Indicates bufferable, cacheable, read-through, and read-back attributes of transaction. This signal is not used in the controller. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: {None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_arprot[2:0]	I	AXI Slave Read Protection. Indicates normal, privileged, or secure protection level of transaction and whether transaction is data access or instruction access. This signal is not used in the controller. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: {None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid

Port Name	I/O	Description
slv_arvalid	I	AXI Slave Read Address Valid. When high, indicates read address and control information is valid and remains stable until arready is high. ■ 0: Address and control information not valid ■ 1: Address and control information valid This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: aux_clk,slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
slv_arvalid_p	I	AXI Slave Read Address Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid

Port Name	I/O	Description
slv_armisc_info[(SLV_MISC_INFO_PW-1):0]	I	AXI Slave Read Transaction Associated Miscellaneous Information. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. You must set all reserved bits to '0'. ■ [4:0]: TLP's TYPE ■ [5]: Serialize NP Requests ■ [6]: Reserved ■ [7]: Reserved ■ [8]: TLP's NS bit ■ [9]: TLP's RO bit ■ [12:10]: TLP's TC bits ■ [20:13]: TLP's MSG code ■ [21]: AXI transaction is a DBI access. This is for SHARED DBI mode only. ■ [24:22]: TLP's function number. Function numbering starts at '0'. Not used when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. For more information, see slv_armisc_info_func_num. This field is only present when multifunction support (CX_NFUNC > 1) is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: None,slv_aclk Ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_armisc_info_tlpprfx[(PRFX_WIDTH-1):0]	I	AXI Slave Read Request TLP Prefixes. The field [31:0] represents the first prefix to be transmitted. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that slv_armisc_info_tlpprfx[31:0] has prefix byte #0 in the lower byte position of the dword slv_armisc_info_tlpprfx[31:0]. That is, FMT =bits 7:5, and Type =bits 4:0. Your application is responsible for passing Local and End-End prefixes in the correct order for transmission. Local prefixes must precede End-End prefixes. Any parity bits that your application appends (when RAS data protection is enabled) to the most significant bits are routed but not checked by the controller. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_TLP_PREFIX_ENABLE) Synchronous To: slv_aclk Ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid

Port Name	I/O	Description
slv_armisc_info_tlpln	I	AXI Slave Read Request TLP Lightweight Notification (LN) bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_LN_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_armisc_info_rid[15:0]	I	AXI Slave Read Request Requester ID. Sets the Requester ID value for AXI slave read requests Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CC_AXI_SLV_MISC_INFO_RID_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: NA Validated by: slv_arvalid
slv_armisc_info_tph[10:0]	I	AXI Slave Read Request TLP Processing Hints. The bits are mapped as follows: ■ [0]: TH (TLP Processing Hint present) ■ [2:1]: PH (TLP Processing Hint) ■ [10:3]: ST (Steering Tag) Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_TPH_ENABLE) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_armisc_info_atu_bypass	I	AXI Slave Read Request Internal ATU Bypass. When set it indicates that this request should not be processed by the internal address translation unit Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_INTERNAL_ATU_ENABLE) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid

Port Name	I/O	Description
slv_armisc_info_nw	I	AT No Write (NW) Bit of the AXI Slave Read Transaction Request. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (ATS_TX_ENABLE) Synchronous To: slv_aclk Ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_armisc_info_ats[1:0]	I	AT Bits of the AXI Slave Read Transaction Request. For more information, see 'PCI-SIG Address Translation Services (ATS)'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (ATS_TX_ENABLE) Synchronous To: slv_aclk Ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_armisc_info_func_num[(PF_WD-1):0]	I	Function Number of the AXI Slave Read Transaction. It is used instead of bits [24:22] of slv_armisc_info when CX_SRIOV_ENABLE =1 or CX_ARI_ENABLE =1. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Function numbering starts at '0'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && ((AMBA_INTERFACE != 0 && (CX_SRIOV_ENABLE CX_ARI_ENABLE))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid

Port Name	I/O	Description
slv_armisc_info_vfunc_active	I	VF Active Indication of the AXI Slave Read Transaction. Indicates the VF that is sending the request. 0: No VF is active and slv_armisc_info_vfunc_num is invalid. A PF is valid and identified by slv_armisc_info_func_num. 1: A VF is active and is identified by slv_armisc_info_vfunc_num. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_SRIOV_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_armisc_info_vfunc_num[(VF_WD-2):0]	I	Virtual Function of the AXI Slave Read Transaction. Identifies the VF that is sending the request. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, slv_armisc_info_vfunc_num = 0 refers to the first VF of the PF that is identified by slv_armisc_info_func_num. For more information of VF numbering, see 'Programming View'. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_SRIOV_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_armisc_info_ido	I	IDO Bit of the AXI Slave Read Transaction. Enables ID-base ordering on outbound requests. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_IDO_ENABLE) Synchronous To: slv_aclk_ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid

Port Name	I/O	Description
slv_aridunq	I	AXI Slave Read Unique ID Indicator. Indicates whether slv_arid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_AXI_UIDI_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_ready	I	AXI Slave Read Ready. Indicates slave can accept read data and response information. ■ 0: Slave not ready ■ 1: Slave ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid
slv_ready_p	I	AXI Slave Read Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid
slv_rid[(SLV_ID_WD-1):0]	O	AXI Slave Read ID. This signal indicates the read response ID which correlates to a request with an identical ID. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid

Port Name	I/O	Description
slv_rdata[(SLV_DATA_WD-1):0]	O	AXI Slave Read Data. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid
slv_ruser[(SLV_R_USER_WIDTH-1):0]	O	AXI Slave Read User. The controller sends all the Slave R channel protection bits through this signal. You can configure the CX_AXI_IF_TYPE_OF_PROT parameter to specify the type of protection. In legacy mode, only slv_rdata is protected. ■ slv_ruser[CC_SLV_BUS_DATA_PROT_WIDTH-1:0] contains the protection bits of slv_rdata ■ slv_ruser[CC_SLV_BUS_DATA_PROT_WIDTH] contains the protection bits of slv_rid ■ slv_ruser[CC_SLV_BUS_DATA_PROT_WIDTH+1] contains the protection bits of {slv_rresp, slv_rlast} ■ slv_ruser[CC_SLV_R_USER_WIDTH-1:CC_SLV_BUS_DATA_PROT_WIDTH+2] contains the protection bits of all slv_rmisc_info signals Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid

Port Name	I/O	Description
slv_rresp[1:0]	O	AXI Slave Read Response. Indicates status of read transaction as follows: ■ 00: OKAY ■ 01: EXOKAY (not supported) ■ 10: SLVERR ■ 11: DECERR EXOKAY response is not supported. The mapping of PCIe completion errors (for non-posted requests) to AXI error responses (SLVERR or DECERR) is detailed in the "AXI Bridge Module" chapter of the <i>Databook</i>. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid
slv_rlast	O	AXI Slave Read Last. Indicates last transfer in read burst. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid
slv_rvalid	O	AXI Slave Read Valid. Indicates that required read data is available and read transfer can complete. ■ 0: Read data not available ■ 1: Read data available Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
slv_rvalid_p	O	AXI Slave Read Valid Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid
slv_rmisc_info[(SLV_RESP_MISC_INFO_PW-1):0]	O	Associated Miscellaneous Information of the AXI Slave Read Transaction Response. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. ■ [0]: Completion timeout bit ■ [1]: EP bit of the TLP ■ [4:2]: TC bits of the TLP ■ [5]: NS bit of the TLP ■ [6]: RO bit of the TLP ■ [9:7]: Completion status field of the TLP ■ [10]: Non-posted request. ■ [13:11]: Function number of the TLP. The bits [13:11] are present only when multifunction support (CX_NFUNC > 1) is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid

Port Name	I/O	Description
slv_rmisc_info_tlpprfx[(PRFX_WIDTH-1):0]	O	AXI Slave Read Response TLP Prefixes. The field [31:0] represents the first prefix received. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX=1), this means that slv_rmisc_info_tlpprfx[31:0] has prefix byte #0 in the lower byte position of the dword slv_rmisc_info_tlpprfx[31:0]. That is, FMT=bits 7:5, and Type =bits 4:0. If no prefix is present for a given TLP, then that DW, including the FMT field is all zeros. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CC_SLV_RESP_TLP_PREFIX_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid
slv_rmisc_info_tlpIn	O	AXI Slave Read Response TLP Lightweight Notification (LN). Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_LN_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid
slv_rmisc_info_ido	O	IDO bit of the AXI Slave Write Transaction Response. Indicates if the response should make use of ID-based ordering. This is an additional signal. It is not a part of the standard AXI interface. For more information, see 'AXI Slave Interface Sideband Signals'. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_IDO_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid

Port Name	I/O	Description
slv_ridunq	O	AXI Slave Read Response Unique ID Indicator. Indicates whether slv_rid is unique in in-flight transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CX_AXI_UIDI_ENABLE) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_rvalid
slv_arready	O	AXI Slave Read Address Ready. Indicates that the slave is ready to accept address and associated control signals. ■ 0: Slave not ready ■ 1: Slave ready Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: slv_aclk Ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_arready_p	O	AXI Slave Read Address Ready Parity bit. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: slv_aclk Ug,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_argos[3:0]	I	AXI Slave Read QoS. This signal is not used internally. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: {None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid

Port Name	I/O	Description
slv_arbar[1:0]	I	AXI Slave Read Barrier Type. You can use this signal to distinguish between normal and transaction barriers, and between different barrier types. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: slv_aclk_ug,{None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_ardomain[1:0]	I	AXI Slave Read Shareability Domain. Indicates the shareability domain to which the transaction applies. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: slv_aclk_ug,{None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_arsnoop[3:0]	I	AXI Slave Read Shareable Transaction Type. You can use this signal to distinguish between the different types of shareable read transactions. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (CC_ACELITE_ENABLE) Synchronous To: slv_aclk_ug,{None},{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_arvalid
slv_csysreq	I	AXI Slave System Low-Power Request. System low-power request from system clock controller for AXI bridge slave to enter low-power state. AXI power modes are not currently supported. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

Port Name	I/O	Description
slv_csysack	O	AXI Slave Low-Power Request Acknowledgment. Indicates acknowledgment from AXI bridge slave to a system low-power request. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: slv_csysreq
slv_cactive	O	AXI Slave Clock Active. Indicates that AXI bridge slave requires a clock signal. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_csysreq
dbi_buser[(DBI_SLV_B_USER_WIDTH-1):0]	O	AXI DBI Slave Write Response User. You can send all the DBI Slave B channel protection bits through this signal. In legacy mode, this signal does not exist. ■ dbi_buser[0] contains the protection bits of dbi_bid ■ dbi_buser[1] contains the protection bits of dbi_bresp Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: slv_bvalid

2.6 CXL Controller RAS Interface Signals

`cxl_ras_stat_rx_uncorr` -
`cxl_ras_stat_rx_hdr_log` -

Table 2-6 CXL Controller RAS Interface Signals

Port Name	I/O	Description
<code>cxl_ras_stat_rx_uncorr[7:0]</code>	I	CXL RAS Statistics Uncorrectable Error Bus from Application Exists: $((CX_CXL_ENABLE \ \&\& \ !CX_CXL_CXS_ENABLE)) \ \&\& \ ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0)$ Synchronous To: <code>core_clk</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
<code>cxl_ras_stat_rx_hdr_log[(CX_CXL_RAS_HDR_LOG_SIZE-1):0]</code>	I	CXL RAS Statistics Header Log Bus from Application Exists: $((CX_CXL_ENABLE \ \&\& \ !CX_CXL_CXS_ENABLE)) \ \&\& \ ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0)$ Synchronous To: <code>core_clk</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: <code>cxl_ras_stat_rx_uncorr</code> is asserted

2.7 CXL Controller RAS Statistics Interface Signals

`cxl_ras_stat_rx_corr` - 

Table 2-7 CXL Controller RAS Statistics Interface Signals

Port Name	I/O	Description
<code>cxl_ras_stat_rx_corr[1:0]</code>	I	CXL RAS Correctable Error Bus from Application Exists: $((CX_CXL_ENABLE \ \&\& \ !CX_CXL_CXS_ENABLE)) \ \&\& \ ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0)$ Synchronous To: <code>core_clk</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.8 CXL Interface Signals

cxl_cache_mst_ack	-	cxl_cache_mst_req
cxl_cache_slv_req	-	cxl_cache_slv_ack
cxl_mst_h2d_req_crd	-	cxl_mst_h2d_req
cxl_mst_h2d0_rsp_crd	-	cxl_mst_h2d0_rsp
cxl_mst_h2d1_rsp_crd	-	cxl_mst_h2d1_rsp
cxl_mst_h2d_data_crd	-	cxl_mst_h2d_data
cxl_slv_d2h0_req	-	cxl_mst_h2d_req_chk
cxl_slv_d2h1_req	-	cxl_mst_h2d0_rsp_chk
cxl_slv_d2h_rsp	-	cxl_mst_h2d1_rsp_chk
cxl_slv_d2h_data	-	cxl_mst_h2d_data_chk
cxl_slv_d2h0_req_chk	-	cxl_slv_d2h0_req_crd
cxl_slv_d2h1_req_chk	-	cxl_slv_d2h1_req_crd
cxl_slv_d2h_rsp_chk	-	cxl_slv_d2h_rsp_crd
cxl_slv_d2h_data_chk	-	cxl_slv_d2h_data_crd
cxl_mem_mst_ack	-	cxl_mem_mst_req
cxl_mem_slv_req	-	cxl_mem_slv_ack
cxl_mst_m2s_req_crd	-	cxl_mst_m2s_req
cxl_mst_m2s_rwd_crd	-	cxl_mst_m2s_rwd
cxl_slv_s2m_ndr	-	cxl_mst_m2s_req_chk
cxl_slv_s2m_drc	-	cxl_mst_m2s_rwd_chk
cxl_slv_s2m_ndr_chk	-	cxl_slv_s2m_ndr_crd
cxl_slv_s2m_drc_chk	-	cxl_slv_s2m_drc_crd

Table 2-8 CXL Interface Signals

Port Name	I/O	Description
cxl_cache_mst_req	O	CXL.Cache Master Channel Initialization Request Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_cache_mst_ack	I	CXL.Cache Master Channel Initialization Acknowledge Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cxl_cache_slv_req	I	CXL.Cache Slave Channel Initialization Request Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_cache_slv_ack	O	CXL.Cache Slave Channel Initialization Acknowledge Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_h2d_req_crd	I	CXL.Cache H2D Request Master Channel Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_h2d0_rsp_crd	I	CXL.Cache H2D Response Master Channel 0 Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_h2d1_rsp_crd	I	CXL.Cache H2D Response Master Channel 1 Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cxl_mst_h2d_data_crd	I	CXL.Cache H2D Data Master Channel Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_h2d_req[(CX_CXL_CACHE_H2D_REQ_SIZE-1):0]	O	CXL.Cache H2D Request Master Channel Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_h2d0_rsp[(CX_CXL_CACHE_H2D_RESP_SIZE-1):0]	O	CXL.Cache H2D Response Master Channel 0 Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_h2d1_rsp[(CX_CXL_CACHE_H2D_RESP_SIZE-1):0]	O	CXL.Cache H2D Response Master Channel 1 Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_h2d_data[(CX_CXL_CACHE_H2D_DATA_SIZE-1):0]	O	CXL.Cache H2D Data Master Channel Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cxl_mst_h2d_req_chk[(CX_CXL_CACHE_H2D_REQ_CHK_SIZE-1):0]	O	CXL.Cache H2D Request Master Channel Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_h2d0_rsp_chk[(CX_CXL_CACHE_H2D_RESP_CHK_SIZE-1):0]	O	CXL.Cache H2D Response Master Channel 0 Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_h2d1_rsp_chk[(CX_CXL_CACHE_H2D_RESP_CHK_SIZE-1):0]	O	CXL.Cache H2D Response Master Channel 1 Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_h2d_data_chk[(CX_CXL_CACHE_H2D_DATA_CHK_SIZE-1):0]	O	CXL.Cache H2D Data Master Channel Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cxl_slv_d2h0_req_crd	O	CXL.Cache D2H Request Slave Channel 0 Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_d2h1_req_crd	O	CXL.Cache D2H Request Slave Channel 1 Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_d2h_rsp_crd	O	CXL.Cache D2H Response Slave Channel Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_d2h_data_crd	O	CXL.Cache D2H Data Slave Channel Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_d2h0_req[(CX_CXL_CACHE_D2H_I_REQ_SIZE-1):0]	I	CXL.Cache D2H Request Slave Channel 0 Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

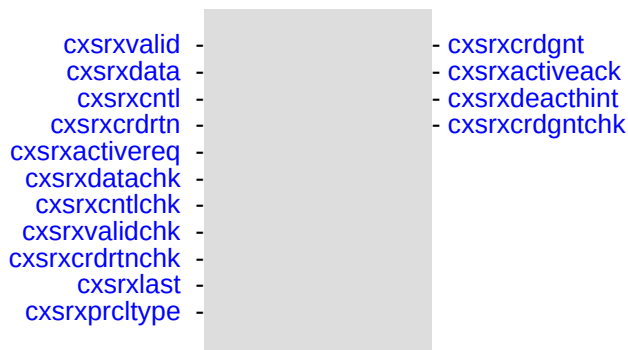
Port Name	I/O	Description
cxl_slv_d2h1_req[(CX_CXL_CACHE_D2H_REQ_SIZE-1):0]	I	CXL.Cache D2H Request Slave Channel 1 Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_d2h_rsp[(CX_CXL_CACHE_D2H_RESP_SIZE-1):0]	I	CXL.Cache D2H Response Slave Channel Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_d2h_data[(CX_CXL_CACHE_D2H_DATA_SIZE-1):0]	I	CXL.Cache D2H Data Slave Channel Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: None,core_clk,perbitclk,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_d2h0_req_chk[(CX_CXL_CACHE_D2H_REQ_CHK_SIZE-1):0]	I	CXL.Cache D2H Request Slave Channel 0 Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_d2h1_req_chk[(CX_CXL_CACHE_D2H_REQ_CHK_SIZE-1):0]	I	CXL.Cache D2H Request Slave Channel 1 Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cxl_slv_d2h_rsp_chk[(CX_CXL_CACHE_D2H_RESP_CHK_SIZE-1):0]	I	CXL.Cache D2H Response Slave Channel Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_d2h_data_chk[(CX_CXL_CACHE_D2H_DATA_CHK_SIZE-1):0]	I	CXL.Cache D2H Data Slave Channel Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: None,perbitclk,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mem_mst_req	O	CXL.Mem Master Channel Initialization Request Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mem_mst_ack	I	CXL.Mem Master Channel Initialization Acknowledge Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mem_slv_req	I	CXL.Mem Slave Channel Initialization Request Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cxl_mem_slv_ack	O	CXL.Mem Slave Channel Initialization Acknowledge Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_m2s_req_crd	I	CXL.Mem M2S Request Master Channel Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_m2s_rwd_crd	I	CXL.Mem M2S Rwd Master Channel Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_m2s_req[(CX_CXL_MEM_M2S_NDR_SIZE-1):0]	O	CXL.Mem M2S Request Master Channel Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_m2s_rwd[(CX_CXL_MEM_M2S_DATA_SIZE-1):0]	O	CXL.Mem M2S Rwd Master Channel Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cxl_mst_m2s_req_chk[(CX_CXL_MEM_M2S_NDR_CHK_SIZE-1):0]	O	CXL.Mem M2S Request Master Channel Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_m2s_rwd_chk[(CX_CXL_MEM_M2S_DATA_CHK_SIZE-1):0]	O	CXL.Mem M2S Rwd Master Channel Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_s2m_ndr_crd	O	CXL.Mem S2M NDR Slave Channel Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_s2m_drc_crd	O	CXL.Mem S2M DRC Slave Channel Credit Grant Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_s2m_ndr[(CX_CXL_MEM_S2M_NDR_SIZE-1):0]	I	CXL.Mem S2M NDR Slave Channel Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cxl_slv_s2m_drc[(CX_CXL_MEM_S2M_DATA_SIZE-1):0]	I	CXL.Mem S2M DRC Slave Channel Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: None,core_clk,perbitclk,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_s2m_ndr_chk[(CX_CXL_MEM_S2M_NDR_CHK_SIZE-1):0]	I	CXL.Mem S2M NDR Slave Channel Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_s2m_drc_chk[(CX_CXL_MEM_S2M_DATA_CHK_SIZE-1):0]	I	CXL.Mem S2M DRC Slave Channel Data Protection Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: None,perbitclk,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal



Port Name	I/O	Description
cxsrvalid	I	Indicates that data flit is valid on CXS Receiver input. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: aux_clk_g,{core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: not validated by any other signal
cxsrdata[(CXSDATAFLITWIDTH-1):0]	I	CXS Receiver data bus. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: aux_clk_g,{core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxsrvalid is asserted
cxsrxcntl[(CXSCNTLWIDTH-1):0]	I	CXS Receiver control bus; contains information on packet start, end, and errors. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: aux_clk_g,{core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxsrvalid is asserted

Port Name	I/O	Description
cxsrxcrdgnt	O	CXS Receiver credit grant; grants a single credit for the Transmitter. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: No Power Domain: PD_VAUX Active State: High Validated by: cxsrxactiveack is asserted
cxsrxcdrtn	I	CXS Receiver credit return; returns a single credit to the Receiver. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: cxsrxvalid is not asserted
cxsrxactiverreq	I	CXS Receiver link activation or deactivation request. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {aux_clk},{cxs_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: not validated by any other signal
cxsrxactiveack	O	CXS Receiver link activation or deactivation acknowledge. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: not validated by any other signal
cxsrxdeacthint	O	CXS Receiver request for link deactivation. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: not validated by any other signal

Port Name	I/O	Description
cxsrxdatachk[(CXSDATACHKWIDTH-1):0]	I	CXS Receiver data bus parity bits. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_RX_DATACHECK > 0) ? 1 : 0) Synchronous To: core_clk,{aux_clk_g},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxsrxvalid is asserted
cxsrxcntlchk[(CXSCNTLCHKWIDTH-1):0]	I	CXS Receiver control bus parity bits. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_RX_DATACHECK > 0) ? 1 : 0) Synchronous To: core_clk,{aux_clk_g},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxsrxvalid is asserted
cxsrxvalidchk[(CXSRXREPWIDTH-1):0]	I	CXS Receiver valid data replication. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_RX_REPLICATION > 0) ? 1 : 0) Synchronous To: CX_CXS_SYNCHRONOUS ? "core_clk" : "cxs_clk" Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: not validated by any other signal
cxsrxcrdrtnchk[(CXSRXREPWIDTH-1):0]	I	CXS Receiver credit return replication. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_RX_REPLICATION > 0) ? 1 : 0) Synchronous To: CX_CXS_SYNCHRONOUS ? "core_clk" : "cxs_clk" Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: not validated by any other signal
cxsrxcrdgrntchk[(CXSRXREPWIDTH-1):0]	O	CXS Receiver credit grant replication. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_RX_REPLICATION > 0) ? 1 : 0) Synchronous To: CX_CXS_SYNCHRONOUS ? "core_clk" : "cxs_clk" Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: not validated by any other signal

Port Name	I/O	Description
cxsrxlast	I	CXS Receiver end of stream indication. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXL_CXS_ENABLE)) Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxsrxvalid is asserted
cxsrxprcltype[2:0]	I	CXS Receiver Protocol Type Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_ENABLE && CX_CCIX_INTERFACE_ENABLE && CX_CXL_CXS_ENABLE)) Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxsrxvalid is asserted

2.10 CXS Tx Interface Signals

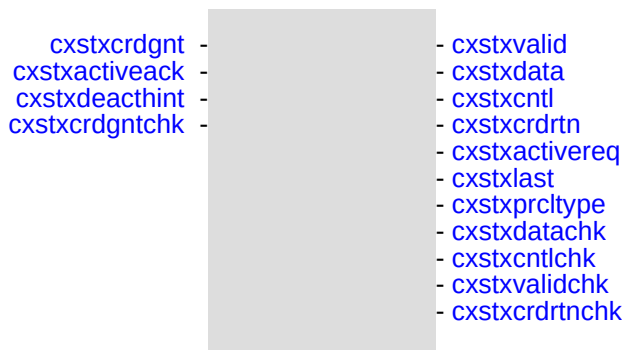


Table 2-10 CXS Tx Interface Signals

Port Name	I/O	Description
cxstxvalid	O	Indicates that data flit is valid on CXS Transmitter output. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: aux_clk_g,{core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: not validated by any other signal
cxstxdata[(CXSDATAFLITWIDTH-1):0]	O	CXS Transmitter data bus. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: aux_clk_g,{core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxstxvalid is asserted
cxstxcntl[(CXSCNTLWIDTH-1):0]	O	CXS Transmitter control bus, contains information on packet start, end, and errors. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: aux_clk_g,{core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxstxvalid is asserted

Port Name	I/O	Description
cxstxcrdgnt	I	CXS Transmitter credit grant; grants a single credit for the transmitter. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: No Power Domain: PD_VAUX Active State: High Validated by: cxstxactiveack is asserted
cxstxcrdrtn	O	CXS Transmitter credit return; returns a single credit to the receiver. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: cxstxvalid is not asserted
cxstxactivereq	O	CXS Transmitter link activation or deactivation request. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: not validated by any other signal
cxstxactiveack	I	CXS Transmitter link activation or deactivation acknowledge. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: No Power Domain: PD_VAUX Active State: High Validated by: not validated by any other signal
cxstxdeacthint	I	CXS Transmitter request for link deactivation. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: not validated by any other signal

Port Name	I/O	Description
cxstxlast	O	CXS Transmitter end of stream indication. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXL_CXS_ENABLE)) Synchronous To: aux_clk_g,{core_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxsrvalid is asserted
cxstxprcltype[2:0]	O	CXS Transmitter Protocol Type Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_ENABLE && CX_CCIX_INTERFACE_ENABLE && CX_CXL_CXS_ENABLE)) Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxstxvalid is asserted
cxstxdatachk[(CXSDATACHKWIDTH-1):0]	O	CXS Transmitter data bus parity bits. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_TX_DATACHECK > 0) ? 1 : 0) Synchronous To: aux_clk_g,{core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxstxvalid is asserted
cxstxcntlchk[(CXSCNTLCHKWIDTH-1):0]	O	CXS Transmitter control bus parity bits. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_TX_DATACHECK > 0) ? 1 : 0) Synchronous To: aux_clk_g,{core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxstxvalid is asserted
cxstxvalidchk[(CXSTXREPWIDTH-1):0]	O	CXS Transmitter valid data replication. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_TX_REPLICATION > 0) ? 1 : 0) Synchronous To: CX_CXS_SYNCHRONOUS ? "core_clk" : "cxs_clk" Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: not validated by any other signal

Port Name	I/O	Description
cxstxcdrtnchk[(CXSTXREPWIDTH-1):0]	O	CXS Transmitter credit return replication. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_TX_REPLICATION > 0) ? 1 : 0) Synchronous To: CX_CXS_SYNCHRONOUS ? "core_clk" : "cxs_clk" Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: not validated by any other signal
cxstxcrdgntchk[(CXSTXREPWIDTH-1):0]	I	CXS Transmitter credit grant replication. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && ((CX_CXS_TX_REPLICATION > 0) ? 1 : 0) Synchronous To: CX_CXS_SYNCHRONOUS ? "core_clk" : "cxs_clk" Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: not validated by any other signal

2.11 XALI0 Interface Signals

client0_addr_align_en	-	xadm_client0_halt
client0_tlp_byte_en	-	
client0_remote_req_id	-	
client0_cpl_byte_cnt	-	
client0_tlp_tc	-	
client0_tlp_attr	-	
client0_tlp_ln	-	
client0_cpl_status	-	
client0_cpl_bcm	-	
client0_tlp_dv	-	
client0_tlp_eot	-	
client0_tlp_bad_eot	-	
client0_tlp_hv	-	
client0_tlp_fmt	-	
client0_tlp_type	-	
client0_tlp_td	-	
client0_tlp_ep	-	
client0_tlp_byte_len	-	
client0_tlp_addr	-	
client0_hdr_prot	-	
client0_tlp_tid	-	
client0_tlp_data	-	
client0_tlp_func_num	-	
client0_tlp_vfunc_num	-	
client0_tlp_vfunc_active	-	
client0_tlp_ats	-	
client0_tlp_nw	-	
client0_tlp_th	-	
client0_tlp_ph	-	
client0_tlp_st	-	
client0_tlp_prfx	-	
client0_tlp_atu_bypass	-	
client0_tlp_dma	-	
client0_tlp_dma_channel	-	

Table 2-11 XALI0 Interface Signals

Port Name	I/O	Description
client0_addr_align_en[(NVC_XALI_EXP-1):0]	I	When client0_addr_align_en is asserted, the controller performs address alignment and generates the first and last byte enables based on the address and number of bytes of the TLP requested from the client interface. When client0_addr_align_en is de-asserted (always for completions), the controller takes the first and last byte enables from client0_tlp_byte_en and does not address-align the data. For more information, see the "Transmit Address Alignment" section in the "Signal Descriptions" chapter of the <i>Databook</i>. When you do not select the Enable ECRC Support (CX_ECRC_ENABLE) option, and your application is generating the ECRC, do not assert client0_addr_align_en for TLPs that contain application-generated ECRC. When the TLP from your application does not contain ECRC, then your application can assert client0_addr_align_en to enable address alignment. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_byte_en[((NVC_XALI_EXP*8)-1):0]	I	Byte enables for the first and last dword of the TLP: ■ [3:0]: Byte enables for the first dword ■ [7:4]: Byte enables for the last dword A TLP can have 'holes' between bytes when the TLP is less than eight bytes (two dwords) long, according to the <i>PCI Express Specification</i> (not all valid bytes need to be contiguous). However, a TLP is not allowed to have holes between bytes when the TLP is more than eight bytes. Not used (set to 0x0) when 'Transmit Address Alignment' is (client0_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1). When sending a message, your application asserts the message code onto these signals. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted. Not valid when client0_addr_align_en is asserted.

Port Name	I/O	Description
client0_remote_req_id[((NVC_XALI_EXP*16)-1):0]	I	Client0 requester ID to be used in the TLP header of a completion TLP. The value must match the requester ID of the corresponding incoming request: ■ [15:8]: Bus number ■ [7:3]: Device number ■ [2:0]: Function number Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_cpl_byte_cnt[((NVC_XALI_EXP*12)-1):0]	I	The value to be used for the Byte Count field of a memory read completion TLP. The value client0_cpl_byte_cnt indicates the number of bytes remaining to be delivered for the request, as defined in the <i>PCI Express Specification</i>. ■ 001h: 1 byte ■ FFFh: 4095 bytes ■ 000h: 4096 bytes When you select the TRGT_CPL_LUT_EN configuration option, the controller does not use client0_cpl_byte_cnt for completions. Instead, the controller takes the current Byte Count value from the target completion LUT entry specified the client0/1/2_cpl_lookup_id input. For more information, see 'Target Completion Lookup Table'. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted. Only used when XALI0 is connected to a completer.
client0_tlp_tc[((NVC_XALI_EXP*3)-1):0]	I	The value to be used for the Traffic Class (TC) field of the TLP header (000b'111b) on XALI0. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted

Port Name	I/O	Description
client0_tlp_attr[((NVC_XALI_EXP*ATTR_WD)-1):0]	I	Attributes field for the TLP header ■ Bit 0: No Snoop ■ Bit 1: Relaxed Ordering ■ Bit 2: IDO attribute When you select the TRGT_CPL_LUT_EN configuration option, the controller does not use client0_tlp_attr for completions. Instead, the controller takes the attr value from the target completion LUT entry specified the client0/1/2_cpl_lookup_id input. For more information, see 'Target Completion Lookup Table'. ■ Width is three bits when CX_IDO_ENABLE =1 ■ Width is two bits when CX_IDO_ENABLE =0 Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_in[(NVC_XALI_EXP-1):0]	I	This signal indicates that the controller can accept LN Read, LN Write or LN Completion, or special zero-Length LN Reads or LN Writes. Exists: !(AMBA_INTERFACE!=0) && (CX_LN_ENABLE) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_cpl_status[((NVC_XALI_EXP*3)-1):0]	I	The value to be used in the completion Status field of a completion TLP header on XALI0: ■ 000b: Successful completion ■ 001b: Unsupported request ■ 010b: Configuration request retry status ■ 100b: Completer abort Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted. Only used when XALI0 is connected to a completer.

Port Name	I/O	Description
client0_cpl_bcm[(NVC_XALI_EXP-1):0]	I	Byte Count Modified field on XALI0. Must be hardwired to '0'. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_dv[(NVC_XALI_EXP-1):0]	I	Indicates that the TLP payload data client0_tlp_data is valid. Your application must not assert client0_tlp_dv for TLPs that do not include a payload, such as memory read requests. Your application must present the first payload data the same cycle with the TLP header. Exists: !(AMBA_INTERFACE!=0) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
client0_tlp_eot[(NVC_XALI_EXP-1):0]	I	Indicates the end of TLP payload data for a TLP that includes a payload. For a TLP that does not include a payload (such as a memory read request), your application must assert and de-assert client0_tlp_eot at the same time as client0_tlp_hv. You must not assert it at any other time. Exists: !(AMBA_INTERFACE!=0) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
client0_tlp_bad_eot[(NVC_XALI_EXP-1):0]]	I	Indicates that the current TLP must be nullified. To nullify a TLP, your application must assert client0_tlp_bad_eot in the same cycle as client_tlp_eot. When client0_tlp_bad_eot is asserted, the controller nullifies the TLP (inverts the LCRC and inserts EDB as the final framing symbol in the transmitted TLP). Your application can use client0_tlp_bad_eot to nullify a TLP because of a known error condition, or to abort the current transaction when the current transaction is being blocked (for example, due to lack of FC credits): ■ To cancel a request that has not been granted (xadm_client0_halt has never been de-asserted for the transfer), the CLIENT_PULLBACK feature from coreConsultant must be selected. In addition, the following signals must be driven as follows. Assert client0_tlp_bad_eot and client0_tlp_eot for one clock cycle while xadm_client0_halt is asserted (indicating that the request has not been granted), then de-assert client0_tlp_hv, client0_tlp_dv, client0_tlp_bad_eot, and client0_tlp_eot to terminate the request. Without the CLIENT_PULLBACK feature, your application has to send the full packet and mark the EOT and BAD_EOT at the end. With the CLIENT_PULLBACK feature, your application can mark EOT and BAD_EOT when it realizes that it needs to nullify the TLP, without having to present the full packet. ■ To cancel a request that is in progress (xadm_client0_halt has been de-asserted for the transfer), assert client0_tlp_bad_eot and client0_tlp_eot for one clock cycle while xadm_client0_halt is de-asserted (indicating that the request has been granted), then de-assert client0_tlp_hv, client0_tlp_dv, client0_tlp_bad_eot, and client0_tlp_eot to terminate the request. In this case, the controller nullifies the TLP as described in this section. ■ When the TLP is a one-cycle request (client0_tlp_hv and client0_tlp_eot asserted in the same cycle) and client0_tlp_bad_eot is also asserted, this request is ignored. Sending a nullified TLP is not very useful because the link partner has to ignore it anyway. Your application should realize that this request is nullified (in this case ignored) and de-assert client0_tlp_hv and client0_tlp_eot while keeping client0_tlp_bad_eot asserted. ■ Nullify Atomic Ops TLP is not supported. Exists: !(AMBA_INTERFACE!=0) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_eot is asserted
client0_tlp_hv[(NVC_XALI_EXP-1):0]	I	Indicates that the TLP header data the TLP header ports is valid. The header data must remain valid until client0_tlp_hv is de-asserted. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
client0_tlp_fmt[((NVC_XALI_EXP*2)-1):0]	I	The value to be used for the Format field of the TLP header on XALI0: ■ 00b: 3 dword header, no data ■ 01b: 4 dword header, no data ■ 10b: 3 dword header, with data ■ 11b: 4 dword header, with data Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_type[((NVC_XALI_EXP*5)-1):0]	I	The value to be used for the Type field of the client0 TLP header, as defined in the <i>PCI Express Specification</i>. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_td[(NVC_XALI_EXP-1):0]	I	TLP Digest bit on XALI0. It is expected that the ECRC is not present, and the controller generates and inserts it. You should set this input to 0. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_ep[(NVC_XALI_EXP-1):0]	I	When asserted, the controller sets the Poisoned TLP (EP) bit in the TLP header on XALI0. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted

Port Name	I/O	Description
client0_tlp_byte_len[((NVC_XALI_EXP*13)-1):0]	I	The number of bytes in the TLP Payload (not including ECRC). The value should be in the range 0x0000 (0) to 0x1000 (4096) When 'Transmit Address Alignment' is (client0_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1), this must be the full byte length of the request. When transmit alignment is off (client0_addr_align_en=0), bits [1:0] must be set to '00'. The byte length value must be rounded up to the next dword boundary. For example, when the request is five bytes long, the value of bits [12:0] must be eight. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_addr[((NVC_XALI_EXP*64)-1):0]	I	The Address field for the TLP header. The TLP address can be 64 or 32 bits. When sending completions, the controller uses client0_tlp_addr[6:0] for the Lower Address field of the completion TLP. The bits in this address bus map directly to the address bits in the TLP header. The third and fourth dwords of the header are mapped as follows: - MEM (32-bit) / IO / CFG have a 3-dword header. The third dword (bytes 8-11) =client0_tlp_addr[31:0]. Therefore, bits [7:0] are mapped to byte 11 which is the lowest eight bits of the TLP header address. For more information, see the 'Endianness' advanced information chapter. - MEM (64-bit) / MSG have a 4-dword header. The third dword (bytes 8-11) =client0_tlp_addr[63:32]. The fourth dword (bytes 12-15) =client0_tlp_addr[31:0]. Therefore, bits [7:0] are mapped to byte 15 which is the lowest eight bits of the TLP header address. For more information, see the 'Endianness' advanced information chapter. The base width of client0_tlp_addr is 64 bits. When 'Transmit Address Alignment' is (client0_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1), this address is the full byte address of the first enabled byte of the request. That is, the address is byte-aligned. When transmit alignment is off (client0_addr_align_en=0), bits [1:0] must be '00', that is, the address must be dword-aligned. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted

Port Name	I/O	Description
client0_hdr_prot[((NVC_XALI_EXP*CLIENT_HDR_PROT_WD)-1):0]	I	ECC or parity bits (protection code) for the XALI0 header. Your application must calculate the protection code over all of the XALI0 header inputs in this order: client0_tlp_th, client0_tlp_st, client0_tlp_ph, client0_tlp_vfunc_active, client0_tlp_vfunc_num, client0_tlp_ln, client0_tlp_func_num, client0_tlp_attr, client0_addr_align_en, client0_tlp_byte_en, client0_remote_req_id, client0_cpl_status, client0_cpl_bcm, client0_cpl_byte_cnt, client0_tlp_tid, client0_tlp_byte_len, client0_tlp_ep, client0_tlp_td, client0_tlp_tc, client0_tlp_type, client0_tlp_fmt, client0_tlp_addr[63:0]. Exists: !(AMBA_INTERFACE!=0) && (((CX_RASDP==0)? 0: 1)) Synchronous To: core_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_tid[((NVC_XALI_EXP*TAG_SIZE)-1):0]	I	Tag field for the TLP header. Your application must provide a valid Tag client0_tlp_tid for all posted and non-posted requests. The value of the CX_MAX_TAG configuration option determines the maximum number of outstanding Client requests. Your application is expected to manage the Tag for every posted and non-posted request. For completions, the client0_tlp_tid value must be the Tag from the corresponding request. When you select the TRGT_CPL_LUT_EN configuration option, the controller does not use client0_tlp_tid for completions. Instead, the controller takes the tag value from the target completion LUT entry. For more information, see 'Target Completion Lookup Table'. Exists: !(AMBA_INTERFACE!=0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted

Port Name	I/O	Description
client0_tlp_data[((NVC_XALI_EXP*DW_W_PAR)-1):0]	I	The data payload for the TLP. The data is in little endian format. The first transmitted payload byte is in [7:0]. The base width of client0_tlp_data is your application datapath width, which is automatically determined according to the selected operating frequency and number of lanes. ■ For a 32-bit controller, the base width is 32 (client0_tlp_data[31:0]) ■ For a 64-bit controller, the base width is 64 (client0_tlp_data[63:0]) ■ For a 128-bit controller, the base width is 128 (client0_tlp_data[127:0]) ■ For a 256-bit controller, the base width is 256 (client0_tlp_data[255:0]) When 'Transmit Address Alignment' is (client0_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1), the first enabled byte of the request must be bits [7:0] of this data bus. The controller up-shifts the data in the transmitted TLP based the lower two bits of the byte address. Your application must append protection code bits to the most significant bits of this bus when ECC or parity datapath protection is used. Exists: !(AMBA_INTERFACE!=0) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_dv is asserted
client0_tlp_func_num[((NVC_XALI_EXP*PF_WD)-1):0]	I	Indicates from which physical function (PF) the request is coming. The controller uses client0_tlp_func_num to form the requester ID for requests and the completer ID for completions. Function numbering starts at '0'. For a single-function device, set client0_tlp_func_num to 000b. Exists: !(AMBA_INTERFACE!=0) Synchronous To: None,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted

Port Name	I/O	Description
client0_tlp_vfunc_num[((NVC_XALI_EXP* (VF_WD-1))-1):0]	I	Indicates from which virtual function (VF) the request is coming. The controller uses this signal to form the requester ID for requests and the completer ID for completions. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, client0_tlp_vfunc_num=0 refers to the first VF of the PF that is identified by client0_tlp_func_num. For more information of VF numbering, see 'Programming View'. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{CX_SRIOV_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv and client0_tlp_vfunc_active are asserted
client0_tlp_vfunc_active[(NVC_XALI_EXP -1):0]	I	Indicates that the TLP request is coming from a VF (Virtual function). 0: No VF is active and client0_tlp_vfunc_num is invalid. A PF is valid and identified by client0_tlp_func_num. 1: A VF is active and is identified by client0_tlp_vfunc_num. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{CX_SRIOV_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_ats[((NVC_XALI_EXP*2)-1):0]	I	AT field in the TLP header on XALI0. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{ATS_TX_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_nw[(NVC_XALI_EXP-1):0]	I	NW field in the TLP header on XALI0. In configuration with DMA this pin is reserved. You must set this input to '0'. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{ATS_TX_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted

Port Name	I/O	Description
client0_tlp_th[(NVC_XALI_EXP-1):0]	I	TH field in the TLP header on XALI0. Exists: $!(\text{AMBA_INTERFACE}!=0) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: None,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_ph[((NVC_XALI_EXP*2)-1):0]	I	PH field in the TLP header on XALI0. Exists: $!(\text{AMBA_INTERFACE}!=0) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: None,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_st[((NVC_XALI_EXP*8)-1):0]	I	ST field in the TLP header on XALI0. Exists: $!(\text{AMBA_INTERFACE}!=0) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: None,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
xadm_client0_halt[(NVC_XALI_EXP-1):0]	O	The controller asserts xadm_client0_halt when it is not ready to process a transaction. The header and payload input for Client0 must remain unchanged while xadm_client0_halt is asserted. The conditions that cause the controller to assert xadm_client0_halt include: ■ The XADM is busy servicing one of the other transmit client interfaces. ■ There are not enough Flow Control credits available to send the requested TLP type (P, non-posted, or CPL). ■ The LTSSM is not ready. ■ A retry is in progress. ■ ECRC or LCRC insertion is in progress. <i>Note:</i> xadm_client0_halt is asserted by default. For more information, see the timing diagrams in "Signal Interfaces" chapter of the <i>Databook</i>. Exists: $!(\text{AMBA_INTERFACE}!=0)$ Synchronous To: aux_clk,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
client0_tlp_prfx[((NVC_XALI_EXP*PRFX_DW)-1):0]	I	TLP prefixes. The field [31:0] represents the first prefix to be transmitted. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that client0_tlp_prfx[31:0] has prefix byte #0 in the lower byte position of the dword client0_tlp_prfx[31:0]. That is, FMT =bits 7:5, and Type =bits 4:0. Your application is responsible for passing Local and End-End prefixes in the correct order for transmission. Local prefixes must precede End-End prefixes. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (!(AMBA_INTERFACE!=0)) && (CX_TLP_PREFIX_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_atu_bypass[(NVC_XALI_EXP-1):0]	I	Internal ATU Bypass for XALI0. When set it indicates that this transaction should not be processed by the internal address translation unit Exists: (!(AMBA_INTERFACE!=0)) && (CX_INTERNAL_ATU_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted
client0_tlp_dma[((NVC_XALI_EXP*DC_DMA_TYPE_WIDTH)-1):0]	I	Specifies the type of TLP you are supplying the XALI0 interface. Bit encoding is as follows: ■ 000: Reserved ■ 001: DMA Write channel MRd completion ■ 010: Reserved ■ 011: Reserved ■ 100: Reserved ■ 101: DMA Write channel LL Element MRd completion ■ 110: Reserved ■ 111: DMA Read channel LL Element MRd completion Exists: (!(AMBA_INTERFACE!=0)) && (CC_DMA_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted

Port Name	I/O	Description
client0_tlp_dma_channel[((NVC_XALI_EXP*CC_NUM_DMA_MAX_CHAN_WIDTH)-1):0]	I	You must set this input to the DMA channel number. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CC_DMA_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted

2.12 XALI1 Interface Signals

client1_addr_align_en	-	xadm_client1_halt
client1_tlp_byte_en	-	
client1_remote_req_id	-	
client1_cpl_status	-	
client1_cpl_bcm	-	
client1_cpl_byte_cnt	-	
client1_tlp_dv	-	
client1_tlp_eot	-	
client1_tlp_bad_eot	-	
client1_tlp_hv	-	
client1_tlp_fmt	-	
client1_tlp_type	-	
client1_tlp_tc	-	
client1_tlp_td	-	
client1_tlp_ep	-	
client1_tlp_attr	-	
client1_tlp_ln	-	
client1_tlp_byte_len	-	
client1_tlp_tid	-	
client1_tlp_addr	-	
client1_hdr_prot	-	
client1_tlp_data	-	
client1_tlp_func_num	-	
client1_tlp_vfunc_num	-	
client1_tlp_vfunc_active	-	
client1_tlp_ats	-	
client1_tlp_nw	-	
client1_tlp_th	-	
client1_tlp_ph	-	
client1_tlp_st	-	
client1_tlp_prfx	-	
client1_tlp_atu_bypass	-	

Table 2-12 XALI1 Interface Signals

Port Name	I/O	Description
client1_addr_align_en[(NVC_XALI_EXP-1):0]	I	When client1_addr_align_en is asserted, the controller performs address alignment and generates the first and last byte enables based on the address and number of bytes of the TLP requested from the client interface. When client1_addr_align_en is de-asserted (always for completions), the controller takes the first and last byte enables from client1_tlp_byte_en and does not address-align the data. For more information, see the "Transmit Address Alignment" section in the "Signal Descriptions" chapter of the <i>Databook</i>. When you do not select the Enable ECRC Support (CX_ECRC_ENABLE) option, and your application is generating the ECRC, do not assert client1_addr_align_en for TLPs that contain application-generated ECRC. When the TLP from your application does not contain ECRC, then your application can assert client1_addr_align_en to enable address alignment. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_byte_en[(NVC_XALI_EXP*8)-1]:0]	I	Byte enables for the first and last dword of the TLP: ■ [3:0]: Byte enables for the first dword ■ [7:4]: Byte enables for the last dword A TLP can have 'holes' between bytes when the TLP is less than eight bytes (2 dwords) long, according to the <i>PCI Express Specification</i> (not all valid bytes need to be contiguous). However, a TLP is not allowed to have holes between bytes when the TLP is more than eight bytes. Not used (set to 0x0) when 'Transmit Address Alignment' is (client1_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1). When sending a message, your application asserts the message code onto these signals. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted. Not valid when client1_addr_align_en is asserted.

Port Name	I/O	Description
client1_remote_req_id[((NVC_XALI_EXP*16)-1):0]	I	Client1 requester ID to be used in the TLP header of a completion TLP. The value must match the requester ID of the corresponding incoming request: ■ [15:8]: Bus number ■ [7:3]: Device number ■ [2:0]: Function number Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_cpl_status[((NVC_XALI_EXP*3)-1):0]	I	The value to be used in the completion Status field of a completion TLP header on XALI1: ■ 000b: Successful completion ■ 001b: Unsupported request ■ 010b: Configuration request retry status ■ 100b: Completer abort Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted. Only used when XALI1 is connected to a completer.
client1_cpl_bcm[((NVC_XALI_EXP-1):0]	I	Byte Count Modified field on XALI1. Must be hardwired to '0'. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted

Port Name	I/O	Description
client1_cpl_byte_cnt[((NVC_XALI_EXP*12)-1):0]	I	The value to be used for the Byte Count field of a memory read completion TLP. The value client1_cpl_byte_cnt indicates the number of bytes remaining to be delivered for the request, as defined in the PCI Express Specification: ■ 001h: 1 byte ■ FFFh: 4095 bytes ■ 000h: 4096 bytes When you select the TRGT_CPL_LUT_EN configuration option, the controller does not use client1_cpl_byte_cnt for completions. Instead, the controller takes the current Byte Count value from the target completion LUT entry specified the client1/1/2_cpl_lookup_id input. For more information, see 'Target Completion Lookup Table'. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted. Only used when XALI1 is connected to a completer.
client1_tlp_dv[(NVC_XALI_EXP-1):0]	I	Indicates that the TLP payload data client1_tlp_data is valid. Your application must not assert client1_tlp_dv for TLPs that do not include a payload, such as memory read requests. Your application must present the first payload data the same cycle with the TLP header. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
client1_tlp_eot[(NVC_XALI_EXP-1):0]	I	Indicates the end of TLP payload data for a TLP that includes a payload. For a TLP that does not include a payload (such as a memory read request), your application must assert and de-assert client1_tlp_eot at the same time as client1_tlp_hv. You must not assert it at any other time. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
client1_tlp_bad_eot[(NVC_XALI_EXP-1):0]	I	Indicates that the current TLP must be nullified. To nullify a TLP, your application must assert client1_tlp_bad_eot in the same cycle as client_tlp_eot. When client1_tlp_bad_eot is asserted, the controller nullifies the TLP (inverts the LCRC and inserts EDB as the final framing symbol in the transmitted TLP). Your application can use client1_tlp_bad_eot to nullify a TLP because of a known error condition, or to abort the current transaction when the current transaction is being blocked (for example, due to lack of FC credits): ■ To cancel a request that has not been granted (xadm_client1_halt has never been de-asserted for the transfer), the CLIENT_PULLBACK feature from coreConsultant must be selected. In addition, the following signals must be driven as follows. Assert client1_tlp_bad_eot and client1_tlp_eot for one clock cycle while xadm_client1_halt is asserted (indicating that the request has not been granted), then de-assert client1_tlp_hv, client1_tlp_dv, client1_tlp_bad_eot, and client1_tlp_eot to terminate the request. Without the CLIENT_PULLBACK feature, your application has to send the full packet and mark the EOT and BAD_EOT at the end. With the CLIENT_PULLBACK feature, your application can mark EOT and BAD_EOT when it realizes that it needs to nullify the TLP, without having to present the full packet. ■ To cancel a request that is in progress (xadm_client1_halt has been de-asserted for the transfer), assert client1_tlp_bad_eot and client1_tlp_eot for one clock cycle while xadm_client1_halt is de-asserted (indicating that the request has been granted), then de-assert client1_tlp_hv, client1_tlp_dv, client1_tlp_bad_eot, and client1_tlp_eot to terminate the request. In this case, the controller nullifies the TLP as described in this section. ■ When the TLP is a one-cycle request (client1_tlp_hv and client1_tlp_eot asserted in the same cycle) and client1_tlp_bad_eot is also asserted, this request is ignored. Sending a nullified TLP is not very useful because the link partner has to ignore it anyway. Your application should realize that this request is nullified (in this case ignored) and de-assert client1_tlp_hv and client1_tlp_eot while keeping client1_tlp_bad_eot asserted. ■ Nullify Atomic Ops TLP is not supported. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_eot is asserted
client1_tlp_hv[(NVC_XALI_EXP-1):0]	I	Indicates that the TLP header data the TLP header ports is valid. The header data must remain valid until client1_tlp_hv is de-asserted. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) Synchronous To: aux_clk,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
client1_tlp_fmt[((NVC_XALI_EXP*2)-1):0]	I	The value to be used for the Format field of the TLP header on XALI1: ■ 00b: 3 dword header, no data ■ 01b: 4 dword header, no data ■ 10b: 3 dword header, with data ■ 11b: 4 dword header, with data Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_type[((NVC_XALI_EXP*5)-1):0]	I	The value to be used for the Type field of the client1 TLP header, as defined in the <i>PCI Express Specification</i>. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_tc[((NVC_XALI_EXP*3)-1):0]	I	The value to be used for the Traffic Class (TC) field of the TLP header (000b'111b) on XALI1. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_td[(NVC_XALI_EXP-1):0]	I	TLP Digest bit on XALI1. It is expected that the ECRC is not present, and the controller generates and inserts it. You should set this input to 0. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted

Port Name	I/O	Description
client1_tlp_ep[(NVC_XALI_EXP-1):0]	I	When asserted, the controller sets the Poisoned TLP (EP) bit in the TLP header on XALI1. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_attr[(((NVC_XALI_EXP*ATTR_WD)-1):0)]	I	Attributes field for the TLP header ■ Bit 0: No Snoop ■ Bit 1: Relaxed Ordering ■ Bit 2: IDO attribute ■ Width is three bits when CX_IDO_ENABLE =1 ■ Width is two bits when CX_IDO_ENABLE =0 When you select the TRGT_CPL_LUT_EN configuration option, the controller does not use client1_tlp_attr for completions. Instead, the controller takes the attr value from the target completion LUT entry specified the client2/1/2_cpl_lookup_id input. For more information, see 'Target Completion Lookup Table'. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_ln[(NVC_XALI_EXP-1):0]	I	Description: This signal indicates that the controller can accept LN Read, LN Write or LN Completion, or special zero-Length LN Reads or LN Writes, on XALI1 Interface. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT1_POPULATED}) \ \&\& \ (\text{CX_LN_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted

Port Name	I/O	Description
client1_tlp_byte_len[((NVC_XALI_EXP*13)-1):0]	I	The number of bytes in the TLP (not including ECRC). The value should be in the range 0x0000 (0) to 0x1000 (4096) When 'Transmit Address Alignment' is (client1_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1), this must be the full byte length of the request. When transmit alignment is off (client1_addr_align_en=0), bits [1:0] must be set to '00'. The byte length value must be rounded up to the next dword boundary. For example, when the request is five bytes long, the value of bits [12:0] must be eight. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_tid[((NVC_XALI_EXP*TAG_SIZE)-1):0]	I	Tag field for the TLP header. Your application must provide a valid Tag client1_tlp_tid for all posted and non-posted requests. The value of the CX_MAX_TAG configuration option determines the maximum number of outstanding Client requests. Your application is expected to manage the Tag for every posted and non-posted request. For completions, the client1_tlp_tid value must be the Tag from the corresponding request. When you select the TRGT_CPL_LUT_EN configuration option, the controller does not use client1_tlp_tid for completions. Instead, the controller takes the tag value from the target completion LUT entry. For more information, see 'Target Completion Lookup Table'. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted

Port Name	I/O	Description
client1_tlp_addr[((NVC_XALI_EXP*64)-1):0]	I	The Address field for the TLP header. The TLP address can be 64 or 32 bits. When sending completions, the controller uses client1_tlp_addr[6:0] for the Lower Address field of the completion TLP. The bits in this address bus map directly to the address bits in the TLP header. The third and fourth dwords of the header are mapped as follows: - MEM (32-bit) / IO / CFG have a 3-dword header. Third dword (bytes 8-11) =client1_tlp_addr[31:0]. Therefore, bits [7:0] are mapped to byte 11 which is the lowest eight bits of the TLP header address. For more information, see the 'Endianness' advanced information chapter. - MEM (64-bit) / MSG have a 4-dword header. Third dword (bytes 8-11) =client1_tlp_addr[63:32]. Fourth dword (bytes 12-15) =client1_tlp_addr[31:0]. Therefore, bits [7:0] are mapped to byte 15 which is the lowest eight bits of the TLP header address. For more information, see the 'Endianness' advanced information chapter. The base width of client1_tlp_addr is 64 bits. When 'Transmit Address Alignment' is (client1_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1), this address is the full byte address of the first enabled byte of the request. That is, the address is byte-aligned. When transmit alignment is off (client1_addr_align_en=0), bits [1:0] must be '00', that is, the address must be dword-aligned. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_hdr_prot[((NVC_XALI_EXP*CLIENT_HDR_PROT_WD)-1):0]	I	ECC or parity bits (protection code) for the XALI1 header. Your application must calculate the protection code over all of the XALI1 header inputs in this order: client1_tlp_th, client1_tlp_st, client1_tlp_ph, client1_tlp_vfunc_active, client1_tlp_vfunc_num, client1_tlp_ln, client1_tlp_func_num, client1_tlp_attr, client1_addr_align_en, client1_tlp_byte_en, client1_remote_req_id, client1_cpl_status, client1_cpl_bcm, client1_cpl_byte_cnt, client1_tlp_tid, client1_tlp_byte_len, client1_tlp_ep, client1_tlp_td, client1_tlp_tc, client1_tlp_type, client1_tlp_fmt, client1_tlp_addr[63:0]. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) && (((CX_RASDP==0)? 0: 1)) Synchronous To: core_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted

Port Name	I/O	Description
client1_tlp_data[((NVC_XALI_EXP*DW_W _PAR)-1):0]	I	The data payload for the TLP. The data is in little endian format. The first transmitted payload byte is in [7:0]. The base width of client1_tlp_data is your application datapath width, which is automatically determined according to the selected operating frequency and number of lanes. ■ For a 32-bit controller, the base width is 32 (client1_tlp_data[31:0]) ■ For a 64-bit controller, the base width is 64 (client1_tlp_data[63:0]) ■ For a 128-bit controller, the base width is 128 (client1_tlp_data[127:0]) ■ For a 256-bit controller, the base width is 256 (client1_tlp_data[255:0]) When 'Transmit Address Alignment' is (client1_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1), the first enabled byte of the request must be bits [7:0] of this data bus. The controller up-shifts the data in the transmitted TLP based the lower two bits of the byte address. Your application must append protection code bits to the most significant bits of this bus when ECC or parity datapath protection is used. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_dv is asserted
client1_tlp_func_num[((NVC_XALI_EXP*P F_WD)-1):0]	I	Indicates from which physical function (PF) the request is coming. The controller uses client1_tlp_func_num to form the requester ID for requests and the completer ID for completions. Function numbering starts at '0'. For a single-function device, set client1_tlp_func_num to 000b. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) Synchronous To: None,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted

Port Name	I/O	Description
client1_tlp_vfunc_num[((NVC_XALI_EXP* (VF_WD-1))-1):0]	I	Indicates from which virtual function (VF) the request is coming. The controller uses this signal to form the requester ID for requests and the completer ID for completions. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, client1_tlp_vfunc_num=0 refers to the first VF of the PF that is identified by client1_tlp_func_num. For more information of VF numbering, see 'Programming View'. Exists: $!(\text{AMBA_INTERFACE}!=0) \ \&\& \ (\text{CLIENT1_POPULATED}) \ \&\& \ (\text{CX_SRIOV_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv and client1_tlp_vfunc_active are asserted
client1_tlp_vfunc_active[(NVC_XALI_EXP-1):0]	I	Indicates that the TLP request is coming from a VF (Virtual function). 0: No VF is active and client1_tlp_vfunc_num is invalid. A PF is valid and identified by client1_tlp_func_num. 1: A VF is active and is identified by client1_tlp_vfunc_num. Exists: $!(\text{AMBA_INTERFACE}!=0) \ \&\& \ (\text{CLIENT1_POPULATED}) \ \&\& \ (\text{CX_SRIOV_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_ats[((NVC_XALI_EXP*2)-1):0]	I	AT field in the TLP header on XALI1. Exists: $!(\text{AMBA_INTERFACE}!=0) \ \&\& \ (\text{CLIENT1_POPULATED}) \ \&\& \ (\text{ATS_TX_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_nw[(NVC_XALI_EXP-1):0]	I	NW field in the TLP header on XALI1. Exists: $!(\text{AMBA_INTERFACE}!=0) \ \&\& \ (\text{CLIENT1_POPULATED}) \ \&\& \ (\text{ATS_TX_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted

Port Name	I/O	Description
client1_tlp_th[(NVC_XALI_EXP-1):0]	I	TH field in the TLP header on XALI1. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CLIENT1_POPULATED}) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_ph[((NVC_XALI_EXP*2)-1):0]	I	PH field in the TLP header on XALI1. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CLIENT1_POPULATED}) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_st[((NVC_XALI_EXP*8)-1):0]	I	ST field in the TLP header on XALI1. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CLIENT1_POPULATED}) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
xadm_client1_halt[(NVC_XALI_EXP-1):0]	O	The controller asserts xadm_client1_halt when it is not ready to process a transaction. The header and payload input for client1 must remain unchanged while xadm_client1_halt is asserted. The conditions that cause the controller to assert xadm_client1_halt include: ■ The XADM is busy servicing one of the other transmit client interfaces. ■ There are not enough Flow Control credits available to send the requested TLP type (P, non-posted, or CPL). ■ The LTSSM is not ready. ■ A retry is in progress. ■ ECRC or LCRC insertion is in progress. <i>Note:</i> xadm_client1_halt is asserted by default. For more information, see the timing diagrams in "Signal Interfaces" chapter of the <i>Databook</i>. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CLIENT1_POPULATED})$ Synchronous To: aux_clk,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
client1_tlp_prfx[((NVC_XALI_EXP*PRFX_DW)-1):0]	I	TLP prefixes. The field [31:0] represents the first prefix to be transmitted. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that client1_tlp_prfx[31:0] has prefix byte #0 in the lower byte position of the dword client1_tlp_prfx[31:0]. That is, FMT =bits 7:5, and Type =bits 4:0. Your application is responsible for passing Local and End-End prefixes in the correct order for transmission. Local prefixes must precede End-End prefixes. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) && (CX_TLP_PREFIX_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client1_tlp_hv is asserted
client1_tlp_atu_bypass[(NVC_XALI_EXP-1):0]	I	Internal ATU Bypass for XALI1. When set it indicates that this transaction should not be processed by the internal address translation unit Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) && (CX_INTERNAL_ATU_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted

2.13 XALI2 Interface Signals

client2_addr_align_en	-	xadm_client2_halt
client2_tlp_byte_en	-	
client2_remote_req_id	-	
client2_cpl_status	-	
client2_cpl_bcm	-	
client2_cpl_byte_cnt	-	
client2_tlp_dv	-	
client2_tlp_eot	-	
client2_tlp_bad_eot	-	
client2_tlp_hv	-	
client2_tlp_fmt	-	
client2_tlp_type	-	
client2_tlp_tc	-	
client2_tlp_td	-	
client2_tlp_ep	-	
client2_tlp_attr	-	
client2_tlp_ln	-	
client2_tlp_byte_len	-	
client2_tlp_tid	-	
client2_tlp_addr	-	
client2_hdr_prot	-	
client2_tlp_data	-	
client2_tlp_func_num	-	
client2_tlp_vfunc_num	-	
client2_tlp_vfunc_active	-	
client2_tlp_ats	-	
client2_tlp_nw	-	
client2_tlp_th	-	
client2_tlp_ph	-	
client2_tlp_st	-	
client2_tlp_prfx	-	
client2_tlp_atu_bypass	-	

Table 2-13 XALI2 Interface Signals

Port Name	I/O	Description
client2_addr_align_en[(NVC_XALI_EXP-1):0]	I	When client2_addr_align_en is asserted, the controller performs address alignment and generates the first and last byte enables based on the address and number of bytes of the TLP requested from the client interface. When client2_addr_align_en is de-asserted (always for completions), the controller takes the first and last byte enables from client2_tlp_byte_en and does not address-align the data. For more information, see the "Transmit Address Alignment" section in the "Signal Descriptions" chapter of the <i>Databook</i>. When you do not select the Enable ECRC Support (CX_ECRC_ENABLE) option, and your application is generating the ECRC, do not assert client2_addr_align_en for TLPs that contain application-generated ECRC. When the TLP from your application does not contain ECRC, then your application can assert client2_addr_align_en to enable address alignment. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_byte_en[(NVC_XALI_EXP*8)-1]:0]	I	Byte enables for the first and last dword of the TLP: ■ [3:0]: Byte enables for the first dword ■ [7:4]: Byte enables for the last dword A TLP can have 'holes' between bytes when the TLP is less than eight bytes (2 dwords) long, according to the <i>PCI Express Specification</i> (not all valid bytes need to be contiguous). However, a TLP is not allowed to have holes between bytes when the TLP is more than eight bytes. Not used (set to 0x0) when 'Transmit Address Alignment' is (client2_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1). When sending a message, your application asserts the message code onto these signals. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted. Not valid when client2_addr_align_en is asserted.

Port Name	I/O	Description
client2_remote_req_id[((NVC_XALI_EXP*16)-1):0]	I	Client2 requester ID to be used in the TLP header of a completion TLP. The value must match the requester ID of the corresponding incoming request: ■ [15:8]: Bus number ■ [7:3]: Device number ■ [2:0]: Function number Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_cpl_status[((NVC_XALI_EXP*3)-1):0]	I	The value to be used in the completion Status field of a completion TLP header: ■ 000b: Successful completion ■ 001b: Unsupported request ■ 010b: Configuration request retry status ■ 100b: Completer abort Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted. Only used when XALI2 is connected to a completer.
client2_cpl_bcm[(NVC_XALI_EXP-1):0]	I	Byte Count Modified field on XALI2. Must be hardwired to '0'. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted

Port Name	I/O	Description
client2_cpl_byte_cnt[((NVC_XALI_EXP*12)-1):0]	I	The value to be used for the Byte Count field of a memory read completion TLP. The value client2_cpl_byte_cnt indicates the number of bytes remaining to be delivered for the request, as defined in the PCI Express Specification: ■ 001h: 1 byte ■ FFFh: 4095 bytes ■ 000h: 4096 bytes When you select the TRGT_CPL_LUT_EN configuration option, the controller does not use client2_cpl_byte_cnt for completions. Instead, the controller takes the current Byte Count value from the target completion LUT entry specified the client2/1/2_cpl_lookup_id input. For more information, see 'Target Completion Lookup Table'. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted. Only used when XALI2 is connected to a completer.
client2_tlp_dv[(NVC_XALI_EXP-1):0]	I	Indicates that the TLP payload data client2_tlp_data is valid. Your application must not assert client2_tlp_dv for TLPs that do not include a payload, such as memory read requests. Your application must present the first payload data the same cycle with the TLP header. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
client2_tlp_eot[(NVC_XALI_EXP-1):0]	I	Indicates the end of TLP payload data for a TLP that includes a payload. For a TLP that does not include a payload (such as a memory read request), your application must assert and de-assert client2_tlp_eot at the same time as client2_tlp_hv. You must not assert it at any other time. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
client2_tlp_bad_eot[(NVC_XALI_EXP-1):0]]	I	Indicates that the current TLP must be nullified. To nullify a TLP, your application must assert client2_tlp_bad_eot in the same cycle as client_tlp_eot. When client2_tlp_bad_eot is asserted, the controller nullifies the TLP (inverts the LCRC and inserts EDB as the final framing symbol in the transmitted TLP). Your application can use client2_tlp_bad_eot to nullify a TLP because of a known error condition, or to abort the current transaction when the current transaction is being blocked (for example, due to lack of FC credits): ■ To cancel a request that has not been granted (xadm_client2_halt has never been de-asserted for the transfer), the CLIENT_PULLBACK feature from coreConsultant must be selected. In addition, the following signals must be driven as follows. Assert client2_tlp_bad_eot and client2_tlp_eot for one clock cycle while xadm_client2_halt is asserted (indicating that the request has not been granted), then de-assert client2_tlp_hv, client2_tlp_dv, client2_tlp_bad_eot, and client2_tlp_eot to terminate the request. Without the CLIENT_PULLBACK feature, your application has to send the full packet and mark the EOT and BAD_EOT at the end. With the CLIENT_PULLBACK feature, your application can mark EOT and BAD_EOT when it realizes that it needs to nullify the TLP, without having to present the full packet. ■ To cancel a request that is in progress (xadm_client2_halt has been de-asserted for the transfer), assert client2_tlp_bad_eot and client2_tlp_eot for one clock cycle while xadm_client2_halt is de-asserted (indicating that the request has been granted), then de-assert client2_tlp_hv, client2_tlp_dv, client2_tlp_bad_eot, and client2_tlp_eot to terminate the request. In this case, the controller nullifies the TLP as described in this section. ■ When the TLP is a one-cycle request (client2_tlp_hv and client2_tlp_eot asserted in the same cycle) and client2_tlp_bad_eot is also asserted, this request is ignored. Sending a nullified TLP is not very useful because the link partner has to ignore it anyway. Your application should realize that this request is nullified (in this case ignored) and de-assert client2_tlp_hv and client2_tlp_eot while keeping client2_tlp_bad_eot asserted. ■ Nullify Atomic Ops TLP is not supported. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_eot is asserted
client2_tlp_hv[(NVC_XALI_EXP-1):0]	I	Indicates that the TLP header data the TLP header ports is valid. The header data must remain valid until client2_tlp_hv is de-asserted. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) Synchronous To: aux_clk,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
client2_tlp_fmt[((NVC_XALI_EXP*2)-1):0]	I	The value to be used for the Format field of the TLP header: ■ 00b: 3 dword header, no data ■ 01b: 4 dword header, no data ■ 10b: 3 dword header, with data ■ 11b: 4 dword header, with data Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_type[((NVC_XALI_EXP*5)-1):0]	I	The value to be used for the Type field of the client2 TLP header, as defined in the <i>PCI Express Specification</i>. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_tc[((NVC_XALI_EXP*3)-1):0]	I	The value to be used for the Traffic Class (TC) field of the TLP header (000b'111b) on XALI2. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_td[(NVC_XALI_EXP-1):0]	I	TLP Digest bit. It is expected that the ECRC is not present, and the controller generates and inserts it. You should set this input to 0. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted

Port Name	I/O	Description
client2_tlp_ep[(NVC_XALI_EXP-1):0]	I	When asserted, the controller sets the Poisoned TLP (EP) bit in the TLP header on XALI2. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_attr[(((NVC_XALI_EXP*ATTR_WD)-1):0)]	I	Attributes field for the TLP header ■ Bit 0: No Snoop ■ Bit 1: Relaxed Ordering ■ Bit 2: IDO attribute ■ Width is three bits when CX_IDO_ENABLE =1 ■ Width is two bits when CX_IDO_ENABLE =0 When you select the TRGT_CPL_LUT_EN configuration option, the controller does not use client2_tlp_attr for completions. Instead, the controller takes the attr value from the target completion LUT entry specified the client2/1/2_cpl_lookup_id input. For more information, see 'Target Completion Lookup Table'. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_ln[(NVC_XALI_EXP-1):0]	I	Description: This signal indicates that the controller can handle LN Read, LN Write or LN Completion, or special zero-Length LN Reads or LN Writes, on XALI2 Interface. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CLIENT2_POPULATED}) \ \&\& \ (\text{CX_LN_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted

Port Name	I/O	Description
client2_tlp_byte_len[((NVC_XALI_EXP*13)-1):0]	I	The number of bytes in the TLP (not including ECRC). The value should be in the range 0x0000 (0) to 0x1000 (4096) When 'Transmit Address Alignment' is (client2_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1), this must be the full byte length of the request. When transmit alignment is off (client2_addr_align_en=0), bits [1:0] must be set to '00'. The byte length value must be rounded up to the next dword boundary. For example, when the request is five bytes long, the value of bits [12:0] must be eight. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_tid[((NVC_XALI_EXP*TAG_SIZE)-1):0]	I	Tag field for the TLP header. Your application must provide a valid Tag client2_tlp_tid for all posted and non-posted requests. The value of the CX_MAX_TAG configuration option determines the maximum number of outstanding Client requests. Your application is expected to manage the Tag for every posted and non-posted request. For completions, the client2_tlp_tid value must be the Tag from the corresponding request. When you select the TRGT_CPL_LUT_EN configuration option, the controller does not use client2_tlp_tid for completions. Instead, the controller takes the tag value from the target completion LUT entry. For more information, see 'Target Completion Lookup Table'. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted

Port Name	I/O	Description
client2_tlp_addr[((NVC_XALI_EXP*64)-1):0]	I	The Address field for the TLP header. The TLP address can be 64 or 32 bits. When sending completions, the controller uses client2_tlp_addr[6:0] for the Lower Address field of the completion TLP. The bits in this address bus map directly to the address bits in the TLP header. The third and fourth dwords of the header are mapped as follows: ■ MEM (32-bit) / IO / CFG have a 3-dword header. Third dword (bytes 8-11) =client2_tlp_addr[31:0]. Therefore, bits [7:0] are mapped to byte 11 which is the lowest eight bits of the TLP header address. For more information, see the 'Endianness' advanced information chapter. ■ MEM (64-bit) / MSG have a 4-dword header. Third dword (bytes 8-11) =client2_tlp_addr[63:32]. Fourth dword (bytes 12-15) =client2_tlp_addr[31:0]. Therefore, bits [7:0] are mapped to byte 15 which is the lowest eight bits of the TLP header address. For more information, see the 'Endianness' advanced information chapter. The base width of client2_tlp_addr is 64 bits. When 'Transmit Address Alignment' is (client2_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1), this address is the full byte address of the first enabled byte of the request. That is, the address is byte-aligned. When transmit alignment is off (client2_addr_align_en=0), bits [1:0] must be '00', that is, the address must be dword-aligned. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_hdr_prot[((NVC_XALI_EXP*CLIENT_HDR_PROT_WD)-1):0]	I	ECC or parity bits (protection code) for the XALI2 header. Your application must calculate the protection code over all of the XALI2 header inputs in this order: client2_tlp_th, client2_tlp_st, client2_tlp_ph, client2_tlp_vfunc_active, client2_tlp_vfunc_num, client2_tlp_ln, client2_tlp_func_num, client2_tlp_attr, client2_addr_align_en, client2_tlp_byte_en, client2_remote_req_id, client2_cpl_status, client2_cpl_bcm, client2_cpl_byte_cnt, client2_tlp_tid, client2_tlp_byte_len, client2_tlp_ep, client2_tlp_td, client2_tlp_tc, client2_tlp_type, client2_tlp_fmt, client2_tlp_addr[63:0]. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) && (((CX_RASDP==0)? 0: 1)) Synchronous To: core_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted

Port Name	I/O	Description
client2_tlp_data[((NVC_XALI_EXP*DW_W _PAR)-1):0]	I	The data payload for the TLP. The data is in little endian format. The first transmitted payload byte is in [7:0]. The base width of client2_tlp_data is your application datapath width, which is automatically determined according to the selected operating frequency and number of lanes. ■ For a 32-bit controller, the base width is 32 (client2_tlp_data[31:0]) ■ For a 64-bit controller, the base width is 64 (client2_tlp_data[63:0]) ■ For a 128-bit controller, the base width is 128 (client2_tlp_data[127:0]) ■ For a 256-bit controller, the base width is 256 (client2_tlp_data[255:0]) When 'Transmit Address Alignment' is (client2_addr_align_en =1 and GLOB_ADDR_ALIGN_EN =1), the first enabled byte of the request must be bits [7:0] of this data bus. The controller up-shifts the data in the transmitted TLP based the lower two bits of the byte address. Your application must append protection code bits to the most significant bits of this bus when ECC or parity datapath protection is used. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_dv is asserted
client2_tlp_func_num[((NVC_XALI_EXP*P F_WD)-1):0]	I	Indicates from which physical function (PF) the request is coming. The controller uses client2_tlp_func_num to form the requester ID for requests and the completer ID for completions. Function numbering starts at '0'. For a single-function device, set client2_tlp_func_num to 000b. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) Synchronous To: None,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted

Port Name	I/O	Description
client2_tlp_vfunc_num[((NVC_XALI_EXP* (VF_WD-1))-1):0]	I	Indicates from which virtual function (VF) the request is coming. The controller uses this signal to form the requester ID for requests and the completer ID for completions. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, client2_tlp_vfunc_num=0 refers to the first VF of the PF that is identified by client2_tlp_func_num. For more information of VF numbering, see 'Programming View'. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CLIENT2_POPULATED}) \ \&\& \ (\text{CX_SRIOV_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv and client2_tlp_vfunc_active are asserted
client2_tlp_vfunc_active[(NVC_XALI_EXP-1):0]	I	Indicates that the TLP request is coming from a VF (Virtual function). 0: No VF is active and client2_tlp_vfunc_num is invalid. A PF is valid and identified by client2_tlp_func_num. 1: A VF is active and is identified by client2_tlp_vfunc_num. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CLIENT2_POPULATED}) \ \&\& \ (\text{CX_SRIOV_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_ats[((NVC_XALI_EXP*2)-1):0]	I	AT field in the TLP header on XALI2. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CLIENT2_POPULATED}) \ \&\& \ (\text{ATS_TX_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_nw[(NVC_XALI_EXP-1):0]	I	NW field in the TLP header on XALI2. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{CLIENT2_POPULATED}) \ \&\& \ (\text{ATS_TX_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted

Port Name	I/O	Description
client2_tlp_th[(NVC_XALI_EXP-1):0]	I	TH field in the TLP header on XALI2. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT2_POPULATED}) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_ph[((NVC_XALI_EXP*2)-1):0]	I	PH field in the TLP header on XALI2. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT2_POPULATED}) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_st[((NVC_XALI_EXP*8)-1):0]	I	ST field in the TLP header on XALI2. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT2_POPULATED}) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
xadm_client2_halt[(NVC_XALI_EXP-1):0]	O	The controller asserts xadm_client2_halt when it is not ready to process a transaction. The header and payload input for client2 must remain unchanged while xadm_client2_halt is asserted. The conditions that cause the controller to assert xadm_client2_halt include: ■ The XADM is busy servicing one of the other transmit client interfaces. ■ There are not enough Flow Control credits available to send the requested TLP type (P, non-posted, or CPL). ■ The LTSSM is not ready. ■ A retry is in progress. ■ ECRC or LCRC insertion is in progress. <i>Note:</i> xadm_client2_halt is asserted by default. For more information, see the timing diagrams in "Signal Interfaces" chapter of the <i>Databook</i>. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{CLIENT2_POPULATED})$ Synchronous To: aux_clk,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
client2_tlp_prfx[((NVC_XALI_EXP*PRFX_DW)-1):0]	I	TLP prefixes. The field [31:0] represents the first prefix to be transmitted. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that client2_tlp_prfx[31:0] has prefix byte #0 in the lower byte position of the dword client2_tlp_prfx[31:0]. That is, FMT =bits 7:5, and Type =bits 4:0. Your application is responsible for passing Local and End-End prefixes in the correct order for transmission. Local prefixes must precede End-End prefixes. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) && (CX_TLP_PREFIX_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client2_tlp_hv is asserted
client2_tlp_atu_bypass[(NVC_XALI_EXP-1):0]	I	Internal ATU Bypass for XALI2. When set it indicates that this transaction should not be processed by the internal address translation unit Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) && (CX_INTERNAL_ATU_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: client0_tlp_hv is asserted

2.14 XALI_CCIX Interface Signals

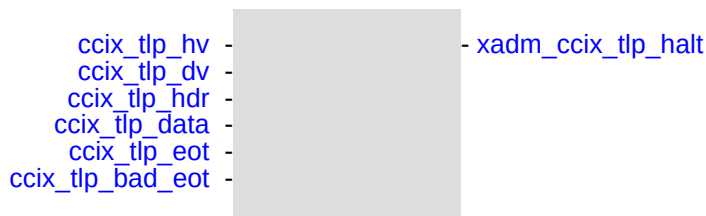


Table 2-14 XALI_CCIX Interface Signals

Port Name	I/O	Description
ccix_tlp_hv	I	Indicates that the TLP header data in the TLP header ports is valid. The header data must remain valid until ccix_tlp_hv is de-asserted. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: aux_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ccix_tlp_dv	I	Indicates that the TLP payload data ccix_tlp_data is valid. Your application must not assert ccix_tlp_dv for TLPs that do not include a payload. Your application must present the first payload data the same cycle with the TLP header. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ccix_tlp_hdr[(CX_CCIX_HDR_WD-1):0]	I	Header for the TLP being presented in the CCIX interface. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: ccix_tlp_hv is asserted

Port Name	I/O	Description
ccix_tlp_data[(CX_CCIX_DATA_WD-1):0]	I	Payload data in CCIX interfaces. Your application must not assert <code>ccix_tlp_dv</code> for TLPs that do not include a payload. Your application must present the first payload data the same cycle with the TLP header. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
ccix_tlp_eot	I	Indicates the last segment of the TLP in the CCIX dedicated interface. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: <code>ccix_tlp_hv</code> or <code>ccix_tlp_dv</code> is asserted
ccix_tlp_bad_eot	I	When asserted causes the TLP being sent to be nullified. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: <code>ccix_tlp_eot</code> is asserted
xadm_ccix_tlp_halt	O	Halt the CCIX interface. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: aux_clk,aux_clk_g,{core_clk} Registered: No Power Domain: PD_VAUX Active State: High Validated by: not validated by any other signal

2.15 ADM adaptor Interface Signals

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xadm_rfc_in -> radm_rfc_out
rtlh_radm_in -> xadm_xtlh_out
cx_pl_ltssm_emu_in -> cx_pl_ltssm_emu_out

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Table 2-15 ADM adaptor Interface Signals

Port Name	I/O	Description
xadm_rfc_in[(XADM_RFC_IN_WD-1):0]	I	xadm input FC received from radm. Exists: CX_ADM_ADAPTOR_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_xtlh_out[(XADM_XTLH_OUT_WD-1):0]	O	xadm transmits TLP traffic to radm. Exists: CX_ADM_ADAPTOR_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
rtlh_radm_in[(RTLH_RAMD_IN_WD-1):0]	I	radm receives TLP traffic from xadm. Exists: CX_ADM_ADAPTOR_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_rfc_out[(RADM_RFC_OUT_WD-1):0]	O	radm output FC transmitted to xadm. Exists: CX_ADM_ADAPTOR_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cx_pl_ltssm_emu_in[(LTSSM_EMU_IN_WD-1):0]	I	LTSSM emu input signals. Exists: CX_ADM_ADAPTOR_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cx_pl_ltssm_emu_out[(LTSSM_EMU_OUT_WD-1):0]	O	LTSSM emu output signals Exists: CX_ADM_ADAPTOR_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.16 Bypass Interface Signals

- radm_bypass_data
- radm_bypass_dwen
- radm_bypass_dv
- radm_bypass_hv
- radm_bypass_eot
- radm_bypass_dllp_abort
- radm_bypass_tlp_abort
- radm_bypass_ecrc_err
- radm_bypass_hdr_prot
- radm_bypass_fmt
- radm_bypass_type
- radm_bypass_tc
- radm_bypass_attr
- radm_bypass_ln
- radm_bypass_reqid
- radm_bypass_tag
- radm_bypass_func_num
- radm_bypass_vfunc_num
- radm_bypass_vfunc_active
- radm_bypass_td
- radm_bypass_poisoned
- radm_bypass_dw_len
- radm_bypass_first_be
- radm_bypass_last_be
- radm_bypass_addr
- radm_bypass_rom_in_range
- radm_bypass_io_req_in_range
- radm_bypass_in_membar_range
- radm_bypass_bcm
- radm_bypass_cpl_last
- radm_bypass_cmpltr_id
- radm_bypass_byte_cnt
- radm_bypass_cpl_status
- radm_bypass_ats
- radm_bypass_th
- radm_bypass_ph
- radm_bypass_st
- radm_bypass_tlp_prfx

Table 2-16 Bypass Interface Signals

Port Name	I/O	Description
radm_bypass_data[(DW_W_PAR-1):0]	O	The payload data from the received TLP. The data is in little endian format. The first received payload byte is in [7:0]. The controller appends protection code bits to the most significant bits of this bus when ECC or parity datapath protection is used. Exists: $\neg(\text{AMBA_INTERFACE} \neq 0) \ \&\& \ ((\text{CX_RADMQ_MODE} == 2))$ Synchronous To: core_clk, {radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_dv is asserted

Port Name	I/O	Description
radm_bypass_dwen[(NW-1):0]	O	The databus dword enables identify the location of the last dword of the TLP the data bus (radm_bypass_data). The width and usage of radm_bypass_dwen depend the data width of the controller. For a 32-bit controller, radm_bypass_dwen is not used. For a 64-bit controller, NW =2. The radm_bypass_dwen[1:0] encoding is: ■ 01: Last dword at radm_bypass_data[31:0]. ■ 11: Last dword at radm_bypass_data[63:32]. For a 128-bit controller, NW =4. The radm_bypass_dwen[3:0] encoding is: ■ 0001: Last dword at radm_bypass_data[31:0]. ■ 0011: Last dword at radm_bypass_data[63:32]. ■ 0111: Last dword at radm_bypass_data[95:64]. ■ 1111: Last dword at radm_bypass_data[127:96]. For a 256-bit controller, NW =8. The radm_bypass_dwen[7:0] encoding is: ■ 00000001: Last dword at radm_bypass_data[31:0]. ■ 00000011: Last dword at radm_bypass_data[63:32]. ■ 00000111: Last dword at radm_bypass_data[95:64]. ■ 00001111: Last dword at radm_bypass_data[127:96]. ■ 00011111: Last dword at radm_bypass_data[159:128]. ■ 00111111: Last dword at radm_bypass_data[191:160]. ■ 01111111: Last dword at radm_bypass_data[223:192]. ■ 11111111: Last dword at radm_bypass_data[255:224]. All other encodings are not generated by the controller. Exists: (!(AMBA_INTERFACE!=0)) && ((CX_RADMQ_MODE==2)) Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_eot is asserted
radm_bypass_dv[(NHQ-1):0]	O	Indicates valid received payload data radm_bypass_data. Exists: (!(AMBA_INTERFACE!=0)) && ((CX_RADMQ_MODE==2)) Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
radm_bypass_hv[(NHQ-1):0]	O	Indicates that the header and prefix information radm_bypass_hdr is valid. The controller asserts radm_bypass_hv for one clock cycle. For bypassed queues, your application must capture the header information as soon as the radm_bypass_hv is asserted. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}=2))$ Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_bypass_eot[(NHQ-1):0]	O	Indicates the last cycle of valid payload data radm_bypass_data for the received TLP. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}=2))$ Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv or radm_bypass_dv is asserted
radm_bypass_dllp_abort[(NHQ-1):0]	O	Indicates to your application to drop the TLP on RBYP because of a Data Link Layer error such as LCRC or otherwise. You can expect the DLLP to be replayed. For more information, see the 'Advanced Error Handling For Received TLPs' advanced information chapter. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}=2))$ Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_eot
radm_bypass_tlp_abort[(NHQ-1):0]	O	Indicates to your application to drop the TLP because of malformed TLP on RBYP, ECRC error, or completion lookup failures (such as TAG or requester ID (RID) mismatches). You should not expect the TLP to be replayed. For more information, see the 'Advanced Error Handling For Received TLPs' advanced information chapter. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}=2))$ Synchronous To: None,aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_eot, radm_bypass_hv

Port Name	I/O	Description
radm_bypass_ecrc_err[(NHQ-1):0]	O	Indicates to your application to drop the TLP because of an ECRC error in the received TLP on RBYP. You should not expect the TLP to be replayed. For more information, see the 'Advanced Error Handling For Received TLPs' advanced information chapter. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_eot
radm_bypass_hdr_prot[((NHQ*RASDP_BYPASS_HDR_PROT_WD)-1):0]	O	ECC or parity bits (protection code) generated by the controller for the TLP header (bypass mode). Your application must calculate the protection code over all of the XALI0 header inputs in this order: radm_bypass_ats, radm_bypass_th, radm_bypass_st, radm_bypass_ph, radm_bypass_ln, radm_bypass_vfunc_active, radm_bypass_vfunc_num, radm_bypass_last_be, radm_bypass_cpl_last, radm_bypass_poisoned, radm_bypass_td, radm_bypass_addr, radm_bypass_first_be, radm_bypass_io_req_in_range, radm_bypass_rom_in_range, radm_bypass_in_membar_range, radm_bypass_dw_len, radm_bypass_cpl_status, radm_bypass_func_num, radm_bypass_tag, radm_bypass_reqid, radm_bypass_attr, radm_bypass_tc, radm_bypass_type, radm_bypass_fmt. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2)) \ \&\& \ (((\text{CX_RASDP}==0)? 0: 1))$ Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: radm_bypass_hv is asserted
radm_bypass_fmt[((NHQ*2)-1):0]	O	The Format field in the received TLP header on RBYP. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_type[((NHQ*5)-1):0]	O	The Type field in the received TLP header on RBYP. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted

Port Name	I/O	Description
radm_bypass_tc[((NHQ*3)-1):0]	O	The Traffic Class (TC) field in the received TLP header on RBYP. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_attr[((NHQ*ATTR_WD)-1):0]	O	The Attributes field in the received TLP header on RBYP. - Width is three bits when IDO_ENABLE =1 - Width is two bits when CX_IDO_ENABLE =0 Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_ln[(NHQ-1):0]	O	The Lightweight Notification (LN) field in the received TLP header on RBYP. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2)) \ \&\& \ (\text{CX_LN_ENABLE})$ Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_reqid[((NHQ*16)-1):0]	O	The requester ID in the received TLP header on RBYP. [15:8]: Bus number [7:3]: Device number [2:0]: Function number Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted

Port Name	I/O	Description
radm_bypass_tag[((NHQ*TAG_SIZE)-1):0]	O	The Tag field in the received TLP header on RBYP. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_func_num[((NHQ*PF_WD)-1):0]	O	The function number of the incoming TLP on RBYP. Function numbering starts at '0'. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_vfunc_num[((NHQ*(VF_WD-1))-1):0]	O	Indicated which virtual function (VF) has been targeted by the received TLP. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, radm_bypass_vfunc_num=0 refers to the first VF of the PF that is identified by radm_bypass_func_num. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2)) \ \&\& \ (\text{CX_SRIOV_ENABLE})$ Synchronous To: aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv and radm_bypass_vfunc_active are asserted
radm_bypass_vfunc_active[(NHQ-1):0]	O	Indicates that the received TLP has been targeted to a virtual function (VF). 0: No VF is active and radm_bypass_vfunc_num is invalid. A PF is valid and identified by radm_bypass_func_num. 1: A VF is active and is identified by radm_bypass_vfunc_num. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2)) \ \&\& \ (\text{CX_SRIOV_ENABLE})$ Synchronous To: aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted

Port Name	I/O	Description
radm_bypass_td[(NHQ-1):0]	O	The TLP Digest (TD) bit in the received TLP header on RBYP. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_poisoned[(NHQ-1):0]	O	The Poisoned (EP) bit in the received TLP header on RBYP. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_dw_len[((NHQ*10)-1):0]	O	The Length field (length of TLP in dwords) in the received TLP header on RBYP. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_first_be[((NHQ*4)-1):0]	O	The first dword byte enable field in the received TLP header on RBYP. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_last_be[((NHQ*4)-1):0]	O	The last dword byte enable field in the received TLP header on RBYP. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted

Port Name	I/O	Description
radm_bypass_addr[((NHQ*FLT_Q_ADDR_WIDTH)-1):0]	O	The Address in the received TLP header on RBYP. The bits in this address bus map directly to the address bits in the TLP header. The third and fourth dwords of the header are mapped as follows: ■ MEM (32-bit) / IO / CFG have a 3-dword header. Then radm_bypass_addr[31:0] =Third dword (bytes 8-11). Therefore, bits [7:0] are mapped to byte 11 which is the lowest eight bits of the TLP header address. For more information, see the 'Endianness' advanced information chapter. ■ MEM (64-bit) / MSG have a 4-dword header. Then radm_bypass_addr[63:32] =Third dword (bytes 8-11) and radm_bypass_addr[31:0] =Fourth dword (bytes 12-15). Therefore, bits [7:0] are mapped to byte 15 which is the lowest eight bits of the TLP header address. For more information, see the 'Endianness' advanced information chapter. Exists: (!(AMBA_INTERFACE!=0)) && ((CX_RADMQ_MODE==2)) Synchronous To: None,aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_rom_in_range[(NHQ-1):0]	O	Indicates that the target address in the received request TLP in range of the expansion ROM. Exists: (!(AMBA_INTERFACE!=0)) && ((CX_RADMQ_MODE==2)) Synchronous To: None,aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_io_req_in_range[(NHQ-1):0]	O	Indicates that the target address in the received request TLP is in the I/O address range. Exists: (!(AMBA_INTERFACE!=0)) && ((CX_RADMQ_MODE==2)) Synchronous To: None,aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted

Port Name	I/O	Description
radm_bypass_in_membar_range[((NHQ*3)-1):0]	O	Indicates which of the configured BARs contains the target address in the received request TLP: ■ 000b: Target address in range of BAR 0. ■ 001b: Target address in range of BAR 1. ■ 010b: Target address in range of BAR 2. ■ 011b: Target address in range of BAR 3. ■ 100b: Target address in range of BAR 4. ■ 101b: Target address in range of BAR 5. ■ 110b: Reserved ■ 111b: Target address not in range of any BAR. When radm_bypass_rom_in_range =0 and radm_bypass_io_req_in_range =0, then radm_bypass_in_membar_range also indicates that the BAR associated with the target address is a memory BAR. Exists: (!(AMBA_INTERFACE!=0)) && ((CX_RADMQ_MODE==2)) Synchronous To: None,aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_bcm[(NHQ-1):0]	O	Byte Count Modified (BCM) bit from the received TLP. Exists: (!(AMBA_INTERFACE!=0)) && ((CX_RADMQ_MODE==2)) Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_cpl_last[(NHQ-1):0]	O	Indicates the last completion TLP of a split completion transaction. For non-split completion transactions, the controller asserts radm_bypass_cpl_last for each completion TLP. In either case, the controller asserts radm_bypass_cpl_last in the same cycle as radm_bypass_hv. Exists: (!(AMBA_INTERFACE!=0)) && ((CX_RADMQ_MODE==2)) Synchronous To: None,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted

Port Name	I/O	Description
radm_bypass_cmpltr_id[((NHQ*16)-1):0]	O	Completer ID field from the header of a received completion TLP on RBYP. ■ [15:8]: Bus number ■ [7:3]: Device number ■ [2:0]: Function number Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ ((\text{CX_RADMQ_MODE}=2))$ Synchronous To: None,aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_byte_cnt[((NHQ*12)-1):0]	O	Byte Count field from the received TLP. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ ((\text{CX_RADMQ_MODE}=2))$ Synchronous To: None,aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_cpl_status[((NHQ*3)-1):0]	O	Completion Status field from the header of a received completion TLP. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ ((\text{CX_RADMQ_MODE}=2))$ Synchronous To: None,aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_ats[((NHQ*2)-1):0]	O	AT field in the TLP header on RBYP. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ ((\text{CX_RADMQ_MODE}=2)) \ \&\& \ (\text{ATS_RX_ENABLE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_th[(NHQ-1):0]	O	The TH field in the received TLP header on RBYP. Exists: $(!(\text{AMBA_INTERFACE}=0)) \ \&\& \ ((\text{CX_RADMQ_MODE}=2)) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted

Port Name	I/O	Description
radm_bypass_ph[((NHQ*2)-1):0]	O	The PH field in the received TLP header on RBYP. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}=2) \ \&\& \ (\text{CX_TPH_ENABLE}))$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_st[((NHQ*8)-1):0]	O	The ST field in the received TLP header on RBYP. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}=2) \ \&\& \ (\text{CX_TPH_ENABLE}))$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted
radm_bypass_tlp_prfx[((NHQ*PRFX_W_PAR)-1):0]	O	Receive TLP prefixes. The field [31:0] represents the first prefix to be received. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that radm_bypass_tlp_prfx[31:0] has prefix byte #0 in the lower byte position of the dword radm_bypass_tlp_prfx[31:0]. That is, FMT =bits 7:5, and Type =bits 4:0. If no prefix is present for a given TLP, then that DW, including the FMT field is all zeros. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: $\neg(\text{AMBA_INTERFACE}=0) \ \&\& \ ((\text{CX_RADMQ_MODE}=2) \ \&\& \ (\text{CX_TLP_PREFIX_ENABLE}))$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_bypass_hv is asserted

2.17 RTRGT1 Interface Signals

trgt1_radm_halt	-	radm_trgt1_dma_grant
d_trgt1_wreq_cmplt_dma	-	radm_trgt1_dv
d_trgt1_wreq_cmplt_dma_channel	-	radm_trgt1_hv
d_trgt1_wreq_err_dma	-	radm_trgt1_eot
trgt1_radm_dma_pkt_halt	-	radm_trgt1_tlp_abort
trgt1_radm_pkt_halt	-	radm_trgt1_dllp_abort
		radm_trgt1_ecrc_err
		radm_trgt1_data
		radm_trgt1_tlp_prfx
		radm_trgt1_dwen
		radm_trgt1_hdr_prot
		radm_trgt1_fmt
		radm_trgt1_type
		radm_trgt1_tc
		radm_trgt1_attr
		radm_trgt1_ln
		radm_trgt1_reqid
		radm_trgt1_tag
		radm_trgt1_func_num
		radm_trgt1_vfunc_num
		radm_trgt1_vfunc_active
		radm_trgt1_td
		radm_trgt1_poisoned
		radm_trgt1_dw_len
		radm_trgt1_first_be
		radm_trgt1_last_be
		radm_trgt1_addr
		radm_trgt1_hdr_uppr_bytes
		radm_trgt1_hdr_uppr_bytes_valid
		radm_trgt1_rom_in_range
		radm_trgt1_in_membar_range
		radm_trgt1_io_req_in_range
		radm_trgt1_cpl_status
		radm_trgt1_atc
		radm_trgt1_nw
		radm_trgt1_th
		radm_trgt1_ph
		radm_trgt1_st
		d_trgt1_req_dma
		d_trgt1_req_dma_channel
		radm_trgt1_bcm
		radm_trgt1_byte_cnt
		radm_trgt1_cpl_last
		radm_trgt1_cmplt_id
		radm_trgt1_vc
		radm_grant_tlp_type
		radm_trgt1_atu_sloc_match
		radm_trgt1_atu_cbuf_err

Table 2-17 RTRGT1 Interface Signals

Port Name	I/O	Description
radm_trgt1_dma_grant	O	Indicates that a particular non-posted DMA transaction has been granted to output. This grant signal is a pulse. It is used together with trgt1_radm_dma_pkt_halt to control the amount of transactions that the DMA outputs Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ ((\text{CX_RADMQ_MODE}==2)) \ \&\& \ (\text{CC_DMA_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
trgt1_radm_halt	I	Flow control input signal. When trgt1_radm_halt is asserted, the controller stops streaming the next valid data from the queue inside the RADM. The data bus (radm_trgt1_data) remains unchanged while trgt1_radm_halt is asserted. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_trgt1_dv	O	Indicates that the received TLP data is valid on radm_trgt1_data. For more information on timing, see the Receive Request Interface (TRGT1) section in the Signal Interfaces chapter of the <i>Databook</i>. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_trgt1_hv	O	Indicates that the received TLP header (and prefix) is valid on the corresponding radm_trgt1_* outputs. For more information on timing, see the Receive Request Interface (TRGT1) section in the Signal Interfaces chapter of the <i>Databook</i>. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
radm_trgt1_eot	O	Indicates the last cycle of valid data for the received TLP. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv or radm_trgt1_dv is asserted
radm_trgt1_tlp_abort	O	Indicates to your application to drop the TLP because of malformed TLP on TRGT1, ECRC error, or completion lookup failures (such as TAG or requester ID (RID) mismatches). You should not expect the TLP to be replayed. For more information, see the 'Advanced Error Handling For Received TLPs' advanced information chapter. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_eot, radm_trgt1_hv
radm_trgt1_dllp_abort	O	Indicates to your application to drop the TLP on TRGT1 because of a Data Link Layer error such as LCRC or otherwise. You can expect the DLLP to be replayed. For more information, see the 'Advanced Error Handling For Received TLPs' advanced information chapter. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_eot
radm_trgt1_ecrc_err	O	Indicates to your application to drop the TLP because of an ECRC error in the received TLP on TRGT1. You should not expect the TLP to be replayed. For more information, see the 'Advanced Error Handling For Received TLPs' advanced information chapter. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_eot

Port Name	I/O	Description
radm_trgt1_data[(TRGT_DATA_WD-1):0]	O	Received TLP payload data from the upstream component to your application client. The data is in little endian format. The first received payload byte is in [7:0]. The controller appends protection code bits to the most significant bits of this bus when ECC or parity datapath protection is used. Exists: $\neg(\text{AMBA_INTERFACE}=0)$ && (TRGT1_POPULATE) Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_dv is asserted
radm_trgt1_tlp_prfx[(PRFX_W_PAR-1):0]	O	Receive TLP prefixes. The field [31:0] represents the first prefix to be received. The optional TLP prefixes are implemented as little endian. Using the example of a controller with just one prefix (CX_NPRFX =1), this means that radm_trgt1_tlp_prfx[31:0] has prefix byte #0 in the lower byte position of the dword radm_trgt1_tlp_prfx[31:0]. That is, FMT =bits 7:5, and Type =bits 4:0. If no prefix is present for a given TLP, then that DW, including the FMT field is all zeros. The controller passes along and appends to the prefix, any protection codes it receives, when datapath protection is enabled. Exists: $\neg(\text{AMBA_INTERFACE}=0)$ && (TRGT1_POPULATE) && (CX_TLP_PREFIX_ENABLE) Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted

Port Name	I/O	Description
radm_trgt1_dwen[(NW-1):0]	O	The databus dword enables identify the location of the last dword of the TLP the data bus (radm_trgt1_data). The width and usage of radm_trgt1_dwen depend the data width of the controller. For a 32-bit controller, radm_trgt1_dwen is not used. For a 64-bit controller, NW =2. The radm_trgt1_dwen[1:0] encoding is: ■ 01: Last dword at radm_trgt1_data[31:0]. ■ 11: Last dword at radm_trgt1_data[63:32]. For a 128-bit controller, NW =4. The radm_trgt1_dwen[3:0] encoding is: ■ 0001: Last dword at radm_trgt1_data[31:0]. ■ 0011: Last dword at radm_trgt1_data[63:32]. ■ 0111: Last dword at radm_trgt1_data[95:64]. ■ 1111: Last dword at radm_trgt1_data[127:96]. For a 256-bit controller, NW =8. The radm_trgt1_dwen[7:0] encoding is: ■ 00000001: Last dword at radm_trgt1_data[31:0]. ■ 00000011: Last dword at radm_trgt1_data[63:32]. ■ 00000111: Last dword at radm_trgt1_data[95:64]. ■ 00001111: Last dword at radm_trgt1_data[127:96]. ■ 00011111: Last dword at radm_trgt1_data[159:128]. ■ 00111111: Last dword at radm_trgt1_data[191:160]. ■ 01111111: Last dword at radm_trgt1_data[223:192]. ■ 11111111: Last dword at radm_trgt1_data[255:224]. For a 512-bit controller, NW =16. The radm_trgt1_dwen[15:0] encoding is: ■ 00000001_11111111: Last dword at radm_trgt1_data[287:256]. ■ 00000011_11111111: Last dword at radm_trgt1_data[319:288]. ■ 00000111_11111111: Last dword at radm_trgt1_data[351:320]. ■ 00001111_11111111: Last dword at radm_trgt1_data[383:352]. ■ 00011111_11111111: Last dword at radm_trgt1_data[415:384]. ■ 00111111_11111111: Last dword at radm_trgt1_data[447:416]. ■ 01111111_11111111: Last dword at radm_trgt1_data[479:448]. ■ 11111111_11111111: Last dword at radm_trgt1_data[511:480]. All other encodings are not generated by the controller. Exists: (!(AMBA_INTERFACE!=0)) && (TRGT1_POPULATE) Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_eot is asserted and NW > 1

Port Name	I/O	Description
radm_trgt1_hdr_prot[(RASDP_TRGT1_HDR_PROT_WD-1):0]	O	ECC or parity bits (protection code) generated by the controller for the TLP header. Your application must calculate the protection code over all of the target1 header inputs in this order: radm_trgt1_hdr_uppr_bytes, radm_trgt1_hdr_uppr_bytes_valid, radm_trgt1_atl, radm_trgt1_th, radm_trgt1_st, radm_trgt1_ph, radm_trgt1_ln, radm_trgt1_vfunc_active, radm_trgt1_vfunc_num, radm_trgt1_last_be, radm_trgt1_cpl_last, radm_trgt1_poisoned, radm_trgt1_td, radm_trgt1_addr, radm_trgt1_first_be, radm_trgt1_io_req_in_range, radm_trgt1_rom_in_range, radm_trgt1_in_membar_range, radm_trgt1_dw_len, radm_trgt1_cpl_status, radm_trgt1_func_num, radm_trgt1_tag, radm_trgt1_reqid, radm_trgt1_attr, radm_trgt1_tc, radm_trgt1_type, radm_trgt1_fmt. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (((\text{CX_RASDP}==0)? \ 0: \ 1))$ Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: radm_trgt1_hv is asserted
radm_trgt1_fmt[1:0]	O	The Format field in the received TLP header on TRGT1. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_type[4:0]	O	The Type field in the received TLP header on TRGT1. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_tc[2:0]	O	The Traffic Class (TC) field in the received TLP header on TRGT1. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted

Port Name	I/O	Description
radm_trgt1_attr[(ATTR_WD-1):0]	O	The Attributes field in the received TLP header on TRGT1. ■ Width is three bits when CX_IDO_ENABLE =1 ■ Width is two bits when CX_IDO_ENABLE =0 Exists: (!(AMBA_INTERFACE!=0)) && (TRGT1_POPULATE) Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_ln	O	The Lightweight Notification (LN) field in the received TLP header on TRGT1. Exists: (!(AMBA_INTERFACE!=0)) && (TRGT1_POPULATE) && (CX_LN_ENABLE) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_reqid[15:0]	O	The requester ID in the received TLP header on TRGT1. ■ [15:8]: Bus number ■ [7:3]: Device number ■ [2:0]: Function number Exists: (!(AMBA_INTERFACE!=0)) && (TRGT1_POPULATE) Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_tag[(TAG_SIZE-1):0]	O	The Tag field in the received TLP header on TRGT1. Exists: (!(AMBA_INTERFACE!=0)) && (TRGT1_POPULATE) Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted

Port Name	I/O	Description
radm_trgt1_func_num[(PF_WD-1):0]	O	The function number of the incoming TLP on TRGT1. Function numbering starts at '0'. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_vfunc_num[(VF_WD-2):0]	O	Indicates which virtual function (VF) has been targeted by the received request. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, radm_trgt1_vfunc_num=0 refers to the first VF of the PF that is identified by radm_trgt1_func_num. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{CX_SRIOV_ENABLE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: radm_trgt1_hv and radm_trgt1_vfunc_active are asserted
radm_trgt1_vfunc_active	O	Indicates that the received request matched a virtual function (VF) BAR. 0: No VF is active and radm_trgt1_vfunc_num is invalid. A PF is valid and identified by radm_trgt1_func_num. 1: A VF is active and is identified by radm_trgt1_vfunc_num. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{CX_SRIOV_ENABLE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_td	O	The TLP Digest (TD) bit in the received TLP header on TRGT1. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted

Port Name	I/O	Description
radm_trgt1_poisoned	O	The Poisoned (EP) bit in the received TLP header on TRGT1. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_dw_len[9:0]	O	The Length field (length of TLP in dwords) in the received TLP header on TRGT1. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_first_be[3:0]	O	The first dword byte enable field in the received TLP header on TRGT1. When a TLP is a message, radm_trgt1_first_be is overlaid with message_code[3:0] Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_last_be[3:0]	O	The last dword byte enable field in the received TLP header on TRGT1. When a TLP is a message, radm_trgt1_last_be is overlaid with message_code[7:4] Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted

Port Name	I/O	Description
radm_trgt1_addr[(ADDR_WIDTH-1):0]	O	The Address in the received TLP header on TRGT1. The bits in this address bus map directly to the address bits in the TLP header. The third and fourth dwords of the header are mapped as follows: - MEM (32-bit) / IO / CFG have a 3-dword header. Then radm_trgt1_addr[31:0] =Third dword (bytes 8-11). Therefore, bits [7:0] are mapped to byte 11 which is the lowest eight bits of the TLP header address. For more information, see the 'Endianness' advanced information chapter. - MEM (64-bit) / MSG have a 4-dword header. Then radm_trgt1_addr[63:32] =Third dword (bytes 8-11) and radm_trgt1_addr[31:0] =Fourth dword (bytes 12-15). Therefore, bits [7:0] are mapped to byte 15 which is the lowest eight bits of the TLP header address. For more information, see the 'Endianness' advanced information chapter. Exists: (!(AMBA_INTERFACE!=0)) && (TRGT1_POPULATE) Synchronous To: {core_clk},{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_hdr_uppr_bytes[(ADDR_WIDTH-1):0]	O	The raw upper bytes in the received TLP header on TRGT1. The upper bytes in the received TLP header on TRGT1. For a 3-dword header these are bytes 8 to 11, for a 4-dword header they are bytes 8-15. When radm_trgt1_hdr_uppr_bytes_valid is asserted, then radm_trgt1_hdr_uppr_bytes has valid Msg data. These bytes of the TLP header are mapped as follows: - MEM (32-bit) / IO / CFG have a 3-dword header. Then radm_trgt1_hdr_uppr_bytes[31:0] = bytes 8-11. Therefore, bits [7:0] are mapped to byte 8 of the TLP header. - MEM (64-bit) / MSG have a 4-dword header. Then radm_trgt1_hdr_uppr_bytes[63:32] = bytes 12-15 and radm_trgt1_hdr_uppr_bytes[31:0] = bytes 8-11. Therefore, bits [7:0] are mapped to byte 8 of the TLP header while bits 39-32 are mapped to byte 12, and so on. <i>Note:</i> For radm_trgt1_hdr_uppr_bytes[63:32] to exist, FLT_Q_ADDR_WIDTH must have a value of 64. If 32 < FLT_Q_ADDR_WIDTH < 64, then radm_trgt1_hdr_uppr_bytes[63:32] = 0; that is, radm_trgt1_hdr_uppr_bytes[63:32] contains TLP Header data only if FLT_Q_ADDR_WIDTH = 64. Your application can ensure FLT_Q_ADDR_WIDTH = 64 by setting VENDOR_MESSAGE_SUPPORT = 1. Exists: (!(AMBA_INTERFACE!=0)) && (TRGT1_POPULATE) Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted

Port Name	I/O	Description
radm_trgt1_hdr_uppr_bytes_valid	O	Indicates that the transaction being presented on the TRGT1 interface has been successfully translated in the iATU from a TLP received in the PCIe controller. radm_trgt1_hdr_uppr_bytes has valid Msg data when this signal is asserted. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ ((\text{CX_RADMQ_MODE}==2)) \ \&\& \ (\text{CX_INTERNAL_ATU_ENABLE})$ Synchronous To: None,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_rom_in_range	O	Indicates that the target address in the received TLP in range of the expansion ROM. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_in_membar_range[2:0]	O	Indicates which of the configured BARs contains the target address in the received TLP: ■ 000b: Target address in range of BAR 0. ■ 001b: Target address in range of BAR 1. ■ 010b: Target address in range of BAR 2. ■ 011b: Target address in range of BAR 3. ■ 100b: Target address in range of BAR 4. ■ 101b: Target address in range of BAR 5. ■ 110b: Reserved ■ 111b: Target address not in range of any BAR. When radm_trgt1_rom_in_range =0 and radm_trgt1_io_req_in_range =0, then radm_trgt1_in_membar_range also indicates that the BAR associated with the target address is a memory BAR. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted

Port Name	I/O	Description
radm_trgt1_io_req_in_range	O	Indicates that the target address in the received TLP is in the I/O address range. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_cpl_status[2:0]	O	When the received TLP is a completion: completion Status field from the header of a received completion TLP. When the received TLP is a request: What the completion status for this request should be, according to internal filtering rules. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted and completion is not in bypass mode
radm_trgt1_ats[1:0]	O	AT field in the TLP header on TRGT1. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{ATS_RX_ENABLE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_nw	O	NW field in the TLP header on TRGT1. The NW bit in radm_trgt1_addr is forced to '0' by the controller, after the controller copies its value to radm_trgt1_nw. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{ATS_RX_ENABLE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted

Port Name	I/O	Description
radm_trgt1_th	O	The TH field in the received TLP header on TRGT1. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_ph[1:0]	O	The PH field in the received TLP header on TRGT1. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_st[7:0]	O	The ST field in the received TLP header on TRGT1. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{CX_TPH_ENABLE})$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
d_trgt1_req_dma[(DC_DMA_TYPE_WIDTH-1):0]	O	Specifies the source of the request the RTRGT1 Interface. Bit encoding as follows: ■ 000: Non-DMA request ■ 001: DMA Write channel MRd request ■ 010: Non-DMA completion ■ 011: DMA Read channel MWr request ■ 100: Reserved ■ 101: DMA Write channel LL Element MRd request ■ 110: Reserved ■ 111: DMA Read channel LL Element MRd request For more information, see 'Native DMA (No AXI Bridge)'. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{CC_DMA_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted

Port Name	I/O	Description
d_trgt1_req_dma_channel[(CC_NUM_DMA_MAX_CHAN_WIDTH-1):0]	O	DMA Channel number. For more information, see 'Native DMA (No AXI Bridge)'. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{CC_DMA_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
d_trgt1_wreq_cmplt_dma	I	Your application must pulse this signal every time a DMA Read Transfer MWr is successfully written to local memory. The DMA also uses this signal to sample d_trgt1_wreq_err_dma. For more information, see 'Native DMA (No AXI Bridge)'. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{CC_DMA_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High (pulse) Validated by: Not validated by another signal.
d_trgt1_wreq_cmplt_dma_channel[(CC_NUM_DMA_RD_CHAN_WIDTH-1):0]	I	DMA Channel Number. You must set this input to the DMA channel number. For more information, see 'Native DMA (No AXI Bridge)'. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{CC_DMA_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: d_trgt1_wreq_cmplt_dma is asserted
d_trgt1_wreq_err_dma	I	Specifies that an error occurred in your application during a DMA Read Transfer MWr to local memory. When an error occurs the DMA halts the Read Transfer and generates an interrupt (if enabled). For more information, see 'Native DMA (No AXI Bridge)'. Exists: $!(\text{AMBA_INTERFACE}=0) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ (\text{CC_DMA_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: d_trgt1_wreq_cmplt_dma is asserted

Port Name	I/O	Description
radm_trgt1_bcm	O	Byte Count Modified (BCM) bit from the header of a received completion TLP. The BCM bit is not applicable for an Endpoint device. However, the controller does provide the value of the BCM bit radm_trgt1_bcm. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted and the completion is not in bypass mode
radm_trgt1_byte_cnt[11:0]	O	Byte Count field from the header of a received completion TLP. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted and completion is not in bypass mode
radm_trgt1_cpl_last	O	Indicates the last completion TLP of a split completion transaction. For non-split completion transactions, the controller asserts radm_trgt1_cpl_last for each completion TLP. In either case, the controller asserts radm_trgt1_cpl_last in the same cycle as radm_trgt1_hv. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted and completion is not in bypass mode

Port Name	I/O	Description
radm_trgt1_cmpltr_id[15:0]	O	Completer ID field from the header of a received completion TLP on TRGT1. ■ [15:8]: Bus number ■ [7:3]: Device number ■ [2:0]: Function number Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted and completion is not in bypass mode
trgt1_radm_dma_pkt_halt	I	Halts the transfer of non-posted packets from DMA. Exists: $(!(\text{AMBA_INTERFACE}!=0)) \ \&\& \ (\text{TRGT1_POPULATE}) \ \&\& \ ((\text{CX_RADMQ_MODE}==2)) \ \&\& \ (\text{CC_DMA_ENABLE})$ Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_trgt1_vc[2:0]	O	Virtual Channel Number of a received TLP on TRGT1. ■ [2:0]: Virtual Channel number Exists: $(\text{TRGT1_POPULATE}) \ \&\& \ ((\text{CX_RADMQ_MODE}==2))$ Synchronous To: {None},{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted and completion is not in bypass mode

Port Name	I/O	Description
radm_grant_tlp_type[((NVC*3)-1):0]	O	Indicates that a particular VC and type transaction has been granted to output from the receive queue. There is one bit for each TLP type for each configured VC: ■ 0: Grant posted TLPs for VC0 ■ 1: Grant non-posted TLPs for VC0 ■ 2: Grant completion TLPs for VC0 ■ 3: Grant posted TLPs for VC1 ■ ■ up to the number of configured VCs. This grant signal is a pulse. It is used together with trgt1_radm_pkt_halt to control the amount of transactions that the controller's output queue outputs. Exists: ((AMBA_POPULATED==0 CX_CCIX_INTERFACE_ENABLE==1)) && ((CX_RADMQ_MODE==2)) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: The TLP type is in not in bypass mode
trgt1_radm_pkt_halt[((NVC*3)-1):0]	I	Halts the transfer of packets from individual queues. There is one bit of trgt1_radm_pkt_halt for each TLP type for each configured VC: ■ 0: Halt posted TLPs for VC0 ■ 1: Halt non-posted TLPs for VC0 ■ 2: Halt completion TLPs for VC0 ■ 3: Halt posted TLPs for VC1 ■ up to the number of configured VCs. For more information, see 'RTRGT1 Protocol Rules and Example Transaction'. Exists: ((AMBA_POPULATED==0 CX_CCIX_INTERFACE_ENABLE==1)) && ((CX_RADMQ_MODE==2)) && (TRGT1_POPULATE) Synchronous To: aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: The TLP type is in not in bypass mode

Port Name	I/O	Description
radm_trgt1_atu_sloc_match[(ATU_IN_MIN1-1):0]	O	ATU Single Location match indication per ATU region. Set for 1 core_clk period pulse when the ATU region matched a received VDM Single Location Address translation. Exists: CX_ATU_SLOC_CBUF_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted
radm_trgt1_atu_cbuf_err[(ATU_IN_MIN1-1):0]	O	ATU Error indication per ATU region. Set for 1 core_clk period pulse when the ATU region matched received VDM size (payload + 3rd and 4th DW of Header) is greater than the programmed Circular Buffer Increment size (CBUF_INCR) for Single Location Address translation. Exists: CX_ATU_SLOC_CBUF_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_trgt1_hv is asserted

2.18 TRGT1_CCIX Interface Signals

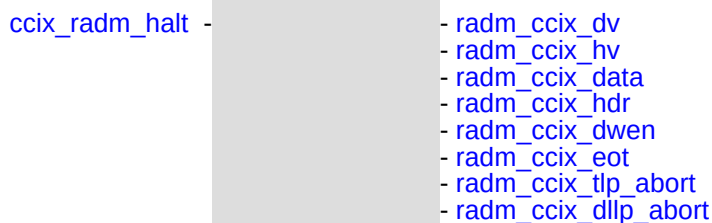


Table 2-18 TRGT1_CCIX Interface Signals

Port Name	I/O	Description
<code>radm_ccix_dv</code>	O	Indicates the data is valid in the CCIX receive data bus. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: <code>radm_clk_g</code> Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
<code>radm_ccix_hv</code>	O	Indicates header is valid in the CCIX receive header bus. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: <code>radm_clk_g</code> Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
<code>radm_ccix_data[(CX_CCIX_DATA_WD-1):0]</code>	O	Data bus in the CCIX receive interface. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: <code>radm_clk_g</code> Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: <code>radm_ccix_dv</code> is asserted

Port Name	I/O	Description
radm_ccix_hdr[(CX_CCIX_HDR_WD-1):0]	O	Header bus in the CCIX receive interface. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: radm_clk_g Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: radm_ccix_hv is asserted
ccix_radm_halt	I	When asserted, this signal causes the controller to stop streaming information in the CCIX receive interface. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: radm_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_ccix_dwen[(NW-1):0]	O	DWord enable for received CCIX data. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: radm_clk_g Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: radm_ccix_dv is asserted
radm_ccix_eot	O	Signals the end of the TLP being delivered in the CCIX receive interface. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: radm_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
radm_ccix_tlp_abort	O	Indication for the application to drop the TLP received in the CCIX receive interface because of the malformed TLP or ECRC error. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: radm_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: radm_ccix_hv, radm_ccix_eot
radm_ccix_dllp_abort	O	Indication for the application to drop the TLP received in the CCIX receive interface because of the data link layer error such as LCRC. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) && (!CX_CXS_ENABLE) Synchronous To: radm_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: radm_ccix_eot

2.19 Clock and Reset (APM) Signals

perst_n	-	pm_req_sticky_rst
app_clk_req_n	-	pm_req_core_rst
phy_clk_req_n	-	pm_req_non_sticky_rst
ack_sticky_rst	-	pm_sel_aux_clk
ack_core_rst	-	pm_en_core_clk
ack_non_sticky_rst	-	pm_req_phy_rst
ret_sticky_rst_n	-	pm_req_phy_perst
ret_non_sticky_rst_n	-	pm_div_max_pclk_ratio
ret_core_rst_n	-	pm_req_retention_rst
ret_core_pl_rst_n	-	

Table 2-19 Clock and Reset (APM) Signals

Port Name	I/O	Description
perst_n	I	Indicates when the main power supply is within its tolerated voltage range and is stable. This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. The minimum duration the PCIe Power Management Controller can handle is $(6 \cdot \text{DEFAULT_AUX_CLK_FREQ}/62.5) + 10$ in slow aux_clk cycles. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
app_clk_req_n	I	Indicates that the application logic is ready to have reference clock removed. In designs which support reference clock removal through either L1 PM Sub-states or L1 CPM, the application should set this signal to 1'b1 if it supports reference clock removal. If the application does not want to remove reference clock it should set this signal to 1'b0. This signal should be asserted or de-asserted when L1 CPM or L1 Sub-states are being configured, it should not be changed dynamically during L1. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

Port Name	I/O	Description
phy_clk_req_n	I	Acknowledge from the PHY that it is ready to have reference clock removed. In designs that support L1 PM Sub-states or L1 CPM, there is a handshake with the PHY prior to reference clock removal. This signal should be connected to the acknowledge signal from the PHY. If the PHY does not want to remove the reference clock it should set this signal to 1'b0. This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
pm_req_sticky_rst	O	Request from controller's PMC to the DWC_pcie_clkrst.v to assert the sticky_rst_n input. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_req_core_rst	O	Request from controller's PMC to the DWC_pcie_clkrst.v to assert the core_rst_n input. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_req_non_sticky_rst	O	Request from controller's PMC to the DWC_pcie_clkrst.v to assert the non_sticky_rst_n input. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pm_sel_aux_clk	O	This signal switches the source of aux_clk from core_clk to the low speed clock. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_en_core_clk	O	This signal is used to gate off the core_clk in low power states. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_req_phy_rst	O	Request from controller's PMC to the DWC_pcie_clkrst.v to assert the phy_rst_n input. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_req_phy_perst	O	Request from controller's PMC to the DWC_pcie_clkrst.v to assert the phy_perst_n output. Application can use this signal as an interrupt to implement custom reset policies. This can be useful, for example, in the presence of bifurcated links, or to stage refclk removal to the PMA. Refer to your PHY vendor for specific guidelines during cold and warm resets. Exists: Always Synchronous To: aux_clk_g,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pm_div_max_pclk_ratio[6:0]	O	The clock divider conversion ratio. The clock divider converts from MAXPCLK to PCLK based on this ratio. Exists: (CX_PIPE_PCLK_MODE==1) Synchronous To: None,perbitclk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ack_sticky_rst	I	Acknowledgment from DWC_pcie_clkrst.v for the corresponding sticky reset request. The sticky reset is applied by the clock and reset generation logic. Exists: CX_ENHANCED_PM_EN Synchronous To: core_clk_ug,pipe_clk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ack_core_rst	I	Acknowledgment from DWC_pcie_clkrst.v for the corresponding controller reset request. The controller reset is applied by the clock and reset generation logic. Exists: CX_ENHANCED_PM_EN Synchronous To: core_clk_ug,pipe_clk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ack_non_sticky_rst	I	Acknowledgment from DWC_pcie_clkrst.v for the corresponding non-sticky reset request. The non-sticky reset is applied by the clock and reset generation logic. Exists: CX_ENHANCED_PM_EN Synchronous To: core_clk_ug,pipe_clk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pm_req_retention_rst	O	Request from controller's PMC to the DWC_pcie_clkrst.v to assert the reset to modules containing L1 power gating retention registers. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section, and the "Reset Requirements" section in the Architecture chapter of the <i>Databook</i>. Exists: CX_L1_PG_ENABLE Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ret_sticky_rst_n	I	Resets all sticky bit registers in the retention (L1 power gating) register based modules of configuration register space. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section, and the "Reset Requirements" section in the Architecture chapter of the <i>Databook</i>. Exists: CX_L1_PG_ENABLE Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
ret_non_sticky_rst_n	I	Resets all non-sticky bit registers in the retention (L1 power gating) register based modules of configuration register space. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section, and the "Reset Requirements" section in the Architecture chapter of the <i>Databook</i>. Exists: CX_L1_PG_ENABLE Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

Port Name	I/O	Description
ret_core_rst_n	I	Resets retention (L1 power gating) register based modules of the controller logic. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section, and the "Reset Requirements" section in the Architecture chapter of the <i>Databook</i>. Asynchronous assertion with synchronous de-assertion. Exists: CX_L1_PG_ENABLE Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
ret_core_pl_rst_n	I	Resets retention (L1 power gating) register based modules of the controller logic for layer1. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section, and the "Reset Requirements" section in the Architecture chapter of the <i>Databook</i>. Asynchronous assertion with synchronous de-assertion. Exists: (CX_L1_PG_ENABLE) && (CX_FREQ_STEP_DL_EN) Synchronous To: None_as Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

2.20 Clock and Reset Signals

ret_pipe_clk - en_aux_clk_g
ret_pipe_rst_n - en_radm_clk_g
ret_core_clk_ug - radm_idle
ret_core_pl_clk_ug - training_rst_n
pipe_clk
pipe_msgbus_rst_n
pipe_rx_clk
pipe_rx_rst_n
cxs_clk
cxs_rst_n
core_clk
ret_core_clk
core_clk_ug
core_pl_clk
core_pl_clk_ug
ret_core_pl_clk
aux_clk
aux_clk_g
radm_clk_g
pwr_rst_n
sticky_rst_n
non_sticky_rst_n
core_rst_n
pipe_rst_n
core_pl_rst_n
app_init_rst
aux_clk_active
slv_pwr_rst_n
mstr_pwr_rst_n
dbi_pwr_rst_n
cxi_clk
cxi_rst_n

Table 2-20 Clock and Reset Signals

Port Name	I/O	Description
ret_pipe_clk	I	This clock is equivalent to pipe_clk but it is routed to modules being retained in L1.2 Exists: ((CX_CPCIE_ENABLE)) && (CX_RET_PIPE_CLK_EN) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ret_pipe_rst_n	I	Resets controller logic clocked on ret_pipe_clk. Reset logic should assert ret_pipe_rst_n at same time as core_rst_n and de-assert synchronous to pipe_clk. Exists: ((CX_CPCIE_ENABLE)) && (CX_RET_PIPE_CLK_EN) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
ret_core_clk_ug	I	The ungated version of the primary clock input to the controller, for modules retained during L1 power gating. The ret_core_clk_ug frequency is the same as core_clk_ug. Exists: ((CX_CPCIE_ENABLE)) && (CX_RET_CORE_CLK_UG_EN) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ret_core_pl_clk_ug	I	The ungated version of the primary clock input to the controller for physical layer, for modules retained during L1 power gating. The ret_core_pl_clk_ug frequency is the same as core_pl_clk_ug. Exists: ((CX_CPCIE_ENABLE)) && (CX_RET_CORE_PL_CLK_UG_EN) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pipe_clk	I	PIPE clock from external PHY. Used by the controller to clock the PIPE. The PHY TX PLL generates pipe_clk from the platform reference clock. The PHY adapts and re-times RX data to pipe_clk. The PIPE clock rate is 62.5 MHz, 125 MHz, 250 MHz, or 500 MHz, depending on datapath configuration and symbol/datapath width at the PHY interface. Exists: ((CX_CPCIE_ENABLE)) && (CX_FREQ_STEP_EN) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pipe_msgbus_rst_n	I	Resets controller logic clocked on pipe_clk. Reset logic should assert pipe_msgbus_rst_n at same time as phy_rst_n and de-assert synchronous to pipe_clk. Exists: ((CX_CPCIE_ENABLE)) && (CX_PIPE5_SUPPORT) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pipe_rx_clk[(NL-1):0]	I	Recovered clock from external PHY. This clock signal is recovered clock used for phy_mac_rxddata in the SerDes architecture. Exists: ((CX_CPCIE_ENABLE)) && (CX_PIPE_SERDES_ARCH) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pipe_rx_rst_n[(NL-1):0]	I	Resets controller logic clocked on pipe_rx_clk. Reset logic should assert pipe_rx_rst_n at same time as core_rst_n and de-assert synchronous to corresponding pipe_rx_clk. Exists: ((CX_CPCIE_ENABLE)) && (CX_PIPE_SERDES_ARCH) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
cxs_clk	I	CXS Interface clock signal Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && (!CX_CXS_SYNCHRONOUS) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxs_rst_n	I	CXS Interface asynchronous active low reset signal Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && (!CX_CXS_SYNCHRONOUS) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

Port Name	I/O	Description
core_clk	I	The gated version of the primary clock input to the controller. It is assumed that all input signals except resets are synchronous to this clock. Depending the controller configuration and Gen2/Gen3 mode, the core_clk frequency is 62.5 MHz, 125 MHz, 250 MHz, or 500 MHz if Base PCIe. You can gate this clock during the L1 and L2 low power states. Exists: Always Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ret_core_clk	I	The gated version of the primary clock input to the controller, for modules retained during L1 power gating. The ret_core_clk frequency is the same as core_clk. You can gate this clock during the L1 and L2 low power states under the same conditions as core_clk. Exists: CX_L1_PG_ENABLE Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
core_clk_ug	I	The ungated version of the primary clock input to the controller. Depending the controller configuration and Gen2/Gen3 mode, the core_clk_ug frequency is 62.5 MHz, 125 MHz, 250 MHz, or 500 MHz. ■ You must not gate this during the L1 and L2 low power states. ■ It is used in the PCIe controller to sample phy_mac_phystatus. Exists: (CX_S_CPCIE_MODE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
core_pl_clk	I	The gated version of the primary clock input to the controller for physical layer. It is assumed that all input signals except resets are synchronous to this clock. Depending the controller configuration and Gen2/Gen3 mode, the core_clk frequency is 62.5 MHz, 125 MHz, 250 MHz, or 500 MHz if Base PCIe. You can gate this clock during the L1 and L2 low power states. Exists: ((CX_S_CPCIE_MODE)) && (CX_FREQ_STEP_DL_EN) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
core_pl_clk_ug	I	The ungated version of the primary clock input to the controller for physical layer. Depending the controller configuration and Gen2/Gen3 mode, the core_pl_clk_ug frequency is 62.5 MHz, 125 MHz, 250 MHz, or 500 MHz. ■ You must not gate this during the L1 and L2 low power states. ■ It is used in the PCIe controller to sample phy_mac_phystatus. Exists: ((CX_S_CPCIE_MODE)) && (CX_FREQ_STEP_DL_EN) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ret_core_pl_clk	I	The gated version of the primary clock input to the controller for physical layer, for modules retained during L1 power gating. The ret_core_pl_clk frequency is the same as core_pl_clk. You can gate this clock during the L1 and L2 low power states under the same conditions as core_plclk. Exists: ((CX_S_CPCIE_MODE)) && (CX_FREQ_STEP_DL_EN) && (CX_L1_PG_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
aux_clk	I	Auxiliary clock to the PMC domain. The partitioning of the controller enables some functions to operate aux_clk in certain power management states. The PMC is partitioned to run aux_clk in L2 states, if supported by your application. In normal operation, the aux_clk and core_clk inputs are assumed to be equivalent and the physical design implementation must assume they are the same in terms of clock tree matching and skew. For power management enabled solutions, it is the responsibility of your application designer to manage any clock switching required to gate core_clk and switch aux_clk from the normal operating frequency to the slower clock rate used in the low power states. Exists: Always Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
aux_clk_g	I	Gated version of aux_clk. Used in CDM. Exists: Always Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
en_aux_clk_g	O	Enable signal used by DWC_pcie_clkrst.v to create aux_clk_g from aux_clk. Exists: Always Synchronous To: {None},{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_clk_g	I	Gated version of core_clk. Used in RADM. Exists: Always Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
en_radm_clk_g	O	Enable signal used by DWC_pcie_clkrst.v to create radm_clk_g from core_clk. The controller de-asserts the en_radm_clk_g output when there is no Rx traffic, Rx queues and pre/post-queue pipelines are empty, RADM completion LUT is empty, and there are no FLR actions pending. You must set the RADM_CLK_GATING_EN field in the CLOCK_GATING_CTRL_OFF register to enable this functionality; otherwise the en_radm_clk_g output is always set to '1'. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_idle	O	RADM activity status signal. The controller creates the en_radm_clk_g output by gating this signal with the output of the RADM_CLK_GATING_EN field in the CLOCK_GATING_CTRL_OFF register. For debug purposes only. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pwr_rst_n	I	Resets the PMC module. The pwr_rst_n signal is used as the 'cold reset' applied to the controller power-up. It must be asserted (asynchronously) following your application of auxiliary power or following your application of main power if auxiliary power is not available to the device. The pwr_rst_n input resets all registers in the aux_clk domain, including sticky bits. It must be de-asserted synchronously. Exists: Always Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
sticky_rst_n	I	Resets all sticky bit registers in the configuration register space. Exists: Always Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

Port Name	I/O	Description
non_sticky_rst_n	I	Resets all non-sticky bit registers in the configuration register space. Exists: Always Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
core_rst_n	I	Resets the controller, except for the PMC module. Upon initial power-up, core_rst_n is asserted in order to reset the non-auxiliary power domain logic. Reset logic should assert core_rst_n whenever link_req_rst_not is transitioned from high to low. It is recommended that you reset your application logic together with the controller. Asynchronous assertion with synchronous de-assertion. Exists: Always Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
pipe_rst_n	I	Resets controller logic clocked on pipe_clk. Reset logic should assert pipe_rst_n at same time as core_rst_n and de-assert synchronous to pipe_clk. Exists: CX_FREQ_STEP_EN Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
core_pl_rst_n	I	Resets the controller, except for the PMC module. Upon initial power-up, core_rst_n is asserted in order to reset the non-auxiliary power domain logic. Reset logic should assert core_pl_rst_n whenever link_req_rst_not is transitioned from high to low. It is recommended that you reset your application logic together with the controller. Asynchronous assertion with synchronous de-assertion. Exists: CX_FREQ_STEP_DL_EN Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

Port Name	I/O	Description
app_init_rst	I	You should set this input to '0'. Note: This signal is not used by the controller to set the SBR field in the BRIDGE_CTRL_INT_PIN_INT_LINE_REG register. During the transition from DL_Active to DL_inactive, assertion of app_init_rst signal indicates a Surprise Down Error, but setting of SBR field in the BRIDGE_CTRL_INT_PIN_INT_LINE_REG register through DBI does not trigger a Surprise Down Error. Exists: Always Synchronous To: aux_clk,aux_clk_g,core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
training_rst_n	O	Hot reset from upstream component. When the controller LTSSM receives two consecutive TS1 ordered sets with the hot_reset bit asserted, it asserts training_rst_n for one clock cycle. This signal is only kept for legacy purposes. You should use the link_req_rst_not signal to reset the controller after a 'hot reset' request. For more information on hot reset, see 'Generating and Processing Hot Resets (Training Resets)' in the <i>Databook</i>. Exists: Always Synchronous To: core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
aux_clk_active	I	Indicates that your external clock logic has switched the aux_clk input from pipe_clk to the platform auxiliary clock. Exists: Always Synchronous To: aux_clk_g,core_clk_ug,pipe_clk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
slv_pwr_rst_n	I	pwr_rst_n signal de-asserted synchronous to the slv_aclk Exists: CX_SLV_ISO_EN Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

Port Name	I/O	Description
mstr_pwr_rst_n	I	pwr_rst_n signal de-asserted synchronous to the mstr_aclk Exists: CX_MSTR_ISO_EN Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
dbi_pwr_rst_n	I	pwr_rst_n signal de-asserted synchronous to the dbi_aclk Exists: CX_DBI_ISO_EN Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
cxl_clk	I	CXS Message Channel Interface clock signal Exists: (CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_rst_n	I	Message Channel Interface asynchronous active low reset signal Exists: (CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

2.21 DBI Interface Signals

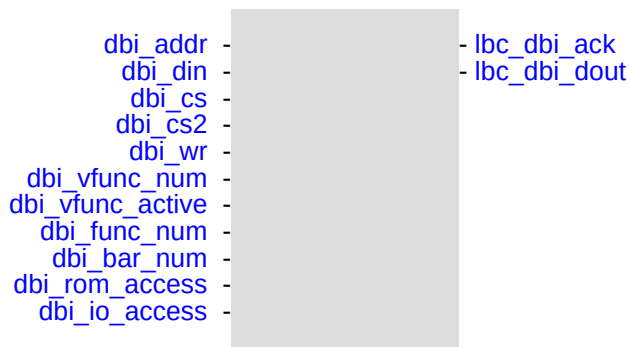


Table 2-21 DBI Interface Signals

Port Name	I/O	Description
dbi_addr[31:0]	I	Address of the configuration register for the current DBI access: ■ [31:19]: Not used ■ [18:16]: Function number when (CX_SRIOV_ENABLE or CX_ARI_ENABLE)=0 ■ [15:12]: Not used ■ [11:2]: Register address, must be dword-aligned. ■ [1]: Not used ■ [0]: Target of DBI access: 0 to access internal register; 1 to access external register the ELBI You can use an address larger than 12 bits dbi_addr when you access the ELBI interface instead of the Configuration Registers. You can use up to 32 bits if the controller has been configured for a 32-bit ELBI address width with the parameter CX_LBC_EXT_AW. For more information, see 'Local Bus Controller (LBC) and Data Bus Interface (DBI)'. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_cs is asserted
dbi_din[31:0]	I	Write data bus to the selected configuration register. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_cs is asserted

Port Name	I/O	Description
dbi_cs	I	Chip select input to access the CDM or ELBI. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) Synchronous To: aux_clk,core_clk Ug,pipe_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
dbi_cs2	I	Additional chip select that enables writing to BAR mask registers. To write to a BAR mask register, your application must assert dbi_cs2 in addition to dbi_cs. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_cs is asserted
dbi_wr[3:0]	I	Indicates the configuration register access type (read or write). For writes, dbi_wr also indicates the byte enables: ■ 0000b: Read ■ 0001b: Write byte 0 ■ 0010b: Write byte 1 ■ 0100b: Write byte 2 ■ 1000b: Write byte 3 ■ 1111b: Write all bytes Combinations of byte enables (for example, 0101b) are also valid. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_cs is asserted
lbc_dbi_ack	O	Indicates that the requested read or write operation to the selected configuration register is complete. Accessing a non-existent register or writing to a read only register is considered an erroneous access. The signal is asserted for one clock cycle instead of two cycles. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
lbc_dbi_dout[31:0]	O	Read data bus from the selected configuration register. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_dbi_ack is asserted
dbi_vfunc_num[(VF_WD-2):0]	I	Indicates which VF is being accessed through the DBI. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, dbi_vfunc_num=0 refers to the first VF of the PF that is identified by dbi_func_num. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) && (CX_SRIOV_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_cs and dvi_vfunc_active are asserted
dbi_vfunc_active	I	Indicates that a VF is being accessed through the DBI. 0: No VF is active and dbi_vfunc_num is invalid. A PF is valid and identified by dbi_func_num. 1: A VF is active and is identified by dbi_vfunc_num. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) && (CX_SRIOV_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_cs is asserted

Port Name	I/O	Description
dbi_func_num[(PF_WD-1):0]	I	The (physical) function number of the current DBI access, when (CX_SRIOV_ENABLE or CX_ARI_ENABLE) =1. ■ 00000b: Function 0 ■ 00001b: Function 1 ■ 00010b: Function 2 ■ 00011b: Function 3 ■ 00100b: Function 4 ■ 00101b: Function 5 ■ 00110b: Function 6 ■ 00111b: Function 7 ■ 00111b: Function 7 ■ ■ 11111b: Function 31 Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) && (SRIOV_OR_DBI_MFBAR_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_cs is asserted
dbi_bar_num[2:0]	I	The BAR number of the current DBI access to the ELBI. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) && (DBI_MULTI_FUNC_BAR_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_cs is asserted
dbi_rom_access	I	Indicates that the current DBI access to the ELBI is for expansion ROM. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) && (DBI_MULTI_FUNC_BAR_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_cs is asserted

Port Name	I/O	Description
dbi_io_access	I	Indicates that the current DBI access to the ELBI is an I/O access. Exists: (!DBI_4SLAVE_POPULATED) && (!SHARED_DBI_ENABLED) && (DBI_MULTI_FUNC_BAR_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: dbi_cs is asserted

2.22 ELBI Interface Signals

ext_lbc_ack -
 ext_lbc_din -

- lbc_ext_addr
 - lbc_ext_dout
 - lbc_ext_cs
 - lbc_ext_wr
 - lbc_ext_rd
 - lbc_ext_dbi_access
 - lbc_ext_cxl_mbar0_access
 - lbc_ext_rom_access
 - lbc_ext_io_access
 - lbc_ext_bar_num
 - lbc_ext_vfunc_num
 - lbc_ext_vfunc_active

Table 2-22 ELBI Interface Signals

Port Name	I/O	Description
ext_lbc_ack[(NF-1):0]	I	Indicates that the requested read or write operation to an external register block is complete. There is no time interval requirement between the request (when lbc_ext_cs is asserted) and the acknowledgement (when ext_lbc_ack is asserted). However, to avoid a lockup condition in the event that the register block never asserts ext_lbc_ack, the external logic can implement a timeout mechanism. The width of the ext_lbc_ack signal is equal to the number of functions in your controller configuration (CX_NFUNC). That is, there is one ext_lbc_ack bit for each configured function. This signal should never be hard-wired to 1'b1, even if the ELBI is not being used. This signal should only be high in response to lbc_ext_cs being asserted. Your application must assert ext_lbc_ack for two clock cycles, except for an error response (accessing a non-existent register or writing to a read-only register), when ext_lbc_ack must be asserted for one clock cycle. Exists: !(CX_IS_RC && !CX_CXL_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_ext_cs is asserted.
ext_lbc_din[(((CX_LBC_NW*32)*NF)-1):0]	I	Data bus from the external register block. Depending on value of CX_LBC_NW, there are 32/64/128 bits of ext_lbc_din for each function in your controller configuration. Exists: !(CX_IS_RC && !CX_CXL_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ext_lbc_ack is asserted

Port Name	I/O	Description
lbc_ext_addr[(LBC_EXT_AW-1):0]	O	Address bus to the external register block. The width of the address bus is the value you select for the CX_LBC_EXT_AW parameter. Exists: !(CX_IS_RC && !CX_CXL_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_ext_cs is asserted
lbc_ext_dout[((CX_LBC_NW*32)-1):0]	O	Write data bus to the external register block, driven to all functions in a multi-function configuration. Exists: !(CX_IS_RC && !CX_CXL_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_ext_cs is asserted
lbc_ext_cs[(NF-1):0]	O	The controller asserts lbc_ext_cs when a received TLP for a read or write request has an address in the range of your application device, as determined by the BAR configuration. The width of the lbc_ext_cs signal is equal to the number of functions in your controller configuration (CX_NFUNC). That is, there is one lbc_ext_cs bit for each configured function. The controller de-asserts lbc_ext_cs only after the external register block acknowledges completion of the access by asserting the corresponding bit of ext_lbc_ack. Exists: !(CX_IS_RC && !CX_CXL_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
lbc_ext_wr[((4*CX_LBC_NW)-1):0]	O	Indicates when the external register access is a read or a write. For writes, lbc_ext_wr also indicates the byte enables. When CX_LBC_NW = 1: ■ 0000b: Read ■ 0001b: Write byte 0 ■ 0010b: Write byte 1 ■ 0100b: Write byte 2 ■ 1000b: Write byte 3 ■ 1111b: Write all bytes When CX_LBC_NW = 2: ■ 00000000b: Read ■ 00000001b: Write byte 0 ■ 00000010b: Write byte 1 ■ 00000100b: Write byte 2 ■ 00001000b: Write byte 3 ■ 00010000b: Write byte 4 ■ 00100000b: Write byte 5 ■ 01000000b: Write byte 6 ■ 10000000b: Write byte 7 ■ 11111111b: Write all bytes Combinations of byte enables (for example, 0011b) are also valid. Exists: !(CX_IS_RC && !CX_CXL_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_ext_cs is asserted
lbc_ext_rd[((4*CX_LBC_NW)-1):0]	O	Indicates which byte lanes are valid for an ELBI read. When CX_LBC_NW = 2: ■ 00000000b: Write ■ 00001111b: Read lower DWORD ■ 11110000b: Read upper DWORD ■ 11111111b: Read all bytes Only the three combinations of byte enables mentioned here are valid. Exists: !(CX_IS_RC && !CX_CXL_ENABLE)) && (CX_UNINVERTED_ELBI_DATA) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_ext_cs is asserted

Port Name	I/O	Description
lbc_ext_dbi_access	O	Indicates that the current ELBI access is initiated using DBI. Exists: $\neg(CX_IS_RC \ \&\& \ \neg CX_CXL_ENABLE)$ Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_ext_cs is asserted
lbc_ext_cxl_mbar0_access	O	Indicates that the current ELBI access is for CXL MEMBAR0. Exists: $\neg(CX_IS_RC \ \&\& \ \neg CX_CXL_ENABLE) \ \&\& \ (CX_CXL_ENABLE)$ Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_ext_cs is asserted
lbc_ext_rom_access	O	Indicates that the current ELBI access is for expansion ROM. Exists: $\neg(CX_IS_RC \ \&\& \ \neg CX_CXL_ENABLE)$ Synchronous To: None, {aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_ext_cs is asserted
lbc_ext_io_access	O	Indicates that the current ELBI access is an I/O access. Exists: $\neg(CX_IS_RC \ \&\& \ \neg CX_CXL_ENABLE)$ Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_ext_cs is asserted

Port Name	I/O	Description
lbc_ext_bar_num[2:0]	O	The BAR number of the current ELBI access: ■ 000b: BAR 0 ■ 001b: BAR 1 ■ 010b: BAR 2 ■ 011b: BAR 3 ■ 100b: BAR 4 ■ 101b: BAR 5 ■ 110b is not used. ■ 111b is used to indicate a configuration access. This is designed for an application with application-specific configuration registers such as vendor capability. lbc_ext_bar_num is set to 3'b111 for all DBI accesses. Exists: !(CX_IS_RC && !CX_CXL_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_ext_cs is asserted
lbc_ext_vfunc_num[(VF_WD-2):0]	O	Identifies the virtual function (VF) inside the controller, which has been targeted. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, lbc_ext_vfunc_num=0 refers to the first VF of the PF that is identified by lbc_ext_func_num. For more information of VF numbering, see 'Programming View'. Exists: CX_SRIOV_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: lbc_ext_cs and lbc_ext_vfunc_active are asserted
lbc_ext_vfunc_active	O	Indicates that the received TLP has been targeted to a virtual function (VF) inside the controller, and is identified by lbc_ext_vfunc_num. Exists: CX_SRIOV_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_ext_cs is asserted

2.23 Master Completion Asynchronous Clock Crossing FIFO RAM Signals (AXI Master #2)

ibh_mcpl_a2c_cdc_ram_doutb -

- ibh_mcpl_a2c_cdc_ram_addra
- ibh_mcpl_a2c_cdc_ram_addra_par
- ibh_mcpl_a2c_cdc_ram_addrb
- ibh_mcpl_a2c_cdc_ram_addrb_par
- ibh_mcpl_a2c_cdc_ram_wea
- ibh_mcpl_a2c_cdc_ram_enb
- ibh_mcpl_a2c_cdc_ram_dina

Table 2-23 Master Completion Asynchronous Clock Crossing FIFO RAM Signals (AXI Master #2)

Port Name	I/O	Description
ibh_mcpl_a2c_cdc_ram_addra[(CC_IB_M CPL_CDC_RAM_ADDR_WD-1):0]	O	ibh_mcpl_a2c_cdc_ram_waddr. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_mcpl_a2c_cdc_ram_addra_par[(CC_I B_MCPL_CDC_RAM_ADDRP_WD-1):0]	O	ibh_mcpl_a2c_cdc_ram_waddr. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_mcpl_a2c_cdc_ram_addrb[(CC_IB_M CPL_CDC_RAM_ADDR_WD-1):0]	O	ibh_mcpl_a2c_cdc_ram_raddr. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ibh_mcpl_a2c_cdc_ram_addrb_par[(CC_IB_M_CPL_CDC_RAM_ADDRP_WD-1):0]	O	ibh_mcpl_a2c_cdc_ram_raddr. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_mcpl_a2c_cdc_ram_wea	O	ibh_mcpl_a2c_cdc_ram_wen. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_mcpl_a2c_cdc_ram_enb	O	ibh_mcpl_a2c_cdc_ram_ren. Port B Read Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_mcpl_a2c_cdc_ram_doutb[(CC_IB_M_CPL_CDC_RAM_DATA_WD-1):0]	I	ibh_mcpl_a2c_cdc_ram_rdata. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ibh_mcpl_a2c_cdc_ram_dina[(CC_IB_MC_PL_CDC_RAM_DATA_WD-1):0]	O	ibh_mcpl_a2c_cdc_ram_wdata. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.24 Master Completion Asynchronous Clock Crossing FIFO RAM Signals (AXI)

ib_mcpl_a2c_cdc_ram_doutb -

- ib_mcpl_a2c_cdc_ram_addra
- ib_mcpl_a2c_cdc_ram_addra_par
- ib_mcpl_a2c_cdc_ram_addrb
- ib_mcpl_a2c_cdc_ram_addrb_par
- ib_mcpl_a2c_cdc_ram_wea
- ib_mcpl_a2c_cdc_ram_enb
- ib_mcpl_a2c_cdc_ram_dina

Table 2-24 Master Completion Asynchronous Clock Crossing FIFO RAM Signals (AXI)

Port Name	I/O	Description
ib_mcpl_a2c_cdc_ram_addra[(CC_IB_MCPL_CDC_RAM_ADDR_WD-1):0]	O	ib_mcpl_a2c_cdc_ram_waddr. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_mcpl_a2c_cdc_ram_addra_par[(CC_IB_MCPL_CDC_RAM_ADDRP_WD-1):0]	O	ib_mcpl_a2c_cdc_ram_waddr. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0)))? 0: 1) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_mcpl_a2c_cdc_ram_addrb[(CC_IB_MCPL_CDC_RAM_ADDR_WD-1):0]	O	ib_mcpl_a2c_cdc_ram_raddr. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_mcpl_a2c_cdc_ram_addrb_par[(CC_IB_MCPL_CDC_RAM_ADDRP_WD-1):0]	O	ib_mcpl_a2c_cdc_ram_raddr. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_mcpl_a2c_cdc_ram_wea	O	ib_mcpl_a2c_cdc_ram_wen. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: mstr_ack Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_mcpl_a2c_cdc_ram_enb	O	ib_mcpl_a2c_cdc_ram_ren. Port B Read Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_mcpl_a2c_cdc_ram_doutb[(CC_IB_MCPL_CDC_RAM_DATA_WD-1):0]	I	ib_mcpl_a2c_cdc_ram_rdata. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_mcpl_a2c_cdc_ram_dina[(CC_IB_MCP_L_CDC_RAM_DATA_WD-1):0]	O	ib_mcpl_a2c_cdc_ram_wdata. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: None,perbitclk,{mstr_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.25 Master Completion Buffer RAM Signals (AXI Master #2)

ibh_mcpl_sb_ram_doutb -

- ibh_mcpl_sb_ram_addra
- ibh_mcpl_sb_ram_addra_par
- ibh_mcpl_sb_ram_addrb
- ibh_mcpl_sb_ram_addrb_par
- ibh_mcpl_sb_ram_dina
- ibh_mcpl_sb_ram_enb
- ibh_mcpl_sb_ram_wea

Table 2-25 Master Completion Buffer RAM Signals (AXI Master #2)

Port Name	I/O	Description
ibh_mcpl_sb_ram_addra[((CC_MSTR_CPL_SEG_N_SLICES*CC_IB_MCPL_SEG_BUF_RAM_ADDR_WD)-1):0]	O	High Priority Master Completion Buffer RAM waddr. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_mcpl_sb_ram_addra_par[((CC_MSTR_CPL_SEG_N_SLICES*CC_IB_MCPL_SEG_BUF_RAM_ADDRP_WD)-1):0]	O	High Priority Master Completion Buffer RAM waddr. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_mcpl_sb_ram_addrb[((CC_MSTR_CPL_SEG_N_SLICES*CC_IB_MCPL_SEG_BUF_RAM_ADDR_WD)-1):0]	O	High Priority Master Completion Buffer RAM raddr. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ibh_mcpl_sb_ram_addrb_par[((CC_MSTR_CPL_SEG_N_SLICES*CC_IB_MCPL_SEG_BUF_RAM_ADDRP_WD)-1):0]	O	High Priority Master Completion Buffer RAM raddr. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_mcpl_sb_ram_dina[((CC_MSTR_CPL_SEG_N_SLICES*CC_MSTR_CPL_SEG_BUF_SLICE_WD)-1):0]	O	High Priority Master Completion Buffer RAM wdata. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_mcpl_sb_ram_doutb[((CC_MSTR_CPL_SEG_N_SLICES*CC_MSTR_CPL_SEG_BUF_SLICE_WD)-1):0]	I	High Priority Master Completion Buffer RAM rdata. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_mcpl_sb_ram_enb[((CC_MSTR_CPL_SEG_N_SLICES-1):0]	O	High Priority Master Completion Buffer RAM ren. Port B read Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ibh_mcpl_sb_ram_wea[(CC_MSTR_CPL_SEG_N_SLICES-1):0]	O	High Priority Master Completion Buffer RAM wen. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.26 Master Completion Buffer RAM Signals (AXI)

ib_mcpl_sb_ram_doutb -

- ib_mcpl_sb_ram_addra
- ib_mcpl_sb_ram_addra_par
- ib_mcpl_sb_ram_addrb
- ib_mcpl_sb_ram_addrb_par
- ib_mcpl_sb_ram_dina
- ib_mcpl_sb_ram_enb
- ib_mcpl_sb_ram_wea

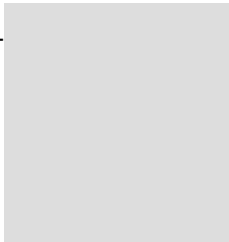
Table 2-26 Master Completion Buffer RAM Signals (AXI)

Port Name	I/O	Description
ib_mcpl_sb_ram_addra[(((CC_MSTR_CPL_SEG_N_SLICES*CC_IB_MCPL_SEG_BUF_RAM_ADDR_WD)-1):0]	O	Master Completion Buffer RAM waddr. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_mcpl_sb_ram_addra_par[(((CC_MSTR_CPL_SEG_N_SLICES*CC_IB_MCPL_SEG_BUF_RAM_ADDRP_WD)-1):0]	O	Master Completion Buffer RAM waddr. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_mcpl_sb_ram_addrb[(((CC_MSTR_CPL_SEG_N_SLICES*CC_IB_MCPL_SEG_BUF_RAM_ADDR_WD)-1):0]	O	Master Completion Buffer RAM raddr. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_mcpl_sb_ram_addrb_par[((CC_MSTR_CPL_SEG_N_SLICES*CC_IB_MCPL_SEG_BUF_RAM_ADDRP_WD)-1):0]	O	Master Completion Buffer RAM raddr. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_mcpl_sb_ram_dina[((CC_MSTR_CPL_SEG_N_SLICES*CC_MSTR_CPL_SEG_BUF_SLICE_WD)-1):0]	O	Master Completion Buffer RAM wdata. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None,perbitclk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_mcpl_sb_ram_doutb[((CC_MSTR_CPL_SEG_N_SLICES*CC_MSTR_CPL_SEG_BUF_SLICE_WD)-1):0]	I	Master Completion Buffer RAM rdata. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_mcpl_sb_ram_enb[(CC_MSTR_CPL_SEG_N_SLICES-1):0]	O	Master Completion Buffer RAM ren. Port B read Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_mcpl_sb_ram_wea[(CC_MSTR_CPL_SEG_N_SLICES-1):0]	O	Master Completion Buffer RAM wen. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.27 Master Non-posted Request Buffer RAM Signals (AXI)

ib_rreq_ordr_ram_douthb -  -

- ib_rreq_ordr_ram_addra
- ib_rreq_ordr_ram_addra_par
- ib_rreq_ordr_ram_addrb
- ib_rreq_ordr_ram_addrb_par
- ib_rreq_ordr_ram_wea
- ib_rreq_ordr_ram_enb
- ib_rreq_ordr_ram_dina

Table 2-27 Master Non-posted Request Buffer RAM Signals (AXI)

Port Name	I/O	Description
ib_rreq_ordr_ram_addra[(CC_IB_RD_REQ_ORDR_RAM_ADDR_WD-1):0]	O	Master Non-posted Request Buffer RAM. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_rreq_ordr_ram_addra_par[(CC_IB_RD_REQ_ORDR_RAM_ADDRP_WD-1):0]	O	Master Non-posted Request Buffer RAM. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_rreq_ordr_ram_addrb[(CC_IB_RD_REQ_ORDR_RAM_ADDR_WD-1):0]	O	Master Non-posted Request Buffer RAM. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_rreq_ordr_ram_addrb_par[(CC_IB_RD_REQ_ORDR_RAM_ADDRP_WD-1):0]	O	Master Non-posted Request Buffer RAM. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_rreq_ordr_ram_wea	O	Master Non-posted Request Buffer RAM. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_rreq_ordr_ram_enb	O	Master Non-posted Request Buffer RAM. Port B enable Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_rreq_ordr_ram_dina[(CC_IB_RD_REQ_ORDR_RAM_DATA_WD-1):0]	O	Master Non-posted Request Buffer RAM. Port A write data Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None,perbitclk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_rreq_ordr_ram_doutb[(CC_IB_RD_REQ_ORDR_RAM_DATA_WD-1):0]	I	Master Non-posted Request Buffer RAM.Port B read data Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.28 Master Posted Tracker Data Buffer RAM Signals (AXI Master #2)



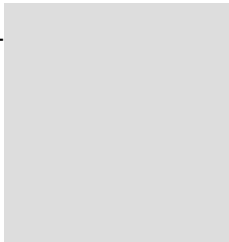
Table 2-28 Master Posted Tracker Data Buffer RAM Signals (AXI Master #2)

Port Name	I/O	Description
<code>ibh_wreq_ptrk_data_ram_addra[(CC_IB_WR_REQ_PTRK_DATA_RAM_ADDR_WD-1):0]</code>	O	Inbound Posted High Priority Tracker Data Buffer RAM waddr. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) Synchronous To: <code>core_clk</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
<code>ibh_wreq_ptrk_data_ram_addra_par[(CC_IB_WR_REQ_PTRK_DATA_RAM_ADDRP_WD-1):0]</code>	O	Inbound Posted High Priority Tracker Data Buffer RAM waddr. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: <code>core_clk</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
<code>ibh_wreq_ptrk_data_ram_addrb[(CC_IB_WR_REQ_PTRK_DATA_RAM_ADDR_WD-1):0]</code>	O	Inbound Posted High Priority Tracker Data Buffer RAM raddr. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) Synchronous To: <code>core_clk</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ibh_wreq_ptrk_data_ram_addrb_par[(CC_IB_WR_REQ_PTRK_DATA_RAM_ADDRP_WD-1):0]	O	Inbound Posted High Priority Tracker Data Buffer RAM raddr. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_ptrk_data_ram_dina[(CC_IB_WR_REQ_PTRK_DATA_RAM_DATA_WD-1):0]	O	Inbound Posted High Priority Tracker Data Buffer RAM wdata. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_ptrk_data_ram_doutb[(CC_IB_WR_REQ_PTRK_DATA_RAM_DATA_WD-1):0]	I	Inbound Posted High Priority Tracker Data Buffer RAM rdata. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_ptrk_data_ram_enb	O	Inbound Posted High Priority Tracker Data Buffer RAM ren. Port B read Enable. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ibh_wreq_ptrk_data_ram_wea	O	Inbound Posted High Priority Tracker Data Buffer RAM wen. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.29 Master Posted Tracker Data Buffer RAM Signals (AXI)

ib_wreq_ptrk_data_ram_douth -  -

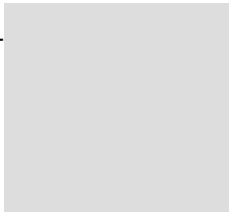
- ib_wreq_ptrk_data_ram_addra
- ib_wreq_ptrk_data_ram_addra_par
- ib_wreq_ptrk_data_ram_addrb
- ib_wreq_ptrk_data_ram_addrb_par
- ib_wreq_ptrk_data_ram_dina
- ib_wreq_ptrk_data_ram_enb
- ib_wreq_ptrk_data_ram_wea

Table 2-29 Master Posted Tracker Data Buffer RAM Signals (AXI)

Port Name	I/O	Description
ib_wreq_ptrk_data_ram_addra[(CC_IB_W R_REQ_PTRK_DATA_RAM_ADDR_WD-1):0]	O	Inbound Posted Tracker Data Buffer RAM waddr. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_data_ram_addra_par[(CC_I B_WR_REQ_PTRK_DATA_RAM_ADDRP _WD-1):0]	O	Inbound Posted Tracker Data Buffer RAM waddr. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_data_ram_addrb[(CC_IB_W R_REQ_PTRK_DATA_RAM_ADDR_WD-1):0]	O	Inbound Posted Tracker Data Buffer RAM raddr. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_wreq_ptrk_data_ram_addrb_par[(CC_IB_WR_REQ_PTRK_DATA_RAM_ADDRP_WD-1):0]	O	Inbound Posted Tracker Data Buffer RAM raddr. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_data_ram_dina[(CC_IB_WR_REQ_PTRK_DATA_RAM_DATA_WD-1):0]	O	Inbound Posted Tracker Data Buffer RAM wdata. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_data_ram_doutb[(CC_IB_WR_REQ_PTRK_DATA_RAM_DATA_WD-1):0]	I	Inbound Posted Tracker Data Buffer RAM rdata. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_data_ram_enb	O	Inbound Posted Tracker Data Buffer RAM ren. Port B read Enable. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_data_ram_wea	O	Inbound Posted Tracker Data Buffer RAM wen. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.30 Master Posted Tracker Header Buffer RAM Signals (AXI Master #2)

ibh_wreq_ptrk_hdr_ram_doutb -  -

- ibh_wreq_ptrk_hdr_ram_addra
- ibh_wreq_ptrk_hdr_ram_addra_par
- ibh_wreq_ptrk_hdr_ram_addrb
- ibh_wreq_ptrk_hdr_ram_addrb_par
- ibh_wreq_ptrk_hdr_ram_dina
- ibh_wreq_ptrk_hdr_ram_wea

Table 2-30 Master Posted Tracker Header Buffer RAM Signals (AXI Master #2)

Port Name	I/O	Description
ibh_wreq_ptrk_hdr_ram_addra[(CC_IB_W R_REQ_PTRK_HDR_RAM_ADDR_WD-1):0]	O	Inbound Posted High Priority Tracker Header Buffer RAM waddr. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_ptrk_hdr_ram_addra_par[(CC_I B_WR_REQ_PTRK_HDR_RAM_ADDRP _WD-1):0]	O	Inbound Posted High Priority Tracker Header Buffer RAM waddr. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_ptrk_hdr_ram_addrb[(CC_IB_W R_REQ_PTRK_HDR_RAM_ADDR_WD-1):0]	O	Inbound Posted High Priority Tracker Header Buffer RAM raddr. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ibh_wreq_ptrk_hdr_ram_addrb_par[(CC_IB_WR_REQ_PTRK_HDR_RAM_ADDRP_WD-1):0]	O	Inbound Posted High Priority Tracker Header Buffer RAM raddr. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_ptrk_hdr_ram_dina[(CC_IB_WR_REQ_PTRK_HDR_RAM_DATA_WD-1):0]	O	Inbound Posted High Priority Tracker Header Buffer RAM wdata. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_ptrk_hdr_ram_doutb[(CC_IB_WR_REQ_PTRK_HDR_RAM_DATA_WD-1):0]	I	Inbound Posted High Priority Tracker Header Buffer RAM rdata. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_ptrk_hdr_ram_wea	O	Inbound Posted High Priority Tracker Header Buffer RAM wen. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.31 Master Posted Tracker Header Buffer RAM Signals (AXI)

ib_wreq_ptrk_hdr_ram_doutb -

- ib_wreq_ptrk_hdr_ram_addra
- ib_wreq_ptrk_hdr_ram_addra_par
- ib_wreq_ptrk_hdr_ram_addrb
- ib_wreq_ptrk_hdr_ram_addrb_par
- ib_wreq_ptrk_hdr_ram_dina
- ib_wreq_ptrk_hdr_ram_enb
- ib_wreq_ptrk_hdr_ram_wea

Table 2-31 Master Posted Tracker Header Buffer RAM Signals (AXI)

Port Name	I/O	Description
ib_wreq_ptrk_hdr_ram_addra[(CC_IB_WR_REQ_PTRK_HDR_RAM_ADDR_WD-1):0]	O	Inbound Posted Tracker Header Buffer RAM waddr. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_hdr_ram_addra_par[(CC_IB_WR_REQ_PTRK_HDR_RAM_ADDRP_WD-1):0]	O	Inbound Posted Tracker Header Buffer RAM waddr. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_hdr_ram_addrb[(CC_IB_WR_REQ_PTRK_HDR_RAM_ADDR_WD-1):0]	O	Inbound Posted Tracker Header Buffer RAM raddr. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_wreq_ptrk_hdr_ram_addrb_par[(CC_IB_WR_REQ_PTRK_HDR_RAM_ADDRP_WD-1):0]	O	Inbound Posted Tracker Header Buffer RAM raddr. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_hdr_ram_dina[(CC_IB_WR_REQ_PTRK_HDR_RAM_DATA_WD-1):0]	O	Inbound Posted Tracker Header Buffer RAM wdata. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: None,perbitclk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_hdr_ram_doutb[(CC_IB_WR_REQ_PTRK_HDR_RAM_DATA_WD-1):0]	I	Inbound Posted Tracker Header Buffer RAM rdata. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_hdr_ram_enb	O	Inbound Posted Tracker Header Buffer RAM ren. Port B read Enable. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_ptrk_hdr_ram_wea	O	Inbound Posted Tracker Header Buffer RAM wen. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (CC_IB_WREQ_PTRK_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.32 Master Read Queue Request Formation CDC Buffer RAM Signals (AXI Master #2)

ibh_rreq_c2a_cdc_ram_doutb -

- ibh_rreq_c2a_cdc_ram_addra
- ibh_rreq_c2a_cdc_ram_addra_par
- ibh_rreq_c2a_cdc_ram_addrb
- ibh_rreq_c2a_cdc_ram_addrb_par
- ibh_rreq_c2a_cdc_ram_wea
- ibh_rreq_c2a_cdc_ram_enb
- ibh_rreq_c2a_cdc_ram_dina

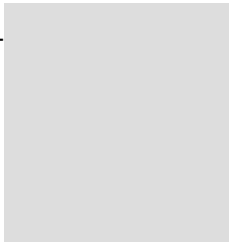
Table 2-32 Master Read Queue Request Formation CDC Buffer RAM Signals (AXI Master #2)

Port Name	I/O	Description
ibh_rreq_c2a_cdc_ram_addra[(CC_IB_RD_REQ_CDC_RAM_ADDR_WD-1):0]	O	High Priority Master Read Queue Request Formation CDC Buffer RAM. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_rreq_c2a_cdc_ram_addra_par[(CC_IB_RD_REQ_CDC_RAM_ADDRP_WD-1):0]	O	High Priority Master Read Queue Request Formation CDC Buffer RAM. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ibh_rreq_c2a_cdc_ram_addrb[(CC_IB_RD_REQ_CDC_RAM_ADDR_WD-1):0]	O	High Priority Master Read Queue Request Formation CDC Buffer RAM. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_rreq_c2a_cdc_ram_addrb_par[(CC_IB_RD_REQ_CDC_RAM_ADDRP_WD-1):0]	O	High Priority Master Read Queue Request Formation CDC Buffer RAM. Port B Read Address Parity Bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_rreq_c2a_cdc_ram_wea	O	High Priority Master Read Queue Request Formation CDC Buffer RAM. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ibh_rreq_c2a_cdc_ram_enb	O	High Priority Master Read Queue Request Formation CDC Buffer RAM. Port B read Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_rreq_c2a_cdc_ram_dina[(CC_IB_RD_REQ_CDC_RAM_DATA_WD-1):0]	O	High Priority Master Read Queue Request Formation CDC Buffer RAM. Port A Data write. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_rreq_c2a_cdc_ram_doutb[(CC_IB_RD_REQ_CDC_RAM_DATA_WD-1):0]	I	High Priority Master Read Queue Request Formation CDC Buffer RAM. Port B Data read. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.33 Master Read Queue Request Formation CDC Buffer RAM Signals (AXI)

ib_rreq_c2a_cdc_ram_douth -  -

- ib_rreq_c2a_cdc_ram_addra
- ib_rreq_c2a_cdc_ram_addra_par
- ib_rreq_c2a_cdc_ram_addrb
- ib_rreq_c2a_cdc_ram_addrb_par
- ib_rreq_c2a_cdc_ram_wea
- ib_rreq_c2a_cdc_ram_enb
- ib_rreq_c2a_cdc_ram_dina

Table 2-33 Master Read Queue Request Formation CDC Buffer RAM Signals (AXI)

Port Name	I/O	Description
ib_rreq_c2a_cdc_ram_addra[(CC_IB_RD_REQ_CDC_RAM_ADDR_WD-1):0]	O	Master Read Queue Request Formation CDC Buffer RAM. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_rreq_c2a_cdc_ram_addra_par[(CC_IB_RD_REQ_CDC_RAM_ADDRP_WD-1):0]	O	Master Read Queue Request Formation CDC Buffer RAM. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_rreq_c2a_cdc_ram_addrb[(CC_IB_RD_REQ_CDC_RAM_ADDR_WD-1):0]	O	Master Read Queue Request Formation CDC Buffer RAM. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_rreq_c2a_cdc_ram_addrb_par[(CC_IB_RD_REQ_CDC_RAM_ADDRP_WD-1):0]	O	Master Read Queue Request Formation CDC Buffer RAM. Port B Read Address Parity Bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_rreq_c2a_cdc_ram_wea	O	Master Read Queue Request Formation CDC Buffer RAM. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_rreq_c2a_cdc_ram_enb	O	Master Read Queue Request Formation CDC Buffer RAM. Port B read Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_rreq_c2a_cdc_ram_dina[(CC_IB_RD_REQ_CDC_RAM_DATA_WD-1):0]	O	Master Read Queue Request Formation CDC Buffer RAM. Port A Data write. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: None,perbitclk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_rreq_c2a_cdc_ram_doutb[(CC_IB_RD_REQ_CDC_RAM_DATA_WD-1):0]	I	Master Read Queue Request Formation CDC Buffer RAM. Port B Data read. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.34 Master Write Queue Request Formation CDC Buffer RAM Signals (AXI Master #2)

ibh_wreq_c2a_cdc_ram_doutb -

- ibh_wreq_c2a_cdc_ram_addra
- ibh_wreq_c2a_cdc_ram_addra_par
- ibh_wreq_c2a_cdc_ram_addrb
- ibh_wreq_c2a_cdc_ram_addrb_par
- ibh_wreq_c2a_cdc_ram_wea
- ibh_wreq_c2a_cdc_ram_enb
- ibh_wreq_c2a_cdc_ram_dina

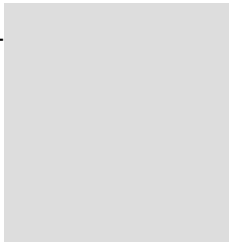
Table 2-34 Master Write Queue Request Formation CDC Buffer RAM Signals (AXI Master #2)

Port Name	I/O	Description
ibh_wreq_c2a_cdc_ram_addra[(CC_IB_WR_REQ_CDC_RAM_ADDR_WD-1):0]	O	High Priority Master Write Queue Request Formation CDC Buffer RAM. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_c2a_cdc_ram_addra_par[(CC_IB_WR_REQ_CDC_RAM_ADDRP_WD-1):0]	O	High Priority Master Write Queue Request Formation CDC Buffer RAM. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ibh_wreq_c2a_cdc_ram_addrb[(CC_IB_W R_REQ_CDC_RAM_ADDR_WD-1):0]	O	High Priority Master Write Queue Request Formation CDC Buffer RAM. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_c2a_cdc_ram_addrb_par[(CC_I B_WR_REQ_CDC_RAM_ADDRP_WD-1): 0]	O	High Priority Master Write Queue Request Formation CDC Buffer RAM. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_c2a_cdc_ram_wea	O	High Priority Master Write Queue Request Formation CDC Buffer RAM. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ibh_wreq_c2a_cdc_ram_enb	O	High Priority Master Write Queue Request Formation CDC Buffer RAM. Port B read Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_c2a_cdc_ram_dina[(CC_IB_WR_REQ_CDC_RAM_DATA_WD-1):0]	O	High Priority Master Write Queue Request Formation CDC Buffer RAM. Port A Data write. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ibh_wreq_c2a_cdc_ram_doutb[(CC_IB_WR_REQ_CDC_RAM_DATA_WD-1):0]	I	High Priority Master Write Queue Request Formation CDC Buffer RAM. Port B Data read. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (CC_HP_MASTER) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.35 Master Write Queue Request Formation CDC Buffer RAM Signals (AXI)

ib_wreq_c2a_cdc_ram_douth -  -

- ib_wreq_c2a_cdc_ram_addra
- ib_wreq_c2a_cdc_ram_addra_par
- ib_wreq_c2a_cdc_ram_addrb
- ib_wreq_c2a_cdc_ram_addrb_par
- ib_wreq_c2a_cdc_ram_wea
- ib_wreq_c2a_cdc_ram_enb
- ib_wreq_c2a_cdc_ram_dina

Table 2-35 Master Write Queue Request Formation CDC Buffer RAM Signals (AXI)

Port Name	I/O	Description
ib_wreq_c2a_cdc_ram_addra[(CC_IB_W R_REQ_CDC_RAM_ADDR_WD-1):0]	O	Master Write Queue Request Formation CDC Buffer RAM. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_c2a_cdc_ram_addra_par[(CC_IB _WR_REQ_CDC_RAM_ADDRP_WD-1):0]	O	Master Write Queue Request Formation CDC Buffer RAM. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_c2a_cdc_ram_addrb[(CC_IB_W R_REQ_CDC_RAM_ADDR_WD-1):0]	O	Master Write Queue Request Formation CDC Buffer RAM. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_wreq_c2a_cdc_ram_addrb_par[(CC_IB_WR_REQ_CDC_RAM_ADDRP_WD-1):0]	O	Master Write Queue Request Formation CDC Buffer RAM. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_c2a_cdc_ram_wea	O	Master Write Queue Request Formation CDC Buffer RAM. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_c2a_cdc_ram_enb	O	Master Write Queue Request Formation CDC Buffer RAM. Port B read Enable. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ib_wreq_c2a_cdc_ram_dina[(CC_IB_WR_REQ_CDC_RAM_DATA_WD-1):0]	O	Master Write Queue Request Formation CDC Buffer RAM. Port A Data write. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: None,perbitclk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ib_wreq_c2a_cdc_ram_doutb[(CC_IB_WR_REQ_CDC_RAM_DATA_WD-1):0]	I	Master Write Queue Request Formation CDC Buffer RAM. Port B Data read. Exists: (AMBA_INTERFACE!=0) && (MASTER_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MSTR_CLK_DIFF_ENABLE) Synchronous To: mstr_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.36 Slave Completion Composer Segmented Buffer RAM Signals (AXI)

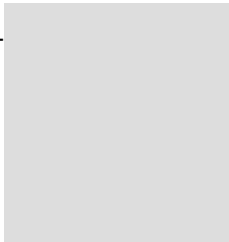
ob_ccmp_data_ram_douth -  - ob_ccmp_data_ram_addra
 - ob_ccmp_data_ram_addra_par
 - ob_ccmp_data_ram_wea
 - ob_ccmp_data_ram_dina
 - ob_ccmp_data_ram_addrb
 - ob_ccmp_data_ram_addrb_par
 - ob_ccmp_data_ram_enb

Table 2-36 Slave Completion Composer Segmented Buffer RAM Signals (AXI)

Port Name	I/O	Description
ob_ccmp_data_ram_addra[(OB_CCMP_DATA_RAM_ADDR_WD-1):0]	O	Slave Response Composer RAM. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_ccmp_data_ram_addra_par[(OB_CCMP_DATA_RAM_ADDRP_WD-1):0]	O	Slave Response Composer RAM. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_ccmp_data_ram_wea	O	Slave Response Composer RAM. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ob_ccmp_data_ram_dina[(OB_CCMP_DATA_RAM_DATA_WD-1):0]	O	Slave Response Composer RAM. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_ccmp_data_ram_addrb[(OB_CCMP_DATA_RAM_ADDR_WD-1):0]	O	Slave Response Composer RAM. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_ccmp_data_ram_addrb_par[(OB_CCMP_DATA_RAM_ADDRP_WD-1):0]	O	Slave Response Composer RAM. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0)))? 0: 1) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_ccmp_data_ram_enb	O	Slave Response Composer RAM. Port B Read Enable. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ob_ccmp_data_ram_douth[(OB_CCMP_DATA_RAM_DATA_WD-1):0]	I	Slave Response Composer RAM. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.37 Slave Non-Posted TLP Decomposer Header/Data FIFO RAM Signals (AXI)

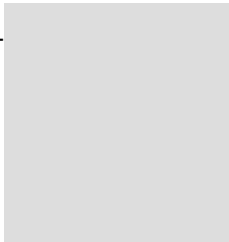
ob_npdcmp_ram_doutb -  - ob_npdcmp_ram_addra
 - ob_npdcmp_ram_addra_par
 - ob_npdcmp_ram_wea
 - ob_npdcmp_ram_dina
 - ob_npdcmp_ram_addrb
 - ob_npdcmp_ram_addrb_par
 - ob_npdcmp_ram_enb

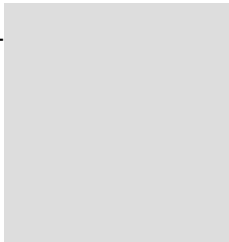
Table 2-37 Slave Non-Posted TLP Decomposer Header/Data FIFO RAM Signals (AXI)

Port Name	I/O	Description
ob_npdcmp_ram_addra[(OB_NPDCMP_RAM_ADDR_WD-1):0]	O	Slave Non-Posted TLP Decomposer Header/Data FIFO RAM. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_npdcmp_ram_addra_par[(OB_NPDCMP_RAM_ADDR_WD-1):0]	O	Slave Non-Posted TLP Decomposer Header/Data FIFO RAM. Port B Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_npdcmp_ram_wea	O	Slave Non-Posted TLP Decomposer Header/Data FIFO RAM. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ob_npdcmp_ram_dina[(OB_NPDCMP_RAM_DATA_WD-1):0]	O	Slave Non-Posted TLP Decomposer Header/Data FIFO RAM. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_npdcmp_ram_addrb[(OB_NPDCMP_RAM_ADDR_WD-1):0]	O	Slave Non-Posted TLP Decomposer Header/Data FIFO RAM. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_npdcmp_ram_addrb_par[(OB_NPDCMP_RAM_ADDRP_WD-1):0]	O	Slave Non-Posted TLP Decomposer Header/Data FIFO RAM. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0)))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_npdcmp_ram_enb	O	Slave Non-Posted TLP Decomposer Header/Data FIFO RAM. Port B Read Enable. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ob_npdcmp_ram_doutb[(OB_NPDCMP_RAM_DATA_WD-1):0]	I	Slave Non-Posted TLP Decomposer Header/Data FIFO RAM. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.38 Slave Non-Posted Write Set-Aside Buffer RAM Signals (AXI)

slv_npw_sab_ram_douthb -  -

- slv_npw_sab_ram_addra
- slv_npw_sab_ram_addra_par
- slv_npw_sab_ram_wea
- slv_npw_sab_ram_dina
- slv_npw_sab_ram_addrb
- slv_npw_sab_ram_addrb_par
- slv_npw_sab_ram_enb

Table 2-38 Slave Non-Posted Write Set-Aside Buffer RAM Signals (AXI)

Port Name	I/O	Description
slv_npw_sab_ram_addra[(SLV_NPW_SAB_RAM_ADDR_WD-1):0]	O	Slave Non-Posted Write Set-Aside RAM. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
slv_npw_sab_ram_addra_par[(SLV_NPW_SAB_RAM_ADDRP_WD-1):0]	O	Slave Non-Posted Write Set-Aside RAM. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
slv_npw_sab_ram_wea	O	Slave Non-Posted Write Set-Aside RAM. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
slv_npw_sab_ram_dina[(SLV_NPW_SAB_RAM_DATA_WD-1):0]	O	Slave Non-Posted Write Set-Aside RAM. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
slv_npw_sab_ram_addrb[(SLV_NPW_SA_B_RAM_ADDR_WD-1):0]	O	Slave Non-Posted Write Set-Aside RAM. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
slv_npw_sab_ram_addrb_par[(SLV_NPW_SAB_RAM_ADDRP_WD-1):0]	O	Slave Non-Posted Write Set-Aside RAM. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
slv_npw_sab_ram_enb	O	Slave Non-Posted Write Set-Aside RAM. Port B Read Enable. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
slv_npw_sab_ram_doutb[(SLV_NPW_SA B_RAM_DATA_WD-1):0]	I	Slave Non-Posted Write Set-Aside RAM. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.39 Slave Posted TLP Decomposer Data FIFO RAM Signals (AXI)

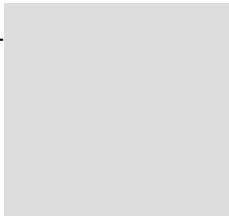
ob_pdcmp_data_ram_douth -  - ob_pdcmp_data_ram_addra
 - ob_pdcmp_data_ram_addra_par
 - ob_pdcmp_data_ram_wea
 - ob_pdcmp_data_ram_dina
 - ob_pdcmp_data_ram_addrb
 - ob_pdcmp_data_ram_addrb_par

Table 2-39 Slave Posted TLP Decomposer Data FIFO RAM Signals (AXI)

Port Name	I/O	Description
ob_pdcmp_data_ram_addra[(OB_PDCMP_DATA_RAM_ADDR_WD-1):0]	O	Slave Posted TLP Decomposer Data FIFO RAM. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_pdcmp_data_ram_addra_par[(OB_PDCMP_DATA_RAM_ADDRP_WD-1):0]	O	Slave Posted TLP Decomposer Data FIFO RAM. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0)))? 0: 1) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_pdcmp_data_ram_wea	O	Slave Posted TLP Decomposer Data FIFO RAM. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ob_pdcmp_data_ram_dina[(OB_PDCMP_DATA_RAM_DATA_WD-1):0]	O	Slave Posted TLP Decomposer Data FIFO RAM. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_pdcmp_data_ram_addrb[(OB_PDCMP_DATA_RAM_ADDR_WD-1):0]	O	Slave Posted TLP Decomposer Data FIFO RAM. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_pdcmp_data_ram_addrb_par[(OB_PDCMP_DATA_RAM_ADDRP_WD-1):0]	O	Slave Posted TLP Decomposer Data FIFO RAM. Port B Read Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0)))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_pdcmp_data_ram_doutb[(OB_PDCMP_DATA_RAM_DATA_WD-1):0]	I	Slave Posted TLP Decomposer Data FIFO RAM. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.40 Slave Posted TLP Decomposer Header FIFO RAM Signals (AXI)

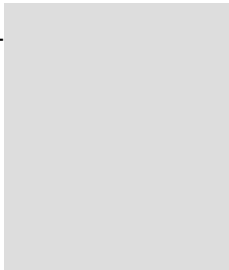
ob_pdcmp_hdr_ram_doutb -  - ob_pdcmp_hdr_ram_addra
 - ob_pdcmp_hdr_ram_addra_par
 - ob_pdcmp_hdr_ram_wea
 - ob_pdcmp_hdr_ram_dina
 - ob_pdcmp_hdr_ram_addrb
 - ob_pdcmp_hdr_ram_addrb_par
 - ob_pdcmp_hdr_ram_enb
 - ob_pdcmp_data_ram_enb

Table 2-40 Slave Posted TLP Decomposer Header FIFO RAM Signals (AXI)

Port Name	I/O	Description
ob_pdcmp_hdr_ram_addra[(OB_PDCMP_HDR_RAM_ADDR_WD-1):0]	O	Slave Posted TLP Decomposer Header FIFO RAM. Port A Write Address. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_pdcmp_hdr_ram_addra_par[(OB_PDCMP_HDR_RAM_ADDRP_WD-1):0]	O	Slave Posted TLP Decomposer Header FIFO RAM. Port A Write Address Parity bits. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0)))? 0: 1) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_pdcmp_hdr_ram_wea	O	Slave Posted TLP Decomposer Header FIFO RAM. Port A Write Enable. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: slv_aclk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ob_pdcmp_hdr_ram_dina[(OB_PDCMP_HDR_RAM_DATA_WD-1):0]	O	Slave Posted TLP Decomposer Header FIFO RAM. Port A Write Data. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: None,perbitclk,{slv_aclk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_pdcmp_hdr_ram_addrb[(OB_PDCMP_HDR_RAM_ADDR_WD-1):0]	O	Slave Posted TLP Decomposer Header FIFO RAM. Port B Read Address. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_pdcmp_hdr_ram_addrb_par[(OB_PDCMP_HDR_RAM_ADDRP_WD-1):0]	O	Slave Posted TLP Decomposer Header FIFO RAM. Port B Read Address Parity bits Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_pdcmp_hdr_ram_enb	O	Slave Posted TLP Decomposer Header FIFO RAM. Port B Read Enable. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ob_pdcmp_hdr_ram_doutb[(OB_PDCMP_HDR_RAM_DATA_WD-1):0]	I	Slave Posted TLP Decomposer Header FIFO RAM. Port B Read Data. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ob_pdcmp_data_ram_enb	O	Outbound Posted TLP Decomposer Data FIFO RAM. Port B Read Enable. Exists: (AMBA_INTERFACE!=0) && (SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.41 CXL Retry Buffer FIFO RAM Signals

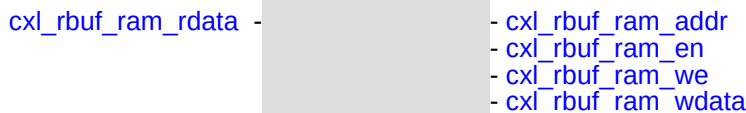


Table 2-41 CXL Retry Buffer FIFO RAM Signals

Port Name	I/O	Description
cxl_rbuf_ram_addr[(CX_CXL_RBUF_RAM_ADDRW-1):0]	O	CXL Retry Buffer RAM - Address. Exists: CX_CXL_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_rbuf_ram_en is asserted
cxl_rbuf_ram_en	O	CXL Retry Buffer RAM - Memory Enable. Exists: CX_CXL_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_rbuf_ram_we	O	CXL Retry Buffer RAM - Write Enable. Exists: CX_CXL_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_rbuf_ram_wdata[(CX_CXL_RBUF_RAM_DATAW-1):0]	O	CXL Retry Buffer RAM - Write Data. Exists: CX_CXL_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_rbuf_ram_en and cxl_rbuf_ram_we are asserted

Port Name	I/O	Description
cxl_rbuf_ram_rdata[(CX_CXL_RBUF_RAM_DATAW-1):0]	I	CXL Retry Buffer RAM - Read Data. Exists: CX_CXL_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_rbuf_ram_en is asserted

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cxl_mst_hreq_drsp_ram0_rdata	-	cxl_mst_hreq_drsp_ram0_we
cxl_mst_hreq_drsp_ram1_rdata	-	cxl_mst_hreq_drsp_ram0_wr_addr
cxl_mst_hrsp_dreq_ram0_rdata	-	cxl_mst_hreq_drsp_ram0_wr_addr_par
cxl_mst_hrsp_dreq_ram1_rdata	-	cxl_mst_hreq_drsp_ram0_wdata
cxl_mst_hrsp_dreq_ram2_rdata	-	cxl_mst_hreq_drsp_ram0_rd_addr
cxl_mst_hrsp_dreq_ram3_rdata	-	cxl_mst_hreq_drsp_ram0_rd_addr_par
cxl_mst_hd_data_ram_rdata	-	cxl_mst_hreq_drsp_ram0_rd
cxl_mst_ms_ndr_ram0_rdata	-	cxl_mst_hreq_drsp_ram1_we
cxl_mst_ms_ndr_ram1_rdata	-	cxl_mst_hreq_drsp_ram1_wr_addr
cxl_mst_ms_data_ram_rdata	-	cxl_mst_hreq_drsp_ram1_wr_addr_par
cxl_slv_hd_data_ram_rdata	-	cxl_mst_hreq_drsp_ram1_wdata
cxl_slv_ms_data_ram_rdata	-	cxl_mst_hreq_drsp_ram1_rd_addr
		cxl_mst_hreq_drsp_ram1_rd_addr_par
		cxl_mst_hreq_drsp_ram1_rd
		cxl_mst_hrsp_dreq_ram0_we
		cxl_mst_hrsp_dreq_ram0_wr_addr
		cxl_mst_hrsp_dreq_ram0_wr_addr_par
		cxl_mst_hrsp_dreq_ram0_wdata
		cxl_mst_hrsp_dreq_ram0_rd_addr
		cxl_mst_hrsp_dreq_ram0_rd_addr_par
		cxl_mst_hrsp_dreq_ram0_rd
		cxl_mst_hrsp_dreq_ram1_we
		cxl_mst_hrsp_dreq_ram1_wr_addr
		cxl_mst_hrsp_dreq_ram1_wr_addr_par
		cxl_mst_hrsp_dreq_ram1_wdata
		cxl_mst_hrsp_dreq_ram1_rd_addr
		cxl_mst_hrsp_dreq_ram1_rd_addr_par
		cxl_mst_hrsp_dreq_ram1_rd
		cxl_mst_hrsp_dreq_ram2_we
		cxl_mst_hrsp_dreq_ram2_wr_addr
		cxl_mst_hrsp_dreq_ram2_wr_addr_par
		cxl_mst_hrsp_dreq_ram2_wdata
		cxl_mst_hrsp_dreq_ram2_rd_addr
		cxl_mst_hrsp_dreq_ram2_rd_addr_par
		cxl_mst_hrsp_dreq_ram2_rd
		cxl_mst_hrsp_dreq_ram3_we
		cxl_mst_hrsp_dreq_ram3_wr_addr
		cxl_mst_hrsp_dreq_ram3_wr_addr_par
		cxl_mst_hrsp_dreq_ram3_wdata
		cxl_mst_hrsp_dreq_ram3_rd_addr
		cxl_mst_hrsp_dreq_ram3_rd_addr_par
		cxl_mst_hrsp_dreq_ram3_rd
		cxl_mst_hd_data_ram_we
		cxl_mst_hd_data_ram_wr_addr
		cxl_mst_hd_data_ram_wr_addr_par
		cxl_mst_hd_data_ram_wdata
		cxl_mst_hd_data_ram_rd_addr
		cxl_mst_hd_data_ram_rd_addr_par
		cxl_mst_hd_data_ram_rd
		cxl_mst_ms_ndr_ram0_we
		cxl_mst_ms_ndr_ram0_wr_addr
		cxl_mst_ms_ndr_ram0_wr_addr_par
		cxl_mst_ms_ndr_ram0_wdata
		cxl_mst_ms_ndr_ram0_rd_addr
		cxl_mst_ms_ndr_ram0_rd_addr_par
		cxl_mst_ms_ndr_ram0_rd
		cxl_mst_ms_ndr_ram1_we
		cxl_mst_ms_ndr_ram1_wr_addr
		cxl_mst_ms_ndr_ram1_wr_addr_par

- cxl_mst_ms_ndr_ram1_wdata
- cxl_mst_ms_ndr_ram1_rd_addr
- cxl_mst_ms_ndr_ram1_rd_addr_par
- cxl_mst_ms_ndr_ram1_rd
- cxl_mst_ms_data_ram_we
- cxl_mst_ms_data_ram_wr_addr
- cxl_mst_ms_data_ram_wr_addr_par
- cxl_mst_ms_data_ram_wdata
- cxl_mst_ms_data_ram_rd_addr
- cxl_mst_ms_data_ram_rd_addr_par
- cxl_mst_ms_data_ram_rd
- cxl_slv_hd_data_ram_we
- cxl_slv_hd_data_ram_wr_addr
- cxl_slv_hd_data_ram_wr_addr_par
- cxl_slv_hd_data_ram_wdata
- cxl_slv_hd_data_ram_rd_addr
- cxl_slv_hd_data_ram_rd_addr_par
- cxl_slv_hd_data_ram_rd
- cxl_slv_ms_data_ram_we
- cxl_slv_ms_data_ram_wr_addr
- cxl_slv_ms_data_ram_wr_addr_par
- cxl_slv_ms_data_ram_wdata
- cxl_slv_ms_data_ram_rd_addr
- cxl_slv_ms_data_ram_rd_addr_par
- cxl_slv_ms_data_ram_rd

Table 2-42 CXL RX FIFO RAM Signals

Port Name	I/O	Description
cxl_mst_hreq_drsp_ram0_we	O	CXL Master H2D Request or D2H Response Message Channel 0 - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hreq_drsp_ram0_wr_addr[(HREQ_DRSP_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Request or D2H Response Message Channel 0 - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram0_we is asserted

Port Name	I/O	Description
cxl_mst_hreq_drsp_ram0_wr_addr_par	O	CXL Master H2D Request or D2H Response Message Channel 0 - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram0_we is asserted
cxl_mst_hreq_drsp_ram0_wdata[(HREQ_DRSP_RX_RAM_DATAW-1):0]	O	CXL Master H2D Request or D2H Response Message Channel 0 - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram0_we is asserted
cxl_mst_hreq_drsp_ram0_rd_addr[(HREQ_DRSP_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Request or D2H Response Message Channel 0 - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram0_rd is asserted
cxl_mst_hreq_drsp_ram0_rd_addr_par	O	CXL Master H2D Request or D2H Response Message Channel 0 - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram0_rd is asserted

Port Name	I/O	Description
cxl_mst_hreq_drsp_ram0_rd	O	CXL Master H2D Request or D2H Response Message Channel 0 - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hreq_drsp_ram0_rdata[(HREQ_DRSP_RX_RAM_DATAW-1):0]	I	CXL Master H2D Request or D2H Response Message Channel 0 - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram0_rd is asserted
cxl_mst_hreq_drsp_ram1_we	O	CXL Master H2D Requestor D2H Response Message Channel 1 - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hreq_drsp_ram1_wr_addr[(HREQ_DRSP_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Request or D2H Response Message Channel 1 - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram1_we is asserted

Port Name	I/O	Description
cxl_mst_hreq_drsp_ram1_wr_addr_par	O	CXL Master H2D Request or D2H Response Message Channel 1 - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram1_we is asserted
cxl_mst_hreq_drsp_ram1_wdata[(HREQ_DRSP_RX_RAM_DATAW-1):0]	O	CXL Master H2D Request or D2H Response Message Channel 1 - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram1_we is asserted
cxl_mst_hreq_drsp_ram1_rd_addr[(HREQ_DRSP_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Request or D2H Response Message Channel 1 - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram1_rd is asserted
cxl_mst_hreq_drsp_ram1_rd_addr_par	O	CXL Master H2D Request or D2H Response Message Channel 1 - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram1_rd is asserted

Port Name	I/O	Description
cxl_mst_hreq_drsp_ram1_rd	O	CXL Master H2D Request or D2H Response Message Channel 1 - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hreq_drsp_ram1_rdata[(HREQ_DRSP_RX_RAM_DATAW-1):0]	I	CXL Master H2D Request or D2H Response Message Channel 1 - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hreq_drsp_ram1_rd is asserted
cxl_mst_hrsp_dreq_ram0_we	O	CXL Master H2D Response or D2H Request Message Channel 0 - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hrsp_dreq_ram0_wr_addr[(HRSP_DREQ_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 0 - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram0_we is asserted

Port Name	I/O	Description
cxl_mst_hrsp_dreq_ram0_wr_addr_par	O	CXL Master H2D Response or D2H Request Message Channel 0 - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram0_we is asserted
cxl_mst_hrsp_dreq_ram0_wdata[(HRSP_DREQ_RX_RAM_DATAW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 0 - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram0_we is asserted
cxl_mst_hrsp_dreq_ram0_rd_addr[(HRSP_DREQ_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 0 - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram0_rd is asserted
cxl_mst_hrsp_dreq_ram0_rd_addr_par	O	CXL Master H2D Response or D2H Request Message Channel 0 - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram0_rd is asserted

Port Name	I/O	Description
cxl_mst_hrsp_dreq_ram0_rd	O	CXL Master H2D Response or D2H Request Message Channel 0 - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hrsp_dreq_ram0_rdata[(HRSP_DREQ_RX_RAM_DATAW-1):0]	I	CXL Master H2D Response or D2H Request Message Channel 0 - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram0_rd is asserted
cxl_mst_hrsp_dreq_ram1_we	O	CXL Master H2D Response or D2H Request Message Channel 1 - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hrsp_dreq_ram1_wr_addr[(HRSP_DREQ_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 1 - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram1_we is asserted

Port Name	I/O	Description
cxl_mst_hrsp_dreq_ram1_wr_addr_par	O	CXL Master H2D Response or D2H Request Message Channel 1 - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram1_we is asserted
cxl_mst_hrsp_dreq_ram1_wdata[(HRSP_DREQ_RX_RAM_DATAW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 1 - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram1_we is asserted
cxl_mst_hrsp_dreq_ram1_rd_addr[(HRSP_DREQ_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 1 - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram1_rd is asserted
cxl_mst_hrsp_dreq_ram1_rd_addr_par	O	CXL Master H2D Response or D2H Request Message Channel 1 - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram1_rd is asserted

Port Name	I/O	Description
cxl_mst_hrsp_dreq_ram1_rd	O	CXL Master H2D Response or D2H Request Message Channel 1 - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hrsp_dreq_ram1_rdata[(HRSP_DREQ_RX_RAM_DATAW-1):0]	I	CXL Master H2D Response or D2H Request Message Channel 1 - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram1_rd is asserted
cxl_mst_hrsp_dreq_ram2_we	O	CXL Master H2D Response or D2H Request Message Channel 2 - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hrsp_dreq_ram2_wr_addr[(HRSP_DREQ_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 2 - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram2_we is asserted

Port Name	I/O	Description
cxl_mst_hrsp_dreq_ram2_wr_addr_par	O	CXL Master H2D Response or D2H Request Message Channel 2 - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram2_we is asserted
cxl_mst_hrsp_dreq_ram2_wdata[(HRSP_DREQ_RX_RAM_DATAW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 2 - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram2_we is asserted
cxl_mst_hrsp_dreq_ram2_rd_addr[(HRSP_DREQ_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 2 - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram2_rd is asserted
cxl_mst_hrsp_dreq_ram2_rd_addr_par	O	CXL Master H2D Response or D2H Request Message Channel 2 - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram2_rd is asserted

Port Name	I/O	Description
cxl_mst_hrsp_dreq_ram2_rd	O	CXL Master H2D Response or D2H Request Message Channel 2 - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hrsp_dreq_ram2_rdata[(HRSP_DREQ_RX_RAM_DATAW-1):0]	I	CXL Master H2D Response or D2H Request Message Channel 2 - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram2_rd is asserted
cxl_mst_hrsp_dreq_ram3_we	O	CXL Master H2D Response or D2H Request Message Channel 3 - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hrsp_dreq_ram3_wr_addr[(HRSP_DREQ_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 3 - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram3_we is asserted

Port Name	I/O	Description
cxl_mst_hrsp_dreq_ram3_wr_addr_par	O	CXL Master H2D Response or D2H Request Message Channel 3 - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram3_we is asserted
cxl_mst_hrsp_dreq_ram3_wdata[(HRSP_DREQ_RX_RAM_DATAW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 3 - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram3_we is asserted
cxl_mst_hrsp_dreq_ram3_rd_addr[(HRSP_DREQ_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Response or D2H Request Message Channel 3 - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram3_rd is asserted
cxl_mst_hrsp_dreq_ram3_rd_addr_par	O	CXL Master H2D Response or D2H Request Message Channel 3 - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram3_rd is asserted

Port Name	I/O	Description
cxl_mst_hrsp_dreq_ram3_rd	O	CXL Master H2D Response or D2H Request Message Channel 3 - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hrsp_dreq_ram3_rdata[(HRSP_DREQ_RX_RAM_DATAW-1):0]	I	CXL Master H2D Response or D2H Request Message Channel 3 - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hrsp_dreq_ram3_rd is asserted
cxl_mst_hd_data_ram_we	O	CXL Master H2D Data or D2H Data Message Channel 0 - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hd_data_ram_wr_addr[(HD_DATA_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Data or D2H Data Message Channel 0 - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hd_data_ram_we is asserted

Port Name	I/O	Description
cxl_mst_hd_data_ram_wr_addr_par	O	CXL Master H2D Data or D2H Data Message Channel 0 - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hd_data_ram_we is asserted
cxl_mst_hd_data_ram_wdata[(HD_DATA_RX_RAM_DATAW-1):0]	O	CXL Master H2D Data or D2H Data Message Channel 0 - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hd_data_ram_we is asserted
cxl_mst_hd_data_ram_rd_addr[(HD_DATA_RX_RAM_ADDRW-1):0]	O	CXL Master H2D Data or D2H Data Message Channel 0 - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hd_data_ram_rd is asserted
cxl_mst_hd_data_ram_rd_addr_par	O	CXL Master H2D Data or D2H Data Message Channel 0 - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hd_data_ram_rd is asserted

Port Name	I/O	Description
cxl_mst_hd_data_ram_rd	O	CXL Master H2D Data or D2H Data Message Channel 0 - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_hd_data_ram_rdata[(HD_DATA_RX_RAM_DATAW-1):0]	I	CXL Master H2D Data or D2H Data Message Channel 0 - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_hd_data_ram_rd is asserted
cxl_mst_ms_ndr_ram0_we	O	CXL Master M2S Request Only or S2M Response only Message Channel 0 - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_ms_ndr_ram0_wr_addr[(MS_NDR_RX_RAM_ADDRW-1):0]	O	CXL Master M2S Request Only or S2M Response only Message Channel 0 - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram0_we is asserted

Port Name	I/O	Description
cxl_mst_ms_ndr_ram0_wr_addr_par	O	CXL Master M2S Request Only or S2M Response only Message Channel 0 - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram0_we is asserted
cxl_mst_ms_ndr_ram0_wdata[(MS_NDR_RX_RAM_DATAW-1):0]	O	CXL Master M2S Request Only or S2M Response only Message Channel 0 - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram0_we is asserted
cxl_mst_ms_ndr_ram0_rd_addr[(MS_NDR_RX_RAM_ADDRW-1):0]	O	CXL Master M2S Request Only or S2M Response only Message Channel 0 - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram0_rd is asserted
cxl_mst_ms_ndr_ram0_rd_addr_par	O	CXL Master M2S Request Only or S2M Response only Message Channel 0 - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram0_rd is asserted

Port Name	I/O	Description
cxl_mst_ms_ndr_ram0_rd	O	CXL Master M2S Request Only or S2M Response only Message Channel 0 - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_ms_ndr_ram0_rdata[(MS_NDR_RX_RAM_DATAW-1):0]	I	CXL Master M2S Request Only or S2M Response only Message Channel 0 - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram0_rd is asserted
cxl_mst_ms_ndr_ram1_we	O	CXL Master M2S Request Only or S2M Response only Message Channel 1 - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_ms_ndr_ram1_wr_addr[(MS_NDR_RX_RAM_ADDRW-1):0]	O	CXL Master M2S Request Only or S2M Response only Message Channel 1 - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram1_we is asserted

Port Name	I/O	Description
cxl_mst_ms_ndr_ram1_wr_addr_par	O	CXL Master M2S Request Only or S2M Response only Message Channel 1 - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram1_we is asserted
cxl_mst_ms_ndr_ram1_wdata[(MS_NDR_RX_RAM_DATAW-1):0]	O	CXL Master M2S Request Only or S2M Response only Message Channel 1 - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram1_we is asserted
cxl_mst_ms_ndr_ram1_rd_addr[(MS_NDR_RX_RAM_ADDRW-1):0]	O	CXL Master M2S Request Only or S2M Response only Message Channel 1 - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram1_rd is asserted
cxl_mst_ms_ndr_ram1_rd_addr_par	O	CXL Master M2S Request Only or S2M Response only Message Channel 1 - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram1_rd is asserted

Port Name	I/O	Description
cxl_mst_ms_ndr_ram1_rd	O	CXL Master M2S Request Only or S2M Response only Message Channel 1 - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_ms_ndr_ram1_rdata[(MS_NDR_RX_RAM_DATAW-1):0]	I	CXL Master M2S Request Only or S2M Response only Message Channel 1 - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_ndr_ram1_rd is asserted
cxl_mst_ms_data_ram_we	O	CXL Master M2S RdW or S2M Data Message Channel 0 - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_ms_data_ram_wr_addr[(MS_DATA_RX_RAM_ADDRW-1):0]	O	CXL Master M2S RdW or S2M Data Message Channel 0 - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_data_ram_we is asserted

Port Name	I/O	Description
cxl_mst_ms_data_ram_wr_addr_par	O	CXL Master M2S RdW or S2M Data Message Channel 0 - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_data_ram_we is asserted
cxl_mst_ms_data_ram_wdata[(MS_DATA_RX_RAM_DATAW-1):0]	O	CXL Master M2S RdW or S2M Data Message Channel 0 - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_data_ram_we is asserted
cxl_mst_ms_data_ram_rd_addr[(MS_DATA_RX_RAM_ADDRW-1):0]	O	CXL Master M2S RdW or S2M Data Message Channel 0 - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_data_ram_rd is asserted
cxl_mst_ms_data_ram_rd_addr_par	O	CXL Master M2S RdW or S2M Data Message Channel 0 - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_data_ram_rd is asserted

Port Name	I/O	Description
cxl_mst_ms_data_ram_rd	O	CXL Master M2S RdD or S2M Data Message Channel 0 - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_mst_ms_data_ram_rdata[(MS_DATA_RX_RAM_DATAW-1):0]	I	CXL Master M2S RdD or S2M Data Message Channel 0 - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_mst_ms_data_ram_rd is asserted
cxl_slv_hd_data_ram_we	O	CXL Slave H2D/D2H Data RAM - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: None,core_clk,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_hd_data_ram_wr_addr[(HD_DATA_RAM_ADDRW-1):0]	O	CXL Slave H2D/D2H Data RAM - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_hd_data_ram_we is asserted
cxl_slv_hd_data_ram_wr_addr_par	O	CXL Slave H2D/D2H Data RAM - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_hd_data_ram_we is asserted

Port Name	I/O	Description
cxl_slv_hd_data_ram_wdata[(HD_DATA_RAM_DATAW-1):0]	O	CXL Slave H2D/D2H Data RAM - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: None,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_hd_data_ram_we is asserted
cxl_slv_hd_data_ram_rd_addr[(HD_DATA_RAM_ADDRW-1):0]	O	CXL Slave H2D/D2H Data RAM - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_hd_data_ram_rd is asserted
cxl_slv_hd_data_ram_rd_addr_par	O	CXL Slave H2D/D2H Data RAM - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,core_clk,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_hd_data_ram_rd is asserted
cxl_slv_hd_data_ram_rd	O	CXL Slave H2D/D2H Data RAM - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: core_clk,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_hd_data_ram_rdata[(HD_DATA_RAM_DATAW-1):0]	I	CXL Slave H2D/D2H Data RAM - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_CACHE_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_hd_data_ram_rd is asserted

Port Name	I/O	Description
cxl_slv_ms_data_ram_we	O	CXL Slave M2S/S2M Data RAM - Write Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: None,core_clk,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_ms_data_ram_wr_addr[(MS_DATA_RAM_ADDRW-1):0]	O	CXL Slave M2S/S2M Data RAM - Write Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_ms_data_ram_we is asserted
cxl_slv_ms_data_ram_wr_addr_par	O	CXL Slave M2S/S2M Data RAM - Write Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_ms_data_ram_we is asserted
cxl_slv_ms_data_ram_wdata[(MS_DATA_RAM_DATAW-1):0]	O	CXL Slave M2S/S2M Data RAM - Write Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: None,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_ms_data_ram_we is asserted
cxl_slv_ms_data_ram_rd_addr[(MS_DATA_RAM_ADDRW-1):0]	O	CXL Slave M2S/S2M Data RAM - Read Address. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_ms_data_ram_rd is asserted

Port Name	I/O	Description
cxl_slv_ms_data_ram_rd_addr_par	O	CXL Slave M2S/S2M Data RAM - Read Address Parity. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) && ((CX_CXL_CHI_DATACHK > 0) ? CX_CXL_ENABLE : 0) Synchronous To: aux_clk_g,core_clk,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_ms_data_ram_rd is asserted
cxl_slv_ms_data_ram_rd	O	CXL Slave M2S/S2M Data RAM - Read Enable. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: core_clk,{cxl_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_slv_ms_data_ram_rdata[(MS_DATA_RAM_DATAW-1):0]	I	CXL Slave M2S/S2M Data RAM - Read Data. Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MEM_EN) Synchronous To: cxl_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cxl_slv_ms_data_ram_rd is asserted

2.43 CXS Rx FIFO RAM Signals

cxs_rxram_douth -

- cxs_rxram_ena
- cxs_rxram_wea
- cxs_rxram_addra
- cxs_rxram_dina
- cxs_rxram_enb
- cxs_rxram_addrb
- cxs_rxram_addra_par
- cxs_rxram_addrb_par

Table 2-43 CXS Rx FIFO RAM Signals

Port Name	I/O	Description
cxs_rxram_ena	O	Port A Enable of CXS Receiver Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: aux_clk,aux_clk_g,{core_clk},{cxs_clk} Registered: No Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxs_rxram_wea	O	Port B Write Enable of CXS Receiver Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: aux_clk,aux_clk_g,{core_clk},{cxs_clk} Registered: No Power Domain: PD_VAUX Active State: High Validated by: cxs_rxram_ena is asserted
cxs_rxram_addra[(CXS_RX_BUFF_ADDR W-1):0]	O	Port A Write Address of CXS Receiver Buffer. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: cxs_rxram_ena and cxs_rxram_wea are asserted
cxs_rxram_dina[(CXS_RX_BUFF_DATAW -1):0]	O	Port A Write Data of CXS Receiver Buffer. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: None,aux_clk_g,perbitclk,{core_clk},{cxs_clk} Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: cxs_rxram_ena and cxs_rxram_wea are asserted

Port Name	I/O	Description
cxs_rxram_enb	O	Port B Enable of CXS Receiver Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: aux_clk,aux_clk_g,{core_clk} Registered: No Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxs_rxram_addrb[(CXS_RX_BUFF_ADDR W-1):0]	O	Port B Read Address of CXS Receiver Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: core_clk Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: cxs_rxram_enb is asserted
cxs_rxram_doutb[(CXS_RX_BUFF_DATA W-1):0]	I	Port B Read Data of CXS Receiver Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxs_rxram_enb is asserted
cxs_rxram_addra_par	O	Write Address Parity of CXS Receiver Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk,aux_clk_g,{core_clk},{cxs_clk} Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: cxs_rxram_ena and cxs_rxram_wea are asserted
cxs_rxram_addrb_par	O	Read Address Parity of CXS Receiver Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk,aux_clk_g,{core_clk} Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: cxs_rxram_enb is asserted

2.44 CXS Tx FIFO RAM Signals

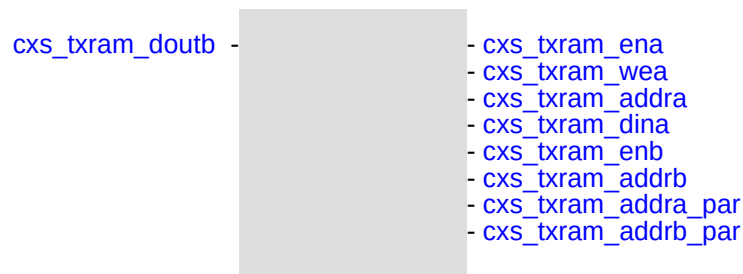


Table 2-44 CXS Tx FIFO RAM Signals

Port Name	I/O	Description
<code>cxs_txram_ena</code>	O	Port A Enable of CXS Transmitter Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: <code>aux_clk_g</code> , { <code>core_clk</code> } Registered: No Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
<code>cxs_txram_wea</code>	O	Port B Write Enable of CXS Transmitter Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: <code>aux_clk_g</code> , { <code>core_clk</code> } Registered: No Power Domain: PD_VAUX Active State: High Validated by: <code>cxs_txram_ena</code> is asserted
<code>cxs_txram_addra</code> [(CX_TX_BUFF_ADDRW-1):0]	O	Port A Write Address of CXS Transmitter Buffer. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: <code>core_clk</code> Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: <code>cxs_txram_ena</code> and <code>cxs_txram_wea</code> are asserted
<code>cxs_txram_dina</code> [(CX_TX_BUFF_DATAW-1):0]	O	Port A Write Data of CXS Transmitter Buffer. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: None, <code>aux_clk_g</code> , <code>perbitclk</code> , { <code>core_clk</code> } Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: <code>cxs_txram_ena</code> and <code>cxs_txram_wea</code> are asserted

Port Name	I/O	Description
cxs_txram_enb	O	Port B Enable of CXS Transmitter Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: aux_clk_g,{core_clk},{cxs_clk} Registered: No Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxs_txram_addrb[(CXS_TX_BUFF_ADDR W-1):0]	O	Port B Read Address of CXS Transmitter Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: cxs_txram_enb is asserted
cxs_txram_doutb[(CXS_TX_BUFF_DATA W-1):0]	I	Port B Read Data of CXS Transmitter Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) Synchronous To: {core_clk},{cxs_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: cxs_txram_enb is asserted
cxs_txram_addra_par	O	Write Address Parity of CXS Transmitter Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0)))? 0: 1) Synchronous To: aux_clk_g,{core_clk} Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: cxs_txram_ena and cxs_txram_wea are asserted
cxs_txram_addrb_par	O	Read Address Parity of CXS Transmitter Buffer RAM. Exists: ((CX_CPCIE_ENABLE)) && (CX_CXS_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0)))? 0: 1) Synchronous To: aux_clk_g,{core_clk},{cxs_clk} Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: cxs_txram_enb is asserted

2.45 Receive Data Queue RAM Signals

p_dataq_dataout -

- p_alt_dataq_enb
- p_dataq_addra
- p_dataq_addrb
- p_dataq_addra_par
- p_dataq_addrb_par
- p_dataq_err_synd
- p_dataq_err_addr
- p_dataq_ce
- p_dataq_ue
- p_dataq_datain
- p_dataq_ena
- p_dataq_enb
- p_dataq_wea

Table 2-45 Receive Data Queue RAM Signals

Port Name	I/O	Description
p_alt_dataq_enb[(RADM_Q_D_CTRLBIT S-1):0]	O	Alternative Port B read enables to the data queue buffers. Because there are 2 buffer queues, but only one queue is read at a time, only one enable is asserted at a time. This ensures that an empty RAM cannot be read. Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (ARC_256_P2) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
p_dataq_dataout[(RADM_Q_DATABITS_O -1):0]	I	Read data from port B of the data queue buffer. The width of the read data bus is automatically set according to the datapath width and number of VCs, as follows: Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: p_dataq_enb is asserted

Port Name	I/O	Description
p_dataq_addra[(RADM_PQ_D_ADDRBIT S-1):0]	O	Address to port A of the data queue buffer. Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: p_dataq_ena or p_dataq_wea is asserted
p_dataq_addrb[(RADM_PQ_D_ADDRBIT S-1):0]	O	Address to port B of the data queue buffer. The width is the same as the p_dataq_addra width. Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: p_dataq_enb is asserted
p_dataq_addra_par[(NDQ-1):0]	O	Parity for address to port A of the data queue buffer. Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0)) ? 0: 1) Synchronous To: aux_clk_g,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
p_dataq_addrb_par[(NDQ-1):0]	O	Parity for address to port B of the data queue buffer. Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0)) ? 0: 1) Synchronous To: aux_clk_g,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
p_dataq_err_synd[((NDQ*RASDP_DATAQ_ERR_SYND_WD)-1):0]	O	Error syndrome from data queue RAM Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
p_dataq_err_addr[(RADM_PQ_D_ADDRBITS-1):0]	O	Error address from data queue RAM Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
p_dataq_ce[(NDQ-1):0]	O	Corrected error indication from data queue RAM Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
p_dataq_ue[(NDQ-1):0]	O	Uncorrectable error indication from data queue RAM Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
p_dataq_datain[(RADM_Q_DATABITS-1):0]	O	Write data to port A of the data queue buffer. The width is the same as the p_dataq_dataout width. Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: None,aux_clk_g,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: p_dataq_wea is asserted
p_dataq_ena[(RADM_Q_D_CTRLBITS-1):0]	O	Port A select to the data queue buffer. The controller asserts this at the same time as p_dataq_wea. Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
p_dataq_enb[(RADM_Q_D_CTRLBITS-1):0]	O	Port B read enable to the data queue buffer. Always '1' even when the read port is idle. Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
p_dataq_wea[(RADM_Q_D_CTRLBITS-1):0]	O	Port A write enable to the data queue buffer. Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.46 Receive Formation Queue RAM Signals

formqram_radm_doutb -

- radm_formqram_addra
- radm_formqram_addra_par
- radm_formqram_ena
- radm_formqram_wea
- radm_formqram_dina
- radm_formqram_addrb
- radm_formqram_addrb_par
- radm_formqram_enb
- radm_formqram_err_addr
- radm_formqram_err_synd
- radm_formqram_ce
- radm_formqram_ue

Table 2-46 Receive Formation Queue RAM Signals

Port Name	I/O	Description
radm_formqram_addra[((NQW*RAM_PW)-1):0]	O	Address buses to port A (write port) of the formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: SNPS_RSVDPARAM_72 Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_formqram_ena or radm_formqram_wea is asserted
radm_formqram_addra_par[(NQW-1):0]	O	Parity bit for address buses to port A (write port) of the formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: (SNPS_RSVDPARAM_72) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_formqram_ena or radm_formqram_wea is asserted
radm_formqram_ena[(NQW-1):0]	O	Port A selects to the formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: SNPS_RSVDPARAM_72 Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
radm_formqram_wea[(NQW-1):0]	O	Port A write enables to the formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: SNPS_RSVDPARAM_72 Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_formqram_dina[((NQW*RAM_WD)-1):0]	O	Write data to port A of the formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: SNPS_RSVDPARAM_72 Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_formqram_wea is asserted
radm_formqram_addrb[((NQW*RAM_PW)-1):0]	O	Address buses to port B (read port) of the formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: SNPS_RSVDPARAM_72 Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_formqram_enb is asserted
radm_formqram_addrb_par[(NQW-1):0]	O	Parity bit for address buses to port B (read port) of the formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: (SNPS_RSVDPARAM_72) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_formqram_enb is asserted
radm_formqram_enb[(NQW-1):0]	O	Port B selects to the formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: SNPS_RSVDPARAM_72 Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
formqram_radm_doutb[((NQW*RAM_WD)-1):0]	I	Read data from port B of the formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: SNPS_RSVDPARAM_72 Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_formqram_enb is asserted
radm_formqram_err_addr[((NQW*RAM_PW)-1):0]	O	Logged error address buses to port B (read port) of the formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: (SNPS_RSVDPARAM_72) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_formqram_enb is asserted
radm_formqram_err_synd[((NQW*RASDP_FORMQ_ERR_SYND_WD)-1):0]	O	Error syndrome for radm formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: (SNPS_RSVDPARAM_72) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_formqram_enb is asserted
radm_formqram_ce[(NQW-1):0]	O	Corrected error indication from radm formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: (SNPS_RSVDPARAM_72) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_formqram_enb is asserted

Port Name	I/O	Description
radm_formqram_ue[(NQW-1):0]	O	Uncorrectable error indication from radm formation queue RAMs, concatenated from RAM 0 to RAM CX_RADM_FORMQ_NQW-1. Exists: (SNPS_RSVDPARAM_72) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_formqram_enb is asserted

2.47 Receive Header Queue RAM Signals

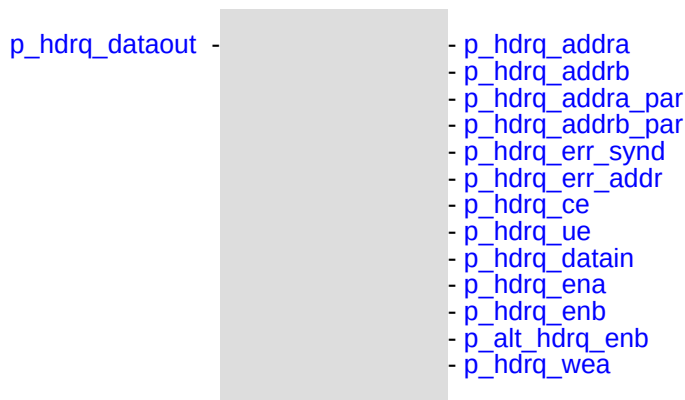


Table 2-47 Receive Header Queue RAM Signals

Port Name	I/O	Description
<code>p_hdrq_dataout[(RADM_PQ_H_DATABITS_O-1):0]</code>	I	Read data from port B of the header queue buffer. Exists: <code>CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)</code> Synchronous To: <code>radm_clk_g</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: <code>p_hdrq_enb</code> is asserted
<code>p_hdrq_addra[(RADM_PQ_H_ADDRBITS-1):0]</code>	O	Address to port A of the header queue buffer. Exists: <code>CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)</code> Synchronous To: <code>radm_clk_g</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: <code>p_hdrq_ena</code> or <code>p_hdrq_wea</code> is asserted

Port Name	I/O	Description
p_hdrq_addrb[(RADM_PQ_H_ADDRBITS-1):0]	O	Address to port B of the header queue buffer. The width is the same as the p_hdrq_addra width. Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: p_hdrq_enb is asserted
p_hdrq_addra_par[(NHQ-1):0]	O	Parity for address to port A of the header queue buffer Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
p_hdrq_addrb_par[(NHQ-1):0]	O	Parity for address to port B of the header queue buffer Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
p_hdrq_err_synd[(RASDP_HDRQ_ERR_SYND_WD-1):0]	O	Error syndrome from header queue RAM Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
p_hdrq_err_addr[(RADM_SBUF_HDRQ_ERR_ADDR_WD-1):0]	O	Error address from header queue RAM Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
p_hdrq_ce[(NHQ-1):0]	O	Corrected error indication from header queue RAM Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
p_hdrq_ue[(NHQ-1):0]	O	Uncorrectable error indication from header queue RAM Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
p_hdrq_datain[(RADM_PQ_H_DATABITS-1):0]	O	Write data to port A of the header queue buffer. The width is the same as the p_hdrq_dataout width. Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: None,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: p_hdrq_wea is asserted
p_hdrq_ena[(RADM_Q_H_CTRLBITS-1):0]	O	Port A select to the header queue buffer. The controller asserts this at the same time as p_hdrq_wea. Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
p_hdrq_enb[(RADM_Q_H_CTRLBITS-1):0]	O	Port B read enable to the header queue buffer. Always '1' even when the read port is idle. Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
p_alt_hdrq_enb[(RADM_Q_H_CTRLBITS-1):0]	O	Alternative Port B read enables to the header queue buffers. Because there are 2 buffer queues, but only one queue is read at a time, only one enable is asserted at a time. This ensures that an empty RAM cannot be read. Exists: (CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4)) && (ARC_256_P2) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
p_hdrq_wea[(RADM_Q_H_CTRLBITS-1):0]	O	Port A write enable to the header queue buffer. Exists: CX_RADMQ_MODE==0 ? (RADM_P_QMODE_VC0!=4 RADM_NP_QMODE_VC0!=4 RADM_CPL_QMODE_VC0!=4) : CX_RADMQ_MODE==2 (CX_RADMQ_MODE==1 && RADM_P_QMODE_VC0!=4) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.48 Transmit Retry Buffer RAM Signals

retryram_xdlh_data -

- xdlh_retryram_addr
- xdlh_retryram_addr_par
- xdlh_retryram_err_synd
- xdlh_retryram_err_addr
- xdlh_retryram_ce
- xdlh_retryram_ue
- xdlh_retryram_data
- xdlh_retryram_we
- xdlh_retryram_en
- xdlh_retrysotram_err_synd
- xdlh_retrysotram_err_addr
- xdlh_retrysotram_ce
- xdlh_retrysotram_ue

Table 2-48 Transmit Retry Buffer RAM Signals

Port Name	I/O	Description
xdlh_retryram_addr[(RBUF_PW-1):0]	O	Address to the retry buffer RAM. The width of the address bus (RBUF_PW) is automatically set as a function of the retry buffer depth, which is automatically calculated, or set explicitly by you if you disable autosizing. Exists: CX_CXPL_EN Synchronous To: aux_clk,aux_clk_g,core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: xdlh_retryram_we or xdlh_retryram_en is asserted
xdlh_retryram_addr_par	O	parity bit of Address to the retry buffer RAM. Exists: (CX_CXPL_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: xdlh_retryram_we or xdlh_retryram_en is asserted
xdlh_retryram_err_synd[(RASDP_RBUF_ERR_SYND_WD-1):0]	O	Error syndrome from retry RAM Exists: (CX_CXPL_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: xdlh_retryram_we or xdlh_retryram_en is asserted

Port Name	I/O	Description
xdlh_retryram_err_addr[(RBUF_PW-1):0]	O	Retry RAM address where a protection error has been detected Exists: (CX_CXPL_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: xdlh_retryram_ce or xdlh_retryram_ue is asserted
xdlh_retryram_ce	O	Corrected error indication from retry RAM Exists: (CX_CXPL_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xdlh_retryram_ue	O	Uncorrectable error indication from retry RAM Exists: (CX_CXPL_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xdlh_retryram_data[(RBUF_WIDTH-1):0]	O	Write data to the retry buffer RAM. Exists: CX_CXPL_EN Synchronous To: None,aux_clk,aux_clk_g,core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: xdlh_retryram_we is asserted
xdlh_retryram_we	O	Write enable to the retry buffer RAM. Exists: CX_CXPL_EN Synchronous To: aux_clk,core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
xdlh_retryram_en	O	Chip enable to the retry buffer RAM. Exists: CX_CXPL_EN Synchronous To: aux_clk,core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
retryram_xdlh_data[(RBUF_WIDTH-1):0]	I	Read data from the retry buffer RAM. The width is the same as the xdlh_retryram_data width. Exists: CX_CXPL_EN Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xdlh_retrysotram_err_synd[(RASDP_SOT_BUF_ERR_SYND_WD-1):0]	O	Error syndrome from retry SOT RAM Exists: (CX_CXPL_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: xdlh_retrysotram_we or xdlh_retrysotram_en is asserted
xdlh_retrysotram_err_addr[(SOTBUF_PW-1):0]	O	Retry SOT RAM address where a protection error has been detected Exists: (CX_CXPL_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: xdlh_retrysotram_ce or xdlh_retrysotram_ue is asserted
xdlh_retrysotram_ce	O	Corrected error indication from retry SOT RAM Exists: (CX_CXPL_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
xdlh_retrysotram_ue	O	Uncorrectable error indication from retry SOT RAM Exists: (CX_CXPL_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) && (CX_RAS_EXT_IF) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.49 Transmit Retry SOT Buffer RAM Signals

retrysotram_xdlh_data -

- xdlh_retrysotram_waddr
- xdlh_retrysotram_raddr
- xdlh_retrysotram_waddr_par
- xdlh_retrysotram_raddr_par
- xdlh_retrysotram_data
- xdlh_retrysotram_we
- xdlh_retrysotram_en

Table 2-49 Transmit Retry SOT Buffer RAM Signals

Port Name	I/O	Description
xdlh_retrysotram_waddr[(SOTBUF_PW-1):0]	O	Write address for the SOT buffer RAM. The width of the address bus (SOTBUF_PW) is automatically set as a function of the SOT buffer depth, which is automatically calculated. Exists: CX_CXPL_EN Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: xdlh_retrysotram_we is asserted
xdlh_retrysotram_raddr[(SOTBUF_PW-1):0]	O	Read address for the SOT buffer RAM. The width of the address bus (SOTBUF_PW) is automatically set as a function of the SOT buffer depth, which is automatically calculated. Exists: CX_CXPL_EN Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: xdlh_retrysotram_en is asserted
xdlh_retrysotram_waddr_par	O	Parity for SOT buffer RAM write address. Exists: (CX_CXPL_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
xdlh_retrysotram_raddr_par	O	Parity for SOT buffer RAM read address. Exists: (CX_CXPL_EN) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xdlh_retrysotram_data[(SOTBUF_WD-1):0]	O	Write data to the SOT buffer RAM. The width of the write data bus is automatically set to the width of the retry buffer address bus. (Each location in the SOT buffer is used to store a retry buffer address.) Exists: CX_CXPL_EN Synchronous To: None,aux_clk,aux_clk_g,core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: xdlh_retrysotram_we is asserted
xdlh_retrysotram_we	O	Write enable to the SOT buffer RAM. Exists: CX_CXPL_EN Synchronous To: aux_clk,aux_clk_g,core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xdlh_retrysotram_en	O	Read enable to the SOT buffer RAM. The controller asserts the SOT buffer enable when replay is in progress and SOT read data is required. This signal is low when the SOT buffer is not accessed, thus saving power. Exists: CX_CXPL_EN Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
retrysotram_xdlh_data[(SOTBUF_WD-1):0]]	I	Read data from the SOT buffer RAM. The width is the same as the xdlh_retrysotram_data width. Exists: CX_CXPL_EN Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.50 MSI-X PBA RAM Signals

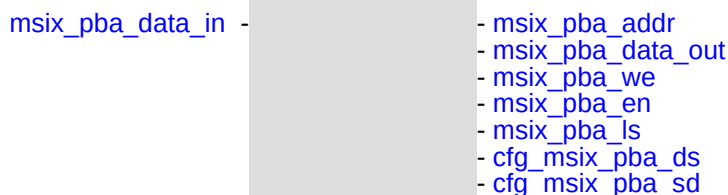


Table 2-50 MSI-X PBA RAM Signals

Port Name	I/O	Description
<code>msix_pba_addr[(MSIX_PBA_PW-1):0]</code>	O	Address to the MSI-X PBA RAM. The width of the address bus (MSIX_PBA_PW) is automatically set as a function of the total PBA depth across all functions. Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: <code>aux_clk_g</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: <code>msix_pba_we[7:0]</code> or <code>msix_pba_en</code> is asserted
<code>msix_pba_data_out[63:0]</code>	O	Write data to the MSI-X PBA RAM. The width RAM is always 64 bits. Each Dword of data is qualified by the corresponding <code>msix_pba_we[7:0]</code> Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: <code>aux_clk_g</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: <code>msix_pba_we[7:0]</code> is asserted
<code>msix_pba_we[7:0]</code>	O	Write data enable to the MSI-X PBA RAM. Each byte of data is qualified by the corresponding <code>msix_pba_we[7:0]</code> Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: <code>aux_clk_g</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
msix_pba_en	O	Read data enable to the MSI-X PBA RAM. Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
msix_pba_data_in[63:0]	I	Read data from the MSI-X PBA RAM. The width RAM is always 64 bits. Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
msix_pba_ls	O	MSIX PBA RAM light sleep control Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_msix_pba_ds	O	MSIX PBA RAM deep sleep control Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_msix_pba_sd	O	MSIX PBA RAM shutdown control Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.51 MSI-X Table RAM Signals

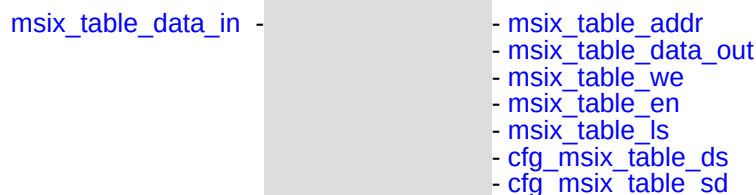


Table 2-51 MSI-X Table RAM Signals

Port Name	I/O	Description
msix_table_addr[(MSIX_TABLE_PW-1):0]	O	Address to the MSI-X Table RAM. The width of the address bus (MSIX_TABLE_PW) is automatically set as a function of the total table depth across all functions. Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: msix_table_we[16:0] or msix_table_en is asserted
msix_table_data_out[130:0]	O	Write data to the MSI-X Table RAM. The width RAM is always 128 bits for vector information and 3 bits for traffic class. Each Dword of data is qualified by the corresponding msix_table_we[16:0] Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: msix_table_we[16:0] is asserted
msix_table_we[16:0]	O	Write data enable to the MSI-X Table RAM. Each byte of data is qualified by the corresponding msix_table_we[16:0] Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
msix_table_en	O	Chip enable to the MSI-X Table RAM. Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
msix_table_data_in[130:0]	I	Read data from the MSI-X Table RAM. The width RAM is always 128 bits for vector data and 3 bits for traffic class. Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
msix_table_ls	O	MSIX table RAM light sleep control Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_msix_table_ds	O	MSIX table RAM deep sleep control Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_msix_table_sd	O	MSIX table RAM shutdown control Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.52 DMA Context Memory RAM Signals

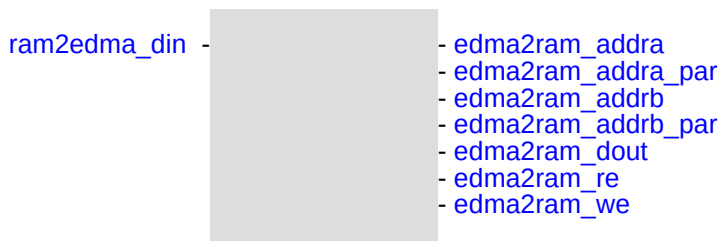


Table 2-52 DMA Context Memory RAM Signals

Port Name	I/O	Description
<code>ram2edma_din[(DMA_CTX_RAM_DATA_WIDTH-1):0]</code>	I	Read data from the DMA Context Memory RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: <code>aux_clk_g</code> Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: <code>edma2ram_re</code> is asserted
<code>edma2ram_addra[(DMA_CTX_RAM_ADDR_WIDTH-1):0]</code>	O	Write address to the DMA Context Memory RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: <code>core_clk</code> , { <code>aux_clk_g</code> } Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: <code>edma2ram_we</code> is asserted
<code>edma2ram_addra_par[(DMA_CTX_RAM_ADDRP_WIDTH-1):0]</code>	O	Write address to the DMA Context Memory RAM. Port A Write Address Parity bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: <code>aux_clk_g</code> , { <code>core_clk</code> } Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
<code>edma2ram_addrb[(DMA_CTX_RAM_ADDR_WIDTH-1):0]</code>	O	Read address to the DMA Context Memory RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: <code>core_clk</code> , { <code>aux_clk_g</code> } Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: <code>edma2ram_re</code> is asserted

Port Name	I/O	Description
edma2ram_addrb_par[(DMA_CTX_RAM_ADDRP_WIDTH-1):0]	O	Read address to the DMA Context Memory RAM. Port B Read Address Parity bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma2ram_dout[(DMA_CTX_RAM_DATA_WIDTH-1):0]	O	Write data to the DMA Context Memory RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma2ram_we is asserted
edma2ram_re	O	Read enable to the DMA Context Memory RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma2ram_we	O	Write enable to the DMA Context Memory RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.53 DMA Handshake Signals

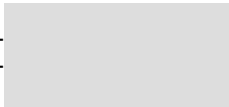
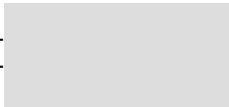
app_wdxfer_go_togg -  - dma_wdxfer_done_togg
 app_rdxfer_go_togg -  - dma_rdxfer_done_togg

Table 2-53 DMA Handshake Signals

Port Name	I/O	Description
app_wdxfer_go_togg[(CC_NUM_DMA_WR_CHAN-1):0]	I	DMA Write engine descriptor transfer go toggle signal. One bit for each implemented DMA write channel. This signal is ignored whenever a DMA channel operates in non-linked mode. Exists: CC_DMA_HSHAKE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
app_rdxfer_go_togg[(CC_NUM_DMA_RD_CHAN-1):0]	I	DMA read engine descriptor transfer go toggle signal. One bit for each implemented DMA read channel. This signal is ignored whenever a DMA channel operates in non-linked mode. Exists: CC_DMA_HSHAKE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
dma_wdxfer_done_togg[(CC_NUM_DMA_WR_CHAN-1):0]	O	DMA write engine descriptor transfer done toggle signal. One bit for each implemented DMA write channel. After a doorbell command, this signal is de-asserted. When a channel is operating in non-linked list mode this signal does not toggles Exists: CC_DMA_HSHAKE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

Port Name	I/O	Description
<code>dma_rdxfer_done_togg[(CC_NUM_DMA_RD_CHAN-1):0]</code>	O	DMA read engine descriptor transfer done toggle signal. One bit for each implemented DMA read channel. After a doorbell command, this signal is de-asserted. When a channel is operating in non-linked list mode this signal does not toggles Exists: CC_DMA_HSHAKE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

2.54 DMA Read Buffer RAM Signals

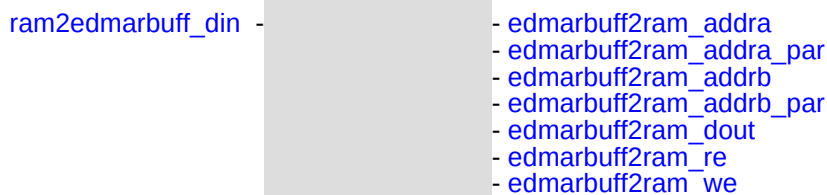


Table 2-54 DMA Read Buffer RAM Signals

Port Name	I/O	Description
<code>ram2edmarbuff_din[(DMA_SEG_BUF_DATA_WIDTH-1):0]</code>	I	Read data from the DMA Read Buffer RAM. Exists: CC_DMA_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: CX_DMA_PF_ENABLE ? "Not validated by another signal" : "edmarbuff2ram_re is asserted"
<code>edmarbuff2ram_addra[(DMA_SEG_BUF_NW_ADDR_WIDTH-1):0]</code>	O	Write address to the DMA Read Buffer RAM. Exists: CC_DMA_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edmarbuff2ram_we is asserted
<code>edmarbuff2ram_addra_par[(DMA_SEG_BUF_NW_ADDR_WIDTH-1):0]</code>	O	Write address to the DMA Read Buffer RAM. Port A Write Address Parity bits. Exists: (CC_DMA_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
<code>edmarbuff2ram_addrb[(DMA_SEG_BUF_NW_ADDR_WIDTH-1):0]</code>	O	Read address to the DMA Read Buffer RAM. Exists: CC_DMA_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edmarbuff2ram_re is asserted

Port Name	I/O	Description
edmarbuff2ram_addrb_par[(DMA_SEG_BUF_NW_ADDRP_WIDTH-1):0]	O	Read address to the DMA Read Buffer RAM. Port B Read Address Parity bits. Exists: (CC_DMA_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edmarbuff2ram_dout[(DMA_SEG_BUF_DATA_WIDTH-1):0]	O	Write data to the DMA Read Buffer RAM. Exists: CC_DMA_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edmarbuff2ram_we is asserted
edmarbuff2ram_re	O	Read enable to the DMA Read Buffer RAM. Exists: CC_DMA_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edmarbuff2ram_we	O	Write enable to the DMA Read Buffer RAM. Exists: CC_DMA_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.55 DMA Read Engine Completion to Memory Write LUT RAM Signals

edma_rd_eng_c2w_lut_din_ram -

- edma_rd_eng_c2w_lut_addra_ram
- edma_rd_eng_c2w_lut_addrb_ram
- edma_rd_eng_c2w_lut_dout_ram
- edma_rd_eng_c2w_lut_we_ram
- edma_rd_eng_c2w_lut_re_ram
- edma_rd_eng_c2w_lut_addra_par_ram
- edma_rd_eng_c2w_lut_addrb_par_ram
- edma_rd_eng_ovrl_addra_par_ram
- edma_rd_eng_ovrl_addrb_par_ram
- edma_rd_eng_ll_addra_par_ram
- edma_rd_eng_ll_addrb_par_ram
- edma_rd_eng_msi_addra_par_ram
- edma_rd_eng_msi_addrb_par_ram
- edma_rd_eng_stsh_lut_addra_par_ram
- edma_rd_eng_stsh_lut_addrb_par_ram
- edma_wr_eng_c2w_lut_addra_par_ram
- edma_wr_eng_c2w_lut_addrb_par_ram
- edma_wr_eng_ovrl_addra_par_ram
- edma_wr_eng_ovrl_addrb_par_ram
- edma_wr_eng_ll_addra_par_ram
- edma_wr_eng_ll_addrb_par_ram
- edma_wr_eng_msi_addra_par_ram
- edma_wr_eng_msi_addrb_par_ram
- edma_wr_eng_stsh_lut_addra_par_ram
- edma_wr_eng_stsh_lut_addrb_par_ram

Table 2-55 DMA Read Engine Completion to Memory Write LUT RAM Signals

Port Name	I/O	Description
edma_rd_eng_c2w_lut_addra_ram[(CC_I F_RD_CTXC2W_RAM_PW-1):0]	O	Write address to the DMA Read Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_c2w_lut_we_ram is asserted
edma_rd_eng_c2w_lut_addrb_ram[(CC_I F_RD_CTXC2W_RAM_PW-1):0]	O	Read address to the DMA Read Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_c2w_lut_re_ram is asserted

Port Name	I/O	Description
edma_rd_eng_c2w_lut_dout_ram[(CC_IF_RD_CTXC2W_RAM_WD-1):0]	O	Write data to the DMA Read Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_c2w_lut_we_ram is asserted
edma_rd_eng_c2w_lut_we_ram	O	Write enable to the DMA Read Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_c2w_lut_re_ram	O	Read enable to the DMA Read Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_c2w_lut_din_ram[(CC_IF_RD_CTXC2W_RAM_WD-1):0]	I	Read data from the DMA Read Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_c2w_lut_addra_par_ram[(C_C_IF_RD_CTXC2W_RAM_PW_PAR-1):0]	O	Write address to the DMA Read Engine Completion to Memory Write LUT RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_rd_eng_c2w_lut_addrb_par_ram[(C C_IF_RD_CTXC2W_RAM_PW_PAR-1):0]	O	Read address to the DMA Read Engine Completion to Memory Write LUT RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_ovrl_addra_par_ram[(CC_I F_RDCTX_LLQ_OVERLAY_RAM_PW_PA R-1):0]	O	Write address to the DMA Read Engine Descriptors Queue RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_ovrl_addrb_par_ram[(CC_I F_RDCTX_LLQ_OVERLAY_RAM_PW_PA R-1):0]	O	Read address to the DMA Read Engine Descriptors Queue RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_ll_addra_par_ram[(CC_IF_ RDCTRL_LL_RAM_PW_PAR-1):0]	O	Write address to the DMA Read Engine Linked List Pointers RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_rd_eng_ll_addrb_par_ram[(CC_IF_RDCTRL_LL_RAM_PW_PAR-1):0]	O	Read address to the DMA Read Engine Linked List Pointers RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_msi_addra_par_ram[(CC_IF_RD_MSI_RAM_PW_PAR-1):0]	O	Write address to the DMA Read Engine MSI Pointers RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_msi_addrb_par_ram[(CC_IF_RD_MSI_RAM_PW_PAR-1):0]	O	Read address to the DMA Read Engine MSI Pointers RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_stsh_lut_addra_par_ram[(CC_IF_RD_CTXSTSH_RAM_PW_PAR-1):0]	O	Write address to the DMA Read Engine Status Handler LUT RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_rd_eng_stsh_lut_addrb_par_ram[(CC_IF_RD_CTXSTSH_RAM_PW_PAR-1):0]	O	Read address to the DMA Read Engine Status Handler LUT RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_c2w_lut_addra_par_ram[(CC_IF_WR_CTXC2W_RAM_PW_PAR-1):0]	O	Write address to the DMA Write Engine Completion to Memory Write LUT RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_c2w_lut_addrb_par_ram[(CC_IF_WR_CTXC2W_RAM_PW_PAR-1):0]	O	Read address to the DMA Write Engine Completion to Memory Write LUT RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_ovrl_addra_par_ram[(CC_IF_WRCTX_LLQ_OVERLAY_RAM_PW_PAR-1):0]	O	Write address to the DMA Write Engine Descriptors Queue RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_wr_eng_ovrl_addrb_par_ram[(CC_IF_WRCTX_LLQ_OVERLAY_RAM_PW_PAR-1):0]	O	Read address to the DMA Write Engine Descriptors Queue RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_ll_addra_par_ram[(CC_IF_WRCTRL_LL_RAM_PW_PAR-1):0]	O	Write address to the DMA Write Engine Linked List Pointers RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_ll_addrb_par_ram[(CC_IF_WRCTRL_LL_RAM_PW_PAR-1):0]	O	Read address to the DMA Write Engine Linked List Pointers RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_msi_addra_par_ram[(CC_IF_WR_MSI_RAM_PW_PAR-1):0]	O	Write address to the DMA Write Engine MSI Pointers RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_wr_eng_msi_addrb_par_ram[(CC_IF_WR_MSI_RAM_PW_PAR-1):0]	O	Read address to the DMA Write Engine MSI Pointers RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_stsh_lut_addrb_par_ram[(CC_IF_WR_CTXSTSH_RAM_PW_PAR-1):0]	O	Write address to the DMA Write Engine Status Handler LUT RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_stsh_lut_addrb_par_ram[(CC_IF_WR_CTXSTSH_RAM_PW_PAR-1):0]	O	Read address to the DMA Write Engine Status Handler LUT RAM Parity Bits. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.56 DMA Read Engine Descriptors Queue RAM Signals

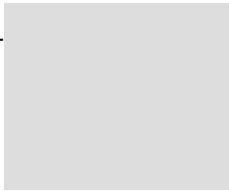
edma_rd_eng_ovrl_din_ram -  - edma_rd_eng_ovrl_addra_ram
 - edma_rd_eng_ovrl_addrb_ram
 - edma_rd_eng_ovrl_dout_ram
 - edma_rd_eng_ovrl_we_ram
 - edma_rd_eng_ovrl_re_ram

Table 2-56 DMA Read Engine Descriptors Queue RAM Signals

Port Name	I/O	Description
edma_rd_eng_ovrl_addra_ram[(CC_IF_R DCTX_LLQ_OVERLAY_RAM_PW-1):0]	O	Write address to the DMA Read Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_ovrl_we_ram is asserted
edma_rd_eng_ovrl_addrb_ram[(CC_IF_R DCTX_LLQ_OVERLAY_RAM_PW-1):0]	O	Read address to the DMA Read Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_ovrl_re_ram is asserted
edma_rd_eng_ovrl_dout_ram[(CC_IF_RD CTX_LLQ_OVERLAY_RAM_WD-1):0]	O	Write data to the DMA Read Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_ovrl_we_ram is asserted
edma_rd_eng_ovrl_we_ram	O	Write enable to the DMA Read Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_rd_eng_ovrl_re_ram	O	Read enable to the DMA Read Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_ovrl_din_ram[(CC_IF_RDC_TX_LLQ_OVERLAY_RAM_WD-1):0]	I	Read data from the DMA Read Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.57 DMA Read Engine Linked List Pointer RAM Signals

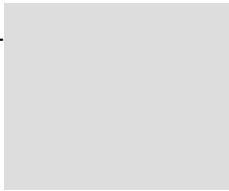
edma_rd_eng_ll_din_ram -  - edma_rd_eng_ll_addra_ram
 - edma_rd_eng_ll_addrb_ram
 - edma_rd_eng_ll_dout_ram
 - edma_rd_eng_ll_we_ram
 - edma_rd_eng_ll_re_ram

Table 2-57 DMA Read Engine Linked List Pointer RAM Signals

Port Name	I/O	Description
edma_rd_eng_ll_addra_ram[(CC_IF_RDC_TRL_LL_RAM_PW-1):0]	O	Write address to the DMA Read Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_ll_we_ram is asserted
edma_rd_eng_ll_addrb_ram[(CC_IF_RDC_TRL_LL_RAM_PW-1):0]	O	Read address to the DMA Read Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_ll_re_ram is asserted
edma_rd_eng_ll_dout_ram[(CC_IF_RDCTRL_LL_RAM_WD-1):0]	O	Write data to the DMA Read Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_ll_we_ram is asserted
edma_rd_eng_ll_we_ram	O	Write enable to the DMA Read Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_rd_eng_ll_re_ram	O	Read enable to the DMA Read Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_ll_din_ram[(CC_IF_RDCTRL_LL_RAM_WD-1):0]	I	Read data from the DMA Read Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.58 DMA Read Engine MSI Pointers RAM Signals

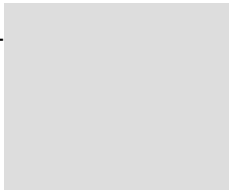
edma_rd_eng_msi_din_ram -  - edma_rd_eng_msi_addra_ram
 - edma_rd_eng_msi_addrb_ram
 - edma_rd_eng_msi_dout_ram
 - edma_rd_eng_msi_we_ram
 - edma_rd_eng_msi_re_ram

Table 2-58 DMA Read Engine MSI Pointers RAM Signals

Port Name	I/O	Description
edma_rd_eng_msi_addra_ram[(CC_IF_RD_MSI_RAM_PW-1):0]	O	Write address to the DMA Read Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_msi_we_ram is asserted
edma_rd_eng_msi_addrb_ram[(CC_IF_RD_MSI_RAM_PW-1):0]	O	Read address to the DMA Read Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_msi_re_ram is asserted
edma_rd_eng_msi_dout_ram[(CC_IF_RD_MSI_RAM_WD-1):0]	O	Write data to the DMA Read Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_msi_we_ram is asserted
edma_rd_eng_msi_we_ram	O	Write enable to the DMA Read Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_rd_eng_msi_re_ram	O	Read enable to the DMA Read Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_msi_din_ram[(CC_IF_RD_MSI_RAM_WD-1):0]	I	Read data from the DMA Read Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.59 DMA Read Engine Status Handler LUT RAM Signals

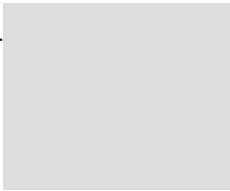
edma_rd_eng_stsh_lut_din_ram -  - edma_rd_eng_stsh_lut_addra_ram
 - edma_rd_eng_stsh_lut_addrb_ram
 - edma_rd_eng_stsh_lut_dout_ram
 - edma_rd_eng_stsh_lut_we_ram
 - edma_rd_eng_stsh_lut_re_ram

Table 2-59 DMA Read Engine Status Handler LUT RAM Signals

Port Name	I/O	Description
edma_rd_eng_stsh_lut_addra_ram[(CC_IF_RD_CTXSTSH_RAM_PW-1):0]	O	Write address to the DMA Read Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_stsh_lut_we_ram is asserted
edma_rd_eng_stsh_lut_addrb_ram[(CC_IF_RD_CTXSTSH_RAM_PW-1):0]	O	Read address to the DMA Read Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_stsh_lut_re_ram is asserted
edma_rd_eng_stsh_lut_dout_ram[(CC_IF_RD_CTXSTSH_RAM_WD-1):0]	O	Write data to the DMA Read Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_rd_eng_stsh_lut_we_ram is asserted
edma_rd_eng_stsh_lut_we_ram	O	Write enable to the DMA Read Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_rd_eng_stsh_lut_re_ram	O	Read enable to the DMA Read Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_rd_eng_stsh_lut_din_ram[(CC_IF_RD_CTXSTSH_RAM_WD-1):0]	I	Read data from the DMA Read Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.60 DMA Write Engine Completion to Memory Write LUT RAM Signals

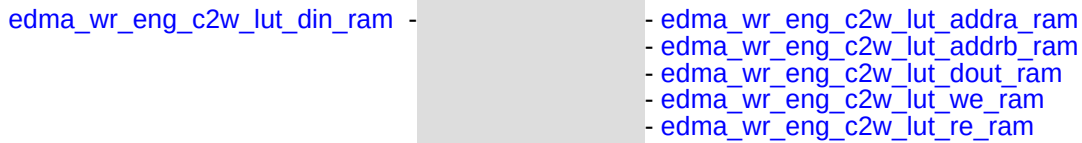


Table 2-60 DMA Write Engine Completion to Memory Write LUT RAM Signals

Port Name	I/O	Description
edma_wr_eng_c2w_lut_addra_ram[(CC_IF_WR_CTXC2W_RAM_PW-1):0]	O	Write address to the DMA Write Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_c2w_lut_we_ram is asserted
edma_wr_eng_c2w_lut_addrb_ram[(CC_IF_WR_CTXC2W_RAM_PW-1):0]	O	Read address to the DMA Write Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_c2w_lut_re_ram is asserted
edma_wr_eng_c2w_lut_dout_ram[(CC_IF_WR_CTXC2W_RAM_WD-1):0]	O	Write data to the DMA Write Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_c2w_lut_we_ram is asserted
edma_wr_eng_c2w_lut_we_ram	O	Write enable to the DMA Write Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_wr_eng_c2w_lut_re_ram	O	Read enable to the DMA Write Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_c2w_lut_din_ram[(CC_IF_WR_CTXC2W_RAM_WD-1):0]	I	Read data from the DMA Write Engine Completion to Memory Write LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.61 DMA Write Engine Descriptors Queue RAM Signals

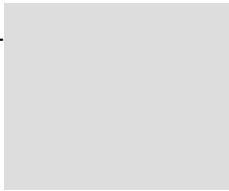
edma_wr_eng_ovrl_din_ram -  - edma_wr_eng_ovrl_addrb_ram
 - edma_wr_eng_ovrl_addrb_ram
 - edma_wr_eng_ovrl_dout_ram
 - edma_wr_eng_ovrl_we_ram
 - edma_wr_eng_ovrl_re_ram

Table 2-61 DMA Write Engine Descriptors Queue RAM Signals

Port Name	I/O	Description
edma_wr_eng_ovrl_addrb_ram[(CC_IF_W RCTX_LLQ_OVERLAY_RAM_PW-1):0]	O	Write address to the DMA Write Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_ovrl_we_ram is asserted
edma_wr_eng_ovrl_addrb_ram[(CC_IF_W RCTX_LLQ_OVERLAY_RAM_PW-1):0]	O	Read address to the DMA Write Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_ovrl_re_ram is asserted
edma_wr_eng_ovrl_dout_ram[(CC_IF_W RCTX_LLQ_OVERLAY_RAM_WD-1):0]	O	Write data to the DMA Write Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_ovrl_we_ram is asserted
edma_wr_eng_ovrl_we_ram	O	Write enable to the DMA Write Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_wr_eng_ovrl_re_ram	O	Read enable to the DMA Write Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_ovrl_din_ram[(CC_IF_WR CTX_LLQ_OVERLAY_RAM_WD-1):0]	I	Read data from the DMA Write Engine Descriptors Queue RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.62 DMA Write Engine Linked List Pointer RAM Signals

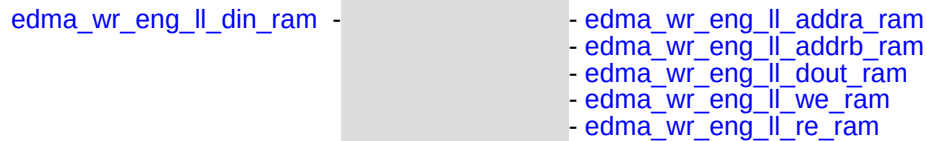


Table 2-62 DMA Write Engine Linked List Pointer RAM Signals

Port Name	I/O	Description
edma_wr_eng_ll_addra_ram[(CC_IF_WR_CTRL_LL_RAM_PW-1):0]	O	Write address to the DMA Write Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_ll_we_ram is asserted
edma_wr_eng_ll_addrb_ram[(CC_IF_WR_CTRL_LL_RAM_PW-1):0]	O	Read address to the DMA Write Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_ll_re_ram is asserted
edma_wr_eng_ll_dout_ram[(CC_IF_WRC_TRL_LL_RAM_WD-1):0]	O	Write data to the DMA Write Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_ll_we_ram is asserted
edma_wr_eng_ll_we_ram	O	Write enable to the DMA Write Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_wr_eng_ll_re_ram	O	Read enable to the DMA Write Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_ll_din_ram[(CC_IF_WRCT RL_LL_RAM_WD-1):0]	I	Read data from the DMA Write Engine Linked List Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.63 DMA Write Engine MSI Pointers RAM Signals

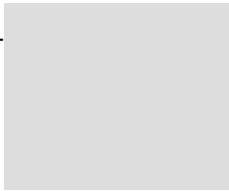
edma_wr_eng_msi_din_ram -  - edma_wr_eng_msi_addra_ram
 - edma_wr_eng_msi_addrb_ram
 - edma_wr_eng_msi_dout_ram
 - edma_wr_eng_msi_we_ram
 - edma_wr_eng_msi_re_ram

Table 2-63 DMA Write Engine MSI Pointers RAM Signals

Port Name	I/O	Description
edma_wr_eng_msi_addra_ram[(CC_IF_W R_MSI_RAM_PW-1):0]	O	Write address to the DMA Write Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_msi_we_ram is asserted
edma_wr_eng_msi_addrb_ram[(CC_IF_W R_MSI_RAM_PW-1):0]	O	Read address to the DMA Write Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_msi_re_ram is asserted
edma_wr_eng_msi_dout_ram[(CC_IF_W R_MSI_RAM_WD-1):0]	O	Write data to the DMA Write Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_msi_we_ram is asserted
edma_wr_eng_msi_we_ram	O	Write enable to the DMA Write Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_wr_eng_msi_re_ram	O	Read enable to the DMA Write Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_msi_din_ram[(CC_IF_WR_MSI_RAM_WD-1):0]	I	Read data from the DMA Write Engine MSI Pointers RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.64 DMA Write Engine Status Handler LUT RAM Signals

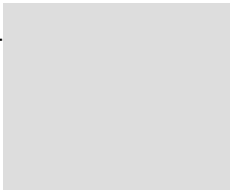
edma_wr_eng_stsh_lut_din_ram -  - edma_wr_eng_stsh_lut_addra_ram
 - edma_wr_eng_stsh_lut_addrb_ram
 - edma_wr_eng_stsh_lut_dout_ram
 - edma_wr_eng_stsh_lut_we_ram
 - edma_wr_eng_stsh_lut_re_ram

Table 2-64 DMA Write Engine Status Handler LUT RAM Signals

Port Name	I/O	Description
edma_wr_eng_stsh_lut_addra_ram[(CC_IF_WR_CTXSTSH_RAM_PW-1):0]	O	Write address to the DMA Write Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_stsh_lut_we_ram is asserted
edma_wr_eng_stsh_lut_addrb_ram[(CC_IF_WR_CTXSTSH_RAM_PW-1):0]	O	Read address to the DMA Write Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_stsh_lut_re_ram is asserted
edma_wr_eng_stsh_lut_dout_ram[(CC_IF_WR_CTXSTSH_RAM_WD-1):0]	O	Write data to the DMA Write Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: edma_wr_eng_stsh_lut_we_ram is asserted
edma_wr_eng_stsh_lut_we_ram	O	Write enable to the DMA Write Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
edma_wr_eng_stsh_lut_re_ram	O	Read enable to the DMA Write Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_wr_eng_stsh_lut_din_ram[(CC_IF_WR_CTXSTSH_RAM_WD-1):0]	I	Read data from the DMA Write Engine Status Handler LUT RAM. Exists: (CC_DMA_ENABLE) && (!CX_DMA_PF_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.65 MSI Interface Signals

ven_msi_req	-	ven_msi_grant
ven_msi_func_num	-	cfg_msi_en
ven_msi_vfunc_num	-	cfg_vf_msi_en
ven_msi_vfunc_active	-	cfg_msi_mask
ven_msi_tc	-	cfg_vf_msi_mask
ven_msi_vector	-	cfg_msi_addr
cfg_msi_pending	-	cfg_msi_data
cfg_vf_msi_pending	-	cfg_msi_64
		cfg_multi_msi_en
		cfg_msi_ext_data_en
		cfg_vf_msi_addr
		cfg_vf_msi_data
		cfg_vf_msi_64
		cfg_vf_multi_msi_en
		cfg_vf_msi_ext_data_en

Table 2-65 MSI Interface Signals

Port Name	I/O	Description
ven_msi_req	I	Request from your application to send an MSI when MSI is enabled. When MSI-X is enabled instead of MSI, assertion of ven_msi_req causes the controller to generate an MSI-X message. Once asserted, ven_msi_req must remain asserted until the controller asserts ven_msi_grant. Exists: Always Synchronous To: aux_clk,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ven_msi_func_num[(PF_WD-1):0]	I	The function number of the MSI request. Function numbering starts at '0'. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msi_req is asserted

Port Name	I/O	Description
ven_msi_vfunc_num[(VF_WD-2):0]	I	Identifies the VF which is making the MSI request. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, ven_msi_vfunc_num=0 refers to the first VF of the PF that is identified by ven_msi_func_num. For more information of VF numbering, see 'Programming View'. When the "External Virtual Function Registers" feature is enabled (CX_EXTENSIBLE_VFUNC=1): ■ In the case of MSI, this I/F cannot be used for external VFs; you must use XALI or AXI slave I/F. It can be used for internal VFs. ■ In the case of MSI-X, this I/F can be used for internal and external VFs. Exists: CX_SRIOV_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: ven_msi_vfunc_active is asserted
ven_msi_vfunc_active	I	Indicates that the MSI request is coming from a VF. 0: No VF is active and ven_msi_vfunc_num is invalid. A PF is valid and identified by ven_msi_func_num. 1: A VF is active and is identified by ven_msi_vfunc_num. Exists: CX_SRIOV_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ven_msi_tc[2:0]	I	Traffic Class of the MSI request, valid when ven_msi_req is asserted. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msi_req is asserted
ven_msi_vector[4:0]	I	Used to modulate the lower five bits of the MSI Data register when multiple message mode is enabled. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msi_req is asserted. Valid only when multiple message mode is enabled for the device through the MSI Control register.

Port Name	I/O	Description
ven_msi_grant	O	One-cycle pulse that indicates that the controller has accepted the request to send an MSI. After asserting ven_msi_grant for one cycle, the controller does not wait for ven_msi_req to be de-asserted then reasserted to generate another MSI. When ven_msi_req remains asserted after the controller asserts ven_msi_grant for one cycle, the controller generates another MSI. Exists: Always Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_msi_en[(NF-1):0]	O	Indicates that MSI is enabled (INTx message is not sent), one bit per configured function. Exists: Always Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vf_msi_en[(INT_NVF-1):0]	O	Indicates when the VF MSI Capability structure is enabled for each configured function. Each bit field corresponds to one of the NVF virtual functions. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. Exists: (CX_SRIOV_ENABLE) && (VF_MSI_CAP_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_msi_mask[((32*NF)-1):0]	O	Contents of the Per Vector Mask register in the MSI Capability structure. For each bit that is set, the function is prohibited from sending the associated message. Exists: MSI_PVM_EN Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_msi_pending[((32*Nf)-1):0]	I	Indication from application about which functions have a pending associated message. Used to determine the value of the Vector Interrupt Pending register in the MSI Capability structure. Exists: MSI_PVM_EN Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_vf_msi_mask[((32*INT_NVf)-1):0]	O	Contents of the Per Vector Mask register in the VF MSI Capability structure. For each bit that is set, the function is prohibited from sending the associated message. Each bit field corresponds to one of the NVF virtual functions. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. Exists: (MSI_PVM_EN) && (CX_SRIOV_ENABLE) && (VF_MSI_CAP_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_vf_msi_pending[((32*INT_NVf)-1):0]	I	Contents of the Per Vector Pending register in the VF MSI Capability structure. For each bit that is set, the function has a pending associated message. Each bit field corresponds to one of the NVF virtual functions. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. Exists: (MSI_PVM_EN) && (CX_SRIOV_ENABLE) && (VF_MSI_CAP_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_msi_addr[((64*Nf)-1):0]	O	Contents of the MSI Lower 32 Bits and Upper 32 Bits Address registers in the MSI Capability structure. There are 64 bits of cfg_msi_addr for each configured function. Exists: MSI_IO Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_msi_data[((32*Nf)-1):0]	O	Contents of the MSI Data register in the MSI Capability structure. There are 32 bits of cfg_msi_data for each configured function. Exists: MSI_IO Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_msi_64[(Nf-1):0]	O	The 64-bit Address Capable bit of the MSI Control register in the MSI Capability structure, one bit for each configured function. Exists: MSI_IO Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_multi_msi_en[((3*Nf)-1):0]	O	The Multiple Message Enabled field of the MSI Control register in the MSI Capability structure. There are 3 bits of cfg_multi_msi_en for each configured function. Exists: MSI_IO Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_msi_ext_data_en[(Nf-1):0]	O	Indicates to application if extended message data for MSI is enabled or not for each physical function. Exists: MSI_IO Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_vf_msi_addr[((64*INT_NVF)-1):0]	O	Contents of the MSI Lower 32 Bits and Upper 32 Bits Address registers in the VF MSI Capability structure. There are 64 bits of cfg_vf_msi_addr for each configured function. Each bit field corresponds to one of the NVF virtual functions. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. Exists: (MSI_IO) && (CX_SRIOV_ENABLE) && (VF_MSI_CAP_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_vf_msi_data[((32*INT_NVF)-1):0]	O	Contents of the MSI Data register in the VF MSI Capability structure. There are 32 bits of cfg_vf_msi_data for each configured function. Each bit field corresponds to one of the NVF virtual functions. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. Exists: (MSI_IO) && (CX_SRIOV_ENABLE) && (VF_MSI_CAP_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_vf_msi_64[(INT_NVF)-1:0]	O	The 64-bit Address Capable bit of the MSI Control register in the VF MSI Capability structure, one bit for each configured function. Each bit field corresponds to one of the NVF virtual functions. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. Exists: (MSI_IO) && (CX_SRIOV_ENABLE) && (VF_MSI_CAP_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: high Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_vf_multi_msi_en[((3*INT_NVF)-1):0]	O	The Multiple Message Capable field of the MSI Control register in the VF MSI Capability structure. There are 3 bits of cfg_vf_multi_msi_en for each configured function. Each bit field corresponds to one of the NVF virtual functions. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. Exists: (MSI_IO) && (CX_SRIOV_ENABLE) && (VF_MSI_CAP_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_vf_msi_ext_data_en[(INT_NVF-1):0]	O	Indicates to application if extended message data for MSI is enabled or not for each virtual function. Exists: (MSI_IO) && (CX_SRIOV_ENABLE) && (VF_MSI_CAP_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.66 MSI-X Interface Signals

msix_addr -
msix_data -

- cfg_msix_en
- cfg_msix_func_mask
- cfg_vf_msix_en
- cfg_vf_msix_func_mask
- cfg_vf_msix_table_size
- cfg_vf_msix_table_bir
- cfg_vf_msix_table_offset
- cfg_vf_msix_pba_bir
- cfg_vf_msix_pba_offset
- cfg_msix_table_size
- cfg_msix_table_bir
- cfg_msix_table_offset
- cfg_msix_pba_bir
- cfg_msix_pba_offset

Table 2-66 MSI-X Interface Signals

Port Name	I/O	Description
msix_addr[63:0]	I	The address value for the MSI-X. Exists: (MSIX_CAP_ENABLE) && (!MSIX_TABLE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msi_req is asserted
msix_data[31:0]	I	The data value for the MSI-X. Exists: (MSIX_CAP_ENABLE) && (!MSIX_TABLE_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msi_req is asserted
cfg_msix_en[(NF-1):0]	O	The MSI-X Enable bit of the MSI-X Control register in the MSI-X Capability structure. There is 1 bit of cfg_msix_en for each configured function. Exists: (MSIX_CAP_ENABLE) && (!MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

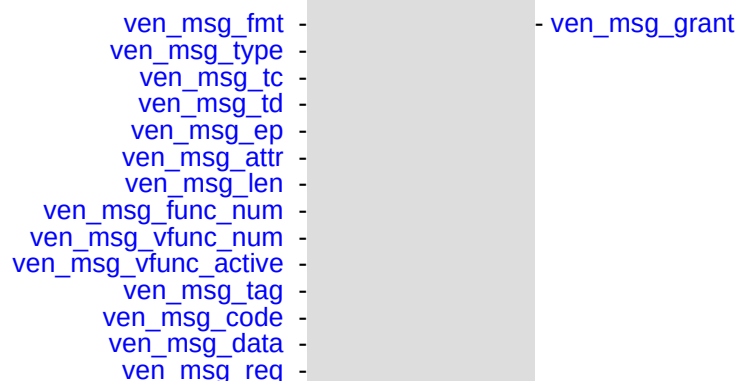
Port Name	I/O	Description
cfg_msix_func_mask[(NF-1):0]	O	The function Mask bit of the MSI-X Control register in the MSI-X Capability structure. There is 1 bit of cfg_msix_func_mask for each configured function. Exists: (MSIX_CAP_ENABLE) && (!MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vf_msix_en[(INT_NVF-1):0]	O	MSIX_EN bits from the Message Control Register in the MSI-X Capability register of each VF. The application needs to ensure that the corresponding bit is set before requesting to send a message for the particular VF. Each bit field corresponds to one of the NVF virtual functions. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. Exists: (MSIX_CAP_ENABLE) && (CX_SRIOV_ENABLE) && (!MSIX_TABLE_EN) && (VF_MSIX_CAP_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vf_msix_func_mask[(INT_NVF-1):0]	O	Function mask bits from the Message Control Register in the MSI-X Capability register of each VF. Each bit field corresponds to one of the NVF virtual functions. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. Exists: (MSIX_CAP_ENABLE) && (CX_SRIOV_ENABLE) && (!MSIX_TABLE_EN) && (VF_MSIX_CAP_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_vf_msix_table_size[((11*NF)-1):0]	O	The MSI-X Table Size field of the VF MSI-X Control register in the MSI-X Capability structure. There are 11 bits of cfg_vf_msix_table_size for each configured function. Exists: (MSIX_CAP_ENABLE) && (CX_SRIOV_ENABLE) && (!MSIX_TABLE_EN) && (VF_MSIX_CAP_ENABLE) && (MSIX_IO) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vf_msix_table_bir[((3*NF)-1):0]	O	Table BAR Indicator Register (BIR) field of the MSI-X Table Offset and BIR register in the VF MSI-X Capability structure. There are 3 bits of cfg_vf_msix_table_bir for each configured function. Exists: (MSIX_CAP_ENABLE) && (CX_SRIOV_ENABLE) && (!MSIX_TABLE_EN) && (VF_MSIX_CAP_ENABLE) && (MSIX_IO) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_vf_msix_table_offset[((29*NF)-1):0]	O	Table Offset field of the MSI-X Table Offset and BIR register in the VF MSI-X Capability structure. There are 29 bits of cfg_vf_msix_table_offset for each configured function. Exists: (MSIX_CAP_ENABLE) && (CX_SRIOV_ENABLE) && (!MSIX_TABLE_EN) && (VF_MSIX_CAP_ENABLE) && (MSIX_IO) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_vf_msix_pba_bir[((3*NF)-1):0]	O	PBA BAR Indicator Register (BIR) field of the MSI-X PBA Offset and BIR register in the VF MSI-X Capability structure. There are 3 bits of cfg_vf_msix_pba_bir for each configured function. Exists: (MSIX_CAP_ENABLE) && (CX_SRIOV_ENABLE) && (!MSIX_TABLE_EN) && (VF_MSIX_CAP_ENABLE) && (MSIX_IO) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_vf_msix_pba_offset[((29*NF)-1):0]	O	PBA Offset field of the MSI-X PBA Offset and BIR register in the VF MSI-X Capability structure. There are 29 bits of cfg_vf_msix_pba_offset for each configured function. Exists: (MSIX_CAP_ENABLE) && (CX_SRIOV_ENABLE) && (!MSIX_TABLE_EN) && (VF_MSIX_CAP_ENABLE) && (MSIX_IO) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_msix_table_size[((11*NF)-1):0]	O	The MSI-X Table Size field of the MSI-X Control register in the MSI-X Capability structure. There are 11 bits of cfg_msix_table_size for each configured function. Exists: (MSIX_CAP_ENABLE) && (MSIX_IO) && (!MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_msix_table_bir[((3*NF)-1):0]	O	Table BAR Indicator Register (BIR) field of the MSI-X Table Offset and BIR register in the MSI-X Capability structure. There are 3 bits of cfg_msix_table_bir for each configured function. Exists: (MSIX_CAP_ENABLE) && (MSIX_IO) && (!MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_msix_table_offset[((29*NF)-1):0]	O	Table Offset field of the MSI-X Table Offset and BIR register in the MSI-X Capability structure. There are 29 bits of cfg_msix_table_offset for each configured function. Exists: (MSIX_CAP_ENABLE) && (MSIX_IO) && (!MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_msix_pba_bir[((3*NF)-1):0]	O	PBA BIR field of the MSI-X PBA Offset and BIR register in the MSI-X Capability structure. There are 3 bits of cfg_msix_pba_bir for each configured function. Exists: (MSIX_CAP_ENABLE) && (MSIX_IO) && (!MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_msix_pba_offset[((29*NF)-1):0]	O	PBA Offset field of the MSI-X PBA Offset and BIR register in the MSI-X Capability structure. There are 29 bits of cfg_msix_pba_offset for each configured function. Exists: (MSIX_CAP_ENABLE) && (MSIX_IO) && (!MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.67 VMI Signals



ven_msg_fmt
 ven_msg_type
 ven_msg_tc
 ven_msg_td
 ven_msg_ep
 ven_msg_attr
 ven_msg_len
 ven_msg_func_num
 ven_msg_vfunc_num
 ven_msg_vfunc_active
 ven_msg_tag
 ven_msg_code
 ven_msg_data
 ven_msg_req

- ven_msg_grant

Table 2-67 VMI Signals

Port Name	I/O	Description
ven_msg_fmt[1:0]	I	The Format field for the vendor-defined Message TLP. Should be set to 0x1. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted
ven_msg_type[4:0]	I	The Type field for the vendor-defined Message TLP. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted
ven_msg_tc[2:0]	I	The Traffic Class field for the vendor-defined Message TLP. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted

Port Name	I/O	Description
ven_msg_td	I	The TLP Digest (TD) bit for the vendor-defined Message TLP, valid when ven_msg_req is asserted. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted
ven_msg_ep	I	The Poisoned TLP (EP) bit for the vendor-defined Message TLP. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted
ven_msg_attr[(ATTR_WD-1):0]	I	The Attributes field for the vendor-defined Message TLP. - Width is three bits when CX_IDO_ENABLE =1 - Width is two bits when CX_IDO_ENABLE =0 Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted
ven_msg_len[9:0]	I	The Length field for the vendor-defined Message TLP (indicates length of data payload in dwords). Should be set to 0x0. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted
ven_msg_func_num[(PF_WD-1):0]	I	Function Number for the vendor-defined Message TLP. Function numbering starts at '0'. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted

Port Name	I/O	Description
ven_msg_vfunc_num[(VF_WD-2):0]	I	Number of the virtual function accessing the VMI interface. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, ven_msg_vfunc_num=0 refers to the first VF of the PF that is identified by ven_msg_func_num. For more information of VF numbering, see 'Programming View'. Exists: CX_SRIOV_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: ven_msg_req and ven_msg_vfunc_active are asserted
ven_msg_vfunc_active	I	Indicates that a VF is accessing the VMI. 0: No VF is active and ven_msg_vfunc_num is invalid. A PF is valid and identified by ven_msg_func_num. 1: A VF is active and is identified by ven_msg_vfunc_num. Exists: CX_SRIOV_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted
ven_msg_tag[(TAG_SIZE-1):0]	I	Tag for the vendor-defined Message TLP. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted
ven_msg_code[7:0]	I	The Message Code for the vendor-defined Message TLP. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted

Port Name	I/O	Description
ven_msg_data[63:0]	I	Third and fourth dwords of the Vendor Defined Message header where: ■ Bytes 8-11 (third header dword) =ven_msg_data[63:32] ■ Bytes 12-15 (fourth header dword) =ven_msg_data[31:0], where ven_msg_data[7:0] =byte 15 For more information, see the 'Endianness' advanced information chapter. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: ven_msg_req is asserted
ven_msg_req	I	Request from your application to send a vendor-defined Message. Once asserted, ven_msg_req must remain asserted until the controller asserts ven_msg_grant. Exists: Always Synchronous To: aux_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ven_msg_grant	O	One-cycle pulse that indicates that the controller has accepted the request to send the vendor-defined Message. Exists: Always Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.68 Power Budgeting Signals

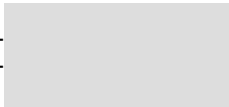
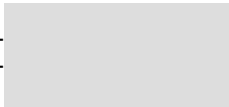
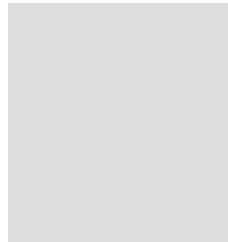
[cfg_pwr_budget_data_reg](#) -  - [cfg_pwr_budget_data_sel_reg](#)
[cfg_pwr_budget_func_num](#) -  - [cfg_pwr_budget_sel](#)

Table 2-68 Power Budgeting Signals

Port Name	I/O	Description
cfg_pwr_budget_data_reg[31:0]	I	New Data Register value. Data needs to be held. Exists: PWR_BUDGET_CAP_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_pwr_budget_func_num[(PF_WD-1):0]	I	Function # of data register above (because this capability is per function). Function numbering starts at '0'. Exists: PWR_BUDGET_CAP_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_pwr_budget_data_sel_reg[7:0]	O	Contents of the Data Select Register. Exists: PWR_BUDGET_CAP_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_pwr_budget_sel[(NF-1):0]	O	One cycle pulse signal indicates new contents in Data Select Register. Something changed the value of Data Select Register, so your application should take a look at the new value and provide the updated Data Register value corresponding to it. Exists: PWR_BUDGET_CAP_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.69 SII: Detect Poisoned CfgWr Signals



- radm_rcvd_cfg0wr_poisoned
- radm_rcvd_cfg1wr_poisoned
- radm_rcvd_cfgwr_poisoned_func_num
- radm_rcvd_cfgwr_poisoned_vfunc_num
- radm_rcvd_cfgwr_poisoned_is_vf
- radm_rcvd_cfgwr_poisoned_ext_reg_num
- radm_rcvd_cfgwr_poisoned_reg_num

Table 2-69 SII: Detect Poisoned CfgWr Signals

Port Name	I/O	Description
radm_rcvd_cfg0wr_poisoned[(FX_TLP-1):0]	O	Indicates that a Poisoned CfgWr Type 0 has been received. Exists: CX_DETECT_POISONED_CFGWR Synchronous To: aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (One-clock-cycle pulse) Validated by: Not validated by another signal
radm_rcvd_cfg1wr_poisoned[(FX_TLP-1):0]	O	Indicates that a Poisoned CfgWr Type 1 has been received. Exists: CX_DETECT_POISONED_CFGWR Synchronous To: aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (One-clock-cycle pulse) Validated by: Not validated by another signal
radm_rcvd_cfgwr_poisoned_func_num[((FX_TLP*PF_WD)-1):0]	O	If CX_SRIOV_ENABLE is defined and radm_rcvd_cfgwr_poisoned_is_vf is set, indicates the PF of the PF/VF on which the poisoned CfgWr has been received. Otherwise, indicates the Function on which the poisoned CfgWr has been received. Exists: CX_DETECT_POISONED_CFGWR Synchronous To: aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_rcvd_cfg0wr_poisoned or radmrcvd_cfg1wr_poisoned

Port Name	I/O	Description
radm_rcvd_cfgwr_poisoned_vfunc_num[((FX_TLP*(VF_WD-1))-1):0]	O	If radm_rcvd_cfgwr_poisoned_is_vf is set, indicates the VF of the PF/VF on which the poisoned CfgWr has been received. VF Numbering starts from 0. Exists: (CX_DETECT_POISONED_CFGWR) && (CX_SRIOV_ENABLE) Synchronous To: aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_rcvd_cfgwr_poisoned_is_vf
radm_rcvd_cfgwr_poisoned_is_vf[(FX_TLP-1):0]	O	If sets, indicates that the received poisoned CfgWr is for a Virtual Function. The PF/VF information is represented by radm_rcvd_cfgwr_poisoned_func_num/radm_rcvd_cfgwr_poisoned_vfunc_num. Exists: (CX_DETECT_POISONED_CFGWR) && (CX_SRIOV_ENABLE) Synchronous To: aux_clk,aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_rcvd_cfg0wr_poisoned or radmrcvd_cfg1wr_poisoned
radm_rcvd_cfgwr_poisoned_ext_reg_num[((FX_TLP*4)-1):0]	O	Extended Register Number field of the received poisoned CfgWr. Exists: CX_DETECT_POISONED_CFGWR Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_rcvd_cfg0wr_poisoned or radmrcvd_cfg1wr_poisoned
radm_rcvd_cfgwr_poisoned_reg_num[((FX_TLP*6)-1):0]	O	Register Number field of the received poisoned CfgWr. Exists: CX_DETECT_POISONED_CFGWR Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_rcvd_cfg0wr_poisoned or radmrcvd_cfg1wr_poisoned

2.70 SII: AER Control Signals

app_dpc_triggered
app_dpc_trig_en
app_dpc_trig_status

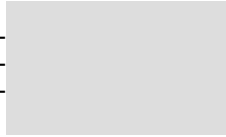


Table 2-70 SII: AER Control Signals

Port Name	I/O	Description
app_dpc_triggered	I	DPC triggered indication from application. A low to high transition indicates that DPC has been triggered and by definition the Port is in DPC. DPC is event triggered. It is expected that this signal is used to update the corresponding DPC Status register bit field in the external DPC capability. Exists: CX_EXTDPC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
app_dpc_trig_en[1:0]	I	DPC trigger Enable from the application and should be taken from the corresponding external DPC Control register field, DPC Trigger Enable. Encodings are: ■ 00b: DPC is disabled. ■ 01b: DPC is enabled and is triggered when the Downstream Port detects an unmasked uncorrectable error or when the Downstream Port receives an ERR_FATAL Message. ■ 10b: DPC is enabled and is triggered when the Downstream Port detects an unmasked uncorrectable error or when the Downstream Port receives an ERR_NONFATAL or ERR_FATAL Message. ■ 11b: Reserved. Exists: CX_EXTDPC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
app_dpc_trig_status	I	DPC trigger status from the application and should be taken from the corresponding external DPC Status register field, DPC Trigger Status. Exists: CX_EXTDPC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

2.71 SII: CCIX Configuration/Control/Status Signals

ccix_tlp_disable -

- cfg_ccix_esm_enable
- cfg_ccix_esm_data_rate0
- cfg_ccix_esm_data_rate1
- cfg_ccix_opt_tlp_sup
- cfg_ccix_vc_resource
- cfg_ccix_opt_tlp_en
- cfg_ccix_tc_enable

Table 2-71 SII: CCIX Configuration/Control/Status Signals

Port Name	I/O	Description
cfg_ccix_esm_enable	O	Contents of the ESM Enable field of ESM Control Register. <i>Note:</i> The DSP application must program (set or reset) this bit in the remote USP first, before programming it locally. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_ESM_SUPPORT) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ccix_esm_data_rate0[6:0]	O	Contents of the ESM Data Rate0 field of ESM Control Register. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_ESM_SUPPORT) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ccix_esm_data_rate1[6:0]	O	Contents of the ESM Data Rate1 field of ESM Control Register. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_ESM_SUPPORT) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_ccix_opt_tlp_sup	O	CCIX optimized TLP format supported field of CCIX Transaction Layer Capabilities Register. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ccix_vc_resource[2:0]	O	CCIX VC Resource field of CCIX Transaction Layer Capabilities Register. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_ccix_opt_tlp_en	O	Enable CCIX optimized TLP Generation and Reception CCIX Transaction Layer Control Register. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ccix_tc_enable[7:0]	O	A bus that indicates which of the TCs are mapped to CCIX VC. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ccix_tlp_disable	I	Disable the support of CCIX TLP. This is a boot strap signal and it needs to be valid and static during reset. Exists: ((CX_CPCIE_ENABLE)) && (CX_CCIX_ENABLE) && (CX_CCIX_INTERFACE_ENABLE) Synchronous To: core_clk,radm_clk_g,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.72 SII: Configuration Information Signals

prs_res_failure	-	cfg_10b_tag_req_en
prs_uprgi	-	cfg_vf_en
prs_stopped	-	cfg_num_vf
cdm_reg_chk_test_en	-	cfg_start_vfi
		cfg_vf_10b_tag_req_en
		cfg_tph_req_en
		cfg_pf_tph_st_mode
		cfg_pf_pasid_en
		cfg_pf_pasid_execute_perm_en
		cfg_pf_pasid_priv_mode_en
		cfg_vf_tph_req_en
		cfg_vf_tph_st_mode
		cfg_vf_pasid_en
		cfg_vf_pasid_execute_perm_en
		cfg_vf_pasid_priv_mode_en
		cfg_atomic_req_en
		cfg_atomic_egress_block
		cfg_obff_en
		cfg_ltr_m_en
		cfg_ltr_max_latency
		cfg_disable_ltr_clr_msg
		cfg_bar0_start
		cfg_bar1_start
		cfg_bar0_limit
		cfg_bar1_limit
		cfg_bar2_start
		cfg_bar2_limit
		cfg_bar3_start
		cfg_bar3_limit
		cfg_bar4_start
		cfg_bar4_limit
		cfg_bar5_start
		cfg_bar5_limit
		cfg_exp_rom_start
		cfg_exp_rom_limit
		cfg_bus_master_en
		cfg_mem_space_en
		cfg_max_rd_req_size
		cfg_max_payload_size
		cfg_ext_tag_en
		cfg_rcb
		radm_cpl_pending_vf
		cfg_pm_no_soft_rst
		cfg_pbus_num
		cfg_pbus_dev_num
		cfg_vc_enable
		cfg_vc_struct_vc_id_map
		cfg_vc_id_vc_struct_map
		cfg_tc_enable
		cfg_tc_struct_vc_map
		cfg_ln_enable
		ln_vf_enable
		cfg_ido_req_en
		cfg_ido_cpl_en
		cfg_no_snoop_en
		cfg_relax_order_en
		cfg_ats_stu
		cfg_ats_cache_en
		cfg_acs_validation_en

- cfg_acs_at_blocking_en
- cfg_acs_p2p_req_redirect_en
- cfg_acs_p2p_compl_redirect_en
- cfg_acs_up_forward_en
- cfg_acs_p2p_egress_ctrl_en
- cfg_acs_p2p_direct_transl_en
- cfg_acs_egress_ctrl_vec
- acs_vf_p2p_req_redirect_en
- acs_vf_p2p_compl_redirect_en
- acs_vf_p2p_egress_ctrl_en
- acs_vf_p2p_direct_transl_en
- acs_vf_egress_ctrl_vec
- cfg_acs_func_grp_en
- cfg_ari_func_grp
- cfg_vf_ari_func_grp
- cfg_prs_enable
- cfg_prs_reset
- cfg_prs_stopped
- cfg_prs_uprgi
- cfg_prs_response_failure
- cfg_prs_outstanding_allocation
- cdm_reg_chk_logic_err
- cdm_reg_chk_cmp_err
- cdm_reg_chk_cmplt
- rbar_ctrl_update
- cfg_rbar_size
- vf_rbar_ctrl_update
- cfg_vf_rbar_size
- cfg_hp_slot_ctrl_access
- cfg_dll_state_chged_en
- cfg_cmd_cpled_int_en
- cfg_hp_int_en
- cfg_pre_det_chged_en
- cfg_mrl_sensor_chged_en
- cfg_pwr_fault_det_en
- cfg_atten_button_pressed_en
- if_timeout_status

Table 2-72 SII: Configuration Information Signals

Port Name	I/O	Description
cfg_10b_tag_req_en[(NF-1):0]	O	Contents of the PCIE_CAP_10BITS_TAG_REQ_EN field in the DEVICE_CONTROL2_DEVICE_STATUS2_REG register. Exists: (!(AMBA_INTERFACE!=0)) && ((CX_10BITS_TAG== 1 && CX_MAX_TAG > 255)? 1: 0) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_vf_en[(NF-1):0]	O	Identifies those physical functions that have virtual functions enabled. Exists: CX_SRIOV_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_num_vf[((16*Nf)-1):0]	O	The number of virtual functions in each physical function. Exists: CX_SRIOV_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_start_vfi[((NF*VFI_WD)-1):0]	O	The index of the first VF in every PF. This bus is a concatenation of NF numbers, each of which is log2NVF bits wide. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview', to index the corresponding bit field for a particular VF in signals such as cfg_vf_bme. For example, if PF0 has 10 VFs, PF1 has 30, PF2 has 50, then cfg_start_vfi would contain the following: ■ {..., 90, 40, 10, 0} If a PF has no VFs associated with it, then there is no entry in the bus for that PF. For example, if PF0 has 10 VFs, PF1 has 0, PF2 has 50, then cfg_start_vfi would contain the following: ■ {..., 60, 10, 10, 0} Your application must correctly interpret that index 10 belongs to PF2 and not of PF1. The cfg_vf_en output identifies those physical functions that have virtual functions enabled and the cfg_num_vf output identifies the number of virtual functions in each physical function. Exists: CX_SRIOV_ENABLE Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_vf_10b_tag_req_en[(NF-1):0]	O	Identifies physical functions for which all their VFs must use 10 bit tag for all Non-Posted Request generated. Exists: (CX_SRIOV_ENABLE) && !(AMBA_INTERFACE!=0) && ((CX_10BITS_TAG== 1 && CX_MAX_TAG > 255)? 1: 0) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_tph_req_en[((2*Nf)-1):0]	O	The 2-bit TPH Requester Enabled field of each TPH Requester Control register. Exists: CX_TPH_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_pf_tph_st_mode[((3*Nf)-1):0]	O	Steering Tag Mode of Operation for Physical Function. Steering Tags are system-specific values used to identify a processing resource that a Requester explicitly targets Exists: CX_TPH_ENABLE Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_pf_pasid_en[(NF-1):0]	O	The value of the PASID Enable field in each PF PASID Control Register. Exists: CX_PASID_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_pf_pasid_execute_perm_en[(NF-1):0]	O	The value of the Execute Permission Enable field in each PF PASID Control Register. Exists: CX_PASID_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_pf_pasid_priv_mode_en[(NF-1):0]	O	The value of the Privileged Mode Enable field in each PF PASID Control Register. Exists: CX_PASID_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vf_tph_req_en[(2*INT_NVF)-1):0]	O	The 2-bit TPH Requester Enabled field of each VF TPH Requester Control register. Each field corresponds to one of the NVF virtual functions. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding field for a particular VF. Exists: (CX_SRIOV_ENABLE) && (VF_TPH_ENABLE_VALUE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vf_tph_st_mode[((3*INT_NVF)-1):0]	O	Steering Tag Mode of Operation for Virtual Function. Steering Tags are system-specific values used to identify a processing resource that a Requester explicitly targets Exists: (CX_SRIOV_ENABLE) && (VF_TPH_ENABLE_VALUE) Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vf_pasid_en[(INT_NVF-1):0]	O	The value of the PASID Enable field in each VF PASID Control Register. Exists: (CX_SRIOV_ENABLE) && (VF_PASID_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_vf_pasid_execute_perm_en[(INT_NV F-1):0]	O	The value of the Execute Permission Enable field in each VF PASID Control Register. Exists: (CX_SRIOV_ENABLE) && (VF_PASID_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vf_pasid_priv_mode_en[(INT_NV F-1):0]	O	The value of the Privileged Mode Enable field in each VF PASID Control Register. Exists: (CX_SRIOV_ENABLE) && (VF_PASID_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_atomic_req_en[(NF-1):0]	O	The AtomicOp Requester Enable field (PCIE_CAP_ATOMIC_REQ_EN) of the Device Control 2 register. Exists: CX_ATOMIC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_atomic_egress_block[(NF-1):0]	O	The AtomicOp Egress Blocking field (PCIE_CAP_ATOMIC_EGRESS_BLK) of the Device Control 2 register. Exists: (CX_ATOMIC_ENABLE) && (CX_ATOMIC_ROUTING_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_obff_en[1:0]	O	The OBFF Enable field of the Device Control 2 register of function 0. Exists: CX_OBFF_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_ltr_m_en	O	The LTR Mechanism Enable field of the Device Control 2 register of function 0. Exists: CX_LTR_M_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ltr_max_latency[((32*NF)-1):0]	O	The concatenated contents of 'LTR Max No-Snoop Latency Register' and 'LTR Max Snoop Latency Register'. The 'LTR Max Snoop Latency' of function 0 occupies the lower 16-bits. Exists: CX_LTR_M_ENABLE Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_disable_ltr_clr_msg	O	Disable the autonomous generation of LTR clear message. Exists: CX_LTR_M_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bar0_start[((64*NF)-1):0]	O	The starting address of BAR 0 (a memory or I/O BAR). There are 64 bits of cfg_bar0_start assigned to each configured function. The actual BAR start address present any 64-bit segment can be either a 64-bit or 32-bit BAR start address. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bar1_start[((32*NF)-1):0]	O	The starting address of BAR 1 (a memory or I/O BAR). There are 32 bits of cfg_bar1_start assigned to each configured function. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_bar0_limit[((64*Nf)-1):0]	O	The end address of BAR 0 (a memory or I/O BAR). There are 64 bits of cfg_bar0_limit assigned to each configured function. The actual BAR end address present any 64-bit segment can be either a 64-bit or 32-bit BAR end address. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bar1_limit[((32*Nf)-1):0]	O	The end address of BAR 1 (a memory or I/O BAR). There are 32 bits of cfg_bar1_limit assigned to each configured function. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bar2_start[((64*Nf)-1):0]	O	The starting address of BAR 2 (a memory or I/O BAR). There are 64 bits of cfg_bar2_start assigned to each configured function. The actual BAR start address present any 64-bit segment can be either a 64-bit or 32-bit BAR start address. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bar2_limit[((64*Nf)-1):0]	O	The end address of BAR 2 (a memory or I/O BAR). There are 64 bits of cfg_bar2_limit assigned to each configured function. The actual BAR end address present any 64-bit segment can be either a 64-bit or 32-bit BAR end address. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_bar3_start[((32*Nf)-1):0]	O	The starting address of BAR 3 (a memory or I/O BAR). There are 32 bits of cfg_bar3_start assigned to each configured function. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bar3_limit[((32*Nf)-1):0]	O	The end address of BAR 3 (a memory or I/O BAR). There are 32 bits of cfg_bar3_limit assigned to each configured function. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bar4_start[((64*Nf)-1):0]	O	The starting address of BAR 4 (a memory or I/O BAR). There are 64 bits of cfg_bar4_start assigned to each configured function. The actual BAR start address present any 64-bit segment can be either a 64-bit or 32-bit BAR start address. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bar4_limit[((64*Nf)-1):0]	O	The end address of BAR 4 (a memory or I/O BAR). There are 64 bits of cfg_bar4_limit assigned to each configured function. The actual BAR end address present any 64-bit segment can be either a 64-bit or 32-bit BAR end address. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_bar5_start[((32*Nf)-1):0]	O	The starting address of BAR 5 (a memory or I/O BAR). There are 32 bits of cfg_bar5_start assigned to each configured function. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bar5_limit[((32*Nf)-1):0]	O	The end address of BAR 5 (a memory or I/O BAR). There are 32 bits of cfg_bar5_limit assigned to each configured function. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_exp_rom_start[((32*Nf)-1):0]	O	The starting address of expansion ROM. There are 32 bits of cfg_exp_rom_start assigned to each configured function. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_exp_rom_limit[((32*Nf)-1):0]	O	The end address of expansion ROM. There are 32 bits of cfg_exp_rom_limit assigned to each configured function. Exists: !(CX_NFUNC>32) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bus_master_en[(Nf-1):0]	O	The state of the bus master enable bit in the PCI-compatible Command register. There is 1 bit of cfg_bus_master_en assigned to each configured function. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_mem_space_en[(NF-1):0]	O	The state of the Memory Space Enable bit in the PCI-compatible Command register. There is 1 bit of cfg_mem_space_en assigned to each configured function. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_max_rd_req_size[((3*NF)-1):0]	O	The value of the Max_Read_Request_Size field in the Device Control register. There are 3 bits of cfg_max_rd_req_size assigned to each configured function. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_max_payload_size[((3*NF)-1):0]	O	The value of the Max_Payload_Size field in the Device Control register. There are 3 bits of cfg_max_payload_size assigned to each configured function. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ext_tag_en[(NF-1):0]	O	When enabled, controller supports up to 8-bit tag values. Exists: Any DEFAULT_EXT_TAG_FIELD_SUPPORTED_n is 1 Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_rcb[(NF-1):0]	O	The value of the RCB bit in the Link Control register. There is 1 bit of cfg_rcb assigned to each configured function. Exists: Always Synchronous To: core_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
radm_cpl_pending_vf[(NVF-1):0]	O	The Transaction Pending field of each VF Device Status Register. Each field corresponds to one of the NVF virtual functions. You can use the <code>cfg_start_vfi</code> output signal or the VF Index in 'Bus Numbering Overview' index the corresponding field for a particular VF. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_pm_no_soft_rst[(NF-1):0]	O	This is the value of the No Soft Reset bit in the Power Management Control and Status Register. When set, you should not reset any controller registers when transitioning from D3hot to D0. Therefore, you should not assert the <code>non_sticky_rst_n</code> and <code>sticky_rst_n</code> inputs. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_pbus_num[(BUSNUM_WD-1):0]	O	The primary bus number assigned to the function. The number of bits depends the value of <code>MULTI_DEVICE_AND_BUS_PER_FUNC_EN</code>: ■ If <code>MULTI_DEVICE_AND_BUS_PER_FUNC_EN</code> =0, there are eight bits of <code>cfg_pbus_num</code> ([7:0]), regardless of the number of functions. ■ If <code>MULTI_DEVICE_AND_BUS_PER_FUNC_EN</code> =1, there are eight bits of <code>cfg_pbus_num</code> for each configured function. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_pbus_dev_num[(DEVNUM_WD-1):0]	O	The device number assigned to the function. The number of bits depends the value of MULTI_DEVICE_AND_BUS_PER_FUNC_EN: ■ If MULTI_DEVICE_AND_BUS_PER_FUNC_EN =0, there are five bits of cfg_pbus_dev_num ([4:0]), regardless of the number of functions. ■ If MULTI_DEVICE_AND_BUS_PER_FUNC_EN =1, there are five bits of cfg_pbus_dev_num for each configured function. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vc_enable[(NVC-1):0]	O	Configured VC Enable. Indicates the VC enable state for each supported VC. NVC represents the number of supported VCs. <i>Note</i>VC0 is always supported and enabled. Exists: VC_ENABLE Synchronous To: None,core_clk,perbitclk,{aux_clk_g} Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vc_struc_vc_id_map[((NVC*3)-1):0]	O	Configured Structural VC to VC Identifier Map. This bus provides the mapping from the structural (physical) VC numbering scheme, that is, the order in which the VCs are declared in the configuration space, to the logical VC identifiers that are assigned by system software. To obtain the logical ID, index this bus with the structural ID. For example, ■ Bits [2:0] return the logical ID for structural VC 0 and always equals to 0. ■ Bits [5:3] return the logical ID for structural VC 1. ■ and so on. Exists: VC_ENABLE Synchronous To: None,core_clk,perbitclk,{aux_clk_g} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_vc_id_vc_struct_map[23:0]	O	Configured VC Identifier to Structural VC Map. This bus provides the mapping of the logical VC identifiers that are defined by the system, to the structural (physical) VC numbering scheme. To obtain the structural ID, index this bus with the logical ID. For example, ■ Bits [2:0] return the structural ID (position) for the VC assigned to VC 0 and always equals to 0. ■ Bits [5:3] return the structural ID (position) for the VC assigned to VC 1. ■ and so on. When a given ID is not assigned, the bus returns 0. Exists: VC_ENABLE Synchronous To: None,core_clk,perbitclk,{aux_clk_g} Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_tc_enable[7:0]	O	Indicates the enabled TCs. TCn is enabled when bit[n] is 1. Exists: VC_ENABLE Synchronous To: None,core_clk,perbitclk,{aux_clk_g} Registered: No Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_tc_struct_vc_map[23:0]	O	TC to Structural VC map. ■ Bits [2:0] return the structural VC number for the TC0 and always equals to 0. ■ Bits [5:3] return the structural VC number for the TC1. ■ and so on. Each structural VC[m] that this signal indicates is enabled by cfg_vc_enable[m]. Exists: VC_ENABLE Synchronous To: None,core_clk,perbitclk,{aux_clk_g} Registered: No Power Domain: PD_VAUX Active State: N/A Validated by: cfg_tc_struct_vc_map[3*n +: 3] is validated by cfg_tc_enable[n].

Port Name	I/O	Description
cfg_ln_enable[(NF-1):0]	O	This signal reflects the contents of the LNR_ENABLE field in the LN Control and Capabilities register in a EP device. Exists: CX_LN_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ln_vf_enable[(INT_NVF-1):0]	O	This signal reflects the contents of the LNR_ENABLE field in the LN Control and Capabilities register, when Virtual Functions are present. Exists: (CX_LN_ENABLE) && (VF_LN_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ido_req_en[(NF-1):0]	O	ID-Based Ordering Requests Enabled. Independent ordering based on Requester/Completer ID Exists: CX_IDO_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ido_cpl_en[(NF-1):0]	O	ID-Based Ordering Completions Enabled. Independent ordering based on Requester/Completer ID Exists: CX_IDO_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_no_snoop_en[(NF-1):0]	O	Contents of the "Enable No Snoop" field (PCIE_CAP_EN_NO_SNOOP) in the "Device Control and Status" register (DEVICE_CONTROL_DEVICE_STATUS) register. Exists: Always Synchronous To: {None},{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_relax_order_en[(NF-1):0]	O	Contents of the "Enable Relaxed Ordering" field (PCIE_CAP_EN_REL_ORDER) in the "Device Control and Status" register (DEVICE_CONTROL_DEVICE_STATUS) register. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ats_stu[((5*Nf)-1):0]	O	Contents of the "Smallest Translation Unit" field (ATU) in the ATS_CAPABILITIES_CTRL_REG register. Exists: CX_ATS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ats_cache_en[(NF-1):0]	O	Contents of the "Cache Enable" (ENABLE) field in the ATS_CAPABILITIES_CTRL_REG register. Exists: CX_ATS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_acs_validation_en	O	Contents of the "ACS Source Validation Enable (V)" field (ACS) in the ACS_CAPABILITIES_CTRL_REG register. When set, the component validates the Bus Number from the requester ID of upstream requests against the secondary/subordinate Bus Numbers. Exists: CX_ACS_ENABLE Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_acs_at_blocking_en	O	Contents of the "ACS Translation Blocking Enable (B)" field (ACS) in the ACS_CAPABILITIES_CTRL_REG register. When set, the component blocks all upstream memory requests whose Address Translation (AT) field is not set to the default value. Exists: CX_ACS_ENABLE Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_acs_p2p_req_redirect_en[(NF-1):0]	O	Contents of the "ACS P2P Request Redirect Enable (R)" field (ACS) in the ACS_CAPABILITIES_CTRL_REG register. It determines when the component redirects peer-to-peer requests upstream. Exists: CX_ACS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_acs_p2p_compl_redirect_en[(NF-1):0]	O	Contents of the "ACS P2P Completion Redirect Enable (C)" field (ACS) in the ACS_CAPABILITIES_CTRL_REG register. It determines when the component redirects peer-to-peer completions upstream. Exists: CX_ACS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_acs_up_forward_en	O	Contents of the "ACS Upstream Forwarding Enable (U)" field (ACS) in the ACS_CAPABILITIES_CTRL_REG register. When set, the component forwards upstream any request or completion TLPs it receives, that had been redirected upstream by a component lower in the hierarchy. Exists: CX_ACS_ENABLE Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_acs_p2p_egress_ctrl_en[(NF-1):0]	O	Contents of the "ACS P2P Egress Control Enable (E)" field (ACS) in the ACS_CAPABILITIES_CTRL_REG register. It determines when to allow, disallow, or redirect peer-to-peer requests. Exists: CX_ACS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_acs_p2p_direct_transl_en[(NF-1):0]	O	Contents of the "ACS Direct Translated P2P Enable (T)" field (ACS) in the ACS_CAPABILITIES_CTRL_REG register. It overrides the ACS P2P Request Redirect and ACS P2P Egress Control mechanisms with peer-to-peer Memory Requests whose Address Translation (AT) field indicates a translated address. Exists: CX_ACS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_acs_egress_ctrl_vec[((ACS_CTRL_VEC_WD*NF)-1):0]	O	Contents of the "ACS Egress Control" Register. The width is defined by ACS_CTRL_VEC_WD, per-function. ACS_CTRL_VEC_WD is the largest egress control vector size of all functions. Exists: (CX_ACS_ENABLE) && ([<functionof> CX_ACS_P2P_EGRESS_CTRL]) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
acs_vf_p2p_req_redirect_en[(INT_NVF-1):0]	O	Contents of the "ACS P2P Request Redirect Enable (R)" field (ACS) in the VF_ACS_CAPABILITIES_CTRL_REG register. It determines when the component redirects peer-to-peer requests upstream. Exists: (CX_ACS_ENABLE) && (VF_ACS_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
acs_vf_p2p_compl_redirect_en[(INT_NVF-1):0]	O	Contents of the "ACS P2P Completion Redirect Enable (C)" field (ACS) in the VF_ACS_CAPABILITIES_CTRL_REG register. It determines when the component redirects peer-to-peer completions upstream. Exists: (CX_ACS_ENABLE) && (VF_ACS_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
acs_vf_p2p_egress_ctrl_en[(INT_NVF-1):0]	O	Contents of the "ACS P2P Egress Control Enable (E)" field (ACS) in the VF_ACS_CAPABILITIES_CTRL_REG register. It determines when to allow, disallow, or redirect peer-to-peer Requests. Exists: (CX_ACS_ENABLE) && (VF_ACS_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
acs_vf_p2p_direct_transl_en[(INT_NVF-1):0]	O	Contents of the "ACS Direct Translated P2P Enable (T)" field (ACS) in the VF_ACS_CAPABILITIES_CTRL_REG register. It overrides the ACS P2P Request Redirect and ACS P2P Egress Control mechanisms with peer-to-peer memory requests whose Address Translation (AT) field indicates a translated address. Exists: (CX_ACS_ENABLE) && (VF_ACS_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
acs_vf_egress_ctrl_vec[((ACS_CTRL_VEC_C_WD*INT_NVF)-1):0]	O	Contents of the "VF ACS Egress Control Vector" Register. The size ACS_CTRL_VEC_WD is the largest egress control vector size of all functions. Exists: (CX_ACS_ENABLE) && (VF_ACS_ENABLE) && ([<functionof> CX_ACS_P2P_EGRESS_CTRL]) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_acs_func_grp_en[(NF-1):0]	O	ACS Function Group Enable, per-PF. Exists: CX_ACS_FUNC_GRP Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ari_func_grp[((3*Nf)-1):0]	O	ARI Function Group[2:0], per-PF. Exists: CX_ACS_FUNC_GRP Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vf_ari_func_grp[((3*INT_NVf)-1):0]	O	Contents of the "VF ARI Function Group [2:0]" settings, per VF. Exists: VF_ACS_FUNC_GRP Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
prs_res_failure[(NF-1):0]	I	Pulse per function(PF) to indicate that the function (PF or associated VF) has received a PRG Response Failure. Sets the PRS_RESP_FAILURE field in the Page Request Status (PRS_CONTROL_STATUS_REG) register. Exists: CX_PRS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
prs_uprgi[(NF-1):0]	I	Pulse per function(PF) to indicate that the function (PF or associated VF) has received a response with Unexpected Page Request Group Index. Sets the PRS_UPRGI field in the Page Request Status (PRS_CONTROL_STATUS_REG) register. Exists: CX_PRS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
prs_stopped[(NF-1):0]	I	Pulse per function(PF) to indicate that the function (PF or associated VF) has completed all previously issued Page Requests and that it has stopped requests for additional pages. Only valid when the PRS enable bit is clear. Sets the PRS_STOPPED field in the Page Request Status Register (PRS_CONTROL_STATUS_REG) register. Exists: CX_PRS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_prs_enable[(NF-1):0]	O	Indicates that the Page Request Interface is allowed to make page requests. This output pin reflects the value of the PRS_ENABLE field in the PRS_CONTROL_STATUS_REG register. Exists: CX_PRS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_prs_reset[(NF-1):0]	O	Pulse to indicate to the application that the associated page request credit counter and pending request state must be cleared. This output pin reflects the value of the PRS_RESET field in the PRS_CONTROL_STATUS_REG register. Exists: CX_PRS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_prs_stopped[(NF-1):0]	O	This output pin reflects the value of the PRS_STOPPED field in the Page Request Status Register (PRS_CONTROL_STATUS_REG) register. Exists: CX_PRS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_prs_uprgi[(NF-1):0]	O	This output pin reflects the value of the PRS_UPRGI field in the Page Request Status (PRS_CONTROL_STATUS_REG) register. Exists: CX_PRS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_prs_response_failure[(NF-1):0]	O	This output pin reflects the value of the PRS_RESP_FAILURE field in the Page Request Status (PRS_CONTROL_STATUS_REG) register. Exists: CX_PRS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_prs_outstanding_allocation[((32*NF)-1):0]	O	Signal to indicate the number of outstanding page request messages the associated Page Request Interface is allowed to issue. This output pin reflects the value of the PRS_OUTSTANDING_ALLOCATION field in the PRS_REQ_ALLOCATION_REG register. Exists: CX_PRS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cdm_reg_chk_test_en	I	Signal to enter test mode in the Register Checking Logic. u_cdm_b is not be written to when this signal is high. Allows a value to be written to u_cdm to compare against the previous value in u_cdm_b Exists: CX_CDM_REG_CHK_EN Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cdm_reg_chk_logic_err	O	Signal to indicate that there is an Error in the Register Checking Logic. Exists: CX_CDM_REG_CHK_EN Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cdm_reg_chk_cmp_err	O	Signal to indicate that the register values read from both CDM's do not match. Exists: CX_CDM_REG_CHK_EN Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cdm_reg_chk_cmplt	O	Signal to indicate that a Register Checking Sequence has Completed. Exists: CX_CDM_REG_CHK_EN Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
rbar_ctrl_update[(NF-1):0]	O	Indicates that a resizable BAR control register has been updated: 1 bit per Physical function. Exists: CX_RBARS_INCLUDED Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_rbar_size[((NF*(6*6))-1):0]	O	BAR size field from each of the resizable BAR control registers, per function. For BARs that are not resizable the corresponding bits in cfg_rbar_size are set to 0. Exists: CX_RBARS_INCLUDED Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
vf_rbar_ctrl_update[(NF-1):0]	O	Indicates that a resizable VF BAR control register has been updated: 1 bit per Physical function. Exists: (CX_RBARS_INCLUDED) && (CX_VF_RBARS_INCLUDED) Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_vf_rbar_size[((NF*(6*6))-1):0]	O	BAR size field from each of the resizable VF BAR control registers, per function. For BARs that are not resizable, the corresponding bits in cfg_vf_rbar_size are set to 0. Exists: (CX_RBARS_INCLUDED) && (CX_VF_RBARS_INCLUDED) Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_hp_slot_ctrl_access[(NF-1):0]	O	Slot Control Accessed. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_dll_state_chged_en[(NF-1):0]	O	Slot Control DLL State Change Enable. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cmd_cpled_int_en[(NF-1):0]	O	Slot Control Command Completed Interrupt Enable. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_hp_int_en[(NF-1):0]	O	Slot Control Hot Plug Interrupt Enable. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_pre_det_chged_en[(NF-1):0]	O	Slot Control Presence Detect Changed Enable. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_mrl_sensor_chged_en[(NF-1):0]	O	Slot Control MRL Sensor Changed Enable. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_pwr_fault_det_en[(NF-1):0]	O	Slot Control Power Fault Detect Enable. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_attn_button_pressed_en[(NF-1):0]	O	Slot Control Attention Button Pressed Enable. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
if_timeout_status	O	Interface timeout status Exists: CX_INTERFACE_TIMER_EN Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.73 SII: CXL Configuration Signals

`cxl_2_0_mbar0_offset` - 

Table 2-73 SII: CXL Configuration Signals

Port Name	I/O	Description
<code>cxl_2_0_mbar0_offset[63:0]</code>	I	CXL 2.0 MEMBAR0 offset in Function 0, BAR0 Exists: (CX_CXL_ENABLE) && (CX_CXL_2_0_ENABLE) Synchronous To: <code>aux_clk_g</code> Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.74 SII: CXL Configuration/Control/Status Signals

- cfg_cxl_dev_mem_enable
- cfg_cxl_dev_range1_mem_size_high
- cfg_cxl_dev_range1_mem_info_valid
- cfg_cxl_dev_range1_mem_active
- cfg_cxl_dev_range1_desired_interleave
- cfg_cxl_dev_range1_mem_size_low
- cfg_cxl_dev_range1_mem_base_high
- cfg_cxl_dev_range1_mem_base_low
- cfg_cxl_dev_range2_mem_size_high
- cfg_cxl_dev_range2_mem_info_valid
- cfg_cxl_dev_range2_mem_active
- cfg_cxl_dev_range2_desired_interleave
- cfg_cxl_dev_range2_mem_size_low
- cfg_cxl_dev_range2_mem_base_high
- cfg_cxl_dev_range2_mem_base_low
- cfg_cxl_dev_cache_enable
- cfg_cxl_dev_cache_sf_coverage
- cfg_cxl_dev_cache_sf_granularity
- cfg_cxl_dev_cache_clean_eviction
- cfg_cxl_dev_initiate_cxl_rst
- cfg_cxl_dev_cxl_rst_mem_clr_enable

Table 2-74 SII: CXL Configuration/Control/Status Signals

Port Name	I/O	Description
cfg_cxl_dev_mem_enable	O	Mem_Enable field of the Flex Bus DVSEC control register Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_range1_mem_size_high[31:0]	O	Corresponds to bits 63:32 of Flex Bus Range 1 memory size Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

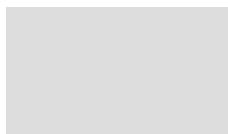
Port Name	I/O	Description
cfg_cxl_dev_range1_mem_info_valid	O	Indicates that the Flex Bus Range 1 Size high and Size Low registers are valid Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_range1_mem_active	O	Indicates that the Flex Bus Range 1 memory is fully initialized and available for software Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_range1_desired_interleave[4:0]	O	Represents the Flex Bus Range 1 memory interleaving desired by the device Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_range1_mem_size_low[3:0]	O	Corresponds to bits 31:28 of Flex Bus Range 1 memory size Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_range1_mem_base_high[31:0]	O	Corresponds to bits 63:32 of Flex Bus Range 1 base in the host address space Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_cxl_dev_range1_mem_base_low[3:0]	O	Corresponds to bits 31:28 of Flex Bus Range 1 base in the host address space Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_range2_mem_size_high[31:0]	O	Corresponds to bits 63:32 of Flex Bus Range 2 memory size Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_range2_mem_info_valid	O	Indicates that the Flex Bus Range 2 Size high and Size Low registers are valid Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_range2_mem_active	O	Indicates that the Flex Bus Range 2 memory is fully initialized and available for software Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_range2_desired_interleave[4:0]	O	Represents the Flex Bus Range 2 memory interleaving desired by the device Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_cxl_dev_range2_mem_size_low[3:0]	O	Corresponds to bits 31:28 of Flex Bus Range 2 memory size Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_range2_mem_base_high[31:0]	O	Corresponds to bits 63:32 of Flex Bus Range 2 base in the host address space Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_range2_mem_base_low[3:0]	O	Corresponds to bits 31:28 of Flex Bus Range 2 base in the host address space Exists: (CX_CXL_ENABLE) && (CX_CXL_MEM_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_cache_enable	O	Cache_Enable field of the Flex Bus DVSEC control register Exists: (CX_CXL_ENABLE) && (CX_CXL_CACHE_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_cache_sf_coverage[4:0]	O	Cache_SF_Coverage field of the Flex Bus DVSEC control register Exists: (CX_CXL_ENABLE) && (CX_CXL_CACHE_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_cxl_dev_cache_sf_granularity[2:0]	O	Cache_SF_Granularity field of the Flex Bus DVSEC control register Exists: (CX_CXL_ENABLE) && (CX_CXL_CACHE_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_cache_clean_eviction	O	Cache_Clean_Eviction field of the Flex Bus DVSEC control register Exists: (CX_CXL_ENABLE) && (CX_CXL_CACHE_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_initiate_cxl_rst	O	Initiate CXL_Reset field of the Flex Bus DVSEC Control2 register Exists: (CX_CXL_ENABLE) && (CX_CXL_RESET_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_cxl_rst_mem_clr_enable	O	CXL_Reset Mem Clear Enable field of the Flex Bus DVSEC Control2 register Exists: (CX_CXL_ENABLE) && (CX_CXL_RESET_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.75 SII: CXL CTS Configuration/Control/Status Signals

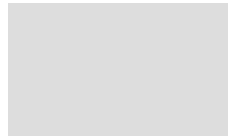


- [cfg_cxl_dev_test_cap_lock](#)
- [cfg_cxl_dev_test_cap_base_low](#)
- [cfg_cxl_dev_test_cap_base_high](#)

Table 2-75 SII: CXL CTS Configuration/Control/Status Signals

Port Name	I/O	Description
cfg_cxl_dev_test_cap_lock	O	TestLock field of the CXL Device Test Cap DVSEC. Software writes 1'b1 to lock the relevant test configuration registers Exists: (CX_CXL_ENABLE) && (CX_CXL_DEVICE_CTS_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_test_cap_base_low[27:0]	O	Bits [31:4] of the base address where the test configuration registers exist Exists: (CX_CXL_ENABLE) && (CX_CXL_DEVICE_CTS_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_test_cap_base_high[31:0]	O	Bits [63:32] of the base address where the test configuration registers exist Exists: (CX_CXL_ENABLE) && (CX_CXL_DEVICE_CTS_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.76 SII: CXL Error Status Signals



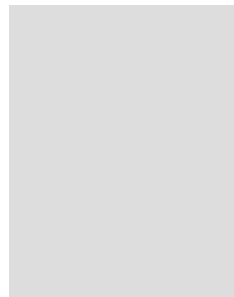
- [cfg_cxl_mbar0_corr_err](#)
 - [cfg_cxl_mbar0_uncorr_err](#)
 - [cfg_cxl_pl_reg_fatal_err](#)

Table 2-76 SII: CXL Error Status Signals

Port Name	I/O	Description
cfg_cxl_mbar0_corr_err[4:0]	O	MEMBAR0_RAS_CORR_ERROR_STATUS_REG_OFF register bits. ■ [0] CRC_Threshold: CRC Threshold Hit ■ [1] Retry_Threshold: Retry Threshold Hit ■ [2] Cache_Poison_Received: Received Poison from the peer on CXL.cache ■ [3] Mem_Poison_Received: Received Poison from the peer on CXL.mem ■ [4] Physical_Layer_Error: Received error indication from Physical Layer Exists: (CX_CXL_ENABLE) && (CX_CXL_ENABLE) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_mbar0_uncorr_err[3:0]	O	MEMBAR0_RAS_UNCOR_ERROR_STATUS_REG_OFF register bits. ■ [0] REINIT_Threshold: REINIT Threshold Hit ■ [1] Rsvd_Encoding_Violation : Received unrecognized encoding ■ [2] Poison_Received : Received Poison from the peer ■ [3] Buffer_Overflow :Indicates Receiver or Transmitter Overflow Exists: (CX_CXL_ENABLE) && (CX_CXL_ENABLE) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_cxl_pl_reg_fatal_err[20:0]	O	CXL_FATAL_ERR_CR_REG_OFF register bits. ■ [0] For EP, or RC, or DM configuration, Retry_Ack_Viral: Retry.Ack control flit is received with Viral status. This is a read only bit. ■ [1] RAS_Error: RAS uncorrectable error detected. ■ [2] LL_Static_Bit_Error: Static bit error is detected in the received LL Control Flit. ■ [3] LRSM_Abort: LRSM FSM reached Abort state. ■ [4] LRSM_Ack_Error: Unexpected Retry.Ack flit is received. Bits[9-5] LL_TX_FlowControl_Error: CXL Transmitter Flow Control Error: ■ [5] Cache Request Credit Overflow ■ [6] Cache Response Credit Overflow ■ [7] Cache Data Credit Overflow ■ [8] Mem NDR/RequestOnly Credit Overflow ■ [9] Mem DRC/RwD Credit Overflow ■ [10] LL_Init_Param_Error: Non-Retry flits received before LLCTRL-INIT.Param flit. ■ [11] LL_Multi_Init_Param: Receipt of an LLCTRL-INIT.Param flit after an LLCTRL-INIT.Param flit has already been received. ■ [12] Retry_Eseq_Overflow: Retry Buffer Eseq Counter Overflow. ■ [13] Retry_CRC_Error: Retry Buffer CRC error. Bits[18-14] LL_RX_FlowControl_Error: CXL Receiver Flow Control Error: ■ [14] Cache Request Credit Return Overflow ■ [15] Cache Response Credit Return Overflow ■ [16] Cache Data Credit Return Overflow ■ [17] Mem NDR/RequestOnly Credit Return Overflow ■ [18] Mem DRC/RwD Credit Return Overflow ■ [19] ULL_CXS_EndError: ULL Transmits CXS Flit with ENDERROR bit set in the Control Bus to trigger Viral Mode. ■ [20] CXS_TX_EndError: If CXS Controller transmits CXS Flit with ENDERROR bit set in the Control Bus then trigger Viral Mode. ■ [21] For EP, or RC, or DM configuration, App_Viral_Request: Set this bit if the application requests to trigger Viral Mode in CXL controller. This is a read only bit. Exists: (CX_CXL_ENABLE) && (CX_CXL_ENABLE) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.77 SII: Debug Signals



- rdlh_link_up
- brdg_slv_xfer_pending
- brdg_dbi_xfer_pending
- edma_xfer_pending
- radm_xfer_pending
- radm_q_not_empty
- radm_qoverflow
- cxpl_debug_info
- cxpl_debug_info_ei

Table 2-77 SII: Debug Signals

Port Name	I/O	Description
rdlh_link_up	O	Data link layer up/down indicator: This status from the flow control initialization state machine indicates that flow control has been initiated and the Data link layer is ready to transmit and receive packets. For multi-VC designs, this signal indicates status for VC0 only. ■ 1: Link is up ■ 0: Link is down Exists: Always Synchronous To: aux_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
brdg_slv_xfer_pending	O	AXI Slave non-DBI transfer pending status. Indicates AXI non-DBI Slave Read or Write transfers are pending, that is, AXI Slave transfers are awaiting a response from the controller. For debugging purposes. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && ((CX_RADMQ_MODE==2)) Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
brdg_dbi_xfer_pending	O	AXI Slave DBI transfer pending status. Indicates AXI DBI Slave Read or Write transfers are pending, that is, AXI Slave transfers are awaiting a response from the controller. For debugging purposes. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && ((CX_RADMQ_MODE==2)) Synchronous To: None,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
edma_xfer_pending	O	eDMA transfer pending status. Indicates eDMA Write or Read Channel transfers are pending, that is, DMA Write or Read Channels have not finished transferring data. For debugging purposes. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && ((CX_RADMQ_MODE==2)) && (CC_DMA_ENABLE) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_xfer_pending	O	Receive request pending status. Indicates Receive TLP requests are pending, that is, requests sent to the RTRGT1 or RTRGT0 interfaces are awaiting a response from your application. For debugging purposes. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && ((CX_RADMQ_MODE==2)) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_q_not_empty[(NVC-1):0]	O	Level indicating that the receive queues contain TLP header/data. There is a 1 bit indication for each virtual channel. Exists: Always Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High indicates data in queues Validated by: Not validated by another signal

Port Name	I/O	Description
radm_qoverflow[(NVC-1):0]	O	Pulse indicating that one or more of the P/NP/CPL receive queues have overflowed. There is a 1-bit indication for each configured virtual channel. You can connect this output to your internal error reporting mechanism. Exists: Always Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High pulse indicates overflow Validated by: Not validated by another signal

Port Name	I/O	Description
cxpl_debug_info[63:0]	O	State of selected internal signals, for debugging purposes only: ■ [63]: smlh_scrambler_disable: Scrambling disabled for the link ■ [62]: smlh_link_disable: TSSM in DISABLE state. Link inoperable ■ [61]: smlh_link_in_training: LTSSM performing link training ■ [60]: smlh_ltssm_in_pollconfig: LTSSM is in Polling.Configuration state ■ [59]: smlh_training_rst_n: LTSSM-negotiated link reset ■ [58:55]: 0000b: Reserved ■ [54]: mac_phy_txdetectrx_loopback: PIPE receiver detect/loopback request ■ [53]: mac_phy_txeleidle[0]: PIPE transmit electrical idle request ■ [52]: mac_phy_txcompliance[0]: PIPE transmit compliance request ■ [51]: app_init_rst: Application request to initiate training reset ■ [50:48]: 000b: Reserved ■ [47:40]: rmlh_ts_link_num: Link number advertised/confirmed by link partner ■ [39:38]: 00b: Reserved ■ [37]: xmtbyte_skip_sent: A skip ordered set has been transmitted ■ [36]: smlh_link_up: LTSSM reports PHY link up or LTSSM is in Loop-back.Active for Loopback Master ■ [35]: smlh_inskip_rcv: Receiver reports skip reception ■ [34]: smlh_ts1_rcvd: TS1 training sequence received (pulse) ■ [33]: smlh_ts2_rcvd: TS2 training sequence received (pulse) ■ [32]: smlh_rcvd_lane_rev: Receiver detected lane reversal ■ [31:28]: smlh_ts_link_ctrl: Link control bits advertised by link partner ■ [27]: smlh_ts_lane_num_is_k237: Currently receiving k237 (PAD) in place of lane number ■ [26]: smlh_ts_link_num_is_k237: Currently receiving k237 (PAD) in place of link number ■ [25]: rplh_rcvd_idle[0]: Receiver is receiving logical idle ■ [24]: rplh_rcvd_idle[1]: 2n symbol is also idle (16bit PHY interface only) ■ [23:8]: mac_phy_txdata: PIPE Transmit data ■ [7:6]: mac_phy_txdatak: PIPE transmit K indication ■ [5:0]: smlh_ltssm_state: LTSSM current state. Encoding is same as the dedicated smlh_ltssm_state output. Exists: Always Synchronous To: None,aux_clk,aux_clk_g,core_pl_clk,perbitclk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cxpl_debug_info_ei[15:0]	O	State of selected internal signals in relation to electrical idle (EI) at the receiver. The encoding of the bits is as follows: <i>Group 1</i> (pulse) - information about received ordered sets: ■ [0]: EIOS detected <i>Group 2</i> (level) - LTSSM is in one of the states that depend rxelecidle =0: ■ [1]: L1 ■ [2]: L2 ■ [3]: RxL0s ■ [4]: Disabled ■ [5]: Detect.Quiet ■ [6]: Polling.Active ■ [7]: Polling.Compliance <i>Group 3</i> (level) - LTSSM is in one of the states that depend rxelecidle =1: ■ [8]: LTSSM is in a transitory state prior to L1 or L2 ■ [9]: LTSSM is in a transitory state prior to Disabled ■ [10]: LTSSM is in Loopback.Active as a Slave at Gen1 ■ [11]: LTSSM is in Polling.Active <i>Group 4</i> (pulse) - LTSSM state transitions with EI inferred: ■ [12]: LTSSM enters Recovery from L0 with EI inferred, first row in base spec Table 4-11 ■ [13]: LTSSM enters Recovery.Speed from Recovery.RcvrCfg with EI inferred, second row in Table 4-11 of <i>PCI Express Specification</i> ■ [14]: EI inferred while LTSSM in Recovery.Speed, third/fourth rows in base spec Table 4-11 ■ [15]: EI inferred while LTSSM in Loopback.Active as a slave, fifth row in base spec Table 4-11 Exists: Always Synchronous To: core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.78 SII: Diagnostic Control Signals

diag_ctrl_bus -  - diag_status_bus

Table 2-78 SII: Diagnostic Control Signals

Port Name	I/O	Description
diag_ctrl_bus[2:0]	I	Diagnostic Control Bus ■ x01: Insert LCRC error by inverting the LSB of LCRC ■ x10: Insert ECRC error by inverting the LSB of ECRC □ The rising edge of these two signals ([1:0]) enable the controller to assert an LCRC or ECRC to the packet that it currently being transferred. □ When LCRC or ECRC error packets are transmitted by the controller, the controller asserts diag_status_bus[lrc_err_asserted] or diag_status_bus[ecrc_err_asserted] to report that the requested action has been completed. This handshake between control and status allows your application to control a specific packet being injected with an CRC or ECRC error. □ The LCRC and ECRC errors are generated by simply inverting the last bit of the LCRC or ECRC value. ■ 1xx: Select Fast Link Mode. □ Sets all internal LTSSM millisecond timers to Fast Mode for speeding up simulation purposes. Forces the LTSSM training (link initialization) to use shorter timeouts and to link up faster. □ The default scaling factor is 1024 for all internal timers. The default scaling factor can be changed using the DEFAULT_FAST_LINK_SCALING_FACTOR parameter or through the FAST_LINK_SCALING_FACTOR field in the TIMER_CTRL_MAX_FUNC_NUM_OFF register. □ Fast Link Mode can also be activated by setting the FAST_LINK_MODE field in the PORT_LINK_CTRL_OFF register. □ For more information, see the "Fast Link Simulation Mode" section in the "Integrating the Controller with the PHY or Application RTL or Verification IP" chapter of the <i>User Guide</i>. Exists: DIAGNOSTIC_ENABLE Synchronous To: aux_clk_g,core_pl_clk,perbitclk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
diag_status_bus[CX_DIAG_STATUS_BUS_WD-1:0]	O	Diagnostic Status Bus. Contains all of the important status signals from each controller module. The following list has the individual diagnostic sub-buses and their corresponding signals in the order from least significant bit to most significant bit (with respect to the entire diagnostic bus group). Therefore, diag_status_bus[0] is rtfngen_incr_amt[0]. The parameter macros are replaced by actual parameter values in the configured IP-XACT and coreConsultant report (DocBook XML and HTML) files. Data Link Layer Diagnostic Signals ■ rtfngen_incr_amt[9*RX_TLP-1:0] = Payload flow control credits consumed. NOTE: hdr credits consumed is always 1 ■ rtfngen_incr_enable[RX_TLP-1:0] = Header flow control credit consumed ■ rtfngen_fctype[2*RX_TLP-1:0] = Flow control type consumed (P=0, NP=1, CPL=2) ■ rtcheck_rtfngen_vc[3*RX_TLP-1:0] = Virtual channel of received TLP ■ xdlh_xtlh_halt = Layer2 is not accepting data to transmit this cycle ■ xtlh_xdlh_data[TRGT_DATA_PROT_WD+(NW*32)-1:0] = Transmit data at the interface between Layer3 and Layer2. If RASDP is enabled, includes the ECC or parity protection code bits. ■ xtlh_xdlh_badeot[XTLH_XDLH_CTL_WD-1:0] = Nullify this transmit TLP (invert CRC, append EDB) ■ xtlh_xdlh_eot[XTLH_XDLH_CTL_WD-1:0] = Transmit End of TLP this cycle ■ xtlh_xdlh_sot[XTLH_XDLH_CTL_WD-1:0] = Transmit Start of TLP this cycle ■ ecrc_err_asserted = End-to-end CRC corrupted for this packet ■ lcrc_err_asserted = Link CRC corrupted for this packet ■ xplh_xdlh_halt = PHY Layer not accepting data this cycle ■ xdlh_xplh_data[32*NW-1:0] = Transmit packet payload (completely framed) ■ xdlh_xplh_sdp[NW-1:0] = Transmit Start of DLLP (per dword) ■ xdlh_xplh_stp[NW-1:0] = Transmit Start of TLP (per dword) ■ xdlh_xplh_eot[NW-1:0] = Transmit end of TLP/DLLP (per dword) ■ rdln_xdlh_req_acknack_seqnum[11:0] = Sequence Number for ACK/NAK DLLP ■ rdln_xdlh_req2send_nack = DataLink Layer request to send NAK ■ rdln_xdlh_req2send_ack_due2dup = Request to send ACK due to duplicate TLP ■ rdln_xdlh_req2send_ack = DataLink Layer request to send ACK

Port Name	I/O	Description
diag_status_bus[CX_DIAG_STATUS_BUS_WD-1:0]...(cont.)	O.	■ rdlh_xdlh_rcvd_acknack_seqnum[11:0] = Sequence number corresponding to NAK/ACK ■ rdlh_xdlh_rcvd_ack = DataLink Layer received ACK DLLP ■ rdlh_xdlh_rcvd_nack = DataLink Layer received NAK DLLP ■ cfg_link_retrain = Software programmed link retrain request ■ rtlh_req_link_retrain = Receive watchdog timer expired, retrain link ■ xdlh_smlh_start_link_retrain = vMax retries attempted, request to retrain link ■ rdlh_rtlh_tlp_dv = Receive data interface from DLL to Transaction Layer valid this cycle (rdlh_rtlh_*) ■ rdlh_rtlh_tlp_eot[NW-1:0] = End of TLP (per dword) ■ rdlh_rtlh_tlp_sot[NW-1:0] = Start of TLP (per dword) MAC Layer Diagnostic Signals ■ rplh_rdlh_pkt_data[32*NW-1:0] = Received packet payload ■ rplh_rdlh_pkt_err[NW-1:0] = Physical Error detected (per dword) ■ rplh_rdlh_pkt_dv = Receive data interface from PHY to DLL valid this cycle (rplh_rdlh_*) ■ rplh_rdlh_pkt_edb[NW-1:0] = Packet terminated with EDB (per dword) ■ rplh_rdlh_pkt_end[NW-1:0] = DLLP/TLP ending (per dword) ■ rplh_rdlh_tlp_start[NW-1:0] = DLLP starting (per dword) ■ rplh_rdlh_dllp_start[NW-1:0] = MAC layer detected runt STP (per dword) ■ rplh_rdlh_nak[NW-1:0] = MAC layer detected runt STP (per dword) ■ rmlh_lanes_rcving[NL-1:0] = lanes active in link training ■ rmlh_rcvd_idle_set = Received EIDLE ordered set, any active lane ■ rplh_rcvd_idle1 = Logical Idle seen for 1+ symbols on all active lanes ■ rplh_rcvd_idle0 = Logical Idle seen for 8+ symbols on all active lanes ■ smlh_rcvd_lane_rev = Receive logic detected logical lane reversal ■ rmlh_ts_link_num_is_k237 = Received Link number (lane 0) is PAD (k237) ■ rmlh_deskew_alignment_err = Deskew logic overflow. Unable to align lanes [level] ■ rmlh_ts_lane_num_is_k237 = Received lane number (lane 0) is PAD (k237) ■ rmlh_ts2_rcvd = At least one active lane received TS2 this cycle ■ rmlh_ts1_rcvd = At least one active lane received TS1 this cycle

Port Name	I/O	Description
diag_status_bus[CX_DIAG_STATUS_BUS_WD-1:0]...(cont..)	O..	■ rmlh_ts_rcv_err = Assert when the received data is not an expected TS symbol or SKP symbol or EIEOS symbol on Lane 0 ■ rmlh_inskip_rcv = Skip character is received this cycle PM Diagnostic Bus ■ xadm_no_fc_credit[NVC-1:0] = No credits of any type are available (per VC). ■ xadm_tlp_pending = No TLP transmit requests currently pending (from internal or external clients). ■ xadm_had_enough_credit[NVC-1:0] = The controller has enough transmit credits of each type (per VC) to meet the PM entry criteria. ■ xdlh_notexpecting_ack = All transmitted TLPs have been acknowledged by the link partner. ■ xdlh_xmt_pme_ack = DataLink layer just transmitted a PME_TO_ACK msg. ■ xdlh_nodllp_pending = There are no pending or in-progress packets going to the PHY. ■ l1sub_state[2:0] = L1 FSM sub-state (Tied to 3'b000 when !CX_L1_SUBSTATES_ENABLE) □ 3'h0: idle state. □ 3'h1: wait for aux_clk_active. □ 3'h2: wait for pclkack. □ 3'h3: wait for clkreq. □ 3'h4: check clkreq_in_n is de-asserted for t_power_off time (only for L1.2, reduces to one cycle for L1.1). □ 3'h5: L1 substate, turn off txcommonmode circuits (L1.2 only) and rx electrical idle detection circuits. □ 3'h6: locally/remotely initiated exit, assert pclkreq, wait for pclkack. □ 3'h7: wait for pclkack when aborting an attempt to enter state 5. RADM Diagnostic Busses. When RX_TLP =2 (256-bit configuration), there are two RADM diagnostic busses because it is possible to receive two TLPs in one clock cycle. RADM Diagnostic Bus 0 <i>The signals from unexpected_cpl_err to flt_q_header_destination are only valid when form_filt_hv/form_filt_eot are asserted.</i> ■ unexpected_cpl_err = Received a completion that was unexpected ■ cpl_ca_err = Received completion with CA status ■ cpl_ur_err = Received completion with UR status ■ flt_q_cpl_last = Final completion for the transaction ■ flt_q_cpl_abort = Current completion is being aborted (trashed) ■ cpl_mlf_err = Malformed completion

Port Name	I/O	Description
diag_status_bus[CX_DIAG_STATUS_BUS_WD-1:0]...(cont...)	O...	■ flt_q_header_cpl_status[2:0] = Indicates the filtering result for the current RX request or the completion status for the current RX received completion. □ 0: =Successful Completion (SC) □ 1: Unsupported request (UR) □ 2: Configuration Retry (CRS) □ 4: Completer abort (CA) ■ flt_q_header_destination[1:0] = Destination interface. □ 0: Drop □ 1: TRGT0 □ 2: TRGT1 □ 3: Reserved ■ form_filt_ecrc_err = This TLP had an ECRC error ■ form_filt_malform_tlp_err = This TLP is malformed ■ form_filt_dllp_err = This TLP has a DataLink Layer Error (for example, LCRC) ■ form_filt_eot = End of TLP received this cycle ■ form_filt_dwen[NW-1:0] = Dword enables ■ form_filt_data[32*NW-1:0] = Packet data from receive Transaction Layer ■ form_filt_dv = Packet is in payload stage ■ form_filt_hdr[127:0] = Header data (size of stored header is configurable) ■ form_filt_hv = Information to the receive packet filter block is valid RADM Diagnostic Bus 1 <i>The signals from unexpected_cpl_err to flt_q_header_destination are only valid when form_filt_hv/form_filt_eot are asserted.</i> ■ unexpected_cpl_err = Received a completion that was unexpected ■ cpl_ca_err = Received completion with CA status ■ cpl_ur_err = Received completion with UR status ■ flt_q_cpl_last = Final completion for the transaction ■ flt_q_cpl_abort = Current completion is being aborted (trashed) ■ cpl_mlf_err = Malformed completion ■ flt_q_header_cpl_status[2:0] = Indicates the filtering result for the current RX request or the completion status for the current RX received completion. □ 0: =Successful Completion (SC) □ 1: Unsupported request (UR) □ 2: Configuration Retry (CRS) □ 4: Completer abort (CA) ■ flt_q_header_destination[1:0] = Destination interface. □ 0: Drop □ 1: TRGT0 □ 2: TRGT1 □ 3: Reserved ■ form_filt_ecrc_err = This TLP had an ECRC error ■ form_filt_malform_tlp_err = This TLP is malformed

Port Name	I/O	Description
diag_status_bus[CX_DIAG_STATUS_BUS_WD-1:0]...(cont....)	O...	■ form_filt_dllp_err = This TLP has a DataLink Layer Error (for example, LCRC) ■ form_filt_eot = End of TLP received this cycle ■ form_filt_dwen[NW-1:0] = Dword enables ■ form_filt_data[32*NW-1:0] = Packet data from receive Transaction Layer ■ form_filt_dv = Packet is in payload stage ■ form_filt_hdr[127:0] = Header data (size of stored header is configurable) ■ form_filt_hv = Information to the receive packet filter block is valid RADM Diagnostic Dual Bus Selector. ■ form_filt_formation = Indicates which TLP was received first by the controller for 256-bit configurations. □ 1: TLP #0 received first, see "RADM Diagnostic Bus 1" □ 0: TLP #1 received first, see "RADM Diagnostic Bus 0" XADM Diagnostic Bus ■ active_grant[NCL+2-1:0] = Currently granted client (1-hot) ■ grant_ack[NCL+2-1:0] = This client has completed his request (1-hot) ■ fc_cds_pass[(NCL+2)*NVC-1:0] = Credit check passed for each request ■ arb_reqs[NCL+2-1:0] = Transmit requests from each client CDM Diagnostic Bus ■ cfg_pbus_num[BUSNUM_WD-1:0] = Bus number; 8 bits per function when MULTI_DEVICE_AND_BUS_PER_FUNC_EN=1, else 8 bits in total. ■ cfg_pbus_dev_num[DEVNUM_WD-1:0] = Device number; 5 bits per function when MULTI_DEVICE_AND_BUS_PER_FUNC_EN=1, else 5 bits in total. ■ xdlh_replay_timeout_err = Retry timer expired [pulse] ■ xdlh_replay_num_rlover_err = Max retries exceeded [pulse] ■ rdllh_bad_dllp_err = Received DLLP with DataLink Layer error [pulse] ■ rdllh_bad_tlp_err = Received TLP with DataLink Layer error [pulse] ■ rdllh_prot_err = DLLP protocol error (out of sequence DLLP) [pulse] ■ rtlh_fc_prot_err = Flow control protocol violation (watchdog timer) [pulse] ■ rmlh_rcvd_err = Received PHY error this cycle [pulse] ■ int_xadm_fc_prot_err = Flow control update protocol violation (opt. checks) ■ radm_cpl_timeout = An request failed to complete in the allotted time ■ radm_qoverflow[NVC-1:0] = Receive queue overflow. Normally happens only when flow control advertisements are ignored

Port Name	I/O	Description
diag_status_bus[CX_DIAG_STATUS_BUS_WD-1:0]...(cont.....)	O... ..	■ radm_unexp_cpl_err[NF-1:0] = Received an unexpected completion [pulse] ■ radm_rcvd_cpl_ur[NF-1:0] = Received a completion with UR status ■ radm_rcvd_cpl_ca[NF-1:0] = Received a completion with CA status ■ radm_rcvd_req_ca[NF-1:0] = Completer aborted request ■ radm_rcvd_req_ur[NF-1:0] = Received a request which device does not support ■ radm_ecrc_err[NF-1:0] = Received a TLP with ECRC error ■ radm_mlf_tlp_err[NF-1:0] = Received a malformed TLP [pulse] ■ radm_rcvd_cpl_poisoned[NF-1:0] = Received a completion with poisoned payload ■ radm_rcvd_wreq_poisoned[NF-1:0] = Received a write with poisoned payload ■ cdm_lbc_ack[NF-1:0] = Local bus Acknowledge ■ lbc_cdm_wr[3:0] = Local bus Write enable (per byte) ■ lbc_cdm_cs[NF-1:0] = Local bus chip select ■ lbc_cdm_data[31:0] = Local bus Data ■ lbc_cdm_addr[31:0] = Local bus Address ■ cfg_sys_err_rc_cor = System Error caused by Correctable Error. (One-clock-cycle pulse) (Tied 0 if not Root Complex) ■ cfg_sys_err_rc_nf = System Error caused by Non-Fatal Error. (One-clock-cycle pulse) (Tied 0 if not Root Complex) ■ cfg_sys_err_rc_f = System Error caused by Fatal Error. (One-clock-cycle pulse) (Tied 0 if not Root Complex) Exists: DIAGNOSTIC_ENABLE Synchronous To: None, aux_clk, core_clk, core_pl_clk, perbitclk, radm_clk_g, ret_core_clk, {aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.79 SII: FRS/DRS Messaging Signals

app_drs_ready -
 app_pf_frs_ready -
 app_vf_frs_ready -

pf_frs_grant
 vf_frs_grant

Table 2-79 SII: FRS/DRS Messaging Signals

Port Name	I/O	Description
app_drs_ready	I	Defers DRS messaging when set to '0'. Exists: CX_RN_DRS_SUPPORTED Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_pf_frs_ready[(NF-1):0]	I	Defers FRS messaging when set to '0'. Exists: CX_RN_FRS_SUPPORTED Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_vf_frs_ready[(INT_NVF-1):0]	I	Defers FRS messaging for a VF when set to '0'. Exists: (CX_RN_FRS_SUPPORTED) && (CX_SRIOV_ENABLE && (CX_INTERNAL_NVFUNC != 0)) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pf_frs_grant[(NF-1):0]	O	Indicator of when an FRS message for this function has been scheduled for transmission. Exists: CX_RN_FRS_SUPPORTED Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High (One clock pulse) Validated by: app_pf_frs_ready asserted

Port Name	I/O	Description
vf_frs_grant[(INT_NVF-1):0]	O	Indicator of when an FRS message for this VF has been scheduled for transmission. Exists: (CX_RN_FRS_SUPPORTED) && (CX_SRIOV_ENABLE && (CX_INTERNAL_NVFUNC != 0)) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High (One clock pulse) Validated by: app_vf_frs_ready asserted

2.80 SII: General Core Control Signals

rx_lane_flip_en	-	cfg_vf_bme
tx_lane_flip_en	-	cxs_ltssm_enable
app_sris_mode	-	cxl_ltssm_enable
app_ltssm_enable	-	cxl_viral_status
cxl_viral_request	-	cxl_viral_ldid
cxl_viral_ldid_request	-	cfg_cxl_dev_disable_caching
app_req_retry_en	-	cfg_cxl_dev_initiate_cache_wr_invl
app_pf_req_retry_en	-	
app_vf_req_retry_en	-	
app_dbi_ro_wr_disable	-	

Table 2-80 SII: General Core Control Signals

Port Name	I/O	Description
rx_lane_flip_en	I	Performs manual lane reversal for receive lanes. For use when automatic lane reversal does not occur because lane 0 is not detected. In most cases, rx_lane_flip_en should be wired to a static value at the chip level. For more information, see 'Lane Reversal and Broken Lanes'. Exists: CX_LANE_FLIP_CTRL_EN Synchronous To: aux_clk,aux_clk_g,core_clk_ug,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
tx_lane_flip_en	I	Performs manual lane reversal for transmit lanes. For use when automatic lane reversal does not occur because lane 0 is not detected. In most cases, tx_lane_flip_en should be wired to a static value at the chip level. For more information, see 'Lane Reversal and Broken Lanes'. Exists: CX_LANE_FLIP_CTRL_EN Synchronous To: aux_clk,aux_clk_g,core_clk_ug,pipe_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_vf_bme[(INT_NVF-1):0]	O	Bus master enable bit from the Control Register in the PCI header of each VF. Each bit field corresponds to one of the NVF virtual functions. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. Exists: (CX_SRIOV_ENABLE) && (INTERNAL_VF_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
app_sris_mode	I	SRIS operating mode: ■ 0b: non-SRIS mode ■ 1b: SRIS mode app_sris_mode should only be changed before releasing app_ltssm_enable. Exists: CX_SRIS_SUPPORT Synchronous To: aux_clk_g, core_clk_ug, core_pl_clk, pipe_clk, ret_core_clk, {core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
app_ltssm_enable	I	Driven low by your application after cold, warm or hot reset to hold the LTSSM in the Detect state until your application is ready for the link training to begin. When your application has finished reprogramming the controller configuration registers using the DBI, it asserts app_ltssm_enable to allow the LTSSM to continue link establishment. Can also be used to delay hot resetting of the controller until you have read out any register status. Cold Reset ■ Optionally hold LTSSM and delay link training, so that you can reprogram some registers through DBI. ❑ Set app_ltssm_enable =0 before your application de-asserts Power-On Reset (power_up_rst_n). Best way is to set app_ltssm_enable =0 at power-up or at assertion of core_rst_n. ❑ Wait for de-assertion of core_rst_n, sticky_rst_n, and non_sticky_rst_n. ❑ Write any register through DBI. ❑ Set app_ltssm_enable =1. ❑ Link training starts. Hot Reset (Link Down Reset) ■ Optionally delay reset of the controller, so that you can read some registers through DBI. ❑ Set app_ltssm_enable =0 immediately (combinatorially) upon falling edge of smlh_req_rst_not. ❑ Keep app_ltssm_enable =0 until bridge finishes "flushing mode". ❑ Read any register through DBI. ❑ Set app_ltssm_enable =1. ❑ Reset of controller begins (sticky_rst is not asserted). ■ Optionally hold LTSSM and delay link training, so that you can reprogram some registers through DBI. ❑ Set app_ltssm_enable =0 immediately (combinatorially) upon falling edge of core_rst_n. ❑ Write any register through DBI. ❑ Set app_ltssm_enable =1. ❑ Link training starts. ■ <i>Note:</i> For Hot Reset, you can do both or any one (delay reset and/or delay link training). If you do both, you must do in order presented. <i>Note:</i> You must only de-assert this signal using one of the recommended timings described in the "Reset Requirements" section in the Architecture chapter of the <i>Databook</i>.

Port Name	I/O	Description
app_ltssm_enable...(cont.)	I.	To do otherwise (that is, de-assert it outside of the Detect LTSSM state) causes the controller to be reset and the LTSSM moves immediately back to the Detect state. This transition is outside of the PCIe Specification and it might cause a PIPE protocol violation. Exists: Always Synchronous To: aux_clk,core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxs_ltssm_enable	O	CXS Reset Controller LTSSM enable Exists: CX_CXS_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_ltssm_enable	O	CXL VLSM Reset Controller LTSSM enable Exists: CX_CXL_ENABLE Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_viral_request	I	CXL Viral Mode Request Exists: (CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE) Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_viral_status	O	CXL Viral Mode Indication Exists: (CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cxl_viral_ldid_request[15:0]	I	CXL Viral Mode LD-ID Request Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MLD_EN) Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by cxl_viral_request
cxl_viral_ldid[15:0]	O	CXL Viral Mode LD-ID Indication Exists: ((CX_CXL_ENABLE && !CX_CXL_CXS_ENABLE)) && (CX_CXL_MLD_EN) Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by cxl_viral_status
app_req_retry_en	I	Provides a capability to defer incoming configuration requests until initialization is complete. When app_req_retry_en is asserted, the controller completes incoming configuration requests with a configuration request retry status. Other incoming requests are not affected. When the Readiness Notification mechanism is supported, DRS messaging is blocked when app_req_retry_en=1. When app_req_retry_en=0 and app_drs_ready=1, the controller autonomously transmits a DRS message when the link transitions from DL_Down to DL_Up. Exists: Always Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (for CRS) Validated by: Not validated by another signal

Port Name	I/O	Description
app_pf_req_retry_en[(NF-1):0]	I	Provides a per Physical Function (PF) capability to defer incoming configuration requests until initialization is complete. When app_pf_req_retry_en is asserted for a certain PF, the controller completes incoming configuration requests with a configuration request retry status; other incoming requests are not affected. When app_req_retry_en is also asserted, the controller completes incoming configuration requests with a configuration request retry status for ALL PFs and VFs, regardless of the status of app_pf_req_retry_en/app_vf_req_retry_en. When the Readiness Notification mechanism is supported, DRS messaging is blocked when app_pf_req_retry_en =1 for at least one PF. The same applies when app_pf_req_retry_en =0 for all PFs, but app_req_retry_en is asserted. When app_pf_req_retry_en =0 for a certain PF, and app_req_retry_en =0, and app_drs_ready =1, the controller autonomously transmits a DRS message when the link transitions from DL_Down to DL_Up. Exists: Always Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (for CRS) Validated by: Not validated by another signal
app_vf_req_retry_en[(NVF-1):0]	I	Provides a per Virtual Function (VF) capability to defer incoming configuration requests until initialization is complete. When app_vf_req_retry_en is asserted for a certain VF, the controller completes incoming configuration requests with a configuration request retry status; other incoming requests are not affected. When app_req_retry_en is also asserted, the controller completes incoming configuration requests with a configuration request retry status for ALL PFs and VFs, regardless of the status of app_pf_req_retry_en/app_vf_req_retry_en. When the Readiness Notification mechanism is supported, DRS messaging is blocked when app_vf_req_retry_en =1 for at least one VF. The same applies when app_vf_req_retry_en =0 for all VFs, but app_req_retry_en is asserted. When app_vf_req_retry_en =0 for a certain VF, and app_req_retry_en =0, and app_drs_ready =1, the controller autonomously transmits a DRS message when the link transitions from DL_Down to DL_Up. Exists: CX_SRIOV_ENABLE Synchronous To: radm_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High (for CRS) Validated by: Not validated by another signal

Port Name	I/O	Description
app_dbi_ro_wr_disable	I	DBI Read-only Write Disable 0: MISC_CONTROL_1_OFF.DBI_RO_WR_EN register field is read-write. 1: MISC_CONTROL_1_OFF.DBI_RO_WR_EN register field is forced to 0 and is read-only. Exists: !CX_PL_REG_DISABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_disable_caching	O	Indicates CXL device Disable Caching control Exists: (CX_CXL_ENABLE) && (CX_CXL_2_0_ENABLE) && (CX_CXL_CACHE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_cxl_dev_initiate_cache_wr_invald	O	Indicates CXL device Initiate Cache Write Back and Invalidation control Exists: (CX_CXL_ENABLE) && (CX_CXL_2_0_ENABLE) && (CX_CXL_CACHE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.81 SII: General Messaging Reception Signals

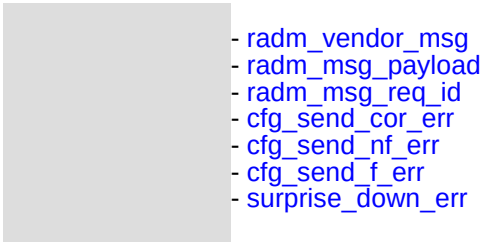


Table 2-81 SII: General Messaging Reception Signals

Port Name	I/O	Description
radm_vendor_msg[(FX_TLP-1):0]	O	One-cycle pulse that indicates the controller received a vendor-defined message. The controller makes the message header available the radm_msg_payload output. When FX_TLP > 1 and when two messages of the same type are received in the same clock cycle (back-to-back), then both bits are asserted. Exists: Always Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
radm_msg_payload[((FX_TLP*64)-1):0]	O	Received message header information. When a <i>vendor-defined message</i> is received (radm_vendor_msg=1), the controller maps radm_msg_payload to the Rx TLP header dwords as follows: When FX_TLP =1 ■ [31:0] = bytes 12-15 (4th dword), where [7:0] =byte 15 ■ [63:32] = bytes 8-11 (3rd dword) When FX_TLP >1 and radm_vendor_msg[1:0] ={01} ■ [31:0] = bytes 12-15 (4th dword) of 1st Message, where [7:0] =byte 15 ■ [63:32] = bytes 8-11 (3rd dword) ■ [127:96] = 'X' When FX_TLP >1 and radm_vendor_msg[1:0] ={10} ■ [95:64] = bytes 12-15 (4th dword) of 1st Message, where [7:0] =byte 15 ■ [127:96] = bytes 8-11 (3rd dword) ■ [63:0] = 'X' When FX_TLP >1 and radm_vendor_msg[1:0] =11 ■ [31:0] = bytes 12-15 (4th dword) of 2nd Message, where [7:0] =byte 15 ■ [63:32] = bytes 8-11 (3rd dword) ■ [95:64] = bytes 12-15 (4th dword) of 1st Message, where [7:0] =byte 15 ■ [127:96] = bytes 8-11 (3rd dword) When any other type of message is received, the controller maps radm_msg_payload to the Rx TLP header dwords as follows: When FX_TLP =1 ■ [31:0] = DW Data of Message ■ [63:32] = '0' When FX_TLP >1 and radm_vendor_msg[1:0] ={01 10} ■ [31:0] = DW Data of 2nd Message ■ [63:32] = '0' ■ [127:96] = 'X' When FX_TLP >1 and radm_vendor_msg[1:0] =11 ■ [31:0] = DW Data of 2nd Message ■ [63:32] = '0' ■ [95:64] = DW Data of 1st Message ■ [127:96] = '0' For more information, see "Routing of Received Messages" and the "Endianness" chapters. Exists: Always Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State:

Port Name	I/O	Description
radm_msg_payload[((FX_TLP*64)-1):0]...(cont.)	O.	High Validated by: Any of {radm_vendor_msg, radm_msg_unlock, pm_req_dwsp_turnoff, radm_msg_obff, radm_msg_idle, radm_msg_cpu_active} is asserted
radm_msg_req_id[((FX_TLP*16)-1):0]	O	The requester ID of the received Message. ■ [15:8]: Bus number ■ [7:3]: Device number ■ [2:0]: Function number Exists: Always Synchronous To: aux_clk_g, core_clk, {radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Any of {radm_vendor_msg, radm_msg_unlock, pm_req_dwsp_turnoff, radm_msg_obff, radm_msg_idle, radm_msg_cpu_active} is asserted
cfg_send_cor_err[(NF-1):0]	O	Sent Correctable Error. Controller has sent a message towards the Root Complex indicating that an Rx TLP that contained an error, and that can be corrected, has been received by the Endpoint Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_send_nf_err[(NF-1):0]	O	Sent Non-Fatal Error. Controller has sent a message towards the Root Complex indicating that an Rx TLP that contained a non-fatal error, and that can not be corrected, has been received by the Endpoint Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_send_f_err[(NF-1):0]	O	Sent Fatal Error. Controller has sent a message towards the Root Complex indicating that an Rx TLP that contained a fatal error, and that can not be corrected, has been received by the Endpoint Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
surprise_down_err[(NF-1):0]	O	Indicates that a surprise down event is occurring in the controller. Exists: SURPRISE_LINK_DOWN_SUPPORTED Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.82 SII: Interrupt Signals



Table 2-82 SII: Interrupt Signals

Port Name	I/O	Description
<code>cfg_vpd_int[(NF-1):0]</code>	O	This pin is set as a one cycle pulse to notify your application to read the VPD registers. The sequence of events for a VPD read cycle is: ■ The controller sends a request (single-cycle pulse of the <code>cfg_vpd_int</code> signal) to your application to read or write vital product data. ■ Your application reads the VPD Control and Capabilities register. ■ The VPD Flag (bit 31) is set to '0' indicating a read request. Your application fetches four bytes of data from the VPD Address location and transfers this to the VPD Data register. ■ Your application sets the VPD Flag bit to '1' indicating the request is complete. The sequence of events for a VPD write cycle is: ■ The controller sends a request (single-cycle pulse of the <code>cfg_vpd_int</code> signal) to your application to read or write vital product data. ■ Your application reads the VPD Control and Capabilities register. ■ The VPD Flag (bit 31) is set to '1' indicating a write request. Your application reads the VPD Data register and transfers this to the location specified by the VPD Address register. ■ Your application sets the VPD Flag bit to '0' indicating the request is complete. Exists: VPD_CAP_ENABLE Synchronous To: <code>aux_clk_g</code> Registered: N/A Power Domain: PD_VAUX Active State: High (pulse) Validated by: Not validated by another signal.

Port Name	I/O	Description
sys_int[(NF-1):0]	I	When sys_int goes from low to high, the controller generates an Assert_INTx Message. When sys_int goes from high to low, the controller generates a Deassert_INTx Message. There is a separate sys_int input bit for each function in your controller configuration. The Interrupt Pin register for the corresponding function determines which INTx Message the controller generates (INTA, INTB, INTC, or INTD). Legacy and native PCIe devices capable of generating an interrupt must support both Assert_INTx/Deassert_INTx and MSI or MSI-X. sys_int is intended to generate a message that emulates the legacy PCI Interrupts. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_link_eq_req_int	O	Interrupt indicating to your application that the Link Equalization Request bit in the Link Status 2 Register has been set and the Link Equalization Request Interrupt Enable (Link Control 3 Register bit 1) is set. Exists: CX_GEN3_MODE != GEN3_DISABLED Synchronous To: core_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
usp_eq_redo_executed_int	O	Interrupt indicating the EQ redo is executed by USP. You can use mac_phy_rate to know the data rate at which the interrupt occurred. You can leave this pin unconnected. Exists: CX_GEN3_MODE != GEN3_DISABLED Synchronous To: core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

Port Name	I/O	Description
edma_int[(((CC_NUM_DMA_RD_CHAN+C_NUM_DMA_WR_CHAN)-1):0]	O	DMA Interrupt. Indicates that the DMA transfer has completed or that an error has occurred. This is a level interrupt. For more information, see 'Interrupts and Error Handling'. Exists: CC_DMA_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
assert_inta_grt	O	The signal assert_inta_grt is a one-clock-cycle pulse that indicates that the controller sent an Assert_INTA Message to the upstream device. Exists: CX_SYS_INT_GRANT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: N/A
assert_intb_grt	O	The signal assert_intb_grt is a one-clock-cycle pulse that indicates that the controller sent an Assert_INTB Message to the upstream device. Exists: CX_SYS_INT_GRANT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: N/A
assert_intc_grt	O	The signal assert_intc_grt is a one-clock-cycle pulse that indicates that the controller sent an Assert_INTC Message to the upstream device. Exists: CX_SYS_INT_GRANT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: N/A
assert_intd_grt	O	The signal assert_intd_grt is a one-clock-cycle pulse that indicates that the controller sent an Assert_INTD Message to the upstream device. Exists: CX_SYS_INT_GRANT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: N/A

Port Name	I/O	Description
deassert_inta_grt	O	The signal deassert_inta_grt is a one-clock-cycle pulse that indicates that the controller sent an Deassert_INTA Message to the upstream device. Exists: CX_SYS_INT_GRANT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: N/A
deassert_intb_grt	O	The signal deassert_intb_grt is a one-clock-cycle pulse that indicates that the controller sent an Deassert_INTB Message to the upstream device. Exists: CX_SYS_INT_GRANT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: N/A
deassert_intc_grt	O	The signal deassert_intc_grt is a one-clock-cycle pulse that indicates that the controller sent an Deassert_INTC Message to the upstream device. Exists: CX_SYS_INT_GRANT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: N/A
deassert_intd_grt	O	The signal deassert_intd_grt is a one-clock-cycle pulse that indicates that the controller sent an Deassert_INTD Message to the upstream device. Exists: CX_SYS_INT_GRANT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: N/A

Port Name	I/O	Description
cfg_int_pin[((8*NF)-1):0]	O	The cfg_int_pin indicates the configured value for the Interrupt Pin Register field in the BRIDGE_CTRL_INT_PIN_INT_LINE register. Exists: CX_SYS_INT_GRANT_EN Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: N/A
cfg_int_disable[(NF-1):0]	O	When high a functions ability to generate INTx messages is Disabled Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_safety_corr	O	The controller asserts cfg_safety_corr when a PCIe protocol correctable error or a Data Path protection correctable error is indicated and the associated mask for that event is not set. Exists: CX_AUTOMOTIVE_ENABLE Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cfg_safety_uncorr	O	The controller asserts cfg_safety_uncorr when any of the internal safety mechanisms reports an uncorrectable error and the associated mask for that event is not set. Exists: CX_AUTOMOTIVE_ENABLE Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

Port Name	I/O	Description
dpc_int	O	DPC Interrupt Status for INTx signalling. If the Port is enabled for level-triggered interrupt signaling using INTx messages, dpc_int is asserted whenever and as long as the following conditions are satisfied: ■ The value of the Interrupt Disable bit in the Command register is 0b. ■ The value of the DPC Interrupt Enable bit is 1b. ■ The value of the DPC Interrupt Status bit is 1b. Exists: CX_DPC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
dpc_msi	O	DPC Interrupt pulse for MSI signalling. If the Port is enabled for edge-triggered interrupt signaling using MSI or MSI-X, dpc_msi is asserted every time the logical AND of the following conditions transitions from FALSE to TRUE: ■ MSI or MSI-X is enabled. ■ The value of the DPC Interrupt Enable bit is 1b. ■ The value of the DPC Interrupt Status bit is 1b. The controller does not check if the associated MSI vector is unmasked. It is up to the application to check whether the vector is masked or unmasked. Exists: CX_DPC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

2.83 SII: Link Reset/Status Signals

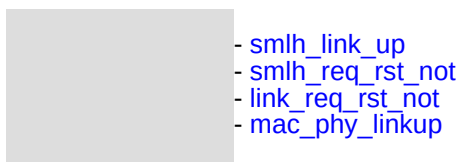


Table 2-83 SII: Link Reset/Status Signals

Port Name	I/O	Description
smlh_link_up	O	PHY Link up/down indicator: ■ 1: Link is up ■ 0: Link is down Exists: Always Synchronous To: core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
smlh_req_rst_not	O	Early version of the link_req_rst_not signal. For more information, see the 'Warm and Hot Resets' section in the Architecture chapter of the <i>Databook</i>. Exists: Always Synchronous To: core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: Low (high-to-low transition) Validated by: Not validated by another signal
link_req_rst_not	O	Reset request because the link has gone down or the controller received a hot-reset request. A low level indicates that the controller is requesting external logic to reset the controller because the PHY link is down. When the AXI bridge module is enabled the link reset request is delayed until all pending AXI transfers have completed. When the AXI Bridge module is not enabled the link reset request is sent immediately. For more information, see the "Hot Reset" section in the Architecture chapter of the <i>Databook</i>. Exists: Always Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_linkup	O	Physical Layer Link up/down flag. ■ 1: up ■ 0: down Exists: (CX_PCIE_MODE == PCIE_OVER_CIO) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.84 SII: LTR Message Generation Signals

app_ltr_msg_req - app_ltr_msg_grant
 app_ltr_msg_latency - app_ltr_latency
 app_ltr_msg_func_num

Table 2-84 SII: LTR Message Generation Signals

Port Name	I/O	Description
app_ltr_msg_req	I	Indicates that your application is requesting to send an LTR message. Once asserted, app_ltr_msg_req must remain asserted until the controller asserts app_ltr_msg_grant. Exists: CX_LTR_M_ENABLE Synchronous To: aux_clk,aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_ltr_msg_grant	O	Indicates that the controller has accepted your request to send an LTR message. Exists: CX_LTR_M_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: app_ltr_msg_req is asserted
app_ltr_msg_latency[31:0]	I	LTR message that your application is requesting to send. The message format is defined in the <i>PCI Express Specification</i>. Exists: CX_LTR_M_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: app_ltr_msg_req is asserted
app_ltr_msg_func_num[(PF_WD-1):0]	I	Function number in your application that is requesting to send an LTR message. Function numbering starts at '0'. Exists: CX_LTR_M_ENABLE Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: app_ltr_msg_req is asserted

Port Name	I/O	Description
app_ltr_latency[31:0]	O	The current LTR values reported and in-use by the device. when LTR is disabled, or when any function is directed to a non-D0 state, if a device had previously reported one or both latency fields with the Requirement bit set, the application must send a new LTR Message with both Requirement bits clear. For more information, see 'Port Logic Registers: Latency Tolerance Reporting (LTR) Registers'. Exists: CX_LTR_M_ENABLE Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.85 SII: MSI-X RAM Debug Signals

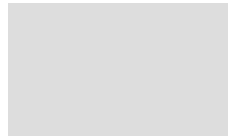
dbg_table -
dbg_pba -



Table 2-85 SII: MSI-X RAM Debug Signals

Port Name	I/O	Description
dbg_table	I	MSIX Table RAM Debug Mode. Use this input to activate the debug mode and allow direct read/write access to the Table. You can also use the MSIX_RAM_CTRL_DBG_TABLE field in MSIX_RAM_CTRL_OFF to activate debug mode. Debug mode turns off the PF/VF/Offset-based addressing into the RAM and maps the entire table linearly from the base address of the BAR (indicated by the BIR) in function 0. When you assert dbg_table and dbg_pba simultaneously, you can access the traffic class (TC) information bits. For more information, see the Interrupts section in the "Controller Operations" chapter of the <i>Databook</i>. Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
dbg_pba	I	MSIX PBA RAM Debug Mode. Use this input to activate the debug mode and allow direct read/write access to the PBA. You can also use the MSIX_RAM_CTRL_DBG_PBA field in MSIX_RAM_CTRL_OFF to activate debug mode. Debug mode turns off the PF/VF/Offset-based addressing into the RAM and maps the entire array linearly from the base address of the BAR (indicated by the BIR) in function 0. When you assert dbg_table and dbg_pba simultaneously, you can access the traffic class (TC) information bits. For more information, see the Interrupts section in the "Controller Operations" chapter of the <i>Databook</i>. Exists: (MSIX_CAP_ENABLE) && (MSIX_TABLE_EN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.86 SII: OBFF Message Reception Signals



- radm_msg_idle
- radm_msg_obff
- radm_msg_cpu_active

Table 2-86 SII: OBFF Message Reception Signals

Port Name	I/O	Description
radm_msg_idle	O	One-clock-cycle pulse that indicates that the controller received an 'IDLE' OBFF message. When FX_TLP > 1 and when two messages of the same type are received in the same clock cycle (back-to-back), then no separate indication is given for the second message. Exists: CX_OBFF_ENABLE Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_msg_obff	O	One-clock-cycle pulse that indicates that the controller received an 'OBFF' OBFF message. When FX_TLP > 1 and when two messages of the same type are received in the same clock cycle (back-to-back), then no separate indication is given for the second message. Exists: CX_OBFF_ENABLE Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_msg_cpu_active	O	One-clock-cycle pulse that indicates that the controller received a 'CPU Active' OBFF message. When FX_TLP > 1 and when two messages of the same type are received in the same clock cycle (back-to-back), then no separate indication is given for the second message. Exists: CX_OBFF_ENABLE Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.87 SII: PM, Unlock, and Error Messages Signals

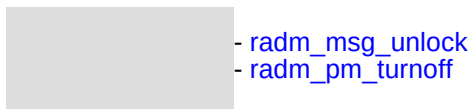


Table 2-87 SII: PM, Unlock, and Error Messages Signals

Port Name	I/O	Description
radm_msg_unlock	O	One-cycle pulse that indicates that the controller received an Unlock message. When FX_TLP > 1 and when two messages of the same type are received in the same clock cycle (back-to-back), then no separate indication is given for the second message. Exists: Always Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_pm_turnoff	O	One-clock-cycle pulse that indicates that the controller received a PME Turnoff message. When FX_TLP > 1 and when two messages of the same type are received in the same clock cycle (back-to-back), then no indication is given for the second message. Exists: Always Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.88 SII: Power Management Signals

outband_pwrup_cmd	-	pm_curnt_state
apps_pm_xmt_pme	-	smih_ltssm_state
apps_pm_vf_xmt_pme	-	wake
sys_aux_pwr_det	-	local_ref_clk_req_n
app_req_entr_l1	-	pm_dstate
app_ready_entr_l23	-	pm_vf_dstate
app_req_exit_l1	-	pm_pme_en
app_xfer_pending	-	pm_linkst_in_l0s
clkreq_in_n	-	pm_linkst_in_l1
app_clk_pm_en	-	pm_linkst_in_l2
test_bypass_lp	-	pm_linkst_l2_exit
ack_en_vmain	-	pm_master_state
app_l1_pwr_off_en	-	pm_slave_state
save_state_ack	-	pm_l1sub_state
restore_state_ack	-	pm_vf_pme_en
app_l1sub_disable	-	pm_status
		pm_vf_status
		aux_pm_en
		pm_linkst_in_l1sub
		cfg_l1sub_en
		dpa_substate_update
		pm_en_vmain_n
		pm_en_vmain
		pm_save_state_req
		pm_restore_state_req
		pm_req_iso_vmain_to_vaux
		pm_req_slv_iso
		pm_req_mstr_iso
		pm_req_dbi_iso
		pm_l1_entry_started
		mac_phy_ltssmstate

Table 2-88 SII: Power Management Signals

Port Name	I/O	Description
outband_pwrup_cmd[(NF-1):0]	I	Wake Up. If PME is enabled and PME support is configured for current PMCSR D-state asserting this signal causes the controller to wake from either L1 or L2 state. When the controller has transitioned back to the L0 state it transmits a PME message and set the PME_Status. There is a separate outband_pwrup_cmd input bit for each function in your controller configuration. This port is functionally identical to apps_pm_xmt_pme. Upon receiving the PME message the root complex should clear the PME_Status and change the D-state back to D0. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
apps_pm_xmt_pme[(NF-1):0]	I	Wake Up. If PME is enabled and PME support is configured for current PMCSR D-state asserting this signal causes the controller to wake from either L1 or L2 state. When the controller has transitioned back to the L0 state it transmits a PME message and set the PME_Status. Upon receiving the PME message the root complex should clear the PME_Status and change the D-state back to D0. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
apps_pm_vf_xmt_pme[(NVF-1):0]	I	Wake Up command to VF. Used by application logic to wake up the PMC state machine from the D3 state. Upon wake-up, the controller sends a PM_PME Message. Needs to be asserted for one clock cycle. Each bit field corresponds to one of the NVF virtual functions. Your application knows how many VFs are assigned to each PF, and by using this knowledge, it can identify the parent PF and the VF number within that PF. For more information, see the VF Index in 'Bus Numbering Overview'. Exists: VF_PM_CAP_ENABLE Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
sys_aux_pwr_det	I	Auxiliary Power Detected. Used to report to the host software that auxiliary power (Vaux) is present. Exists: Always Synchronous To: aux_clk_g,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_curnt_state[2:0]	O	Indicates the current power state. The pm_curnt_state output is intended for debugging purposes, not for system operation. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
smlh_ltssm_state[5:0]	O	Current state of the LTSSM. Encoding is defined as follows: ■ 6'h00: S_DETECT_QUIET ■ 6'h01: S_DETECT_ACT ■ 6'h02: S_POLL_ACTIVE ■ 6'h03: S_POLL_COMPLIANCE ■ 6'h04: S_POLL_CONFIG ■ 6'h05: S_PRE_DETECT_QUIET ■ 6'h06: S_DETECT_WAIT ■ 6'h07: S_CFG_LINKWD_START ■ 6'h08: S_CFG_LINKWD_ACEPT ■ 6'h09: S_CFG_LANENUM_WAI ■ 6'h0A: S_CFG_LANENUM_ACEPT ■ 6'h0B: S_CFG_COMPLETE ■ 6'h0C: S_CFG_IDLE ■ 6'h0D: S_RCVRY_LOCK ■ 6'h0E: S_RCVRY_SPEED ■ 6'h0F: S_RCVRY_RCVRCFG ■ 6'h10: S_RCVRY_IDLE ■ 6'h11: S_L0 ■ 6'h12: S_L0S ■ 6'h13: S_L123_SEND_EIDLE ■ 6'h14: S_L1_IDLE ■ 6'h15: S_L2_IDLE ■ 6'h16: S_L2_WAKE ■ 6'h17: S_DISABLED_ENTRY ■ 6'h18: S_DISABLED_IDLE ■ 6'h19: S_DISABLED ■ 6'h1A: S_LPBK_ENTRY ■ 6'h1B: S_LPBK_ACTIVE ■ 6'h1C: S_LPBK_EXIT ■ 6'h1D: S_LPBK_EXIT_TIMEOUT ■ 6'h1E: S_HOT_RESET_ENTRY ■ 6'h1F: S_HOT_RESET ■ 6'h20: S_RCVRY_EQ0 ■ 6'h21: S_RCVRY_EQ1 ■ 6'h22: S_RCVRY_EQ2 ■ 6'h23: S_RCVRY_EQ3 Exists: Always Synchronous To: core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
wake	O	Wake Up. Wake up from power management unit. The controller generates wake to request the system to restore power and clock when a wakeup event has been detected such as apps_pm_xmt_pme, apps_pm_vf_xmt_pme, or outband_pwrup_cmd. The wake signal is an active high signal and its rising edge should be detected to drive the WAKE# the connector. The wake signal is de-asserted when the main power has been restored. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
local_ref_clk_req_n	O	This signal may be connected to the CLKREQ# driver to negotiate entry into L1 sub-states. When this signal is set to 1 the controller is requesting entry into L1 sub-states and CLKREQ# may be de-asserted. For an Upstream port this signal is also used to request reference clock removal when Clock Power Management is enabled. This signal should only be set to 1 if the PHY is also ready for reference clock removal, the input phy_clk_req_n is a handshake from the PHY, which when set to 1 indicates that the PHY is ready for reference clock removal. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
pm_dstate[(((3*Nf)-1):0]	O	The current power management D-state of the function: ■ 000b: D0 ■ 001b: D1 ■ 010b: D2 ■ 011b: D3 ■ 100b: Uninitialized ■ Other values: Not applicable There are 3 bits of pm_dstate for each configured function. Exists: Always Synchronous To: aux_clk_g,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pm_vf_dstate[((3*NVF)-1):0]	O	The current power management D-state of the virtual function: ■ 000b: D0 ■ 001b: D1 ■ 010b: D2 ■ 011b: D3 ■ 100b: Uninitialized ■ Other values: N/A There are 3 bits of pm_vf_dstate for each configured function. Each bit field corresponds to one of the NVF virtual functions. Your application knows how many VFs are assigned to each PF, and by using this knowledge, it can identify the parent PF and the VF number within that PF. For more information, see the VF Index in 'Bus Numbering Overview'. Exists: VF_PM_CAP_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
pm_pme_en[(NF-1):0]	O	PME Enable bit in the PMCSR. There is 1 bit of pm_pme_en for each configured function. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_linkst_in_l0s	O	Power management is in L0s state. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_linkst_in_l1	O	Power management is in L1 state. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pm_linkst_in_l2	O	Power management is in L2 state. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_linkst_l2_exit	O	Power management is exiting L2 state. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_master_state[(PM_MST_WD-1):0]	O	Power management master FSM state. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_slave_state[(PM_SLV_WD-1):0]	O	Power management slave FSM state. Exists: Always Synchronous To: None,perbitclk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_l1sub_state[2:0]	O	Power management L1 sub-states FSM state. Exists: CX_L1_SUBSTATES_ENABLE Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pm_vf_pme_en[(NVF-1):0]	O	Enable bit in the VF PMCSR. There is 1 bit of pm_pme_en for each configured function. Each bit field corresponds to one of the NVF virtual functions. Your application knows how many VFs are assigned to each PF, and by using this knowledge, it can identify the parent PF and the VF number within that PF. For more information, see the VF Index in 'Bus Numbering Overview'. Exists: VF_PM_CAP_ENABLE Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_status[(NF-1):0]	O	PME Status bit from the PMCSR. There is 1 bit of pm_status for each configured function. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_vf_status[(NVF-1):0]	O	PME Status bit from the VF PMCSR. There is 1 bit of pm_status for each configured function. Each bit field corresponds to one of the NVF virtual functions. Your application knows how many VFs are assigned to each PF, and by using this knowledge, it can identify the parent PF and the VF number within that PF. For more information, see the VF Index in 'Bus Numbering Overview'. Exists: VF_PM_CAP_ENABLE Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
aux_pm_en[(NF-1):0]	O	Auxiliary Power Enable bit in the Device Control register. There is 1 bit of aux_pm_en for each configured function. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
app_req_entr_l1	I	Application request to Enter L1 ASPM state. The app_req_entr_l1 signal is for use by applications that need to control L1 entry instead of using the L1 entry timer as defined in the PCI Express Specification. It is only effective when L1 is enabled. The controller latches this request when in L0 or L0s; to be acted upon later. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Pulse Validated by: Not validated by another signal
app_ready_entr_l23	I	Application Ready to Enter L23. Indication from your application that it is ready to enter the L23 state. The app_ready_entr_l23 signal is provided for applications that must control L23 entry (in case certain tasks must be performed before going into L23). The controller delays sending PM_Enter_L23 (in response to PM_Turn_Off) until this signal becomes active. When this signal has been asserted by the application, it must be kept asserted until L2 entry has completed. Hardwire to 1 for applications that do not require this feature. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_req_exit_l1	I	Application request to Exit L1. Request from your application to exit L1. It is only effective when L1 is enabled. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Pulse Validated by: Not validated by another signal

Port Name	I/O	Description
app_xfer_pending	I	Indicates that your application has transfers pending and prevents the controller from entering L1. If the entry into L1 is already in progress, assertion of app_xfer_pending causes an exit from L1. This is a level signal used to inform the controller about the state of external queues and pipeline stages that contain transactions to be transmitted by the controller. The controller uses this information to determine when to enter/exit L1. When this signal is asserted, it indicates that there are transactions outside the controller that the controller needs to transmit. When de-asserted, it indicates that there are no transactions outside the controller. The controller responds to an assertion on this signal as follows: Prevents generation of requests to enter L1. Triggers exit if already in L1. You can instruct the controller to exit L1 by asserting either or both of app_xfer_pending and app_req_exit_l1. The controller only samples app_req_exit_l1 when the controller is already in the L1 state. Exists: Always Synchronous To: aux_clk_g,core_clk_ug,pipe_clk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
clkreq_in_n	I	Status of the CLKREQ# bidirectional CMOS board-level signal. Used by the controller to determine when to enter and exit L1 Substates when using the CLKREQ#-based mechanism. For more information, see 'L1 Substates'. This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: CX_L1_SUBSTATES_ENABLE Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

Port Name	I/O	Description
pm_linkst_in_l1sub	O	Power management is in L1 substate. Indicates when the link has entered L1 substates. It is used in DWC_pcie_clkrst.v (see 'Clock Generation and Gating Design Example') to ensure that the switching back of aux_clk from AUXCLK to PCLK occurs only after L1 substates have been exited. For L1.2 this signal is de-asserted at the end of the L1.2.Exit state, after the t_power_on constraint has been satisfied. External logic can use the transition high to low on this signal to initiate REFCLK restore. Exists: CX_L1_SUBSTATES_ENABLE Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_l1sub_en	O	Indicates that any of the L1 Substates are enabled in the L1 Substates Control 1 Register. Could be used by your application in a downstream port to determine when not to drive CLKREQ# such as when L1 Substates are not enabled. Exists: CX_L1_SUBSTATES_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_clk_pm_en	I	Clock PM feature enabled by application. Used to inhibit the programming of the Clock PM in Link Control Register. For more information, see 'L1 with Clock PM (L1 with REFCLK removal/PLL Off)'. Exists: Always Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
dpa_substate_update[(NF-1):0]	O	Function: DPA Substate-Updated Indication Signal Bus. The controller asserts it for one core_clk cycle when the Substate Control field of the DPA Control Register has been updated with a new value for the associated function. The controller asserts it when the updated value of the Substate Control field does not match the Substate Status field of the Status Register when the Substate Control Enable bit field of the Status Register is set to "1". <i>Note:</i> Your application logic is required to do the following: ■ Detect the one cycle pulse on the signal for the relevant function. ■ Read the function's substate control register to determine the software's target substate (transition). ■ Read the function's transition latency value, unit, and indicator. Calculate the maximum transition latency for the substate transition. ■ Execute the substate transition and complete it within the maximum transition latency. ■ Update the substate status after completing the transition through a DBI write. The controller updates the substate status during the substate transition if the transition is from a lower power substate to a higher; the status is the higher power substate during the substate transition. Exists: CX_DPA_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
test_bypass_lp	I	Test mode override of isolation enable. Exists: CX_ENHANCED_PM_EN Synchronous To: dbi_aclk,dbi_aclk_ug,slv_aclk,slv_aclk_ug,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ack_en_vmain	I	Acknowledge from power switch responsible for autonomous power-gating in L1.2. The application must set this signal to 0 when pm_en_vmain is set to 0 by the controller to request power-gating in L1.2. The application must set this signal to 1 when pm_en_vmain is set to 1 by the controller to request power to be restored. This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: (CX_ENHANCED_PM_EN) && (CX_L1_PG_ENABLE) Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_l1_pwr_off_en	I	Application permits to gate power to parts of the controller in L1 state. Exists: CX_ENHANCED_PM_EN Synchronous To: None,core_clk_ug,pipe_clk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
save_state_ack	I	Application acknowledges that state retention (L1 power gating) task is completed. Exists: CX_ENHANCED_PM_EN Synchronous To: core_clk_ug,pipe_clk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
restore_state_ack	I	Application acknowledges that saved state has been restored after power-up. Exists: CX_ENHANCED_PM_EN Synchronous To: core_clk_ug,pipe_clk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pm_en_vmain_n	O	Active Low Enable for power switch supplying PD_VMAIN. This signal is set to 1 when the controller is requesting power to be gated in L1.2. The application must set the acknowledge ack_en_vmain to 0 when power has been gated. The controller sets this signal to 0 when it is requesting power to be restored, the application should set ack_en_vmain to 1 when it has restored power. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section of the <i>Databook</i>. Exists: (CX_ENHANCED_PM_EN) && (CX_PSW_EN_ACTIVE_LOW) Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
pm_en_vmain	O	Active High Enable for power switch supplying PD_VMAIN. This signal is set to 0 when the controller is requesting power to be gated in L1.2. The application must set the acknowledge ack_en_vmain to 0 when power has been gated. The controller sets this signal to 1 when it is requesting power to be restored, the application should set ack_en_vmain to 1 when it has restored power. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section of the <i>Databook</i>. Exists: (CX_ENHANCED_PM_EN) && (!CX_PSW_EN_ACTIVE_LOW) Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_save_state_req	O	Request which can be used by your application to store any data which needs to be retained during L1 power gating. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section of the <i>Databook</i>. Exists: CX_ENHANCED_PM_EN Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pm_restore_state_req	O	Request which can be used by your application to restore any data which has been retained prior to L1 power gating. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section of the <i>Databook</i>. Exists: CX_ENHANCED_PM_EN Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_req_iso_vmain_to_vaux	O	Request from PMC to enable isolation on VMAIN to VAUX boundary. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section of the <i>Databook</i>. Exists: CX_ENHANCED_PM_EN Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_req_slv_iso	O	Request from PMC to enable isolation on AXI slave output signals. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section of the <i>Databook</i>. Exists: CX_SLV_ISO_EN Synchronous To: slv_aclk_ug Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_req_mstr_iso	O	Request from PMC to enable isolation on AXI master output signals. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section of the <i>Databook</i>. Exists: CX_MSTR_ISO_EN Synchronous To: mstr_aclk_ug Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pm_req_dbi_iso	O	Request from PMC to enable isolation on AXI DBI slave output signals. For more information, see "Advanced Power Management and Power Domain Gating" in the Power Management section of the <i>Databook</i> . Exists: CX_DBI_ISO_EN Synchronous To: dbi_aclk_ug Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_l1sub_disable	I	The application can set this input to 1'b1 to prevent entry to L1 Sub-states. This pin is used to gate the L1 sub-state enable bits from the L1 PM Substates Control 1 Register. Exists: CX_APP_L1SUB_CONTROL Synchronous To: core_clk_ug,pipe_clk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_l1_entry_started	O	L1 entry process is in progress. Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_ltssmstate[5:0]	O	Indicates the current state of the LTSSM. The encoding is the same as port smlh_ltssm_state. Exists: (CX_PCIE_MODE == PCIE_OVER_CIO) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.89 SII: Precision Time Management Signals

phy_mac_ts_sample	-	ptm_context_valid
phy_mac_ts_ack	-	ptm_responder_rdy_to_validate
phy_mac_ts_value	-	ptm_local_clock
ptm_external_master_strobe	-	mac_phy_ts_req
ptm_external_master_time	-	ptm_trigger_allowed
ptm_auto_update_signal	-	ptm_updating
ptm_manual_update_pulse	-	ptm_clock_updated
		ptm_clock_correction
		ptm_req_response_timeout
		ptm_req_dup_rx
		ptm_req_replay_tx

Table 2-89 SII: Precision Time Management Signals

Port Name	I/O	Description
ptm_context_valid	O	Context Valid. Exists: CX_PTM_ENABLE Synchronous To: core_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ptm_responder_rdy_to_validate	O	PTM Responder Ready to Validate. Exists: CX_PTM_ENABLE Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ptm_local_clock[63:0]	O	Local Clock value. Exists: CX_PTM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Context Valid

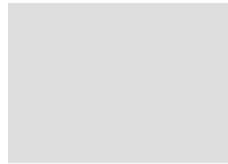
Port Name	I/O	Description
phy_mac_ts_sample	I	When USB4 TMU asserts this signal, the controller captures internal PTM timestamps. Exists: (CX_PTM_ENABLE) && ((CX_PCIE_MODE == PCIE_OVER_CIO)) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_ts_ack	I	USB4 TMU acknowledges controller's request for USB4 TMU timestamps through this signal. When USB4 TMU asserts this signal, the controller can sample phy_mac_ts_value to capture USB4 TMU timestamps. Exists: (CX_PTM_ENABLE) && ((CX_PCIE_MODE == PCIE_OVER_CIO)) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_ts_value[79:0]	I	This signal reflects the USB4 TMU timestamp value. Exists: (CX_PTM_ENABLE) && ((CX_PCIE_MODE == PCIE_OVER_CIO)) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: phy_mac_ts_ack
mac_phy_ts_req	O	The controller asserts this signal to request for USB4 TMU timestamps. The signal mac_phy_ts_req remains asserted until the USB4 TMU asserts phy_mac_ts_sample and phy_mac_ts_ack. Exists: (CX_PTM_ENABLE) && ((CX_PCIE_MODE == PCIE_OVER_CIO)) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ptm_external_master_strobe	I	PTM External Master Time Strobe. Exists: (CX_PTM_ENABLE) && (CX_PTM_EXTERNAL_MASTER_TIME) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ptm_external_master_time[63:0]	I	PTM External Master Time. Exists: (CX_PTM_ENABLE) && (CX_PTM_EXTERNAL_MASTER_TIME) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ptm_auto_update_signal	I	Indicates that the controller should update the PTM Requester Context and Clock automatically every 10ms. Exists: CX_PTM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (Level) Validated by: Not validated by another signal
ptm_manual_update_pulse	I	Indicates that the controller should update the PTM Requester Context and Clock now. Exists: CX_PTM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (Pulse) Validated by: Not validated by another signal
ptm_trigger_allowed	O	Indicates that a PTM Requester manual update trigger is allowed. Exists: CX_PTM_ENABLE Synchronous To: aux_clk,core_clk,ret_core_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
ptm_updating	O	Indicates that a PTM update is in progress. Exists: CX_PTM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ptm_clock_updated	O	Indicates that the controller has updated the Local Clock. Exists: CX_PTM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (Pulse) Validated by: Not validated by another signal
ptm_clock_correction[63:0]	O	Amount by which Local Clock has been corrected. Exists: CX_PTM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ptm_req_response_timeout	O	PTM Requester Response Timeout. Single-cycle pulse indicating 100us timeout occurred while waiting for a PTM Response or PTM ResponseD message. Exists: CX_PTM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
ptm_req_dup_rx	O	PTM Requester Duplicate Received. Single-cycle pulse indicating PTM Requester received a duplicate TLP while, ■ PTM Requester is in the process of updating PTM local clock, or ■ Following calculation of the update, when the PTM context is valid Exists: CX_PTM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (Pulse) Validated by: Not validated by another signal

Port Name	I/O	Description
ptm_req_replay_tx	O	PTM Requester Replay Sent. Single-cycle pulse indicating PTM Requester detected a TLP replay being sent when ResponseD messages are in use while, ■ PTM Requester is in the process of updating PTM local clock, or ■ Following calculation of the update, when the PTM context is valid Exists: CX_PTM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.90 SII: RAS Data Protection Signals



- mstr_rasdp_err_mode
- slv_rasdp_err_mode
- cfg_rasdp_err_mode
- dbi_rasdp_err_mode

Table 2-90 SII: RAS Data Protection Signals

Port Name	I/O	Description
mstr_rasdp_err_mode	O	Indicates (mstr_aclk version) that the controller has entered the RASDP error mode. The controller enters RASDP error mode (if the ERROR_MODE_EN register field =1) upon detection of the first uncorrectable error. During this mode: ■ Controller sets slv_rasdp_err_mode =1 ■ Controller sets mstr_rasdp_err_mode =1 ■ Controller sets dbi_rasdp_err_mode =1 ■ Rx TLPs that are forwarded to your application are not guaranteed to be correct; you must discard them. For more information, see the RAS Data Protection (DP) section in the Controller Operations chapter of the <i>Databook</i>. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (MASTER_POPULATED) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g,{mstr_aclk_ug} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
slv_rasdp_err_mode	O	Indication (slv_aclk version) from the controller that it has entered RASDP error mode. The controller enters RASDP error mode (if the ERROR_MODE_EN register field =1) upon detection of the first uncorrectable error. During this mode: ■ Controller sets mstr_rasdp_error_mode =1 ■ Controller sets slv_rasdp_error_mode =1 ■ Controller sets dbi_rasdp_error_mode =1 ■ Rx TLPs that are forwarded to your application are not guaranteed to be correct; you must discard them. For more information, see the RAS Data Protection (DP) section in the Controller Operations chapter of the <i>Databook</i>. Exists: (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (SLAVE_POPULATED) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g,{slv_aclk_ug} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_rasdp_error_mode	O	Indication from the controller that it has entered RASDP error mode. The controller enters RASDP error mode (if the ERROR_MODE_EN register field =1) upon detection of the first uncorrectable error. During this mode, Rx TLPs that are forwarded to your application are not guaranteed to be correct; you must discard them. Controller sets mstr_rasdp_error_mode and slv_rasdp_error_mode when AXI bridge is used, because this signal is not exposed then. For more information, see the RAS Data Protection (DP) section in the Controller Operations chapter of the <i>Databook</i>. Exists: (!(AMBA_INTERFACE!=0)) && (((CX_RASDP==0)&&(CX_RASDP_RAM_PROT==0))? 0: 1) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
dbi_rasdp_err_mode	O	Indication (dbo_aclk version) from the controller that it has entered RASDP error mode. The controller enters RASDP error mode (if the ERROR_MODE_EN register field =1) upon detection of the first uncorrectable error. During this mode: ■ Controller sets mstr_rasdp_error_mode =1 ■ Controller sets slv_rasdp_error_mode =1 ■ Controller sets dbi_rasdp_error_mode =1 ■ Rx TLPs that are forwarded to your application are not guaranteed to be correct; you must discard them. For more information, see the RAS Data Protection (DP) section in the Controller Operations chapter of the <i>Databook</i>. Exists: (DBI_4SLAVE_POPULATED) && (((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))) && (((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)) Synchronous To: dbi_aclk_ug Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.91 SII: Receive Control/CPL Timeout Signals

app_ph_ca	-	radm_cpl_timeout
app_pd_ca	-	radm_timeout_func_num
app_nph_ca	-	radm_timeout_vfunc_num
app_npd_ca	-	radm_timeout_vfunc_active
app_cplh_ca	-	radm_timeout_vfunc_index
app_cpld_ca	-	radm_timeout_cpl_tc
cxl_dev_cache_invalid	-	radm_timeout_cpl_attr
cxl_dev_cache_wr_invid_cmpl	-	radm_timeout_cpl_ln
cxl_pm_init_cmpl	-	radm_timeout_cpl_len
	-	radm_timeout_cpl_tag
	-	radm_cpl_lut_valid

Note: If your application is using the application credit return indication, it should set app_xfer_pending to 1'b1 until the credits have been returned. This is required because the app_*_ca signals do not trigger exit from low power states.

Table 2-91 SII: Receive Control/CPL Timeout Signals

Port Name	I/O	Description
radm_cpl_timeout	O	Indicates that the completion TLP for a request has not been received within the expected time window. Exists: Always Synchronous To: aux_clk_g, core_clk, {radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_timeout_func_num[(PF_WD-1):0]	O	The function Number of the timed out completion. Function numbering starts at '0'. Exists: Always Synchronous To: core_clk, {radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_cpl_timeout is asserted

Port Name	I/O	Description
radm_timeout_vfunc_num[(VF_WD-2):0]	O	Indicates which virtual function (VF) had a completion timeout. The PCIe SR-IOV specification starts numbering VFs at '1'. To ease timing during synthesis, the PCIe controller starts numbering VFs at '0'. For example, radm_timeout_vfunc_num =0 refers to the first VF of the PF that is identified by radm_timeout_func_num. For more information of VF numbering, see 'Programming View'. Exists: CX_SRIOV_ENABLE Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: radm_cpl_timeout and radm_timeout_vfunc_active are asserted
radm_timeout_vfunc_active	O	Indicates that a virtual function (VF) had a completion timeout. 0: No VF is active and radm_timeout_vfunc_num is invalid. A PF is valid and identified by radm_timeout_func_num. 1: A VF is active and is identified by radm_timeout_vfunc_num. Exists: CX_SRIOV_ENABLE Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_cpl_timeout is asserted
radm_timeout_vfunc_index[(VFI_WD-1):0]	O	Indicates which VF Index had a completion timeout. Your application knows how many VFs are assigned to each PF, and by using this knowledge, it can identify the parent PF and the VF number within that PF. For more information, see the VF Index in 'Bus Numbering Overview'. Exists: (CX_SRIOV_ENABLE) && (CX_EXTENSIBLE_VFUNC) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: radm_cpl_timeout and radm_timeout_vfunc_active are asserted
radm_timeout_cpl_tc[2:0]	O	The Traffic Class of the timed out completion. Exists: Always Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_cpl_timeout is asserted

Port Name	I/O	Description
radm_timeout_cpl_attr[1:0]	O	The Attributes field of the timed out completion. Exists: Always Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_cpl_timeout is asserted
radm_timeout_cpl_ln	O	The Lightweight Notification (LN) field of the timed out completion. Exists: CX_LN_ENABLE Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_cpl_timeout is asserted
radm_timeout_cpl_len[11:0]	O	Length (in bytes) of the timed out completion. For a split completion, it indicates the number of bytes remaining to be delivered when the completion timed out. Exists: Always Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_cpl_timeout is asserted. When RADM_CPL_LUT_STORE_BYTE_CNT is not populated, then this signal is always '0'.
radm_timeout_cpl_tag[(TAG_SIZE-1):0]	O	The Tag field of the timed out completion. Exists: Always Synchronous To: None,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_cpl_timeout is asserted

Port Name	I/O	Description
radm_cpl_lut_valid[(CPL_LUT_DEPTH-1):0]	O	Indicates which entries in the completion lookup table have valid entries stored. To populate this output, you must set the hidden configuration parameter (RADM_TAG_PORT_EN) using the set_configuration_parameter command. You cannot set it when you are using the AXI bridge module. When CX_EXTENSIBLE_VFUNC=1, RADM_TAG_PORT_EN is forced to 1. Exists: RADM_TAG_PORT_EN Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_ph_ca[(APP_CRD_WD-1):0]	I	One-cycle pulse indicating that posted header credits have been returned. In implementations where your application is responsible for queuing the receive data (such as in bypass mode), your application must pulse the app_ph_ca input each time your application consumes a posted header credit so that the controller can update the number of posted header credits available. There is 1 bit of app_ph_ca assigned to each configured virtual channel. Exists: APP_RETURN_CRD_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_pd_ca[(APP_CRD_WD-1):0]	I	One-cycle pulse indicating that posted data credits have been returned. In implementations where your application is responsible for queuing the receive data (such as in bypass mode), your application must pulse the app_pd_ca input each time your application consumes a posted data credit so that the controller can update the number of posted data credits available. Your application must assert app_pd_ca once for each packet, even if the packet contains less than 16 bytes (1 credit) of data. There is 1 bit of app_pd_ca assigned to each configured virtual channel. Exists: APP_RETURN_CRD_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
app_nph_ca[(APP_CRD_WD-1):0]	I	One-cycle pulse indicating that non-posted header credits have been returned. In implementations where your application is responsible for queuing the receive data (such as in bypass mode), your application must pulse the app_nph_ca input each time your application consumes a non-posted header credit so that the controller can update the number of non-posted header credits available. There is 1 bit of app_nph_ca assigned to each configured virtual channel. Exists: APP_RETURN_CRD_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_npd_ca[(APP_CRD_WD-1):0]	I	One-cycle pulse indicating that non-posted data credits have been returned. In implementations where your application is responsible for queuing the receive data (such as in bypass mode), your application must pulse the app_npd_ca input each time your application consumes a non-posted data credit so that the controller can update the number of non-posted data credits available. Your application must assert app_npd_ca once for each packet, even if the packet contains less than 16 bytes (1 credit) of data. There is 1 bit of app_npd_ca assigned to each configured virtual channel. Exists: APP_RETURN_CRD_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_cplh_ca[(APP_CRD_WD-1):0]	I	One-cycle pulse indicating that completion header credits have been returned. In implementations where your application is responsible for queuing the receive data (such as in bypass mode), your application must pulse the app_cplh_ca input each time your application consumes a completion header credit so that the controller can update the number of completion header credits available. There is 1 bit of app_cplh_ca assigned to each configured virtual channel. Exists: APP_RETURN_CRD_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
app_cpld_ca[(APP_CRD_WD-1):0]	I	One-cycle pulse indicating that completion data credits have been returned. In implementations where your application is responsible for queuing the receive data (such as in bypass mode), your application must pulse the app_cpld_ca input each time your application consumes a completion data credit so that the controller can update the number of completion data credits available. Your application must assert app_cpld_ca once for each packet, even if the packet contains less than 16 bytes (1 credit) of data. There is 1 bit of app_cpld_ca assigned to each configured virtual channel. Exists: APP_RETURN_CRD_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_dev_cache_invalid	I	CXL device indicate cache invalid state Exists: (CX_CXL_ENABLE) && (CX_CXL_2_0_ENABLE) && (CX_CXL_CACHE_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_dev_cache_wr_invl_d_cmpl	I	CXL device indicate cache writeback and invalidation is complete state Exists: (CX_CXL_ENABLE) && (CX_CXL_2_0_ENABLE) && (CX_CXL_CACHE_EN) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cxl_pm_init_cmpl	I	CXL PM initialization completion state Exists: (CX_CXL_ENABLE) && (CX_CXL_2_0_ENABLE) Synchronous To: aux_clk_g Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.92 SII: Reset/Status Signals

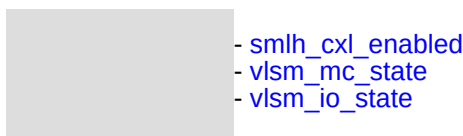


Table 2-92 SII: Reset/Status Signals

Port Name	I/O	Description
smlh_cxl_enabled[5:0]	O	CXL PhyLogical Layer LinkUp Enabled Bit 4,5 is applicable only if Version of CXL selected is 2, else driven 0 ■ bit 0: CXL mode linked up ■ bit 1: CXL.IO enabled ■ bit 2: CXL.MEM enabled ■ bit 3: CXL.CACHE enabled ■ bit 4: CXL 2.0 enabled ■ bit 5: Multi Logic Device enabled Exists: CX_CXL_ENABLE Synchronous To: None,perbitclk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
vlsm_mc_state[3:0]	O	This signal provides the CXL.cache/mem vLSM state. Byte2 Bit - Virtual LSM State Encoding for Status ALMP: ■ 0000: Reset (for Status ALMP) ■ 0000: Reserved (for Request ALMP) ■ 0001: Active ■ 0011: Reserved (for Status ALMP) ■ 0011: Deepest Allowable PM State (for Request ALMP) ■ 0100: IDLE_L1.0 (maps to PCIe L1) ■ 1000: L2 ■ 1011: Retrain (for Status ALMP) ■ 1011: Reserved (for Request ALMP) ■ All other encodings are Reserved. Exists: CX_CXL_ENABLE Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
vlsn_io_state[3:0]	O	This signal provides the CXL.io vLSM state. Byte2 Bit - Virtual LSM State Encoding for Status ALMP: ■ 0000: Reset (for Status ALMP) ■ 0000: Reserved (for Request ALMP) ■ 0001: Active ■ 0011: Reserved (for Status ALMP) ■ 0011: Deepest Allowable PM State (for Request ALMP) ■ 0100: IDLE_L1.0 (maps to PCIe L1) ■ 1000: L2 ■ 1011: Retrain (for Status ALMP) ■ 1011: Reserved (for Request ALMP) ■ All other encodings are Reserved. Exists: CX_CXL_ENABLE Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.93 SII: Transmit Control Signals

app_hdr_valid	-	pm_xtlh_block_tlp
app_hdr_log	-	app_parity_errs
app_tlp_prfx_log	-	app_tlp_errs
app_err_bus	-	xadm_ph_cdts
app_err_advisory	-	xadm_pd_cdts
app_err_func_num	-	xadm_nph_cdts
app_err_vfunc_num	-	xadm_npd_cdts
app_err_vfunc_active	-	xadm_cplh_cdts
app_crd_cpl_grant	-	xadm_cpld_cdts
app_crd_msg_grant	-	xadm_crd_msg_h_req
app_npd_control	-	xadm_crd_msg_d_req
		xadm_crd_cpl_h_req
		xadm_crd_cpl_d_req
		xadm_crd_msg_vc
		xadm_crd_cpl_vc
		xadm_crd_msg_consumed
		xadm_crd_cpl_consumed
		xadm_hcdt_consumed
		xadm_dcdt_consumed
		xadm_data_credit
		xadm_client_tlp_sent

Table 2-93 SII: Transmit Control Signals

Port Name	I/O	Description
pm_xtlh_block_tlp	O	Indicates that your application must stop generating new outgoing request TLPs due to the current power management state. Your application can continue to generate completion TLPs. For more information, see "Outbound TLP Blocking In USP (When AHX/AXI Bridge Not Used)". Exists: Always Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
app_parity_errs[2:0]	O	Indicates that the controller detected a datapath parity error, one bit for each of the following parity errors: ■ app_parity_errs[0]: Parity error at front end of the transmit datapath. ■ app_parity_errs[1]: Parity error at back end of the transmit datapath. ■ app_parity_errs[2]: Parity error the receive datapath. The app_parity_errs signals are one-cycle pulses that indicate the associated parity error occurred; they do not indicate which packet contained the parity error. A suggested usage of the app_parity_errs signals is to register each bit and to provide a control to turn off system notification of parity errors. By doing so, your application can choose to only respond to the first detection of a parity error. The controller performs transmit and receive datapath parity if data path protection is selected. Exists: APP_PAR_ERR_OUT_EN Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_tlp_errs[((NVC*3)-1):0]	O	Indicates that your application has transmitted a wrong TLP on the client interface. That is, the TLP has a wrong tc, or fmt, or type header field on the TLP. There are three bits per VC: ■ app_tlp_errs[3*VCn+0]: Wrong TLP detected on Client0 interface ■ app_tlp_errs[3*VCn+1]: Wrong TLP detected on Client1 interface ■ app_tlp_errs[3*VCn+2]: Wrong TLP detected on Client2 interface Where VCn denotes the virtual channel number ranging from 0 to 7. The app_tlp_errs signals are one-cycle pulses that indicate associated error. Exists: CX_MULTISTREAM Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
xadm_ph_cdts[((NVC*HCRD_WD)-1):0]	O	The amount of posted header buffer space currently available at the receiver at the other end of the link (in units of posted header credits). The controller maintains the current number of credits xadm_ph_cdts as the receiver continues to send UpdateFC DLLPs and your application uses credits by sending new TLPs to be transmitted. There are eight bits of xadm_ph_cdts assigned to each configured virtual channel. Only needed if you want your application to check available credits before submitting TLPs for transmission. Exists: XADM_CRD_EN Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_pd_cdts[((NVC*DCRD_WD)-1):0]	O	The amount of posted data buffer space currently available at the receiver at the other end of the link (in units of posted data credits). The controller maintains the current number of credits xadm_pd_cdts as the receiver continues to send UpdateFC DLLPs and your application uses credits by sending new TLPs to be transmitted. There are twelve bits of xadm_pd_cdts assigned to each configured virtual channel. Only needed if you want your application to check available credits before submitting TLPs for transmission. Exists: XADM_CRD_EN Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_nph_cdts[((NVC*HCRD_WD)-1):0]	O	The amount of non-posted header buffer space currently available at the receiver at the other end of the link (in units of non-posted header credits). The controller maintains the current number of credits xadm_nph_cdts as the receiver continues to send UpdateFC DLLPs and your application uses credits by sending new TLPs to be transmitted. There are eight bits of xadm_nph_cdts assigned to each configured virtual channel. Only needed if you want your application to check available credits before submitting TLPs for transmission. Exists: XADM_CRD_EN Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
xadm_npd_cdts[((NVC*DCRD_WD)-1):0]	O	The amount of non-posted data buffer space currently available at the receiver at the other end of the link (in units of non-posted data credits). The controller maintains the current number of credits xadm_npd_cdts as the receiver continues to send UpdateFC DLLPs and your application uses credits by sending new TLPs to be transmitted. There are twelve bits of xadm_npd_cdts assigned to each configured virtual channel. Only needed if you want your application to check available credits before submitting TLPs for transmission. Exists: XADM_CRD_EN Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_cplh_cdts[((NVC*HCRD_WD)-1):0]	O	The amount of completion header buffer space currently available at the receiver at the other end of the link (in units of completion header credits). The controller maintains the current number of credits xadm_cplh_cdts as the receiver continues to send UpdateFC DLLPs and your application uses credits by sending new TLPs to be transmitted. There are eight bits of xadm_cplh_cdts assigned to each configured virtual channel. Only needed if you want your application to check available credits before submitting TLPs for transmission. Exists: XADM_CRD_EN Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_cpld_cdts[((NVC*DCRD_WD)-1):0]	O	The amount of completion data buffer space currently available at the receiver at the other end of the link (in units of completion data credits). The controller maintains the current number of credits xadm_cpld_cdts as the receiver continues to send UpdateFC DLLPs and your application uses credits by sending new TLPs to be transmitted. There are twelve bits of xadm_cpld_cdts assigned to each configured virtual channel. Only needed if you want your application to check available credits before submitting TLPs for transmission. Exists: XADM_CRD_EN Synchronous To: ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
app_hdr_valid	I	One-clock-cycle pulse indicating that the data app_hdr_log, app_err_bus, app_err_func_num, and app_tlp_prfx_log is valid. Exists: APP_RETURN_ERR_EN Synchronous To: radm_clk_g,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_hdr_log[127:0]	I	The header of the TLP that contained the error indicated app_err_bus, valid when 1. app_hdr_valid is asserted. For Corrected Internal and Uncorrectable Internal errors (app_err_bus[10:9]),and receiver Overflow (app_err_bus[1]), the header information this input is not logged by the controller. 2. app_dpc_err_valid is asserted. The header of the TLP that contained the error indicated on app_dpc_err_bus. When DPC RP PIO exgtensions are supported, it is used to update DPC RP PIO Header Log Register. Exists: APP_RETURN_ERR_EN Synchronous To: perbitclk,radm_clk_g,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: app_hdr_valid or app_dpc_err_valid is asserted
app_tlp_prfx_log[127:0]	I	End-End TLP prefixes of the TLP that contained an error. Valid when 1. app_hdr_valid is asserted. The header of the TLP that contained the error indicated on app_err_bus. For Corrected Internal and Uncorrectable Internal errors (app_err_bus[10:9]),and receiver Overflow (app_err_bus[1]), the header information this input is not logged by the controller. 2. app_dpc_err_valid is asserted. The header of the TLP that contained the error indicated on app_dpc_err_bus. When DPC RP PIO exgtensions are supported, it is used to update DPC RP PIO Header Log Register. Exists: (APP_RETURN_ERR_EN) && (CX_TLP_PREFIX_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: app_hdr_valid is asserted

Port Name	I/O	Description
app_err_bus[(ERR_BUS_WD-1):0]	I	The type of error that your application detected. The controller combines the values the app_err_bus bits with the internally-detected error signals to set the corresponding bit in the Uncorrectable or Correctable Error Status Registers: ■ app_err_bus[0]: Malformed TLP ■ app_err_bus[1]: Receiver Overflow ■ app_err_bus[2]: Unexpected completion ■ app_err_bus[3]: Completer abort ■ app_err_bus[4]: Completion Timeout ■ app_err_bus[5]: Unsupported request ■ app_err_bus[6]: ECRC Check Failed ■ app_err_bus[7]: Poisoned TLP received ■ app_err_bus[8]: Reserved: set this bit to '0'. ■ app_err_bus[9]: Uncorrectable Internal Error ■ app_err_bus[10]: Corrected Internal Error ■ app_err_bus[11]: Reserved: set this bit to '0'. ■ app_err_bus[12]: ACS Violation: only valid when CX_ACS_ENABLE =1 ■ app_err_bus[13]: Poisoned TLP Egress Blocked ■ app_err_bus[14]: Reserved: set this bit to '0'. For more information, see 'Application Error Reporting Interface'. Exists: APP_RETURN_ERR_EN Synchronous To: None,perbitclk,radm_clk_g,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: app_hdr_valid is asserted

Port Name	I/O	Description
app_err_advisory	I	Indicates that your application error is an advisory error. Your application should assert app_err_advisory under either of the following conditions: ■ The controller is configured to mask completion timeout errors, your application is reporting a completion timeout error app_err_bus, and your application intends to resend the request. In such cases the error is an advisory error, as described in PCI Express Specification. When your application does not intend to resend the request, then your application must keep app_err_advisory de-asserted when reporting a completion timeout error. ■ The controller is configured to forward poisoned TLPs to your application and your application is going to treat the poisoned TLP as a normal TLP, as described in PCI Express Specification. Upon receipt of a poisoned TLP, your application must report the error app_err_bus, and either assert app_err_advisory (to indicate an advisory error) or de-assert app_err_advisory (to indicate that your application is dropping the TLP). For more information, see the PCI Express Specification to determine when an application error is an advisory error. To configure the controller, see "Disabling of Error Filtering". Exists: APP_RETURN_ERR_EN Synchronous To: radm_clk_g,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: app_hdr_valid is asserted
app_err_func_num[(PF_WD-1):0]	I	The number of the function that is reporting the error indicated app_err_bus, valid when app_hdr_valid is asserted. Correctable and Uncorrected Internal errors (app_err_bus[10:9]) are not function specific, and are recorded for all physical functions, regardless of the value this bus. Function numbering starts at '0'. Exists: APP_RETURN_ERR_EN Synchronous To: radm_clk_g,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: app_hdr_valid is asserted

Port Name	I/O	Description
app_err_vfunc_num[(VF_WD-2):0]	I	The number of the virtual function that is reporting the error indicated app_err_bus, valid when app_hdr_valid is asserted. Uncorrected Internal errors (app_err_bus[9]) are not recorded for virtual functions. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, app_err_vfunc_num=0 refers to the first VF of the PF that is identified by app_err_func_num. For more information of VF numbering, see 'Programming View'. When CX_EXTENSIBLE_VFUNC=1, this signal is only for internal VFs. Exists: (APP_RETURN_ERR_EN) && (CX_SRIOV_ENABLE && (CX_INTERNAL_NVFUNC != 0)) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: app_err_vfunc_active is asserted
app_err_vfunc_active	I	Indicates the function number in app_err_vfunc_num is valid. Exists: (APP_RETURN_ERR_EN) && (CX_SRIOV_ENABLE && (CX_INTERNAL_NVFUNC != 0)) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: app_hdr_valid is asserted
app_crd_cpl_grant	I	Application grants the use of credits for internally generated completions. Exists: APP_CREDIT_CTRL Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_crd_msg_grant	I	Application grants the use of credits for internally generated messages. Exists: APP_CREDIT_CTRL Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
xadm_crd_msg_h_req	O	The controller requests app for grant to send internally generated msg (consuming header credit). Exists: APP_CREDIT_CTRL Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_crd_msg_d_req	O	The controller requests app for grant to send internally generated msg (consuming data credit). Exists: APP_CREDIT_CTRL Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_crd_cpl_h_req	O	The controller requests app for grant to send internally generated cpl (consuming header credit). Exists: APP_CREDIT_CTRL Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_crd_cpl_d_req	O	The controller requests app for grant to send internally generated cpl (consuming data credit). Exists: APP_CREDIT_CTRL Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_crd_msg_vc[2:0]	O	Indicates the VC id for the msg request. Exists: APP_CREDIT_CTRL Synchronous To: None,{core_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

Port Name	I/O	Description
xadm_crd_cpl_vc[2:0]	O	Indicates the VC id for the cpl request. Exists: APP_CREDIT_CTRL Synchronous To: None,{core_clk} Registered: Yes Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
xadm_crd_msg_consumed	O	Indicates the granted MSG has been consumed. Exists: APP_CREDIT_CTRL Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_crd_cpl_consumed	O	Indicates the granted CPL has been consumed. Exists: APP_CREDIT_CTRL Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_hcdt_consumed[((3*NVC)-1):0]	O	Indicates header credits has been consumed. ■ Bit 2 indicates Posted ■ Bit 1 indicates Non-posted ■ Bit 0 indicates Completion Exists: APP_CREDIT_CTRL Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_dcdt_consumed[((3*NVC)-1):0]	O	Indicates data credits has been consumed. ■ Bit 2 indicates Posted ■ Bit 1 indicates Non-posted ■ Bit 0 indicates Completion Exists: APP_CREDIT_CTRL Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
xadm_data_credit[8:0]	O	Indicates how many data credits has been consumed. Exists: APP_CREDIT_CTRL Synchronous To: core_clk Registered: Yes Power Domain: PD_VAUX Active State: High Validated by: xadm_dcdt_consumed
app_npd_control	I	When asserted, the controller relaxes the check of NPD credits such that it never causes throttling of the client interface for other FC Types. The controller assumes the completion of transmitted NP TLPs check to ensure enough credit was available prior to transmission. You cannot toggle this signal when the traffic is ongoing. Exists: CX_APP_NPD_CONTROL Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xadm_client_tlp_sent[((NVC*7)-1):0]	O	Indicates the TLP transmitted by the controller. Your application can use this information for TLP ordering. This is a one-hot signal. For each VC the bit definition is as follows: ■ Bit0: Internally generated message by MSG_GEN module ■ Bit1: Completion TLP generated by LBC ■ Bit2: Completion TLP transmitted by your application on Client0 ■ Bit3: Non-Posted TLP transmitted by your application on Client1 ■ Bit4: Posted TLP transmitted by your application on Client2 ■ Bit5: Reserved ■ Bit6: Reserved Exists: CX_MULTISTREAM Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.94 Common PIPE Signals

phy_mac_rxelecidle	-	mac_phy_powerdown
phy_mac_phystatus	-	adm_adaptor_powerdown
phy_mac_rxdata	-	mac_phy_txdata
phy_mac_rxdatak	-	mac_phy_txdatak
phy_mac_rxvalid	-	mac_phy_elasticbuffermode
phy_mac_rxstatus	-	mac_phy_txdatavalid
phy_mac_rxstandbystatus	-	mac_phy_txdetectrx_loopback
phy_cfg_status	-	mac_phy_txelecidle
phy_mac_pclkchangeok	-	mac_phy_txcompliance
phy_mac_maxpclkack_n	-	mac_phy_rxpolarity
phy_mac_rxdatavalid	-	mac_phy_width
phy_mac_messagebus	-	mac_phy_serdes_arch
phy_mac_pclkack_n	-	mac_phy_rxwidth
		serdes_pipe_rxready
		serdes_pipe_turnoff_lanes
		mac_phy_pclk_rate
		mac_phy_cclk_rate
		mac_phy_plclk_rate
		mac_phy_rxstandby
		pm_current_data_rate
		mac_phy_pclkchangeack
		mac_phy_maxpclkreq_n
		mac_phy_messagebus
		cfg_phy_control
		mac_phy_pclkreq_n
		mac_phy_rxelecidle_disable
		mac_phy_txcommonmode_disable
		mac_phy_asyncpowerchangeack

Table 2-94 Common PIPE Signals

Port Name	I/O	Description
mac_phy_powerdown[(PDWN_WIDTH-1):0]	O	Power control bits that drive the PHY power state. This signal is shared by all the lanes of the PHY. so you must fan out this signal (through your power-down controller block) to all the corresponding lanes of the PHY. Power states (with the corresponding link states and L1 Substates) are as follows: ■ 0000: P0 (L0): normal ■ 0001: P0s (L0s): low recovery time, power saving. ■ 0010: P1 (L1): longer recovery time, additional power saving. ■ 0011: P2 (L2): lowest power state. The following encodings are used when CX_L1_SUBSTATES_ENABLE = CX_PIPE43_SUPPORT = 1. You can specify the encodings using CX_PIPE43_P1CPM_ENCODING, CX_PIPE43_P1_1_ENCODING, and CX_PIPE43_P1_2_ENCODING parameters. ■ CX_PIPE43_P1CPM_ENCODING: P1.CPM (L1 Substates L1.0): PHY specific powerdown state, instructs PHY to turn off PCLK. Controller stays in this state until PHY asserts phy_mac_phystatus back. ■ CX_PIPE43_P1_1_ENCODING: P1.1 (L1 Substates L1.1): PHY specific powerdown state, instructs PHY to disable Receiver Electrical Idle detection logic. If CX_PIPE43_ASYNC_HS_BYPASS is not set, controller stays in this state until the asynchronous powerdown change handshake completes. ■ CX_PIPE43_P1_2_ENCODING: P1.2 (L1 Substates L1.2): PHY specific powerdown state, instructs PHY to disable Receiver Electrical Idle detection logic and Transmitter Common Mode logic. When CX_PIPE43_ASYNC_HS_BYPASS is not set, the controller stays in this state until the asynchronous powerdown change handshake completes. This is a 4-bit output to support PIPE 4.3 version. When your PHY is not PIPE 4.3 compliant, use the lower bits for connection; the controller drives the MSB bits to '0'. The following encoding is used when CX_P2NOBEACON_ENABLE = 1 and PHY_INTEROP_CTRL_OFF.P2NOBEACON_ENABLE = 1. ■ CX_PIPE43_P2NOBEACON_ENCODING: P2.NoBeacon: When CX_P2NOBEACON_ENABLE = 1, mac_phy_powerdown drives P2.NoBeacon encoding instead of P2 encoding, when the link goes to L2. To configure P2.NoBeacon encoding use CX_PIPE43_P2NOBEACON_ENCODING parameter. The default value of this parameter is 4'b1111. Exists: (CX_CPCIE_ENABLE) Synchronous To: None,perbitclk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
phy_mac_rxelecidle[(NL-1):0]	I	Indicates receiver detection of an Electrical Idle for each lane. This can be signaled asynchronously by the PHY macro. ■ Bit 0 corresponds to lane 0 ■ Bit 1 corresponds to lane 1 ■ and so on, up to the maximum number of lanes. You can set a DFT control point to force this controller input to '1' during scan mode to make the pc_rxelecidle register in the rxidle_squelch module scannable. This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: ((CX_CPCIE_ENABLE)) && (!CX_ADM_ADAPTOR_ENABLE) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_phystatus[(NL-1):0]	I	Communicates completion of PHY functions, including power management transitions, receiver detection, speed change, and EQ evaluation at Gen3 rate. The core_clk should not be shut off before the PHY acknowledges the state change with PHYstatus. This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: (CX_CPCIE_ENABLE) Synchronous To: None,aux_clk,core_clk_ug,perbitclk,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
adm_adaptor_powerdown[(NL-1):0]	O	Emulates completion of PHY functions for Embedded Endpoint solution. This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: ((CX_CPCIE_ENABLE)) && (CX_ADM_ADAPTOR_ENABLE) Synchronous To: CX_FREQ_STEP_EN ? "pipe_clk" : "core_clk" Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
phy_mac_rxdata[(PIPE_DATA_WD-1):0]	I	Parallel received data. For Original PIPE architecture: ■ Bits 7:0 correspond to the first symbol of lane 0. ■ Bits 15:8 correspond to the second symbol of lane 0 (16-bit PHY interface) or the single symbol of lane 1 (8-bit PHY interface). ■ Bits 23:16 correspond to the third symbol of lane 0 (32-bit PHY interface) or the single symbol of lane 2 (8-bit PHY interface) or the first symbol of lane 1 (16-bit PHY interface). ■ Bits 31:24 correspond to the fourth symbol of lane 0 (32-bit PHY interface) or the single symbol of lane 3 (8-bit PHY interface) or the second symbol of lane 1 (16-bit PHY interface). The remaining bits continue similarly. For SerDes architecture: ■ Bits 9:0 correspond to the first symbol of lane 0. ■ Bits 19:10 correspond to the second symbol of lane 0 (20-bit PHY interface) or the single symbol of lane 1 (10-bit PHY interface). ■ Bits 29:20 correspond to the third symbol of lane 0 (40-bit PHY interface) or the single symbol of lane 2 (10-bit PHY interface) or the first symbol of lane 1 (20-bit PHY interface). ■ Bits 39:30 correspond to the fourth symbol of lane 0 (40-bit PHY interface) or the single symbol of lane 3 (10-bit PHY interface) or the second symbol of lane 1 (20-bit PHY interface). The remaining bits continue similarly. Bit 0 is the first bit received through link. For 128b/130b encoded data at Gen3 speed or higher, only 8 bits out of 10-bit slice are used, for example, [7:0] represent byte 0, [9:8] are reserved, [17:10] represent byte 1, and [19:18] are reserved, and so on. Exists: (CX_CPCIE_ENABLE) Synchronous To: aux_clk_g,core_clk_ug,ret_core_clk,{core_clk},{pipe_clk},{pipe_rx_clk[0]},{pipe_rx_clk[10]},{pipe_rx_clk[11]},{pipe_rx_clk[12]},{pipe_rx_clk[13]},{pipe_rx_clk[14]},{pipe_rx_clk[15]},{pipe_rx_clk[1]},{pipe_rx_clk[2]},{pipe_rx_clk[3]},{pipe_rx_clk[4]},{pipe_rx_clk[5]},{pipe_rx_clk[6]},{pipe_rx_clk[7]},{pipe_rx_clk[8]},{pipe_rx_clk[9]} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: phy_mac_rxvalid is asserted, per lane. In SerDes architecture, phy_mac_rxdata is not validated by another signal.

Port Name	I/O	Description
phy_mac_rxdatak[((NL*PHY_NB)-1):0]	I	Control (K character) indicator bits for received data: ■ Bit 0 corresponds to the first symbol of lane 0. ■ Bit 1 corresponds to the second symbol of lane 0 (16-bit PHY interface) or the single symbol of lane 1 (8-bit PHY interface). ■ Bit 2 corresponds to the third symbol of lane 0 (32-bit PHY interface) or the single symbol of lane 2 (8-bit PHY interface) or the first symbol of lane 1 (16-bit PHY interface). ■ Bits 3 corresponds to the fourth symbol of lane 0 (32-bit PHY interface) or the single symbol of lane 3 (8-bit PHY interface) or the second symbol of lane 1 (16-bit PHY interface). The remaining bits continue similarly. Exists: ((CX_CPCIE_ENABLE)) && (!CX_PIPE_SERDES_ARCH) Synchronous To: aux_clk_g,core_clk_ug,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High indicates control symbol, low indicates data symbol. Validated by: phy_mac_rxvalid is asserted, per lane
phy_mac_rxvalid[(NL-1):0]	I	Indicates symbol lock and valid data for each lane. Bit 0 corresponds to lane 0. For SerDes architecture, this signal is used to indicate that pipe_rx_clk is stable. Exists: (CX_CPCIE_ENABLE) Synchronous To: aux_clk_g,core_clk_ug,ret_core_clk,{core_clk},{pipe_clk},{pipe_rx_clk[0]},{pipe_rx_clk[10]},{pipe_rx_clk[11]},{pipe_rx_clk[12]},{pipe_rx_clk[13]},{pipe_rx_clk[14]},{pipe_rx_clk[15]},{pipe_rx_clk[1]},{pipe_rx_clk[2]},{pipe_rx_clk[3]},{pipe_rx_clk[4]},{pipe_rx_clk[5]},{pipe_rx_clk[6]},{pipe_rx_clk[7]},{pipe_rx_clk[8]},{pipe_rx_clk[9]} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
phy_mac_rxstatus[((NL*3)-1):0]	I	Receive status and error codes for each lane. Bits 2:0 correspond to lane 0; bits 5:3 correspond to lane 1, and so on, up to the maximum number of lanes. Encoding is as follows: ■ 000: Received data OK ■ 001: A SKP set has been added. For Gen1/2, a SKP set is one byte. For Gen3, a SKP set is four bytes. ■ 010: A SKP set has been removed. For Gen1/2: A SKP set is one byte. For Gen3: A SKP set is four bytes ■ 011: Receiver detected ■ 100: Decode error. For Gen1/2, this indicates detection of an 8b/10b decode error. For Gen3, this indicates detection of a 128b/130b decode error. ■ 101: Elastic buffer overflow ■ 110: Elastic buffer underflow ■ 111: Receive disparity error. The controller does not report disparity errors at Gen3 speed. The controller supports Bare COM. When the PHY removes SKP symbols it assigns phy_mac_rxstatus[2:0] = 3'b010 aligned to COM for Gen1/2 rate or to phy_mac_rxstartblock=1 for Gen3/4 rate at the same clock cycle. If the controller is at Gen1 or Gen2 rate and if it receives two consecutive SKP OSs (two COMs) with phy_mac_rxstatus = 3'b010 in the same clock cycle, the controller assumes that the phy_max_rxstatus = 3'b010 applies to the first received SKP OS (the lower byte-located COM). For SerDes architecture, the only status applicable is '011: Receiver detected'. Exists: (CX_CPCIE_ENABLE) Synchronous To: aux_clk_g,core_clk_ug,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
phy_mac_rxstandbystatus[(NL-1):0]	I	The PHY uses this signal to indicate its RxStandby state. ■ 0: Active ■ 1: Standby The Synopsys controller uses this signal only when CX_RXSTANDBY_CONTROL[6] =1. Contact your PHY vendor for guidelines on how to properly configure the RxStandby behavior. For more information on how to configure this parameter for each supported Synopsys PHY, see the <i>User Guide</i>. Exists: (CX_CPCIE_ENABLE) Synchronous To: aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
phy_cfg_status[31:0]	I	Input bus that can optionally be used to read PHY status. The phy_cfg_status bus maps to the PHY Status register (PHY_STATUS_REG). For specific bit usage, see 'PHY Status Register'. Exists: (CX_CPCIE_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_txdata[(PIPE_DATA_WD-1):0]	O	Parallel data for transmission. For Original PIPE architecture: ■ Bits 7:0 correspond to the first symbol of lane 0. ■ Bits 15:8 correspond to the second symbol of lane 0 (16-bit PHY interface) or the single symbol of lane 1 (8-bit PHY interface). ■ Bits 23:16 correspond to the third symbol of lane 0 (32-bit PHY interface) or the single symbol of lane 2 (8-bit PHY interface) or the first symbol of lane 1 (16-bit PHY interface). ■ Bits 31:24 correspond to the fourth symbol of lane 0 (32-bit PHY interface) or the single symbol of lane 3 (8-bit PHY interface) or the second symbol of lane 1 (16-bit PHY interface). The remaining bits continue similarly. For SerDes architecture: ■ Bits 9:0 correspond to the first symbol of lane 0. ■ Bits 19:10 correspond to the second symbol of lane 0 (20-bit PHY interface) or the single symbol of lane 1 (10-bit PHY interface). ■ Bits 29:20 correspond to the third symbol of lane 0 (40-bit PHY interface) or the single symbol of lane 2 (10-bit PHY interface) or the first symbol of lane 1 (20-bit PHY interface). ■ Bits 39:30 correspond to the fourth symbol of lane 0 (40-bit PHY interface) or the single symbol of lane 3 (10-bit PHY interface) or the second symbol of lane 1 (20-bit PHY interface). The remaining bits continue similarly. Bit 0 is the first bit to be transmitted through link. For 128b/130b encoded data at Gen3 speed or higher, only 8 bits out of each 10-bit slice are used, for example, [7:0] represent byte 0, [9:8] are reserved, [17:10] represent byte 1, and [19:18] are reserved, and so on. Exists: (CX_CPCIE_ENABLE) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk_ug},{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: mac_phy_txelecidle is de-asserted, per lane

Port Name	I/O	Description
mac_phy_txdatak[((NL*PHY_NB)-1):0]	O	Control (K character) indicator bits for transmitted data: ■ Bit 0 corresponds to the first symbol of lane 0. ■ Bit 1 corresponds to the second symbol of lane 0 (16-bit PHY interface) or the single symbol of lane 1 (8-bit PHY interface). ■ Bit 2 corresponds to the third symbol of lane 0 (32-bit PHY interface) or the single symbol of lane 2 (8-bit PHY interface) or the first symbol of lane 1 (16-bit PHY interface). ■ Bits 3 corresponds to the fourth symbol of lane 0 (32-bit PHY interface) or the single symbol of lane 3 (8-bit PHY interface) or the second symbol of lane 1 (16-bit PHY interface). The remaining bits continue similarly. Exists: ((CX_CPCIE_ENABLE)) && !(CX_PIPE_SERDES_ARCH) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High indicates control symbol, low indicates data symbol. Validated by: mac_phy_txelecidle is de-asserted, per lane
mac_phy_elasticbuffermode	O	Selects Elasticity Buffer operating mode. 0: Nominal Half Full Buffer mode 1: Nominal Empty Buffer Mode Exists: ((CX_CPCIE_ENABLE)) && !(CX_PIPE5_SUPPORT) Synchronous To: aux_clk,{None},{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_txdatavalid[(NL-1):0]	O	This signal allows the PCIe controller to instruct the PHY to ignore the data interface for one or two clock cycles. A value of one indicates that the data is valid and should be used. The PCIe controller asserts this output during mac_phy_txelecidle. If data throttling is enabled, see the "Data Throttling (DT) Support" section in the Product Overview chapter of the <i>Databook</i>. Note: If your PHY does not support the mac_phy_txdatavalid signal you can leave this pin unconnected. Exists: (CX_CPCIE_ENABLE) Synchronous To: aux_clk,aux_clk_g,core_clk,ug,ret_core_clk,{None},{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_txdetectrx_loopback[(NL-1):0]	O	Combined loopback and transmit detect control, as per PIPE specification. Note: mac_phy_txdetectrx_loopback[0] may be connected to PHYs that share mac_phy_txdetectrx_loopback between their lanes only when the PHY asserts phystatus on all lanes, in the same clock cycle, for receiver's detection. For SerDes architecture, this signal is only used for receiver detection. Exists: (CX_CPCIE_ENABLE) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk_ug},{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_txelecidle[((NL*PHY_TXEI_WD)-1):0]	O	Forces transmit output to Electrical Idle for each lane which it is asserted. For PIPE4.4 or below, bit 0 corresponds to lane 0. Bit 1 corresponds to lane 1, and so on, up to the maximum number of lanes. For PIPE5 Original PIPE architecture, only bit 0 is used for lane 0 and bit 3:1 are reserved. Only bit 4 is used for lane 1 and bit 7:5 are reserved. The remaining bits continue similarly. For SerDes architecture, one bit is required per two symbols of interface data. For example, for an eight symbol wide interface, bit 0 would apply to symbols 0 and 1, bit 1 would apply to symbols 2 and 3, bit 2 would apply to symbols 4 and 5, bit 3 would apply to symbol 6 and 7 of lane 0. For narrower interfaces, unused bits of this signal are reserved. Exists: (CX_CPCIE_ENABLE) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk_ug},{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_txcompliance[(NL-1):0]	O	Sets the running disparity to negative. Used when transmitting compliance pattern. Bit 0 corresponds to lane 0; bit 1 corresponds to lane 1, and so on, up to the maximum number of lanes. Exists: ((CX_CPCIE_ENABLE)) && (!CX_PIPE_SERDES_ARCH) Synchronous To: aux_clk,aux_clk_g,core_pl_clk,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_rxpolarity[(NL-1):0]	O	Directs the PHY to perform a polarity inversion the received data the specified lanes. Bit 0 corresponds to lane 0; bit 1 corresponds to lane 1, and so on, up to the maximum number of lanes. 1: Polarity inversion 0: No polarity inversion Exists: ((CX_CPCIE_ENABLE)) && !(CX_PIPE5_SUPPORT) Synchronous To: aux_clk,aux_clk_g,core_clk_ug,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_width[(WIDTH_WIDTH-1):0]	O	Controls the PIPE datapath width. For SerDes architecture, this applies only to the transmit side and mac_phy_rxwidth controls the receive side. For Original PIPE architecture: 0: 8 bits 1: 16 bits 2: 32 bits 3: 64 bits 4: 128 bits For SerDes architecture: 0: 10 bits 1: 20 bits 2: 40 bits 3: 80 bits 4: 160 bits Exists: (CX_CPCIE_ENABLE) Synchronous To: None,aux_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_serdes_arch	O	Indicates whether SerDes architecture is enabled. Exists: ((CX_CPCIE_ENABLE)) && (CX_PIPE5_SUPPORT) Synchronous To: aux_clk,{None} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_rxwidth[(WIDTH_WIDTH-1):0]	O	Controls the PIPE receive datapath width. This signal is only used in the SerDes architecture. ■ 0: 10 bits ■ 1: 20 bits ■ 2: 40 bits ■ 3: 80 bits ■ 4: 160 bits Exists: ((CX_CPCIE_ENABLE)) && (CX_PIPE_SERDES_ARCH) Synchronous To: aux_clk,perbitclk,{None},{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
serdes_pipe_rxready[(NL-1):0]	O	Indicates symbol lock/block alignment is done for each lane. Bit 0 corresponds to lane 0. Exists: ((CX_CPCIE_ENABLE)) && (CX_PIPE_SERDES_ARCH) Synchronous To: aux_clk,aux_clk_g,core_clk,pipe_rx_clk[0],pipe_rx_clk[10],pipe_rx_clk[11],pipe_rx_clk[12],pipe_rx_clk[13],pipe_rx_clk[14],pipe_rx_clk[15],pipe_rx_clk[1],pipe_rx_clk[2],pipe_rx_clk[3],pipe_rx_clk[4],pipe_rx_clk[5],pipe_rx_clk[6],pipe_rx_clk[7],pipe_rx_clk[8],pipe_rx_clk[9],{core_clk_ug},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
serdes_pipe_turnoff_lanes[(NL-1):0]	O	Indicates lanes are turned off. Bit 0 corresponds to lane 0. This signal behavior follows the 'turn off' functionality described in PIPE specification. Exists: ((CX_CPCIE_ENABLE)) && (CX_SERDES_ARCH_LANES_TURNOFF) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_pclk_rate[(P_R_WD-1):0]	O	Controls the PIPE PCLK rate ■ 0: 62.5 MHz ■ 1: 125 MHz ■ 2: 250 MHz ■ 3: 500 MHz ■ 4: 1000 MHz ■ 5: 2000 MHz ■ 6: Reserved ■ 7: Reserved ■ 8: 625 MHz if CX_CCIX_ESM_SUPPORT = 1 ■ 9: 781.25 MHz if CX_CCIX_ESM_SUPPORT = 1 ■ A: 1250 MHz if CX_CCIX_ESM_SUPPORT = 1 ■ B: 1562.5 MHz if CX_CCIX_ESM_SUPPORT = 1 ■ C: 156.25MHz if CX_CCIX_ESM_SUPPORT = 1 ■ D: 195.31MHz if CX_CCIX_ESM_SUPPORT = 1 ■ E: 312.5MHz if CX_CCIX_ESM_SUPPORT = 1 ■ F: 390.62MHz if CX_CCIX_ESM_SUPPORT = 1 Exists: (CX_CPCIE_ENABLE) Synchronous To: None,aux_clk,aux_clk_g,core_pl_clk,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_cclk_rate[2:0]	O	Controls clock frequency ratio between the core_clk and the core_pl_clk. ■ 0: Reserved ■ 1: 1:1 ■ 2: 1:2 ■ 3: 1:4 ■ 4: 1:8 ■ 5: 1:16 ■ 6: Reserved ■ 7: Reserved Exists: ((CX_CPCIE_ENABLE)) && (CX_FREQ_STEP_DL_EN) Synchronous To: None,aux_clk_g,core_pl_clk,perbitclk,{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_plclk_rate[2:0]	O	Controls clock frequency ratio between the core_pl_clk and the pipe_clk. ■ 0: 2:1 ■ 1: 1:1 ■ 2: 1:2 ■ 3: 1:4 ■ 4: 1:8 ■ 5: 1:16 ■ 6: Reserved ■ 7: Reserved Exists: ((CX_CPCIE_ENABLE)) && (CX_FREQ_STEP_DL_EN) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_rxstandby[(NL-1):0]	O	Determines if the PHY RX is active when the PHY is in P0 or P0s. ■ 0: Active ■ 1: Standby The controller asserts the signal as follows: ■ if Reset mac_phy_rxstandby = CX_RXSTANDBY_DEFAULT; ■ else if CX_RXSTANDBY_CONTROL[0] =1 & receiving EIOS & subsequent T TX-IDLE-MIN mac_phy_rxstandby = 1; Note: In Loopback.Active state as Slave at Gen2 data rate or above, EIOSs indicates four EIOSs. In the other cases, EIOSs indicates one EIOS. ■ else if CX_RXSTANDBY_CONTROL[1] =1 & for the duration of rate change (from one cycle before rate signal change to receiving all active lanes' PhyStatus for the rate change request) mac_phy_rxstandby = 1; ■ else if CX_RXSTANDBY_CONTROL[2] =1 & the lane is inactive when LTSSM exiting from Configuration.Idle state mac_phy_rxstandby =1; ■ else if CX_RXSTANDBY_CONTROL[3] =1 & mac_phy_powerdown =(P1 P2) mac_phy_rxstandby =1; ■ else if CX_RXSTANDBY_CONTROL[4] =1 & LTSSM is in RxL0s.Idle state mac_phy_rxstandby =1; ■ else if CX_RXSTANDBY_CONTROL[5] =1 & LTSSM moves from L0 state to Recovery state by Inferring Electrical Idle mac_phy_rxstandby =1; ■ else if phy_mac_rxelecidle =0 mac_phy_rxstandby =0; ■ else mac_phy_rxstandby =mac_phy_rxstandby. ■ If CX_RXSTANDBY_CONTROL[6] =1 □ controller checks phy_mac_rxstandbystatus =0 before changing mac_phy_rxstandby from '0' to '1'. □ controller checks phy_mac_rxstandbystatus =1 before changing mac_phy_rxstandby from '1' to '0'. The behavior of this signal is controlled by the CX_RXSTANDBY_CONTROL and CX_RXSTANDBY_DEFAULT configuration parameters.

Port Name	I/O	Description
mac_phy_rxstandby[(NL-1):0]...(cont.)	O.	Contact your PHY vendor for guidelines on how to properly configure the RxStandby behavior. For more information on how to configure this parameter for each supported Synopsys PHY, see the <i>User Guide</i>. Exists: (CX_CPCIE_ENABLE) Synchronous To: aux_clk,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pm_current_data_rate[2:0]	O	The current link operating rate. ■ 0: 2.5 GT/s signaling rate ■ 1: 5.0 GT/s signaling rate ■ 2: 8.0 GT/s signaling rate for PCIe mode or ESM data rate0 for ESM mode ■ 3: 16.0 GT/s signaling rate for PCIe mode or ESM data rate1 for ESM mode ■ 4: 32.0 GT/s signaling rate Exists: (CX_CPCIE_ENABLE) Synchronous To: core_pl_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
phy_mac_pclkchangeok[(NL-1):0]	I	The PHY asserts this signal when it is ready for the MAC to change the PCLK rate. The PHY only asserts this signal after the MAC has requested a PCLK rate change by changing PCLK Rate. Exists: ((CX_CPCIE_ENABLE)) && ((CX_PIPE_PCLK_MODE==1)) Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_pclkchangeack[(NL-1):0]	O	MAC asserts this signal when a PCLK rate change is complete and stable. After the MAC asserts mac_phy_pclkchangeack, the PHY responds by asserting PhyStatus for one cycle and de-asserting phy_mac_pclkchangeok at the same time as PhyStatus. The controller de-asserts mac_phy_pclkchangeack when phy_mac_pclkchangeok is sampled low. Exists: ((CX_CPCIE_ENABLE)) && ((CX_PIPE_PCLK_MODE==1)) Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_maxpclkreq_n	O	The controller asserts this signal to request the PHY to remove MAXPCLK. The controller de-asserts this signal to request the PHY to enable MAXPCLK. Exists: ((CX_CPCIE_ENABLE)) && (CX_PIPE_PCLK_MODE > 0) Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_maxpclkack_n	I	The PHY asserts this signal indicating the removal of MAXPCLK. The PHY de-asserts this signal to indicate that MAXPCLK is enabled and is stable. Exists: ((CX_CPCIE_ENABLE)) && (CX_PIPE_PCLK_MODE > 0) Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
phy_mac_rxdatavalid[(NL-1):0]	I	This signal allows a PHY to indicate valid data on the PIPE interface. A value of '1' indicates that the data is valid and should be used. If your PHY does not support this signal, then set it to '1'. You should set this signal at the different link rates as follows: ■ Gen1/2: If data throttling is enabled in this mode, then toggle every other or fourth clock, else '1' ■ Gen3 or higher: If data throttling is enabled in this mode, then toggle every other or fourth clock except for regular de-assertion for 128b/130b encoding/decoding, else '1' except for regular de-assertion for 128b/130b encoding/decoding For more information, see the "Data Throttling (DT) Support" section in the Product Overview chapter of the <i>Databook</i>. Clock tolerance compensation is normally done by inserting or removing SKIP symbols from a SKIP ordered set (OS). The controller also supports a non-standard clock tolerance compensation mode at Gen3 speed. In this mode SKIP OS's can be left unaltered in length. Clock tolerance compensation is then achieved by modulating the period of de-assertion of phy_mac_rxdatavalid. Therefore this input might be de-asserted anywhere within a block (and not only at the end of a block), and the low pulses can be up to two cycles long. Exists: ((CX_CPCIE_ENABLE)) && (!CX_ADM_ADAPTOR_ENABLE) && (!CX_PIPE_SERDES_ARCH) Synchronous To: None,aux_clk_g,core_clk_ug,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_messagebus[((NL*8)-1):0]	O	PIPE Compliant MAC driven message bus commands to PHY. Exists: ((CX_CPCIE_ENABLE)) && ((CX_GEN4_SPEED && CX_PIPE44_SUPPORT) CX_PIPE5_SUPPORT) Synchronous To: aux_clk,ret_core_clk_ug,ret_pipe_clk,{core_clk_ug},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_messagebus[((NL*8)-1):0]	I	PIPE Compliant PHY driven message bus commands to MAC. Exists: ((CX_CPCIE_ENABLE)) && ((CX_GEN4_SPEED && CX_PIPE44_SUPPORT) CX_PIPE5_SUPPORT) Synchronous To: ret_core_clk_ug,ret_pipe_clk,{core_clk_ug},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_phy_control[31:0]	O	Output bus that can optionally be used for additional PHY control purposes. The cfg_phy_control bus maps to the PHY Control register (PHY_CONTROL_REG). You can use app_hold_phy_rst to keep your PHY in reset, if you need to perform changes during the power on sequence. <i>Note</i> This bus is reserved when the controller is connected to a Synopsys PHY and must be connected according to the reference integration files available in <workspace>/src/Phy/phy_instance.vh. Exists: Always Synchronous To: aux_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_pclkreq_n[1:0]	O	Request PCLK removal: ■ 00: Do not request PCLK removal ■ 01: Request PCLK removal for executing L1 with Clock PM ■ 10: Request PCLK removal for executing L1 substates ■ 11: Request PCLK removal for executing L1 substates with Clock PM When mac_phy_pclkreq_n[1:0] transitions to a non-zero value, the signal is held constant until phy_mac_pclkack_n =1. Similarly, when mac_phy_pclkreq_n[1:0] transitions back to a zero value the signal is held constant until phy_mac_pclkack_n=0. Exists: !CX_PIPE43_SUPPORT Synchronous To: None,perbitclk,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
phy_mac_pclkack_n	I	PCLK removal request acknowledge. ■ 1: PCLK is removed, PHY is ready for REFCLK removal ■ 0: PCLK is active This signal is synchronized using aux_clk and can be driven/supplied asynchronously to the controller in certain low-power modes. Exists: (CX_L1_SUBSTATES_ENABLE) && (!CX_PIPE43_SUPPORT) Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_rxelecidle_disable	O	Instructs the PHY to disable the receiver Electrical Idle detection logic. For more information, see 'L1 Substates'. Exists: CX_L1_SUBSTATES_ENABLE Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_txcommonmode_disable	O	Instructs the PHY to disable the transmitter Common Mode logic. For more information, see 'L1 Substates'. Exists: CX_L1_SUBSTATES_ENABLE Synchronous To: aux_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_asyncpowerchangeack	O	Acknowledgement of asynchronous phy_mac_phystatus feedback for powerdown change complete in P1.1 or P1.2 state. The acknowledgment handshake mechanism is: ■ PHY asserts phy_mac_phystatus asynchronously when it completes P1.1 or P1.2 powerdown change. ■ MAC asserts mac_phy_asyncpowerchangeack when it detects phy_mac_phystatus high. ■ PHY de-asserts phy_mac_phystatus asynchronously after it detects mac_phy_asyncpowerchangeack high. ■ MAC de-asserts mac_phy_asyncpowerchangeack when it detects phy_mac_phystatus low. Exists: (CX_PIPE43_SUPPORT) && (!CX_PIPE43_ASYNC_HS_BYPASS) Synchronous To: {None},{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.95 Gen2/3-Only PIPE Signals

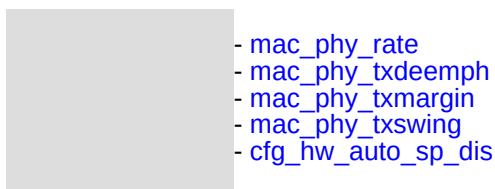


Table 2-95 Gen2/3-Only PIPE Signals

Port Name	I/O	Description
mac_phy_rate[(RATE_WIDTH-1):0]	O	Controls the link signaling rate. ■ 0: Use 2.5 GT/s signaling rate ■ 1: Use 5.0 GT/s signaling rate ■ 2: Use 8.0 GT/s signaling rate for PCIe mode or ESM data rate0 (8.0 GT/s or 16.0 GT/s) for ESM mode ■ 3: Use 16.0 GT/s signaling rate for PCIe mode or ESM data rate1 (20.0 GT/s or 25.0 GT/s) for ESM mode ■ 3: Use 16.0 GT/s signaling rate ■ 4: Use 32.0 GT/s signaling rate PIPE implementations that only support 2.5GT/s signaling rate do not implement this signal. This bus is one bit wide for a Gen2 configured controller. It is two bits wide for a Gen3 configured controller and 3 bits for Gen5 configured controller. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN2_MODE != GEN2_DISABLED) Synchronous To: aux_clk,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_txdeemph[(TX_DEEMPH_WD-1):0]	O	For Gen2 configurations, this two bits wide output selects transmitter de-emphasis as follows: ■ 00: -6 dB de-emphasis at 5GT/s ■ 01: -3.5 dB de-emphasis at 5GT/s ■ 10: No de-emphasis. Note: This is the only possible value when PIPE is configured for low swing (mac_phy_txswing == 1) as per base specification Sect. 4.3.3.6 For Gen3 configurations at Gen3 speed, this (NL*18)-bit output conveys the transmitter equalization coefficients where: ■ [5:0]: C-1 (Pre-cursor) ■ [11:6]: C0 (Cursor) ■ [17:12]: C+1 (Post-cursor). If your PHY has assigned mac_phy_txdeemph[17:0] as {C-1, C0, C+1}, then you must swap C-1 and C+1 to match the mapping of these fields in the controller. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN2_MODE != GEN2_DISABLED) && (!(CX_PIPE5_SUPPORT)) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_txmargin[2:0]	O	Selects transmitter voltage levels: ■ 000: Normal operating range ■ 001: 800-1200 mV for Full swing* or 400-700mV for Half swing ■ 010: Required and vendor defined ■ 011: Required and vendor defined ■ 100: Required and 200-400 mV for Full swing* or 100-200 mV for Half swing* if the last value or vendor defined ■ 101: Optional and 200-400 mV for Full swing* or 100-200 mV for Half swing* if the last value or vendor defined or Reserved if no other values supported ■ 110: Optional and 200-400 mV for Full swing* or 100-200 mV for Half swing* if the last value or vendor defined or Reserved if no other values supported ■ 111: optional and 200-400 mV for Full swing* or 100-200 mV for Half swing* if the last value or Reserved if no other values supported *PIPE2 implementations that only support 2.5GT/s signaling rate do not implement this signal. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN2_MODE != GEN2_DISABLED) && (!(CX_PIPE5_SUPPORT)) Synchronous To: aux_clk,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_txswing	O	Controls the PHY transmitter voltage swing level: ■ 0: Full swing ■ 1: Low swing The controller drives this output from the 'Config PHY Tx Swing' bit in the 'Link Width and Speed Change Control Register'. Implementation of this signal is optional. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN2_MODE != GEN2_DISABLED) && !(CX_PIPE5_SUPPORT) Synchronous To: aux_clk,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cfg_hw_auto_sp_dis	O	Autonomous speed disable. Used in downstream ports only. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN2_MODE != GEN2_DISABLED) Synchronous To: core_clk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.96 Gen3-Only Equalization PIPE Signals

phy_mac_dirfeedback	-	smlh_ltssm_state_rcvry_eq
phy_mac_fomfeedback	-	mac_phy_rxpresethint
phy_mac_local_tx_pset_coef	-	mac_phy_fs
phy_mac_local_tx_coef_valid	-	mac_phy_if
		mac_phy_local_pset_index
		mac_phy_getlocal_pset_coef
		mac_phy_rxeqinprogress
		mac_phy_invalid_req
		mac_phy_rxeqeval
		mac_phy_dirchange
		mac_phy_smlh_coef_rtx

Table 2-96 Gen3-Only Equalization PIPE Signals

Port Name	I/O	Description
smlh_ltssm_state_rcvry_eq	O	This status signal is asserted during all Recovery Equalization states. Exists: CX_GEN3_MODE != GEN3_DISABLED Synchronous To: core_pl_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_rxpresethint[((NL*RX_PSET_WD)-1):0]	O	Provides the RX equalization preset hint for the receiver. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCI_E_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_fs[((NL*TX_FS_WD)-1):0]	O	Provides the FS value advertised by the link partner. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCI_E_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_lf[((NL*TX_FS_WD)-1):0]	O	Provides the LF value advertised by the link partner. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_dirfeedback[((NL*DIRFEEDBACK_WD)-1):0]	I	Provides the link equalization evaluation feedback in the Direction Change format. Feedback is provided for each coefficient: ■ [1:0]: C-1 ■ [3:2]: C0 ■ [5:4]: C+1 The feedback value for each coefficient is encoded as follows: ■ 00: No change ■ 01: Increment by 1 ■ 10: Decrement by 1 ■ 11: Reserved The feedback C0 is ignored by the PCIe controller. You should hardwire this input to a constant value when you are not using Direction Change. This signal is only used at the 8.0 GT/s signaling rate. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT) Synchronous To: ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: phy_mac_phystatus is pulsed, per lane
phy_mac_fomfeedback[((NL*FOMFEEDBACK_WD)-1):0]	I	Provides the link equalization evaluation feedback in the Figure of Merit (FOM) format. The value is encoded as an integer from 0 to 255. You should hardwire this input to a constant value when you are not using FOM. This signal is only used at the 8.0 GT/s signaling rate. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT) Synchronous To: ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: phy_mac_phystatus is pulsed, per lane

Port Name	I/O	Description
phy_mac_local_tx_pset_coef[(NL*TX_COEF_WD)-1]:0]	I	These are the coefficients post-mapping from the preset the mac_phy_local_pset_index after a request mac_phy_getlocal_pset_coef. ■ [5:0]: C-1 ■ [11:6]: C0 ■ [17:12]: C+1 The controller reflects these values mac_phy_txdeemph when it wants to apply this preset. You must externally tie-off this input when CX_GEN3_EQ_PSET_COEF_MAP_MODE ='Programmable Table' 'Dynamic MAC'. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT)) Synchronous To: ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: phy_mac_local_tx_coef_valid
phy_mac_local_tx_coef_valid[(NL-1):0]	I	Single pulse indicating that the phy_mac_local_tx_pset_coef bus correctly represents the coefficients values mapped from the preset the mac_phy_local_pset_index bus You must externally tie-off this input when CX_GEN3_EQ_PSET_COEF_MAP_MODE ='Programmable Table' 'Dynamic MAC'. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT)) Synchronous To: ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_local_pset_index[((NL*PSET_ID_WD)-1):0]	O	The TX equalization preset that the MAC wants the PHY to convert into coefficients. For more information, see mac_phy_getlocal_pset_coef description. You can leave this output unconnected when CX_GEN3_EQ_PSET_COEF_MAP_MODE = 'Programmable Table' 'Dynamic MAC'. For ESM mode the ESM data rate0 corresponds to PCIe mode 8 GT/s and the ESM data rate1 corresponds to PCIe mode 16 GT/s. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_getlocal_pset_coef[(NL-1):0]	O	Single pulse used by MAC to request the PHY to perform a preset-to-co-efficient mapping, from preset mac_phy_local_pset_index, to coefficients phy_mac_local_tx_pset_coef. You can leave this output unconnected when CX_GEN3_EQ_PSET_COEF_MAP_MODE = 'Programmable Table' 'Dynamic MAC'. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_rxeqinprogress[(NL-1):0]	O	This status signal is asserted and remains asserted for the duration of Recovery.Equalization Phase2 when the PCIe controller is the USP (Phase3 when DSP). Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_invalid_req[(NL-1):0]	O	Indicates that the Link Evaluation feedback requested a link partner TX EQ setting that has been out of range. The controller asserts this signal when it detects an out-of-range error. It remains asserted until the next time mac_phy_rxeqeval asserts. <i>Note:</i> If your PHY requires mac_phy_invalid_req to follow the same timing protocol as mac_phy_rxeqeval, you must program GEN3_RELATED_OFF[23] register field to '1'. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT)) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_rxeqeval[(NL-1):0]	O	The PHY starts evaluation of the far end transmitter TX EQ settings while this signal is held high by the controller. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT)) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_dirchange[(NL-1):0]	O	Indicates the PHY to perform Figure Of Merit or Direction Change evaluation. 0: when mac_phy_rxeqeval = 1, the PHY performs Figure Of Merit 1: when mac_phy_rxeqeval = 1, the PHY performs Direction Change <i>Note:</i> If your PHY does not support the mac_phy_dirchange signal you can leave this pin unconnected. <i>Note:</i> If your PHY supports mac_phy_dirchange signal, then your PHY must drive phy_mac_dirfeedback signal value to 0 when mac_phy_dirchange = 0. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_smlh_coef_rtx[((NL*TX_COEF_WD)-1):0]	O	This signal is the coefficients for the remote partner. When not in EQ master state, this signal is de-asserted. You can leave this signal disconnected. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && (CX_EXPORT_REMOTE_TX_COEF_PORT) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

2.97 Gen3-Only Equalization V7 Signals

phy_mac_v7_txeq_fs	-	mac_phy_v7_rxeq_preset
phy_mac_v7_txeq_lf	-	mac_phy_v7_rxeq_lffs
phy_mac_v7_rxeq_done	-	mac_phy_v7_txeq_control
phy_mac_v7_rxeq_adapt_done	-	mac_phy_v7_rxeq_control
phy_mac_v7_txeq_done	-	mac_phy_v7_txeq_preset
phy_mac_v7_txeq_coeff	-	mac_phy_v7_txeq_deemph
phy_mac_v7_rxeq_new_txcoeff	-	mac_phy_v7_rxeq_txpreset
phy_mac_v7_rxeq_lffs_sel	-	

Table 2-97 Gen3-Only Equalization V7 Signals

Port Name	I/O	Description
phy_mac_v7_txeq_fs[(TX_FS_WD-1):0]	I	TX Equalization Full Swing. Value based on characteristics of TX driver. For Gen3 only. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_v7_txeq_lf[(TX_FS_WD-1):0]	I	TX Equalization Low Frequency. Value based on characteristics of TX driver. For Gen3 only. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_v7_rxeq_preset[((NL*RX_PSET_WD)-1):0]	O	RX Equalization Preset. For Gen3 only. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: pipe_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_v7_rxeq_lffs[((NL*TX_FS_WD)-1):0]	O	RX Equalization Remote Transmitter Low Frequency / Full Swing. For Gen3 only. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: pipe_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_v7_rxeq_done[(NL-1):0]	I	RX Equalization Done for mac_phy_v7_rxeq_control. Must set mac_phy_v7_rxeq_control back to 00b when phy_mac_v7_rxeq_done == 1b detected. 0b: RX equalization not done 1b: RX equalization done Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_v7_rxeq_adapt_done[(NL-1):0]	I	RX Equalization Adaptation Done for mac_phy_v7_rxeq_control == 10b and 11b. - if both phy_mac_v7_rxeq_adapt_done == 1b and phy_mac_v7_rxeq_done == 1b, then RX equalization is successfully done. - if phy_mac_v7_rxeq_adapt_done == 0b and phy_mac_v7_rxeq_done == 1b, then perform RX equalization again. 0b: RX equalization not done 1b: RX equalization done Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
phy_mac_v7_txeq_done[(NL-1):0]	I	TX Equalization Done for mac_phy_v7_txeq_control. Must set mac_phy_v7_txeq_control back to 00b when phy_mac_v7_txeq_done == 1b detected. For Gen3 only. ■ 0b: TX equalization not done ■ 1b: TX equalization done Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_v7_txeq_coeff[((NL*TX_COEF_WD)-1):0]	I	TX Equalization Coefficient. Display the current coefficient on TX driver. For Gen3 only. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: phy_mac_txeq_done
phy_mac_v7_rxeq_new_txcoeff[((NL*TX_COEF_WD)-1):0]	I	RX Equalization New TX Coefficient. Valid only when phy_mac_v7_rxeq_done == 1b. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: phy_mac_rxeq_done

Port Name	I/O	Description
phy_mac_v7_rxeq_lffs_sel[(NL-1):0]	I	RX Equalization Remote Transmitter Low Frequency / Full Swing Select. This signal serves two indications. (1) Full swing or low frequency. (2) Coefficient or preset. When phy_mac_rxeq_done = 0b, then mac_phy_v7_rxeq_lffs is either indicating low frequency or full swing: ■ 0b: Full swing ■ 1b: Low frequency When phy_mac_rxeq_done = 1b, then phy_mac_v7_rxeq_new_txcoeff is either indicating coefficient or preset: ■ 0b: Coefficients ■ 1b: Preset Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: phy_mac_rxeq_done
mac_phy_v7_txeq_control[((NL*2)-1):0]	O	TX Equalization Control. Must set back to 00b when phy_mac_v7_txeq_done = 1b detected. For Gen3 only. ■ 00b: Idle ■ 01b: Tx Preset ■ 10b: Tx Coefficients ■ 11b: Tx Query Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: pipe_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_v7_rxeq_control[((NL*2)-1):0]	O	RX Equalization Control. Must set back to 00b when phy_mac_rxeq_done = 1b detected. For Gen3 only. 00b: Idle 01b: Rx Preset 10b: Rx Margining, for Rx EQ phase 2/3 11b: Rx Adaptation, for Rx EQ phase 2/3 bypass Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: pipe_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_v7_txeq_preset[((NL*TX_PSET_WD)-1):0]	O	TX Equalization Preset. Use when mac_phy_v7_txeq_control == 01b. For Gen3 only. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: pipe_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_v7_txeq_deemph[((NL*TX_CRST_WD)-1):0]	O	TX Equalization De-Emphasis. Use when mac_phy_v7_txeq_control == 10b. 3 consecutive PCLK cycles are required to register the new 18-bit TX coefficient. For Gen3 only. 1st cycle for pre-cursor 2nd cycle for main-cursor 3rd cycle for post-cursor Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: pipe_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_v7_rxeq_txpreset[((NL*TX_PSET_WD)-1):0]	O	RX Equalization Remote Transmitter Preset. For Gen3 only. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && (!PHY_IS_PIPE && PHY_IS_V7) Synchronous To: pipe_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.98 Gen3-Only PIPE Signals

phy_mac_rxstartblock	-	mac_phy_txsynheader
phy_mac_rxsynheader	-	mac_phy_txstartblock
phy_mac_localfs	-	mac_phy_encodeddecodebypass
phy_mac_locallf	-	mac_phy_blockaligncontrol

Table 2-98 Gen3-Only PIPE Signals

Port Name	I/O	Description
mac_phy_txsynheader[(NL*2)-1:0]	O	At 8.0 GT/s this is used by the PCIe controller MAC to tell the PHY the value of the Sync header in the current 130b block. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN3_MODE != GEN3_DISABLED) && (!CX_PIPE_SERDES_ARCH) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_txstartblock[(NL-1):0]	O	Only used at the 8.0 GT/s signaling rate. This signal allows the PCIe controller MAC to indicate (to the PHY) the starting cycle mac_phy_txdata for a 128b block. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN3_MODE != GEN3_DISABLED) && (!CX_PIPE_SERDES_ARCH) Synchronous To: aux_clk,aux_clk_g,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
mac_phy_encodeddecodebypass	O	Controls if the PHY does encode and decode operations. ■ 0: Encode/decode performed normally by the PHY ■ 1: Encode/decode bypassed Support of this signal is optional for a PHY. The Synopsys controller always sets this output to 0. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && (!CX_PIPE5_SUPPORT)) Synchronous To: aux_clk,{None} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
mac_phy_blockaligncontrol	O	Enable EIEOS detection in the PHY. EIEOS detection is one of the mechanisms used to control the block aligner in the PHY. The PCIe controller uses this signal at the 8.0 GT/s signaling rate to disable the PHY from searching for a new block alignment. The PHY should ignore this signal in Gen3 loopback slave mode. This output is a level signal and not a pulse. It is not time-sensitive, so you can re-register it if required. <i>Note:</i> For Virtex-7 GTX PHY configurations, EIOS detection is not used and this Block Align Control output directs the PHY to perform internal sync header detection and re-alignment. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN3_MODE != GEN3_DISABLED) && (!CX_PIPE5_SUPPORT) Synchronous To: aux_clk,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_rxstartblock[((NL*PHY_RXSB_WD)-1):0]	I	Only used at the 8.0 GT/s signaling rate. This signal allows the PHY to indicate (to the PCIe controller MAC) the starting cycle phy_mac_rxdata for a 128b block. The starting byte for a 128b block must always start with Bit 0 of the data interface. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN3_MODE != GEN3_DISABLED) && (!CX_PIPE_SERDES_ARCH) Synchronous To: aux_clk_g,core_clk_ug,ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_rxsynheader[((NL*PHY_RXSH_WD)-1):0]	I	At 8.0 GT/s this is used by the PHY to tell the PCIe controller MAC the value of the Sync header in the current 130b block. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN3_MODE != GEN3_DISABLED) && (!CX_PIPE_SERDES_ARCH) Synchronous To: ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
phy_mac_localfs[((NL*TX_FS_WD)-1):0]	I	At 8.0 GT/s rate, it is the Local FS value advertised by the local PHY. This signal is driven to a constant value following the first speed change to Gen3. You must externally tie-off this input when CX_GEN3_EQ_PSET_COEF_MAP_MODE ='Programmable Table'. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCI_E_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT) Synchronous To: ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_locallf[((NL*TX_FS_WD)-1):0]	I	At 8.0 GT/s rate, it is the Local LF value advertised by the local PHY. This signal is driven to a constant value following the first speed change to Gen3. You must externally tie-off this input when CX_GEN3_EQ_PSET_COEF_MAP_MODE ='Programmable Table'. Exists: (CX_GEN3_MODE != GEN3_DISABLED) && ((CX_CPCI_E_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))) && !(CX_PIPE5_SUPPORT) Synchronous To: ret_core_clk,{core_clk},{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.99 Gen4-Only Margining Signals

app_margining_ready -
app_margining_software_ready -

Table 2-99 Gen4-Only Margining Signals

Port Name	I/O	Description
app_margining_ready	I	Margining Ready. Indicates when the PHY or Margining software is ready to accept margining commands. This signal is reflected in Margining Ready field of Margining Port Status register. If the PHY can accept margining commands when the current link speed is 16GT/s, you can set this signal to a fixed value of '1'. For Synopsys Gen4 PHYs, this signal should be tied to '1'. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN4_MODE != GEN4_DISABLED) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
app_margining_software_ready	I	Margining Software Ready. Indicates that the required margining software has performed the required initialization when margining is partially implemented using Device Driver software. This signal is reflected in Margining Software Ready field of Margining Port Status register. If MARGINING_USES_DRIVER_SOFTWARE field of Margining Port Capabilities Register is '0', you can set this signal to a fixed value of '0'. For Synopsys Gen4 PHYs, this signal should be tied to '0'. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN4_MODE != GEN4_DISABLED) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

2.100 Gen4-Only PIPE Signals

phy_mac_pbus -  - mac_phy_mbus

Table 2-100 Gen4-Only PIPE Signals

Port Name	I/O	Description
mac_phy_mbus[((NL*8)-1):0]	O	Synopsys Specific Interface MAC driven Command/Address/Data Bus for RBI write operation into PHY register. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN4_SPEED && (!CX_PIPE44_SUPPORT && !CX_PIPE5_SUPPORT)) Synchronous To: aux_clk,core_clk_ug,{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_mac_pbus[((NL*8)-1):0]	I	Synopsys Specific Interface PHY driven Command/Address/Data Bus for RBI write operation into MAC register. Exists: ((CX_CPCIE_ENABLE)) && (CX_GEN4_SPEED && (!CX_PIPE44_SUPPORT && !CX_PIPE5_SUPPORT)) Synchronous To: core_clk_ug,{pipe_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.101 Function Level Reset (FLR) Signals

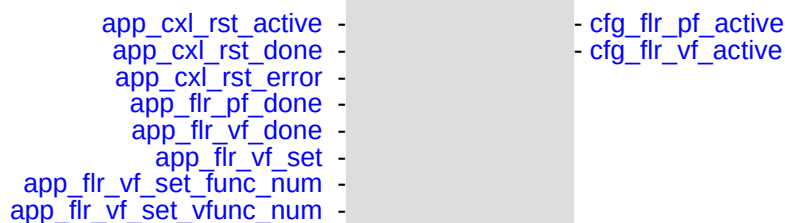


Table 2-101 Function Level Reset (FLR) Signals

Port Name	I/O	Description
app_cxl_rst_active[(NF-1):0]	I	Indicates that the FLR for the following functions are to be triggered. The application must assert the bits for all the CXL functions once <code>cfg_cxl_dev_initiate_cxl_rst</code> is asserted by controller. This should be held atleast till the <code>cfg_flr_pf_active[x]</code> is driven by the controller. Exists: CX_CXL_RESET_EN Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_cxl_rst_done	I	A one cycle pulse to indicate that the CXL_Reset is completed. Exists: CX_CXL_RESET_EN Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_cxl_rst_error	I	Indicates that device has completed CXL_Reset with errors Exists: CX_CXL_RESET_EN Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_flr_pf_active[(NF-1):0]	O	Set when the software initiates FLR a physical function by writing to the "Initiate FLR" register bit of that function. This signal is held asserted until reset of both the application and controller logic has been completed. Exists: CX_FLR_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_flr_pf_done[(NF-1):0]	I	Indicates that FLR a physical function has been completed, which means the application has completed initializing its data structures for the function and there are no more TLPs associated with this function in the application's transmit queue. This signal must be asserted until the controller de-asserts cfg_flr_pf_active for the function that has been reset. Exists: CX_FLR_ENABLE Synchronous To: radm_clk_g,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_flr_vf_active[(NVF-1):0]	O	Set when the software initiates FLR a virtual function by writing to the 'Initiate FLR' register bit of that function. This signal is held asserted until the reset of both the application and internal controller logic has been completed. Each bit position corresponds to one of the NVF virtual functions. You can have multiple VFs in FLR at any given time, and therefore, the controller can asserts multiple bits in output bus. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. When CX_EXTENSIBLE_VFUNC=1, bits [INT_NVF-1:0] represent internal VFs, [NVF-1:INT_NVF] represent external VFs. Exists: (CX_FLR_ENABLE) && (CX_SRIOV_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
app_flr_vf_done[(NVF-1):0]	I	Indicates that FLR a virtual function has been completed, which means the application has completed initializing its data structures for the function and there are no more TLPs associated with this function in the application's transmit queue. This signal is held asserted until the controller de-asserts cfg_flr_vf_active for the function that has been reset. Each bit position corresponds to one of the NVF virtual functions. You can have multiple VFs in FLR at any given time, and therefore, the controller can asserts multiple bits in output bus. You can use the cfg_start_vfi output signal or the VF Index in 'Bus Numbering Overview' index the corresponding bit field for a particular VF. When CX_EXTENSIBLE_VFUNC=1, bits [INT_NVF-1:0] represent internal VFs, [NVF-1:INT_NVF] represent external VFs. Exists: (CX_FLR_ENABLE) && (CX_SRIOV_ENABLE) Synchronous To: aux_clk_g,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_flr_vf_set	I	One-clock-cycle pulse, asserted by your external VF-register application logic, that indicates a '1' has been written to the FLR register of external VFs. This signal affects cfg_flr_vf_active [NVF-1:0]. PF and VF number are specified by app_flr_vf_set_func_num and app_flr_vf_set_vfunc_num. Exists: (CX_FLR_ENABLE) && (CX_SRIOV_ENABLE) && (CX_EXTENSIBLE_VFUNC) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
app_flr_vf_set_func_num[(PF_WD-1):0]	I	Physical function number corresponding to app_flr_vf_set. Function numbering starts at '0'. Exists: (CX_FLR_ENABLE) && (CX_SRIOV_ENABLE) && (CX_EXTENSIBLE_VFUNC) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: app_flr_vf_set is asserted

Port Name	I/O	Description
app_flr_vf_set_vfunc_num[(VF_WD-2):0]	I	Virtual function number corresponding to app_flr_vf_set. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, app_flr_vf_set_vfunc_num=0 refers to the first VF of the PF that is identified by app_flr_vf_set_func_num. Exists: (CX_FLR_ENABLE) && (CX_SRIOV_ENABLE) && (CX_EXTENSIBLE_VFUNC) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: app_flr_vf_set is asserted

2.102 Target Completion LUT Signals

client0_cpl_lookup_id -
 client1_cpl_lookup_id -
 client2_cpl_lookup_id -

- trgt_cpl_timeout
 - trgt_timeout_cpl_func_num
 - trgt_timeout_cpl_vfunc_num
 - trgt_timeout_cpl_vfunc_active
 - trgt_timeout_cpl_tc
 - trgt_timeout_cpl_attr
 - trgt_timeout_cpl_len
 - trgt_timeout_lookup_id
 - trgt_lookup_id
 - trgt_lookup_empty

Table 2-102 Target Completion LUT Signals

Port Name	I/O	Description
client0_cpl_lookup_id[(((NVC_XALI_EXP*LOOKUPID_WD)-1):0]	I	The target completion LUT lookup ID for a completion TLP on XALI0, which must match the trgt_lookup_id value for the corresponding request. When using the optional target completion LUT feature, the controller uses the Byte Count, Tag, Attributes, requester ID, and Address values stored in the target completion LUT for the completion TLP header instead of the values of the corresponding input signals. This signal only needs to be valid when transmitting completions. Exists: (!(AMBA_INTERFACE!=0)) && (TRGT_CPL_LUT_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
client1_cpl_lookup_id[(((NVC_XALI_EXP*LOOKUPID_WD)-1):0]	I	The target completion LUT lookup ID for a completion TLP on XALI1, which must match the trgt_lookup_id value for the corresponding request. When using the optional target completion LUT feature, the controller uses the Byte Count, Tag, Attributes, requester ID, and Address values stored in the target completion LUT for the completion TLP header instead of the values of the corresponding input signals. This signal only needs to be valid when transmitting completions. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT1_POPULATED) && (TRGT_CPL_LUT_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
client2_cpl_lookup_id[((NVC_XALI_EXP*LOOKUPID_WD)-1):0]	I	The target completion LUT lookup ID for a completion TLP on XALI2, which must match the trgt_lookup_id value for the corresponding request. When using the optional target completion LUT feature, the controller uses the Byte Count, Tag, Attributes, requester ID, and Address values stored in the target completion LUT for the completion TLP header instead of the values of the corresponding input signals. This signal only needs to be valid when transmitting completions. Exists: (!(AMBA_INTERFACE!=0)) && (CLIENT2_POPULATED) && (TRGT_CPL_LUT_EN) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
trgt_cpl_timeout	O	Indicates that the application has not generated a completion for an incoming request within the required time interval. Information about the timed out completion is available the trgt_timeout_* outputs listed later. When a completion timeout occurs, the controller does not clear the corresponding entry from the completion lookup table. The completion timeout value is approximately 10 ms. Exists: TRGT_CPL_LUT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High (pulse) Validated by: Not validated by another signal
trgt_timeout_cpl_func_num[(PF_WD-1):0]	O	The function number of the timed out completion. Function numbering starts at '0'. Exists: TRGT_CPL_LUT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: trgt_cpl_timeout is asserted

Port Name	I/O	Description
trgt_timeout_cpl_vfunc_num[(VF_WD-2):0]	O	Indicates which virtual function (VF) timed out. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, trgt_timeout_cpl_vfunc_num=0 refers to the first VF of the PF that is identified by trgt_timeout_cpl_func_num. Exists: (TRGT_CPL_LUT_EN) && (CX_SRIOV_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: trgt_cpl_timeout and trgt_timeout_cpl_vfunc_active is asserted
trgt_timeout_cpl_vfunc_active	O	Indicates that the timeout is coming from a VF (Virtual function). 0: No VF is active and trgt_timeout_cpl_vfunc_num is invalid. A PF is valid and identified by trgt_timeout_cpl_func_num 1: A VF is active and is identified by trgt_timeout_cpl_vfunc_num Exists: (TRGT_CPL_LUT_EN) && (CX_SRIOV_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: trgt_cpl_timeout is asserted
trgt_timeout_cpl_tc[2:0]	O	The TC of the timed out completion. Exists: TRGT_CPL_LUT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: trgt_cpl_timeout is asserted and completion is in store-and-forward mode
trgt_timeout_cpl_attr[1:0]	O	The Attributes value of the timed out completion. Does not include IDO bit. Exists: TRGT_CPL_LUT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: trgt_cpl_timeout is asserted

Port Name	I/O	Description
trgt_timeout_cpl_len[11:0]	O	The Length of the timed out completion. Exists: TRGT_CPL_LUT_EN Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: trgt_cpl_timeout is asserted
trgt_timeout_lookup_id[(LOOKUPID_WD-1):0]	O	The target completion LUT lookup ID of the timed out completion Exists: TRGT_CPL_LUT_EN Synchronous To: None,perbitclk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: trgt_cpl_timeout is asserted
trgt_lookup_id[(LOOKUPID_WD-1):0]	O	The target completion LUT lookup ID for the incoming request TLP. When using the optional target completion lookup table feature, the application must save the lookup ID and assert the same lookup ID client0/1/2_cpl_lookup_id when generating a completion for the request. Exists: TRGT_CPL_LUT_EN Synchronous To: None,perbitclk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
trgt_lookup_empty	O	When this signal is asserted with radm_trgt1_hv, it indicates that the target completion LUT is not full. The application can use this signal to determine a back-pressure mechanism to the controller so that there is not an overflow condition when the number of outstanding non-posted transactions received is more than expected. Exists: TRGT_CPL_LUT_EN Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

2.103 RAS D.E.S. Event Counter RAM Signals

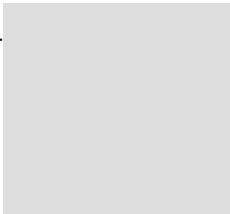
cdm_ras_des_ec_ram_dataout -  - cdm_ras_des_ec_ram_addra
 - cdm_ras_des_ec_ram_addrb
 - cdm_ras_des_ec_ram_datain
 - cdm_ras_des_ec_ram_wea
 - cdm_ras_des_ec_ram_ena
 - cdm_ras_des_ec_ram_enb

Table 2-103 RAS D.E.S. Event Counter RAM Signals

Port Name	I/O	Description
cdm_ras_des_ec_ram_addra[(RASDES_EC_RAM_ADDR_WIDTH-1):0]	O	Port A write address for the RAM in RAS D.E.S. event counter register. Exists: CX_RAS_DES_EC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: cdm_ras_des_ec_ram_wea & cdm_ras_des_ec_ram_ena are asserted
cdm_ras_des_ec_ram_addrb[(RASDES_EC_RAM_ADDR_WIDTH-1):0]	O	Port B read address for the RAM in RAS D.E.S. event counter register. Exists: CX_RAS_DES_EC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: cdm_ras_des_ec_ram_enb is asserted
cdm_ras_des_ec_ram_datain[(RASDES_EC_RAM_DATA_WIDTH-1):0]	O	Port A write data to the RAM in RAS D.E.S. event counter register. Exists: CX_RAS_DES_EC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: cdm_ras_des_ec_ram_wea & cdm_ras_des_ec_ram_ena are asserted
cdm_ras_des_ec_ram_wea	O	Port A write enable for the RAM in RAS D.E.S. event counter register. Exists: CX_RAS_DES_EC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cdm_ras_des_ec_ram_ena	O	Port A access enable for the RAM in RAS D.E.S. event counter register. Exists: CX_RAS_DES_EC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cdm_ras_des_ec_ram_enb	O	Port B access enable for the RAM in RAS D.E.S. event counter register. Exists: CX_RAS_DES_EC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cdm_ras_des_ec_ram_dataout[(RASDES_EC_RAM_DATA_WIDTH-1):0]	I	Port B read data from the RAM that used in RAS D.E.S. event counter register. Exists: CX_RAS_DES_EC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

2.104 RAS D.E.S. Event Counters Debug Signals

- [cdm_ras_des_ec_info_common](#)
 - [cdm_ras_des_ec_info_li](#) (for i = 0; i <= CX_NL)

Table 2-104 RAS D.E.S. Event Counters Debug Signals

Port Name	I/O	Description
cdm_ras_des_ec_info_common[(RASDES_EC_INFO_CM-1):0]	O	Common event signal bus that is used in RAS D.E.S. event counters ■ [176:173]: rtlh_rx_defmemwr_evt: Pulse: Rx Deferrable Memory Write ■ [172:171]: xtlh_tx_defmemwr_evt: Pulse: Tx Deferrable Memory Write ■ [170:167]: rtlh_rx_ccix_tlp_evt: Pulse: Rx CCIX TLP ■ [166:165]: xtlh_tx_ccix_tlp_evt: Pulse: Tx CCIX TLP ■ [164:161]: rtlh_rx_tlpwprefix_evt: Pulse: Rx TLP with Prefix ■ [160:157]: rtlh_rx_atmcop_evt: Pulse: Rx Atomic ■ [156:153]: rtlh_rx_msg_evt: Pulse: Rx Message TLP ■ [152:149]: rtlh_rx_cplwd_evt: Pulse: Rx Completion w data ■ [148:145]: rtlh_rx_cplwod_evt: Pulse: Rx Completion wo data ■ [144:141]: rtlh_rx_iord_evt: Pulse: Rx IO Read ■ [140:137]: rtlh_rx_iowr_evt: Pulse: Rx IO Write ■ [136:133]: rtlh_rx_cfgrd_evt: Pulse: Rx Config Read ■ [132:129]: rtlh_rx_cfgwr_evt: Pulse: Rx Config Write ■ [128:125]: rtlh_rx_memrd_evt: Pulse: Rx Memory Read ■ [124:121]: rtlh_rx_memwr_evt: Pulse: Rx Memory Write ■ [120:119]: xtlh_tx_tlpwprefix_evt: Pulse: Tx TLP with Prefix ■ [118:117]: xtlh_tx_atmcop_evt: Pulse: Tx AtomicOp ■ [116:115]: xtlh_tx_msg_evt: Pulse: Tx Message ■ [114:113]: xtlh_tx_cplwd_evt: Pulse: Tx Completion w data ■ [112:111]: xtlh_tx_cplwod_evt: Pulse: Tx Completion wo data ■ [110:109]: xtlh_tx_iord_evt: Pulse: Tx IO Read ■ [108:107]: xtlh_tx_iowr_evt: Pulse: Tx IO Write ■ [106:105]: xtlh_tx_cfgrd_evt: Pulse: Tx Config Read ■ [104:103]: xtlh_tx_cfgwr_evt: Pulse: Tx Config Write ■ [102:101]: xtlh_tx_memrd_evt: Pulse: Tx Memory Read ■ [100:99]: xtlh_tx_memwr_evt: Pulse: Tx Memory Write ■ [98:95]: rdlh_duplicate_tlp_err_pertlp: Pulse: Rx Duplicate TLP ■ [94]: xtlh_xadm_restore_enable: Pulse: Tx Nullified TLP ■ [93:90]: rdlh_nulified_tlp_err_pertlp : Pulse: Rx Nullified TLP ■ [89:82]: rtlh_rcvd_ufc_perdlp: Pulse: Rx Update FC DLLP ■ [81:74]: rdlh_rcvd_ack_perdlp: Pulse: Rx Ack DLLP ■ [73]: xdlh_update_fc_sent: Pulse: Tx Update FC DLLP ■ [72]: xdlh_ack_sent: Pulse: Tx Ack DLLP

Port Name	I/O	Description
cdm_ras_des_ec_info_common[(RASDES_EC_INFO_CM-1):0]...(cont.)	O.	■ [71]: smlh_lwd_change: Pulse: Link width Change ■ [70]: smlh_spd_change: Pulse: Speed Change ■ [69]: smlh_in_l23: Level: L2 Entry ■ [68]: pm_in_l1_abort: Level: L1.2 abort ■ [67]: pm_in_l1_cpm: Level: L1 Clock PM (L1 with REFCLK removal/PLL Off) ■ [66]: pm_in_l1_short: Level: L1 short duration ■ [65]: pm_in_l12: Level: L1.2 Entry ■ [64]: pm_in_l11: Level: L1.1 Entry ■ [63]: smlh_in_l1: Level: L1 Entry ■ [62]: pm_asnak: Level: ASPM L1 reject ■ [61]: smlh_in_rl0s: Level: Rx L0s Entry ■ [60]: smlh_in_l0s: Level: Tx L0s Entry ■ [59]: smlh_l1_to_recovery: Pulse: L1 to Recovery Entry ■ [58]: smlh_l0_to_recovery: Pulse: L0 to Recovery Entry ■ [57:56]: cfg_cpl_timeout[1:0]: Pulse: Completion Timeout ■ [55:52]: cfg_ca_tlp: Pulse: Completer Abort ■ [51:48]: cfg_ur_tlp: Pulse: Unsupported Request ■ [47:44]: cfg_ecrc_tlp_err: Pulse: ECRC Error ■ [43:40]: cfg_poisoned_tlp: Pulse: Poisoned TLP ■ [39]: rtlh_req_link_retrain: Level: FC Timeout ■ [38]: xdlh_retry_req: Pulse: Retry TLP ■ [37]: xdlh_nak_sent: Pulse: Tx Nak DLLP ■ [36:29]: rdlh_rcvd_nack_perdlp: Pulse: Rx Nak DLLP ■ [28]: xdlh_replay_timeout_err: Pulse: Replay Timeout ■ [27]: xdlh_replay_num_rollover_err: Pulse: Replay_Num Rollover ■ [26:19]: rdlh_bad_dllp_err_perdlp: Pulse: BAD DLLP ■ [18:15]: rdlh_lcrc_tlp_err_pertlp : Pulse: LCRC Error ■ [14:11]: rdlh_bad_tlp_err_pertlp : Pulse: BAD TLP ■ [10]: rmlh_deskew_alignment_err: Level: Deskew Error ■ [9]: rmlh_framing_err: Pulse: Framing Error ■ [8]: smlh_timeout_nfts: Level: FTS Timeout ■ [7]: smlh_rx_rcvry_req: Level: Rx Recovery Request ■ [6]: rmlh_rcvd_err: Pulse: Receiver Error ■ [5]: smlh_idle_inferred_in_l0: Level: Detect EI Infer ■ [4]: Reserved (fixed "0") ■ [3]: Reserved (fixed "0") ■ [2]: Reserved (fixed "0") ■ [1]: Reserved (fixed "0") ■ [0]: Reserved (fixed "0") Exists:

Port Name	I/O	Description
cdm_ras_des_ec_info_common[(RASDES_EC_INFO_CM-1):0]...(cont..)	O..	(CX_RAS_DES_EC_ENABLE) && (CX_RAS_DES_DBGIO_ENABLE) Synchronous To: None,aux_clk,aux_clk_g,core_pl_clk,perbitclk,radm_clk_g,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cdm_ras_des_ec_info_li[(RASDES_EC_INFO_PL-1):0] (for i = 0; i <= CX_NL)	O	Lane<i>i</i> event signal bus that is used in RAS D.E.S. event counters. ■ [12]: rmlh_deskew_ctlskip_err[i*5+4:i*5+3]: Pulse: Margin CRC and Parity Error ■ [11]: rmlh_deskew_ctlskip_err[i*5+2]: Pulse: 2nd Retimer Parity Error ■ [10]: rmlh_deskew_ctlskip_err[i*5+1]: Pulse: 1st Retimer Parity Error ■ [9]: rmlh_deskew_ctlskip_err[i*5]: Pulse: CTL SKP OS Data Parity Error ■ [8]: rmlh_ebuf_rxskipremoved: Pulse: EBUF SKP Del ■ [7]: rmlh_ebuf_rxskipadded: Pulse: EBUF SKP Add ■ [6]: rmlh_rxvalid_deasserted: Pulse: Rx Valid de-assertion ■ [5]: rmlh_sync_header_err: Pulse: SYNC Header Error ■ [4]: rmlh_skip_parity_err: Pulse: SKP OS Parity Error ■ [3]: rmlh_phy_rxdisperror: Pulse: Running Disparity Error ■ [2]: rmlh_phy_rxcodeerror: Pulse: Decode Error ■ [1]: rmlh_ebuf_rxunderflow: Pulse: EBUF underrun ■ [0]: rmlh_ebuf_rxoverflow: Pulse: EBUF Overflow Exists: (CX_RAS_DES_EC_ENABLE) && (CX_RAS_DES_DBGIO_ENABLE) Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

2.105 RAS D.E.S. Silicon Debug Internal Signals

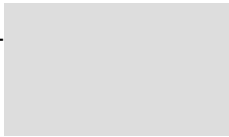
app_ras_des_sd_hold_ltssm -  - cdm_ras_des_sd_info_common
 - cdm_ras_des_sd_info_li (for i = 0; i <= CX_NL)
 - cdm_ras_des_sd_info_vi (for i = 0; i <= CX_NVC)

Table 2-105 RAS D.E.S. Silicon Debug Internal Signals

Port Name	I/O	Description
app_ras_des_sd_hold_ltssm	I	Hold and release LTSSM. For as long as this signal is '1', the controller stays in the current LTSSM. Exists: CX_RAS_DES_SD_ENABLE Synchronous To: core_pl_clk, ret_core_clk, {core_clk} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
cdm_ras_des_sd_info_common[(RASDES_SD_INFO_CM-1):0]	O	Common debug signal bus that is used in RAS D.E.S. silicon debug ■ [78]: init_eq_pending_g4: Level: Equalization sequence Gen5 ■ [77:75]: l1sub_state: Level: L1 sub state ■ [74]: init_eq_pending_g4: Level: Equalization sequence Gen4 ■ [73]: init_eq_pending: Level: Equalization sequence Gen3 ■ [72:61]: xdlh_currt_seqnum [11:0]: Level: Tx TLP SEQ# ■ [60:49]: rdlh_currt_rx_ack_seqnum [11:0]: Level: Rx ACK SEQ# ■ [48]: rdlh_vc0_initfc2_status: Level: Init-FC Flag2 VC0 ■ [47]: rdlh_vc0_initfc1_status: Level: Init-FC Flag1 VC0 ■ [46:45]: rdlh_dlcctrl_state [1:0]: Level: DLCM ■ [44:37]: latched_ts_nfts [7:0]: Level: Latched NFTS ■ [36:34]: ltssm_powerdown [1:0]: Level: PIPE: Power Down ■ [33:18]: smlh_ltssm_variable [15:0]: Level: LTSSM Variable ■ [17]: pm_pme_resend_flag: Pulse: PME Re-Send flag ■ [16]: smlh_lane_reversed: Level: Lane Reversal Operation ■ [15:9]: rmlh_framing_err_ptr [6:0]: Pulse: 1st Framing Error Pointer ■ [8:5]: pm_slave_state [3:0]: Level: PM Internal State (Slave) ■ [4:0]: pm_master_state [4:0]: Level: PM Internal State (Master) Exists: (CX_RAS_DES_SD_ENABLE) && (CX_RAS_DES_DBGIO_ENABLE) Synchronous To: None, aux_clk, core_clk, core_pl_clk, perbitclk, ret_core_clk, {aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

Port Name	I/O	Description
cdm_ras_des_sd_info_li[(RASDES_SD_I NFO_PL-1):0] (for i = 0; i <= CX_NL)	O	Lane<i>i</i> debug signal bus that is used in RAS D.E.S. silicon debug ■ [204:203]: eq_convergence_sts_g5 [1:0]: Level: Equalization convergence information Gen5 ■ [202]: eqpa_violate_rule_123_g5[2]: Level: Rule C Violation Event Status Gen5 ■ [201]: eqpa_violate_rule_123_g5[1]: Level: Rule B Violation Event Status Gen5 ■ [200]: eqpa_violate_rule_123_g5[0]: Level: Rule A Violation Event Status Gen5 ■ [199]: mac_cdm_ras_des_reject_rtx_g5: Level: Receive Reject Coefficient Event status Gen5 ■ [198:191]: phy_cdm_ras_des_fomfeedback_g5: Level: Current Figure of Merit Gen5 ■ [190:185]: mac_cdm_ras_des_coef_ltx_g5[5:0]: Level: Current Local Transmitter Pre Cursor coefficient Gen5 ■ [184:179]: mac_cdm_ras_des_coef_ltx_g5[11:6]: Level: Current Local Transmitter Cursor coefficient Gen5 ■ [178:173]: mac_cdm_ras_des_coef_ltx_g5[17:12]: Level: Current Local Transmitter Post Cursor coefficient Gen5 ■ [172:167]: mac_cdm_ras_des_coef_rtx_g5[5:0]: Level: Current Remote Transmitter Pre Cursor coefficient Gen5 ■ [166:161]: mac_cdm_ras_des_coef_rtx_g5[11:6]: Level: Current Remote Transmitter Cursor coefficient Gen5 ■ [160:155]: mac_cdm_ras_des_coef_rtx_g5[17:12]: Level: Current Remote Transmitter Post Cursor coefficient Gen5 ■ [154:149]: mac_cdm_ras_des_lf_g5: Level: Remote Device LF Gen5 ■ [148:143]: mac_cdm_ras_des_fs_g5: Level: Remote Device FS Gen5 ■ [142:141]: eq_convergence_sts_g4 [1:0]: Level: Equalization convergence information Gen4 ■ [140]: eqpa_violate_rule_123_g4[2]: Level: Rule C Violation Event Status Gen4 ■ [139]: eqpa_violate_rule_123_g4[1]: Level: Rule B Violation Event Status Gen4 ■ [138]: eqpa_violate_rule_123_g4[0]: Level: Rule A Violation Event Status Gen4 ■ [137]: mac_cdm_ras_des_reject_rtx_g4: Level: Receive Reject Coefficient Event status Gen4 ■ [136:129]: phy_cdm_ras_des_fomfeedback_g4: Level: Current Figure of Merit Gen4 ■ [128:126]: 3'b000: Reserved ■ [125:120]: mac_cdm_ras_des_coef_ltx_g4[5:0]: Level: Current Local Transmitter Pre Cursor coefficient Gen4

Port Name	I/O	Description
cdm_ras_des_sd_info_li[(RASDES_SD_I NFO_PL-1):0](for i = 0; i <= CX_NL)...(cont.)	O.	■ [119:114]: mac_cdm_ras_des_coef_ltx_g4[11:6]: Level: Current Local Transmitter Cursor coefficient Gen4 ■ [113:108]: mac_cdm_ras_des_coef_ltx_g4[17:12]: Level: Current Local Transmitter Post Cursor coefficient Gen4 ■ [107:102]: mac_cdm_ras_des_coef_rtx_g4[5:0]: Level: Current Remote Transmitter Pre Cursor coefficient Gen4 ■ [101:96]: mac_cdm_ras_des_coef_rtx_g4[11:6]: Level: Current Remote Transmitter Cursor coefficient Gen4 ■ [95:90]: mac_cdm_ras_des_coef_rtx_g4[17:12]: Level: Current Remote Transmitter Post Cursor coefficient Gen4 ■ [89:84]: mac_cdm_ras_des_lf_g4: Level: Remote Device LF Gen4 ■ [83:78]: mac_cdm_ras_des_fs_g4: Level: Remote Device FS Gen4 ■ [77:76]: eq_convergence_sts [1:0]: Level: Equalization convergence information Gen3 ■ [75]: eqpa_violate_rule_123[2]: Level: Rule C Violation Event Status Gen3 ■ [74]: eqpa_violate_rule_123[1]: Level: Rule B Violation Event Status Gen3 ■ [73]: eqpa_violate_rule_123[0]: Level: Rule A Violation Event Status Gen3 ■ [72]: mac_cdm_ras_des_reject_rtx: Level: Receive Reject Coefficient Event status Gen3 ■ [71:64]: phy_cdm_ras_des_fomfeedback: Level: Current Figure of Merit Gen3 ■ [63:61]: mac_cdm_ras_des_pset_lrx: Level: Current Local Receiver Preset Hint Gen3 ■ [60:55]: mac_cdm_ras_des_coef_ltx[5:0]: Level: Current Local Transmitter Pre Cursor coefficient Gen3 ■ [54:49]: mac_cdm_ras_des_coef_ltx[11:6]: Level: Current Local Transmitter Cursor coefficient Gen3 ■ [48:43]: mac_cdm_ras_des_coef_ltx[17:12]: Level: Current Local Transmitter Post Cursor coefficient Gen3 ■ [42:37]: mac_cdm_ras_des_coef_rtx[5:0]: Level: Current Remote Transmitter Pre Cursor coefficient Gen3 ■ [36:31]: mac_cdm_ras_des_coef_rtx[11:6]: Level: Current Remote Transmitter Cursor coefficient Gen3 ■ [30:25]: mac_cdm_ras_des_coef_rtx[17:12]: Level: Current Remote Transmitter Post Cursor coefficient Gen3 ■ [24:19]: mac_cdm_ras_des_lf: Level: Remote Device LF Gen3 ■ [18:13]: mac_cdm_ras_des_fs: Level: Remote Device FS Gen3

Port Name	I/O	Description
cdm_ras_des_sd_info_li[(RASDES_SD_I NFO_PL-1):0](for i = 0; i <= CX_NL)...(cont..)	O..	■ [12:5]: rmlh_deskew_fifo_ptr: Level: Deskew Pointer ■ [4]: mac_phy_rxpolarity: Level: PIPE: RxPolarity ■ [3]: latched_rxdetected: Level: PIPE: Detect Lane ■ [2]: phy_mac_rxvalid_rxburst: Level: PIPE: RxValid/RxBurst ■ [1]: phy_mac_rxelec_rhx8exit: Level: PIPE: RxElecIdle/RxHi-bern8ExitType1 ■ [0]: mac_phy_txelec_txburst: Level: PIPE: TxElecIdle/TxBurst Exists: (CX_RAS_DES_SD_ENABLE) && (CX_RAS_DES_DBGIO_ENABLE) Synchronous To: None,aux_clk,aux_clk_g,core_clk_ug,core_pl_clk,perbitclk,pipe_clk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

Port Name	I/O	Description
cdm_ras_des_sd_info_vi[(RASDES_SD_I NFO_PV-1):0] (for i = 0; i <= CX_NVC)	O	VCi debug signal bus that is used in RAS D.E.S. silicon debug. ■ [239:228]: rtlh_fc_allctd_cpId: Level: Credit Allocated (CD) VCn ■ [227:220]: rtlh_fc_allctd_cplh: Level: Credit Allocated (CH) VCn ■ [219:208]: rtlh_fc_allctd_npd: Level: Credit Allocated (ND) VCn ■ [207:200]: rtlh_fc_allctd_nph: Level: Credit Allocated (NH) VCn ■ [199:188]: rtlh_fc_allctd_pd: Level: Credit Allocated (PD) VCn ■ [187:180]: rtlh_fc_allctd_ph: Level: Credit Allocated (PH) VCn ■ [179:168]: rtlh_fc_rcvd_cpId: Level: Credit Received (CD) VCn ■ [167:160]: rtlh_fc_rcvd_cplh: Level: Credit Received (CH) VCn ■ [159:148]: rtlh_fc_rcvd_npd: Level: Credit Received (ND) VCn ■ [147:140]: rtlh_fc_rcvd_nph: Level: Credit Received (NH) VCn ■ [139:128]: rtlh_fc_rcvd_pd: Level: Credit Received (PD) VCn ■ [127:120]: rtlh_fc_rcvd_ph: Level: Credit Received (PH) VCn ■ [119:108]: xadm_fc_limit_cpId: Level: Credit Limit (CD) VCn ■ [107:100]: xadm_fc_limit_cplh: Level: Credit Limit (CH) VCn ■ [99:88]: xadm_fc_limit_npd: Level: Credit Limit (ND) VCn ■ [87:80]: xadm_fc_limit_nph: Level: Credit Limit (NH) VCn ■ [79:68]: xadm_fc_limit_pd: Level: Credit Limit (PD) VCn ■ [67:60]: xadm_fc_limit_ph: Level: Credit Limit (PH) VCn ■ [59:48]: xadm_fc_cnsmd_cpId: Level: Credit Consumed (CD) VCn ■ [47:40]: xadm_fc_cnsmd_cplh: Level: Credit Consumed (CH) VCn ■ [39:28]: xadm_fc_cnsmd_npd: Level: Credit Consumed (ND) VCn ■ [27:20]: xadm_fc_cnsmd_nph: Level: Credit Consumed (NH) VCn ■ [19:8]: xadm_fc_cnsmd_pd: Level: Credit Consumed (PD) VCn ■ [7:0]: xadm_fc_cnsmd_ph: Level: Credit Consumed (PH) VCn Exists: (CX_RAS_DES_SD_ENABLE) && (CX_RAS_DES_DBGIO_ENABLE) && (Always) Synchronous To: aux_clk_g,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

2.106 RAS D.E.S. Time-Based Analysis Debug Signals

app_ras_des_tba_ctrl -  - cdm_ras_des_tba_info_common

Table 2-106 RAS D.E.S. Time-Based Analysis Debug Signals

Port Name	I/O	Description
cdm_ras_des_tba_info_common[(RASDES_TBA_INFO_CM-1):0]	O	Common event signal status bus used in RAS D.E.S. time based analysis. Indicates the internal signals that are used in the time-based analysis . The results are in TIME_BASED_ANALYSIS_DATA_REG. Each bit indicates the state that the controller stays in. All signals are level sensitive unless otherwise indicated. ■ [6]: smlh_link_in_training: Config/Recovery ■ [5]: pm_in_l12: L1.2 ■ [4]: pm_in_l11: L1.1 ■ [3]: smlh_in_l1: L1 ■ [2]: smlh_in_l0: L0 ■ [1]: smlh_in_rl0s: Rx L0s ■ [0]: smlh_in_l0s: Tx L0s Exists: (CX_RAS_DES_TBA_ENABLE) && (CX_RAS_DES_DBGIO_ENABLE) Synchronous To: None,aux_clk,core_pl_clk,perbitclk,ret_core_clk,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: See description. Validated by: Not validated by another signal

Port Name	I/O	Description
app_ras_des_tba_ctrl[1:0]	I	Controls the start/end of time based analysis. You must only set the pins to the required value for the duration of one clock cycle. This signal must be 2'b00 while the TIMER_START field in TIME_BASED_ANALYSIS_CONTROL_REG register is controlled by the DBI interface or the accesses from the wire side. ■ 2'b00: No action ■ 2'b01: Start ■ 2'b10: End. This setting is only used when the TIME_BASED_DURATION_SELECT field of TIME_BASED_ANALYSIS_CONTROL_REG is set to "manual control". ■ 2'b11: Reserved These pins also set the contents of the TIMER_START field in TIME_BASED_ANALYSIS_CONTROL_REG register. Exists: CX_RAS_DES_TBA_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

2.107 RAS D.E.S. Time-Based Analysis RAM Signals

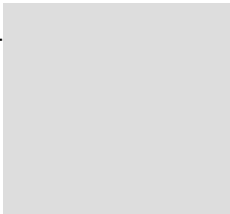
cdm_ras_des_tba_ram_dataout -  - cdm_ras_des_tba_ram_addra
 - cdm_ras_des_tba_ram_addrb
 - cdm_ras_des_tba_ram_datain
 - cdm_ras_des_tba_ram_wea
 - cdm_ras_des_tba_ram_ena
 - cdm_ras_des_tba_ram_enb

Table 2-107 RAS D.E.S. Time-Based Analysis RAM Signals

Port Name	I/O	Description
cdm_ras_des_tba_ram_addra[(RASDES_TBA_RAM_ADDR_WIDTH-1):0]	O	Port A write address for the RAM in RAS D.E.S. time based analysis register. Exists: CX_RAS_DES_TBA_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: cdm_ras_des_tba_ram_wea & cdm_ras_des_tba_ram_ena are asserted
cdm_ras_des_tba_ram_addrb[(RASDES_TBA_RAM_ADDR_WIDTH-1):0]	O	Port B read address for the RAM in RAS D.E.S. time based analysis register. Exists: CX_RAS_DES_TBA_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: cdm_ras_des_tba_ram_enb is asserted
cdm_ras_des_tba_ram_datain[(RASDES_TBA_RAM_DATA_WIDTH-1):0]	O	Port A write data for the RAM in RAS D.E.S. time based analysis register. Exists: CX_RAS_DES_TBA_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: cdm_ras_des_tba_ram_wea & cdm_ras_des_tba_ram_ena are asserted

Port Name	I/O	Description
cdm_ras_des_tba_ram_wea	O	Port A write enable for the RAM in RAS D.E.S. time based analysis register. Exists: CX_RAS_DES_TBA_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cdm_ras_des_tba_ram_ena	O	Port A access enable for the RAM in RAS D.E.S. time based analysis register. Exists: CX_RAS_DES_TBA_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cdm_ras_des_tba_ram_enb	O	Port B access enable for the RAM in RAS D.E.S. time based analysis register. Exists: CX_RAS_DES_TBA_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cdm_ras_des_tba_ram_dataout[(RASDES_TBA_RAM_DATA_WIDTH-1):0]	I	Port B read data from the RAM that used in RAS D.E.S. time based analysis register. Exists: CX_RAS_DES_TBA_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

2.108 Configuration Intercept Interface (CII) Signals

cii_lbc_halt	-	- lbc_cii_hv
cii_lbc_override_en	-	- lbc_cii_hdr_poisoned
cii_lbc_cpl_status	-	- lbc_cii_hdr_type
cii_lbc_override_data	-	- lbc_cii_hdr_first_be
		- lbc_cii_hdr_tag
		- lbc_cii_hdr_req_id
		- lbc_cii_hdr_addr
		- lbc_cii_hdr_bus_num
		- lbc_cii_hdr_dev_num
		- lbc_cii_hdr_func_num
		- lbc_cii_hdr_vfunc_num
		- lbc_cii_hdr_vfunc_active
		- lbc_cii_dv
		- lbc_cii_data

Table 2-108 Configuration Intercept Interface (CII) Signals

Port Name	I/O	Description
lbc_cii_hv	O	Indicates that the header information outputs and lbc_cii_dv signal of CII are valid. For more information, see "Configuration Intercept Controller (CIC) for USP" section in the "Controller Operations" chapter of the <i>Databook</i>. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
lbc_cii_hdr_poisoned	O	The Poisoned (EP) bit in the received TLP header on CII. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted

Port Name	I/O	Description
lbc_cii_hdr_type[4:0]	O	The Type field in the received TLP header on CII. Only the following TLP Type can be appeared on CII. ■ 00100b: Configuration Read/Write Type0 ■ 00101b: Configuration Read/Write Type1 (SRIOV configuration only) Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted
lbc_cii_hdr_first_be[3:0]	O	The first dword byte enable field in the received TLP header on CII. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted
lbc_cii_hdr_tag[(TAG_SIZE-1):0]	O	The Tag field in the received TLP header on CII. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted
lbc_cii_hdr_req_id[15:0]	O	The requester ID in the received TLP header on CII. ■ [15:8]: Bus number ■ [7:3]: Device number ■ [2:0]: Function number Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted

Port Name	I/O	Description
lbc_cii_hdr_addr[11:0]	O	The Register Address in the received TLP header on CII. ■ [11:8]: Extended Register Number[3:0] ■ [7:2]: Register Number[5:0] ■ [1:0]: Reserved Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted
lbc_cii_hdr_bus_num[7:0]	O	The Bus number in the received TLP header on CII. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted
lbc_cii_hdr_dev_num[4:0]	O	The Device number in the received TLP header on CII. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted
lbc_cii_hdr_func_num[(PF_WD-1):0]	O	Identifies the targeted controller physical function (PF). Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted
lbc_cii_hdr_vfunc_num[(VF_WD-2):0]	O	Identifies the virtual function (VF) inside the controller, which has been targeted. Exists: (CX_CONFIG_INTERCEPT_ENABLE) && (CX_SRIOV_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv and lbc_cii_hdr_vfunc_active are asserted

Port Name	I/O	Description
lbc_cii_hdr_vfunc_active	O	Indicates that the received TLP has been targeted to a virtual function (VF) inside the controller, and is identified by lbc_cii_hdr_vfunc_num. Exists: (CX_CONFIG_INTERCEPT_ENABLE) && (CX_SRIOV_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted
lbc_cii_dv	O	Indicates that the data information outputs of CII are valid. This signal is asserted only for a configuration Write request. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted
lbc_cii_data[31:0]	O	Received TLP payload data from the link partner to your application client. The data is in little endian format. The first received payload byte is in [7:0]. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_dv is asserted
cii_lbc_halt	I	Flow control input signal. When cii_lbc_halt is asserted, the controller halts processing of CFG requests for the CDM and ELBI configuration spaces. For more information, see "Configuration Intercept Controller (CIC) for USP" section in the "Controller Operations" chapter of the <i>Databook</i>. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted

Port Name	I/O	Description
cii_lbc_override_en	I	Override enable. When your application logic asserts this input, the controller overrides the CfgWr payload or CfgRd completion using the data supplied by your application logic on cii_lbc_override_data. For more information, see "Configuration Intercept Controller (CIC) for USP" section in the "Controller Operations" chapter of the <i>Databook</i>. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted
cii_lbc_cpl_status[2:0]	I	Override Completion Status. When cii_lbc_override_en is set, the value provided on cii_lbc_cpl_status[2:0] is used as the Completion Status for the resulting completion. This applies to both reads and writes. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv and cii_lbc_override_en is asserted.
cii_lbc_override_data[31:0]	I	Override data. ■ CfgWr: override write data to CDM register with data supplied by your application logic on cii_lbc_override_data. ■ CfgRd: override data payload of the completion TLP with data supplied by your application logic on cii_lbc_override_data. For more information, see "Configuration Intercept Controller (CIC) for USP" section in the "Controller Operations" chapter of the <i>Databook</i>. Exists: CX_CONFIG_INTERCEPT_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: lbc_cii_hv is asserted

2.109 EXTSRIOV: External VFs Access Signals

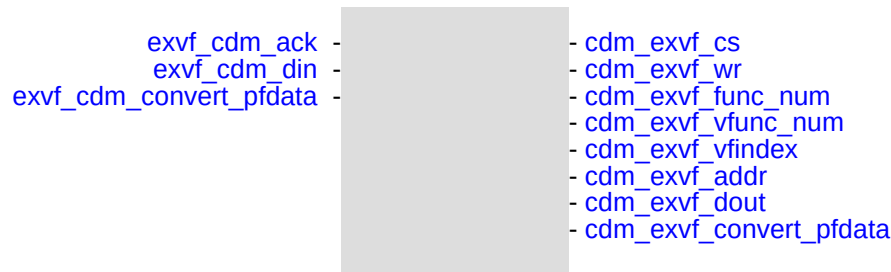


Table 2-109 EXTSRIOV: External VFs Access Signals

Port Name	I/O	Description
cdm_exvf_cs	O	The controller asserts cdm_exvf_cs when it receives a read or write request for external VF registers. The controller de-asserts cdm_exvf_cs only after the external register block acknowledges completion of the access by asserting the exvf_cdm_ack. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cdm_exvf_wr[3:0]	O	Indicates the access type (read or write) for the external VF register being accessed currently. For writes, cdm_exvf_wr also indicates the byte enables. ■ 0000b: Read ■ 0001b: Write byte 0 ■ 0010b: Write byte 1 ■ 0100b: Write byte 2 ■ 1000b: Write byte 3 ■ 1111b: Write all bytes Combinations of byte enables (for example, 0101b) are also valid. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: cdm_exvf_cs is asserted

Port Name	I/O	Description
cdm_exvf_func_num[(PF_WD-1):0]	O	The physical function number of the current external VF access. ■ 00000b: Function 0 ■ 00001b: Function 1 ■ 00010b: Function 2 ■ 00011b: Function 3 ■ 00100b: Function 4 ■ 00101b: Function 5 ■ 00110b: Function 6 ■ 00111b: Function 7 ■ 00111b: Function 7 ■ ■ 11111b: Function 31 Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: cdm_exvf_cs is asserted
cdm_exvf_vfunc_num[(VF_WD-2):0]	O	Indicates the VF being accessed currently. The PCIe SRIOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, cdm_exvf_vfunc_num =0 refers to the first VF of the PF that is identified by cdm_exvf_cs. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: cdm_exvf_cs is asserted
cdm_exvf_vfindx[(VFI_WD-1):0]	O	Indicates VF Index is being accessed currently. Your application knows how many VFs are assigned to each PF, and by using this knowledge, it can identify the parent PF and the VF number within that PF. For more information, see the VF Index in 'Bus Numbering Overview'. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: cdm_exvf_cs is asserted

Port Name	I/O	Description
cdm_exvf_addr[31:0]	O	Address bus to the external VF register block. Reserved fields should be ignored. ■ [31:12]: Reserved ■ [11:8]: Extended Register Number[3:0] ■ [7:2]: Register Number[5:0] ■ [1:0]: Reserved Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: cdm_exvf_cs is asserted
cdm_exvf_dout[31:0]	O	Write data bus to the selected external VF configuration register. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: cdm_exvf_cs is asserted
exvf_cdm_ack	I	Indicates that the requested read or write operation to the selected external VF's configuration register is complete. This signal should be high for two clock cycles in response to cdm_exvf_cs being asserted. That is, it should be inactive when cdm_exvf_cs is inactive. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
exvf_cdm_din[31:0]	I	Read data bus from the selected external VF's configuration register. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: exvf_cdm_ack is asserted

Port Name	I/O	Description
exvf_cdm_convert_pfddata[31:0]	I	Indicates whether the controller should compose the read data of the selected external VF register by copying the contents from the corresponding register of the parent PF or from selected external VF's configuration registers. ■ 1: Compose from corresponding register of parent PF. ■ 0: Use the exvf_cdm_din. In the SR-IOV specification, some VF's registers are defined to have the same value as the corresponding register in the parent PF. Exists: (CX_EXTENSIBLE_VFUNC) && (CX_EXTENSIBLE_CONVERT_IN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: exvf_cdm_ack is asserted
cdm_exvf_convert_pfddata[31:0]	O	Indicates whether the controller should compose the read data of the selected external VF register by copying the contents from the corresponding register of the parent PF or from selected external VF's configuration registers. ■ 1: Compose from the corresponding register of the parent PF. ■ 0: Use the exvf_cdm_din. In the SR-IOV specification, some VF's registers are defined to have the same value as the corresponding register in the parent PF. Exists: (CX_EXTENSIBLE_VFUNC) && (!CX_EXTENSIBLE_CONVERT_IN) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: exvf_cdm_ack is asserted

2.110 EXTSRIOV: VF Error Flag Signals

- radm_rcvd_wreq_poisoned_vf
- radm_rcvd_cpl_poisoned_vf
- radm_rcvd_req_ur_vf
- radm_rcvd_req_ca_vf
- radm_rcvd_cpl_ur_vf
- radm_rcvd_cpl_ca_vf
- radm_rcvd_err_func_num
- radm_rcvd_err_vfunc_num
- radm_rcvd_err_vfindx
- radm_hdr_log_valid_vf
- radm_hdr_log
- radm_tlp_prfx_log
- xtlh_xmt_cpl_ca_vf
- xtlh_xmt_wreq_poisoned_vf
- xtlh_xmt_err_func_num
- xtlh_xmt_err_vfunc_num
- xtlh_xmt_err_vfindx

Table 2-110 EXTSRIOV: VF Error Flag Signals

Port Name	I/O	Description
radm_rcvd_wreq_poisoned_vf[(FX_TLP-1):0]	O	One-clock-cycle pulse that indicates that the controller received a poisoned write request. Sets Poisoned TLP Status in Uncorrectable Error Status and Detected Parity Error in Status Register in application logic. The PF/VF Number is indicated by radm_rcvd_err_func_num and radm_rcvd_err_vfunc_num. This signal is also asserted for an internal VF error. This signal is not asserted for a PF error. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (One-clock-cycle pulse) Validated by: Not validated by another signal
radm_rcvd_cpl_poisoned_vf[(FX_TLP-1):0]	O	One-clock-cycle pulse that indicates that the controller received a poisoned Completion. Set Poisoned TLP Status in Uncorrectable Error Status and Master Data Parity Error/Detected Parity Error in Status Register in application logic. The PF/VF Number is indicated by radm_rcvd_err_func_num and radm_rcvd_err_vfunc_num. This signal is also asserted for an internal VF error. This signal is not asserted for a PF error. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (One-clock-cycle pulse) Validated by: Not validated by another signal

Port Name	I/O	Description
radm_rcvd_req_ur_vf[(FX_TLP-1):0]	O	One-clock-cycle pulse that indicates that the controller received an unsupported request. Set Unsupported Request Error Status in Uncorrectable Error Status in application logic. The PF/VF Number is indicated by radm_rcvd_err_func_num and radm_rcvd_err_vfunc_num. This signal is also asserted for an internal VF error. This signal is not asserted for a PF error. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (One-clock-cycle pulse) Validated by: Not validated by another signal
radm_rcvd_req_ca_vf[(FX_TLP-1):0]	O	One-clock-cycle pulse that indicates that the controller received a completer abort request. Set Completer Abort Error Status in Uncorrectable Error Status in application logic. The PF/VF Number is indicated by radm_rcvd_err_func_num and radm_rcvd_err_vfunc_num. This signal is also asserted for an internal VF error. This signal is not asserted for a PF error.. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g,core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (One-clock-cycle pulse) Validated by: Not validated by another signal
radm_rcvd_cpl_ur_vf[(FX_TLP-1):0]	O	One-clock-cycle pulse that indicates that the controller received a UR status completion. Set Received Master Abort in Status Register in application logic. The PF/VF Number is indicated by radm_rcvd_err_func_num and radm_rcvd_err_vfunc_num. This signal is also asserted for an internal VF error. This signal is not asserted for a PF error. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (One-clock-cycle pulse) Validated by: Not validated by another signal

Port Name	I/O	Description
radm_rcvd_cpl_ca_vf[(FX_TLP-1):0]	O	One-clock-cycle pulse that indicates that the controller received a CA status completion. Set Received Target Abort in Status Register in application logic. The PF/VF Number is indicated by radm_rcvd_err_func_num and radm_rcvd_err_vfunc_num. This signal is also asserted for an internal VF error. This signal is not asserted for a PF error. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (One-clock-cycle pulse). Validated by: Not validated by another signal
radm_rcvd_err_func_num[(((FX_TLP*PF_WD)-1):0)]	O	Indicates the PF number of the following errors. Function numbering starts at '0'. ■ radm_rcvd_wreq_poisoned_vf ■ radm_rcvd_cpl_poisoned_vf ■ radm_rcvd_req_ur_vf ■ radm_rcvd_req_ca_vf ■ radm_rcvd_cpl_ur_vf ■ radm_rcvd_cpl_ca_vf Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: radm_rcvd_wreq_poisoned_vf or radm_rcvd_cpl_poisoned_vf or radm_rcvd_req_ur_vf or radm_rcvd_req_ca_vf or radm_rcvd_cpl_ur_vf or radm_rcvd_cpl_ca_vf

Port Name	I/O	Description
radm_rcvd_err_vfunc_num[((FX_TLP*(VF_WD-1))-1):0]	O	Indicates which virtual function (VF) had following errors. The PCIe SR-IOV specification starts numbering VFs at '1'. To ease timing during synthesis, the PCIe controller starts numbering VFs at '0'. For example, radm_rcvd_err_vfunc_num =0 refers to the first VF of the PF that is identified by radm_rcvd_err_func_num. For more information of VF numbering, see 'Programming View'. ■ radm_rcvd_wreq_poisoned_vf ■ radm_rcvd_cpl_poisoned_vf ■ radm_rcvd_req_ur_vf ■ radm_rcvd_req_ca_vf ■ radm_rcvd_cpl_ur_vf ■ radm_rcvd_cpl_ca_vf Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: radm_rcvd_wreq_poisoned_vf or radm_rcvd_cpl_poisoned_vf or radm_rcvd_req_ur_vf or radm_rcvd_req_ca_vf or radm_rcvd_cpl_ur_vf or radm_rcvd_cpl_ca_vf are asserted
radm_rcvd_err_vfindx[((FX_TLP*VFI_WD)-1):0]	O	Indicates which VF Index had following errors. ■ radm_rcvd_wreq_poisoned_vf ■ radm_rcvd_cpl_poisoned_vf ■ radm_rcvd_req_ur_vf ■ radm_rcvd_req_ca_vf ■ radm_rcvd_cpl_ur_vf ■ radm_rcvd_cpl_ca_vf Your application knows how many VFs are assigned to each PF, and by using this knowledge, it can identify the parent PF and the VF number within that PF. For more information, see the VF Index in 'Bus Numbering Overview'. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk,{radm_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: radm_rcvd_wreq_poisoned_vf or radm_rcvd_cpl_poisoned_vf or radm_rcvd_req_ur_vf or radm_rcvd_req_ca_vf or radm_rcvd_cpl_ur_vf or radm_rcvd_cpl_ca_vf are asserted

Port Name	I/O	Description
radm_hdr_log_valid_vf[(FX_TLP-1):0]	O	Indicate that radm_hdr_log and radm_tlp_prfx_log are valid. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g,radm_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
radm_hdr_log[((FX_TLP*128)-1):0]	O	Indicates Header Log of following errors. ■ radm_rcvd_wreq_poisoned_vf ■ radm_rcvd_cpl_poisoned_vf ■ radm_rcvd_req_ur_vf ■ radm_rcvd_req_ca_vf Exists: CX_EXTENSIBLE_VFUNC Synchronous To: radm_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: radm_hdr_log_valid_vf is asserted
radm_tlp_prfx_log[((FX_TLP*128)-1):0]	O	Indicates TLP Prefix Log of following errors. ■ radm_rcvd_wreq_poisoned_vf ■ radm_rcvd_cpl_poisoned_vf ■ radm_rcvd_req_ur_vf ■ radm_rcvd_req_ca_vf Exists: (CX_EXTENSIBLE_VFUNC) && (CX_TLP_PREFIX_ENABLE) Synchronous To: None,perbitclk,radm_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: radm_hdr_log_valid_vf is asserted
xtlh_xmt_cpl_ca_vf	O	One-clock-cycle pulse that indicates that the controller transmitted a completion with CA status for a VF. Set Signaled Target Abort in Status register in application logic. The PF/VF Number is indicated by xtlh_xmt_err_func_num and xtlh_xmt_err_vfunc_num. This signal is also asserted for an internal VF error. This signal is not asserted for a PF error. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
xtlh_xmt_wreq_poisoned_vf	O	One-clock-cycle pulse that indicates that controller transmitted a memory write with poisoned for a VF. Set Master Data Parity Error in Status register in application logic. The PF/VF Number is indicated by xtlh_xmt_err_func_num and xtlh_xmt_err_vfunc_num. This signal is also asserted for an internal VF error. This signal is not asserted for a PF error. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
xtlh_xmt_err_func_num[(PF_WD-1):0]	O	Indicates PF number of following errors. Function numbering starts at '0'. ■ xtlh_xmt_cpl_ca_vf ■ xtlh_xmt_wreq_poisoned_vf Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
xtlh_xmt_err_vfunc_num[(VF_WD-2):0]	O	Indicates which virtual function (VF) had a following errors. The PCIe SR-IOV specification starts numbering VFs at '1'. To ease timing during synthesis, the PCIe controller starts numbering VFs at '0'. For example, xtlh_xmt_err_vfunc_num =0 refers to the first VF of the PF that is identified by xtlh_xmt_err_func_num. For more information of VF numbering, see 'Programming View'. ■ xtlh_xmt_cpl_ca_vf ■ xtlh_xmt_wreq_poisoned_vf Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

Port Name	I/O	Description
xtlh_xmt_err_vfindx[(VFI_WD-1):0]	O	Indicates which VF Index had following errors. ■ xtlh_xmt_cpl_ca_vf ■ xtlh_xmt_wreq_poisoned_vf Your application knows how many VFs are assigned to each PF, and by using this knowledge, it can identify the parent PF and the VF number within that PF. For more information, see the VF Index in 'Bus Numbering Overview'. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

2.111 EXTSRIOV: PF Registers Value for VF Function Signals

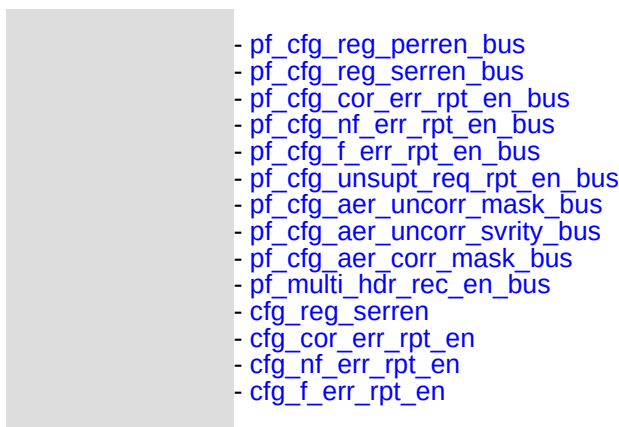


Table 2-111 EXTSRIOV: PF Registers Value for VF Function Signals

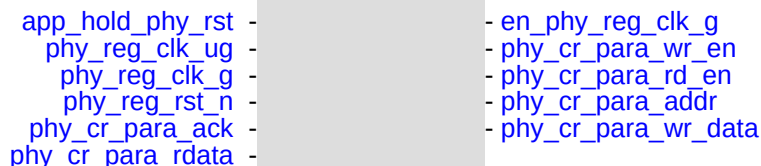
Port Name	I/O	Description
pf_cfg_reg_perren_bus[(NF-1):0]	O	PF's Parity Error Response registers value in Command Register of Type0 Header. You can use this value for setting error status registers of external VFs. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pf_cfg_reg_serren_bus[(NF-1):0]	O	PF's SERR# Enable registers value in Command Register of Type0 Header. You can use this value for setting error status registers of external VFs. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pf_cfg_cor_err_rpt_en_bus[(NF-1):0]	O	PF's Correctable Error Reporting Enable registers value in Device Control Register of PCIe Capability. You can use this value for sending ERR_MSG of external VFs. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pf_cfg_nf_err_rpt_en_bus[(NF-1):0]	O	PF's Non-Fatal Error Reporting Enable registers value in Device Control Register of PCIe Capability. You can use this value for sending ERR_MSG of external VFs. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pf_cfg_f_err_rpt_en_bus[(NF-1):0]	O	PF's Fatal Error Reporting Enable registers value in Device Control Register of PCIe Capability. You can use this value for sending ERR_MSG of external VFs. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pf_cfg_unsupt_req_rpt_en_bus[(NF-1):0]	O	PF's Unsupported Request Reporting Enable registers value in Device Control Register of PCIe Capability. You can use this value for sending ERR_MSG of external VFs. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pf_cfg_aer_uncorr_mask_bus[((32*NF)-1):0]	O	PF's all bit of Uncorrectable Error Mask Register of AER Capability. You can use this value for sending ERR_MSG of external VFs. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
pf_cfg_aer_uncorr_svrity_bus[((32*Nf)-1):0]	O	PF's all bit of Uncorrectable Error Severity Register of AER Capability. You can use this value for sending ERR_MSG of external VFs. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
pf_cfg_aer_corr_mask_bus[((32*Nf)-1):0]	O	PF's all bit of Correctable Error Mask Register of AER Capability. You can use this value for sending ERR_MSG of external VFs. Exists: CX_EXTENSIBLE_VFUNC Synchronous To: None,perbitclk,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
pf_multi_hdr_rec_en_bus[(Nf-1):0]	O	PF's Multiple Header Recording Enable Register in Advanced Error Capabilities and Control Register of AER Capability. You can use this value for Multiple Header LOG if you support Shared Header LOG in external VFs. Exists: (CX_EXTENSIBLE_VFUNC) && (!(CX_SRIOV_ENABLE && (CX_INTERNAL_NVFUNC != 0))) Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_reg_serren[(Nf-1):0]	O	PF's SERR# Enable registers value in Command Register of Type0 Header, for setting error status registers of external VFs. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_cor_err_rpt_en[(NF-1):0]	O	PF's Correctable Error Reporting Enable registers value in Device Control Register of PCIe Capability, for sending ERR_MSG of external VFs. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_nf_err_rpt_en[(NF-1):0]	O	PF's Non-Fatal Error Reporting Enable registers value in Device Control Register of PCIe Capability, for sending ERR_MSG of external VFs. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_f_err_rpt_en[(NF-1):0]	O	PF's Fatal Error Reporting Enable registers value in Device Control Register of PCIe Capability, for sending ERR_MSG of external VFs. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.112 PHY Register Bus Interface Signals



app_hold_phy_rst
 phy_reg_clk_ug
 phy_reg_clk_g
 phy_reg_rst_n
 phy_cr_para_ack
 phy_cr_para_rdata

en_phy_reg_clk_g
 phy_cr_para_wr_en
 phy_cr_para_rd_en
 phy_cr_para_addr
 phy_cr_para_wr_data

Table 2-112 PHY Register Bus Interface Signals

Port Name	I/O	Description
app_hold_phy_rst	I	Set this signal to one before the de-assertion of power on reset to hold the PHY in reset. This can be used to configure your PHY. Synopsys PHYs can be configured through the PHY Viewport if desired. Tie this port to zero if your design does not need to control PHY reset during power up. Exists: Always Synchronous To: aux_clk_g,{aux_clk} Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
phy_reg_clk_ug	I	Ungated clock for the PHY register interface. This clock should be always active when you need to access the PHY registers through the PHY register bus interface (PRBI). The maximum frequency of the clock is 100 MHz. <i>Note:</i> For Synopsys PHYs supporting Control Register (CR) Parallel Interface, this signal should not be connected to the PHY if the PHY registers are accessed through the PRBI. You should connect phy_reg_clk_g to the PHY. Exists: CX_PHY_VIEWPORT_ENABLE Synchronous To: N/A Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal

Port Name	I/O	Description
phy_reg_clk_g	I	This clock is gated by en_phy_reg_clk_g for the PHY Register Bus Interface. <i>Note:</i> For Synopsys PHYs supporting Control Register (CR) Parallel Interface, this clock should be used as cr_para_clk. Exists: CX_PHY_VIEWPORT_ENABLE Synchronous To: N/A Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: Not validated by another signal
en_phy_reg_clk_g	O	Enable phy_reg_clk_g. phy_reg_clk_g should be active while this signal is 1. Exists: CX_PHY_VIEWPORT_ENABLE Synchronous To: phy_reg_clk_ug Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_reg_rst_n	I	Reset for the phy_reg_clk_ug/phy_reg_clk_g domain. The reset logic should assert phy_reg_rst_n at the same time as sticky_rst_n and de-assert synchronous to phy_reg_clk_ug. / <i>Note:</i> For Synopsys PHYs supporting Control Register (CR) Parallel Interface, this signal should not be connected to the PHY. Exists: CX_PHY_VIEWPORT_ENABLE Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
phy_cr_para_ack[(PHY_VPT_NUM-1):0]	I	Control Register (CR) Read/Write acknowledgment. This signal indicates the completion of a CR parallel read/write access. Exists: CX_PHY_VIEWPORT_ENABLE Synchronous To: phy_reg_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
phy_cr_para_rdata[((PHY_VPT_NUM*PHY_VPT_DATA)-1):0]	I	Control Register (CR) read data. Exists: CX_PHY_VIEWPORT_ENABLE Synchronous To: phy_reg_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: phy_cr_para_ack
phy_cr_para_wr_en[(PHY_VPT_NUM-1):0]	O	Control Register (CR) write enable. When this signal is asserted, data on phy_cr_para_wr_data is written to the address referenced on phy_cr_para_addr. This signal is asserted for a single cycle. Exists: CX_PHY_VIEWPORT_ENABLE Synchronous To: phy_reg_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_cr_para_rd_en[(PHY_VPT_NUM-1):0]	O	Control Register (CR) read enable. When this signal is asserted, data is read from the address referenced on phy_cr_para_addr and provided on phy_cr_para_rd_data. This signal is asserted for a single cycle. Exists: CX_PHY_VIEWPORT_ENABLE Synchronous To: phy_reg_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
phy_cr_para_addr[15:0]	O	Control Register (CR) Read/Write address. Exists: CX_PHY_VIEWPORT_ENABLE Synchronous To: phy_reg_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: (phy_cr_para_wr_en) (phy_cr_para_rd_en)
phy_cr_para_wr_data[(PHY_VPT_DATA-1):0]	O	Control Register (CR) write data. Exists: CX_PHY_VIEWPORT_ENABLE Synchronous To: phy_reg_clk_g Registered: N/A Power Domain: PD_VAUX Active State: N/A Validated by: phy_cr_wr_en

2.113 SII: ATS Capability Signals

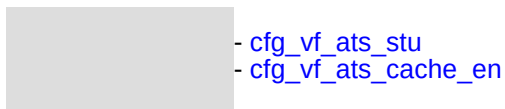


Table 2-113 SII: ATS Capability Signals

Port Name	I/O	Description
cfg_vf_ats_stu[((5*INT_NVF)-1):0]	O	ATS status for VFs Exists: VF_ATS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_vf_ats_cache_en[(INT_NVF-1):0]	O	ATS cache en Exists: VF_ATS_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.114 SII: DPC Signals

cxl_dpc_fmld_triggered	-	rppio_exception_error
cxl_dpc_fmld_trig_reason	-	cpl_lut_hdr_addra
cxl_dpc_fmld_trig_err_src_id	-	cpl_lut_hdr_addrb
app_dpc_rp_busy	-	cpl_lut_hdr_dina
cpl_lut_hdr_doutb	-	cpl_lut_hdr_ena
cpl_lut_prfx_doutb	-	cpl_lut_hdr_enb
app_dpcimpspec_log	-	cpl_lut_hdr_wea
app_flush_ack	-	cpl_lut_prfx_addra
		cpl_lut_prfx_addrb
		cpl_lut_prfx_dina
		cpl_lut_prfx_ena
		cpl_lut_prfx_enb
		cpl_lut_prfx_wea
		core_in_dpc
		cfg_dpc_trig_en
		cfg_dpc_cpl_cntrl
		cfg_dpc_int_en
		cfg_dpc_err_cor_en
		cfg_dpc_sw_trigger
		cfg_dpc_poison_tlp_egr_block_en
		cfg_dpc_dl_active_err_cor_en
		rstctl_core_flush_req

Table 2-114 SII: DPC Signals

Port Name	I/O	Description
cxl_dpc_fmld_triggered	I	Indicates that a recieved TLP owned by FMLD, had triggered DPC. This event sets DPC in the Switch downstream port of CXL controller. Exists: (CX_DPC_ENABLE) && (CX_CXL_IS_SW_MLD_DPC_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

Port Name	I/O	Description
cxl_dpc_fmld_trig_reason[1:0]	I	DPC Trigger reason for FMLD owned TLP. Indicates the reason for DPC triggered by TLP owned by FMLD. 01b: sets when the Downstream Port detects an ERR_NONFATAL Message in a FMLD owned TLP. 10b: sets when the Downstream Port detects an ERR_FATAL Message in a FMLD owned TLP. 00b,11b: Reserved. Exists: (CX_DPC_ENABLE) && (CX_CXL_IS_SW_MLD_DPC_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: validated by cxl_dpc_fmld_triggered
cxl_dpc_fmld_trig_err_src_id[15:0]	I	DPC Trigger related Requestor ID of the recieved Message. Exists: (CX_DPC_ENABLE) && (CX_CXL_IS_SW_MLD_DPC_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: validated by cxl_dpc_fmld_triggered
app_dpc_rp_busy	I	DPC Root Port Busy. When Set, it indicates that the Root Port is busy with internal activity. DPC is event triggered. Used to update the corresponding DPC Status register field. Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cpl_lut_hdr_doutb[127:0]	I	Completion LUT RAM data output for all transmitted non-posted TLP headers Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) Synchronous To: perbitclk,radm_clk_g,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

Port Name	I/O	Description
cpl_lut_prfx_doutb[(DPC_PRFX_LOG_WD-1):0]	I	Completion LUT RAM data output for all transmitted non-posted TLP Prefixes Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) && (CX_TLP_PREFIX_ENABLE) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
rppio_exception_error[(NF-1):0]	O	Exception Error when RP PIO error occurs Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cpl_lut_hdr_addra[(CPL_LUT_PTR_WD-1):0]	O	Completion LUT RAM port A address for all transmitted non-posted TLP headers Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cpl_lut_hdr_addrb[(CPL_LUT_PTR_WD-1):0]	O	Completion LUT RAM port B address for all transmitted non-posted TLP headers Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cpl_lut_hdr_dina[127:0]	O	Completion LUT RAM data input for all transmitted non-posted TLP headers Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

Port Name	I/O	Description
cpl_lut_hdr_ena	O	Completion LUT RAM port A enable for all transmitted non-posted TLP headers Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cpl_lut_hdr_enb	O	Completion LUT RAM port B enable for all transmitted non-posted TLP headers Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cpl_lut_hdr_wea	O	Completion LUT RAM port A write enable for all transmitted non-posted TLP headers Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cpl_lut_prfx_addra[(CPL_LUT_PTR_WD-1):0]	O	Completion LUT RAM port A address for all transmitted non-posted TLP prefixes Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) && (CX_TLP_PREFIX_ENABLE) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cpl_lut_prfx_addrb[(CPL_LUT_PTR_WD-1):0]	O	Completion LUT RAM port B address for all transmitted non-posted TLP prefixes Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) && (CX_TLP_PREFIX_ENABLE) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

Port Name	I/O	Description
cpl_lut_prfx_dina[(DPC_PRFX_LOG_WD-1):0]	O	Completion LUT RAM data input for all transmitted non-posted TLP prefixes Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) && (CX_TLP_PREFIX_ENABLE) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cpl_lut_prfx_ena	O	Completion LUT RAM port A enable for all transmitted non-posted TLP prefixes Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) && (CX_TLP_PREFIX_ENABLE) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cpl_lut_prfx_enb	O	Completion LUT RAM port B enable for all transmitted non-posted TLP prefixes Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) && (CX_TLP_PREFIX_ENABLE) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cpl_lut_prfx_wea	O	Completion LUT RAM port A write enable for all transmitted non-posted TLP prefixes Exists: (CX_DPC_ENABLE) && (CX_DPC_RP_PIO_EXTNS) && (CX_TLP_PREFIX_ENABLE) Synchronous To: radm_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

Port Name	I/O	Description
app_dpc_impspec_log[31:0]	I	Root Port Programmed I/O Implementation-specific information associated with the recorded error which triggered Downstream Port Containment. Used to update DPC RP PIO ImpSpec Log Register. Exists: (CX_DPC_ENABLE) && (CX_RP_PIO_IMPSPEC) Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
core_in_dpc	O	PCIe Controller is in Downstream Port Containment mode. Exists: CX_DPC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cfg_dpc_trig_en[1:0]	O	DPC Trigger Enable. Exists: CX_DPC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cfg_dpc_cpl_cntrl	O	DPC Completion Control. Exists: CX_DPC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cfg_dpc_int_en	O	DPC Interrupt Enable. Exists: CX_DPC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_dpc_err_cor_en	O	DPC ERR_COR Enable Exists: CX_DPC_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cfg_dpc_sw_trigger	O	DPC Software Trigger Exists: CX_DPC_ENABLE Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cfg_dpc_poison_tlp_egr_block_en	O	DPC Poisoned TLP Egress Blocking Enable. Exists: CX_DPC_ENABLE Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
cfg_dpc_dl_active_err_cor_en	O	DPC DL_Active ERR_COR Enable Exists: CX_DPC_ENABLE Synchronous To: None,{aux_clk_g} Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
app_flush_ack	I	Application Flush Acknowledge. When set, application acknowledges that new requests are stalled and all pending transfers in the pipeline have completed. Exists: (!AXI_RADM_SEG_BUF_ENABLE && FLUSH_CNTRL_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
rstctl_core_flush_req	O	Flush request after LTSSM has been directed to DISABLE state. Exists: (!AXI_RADM_SEG_BUF_ENABLE && FLUSH_CNTRL_ENABLE) Synchronous To: core_clk Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

2.115 SII: TLP Bypass Internal Error Reporting Signals

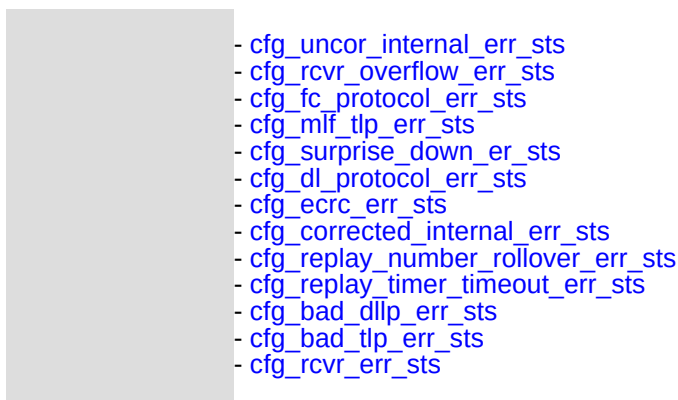


Table 2-115 SII: TLP Bypass Internal Error Reporting Signals

Port Name	I/O	Description
cfg_uncor_internal_err_sts	O	Indication from the controller that the controller has detected an Uncorrectable Internal Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_rcvr_overflow_err_sts	O	Indication from the controller that the controller has detected an Receiver Overflow Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_fc_protocol_err_sts	O	Indication from the controller that the controller has detected an Flow Control Protocol Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_mlf_tlp_err_sts	O	Indication from the controller that the controller detected an Malformed TLP Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_surprise_down_er_sts	O	Indication from the controller that the controller detected an Surprise Down Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_dl_protocol_err_sts	O	Indication from the controller that the controller detected an Data Link Protocol Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_ecrc_err_sts	O	Indication from the controller that the controller detected an ECRC Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_corrected_internal_err_sts	O	Indication from the controller that the controller detected an Corrected Internal Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_replay_number_rollover_err_sts	O	Indication from the controller that the controller detected an REPLAY_NUMBER Rollover Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_replay_timer_timeout_err_sts	O	Indication from the controller that the controller detected an Replay Timer Timeout. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bad_dllp_err_sts	O	Indication from the controller that the controller detected an Bad DLLP Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_bad_tlp_err_sts	O	Indication from the controller that the controller detected an Bad TLP Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
cfg_rcvr_err_sts	O	Indication from the controller that the controller detected an Receiver Error. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.116 PIPE-C Signals

phy_mac_txcready -
 phy_mac_rxcvalid -

Table 2-116 PIPE-C Signals

Port Name	I/O	Description
phy_mac_txcready	I	The USB4 PCIe Adapter asserts this signal to indicate Tx back-pressure. Exists: (CX_PCIE_MODE == PCIE_OVER_CIO) Synchronous To: aux_clk_g,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
phy_mac_rxcvalid	I	The USB4 PCIe Adapter de-asserts this signal to indicate Rx pacing. Exists: (CX_PCIE_MODE == PCIE_OVER_CIO) Synchronous To: core_clk_ug,{core_clk} Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
mac_phy_txcvalid	O	The controller always asserts mac_phy_txcvalid signal. Exists: (CX_PCIE_MODE == PCIE_OVER_CIO) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal
mac_phy_rxcready	O	The controller always asserts mac_phy_rxcread signal. Exists: (CX_PCIE_MODE == PCIE_OVER_CIO) Synchronous To: None Registered: N/A Power Domain: PD_VAUX Active State: High (level) Validated by: Not validated by another signal

2.117 SII: ROM Validation Signals

exp_rom_validation_status_strobe
exp_rom_validation_status
exp_rom_validation_details_strobe
exp_rom_validation_details

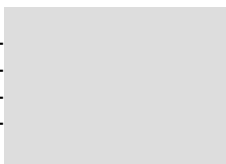


Table 2-117 SII: ROM Validation Signals

Port Name	I/O	Description
exp_rom_validation_status_strobe[(NF-1):0]	I	This strobe sets the Expansion ROM Validation Status field (ROM_BAR_VALIDATION_STATUS) of the Expansion ROM BAR register as specified on the exp_rom_validation_status input signal. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: high Validated by: Not validated by another signal
exp_rom_validation_status[((NF*3)-1):0]	I	You can use this signal to notify the controller about the expansion ROM validation status. On assertion of exp_rom_validation_status_strobe, the controller stores the value of this signal in Expansion ROM Validation Status field (ROM_BAR_VALIDATION_STATUS) of the Expansion ROM BAR register. ■ [000b] Validation not supported ■ [001b] Validation in Progress ■ [010b] Validation Pass Valid contents, trust test is not performed ■ [011b] Validation Pass Valid and trusted contents ■ [100b] Validation Fail Invalid contents ■ [101b] Validation Fail Valid but untrusted contents (For example, Out of Date, Expired or Revoked Certificate) ■ [110b] Warning Pass Validation Passed with implementation-specific warning. Valid contents, trust test is not performed ■ [111b] Warning Pass Validation Passed with implementation-specific warning. Valid and trusted contents Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: high Validated by: exp_rom_validation_status_strobe

Port Name	I/O	Description
exp_rom_validation_details_strobe[(NF-1):0]	I	This strobe sets the Expansion ROM Validation Details field (ROM_BAR_VALIDATION_DETAILS) of the Expansion ROM BAR register as specified on the exp_rom_validation_details input signal. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: high Validated by: Not validated by another signal
exp_rom_validation_details[((NF*4)-1):0]	I	You can use this signal to notify the controller about the expansion ROM validation details. On assertion of exp_rom_validation_details_strobe, the controller stores the value of this signal in Expansion ROM Validation Details field (ROM_BAR_VALIDATION_DETAILS) of the Expansion ROM BAR register. Exists: Always Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: high Validated by: exp_rom_validation_details_strobe

2.118 Emergency Power Reduction Signals

pwrbrk_n -  - cfg_emergency_pwr_red_det

Table 2-118 Emergency Power Reduction Signals

Port Name	I/O	Description
pwrbrk_n	I	Indicates requests to enter and exit the Emergency Power Reduction State. Exists: EMERGENCY_POWER_REDUCTION_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: Low Validated by: Not validated by another signal
cfg_emergency_pwr_red_det[(NF-1):0]	O	Indicates entry and exit to Power reduction state. Exists: EMERGENCY_POWER_REDUCTION_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

2.119 Native PCIe Enclosure Management Signals

npem_cmd_cmplt -
 npem_status -

- cfg_npem_cmd
- npem_enable
- npem_initiate_reset
- npem_ok_control
- npem_locate_control
- npem_fail_control
- npem_rebuild_control
- npem_pfa_control
- npem_hot_spare_control
- npem_critical_array_control
- npem_failed_array_control
- npem_invalid_device_type_control
- npem_disabled_control
- npem_enclosure_specific_control

Table 2-119 Native PCIe Enclosure Management Signals

Port Name	I/O	Description
npem_cmd_cmplt	I	NPEM Command completed Status. Application drives a pulse on this port to indicate that the NPEM command has completed and that the NPEM controller is ready to accept a subsequent command. Controller latches a one in the NPEM Command completed Status register when this signal is released. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_status[7:0]	I	NPEM Enclosure-Specific Status. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
cfg_npem_cmd	O	NPEM Command. When set, this bit indicates that a NPEM command is issued to the NPEM control register by Software. This signal is asserted for 1 clock cycle when a NPEM command is issued by Software regardless of the npem_enable port status. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_enable	O	NPEM Enable. When Set, this bit enables the NPEM capability. When Clear, this bit disables the NPEM capability. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_initiate_reset	O	NPEM Initiate Reset. When Set, this bit initiates the NPEM Reset procedure. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_ok_control	O	NPEM OK Control. When Set, this bit enables the NPEM OK indicator. When Clear, this bit disables the NPEM OK indicator. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_locate_control	O	NPEM Locate Control. When Set, this bit enables the NPEM Locate indicator. When Clear, this bit disables the NPEM Locate indicator. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
npem_fail_control	O	NPEM Fail Control. When Set, this bit enables the NPEM Fail indicator. When Clear, this bit disables the NPEM Fail indicator. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_rebuild_control	O	NPEM Rebuild Control. When Set, this bit enables the NPEM Rebuild indicator. When Clear, this bit disables the NPEM Rebuild indicator. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_pfa_control	O	NPEM PFA Control. When Set, this bit enables the NPEM PFA indicator. When Clear, this bit disables the NPEM PFA indicator. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_hot_spare_control	O	NPEM Hot Spare Control. When Set, this bit enables the NPEM Hot Spare indicator. When Clear, this bit disables the NPEM Hot Spare indicator. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_critical_array_control	O	NPEM Critical Array Control. When Set, this bit enables the NPEM Critical Array indicator. When Clear, this bit disables the NPEM Critical Array indicator. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Port Name	I/O	Description
npem_failed_array_control	O	NPEM Failed Array Control. When Set, this bit enables the NPEM Failed Array indicator. When Clear, this bit disables the NPEM Failed Array indicator. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_invalid_device_type_control	O	NPEM Invalid Device Type Control. When Set, this bit enables the NPEM Invalid Device Type indicator. When Clear, this bit disables the NPEM Invalid Device Type indicator. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_disabled_control	O	NPEM Disabled Array Control. When Set, this bit enables the NPEM Disabled Array indicator. When Clear, this bit disables the NPEM Disabled indicator. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal
npem_enclosure_specific_control[7:0]	O	NPEM Enclosure Specific Control. When Set, these bits enable the NPEM Enclosure Specific indicators. When Clear, these bits disable the NPEM Enclosure Specific indicators. Exists: NPEM_ENABLE Synchronous To: aux_clk_g Registered: N/A Power Domain: PD_VAUX Active State: High Validated by: Not validated by another signal

Register Descriptions

This chapter details all possible registers in the controller. They are arranged hierarchically into maps and blocks (banks). For configurable IP titles, your actual configuration might not contain all of these registers.

Attention: For configurable IP titles, do not use this document to determine the exact attributes of your register map. It is for reference purposes only.

When you configure the controller in coreConsultant, you must access the register attributes for your actual configuration at `workspace/report/ComponentRegisters.html` or `workspace/report/ComponentRegisters.xml` after you have completed the report creation activity. That report comes from the exact same source as this chapter but removes all the registers that are not in your actual configuration. This does not apply to non-configurable IP titles. In addition, all parameter expressions are evaluated to actual values. Therefore, the Offset and Memory Access values might change depending on your actual configuration.

Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

Exists Expressions

These expressions indicate the combination of configuration parameters required for a register, field, or block to exist in the memory map. The expression is only valid in the local context and does not indicate the conditions for existence of the parent. For example, the expression for a bit field in a register assumes that the register exists and does not include the conditions for existence of the register.

Offset

The term *Offset* is synonymous with *Address*.

Memory Access Attributes

The Memory Access attribute is defined as `<ReadBehavior>/<WriteBehavior>` which are defined in the following table.

Table 3-1 Possible Read and Write Behaviors

Read (or Write) Behavior	Description
RC	A read clears this register field.
RS	A read sets this register field.
RM	A read modifies the contents of this register field.
Wo	You can only write to this register once field.
W1C	A write of 1 clears this register field.
W1S	A write of 1 sets this register field.
W1T	A write of 1 toggles this register field.
W0C	A write of 0 clears this register field.
W0S	A write of 0 sets this register field.
W0T	A write of 0 toggles this register field.
WC	Any write clears this register field.
WS	Any write sets this register field.
WM	Any write toggles this register field.
no Read Behavior attribute	You cannot read this register. It is Write-Only.
no Write Behavior attribute	You cannot write to this register. It is Read-Only.

Table 3-2 Memory Access Examples

Memory Access	Description
R	Read-only register field.
W	Write-only register field.
R/W	Read/write register field.
R/W1C	You can read this register field. Writing 1 clears it.
RC/W1C	Reading this register field clears it. Writing 1 clears it.
R/Wo	You can read this register field. You can only write to it once.

Special Optional Attributes

Some register fields might use the following optional attributes.

Table 3-3 Optional Attributes

Attribute	Description
Volatile	As defined by the IP-XACT specification. If true, indicates in the case of a write followed by read, or in the case of two consecutive reads, there is no guarantee as to what is returned by the read on the second transaction or that this return value is consistent with the write or read of the first transaction. The element implies there is some additional mechanism by which this field can acquire new values other than by reads/writes/resets and other access methods known to IP-XACT. For example, when the controller updates the register field contents.
Testable	As defined by the IP-XACT specification. Possible values are unconstrained, untestable, readOnly, writeAsRead, restore. Untestable means that this field is untestable by a simple automated register test. For example, the read-write access of the register is controlled by a pin or another register. readOnly means that you should not write to this register; only read from it. This might apply for a register that modifies the contents of another register.
Reset Mask	As defined by the IP-XACT specification. Indicates that this register field has an unknown reset value. For example, the reset value is set by another register or an input pin; or the register is implemented using RAM.
* Varies	Indicates that the memory access (or reset) attribute (read, write behavior) is not fixed. For example, the read-write access of the register is controlled by a pin or another register. Or when the access depends on some configuration parameter; in this case the post-configuration report in coreConsultant gives the actual access value.

PCI-compatible and PCI Express extended capability registers are organized into a linked list of structures (blocks). The address of each block is configuration-dependent and cannot be computed until configuration time in coreConsultant. In this document, register offsets within each block are specified (**for example B+0x04, where B+ indicates an offset of 0x04 within the block**); but the starting location of each block must be found by traversing the linked list in software after configuration. The host O/S is expected to traverse the linked list at boot time using the pointers in each block, and not the absolute address of the blocks.

The B+ symbol appearing in the Offset value (as in **Offset: B+0x4**) indicates an address offset relative to the capability base address. However, when you configure the core in coreConsultant, all parameters are evaluated completely; and the resulting Offsets in the IP-XACT and coreConsultant GUI reports are absolute, and not relative values.

CX_MEMORY_MAP_VIEW in Memory Access Expressions

You must set the CX_MEMORY_MAP_VIEW parameter (in coreConsultant) to choose the map that you want generated in the IP-XACT XML and DocBook XML files; and HTML reports. The register map as perceived from the remote link partner (1: wire view) is different to that of the local application (0: DBI view). For more details, see the Register Configuration Space chapter in the Databook. Sometimes you will see this parameter in expressions throughout this document. It is included for completeness. As previously mentioned, do not use this document to determine the exact attributes of your register map. It is for reference purposes only. Use the configured version of this document from coreConsultant where all parameters are resolved to hard values.

Register definitions for each component memory map.

Table 3-4 Registers for the DWC_PCIE_USP Memory Map

Register	Offset	Description
PF PCI-Compatible Configuration Space Header Type0 PF_TYPE0_HDRi (for i = 0; i <= CX_NFUNC-1) Exists: Always		
"DEVICE_ID_VENDOR_ID_REG" on page 982	B+0x0	Device ID , RCRB next offset pointer and Vendor ID Register.
"STATUS_COMMAND_REG" on page 985	B+0x4	Status and Command Register.
"CLASS_CODE_REVISION_ID" on page 992	B+0x8	Class Code and Revision ID Register.
"BIST_HEADER_TYPE_LATENCY_CACHE_LINE_SIZE_REG" on page 994	B+0xc	BIST, Header Type, Latency Timer, and Cache Line Size Register.
"BAR0_REG" on page 997	B+0x10	BAR0 Register.
"BAR1_REG" on page 1002	B+0x14	BAR1 Register.
"BAR2_REG" on page 1007	B+0x18	BAR2 Register.
"BAR3_REG" on page 1011	B+0x1c	BAR3 Register.
"BAR4_REG" on page 1015	B+0x20	BAR4 Register.
"BAR5_REG" on page 1019	B+0x24	BAR5 Register.
"CARDBUS_CIS_PTR_REG" on page 1023	B+0x28	CardBus CIS Pointer Register.
"SUBSYSTEM_ID_SUBSYSTEM_VENDOR_ID_REG" on page 1024	B+0x2c	Subsystem ID and Subsystem Vendor ID Register.
"EXP_ROM_BASE_ADDR_REG" on page 1026	B+0x30	Expansion ROM BAR Register.
"PCI_CAP_PTR_REG" on page 1030	B+0x34	Capabilities Pointer Register.
"MAX_LATENCY_MIN_GRANT_INTERRUPT_PIN_INTERRUPT_LINE_REG" on page 1031	B+0x3c	Max_Lat, Min_Gnt, Interrupt Pin, and Interrupt Line Register.
DBI2 Shadow Block: PF PCI-Compatible Configuration Space Header Type0 PF_TYPE0_HDR_DBI2i (for i = 0; i <= CX_NFUNC-1) Exists: Always		
"BAR0_MASK_REG" on page 1034	B+0x10	BAR\$ Mask Register.
"BAR1_MASK_REG" on page 1036	B+0x14	BAR1 Mask Register.
"BAR2_MASK_REG" on page 1038	B+0x18	BAR2 Mask Register.
"BAR3_MASK_REG" on page 1040	B+0x1c	BAR3 Mask Register.

Register	Offset	Description
"BAR4_MASK_REG" on page 1042	B+0x20	BAR4 Mask Register.
"BAR5_MASK_REG" on page 1044	B+0x24	BAR5 Mask Register.
"EXP_ROM_BAR_MASK_REG" on page 1046	B+0x30	Expansion ROM BAR Mask Register.
PF PCI Power Management Capability Structure PF_PM_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2)		
"CAP_ID_NXT_PTR_REG" on page 1049	B+0x0	Power Management Capabilities Register.
"CON_STATUS_REG" on page 1054	B+0x4	Power Management Control and Status Register.
PF MSI Capability Structure PF_MSI_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"PCI_MSI_CAP_ID_NEXT_CTRL_REG" on page 1059	B+0x0	MSI Capability Header and Message Control Register.
"MSI_CAP_OFF_04H_REG" on page 1066	B+0x4	Message Address Register for MSI (Offset 04h).
"MSI_CAP_OFF_08H_REG" on page 1068	B+0x8	Message Address Register for MSI (Offset 08h).
"MSI_CAP_OFF_0CH_REG" on page 1071	B+0xc	Message Address Register for MSI (Offset 0Ch).
"MSI_CAP_OFF_10H_REG" on page 1074	B+0x10	Message Address Register for MSI (Offset 10h).
"MSI_CAP_OFF_14H_REG" on page 1076	B+0x14	Message Address Register for MSI (Offset 14h).
PF PCI Express Capability Structure PF_PCIE_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: Always		
"PCIE_CAP_ID_PCIE_NEXT_CAP_PTR_PCIE_CAP_REG" on page 1078	B+0x0	PCI Express Capabilities, ID, Next Pointer Register.
"DEVICE_CAPABILITIES_REG" on page 1082	B+0x4	Device Capabilities Register.
"DEVICE_CONTROL_DEVICE_STATUS" on page 1091	B+0x8	Device Control and Device Status Register.
"LINK_CAPABILITIES_REG" on page 1103	B+0xc	Link Capabilities Register.
"LINK_CONTROL_LINK_STATUS_REG" on page 1112	B+0x10	Link Control and Link Status Register.
"DEVICE_CAPABILITIES2_REG" on page 1126	B+0x24	Device Capabilities 2 Register.

Register	Offset	Description
“DEVICE_CONTROL2_DEVICE_STATUS2_REG” on page 1138	B+0x28	Device Control 2 and Status 2 Register.
“LINK_CAPABILITIES2_REG” on page 1148	B+0x2c	Link Capabilities 2 Register.
“LINK_CONTROL2_LINK_STATUS2_REG” on page 1153	B+0x30	Link Control 2 and Status 2 Register.
DBI2 Shadow Block: PF PCI Express Capability Structure PF_PCIE_CAP_DBI2i (for i = 0; i <= CX_NFUNC-1) Exists: Always		
“SHADOW_LINK_CAPABILITIES_REG” on page 1169	B+0xc	Shadow Link Capabilities Register.
PF MSI-X Capability Structure PF_MSIX_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
“PCI_MSIX_CAP_ID_NEXT_CTRL_REG” on page 1179	B+0x0	MSI-X Capability ID, Next Pointer, Control Register.
“MSIX_TABLE_OFFSET_REG” on page 1183	B+0x4	MSI-X Table Offset and BIR Register.
“MSIX_PBA_OFFSET_REG” on page 1186	B+0x8	MSI-X PBA Offset and BIR Register.
DBI2 Shadow Block: PF MSI-X Capability Structure PF_MSIX_CAP_DBI2i (for i = 0; i <= CX_NFUNC-1) Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
“SHADOW_PCI_MSIX_CAP_ID_NEXT_CTRL_REG” on page 1190	B+0x0	MSI-X Capability ID, Next Pointer, Control Register.
“SHADOW_MSIX_TABLE_OFFSET_REG” on page 1192	B+0x4	MSI-X Table Offset and BIR Register.
“SHADOW_MSIX_PBA_OFFSET_REG” on page 1195	B+0x8	MSI-X PBA Offset and BIR Register.
VPD Capability Structure PF_VPD_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (VPD_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))		
“VPD_BASE” on page 1198	B+0x0	VPD Control and Capabilities Register.
“DATA_REG” on page 1201	B+0x4	VPD Data Register.

Register	Offset	Description
PF Advanced Error Reporting Capability Structure PF_AER_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"AER_EXT_CAP_HDR_OFF" on page 1203	B+0x0	Advanced Error Reporting Extended Capability Header.
"UNCORR_ERR_STATUS_OFF" on page 1205	B+0x4	Uncorrectable Error Status Register.
"UNCORR_ERR_MASK_OFF" on page 1211	B+0x8	Uncorrectable Error Mask Register.
"UNCORR_ERR_SEV_OFF" on page 1217	B+0xc	Uncorrectable Error Severity Register.
"CORR_ERR_STATUS_OFF" on page 1223	B+0x10	Correctable Error Status Register.
"CORR_ERR_MASK_OFF" on page 1227	B+0x14	Correctable Error Mask Register.
"ADV_ERR_CAP_CTRL_OFF" on page 1230	B+0x18	Advanced Error Capabilities and Control Register.
"HDR_LOG_0_OFF" on page 1234	B+0x1c	Header Log Register 0.
"HDR_LOG_1_OFF" on page 1236	B+0x20	Header Log Register 1.
"HDR_LOG_2_OFF" on page 1238	B+0x24	Header Log Register 2.
"HDR_LOG_3_OFF" on page 1240	B+0x28	Header Log Register 3.
"TLP_PREFIX_LOG_i_OFF (for i = 1; i <= 4)" on page 1242	B+0x38 + i*0x4	TLP Prefix Log Register 1.
Virtual Channel Capability Structure PF0_VC_CAP Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))		
"VC_BASE" on page 1245	B+0x0	VC Extended Capability Header.
"VC_CAPABILITIES_REG_1" on page 1247	B+0x4	Port VC Capability Register 1.
"VC_CAPABILITIES_REG_2" on page 1250	B+0x8	Port VC Capability Register 2.
"VC_STATUS_CONTROL_REG" on page 1252	B+0xc	Port VC Control and Status Register.
"RESOURCE_CAP_REG_VC0" on page 1254	B+0x10	VC Resource Capability Register (0).
"RESOURCE_CON_REG_VC0" on page 1258	B+0x14	VC Resource Control Register (0).
"RESOURCE_STATUS_REG_VCi (for i = 0; i <= 7)" on page 1262	B+0x18 + i*0x4	VC Resource Status Register (0).
"RESOURCE_CAP_REG_VCi (for i = 1; i <= 7)" on page 1264	B+0x1c + i*0x4	VC Resource Capability Register (1).
"RESOURCE_CON_REG_VCi (for i = 1; i <= 7)" on page 1267	B+0x20 + i*0x4	VC Resource Control Register (1).
"VC_ARBI_TABLE_REG0_OFF" on page 1271	B+0x70	VC Arbitration Table entry 0.

Register	Offset	Description
"VC_ARBI_TABLE_REG1_OFF" on page 1272	B+0x74	VC Arbitration Table entry 1.
"VC_ARBI_TABLE_REG2_OFF" on page 1273	B+0x78	VC Arbitration Table entry 2.
"VC_ARBI_TABLE_REG3_OFF" on page 1274	B+0x7c	VC Arbitration Table entry 3.
"VC_ARBI_TABLE_REG4_OFF" on page 1275	B+0x80	VC Arbitration Table entry 4.
"VC_ARBI_TABLE_REG5_OFF" on page 1276	B+0x84	VC Arbitration Table entry 5.
"VC_ARBI_TABLE_REG6_OFF" on page 1277	B+0x88	VC Arbitration Table entry 6.
"VC_ARBI_TABLE_REG7_OFF" on page 1278	B+0x8c	VC Arbitration Table entry 7.
"VC_ARBI_TABLE_REG8_OFF" on page 1279	B+0x90	VC Arbitration Table entry 8.
"VC_ARBI_TABLE_REG9_OFF" on page 1280	B+0x94	VC Arbitration Table entry 9.
"VC_ARBI_TABLE_REG10_OFF" on page 1281	B+0x98	VC Arbitration Table entry 10.
"VC_ARBI_TABLE_REG11_OFF" on page 1282	B+0x9c	VC Arbitration Table entry 11.
"VC_ARBI_TABLE_REG12_OFF" on page 1283	B+0xa0	VC Arbitration Table entry 12.
"VC_ARBI_TABLE_REG13_OFF" on page 1284	B+0xa4	VC Arbitration Table entry 13.
"VC_ARBI_TABLE_REG14_OFF" on page 1285	B+0xa8	VC Arbitration Table entry 14.
"VC_ARBI_TABLE_REG15_OFF" on page 1286	B+0xac	VC Arbitration Table entry 15.
Device Serial Number Capability Structure PF_SN_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (SERIAL_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))		
"SN_BASE" on page 1288	B+0x0	Device Serial Number Extended Capability Header.
"SER_NUM_REG_DW_1" on page 1291	B+0x4	Serial Number 1 Register.
"SER_NUM_REG_DW_2" on page 1292	B+0x8	Serial Number 2 Register.
Power Budgeting Capability Structure PF_PB_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"PB_BASE" on page 1294	B+0x0	Power Budgeting Extended Capability Header.
"PB_DATA_SELECT" on page 1297	B+0x4	Data select Register.

Register	Offset	Description
"DATA_REG_PB" on page 1298	B+0x8	Data Register.
"CAP_REG_PB" on page 1301	B+0xc	Power Budget Capability Register.
PF ARI Capability Structure PF_ARI_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"ARI_BASE" on page 1303	B+0x0	ARI Capability Header.
"CAP_REG" on page 1305	B+0x4	ARI Capability and Control Register.
Secondary PCI Express Capability Structure PF0_SPCIE_CAP Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2))		
"SPCIE_CAP_HEADER_REG" on page 1311	B+0x0	SPCIE Capability Header.
"LINK_CONTROL3_REG" on page 1313	B+0x4	Link Control 3 Register.
"LANE_ERR_STATUS_REG" on page 1316	B+0x8	Lane Error Status Register.
"SPCIE_CAP_OFF_iH_REG (for i = 0C; i <= 28)" on page 1318	B+0xc + i*0x4	Lane Equalization Control Register for lanes 1 and 0.
Physical Layer 16.0 GT/s Extended Capability Structure PF0_PL16G_CAP Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))		
"PL16G_EXT_CAP_HDR_REG" on page 1328	B+0x0	Physical Layer 16.0 GT/s Extended Capability Header.
"PL16G_CAPABILITY_REG" on page 1330	B+0x4	16.0 GT/s Capabilities Register.
"PL16G_CONTROL_REG" on page 1331	B+0x8	16.0 GT/s Control Register.
"PL16G_STATUS_REG" on page 1332	B+0xc	16.0 GT/s Status Register.
"PL16G_LC_DPAR_STATUS_REG" on page 1335	B+0x10	16.0 GT/s Local Data Parity Mismatch Status Register.
"PL16G_FIRST_RETIMER_DPAR_STATUS_REG" on page 1337	B+0x14	16.0 GT/s First Retimer Data Parity Mismatch Status Register.
"PL16G_SECOND_RETIMER_DPAR_STATUS_REG" on page 1339	B+0x18	16.0 GT/s Second Retimer Data Parity Mismatch Status Register.
"PL16G_CAP_OFF_iH_REG (for i = 20; i <= 2C)" on page 1341	B+0x20 + i*0x4	16.0 GT/s Lane Equalization Control Register for Lane 0-3.
Margining Extended Capability Structure PF0_MARGIN_CAP Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))		

Register	Offset	Description
"MARGIN_EXT_CAP_HDR_REG" on page 1351	B+0x0	Margining Extended Capability Header.
"MARGIN_PORT_CAPABILITIES_STATUS_REG" on page 1353	B+0x4	Margining Port Capabilities and Status Register.
"MARGIN_LANE_CNTRL_STATUS _i _REG (for $i = 0; i \leq 15$)" on page 1356	B+0x8 + $i \times 0x4$	Margining Lane Control and Status Register for Lane 0.
Physical Layer 32.0 GT/s Extended Capability Structure PF0_PL32G_CAP Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE < 2))		
"PL32G_EXT_CAP_HDR_REG" on page 1360	B+0x0	Physical Layer 32.0 GT/s Extended Capability Header.
"PL32G_CAPABILITY_REG" on page 1362	B+0x4	32.0 GT/s Capabilities Register.
"PL32G_CONTROL_REG" on page 1365	B+0x8	32.0 GT/s Control Register.
"PL32G_STATUS_REG" on page 1368	B+0xc	32.0 GT/s Status Register.
"PL32G_RCVD_MOD_TS_DATA1_REG" on page 1372	B+0x10	Received Modified TS Data 1 Register.
"PL32G_RCVD_MOD_TS_DATA2_REG" on page 1375	B+0x14	Received Modified TS Data 2 Register.
"PL32G_TX_MOD_TS_DATA1_REG" on page 1378	B+0x18	Transmitted Modified TS Data 1 Register.
"PL32G_TX_MOD_TS_DATA2_REG" on page 1380	B+0x1c	Transmitted Modified TS Data 2 Register.
"PL32G_CAP_OFF_iH_REG (for $i = 20; i \leq 2C$)" on page 1382	B+0x20 + $i \times 0x4$	32.0 GT/s Lane Equalization Control Register for Lane 0-3.
SR-IOV Capability Structure PF_SRIOV_CAP _i (for $i = 0; i \leq CX_NFUNC - 1$) Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE == 0 CX_MEMORY_MAP_PCIE_TYPE == 2))		
"SRIOV_BASE_REG" on page 1388	B+0x0	SR-IOV Capability Header Register.
"CAPABILITIES_REG" on page 1391	B+0x4	SR-IOV Capability Register.
"STATUS_CONTROL_REG" on page 1394	B+0x8	SR-IOV Control and Status Register.
"TOTAL_VFS_INITIAL_VFS_REG" on page 1397	B+0xc	TotalVFs InitialVFs Register.
"SRIOV_NUM_VFS" on page 1399	B+0x10	NumVFs and Function Dependency Link Register.
"SRIOV_VF_OFFSET_POSITION" on page 1402	B+0x14	VF Stride and Offset Register.

Register	Offset	Description
"VF_DEVICE_ID_REG" on page 1405	B+0x18	VF Device ID Register
"SUP_PAGE_SIZES_REG" on page 1407	B+0x1c	Supported Page Sizes.
"SYSTEM_PAGE_SIZE_REG" on page 1408	B+0x20	System Page Size.
"SRIOV_BAR0_REG" on page 1409	B+0x24	VF BAR0.
"SRIOV_BAR1_REG" on page 1411	B+0x28	VF BAR1.
"SRIOV_BAR2_REG" on page 1413	B+0x2c	VF BAR2.
"SRIOV_BAR3_REG" on page 1415	B+0x30	VF BAR3.
"SRIOV_BAR4_REG" on page 1417	B+0x34	VF BAR4.
"SRIOV_BAR5_REG" on page 1419	B+0x38	VF BAR5.
"VF_MIGRATION_STATE_ARRAY_REG" on page 1421	B+0x3c	VF Migration State Array Offset Register.
DBI2 Shadow Block: SR-IOV Capability Structure PF_SRIOV_CAP_DBI2i (for i = 0; i <= CX_NFUNC-1) Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))		
"SHADOW_SRIOV_VF_OFFSET_POSITION" on page 1424	B+0x14	Shadow VF Stride and Offset Register.
"SRIOV_BAR0_MASK_REG" on page 1427	B+0x24	BAR0 and BAR Mask.
"SRIOV_BAR1_MASK_REG" on page 1429	B+0x28	BAR1 and BAR Mask.
"SRIOV_BAR2_MASK_REG" on page 1431	B+0x2c	BAR2 and BAR Mask.
"SRIOV_BAR3_MASK_REG" on page 1433	B+0x30	BAR3 and BAR Mask.
"SRIOV_BAR4_MASK_REG" on page 1435	B+0x34	BAR4 and BAR Mask.
"SRIOV_BAR5_MASK_REG" on page 1437	B+0x38	BAR5 and BAR Mask.
PF TLP Processing Hints Capability Structure PF_TPH_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"TPH_EXT_CAP_HDR_REG" on page 1440	B+0x0	TPH Extended Capability Header.
"TPH_REQ_CAP_REG_REG" on page 1442	B+0x4	TPH Requester Capability Register.
"TPH_REQ_CONTROL_REG_REG" on page 1446	B+0x8	TPH Requester Control Register.
"TPH_ST_TABLE_REG_i (for i = 0; i <= 31)" on page 1449	B+0xc + i*0x4	TPH ST Table Register 0.

Register	Offset	Description
DBI2 Shadow Block: PF TLP Processing Hints Capability Structure PF_TPH_CAP_DBI2i (for i = 0; i <= CX_NFUNC-1) Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"SHADOW_TPH_REQ_CAP_REG_REG" on page 1452	B+0x4	Shadow register TPH Requester Capability Register.
PF Address Translation Services Capability Structure PF_ATS_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_ATS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"ATS_CAP_HDR_REG" on page 1458	B+0x0	ATS Extended Capability Header.
"ATS_CAPABILITIES_CTRL_REG" on page 1460	B+0x4	ATS Requester Capability Register.
PF ACS Capability Structure PF_ACS_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1))		
"ACS_CAP_HDR_REG" on page 1464	B+0x0	Access Control Services Extended Capability Header Register.
"ACS_CAPABILITIES_CTRL_REG" on page 1466	B+0x4	Access Control Services Capability Register.
"ACS_EGRESS_CTRL_VECTOR_REGi (for i = 0; i <= 7)" on page 1474	B+0x8 + i*0x4	Access Control Services Egress Control Vector Register #i
PF PRS Capability Structure PF_PRS_EXT_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))		
"PRS_EXT_CAP_HDR_REG" on page 1477	B+0x0	Page Request Services Capability Header.
"PRS_CONTROL_STATUS_REG" on page 1479	B+0x4	Page Request Services Control and Status Register.
"PRS_REQ_CAPACITY_REG" on page 1484	B+0x8	Page Request Services Outstanding Capacity.
"PRS_REQ_ALLOCATION_REG" on page 1486	B+0xc	Page Request Services Outstanding Allocation.
PF Latency Tolerance Reporting Capability Structure PF0_LTR_CAP Exists: (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))		
"LTR_CAP_HDR_REG" on page 1488	B+0x0	LTR Extended Capability Header.

Register	Offset	Description
"LTR_LATENCY_REG" on page 1490	B+0x4	LTR Max Snoop and No-Snoop Latency Register.
L1 Substates Capability Structure PF0_L1SUB_CAP Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))		
"L1SUB_CAP_HEADER_REG" on page 1493	B+0x0	L1 Substates Extended Capability Header.
"L1SUB_CAPABILITY_REG" on page 1495	B+0x4	L1 Substates Capability Register.
"L1SUB_CONTROL1_REG" on page 1500	B+0x8	L1 Substates Control 1 Register.
"L1SUB_CONTROL2_REG" on page 1503	B+0xc	L1 Substates Control 2 Register.
PF PASID Capability Structure PF_PASID_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"PASID_EXT_CAP_HDR_REG" on page 1507	B+0x0	PASID Extended Capability Header.
"PASID_CAP_CNTRL_REG" on page 1509	B+0x4	PASID Capability Control Register.
PF DPA Capability Structure PF_DPA_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"DPA_EXT_CAP_HDR_REG" on page 1514	B+0x0	DPA Extended Capability Header.
"DPA_CAP_REG" on page 1516	B+0x4	DPA Capability Register.
"DPA_LAT_IND_REG" on page 1520	B+0x8	DPA Latency Indicator Register.
"DPA_STATUS_CNTRL_REG" on page 1521	B+0xc	DPA Status and Control Register.
"DPA_PWR_ALLOC_ARRAY0" on page 1524	B+0x10	DSP Substate Power Allocation Register 3..0.
"DPA_PWR_ALLOC_ARRAY4" on page 1527	B+0x14	DSP Substate Power Allocation Register 7..4.
"DPA_PWR_ALLOC_ARRAY8" on page 1530	B+0x18	DSP Substate Power Allocation Register 11..8.
"DPA_PWR_ALLOC_ARRAY12" on page 1533	B+0x1c	DSP Substate Power Allocation Register 15..12.
"DPA_PWR_ALLOC_ARRAY16" on page 1536	B+0x20	DSP Substate Power Allocation Register 19..16.
"DPA_PWR_ALLOC_ARRAY20" on page 1539	B+0x24	DSP Substate Power Allocation Register 23..20.
"DPA_PWR_ALLOC_ARRAY24" on page 1542	B+0x28	DSP Substate Power Allocation Register 27..24.
"DPA_PWR_ALLOC_ARRAY28" on page 1545	B+0x2c	DSP Substate Power Allocation Register 31..28.
PF RTR Capability Structure PF_RTR_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		

Register	Offset	Description
"RTR_EXT_CAP_HDR_OFF" on page 1549	B+0x0	Readiness Time Reporting PCI Express Extended Capability Header.
"READI_TIME_REPORTING1_OFF" on page 1551	B+0x4	Readiness Time Reporting 1.
"READI_TIME_REPORTING2_OFF" on page 1553	B+0x8	Readiness Time Reporting 2.
DBI2 Shadow Block: PF RTR Capability Structure PF_RTR_CAP_DBI2i (for i = 0; i <= CX_NFUNC-1) Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"SHADOW_READI_TIME_REPORTING1_OFF" on page 1556	B+0x4	Shadow Register Readiness Time Reporting 1.
"SHADOW_READI_TIME_REPORTING2_OFF" on page 1559	B+0x8	Shadow Register Readiness Time Reporting 2.
PF LNR Capability Structure PF_LNR_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"LNR_EXT_CAP_HDR_OFF" on page 1562	B+0x0	LNR Extended Capability Header.
"LNR_CAP_CONTROL_OFF" on page 1564	B+0x4	LNR Control Register and Capability Register.
RAS D.E.S. Capability Structure (VSEC) PF_RAS_DES_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"RAS_DES_CAP_HEADER_REG" on page 1569	B+0x0	Vendor-Specific Extended Capability Header.
"VENDOR_SPECIFIC_HEADER_REG" on page 1572	B+0x4	Vendor-Specific Header.
"EVENT_COUNTER_CONTROL_REG" on page 1574	B+0x8	Event Counter Control.
"EVENT_COUNTER_DATA_REG" on page 1578	B+0xc	Event Counter Data.
"TIME_BASED_ANALYSIS_CONTROL_REG" on page 1580	B+0x10	Time-based Analysis Control.
"TIME_BASED_ANALYSIS_DATA_REG" on page 1584	B+0x14	Time-based Analysis Data.
"TIME_BASED_ANALYSIS_DATA_63_32_REG" on page 1586	B+0x18	Upper 32 bits of Time-based Analysis Data.

Register	Offset	Description
"EINJ_ENABLE_REG" on page 1587	B+0x30	Error Injection Enable.
"EINJ0_CRC_REG" on page 1592	B+0x34	Error Injection Control 0 (CRC Error).
"EINJ1_SEQNUM_REG" on page 1594	B+0x38	Error Injection Control 1 (Sequence Number Error).
"EINJ2_DLLP_REG" on page 1597	B+0x3c	Error Injection Control 2 (DLLP Error).
"EINJ3_SYMBOL_REG" on page 1599	B+0x40	Error Injection Control 3 (Symbol Error).
"EINJ4_FC_REG" on page 1601	B+0x44	Error Injection Control 4 (FC Credit Error).
"EINJ5_SP_TLP_REG" on page 1604	B+0x48	Error Injection Control 5 (Specific TLP Error).
"EINJ6_COMPARE_POINT_Hi_REG (for i = 0; i <= 3)" on page 1606	B+0x4c + i*0x4	Error Injection Control 6 (Compare Point Header DWORD #0).
"EINJ6_COMPARE_VALUE_Hi_REG (for i = 0; i <= 3)" on page 1608	B+0x5c + i*0x4	Error Injection Control 6 (Compare Value Header DWORD #0).
"EINJ6_CHANGE_POINT_Hi_REG (for i = 0; i <= 3)" on page 1609	B+0x6c + i*0x4	Error Injection Control 6 (Change Point Header DWORD #0).
"EINJ6_CHANGE_VALUE_Hi_REG (for i = 0; i <= 3)" on page 1610	B+0x7c + i*0x4	Error Injection Control 6 (Change Value Header DWORD #0).
"EINJ6_TLP_REG" on page 1611	B+0x8c	Error Injection Control 6 (Packet Error).
"SD_CONTROL1_REG" on page 1614	B+0xa0	Silicon Debug Control 1.
"SD_CONTROL2_REG" on page 1617	B+0xa4	Silicon Debug Control 2.
"SD_STATUS_L1LANE_REG" on page 1621	B+0xb0	Silicon Debug Status(Layer1 Per-lane).
"SD_STATUS_L1LTSSM_REG" on page 1625	B+0xb4	Silicon Debug Status(Layer1 LTSSM).
"SD_STATUS_PM_REG" on page 1630	B+0xb8	Silicon Debug Status(PM).
"SD_STATUS_L2_REG" on page 1634	B+0xbc	Silicon Debug Status(Layer2).
"SD_STATUS_L3FC_REG" on page 1637	B+0xc0	Silicon Debug Status(Layer3 FC).
"SD_STATUS_L3_REG" on page 1641	B+0xc4	Silicon Debug Status(Layer3).
"SD_EQ_CONTROL1_REG" on page 1643	B+0xd0	Silicon Debug EQ Control 1.
"SD_EQ_CONTROL2_REG" on page 1647	B+0xd4	Silicon Debug EQ Control 2.
"SD_EQ_CONTROL3_REG" on page 1651	B+0xd8	Silicon Debug EQ Control 3.
"SD_EQ_STATUS1_REG" on page 1654	B+0xe0	Silicon Debug EQ Status 1.
"SD_EQ_STATUS2_REG" on page 1658	B+0xe4	Silicon Debug EQ Status 2.
"SD_EQ_STATUS3_REG" on page 1661	B+0xe8	Silicon Debug EQ Status 3.

Register	Offset	Description
PF RAS Datapath Protection Capability Structure (VSEC) PF_VSECRAS_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"RASDP_EXT_CAP_HDR_OFF" on page 1665	B+0x0	PCIe Extended capability ID, Capability version and Next capability offset.
"RASDP_VENDOR_SPECIFIC_HDR_OFF" on page 1668	B+0x4	Vendor Specific Header.
"RASDP_ERROR_PROT_CTRL_OFF" on page 1670	B+0x8	ECC Error Correction and Control Register.
"RASDP_CORR_COUNTER_CTRL_OFF" on page 1675	B+0xc	Corrected Error (1-bit ECC) Counter Selection and Control Register.
"RASDP_CORR_COUNT_REPORT_OFF" on page 1679	B+0x10	Corrected Error (1-bit ECC) Counter Data Register.
"RASDP_UNCORR_COUNTER_CTRL_OFF" on page 1682	B+0x14	Uncorrected Error (2-bit ECC and parity) Counter Selection and Control Register.
"RASDP_UNCORR_COUNT_REPORT_OFF" on page 1686	B+0x18	Uncorrected Error (2-bit ECC and parity) Counter Data Register.
"RASDP_ERROR_INJ_CTRL_OFF" on page 1689	B+0x1c	Error Injection Control Register.
"RASDP_CORR_ERROR_LOCATION_OFF" on page 1692	B+0x20	Corrected Errors Location Register.
"RASDP_UNCORR_ERROR_LOCATION_OFF" on page 1696	B+0x24	Uncorrected Errors Location Register.
"RASDP_ERROR_MODE_EN_OFF" on page 1700	B+0x28	RASDP error mode enable. The controller enters RASDP error mode (if ERROR_MODE_EN =1) upon detection...
"RASDP_ERROR_MODE_CLEAR_OFF" on page 1702	B+0x2c	RASDP Error Mode Clear Register.
"RASDP_RAM_ADDR_CORR_ERROR_OFF" on page 1703	B+0x30	RAM Address where a corrected error (1-bit ECC) is detected.
"RASDP_RAM_ADDR_UNCORR_ERROR_OFF" on page 1705	B+0x34	RAM Address where an uncorrected error (2-bit ECC) is detected.
PF DLINK Capability Structure PF0_DLINK_CAP Exists: (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2))		
"DATA_LINK_FEATURE_EXT_HDR_OFF" on page 1708	B+0x0	Data Link Feature Extended Capability Header Register.

Register	Offset	Description
"DATA_LINK_FEATURE_CAP_OFF" on page 1710	B+0x4	Data Link Feature Capabilities Register.
"DATA_LINK_FEATURE_STATUS_OFF" on page 1712	B+0x8	Data Link Feature Status Register.
PF PTM Capability Structure PF0_PTM_CAP Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"PTM_EXT_CAP_HDR_OFF" on page 1715	B+0x0	Precision Time Measurement Capability Header.
"PTM_CAP_OFF" on page 1717	B+0x4	PTM Capability Register.
"PTM_CONTROL_OFF" on page 1721	B+0x8	PTM Control Register.
PF PTM Requestor Capability Structure (VSEC) PF_PTM_REQ_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))		
"PTM_REQ_CAP_HDR_OFF" on page 1725	B+0x0	Precision Time Measurement Requester Capability Header (VSEC).
"PTM_REQ_HDR_OFF" on page 1727	B+0x4	Precision Time Measurement Requester Vendor Specific Header.
"PTM_REQ_CONTROL_OFF" on page 1729	B+0x8	PTM Requester Vendor Specific Control Register.
"PTM_REQ_STATUS_OFF" on page 1732	B+0xc	PTM Requester Vendor Specific Status Register.
"PTM_REQ_LOCAL_LSB_OFF" on page 1734	B+0x10	PTM Requester Local Clock LSB.
"PTM_REQ_LOCAL_MSB_OFF" on page 1735	B+0x14	PTM Requester Local Clock MSB.
"PTM_REQ_T1_LSB_OFF" on page 1736	B+0x18	PTM Requester T1 Timestamp LSB.
"PTM_REQ_T1_MSB_OFF" on page 1737	B+0x1c	PTM Requester T1 Timestamp MSB.
"PTM_REQ_T1P_LSB_OFF" on page 1738	B+0x20	PTM Requester T1 Previous Timestamp LSB.
"PTM_REQ_T1P_MSB_OFF" on page 1739	B+0x24	PTM Requester T1 Previous Timestamp MSB.
"PTM_REQ_T4_LSB_OFF" on page 1740	B+0x28	PTM Requester T4 Timestamp LSB.
"PTM_REQ_T4_MSB_OFF" on page 1741	B+0x2c	PTM Requester T4 Timestamp MSB.
"PTM_REQ_T4P_LSB_OFF" on page 1742	B+0x30	PTM Requester T4 Previous Timestamp LSB.
"PTM_REQ_T4P_MSB_OFF" on page 1743	B+0x34	PTM Requester T4 Previous Timestamp MSB.
"PTM_REQ_MASTER_LSB_OFF" on page 1744	B+0x38	>PTM Requester Master Time LSB.
"PTM_REQ_MASTER_MSB_OFF" on page 1745	B+0x3c	PTM Requester Master Time MSB.

Register	Offset	Description
"PTM_REQ_PROP_DELAY_OFF" on page 1746	B+0x40	PTM Requester Propagation Delay.
"PTM_REQ_MASTERT1_LSB_OFF" on page 1747	B+0x44	PTM Requester Master Time at T1 LSB.
"PTM_REQ_MASTERT1_MSB_OFF" on page 1748	B+0x48	PTM Requester Master Time at T1 MSB.
"PTM_REQ_TX_LATENCY_OFF" on page 1749	B+0x4c	PTM Requester TX Latency.
"PTM_REQ_RX_LATENCY_OFF" on page 1751	B+0x50	PTM Requester RX Latency.
"PTM_REQ_CLOCK_CORR_LSB_OFF" on page 1753	B+0x54	PTM Requester Clock Correction LSB.
"PTM_REQ_CLOCK_CORR_MSB_OFF" on page 1754	B+0x58	PTM Requester Clock Correction MSB.
"PTM_REQ_NOM_CLOCK_T_OFF" on page 1755	B+0x5c	PTM Requester Nominal Clock Period.
"PTM_REQ_SCALED_CLOCK_T_OFF" on page 1757	B+0x60	PTM Requester Scaled Clock Period.
"PTM_REQ_LATENCY_REG_SEL_OFF" on page 1759	B+0x64	PTM Requester Latency Viewport Register Select.
"PTM_REQ_USB4_CIO_CTRL_OFF" on page 1761	B+0x68	PTM Requester USB4 Control Register.
"PTM_REQ_TMU_TS_LSB_OFF" on page 1763	B+0x6c	PTM Requester Reference USB4 TMU Timestamp Sample LSB Register.
"PTM_REQ_TMU_TS_MSB_OFF" on page 1764	B+0x70	PTM Requester Reference USB4 TMU Timestamp Sample MSB Register.
CCIX Transport DVSEC Capability Structure PF0_CCIX_TP_CAP Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"CCIX_TP_CAP_HDR_OFF" on page 1766	B+0x0	CCIX Transport DVSEC Capability Header Register.
"CCIX_TP_HDR1_OFF" on page 1768	B+0x4	Designated Vendor Specific Header 1.
"CCIX_TP_CAP_TP_HDR2_OFF" on page 1770	B+0x8	Designated Vendor Specific Header 2 and Transport Capabilities Register.
"ESM_MNDTRY_RATE_CAP_OFF" on page 1776	B+0xc	ESM Mandatory Data Rate Capability Register.
"ESM_STAT_OFF" on page 1781	B+0x14	ESM Status Register.

Register	Offset	Description
"ESM_CNTL_OFF" on page 1783	B+0x18	ESM Control Register.
"ESM_LN_EQ_CNTL_20G_i_OFF (for i = 0; i <= 12)" on page 1790	B+0x1c + i*0x4	ESM Lane Equalization Control Register for 20.0 GT/s for Lanes 0-3.
"ESM_LN_EQ_CNTL_25G_i_OFF (for i = 0; i <= 12)" on page 1797	B+0x2c + i*0x4	ESM Lane Equalization Control Register for 25.0 GT/s for Lanes 0-3.
"CCIX_TL_CAP_OFF" on page 1803	B+0x3c	Transaction Layer Capabilities Register.
"CCIX_TL_CNTL_OFF" on page 1806	B+0x40	Transaction Layer Control Register.
PF CXS Capability Structure (VSEC) PF0_CXS_CAP Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"CXS_CAP_HDR_OFF" on page 1808	B+0x0	CXS Extended Capability Header.
"CXS_VENDOR_SPECIFIC_HEADER_OFF" on page 1810	B+0x4	CXS VSEC Header Register.
"CXS_TX_CTRL_OFF" on page 1811	B+0x8	CXS Transmitter Control Register.
"CXS_RX_CTRL_OFF" on page 1813	B+0xc	CXS Receiver Control Register.
"CXS_ERROR_CTRL_OFF" on page 1814	B+0x10	Error Control Register.
"CXS_ERROR_STAT_OFF" on page 1815	B+0x14	Error Status Register.
PF DMA Capability Structure (VSEC) PF_VSECDMA_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"VSECDMA_EXT_CAP_HDR_OFF" on page 1819	B+0x0	PCIe Extended Capability ID, Capability Version, and Next Capability Offset Register.
"VSECDMA_VENDOR_SPECIFIC_HDR_OFF" on page 1821	B+0x4	Vendor Specific Header Register.
"VSECDMA_DEVICE_INFORMATION_OFF" on page 1823	B+0x8	DMA and related AXI Bridge Implementation Information.
"VSECDMA_NUM_CHAN_OFF" on page 1827	B+0xc	Number of Implemented Channels Register.
"VSECDMA_UNROLL_ADDR_OFFSET_LOW_OFF" on page 1829	B+0x10	DMA Register Map Start Address Offset Low Register.
"VSECDMA_UNROLL_ADDR_OFFSET_HIGH_OFF" on page 1830	B+0x14	DMA Register Map Start Address Offset High Register.
Native PCIe Enclosure Management Extended Capability Structure PF0_NPEM_CAP Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		

Register	Offset	Description
"NPEM_EXT_CAP_HDR_OFF" on page 1832	B+0x0	NPEM Extended Capability Header.
"NPEM_CAP_OFF" on page 1834	B+0x4	NPEM Capability Register.
"NPEM_CTRL_OFF" on page 1840	B+0x8	NPEM Control Register.
"NPEM_STATUS_OFF" on page 1853	B+0xc	NPEM Status Register.
Device 3 Extended Capability Structure PF_DEV3_EXT_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"DEV3_EXT_CAP_HDR_REG" on page 1856	B+0x0	DEV3 Header Register.
"DEVICE_CAPABILITIES3_REG" on page 1858	B+0x4	DEV3 Capabilities Register.
"DEVICE_CONTROL3_REG" on page 1860	B+0x8	DEV3 Control Register.
CXL-RCRB Transport DVSEC Capability Structure PF0_CXL_PORT_CAP Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2))		
"CXL_RCRB_PCIE_EXT_CAP_HDR_OFF" on page 1863	B+0x0	Capability ID, Version ID, Next offset pointer.
"CXL_RCRB_DVSEC_PORT_HDR_1_OFF" on page 1865	B+0x4	DVSEC Header 1: Vendor ID, Revision, Length.
"CXL_RCRB_DVSEC_HDR_2_FLEXBUS_CAP_OFF" on page 1867	B+0x8	DVSEC HEADER 2 and DVSEC Flex Bus Port Capability.
"CXL_RCRB_FLEXBUS_CNTRL_STATUS_OFF" on page 1871	B+0xc	DVSEC Flex Bus Port Control and Status.
"CXL_RCRB_FLEXBUS_RCVD_MOD_TS_PHASE1_1_OFF" on page 1880	B+0x10	DVSEC flexBus Port Received Modified TS data Phase 1.
CXL-RCIEP Transport DVSEC Capability Structure PF0_CXL_DEVICE_CAP Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))		
"CXL_RCIEP_PCIE_EXT_CAP_HDR_OFF" on page 1882	B+0x0	Device Cap ID, Device Cap Version, Device Next Offset.
"CXL_RCIEP_DVSEC_HDR_1_OFF" on page 1884	B+0x4	DVSEC Vendor ID, DVSEC Revision, DVSEC Length.
"CXL_RCIEP_DVSEC_HDR_2_FLEXBUS_CAP_OFF" on page 1886	B+0x8	DVSEC ID,cache Capable, IO capable, Memory capable, Memory HwInit Mode, HDM count and Viral...
"CXL_RCIEP_FLEXBUS_CNTRL_STATUS_OFF" on page 1893	B+0xc	Cache Enable, IO Enable, Memory Enable, Cache SF Coverage, Cache SF Granularity, Cache Clean and...

Register	Offset	Description
"CXL_RCIEP_FLEXBUS_CNTRL_2_STATUS_2_OFF" on page 1899	B+0x10	Disable Caching, Initiate Cache Write Back and Invalidation, Initiate eFLR, Cache Invalid, eFLR...
"CXL_RCIEP_FLEX_BUS_LOCK_OFF" on page 1902	B+0x14	Config Lock.
"CXL_RCIEP_FLEXBUS_R1_SIZE_HIGH_OF F" on page 1904	B+0x18	Memory size high.
"CXL_RCIEP_FLEXBUS_R1_SIZE_LOW_OFF" on page 1905	B+0x1c	Memory Info Valid, Memory Active, Memory Type, Memory Class, Desired Interleave and Memory size...
"CXL_RCIEP_FLEXBUS_R1_BASE_HIGH_O FF" on page 1910	B+0x20	Memory base high.
"CXL_RCIEP_FLEXBUS_R1_BASE_LOW_OF F" on page 1911	B+0x24	Memory base Low.
"CXL_RCIEP_FLEXBUS_R2_SIZE_HIGH_OF F" on page 1913	B+0x28	Memory Size High.
"CXL_RCIEP_FLEXBUS_R2_SIZE_LOW_OFF" on page 1914	B+0x2c	Memory Info Valid, Memory Active, Memory Type, Memory Class, Desired Interleave and Memory size...
"CXL_RCIEP_FLEXBUS_R2_BASE_HIGH_O FF" on page 1919	B+0x30	Memory base Low.
"CXL_RCIEP_FLEXBUS_R2_BASE_LOW_OF F" on page 1920	B+0x34	Memory base high.
CXL DEVICE TEST DVSEC Capability Structure PF0_CXL_DEVICE_TEST_CAP_CAP Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))		
"CXL_TEST_PCIE_EXT_CAP_HDR_OFF" on page 1922	B+0x0	Device Test Capability ID, Device Test Version ID and Device Test Next offset pointer.
"CXL_TEST_DVSEC_HDR_1_OFF" on page 1925	B+0x4	DVSEC Device TEST Header 1: Vendor ID, Revision, Length.
"CXL_TEST_DVSEC_HDR_2_TEST_LOCK_O FF" on page 1927	B+0x8	DVSEC DEVICE TEST ID and TEST LOCK Capability.
"CXL_TEST_DVSEC_CAPABILITY_OFF" on page 1929	B+0xc	DVSEC CXL TEST CAPABILITY 1.
"CXL_TEST_DVSEC_CAPABILITY_2_OFF" on page 1938	B+0x10	CACHE SIZE and CACHE SIZE UNITS.
"CXL_TEST_CONFIG_BASE_LOW_OFF" on page 1940	B+0x14	DVSEC CXL TEST CONFIGURATION BASE LOW.
"CXL_TEST_CONFIG_BASE_HIGH_OFF" on page 1942	B+0x18	DVSEC CXL TEST CONFIGURATION BASE REGISTER.

Register	Offset	Description
Resizable BAR Capability Structure PF_RESBAR_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"RESBAR_CAP_HDR_REG" on page 1944	B+0x0	Resizable BAR Capability Header.
"RESBAR_CAP_REG_0_REG" on page 1947	B+0x4	Resizable BAR0 Capability Register.
"RESBAR_CAP_REG_1_REG" on page 1963	B+0xc	Resizable BAR1 Capability Register.
"RESBAR_CAP_REG_2_REG" on page 1979	B+0x14	Resizable BAR2 Capability Register.
"RESBAR_CAP_REG_3_REG" on page 1995	B+0x1c	Resizable BAR3 Capability Register.
"RESBAR_CAP_REG_4_REG" on page 2011	B+0x24	Resizable BAR4 Capability Register.
"RESBAR_CAP_REG_5_REG" on page 2027	B+0x2c	Resizable BAR5 Capability Register.
"RESBAR_CTRL_REG_0_REG" on page 2043	B+0x8	Resizable BAR0 Control Register.
"RESBAR_CTRL_REG_1_REG" on page 2054	B+0x10	Resizable BAR1 Control Register.
"RESBAR_CTRL_REG_2_REG" on page 2065	B+0x18	Resizable BAR2 Control Register.
"RESBAR_CTRL_REG_3_REG" on page 2076	B+0x20	Resizable BAR3 Control Register.
"RESBAR_CTRL_REG_4_REG" on page 2087	B+0x28	Resizable BAR4 Control Register.
"RESBAR_CTRL_REG_5_REG" on page 2098	B+0x30	Resizable BAR5 Control Register.
VF Resizable BAR Capability Structure PF_VF_RESBAR_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"VF_RESBAR_CAP_HDR_REG" on page 2110	B+0x0	Resizable BAR Capability Header.
"VF_RESBAR_CAP_REG_0_REG" on page 2112	B+0x4	Resizable BAR0 Capability Register.
"VF_RESBAR_CAP_REG_1_REG" on page 2128	B+0xc	Resizable BAR1 Capability Register.
"VF_RESBAR_CAP_REG_2_REG" on page 2144	B+0x14	Resizable BAR2 Capability Register.
"VF_RESBAR_CAP_REG_3_REG" on page 2160	B+0x1c	Resizable BAR3 Capability Register.
"VF_RESBAR_CAP_REG_4_REG" on page 2176	B+0x24	Resizable BAR4 Capability Register.
"VF_RESBAR_CAP_REG_5_REG" on page 2192	B+0x2c	Resizable BAR5 Capability Register.

Register	Offset	Description
"VF_RESBAR_CTRL_REG_0_REG" on page 2208	B+0x8	Resizable BAR0 Control Register.
"VF_RESBAR_CTRL_REG_1_REG" on page 2219	B+0x10	Resizable BAR1 Control Register.
"VF_RESBAR_CTRL_REG_2_REG" on page 2230	B+0x18	Resizable BAR2 Control Register.
"VF_RESBAR_CTRL_REG_3_REG" on page 2241	B+0x20	Resizable BAR3 Control Register.
"VF_RESBAR_CTRL_REG_4_REG" on page 2252	B+0x28	Resizable BAR4 Control Register.
"VF_RESBAR_CTRL_REG_5_REG" on page 2263	B+0x30	Resizable BAR5 Control Register.
Port Logic PF_PORT_LOGIC _i (for i = 0; i ≤ CX_NFUNC-1) Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))		
"ACK_LATENCY_TIMER_OFF" on page 2275	B+0x0	Ack Latency Timer and Replay Timer Register.
"VENDOR_SPEC_DLLP_OFF" on page 2277	B+0x4	Vendor Specific DLLP Register.
"PORT_FORCE_OFF" on page 2278	B+0x8	Port Force Link Register.
"ACK_F_ASPM_CTRL_OFF" on page 2281	B+0xc	Ack Frequency and L0-L1 ASPM Control Register.
"PORT_LINK_CTRL_OFF" on page 2285	B+0x10	Port Link Control Register.
"LANE_SKEW_OFF" on page 2291	B+0x14	Lane Skew Register.
"TIMER_CTRL_MAX_FUNC_NUM_OFF" on page 2295	B+0x18	Timer Control and Max Function Number Register.
"SYMBOL_TIMER_FILTER_1_OFF" on page 2298	B+0x1c	Symbol Timer Register and Filter Mask 1 Register.
"FILTER_MASK_2_OFF" on page 2303	B+0x20	Filter Mask 2 Register.
Reserved	0x724 to 0x728	
"PL_DEBUG0_OFF" on page 2306	B+0x28	Debug Register 0.
"PL_DEBUG1_OFF" on page 2307	B+0x2c	Debug Register 1.
"TX_P_FC_CREDIT_STATUS_OFF" on page 2308	B+0x30	Transmit Posted FC Credit Status.
"TX_NP_FC_CREDIT_STATUS_OFF" on page 2310	B+0x34	Transmit Non-Posted FC Credit Status.

Register	Offset	Description
"TX_CPL_FC_CREDIT_STATUS_OFF" on page 2312	B+0x38	Transmit Completion FC Credit Status
"QUEUE_STATUS_OFF" on page 2314	B+0x3c	Queue Status.
"VC_TX_ARBI_1_OFF" on page 2318	B+0x40	VC Transmit Arbitration Register 1.
"VC_TX_ARBI_2_OFF" on page 2321	B+0x44	VC Transmit Arbitration Register 2.
"VC0_P_RX_Q_CTRL_OFF" on page 2324	B+0x48	Segmented-Buffer VC0 Posted Receive Queue Control.
"VC0_NP_RX_Q_CTRL_OFF" on page 2328	B+0x4c	Segmented-Buffer VC0 Non-Posted Receive Queue Control.
"VC0_CPL_RX_Q_CTRL_OFF" on page 2332	B+0x50	Segmented-Buffer VC0 Completion Receive Queue Control.
"VCi_P_RX_Q_CTRL_OFF (for i = 1; i <= CX_NVC-1)" on page 2336	B+0x54 + i*0x4	Segmented-Buffer VC1 Posted Receive Queue Control.
"VCi_NP_RX_Q_CTRL_OFF (for i = 1; i <= CX_NVC-1)" on page 2340	B+0x58 + i*0x4	Segmented-Buffer VC1 Non-Posted Receive Queue Control.
"VCi_CPL_RX_Q_CTRL_OFF (for i = 1; i <= CX_NVC-1)" on page 2344	B+0x5c + i*0x4	Segmented-Buffer VC1 Completion Receive Queue Control.
Reserved	0x760 to 0x80c	
"GEN2_CTRL_OFF" on page 2348	B+0x10c	Link Width and Speed Change Control Register.
"PHY_STATUS_OFF" on page 2360	B+0x110	PHY Status Register.
"PHY_CONTROL_OFF" on page 2361	B+0x114	PHY Control Register.
Reserved	0x818 to 0x81c	
"TRGT_MAP_CTRL_OFF" on page 2362	B+0x11c	Programmable Target Map Control Register.
Reserved	0x820 to 0x88c	
"CLOCK_GATING_CTRL_OFF" on page 2365	B+0x18c	Clock Gating Control Register.
"GEN3_RELATED_OFF" on page 2367	B+0x190	Gen3 Control Register.
"GEN3_EQ_LOCAL_FS_LF_OFF" on page 2380	B+0x194	Gen3 EQ FS and LF Register.
"GEN3_EQ_PSET_COEF_MAP__i (for i = 0; i <= 10)" on page 2383	B+0x198	Gen3 EQ Presets to Coefficients Mapping Register.
"GEN3_EQ_PSET_INDEX_OFF" on page 2386	B+0x19c	Gen3 EQ Preset Index Register.

Register	Offset	Description
Reserved	0x8a0 to 0x8a4	
“GEN3_EQ_COEFF_LEGALITY_STATUS_OF F” on page 2388	B+0x1a4	Gen3 EQ Status Register.
“GEN3_EQ_CONTROL_OFF” on page 2390	B+0x1a8	Gen3 EQ Control Register.
“GEN3_EQ_FB_MODE_DIR_CHANGE_OFF” on page 2399	B+0x1ac	Gen3 EQ Direction Change Feedback Mode Control Register.
Reserved	0x8b0 to 0x8b4	
“ORDER_RULE_CTRL_OFF” on page 2404	B+0x1b4	Order Rule Control Register.
“PIPE_LOOPBACK_CONTROL_OFF” on page 2406	B+0x1b8	PIPE Loopback Control Register.
“MISC_CONTROL_1_OFF” on page 2408	B+0x1bc	DBI Read-Only Write Enable Register.
“MULTI_LANE_CONTROL_OFF” on page 2415	B+0x1c0	UpConfigure Multi-lane Control Register.
“PHY_INTEROP_CTRL_OFF” on page 2418	B+0x1c4	PHY Interoperability Control Register.
“TRGT_CPL_LUT_DELETE_ENTRY_OFF” on page 2423	B+0x1c8	TRGT_CPL_LUT Delete Entry Control register.
“LINK_FLUSH_CONTROL_OFF” on page 2424	B+0x1cc	Link Reset Request Flush Control Register.
“AMBA_ERROR_RESPONSE_DEFAULT_OFF” on page 2426	B+0x1d0	AXI Bridge Slave Error Response Register.
“AMBA_LINK_TIMEOUT_OFF” on page 2431	B+0x1d4	Link Down AXI Bridge Slave Timeout Register.
“AMBA_ORDERING_CTRL_OFF” on page 2433	B+0x1d8	AXI Bridge Ordering Control.
“AMBA_ORDRMGR_WDOG_OFF” on page 2437	B+0x1dc	AHB Ordering Manager Watchdog Timer.
“COHERENCY_CONTROL_1_OFF” on page 2439	B+0x1e0	Cache Coherency Control Register 1.
“COHERENCY_CONTROL_2_OFF” on page 2441	B+0x1e4	Cache Coherency Control Register 2.
“COHERENCY_CONTROL_3_OFF” on page 2442	B+0x1e8	Cache Coherency Control Register 3.
Reserved	0x8ec to 0x8f0	

Register	Offset	Description
"AXI_MSTR_MSG_ADDR_LOW_OFF" on page 2447	B+0x1f0	Lower 32-bits of the Programmable AXI Address.
"AXI_MSTR_MSG_ADDR_HIGH_OFF" on page 2449	B+0x1f4	Upper 32-bits of the Programmable AXI Address.
"PCIE_VERSION_NUMBER_OFF" on page 2450	B+0x1f8	PCIe Controller IIP Release Version Number.
"PCIE_VERSION_TYPE_OFF" on page 2451	B+0x1fc	PCIe Controller IIP Release Version Type.
"IATU_VIEWPORT_OFF" on page 2452	B+0x200	iATU Index Register.
"IATU_REGION_CTRL_1_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 2454	B+0x204	iATU Region Control 1 Register.
"IATU_REGION_CTRL_1_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 2459	+0x904	iATU Region Control 1 Register.
"IATU_REGION_CTRL_2_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 2464	B+0x208	iATU Region Control 2 Register.
"IATU_REGION_CTRL_2_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 2472	+0x908	iATU Region Control 2 Register.
"IATU_LWR_BASE_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 2483	B+0x20c	iATU Lower Base Address Register.
"IATU_LWR_BASE_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 2485	+0x90c	iATU Lower Base Address Register.
"IATU_UPPER_BASE_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 2487	B+0x210	iATU Upper Base Address Register.
"IATU_UPPER_BASE_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 2488	+0x910	iATU Upper Base Address Register.

Register	Offset	Description
"IATU_LIMIT_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 2489	B+0x214	iATU Limit Address Register.
"IATU_LIMIT_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 2491	+0x914	iATU Limit Address Register.
"IATU_LWR_TARGET_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 2495	B+0x218	iATU Lower Target Address Register.
"IATU_LWR_TARGET_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 2497	+0x918	iATU Lower Target Address Register.
"IATU_UPPER_TARGET_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 2499	B+0x21c	iATU Upper Target Address Register.
"IATU_UPPER_TARGET_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 2500	+0x91c	iATU Upper Target Address Register.
"IATU_REGION_CTRL_3_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 2501	B+0x220	iATU Region Control 3 Register.
"IATU_REGION_CTRL_3_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 2503	+0x920	iATU Region Control 3 Register.
"IATU_UPPR_LIMIT_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 2505	B+0x224	iATU Upper Limit Address Register.
"IATU_UPPR_LIMIT_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 2507	+0x924	iATU Upper Limit Address Register.
Reserved	0x928 to 0x930	

Register	Offset	Description
"INTERFACE_TIMER_CONTROL_OFF" on page 2509	B+0x230	Interface Timer Control Register.
"INTERFACE_TIMER_TARGET_OFF" on page 2511	B+0x234	Interface Timer Target Register.
"INTERFACE_TIMER_STATUS_OFF" on page 2512	B+0x238	Interface Timer Status Register.
Reserved	0x93c to 0x940	
"MSIX_ADDRESS_MATCH_LOW_OFF" on page 2517	B+0x240	MSI-X Address Match Low Register.
"MSIX_ADDRESS_MATCH_HIGH_OFF" on page 2519	B+0x244	MSIX Address Match High Register.
"MSIX_DOORBELL_OFF" on page 2520	B+0x248	MSI-X Doorbell Register.
"MSIX_RAM_CTRL_OFF" on page 2522	B+0x24c	MSI-X RAM power mode and debug control register.
Reserved	0x950 to 0x960	
"SAFETY_MASK_OFF" on page 2526	B+0x260	Functional Safety Interrupt Events Mask Register.
"SAFETY_STATUS_OFF" on page 2528	B+0x264	Status for functional safety interrupt events.
Reserved	0x968 to 0xb10	
"PL_APP_BUS_DEV_NUM_STATUS_OFF" on page 2531	B+0x410	Application driven bus and device number register.
Reserved	0xb14 to 0xb1c	
"PCIPM_TRAFFIC_CTRL_OFF" on page 2533	B+0x41c	TLP Traffic during Non-D0 State Control Register.
"PL_CHK_REG_CONTROL_STATUS_OFF" on page 2535	B+0x420	CDM Register Checking Control and Status Register.
"PL_CHK_REG_START_END_OFF" on page 2538	B+0x424	CDM Register Checking First and Last address to check.
"PL_CHK_REG_ERR_ADDR_OFF" on page 2540	B+0x428	CDM Register Checking Error Address.
"PL_CHK_REG_ERR_PF_VF_OFF" on page 2541	B+0x42c	CDM Register Checking error PF and VF Numbers.
"PL_LTR_LATENCY_OFF" on page 2543	B+0x430	LTR Latency Register.

Register	Offset	Description
Reserved	0xb34 to 0xb40	
"AUX_CLK_FREQ_OFF" on page 2546	B+0x440	Auxiliary Clock Frequency Control Register.
"L1_SUBSTATES_OFF" on page 2548	B+0x444	L1 Substates Timing Register.
"POWERDOWN_CTRL_STATUS_OFF" on page 2551	B+0x448	Powerdown Control and Status Register.
"PHY_INTEROP_CTRL_2_OFF" on page 2554	B+0x44c	PHY Interoperability Control 2 Register.
Reserved	0xb50 to 0xb70	
"PHY_VIEWPORT_CTLSTS_OFF" on page 2556	B+0x470	PHY Register Viewport Control and Status Register.
"PHY_VIEWPORT_DATA_OFF" on page 2560	B+0x474	PHY Register Viewport Data Register.
Reserved	0xb78 to 0xb80	
"GEN4_LANE_MARGINING_1_OFF" on page 2561	B+0x480	Gen4 Lane Margining 1 Register.
"GEN4_LANE_MARGINING_2_OFF" on page 2564	B+0x484	Gen4 Lane Margining 2 Register.
"GEN5_LANE_MARGINING_1_OFF" on page 2569	B+0x488	Gen5 Lane Margining 1 Register.
"GEN5_LANE_MARGINING_2_OFF" on page 2573	B+0x48c	Gen5 Lane Margining 2 Register.
"PIPE_RELATED_OFF" on page 2578	B+0x490	PIPE Related Register.
Reserved	0xb94 to 0xc00	
"RX_SERIALIZATION_Q_CTRL_OFF" on page 2581	B+0x500	Receive Serialization Queue Control Register.
Reserved	0xc04 to 0xc20	
"CCIX_CTRL_OFF" on page 2583	B+0x520	CCIX Control Register.
"GEN3_FCOEF_VEC_DEPTH_OFF" on page 2584	B+0x524	Gen3 FOM Coefficients Table Depth Register.
"GEN3_FCOEF_LANE1_0_OFF" on page 2587	B+0x528	Gen3 FOM Coefficients Table Register for Lane 1 and 0.

Register	Offset	Description
"GEN3_FCOEF_LANE3_2_OFF" on page 2591	B+0x52c	Gen3 FOM Coefficients Table Register for Lane 3 and 2.
"GEN3_FCOEF_LANE5_4_OFF" on page 2595	B+0x530	Gen3 FOM Coefficients Table Register for Lane 5 and 4.
"GEN3_FCOEF_LANE7_6_OFF" on page 2599	B+0x534	Gen3 FOM Coefficients Table Register for Lane 7 and 6.
"GEN3_FCOEF_LANE9_8_OFF" on page 2603	B+0x538	Gen3 FOM Coefficients Table Register for Lane 9 and 8.
"GEN3_FCOEF_LANE11_10_OFF" on page 2607	B+0x53c	Gen3 FOM Coefficients Table Register for Lane 11 and 10.
"GEN3_FCOEF_LANE13_12_OFF" on page 2611	B+0x540	Gen3 FOM Coefficients Table Register for Lane 13 and 12.
"GEN3_FCOEF_LANE15_14_OFF" on page 2615	B+0x544	Gen3 FOM Coefficients Table Register for Lane 15 and 14.
"CXL_VLSM_CR_REG_OFF" on page 2619	B+0x548	CXL VLSM CREDIT REGISTER.
"CXL_LRSM_CSR_REG_OFF" on page 2622	B+0x54c	CXL LSRM CSR.
"CXL_LL_CTRL_CR_REG_OFF" on page 2624	B+0x550	CXL Link Layer Control credit.
"CXL_FATAL_ERR_CR_REG_OFF" on page 2626	B+0x554	CXL FATAL ERROR.
"CXL_CACHE_MEM_LL_INJ_OFF" on page 2634	B+0x558	CXL.cache / CXL.mem Link Layer Error Injection Register.
"CXL_IO_LL_INJ_OFF" on page 2638	B+0x55c	CXL.io Link Layer Error Injection Register.
"FLEXBUS_LOG_PHY_ERR_INJ_OFF" on page 2640	B+0x560	Flex Bus Log Phy Protocol Error Injection Register.
"CXL_PORT_LOGIC_VIRAL_STS_OFF" on page 2642	B+0x564	CXL Port Logic Viral Status Register.
Reserved	0xc68 to 0xc70	
"ECAM_BASE_ADDR_LWR_OFF" on page 2644	B+0x570	Lower ECAM Base Address Register.
"ECAM_BASE_ADDR_UPPER_OFF" on page 2645	B+0x574	Upper ECAM Base Address Register.
"ECAM_CTRL_OFF" on page 2646	B+0x578	ECAM Control Register.
"DBI_FUNCTION_BANK_CTRL_REG_OFF" on page 2648	B+0x57c	DBI Function Bank Control Register.

Register	Offset	Description
"UTILITY_OFF" on page 2650	B+0x580	UTILITY register (Reserved).
"AXIG_CTRL_OFF" on page 2651	B+0x584	AXI-G Control Register.
"PM_UTILITY_OFF" on page 2653	B+0x588	PM Shadow of UTILITY register (Reserved).
VF PCI-Compatible Configuration Space Header Type0 VF_PF_TYPE0_HDRi (for i = 0; i <= CX_NFUNC-1) Exists: CX_MEMORY_MAP_PCIE_TYPE!=1		
"VF_DEVICE_ID_VENDOR_ID_REG" on page 2655	B+0x0	Device ID , RCRB next offset pointer and Vendor ID Register.
"VF_STATUS_COMMAND_REG" on page 2657	B+0x4	Status and Command Register.
"VF_CLASS_CODE_REVISION_ID" on page 2665	B+0x8	Class Code and Revision ID Register.
"VF_BIST_HEADER_TYPE_LATENCY_CACHE_LINE_SIZE_REG" on page 2667	B+0xc	BIST, Header Type, Latency Timer, and Cache Line Size Register.
"VF_BAR0_REG" on page 2670	B+0x10	BAR0 Register.
"VF_BAR1_REG" on page 2674	B+0x14	BAR1 Register.
"VF_BAR2_REG" on page 2678	B+0x18	BAR2 Register.
"VF_BAR3_REG" on page 2681	B+0x1c	BAR3 Register.
"VF_BAR4_REG" on page 2684	B+0x20	BAR4 Register.
"VF_BAR5_REG" on page 2687	B+0x24	BAR5 Register.
"VF_CARDBUS_CIS_PTR_REG" on page 2690	B+0x28	CardBus CIS Pointer Register.
"VF_SUBSYSTEM_ID_SUBSYSTEM_VENDOR_ID_REG" on page 2691	B+0x2c	Subsystem ID and Subsystem Vendor ID Register.
"VF_PCI_CAP_PTR_REG" on page 2692	B+0x34	Capabilities Pointer Register.
"VF_MAX_LATENCY_MIN_GRANT_INTERRUPT_PIN_INTERRUPT_LINE_REG" on page 2693	B+0x3c	Max_Lat, Min_Gnt, Interrupt Pin, and Interrupt Line Register.
VF PCI Power Management Capability Structure VF_PF_PM_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)		
"VF_CAP_ID_NXT_PTR_REG" on page 2696	B+0x0	Power Management Capabilities Register.
"VF_CON_STATUS_REG" on page 2702	B+0x4	Power Management Control and Status Register.

Register	Offset	Description
VF MSI Capability Structure VF_PF_MSI_CAPi (for i = 0; i <= CX_NVFUNC-1) Exists: ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)		
"VF_PCI_MSI_CAP_ID_NEXT_CTRL_REG" on page 2707	B+0x0	MSI Capability Header and Message Control Register.
"VF_MSI_CAP_OFF_04H_REG" on page 2713	B+0x4	Message Address Register for MSI (Offset 04h).
"VF_MSI_CAP_OFF_08H_REG" on page 2715	B+0x8	Message Address Register for MSI (Offset 08h).
"VF_MSI_CAP_OFF_0CH_REG" on page 2718	B+0xc	Message Address Register for MSI (Offset 0Ch).
"VF_MSI_CAP_OFF_10H_REG" on page 2721	B+0x10	Message Address Register for MSI (Offset 10h).
"VF_MSI_CAP_OFF_14H_REG" on page 2723	B+0x14	Message Address Register for MSI (Offset 14h).
VF PCI Express Capability Structure VF_PF_PCIE_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: CX_MEMORY_MAP_PCIE_TYPE!=1		
"VF_PCIE_CAP_ID_PCIE_NEXT_CAP_PTR_PCIE_CAP_REG" on page 2725	B+0x0	PCI Express Capabilities, ID, Next Pointer Register.
"VF_DEVICE_CAPABILITIES_REG" on page 2729	B+0x4	Device Capabilities Register.
"VF_DEVICE_CONTROL_DEVICE_STATUS" on page 2736	B+0x8	Device Control and Device Status Register.
"VF_LINK_CAPABILITIES_REG" on page 2746	B+0xc	Link Capabilities Register.
"VF_LINK_CONTROL_LINK_STATUS_REG" on page 2755	B+0x10	Link Control and Link Status Register.
"VF_DEVICE_CAPABILITIES2_REG" on page 2769	B+0x24	Device Capabilities 2 Register.
"VF_DEVICE_CONTROL2_DEVICE_STATUS2_REG" on page 2781	B+0x28	Device Control 2 and Status 2 Register.
"VF_LINK_CAPABILITIES2_REG" on page 2789	B+0x2c	Link Capabilities 2 Register.
"VF_LINK_CONTROL2_LINK_STATUS2_REG" on page 2793	B+0x30	Link Control 2 and Status 2 Register.
VF MSI-X Capability Structure VF_PF_MSIX_CAPi (for i = 0; i <= CX_NVFUNC-1) Exists: ((VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)		

Register	Offset	Description
"VF_PCI_MSIX_CAP_ID_NEXT_CTRL_REG" on page 2807	B+0x0	MSI-X Capability ID, Next Pointer, Control Register.
"VF_MSIX_TABLE_OFFSET_REG" on page 2810	B+0x4	MSI-X Table Offset and BIR Register.
"VF_MSIX_PBA_OFFSET_REG" on page 2813	B+0x8	MSI-X PBA Offset and BIR Register.
VF Advanced Error Reporting Capability Structure VF_PF_AER_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)		
"VF_AER_EXT_CAP_HDR_OFF" on page 2817	B+0x0	Advanced Error Reporting Extended Capability Header.
"VF_UNCORR_ERR_STATUS_OFF" on page 2819	B+0x4	Uncorrectable Error Status Register.
"VF_UNCORR_ERR_MASK_OFF" on page 2826	B+0x8	Uncorrectable Error Mask Register.
"VF_UNCORR_ERR_SEV_OFF" on page 2831	B+0xc	Uncorrectable Error Severity Register.
"VF_CORR_ERR_STATUS_OFF" on page 2837	B+0x10	Correctable Error Status Register.
"VF_CORR_ERR_MASK_OFF" on page 2841	B+0x14	Correctable Error Mask Register.
"VF_ADV_ERR_CAP_CTRL_OFF" on page 2844	B+0x18	Advanced Error Capabilities and Control Register.
"VF_HDR_LOG_0_OFF" on page 2848	B+0x1c	Header Log Register 0.
"VF_HDR_LOG_1_OFF" on page 2850	B+0x20	Header Log Register 1.
"VF_HDR_LOG_2_OFF" on page 2852	B+0x24	Header Log Register 2.
"VF_HDR_LOG_3_OFF" on page 2854	B+0x28	Header Log Register 3.
"VF_TLP_PREFIX_LOG_i_OFF (for i = 1; i <= 4)" on page 2856	B+0x38 + i*0x4	TLP Prefix Log Register 1.
VF ARI Capability Structure VF_PF_ARI_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: CX_MEMORY_MAP_PCIE_TYPE!=1		
"VF_ARI_BASE" on page 2859	B+0x0	ARI Capability Header.
"VF_CAP_REG" on page 2861	B+0x4	ARI Capability and Control Register.

Register	Offset	Description
VF Access Control Services Capability Structure VF_PF_ACS_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1)		
"VF_ACS_CAP_HDR_REG" on page 2865	B+0x0	Access Control Services Extended Capability Header Register.
"VF_ACS_CAPABILITIES_CTRL_REG" on page 2867	B+0x4	Access Control Services Capability Register.
"VF_ACS_EGRESS_CTRL_VECTOR_REGi" (for i = 0; i <= 7) on page 2873	B+0x8 + i*0x4	Access Control Services Egress Control Vector Register #i
VF Address Translation Services Capability Structure VF_PF_ATS_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: ((VF_ATS_ENABLE EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)		
"VF_ATS_CAP_HDR_REG" on page 2876	B+0x0	ATS Extended Capability Header.
"VF_ATS_CAPABILITIES_CTRL_REG" on page 2878	B+0x4	ATS Requester Capability Register.
VF LNR Capability Structure VF_PF_LNR_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)		
"VF_LNR_EXT_CAP_HDR_OFF" on page 2882	B+0x0	LNR Extended Capability Header.
"VF_LNR_CAP_CONTROL_OFF" on page 2884	B+0x4	LNR Control Register and Capability Register.
VF PASID Capability Structure VF_PF_PASID_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)		
"VF_PASID_EXT_CAP_HDR_REG" on page 2889	B+0x0	PASID Extended Capability Header.
"VF_PASID_CAP_CNTRL_REG" on page 2891	B+0x4	PASID Capability Control Register.
VF RTR Capability Structure VF_PF_RTR_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: ((VF_RTR_ENABLE EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)		
"VF_RTR_EXT_CAP_HDR_OFF" on page 2896	B+0x0	Readiness Time Reporting PCI Express Extended Capability Header.

Register	Offset	Description
"VF_READI_TIME_REPORTING1_OFF" on page 2898	B+0x4	Readiness Time Reporting 1.
"VF_READI_TIME_REPORTING2_OFF" on page 2901	B+0x8	Readiness Time Reporting 2.
VF TLP Processing Hints Capability Structure VF_PF_TPH_CAPi (for i = 0; i <= CX_NFUNC-1) Exists: ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)		
"VF_TPH_EXT_CAP_HDR_REG" on page 2904	B+0x0	TPH Extended Capability Header.
"VF_TPH_REQ_CAP_REG_REG" on page 2906	B+0x4	TPH Requester Capability Register.
"VF_TPH_REQ_CONTROL_REG_REG" on page 2910	B+0x8	TPH Requester Control Register.
"VF_TPH_ST_TABLE_REG_i (for i = 0; i <= 31)" on page 2913	B+0xc + i*0x4	TPH ST Table Register 0.
CXL-RCRB MEMBAR0 Capability Structure PF0_MEMBAR0_HEADER_CAP Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)		
"MEMBAR0_HEADER_CAP_REG_OFF" on page 2916	B+0x0	CXL Capability Header Register of CXL.cache and CXL.mem Registers.
"MEMBAR0_RAS_CAP_REG_OFF" on page 2919	B+0x4	CXL RAS Capability Header.
"MEMBAR0_SECURITY_CAP_REG_OFF" on page 2921	B+0x8	CXL Security Capability Header.
"MEMBAR0_LINK_CAP_REG_OFF" on page 2923	B+0xc	CXL Link Capability Header.
CXL-RCRB MEMBAR0 RAS Capability Structure PF0_MEMBAR0_RAS_CAP Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)		
"MEMBAR0_RAS_UNCOR_ERROR_STATUS_REG_OFF" on page 2926	B+0x0	CXL RCRB MEMBAR0 Uncorrectable Error Status Register.
"MEMBAR0_RAS_UNCOR_ERROR_MASK_REG_OFF" on page 2931	B+0x4	CXL RCRB MEMBAR0 Uncorrectable Error Mask Register.
"MEMBAR0_RAS_UNCOR_ERROR_SEVERITY_REG_OFF" on page 2935	B+0x8	CXL RCRB MEMBAR0 Uncorrectable Error Severity Register.
"MEMBAR0_RAS_CORR_ERROR_STATUS_REG_OFF" on page 2939	B+0xc	CXL RCRB MEMBAR0 Correctable Error Status Register.

Register	Offset	Description
“MEMBAR0_RAS_CORR_ERROR_MASK_REG_OFF” on page 2942	B+0x10	CXL RCRB MEMBAR0 Correctable Error Mask Register.
“MEMBAR0_RAS_ERR_CAP_CNTRL_REG_OFF” on page 2944	B+0x14	CXL RCRB MEMBAR0 ERROR CAPABILITIES AND CONTROL STATUS.
“MEMBAR0_RAS_HEADER_LOG_REG_00_OFF” on page 2947	B+0x18	CXL RCRB MEMBAR0 HEADER LOG REGISTER 18H.
“MEMBAR0_RAS_HEADER_LOG_REG_01_OFF” on page 2948	B+0x1c	CXL RCRB MEMBAR0 HEADER LOG REGISTER 1CH.
“MEMBAR0_RAS_HEADER_LOG_REG_02_OFF” on page 2949	B+0x20	CXL RCRB MEMBAR0 HEADER LOG REGISTER 20H.
“MEMBAR0_RAS_HEADER_LOG_REG_03_OFF” on page 2950	B+0x24	CXL RCRB MEMBAR0 HEADER LOG REGISTER 24H.
“MEMBAR0_RAS_HEADER_LOG_REG_04_OFF” on page 2951	B+0x28	CXL RCRB MEMBAR0 HEADER LOG REGISTER 28H.
“MEMBAR0_RAS_HEADER_LOG_REG_05_OFF” on page 2952	B+0x2c	CXL RCRB MEMBAR0 HEADER LOG REGISTER 2CH.
“MEMBAR0_RAS_HEADER_LOG_REG_06_OFF” on page 2953	B+0x30	CXL RCRB MEMBAR0 HEADER LOG REGISTER 30H.
“MEMBAR0_RAS_HEADER_LOG_REG_07_OFF” on page 2954	B+0x34	CXL RCRB MEMBAR0 HEADER LOG REGISTER 34H.
“MEMBAR0_RAS_HEADER_LOG_REG_08_OFF” on page 2955	B+0x38	CXL RCRB MEMBAR0 HEADER LOG REGISTER 38H.
“MEMBAR0_RAS_HEADER_LOG_REG_09_OFF” on page 2956	B+0x3c	CXL RCRB MEMBAR0 HEADER LOG REGISTER 3CH.
“MEMBAR0_RAS_HEADER_LOG_REG_10_OFF” on page 2957	B+0x40	CXL RCRB MEMBAR0 HEADER LOG REGISTER 40H.
“MEMBAR0_RAS_HEADER_LOG_REG_11_OFF” on page 2958	B+0x44	CXL RCRB MEMBAR0 HEADER LOG REGISTER 44H.
“MEMBAR0_RAS_HEADER_LOG_REG_12_OFF” on page 2959	B+0x48	CXL RCRB MEMBAR0 HEADER LOG REGISTER 48H.
“MEMBAR0_RAS_HEADER_LOG_REG_13_OFF” on page 2960	B+0x4c	CXL RCRB MEMBAR0 HEADER LOG REGISTER 4CH.
“MEMBAR0_RAS_HEADER_LOG_REG_14_OFF” on page 2961	B+0x50	CXL RCRB MEMBAR0 HEADER LOG REGISTER 50H.
“MEMBAR0_RAS_HEADER_LOG_REG_15_OFF” on page 2962	B+0x54	CXL RCRB MEMBAR0 HEADER LOG REGISTER 54H.

Register	Offset	Description
CXL-RCRB MEMBAR0 Link Capability Structure PF0_MEMBAR0_LINK_CAP Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)		
"MEMBAR0_LINK_LAYER_CAP_REG_OFF" on page 2964	B+0x0	CXL LINK LAYER CAPABILITY REG.
"MEMBAR0_LINK_LAYER_CAP_2_REG_OFF" on page 2966	B+0x4	CXL LINK LAYER CAPABILITY Second REG.
"MEMBAR0_LINK_CNTRL_STATUS_REG_OFF" on page 2968	B+0x8	CXL LINK LAYER CONTROL AND STATUS REG.
"MEMBAR0_LINK_RX_CR_CNTRL_REG_OFF" on page 2970	B+0x10	CXL LINK RX CREDIT CONTROL REG.
"MEMBAR0_LINK_RX_CR_CNTRL_2_REG_OFF" on page 2972	B+0x14	CXL LINK RX CREDIT CONTROL Second REG.
"MEMBAR0_LINK_RX_CR_RTN_STATUS_REG_OFF" on page 2974	B+0x18	CXL LINK RX CREDIT STATUS REG.
"MEMBAR0_LINK_RX_CR_RTN_STATUS_2_REG_OFF" on page 2976	B+0x1c	CXL LINK RX CREDIT STATUS Second REG.
"MEMBAR0_LINK_TX_CR_STATUS_REG_OFF" on page 2978	B+0x20	CXL LAYER TX CREDIT STATUS.
"MEMBAR0_LINK_TX_CR_STATUS_2_REG_OFF" on page 2980	B+0x24	CXL LAYER TX CREDIT STATUS Second REG.
"MEMBAR0_LINK_ACK_TMR_CNTRL_REG_OFF" on page 2982	B+0x28	CXL LINK ACK TIMER CONTROL REG.
"MEMBAR0_LINK_DEFEATURE_REG_OFF" on page 2984	B+0x30	CXL LINK DEFEATURE REG.
CXL-RCRB MEMBAR0 ARB/ MUX Capability Structure PF0_MEMBAR0_ARB_MUX_CAP Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)		
"MEMBAR0_ARB_MUX_CXL_IO_REG_OFF" on page 2987	B+0x0	ARB/MUX Arbitration Control Register for CXL.io.
"MEMBAR0_ARB_MUX_CXL_CACHE_MEM_REG_OFF" on page 2989	B+0x40	ARB/MUX Arbitration Control Register for CXL.cache and CXL.mem.
ATU Unroll Structure PF0_ATU_CAP Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1))		

Register	Offset	Description
"IATU_REGION_CTRL_1_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 2992	B+i*2*0x100+0x0	iATU Region Control 1 Register.
"IATU_REGION_CTRL_2_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 2997	B+i*2*0x100+0x4	iATU Region Control 2 Register.
"IATU_LWR_BASE_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 3004	B+i*2*0x100+0x8	iATU Lower Base Address Register.
"IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 3006	B+i*2*0x100+0xc	iATU Upper Base Address Register.
"IATU_LIMIT_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 3007	B+i*2*0x100+0x10	iATU Limit Address Register.
"IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 3009	B+i*2*0x100+0x14	iATU Lower Target Address Register.
"IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 3011	B+i*2*0x100+0x18	iATU Upper Target Address Register.
"IATU_REGION_CTRL_3_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 3012	B+i*2*0x100+0x1c	iATU Region Control 3 Register.
"IATU_UPPR_LIMIT_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)" on page 3014	B+i*2*0x100+0x20	iATU Upper Limit Address Register.
"IATU_REGION_CTRL_1_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 3016	B+i*2*0x100+0x0+0x100	iATU Region Control 1 Register.

Register	Offset	Description
"IATU_REGION_CTRL_2_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 3021	B+i*2*0x100+0x4+0x100	iATU Region Control 2 Register.
"IATU_LWR_BASE_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 3032	B+i*2*0x100+0x8+0x100	iATU Lower Base Address Register.
"IATU_UPPER_BASE_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 3034	B+i*2*0x100+0xc+0x100	iATU Upper Base Address Register.
"IATU_LIMIT_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 3035	B+i*2*0x100+0x10+0x100	iATU Limit Address Register.
"IATU_LWR_TARGET_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 3039	B+i*2*0x100+0x14+0x100	iATU Lower Target Address Register.
"IATU_UPPER_TARGET_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 3041	B+i*2*0x100+0x18+0x100	iATU Upper Target Address Register.
"IATU_REGION_CTRL_3_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 3042	B+i*2*0x100+0x1c+0x100	iATU Region Control 3 Register.
"IATU_UPPER_LIMIT_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)" on page 3044	B+i*2*0x100+0x20+0x100	iATU Upper Limit Address Register.
DMA Unroll Structure PF0_DMA_CAP Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1))		
"DMA_CTRL_DATA_ARB_PRIOR_OFF" on page 3047	B+0x0	DMA Arbitration Scheme for TRGT1 Interface.
"DMA_CTRL_OFF" on page 3050	B+0x8	DMA Number of Channels Register.
"DMA_WRITE_ENGINE_EN_OFF" on page 3053	B+0xc	DMA Write Engine Enable Register.

Register	Offset	Description
"DMA_WRITE_DOORBELL_OFF" on page 3060	B+0x10	DMA Write Doorbell Register.
"DMA_WRITE_ENGINE_PF_OFF" on page 3063	B+0x14	DMA Write Engine Pre-Fetch Register.
"DMA_WRITE_CHANNEL_ARB_WEIGHT_LO W_OFF" on page 3065	B+0x18	DMA Write Engine Channel Arbitration Weight Low Register.
"DMA_WRITE_CHANNEL_ARB_WEIGHT_HI GH_OFF" on page 3068	B+0x1c	DMA Write Engine Channel Arbitration Weight High Register.
"DMA_READ_ENGINE_EN_OFF" on page 3071	B+0x2c	DMA Read Engine Enable Register.
"DMA_READ_DOORBELL_OFF" on page 3078	B+0x30	DMA Read Doorbell Register.
"DMA_READ_ENGINE_PF_OFF" on page 3081	B+0x34	DMA Write Engine Pre-Fetch Register.
"DMA_READ_CHANNEL_ARB_WEIGHT_LO W_OFF" on page 3083	B+0x38	DMA Read Engine Channel Arbitration Weight Low Register.
"DMA_READ_CHANNEL_ARB_WEIGHT_HI G_H_OFF" on page 3086	B+0x3c	DMA Read Engine Channel Arbitration Weight High Register.
"DMA_WRITE_INT_STATUS_OFF" on page 3089	B+0x4c	DMA Write Interrupt Status Register.
"DMA_WRITE_INT_MASK_OFF" on page 3092	B+0x54	DMA Write Interrupt Mask Register.
"DMA_WRITE_INT_CLEAR_OFF" on page 3095	B+0x58	DMA Write Interrupt Clear Register.
"DMA_WRITE_ERR_STATUS_OFF" on page 3097	B+0x5c	DMA Write Error Status Register.
"DMA_WRITE_DONE_IMWR_LOW_OFF" on page 3100	B+0x60	DMA Write Done IMWr Address Low Register.
"DMA_WRITE_DONE_IMWR_HIGH_OFF" on page 3101	B+0x64	DMA Write Done IMWr Interrupt Address High Register.
"DMA_WRITE_ABORT_IMWR_LOW_OFF" on page 3102	B+0x68	DMA Write Abort IMWr Address Low Register.
"DMA_WRITE_ABORT_IMWR_HIGH_OFF" on page 3103	B+0x6c	DMA Write Abort IMWr Address High Register.
"DMA_WRITE_CH01_IMWR_DATA_OFF" on page 3104	B+0x70	DMA Write Channel 1 and 0 IMWr Data Register.

Register	Offset	Description
"DMA_WRITE_CH23_IMWR_DATA_OFF" on page 3106	B+0x74	DMA Write Channel 3 and 2 IMWr Data Register.
"DMA_WRITE_CH45_IMWR_DATA_OFF" on page 3108	B+0x78	DMA Write Channel 5 and 4 IMWr Data Register.
"DMA_WRITE_CH67_IMWR_DATA_OFF" on page 3110	B+0x7c	DMA Write Channel 7 and 6 IMWr Data Register.
"DMA_WRITE_LINKED_LIST_ERR_EN_OFF" on page 3112	B+0x90	DMA Write Linked List Error Enable Register.
"DMA_READ_INT_STATUS_OFF" on page 3115	B+0xa0	DMA Read Interrupt Status Register.
"DMA_READ_INT_MASK_OFF" on page 3118	B+0xa8	DMA Read Interrupt Mask Register.
"DMA_READ_INT_CLEAR_OFF" on page 3121	B+0xac	DMA Read Interrupt Clear Register.
"DMA_READ_ERR_STATUS_LOW_OFF" on page 3123	B+0xb4	DMA Read Error Status Low Register.
"DMA_READ_ERR_STATUS_HIGH_OFF" on page 3126	B+0xb8	DMA Read Error Status High Register.
"DMA_READ_LINKED_LIST_ERR_EN_OFF" on page 3129	B+0xc4	DMA Read Linked List Error Enable Register.
"DMA_READ_DONE_IMWR_LOW_OFF" on page 3132	B+0xcc	DMA Read Done IMWr Address Low Register.
"DMA_READ_DONE_IMWR_HIGH_OFF" on page 3133	B+0xd0	DMA Read Done IMWr Address High Register.
"DMA_READ_ABORT_IMWR_LOW_OFF" on page 3134	B+0xd4	DMA Read Abort IMWr Address Low Register.
"DMA_READ_ABORT_IMWR_HIGH_OFF" on page 3135	B+0xd8	DMA Read Abort IMWr Address High Register.
"DMA_READ_CH01_IMWR_DATA_OFF" on page 3136	B+0xdc	DMA Read Channel 1 and 0 IMWr Data Register.
"DMA_READ_CH23_IMWR_DATA_OFF" on page 3138	B+0xe0	DMA Read Channel 3 and 2 IMWr Data Register.
"DMA_READ_CH45_IMWR_DATA_OFF" on page 3140	B+0xe4	DMA Read Channel 5 and 4 IMWr Data Register.
"DMA_READ_CH67_IMWR_DATA_OFF" on page 3142	B+0xe8	DMA Read Channel 7 and 6 IMWr Data Register.
"DMA_WRITE_ENGINE_CHGROUP_OFF" on page 3144	B+0x100	DMA Write Engine Channel Grouping Control Register.

Register	Offset	Description
"DMA_READ_ENGINE_CHGROUP_OFF" on page 3146	B+0x104	DMA Read Engine Channel Grouping Control Register.
"DMA_WRITE_ENGINE_HSHAKE_CNT_LOW_OFF" on page 3148	B+0x108	DMA Write Engine Handshake Counter Channel 0/1/2/3 Register.
"DMA_WRITE_ENGINE_HSHAKE_CNT_HIGH_OFF" on page 3151	B+0x10c	DMA Write Engine Handshake Counter Channel 4/5/6/7 Register.
"DMA_READ_ENGINE_HSHAKE_CNT_LOW_OFF" on page 3154	B+0x118	DMA Read Engine Handshake Counter Channel 0/1/2/3 Register.
"DMA_READ_ENGINE_HSHAKE_CNT_HIGH_OFF" on page 3157	B+0x11c	DMA Read Engine Handshake Counter Channel 4/5/6/7 Register.
"DMA_WRITE_CHi_PWR_EN_OFF (for i = 0; i <= 7)" on page 3160	B+0x128 + i*0x4	DMA Write Channel 0 Power Enable Register.
"DMA_READ_CHi_PWR_EN_OFF (for i = 0; i <= 7)" on page 3162	B+0x168 + i*0x4	DMA Read Channel 0 Power Enable Register.
"DMA_CH_CONTROL1_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3164	B+i*2*0x100+0x200	DMA Write Channel Control 1 Register.
"DMA_CH_CONTROL2_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3179	B+i*2*0x100+0x204	DMA Write Channel Control 2 Register.
"DMA_TRANSFER_SIZE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3184	B+i*2*0x100+0x208	DMA Write Transfer Size Register.
"DMA_SAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3186	B+i*2*0x100+0x20c	DMA Write SAR Low Register.
"DMA_SAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3188	B+i*2*0x100+0x210	DMA Write SAR High Register.
"DMA_DAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3190	B+i*2*0x100+0x214	DMA Write DAR Low Register.
"DMA_DAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3192	B+i*2*0x100+0x218	DMA Write DAR High Register.
"DMA_LL_P_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3194	B+i*2*0x100+0x21c	DMA Write Linked List Pointer Low Register.
"DMA_LL_P_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3196	B+i*2*0x100+0x220	DMA Write Linked List Pointer High Register.

Register	Offset	Description
"DMA_CH_CONTROL1_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3197	B+i*2*0x100+0x200+0x100	DMA Read Channel Control 1 Register.
"DMA_CH_CONTROL2_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3212	B+i*2*0x100+0x204+0x100	DMA Read Channel Control 2 Register.
"DMA_TRANSFER_SIZE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3218	B+i*2*0x100+0x208+0x100	DMA Read Transfer Size Register.
"DMA_SAR_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3220	B+i*2*0x100+0x20c+0x100	DMA Read SAR Low Register.
"DMA_SAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3222	B+i*2*0x100+0x210+0x100	DMA Read SAR High Register.
"DMA_DAR_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3224	B+i*2*0x100+0x214+0x100	DMA Read DAR Low Register.
"DMA_DAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3226	B+i*2*0x100+0x218+0x100	DMA Read DAR High Register.
"DMA_LL_P_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3228	B+i*2*0x100+0x21c+0x100	DMA Read Linked List Pointer Low Register.
"DMA_LL_P_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3231	B+i*2*0x100+0x220+0x100	DMA Read Linked List Pointer High Register.
HDMA Unroll Structure PF0_HDMA_CAP Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1))		
"HDMA_EN_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3233	B+i*2*pow(2,(CX_DMA_REG_CHADDR_SPACE+8))+0x0	HDMA Write Channel Enable Register.

Register	Offset	Description
"HDMA_DOORBELL_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3234	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x4$	HDMA Write Channel Doorbell Register.
"HDMA_ELEM_PF_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3236	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x8$	HDMA Write Channel Prefetch Register.
"HDMA_HANDSHAKE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3238	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0xc$	HDMA Write Channel Handshake Register.
"HDMA_LLP_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3240	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x10$	HDMA Write Channel Linked List Pointer Low Register.
"HDMA_LLP_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3242	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x14$	HDMA Write Channel Linked List Pointer High Register.
"HDMA_CYCLE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3244	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x18$	HDMA Write Channel Producer-Consumer Cycle Synchronization Register.
"HDMA_XFERSIZE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3246	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x1c$	HDMA Write Channel Transfer Size Register.

Register	Offset	Description
"HDMA_SAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3248	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x20$	HDMA Write Channel SAR Low Register.
"HDMA_SAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3250	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x24$	HDMA Write Channel SAR High Register.
"HDMA_DAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3252	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x28$	HDMA Write Channel DAR Low Register.
"HDMA_DAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3254	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x2c$	HDMA Write Channel DAR High Register.
"HDMA_WATERMARK_EN_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3256	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x30$	HDMA Write Channel Linked-list Watermark Enable Register.
"HDMA_CONTROL1_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3258	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x34$	HDMA Write Channel Control Settings 1 Register.
"HDMA_FUNC_NUM_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3263	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x38$	HDMA Write Channel Function Number Register.

Register	Offset	Description
"HDMA_QOS_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3267	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x3c$	HDMA Write Channel QoS Settings Register.
"HDMA_STATUS_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3270	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x80$	HDMA Write Channel Status Register.
"HDMA_INT_STATUS_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3271	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x84$	HDMA Write Channel Interrupt Status Register.
"HDMA_INT_SETUP_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3275	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x88$	HDMA Write Channel Interrupt Setup Register.
"HDMA_INT_CLEAR_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3280	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x8c$	HDMA Write Channel Interrupt Clear Register.
"HDMA_MSI_STOP_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3282	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x90$	HDMA Read Stop Remote Interrupt Address Low Register.
"HDMA_MSI_STOP_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3284	$B+i*2^{pow}(2,(CX_D_MAREG_CHADDR_SPACE+8))+0x94$	HDMA Read Stop Remote Interrupt Address High Register.

Register	Offset	Description
"HDMA_MSI_WATERMARK_LOW_OFF_WRC H_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3286	B+i*2*pow(2,(CX_D MAREG_ CHADDR _SPACE+ 8))+0x98	HDMA Read Watermark Remote Interrupt Address Low Register.
"HDMA_MSI_WATERMARK_HIGH_OFF_WR CH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3288	B+i*2*pow(2,(CX_D MAREG_ CHADDR _SPACE+ 8))+0x9c	HDMA Read Watermark Remote Interrupt Address High Register.
"HDMA_MSI_ABORT_LOW_OFF_WRC H_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3290	B+i*2*pow(2,(CX_D MAREG_ CHADDR _SPACE+ 8))+0xa0	HDMA Read Abort Remote Interrupt Address Low Register.
"HDMA_MSI_ABORT_HIGH_OFF_WRC H_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3292	B+i*2*pow(2,(CX_D MAREG_ CHADDR _SPACE+ 8))+0xa4	HDMA Read Abort Remote Interrupt Address High Register.
"HDMA_MSI_MSGD_OFF_WRC H_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)" on page 3294	B+i*2*pow(2,(CX_D MAREG_ CHADDR _SPACE+ 8))+0xa8	HDMA Write Channel Remote Interrupt Data Register.
"HDMA_EN_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3295	B+i*2*pow(2,(CX_D MAREG_ CHADDR _SPACE+ 8))+0x0+p ow(2,(CX_ DMAREG_ _CHADD R_SPACE +8))	HDMA Read Channel Enable.

Register	Offset	Description
"HDMA_DOORBELL_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3296	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x4+pow(2,(CX_DMAREG_CHADDR_R_SPACE+8))}$	HDMA Read Channel Doorbell Register.
"HDMA_ELEM_PF_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3298	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x8+pow(2,(CX_DMAREG_CHADDR_R_SPACE+8))}$	HDMA Read Channel Prefetch Register.
"HDMA_HANDSHAKE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3300	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0xc+pow(2,(CX_DMAREG_CHADDR_R_SPACE+8))}$	HDMA Read Channel Handshake Register.
"HDMA_LL_P_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3302	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x10+pow(2,(CX_DMAREG_CHADDR_R_SPACE+8))}$	HDMA Read Channel Linked List Pointer Low Register.

Register	Offset	Description
"HDMA_LLH_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3304	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x14+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel Linked List Pointer High Register.
"HDMA_CYCLE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3306	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x18+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel Producer-Consumer Cycle Synchronization Register.
"HDMA_XFERSIZE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3308	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x1c+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel Transfer Size Register.
"HDMA_SAR_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3310	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x20+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel SAR Low Register.

Register	Offset	Description
"HDMA_SAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3312	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x24+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel SAR High Register.
"HDMA_DAR_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3314	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x28+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel DAR Low Register.
"HDMA_DAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3316	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x2c+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel DAR High Register.
"HDMA_WATERMARK_EN_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3318	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x30+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel Linked-list Watermark Enable Register.

Register	Offset	Description
"HDMA_CONTROL1_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3321	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x34+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel Control Settings 1 Register.
"HDMA_FUNC_NUM_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3326	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x38+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel Function Number Register.
"HDMA_QOS_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3330	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x3c+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel QoS Settings Register.
"HDMA_STATUS_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3333	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x80+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel Status Register.

Register	Offset	Description
"HDMA_INT_STATUS_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3334	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x84+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel Interrupt Status Register.
"HDMA_INT_SETUP_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3338	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x88+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel Interrupt Setup Register.
"HDMA_INT_CLEAR_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3343	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x8c+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Channel Interrupt Clear Register.
"HDMA_MSI_STOP_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3345	$B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x90+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$	HDMA Read Stop Remote Interrupt Address Low Register.

Register	Offset	Description
"HDMA_MSI_STOP_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3347	$B+i*2*\text{pow}(2,(\text{CX_D_MAREG_CHADDR_SPACE}+8))+0x94+\text{pow}(2,(\text{CX_DMAREG_CHADDR_SPACE}+8))$	HDMA Read Stop Remote Interrupt Address High Register.
"HDMA_MSI_WATERMARK_LOW_OFF_RDC H_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3349	$B+i*2*\text{pow}(2,(\text{CX_D_MAREG_CHADDR_SPACE}+8))+0x98+\text{pow}(2,(\text{CX_DMAREG_CHADDR_SPACE}+8))$	HDMA Read Watermark Remote Interrupt Address Low Register.
"HDMA_MSI_WATERMARK_HIGH_OFF_RDC H_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3351	$B+i*2*\text{pow}(2,(\text{CX_D_MAREG_CHADDR_SPACE}+8))+0x9c+\text{pow}(2,(\text{CX_DMAREG_CHADDR_SPACE}+8))$	HDMA Read Watermark Remote Interrupt Address High Register.
"HDMA_MSI_ABORT_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)" on page 3353	$B+i*2*\text{pow}(2,(\text{CX_D_MAREG_CHADDR_SPACE}+8))+0xa0+\text{pow}(2,(\text{CX_DMAREG_CHADDR_SPACE}+8))$	HDMA Read Abort Remote Interrupt Address Low Register.

Register	Offset	Description
“ HDMA_MSI_ABORT_HIGH_OFF_RDCH_i (for $i = 0; i \leq CC_NUM_DMA_RD_CHAN-1$)” on page 3355	$B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0xa4+pow(2,(CX_DMAREG_CHADDR_SPACE+8))$	HDMA Read Abort Remote Interrupt Address High.
“ HDMA_MSI_MSGD_OFF_RDCH_i (for $i = 0; i \leq CC_NUM_DMA_RD_CHAN-1$)” on page 3356	$B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0xa8+pow(2,(CX_DMAREG_CHADDR_SPACE+8))$	HDMA Read Channel Remote Interrupt Data Register.

3.1 PF_TYPE0_HDRi Registers

3.1.1 **DEVICE_ID_VENDOR_ID_REG**

- **Name:** Device ID , RCRB next offset pointer and Vendor ID Register.
- **Description:** This register holds the device ID, next offset pointer for RCRB mode and vendor ID.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** Always

PCI_TYPE0_DEVICE_ID	31:16
PCI_TYPE0_VENDOR_ID	15:0

Table 3-5 Fields for Register: DEVICE_ID_VENDOR_ID_REG

Bits	Name	Memory Access	Description
31:16	PCI_TYPE0_DEVICE_ID	* Varies	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB VERSION[19:16] & RCRB Next Offset Pointer[31:20]. ■ The two CXL based fields will be enabled when target protocol CXL is enabled and the memorymap region is RCRB. □ The Version number for the Root Complex Register Block (RCRB). This register space was reserved for RCRB memory mapped region in Type0 and Type1 Configuration Header capability. □ The RCRB Next offset pointer register identifies the offset value of the next register field. This register space was reserved for RCRB mode in Type0 and Type1. 0: DEVICE_ID [31:16] ■ The Device ID register identifies the particular Function for PCIe Type0 and Type1 configuration header. This identifier is allocated by the vendor. <i>Databook.</i> <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && !SNPS_RSVDPARAM_11) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: CX_DEVICE_ID_0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15:0	PCI_TYPE0_VENDOR_ID	* Varies	Vendor ID. The Vendor ID register identifies the manufacturer of the Function. Valid vendor identifiers are allocated by the PCI-SIG to ensure uniqueness. It is not permitted to populate this register with a value of FFFFh, which is an invalid value for Vendor ID. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && !SNPS_RSVDPARAM_11) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1) ? 0 : CX_VENDOR_ID_0 } Exists: Always Testable: wire ? unconstrained : writeAsRead

3.1.2 STATUS_COMMAND_REG

- **Name:** Status and Command Register.
- **Description:** This register provides the status and controls the behavior of a function.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** Always

DETECTED_PARITY_ERR	31
SIGNALLED_SYS_ERR	30
RCVD_MASTER_ABORT	29
RCVD_TARGET_ABORT	28
SIGNALLED_TARGET_ABORT	27
DEV_SEL_TIMING	26:25
MASTER_DPE	24
FAST_B2B_CAP	23
RSVDP_22	22
FAST_66MHZ_CAP	21
CAP_LIST	20
INT_STATUS	19
RSVDP_17	18:17
IMM_READINESS	16
PCI_TYPE_RESERV	15:11
PCI_TYPE0_INT_EN	10
RSVDP_9	9
PCI_TYPE0_SERREN	8
PCI_TYPE_IDSEL_STEPPING	7
PCI_TYPE0_PARITY_ERR_EN	6
PCI_TYPE_VGA_PALETTE_SNOOP	5
PCI_TYPE_MWI_ENABLE	4
PCI_TYPE0_SPECIAL_CYCLE_OPERATION	3
PCI_TYPE0_BUS_MASTER_EN	2
PCI_TYPE0_MEM_SPACE_EN	1
PCI_TYPE0_IO_EN	0

Table 3-6 Fields for Register: STATUS_COMMAND_REG

Bits	Name	Memory Access	Description
31	DETECTED_PARITY_ERR	R/W1C	Detected Parity Error. Values: ■ 0x1 (SET): This bit is set by a Function whenever it receives a Poisoned TLP, regardless of the state the Parity Error Response bit in the Command register. Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
30	SIGNALED_SYS_ERR	R/W1C	Signaled System Error. Values: 0x1 (SET): This bit is set when a Function sends an ERR_FATAL or ERR_NONFATAL message, and the SERR# Enable bit in the Command register is 1b. For Functions that do not send ERR_FATAL or ERR_NONFATAL messages, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true
29	RCVD_MASTER_ABORT	R/W1C	Received Master Abort. Values: 0x1 (SET): This bit is set when a Requester receives a Completion with Unsupported Request Completion Status. For Functions that do not make Non-Posted Requests on their own behalf, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true
28	RCVD_TARGET_ABORT	R/W1C	Received Target Abort. Values: 0x1 (SET): This bit is set when a Requester receives a Completion with Completer Abort Completion Status. For Functions that do not make Non-Posted Requests on their own behalf, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true
27	SIGNALED_TARGET_ABORT	R/W1C	Signaled Target Abort. Values: 0x1 (SET): This bit is set when a Function completes a Posted or Non-Posted Request as a Completer Abort error. The controller hardwires this bit to 0b for Functions that do not signal Completer Abort. Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
26:25	DEV_SEL_TIMING	R	DEVSEL Timing. This field was originally described in the PCI Local Bus Specification. Its functionality does not apply to PCI Express. The controller hardwires this field to 00b. Value After Reset: 0x0 Exists: Always
24	MASTER_DPE	R/W1C	Master Data Parity Error. This bit is set by a Function if the Parity Error Response bit in the Command register is 1b and either of the following two conditions occurs: ■ Function receives a Poisoned Completion ■ Function transmits a Poisoned Request If the Parity Error Response bit is 0b, this bit is never set. Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true
23	FAST_B2B_CAP	R	Fast Back to Back Transaction Capable. This bit was originally described in the PCI Local Bus Specification. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: Always
22	RSVDP_22	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: Always
21	FAST_66MHZ_CAP	R	66MHz Capable. This bit was originally described in the PCI Local Bus Specification. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: Always
20	CAP_LIST	R	Capabilities List. Indicates the presence of an Extended Capability list item. Since all PCI Express device Functions are required to implement the PCI Express Capability structure, the controller hardwires this bit to 1b. Value After Reset: 0x1 Exists: Always

Bits	Name	Memory Access	Description
19	INT_STATUS	R	Emulation interrupt pending. When set, indicates that an INTx emulation interrupt is pending internally in the Function. Setting the Interrupt Disable bit has no effect on the state of this bit. For Functions that do not generate INTx interrupts, the controller hardwires this bit to 0b. Value After Reset: * Varies Exists: Always Testable: writeAsRead Reset Mask: 0x0 Volatile: true
18:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: Always
16	IMM_READINESS	* Varies	Immediate Readiness. This optional bit, when set, indicates the Function is guaranteed to be ready to successfully complete valid configuration accesses at any time following any reset that the host is capable of issuing Configuration Requests to this Function. When this bit is set, for accesses to this Function, software is exempt from all requirements to delay configuration accesses following any type of reset, including but not limited to the timing requirements defined in "PCI Express Reset - Rules" section of the <i>PCI Express Base Specification</i>. It is permitted that system software/firmware provide mechanisms that supersede the indication provided by this bit, however such software/firmware mechanisms are outside the scope of <i>PCI Express Base Specification</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(Sticky) else R(Sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_RN_IMM_VALUE_0 Exists: CX_RN_IMM_EN Testable: wire ? unconstrained : writeAsRead
15:11	PCI_TYPE_RESERV	R	Reserved. Value After Reset: 0x0 Exists: Always

Bits	Name	Memory Access	Description
10	PCI_TYPE0_INT_EN	R/W	Interrupt Disable. Controls the ability of a Function to generate INTx emulation interrupts. When set, Functions are prevented from asserting INTx interrupts. Note: Any INTx emulation interrupts already asserted by the Function must be de-asserted when this bit is Set. INTx interrupts use virtual wires that must, if asserted, be de-asserted using the appropriate Deassert_INTx message(s) when this bit is set. Only the INTx virtual wire interrupt(s) associated with the Function(s) for which this bit is set are affected. For functions that generate INTx interrupts, this bit is required. For functions that do not generate INTx interrupts, this bit is optional. Value After Reset: 0x0 Exists: Always
9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: Always
8	PCI_TYPE0_SERREN	R/W	SERR# Enable. When set, this bit enables reporting upstream of Non-fatal and Fatal errors detected by the Function. Note: The errors are reported if enabled either through this bit or through the PCI Express specific bits in the Device Control register. Value After Reset: 0x0 Exists: Always
7	PCI_TYPE_IDSEL_STEPPING	R	IDSEL Stepping/Wait Cycle Control. This bit was originally described in the <i>PCI Local Bus Specification</i>. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: Always
6	PCI_TYPE0_PARITY_ERR_EN	R/W	Parity Error Response. This bit controls the logging of poisoned TLPs in the Master Data Parity Error bit in the Status register. Value After Reset: 0x0 Exists: Always

Bits	Name	Memory Access	Description
5	PCI_TYPE_VGA_PALETTE_SNOOP	R	VGA Palette Snoop. This bit was originally described in the <i>PCI Local Bus Specification</i> and the <i>PCI-to-PCI Bridge architecture specification</i> . Its functionality does not apply to PCI Express, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: Always
4	PCI_TYPE_MWI_ENABLE	R	Memory Write and Invalidate. This bit was originally described in the <i>PCI Local Bus Specification</i> and the <i>PCI-to-PCI Bridge architecture specification</i> . Its functionality does not apply to PCI Express, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: Always
3	PCI_TYPE0_SPECIAL_CYCLE_OPERATION	R	Special Cycle Enable. This bit was originally described in the <i>PCI Local Bus Specification</i> . Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: Always
2	PCI_TYPE0_BUS_MASTER_EN	* Varies	Bus Master Enable. Controls the ability of a Function to issue Memory and I/O Read/Write requests. ■ When this bit is set, the Function is allowed to issue Memory or I/O Requests. ■ When this bit is clear, the Function is not allowed to issue any Memory or I/O Requests. Requests other than Memory or I/O Requests are not controlled by this bit. <i>Note:</i> MSI/MSI-X interrupt Messages are in-band memory writes, setting the Bus Master Enable bit to 0b disables MSI/MSI-X interrupt Messages as well. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: Always Memory Access: "(CX_MEMORY_MAP_VIEW==0 && !SNPS_RSVDPARAM_11) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && SNPS_RSVDPARAM_11) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==2 && !SNPS_RSVDPARAM_11) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
1	PCI_TYPE0_MEM_SPACE_EN	* Varies	Memory Space Enable. Controls a Function's response to Memory Space accesses. ■ When this bit is set, the Function is enabled to decode the address and further process Memory Space accesses. ■ When this bit is clear, all received Memory Space accesses are caused to be handled as Unsupported Requests. For a Function does not support Memory Space accesses, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: !has_mem_bar ? RO : RW ■ Dbi: !has_mem_bar ? RO : RW Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true
0	PCI_TYPE0_IO_EN	* Varies	IO Space Enable. Controls a Function's response to I/O Space accesses. ■ When this bit is set, the Function is enabled to decode the address and further process I/O Space accesses. ■ When this bit is clear, all received I/O accesses are caused to be handled as Unsupported Requests. For a Function that does not support I/O Space accesses, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: !has_io_bar ? RO : RW ■ Dbi: !has_io_bar ? RO : RW and more specifically: ((MEM0_SPACE_DECODER_\$pf && BAR0_ENABLED_\$pf) (MEM1_SPACE_DECODER_\$pf && BAR1_ENABLED_\$pf) (MEM2_SPACE_DECODER_\$pf && BAR2_ENABLED_\$pf) (MEM3_SPACE_DECODER_\$pf && BAR3_ENABLED_\$pf) (MEM4_SPACE_DECODER_\$pf && BAR4_ENABLED_\$pf) (MEM5_SPACE_DECODER_\$pf && BAR5_ENABLED_\$pf)) ? R/W : R Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true

3.1.3 CLASS_CODE_REVISION_ID

- **Name:** Class Code and Revision ID Register.
- **Description:** This register specifies the class code and revision ID of a function.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** Always

31:24	BASE_CLASS_CODE
23:16	SUBCLASS_CODE
15:8	PROGRAM_INTERFACE
7:0	REVISION_ID

Table 3-7 Fields for Register: CLASS_CODE_REVISION_ID

Bits	Name	Memory Access	Description
31:24	BASE_CLASS_CODE	* Varies	Base Class Code. A code that broadly classifies the type of operation the Function performs. Encodings for base class, are provided in the PCI Code and ID Assignment Specification. All unspecified encodings are Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: BASE_CLASS_CODE_0 Exists: Always Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
23:16	SUBCLASS_CODE	* Varies	Sub-Class Code. Specifies a base class sub-class, which identifies more specifically the operation of the Function. Encodings for sub-class are provided in the PCI Code and ID Assignment Specification. All unspecified encodings are Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: SUB_CLASS_CODE_0 Exists: Always Testable: wire ? unconstrained : writeAsRead
15:8	PROGRAM_INTERFACE	* Varies	Programming Interface. This field identifies a specific register-level programming interface (if any) so that device independent software can interact with the Function. Encodings for interface are provided in the PCI Code and ID Assignment Specification. All unspecified encodings are Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: IF_CODE_0 Exists: Always Testable: wire ? unconstrained : writeAsRead
7:0	REVISION_ID	* Varies	Revision ID. The value in this register specifies a Function specific revision identifier. The value is chosen by the vendor. Zero is an acceptable value. The Revision ID should be viewed as a vendor defined extension to the Device ID. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_REVISION_ID_0 Exists: Always Testable: wire ? unconstrained : writeAsRead

3.1.4 **BIST_HEADER_TYPE_LATENCY_CACHE_LINE_SIZE_REG**

- **Name:** BIST, Header Type, Latency Timer, and Cache Line Size Register.
- **Description:** This register provides the status and controls BIST. It also holds information regarding the header layout, latency timer, and cache line size.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** Always

BIST	31:24
MULTI_FUNC	23
HEADER_TYPE	22:16
LATENCY_MASTER_TIMER	15:8
CACHE_LINE_SIZE	7:0

Table 3-8 Fields for Register: BIST_HEADER_TYPE_LATENCY_CACHE_LINE_SIZE_REG

Bits	Name	Memory Access	Description
31:24	BIST	R	BIST. This register is used for control and status of BIST. For Functions that do not support BIST the controller hardwires the register to 00h. A Function whose BIST is invoked must not prevent normal operation of the PCI Express Link. Bit descriptions: ■ [31]: BIST Capable. When Set, this bit indicates that the Function supports BIST. When Clear, the Function does not support BIST. ■ [30]: Start BIST. If BIST Capable is Set, Set this bit to invoke BIST. The Function resets the bit when BIST is complete. Software is permitted to fail the device if this bit is not Clear (BIST is not complete) 2 seconds after it had been Set. Writing this bit to 0b has no effect. This bit must be hardwired to 0b if BIST Capable is Clear. ■ [29:28]: Reserved. ■ [27:24]: Completion Code. This field encodes the status of the most recent test. A value of 0000b means that the Function has passed its test. Non-zero values mean the Function failed. Function-specific failure codes can be encoded in the non-zero values. This field's value is only meaningful when BIST Capable is Set and Start BIST is Clear. This field must be hardwired to 0000b if BIST Capable is clear. Values: ■ 0x0 (Min_field_value): Zero value ■ 0xff (Max_field_value): Max value Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
23	MULTI_FUNC	* Varies	Multi-Function Device. Except where stated otherwise, it is recommended that this bit be set if there are multiple Functions, and clear if there is only one Function. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): Indicates that the Device may contain multiple Functions, but not necessarily. Software is permitted to probe for Functions other than Function 0 ■ 0x0 (CLEAR): Software must not probe for Functions other than Function 0 unless explicitly indicated by another mechanism, such as an ARI or SR-IOV Capability structure. Value After Reset: {(CX_NFUNC > 1)? 1 : 0} Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: wire ? unconstrained : writeAsRead
22:16	HEADER_TYPE	R	Header Layout. This field identifies the layout of the second part of the predefined header. The controller uses 000 0000b encoding. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<=2
15:8	LATENCY_MASTER_TIMER	R	Latency Timer. The Latency Timer was originally described in the PCI Local Bus Specification and the PCI-to-PCI Bridge Architecture Specification. Its functionality does not apply to PCI Express. The controller hardwires this register to 00h. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1
7:0	CACHE_LINE_SIZE	R/W	Cache Line Size. The Cache Line Size register is programmed by the system firmware or the operating system to system cache line size. However, legacy conventional PCI software may not always be able to program this register correctly especially in the case of Hot-Plug devices. This read-write register is implemented for legacy compatibility purposes but has no effect on any PCI Express device behavior. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<=2

3.1.5 BAR0_REG

- **Name:** BAR0 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR0_MASK_REG.PCI_TYPE0_BAR0_ENABLED)

BAR0_START	31:4
BAR0_PREFETCH	3
BAR0_TYPE	2:1
BAR0_MEM_IO	0

Table 3-9 Fields for Register: BAR0_REG

Bits	Name	Memory Access	Description
31:4	BAR0_START	* Varies	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR0[31:4]. ■ The MEMBAR0 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. ■ This is 32bits field, we use this to write MEMBAR0 address into RCRB space. MEMBAR0 registers should be able to access BAR0_REG offset 10h of PCIe memory-mapped space. 0: BAR0_START. BAR0 Base Address. ■ Memory Space: Base Address. ■ IO Space: bits[31:2] are used to map the function into IO space/Base Address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R(Sticky)/W(Sticky) if enabled else R(Sticky) ■ Dbi: R(Sticky)/W(Sticky) if enabled else R(Sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	BAR0_PREFETCH	* Varies	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR0[3]. The MEMBAR0 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. This is 32bits field, we use this to write MEMBAR0 address into RCRB space. MEMBAR0 registers should be able to access BAR0_REG offset 10h of PCIe memory-mapped space. 0: BAR0 Prefetchable. Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must be clear otherwise. IO Space: Not applicable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE!=1 && CX_DBI_RO_WR_EN && DEFAULT_BAR0_ENABLED \$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE!=1) ? R : (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R : R <i>Note:</i> This register field is sticky. Value After Reset: PREFETCHABLE0_0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2:1	BAR0_TYPE	* Varies	- The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR0[2:1]. - The MEMBAR0 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. - This is 32bits field, we use this to write MEMBAR0 address into RCRB space. MEMBAR0 registers should be able to access BAR0_REG offset 10h of PCIe memory-mapped space. 0: BAR0 Type. - Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. - IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. <i>Note:</i> The access attributes of this field are as follows: - Wire: R (sticky) - Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE!=1 && CX_DBI_RO_WR_EN && DEFAULT_BAR0_ENABLED_\$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R : R <i>Note</i>
2:1...(cont.)	BAR0_TYPE.	* Varies	: This register field is sticky. Values: 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. 0x1 (RSVD_1): Reserved. 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. 0x3 (RSVD_2): Reserved. Value After Reset: BAR0_TYPE_0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR0_MEM_IO	* Varies	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR0[0]. ■ The MEMBAR0 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. ■ This is 32bits field, we use this to write MEMBAR0 address into RCRB space. MEMBAR0 registers should be able to access BAR0_REG offset 10h of PCIe memory-mapped space. 0: BAR0 Memory Space Indicator. ■ This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. ■ Base Address registers that map to I/O Space must return a 1b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE!=1 && CX_DBI_RO_WR_EN && DEFAULT_BAR0_ENABLED \$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE!=1) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: MEM0_SPACE_DECODER_0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.1.6 BAR1_REG

- **Name:** BAR1 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR1_MASK_REG.PCI_TYPE0_BAR1_ENABLED || BAR0_TYPE_n ==2)

BAR1_START	31:4
BAR1_PREFETCH	3
BAR1_TYPE	2:1
BAR1_MEM_IO	0

Table 3-10 Fields for Register: BAR1_REG

Bits	Name	Memory Access	Description
31:4	BAR1_START	* Varies	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR1[31:4]. ■ The MEMBAR1 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. ■ This is 32bits field, we use this to write MEMBAR1 address into RCRB space. MEMBAR1 registers should be able to access BAR1_REG offset 14h of PCIe memory-mapped space. 0: BAR1 Base Address. ■ Memory Space: Base Address. ■ IO Space: bits[31:2] are used to map the function into IO space/Base Address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R(Sticky)/W(Sticky) if enabled else R(Sticky) ■ Dbi: R(Sticky)/W(Sticky) if enabled else R(Sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	BAR1_PREFETCH	* Varies	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR1[3]. The MEMBAR1 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. This is 32bits field, we use this to write MEMBAR1 address into RCRB space. MEMBAR1 registers should be able to access BAR1_REG offset 14h of PCIE memory-mapped space. 0: BAR1 Prefetchable. Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must be clear otherwise. IO Space: Not applicable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (BAR0_TYPE_0 == 2 && CX_MEMORY_MAP_VIEW != 1) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE != 1 && CX_DBI_RO_WR_EN && DEFAULT_BAR1_ENABLED_0pf) ? R/W : (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE != 1) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE == 1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: {(BAR0_TYPE_0 != 2) ? (PREFETCHABLE1_0) : 0} Exists: CX_MEMORY_MAP_PCIE_TYPE != 1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2:1	BAR1_TYPE	* Varies	- The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR1[2:1]. - The MEMBAR1 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. - This is 32bits field, we use this to write MEMBAR1 address into RCRB space. MEMBAR1 registers should be able to access BAR1_REG offset 14h of PCIE memory-mapped space. 0: BAR1 Type. - Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. - IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. <i>Note:</i> The access attributes of this field are as follows: - Wire: R (sticky)
2:1...(cont.)	BAR1_TYPE.	* Varies	- Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (BAR0_TYPE_\$pf == 2 && CX_MEMORY_MAP_VIEW != 1) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE != 1 && CX_DBI_RO_WR_EN && DEFAULT_BAR1_ENABLED_\$pf) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE != 1) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE == 1) ? R: R <i>Note:</i> This register field is sticky. Values: 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. 0x1 (RSVD_1): Reserved. 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. 0x3 (RSVD_2): Reserved. Value After Reset: {(BAR0_TYPE_0 != 2) ? (BAR1_TYPE_0) : 0} Exists: CX_MEMORY_MAP_PCIE_TYPE != 1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR1_MEM_IO	* Varies	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR1[0]. ■ The MEMBAR1 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. ■ This is 32bits field, we use this to write MEMBAR1 address into RCRB space. MEMBAR1 registers should be able to access BAR1_REG offset 14h of PCIE memory-mapped space. 0: BAR1 Memory Space Indicator. ■ This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. ■ Base Address registers that map to I/O Space must return a 1b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (BAR0_TYPE_0pf == 2 && CX_MEMORY_MAP_VIEW != 1) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE != 1 && CX_DBI_RO_WR_EN && DEFAULT_BAR1_ENABLED_0pf) ? R/W : (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE != 1) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE == 1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: {(BAR0_TYPE_0 != 2) ? (MEM1_SPACE_DECODER_0) : 0} Exists: CX_MEMORY_MAP_PCIE_TYPE != 1 Testable: writeAsRead Volatile: true

3.1.7 BAR2_REG

- **Name:** BAR2 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR2_MASK_REG.PCI_TYPE0_BAR2_ENABLED)

31:4	3	2:1	0
BAR2_START	BAR2_PREFETCH	BAR2_TYPE	BAR2_MEM_IO

Table 3-11 Fields for Register: BAR2_REG

Bits	Name	Memory Access	Description
31:4	BAR2_START	* Varies	BAR2 Base Address. ■ Memory Space: Base Address. ■ IO Space: bits[31:2] are used to map the function into IO space/Base. Address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R(Sticky)/W(Sticky) if enabled else R(Sticky) ■ Dbi: R(Sticky)/W(Sticky) if enabled else R(Sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	BAR2_PREFETCH	* Varies	BAR2 Prefetchable. ■ Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must be clear otherwise. ■ IO Space: Not applicable <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE!=1 && CX_DBI_RO_WR_EN && DEFAULT_BAR2_ENABLED \$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE!=1) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: PREFETCHABLE2_0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2:1	BAR2_TYPE	* Varies	BAR2 Type. - Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. - IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. <i>Note:</i> The access attributes of this field are as follows: - Wire: R (sticky) - Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE!=1 && CX_DBI_RO_WR_EN && DEFAULT_BAR2_ENABLED \$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R: R <i>Note:</i> This register field is sticky. Values: 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. 0x1 (RSVD_1): Reserved. 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. 0x3 (RSVD_2): Reserved. Value After Reset: BAR2_TYPE_0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR2_MEM_IO	* Varies	BAR2 Memory Space Indicator. This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. Base Address registers that map to I/O Space must return a 1b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE!=1 && CX_DBI_RO_WR_EN && DEFAULT_BAR2_ENABLED \$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE!=1) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: MEM2_SPACE_DECODER_0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.1.8 BAR3_REG

- **Name:** BAR3 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR3_MASK_REG.PCI_TYPE0_BAR3_ENABLED || BAR2_TYPE_n == 2)

31:4	3	2:1	0
BAR3_START	BAR3_PREFETCH	BAR3_TYPE	BAR3_MEM_IO

Table 3-12 Fields for Register: BAR3_REG

Bits	Name	Memory Access	Description
31:4	BAR3_START	* Varies	BAR3 Base Address. ■ Memory Space: Base Address. ■ IO Space: bits[31:2] are used to map the function into IO space/Base Address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R(Sticky)/W(Sticky) if enabled else R(Sticky) ■ Dbi: R(Sticky)/W(Sticky) if enabled else R(Sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	BAR3_PREFETCH	* Varies	BAR3 Prefetchable. ■ Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must be clear otherwise. ■ IO Space: Not applicable <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (BAR2_TYPE_0pf == 2 && CX_MEMORY_MAP_VIEW != 1) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE != 1 && CX_DBI_RO_WR_EN && DEFAULT_BAR3_ENABLED_0pf) ? R/W : (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE != 1) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE == 1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: {(BAR2_TYPE_0 != 2) ? (PREFETCHABLE3_0) : 0} Exists: CX_MEMORY_MAP_PCIE_TYPE != 1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2:1	BAR3_TYPE	* Varies	BAR3 Type. - Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. - IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. <i>Note:</i> The access attributes of this field are as follows: - Wire: R (sticky) - Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (BAR2_TYPE_\$pf == 2 && CX_MEMORY_MAP_VIEW != 1) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE != 1 && CX_DBI_RO_WR_EN && DEFAULT_BAR3_ENABLED_\$pf) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE != 1) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE == 1) ? R: R <i>Note:</i> This register field is sticky. Values: 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. 0x1 (RSVD_1): Reserved. 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. 0x3 (RSVD_2): Reserved. Value After Reset: {(BAR2_TYPE_0 != 2) ? (BAR3_TYPE_0) : 0} Exists: CX_MEMORY_MAP_PCIE_TYPE != 1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR3_MEM_IO	* Varies	BAR3 Memory Space Indicator. This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. Base Address registers that map to I/O Space must return a 1b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (BAR2_TYPE_0pf == 2 && CX_MEMORY_MAP_VIEW != 1) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE != 1 && CX_DBI_RO_WR_EN && DEFAULT_BAR3_ENABLED_0pf) ? R/W : (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE != 1) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE == 1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: {(BAR2_TYPE_0 != 2) ? (MEM3_SPACE_DECODER_0) : 0} Exists: CX_MEMORY_MAP_PCIE_TYPE != 1 Testable: writeAsRead Volatile: true

3.1.9 BAR4_REG

- **Name:** BAR4 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR4_MASK_REG.PCI_TYPE0_BAR4_ENABLED)

31:4	3	2:1	0
BAR4_START	BAR4_PREFETCH	BAR4_TYPE	BAR4_MEM_IO

Table 3-13 Fields for Register: BAR4_REG

Bits	Name	Memory Access	Description
31:4	BAR4_START	* Varies	BAR4 Base Address. ■ Memory Space: Base Address. ■ IO Space: bits[31:2] are used to map the function into IO space/Base. Address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R(Sticky)/W(Sticky) if enabled else R(Sticky) ■ Dbi: R(Sticky)/W(Sticky) if enabled else R(Sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	BAR4_PREFETCH	* Varies	BAR4 Prefetchable. ■ Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must be clear otherwise. ■ IO Space: Not applicable <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE!=1 && CX_DBI_RO_WR_EN && DEFAULT_BAR4_ENABLED \$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE!=1) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: PREFETCHABLE4_0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2:1	BAR4_TYPE	* Varies	BAR4 Type. - Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. - IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. <i>Note:</i> The access attributes of this field are as follows: - Wire: R (sticky) - Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE!=1 && CX_DBI_RO_WR_EN && DEFAULT_BAR4_ENABLED \$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R: R <i>Note:</i> This register field is sticky. Values: 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. 0x1 (RSVD_1): Reserved. 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. 0x3 (RSVD_2): Reserved. Value After Reset: BAR4_TYPE_0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR4_MEM_IO	* Varies	BAR4 Memory Space Indicator. This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. Base Address registers that map to I/O Space must return a 1b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE!=1 && CX_DBI_RO_WR_EN && DEFAULT_BAR4_ENABLED \$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE!=1) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE==1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: MEM4_SPACE_DECODER_0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.1.10 BAR5_REG

- **Name:** BAR5 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR5_MASK_REG.PCI_TYPE0_BAR5_ENABLED || BAR4_-TYPE_n ==2)

31:4	3	2:1	0
BAR5_START	BAR5_PREFETCH	BAR5_TYPE	BAR5_MEM_IO

Table 3-14 Fields for Register: BAR5_REG

Bits	Name	Memory Access	Description
31:4	BAR5_START	* Varies	BAR5 Base Address. ■ Memory Space: Base Address. ■ IO Space: bits[31:2] are used to map the function into IO space/Base. Address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R(Sticky)/W(Sticky) if enabled else R(Sticky) ■ Dbi: R(Sticky)/W(Sticky) if enabled else R(Sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	BAR5_PREFETCH	* Varies	BAR5 Prefetchable. ■ Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must be clear otherwise. ■ IO Space: Not applicable <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (BAR4_TYPE_0 == 2 && CX_MEMORY_MAP_VIEW != 1) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE != 1 && CX_DBI_RO_WR_EN && DEFAULT_BAR5_ENABLED_0) ? R/W : (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE != 1) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE == 1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: {(BAR4_TYPE_0 != 2) ? (PREFETCHABLE5_0) : 0} Exists: CX_MEMORY_MAP_PCIE_TYPE != 1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2:1	BAR5_TYPE	* Varies	BAR5 Type. - Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. - IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. <i>Note:</i> The access attributes of this field are as follows: - Wire: R (sticky) - Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (BAR4_TYPE_\$pf == 2 && CX_MEMORY_MAP_VIEW != 1) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE != 1 && CX_DBI_RO_WR_EN && DEFAULT_BAR5_ENABLED_\$pf) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE != 1) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE == 1) ? R: R <i>Note:</i> This register field is sticky. Values: 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. 0x1 (RSVD_1): Reserved. 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. 0x3 (RSVD_2): Reserved. Value After Reset: {(BAR4_TYPE_0 != 2) ? (BAR5_TYPE_0) : 0} Exists: CX_MEMORY_MAP_PCIE_TYPE != 1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR5_MEM_IO	* Varies	BAR5 Memory Space Indicator. This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. Base Address registers that map to I/O Space must return a 1b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (BAR_ENABLED == 1) then (if [DBI_RO_WR_EN == 1] then R(Sticky)/W(Sticky) else R(Sticky)) else RO(Sticky) and more specifically: (BAR4_TYPE_0pf == 2 && CX_MEMORY_MAP_VIEW != 1) ? R/W: (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE != 1 && CX_DBI_RO_WR_EN && DEFAULT_BAR5_ENABLED_0pf) ? R/W : (CX_MEMORY_MAP_VIEW == 0 && CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE != 1) ? R: (CX_MEMORY_MAP_VIEW == 1 && CX_MEMORY_MAP_PCIE_TYPE == 1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: {(BAR4_TYPE_0 != 2) ? (MEM5_SPACE_DECODER_0) : 0} Exists: CX_MEMORY_MAP_PCIE_TYPE != 1 Testable: writeAsRead Volatile: true

3.1.11 CARDBUS_CIS_PTR_REG

- **Name:** CardBus CIS Pointer Register.
- **Description:** This register holds the CardBus CIS pointer.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** Always

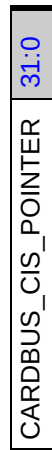


Table 3-15 Fields for Register: CARDBUS_CIS_PTR_REG

Bits	Name	Memory Access	Description
31:0	CARDBUS_CIS_POINTER	* Varies	CardBus CIS Pointer. Its functionality does not apply to PCI Express. It must be hardwired to 0000 0000h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CARDBUS_CIS_PTR_0 Exists: (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0) Testable: wire ? unconstrained : writeAsRead

3.1.12 SUBSYSTEM_ID_SUBSYSTEM_VENDOR_ID_REG

- **Name:** Subsystem ID and Subsystem Vendor ID Register.
- **Description:** These registers are used to uniquely identify the add-in card or subsystem where the PCI Express component resides. They provide a mechanism for vendors to distinguish their products from one another even though the assemblies may have the same PCI Express component on them (and, therefore, the same Vendor ID and Device ID).
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** Always



Table 3-16 Fields for Register: SUBSYSTEM_ID_SUBSYSTEM_VENDOR_ID_REG

Bits	Name	Memory Access	Description
31:16	SUBSYS_DEV_ID	* Varies	Subsystem ID. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: SUBSYS_DEV_ID_0 Exists: Always Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15:0	SUBSYS_VENDOR_ID	* Varies	Subsystem Vendor ID. Subsystem Vendor IDs can be obtained from the PCI SIG and are used to identify the vendor of the add-in card or subsystem. Values for the Subsystem ID are vendor-specific. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: SUBSYS_VENDOR_ID_0 Exists: Always Testable: wire ? unconstrained : writeAsRead

3.1.13 EXP_ROM_BASE_ADDR_REG

- **Name:** Expansion ROM BAR Register.
- **Description:** This register handles the base address and size information for this expansion ROM.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** Always

31:11	EXP_ROM_BASE_ADDRESS
10:8	RSVDP_8
7:4	ROM_BAR_VALIDATION_DETAILS
3:1	ROM_BAR_VALIDATION_STATUS
0	ROM_BAR_ENABLE

Table 3-17 Fields for Register: EXP_ROM_BASE_ADDR_REG

Bits	Name	Memory Access	Description
31:11	EXP_ROM_BASE_ADDRESS	* Varies	Expansion ROM Base Address. Upper 21 bits of the Expansion ROM base address. The number of bits (out of these 21) that a Function actually implements depends on how much address space the Function requires. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: if (EXP_ROM_BAR_MASK_REG.ROM_BAR_ENABLED == 1) then R/W else R ■ Dbi: if (EXP_ROM_BAR_MASK_REG.ROM_BAR_ENABLED == 1) then R/W else R Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
10:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1
7:4	ROM_BAR_VALIDATION_DETAILS	* Varies	Expansion ROM Validation Details. The field contains optional, implementation-specific details associated with Expansion ROM Validation. ■ If validation is in progress (Expansion ROM Validation Status is 001b), non-zero values of this field represent implementation-specific indications of the phase of the validation progress (for example, 50% complete). The value 0000b indicates that no validation progress information is provided. ■ If validation is completed (Expansion ROM Validation Status 010b to 111b inclusive), non-zero values in this field represent additional implementation-specific information. The value 0000b indicates that no information is provided. ■ When validation is supported and this field is not implemented, this field must be hardwired to 0000b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: if (EXP_ROM_BAR_MASK_REG.ROM_BAR_ENABLED == 1) then R/W else R ■ Dbi: if (EXP_ROM_BAR_MASK_REG.ROM_BAR_ENABLED == 1 && DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3:1	ROM_BAR_VALIDATION_STATUS	* Varies	Expansion ROM Validation Status. When this field is non-zero, it indicates the status of hardware validation of the Expansion ROM contents. ■ If the Function does not support validation, this field must be hardwired to 000b. ■ It is optional whether an implementation is capable of returning Validation Status values 011b, 101b, 110b, or 111b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: if (EXP_ROM_BAR_MASK_REG.ROM_BAR_ENABLED == 1) then R/W else R ■ Dbi: if (EXP_ROM_BAR_MASK_REG.ROM_BAR_ENABLED == 1 && DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ZERO): Validation not supported. ■ 0x1 (ONE): Validation in Progress. ■ 0x2 (TWO): Validation Pass Valid contents, trust test was not performed. ■ 0x3 (THREE): Validation Pass Valid and trusted contents. ■ 0x4 (FOUR): Validation Fail Invalid contents. ■ 0x5 (FIVE): Validation Fail Valid but untrusted contents (For example, Out of Date, Expired or Revoked Certificate). ■ 0x6 (SIX): Warning Pass Validation Passed with implementation-specific warning. Valid contents, trust test was not performed. ■ 0x7 (SEVEN): Warning Pass Validation Passed with implementation-specific warning. Valid and trusted contents. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	ROM_BAR_ENABLE	* Varies	Expansion ROM Enable. This bit controls whether or not the Function accepts accesses to its expansion ROM. The Memory Space Enable bit in the Command register has precedence over the Expansion ROM Enable bit. A Function must claim accesses to its expansion ROM only if both the Memory Space Enable bit and the Expansion ROM Enable bit are set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: if (EXP_ROM_BAR_MASK_REG.ROM_BAR_ENABLED == 1) then R/W else R ■ Dbi: if (EXP_ROM_BAR_MASK_REG.ROM_BAR_ENABLED == 1) then R/W else R Values: ■ 0x1 (ONE): When the bit is 1b, address decoding is enabled using the parameters in the other part of the Expansion ROM Base Address register. ■ 0x0 (ZERO): When this bit is 0b, the Function's expansion ROM address space is disabled. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.1.14 PCI_CAP_PTR_REG

- **Name:** Capabilities Pointer Register.
- **Description:** This register is used to point to a linked list of capabilities implemented by a Function.
- **Size:** 32 bits
- **Offset:** B+0x34
- **Exists:** Always

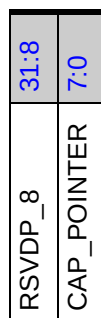


Table 3-18 Fields for Register: PCI_CAP_PTR_REG

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: Always
7:0	CAP_POINTER	* Varies	Capabilities Pointer. This register points to a valid capability structure. Either this structure is the PCI Express Capability structure, or a subsequent list item points to the PCI Express Capability structure. The bottom two bits are reserved, the controller sets it to 00b. Software must mask these bits off before using this register as a pointer in Configuration Space to the first entry of a linked list of new capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: Always Testable: wire ? unconstrained : writeAsRead

3.1.15 MAX_LATENCY_MIN_GRANT_INTERRUPT_PIN_INTERRUPT_LINE_REG

- **Name:** Max_Lat, Min_Gnt, Interrupt Pin, and Interrupt Line Register.
- **Description:** The Interrupt Line register communicates interrupt line routing information. The Interrupt Pin register identifies the legacy interrupt Message(s) the Function uses.
- **Size:** 32 bits
- **Offset:** B+0x3c
- **Exists:** Always

RSVDP_16	31:16
INT_PIN	15:8
INT_LINE	7:0

Table 3-19 Fields for Register: MAX_LATENCY_MIN_GRANT_INTERRUPT_PIN_INTERRUPT_LINE_REG

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: Always

Bits	Name	Memory Access	Description
15:8	INT_PIN	* Varies	Interrupt Pin. The Interrupt Pin register identifies the legacy interrupt Message(s) the Function uses. All encodings other than the defined encodings are reserved. PCI Express defines one legacy interrupt Message for a single Function device and up to four legacy interrupt Messages for a multi-Function device. For a single Function device, only INTA may be used. Any Function on a multi-Function device can use any of the INTx Messages. If a device implements a single legacy interrupt Message, it must be INTA; if it implements two legacy interrupt Messages, they must be INTA and INTB; and so forth. For a multi-Function device, all Functions may use the same INTx Message or each may have its own (up to a maximum of four Functions) or any combination thereof. A single Function can never generate an interrupt request on more than one INTx Message. Note: The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky. Values: ■ 0x1 (INTA): Map to legacy interrupt Messages for INTA ■ 0x2 (INTB): Map to legacy interrupt Messages for INTB ■ 0x3 (INTC): Map to legacy interrupt Messages for INTC ■ 0x4 (INTD): Map to legacy interrupt Messages for INTD ■ 0x0 (NO_INT): Indicates that the Function uses no legacy interrupt Message(s). Value After Reset: INT_PIN_MAPPING_0 Exists: Always Testable: wire ? unconstrained : writeAsRead
7:0	INT_LINE	R/W	Interrupt Line. The Interrupt Line register communicates interrupt line routing information. The register must be implemented by any Function that uses an interrupt pin. Values in this register are programmed by system software and are system architecture specific. The Function itself does not use this value; rather the value in this register is used by device drivers and operating systems. Value After Reset: 0xff Exists: Always

3.2 PF_TYPE0_HDR_DBI2i Registers

3.2.1 BAR0_MASK_REG

- **Name:** BAR\$ Mask Register.
- **Description:** This register is the mask for BAR\$_REG. If implemented, it exists as a shadow register at the BAR\$_REG address. Normally, the BAR masks are used for indicating the amount of address space that each BAR requests from host software. The BAR masks determine which bits in each BAR are non-writable by host software, which determines the size of the address space claimed by each BAR. The BAR mask values indicate the range of low-order bits, in each implemented BAR, not to use for address matching. The BAR mask value also indicates the range of low-order bits in the BAR that cannot be written from the host. The application can write to all BAR bits to allow setting of memory, I/O, and other standard BAR options.
Your local CPU can change the mask at runtime using the DBI. The mask register is invisible to the PCIe wire but visible to your local CPU through the DBI. You cannot read the mask register but you can write to it. It is accessed by asserting dbi_cs2 and dbi_cs. If you only assert dbi_cs then you will access the BAR which is the primary register at that location. Use CS2 instead of dbi_cs2 when you are using the AHB/AXI bridge.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (BAR0_ENABLED_n && BAR0_SIZING_SCHEME_n==1)

31:1	0
PCI_TYPE0_BAR0_MASK	PCI_TYPE0_BAR0_ENABLED

Table 3-20 Fields for Register: BAR0_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_TYPE0_BAR0_MASK	* Varies	- The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR0[31:1]. - The MEMBAR0 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. - This is 32bits field, we use this to write MEMBAR0 address into RCRB space. MEMBAR0 registers should be able to access BAR0_REG offset 10h of PCIE memory-mapped space. 0: BAR0 Mask. <i>Note:</i> The access attributes of this field are as follows: - Wire: No access - Dbi: No access - Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: BAR0_MASK_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"write-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"write-only\""
0	PCI_TYPE0_BAR0_ENABLED	W	- The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR0[0]. - The MEMBAR0 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. - This is 32bits field, we use this to write MEMBAR0 address into RCRB space. MEMBAR0 registers should be able to access BAR0_REG offset 10h of PCIE memory-mapped space. 0: BAR0 Mask Enabled. <i>Note:</i> This register field is sticky. Value After Reset: BAR0_ENABLED_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable

3.2.2 BAR1_MASK_REG

- **Name:** BAR1 Mask Register.
- **Description:** This register is the mask for BAR\$_REG. If implemented, it exists as a shadow register at the BAR\$_REG address. Normally, the BAR masks are used for indicating the amount of address space that each BAR requests from host software. The BAR masks determine which bits in each BAR are non-writable by host software, which determines the size of the address space claimed by each BAR. The BAR mask values indicate the range of low-order bits, in each implemented BAR, not to use for address matching. The BAR mask value also indicates the range of low-order bits in the BAR that cannot be written from the host. The application can write to all BAR bits to allow setting of memory, I/O, and other standard BAR options.
Your local CPU can change the mask at runtime using the DBI. The mask register is invisible to the PCIe wire but visible to your local CPU through the DBI. You cannot read the mask register but you can write to it. It is accessed by asserting dbi_cs2 and dbi_cs. If you only assert dbi_cs then you will access the BAR which is the primary register at that location. Use CS2 instead of dbi_cs2 when you are using the AHB/AXI bridge.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** ((BAR1_ENABLED_n && BAR1_TYPE_n==0 && BAR1_SIZING_SCHEME_n==1) || (BAR0_ENABLED_n && BAR0_TYPE_n==2 && BAR0_SIZING_SCHEME_n==1))

31:1	0
PCI_TYPE0_BAR1_MASK	PCI_TYPE0_BAR1_ENABLED

Table 3-21 Fields for Register: BAR1_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_TYPE0_BAR1_MASK	* Varies	The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR1[31:0]. ■ The MEMBAR1 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. ■ This is 32bits field, we use this to write MEMBAR1 address into RCRB space. MEMBAR1 registers should be able to access BAR1_REG offset 14h of PCIE memory-mapped space. 0: BAR1 Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: BAR1_MASK_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"write-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"write-only\""
0	PCI_TYPE0_BAR1_ENABLED	W	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR1[0]. ■ The MEMBAR1 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. ■ This is 32bits field, we use this to write MEMBAR1 address into RCRB space. MEMBAR1 registers should be able to access BAR1_REG offset 14h of PCIE memory-mapped space. 0: BAR1 Mask Enabled. <i>Note:</i> This register field is sticky. Value After Reset: BAR1_ENABLED_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable

3.2.3 BAR2_MASK_REG

- **Name:** BAR2 Mask Register.
- **Description:** This register is the mask for BAR2_REG. If implemented, it exists as a shadow register at the BAR\$_REG address. Normally, the BAR masks are used for indicating the amount of address space that each BAR requests from host software. The BAR masks determine which bits in each BAR are non-writable by host software, which determines the size of the address space claimed by each BAR. The BAR mask values indicate the range of low-order bits, in each implemented BAR, not to use for address matching. The BAR mask value also indicates the range of low-order bits in the BAR that cannot be written from the host. The application can write to all BAR bits to allow setting of memory, I/O, and other standard BAR options.
Your local CPU can change the mask at runtime using the DBI. The mask register is invisible to the PCIe wire but visible to your local CPU through the DBI. You cannot read the mask register but you can write to it. It is accessed by asserting dbi_cs2 and dbi_cs. If you only assert dbi_cs then you will access the BAR which is the primary register at that location. Use CS2 instead of dbi_cs2 when you are using the AHB/AXI bridge.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (BAR2_ENABLED_n && BAR2_SIZING_SCHEME_n==1)

31:1	0
PCI_TYPE0_BAR2_MASK	PCI_TYPE0_BAR2_ENABLED

Table 3-22 Fields for Register: BAR2_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_TYPE0_BAR2_MASK	* Varies	BAR2 Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: BAR2_MASK_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \\"write-only\\" : (CX_MEMORY_MAP_VIEW==1) ? \\"read-only\\" : \\"write-only\\""
0	PCI_TYPE0_BAR2_ENABLED	W	BAR2 Mask Enabled. <i>Note:</i> This register field is sticky. Value After Reset: BAR2_ENABLED_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable

3.2.4 BAR3_MASK_REG

- **Name:** BAR3 Mask Register.
- **Description:** This register is the mask for BAR3_REG. If implemented, it exists as a shadow register at the BAR\$_REG address. Normally, the BAR masks are used for indicating the amount of address space that each BAR requests from host software. The BAR masks determine which bits in each BAR are non-writable by host software, which determines the size of the address space claimed by each BAR. The BAR mask values indicate the range of low-order bits, in each implemented BAR, not to use for address matching. The BAR mask value also indicates the range of low-order bits in the BAR that cannot be written from the host. The application can write to all BAR bits to allow setting of memory, I/O, and other standard BAR options.
Your local CPU can change the mask at runtime using the DBI. The mask register is invisible to the PCIe wire but visible to your local CPU through the DBI. You cannot read the mask register but you can write to it. It is accessed by asserting dbi_cs2 and dbi_cs. If you only assert dbi_cs then you will access the BAR which is the primary register at that location. Use CS2 instead of dbi_cs2 when you are using the AHB/AXI bridge.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** ((BAR3_ENABLED_n && BAR3_TYPE_n==0 && BAR3_SIZING_SCHEME_n==1) || (BAR2_ENABLED_n && BAR2_TYPE_n==2 && BAR2_SIZING_SCHEME_n==1))

31:1	0
PCI_TYPE0_BAR3_MASK	PCI_TYPE0_BAR3_ENABLED

Table 3-23 Fields for Register: BAR3_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_TYPE0_BAR3_MASK	* Varies	BAR3 Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: BAR3_MASK_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \\"write-only\\" : (CX_MEMORY_MAP_VIEW==1) ? \\"read-only\\" : \\"write-only\\""
0	PCI_TYPE0_BAR3_ENABLED	W	BAR3 Mask Enabled. <i>Note:</i> This register field is sticky. Value After Reset: BAR3_ENABLED_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable

3.2.5 BAR4_MASK_REG

- **Name:** BAR4 Mask Register.
- **Description:** This register is the mask for BAR4_REG. If implemented, it exists as a shadow register at the BAR\$_REG address. Normally, the BAR masks are used for indicating the amount of address space that each BAR requests from host software. The BAR masks determine which bits in each BAR are non-writable by host software, which determines the size of the address space claimed by each BAR. The BAR mask values indicate the range of low-order bits, in each implemented BAR, not to use for address matching. The BAR mask value also indicates the range of low-order bits in the BAR that cannot be written from the host. The application can write to all BAR bits to allow setting of memory, I/O, and other standard BAR options.
Your local CPU can change the mask at runtime using the DBI. The mask register is invisible to the PCIe wire but visible to your local CPU through the DBI. You cannot read the mask register but you can write to it. It is accessed by asserting dbi_cs2 and dbi_cs. If you only assert dbi_cs then you will access the BAR which is the primary register at that location. Use CS2 instead of dbi_cs2 when you are using the AHB/AXI bridge.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (BAR4_ENABLED_n && BAR4_SIZING_SCHEME_n==1)

31:1	0
PCI_TYPE0_BAR4_MASK	PCI_TYPE0_BAR4_ENABLED

Table 3-24 Fields for Register: BAR4_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_TYPE0_BAR4_MASK	* Varies	BAR4 Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: BAR4_MASK_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \\"write-only\\" : (CX_MEMORY_MAP_VIEW==1) ? \\"read-only\\" : \\"write-only\\""
0	PCI_TYPE0_BAR4_ENABLED	W	BAR4 Mask Enabled. <i>Note:</i> This register field is sticky. Value After Reset: BAR4_ENABLED_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable

3.2.6 BAR5_MASK_REG

- **Name:** BAR5 Mask Register.
- **Description:** This register is the mask for BAR5_REG. If implemented, it exists as a shadow register at the BAR\$_REG address. Normally, the BAR masks are used for indicating the amount of address space that each BAR requests from host software. The BAR masks determine which bits in each BAR are non-writable by host software, which determines the size of the address space claimed by each BAR. The BAR mask values indicate the range of low-order bits, in each implemented BAR, not to use for address matching. The BAR mask value also indicates the range of low-order bits in the BAR that cannot be written from the host. The application can write to all BAR bits to allow setting of memory, I/O, and other standard BAR options.
Your local CPU can change the mask at runtime using the DBI. The mask register is invisible to the PCIe wire but visible to your local CPU through the DBI. You cannot read the mask register but you can write to it. It is accessed by asserting dbi_cs2 and dbi_cs. If you only assert dbi_cs then you will access the BAR which is the primary register at that location. Use CS2 instead of dbi_cs2 when you are using the AHB/AXI bridge.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** ((BAR5_ENABLED_n && BAR5_TYPE_n==0 && BAR5_SIZING_SCHEME_n==1) || (BAR4_ENABLED_n && BAR4_TYPE_n==2 && BAR4_SIZING_SCHEME_n==1))

31:1	0
PCI_TYPE0_BAR5_MASK	PCI_TYPE0_BAR5_ENABLED

Table 3-25 Fields for Register: BAR5_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_TYPE0_BAR5_MASK	* Varies	BAR5 Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: BAR5_MASK_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \\"write-only\\" : (CX_MEMORY_MAP_VIEW==1) ? \\"read-only\\" : \\"write-only\\""
0	PCI_TYPE0_BAR5_ENABLED	W	BAR5 Mask Enabled. <i>Note:</i> This register field is sticky. Value After Reset: BAR5_ENABLED_n Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable

3.2.7 EXP_ROM_BAR_MASK_REG

- **Name:** Expansion ROM BAR Mask Register.
- **Description:** This register is the mask for EXP_ROM_BASE_ADDR_REG register. If implemented, it exists as a shadow register at EXP_ROM_BAR_MASK_REG address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to this register.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** Always

31:1	ROM_MASK
0	ROM_BAR_ENABLED

Table 3-26 Fields for Register: EXP_ROM_BAR_MASK_REG

Bits	Name	Memory Access	Description
31:1	ROM_MASK	W	Expansion ROM Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if ROM_BAR_ENABLED && ROM_MASK_WRITABLE then W <i>Note:</i> This register field is sticky. Value After Reset: 0xffff Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable

Bits	Name	Memory Access	Description
0	ROM_BAR_ENABLED	W	Expansion ROM Bar Mask Register Enabled. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if ROM_MASK_WRITABLE then W <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_ROM_BAR_ENABLED_0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable

3.3 PF_PM_CAPI Registers

3.3.1 CAP_ID_NXT_PTR_REG

- **Name:** Power Management Capabilities Register.
- **Description:** This register provides information regarding the Power Management Capabilities.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_MEMORY_MAP_PCIE_TYPE<=2)

PME_SUPPORT	31:27
D2_SUPPORT	26
D1_SUPPORT	25
AUX_CURR	24:22
DSI	21
PME_IMM_READI_RETURN_DO	20
PME_CLK	19
PM_SPEC_VER	18:16
PM_NEXT_POINTER	15:8
PM_CAP_ID	7:0

Table 3-27 Fields for Register: CAP_ID_NXT_PTR_REG

Bits	Name	Memory Access	Description
31:27	PME_SUPPORT	* Varies	PME_Support. This 5-bit field indicates the power states in which the function may generate a PME and/or forward PME messages. A value of 0b for any bit indicates that the function is not capable of asserting PME while in that power state. ■ bit(27) X XXX1b - PME can be generated from D0 ■ bit(28) X XX1Xb - PME can be generated from D1 ■ bit(29) X X1XXb - PME can be generated from D2 ■ bit(30) X 1XXXb - PME can be generated from D3hot ■ bit(31) 1 XXXXb - PME can be generated from D3cold Bit 31 (PME can be asserted from D3cold) represents a special case. Functions that set this bit require some sort of auxiliary power source. Implementation specific mechanisms are recommended to validate that the power source is available before setting this bit. The read value from this field is the write value && (sys_aux_pwr_det, 1'b1, D2_SUPPORT, D1_SUPPORT, 1'b1), where D1_SUPPORT and D2_SUPPORT are fields in this register. The reset value PME_SUPPORT_n && (sys_aux_pwr_det, 1'b1, D2_SUPPORT, D1_SUPPORT, 1'b1), where PME_SUPPORT_n is a configuration parameter. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
26	D2_SUPPORT	* Varies	D2_Support. If this bit is set, this function supports the D2 Power Management state. Functions that do not support D2 must always return a value of 0b for this bit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: D2_SUPPORT_0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
25	D1_SUPPORT	* Varies	D1_Support. If this bit is set, this function supports the D1 Power Management state. Functions that do not support D1 must always return a value of 0b for this bit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: D1_SUPPORT_0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: wire ? unconstrained : writeAsRead
24:22	AUX_CURR	* Varies	Aux_Current. This 3 bit field reports the Vaux auxiliary current requirements for the function. If this function implements the Data Register, the controller hardwires this field to 000b. If PME_Support is 0 xxxb (PME assertion from D3cold is not supported), the controller hardwires this field to 0000b. For functions where PME_Support is 1 xxxb (PME assertion from D3cold is supported), and which do not implement the Data field, the encodings defined in Values: apply: <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Values: ■ 0x7 (_370mA_): 375mA Vaux Max. Current Required ■ 0x6 (_320mA_): 320mA Vaux Max. Current Required ■ 0x5 (_270mA_): 270mA Vaux Max. Current Required ■ 0x4 (_220mA_): 220mA Vaux Max. Current Required ■ 0x3 (_160mA_): 160mA Vaux Max. Current Required ■ 0x2 (_100mA_): 100mA Vaux Max. Current Required ■ 0x1 (_055mA_): 55mA Vaux Max. Current Required ■ 0x0 (SELF_): 0 self powered Value After Reset: AUX_CURRENT_0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
21	DSI	* Varies	Device Specific Initialization. The DSI bit indicates whether special initialization of this function is required. When set, indicates that the function requires a device specific initialization sequence following a transition to the D0uninitialized state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: DEV_SPEC_INIT_0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: wire ? unconstrained : writeAsRead
20	PME_IMM_READI_RETURN_DO	* Varies	Immediate_Readiness_on_Return_to_D0. If this bit is a '1', this function is guaranteed to be ready to successfully complete valid accesses immediately after being set to D0. These accesses include configuration cycles, and if the function returns to D0initialized, they also include Memory and I/O Cycles. When this bit is '1', for accesses to this function, software is exempt from all requirements to delay accesses following a transition to D0, including but not limited to the 10ms delay; the delays described in Section 5.9 of <i>PCI Express Base Specification</i>. How this guarantee is established is beyond the scope of this document. It is permitted that your system software/firmware provides mechanisms that supersede the indication provided by this bit, however such software/firmware mechanisms are outside the scope of this document. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: CX_RN_IMM_D0_VALUE_0 Exists: CX_RN_IMM_EN && (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: wire ? unconstrained : writeAsRead
19	PME_CLK	R	PME Clock. Does not apply to PCI Express, the controller hardwires it to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2)

Bits	Name	Memory Access	Description
18:16	PM_SPEC_VER	* Varies	Version. This field provides the Power Management specification version. The controller hardwires this field to 011b for functions compliant to <i>PCI Express Base Specification, Revision 4.0, Version 1.0</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: 0x3 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: wire ? unconstrained : writeAsRead
15:8	PM_NEXT_POINTER	* Varies	Next Capability Pointer. This field provides an offset into the function's configuration space pointing to the location of next item in the capabilities list. If there are no additional items in the capabilities list, this field is set to 00h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: wire ? unconstrained : writeAsRead
7:0	PM_CAP_ID	R	Capability ID. This field returns 01h to indicate that this is the PCI Power Management Capability. Each function may have only one item in its capability list with Capability ID set to 01h. Value After Reset: 0x1 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2)

3.3.2 CON_STATUS_REG

- **Name:** Power Management Control and Status Register.
- **Description:** This register is used to manage the PCI function's power management state as well as to enable/monitor PMEs.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_MEMORY_MAP_PCIE_TYPE<=2)

31:24	DATA_REG_ADD_INFO
23	BUS_PWR_CLK_CON_EN
22	B2_B3_SUPPORT
21:16	RSVDP_16
15	PME_STATUS
14:13	DATA_SCALE
12:9	DATA_SELECT
8	PME_ENABLE
7:4	RSVDP_4
3	NO_SOFT_RST
2	RSVDP_2
1:0	POWER_STATE

Table 3-28 Fields for Register: CON_STATUS_REG

Bits	Name	Memory Access	Description
31:24	DATA_REG_ADD_INFO	R	Data. This field is used to report the state dependent data requested by the Data_Select field. The value of this field is scaled by the value reported by the Data_Scale field. Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2)
23	BUS_PWR_CLK_CON_EN	R	Bus Power/Clock Control Enable. If this field is set, Bus Power/Clock Control is Enable. Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2)
22	B2_B3_SUPPORT	R	B2B3 Support for D3hot. If this field is set, B2B3 support for D3hot is available. Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2)
21:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2)

Bits	Name	Memory Access	Description
15	PME_STATUS	R/W1C	PME_Status. This bit is set when the function normally generates a PME signal. The value of this bit is not affected by the value of the PME_En bit. If PME_Support bit 31 of the Power Management Capabilities register is clear, this bit is permitted to be hardwired to 0b. Functions that consume Aux power must preserve the value of this sticky register when Aux power is available. In such functions, this register value is not modified by Conventional Reset or FLR. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: writeAsRead Volatile: true
14:13	DATA_SCALE	R	Data_Scale. This field indicates the scaling factor to be used when interpreting the value of the Data field. The value and meaning of this field varies depending on which data value has been selected by the Data_Select field. For more information, see 7.5.2.3 section of <i>PCI Express Base Specification</i>. Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2)
12:9	DATA_SELECT	R	Data_Select. This 4-bit field is used to select which data is to be reported through the Data and Data_Scale field. If the Data field is not implemented, this field must be hardwired to 0000b. Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2)
8	PME_ENABLE	R/W	PME_En. ■ When set, the function is permitted to generate a PME. ■ When clear, the function is not permitted to generate a PME. If PME_Support is 1 xxxx (PME generation from D3cold) or the function consumes Aux power and Aux power is available this bit is RWS and the bit is not modified by Conventional Reset or FLR. If PME_Support is 0 xxxx, this field is not sticky (RW). If PME_Support is 0 0000b, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
7:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2)
3	NO_SOFT_RST	* Varies	No_Soft_Reset. This bit indicates the state of the function after writing the PowerState field to transition the function from D3hot to D0. ■ When set, this transition preserves internal function state. The function is in D0Active and no additional software intervention is required. ■ When clear, this transition results in undefined internal function state. Regardless of this bit, functions that transition from D3hot to D0 by Fundamental Reset will return to D0Uninitialized with only PME context preserved if PME is supported and enabled. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_NO_SOFT_RESET_0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: wire ? unconstrained : writeAsRead
2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2)

Bits	Name	Memory Access	Description
1:0	POWER_STATE	* Varies	PowerState. This 2-bit field is used both to determine the current power state of a function and to set the function into a new power state. You can write to this register; however, the read-back value is the actual power state, not the write value. If you attempt to write an unsupported, optional state to this field, the write operation completes normally; however, the data is discarded and no state change occurs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (D0): D0 power state ■ 0x1 (D1): D1 power state ■ 0x2 (D2): D2 power state ■ 0x3 (D3hot): D3hot D3hot power state Value After Reset: * Varies Exists: (CX_MEMORY_MAP_PCIE_TYPE<=2) Testable: writeAsRead Reset Mask: 0x0 Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""

3.4 PF_MSI_CAPi Registers

3.4.1 PCI_MSI_CAP_ID_NEXT_CTRL_REG

- **Name:** MSI Capability Header and Message Control Register.
- **Description:** This register holds MSI Capability Header information and controls the MSI behaviour.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:27	RSVDP_27
26	PCI_MSI_EXT_DATA_EN
25	PCI_MSI_EXT_DATA_CAP
24	PCI_PVM_SUPPORT
23	PCI_MSI_64_BIT_ADDR_CAP
22:20	PCI_MSI_MULTIPLE_MSG_EN
19:17	PCI_MSI_MULTIPLE_MSG_CAP
16	PCI_MSI_ENABLE
15:8	PCI_MSI_CAP_NEXT_OFFSET
7:0	PCI_MSI_CAP_ID

Table 3-29 Fields for Register: PCI_MSI_CAP_ID_NEXT_CTRL_REG

Bits	Name	Memory Access	Description
31:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
26	PCI_MSI_EXT_DATA_EN	* Varies	Extended Message Data Enable. ■ If set, the function is enabled to provide Extended Message Data. ■ If clear, the function is not enabled to provide Extended Message Data. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PCI_MSI_CAP_ID_NEXT_CTRL_REG.PCI_MSI_EXT_DATA_CAP ? RW : RO ■ Dbi: PCI_MSI_CAP_ID_NEXT_CTRL_REG.PCI_MSI_EXT_DATA_CAP ? RW : RO Value After Reset: 0x0 Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
25	PCI_MSI_EXT_DATA_CAP	* Varies	Extended Message Data Capable. ■ If set, the function is capable of providing Extended Message Data. ■ If clear, the function does not support providing Extended Message Data. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_EXT_MSI_DATA_CAPABLE Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
24	PCI_PVM_SUPPORT	R	Per-Vector Masking Capable. ■ If set, the function supports MSI Per-Vector Masking. ■ If clear, the function does not support MSI Per-Vector Masking. This bit must be set if the function is a PF or VF within an SR-IOV Device. Value After Reset: MSI_PVM_EN_VALUE Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
23	PCI_MSI_64_BIT_ADDR_CAP	* Varies	64 bit address capable. This bit must be set if the function is a PCI Express Endpoint. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (set_0): If set, the function is capable of sending a 64-bit message address. ■ 0x1 (set_1): If clear, the function is not capable of sending a 64-bit message address. Value After Reset: MSI_64_EN Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
22:20	PCI_MSI_MULTIPLE_MSG_EN	* Varies	Multiple Message Enable. Software writes to this field to indicate the number of allocated vectors (equal to or less than the number of requested vectors). The number of allocated vectors is aligned to a power of two. If a function requests four vectors (indicated by a Multiple Message Capable encoding of 010b), system software can allocate either four, two, or one vector by writing a 010b, 001b, or 000b to this field, respectively. When MSI is enabled, a function will be allocated at least 1 vector. All encodings other than the defined encodings are reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (_1_VECTOR): 1 vector allocated ■ 0x1 (_2_VECTOR): 2 vectors allocated ■ 0x2 (_4_VECTOR): 4 vectors allocated ■ 0x3 (_8_VECTOR): 8 vectors allocated ■ 0x4 (_16_VECTOR): 16 vectors allocated ■ 0x5 (_32_VECTOR): 32 vectors allocated Value After Reset: 0x0 Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && MSI_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !MSI_CAP_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && MSI_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !MSI_CAP_ENABLE) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19:17	PCI_MSI_MULTIPLE_MSG_CAP	* Varies	Multiple Message Capable. System software reads this field to determine the number of requested vectors. The number of requested vectors must be aligned to a power of two (if a function requires three vectors, it requests four by initializing this field to 010b). All encodings other than the defined encodings are reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_1_VECTOR): 1 vector requested ■ 0x1 (_2_VECTOR): 2 vectors requested ■ 0x2 (_4_VECTOR): 4 vectors requested ■ 0x3 (_8_VECTOR): 8 vectors requested ■ 0x4 (_16_VECTOR): 16 vectors requested ■ 0x5 (_32_VECTOR): 32 vectors requested Value After Reset: DEFAULT_MULTI_MSI_CAPABLE_0 Exists: MSI_CAP_ENABLE && (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
16	PCI_MSI_ENABLE	* Varies	MSI Enable. ■ If set and the MSI-X Enable bit in the MSI-X Message Control register is clear, the function is permitted to use MSI to request service and is prohibited from using INTx interrupts. System configuration software sets this bit to enable MSI. A device driver is prohibited from writing this bit to mask a function's service request. For more information on control of INTx interrupts, see section 7.5.1.1 of <i>PCI Express Base Specification</i>. ■ If clear, the function is prohibited from using MSI to request service. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && MSI_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !MSI_CAP_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && MSI_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !MSI_CAP_ENABLE) ? \"read-only\" : \"read-only\""
15:8	PCI_MSI_CAP_NEXT_OFFSET	* Varies	Next Capability Pointer. This field contains the offset to the next PCI Capability structure or 00h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
7:0	PCI_MSI_CAP_ID	R	Capability ID. Indicates the MSI Capability structure. This field returns a Capability ID of 05h indicating that this is an MSI Capability structure. Value After Reset: 0x5 Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.4.2 MSI_CAP_OFF_04H_REG

- **Name:** Message Address Register for MSI (Offset 04h).
- **Description:** This register holds the system specified message address for an MSI transaction.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 | | CX_MEMORY_MAP_PCIE_TYPE==0))

31:2	
PCI_MSI_CAP_OFF_04H	
1:0	
RSVDP_0	

Table 3-30 Fields for Register: MSI_CAP_OFF_04H_REG

Bits	Name	Memory Access	Description
31:2	PCI_MSI_CAP_OFF_04H	* Varies	Message Address - System-specified message address. If the Message Enable bit (bit 16 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG register) is set, the contents of this field specify the DWORD-aligned address (Address[31:02]) for the MSI transaction. Address[1:0] are set to 00b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && MSI_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !MSI_CAP_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && MSI_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !MSI_CAP_ENABLE) ? \"read-only\" : (MSI_CAP_ENABLE) ? \"read-write\" : \"read-only\" "
1:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.4.3 MSI_CAP_OFF_08H_REG

- **Name:** Message Address Register for MSI (Offset 08h).
- **Description:** For a function that supports a 32-bit message address,
 - bits[31:16] of this register represent the Extended Message Data, and
 - bits[15:0] of this register represent the Message Data

For a function that supports a 64-bit message address (bit 23 in PCI_MSI_CAP_ID_NEXT_CTRL_REG register set), this register represents the Message Upper Address Register for MSI (Offset 08h). It specifies the Message Upper Address (System-specified message upper address). This register is required for PCI Express Endpoints and is optional for other function types. If the Message Enable bit (bit 0 of the Message Control register) is set, the contents of this register (if non-zero) specify the upper 32-bits of a 64-bit message address (Address[63:32]). If the contents of this register are zero, the Function uses the 32 bit address specified by the Message Address register.

- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	PCI_MSI_CAP_OFF_0AH
15:0	PCI_MSI_CAP_OFF_08H

Table 3-31 Fields for Register: MSI_CAP_OFF_08H_REG

Bits	Name	Memory Access	Description
31:16	PCI_MSI_CAP_OFF_0AH	* Varies	For a function that supports a 32-bit message address, this field contains Extended Message Data (System-specified message data). For the MSI Capability structures without per-vector masking, it must be implemented if the Extended Message Data Capable bit is set; otherwise, it is outside the MSI Capability structure and undefined. For the MSI Capability structures with Per-vector Masking, it must be implemented if the Extended Message Data Capable bit is set; otherwise, it is RsvdP. If the Extended Message Data Enable bit (bit 26 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set, the DWORD Memory Write transaction uses Extended Message Data for the upper 16 bits; otherwise, it uses 0000h for the upper 16 bits. For a function that supports a 64-bit message address, it contains upper 16 bits of the Message Upper Address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PCI_MSI_64_BIT_ADDR_CAP `DEFAULT_EXT_MSI_DATA_CAPABLE ? R/W : R</li> <li>■ Dbi: PCI_MSI_64_BIT_ADDR_CAP    `DEFAULT_EXT_MSI_DATA_CAPABLE ? R/W : R and more specifically: (CX_MEMORY_MAP_VIEW==0 &&(MSI_64_EN DEFAULT_EXT_MSI_DATA_CAPABLE)) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R : (CX_MEMORY_MAP_VIEW==1 && (MSI_64_EN DEFAULT_EXT_MSI_DATA_CAPABLE)) ? R/W : (CX_MEMORY_MAP_VIEW==1) ? R : (MSI_64_EN DEFAULT_EXT_MSI_DATA_CAPABLE) ? R/W : R Value After Reset: 0x0 Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
15:0	PCI_MSI_CAP_OFF_08H	* Varies	For a function that supports a 32-bit message address, this field contains Message Data (System-specified message data). If the Message Enable bit (bit 16 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set, the function sends a DWORD Memory Write transaction using Message Data for the lower 16 bits. All 4 Byte Enables are set. The Multiple Message Enable field (bits 22:20 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG) defines the number of low order message data bits the function is permitted to modify to generate its system software allocated vectors. For example, a Multiple Message Enable encoding of 010b indicates the function has been allocated four vectors and is permitted to modify message data bits 1 and 0 (a function modifies the lower message data bits to generate the allocated number of vectors). If the Multiple Message Enable field is 000b, the Function is not permitted to modify the message data. For a function that supports a 64-bit message address, it contains lower 16 bits of the Message Upper Address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && MSI_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !MSI_CAP_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && MSI_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !MSI_CAP_ENABLE) ? \"read-only\" : (MSI_CAP_ENABLE) ? \"read-write\" : \"read-only\" "

3.4.4 MSI_CAP_OFF_0CH_REG

- **Name:** Message Address Register for MSI (Offset 0Ch).
- **Description:** For a function that supports a 32-bit message address, this register contains the Mask Bits when the Per-Vector Masking Capable bit (bit 24 of PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set. For a function that supports a 64-bit message address, this register contains Message Data.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	PCI_MSI_CAP_OFF_0EH
15:0	PCI_MSI_CAP_OFF_0CH

Table 3-32 Fields for Register: MSI_CAP_OFF_0CH_REG

Bits	Name	Memory Access	Description
31:16	PCI_MSI_CAP_OFF_0EH	* Varies	For a function that supports a 32-bit message address, this field contains the upper Mask Bits when the Per Vector Masking Capable bit (PCI_MSI_CAP_ID_NEXT_CTRL_REG.PCI_PVM_SUPPORT) is set. For a function that supports a 64-bit message address, this field contains Message Data (System-specified message data). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: (!MSI_64_EN && MSI_PVM_EN_VALUE) ? RW: MSI_64_EN && DEFAULT_EXT_MSI_DATA_CAPABLE ? RW : RO ■ Dbi: (!MSI_64_EN && MSI_PVM_EN_VALUE) ? RW: MSI_64_EN && DEFAULT_EXT_MSI_DATA_CAPABLE ? RW : RO and more specifically: (CX_MEMORY_MAP_VIEW==0 && MSI_64_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1 && MSI_64_EN) ? R/W : (CX_MEMORY_MAP_VIEW==1) ? R: (MSI_64_EN) ? R/W : R Value After Reset: 0x0 Exists: (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
15:0	PCI_MSI_CAP_OFF_0CH	* Varies	For a function that supports a 32-bit message address, this field contains the lower Mask Bits when the Per Vector Masking Capable bit (PCI_MSI_CAP_ID_NEXT_CTRL_REG.PCI_PVM_SUPPORT) is set. For a function that supports a 64-bit message address, this field contains Message Data (System-specified message data). If the Message Enable bit (bit 16 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set, the function sends a DWORD Memory Write transaction using Message Data for the lower 16 bits. All 4 Byte Enables are set. The Multiple Message Enable field (bits 22:20 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG) defines the number of low order message data bits the function is permitted to modify to generate its system software allocated vectors. For example, a Multiple Message Enable encoding of 010b indicates the function has been allocated four vectors and is permitted to modify message data bits 1 and 0 (a function modifies the lower message data bits to generate the allocated number of vectors). If the Multiple Message Enable field is 000b, the Function is not permitted to modify the message data. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PCI_MSI_64_BIT_ADDR_CAP MSI_PVM_EN ? R/W : R ■ Dbi: PCI_MSI_64_BIT_ADDR_CAP MSI_PVM_EN ? R/W : R and more specifically: (CX_MEMORY_MAP_VIEW==0 && (MSI_64_EN MSI_PVM_EN)) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !MSI_CAP_ENABLE) ? R: (CX_MEMORY_MAP_VIEW==1 && (MSI_64_EN MSI_PVM_EN)) ? R/W : (CX_MEMORY_MAP_VIEW==1 && !MSI_CAP_ENABLE) ? R: (MSI_64_EN MSI_PVM_EN) ? R/W : R Value After Reset: 0x0 Exists: (MSI_64_EN MSI_PVM_EN) && (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.4.5 MSI_CAP_OFF_10H_REG

- **Name:** Message Address Register for MSI (Offset 10h).
- **Description:** For a function that supports a 32-bit message address, this register contains the Pending Bits when the Per-Vector Masking Capable bit (bit 24 of PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set. For a function that supports a 64-bit message address, this register contains the Mask Bits when the Per-Vector Masking Capable bit (bit 24 of PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 | | CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-33 Fields for Register: MSI_CAP_OFF_10H_REG

Bits	Name	Memory Access	Description
31:0	PCI_MSI_CAP_OFF_10H	* Varies	Used for MSI when the Per Vector Masking Capable bit (PCI_MSI_CAP_ID_NEXT_CTRL_REG.PCI_PVM_SUPPORT) is set. For 32-bit contains Pending Bits. For 64-bit, contains Mask Bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PCI_MSI_64_BIT_ADDR_CAP && MSI_PVM_EN ? R/W : R ■ Dbi: PCI_MSI_64_BIT_ADDR_CAP && MSI_PVM_EN ? R/W : R and more specifically: (CX_MEMORY_MAP_VIEW==0 && MSI_CAP_ENABLE && MSI_PVM_EN_VALUE && MSI_64_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !MSI_64_EN) ? R: (CX_MEMORY_MAP_VIEW==1 && MSI_64_EN && MSI_PVM_EN_VALUE && MSI_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==1 && !MSI_64_EN) ? R: (MSI_64_EN && MSI_PVM_EN_VALUE && MSI_CAP_ENABLE) ? R/W : R Value After Reset: 0x0 Exists: MSI_PVM_EN && (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.4.6 MSI_CAP_OFF_14H_REG

- **Name:** Message Address Register for MSI (Offset 14h).
- **Description:** Pending Bits Register for MSI. This register is used for a function that supports a 64-bit message address when the Per-Vector Masking Capable bit (bit 24 of PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-34 Fields for Register: MSI_CAP_OFF_14H_REG

Bits	Name	Memory Access	Description
31:0	PCI_MSI_CAP_OFF_14H	R	Pending Bits. For each pending bit that is set, the function has a pending associated message. Value After Reset: 0x0 Exists: MSI_64_EN && MSI_PVM_EN && (MSI_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.5 PF_PCIE_CAPI Registers

3.5.1 PCIE_CAP_ID_PCIE_NEXT_CAP_PTR_PCIE_CAP_REG

- **Name:** PCI Express Capabilities, ID, Next Pointer Register.
- **Description:** This is the PCI Express Capabilities, ID, and Next Pointer Register.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** Always

RSVDP_31	31
RSVD	30
PCIE_INT_MSG_NUM	29:25
PCIE_SLOT_IMP	24
PCIE_DEV_PORT_TYPE	23:20
PCIE_CAP_REG	19:16
PCIE_CAP_NEXT_PTR	15:8
PCIE_CAP_ID	7:0

Table 3-35 Fields for Register: PCIE_CAP_ID_PCIE_NEXT_CAP_PTR_PCIE_CAP_REG

Bits	Name	Memory Access	Description
31	RSVDP_31	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: Always
30	RSVD	R	Reserved. Value After Reset: 0x0 Exists: Always

Bits	Name	Memory Access	Description
29:25	PCIE_INT_MSG_NUM	* Varies	PCIE Interrupt Message Number. Interrupt Message Number. This field indicates which MSI/MSI-X vector is used for the interrupt message generated in association with any of the status bits of this Capability structure. For MSI, the value in this field indicates the offset between the base Message Data and the interrupt message that is generated. Hardware is required to update this field so that it is correct if the number of MSI Messages assigned to the Function changes when software writes to the Multiple Message Enable field in the MSI Message Control register. For MSI-X, the value in this field indicates which MSI-X Table entry is used to generate the interrupt message. The entry must be one of the first 32 entries even if the Function implements more than 32 entries. For a given MSI-X implementation, the entry must remain constant. If both MSI and MSI-X are implemented, they are permitted to use different vectors, though software is permitted to enable only one mechanism at a time. If MSI-X is enabled, the value in this field must indicate the vector for MSI-X. If MSI is enabled or neither is enabled, the value in this field must indicate the vector for MSI. If software enables both MSI and MSI-X at the same time, the value in this field is undefined. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : PCIE_CAP_INT_MSG_NUM_0} Exists: Always Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
24	PCIE_SLOT_IMP	* Varies	Slot Implemented. When set, this bit indicates that the Link associated with this Port is connected to a slot (as compared to being connected to a system-integrated device or being disabled). This bit is valid for Downstream Ports. This bit is undefined for Upstream Ports. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : SLOT_IMPLEMENTED_0} Exists: Always Testable: wire ? unconstrained : writeAsRead
23:20	PCIE_DEV_PORT_TYPE	R	Device/Port Type. Indicates the specific type of this PCI Express function. <i>Note:</i> Different functions in a Multi-Function Device can generally be of different types. Defined encodings for functions that implement a Type 00h PCI Configuration Space header are: Defined encodings for functions that implement a Type 01h PCI Configuration Space header are: All other encodings are Reserved. <i>Note:</i> Different Endpoint types have notably different requirements in Section 1.3.2 of <i>PCI Express Base Specification</i> regarding I/O resources, Extended Configuration Space, and other capabilities. Values: ■ 0x0 (PCIE_EP): PCI Express Endpoint ■ 0x1 (PCIE_LEGACY_EP): Legacy PCI Express Endpoint ■ 0x4 (ROOT_PORT_PCIE_RC): Root Port of PCI Express Root Complex ■ 0x5 (USP_PCIE_SWITCH): Upstream Port of PCI Express Switch ■ 0x6 (DSP_PCIE_SWITCH): Downstream Port of PCI Express Switch Value After Reset: * Varies Exists: Always Testable: untestable Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
19:16	PCIE_CAP_REG	R	Capability Version. Indicates PCI-SIG defined PCI Express Capability structure version number. A version of the specification that changes the PCI Express Capability structure in a way that is not otherwise identifiable (for example, through a new Capability field) is permitted to increment this field. All such changes to the PCI Express Capability structure must be software-compatible. Software must check for Capability Version numbers that are greater than or equal to the highest number defined when the software is written, as functions reporting any such Capability Version numbers will contain a PCI Express Capability structure that is compatible with that piece of software. The controller hardwires this field to 2h for functions compliant to <i>PCI Express Base Specification, Revision 4.0, Version 1.0</i>. <i>Note:</i> This register field is sticky. Value After Reset: 0x2 Exists: Always
15:8	PCIE_CAP_NEXT_PTR	* Varies	Next Capability Pointer. This field contains the offset to the next PCI Capability structure or 00h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: Always Testable: wire ? unconstrained : writeAsRead
7:0	PCIE_CAP_ID	R	Capability ID. Indicates the PCI Express Capability structure. This field must return a Capability ID of 10h indicating that this is a PCI Express Capability structure. Value After Reset: 0x10 Exists: Always

3.5.2 DEVICE_CAPABILITIES_REG

- **Name:** Device Capabilities Register.
- **Description:** The Device Capabilities register identifies PCI Express device function specific capabilities.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** Always

RSVDP_29	31:29
PCIE_CAP_FLR_CAP	28
PCIE_CAP_CAP_SLOT_PWR_LMT_SCALE	27:26
PCIE_CAP_CAP_SLOT_PWR_LMT_VALUE	25:18
RSVDP_16	17:16
PCIE_CAP_ROLE_BASED_ERR_REPORT	15
RSVDP_12	14:12
PCIE_CAP_EP_L1_ACCPT_LATENCY	11:9
PCIE_CAP_EP_L0S_ACCPT_LATENCY	8:6
PCIE_CAP_EXT_TAG_SUPP	5
PCIE_CAP_PHANTOM_FUNC_SUPPORT	4:3
PCIE_CAP_MAX_PAYLOAD_SIZE	2:0

Table 3-36 Fields for Register: DEVICE_CAPABILITIES_REG

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: Always

Bits	Name	Memory Access	Description
28	PCIE_CAP_FLR_CAP	* Varies	Function Level Reset Capability. A value of 1b indicates the function supports the optional Function Level Reset mechanism described in section 6.6.2 of the <i>PCI Express Base Specification</i>. This bit applies to Endpoints only. For all other function types the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_PCIE_TYPE == 1 && 1) ? R : (CX_MEMORY_MAP_VIEW == 0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW == 0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW == 1) ? R : R <i>Note:</i> This register field is sticky. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE == 1) ? 0 : CX_FLR_ENABLE_VALUE} Exists: Always Testable: wire ? unconstrained : writeAsRead
27:26	PCIE_CAP_CAP_SLOT_PWR_LMT_SCALE	R	Captured Slot Power Limit Scale. Captured Slot Power Limit Scale (Upstream Ports only). Specifies the scale used for the Slot Power Limit Value. This value is set by the Set_Slot_Power_Limit Message or hardwired to 00b (for more details, see section 6.9 of <i>PCI Express Base Specification</i>). Values: ■ 0x0 (_1.0X): 1.0x ■ 0x1 (_0.1X): 0.1x ■ 0x2 (_0.01X): 0.01x ■ 0x3 (_0.001X): 0.001x Value After Reset: * Varies Exists: Always Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
25:18	PCIE_CAP_CAP_SLOT_PWR_LIMIT_VALUE	R	Captured Slot Power Limit Value. Captured Slot Power Limit Value (Upstream Ports only). In combination with the Captured Slot Power Limit Scale value, specifies the upper limit on power available to the adapter. Power limit (in Watts) is calculated by multiplying the value in this field by the value in the Captured Slot Power Limit Scale field except when the Captured Slot Power Limit Scale field equals 00b (1.0x) and the Captured Slot Power Limit Value exceeds EFh, then alternative encodings are used (for more details, see section 7.5.3.9 of <i>PCI Express Base Specification</i>). This value is set by the Set_Slot_Power_Limit Message or hardwired to 00h (for more details, see section 6.9 of <i>PCI Express Base Specification</i>). Value After Reset: * Varies Exists: Always Testable: writeAsRead Reset Mask: 0x0 Volatile: true
17:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: Always
15	PCIE_CAP_ROLE_BASED_ERROR_REPORT	* Varies	Role-Based Error Reporting. When set, this bit indicates that the function implements the functionality originally defined in the Error Reporting ECN for <i>PCI Express Base Specification, Revision 1.0a</i>, and later incorporated into <i>PCI Express Base Specification, Revision 1.1</i>. This bit must be set by all functions conforming to the ECN, <i>PCI Express Base Specification, Revision 1.1</i>, or subsequent <i>PCI Express Base Specification</i> revisions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_PCIE_TYPE == 1 && 1) ? R : (CX_MEMORY_MAP_VIEW == 0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW == 0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW == 1) ? R : R <i>Note:</i> This register field is sticky. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_MEMORY_MAP_POSITION) ? "0x0" : "0x1"} Exists: Always Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
14:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: Always
11:9	PCIE_CAP_EP_L1_ACCPT_LATENCY	* Varies	Endpoint L1 Acceptable Latency. This field indicates the acceptable latency that an Endpoint can withstand due to the transition from L1 state to the L0 state. It is essentially an indirect measure of the Endpoint's internal buffering. Power management software uses the reported L1 Acceptable Latency number to compare against the L1 Exit Latencies reported (see below) by all components comprising the data path from this Endpoint to the Root Complex Root Port to determine whether ASPM L1 entry can be used with no loss of performance. For functions other than Endpoints, this field is Reserved and the controller hardwires it to 000b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W (Sticky) else R(Sticky) and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MAX_1US): Maximum of 1 us ■ 0x1 (MAX_2US): Maximum of 2 us ■ 0x2 (MAX_4US): Maximum of 4 us ■ 0x3 (MAX_8US): Maximum of 8 us ■ 0x4 (MAX_16US): Maximum of 16 us ■ 0x5 (MAX_32US): Maximum of 32 us ■ 0x6 (MAX_64US): Maximum of 64 us ■ 0x7 (NO_LIMIT): No limit Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1) ? 0 : DEFAULT_EP_L1_ACCPT_LATENCY} Exists: Always Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
8:6	PCIE_CAP_EP_L0S_ACCPT_LATENCY	* Varies	Endpoint L0s Acceptable Latency. This field indicates the acceptable total latency that an Endpoint can withstand due to the transition from L0s state to the L0 state. It is essentially an indirect measure of the Endpoint's internal buffering. Power management software uses the reported L0s Acceptable Latency number to compare against the L0s exit latencies reported by all components comprising the data path from this Endpoint to the Root Complex Root Port to determine whether ASPM L0s entry can be used with no loss of performance. For functions other than Endpoints, this field is Reserved and the controller hardwires it to 000b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MAX_64NS): Maximum of 64 ns ■ 0x1 (MAX_128NS): Maximum of 128 ns ■ 0x2 (MAX_256NS): Maximum of 256 ns ■ 0x3 (MAX_512NS): Maximum of 512 ns ■ 0x4 (MAX_1US): Maximum of 1 us ■ 0x5 (MAX_2US): Maximum of 2 us ■ 0x6 (MAX_4US): Maximum of 4 us ■ 0x7 (NO_LIMIT): No limit Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1) ? 0 : DEFAULT_EP_L0S_ACCPT_LATENCY} Exists: Always Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
5	PCIE_CAP_EXT_TAG_SUPP	* Varies	Extended Tag Field Supported. This bit, in combination with the 10-Bit Tag Requester Supported bit in the Device Capabilities 2 register, indicates the maximum supported size of the Tag field as a Requester. This bit must be set if the 10-Bit Tag Requester Supported bit is set. <i>Note:</i> 8-bit Tag field generation must be enabled by the Extended Tag Field Enable bit in the Device Control register of the Requester Function before 8-bit Tags can be generated by the Requester. See Section 2.2.6.2 of <i>PCI Express Base Specification</i> for interactions with enabling the use of 10-Bit Tags. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_PCIE_TYPE == 1 && 1) ? R : (CX_MEMORY_MAP_VIEW == 0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW == 0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW == 1) ? R : R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_5b_TAG): 5-bit Tag field supported ■ 0x1 (_8b_TAG): 8-bit Tag field supported Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_MEMORY_MAP_POSITION) ? 0 : DEFAULT_EXT_TAG_FIELD_SUPPORTED_0} Exists: Always Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
4:3	PCIE_CAP_PHANTOM_FUNC_SUPPORT	* Varies	Phantom Functions Supported. This field indicates the support for use of unclaimed function numbers to extend the number of outstanding transactions allowed by logically combining unclaimed function numbers (called Phantom Functions) with the Tag identifier (see Section 2.2.6.2 of <i>PCI Express Base Specification</i> for a description of Tag Extensions). With every Function in an ARI Device, the Phantom Functions Supported field must be set to 00b. The remainder of this field description applies only to non-ARI Multi-Function Devices. This field indicates the number of most significant bits of the Function Number portion of Requester ID that are logically combined with the Tag identifier. <i>Note:</i> Phantom Function support for the function must be enabled by the Phantom Functions Enable field in the Device Control register before the Function is permitted to use the Function Number field in the Requester ID for Phantom Functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NO_PHNATOM_FUNC): No Function Number bits are used for Phantom Functions. Multi-Function Devices are permitted to implement up to 8 independent functions.

Bits	Name	Memory Access	Description
4:3...(cont.)	PCIE_CAP_PHANTOM_FUNC_SUPPORT.	* Varies	■ 0x1 (MAX_1_PHANTOM_PER_FUNC): The most significant bit of the Function number in Requester ID is used for Phantom Functions; a Multi-Function Device is permitted to implement Functions 0-3. Functions 0, 1, 2, and 3 are permitted to use Function Numbers 4, 5, 6, and 7 respectively as Phantom Functions. ■ 0x2 (MAX_3_PHANTOM_PER_FUNC): The two most significant bits of Function Number in Requester ID are used for Phantom Functions; a Multi-Function Device is permitted to implement Functions 0-1. Function 0 is permitted to use Function Numbers 2, 4, and 6 for Phantom Functions. Function 1 is permitted to use Function Numbers 3, 5, and 7 as Phantom Functions. ■ 0x3 (SINGLE_FUNC_MAX_7_PHANTOM_FUNC): All 3 bits of Function Number in Requester ID used for Phantom Functions. The device must have a single Function 0 that is permitted to use all other Function Numbers as Phantom Functions. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : DEFAULT_PHANTOM_FUNC_SUPPORTED_0} Exists: Always Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
2:0	PCIE_CAP_MAX_PAYLOAD_SIZE	* Varies	Max_Payload_Size Supported. This field indicates the maximum payload size that the function can support for TLPs. All encodings other than the defined encodings are reserved. The functions of a Multi-Function Device are permitted to report different values for this field. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_PCIE_TYPE == 1 && 1) ? R : (CX_PCIE_OVER_CIO_ENABLE) ? R : (CX_MEMORY_MAP_VIEW == 0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW == 0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW == 1) ? R : R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MAX_128B_SIZE): 128 bytes max payload size ■ 0x1 (MAX_256B_SIZE): 256 bytes max payload size ■ 0x2 (MAX_512B_SIZE): 512 bytes max payload size ■ 0x3 (MAX_1024B_SIZE): 1024 bytes max payload size ■ 0x4 (MAX_2048B_SIZE): 2048 bytes max payload size ■ 0x5 (MAX_4096B_SIZE): 4096 bytes max payload size Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE == 1 && !CX_MEMORY_MAP_POSITION) ? "0x0" : ((CX_MAX_MTU == 128) ? "0x0" : (CX_MAX_MTU == 256) ? "0x1" : (CX_MAX_MTU == 512) ? "0x2" : (CX_MAX_MTU == 1024) ? "0x3" : (CX_MAX_MTU == 2048) ? "0x4" : (CX_MAX_MTU == 4096) ? "0x5" : "0x0")} Exists: Always Testable: wire ? unconstrained : writeAsRead

3.5.3 DEVICE_CONTROL_DEVICE_STATUS

- **Name:** Device Control and Device Status Register.
- **Description:** This register controls PCI Express device specific parameters and provides information about PCI Express device (function) specific parameters.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** Always

RSVDP_23	31:23
PCIE_CAP_EMERGENCY_PWR_RED_DETECTED	22
PCIE_CAP_TRANS_PENDING	21
PCIE_CAP_AUX_POWER_DETECTED	20
PCIE_CAP_UNSUPPORTED_REQ_DETECTED	19
PCIE_CAP_FATAL_ERR_DETECTED	18
PCIE_CAP_NON_FATAL_ERR_DETECTED	17
PCIE_CAP_CORR_ERR_DETECTED	16
PCIE_CAP_INITIATE_FLR	15
PCIE_CAP_MAX_READ_REQ_SIZE	14:12
PCIE_CAP_EN_NO_SNOOP	11
PCIE_CAP_AUX_POWER_PM_EN	10
PCIE_CAP_PHANTOM_FUNC_EN	9
PCIE_CAP_EXT_TAG_EN	8
PCIE_CAP_MAX_PAYLOAD_SIZE_CS	7:5
PCIE_CAP_EN_REL_ORDER	4
PCIE_CAP_UNSUPPORTED_REQ_REP_EN	3
PCIE_CAP_FATAL_ERR_REPORT_EN	2
PCIE_CAP_NON_FATAL_ERR_REPORT_EN	1
PCIE_CAP_CORR_ERR_REPORT_EN	0

Table 3-37 Fields for Register: DEVICE_CONTROL_DEVICE_STATUS

Bits	Name	Memory Access	Description
31:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: Always

Bits	Name	Memory Access	Description
22	PCIE_CAP_EMERGENCY_PWR_RED_DETECTED	R	Emergency Power Reduction Detected. When set, this bit indicates that the function is in the Emergency Power Reduction State. Values: ■ 0x0 (CLEAR): Function is not in the Emergency Power Reduction State. ■ 0x1 (SET): Function is in the Emergency Power Reduction State. Value After Reset: 0x0 Exists: EMERGENCY_POWER_REDUCTION_ENABLE Testable: untestable Volatile: true
21	PCIE_CAP_TRANS_PENDING	R	Transactions Pending. Endpoints: When set, this bit indicates that the function has issued Non-Posted Requests that have not been completed. A Function reports this bit cleared only when all outstanding Non-Posted Requests have completed or have been terminated by the Completion Timeout mechanism. This bit must also be cleared upon the completion of an FLR. Value After Reset: * Varies Exists: Always Testable: writeAsRead Reset Mask: 0x0 Volatile: true
20	PCIE_CAP_AUX_POWER_DETECTED	R	AUX Power Detected. Functions that require Aux power report this bit as set if Aux power is detected by the function. This bit is derived by sampling the sys_aux_pwr_det input. Value After Reset: * Varies Exists: Always Testable: writeAsRead Reset Mask: 0x0 Volatile: true
19	PCIE_CAP_UNSUPPORTED_REQ_DETECTED	R/W1C	Unsupported Request Detected. This bit indicates that the function received an Unsupported Request. Errors are logged in this register regardless of whether error reporting is enabled or not in the Device Control register. For a Multi-Function Device, each function indicates status of errors as perceived by the respective function. Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
18	PCIE_CAP_FATAL_ERR_DETECTED	R/W1C	Fatal Error Detected. This bit indicates status of Fatal errors detected. Errors are logged in this register regardless of whether error reporting is enabled or not in the Device Control register. For a Multi-Function device, each function indicates status of errors as perceived by the respective Function. For Functions supporting Advanced Error Handling, errors are logged in this register regardless of the settings of the Uncorrectable Error Mask register. Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true
17	PCIE_CAP_NON_FATAL_ERR_DETECTED	R/W1C	Non-Fatal Error Detected. This bit indicates status of Non-fatal errors detected. Errors are logged in this register regardless of whether error reporting is enabled or not in the Device Control register. For a Multi-Function device, each function indicates status of errors as perceived by the respective Function. For functions supporting Advanced Error Handling, errors are logged in this register regardless of the settings of the Uncorrectable Error Mask register. Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true
16	PCIE_CAP_CORR_ERR_DETECTED	R/W1C	Correctable Error Detected. This bit indicates status of correctable errors detected. Errors are logged in this register regardless of whether error reporting is enabled or not in the Device Control register. For a Multi-Function device, each function indicates status of errors as perceived by the respective function. For functions supporting Advanced Error Handling, errors are logged in this register regardless of the settings of the Correctable Error Mask register. Value After Reset: 0x0 Exists: Always Testable: writeAsRead Volatile: true
15	PCIE_CAP_INITIATE_FLR	W	Initiate Function Level Reset (applicable to Endpoints with Function Level Reset Capability set to 1b). A write of 1b initiates Function Level Reset to the Function. The value read by software from this bit is always 0b. Value After Reset: 0x0 Exists: (CX_FLR_ENABLE && !CX_PCIE_BRIDGE) Testable: {(CX_EXTENSIBLE_VFUNC) ? "untestable" : "unconstrained" }

Bits	Name	Memory Access	Description
14:12	PCIE_CAP_MAX_READ_REQ_SIZE	* Varies	Max_Read_Request_Size. This field sets the maximum Read Request size for the function as a Requester. The function must not generate Read Requests with a size exceeding the set value. For functions that do not generate Read Requests larger than 128 bytes and functions that do not generate Read Requests on their own behalf, the controller implements this field as Read Only (RO) with a value of 000b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (MAX_128B_SIZE): 128 bytes maximum Read Request size ■ 0x1 (MAX_256B_SIZE): 256 bytes maximum Read Request size ■ 0x2 (MAX_512B_SIZE): 512 bytes maximum Read Request size ■ 0x3 (MAX_1024B_SIZE): 1024 bytes maximum Read Request size ■ 0x4 (MAX_2048B_SIZE): 2048 bytes maximum Read Request size ■ 0x5 (MAX_4096B_SIZE): 4096 bytes maximum Read Request size ■ 0x6 (RESERVED1): RESERVED ■ 0x7 (RESERVED2): RESERVED Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? "0x0" : "0x2"} Exists: Always Memory Access: "(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
11	PCIE_CAP_EN_NO_SNOOP	* Varies	Enable No Snoop. If this bit is set, the function is permitted to Set the No Snoop bit in the Requester Attributes of transactions it initiates that do not require hardware enforced cache coherency (see section 2.2.6.5 in <i>PCI Express Base Specification</i>). <i>Note:</i> Setting this bit to 1b should not cause a function to set the No Snoop attribute on all transactions that it initiates. Even when this bit is set, a function is only permitted to set the No Snoop attribute on a transaction when it can guarantee that the address of the transaction is not stored in any cache in the system. The controller hardwires this bit 0b if a function would never set the No Snoop attribute in transactions it initiates. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : DEFAULT_NO_SNOOP_SUPPORTED_0} Exists: Always Memory Access: "(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==0 && DEFAULT_NO_SNOOP_SUPPORTED_0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !DEFAULT_NO_SNOOP_SUPPORTED_0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && DEFAULT_NO_SNOOP_SUPPORTED_0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !DEFAULT_NO_SNOOP_SUPPORTED_0) ? \"read-only\" : (DEFAULT_NO_SNOOP_SUPPORTED_0) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
10	PCIE_CAP_AUX_POWER_PM_EN	R/W	Aux Power PM Enable. This bit is derived by sampling the sys_aux_pwr_det input. When set this bit, enables a function to draw Aux power independent of PME Aux power. Functions that require Aux power on legacy operating systems should continue to indicate PME Aux power requirements. Aux power is allocated as requested in the Aux_Current field of the Power Management Capabilities register (PMC), independent of the PME_En bit in the Power Management Control/Status register (PMCSR). For Multi-Function devices, a component is allowed to draw Aux power if at least one of the functions has this bit set.<i>Note:</i> Functions that consume Aux power must preserve the value of this sticky register when Aux power is available. In such functions, this bit is not modified by Conventional Reset. For functions that do not implement this capability, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: Always Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
9	PCIE_CAP_PHANTOM_FUNC_EN	* Varies	Phantom Functions Enable. This bit, in combination with the 10-Bit Tag Requester Enable bit in the Device Control 2 register, determines how many Tag field bits a Requester is permitted to use. When the 10-Bit Tag Requester Enable bit is clear, ■ If this bit is set, it enables a function to use unclaimed functions as Phantom functions to extend the number of outstanding transaction identifiers ■ If this bit is clear, the function is not allowed to use Phantom functions For more information, see section 2.2.6.2 of <i>PCI Express Base Specification</i>. Software should not change the value of this bit while the function has outstanding Non-Posted Requests; otherwise, the result is undefined. For functions that do not implement this capability, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: DEVICE_CAPABILITIES_REG.PCIE_CAP_PHANTOM_FUNC_SUPPORT ? RW : RO ■ Dbi: DEVICE_CAPABILITIES_REG.PCIE_CAP_PHANTOM_FUNC_SUPPORT ? RW : RO and more specifically: (CX_MEMORY_MAP_VIEW==0 && DEFAULT_PHANTOM_FUNC_SUPPORTED_\$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !DEFAULT_PHANTOM_FUNC_SUPPORTED_\$pf) ? R : (CX_MEMORY_MAP_VIEW==1 && DEFAULT_PHANTOM_FUNC_SUPPORTED_\$pf) ? R/W : (CX_MEMORY_MAP_VIEW==1 && !DEFAULT_PHANTOM_FUNC_SUPPORTED_\$pf) ? R : (DEFAULT_PHANTOM_FUNC_SUPPORTED_\$pf) ? R/W : R Value After Reset: * Varies Exists: Always Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
8	PCIE_CAP_EXT_TAG_EN	* Varies	Extended Tag Field Enable. This bit, in combination with the 10-Bit Tag Requester Enable bit in the Device Control 2 register, determines how many Tag field bits a Requester is permitted to use. When the 10-Bit Tag Requester Enable bit is clear, ■ If the Extended Tag Field Enable bit is set, the function is permitted to use an 8-bit Tag field as a Requester ■ If the Extended Tag Field Enable bit is clear, the Function is restricted to a 5-bit Tag field See section 2.2.6.2 of <i>PCI Express Base Specification</i> for required behavior when the 10-Bit Tag Requester Enable bit is set. If software changes the value of the Extended Tag Field Enable bit while the function has outstanding Non-Posted Requests, the result is undefined. For functions that do not implement this capability, the controller hardwires this bit to 0b. Note: The access attributes of this field are as follows: ■ Wire: DEVICE_CAPABILITIES_REG.PCIE_CAP_EXT_TAG_SUPP ? RW : RO ■ Dbi: DEVICE_CAPABILITIES_REG.PCIE_CAP_EXT_TAG_SUPP ? RW : RO and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && DEFAULT_EXT_TAG_FIELD_SUPPORTED_\$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !DEFAULT_EXT_TAG_FIELD_SUPPORTED_\$pf) ? R : (CX_MEMORY_MAP_VIEW==1 && DEFAULT_EXT_TAG_FIELD_SUPPORTED_\$pf) ? R/W : (CX_MEMORY_MAP_VIEW==1 && !DEFAULT_EXT_TAG_FIELD_SUPPORTED_\$pf) ? R : (DEFAULT_EXT_TAG_FIELD_SUPPORTED_\$pf) ? R/W : R Value After Reset: * Varies Exists: Always Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
7:5	PCIE_CAP_MAX_PAYLOAD_SIZE_CS	* Varies	Max_Payload_Size. This field sets maximum TLP payload size for the Function. As a Receiver, the Function must handle TLPs as large as the set value. As a Transmitter, the Function must not generate TLPs exceeding the set value. Permissible values that can be programmed are indicated by the Max_Payload_Size Supported field (PCIE_CAP_MAX_PAYLOAD_SIZE) in the Device Capabilities (DEVICE_CAPABILITIES_REG) register (for more details, see section 7.5.3.3 of <i>PCI Express Base Specification</i>). This field sets the maximum Read Request size for the function as a Requester. The function must not generate Read Requests with a size exceeding the set value. For Functions that support only the 128-byte max payload size, the controller hardwires this field to 000b. System software is not required to program the same value for this field for all the Functions of a Multi-Function device (for more details, see section 2.2.2 of <i>PCI Express Base Specification</i>). For ARI Devices, Max_Payload_Size is determined solely by the setting in Function0. The settings in the other Functions always return whatever value software programmed for each, but otherwise are ignored by the component. Note: The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (MAX_128B_SIZE): 128 bytes maximum Read Request size ■ 0x1 (MAX_256B_SIZE): 256 bytes maximum Read Request size ■ 0x2 (MAX_512B_SIZE): 512 bytes maximum Read Request size ■ 0x3 (MAX_1024B_SIZE): 1024 bytes maximum Read Request size ■ 0x4 (MAX_2048B_SIZE): 2048 bytes maximum Read Request size
7:5...(cont.)	PCIE_CAP_MAX_PAYLOAD_SIZE_CS.	* Varies	■ 0x5 (MAX_4096B_SIZE): 4096 bytes maximum Read Request size ■ 0x6 (RESERVED1): RESERVED ■ 0x7 (RESERVED2): RESERVED Value After Reset: 0x0 Exists: Always Memory Access: "(CX_PCIE_OVER_CIO_ENABLE (!CX_MEMORY_MAP_POSITION && CX_MEMORY_MAP_PCIE_TYPE==1)) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
4	PCIE_CAP_EN_REL_ORDER	* Varies	Enable Relaxed Ordering. If this bit is set, the function is permitted to set the Relaxed Ordering bit in the Attributes field of transactions it initiates that do not require strong write ordering (for more details, see section 2.2.6.4 and section 2.4 of <i>PCI Express Base Specification</i>). For a function that never sets the Relaxed Ordering attribute in transactions it initiates as a Requester, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : (CX_PCIE_BRIDGE_ENABLE ? 0 : 1) } Exists: Always Memory Access: "(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==0 && !CX_PCIE_BRIDGE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && CX_PCIE_BRIDGE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && !CX_PCIE_BRIDGE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && CX_PCIE_BRIDGE) ? \"read-only\" : (!CX_PCIE_BRIDGE)? \"read-write\" : \"read-only\" "
3	PCIE_CAP_UNSUPPORT_REQ_REP_EN	* Varies	Unsupported Request Reporting Enable. This bit, in conjunction with other bits, controls the signaling of Unsupported Request Errors by sending error Messages (for more details, see section 6.2.5 and section 6.2.6 of <i>PCI Express Base Specification</i>). For a Multi-Function Device, this bit controls error reporting for each Function from point-of-view of the respective Function. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: Always Memory Access: "(CX_MEMORY_MAP_POSITION==0 && CX_MEMORY_MAP_PCIE_TYPE==1) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
2	PCIE_CAP_FATAL_ERR_REPORT_EN	* Varies	Fatal Error Reporting Enable. This bit, in conjunction with other bits, controls sending ERR_FATAL Messages (for more details, see section 6.2.5 and section 6.2.6 of <i>PCI Express Base Specification</i>). For a Multi-Function device, this bit controls error reporting for each function from point-of-view of the respective function. For a Root Port, the reporting of Fatal errors is internal to the root. No external ERR_FATAL Message is generated. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: Always Memory Access: "(CX_MEMORY_MAP_POSITION==0 && CX_MEMORY_MAP_PCIE_TYPE==1) ? \"read-only\" : \"read-write\" "
1	PCIE_CAP_NON_FATAL_ERR_REPORT_EN	* Varies	Non-Fatal Error Reporting Enable. This bit, in conjunction with other bits, controls sending ERR_NONFATAL Messages (for more details, see section 6.2.5 and Section 6.2.6 of <i>PCI Express Base Specification</i>). For a Multi-Function Device, this bit controls error reporting for each function from point-of-view of the respective Function. For a Root Port, the reporting of Non-fatal errors is internal to the root. No external ERR_NONFATAL Message is generated. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: Always Memory Access: "(CX_MEMORY_MAP_POSITION==0 && CX_MEMORY_MAP_PCIE_TYPE==1) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
0	PCIE_CAP_CORR_ERR_REPOR T_EN	* Varies	Correctable Error Reporting Enable. This bit, in conjunction with other bits, controls sending ERR_COR Messages (for more details, see section 6.2.5, section 6.2.6, and section 6.2.10.2 of <i>PCI Express Base Specification</i>). For a Multi-Function device, this bit controls error reporting for each function from point-of-view of the respective function. For a Root Port, the reporting of correctable errors is internal to the root. No external ERR_COR Message is generated. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: Always Memory Access: "(CX_MEMORY_MAP_POSITION==0 && CX_MEMORY_MAP_PCIE_TYPE==1) ? \"read-only\" : \"read-write\" "

3.5.4 LINK_CAPABILITIES_REG

- **Name:** Link Capabilities Register.
- **Description:** The Link Capabilities register identifies PCI Express Link specific capabilities and CXL-RCRB link specific capabilities.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** Always

PCIE_CAP_PORT_NUM	31:24
RSVDP_23	23
PCIE_CAP_ASPM_OPT_COMPLIANCE	22
PCIE_CAP_LINK_BW_NOT_CAP	21
PCIE_CAP_DLL_ACTIVE_REP_CAP	20
PCIE_CAP_SURPRISE_DOWN_ERR_REP_CAP	19
PCIE_CAP_CLOCK_POWER_MAN	18
PCIE_CAP_L1_EXIT_LATENCY	17:15
PCIE_CAP_L0S_EXIT_LATENCY	14:12
PCIE_CAP_ACTIVE_STATE_LINK_PM_SUPPORT	11:10
PCIE_CAP_MAX_LINK_WIDTH	9:4
PCIE_CAP_MAX_LINK_SPEED	3:0

Table 3-38 Fields for Register: LINK_CAPABILITIES_REG

Bits	Name	Memory Access	Description
31:24	PCIE_CAP_PORT_NUM	* Varies	Port Number. This field indicates the PCI Express Port number for the given PCI Express Link. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Value After Reset: PORT_NUM Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2
22	PCIE_CAP_ASPM_OPT_COMPLIANCE	* Varies	ASPM Optionality Compliance. This field must be set to 1b in all functions. Components implemented against certain earlier versions of this specification will have this bit set to 0b. Software is permitted to use the value of this bit to help determine whether to enable ASPM or whether to run ASPM compliance tests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Value After Reset: ASPM_OPTIONALITY_COMPLIANCE Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: wire ? unconstrained : writeAsRead
21	PCIE_CAP_LINK_BW_NOT_CAP	R	Link Bandwidth Notification Capability. A value of 1b indicates support for the Link Bandwidth Notification status and interrupt mechanisms. This capability is required for all Root Ports and Switch Downstream Ports supporting Links wider than x1 and/or multiple Link speeds. This field is not applicable and is Reserved for Endpoints, PCI Express to PCI/PCI-X bridges, and Upstream Ports of Switches. For functions that do not implement the Link Bandwidth Notification Capability the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2
20	PCIE_CAP_DLL_ACTIVE_REP_CAP	R	Data Link Layer Link Active Reporting Capable. For a Downstream Port, the controller hardwires this bit to 1b if the component supports the optional capability of reporting the DL_Active state of the Data Link Control and Management State Machine. For a hot-plug capable Downstream Port (as indicated by the Hot-Plug Capable bit of the Slot Capabilities register) or a Downstream Port that supports Link speeds greater than 5.0 GT/s, the controller hardwires this bit to 1b. For Upstream Ports and components that do not support this optional capability, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2

Bits	Name	Memory Access	Description
19	PCIE_CAP_SURPRISE_DOWN_ERROR_REP_CAP	R	Surprise Down Error Reporting Capable. For a Downstream Port, this bit must be set if the component supports the optional capability of detecting and reporting a Surprise Down error condition. For Upstream Ports and components that do not support this optional capability, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2
18	PCIE_CAP_CLOCK_POWER_MGMT	* Varies	Clock Power Management. For Upstream Ports, a value of 1b in this bit indicates that the component tolerates the removal of any reference clock(s) through the "clock request" (CLKREQ#) mechanism when the Link is in the L1 and L2/L3 Ready Link states. A value of 0b indicates the component does not have this capability and that reference clock(s) must not be removed in these Link states. L1 PM Substates defines other semantics for the CLKREQ# signal, which are managed independently of Clock Power Management. This Capability is applicable only in form factors that support "clock request" (CLKREQ#) capability. For a Multi-Function device associated with an Upstream Port, each Function indicates its capability independently. Power Management configuration software must only permit reference clock removal if all functions of the Multi-Function device indicate a 1b in this bit. For ARI Devices, all Functions must indicate the same value in this bit. For Downstream Ports, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_CLK_PM_CAP Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
17:15	PCIE_CAP_L1_EXIT_LATENCY	* Varies	L1 Exit Latency. This field indicates the L1 exit latency for the given PCI Express Link. The value reported indicates the length of time this Port requires to complete transition from ASPM L1 to L0. If ASPM L1 is not supported, the value is undefined. <i>Note:</i> Exit latencies may be influenced by PCI Express reference clock configuration depending upon whether a component uses a common or separate reference clock. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all functions. There are two each of these register fields, this one and a shadow one at the same address. The Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG) determines which one is used by the controller and which one is accessed by a read request. Common Clock operation is supported (possible) in the controller when one or more of the following expressions is true: ■ CX_NFTS !=CX_COMM_NFTS ■ DEFAULT_L0S_EXIT_LATENCY !=DEFAULT_COMM_L0S_EXIT_LATENCY ■ DEFAULT_L1_EXIT_LATENCY !=DEFAULT_COMM_L1_EXIT_LATENCY Common Clock operation is enabled in the controller when you set the Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG). The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the shadow field at this location. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky)

Bits	Name	Memory Access	Description
17:15...(cont.)	PCIE_CAP_L1_EXIT_LATENCY.	* Varies	■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: (CX_MEMORY_MAP_VIEW==2 && CX_COMM_CLK_EN) ? R/W : R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (LESS_THAN_1US): Less than 1us ■ 0x1 (_1US_TO_2US): 1 us to less than 2 us ■ 0x2 (_2US_TO_4US): 2 us to less than 4 us ■ 0x3 (_4US_TO_8US): 4 us to less than 8 us ■ 0x4 (_8US_TO_16US): 8 us to less than 16 us ■ 0x5 (_16US_TO_32US): 16 us to less than 32 us ■ 0x6 (_32US_TO_64US): 32 us to 64 us ■ 0x7 (GREATER_THAN_64US): More than 64 us Value After Reset: DEFAULT_L1_EXIT_LATENCY Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
14:12	PCIE_CAP_L0S_EXIT_LATENCY	* Varies	L0s Exit Latency. This field indicates the L0s exit latency for the given PCI Express Link. The value reported indicates the length of time this Port requires to complete transition from L0s to L0. If L0s is not supported, the value is undefined; however, see the Implementation Note "Potential Issues With Legacy Software When L0s is Not Supported" in section 5.4.1.1 of <i>PCI Express Base Specification</i> for the recommended value. <i>Note:</i> Exit latencies may be influenced by PCI Express reference clock configuration depending upon whether a component uses a common or separate reference clock. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all functions. There are two each of these register fields, this one and a shadow one at the same address. The Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG) determines which one is used by the controller and which one is accessed by a read request. Common Clock operation is supported (possible) in the controller when one or more of the following expressions is true: ■ CX_NFTS !=CX_COMM_NFTS ■ DEFAULT_L0S_EXIT_LATENCY !=DEFAULT_COMM_L0S_EXIT_LATENCY ■ DEFAULT_L1_EXIT_LATENCY !=DEFAULT_COMM_L1_EXIT_LATENCY Common Clock operation is enabled in the controller when you set the Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG). The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the shadow field at this location. <i>Note</i>

Bits	Name	Memory Access	Description
14:12...(cont.)	PCIE_CAP_L0S_EXIT_LATENCY.	* Varies	: The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: (CX_MEMORY_MAP_VIEW==2 && CX_COMM_CLK_EN) ? R/W : R Note: This register field is sticky. Values: ■ 0x0 (LESS_THAN_64NS): Less than 64 ns ■ 0x1 (_64NS_TO_128NS): 64 ns to less than 128 ns ■ 0x2 (_128NS_TO_256NS): 128 ns to less than 256 ns ■ 0x3 (_256NS_TO_512NS): 256 ns to less than 512 ns ■ 0x4 (_512NS_TO_1US): 512 ns to less than 1 us ■ 0x5 (_1US_TO_2US): 1 us to less than 2 us ■ 0x6 (_2US_TO_4US): 2 us to 4 us ■ 0x7 (GREATER_THAN_4US): More than 4 us Value After Reset: DEFAULT_L0S_EXIT_LATENCY Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Volatile: true
11:10	PCIE_CAP_ACTIVE_STATE_LINK_PM_SUPPORT	* Varies	Active State Power Management (ASPM) Support. This field indicates the level of ASPM supported on the given PCI Express Link. For more information on ASPM support requirements, see section 5.4.1 of <i>PCI Express Base Specification</i>. Multi-Function devices associated with an Upstream Port must report the same value in this field for all functions. Note: The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_PCIE_OVER_CIO_ENABLE) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Note: This register field is sticky. Values: ■ 0x0 (NO_ASPM_SUP): No ASPM Support ■ 0x1 (L0S_SUP): L0s Supported ■ 0x2 (L1_SUP): L1 Supported ■ 0x3 (L0S_L1_SUP): L0s and L1 Supported Value After Reset: AS_LINK_PM_SUPT Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
9:4	PCIE_CAP_MAX_LINK_WIDTH	* Varies	Maximum Link Width. This field indicates the maximum Link width (xN - corresponding to N Lanes) implemented by the component. This value is permitted to exceed the number of Lanes routed to the slot (Downstream Port), adapter connector (Upstream Port), or in the case of component-to-component connections, the actual wired connection width. All encodings other than the defined encodings are reserved. Multi-Function devices associated with an Upstream Port must report the same value in this field for all functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_PCIE_OVER_CIO_ENABLE) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R: R <i>Note:</i> This register field is sticky. Values: ■ 0x1 (X1): x1 ■ 0x2 (X2): x2 ■ 0x4 (X4): x4 ■ 0x8 (X8): x8 ■ 0xc (X12): x12 ■ 0x10 (X16): x16 ■ 0x20 (X32): x32 Value After Reset: CX_NL Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
3:0	PCIE_CAP_MAX_LINK_SPEED	* Varies	Max Link Speed. This field indicates the maximum Link speed of the associated Port. The encoded value specifies a bit location in the Supported Link Speeds Vector (in the Link Capabilities 2 register) that corresponds to the maximum Link speed. All encodings other than the defined encodings are reserved. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && !CX_PCIE_OVER_CIO_ENABLED) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SUP_LINK_SPEED_FIELD_BIT_0): Supported Link Speeds Vector field bit 0 ■ 0x2 (SUP_LINK_SPEED_FIELD_BIT_1): Supported Link Speeds Vector field bit 1 ■ 0x3 (SUP_LINK_SPEED_FIELD_BIT_2): Supported Link Speeds Vector field bit 2 ■ 0x4 (SUP_LINK_SPEED_FIELD_BIT_3): Supported Link Speeds Vector field bit 3 ■ 0x5 (SUP_LINK_SPEED_FIELD_BIT_4): Supported Link Speeds Vector field bit 4 ■ 0x6 (SUP_LINK_SPEED_FIELD_BIT_5): Supported Link Speeds Vector field bit 5 ■ 0x7 (SUP_LINK_SPEED_FIELD_BIT_6): Supported Link Speeds Vector field bit 6 Value After Reset: MAX_LINK_SP Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: wire ? unconstrained : writeAsRead

3.5.5 LINK_CONTROL_LINK_STATUS_REG

- **Name:** Link Control and Link Status Register.
- **Description:** This register controls and provides information about PCI Express Link specific parameters as well as RCRB link associated parameters.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** Always

PCIE_CAP_LINK_AUTO_BW_STATUS	31
PCIE_CAP_LINK_BW_MAN_STATUS	30
PCIE_CAP_DLL_ACTIVE	29
PCIE_CAP_SLOT_CLK_CONFIG	28
PCIE_CAP_LINK_TRAINING	27
RSVDP_26	26
PCIE_CAP_NEGO_LINK_WIDTH	25:20
PCIE_CAP_LINK_SPEED	19:16
PCIE_CAP_DRS_SIGNALING_CONTROL	15:14
RSVDP_12	13:12
PCIE_CAP_LINK_AUTO_BW_INT_EN	11
PCIE_CAP_LINK_BW_MAN_INT_EN	10
PCIE_CAP_HW_AUTO_WIDTH_DISABLE	9
PCIE_CAP_EN_CLK_POWER_MAN	8
PCIE_CAP_EXTENDED_SYNC	7
PCIE_CAP_COMMON_CLK_CONFIG	6
PCIE_CAP_RETRAIN_LINK	5
PCIE_CAP_LINK_DISABLE	4
PCIE_CAP_RCB	3
RSVDP_2	2
PCIE_CAP_ACTIVE_STATE_LINK_PM_CONTROL	1:0

Table 3-39 Fields for Register: LINK_CONTROL_LINK_STATUS_REG

Bits	Name	Memory Access	Description
31	PCIE_CAP_LINK_AUTO_BW_STATUS	* Varies	Link Autonomous Bandwidth Status. This bit is set by hardware to indicate that hardware has autonomously changed Link speed or width, without the Port transitioning through DL_Down status, for reasons other than to attempt to correct unreliable Link operation. This bit must be set if the Physical Layer reports a speed or width change was initiated by the Downstream component that was indicated as an autonomous change. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG. This bit is not applicable and is Reserved for Endpoints, PCI Express-to-PCI/PCI-X bridges, and Upstream Ports of Switches. For functions that do not implement the Link Bandwidth Notification Capability, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: R Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
30	PCIE_CAP_LINK_BW_MAN_STAT US	* Varies	Link Bandwidth Management Status. This bit is set by hardware to indicate that either of the following has occurred without the Port transitioning through DL_Down status: - A Link retraining has completed following a write of 1b to the Retrain Link bit. <i>Note:</i> This bit is set following any write of 1b to the Retrain Link bit, including when the Link is in the process of retraining for some other reason. Hardware has changed Link speed or width to attempt to correct unreliable Link operation, either through an LTSSM timeout or a higher level process. This bit must be set if the Physical Layer reports a speed or width change was initiated by the Downstream component that was not indicated as an autonomous change. This bit is not applicable and is Reserved for Endpoints, PCI Express-to-PCI/PCI-X bridges, and Upstream Ports of Switches. For functions that do not implement the Link Bandwidth Notification Capability, the controller hardwires this bit to 0b. The default value of this bit is 0b. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: R Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
29	PCIE_CAP_DLL_ACTIVE	R	Data Link Layer Link Active. This bit indicates the status of the Data Link Control and Management State Machine. It returns a 1b to indicate the DL_Active state, 0b otherwise. This bit must be implemented if the Data Link Layer Link Active Reporting Capable bit is 1b. Otherwise, the controller hardwires it to 0b. Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<2)

Bits	Name	Memory Access	Description
28	PCIE_CAP_SLOT_CLK_CONFIG	* Varies	Slot Clock Configuration. This bit indicates that the component uses the same physical reference clock that the platform provides on the connector. If the device uses an independent clock irrespective of the presence of a reference clock on the connector, this bit must be clear. For a Multi-Function Device, each Function must report the same value for this bit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Value After Reset: SLOT_CLK_CONFIG Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: wire ? unconstrained : writeAsRead
27	PCIE_CAP_LINK_TRAINING	* Varies	Link Training. This read-only bit indicates that the Physical Layer LTSSM is in the Configuration or Recovery state, or that 1b was written to the Retrain Link bit but Link training has not yet begun. Hardware clears this bit when the LTSSM exits the Configuration/Recovery state. This bit is not applicable and Reserved for Endpoints, PCI Express to PCI/PCI-X bridges, and Upstream Ports of Switches, and the controller hardwires it to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: R Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Reset Mask: 0x0 Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2

Bits	Name	Memory Access	Description
25:20	PCIE_CAP_NEGO_LINK_WIDTH	R	Negotiated Link Width. This field indicates the negotiated width of the given PCI Express Link. All encodings other than the defined encodings are reserved. The value in this field is undefined when the Link is not up. Values: ■ 0x1 (X1): x1 ■ 0x2 (X2): x2 ■ 0x4 (X4): x4 ■ 0x8 (X8): x8 ■ 0xc (X12): x12 ■ 0x10 (X16): x16 ■ 0x20 (X32): x32 Value After Reset: * Varies Exists: (CX_MEMORY_MAP_PCIE_TYPE<2) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
19:16	PCIE_CAP_LINK_SPEED	R	Current Link Speed. This field indicates the negotiated Link speed of the given PCI Express Link. The encoded value specifies a bit location in the Supported Link Speeds Vector (in the Link Capabilities 2 register) that corresponds to the current Link speed. All encodings other than the defined encodings are reserved. The value in this field is undefined when the Link is not up. Values: ■ 0x1 (SUP_LINK_SPEED_FIELD_BIT_0): Supported Link Speeds Vector field bit 0 ■ 0x2 (SUP_LINK_SPEED_FIELD_BIT_1): Supported Link Speeds Vector field bit 1 ■ 0x3 (SUP_LINK_SPEED_FIELD_BIT_2): Supported Link Speeds Vector field bit 2 ■ 0x4 (SUP_LINK_SPEED_FIELD_BIT_3): Supported Link Speeds Vector field bit 3 ■ 0x5 (SUP_LINK_SPEED_FIELD_BIT_4): Supported Link Speeds Vector field bit 4 ■ 0x6 (SUP_LINK_SPEED_FIELD_BIT_5): Supported Link Speeds Vector field bit 5 ■ 0x7 (SUP_LINK_SPEED_FIELD_BIT_6): Supported Link Speeds Vector field bit 6 Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
15:14	PCIE_CAP_DRS_SIGNALING_CONTROL	R	DRS Signaling Control. Indicates the mechanism used to report reception of a DRS message. Must be implemented for Downstream Ports with the DRS Supported bit Set in the Link Capabilities 2 Register. Encodings are: If DRS Supported is set, receiving a DRS Message will set DRS Message Received in the Link Status 2 Register but will otherwise have no effect If the DRS Message Received bit in the Link Status 2 Register transitions from 0 to 1, and either MSI or MSI-X is enabled, an MSI or MSI-X interrupt is generated using the vector in Interrupt Message Number (section 7.5.3.2) If the DRS Message Received bit in the Link Status 2 Register transitions from 0 to 1, the Port must send an FRS Message Upstream with the FRS Reason field set to DRS Message Received. Behavior is undefined if this field is set to 10b and the FRS Supported bit in the Device Capabilities 2 Register is Clear. Behavior is undefined if this field is set to 11b. For Downstream Ports with the DRS Supported bit clear in the Link Capabilities 2 register, the controller hardwires this field to 00b. This field is Reserved for Upstream Ports. Values: ■ 0x0 (DRS_NOT_REPORTED): DRS not Reported ■ 0x1 (DRS_INTERRUPT_EN): DRS Interrupt Enabled ■ 0x2 (FRS_EN): DRS to FRS Signaling Enabled Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2
13:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2

Bits	Name	Memory Access	Description
11	PCIE_CAP_LINK_AUTO_BW_INT_EN	* Varies	Link Autonomous Bandwidth Management Interrupt Enable. When set, this bit enables the generation of an interrupt to indicate that the Link Autonomous Bandwidth Status bit has been set. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG. This bit is not applicable and is Reserved for Endpoints, PCI Express-to-PCI/PCI-X bridges, and Upstream Ports of Switches. For functions that do not implement the Link Bandwidth Notification Capability, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: LINK_CAPABILITIES_REG.PCIE_CAP_LINK_BW_NOT_CAP ? RW : RO ■ Dbi: LINK_CAPABILITIES_REG.PCIE_CAP_LINK_BW_NOT_CAP ? RW : RO Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Reset Mask: 0x0 Volatile: true
10	PCIE_CAP_LINK_BW_MAN_INT_EN	* Varies	Link Bandwidth Management Interrupt Enable. When set, this bit enables the generation of an interrupt to indicate that the Link Bandwidth Management Status bit has been set. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG. This bit is not applicable and is Reserved for Endpoints, PCI Express-to-PCI/PCI-X bridges, and Upstream Ports of Switches. For functions that do not implement the Link Bandwidth Notification Capability, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: LINK_CAPABILITIES_REG.PCIE_CAP_LINK_BW_NOT_CAP ? RW : RO ■ Dbi: LINK_CAPABILITIES_REG.PCIE_CAP_LINK_BW_NOT_CAP ? RW : RO Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
9	PCIE_CAP_HW_AUTO_WIDTH_DISABLE	* Varies	Hardware Autonomous Width Disable. When set, this bit disables hardware from changing the Link width for reasons other than attempting to correct unreliable Link operation by reducing Link width. For a Multi-Function Device associated with an Upstream Port, the bit in Function 0 is of type RW, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is of type RsvdP. For components that do not implement the ability autonomously to change Link width, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Memory Access: "(CX_MEMORY_MAP_VIEW==0 && 0==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && 0==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : (0==0) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	PCIE_CAP_EN_CLK_POWER_MAN	* Varies	Enable Clock Power Management. Applicable only for Upstream Ports and with form factors that support a "Clock Request" (CLKREQ#) mechanism, this bit operates as follows: For a non-ARI Multi-Function Device, power-management-configuration software must only Set this bit if all Functions of the Multi-Function Device indicate a 1b in the Clock Power Management bit of the Link Capabilities register. The component is permitted to use the CLKREQ# signal to power manage Link clock only if this bit is Set for all Functions. For ARI Devices, Clock Power Management is enabled solely by the setting in Function 0. The settings in the other Functions always return whatever value software programmed for each, but otherwise are ignored by the component. The CLKREQ# signal may also be controlled through the L1 PM Substates mechanism. Such control is not affected by the setting of this bit. For Downstream Ports and components that do not support Clock Power Management (as indicated by a 0b value in the Clock Power Management bit of the Link Capabilities register), the controller hardwires this bit to 0b. The write value is gated with the PCIE_CAP_CLOCK_POWER_MAN field in LINK_CAPABILITIES_REG. Note: The access attributes of this field are as follows: ■ Wire: LINK_CAPABILITIES_REG.PCIE_CAP_CLOCK_POWER_MAN ? RWS : ROS ■ Dbi: LINK_CAPABILITIES_REG.PCIE_CAP_CLOCK_POWER_MAN ? RWS : ROS Note: This register field is sticky. Values: ■ 0x0 (PM_DISABLE_LOW_CLKREQ): Clock power management is disabled and device must hold CLKREQ# signal low. ■ 0x1 (CLKREQ_IN_USE): When this bit is set, the device is permitted to use CLKREQ# signal to power manage Link clock according to protocol defined in appropriate form factor specification. Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
7	PCIE_CAP_EXTENDED_SYNC	R/W	Extended Synch. When set, this bit forces the transmission of additional Ordered Sets when exiting the L0s state (see section 4.2.4.5 of <i>PCI Express Base Specification</i>) and when in the Recovery state (see section 4.2.6.4.1 of <i>PCI Express Base Specification</i>). This mode provides external devices (for example, logic analyzers) monitoring the Link time to achieve bit and Symbol lock before the Link enters the L0 state and resumes communication. For Multi-Function devices if any function has this bit set, then the component must transmit the additional Ordered Sets when exiting L0s or when in Recovery. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Volatile: true
6	PCIE_CAP_COMMON_CLK_CONFIG	R/W	Common Clock Configuration. When set, this bit indicates that this component and the component at the opposite end of this Link are operating with a distributed common reference clock. A value of 0b indicates that this component and the component at the opposite end of this Link are operating with asynchronous reference clock. For non-ARI Multi-Function Devices, software must program the same value for this bit in all Functions. If not all Functions are Set, then the component must as a whole assume that its reference clock is not common with the Upstream component. For ARI Devices, Common Clock Configuration is determined solely by the setting in Function 0. The settings in the other Functions always return whatever value software programmed for each, but otherwise are ignored by the component. Components utilize this common clock configuration information to report the correct L0s and L1 Exit Latencies. After changing the value in this bit in both components on a Link, software must trigger the Link to retrain by writing a 1b to the Retrain Link bit of the Downstream Port. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2

Bits	Name	Memory Access	Description
5	PCIE_CAP_RETRAIN_LINK	* Varies	Retrain Link. A write of 1b to this bit initiates Link retraining by directing the Physical Layer LTSSM to the Recovery state. If the LTSSM is already in Recovery or Configuration, re-entering Recovery is permitted but not required. If the Port is in DPC when a write of 1b to this bit occurs, the result is undefined. Reads of this bit always return 0b. It is permitted to write 1b to this bit while simultaneously writing modified values to other fields in this register. If the LTSSM is not already in Recovery or Configuration, the resulting Link training must use the modified values. If the LTSSM is already in Recovery or Configuration, the modified values are not required to affect the Link training that's already in progress. This bit is not applicable and is Reserved for Endpoints, PCI Express to PCI/PCI-X bridges, and Upstream Ports of Switches. This bit always returns 0b when read. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: see description ■ Dbi: see description Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
4	PCIE_CAP_LINK_DISABLE	* Varies	Link Disable. This bit disables the Link by directing the LTSSM to the Disabled state when set; this bit is Reserved on Endpoints, PCI Express to PCI/PCI-X bridges, and Upstream Ports of Switches. Writes to this bit are immediately reflected in the value read from the bit, regardless of actual Link state. After clearing this bit, software must honor timing requirements defined in Section 6.6.1 with respect to the first Configuration Read following a Conventional Reset. In a DSP that supports crosslink, the controller gates the write value with the CROSS_LINK_EN field in PORT_LINK_CTRL_OFF. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: CX_CROSSLINK_ENABLE=1 && PORT_LINK_CTRL_OFF.CROSS_LINK_EN=1 CX_CROSSLINK_ENABLE=0 && dsp=1 ? RW : RO ■ Dbi: CX_CROSSLINK_ENABLE=1 && PORT_LINK_CTRL_OFF.CROSS_LINK_EN=1 CX_CROSSLINK_ENABLE=0 && dsp=1? RW : RO Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
3	PCIE_CAP_RCB	* Varies	Read Completion Boundary (RCB). <i>Endpoints and Bridges:</i> Optionally set by configuration software to indicate the RCB value of the Root Port Upstream from the Endpoint or Bridge. Refer to Section 2.3.1.1 of <i>PCI Express Base Specification</i> for the definition of the parameter RCB is same as Root Port. Configuration software must only set this bit if the Root Port Upstream from the Endpoint or Bridge reports an RCB value of 128 bytes (a value of 1b in the Read Completion Boundary bit). For functions that do not implement this feature, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (_64B): 64 byte ■ 0x1 (_128B): 128 byte Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_RCB_SUPPORT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_RCB_SUPPORT) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_RCB_SUPPORT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_RCB_SUPPORT) ? \"read-only\" : (CX_RCB_SUPPORT) ? \"read-write\" : \"read-only\""
2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2

Bits	Name	Memory Access	Description
1:0	PCIE_CAP_ACTIVE_STATE_LINK_PM_CONTROL	R/W	Active State Power Management (ASPM) Control. This field controls the level of ASPM enabled on the given PCI Express Link. See section 5.4.1.3 of <i>PCI Express Base Specification</i> for requirements on when and how to enable ASPM. Note: "L0s Entry Enabled" enables the Transmitter to enter L0s. If L0s is supported, the Receiver must be capable of entering L0s even when the Transmitter is disabled from entering L0s (00b or 10b). ASPM L1 must be enabled by software in the Upstream component on a Link prior to enabling ASPM L1 in the Downstream component on that Link. When disabling ASPM L1, software must disable ASPM L1 in the Downstream component on a Link prior to disabling ASPM L1 in the Upstream component on that Link. ASPM L1 must only be enabled on the Downstream component if both components on a Link support ASPM L1. For Multi-Function Devices (including ARI Devices), it is recommended that software program the same value for this field in all Functions. For non-ARI Multi-Function Devices, only capabilities enabled in all Functions are enabled for the component as a whole. For ARI Devices, ASPM Control is determined solely by the setting in Function0, regardless of Function 0's D-state. The settings in the other Functions always return whatever value software programmed for each, but otherwise are ignored by the component. Software must not enable L0s in either direction on a given Link unless components on both sides of the Link each support L0s; otherwise, the result is undefined. Values: ■ 0x0 (DISABLED): Disabled ■ 0x1 (L0S_ENTRY_EN): L0s Entry Enabled ■ 0x2 (L1_ENTRY_EN): L1 Entry Enabled ■ 0x3 (L0S_L1_ENTRY_EN): L0s and L1 Entry Enabled Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2

3.5.6 DEVICE_CAPABILITIES2_REG

- **Name:** Device Capabilities 2 Register.
- **Description:** This register identifies PCI Express device specific capabilities; in addition to the Device Capabilities Register.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** Always

PCIE_CAP_FRS_SUPPORTED	31
PCIE_CAP_DMWR_LEN_SUPP	30:29
PCIE_CAP_DMWR_CPL_SUPP	28
RSVDP_27	27
PCIE_CAP2_EMERGENCY_PWR_RED_INIT_REQUIRED	26
PCIE_CAP2_EMERGENCY_PWR_RED_SUPPORT	25:24
PCIE_CAP2_CFG_MAX_END2END_TLP_PRFXS	23:22
PCIE_CAP2_CFG_END2END_TLP_PRFX_SUPPORT	21
PCIE_CAP2_CFG_EXTND_FMT_SUPPORT	20
PCIE_CAP_OBFF_SUPPORT	19:18
PCIE_CAP2_10_BIT_TAG_REQ_SUPPORT	17
PCIE_CAP2_10_BIT_TAG_COMP_SUPPORT	16
PCIE_CAP2_LN_SYS_CLS	15:14
PCIE_CAP_TPH_CMPLT_SUPPORT_1	13
PCIE_CAP_TPH_CMPLT_SUPPORT_0	12
PCIE_CAP_LTR_SUPP	11
PCIE_CAP_NO_RO_EN_PR2PR_PAR	10
PCIE_CAP_128_CAS_CPL_SUPP	9
PCIE_CAP_64_ATOMIC_CPL_SUPP	8
PCIE_CAP_32_ATOMIC_CPL_SUPP	7
PCIE_CAP_ATOMIC_ROUTING_SUPPORT	6
PCIE_CAP_ARI_FORWARD_SUPPORT	5
PCIE_CAP_CPL_TIMEOUT_DISABLE_SUPPORT	4
PCIE_CAP_CPL_TIMEOUT_RANGE	3:0

Table 3-40 Fields for Register: DEVICE_CAPABILITIES2_REG

Bits	Name	Memory Access	Description
31	PCIE_CAP_FRS_SUPPORTED	* Varies	FRS Supported. When set, indicates support for the optional Function Readiness Status (FRS) capability. Must be set for all Functions that support generation or reception capabilities of FRS Messages. Must not be set by Switch Functions that do not generate FRS Messages on their own behalf. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_RN_FRS_VALUE_0} Exists: CX_RN_FRS_SUPPORTED Testable: wire ? unconstrained : writeAsRead
30:29	PCIE_CAP_DMWR_LEN_SUPP	R	Deferrable Memory Write (DMWr) Lengths Supported. Applicable to Functions with either the DMWr Routing Supported bit Set or the DMWR Completer Supported bit Set (or both). This field indicates the largest DMWr TLP that this Function can receive. When applicable, all Functions in a Multi-Function Device associated with an Upstream Port must report the same value in this field. This field is RsvdP if both DMWr Completer Supported and DMWr Routing Supported are Clear. For more information, see Section 6.30 of <i>PCI Express Base Specification</i>. Values: ■ 0x0 (up_to_64_bytes): 0 up to 64 bytes ■ 0x1 (up_to_128_bytes): 1 up to 128 bytes Value After Reset: {((CX_DEF_MEM_WR_CPL_EN==1) (CX_DEF_MEM_WR_ROUTING_EN==1)) ? CX_DEF_MEM_WR_LEN_SUPP : 0} Exists: Always

Bits	Name	Memory Access	Description
28	PCIE_CAP_DMWR_CPL_SUPP	R	Deferrable Memory Write (DMWr) Completer Supported. Applicable to Functions with Memory Space BARs as well as all Root Ports; This bit must be Set if the Function can serve as a DMWr Completer. For more information, see Section 6.30 of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_DEF_MEM_WR_CPL_EN} Exists: Always
27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: Always
26	PCIE_CAP2_EMERGENCY_PWR_RED_INIT_REQUIRED	* Varies	Emergency Power Reduction Initialization Required. Indicates if Software initialization upon exit from the Emergency Power Reduction State is required. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(Sticky) else R(Sticky) Values: ■ 0x0 (CLEAR): The Function requires no software intervention to return to normal operation upon exit from the Emergency Power Reduction State. ■ 0x1 (SET): The Function requires complete or partial initialization upon exit from the Emergency Power Reduction State. Value After Reset: 0x0 Exists: EMERGENCY_POWER_REDUCTION_ENABLE Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25:24	PCIE_CAP2_EMERGENCY_PWR_RED_SUPPORT	* Varies	Emergency Power Reduction Supported. Indicates support level of the optional Emergency Power Reduction State feature. For Multi-Function Devices associated with an Upstream Port, all Functions that report a non-zero value for this field, must report the same non-zero value for this field. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(Sticky) else R(Sticky) Values: ■ 0x0 (NO_SUP): Emergency Power Reduction State not supported. ■ 0x1 (SUP_01b): Emergency Power Reduction State is supported and is triggered by Device Specific mechanism(s). ■ 0x2 (SUP_10b): Emergency Power Reduction State is supported and is triggered either by the mechanism defined in the corresponding Form Factor specification or by Device Specific mechanism(s). ■ 0x3 (RSVD_4): Reserved Value After Reset: 0x0 Exists: EMERGENCY_POWER_REDUCTION_ENABLE Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23:22	PCIE_CAP2_CFG_MAX_END2END_TLP_PRFXS	* Varies	Max End-End TLP Prefixes. Indicates the maximum number of End-End TLP Prefixes supported by this Function. For more information, see Section 2.2.10.2 of <i>PCI Express Base Specification</i>. If End-End TLP Prefix Supported is clear, this field is RsvdP. Different Root Ports that have the End-End TLP Prefix Supported bit set are permitted to report different values for this field. For Switches where End-End TLP Prefix Supported is set, this field must be 00b indicating support for up to four End-End TLP Prefixes. Note: The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R Note: This register field is sticky. Values: ■ 0x1 (ONE_END_END_TLP_PREFIX): 1 End-End TLP Prefix ■ 0x2 (TWO_END_END_TLP_PREFIX): 2 End-End TLP Prefixes ■ 0x3 (THREE_END_END_TLP_PREFIX): 3 End-End TLP Prefixes ■ 0x0 (FOUR_END_END_TLP_PREFIX): 4 End-End TLP Prefixes Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_MAX_END2END_TLP_PRFXS} Exists: CX_TLP_PREFIX_ENABLE Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
21	PCIE_CAP2_CFG_END2END_TLP_PRFX_SUPPORT	* Varies	End-End TLP Prefix Supported. Indicates whether End-End TLP Prefix support is offered by a Function. Values are: All Ports of a Switch must have the same value for this bit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NO_SUP): No Support ■ 0x1 (SUP): Support is provided to receive TLPs containing End-End TLP Prefixes. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_TLP_PREFIX_ENABLE_VALUE} Exists: CX_TLP_PREFIX_ENABLE Testable: wire ? unconstrained : writeAsRead
20	PCIE_CAP2_CFG_EXTND_FMT_SUPPORT	* Varies	Extended Fmt Field Supported. If set, the Function supports the 3-bit definition of the Fmt field. If clear, the Function supports a 2-bit definition of the Fmt field. For more information, see section 2.2 of <i>PCI Express Base Specification</i>. Must be set for Functions that support End-End TLP Prefixes. All Functions in an Upstream Port must have the same value for this bit. Each Downstream Port of a component may have a different value for this bit. It is strongly recommended that Functions support the 3-bit definition of the Fmt field. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? "0x0" : "0x1"} Exists: CX_TLP_PREFIX_ENABLE Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:18	PCIE_CAP_OBFF_SUPPORT	* Varies	OBFF Supported. This field indicates if OBFF is supported and, if so, what signaling mechanism is used. The value reported in this field must indicate support for WAKE# signaling only if: ■ for a Downstream Port, driving the WAKE# signal for OBFF is supported and the connector or component connected Downstream is known to receive that same WAKE# signal ■ for an Upstream Port, receiving the WAKE# signal for OBFF is supported and, if the component is on an add-in-card, that the component is connected to the WAKE# signal on the connector. Root Ports, Switch Ports, and Endpoints are permitted to implement this capability. For a Multi-Function Device associated with an Upstream Port, each Function must report the same value for this field. For Bridges and Ports that do not implement this capability, the controller hardwires this field to 00b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(Sticky) else R(Sticky) and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NOT_SUP): OBFF Not Supported ■ 0x1 (SUP_USING_MSG): OBFF supported using Message signaling only ■ 0x2 (SUP_USING_WAKE): OBFF supported using WAKE# signaling only ■ 0x3 (SUP_USING_MSG_AND_WAKE): OBFF supported using WAKE# and Message signaling Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_OBFF_SUPPORT} Exists: CX_OBFF_SUPPORT Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
17	PCIE_CAP2_10_BIT_TAG_REQ_SUPPORT	R	10-Bit Tag Requester Supported. If this bit is set, the Function supports 10-Bit Tag Requester capability; otherwise, the Function does not. This bit must not be set if the 10-Bit Tag Completer Supported bit is clear. <i>Note:</i> 10-Bit Tag field generation must be enabled by the 10-Bit Tag Requester Enable bit in the Device Control 2 register of the Requester Function before 10-Bit Tags can be generated by the Requester. For more information, see section 2.2.6.2. of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_10BITS_TAG_REQ_VALUE} Exists: Always
16	PCIE_CAP2_10_BIT_TAG_COMP_SUPPORT	R	10-Bit Tag Completer Supported. If this bit is set, the Function supports 10-Bit Tag Completer capability; otherwise, the Function does not. For more information, see section 2.2.6.2. of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_10BITS_TAG_VALUE} Exists: Always
15:14	PCIE_CAP2_LN_SYS_CLS	R	LN System CLS. Applicable only to Root Ports and RCRBs; must be 00b for all other Function types. This field indicates if the Root Port or RCRB supports LN protocol as an LN Completer, and if so, what cacheline size is in effect. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (LN_COMPLETER_NO_SUP): LN Completer either not supported or not in effect ■ 0x1 (LN_COMPLETER_64B_CACHE): LN Completer with 64-byte cachelines in effect ■ 0x2 (LN_COMPLETER_128B_CACHE): LN Completer with 128-byte cachelines in effect Value After Reset: 0x0 Exists: Always
13	PCIE_CAP_TPH_CMPLT_SUPP_ORT_1	R	TPH Completer Supported Bit 1. Value After Reset: 0x0 Exists: Always

Bits	Name	Memory Access	Description
12	PCIE_CAP_TPH_CMPLT_SUPP0RT_0	R	TPH Completer Supported Bit 0. Value of this bit along with TPH Completer Supported Bit 1 indicates Completer support for TPH or Extended TPH. Applicable only to Root Ports and Endpoints. For all other Functions, this field is Reserved. For more information, see section 6.17 of <i>PCI Express Base Specification</i>. Values: 0x0 (TPH_EXT_TPH_CMPL_NOT_SUP_OR_RSVD): TPH and Extended TPH Completer not supported (if TPH Completer Supported Bit is 0) or Reserved ((if TPH Completer Supported Bit is 1). 0x1 (TPH_SUP_EXT_TPH_NOT_SUP_OR_BOTH_SUPP): TPH Completer supported; Extended TPH Completer not supported (if TPH Completer Supported Bit is 0) or Both TPH and Extended TPH Completer supported ((if TPH Completer Supported Bit is 1). Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_TPH_ENABLE_VALUE} Exists: Always
11	PCIE_CAP_LTR_SUPP	* Varies	LTR Mechanism Supported. A value of 1b indicates support for the optional Latency Tolerance Reporting (LTR) mechanism. Root Ports, Switches and Endpoints are permitted to implement this capability. For a Multi-Function Device associated with an Upstream Port, each Function must report the same value for this bit. For Bridges and other Functions that do not implement this capability, the controller hardwires this bit to 0b. Note: The access attributes of this field are as follows: - Wire: R (sticky) - Dbi: if (DBI_RO_WR_EN == 1) then R/W(Sticky) else R(Sticky) and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R Note: This register field is sticky. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_LTR_M_ENABLE_VALUE} Exists: CX_LTR_M_ENABLE_VALUE Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
10	PCIE_CAP_NO_RO_EN_PR2PR_PAR	R	No RO-enabled PR-PR Passing. If this bit is set, the routing element never carries out the passing permitted by Table 2-39 of <i>PCI Express Base Specification</i> entry A2b that is associated with the Relaxed Ordering Attribute field being Set. This bit applies only for Switches and RCs that support peer-to-peer traffic between Root Ports. This bit applies only to Posted Requests being forwarded through the Switch or RC and does not apply to traffic originating or terminating within the Switch or RC itself. All Ports on a Switch or RC must report the same value for this bit. For all other functions, this bit must be 0b. Value After Reset: 0x0 Exists: Always
9	PCIE_CAP_128_CAS_CPL_SUPP	R	128-bit CAS Completer Supported. Applicable to Functions with Memory Space BARs as well as all Root Ports; must be 0b otherwise. This bit must be set to 1b if the Function supports this optional capability. For more information, see section 6.15 of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_ATOMIC_128_CAS_EN} Exists: Always
8	PCIE_CAP_64_ATOMIC_CPL_SUPP	R	64-bit AtomicOp Completer Supported. Applicable to Functions with Memory Space BARs as well as all Root Ports; must be 0b otherwise. Includes FetchAdd, Swap, and CAS AtomicOps. This bit must be set to 1b if the Function supports this optional capability. For more information on additional RC requirements, see section 6.15.3.1 of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_ATOMIC_64_CPL_EN} Exists: Always
7	PCIE_CAP_32_ATOMIC_CPL_SUPP	R	32-bit AtomicOp Completer Supported. Applicable to Functions with Memory Space BARs as well as all Root Ports; must be 0b otherwise. Includes FetchAdd, Swap, and CAS AtomicOps. This bit must be set to 1b if the Function supports this optional capability. For more information on additional RC requirements, see section 6.15.3.1 of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_ATOMIC_32_CPL_EN} Exists: Always

Bits	Name	Memory Access	Description
6	PCIE_CAP_ATOMIC_ROUTING_SUPP	R	AtomicOp Routing Supported. Applicable only to Switch Upstream Ports, Switch Downstream Ports, and Root Ports; must be 0b for other Function types. This bit must be set to 1b if the Port supports this optional capability. For more information, see section 6.15 of <i>PCI Express Base Specification</i>. Value After Reset: 0x0 Exists: Always
5	PCIE_CAP_ARI_FORWARD_SUPPORT	R	ARI Forwarding Supported. Applicable only to Switch Downstream Ports and Root Ports; must be 0b for other Function types. This bit must be set to 1b if a Switch Downstream Port or Root Port supports this optional capability. For more information, see section 6.13 of <i>PCI Express Base Specification</i>. Value After Reset: 0x0 Exists: Always
4	PCIE_CAP_CPL_TIMEOUT_DISABLE_SUPPORT	R	Completion Timeout Disable Supported. A value of 1b indicates support for the Completion Timeout Disable mechanism. The Completion Timeout Disable mechanism is required for Endpoints that issue Requests on their own behalf and PCI Express to PCI/PCI-X Bridges that take ownership of Requests issued on PCI Express. This mechanism is optional for Root Ports. For all other Functions this field is Reserved and the controller hardwires this bit to 0b. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? "0x0" : "0x1"} Exists: Always

Bits	Name	Memory Access	Description
3:0	PCIE_CAP_CPL_TIMEOUT_RANGE	R	Completion Timeout Ranges Supported. This field indicates device Function support for the optional Completion Timeout programmability mechanism. This mechanism allows system software to modify the Completion Timeout value. This field is applicable only to Root Ports, Endpoints that issue Requests on their own behalf, and PCI Express to PCI/PCI-X Bridges that take ownership of Requests issued on PCI Express. For all other Functions this field is Reserved and must be hardwired to 0000b. Four time value ranges are defined: ■ Range A: 50 us to 10 ms ■ Range B: 10 ms to 250 ms ■ Range C: 250 ms to 4 s ■ Range D: 4 s to 64 s Bits are set according to the list below to show timeout value ranges supported. All encodings other than the defined encodings are reserved. It is strongly recommended that the Completion Timeout mechanism not expire in less than 10 ms. Values: ■ 0x0 (NOT_SUP): Completion Timeout programming not supported, the Function must implement a timeout value in the range 50 us to 50 ms. ■ 0x1 (RANGE_A): Range A ■ 0x2 (RANGE_B): Range B ■ 0x3 (RANGE_A_B): Ranges A and B ■ 0x6 (RANGE_B_C): Ranges B and C ■ 0x7 (RANGE_A_B_C): Ranges A, B, and C ■ 0xe (RANGE_B_C_D): Ranges B, C, and D ■ 0xf (RANGE_A_B_C_D): Ranges A, B, C, and D Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? "0x0" : ((CX_CPL_TO_RANGES_ENABLE==1) ? 0xf : 0)} Exists: Always

3.5.7 DEVICE_CONTROL2_DEVICE_STATUS2_REG

- **Name:** Device Control 2 and Status 2 Register.
- **Description:** This register controls PCI Express device specific parameters and provides information about PCI Express device (function) specific parameters; in addition to the Device Control and Device Status Register.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** Always

31:16	RSVDP_16
15	PCIE_CTRL2_CFG_END2END_TLP_PFX_BLK
14:13	PCIE_CAP_OBFF_EN
12	PCIE_CAP_10BITS_TAG_REQ_EN
11	PCIE_CAP_EMERGENCY_PWR_RED_REQUEST
10	PCIE_CAP_LTR_EN
9	PCIE_CAP_IDO_CPL_EN
8	PCIE_CAP_IDO_REQ_EN
7	PCIE_CAP_ATOMIC_EGRESS_BLK
6	PCIE_CAP_ATOMIC_REQ_EN
5	PCIE_CAP_ARI_FORWARD_SUPPORT_CS
4	PCIE_CAP_CPL_TIMEOUT_DISABLE
3:0	PCIE_CAP_CPL_TIMEOUT_VALUE

Table 3-41 Fields for Register: DEVICE_CONTROL2_DEVICE_STATUS2_REG

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_TLP_PREFIX_ENABLE

Bits	Name	Memory Access	Description
15	PCIE_CTRL2_CFG_END2END_TLP_PFX_BLK	R	End-End TLP Prefix Blocking. Controls whether the routing function is permitted to forward TLPs containing an End-End TLP Prefix. This bit affects TLPs that exit the Switch/Root Complex using the associated Port. It does not affect TLPs forwarded internally within the Switch/Root Complex. It does not affect TLPs that enter through the associated Port, that originate in the associated Port or originate in a Root Complex Integrated Device integrated with the associated Port. Blocked TLPs are reported by the TLP Prefix Blocked Error. This bit is hardwired to 1b in Root Ports that support End-End TLP Prefixes but do not support forwarding of End-End TLP Prefixes. This bit is applicable to Root Ports and Switch Ports where the End-End TLP Prefix Supported bit is set. This bit is not applicable and is RsvdP in all other cases. Values: ■ 0x0 (FORWARDING_EN): Forwarding Enabled, Function is permitted to send TLPs with End-End TLP Prefixes. ■ 0x1 (FORWARDING_BLK): Forwarding Blocked, Function is not permitted to send TLPs with End-End TLP Prefixes. Value After Reset: 0x0 Exists: CX_TLP_PREFIX_ENABLE

Bits	Name	Memory Access	Description
14:13	PCIE_CAP_OBFF_EN	* Varies	OBFF Enable. This field enables the OBFF mechanism and selects the signaling method. For more information on these encodings, see section 6.19 of <i>PCI Express Base Specification</i>. This field is required for all Ports that support the OBFF Capability. For a Multi-Function Device associated with an Upstream Port of a Device that implements OBFF, the field in Function 0 is of type RW, and only Function 0 controls the Component's behavior. In all other Functions of that Device, this field is of type RsvdP. Ports that do not implement OBFF are permitted to hardwire this field to 00b Note: RW for function #0 and RsvdP for all other functions. Note: The access attributes of this field are as follows: ■ Wire: if (pf=0 && DEVICE_CAPABILITIES2_REG.PCIE_CAP_OBFF_SUPPORT) then R/W else R ■ Dbi: if (pf=0 && DEVICE_CAPABILITIES2_REG.PCIE_CAP_OBFF_SUPPORT) then R/W else R Values: ■ 0x0 (DISABLED): Disabled ■ 0x1 (EN_USING_MSG_SIG_A): Enabled using Message signaling [Variation A] ■ 0x2 (EN_USING_MSG_SIG_B): Enabled using Message signaling [Variation B] ■ 0x3 (EN_USING_WAKE_SIG): Enabled using WAKE# signaling Value After Reset: 0x0 Exists: CX_OBFF_ENABLE Testable: writeAsRead

Bits	Name	Memory Access	Description
12	PCIE_CAP_10BITS_TAG_REQ_EN	* Varies	10-Bit Tag Requester Enable. 10-Bit Tag Requester Enable. This bit, in combination with the Extended Tag Field Enable bit in the Device Control register, determines how many Tag field bits a Requester is permitted to use. When the 10-Bit Tag Requester Enable bit is Set, the Requester is permitted to use 10-Bit Tags. For more information, see section 2.2.6.2 of <i>PCI Express Base Specification</i>. If software changes the value of this bit while the Function has outstanding Non-Posted Requests, the result is undefined. Functions that do not implement 10-Bit Tag Requester capability are permitted to hardwire this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : 0} Exists: CX_10BITS_TAG_REQ Memory Access: "(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? \"read-only\" : \"read-write\""
11	PCIE_CAP_EMERGENCY_PWR_RED_REQUEST	* Varies	Emergency Power Reduction Request. This bit is implemented in the lowest numbered Function associated with an Upstream Port that has a non-zero value in the Emergency Power Reduction Supported field. This bit is RsvdP in all other Functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (CLEAR): If Clear these Functions must exit the Emergency Power Reduction State if no other reasons exist to preclude exiting this state. ■ 0x1 (SET): If Set, all Functions in the component that support Emergency Power Reduction State must enter the Emergency Power Reduction State. Value After Reset: 0x0 Exists: EMERGENCY_POWER_REDUCTION_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && 0==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && 0==0) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
10	PCIE_CAP_LTR_EN	* Varies	LTR Mechanism Enable. When set to 1b, this bit enables Upstream Ports to send LTR messages and Downstream Ports to process LTR Messages. For a Multi-Function Device associated with an Upstream Port of a device that implements LTR, the bit in Function 0 is RW, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is RsvdP. Functions that do not implement the LTR mechanism are permitted to hardwire this bit to 0b. For Downstream Ports, this bit must be reset to the default value if the Port goes to DL_Down status. The write value is gated with the PCIE_CAP_LTR_SUPP field of DEVICE_CAPABILITIES2_REG. <i>Note:</i> RW for function #0 and RsvdP for all other functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: if (pf=0 && DEVICE_CAPABILITIES2_REG.PCIE_CAP_LTR_SUPP) then R/W else R ■ Dbi: if (pf=0 && DEVICE_CAPABILITIES2_REG.PCIE_CAP_LTR_SUPP) then R/W else R Value After Reset: 0x0 Exists: CX_LTR_M_ENABLE Testable: writeAsRead
9	PCIE_CAP_IDO_CPL_EN	* Varies	IDO Completion Enable. If this bit is set, the Function is permitted to set the ID-Based Ordering (IDO) bit (Attr[2]) of Completions it returns (for more details, see section 2.2.6.3 and Section 2.4 of <i>PCI Express Base Specification</i>). Endpoints and Root Ports are permitted to implement this capability. A Function is permitted to hardwire this bit to 0b if it never sets the IDO attribute in Completions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_IDO_ENABLE Memory Access: "(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
8	PCIE_CAP_IDO_REQ_EN	* Varies	IDO Request Enable. If this bit is set, the Function is permitted to set the ID-Based Ordering (IDO) bit (Attr[2]) of Requests it initiates (for more details, see section 2.2.6.3 and section 2.4 of <i>PCI Express Base Specification</i>). Endpoints and Root Ports are permitted to implement this capability. A Function is permitted to hardwire this bit to 0b if it never sets the IDO attribute in Requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_IDO_ENABLE && CC_DEVICE_TYPE!=3) Memory Access: "(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? \"read-only\" : \"read-write\" "
7	PCIE_CAP_ATOMIC_EGRESS_BLOCK	R	AtomicOp Egress Blocking. AtomicOp Egress Blocking. Applicable and mandatory for Switch Upstream Ports, Switch Downstream Ports, and Root Ports that implement AtomicOp routing capability; otherwise the controller hardwires this bit to 0b. When this bit is set, AtomicOp Requests that target going out this Egress Port must be blocked. For more information, see section 6.15.2 of <i>PCI Express Base Specification</i>. Reflected on the cfg_atomic_egress_block output. Value After Reset: 0x0 Exists: CX_ATOMIC_ENABLE

Bits	Name	Memory Access	Description
6	PCIE_CAP_ATOMIC_REQ_EN	* Varies	AtomicOp Requester Enable. Applicable only to Endpoints and Root Ports; the controller hardwires this bit to 0b for other Function types. The Function is allowed to initiate AtomicOp Requests only if this bit and the Bus Master Enable bit in the Command register are both set. This bit is required to be RW if the Endpoint or Root Port is capable of initiating AtomicOp Requests, but otherwise is permitted to be hardwired to 0b. This bit does not serve as a capability bit. This bit is permitted to be RW even if no AtomicOp Requester capabilities are supported by the Endpoint or Root Port. Reflected on the <code>cfg_atomic_req_en</code> output. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_ATOMIC_ENABLE Memory Access: "(CX_MEMORY_MAP_POSITION==0 && CX_MEMORY_MAP_PCIE_TYPE==1) ? \"read-only\" : \"read-write\" "
5	PCIE_CAP_ARI_FORWARD_SUP PORT_CS	R	ARI Forwarding Enable. When set, the Downstream Port disables its traditional Device Number field being 0 enforcement when turning a Type 1 Configuration Request into a Type 0 Configuration Request, permitting access to Extended Functions in an ARI Device immediately below the Port. For more information, see Section 6.13 of <i>PCI Express Base Specification</i>. Value After Reset: 0x0 Exists: Always

Bits	Name	Memory Access	Description
4	PCIE_CAP_CPL_TIMEOUT_DISABLE	* Varies	Completion Timeout Disable. When set, this bit disables the Completion Timeout mechanism. This bit is required for all Functions that support the Completion Timeout Disable Capability. Functions that do not support this optional capability are permitted to hardwire this bit to 0b. Software is permitted to set or clear this bit at any time. When set, the Completion Timeout detection mechanism is disabled. If there are outstanding Requests when the bit is cleared, it is permitted but not required for hardware to apply the completion timeout mechanism to the outstanding Requests. If this is done, it is permitted to base the start time for each Request on either the time this bit was cleared or the time each Request was issued. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : 0} Exists: Always Memory Access: "(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
3:0	PCIE_CAP_CPL_TIMEOUT_VALUE	* Varies	Completion Timeout Value. In device Functions that support Completion Timeout programmability, this field allows system software to modify the Completion Timeout value. This field is applicable to Root Ports, Endpoints that issue Requests on their own behalf, and PCI Express to PCI/PCI-X Bridges that take ownership of Requests issued on PCI Express. For all other Functions this field is Reserved and controller hardwires it to 0000b. A Function that does not support this optional capability must hardwire this field to 0000b and is required to implement a timeout value in the range 50 us to 50 ms. Functions that support Completion Timeout programmability must support the values given below corresponding to the programmability ranges indicated in the Completion Timeout Ranges Supported field. All encodings other than the defined encodings are reserved. It is strongly recommended that the Completion Timeout mechanism not expire in less than 10 ms. Values available if Range A (50 us to 10 ms) programmability range is supported: Values available if Range B (10 ms to 250 ms) programmability range is supported: Values available if Range C (250 ms to 4 s) programmability range is supported: Values available if the Range D (4 s to 64 s) programmability range is supported: Software is permitted to change the value in this field at any time. For Requests already pending when the Completion Timeout Value is changed, hardware is permitted to use either the new or the old value for the outstanding Requests, and is permitted to base the start time for each Request either on when this value was changed or on when each request was issued. <i>Note</i>

Bits	Name	Memory Access	Description
3:0...(cont.)	PCIE_CAP_CPL_TIMEOUT_VALUE.	* Varies	: The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (DEFAULT): Default range: 50 us to 50 ms ■ 0x1 (_50_TO_100_US): 50 us to 100 us ■ 0x2 (_1_TO_10_MS): 1 ms to 10 ms ■ 0x5 (_16_TO_55_MS): 16 ms to 55 ms ■ 0x6 (_65_TO_210_MS): 65 ms to 210 ms ■ 0x9 (_260_TO_900_MS): 260 ms to 900 ms ■ 0xa (_1_TO_3_5_S): 1 s to 3.5 s ■ 0xd (_4_TO_13_S): 4 s to 13 s ■ 0xe (_17_TO_64_S): 17 s to 64 s Value After Reset: 0x0 Exists: Always Memory Access: "(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? \"read-only\" : (CC_DEVICE_TYPE == 3 && !CX_MEMORY_MAP_POSITION) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==0 && CX_CPL_TO_RANGES_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_CPL_TO_RANGES_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_CPL_TO_RANGES_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_CPL_TO_RANGES_ENABLE) ? \"read-only\" : (CX_CPL_TO_RANGES_ENABLE)? \"read-write\": \"read-only\" "

3.5.8 LINK_CAPABILITIES2_REG

- **Name:** Link Capabilities 2 Register.
- **Description:** This register identifies PCI Express Link specific capabilities and CXL-RCRB link specific capabilities; in addition to the Link Capabilities Register.
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** Always

DRS_SUPPORTED	31
RSVDP_25	30:25
PCIE_CAP_TWO_RETIMERS_PRE_DET_SUPPORT	24
PCIE_CAP_RETIMER_PRE_DET_SUPPORT	23
RSVDP_9	22:9
PCIE_CAP_CROSS_LINK_SUPPORT	8
PCIE_CAP_SUPPORT_LINK_SPEED_VECTOR	7:1
RSVDP_0	0

Table 3-42 Fields for Register: LINK_CAPABILITIES2_REG

Bits	Name	Memory Access	Description
31	DRS_SUPPORTED	* Varies	DRS Supported. When set, indicates support for the optional Device Readiness Status (DRS) capability. Must be Set in Downstream Ports that support DRS. Must be Set in Downstream Ports that support FRS. For Upstream Ports that support DRS, it is strongly recommended that this bit be Set in Function 0. For all other Functions associated with an Upstream Port, this bit must be Clear.¹²⁷ Must be Clear in Functions that are not associated with a Port. RsvdP in all other Functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE<2 && CX_DBI_RO_WR_EN && \$pf==0) ? R/W : (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE==2 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE==2) ? R : R Value After Reset: {(CX_RN_DRS_SUPPORTED && 0==0) ? 1 : 0} Exists: CX_RN_DRS_SUPPORTED && (CX_MEMORY_MAP_PCIE_TYPE<2) Testable: wire ? unconstrained : writeAsRead
30:25	RSVDP_25	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2

Bits	Name	Memory Access	Description
24	PCIE_CAP_TWO_RETIMERS_PRE_DET_SUPPORT	* Varies	Two Retimers Presence Detect Supported. When set to 1b, this bit indicates that the associated Port supports detection and reporting of two Retimers presence. This bit must be set to 1b in a Port when the Supported Link Speeds Vector of the Link Capabilities 2 register indicates support for a Link speed of 16.0 GT/s or higher. It is permitted to be set to 1b regardless of the supported Link speeds if the Retimer Presence Detect Supported bit is also set to 1b. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all Functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: 0x1 Exists: CX_GEN4_SPEED && CX_PL16G_ENABLE && CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Volatile: true
23	PCIE_CAP_RETIMER_PRE_DET_SUPPORT	* Varies	Retimer Presence Detect Supported. When set to 1b, this bit indicates that the associated Port supports detection and reporting of Retimer presence. This bit must be set to 1b in a Port when the Supported Link Speeds Vector of the Link Capabilities 2 register indicates support for a Link speed of 16.0 GT/s or higher. It is permitted to be set to 1b regardless of the supported Link speeds. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all Functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE<2 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && CX_MEMORY_MAP_PCIE_TYPE==2 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1 && CX_MEMORY_MAP_PCIE_TYPE==2) ? R : R Value After Reset: 0x1 Exists: CX_GEN4_SPEED && CX_PL16G_ENABLE && CX_MEMORY_MAP_PCIE_TYPE<2 Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
22:9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2
8	PCIE_CAP_CROSS_LINK_SUPPORT	R	Crosslink Supported. When set to 1b, this bit indicates that the associated Port supports crosslinks (for more details, see section 4.2.6.3.1 of <i>PCI Express Base Specification</i>). When set to 0b on a Port that supports Link speeds of 8.0 GT/s or higher, this bit indicates that the associated Port does not support crosslinks. When set to 0b on a Port that only supports Link speeds of 2.5 GT/s or 5.0 GT/s, this bit provides no information regarding the Port's level of crosslink support. It is recommended that this bit be Set in any Port that supports crosslinks even though doing so is only required for Ports that also support operating at 8.0 GT/s or higher Link speeds. <i>Note:</i> Software should use this bit when referencing fields whose definition depends on whether or not the Port supports crosslinks (for more details, see section 7.7.3.4 of <i>PCI Express Base Specification</i>). Multi-Function Devices associated with an Upstream Port must report the same value in this field for all Functions. Value After Reset: CX_CROSSLINK_ENABLE Exists: CX_MEMORY_MAP_PCIE_TYPE<2

Bits	Name	Memory Access	Description
7:1	PCIE_CAP_SUPPORT_LINK_SPEED_VECTOR	R	Supported Link Speeds Vector. This field indicates the supported Link speed(s) of the associated Port. For each bit, a value of 1b indicates that the corresponding Link speed is supported; otherwise, the Link speed is not supported. For more information, see section 8.2.1 of <i>PCI Express Base Specification</i>. Bit definitions within this field are: ■ 0: 2.5 GT/s ■ 1: 5.0 GT/s ■ 2: 8.0 GT/s ■ 3: 16.0 GT/s ■ 4: 32.0 GT/s ■ 6:5 RsvdP Multi-Function Devices associated with an Upstream Port must report the same value in this field for all Functions. This field has a default of (PCIE_CAP_MAX_LINK_SPEED == 0101) ? 0011111 : (PCIE_CAP_MAX_LINK_SPEED == 0100) ? 0001111 : (PCIE_CAP_MAX_LINK_SPEED == 0011) ? 0000111 : (PCIE_CAP_MAX_LINK_SPEED == 0010) ? 0000011 : 0000001 where PCIE_CAP_MAX_LINK_SPEED is a field in the LINK_CAPABILITIES_REG register. Values: ■ 0x0 (ini_val): Zero value. ■ 0x7f (def_last_val): last default value. Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 Testable: writeAsRead Reset Mask: 0x0 Volatile: true
0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2

3.5.9 LINK_CONTROL2_LINK_STATUS2_REG

- **Name:** Link Control 2 and Status 2 Register.
- **Description:** This register controls and provides information about PCI Express Link specific parameters as well as RCRB link associated parameters; in addition to the Link Control and Link Status Register.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** Always

31	DRS_MESSAGE_RECEIVED
30:28	DOWNSTREAM_COMPO_PRESENCE
27:26	RSVDP_26
25:24	PCIE_CAP_CROSSLINK_RESOLUTION
23	PCIE_CAP_TWO_RETIMERS_PRE_DET
22	PCIE_CAP_RETIMER_PRE_DET
21	PCIE_CAP_LINK_EQ_REQ
20	PCIE_CAP_EQ_CPL_P3
19	PCIE_CAP_EQ_CPL_P2
18	PCIE_CAP_EQ_CPL_P1
17	PCIE_CAP_EQ_CPL
16	PCIE_CAP_CURR_DEEMPHASIS
15:12	PCIE_CAP_COMPLIANCE_PRESET
11	PCIE_CAP_COMPLIANCE_SOS
10	PCIE_CAP_ENTER_MODIFIED_COMPLIANCE
9:7	PCIE_CAP_TX_MARGIN
6	PCIE_CAP_SEL_DEEMPHASIS
5	PCIE_CAP_HW_AUTO_SPEED_DISABLE
4	PCIE_CAP_ENTER_COMPLIANCE
3:0	PCIE_CAP_TARGET_LINK_SPEED

Table 3-43 Fields for Register: LINK_CONTROL2_LINK_STATUS2_REG

Bits	Name	Memory Access	Description
31	DRS_MESSAGE_RECEIVED	R	DRS Message Received. This bit must be set whenever the Port receives a DRS Message. This bit must be cleared in DL_Down. This bit must be implemented in any Downstream Port where the DRS Supported bit is Set in the Link Capabilities 2 register. This bit is RsvdZ for all other Functions. Value After Reset: 0x0 Exists: (CX_RN_DRS_SUPPORTED && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
30:28	DOWNSTREAM_COMPO_PRESENCE	R	Downstream Component Presence. This field indicates the presence and DRS status for the Downstream Component, if any, connected to the Link. All encodings other than the defined encodings are reserved. Component Presence state must be determined by the logical "OR" of the Physical Layer in-band presence detect mechanism and, if present, any out-of-band presence detect mechanism implemented for the Link. If no out-of-band presence detect mechanism is implemented, then Component Presence state must be determined solely by the Physical Layer in-band presence detect mechanism. This field must be implemented in any Downstream Port where the DRS Supported bit is Set in the Link Capabilities 2 register. This field is RsvdZ for all other Functions. Values: ■ 0x0 (LINK_DN_NOT_DETERMINED): Link Down - Presence Not Determined ■ 0x1 (LINK_DN_DSP_NOT_PRESENT): Link Down - Component Not Present indicates the Downstream Port (DP) has determined that a Downstream Component is not present ■ 0x2 (LINK_DN_DLL_NOT_ACTIVE): Link Down - Component Present indicates the DP has determined that a Downstream Component is present, but the Data Link Layer is not active ■ 0x4 (LINK_UP_DRS_NOT_RECEIVED): Link Up - Component Present indicates the DP has determined that a Downstream Component is present, but no DRS Message has been received since the Data Link Layer became active ■ 0x5 (LINK_UP_DRS_RECEIVED): Link Up - Component Present and DRS Received indicates the DP has received a DRS Message since the Data Link Layer became active Value After Reset: 0x0 Exists: (CX_RN_DRS_SUPPORTED && (CX_MEMORY_MAP_PCIE_TYPE<2))
27:26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
25:24	PCIE_CAP_CROSSLINK_RESOLUTION	R	Crosslink Resolution. This field indicates the state of the Crosslink negotiation. It must be implemented if Crosslink Supported is Set and the Port supports 16.0 GT/s or higher data rate. It is permitted to be implemented in all other Ports. If Crosslink Supported is clear, the controller hardwires this field to 01b or 10b. Once a value of 01b or 10b is returned in this field, that value must continue to be returned while the Link is Up. Values: ■ 0x0 (CROSSLINK_RESOLUTION_NOT_SUP): Crosslink Resolution is not supported. No information is provided regarding the status of the Crosslink negotiation. ■ 0x1 (CROSSLINK_NEGOTIATION_RESOLVED_USP): Crosslink negotiation resolved as an Upstream Port. ■ 0x2 (CROSSLINK_NEGOTIATION_RESOLVED_DSP): Crosslink negotiation resolved as a Downstream Port. ■ 0x3 (CROSSLINK_NEGOTIATION_NOT_CMPL): Crosslink negotiation is not completed. Value After Reset: {(CX_CROSSLINK_ENABLE) ? 0x3 : 0x1} Exists: (CX_GEN4_SPEED && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: writeAsRead Volatile: true
23	PCIE_CAP_TWO_RETIMERS_PRE_DET	R	Two Retimers Presence Detected. When set to 1b, this bit indicates that two Retimers were present during the most recent Link negotiation. For more information, see section 4.2.6.3.5.1 of <i>PCI Express Base Specification</i>. This bit is required for Ports that have the Two Retimers Presence Detect Supported bit of the Link Capabilities 2 register set to 1b. Ports that have the Two Retimers Presence Detect Supported bit set to 0b are permitted to hardwire this bit to 0b. For Multi-Function Devices associated with an Upstream Port, this bit must be implemented in Function 0 and RsvdZ in all other Functions. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN4_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
22	PCIE_CAP_RETIMER_PRE_DET	R	Retimer Presence Detected. When set to 1b, this bit indicates that a Retimer was present during the most recent Link negotiation. For more information, see section 4.2.6.3.5.1 of <i>PCI Express Base Specification</i>. This bit is required for Ports that have the Retimer Presence Detect Supported bit of the Link Capabilities 2 register set to 1b. For Ports that have the Retimer Presence Detect Supported bit set to 0b, the controller hardwires this bit to 0b. For Multi-Function Devices associated with an Upstream Port, this bit must be implemented in Function 0 and is RsvdZ in all other Functions. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN4_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
21	PCIE_CAP_LINK_EQ_REQ	R/W1C	Link Equalization Request 8.0 GT/s. This bit is set by hardware to request the 8.0 GT/s Link equalization process to be performed on the Link. For more information, see sections 4.2.3 and 4.2.6.4.2 of <i>PCI Express Base Specification</i>. For Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. For components that only support speeds below 8.0 GT/s, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
20	PCIE_CAP_EQ_CPL_P3	R	EEqualization 8.0 GT/s Phase 3 Successful. When set to 1b, this bit indicates that Phase 3 of the 8.0 GT/s Transmitter Equalization procedure has successfully completed. Details of the Transmitter Equalization process and when this bit needs to be set to 1b is provided in section 4.2.6.4.2 of <i>PCI Express Base Specification</i>. For Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. For components that only support speeds below 8.0 GT/s, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
19	PCIE_CAP_EQ_CPL_P2	R	Equalization 8.0 GT/s Phase 2 Successful. When set to 1b, this bit indicates that Phase 2 of the 8.0 GT/s Transmitter Equalization procedure has successfully completed. Details of the Transmitter Equalization process and when this bit needs to be set to 1b is provided in section 4.2.6.4.2 of <i>PCI Express Base Specification</i>. For Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. For components that only support speeds below 8.0 GT/s, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
18	PCIE_CAP_EQ_CPL_P1	R	Equalization 8.0 GT/s Phase 1 Successful. When set to 1b, this bit indicates that Phase 1 of the 8.0 GT/s Transmitter Equalization procedure has successfully completed. Details of the Transmitter Equalization process and when this bit needs to be set to 1b is provided in section 4.2.6.4.2 of <i>PCI Express Base Specification</i>. For Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. For components that only support speeds below 8.0 GT/s, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
17	PCIE_CAP_EQ_CPL	R	Equalization 8.0 GT/s Complete. When set to 1b, this bit indicates that the Transmitter Equalization procedure at the 8.0 GT/s data rate has completed. Details of the Transmitter Equalization process and when this bit needs to be set to 1b is provided in section 4.2.6.4.2 of <i>PCI Express Base Specification</i>. For Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. For components that only support speeds below 8.0 GT/s, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
16	PCIE_CAP_CURR_DEEMPHASIS	R	Current De-emphasis Level. When the Link is operating at 5.0 GT/s speed, this bit reflects the level of de-emphasis. The value in this bit is undefined when the Link is not operating at 5.0 GT/s speed. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 0b. For components that support speeds greater than 2.5 GT/s, Multi-Function Devices associated with an Upstream Port must report the same value in this field for all Functions of the Port. In C-PCIe mode, its contents are derived by sampling the PIPE. Values: ■ 0x1 (_3_5DB): -3.5 dB ■ 0x0 (_6DB): -6 dB Value After Reset: 0x1 Exists: (CX_GEN2_SPEED && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:12	PCIE_CAP_COMPLIANCE_PRES ET	* Varies	Compliance Preset/De-emphasis. ■ For 8.0 GT/s and higher Data Rate: This field sets the Transmitter Preset in Polling.Compliance state if the entry occurred due to the Enter Compliance bit being 1b. The encodings are defined in section 4.2.3.2 of <i>PCI Express Base Specification</i> . Results are undefined if a reserved preset encoding is used when entering Polling.Compliance in this way. ■ For 5.0 GT/s Data Rate: This field sets the de-emphasis level in Polling.Compliance state if the entry occurred due to the Enter Compliance bit being 1b. ■ When the Link is operating at 2.5 GT/s, the setting of this field has no effect. Components that support only 2.5 GT/s speed are permitted to hardwire this field to 0000b. ■ For a Multi-Function Device associated with an Upstream Port, the field in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this field is of type RsvdP. ■ This field is intended for debug and compliance testing purposes. System firmware and software is allowed to modify this field only during debug or compliance testing. In all other cases, the system must ensure that this field is set to the default value. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (_3_5_DB): -3.5 dB (for 5.0 GT/s Data Rate) ■ 0x0 (_6_DB): -6 dB (for 5.0 GT/s Data Rate) Value After Reset: 0x0 Exists:
15:12...(cont.)	PCIE_CAP_COMPLIANCE_PRES ET.	* Varies	(CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && 0 == 0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && 0 != 0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && 0 == 0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && 0 != 0) ? \"read-only\" : (0 == 0)? \"read-write\": \"read-only\" "

Bits	Name	Memory Access	Description
11	PCIE_CAP_COMPLIANCE_SOS	* Varies	Compliance SOS. When set to 1b, the LTSSM is required to send SKP Ordered Sets between sequences when sending the Compliance Pattern or Modified Compliance Pattern. For a Multi-Function Device associated with an Upstream Port, the bit in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is of type RsvdP. This bit is applicable when the Link is operating at 2.5 GT/s or 5.0 GT/s data rates only. For components that support only the 2.5 GT/s speed, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && 0 == 0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && 0 != 0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && 0 == 0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && 0 != 0) ? \"read-only\" : (0 == 0)? \"read-write\": \"read-only\" "

Bits	Name	Memory Access	Description
10	PCIE_CAP_ENTER_MODIFIED_COMPLIANCE	* Varies	Enter Modified Compliance. When this bit is set to 1b, the device transmits Modified Compliance Pattern if the LTSSM enters Polling.Compliance substate. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 0b. For a Multi-Function Device associated with an Upstream Port, the bit in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is of type RsvdP. This bit is intended for debug, compliance testing purposes only. System firmware and software is allowed to modify this bit only during debug or compliance testing. In all other cases, the system must ensure that this bit is set to the default value. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==0 && 0 == 0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && 0 != 0) ? \"read-only\": (CX_MEMORY_MAP_VIEW==1 && 0 == 0) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && 0 != 0) ? \"read-only\": (0 == 0)? \"read-write\": \"read-only\" "

Bits	Name	Memory Access	Description
9:7	PCIE_CAP_TX_MARGIN	R/W	Transmit Margin, This field controls the value of the non-deemphasized voltage level at the Transmitter pins. This field is reset to 000b on entry to the LTSSM Polling.Configuration substate (see Chapter 4 of <i>PCI Express Base Specification</i> for details of how the Transmitter voltage level is determined in various states). 001b-111b: As defined in Section 8.3.4 not all encodings are required to be implemented. For a Multi-Function Device associated with an Upstream Port, the field in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this field is of type RsvdP. For components that support only the 2.5 GT/s speed, the controller hardwires this bit to 000b. This field is intended for debug, compliance testing purposes only. System firmware and software is allowed to modify this field only during debug or compliance testing. In all other cases, the system must ensure that this field is set to the default value. <i>Note:</i> This register field is sticky. Values: 0x0 (NORMAL_RANGE): Normal operating range Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
6	PCIE_CAP_SEL_DEEMPHASIS	R	Selectable De-emphasis. When the Link is operating at 5.0 GT/s speed, this bit is used to control the transmit de-emphasis of the link in specific situations. For more information, see section 4.2.6 of <i>PCI Express Base Specification</i>. When the Link is not operating at 5.0 GT/s speed, the setting of this bit has no effect. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 0b. This bit is not applicable and Reserved for Endpoints, PCI Express to PCI/PCI-X bridges, and Upstream Ports of Switches. <i>Note:</i> This register field is sticky. Values: 0x1 (_3_5_DB): -3.5 dB 0x0 (_6_DB): -6 dB Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
5	PCIE_CAP_HW_AUTO_SPEED_DISABLE	* Varies	Hardware Autonomous Speed Disable. When set, this bit disables hardware from changing the Link speed for device-specific reasons other than attempting to correct unreliable Link operation by reducing Link speed. Initial transition to the highest supported common link speed is not blocked by this bit. For a Multi-Function Device associated with an Upstream Port, the bit in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is of type RsvdP. Functions that do not implement the associated mechanism are permitted to hardwire this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && 0 == 0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && 0 != 0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && 0 == 0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && 0 != 0) ? \"read-only\" : (0 == 0)? \"read-write\": \"read-only\" "

Bits	Name	Memory Access	Description
4	PCIE_CAP_ENTER_COMPLIANCE	* Varies	Enter Compliance. Software is permitted to force a Link to enter Compliance mode (at the speed indicated in the Target Link Speed field and the de-emphasis/preset level indicated by the Compliance Preset/De-emphasis field) by setting this bit to 1b in both components on a Link and then initiating a hot reset on the Link. Default value of this bit following Fundamental Reset is 0b. For a Multi-Function Device associated with an Upstream Port, the bit in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is of type RsvdP. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 0b. This bit is intended for debug, compliance testing purposes only. System firmware and software is allowed to modify this bit only during debug or compliance testing. In all other cases, the system must ensure that this bit is set to the default value. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
3:0	PCIE_CAP_TARGET_LINK_SPEED	* Varies	Target Link Speed. For Downstream Ports, this field sets an upper limit on Link operational speed by restricting the values advertised by the Upstream component in its training sequences. The encoded value specifies a bit location in the Supported Link Speeds Vector (in the Link Capabilities 2 register) that corresponds to the desired target Link speed. All encodings other than the defined encodings are reserved. If a value is written to this field that does not correspond to a supported speed (as indicated by the Supported Link Speeds Vector), the result is undefined. If either of the Enter Compliance or Enter Modified Compliance bits are implemented, then this field must also be implemented. The default value of this field is the highest Link speed supported by the component (as reported in the Max Link Speed field of the Link Capabilities register) unless the corresponding platform/form factor requires a different default value. For both Upstream and Downstream Ports, this field is used to set the target compliance mode speed when software is using the Enter Compliance bit to force a Link into compliance mode. For Upstream Ports, if the Enter Compliance bit is Clear, this field is permitted to have no effect. For a Multi-Function Device associated with an Upstream Port, the field in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this field is of type RsvdP. Components that support only the 2.5 GT/s speed are permitted to hardwire this field to 0000b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky)

Bits	Name	Memory Access	Description
3:0...(cont.)	PCIE_CAP_TARGET_LINK_SPEED.	* Varies	■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SUP_LINK_SPEED_VECTOR_BIT_0): Supported Link Speeds Vector field bit 0 ■ 0x2 (SUP_LINK_SPEED_VECTOR_BIT_1): Supported Link Speeds Vector field bit 1 ■ 0x3 (SUP_LINK_SPEED_VECTOR_BIT_2): Supported Link Speeds Vector field bit 2 ■ 0x4 (SUP_LINK_SPEED_VECTOR_BIT_3): Supported Link Speeds Vector field bit 3 ■ 0x5 (SUP_LINK_SPEED_VECTOR_BIT_4): Supported Link Speeds Vector field bit 4 ■ 0x6 (SUP_LINK_SPEED_VECTOR_BIT_5): Supported Link Speeds Vector field bit 5 ■ 0x7 (SUP_LINK_SPEED_VECTOR_BIT_6): Supported Link Speeds Vector field bit 6 Value After Reset: {(0==0) ? MAX_LINK_SP : 0} Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: {(SNPS_RSVDPARAM_25) ? "writeAsRead" : "unconstrained" } Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==0 && 0==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\":(CX_MEMORY_MAP_VIEW==1 && 0==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\": (0==0) ? \"read-write\" : \"read-only\" "

3.6 PF_PCIE_CAP_DBI2i Registers

3.6.1 SHADOW_LINK_CAPABILITIES_REG

- **Name:** Shadow Link Capabilities Register.
- **Description:** This is a 7-bit register at the same address as LINK_CAPABILITIES_REG, and it only exists when one or more of the following expressions are true:
 - CX_NFTS !=CX_COMM_NFTS
 - DEFAULT_L0S_EXIT_LATENCY !=DEFAULT_COMM_L0S_EXIT_LATENCY
 - DEFAULT_L1_EXIT_LATENCY !=DEFAULT_COMM_L1_EXIT_LATENCY

The register has only two fields in RTL. All of the other fields here do not exist in RTL and accesses to them are redirected to the corresponding fields in the parent register (LINK_CAPABILITIES_REG). They are modeled here to make the shadow register (model) 32 bits wide which is a tool requirement.

- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** Always

PCIE_CAP_PORT_NUM	31:24
RSVDP_23	23
PCIE_CAP_ASPM_OPT_COMPLIANCE	22
PCIE_CAP_LINK_BW_NOT_CAP	21
PCIE_CAP_DLL_ACTIVE_REP_CAP	20
PCIE_CAP_SURPRISE_DOWN_ERR_REP_CAP	19
PCIE_CAP_CLOCK_POWER_MAN	18
SHADOW_PCIE_CAP_L1_EXIT_LATENCY	17:15
SHADOW_PCIE_CAP_L0S_EXIT_LATENCY	14:12
PCIE_CAP_ACTIVE_STATE_LINK_PM_SUPPORT	11:10
PCIE_CAP_MAX_LINK_WIDTH	9:4
PCIE_CAP_MAX_LINK_SPEED	3:0

Table 3-44 Fields for Register: SHADOW_LINK_CAPABILITIES_REG

Bits	Name	Memory Access	Description
31:24	PCIE_CAP_PORT_NUM	* Varies	Shadow Port Number. This field indicates the PCI Express Port number for the given PCI Express Link. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Value After Reset: PORT_NUM Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: untestable
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2)
22	PCIE_CAP_ASPM_OPT_COMPLIANCE	* Varies	Shadow ASPM Optionality Compliance. This field must be set to 1b in all functions. Components implemented against certain earlier versions of this specification will have this bit set to 0b. Software is permitted to use the value of this bit to help determine whether to enable ASPM or whether to run ASPM compliance tests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Value After Reset: ASPM_OPTIONALITY_COMPLIANCE Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: untestable

Bits	Name	Memory Access	Description
21	PCIE_CAP_LINK_BW_NOT_CAP	R	Shadow Link Bandwidth Notification Capable. A value of 1b indicates support for the Link Bandwidth Notification status and interrupt mechanisms. This capability is required for all Root Ports and Switch Downstream Ports supporting Links wider than x1 and/or multiple Link speeds. This field is not applicable and is Reserved for Endpoints, PCI Express to PCI/PCI-X bridges, and Upstream Ports of Switches. For functions that do not implement the Link Bandwidth Notification Capability the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: untestable
20	PCIE_CAP_DLL_ACTIVE_REP_CAP	R	Shadow Data Link Layer Link Active Reporting Capable. For a Downstream Port, the controller hardwires this bit to 1b if the component supports the optional capability of reporting the DL_Active state of the Data Link Control and Management State Machine. For a hot-plug capable Downstream Port (as indicated by the Hot-Plug Capable bit of the Slot Capabilities register) or a Downstream Port that supports Link speeds greater than 5.0 GT/s, the controller hardwires this bit to 1b. For Upstream Ports and components that do not support this optional capability, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: untestable
19	PCIE_CAP_SURPRISE_DOWN_ERR_REP_CAP	R	Shadow Surprise Down Error Reporting Capable. For a Downstream Port, this bit must be set if the component supports the optional capability of detecting and reporting a Surprise Down error condition. For Upstream Ports and components that do not support this optional capability, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: untestable

Bits	Name	Memory Access	Description
18	PCIE_CAP_CLOCK_POWER_MAN	* Varies	Shadow Clock Power Management. For Upstream Ports, a value of 1b in this bit indicates that the component tolerates the removal of any reference clock(s) through the "clock request" (CLKREQ#) mechanism when the Link is in the L1 and L2/L3 Ready Link states. A value of 0b indicates the component does not have this capability and that reference clock(s) must not be removed in these Link states. L1 PM Substates defines other semantics for the CLKREQ# signal, which are managed independently of Clock Power Management. This Capability is applicable only in form factors that support "clock request" (CLKREQ#) capability. For a Multi-Function device associated with an Upstream Port, each Function indicates its capability independently. Power Management configuration software must only permit reference clock removal if all functions of the Multi-Function device indicate a 1b in this bit. For ARI Devices, all Functions must indicate the same value in this bit. For Downstream Ports, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_CLK_PM_CAP Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: untestable

Bits	Name	Memory Access	Description
17:15	SHADOW_PCIE_CAP_L1_EXIT_LATENCY	* Varies	Shadow L1 Exit Latency. There are two each of these register fields, this one and a shadow one at the same address. The Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG) determines which one is used by the controller and which one is accessed by a read request. Common Clock operation is supported (possible) in the controller when one or more of the following expressions is true: ■ CX_NFTS !=CX_COMM_NFTS ■ DEFAULT_L0S_EXIT_LATENCY !=DEFAULT_COMM_L0S_EXIT_LATENCY ■ DEFAULT_L1_EXIT_LATENCY !=DEFAULT_COMM_L1_EXIT_LATENCY Common Clock operation is enabled in the controller when you set the Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG). The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the shadow field at this location. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R ■ Dbi2: R/W Value After Reset: {(CX_COMM_CLK_EN == 1) ? (DEFAULT_COMM_L1_EXIT_LATENCY) : (DEFAULT_L1_EXIT_LATENCY) } Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: untestable Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\": (CX_MEMORY_MAP_VIEW==1) ? \"read-only\": (CX_MEMORY_MAP_VIEW==2 && CX_COMM_CLK_EN && CX_DBI_RO_WR_EN) ? \"read-write\": \"read-only\" "

Bits	Name	Memory Access	Description
14:12	SHADOW_PCIE_CAP_L0S_EXIT_LATENCY	* Varies	Shadow L0s Exit Latency. There are two each of these register fields, this one and a shadow one at the same address. The Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG) determines which one is used by the controller and which one is accessed by a read request. Common Clock operation is supported (possible) in the controller when one or more of the following expressions is true: ■ CX_NFTS !=CX_COMM_NFTS ■ DEFAULT_L0S_EXIT_LATENCY !=DEFAULT_COMM_L0S_EXIT_LATENCY ■ DEFAULT_L1_EXIT_LATENCY !=DEFAULT_COMM_L1_EXIT_LATENCY Common Clock operation is enabled in the controller when you set the Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG). The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the shadow field at this location. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R ■ Dbi2: R/W Value After Reset: {(CX_COMM_CLK_EN == 1) ? (DEFAULT_COMM_L0S_EXIT_LATENCY) : (DEFAULT_L0S_EXIT_LATENCY) } Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: untestable Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\": (CX_MEMORY_MAP_VIEW==1) ? \"read-only\": (CX_MEMORY_MAP_VIEW==2 && CX_COMM_CLK_EN && CX_DBI_RO_WR_EN) ? \"read-write\": \"read-only\" "

Bits	Name	Memory Access	Description
11:10	PCIE_CAP_ACTIVE_STATE_LINK_PM_SUPPORT	* Varies	Shadow Level of ASPM (Active State Power Management) Support. This field indicates the level of ASPM supported on the given PCI Express Link. For more information on ASPM support requirements, see section 5.4.1 of <i>PCI Express Base Specification</i>. Multi-Function devices associated with an Upstream Port must report the same value in this field for all functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_PCIE_OVER_CIO_ENABLE) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NO_ASPM_SUP): No ASPM Support ■ 0x1 (L0S_SUP): L0s Supported ■ 0x2 (L1_SUP): L1 Supported ■ 0x3 (L0S_L1_SUP): L0s and L1 Supported Value After Reset: AS_LINK_PM_SUPT Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: untestable

Bits	Name	Memory Access	Description
9:4	PCIE_CAP_MAX_LINK_WIDTH	* Varies	Shadow Maximum Link Width. This field indicates the maximum Link width (xN - corresponding to N Lanes) implemented by the component. This value is permitted to exceed the number of Lanes routed to the slot (Downstream Port), adapter connector (Upstream Port), or in the case of component-to-component connections, the actual wired connection width. All encodings other than the defined encodings are reserved. Multi-Function devices associated with an Upstream Port must report the same value in this field for all functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_PCIE_OVER_CIO_ENABLE) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Values: ■ 0x1 (X1): x1 ■ 0x2 (X2): x2 ■ 0x4 (X4): x4 ■ 0x8 (X8): x8 ■ 0xc (X12): x12 ■ 0x10 (X16): x16 ■ 0x20 (X32): x32 Value After Reset: CX_NL Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: untestable

Bits	Name	Memory Access	Description
3:0	PCIE_CAP_MAX_LINK_SPEED	* Varies	Shadow Maximum Link Speed. This field indicates the maximum Link speed of the associated Port. The encoded value specifies a bit location in the Supported Link Speeds Vector (in the Link Capabilities 2 register) that corresponds to the maximum Link speed. All encodings other than the defined encodings are reserved. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_PCIE_OVER_CIO_ENABLE) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SUP_LINK_SPEED_FIELD_BIT_0): Supported Link Speeds Vector field bit 0 ■ 0x2 (SUP_LINK_SPEED_FIELD_BIT_1): Supported Link Speeds Vector field bit 1 ■ 0x3 (SUP_LINK_SPEED_FIELD_BIT_2): Supported Link Speeds Vector field bit 2 ■ 0x4 (SUP_LINK_SPEED_FIELD_BIT_3): Supported Link Speeds Vector field bit 3 ■ 0x5 (SUP_LINK_SPEED_FIELD_BIT_4): Supported Link Speeds Vector field bit 4 ■ 0x6 (SUP_LINK_SPEED_FIELD_BIT_5): Supported Link Speeds Vector field bit 5 ■ 0x7 (SUP_LINK_SPEED_FIELD_BIT_6): Supported Link Speeds Vector field bit 6 Value After Reset: MAX_LINK_SP Exists: (CX_COMM_CLK_EN && CX_MEMORY_MAP_PCIE_TYPE<2) Testable: untestable

3.7 PF_MSIX_CAPI Registers

3.7.1 PCI_MSIX_CAP_ID_NEXT_CTRL_REG

- **Name:** MSI-X Capability ID, Next Pointer, Control Register.
- **Description:** This Register holds MSI-X Capability ID, Next Capability pointer. It also controls the MSI-X behaviour.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	PCI_MSIX_ENABLE
30	PCI_MSIX_FUNCTION_MASK
29:27	RSVDP_27
26:16	PCI_MSIX_TABLE_SIZE
15:8	PCI_MSIX_CAP_NEXT_OFFSET
7:0	PCI_MSIX_CAP_ID

Table 3-45 Fields for Register: PCI_MSIX_CAP_ID_NEXT_CTRL_REG

Bits	Name	Memory Access	Description
31	PCI_MSIX_ENABLE	R/W	MSI-X Enable. If Set and the MSI Enable bit in the MSI Message Control Register for MSI is Clear, the Function is permitted to use MSI-X to request service and is prohibited from using INTx interrupts (if implemented). System configuration software Sets this bit to enable MSI-X. A device driver is prohibited from writing this bit to mask a Function's service request. If Clear, the Function is prohibited from using MSI-X to request service. Value After Reset: 0x0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
30	PCI_MSIX_FUNCTION_MASK	* Varies	Function Mask. If Set, all of the vectors associated with the Function are masked, regardless of their per-vector Mask bit values. If Clear, each vector's Mask bit determines whether the vector is masked or not. Setting or Clearing the MSI-X Function Mask bit has no effect on the value of the per-vector Mask bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && MSIX_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !MSIX_CAP_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && MSIX_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !MSIX_CAP_ENABLE) ? \"read-only\" : \"read-only\""
29:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
26:16	PCI_MSIX_TABLE_SIZE	* Varies	MSI-X Table Size. System software reads this field to determine the MSI-X Table Size N, which is encoded as N-1. For example, a returned value of 000 0000 0011b indicates a table size of 4. SRIOV Note: All VFs in a single PF have the same value for "MSI-X Table Size" (PCI_MSIX_TABLE_SIZE field in SHADOW_PCI_MSIX_CAP_ID_NEXT_CTRL_REG). To write this common value, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PCI_MSIX_TABLE_SIZE field in the PF PCI_MSIX_CAP_ID_NEXT_CTRL_REG register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: (MSIX_CAP_ENABLE && CX_DBI_RO_WR_EN) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: MSIX_TABLE_SIZE_0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:8	PCI_MSIX_CAP_NEXT_OFFSET	* Varies	MSI-X Next Capability Pointer. This field contains the offset to the next PCI Capability structure or 00h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
7:0	PCI_MSIX_CAP_ID	R	MSI-X Capability ID. This field indicates the MSI-X Capability structure. This field must return a Capability ID of 11h indicating that this is an MSI-X Capability structure. Value After Reset: 0x11 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.7.2 MSIX_TABLE_OFFSET_REG

- **Name:** MSI-X Table Offset and BIR Register.
- **Description:** This register provides Table BIR and MSI-x Table offset select.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0))

31:3	2:0
PCI_MSIX_TABLE_OFFSET	PCI_MSIX_BIR

Table 3-46 Fields for Register: MSIX_TABLE_OFFSET_REG

Bits	Name	Memory Access	Description
31:3	PCI_MSIX_TABLE_OFFSET	* Varies	MSI-X Table Offset. Used as an offset from the address contained by one of the Function's Base Address Registers to point to the base of the MSI-X Table. The lower 3 Table BIR bits are masked off (set to zero) by software to form a 32-bit QWORD-aligned offset. SRIOV Note: All VFs in a single PF have the same value for "MSI-X Table Offset" (PCI_MSIX_TABLE_OFFSET field in SHADOW_MSIX_TABLE_OFFSET_REG). To write this common value, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PCI_MSIX_TABLE_OFFSET field in the PF MSIX_TABLE_OFFSET_REG register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : (MSIX_CAP_ENABLE && CX_DBI_RO_WR_EN) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: MSIX_TABLE_OFFSET_0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
2:0	PCI_MSIX_BIR	* Varies	MSI-X Table BAR Indicator Register Field. This field indicates which one of a Function's Base Address Registers , located beginning at 10h in Configuration Space, or entry in the Enhanced Allocation capability with a matching BEI , is used to map the Function's MSI-X Table into Memory Space. All encodings other than the defined encodings are reserved. For a 64-bit Base Address Register , the Table BIR indicates the lower DWORD. For Functions with Type 1 Configuration Space headers, BIR values 2 through 5 are also Reserved. SRIOV Note: All VFs in a single PF have the same value for "MSI-X Table BAR Indicator Register" (PCI_MSIX_BIR field in SHADOW_MSIX_TABLE_OFFSET_REG). To write this common value, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PCI_MSIX_BIR field in the PF MSIX_TABLE_OFFSET_REG register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: (MSIX_CAP_ENABLE && CX_DBI_RO_WR_EN) ? R/W : R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BAR_10): Base Address Register 10h ■ 0x1 (BAR_14): Base Address Register 14h ■ 0x2 (BAR_18): Base Address Register 18h ■ 0x3 (BAR_1C): Base Address Register 1Ch ■ 0x4 (BAR_20): Base Address Register 20h ■ 0x5 (BAR_24): Base Address Register 24h Value After Reset: MSIX_TABLE_BIR_0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.7.3 MSIX_PBA_OFFSET_REG

- **Name:** MSI-X PBA Offset and BIR Register.
- **Description:** This register provides PBA Offset and PBA BIR value.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 | | CX_MEMORY_MAP_PCIE_TYPE==0))

31:3	
PCI_MSIX_PBA_OFFSET	
2:0	
PCI_MSIX_PBA_BIR	

Table 3-47 Fields for Register: **MSIX_PBA_OFFSET_REG**

Bits	Name	Memory Access	Description
31:3	PCI_MSIX_PBA_OFFSET	* Varies	MSI-X PBA Offset. Used as an offset from the address contained by one of the Function's Base Address Registers to point to the base of the MSI-X PBA. The lower 3 PBA BIR bits are masked off (set to zero) by software to form a 32-bit QWORD-aligned offset. SRIOV Note: All VFs in a single PF have the same value for "MSI-X PBA Offset" (PCI_MSIX_PBA_OFFSET field in SHADOW_MSIX_PBA_OFFSET_REG). To write this common value, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PCI_MSIX_PBA_OFFSET field in the PF MSIX_PBA_OFFSET_REG register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : (MSIX_CAP_ENABLE && CX_DBI_RO_WR_EN) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: MSIX_PBA_OFFSET_0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
2:0	PCI_MSIX_PBA_BIR	* Varies	MSI-X PBA BIR. This field indicates which one of a Function's Base Address Registers , located beginning at 10h in Configuration Space, or entry in the Enhanced Allocation capability with a matching BEI, is used to map the Function's MSI-X PBA into Memory Space. The PBA BIR value definitions are identical to those for the Table BIR . SRIOV Note: All VFs in a single PF have the same value for "MSI-X PBA BIR" (PCI_MSIX_PBA_BIR field in SHADOW_MSIX_PBA_OFFSET_REG). To write this common value, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PCI_MSIX_PBA_BIR field in the PF MSIX_PBA_OFFSET_REG register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : (MSIX_CAP_ENABLE && CX_DBI_RO_WR_EN) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: MSIX_PBA_BIR_0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.8 PF_MSIX_CAP_DBI2i Registers

3.8.1 SHADOW_PCI_MSIX_CAP_ID_NEXT_CTRL_REG

- **Name:** MSI-X Capability ID, Next Pointer, Control Register.
- **Description:** This is the shadow register of the MSI-X Capability ID, Next Pointer, and Control Register.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:27	26:16	15:0
PCI_MSIX_RESERVED1	PCI_MSIX_TABLE_SIZE	PCI_MSIX_RESERVED0

Table 3-48 Fields for Register: SHADOW_PCI_MSIX_CAP_ID_NEXT_CTRL_REG

Bits	Name	Memory Access	Description
31:27	PCI_MSIX_RESERVED1	R	reserved field 1 in the shadow register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable

Bits	Name	Memory Access	Description
26:16	PCI_MSIX_TABLE_SIZE	* Varies	MSI-X Table Size in the shadow register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: (CX_MEMORY_MAP_VIEW==2 && MSIX_CAP_ENABLE && CX_DBI_RO_WR_EN) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: MSIX_TABLE_SIZE_0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true
15:0	PCI_MSIX_RESERVED0	R	reserved field 0 in the shadow register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable

3.8.2 SHADOW_MSIX_TABLE_OFFSET_REG

- **Name:** MSI-X Table Offset and BIR Register.
- **Description:** This register provides Table BIR and MSI-x Table offset select.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0))

31:3	
PCI_MSIX_TABLE_OFFSET	
2:0	
PCI_MSIX_BIR	

Table 3-49 Fields for Register: SHADOW_MSIX_TABLE_OFFSET_REG

Bits	Name	Memory Access	Description
31:3	PCI_MSIX_TABLE_OFFSET	* Varies	MSI-X Table Offset. Used as an offset from the address contained by one of the Function's Base Address Registers to point to the base of the MSI-X Table. The lower 3 Table BIR bits are masked off (set to zero) by software to form a 32-bit QWORD-aligned offset. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1 && MSIX_CAP_ENABLE=1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R <i>Note:</i> This register field is sticky. Value After Reset: VF_MSIX_TABLE_OFFSET_0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true

Bits	Name	Memory Access	Description
2:0	PCI_MSIX_BIR	* Varies	MSI-X Table BAR Indicator Register Field. This field indicates which one of a Function's Base Address Registers , located beginning at 10h in Configuration Space, or entry in the Enhanced Allocation capability with a matching BEI , is used to map the Function's MSI-X Table into Memory Space. All encodings other than the defined encodings are reserved. For a 64-bit Base Address Register , the Table BIR indicates the lower DWORD. For Functions with Type 1 Configuration Space headers, BIR values 2 through 5 are also Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1 && MSIX_CAP_ENABLE=1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BAR_10): Base Address Register 10h ■ 0x1 (BAR_14): Base Address Register 14h ■ 0x2 (BAR_18): Base Address Register 18h ■ 0x3 (BAR_1C): Base Address Register 1Ch ■ 0x4 (BAR_20): Base Address Register 20h ■ 0x5 (BAR_24): Base Address Register 24h Value After Reset: VF_MSIX_TABLE_BIR_0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true

3.8.3 SHADOW_MSIX_PBA_OFFSET_REG

- **Name:** MSI-X PBA Offset and BIR Register.
- **Description:** This register provides PBA Offset and PBA BIR value.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0))

31:3	2:0
PCI_MSIX_PBA_OFFSET	PCI_MSIX_PBA_BIR

Table 3-50 Fields for Register: SHADOW_MSIX_PBA_OFFSET_REG

Bits	Name	Memory Access	Description
31:3	PCI_MSIX_PBA_OFFSET	* Varies	MSI-X PBA Offset. Used as an offset from the address contained by one of the Function's Base Address Registers to point to the base of the MSI-X PBA. The lower 3 PBA BIR bits are masked off (set to zero) by software to form a 32-bit QWORD-aligned offset. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1 && MSIX_CAP_ENABLE=1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Value After Reset: VF_MSIX_PBA_OFFSET_0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true
2:0	PCI_MSIX_PBA_BIR	* Varies	MSI-X PBA BIR. This field indicates which one of a Function's Base Address Registers , located beginning at 10h in Configuration Space, or entry in the Enhanced Allocation capability with a matching BEI, is used to map the Function's MSI-X PBA into Memory Space. The PBA BIR value definitions are identical to those for the Table BIR . <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1 && MSIX_CAP_ENABLE=1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Value After Reset: VF_MSIX_PBA_BIR_0 Exists: (MSIX_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true

3.9 PF_VPD_CAPi Registers

3.9.1 VPD_BASE

- **Name:** VPD Control and Capabilities Register.
- **Description:** VPD BASE Register provides Capability Id, VPD Flag, VPD next offset value, and VPD address.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (VPD_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

VPD_FLAG	31
VPD_ADDRESS	30:16
VPD_NEXT_OFFSET	15:8
VPD_PCIE_EXTENDED_CAP_ID	7:0

Table 3-51 Fields for Register: VPD_BASE

Bits	Name	Memory Access	Description
31	VPD_FLAG	* Varies	VPD Flag. The Flag bit is always written along with VPD Address . The value of Flag indicates the direction of transfer being requested (0b = read, 1b = write). When the transfer is complete, the Flag bit value changes to indicate completion (1b = read complete, 0b = write complete). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (VPD_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && VPD_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !VPD_CAP_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && VPD_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !VPD_CAP_ENABLE) ? \"read-only\" : (VPD_CAP_ENABLE) ? \"read-write\" : \"read-only\" "
30:16	VPD_ADDRESS	* Varies	VPD Address. DWORD-aligned byte address of the VPD to be accessed. Behavior is undefined if the low 2 bits of this field are non-zero. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (VPD_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && VPD_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !VPD_CAP_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && VPD_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !VPD_CAP_ENABLE) ? \"read-only\" : (VPD_CAP_ENABLE) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
15:8	VPD_NEXT_OFFSET	* Varies	VPD Pointer to Next Capability. Pointer to the next capability structure, or 00h if this is the last structure in the Capability List. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VPD_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: Pointer value calculated at configuration time. Exists: (VPD_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
7:0	VPD_PCIE_EXTENDED_CAP_ID	R	VPD Extended Capability ID. Capability Id for Vital Product data Capability is 03h. Value After Reset: 0x3 Exists: (VPD_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.9.2 DATA_REG

- **Name:** VPD Data Register.
- **Description:** This Register holds the VPD data.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (VPD_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

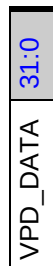


Table 3-52 Fields for Register: DATA_REG

Bits	Name	Memory Access	Description
31:0	VPD_DATA	R/W	VPD Data. VPD Data can be read through this register. The least significant byte of this register (at offset 04h in this capability structure) corresponds to the byte of VPD at the address specified by VPD Address . Behavior is undefined for any read or write of this register with Byte Enables other than 1111b. Value After Reset: 0x0 Exists: (VPD_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.10 PF_AER_CAPi Registers

3.10.1 AER_EXT_CAP_HDR_OFF

- **Name:** Advanced Error Reporting Extended Capability Header.
- **Description:** Advanced Error Reporting Extended Capability Header provides information about Capability ID, Version, and next offset.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	CAP_ID

Table 3-53 Fields for Register: AER_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field must be 2h if the End-End TLP Prefix Supported bit is set and must be 1h or 2h otherwise. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x2 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	CAP_ID	* Varies	AER Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. The Extended Capability ID for the Advanced Error Reporting Capability is 0001h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.10.2 UNCORR_ERR_STATUS_OFF

- **Name:** Uncorrectable Error Status Register.
- **Description:** The Uncorrectable Error Status Register (Offset 04h) indicates error detection status of individual errors on a PCI Express device Function. An individual error status bit that is Set indicates that a particular error was detected; software may clear an error status by writing a 1b to the respective bit. Register bits not implemented by the Function are hardwired to 0b.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_28	31:28
DMWR_EGRESS_BLOCKED_ERR_STATUS	27
PSND_TLP_EGRSS_BLOCKED_ERR_STATUS	26
TLP_PREFX_BLOCKED_ERR_STATUS	25
ATOMIC_EGRESS_BLOCKED_ERR_STATUS	24
RSVDP_23	23
INTERNAL_ERR_STATUS	22
ACS_VIOLATION_STATUS	21
UNSUPPORTED_REQ_ERR_STATUS	20
ECRC_ERR_STATUS	19
MALF_TLP_ERR_STATUS	18
REC_OVERFLOW_ERR_STATUS	17
UNEXP_CMPLT_ERR_STATUS	16
CMPLT_ABORT_ERR_STATUS	15
CMPLT_TIMEOUT_ERR_STATUS	14
FC_PROTOCOL_ERR_STATUS	13
POIS_TLP_ERR_STATUS	12
RSVDP_6	11:6
SURPRISE_DOWN_ERR_STATUS	5
DL_PROTOCOL_ERR_STATUS	4
RSVDP_0	3:0

Table 3-54 Fields for Register: UNCORR_ERR_STATUS_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27	DMWR_EGRESS_BLOCKED_ERR_STATUS	R/W1C	Deferrable Memory Write Egress Blocked Status. Status bit for Deferrable Memory Write Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: CX_DEF_MEM_WR_ROUTING_SUP && CC_DEVICE_TYPE==3 && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
26	PSND_TLP_EGRSS_BLOCKED_ERR_STATUS	R/W1C	Poisoned TLP Egress Blocked Status. This field provides status of Poisoned TLP Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: AER_ENABLE && CX_DPC_ENABLE && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
25	TLP_PRFX_BLOCKED_ERR_STATUS	R/W1C	TLP Prefix Blocked Error Status. Status bit for TLP Prefix Blocked Error. <i>Note:</i> Not supported. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_TLP_PREFIX_ENABLE && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
24	ATOMIC_EGRESS_BLOCKED_ERR_STATUS	R/W1C	AtomicOp Egress Blocked Status. Status bit for AtomicOp Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: ATOMIC_ROUTING_SUP && CC_DEVICE_TYPE==3 && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
22	INTERNAL_ERR_STATUS	R/W1C	Uncorrectable Internal Error Status. This field gives status of the Uncorrectable Internal Error. The controller sets this bit when your application asserts app_err_bus[9]. It does not set this bit when it detects internal uncorrectable internal errors such as parity and ECC failures. You should use the outputs from these errors to drive the app_err_bus[9] input. For more information, see the "Data Integrity (Wire, Datapath, and RAM Protection)" section in the <i>Databook</i> . This register field is reserved for VFs. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_INTERNAL_ERR_REPORTING && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
21	ACS_VIOLATION_STATUS	R/W1C	ACS Violation Status. ACS Violations may occur due to either hardware or software defects/failures. The ACS Violation Status bit provides positive identification of the error. The controller sets this bit when your application asserts app_err_bus[12]. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_ACS_ENABLE && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
20	UNSUPPORTED_REQ_ERR_STATUS	R/W1C	Unsupported Request Error Status. This field represents status of Unsupported Request Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
19	ECRC_ERR_STATUS	R/W1C	ECRC Error Status. This field represents status of ECRC Error. <i>Note:</i> If CX_ECRC_ENABLE=0 the register field always reads 0. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
18	MALF_TLP_ERR_STATUS	R/W1C	Malformed TLP Status. This field represents status of Malformed TLP. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
17	REC_OVERFLOW_ERR_STATUS	R/W1C	Receiver Overflow Status. Status bit for Receiver Overflow. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
16	UNEXP_CMPLT_ERR_STATUS	R/W1C	Unexpected Completion Status. Status bit for Unexpected Completion. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
15	CMPLT_ABORT_ERR_STATUS	R/W1C	Completer Abort Status. Status bit for Completer Abort. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
14	CMPLT_TIMEOUT_ERR_STATUS	R/W1C	Completion Timeout Status. Status for Completion Timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
13	FC_PROTOCOL_ERR_STATUS	R/W1C	Flow Control Protocol Error Status. Status bit for Flow Control Protocol Error. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
12	POIS_TLP_ERR_STATUS	R/W1C	Poisoned TLP Status. Status bit for Poisoned TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
11:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5	SURPRISE_DOWN_ERR_STATUS	R/W1C	Surprise Down Error Status (Optional). Status bit for Surprise Down Error. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
4	DL_PROTOCOL_ERR_STATUS	R/W1C	Data Link Protocol Error Status. Status bit for Data Link Protocol Error. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.10.3 UNCORR_ERR_MASK_OFF

- **Name:** Uncorrectable Error Mask Register.
- **Description:** The Uncorrectable Error Mask Register controls reporting of individual errors by the device Function to the PCI Express Root Complex through a PCI Express error Message. A masked error (respective bit Set in the mask register) is not recorded or reported in the Header Log, TLP Prefix Log, or First Error Pointer, and is not reported to the PCI Express Root Complex by this Function. There is a mask bit per error bit of the Uncorrectable Error Status register. Register fields for bits not implemented by the Function are hardwired to 0b.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_28	31:28
DMWR_EGRESS_BLOCKED_ERR_MASK	27
RSVDP_26	26
TLP_PREFIX_BLOCKED_ERR_MASK	25
ATOMIC_EGRESS_BLOCKED_ERR_MASK	24
RSVDP_23	23
INTERNAL_ERR_MASK	22
ACS_VIOLATION_MASK	21
UNSUPPORTED_REQ_ERR_MASK	20
ECRC_ERR_MASK	19
MALF_TLP_ERR_MASK	18
REC_OVERFLOW_ERR_MASK	17
UNEXP_CMPLT_ERR_MASK	16
CMPLT_ABORT_ERR_MASK	15
CMPLT_TIMEOUT_ERR_MASK	14
FC_PROTOCOL_ERR_MASK	13
POIS_TLP_ERR_MASK	12
RSVDP_6	11:6
SURPRISE_DOWN_ERR_MASK	5
DL_PROTOCOL_ERR_MASK	4
RSVDP_0	3:0

Table 3-55 Fields for Register: UNCORR_ERR_MASK_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27	DMWR_EGRESS_BLOCKED_ERR_MASK	R	Deferrable Memory Write Egress Block Mask (Optional). Mask bit for Deferrable Memory Write Egress Block Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
25	TLP_PRFX_BLOCKED_ERR_MASK	R	TLP Prefix Blocked Error Mask. Mask bit for TLP Prefix Blocked Error. <i>Note:</i> Not supported. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
24	ATOMIC_EGRESS_BLOCKED_ERR_MASK	R	AtomicOp Egress Block Mask (Optional). Mask bit for AtomicOp Egress Block Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
22	INTERNAL_ERR_MASK	R/W	Uncorrectable Internal Error Mask (Optional). Mask bit for Uncorrectable Internal Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: CX_INTERNAL_ERR_REPORTING && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
21	ACS_VIOLATION_MASK	* Varies	ACS Violation Mask. Setting the ACS Violation Mask bit disables error logging and signaling for ACS Violation errors. The bit is Read-Only Zero for upstream ports, when ACS P2P Egress Control Enable is not set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: if (acs_viol_svrity_mask_wr_en == 1) then R/W (Sticky) else R(Sticky) ■ Dbi: if (acs_viol_svrity_mask_wr_en == 1) then R/W (Sticky) else R(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_ACS_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_ACS_ENABLE) ? R : (CX_MEMORY_MAP_VIEW==1 && CX_ACS_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==1 && !CX_ACS_ENABLE) ? R : (CX_ACS_ENABLE) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_ACS_ENABLE && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
20	UNSUPPORTED_REQ_ERR_MASK	R/W	Unsupported Request Error Mask. Mask bit for Unsupported Request Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
19	ECRC_ERR_MASK	* Varies	ECRC Error Mask (Optional). Mask bit for ECRC Error. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_ECRC_ENABLE && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && DEFAULT_ECRC_CHK_CAP_0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !DEFAULT_ECRC_CHK_CAP_0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && DEFAULT_ECRC_CHK_CAP_0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !DEFAULT_ECRC_CHK_CAP_0) ? \"read-only\" : (DEFAULT_ECRC_CHK_CAP_0) ? \"read-write\" : \"read-only\""
18	MALF_TLP_ERR_MASK	R/W	Malformed TLP Mask. Mask bit for Malformed TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
17	REC_OVERFLOW_ERR_MASK	R/W	Receiver Overflow Mask (Optional). This field represents Receiver Overflow Mask. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
16	UNEXP_CMPLT_ERR_MASK	R/W	Unexpected Completion Mask. Mask bit for Unexpected Completion Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15	CMPLT_ABORT_ERR_MASK	R/W	Completer Abort Error Mask (Optional). Mask bit for Completer Abort Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
14	CMPLT_TIMEOUT_ERR_MASK	R/W	Completion Timeout Error Mask. Mask bit for Completion Timeout Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13	FC_PROTOCOL_ERR_MASK	R/W	Flow Control Protocol Error Mask. Mask bit for Flow Control Protocol Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
12	POIS_TLP_ERR_MASK	R/W	Poisoned TLP Error Mask. Mask bit for Poisoned TLP Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5	SURPRISE_DOWN_ERR_MASK	R	Surprise Down Error Mask. Mask bit for Surprise Down Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4	DL_PROTOCOL_ERR_MASK	R/W	Data Link Protocol Error Mask. This field informs whether Data Link Protocol Error is masked or not. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.10.4 UNCORR_ERR_SEV_OFF

- **Name:** Uncorrectable Error Severity Register.
- **Description:** The Uncorrectable Error Severity Register controls whether an individual error is reported as a Non-fatal or Fatal error. An error is reported as fatal when the corresponding error bit in the severity register is Set. If the bit is Clear, the corresponding error is considered non-fatal.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_28	31:28
DMWR_EGRESS_BLOCKED_ERR_SEVERITY	27
PSND_TLP_EGRSS_BLOCKED_ERR_SEVERITY	26
TLP_PRFX_BLOCKED_ERR_SEVERITY	25
ATOMIC_EGRESS_BLOCKED_ERR_SEVERITY	24
RSVDP_23	23
INTERNAL_ERR_SEVERITY	22
ACS_VIOLATION_SEVERITY	21
UNSUPPORTED_REQ_ERR_SEVERITY	20
ECRC_ERR_SEVERITY	19
MALF_TLP_ERR_SEVERITY	18
REC_OVERFLOW_ERR_SEVERITY	17
UNEXP_CMPLT_ERR_SEVERITY	16
CMPLT_ABORT_ERR_SEVERITY	15
CMPLT_TIMEOUT_ERR_SEVERITY	14
FC_PROTOCOL_ERR_SEVERITY	13
POIS_TLP_ERR_SEVERITY	12
RSVDP_6	11:6
SURPRISE_DOWN_ERR_SVRITY	5
DL_PROTOCOL_ERR_SEVERITY	4
RSVDP_0	3:0

Table 3-56 Fields for Register: UNCORR_ERR_SEV_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27	DMWR_EGRESS_BLOCKED_ERR_SEVERITY	R	Deferrable Memory Write Egress Blocked Severity (Optional). Severity bit for Deferrable Memory Write Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
26	PSND_TLP_EGRSS_BLOCKED_ERR_SEVERITY	R/W	Poisoned TLP Egress Blocked Severity. Severity bit for Poisoned TLP Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: AER_ENABLE && CX_DPC_ENABLE && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
25	TLP_PRFX_BLOCKED_ERR_SEVERITY	R	TLP Prefix Blocked Error Severity (Optional). Severity bit for TLP Prefix Blocked Error. <i>Note:</i> Not supported. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
24	ATOMIC_EGRESS_BLOCKED_ERR_SEVERITY	R	AtomicOp Egress Blocked Severity (Optional). Severity bit for AtomicOp Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: ATOMIC_ROUTING_SUP && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
22	INTERNAL_ERR_SEVERITY	R/W	Uncorrectable Internal Error Severity (Optional). Severity bit for Uncorrectable Internal Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: CX_INTERNAL_ERR_REPORTING && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
21	ACS_VIOLATION_SEVERITY	* Varies	ACS Violation Severity. Setting the ACS Violation Severity bit sets ACS Violation error severity to Fatal. The bit is Read-Only Zero for upstream ports, when ACS P2P Egress Control Enable is not set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: if (acs_viol_svrity_mask_wr_en == 1) then R/W (Sticky) else R(Sticky) ■ Dbi: if (acs_viol_svrity_mask_wr_en == 1) then R/W (Sticky) else R(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_ACS_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_ACS_ENABLE) ? R : (CX_MEMORY_MAP_VIEW==1 && CX_ACS_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==1 && !CX_ACS_ENABLE) ? R : (CX_ACS_ENABLE) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_ACS_ENABLE && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
20	UNSUPPORTED_REQ_ERR_SEVERITY	R/W	Unsupported Request Error Severity. Severity bit for Unsupported Request Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
19	ECRC_ERR_SEVERITY	* Varies	ECRC Error Severity (Optional). Severity bit for ECRC Error. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_ECRC_ENABLE && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && DEFAULT_ECRC_CHK_CAP_0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !DEFAULT_ECRC_CHK_CAP_0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && DEFAULT_ECRC_CHK_CAP_0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !DEFAULT_ECRC_CHK_CAP_0) ? \"read-only\" : (DEFAULT_ECRC_CHK_CAP_0) ? \"read-write\" : \"read-only\""
18	MALF_TLP_ERR_SEVERITY	R/W	Malformed TLP Severity. Severity bit for Malformed TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
17	REC_OVERFLOW_ERR_SEVERITY	R/W	Receiver Overflow Error Severity (Optional). Severity bit for Receiver Overflow Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
16	UNEXP_CMPLT_ERR_SEVERITY	R/W	Unexpected Completion Error Severity. Severity bit for Unexpected Completion Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15	CMPLT_ABORT_ERR_SEVERITY	R/W	Completer Abort Error Severity (Optional). Severity bit for Completer Abort Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
14	CMPLT_TIMEOUT_ERR_SEVERITY	R/W	Completion Timeout Error Severity. Severity bit for Completion Timeout Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13	FC_PROTOCOL_ERR_SEVERITY	R/W	Flow Control Protocol Error Severity (Optional). Severity bit for Flow Control Protocol Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
12	POIS_TLP_ERR_SEVERITY	R/W	Poisoned TLP Severity. Severity bit for Poisoned TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
5	SURPRISE_DOWN_ERR_SVRITY	R	Surprise Down Error Severity (Optional). Severity bit for Surprise Down Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4	DL_PROTOCOL_ERR_SEVERITY	R/W	Data Link Protocol Error Severity. Severity bit for Data Link Protocol Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.10.5 CORR_ERR_STATUS_OFF

- **Name:** Correctable Error Status Register.
- **Description:** The Correctable Error Status register reports error status of individual correctable error sources on a PCI Express device Function. When an individual error status bit is Set, it indicates that a particular error occurred; software may clear an error status by writing a 1b to the respective bit. Register bits not implemented by the Function are hardwired to 0b by the controller.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	RSVDP_16
15	HEADER_LOG_OVERFLOW_STATUS
14	CORRECTED_INT_ERR_STATUS
13	ADVISORY_NON_FATAL_ERR_STATUS
12	RPL_TIMER_TIMEOUT_STATUS
11:9	RSVDP_9
8	REPLAY_NO_ROLEOVER_STATUS
7	BAD_DLLP_STATUS
6	BAD_TLP_STATUS
5:1	RSVDP_1
0	RX_ERR_STATUS

Table 3-57 Fields for Register: CORR_ERR_STATUS_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15	HEADER_LOG_OVERFLOW_STATUS	R/W1C	Header Log Overflow Error Status (Optional). This field provides status of Header Log Overflow Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
14	CORRECTED_INT_ERR_STATUS	R/W1C	Corrected Internal Error Status (Optional). This field provides status of Corrected Internal Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_INTERNAL_ERR_REPORTING && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
13	ADVISORY_NON_FATAL_ERR_STATUS	R/W1C	Advisory Non-Fatal Error Status. Status bit for Advisory Non-Fatal Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
12	RPL_TIMER_TIMEOUT_STATUS	R/W1C	Replay Timer Timeout Status. Status bit for Replay Timer Timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
11:9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
8	REPLAY_NO_ROLEOVER_STATUS	R/W1C	REPLAY_NUM Rollover Status. Status bit for REPLAY_NUM Rollover. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7	BAD_DLLP_STATUS	R/W1C	Bad DLLP Status. Status bit for Bad DLLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
6	BAD_TLP_STATUS	R/W1C	Bad TLP Status. Status bit for Bad TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
5:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
0	RX_ERR_STATUS	R/W1C	Receiver Error Status (Optional). This field provides status of Receiver Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.10.6 CORR_ERR_MASK_OFF

- **Name:** Correctable Error Mask Register.
- **Description:** The Correctable Error Mask Register controls reporting of individual correctable errors by this Function to the PCI Express Root Complex through a PCI Express error Message. A masked error (respective bit Set in the mask register) is not reported to the PCI Express Root Complex by this Function. There is a mask bit per error bit in the Correctable Error Status register. Register fields for bits not implemented by the Function are hardwired to 0b by the controller.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_16	31:16
HEADER_LOG_OVERFLOW_MASK	15
CORRECTED_INT_ERR_MASK	14
ADVISORY_NON_FATAL_ERR_MASK	13
RPL_TIMER_TIMEOUT_MASK	12
RSVDP_9	11:9
REPLAY_NO_ROLEOVER_MASK	8
BAD_DLLP_MASK	7
BAD_TLP_MASK	6
RSVDP_1	5:1
RX_ERR_MASK	0

Table 3-58 Fields for Register: CORR_ERR_MASK_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15	HEADER_LOG_OVERFLOW_MASK	R/W	Header Log Overflow Error Mask (Optional). Masking bit for Header Log Overflow Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
14	CORRECTED_INT_ERR_MASK	R/W	Corrected Internal Error Mask (Optional). Masking bit for Corrected Internal Error Mask. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: CX_INTERNAL_ERR_REPORTING && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13	ADVISORY_NON_FATAL_ERR_MASK	R/W	Advisory Non-Fatal Error Mask. Masking bit for Advisory Non-Fatal Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
12	RPL_TIMER_TIMEOUT_MASK	R/W	Replay Timer Timeout Mask. Masking bit for Replay Timer Timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
8	REPLAY_NO_ROLEOVER_MASK	R/W	REPLAY_NUM Rollover Mask. Masking bit for REPLAY_NUM Rollover. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7	BAD_DLLP_MASK	R/W	Bad DLLP Mask. Masking bit for Bad DLLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
6	BAD_TLP_MASK	R/W	Bad TLP Mask. Masking bit for Bad TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	RX_ERR_MASK	R/W	Receiver Error Mask (Optional). Masking bit for Receiver Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.10.7 ADV_ERR_CAP_CTRL_OFF

- **Name:** Advanced Error Capabilities and Control Register.
- **Description:** Advanced Error Capabilities and Control Register provides information whether the individual capability is supported or not. If the capability is supported then it is enabled or not.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:13	RSVDP_13
12	CTO_PRFX_HDR_LOG_CAP
11	TLP_PRFX_LOG_PRESENT
10	MULTIPLE_HEADER_EN
9	MULTIPLE_HEADER_CAP
8	ECRC_CHECK_EN
7	ECRC_CHECK_CAP
6	ECRC_GEN_EN
5	ECRC_GEN_CAP
4:0	FIRST_ERR_POINTER

Table 3-59 Fields for Register: ADV_ERR_CAP_CTRL_OFF

Bits	Name	Memory Access	Description
31:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
12	CTO_PRFX_HDR_LOG_CAP	R	Completion Timeout Prefix/Header Log Capable. If Set, this bit indicates that the Function records the prefix/header of Request TLPs that experience a Completion Timeout error. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
11	TLP_PRFX_LOG_PRESENT	R	TLP Prefix Log Present. If Set and the First Error Pointer is valid, indicates that the TLP Prefix Log register contains valid information. If Clear or if First Error Pointer is invalid, the TLP Prefix Log register is undefined. Default value of this bit is 0. This bit is RsvdP if the End-End TLP Prefix Supported bit is Clear. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_TLP_PREFIX_ENABLE && (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
10	MULTIPLE_HEADER_EN	* Varies	Multiple Header Recording Enable. When Set, this bit enables the Function to record more than one error header. Functions that do not implement the associated mechanism are permitted to hardwire this bit to 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_HDR_LOG_DEPTH_0>=2) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_HDR_LOG_DEPTH_0>=2) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : (CX_HDR_LOG_DEPTH_0>=2) ? \"read-write\" : \"read-only\""
9	MULTIPLE_HEADER_CAP	R	Multiple Header Recording Capable. If Set, this bit indicates that the Function is capable of recording more than one error header. <i>Note:</i> This register field is sticky. Value After Reset: "(CX_HDR_LOG_DEPTH_0 > 1) ? 1 : 0" Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
8	ECRC_CHECK_EN	R/W	ECRC Check Enable. When Set, ECRC checking is enabled. Functions that do not implement the associated mechanism are permitted to hardwire this bit to 0b. Default value of this bit is 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7	ECRC_CHECK_CAP	R	ECRC Check Capable. If Set, this bit indicates that the Function is capable of checking ECRC. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_ECRC_CHK_CAP_0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
6	ECRC_GEN_EN	R/W	ECRC Generation Enable. When Set, ECRC generation is enabled. Functions that do not implement the associated mechanism are permitted to hardwire this bit to 0b. Default value of this bit is 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5	ECRC_GEN_CAP	R	ECRC Generation Capable. If Set, this bit indicates that the Function is capable of generating ECRC. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_ECRC_GEN_CAP_0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:0	FIRST_ERR_POINTER	R	First Error Pointer. The First Error Pointer is a field that identifies the bit position of the first error reported in the Uncorrectable Error Status register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.10.8 HDR_LOG_0_OFF

- **Name:** Header Log Register 0.
- **Description:** The Header Log Register 0 contains the header for the TLP corresponding to a detected error; The header is captured such that, when read using DW accesses, the fields of the header are laid out in the same way the headers are presented in the specification. Therefore, byte 0 of the header is located in byte 3 of the Header Log Register 0, byte 1 of the header is in byte 2 of the Header Log Register 0, and so forth. For 12-byte headers, only bytes 0 through 11 of the Header Log Register are used and values in bytes 12 through 15 are undefined. In certain cases where a Malformed TLP is reported, the Header Log Register may contain TLP Prefix information.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	FIRST_DWORD_FOURTH_BYTE
23:16	FIRST_DWORD_THIRD_BYTE
15:8	FIRST_DWORD_SECOND_BYTE
7:0	FIRST_DWORD_FIRST_BYTE

Table 3-60 Fields for Register: HDR_LOG_0_OFF

Bits	Name	Memory Access	Description
31:24	FIRST_DWORD_FOURTH_BYTE	R	Byte 3 of Header log register of First 32-bit Data Word. This field represents fourth byte of First DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:16	FIRST_DWORD_THIRD_BYTE	R	Byte 2 of Header log register of First 32-bit Data Word. This field represents third byte of First DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
15:8	FIRST_DWORD_SECOND_BYTE	R	Byte 1 of Header log register of First 32-bit Data Word. This field represents second byte of First DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7:0	FIRST_DWORD_FIRST_BYTE	R	Byte 0 of Header log register of First 32-bit Data Word. This field represents first byte of First DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.10.9 HDR_LOG_1_OFF

- **Name:** Header Log Register 1.
- **Description:** The Header Log Register 1 contains the header for the TLP corresponding to a detected error; The header is captured such that, when read using DW accesses, the fields of the header are laid out in the same way the headers are presented in the specification. Therefore, byte 0 of the header is located in byte 3 of the Header Log Register 1, byte 1 of the header is in byte 2 of the Header Log Register 1 and so forth. For 12-byte headers, only bytes 0 through 11 of the Header Log Register are used and values in bytes 12 through 15 are undefined. In certain cases where a Malformed TLP is reported, the Header Log Register may contain TLP Prefix information.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	SECOND_DWORD_FOURTH_BYTE
23:16	SECOND_DWORD_THIRD_BYTE
15:8	SECOND_DWORD_SECOND_BYTE
7:0	SECOND_DWORD_FIRST_BYTE

Table 3-61 Fields for Register: HDR_LOG_1_OFF

Bits	Name	Memory Access	Description
31:24	SECOND_DWORD_FOURTH_BYTE	R	Byte 3 of Header log register of Second 32-bit Data Word. This field represents fourth byte of Second DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:16	SECOND_DWORD_THIRD_BYTE	R	Byte 2 of Header log register of Second 32-bit Data Word. This field represents third byte of Second DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
15:8	SECOND_DWORD_SECOND_BYTE	R	Byte 1 of Header log register of Second 32-bit Data Word. This field represents second byte of Second DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7:0	SECOND_DWORD_FIRST_BYTE	R	Byte 0 of Header log register of Second 32-bit Data Word. This field represents first byte of Second DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.10.10 HDR_LOG_2_OFF

- **Name:** Header Log Register 2.
- **Description:** The Header Log Register 2 contains the header for the TLP corresponding to a detected error; The header is captured such that, when read using DW accesses, the fields of the header are laid out in the same way the headers are presented in the specification. Therefore, byte 0 of the header is located in byte 3 of the Header Log Register 2, byte 1 of the header is in byte 2 of the Header Log Register 2 and so forth. For 12-byte headers, only bytes 0 through 11 of the Header Log Register are used and values in bytes 12 through 15 are undefined. In certain cases where a Malformed TLP is reported, the Header Log Register may contain TLP Prefix information.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	THIRD_DWORD_FOURTH_BYTE
23:16	THIRD_DWORD_THIRD_BYTE
15:8	THIRD_DWORD_SECOND_BYTE
7:0	THIRD_DWORD_FIRST_BYTE

Table 3-62 Fields for Register: HDR_LOG_2_OFF

Bits	Name	Memory Access	Description
31:24	THIRD_DWORD_FOURTH_BYTE	R	Byte 3 of Header log register of Third 32-bit Data Word. This field represents fourth byte of Third DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:16	THIRD_DWORD_THIRD_BYTE	R	Byte 2 of Header log register of Third 32-bit Data Word. This field represents third byte of Third DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
15:8	THIRD_DWORD_SECOND_BYTE	R	Byte 1 of Header log register of Third 32-bit Data Word. This field represents second byte of Third DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7:0	THIRD_DWORD_FIRST_BYTE	R	Byte 0 of Header log register of Third 32-bit Data Word. This field represents first byte of Third DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.10.11 HDR_LOG_3_OFF

- **Name:** Header Log Register 3.
- **Description:** The Header Log Register 3 contains the header for the TLP corresponding to a detected error; The header is captured such that, when read using DW accesses, the fields of the header are laid out in the same way the headers are presented in the specification. Therefore, byte 0 of the header is located in byte 3 of the Header Log Register 3, byte 1 of the header is in byte 2 of the Header Log Register 3 and so forth. For 12-byte headers, only bytes 0 through 11 of the Header Log Register are used and values in bytes 12 through 15 are undefined. In certain cases where a Malformed TLP is reported, the Header Log Register may contain TLP Prefix information.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	FOURTH_DWORD_FOURTH_BYTE
23:16	FOURTH_DWORD_THIRD_BYTE
15:8	FOURTH_DWORD_SECOND_BYTE
7:0	FOURTH_DWORD_FIRST_BYTE

Table 3-63 Fields for Register: HDR_LOG_3_OFF

Bits	Name	Memory Access	Description
31:24	FOURTH_DWORD_FOURTH_BYTE	R	Byte 3 of Header log register of Fourth 32-bit Data Word. This field represents fourth byte of Fourth DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:16	FOURTH_DWORD_THIRD_BYTE	R	Byte 2 of Header log register of Fourth 32-bit Data Word. This field represents third byte of Fourth DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
15:8	FOURTH_DWORD_SECOND_BYTE	R	Byte 1 of Header log register of Fourth 32-bit Data Word. This field represents second byte of Fourth DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7:0	FOURTH_DWORD_FIRST_BYTE	R	Byte 0 of Header log register of Fourth 32-bit Data Word. This field represents first byte of Fourth DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.10.12 TLP_PREFIX_LOG_i_OFF (for i = 1; i <= 4)

- **Name:** TLP Prefix Log Register 1.
- **Description:** The First TLP Prefix Log Register contains the first End-End TLP Prefix from the TLP corresponding to the detected error. The TLP Prefix Log Register is only meaningful when the TLP Prefix Log Present bit is Set. The TLP Prefixes are captured such that, when read using DW accesses, the fields of the TLP Prefix are laid out in the same way the fields of the TLP Prefix are described. Therefore, byte 0 of a TLP Prefix is located in byte 3 of the associated TLP Prefix Log Register; byte 1 of a TLP Prefix is located in byte 2; and so forth. The TLP Prefix Log Registers beyond the number supported by the Function are hardwired to zero by controller. If the End-End TLP Prefix Supported bit is Clear, the TLP Prefix Log Register is not required to be implemented.
- **Size:** 32 bits
- **Offset:** B+0x38 + i*0x4
- **Exists:** (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

CFG_TLP_PFX_LOG_1_FOURTH_BYTE	31:24
CFG_TLP_PFX_LOG_1_THIRD_BYTE	23:16
CFG_TLP_PFX_LOG_1_SECOND_BYTE	15:8
CFG_TLP_PFX_LOG_1_FIRST_BYTE	7:0

Table 3-64 Fields for Register: TLP_PREFIX_LOG_i_OFF (for i = 1; i <= 4)

Bits	Name	Memory Access	Description
31:24	CFG_TLP_PFX_LOG_1_FOURTH_BYTE	R	Byte 3 of Error TLP Prefix Log 1. This field contains fourth byte of First DW of TLP Prefix. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
23:16	CFG_TLP_PFX_LOG_1_THIRD_BYTE	R	Byte 2 of Error TLP Prefix Log 1. This field contains third byte of First DW of TLP Prefix. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
15:8	CFG_TLP_PFX_LOG_1_SECOND_BYTE	R	Byte 1 of Error TLP Prefix Log 1. This field contains second byte of First DW of TLP Prefix. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7:0	CFG_TLP_PFX_LOG_1_FIRST_BYTE	R	Byte 0 of Error TLP Prefix Log 1. This field contains first byte of First DW of TLP Prefix. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AER_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.11 PF0_VC_CAP Registers

3.11.1 VC_BASE

- **Name:** VC Extended Capability Header.
- **Description:** This Register provides Capability Id, Capability version, and next offset of VC Extended Capability Structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:20	VC_NEXT_OFFSET
19:16	VC_CAP_VERSION
15:0	VC_PCIE_EXTENDED_CAP_ID

Table 3-65 Fields for Register: VC_BASE

Bits	Name	Memory Access	Description
31:20	VC_NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	VC_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field depends on the version of specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
15:0	VC_PCIE_EXTENDED_CAP_ID	* Varies	VC Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. Extended Capability ID for the Virtual Channel Extended Capability is either 0002h or 0009h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {(0==0) ? 0x2 : 0} Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

3.11.2 VC_CAPABILITIES_REG_1

- **Name:** Port VC Capability Register 1.
- **Description:** The Port VC Capability Register 1 describes the configuration of the Virtual Channels associated with a PCI Express Port.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && \$pf==0

31:12	RSVDP_12
11:10	VC_PORT_ARBI_TBL_ENTRY_SIZE
9:8	VC_REFERENCE_CLOCK
7	RSVDP_7
6:4	VC_LOW_PRI_EXT_VC_CNT
3	RSVDP_3
2:0	VC_EXT_VC_CNT

Table 3-66 Fields for Register: VC_CAPABILITIES_REG_1

Bits	Name	Memory Access	Description
31:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
11:10	VC_PORT_ARBI_TBL_ENTRY_SIZE	R	Port Arbitration Table Entry Size. This field indicates the size (in bits) of Port Arbitration table entry in the Function. This field is valid only for RCRBs, Switch Ports, and Root Ports that support peer-to-peer traffic. It is not valid and must be hardwired to 00b for Root Ports that do not support peer-to-peer traffic and Endpoints. Values: ■ 0x0 (_1b_size): The size of Port Arbitration table entry is 1 bit. ■ 0x1 (_2b_size): The size of Port Arbitration table entry is 2 bit. ■ 0x2 (_3b_size): The size of Port Arbitration table entry is 4 bit. ■ 0x3 (_4b_size): The size of Port Arbitration table entry is 8 bit. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
9:8	VC_REFERENCE_CLOCK	R	Reference Clock. This field indicates the reference clock for Virtual Channels that support time-based WRR Port Arbitration. This field is valid for RCRBs, Switch Ports, and Root Ports that support peer-to-peer traffic. It is not valid for Root Ports that do not support peer-to-peer traffic, Endpoints, and Switches or Root Complexes not implementing WRR, and must be hardwired to 00b by the controller. Other values are reserved. Values: ■ 0x0 (_100ns_REF_CLK): 100ns reference clock. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
6:4	VC_LOW_PRI_EXT_VC_CNT	* Varies	Low Priority Extended VC Count. This field indicates the number of (extended) Virtual Channels in addition to the default VC belonging to the low-priority VC (LPVC) group that has the lowest priority with respect to other VC resources in a strict-priority VC Arbitration. This field is valid for all Functions. The minimum value of this field is 000b and the maximum value is Extended VC Count . <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_LOW_PRI_EXT_VC_CNT Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
2:0	VC_EXT_VC_CNT	R	Extended VC Count. Indicates the number of (extended) Virtual Channels in addition to the default VC supported by the device. This field is valid for all Functions. The minimum value of this field is 000b (for devices that only support the default VC). The maximum value is 7. Value After Reset: {CX_NVC - 1} Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

3.11.3 VC_CAPABILITIES_REG_2

- **Name:** Port VC Capability Register 2.
- **Description:** The Port VC Capability Register 2 provides further information about the configuration of the Virtual Channels associated with a PCI Express Port.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && \$pf==0

31:24	VC_ARBI_TABLE_OFFSET
23:4	RSVDP_4
3:0	VC_ARBI_CAP

Table 3-67 Fields for Register: VC_CAPABILITIES_REG_2

Bits	Name	Memory Access	Description
31:24	VC_ARBI_TABLE_OFFSET	R	VC Arbitration Table Offset. This field indicates the location of the VC Arbitration Table. This field is valid for all Functions. This field contains the zero-based offset of the table in DQWORDS (16 bytes) from the base address of the Virtual Channel Extended Capability structure. A value of '0' indicates that the table is not present. Value After Reset: DEFAULT_VC_ARB_TABLE_OFFSET Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
23:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
3:0	VC_ARBI_CAP	* Varies	VC Arbitration Capability. This field indicates the types of VC Arbitration supported by the Function for the LPVC group. This field is valid for all Functions that report a Low Priority Extended VC Count field greater than 0. For all other Functions, this field must be hardwired to 00h. Each Bit Location within this field corresponds to a VC Arbitration Capability defined below. When more than 1 bit in this field is Set, it indicates that the Port can be configured to provide different VC arbitration services. Defined bit positions are: ■ Bit 0: Weighted Round Robin (WRR) arbitration with 16 phases ■ Bit 1: WRR arbitration with 32 phases ■ Bit 2: WRR arbitration with 64 phases ■ Bit 3: WRR arbitration with 128 phases ■ Bit 4-7: Reserved <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (enum_value_bit0): Zero value. ■ 0xf (enum_value_bit4): Max value. Value After Reset: DEFAULT_VC_ARB_32 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

3.11.4 VC_STATUS_CONTROL_REG

- **Name:** Port VC Control and Status Register.
- **Description:** This register provides the status and controls the behaviour of the extended capability.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && \$pf==0

RSVDP_17	31:17
VC_ARBI_TABLE_STATUS	16
RSVDP_4	15:4
VC_ARBI_SELECT	3:1
VC_LOAD_VC_ARBI_TABLE	0

Table 3-68 Fields for Register: VC_STATUS_CONTROL_REG

Bits	Name	Memory Access	Description
31:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
16	VC_ARBI_TABLE_STATUS	R	VC Arbitration Table Status. Indicates the coherency status of the VC Arbitration Table. This bit is valid for all Functions when the selected VC uses the VC Arbitration Table. ■ This bit is Set by hardware when any entry of the VC Arbitration Table is written by software. ■ This bit is Cleared by hardware when hardware finishes loading values stored in the VC Arbitration Table after software sets the Load VC Arbitration Table bit in the Port VC Control Register . Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: {(CX_XADM_VC_ARB_WRR==1) ? "writeAsRead" : "unconstrained" }

Bits	Name	Memory Access	Description
15:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
3:1	VC_ARBI_SELECT	R/W	VC Arbitration Select. Used by software to configure the VC arbitration by selecting one of the supported VC Arbitration schemes indicated by the VC Arbitration Capability field in the Port VC Capability Register 2 . This field is valid for all Functions. The permissible values of this field are numbers corresponding to one of the asserted bits in the VC Arbitration Capability field. This field cannot be modified when more than one VC in the LPVC group is enabled. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
0	VC_LOAD_VC_ARBI_TABLE	W	Requests Hardware to Load VC Arbitration Table. Used by software to update the VC Arbitration Table. This bit is valid for all Functions when the selected VC Arbitration uses the VC Arbitration Table. Software sets this bit to request hardware to apply new values programmed into VC Arbitration Table; clearing this bit has no effect. Software checks the VC Arbitration Table Status bit to confirm that new values stored in the VC Arbitration Table are latched by the VC arbitration logic. This bit always returns 0b when read. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

3.11.5 RESOURCE_CAP_REG_VC0

- **Name:** VC Resource Capability Register (0).
- **Description:** The VC Resource Capability Register describes the capabilities and configuration of a particular Virtual Channel (0) resource.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && \$pf==0

31:24	VC_PORT_ARBI_TABLE_VC0
23:22	RSVDP_22
21:16	VC_MAX_TIME_SLOT_VC0
15	VC_REJECT_SNOOP_TRANS_VC0
14:8	RSVDP_8
7:0	VC_PORT_ARBI_CAP_VC0

Table 3-69 Fields for Register: RESOURCE_CAP_REG_VC0

Bits	Name	Memory Access	Description
31:24	VC_PORT_ARBI_TABLE_VC0	R	Port Arbitration Table Offset. This field indicates the location of the Port Arbitration Table associated with the VC resource. This field is valid for all Switch Ports, Root Ports that support peer-to-peer traffic, and RCRBs, but is not valid for Endpoints or Root Ports that do not support peer-to-peer traffic. This field contains the zero-based offset of the table in DQWORDS (16 bytes) from the base address of the Virtual Channel Extended Capability structure. A value of 00h indicates that the table is not present. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
23:22	RSVDP_22	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
21:16	VC_MAX_TIME_SLOT_VC0	* Varies	Maximum Time Slots-1 supported. Indicates the maximum number of time slots (minus one) that the VC resource is capable of supporting when it is configured for time-based WRR Port Arbitration. For example, a value 000 0000b in this field indicates the supported maximum number of time slots is 1 and a value of 111 1111b indicates the supported maximum number of time slots is 128. This field is valid for all Switch Ports, Root Ports that support peer-to-peer traffic, and RCRBs, but is not valid for Endpoints or Root Ports that do not support peer-to-peer traffic. In addition, this field is valid only when the Port Arbitration Capability field indicates that the VC resource supports time-based WRR Port Arbitration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: R Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	VC_REJECT_SNOOP_TRANS_VC0	* Varies	Reject Snoop Transactions. This bit is valid for Root Ports and RCRB; it is not valid for Endpoints or Switch Ports. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: R Values: ■ 0x1 (SET): Any transaction for which the No Snoop attribute is applicable but is not Set within the TLP header is permitted to be rejected as an Unsupported Request. ■ 0x0 (CLEAR): Transactions with or without the No Snoop bit Set within the TLP header are allowed on this VC. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
7:0	VC_PORT_ARBI_CAP_VC0	R	Port Arbitration Capability. This field indicates types of Port Arbitration supported by the VC resource. This field is valid for all Switch Ports, Root Ports that support peer-to-peer traffic, and RCRBs, but not for Endpoints or Root Ports that do not support peer-to-peer traffic. Each Bit Location within this field corresponds to a Port Arbitration Capability defined below. When more than 1 bit in this field is Set, it indicates that the VC resource can be configured to provide different arbitration services. Software selects among these capabilities by writing to the Port Arbitration Select field. Defined bit positions are : ■ Bit 0: Non-configurable hardware-fixed arbitration scheme. For example, Round Robin (RR) ■ Bit 1: Weighted Round Robin (WRR) arbitration with 32 phases ■ Bit 2: WRR arbitration with 64 phases ■ Bit 3: WRR arbitration with 128 phases ■ Bit 4: Time-based WRR with 128 phases ■ Bit 5: WRR arbitration with 256 phases ■ Bits 6-7: Reserved Values: ■ 0x0 (bit_0): Zero value. ■ 0xff (bit_max): Zero value. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

3.11.6 RESOURCE_CON_REG_VC0

- **Name:** VC Resource Control Register (0).
- **Description:** This Register provides VC Enable, VC ID, port Arbitration table and select, and so on for VC (0).
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && \$pf==0

VC_ENABLE_VC0	31
RSVDP_27	30:27
VC_ID_VC	26:24
RSVDP_18	23:18
VC_PORT_ARBI_SELECT_VC0	17
VC_LOAD_PORT_ARBI_TABLE_VC0	16
RSVDP_8	15:8
VC_TC_MAP_VC0_BIT1	7:1
VC_TC_MAP_VC0	0

Table 3-70 Fields for Register: RESOURCE_CON_REG_VC0

Bits	Name	Memory Access	Description
31	VC_ENABLE_VC0	R	VC Enable. This bit is valid for all Functions. Software must use the VC Negotiation Pending bit to check whether the VC negotiation is complete. Default value of this bit is 1b for the first VC resource and is 0b for other VC resource(s). Note: 1. This bit is hardwired to 1b for the default VC (VC0), that is, writing to this bit has no effect for VC0. 2. To enable a Virtual Channel, the VC Enable bits for that Virtual Channel must be Set in both components on a Link. 3. To disable a Virtual Channel, the VC Enable bits for that Virtual Channel must be cleared in both components on a Link. 4. Software must ensure that no traffic is using a Virtual Channel at the time it is disabled. 5. Software must fully disable a Virtual Channel in both components on a Link before re-enabling the Virtual Channel. Values: ■ 0x1 (SET): Enables a Virtual Channel (see Note 1 for exceptions). ■ 0x0 (CLEAR): Disables a Virtual Channel Value After Reset: 0x1 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
30:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
26:24	VC_ID_VC	R	VC ID. This field assigns a VC ID to the VC resource (see note for exceptions). This field is valid for all Functions. This field cannot be modified when the VC is already enabled.<i>Note:</i> For the first VC resource (default VC), this field is read-only and must be hardwired to 000b. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
23:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
17	VC_PORT_ARBI_SELECT_VC0	R	Port Arbitration Select. This field configures the VC resource to provide a particular Port Arbitration service. This field is valid for RCRBs, Root Ports that support peer-to-peer traffic, and Switch Ports, but is not valid for Endpoints or Root Ports that do not support peer-to-peer traffic. The permissible value of this field is a number corresponding to one of the asserted bits in the Port Arbitration Capability field of the VC resource. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
16	VC_LOAD_PORT_ARBI_TABLE_VC0	W	Load Port Arbitration Table. When Set, this bit updates the Port Arbitration logic from the Port Arbitration Table for the VC resource. This bit is valid for all Switch Ports, Root Ports that support peer-to-peer traffic, and RCRBs, but is not valid for Endpoints or Root Ports that do not support peer-to-peer traffic. In addition, this bit is only valid when the Port Arbitration Table is used by the selected Port Arbitration scheme (that is indicated by a Set bit in the Port Arbitration Capability field selected by Port Arbitration Select). Software sets this bit to signal hardware to update Port Arbitration logic with new values stored in Port Arbitration Table ; clearing this bit has no effect. Software uses the Port Arbitration Table Status bit to confirm whether the new values of Port Arbitration Table are completely latched by the arbitration logic. This bit always returns 0b when read. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
7:1	VC_TC_MAP_VC0_BIT1	R/W	Bits 7:1 of TC to VC Mapping. This field indicates the TCs that are mapped to the VC resource. This field is valid for all Functions. Bit locations within this field correspond to TC values. For example, when bit 7 is Set in this field, TC7 is mapped to this VC resource. When more than 1 bit in this field is Set, it indicates that multiple TCs are mapped to the VC resource. In order to remove one or more TCs from the TC/VC Map of an enabled VC, software must ensure that no new or outstanding transactions with the TC labels are targeted at the given Link. Value After Reset: 0x7f Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
0	VC_TC_MAP_VC0	R	Bit 0 of TC to VC Mapping. Bit 0 of this field is read-only. It must be Set for the default VC0 and Clear for all other enabled VCs. Value After Reset: 0x1 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

3.11.7 RESOURCE_STATUS_REG_VCi (for i = 0; i <= 7)

- **Name:** VC Resource Status Register (0).
- **Description:** This Register provides VC Port Arbitration table status and VC Negotiation Pending status for VC(0).
- **Size:** 32 bits
- **Offset:** B+0x18 + i*0x4
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && \$pf==0

31:18	RSVDP_18
17	VC_NEGO_PENDING_VC0
16	VC_PORT_ARBI_TABLE_STATUS_VC0
15:0	RSVDP_0

Table 3-71 Fields for Register: RESOURCE_STATUS_REG_VCi (for i = 0; i <= 7)

Bits	Name	Memory Access	Description
31:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
17	VC_NEGO_PENDING_VC0	R	VC Negotiation Pending. This bit indicates whether the Virtual Channel negotiation (initialization or disabling) is in pending state. This bit is valid for all Functions. The value of this bit is defined only when the Link is in the DL_Active state and the Virtual Channel is enabled (its VC Enable bit is Set). For VC0, this bit is permitted to be hardwired to 0b. Before using a Virtual Channel, software must check whether the VC Negotiation Pending bits for that Virtual Channel are Clear in both components on the Link. Values: ■ 0x1 (SET): When this bit is Set by hardware, it indicates that the VC resource has not completed the process of negotiation. ■ 0x0 (CLEAR): This bit is Cleared by hardware after the VC negotiation is complete (on exit from the FC_INIT2 state). Value After Reset: * Varies Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
16	VC_PORT_ARBI_TABLE_STATUS_VC0	R	Port Arbitration Table Status. This field indicates the coherency status of the Port Arbitration Table associated with the VC resource. This field is valid for RCRBs, Root Ports that support peer-to-peer traffic, and Switch Ports, but is not valid for Endpoints or Root Ports that do not support peer-to-peer traffic. In addition, this bit is valid only when the Port Arbitration Table is used by the selected Port Arbitration for the VC resource. Values: ■ 0x1 (SET): This field is Set by hardware when any entry of the Port Arbitration Table is written to by the software. ■ 0x0 (CLEAR): This field is Cleared by hardware when hardware finishes loading values stored in the Port Arbitration Table after software sets the Load Port Arbitration Table bit. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
15:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

3.11.8 RESOURCE_CAP_REG_VCi (for i = 1; i <= 7)

- **Name:** VC Resource Capability Register (1).
- **Description:** The VC Resource Capability Register describes the capabilities and configuration of a particular Virtual Channel (1) resource.
- **Size:** 32 bits
- **Offset:** B+0x1c + i*0x4
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1 && \$pf==0)

31:24	VC_PORT_ARBI_TABLE_VC1
23:22	RSVDP_22
21:16	VC_MAX_TIME_SLOT_VC1
15	VC_REJECT_SNOOP_TRANS_VC1
14:8	RSVDP_8
7:0	VC_PORT_ARBI_CAP_VC1

Table 3-72 Fields for Register: RESOURCE_CAP_REG_VCi (for i = 1; i <= 7)

Bits	Name	Memory Access	Description
31:24	VC_PORT_ARBI_TABLE_VC1	R	VC1 Port Arbitration Table Offset. This field indicates the location of the Port Arbitration Table associated with the VC resource. This field is valid for all Switch Ports, Root Ports that support peer-to-peer traffic, and RCRBs, but is not valid for Endpoints or Root Ports that do not support peer-to-peer traffic. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)
23:22	RSVDP_22	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)

Bits	Name	Memory Access	Description
21:16	VC_MAX_TIME_SLOT_VC1	* Varies	VC1 Maximum Time Slots-1 supported. This field indicates the maximum number of time slots (minus one) that the VC resource is capable of supporting when it is configured for time-based WRR Port Arbitration. For example, a value 000 0000b in this field indicates the supported maximum number of time slots is 1 and a value of 111 1111b indicates the supported maximum number of time slots is 128. This field is valid for all Switch Ports, Root Ports that support peer-to-peer traffic, and RCRBs, but is not valid for Endpoints or Root Ports that do not support peer-to-peer traffic. In addition, this field is valid only when the Port Arbitration Capability field indicates that the VC resource supports time-based WRR Port Arbitration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: R Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
15	VC_REJECT_SNOOP_TRANS_VC1	* Varies	VC1 Reject Snoop Transactions. This bit is valid for Root Ports and RCRB; it is not valid for Endpoints or Switch Ports. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: R Values: ■ 0x1 (SET): Any transaction for which the No Snoop attribute is applicable but is not Set within the TLP header is permitted to be rejected as an Unsupported Request. ■ 0x0 (CLEAR): Transactions with or without the No Snoop bit Set within the TLP header are allowed on this VC. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
14:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)
7:0	VC_PORT_ARBI_CAP_VC1	R	VC1 Port Arbitration Capability. This field indicates types of Port Arbitration supported by the VC resource. This field is valid for all Switch Ports, Root Ports that support peer-to-peer traffic, and RCRBs, but not for Endpoints or Root Ports that do not support peer-to-peer traffic. Each Bit Location within this field corresponds to a Port Arbitration Capability defined below. When more than 1 bit in this field is Set, it indicates that the VC resource can be configured to provide different arbitration services. Software selects among these capabilities by writing to the Port Arbitration Select field. Defined bit positions are: ■ Bit 0: Non-configurable hardware-fixed arbitration scheme. For example, Round Robin (RR) ■ Bit 1: Weighted Round Robin (WRR) arbitration with 32 phases ■ Bit 2: WRR arbitration with 64 phases ■ Bit 3: WRR arbitration with 128 phases ■ Bit 4: Time-based WRR with 128 phases ■ Bit 5: WRR arbitration with 256 phases ■ Bits 6-7: Reserved. Values: ■ 0x0 (bit_min_val): Zero value. ■ 0xff (bit_max_val): Max value. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)

3.11.9 RESOURCE_CON_REG_VCi (for i = 1; i <= 7)

- **Name:** VC Resource Control Register (1).
- **Description:** This Register provides VC Enable, VC ID, port Arbitration table and select, and so on for VC (1).
- **Size:** 32 bits
- **Offset:** B+0x20 + i*0x4
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1&& \$pf==0)

VC_ENABLE_VC1	31
RSVDP_27	30:27
VC_ID_VC1	26:24
RSVDP_20	23:20
VC_PORT_ARBI_SELECT_VC1	19:17
VC_LOAD_PORT_ARBI_TABLE_VC1	16
RSVDP_8	15:8
VC_TC_MAP_VC1_BIT1	7:1
VC_TC_MAP_VC1	0

Table 3-73 Fields for Register: RESOURCE_CON_REG_VCi (for i = 1; i <= 7)

Bits	Name	Memory Access	Description
31	VC_ENABLE_VC1	R/W	VC1 VC Enable. This bit, when Set, enables a Virtual Channel (note 1 for exceptions). The Virtual Channel is disabled when this bit is cleared. This bit is valid for all Functions. Software must use the VC Negotiation Pending bit to check whether the VC negotiation is complete. Default value of this bit is 1b for the first VC resource and is 0b for other VC resource(s). Note: 1. This bit is hardwired to 1b for the default VC (VC0), that is, writing to this bit has no effect for VC0. 2. To enable a Virtual Channel, the VC Enable bits for that Virtual Channel must be Set in both components on a Link. 3. To disable a Virtual Channel, the VC Enable bits for that Virtual Channel must be cleared in both components on a Link. 4. Software must ensure that no traffic is using a Virtual Channel at the time it is disabled. 5. Software must fully disable a Virtual Channel in both components on a Link before re-enabling the Virtual Channel. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)
30:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)
26:24	VC_ID_VC1	R/W	VC1 VC ID. This field assigns a VC ID to the VC resource (see note for exceptions). This field is valid for all Functions. This field cannot be modified when the VC is already enabled. For the first VC resource (default VC), this field is read-only and must be hardwired to 000b by the controller. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1) Testable: writeAsRead
23:20	RSVDP_20	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)

Bits	Name	Memory Access	Description
19:17	VC_PORT_ARBI_SELECT_VC1	R	VC1 Port Arbitration Select. This field configures the VC resource to provide a particular Port Arbitration service. This field is valid for RCRBs, Root Ports that support peer-to-peer traffic, and Switch Ports, but is not valid for Endpoints or Root Ports that do not support peer-to-peer traffic. The permissible value of this field is a number corresponding to one of the asserted bits in the Port Arbitration Capability field of the VC resource. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)
16	VC_LOAD_PORT_ARBI_TABLE_VC1	W	VC1 Load Port Arbitration Table. This bit is valid for all Switch Ports, Root Ports that support peer-to-peer traffic, and RCRBs, but is not valid for Endpoints or Root Ports that do not support peer-to-peer traffic. In addition, this bit is only valid when the Port Arbitration Table is used by the selected Port Arbitration scheme (that is indicated by a Set bit in the Port Arbitration Capability field selected by Port Arbitration Select). Software sets this bit to signal hardware to update Port Arbitration logic with new values stored in Port Arbitration Table; clearing this bit has no effect. Software uses the Port Arbitration Table Status bit to confirm whether the new values of Port Arbitration Table are completely latched by the arbitration logic. This bit always returns 0b when read. Values: ■ 0x1 (SET): When Set, this bit updates the Port Arbitration logic from the Port Arbitration Table for the VC resource. ■ 0x0 (CLEAR): Clearing this bit has no effect. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)

Bits	Name	Memory Access	Description
7:1	VC_TC_MAP_VC1_BIT1	R/W	VC1 Bits 7:1 of TC to VC Mapping. This field indicates the TCs that are mapped to the VC resource. Bit locations within this field correspond to TC values. For example, when bit 7 is Set in this field, TC7 is mapped to this VC resource. When more than 1 bit in this field is Set, it indicates that multiple TCs are mapped to the VC resource. In order to remove one or more TCs from the TC/VC Map of an enabled VC, software must ensure that no new or outstanding transactions with the TC labels are targeted at the given Link. Default value of this field is FFh for the first VC resource and is 00h for other VC resources. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)
0	VC_TC_MAP_VC1	R	VC1 Bit 0 of TC to VC Mapping. This field indicates the TCs that are mapped to the VC resource. Bit 0 is read-only. It must be Set for the default VC0 and Clear for all other enabled VCs. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_NVC>1)

3.11.10 VC_ARBI_TABLE_REG0_OFF

- **Name:** VC Arbitration Table entry 0.
- **Description:** This register contains the VC Arbitration Table entry for phases 7 down to 0.
- **Size:** 32 bits
- **Offset:** B+0x70
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

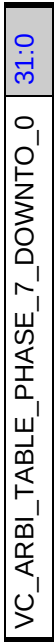


Table 3-74 Fields for Register: VC_ARBI_TABLE_REG0_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_7_DOWNT0_0	R/W	VC Arbitration Table entry. This field contains the phases from 7 to 0. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.11 VC_ARBI_TABLE_REG1_OFF

- **Name:** VC Arbitration Table entry 1.
- **Description:** This register contains the VC Arbitration Table entry for phases 15 down to 8.
- **Size:** 32 bits
- **Offset:** B+0x74
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_15_DOWNT0_8 31:0

Table 3-75 Fields for Register: VC_ARBI_TABLE_REG1_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_15_DOWNT0_8	R/W	VC Arbitration Table entry. This field contains the phases from 15 to 8. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.12 VC_ARBI_TABLE_REG2_OFF

- **Name:** VC Arbitration Table entry 2.
- **Description:** This register contains the VC Arbitration Table entry for phases 23 down to 16.
- **Size:** 32 bits
- **Offset:** B+0x78
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_23_DOWNT0_16

31:0

Table 3-76 Fields for Register: VC_ARBI_TABLE_REG2_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_23_DOWNT0_16	R/W	VC Arbitration Table entry. This field contains the phases from 23 to 16. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.13 VC_ARBI_TABLE_REG3_OFF

- **Name:** VC Arbitration Table entry 3.
- **Description:** This register contains the VC Arbitration Table entry for phases 31 down to 24.
- **Size:** 32 bits
- **Offset:** B+0x7c
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_31_DOWNT0_24

Table 3-77 Fields for Register: VC_ARBI_TABLE_REG3_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_31_DOWNT0_24	R/W	VC Arbitration Table entry. This field contains the phases from 31 down to 24. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.14 VC_ARBI_TABLE_REG4_OFF

- **Name:** VC Arbitration Table entry 4.
- **Description:** This register contains the VC Arbitration Table entry for phases 39 down to 32.
- **Size:** 32 bits
- **Offset:** B+0x80
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_39_DOWNT0_32

Table 3-78 Fields for Register: VC_ARBI_TABLE_REG4_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_39_DOWNT0_32	R/W	VC Arbitration Table entry. This field contains the phases from 39 to 32. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.15 VC_ARBI_TABLE_REG5_OFF

- **Name:** VC Arbitration Table entry 5.
- **Description:** This register contains the VC Arbitration Table entry for phases 47 down to 40.
- **Size:** 32 bits
- **Offset:** B+0x84
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_47_DOWNT0_40

Table 3-79 Fields for Register: VC_ARBI_TABLE_REG5_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_47_DOWNT0_40	R/W	VC Arbitration Table entry. This field contains the phases from 47 to 40. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.16 VC_ARBI_TABLE_REG6_OFF

- **Name:** VC Arbitration Table entry 6.
- **Description:** This register contains the VC Arbitration Table entry for phases 55 down to 48.
- **Size:** 32 bits
- **Offset:** B+0x88
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_55_DOWNT0_48

Table 3-80 Fields for Register: VC_ARBI_TABLE_REG6_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_55_DOWNT0_48	R/W	VC Arbitration Table entry. This field contains the phases from 55 to 48. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.17 VC_ARBI_TABLE_REG7_OFF

- **Name:** VC Arbitration Table entry 7.
- **Description:** This register contains the VC Arbitration Table entry for phases 63 down to 56.
- **Size:** 32 bits
- **Offset:** B+0x8c
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_63_DOWNT0_56

Table 3-81 Fields for Register: VC_ARBI_TABLE_REG7_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_63_DOWNT0_56	R/W	VC Arbitration Table entry. This field contains the phases from 63 to 56. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.18 VC_ARBI_TABLE_REG8_OFF

- **Name:** VC Arbitration Table entry 8.
- **Description:** This register contains the VC Arbitration Table entry for phases 71 downto 64.
- **Size:** 32 bits
- **Offset:** B+0x90
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_71_DOWNT0_64

31:0

Table 3-82 Fields for Register: VC_ARBI_TABLE_REG8_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_71_DOWNT0_64	R/W	VC Arbitration Table entry. This field contains the phases from 71 to 64. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.19 VC_ARBI_TABLE_REG9_OFF

- **Name:** VC Arbitration Table entry 9.
- **Description:** This register contains the VC Arbitration Table entry for phases 79 down to 72.
- **Size:** 32 bits
- **Offset:** B+0x94
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_79_DOWNTO_72

31:0

Table 3-83 Fields for Register: VC_ARBI_TABLE_REG9_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_79_DOWNTO_72	R/W	VC Arbitration Table entry. This field contains the phases from 79 to 72. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.20 VC_ARBI_TABLE_REG10_OFF

- **Name:** VC Arbitration Table entry 10.
- **Description:** This register contains the VC Arbitration Table entry for phases 87 down to 80.
- **Size:** 32 bits
- **Offset:** B+0x98
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

 31:0
VC_ARBI_TABLE_PHASE_87_DOWNT0_80

Table 3-84 Fields for Register: VC_ARBI_TABLE_REG10_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_87_DOWNT0_80	R/W	VC Arbitration Table entry. This field contains the phases from 87 to 80. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.21 VC_ARBI_TABLE_REG11_OFF

- **Name:** VC Arbitration Table entry 11.
- **Description:** This register contains the VC Arbitration Table entry for phases 95 down to 88.
- **Size:** 32 bits
- **Offset:** B+0x9c
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_95_DOWNT0_88

Table 3-85 Fields for Register: VC_ARBI_TABLE_REG11_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_95_DOWNT0_88	R/W	VC Arbitration Table entry. This field contains the phases from 95 to 88. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.22 VC_ARBI_TABLE_REG12_OFF

- **Name:** VC Arbitration Table entry 12.
- **Description:** This register contains the VC Arbitration Table entry for phases 103 downto 96.
- **Size:** 32 bits
- **Offset:** B+0xa0
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0



Table 3-86 Fields for Register: VC_ARBI_TABLE_REG12_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_103_DOWNT0_96	R/W	VC Arbitration Table entry. This field contains the phases from 103 to 96. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.23 VC_ARBI_TABLE_REG13_OFF

- **Name:** VC Arbitration Table entry 13.
- **Description:** This register contains the VC Arbitration Table entry for phases 111 down to 104.
- **Size:** 32 bits
- **Offset:** B+0xa4
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_111_DOWNT0_104

Table 3-87 Fields for Register: VC_ARBI_TABLE_REG13_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_111_DOWNT0_104	R/W	VC Arbitration Table entry. This field contains the phases from 111 to 104. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.24 VC_ARBI_TABLE_REG14_OFF

- **Name:** VC Arbitration Table entry 14.
- **Description:** This register contains the VC Arbitration Table entry for phases 119 down to 112.
- **Size:** 32 bits
- **Offset:** B+0xa8
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

VC_ARBI_TABLE_PHASE_119_DOWNT0_112

Table 3-88 Fields for Register: VC_ARBI_TABLE_REG14_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_119_DOWNT0_112	R/W	VC Arbitration Table entry. This field contains the phases from 119 to 112. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.11.25 VC_ARBI_TABLE_REG15_OFF

- **Name:** VC Arbitration Table entry 15.
- **Description:** This register contains the VC Arbitration Table entry for phases 127 down to 120.
- **Size:** 32 bits
- **Offset:** B+0xac
- **Exists:** (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR && \$pf==0

 31:0
VC_ARBI_TABLE_PHASE_127_DOWNT0_120

Table 3-89 Fields for Register: VC_ARBI_TABLE_REG15_OFF

Bits	Name	Memory Access	Description
31:0	VC_ARBI_TABLE_PHASE_127_DOWNT0_120	R/W	VC Arbitration Table entry. This field contains the phases from 127 to 120. Each phase is represented on 4 bits. Bit [2:0] of each phase contains the VC ID of the scheduled Virtual Channel for this phase. Bit 3 of each phase is reserved. Value After Reset: 0x0 Exists: (VC_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_XADM_VC_ARB_WRR==1

3.12 PF_SN_CAPi Registers

3.12.1 SN_BASE

- **Name:** Device Serial Number Extended Capability Header.
- **Description:** The Device Serial Number Extended Capability is an optional Extended Capability that may be implemented by any PCI Express device Function. This register provides Capability Id, Capability version, and Next offset of Capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (SERIAL_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

SN_NEXT_OFFSET	31:20
SN_CAP_VERSION	19:16
SN_PCIE_EXTENDED_CAP_ID	15:0

Table 3-90 Fields for Register: SN_BASE

Bits	Name	Memory Access	Description
31:20	SN_NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (SERIAL_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
19:16	SN_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field is depends on the version of specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (SERIAL_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15:0	SN_PCIE_EXTENDED_CAP_ID	* Varies	Serial Number Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. Extended Capability ID for the Device Serial Number Extended Capability is 0003h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x3 Exists: (SERIAL_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

3.12.2 SER_NUM_REG_DW_1

- **Name:** Serial Number 1 Register.
- **Description:** The Serial Number register is a 64-bit field that contains the IEEE defined 64-bit extended unique identifier.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (SERIAL_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

SN_SER_NUM_REG_1_DW 31:0

Table 3-91 Fields for Register: SER_NUM_REG_DW_1

Bits	Name	Memory Access	Description
31:0	SN_SER_NUM_REG_1_DW	* Varies	IEEE 64 bit Device Serial Number (DW 1). This field contains the IEEE defined 64-bit Extended Unique Identifier [EUI-64]. This identifier includes a 24-bit company id value assigned by IEEE registration authority and a 40-bit extension identifier assigned by the manufacturer. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_SN_DW1 Exists: (SERIAL_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && SERIAL_CAP_ENABLE) ? \\"read-write\\" : (CX_MEMORY_MAP_VIEW==0) ? \\"read-only\\" : (CX_MEMORY_MAP_VIEW==1) ? \\"read-only\\" : \\"read-only\\" "

3.12.3 SER_NUM_REG_DW_2

- **Name:** Serial Number 2 Register.
- **Description:** The Serial Number register is a 64-bit field that contains the IEEE defined 64-bit extended unique identifier.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (SERIAL_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

SN_SER_NUM_REG_2_DW 31:0

Table 3-92 Fields for Register: SER_NUM_REG_DW_2

Bits	Name	Memory Access	Description
31:0	SN_SER_NUM_REG_2_DW	* Varies	IEEE 64 bit Device Serial Number (DW 2). This field contains the IEEE defined 64-bit Extended Unique Identifier [EUI-64]. This identifier includes a 24-bit company id value assigned by IEEE registration authority and a 40-bit extension identifier assigned by the manufacturer. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_SN_DW2 Exists: (SERIAL_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && SERIAL_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

3.13 PF_PB_CAPi Registers

3.13.1 PB_BASE

- **Name:** Power Budgeting Extended Capability Header.
- **Description:** This register provides Capability Id, Capability version, and Next offset of Capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

PB_NEXT_OFFSET	31:20
PB_CAP_VERSION	19:16
PB_PCIE_EXTENDED_CAP_ID	15:0

Table 3-93 Fields for Register: PB_BASE

Bits	Name	Memory Access	Description
31:20	PB_NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
19:16	PB_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field depends on the version of specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15:0	PB_PCIE_EXTENDED_CAP_ID	* Varies	PB Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. Extended Capability ID for the Power Budgeting Extended Capability is 0004h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x4 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.13.2 PB_DATA_SELECT

- **Name:** Data select Register.
- **Description:** The Power Budgeting Data Select Register is an 8-bit read-write register that indexes the Power Budgeting Data reported through the Power Budgeting Data Register and selects the DWORD of Power Budgeting Data that is to appear in the Power Budgeting Data Register.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:8	7:0
RSVDP_8	PB_DATA_SEL

Table 3-94 Fields for Register: PB_DATA_SELECT

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:0	PB_DATA_SEL	R/W	Data Select Register. The Power Budgeting Data Select Register is an 8-bit read-write register that indexes the Power Budgeting Data reported through the Power Budgeting Data Register and selects the DWORD of Power Budgeting Data that is to appear in the Power Budgeting Data Register. Values for this register start at zero to select the first DWORD of Power Budgeting Data; subsequent DWORDs of Power Budgeting Data are selected by increasing index values. Value After Reset: 0x0 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.13.3 DATA_REG_PB

- **Name:** Data Register.
- **Description:** This read-only register returns the DWORD of Power Budgeting Data selected by the Power Budgeting Data Select Register . Each DWORD of the Power Budgeting Data describes the power usage of the device in a particular operating condition. Power Budgeting Data for different operating conditions is not required to be returned in any particular order, as long as incrementing the Power Budgeting Data Select Register causes information for a different operating condition to be returned. If the Power Budgeting Data Select Register contains a value greater than or equal to the number of operating conditions for which the device provides power information, this register must return all zeros. The Base Power and Data Scale fields describe the power usage of the device; the Power Rail , Type , PM State , and PM Sub State fields describe the conditions under which the device has this power usage.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:21	RSVDP_21
20:18	PB_POWER_RAIL_STATE
17:15	PB_TYPE
14:13	PB_PM_STATE
12:10	PB_PM_SUB_STATE
9:8	PB_DATA_SCALE
7:0	PB_BASE_POWER

Table 3-95 Fields for Register: DATA_REG_PB

Bits	Name	Memory Access	Description
31:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
20:18	PB_POWER_RAIL_STATE	R	Power Rail State. This field specifies the thermal load or power rail of the operating condition being described. All encodings other than the defined encodings are reserved. Values: ■ 0x0 (PWR_12V): Power (12V) ■ 0x1 (PWR_3_3V): Power (3.3V) ■ 0x2 (PWR_1_5V_OR_1_8V): Power (1.5V or 1.8V) ■ 0x7 (THERMAL): Thermal Value After Reset: 0x0 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
17:15	PB_TYPE	R	Type of Operating Condition. This field specifies the type of the operating condition being described. All encodings other than the defined encodings are reserved. Values: ■ 0x0 (PME_AUX): PME Aux ■ 0x1 (AUX): Auxiliary ■ 0x2 (IDL): Idle ■ 0x3 (SUSTAINED): Sustained ■ 0x4 (SUSTAINED_PWR_REDUCTION): Sustained - Emergency Power Reduction State ■ 0x5 (MAX_PWR_RREDUCTION): Maximum - Emergency Power Reduction State ■ 0x7 (MAX): Maximum Value After Reset: 0x0 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
14:13	PB_PM_STATE	R	PM State. This field specifies the power management state of the operating condition being described. A device returns 11b in this field and Aux or PME Aux in the Type field to specify the D3 cold PM State. An encoding of 11b along with any other Type field value specifies the D3 hot state. Values: ■ 0x0 (D0): D0 ■ 0x1 (D1): D1 ■ 0x2 (D2): D2 ■ 0x3 (D3): D2 Value After Reset: 0x0 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
12:10	PB_PM_SUB_STATE	R	PM Sub State. This field specifies the power management sub state of the operating condition being described. For other value device specific sub state. Values: 0x0 (DEFAULT): default sub state. Value After Reset: 0x0 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
9:8	PB_DATA_SCALE	R	Data Scale. This field specifies the scale to apply to the Base Power value. The powerconsumption of the device is determined by multiplying the contents of the Base Power field with the value corresponding to the encoding returned by this field, except as noted above. Values: 0x0 (_1.0_X): 1.0x 0x1 (_0.1_X): 0.1x 0x2 (_0.01_X): 0.01x 0x3 (_0.001_X): 0.001x Value After Reset: 0x0 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:0	PB_BASE_POWER	R	Base Power. This field specifies in watts the base power value in the given operating condition. This value must be multiplied by the data scale to produce the actual power consumption value except when the Data Scale field equals 00b (1.0x) and Base Power exceeds EFh. The alternative encodings defined in Values: are used. Values between F3h to FFh are reserved for the value greater than 300W. Values: 0xf0 (_240W_TO_250W): greater than 239 W and less than or equal to 250 W Slot Power Limit. 0xf1 (_251W_TO_275W): greater than 250 W and less than or equal to 275 W Slot Power Limit. 0xf2 (_276W_TO_300W): greater than 275 W and less than or equal to 300 W Slot Power Limit. Value After Reset: 0x0 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.13.4 CAP_REG_PB

- **Name:** Power Budget Capability Register.
- **Description:** This register indicates the power budgeting capabilities of a device.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:1	
RSVDP_1	PB_SYS_ALLOC

Table 3-96 Fields for Register: CAP_REG_PB

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	PB_SYS_ALLOC	* Varies	System Allocated PB. When Set, this bit indicates that the power budget for the device is included within the system power budget. Reported Power Budgeting Data for this device must be ignored by software for power budgeting decisions if this bit is Set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Value After Reset: DEFAULT_PWR_BUDGET_SYS_ALLOC Exists: (PWR_BUDGET_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.14 PF_ARI_CAPi Registers

3.14.1 ARI_BASE

- **Name:** ARI Capability Header.
- **Description:** This register provides Capability Id, Capability version, and Next offset of Capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	ARI_NEXT_OFFSET
19:16	ARI_CAP_VERSION
15:0	ARI_PCIE_EXTENDED_CAP_ID

Table 3-97 Fields for Register: ARI_BASE

Bits	Name	Memory Access	Description
31:20	ARI_NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	ARI_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the capability structure present. This field is depends on the version of specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	ARI_PCIE_EXTENDED_CAP_ID	* Varies	ARI Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the extended capability. PCI Express Extended Capability ID for the ARI Capability is 000Eh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xe Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.14.2 CAP_REG

- **Name:** ARI Capability and Control Register.
- **Description:** This Register provides information regarding supported ARI Capability, and controls that capability.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:23	RSVDP_23
22:20	ARI_FUN_GRP
19:18	RSVDP_18
17	ARI_ACS_FUN_GRP_EN
16	ARI_MFVC_FUN_GRP_EN
15:8	ARI_NEXT_FUN_NUM
7:2	RSVDP_2
1	ARI_ACS_FUN_GRP_CAP
0	ARI_MFVC_FUN_GRP_CAP

Table 3-98 Fields for Register: CAP_REG

Bits	Name	Memory Access	Description
31:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
22:20	ARI_FUN_GRP	* Varies	Function Group. Assigns a Function Group Number to this Function. Must be hardwired to 000b by the controller if in Function 0, the MFVC Function Groups Capability bit and ACS Function Groups Capability bit are both 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: CAP_REG.ARI_ACS_FUN_GRP_CAP ? RW : RO ■ Dbi: CAP_REG.ARI_ACS_FUN_GRP_CAP ? RW : RO and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_ACS_FUNC_GRP_VALUE) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_ACS_FUNC_GRP_VALUE) ? R : (CX_MEMORY_MAP_VIEW==1 && CX_ACS_FUNC_GRP_VALUE) ? R/W : (CX_MEMORY_MAP_VIEW==1 && !CX_ACS_FUNC_GRP_VALUE) ? R : (CX_ACS_FUNC_GRP_VALUE) ? R/W : R Value After Reset: 0x0 Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
19:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
17	ARI_ACS_FUN_GRP_EN	* Varies	ACS Function Groups Enable. Applicable only for Function 0; must be hardwired to 0b for all other Functions. When set, each Function in the ARI Device must associate bits within its Egress Control Vector with Function Group Numbers rather than Function Numbers. Must be hardwired to 0b if the ACS Function Groups Capability bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: if (CAP_REG.ARI_ACS_FUN_GRP_CAP && pf==0) then R/W else R ■ Dbi: if (CAP_REG.ARI_ACS_FUN_GRP_CAP && pf==0) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && \$pf==0 && CX_ACS_FUNC_GRP_VALUE)? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_ACS_FUNC_GRP_VALUE)? R: (CX_MEMORY_MAP_VIEW==1 && \$pf==0 && CX_ACS_FUNC_GRP_VALUE)? R/W: (CX_MEMORY_MAP_VIEW==1 && !CX_ACS_FUNC_GRP_VALUE)? R: (\$pf==0 && CX_ACS_FUNC_GRP_VALUE)? R/W: R Value After Reset: 0x0 Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
16	ARI_MFVC_FUN_GRP_EN	R	MFVC Function Groups Enable. Applicable only for Function 0; must be hardwired to 0b for all other Functions. When set, the ARI Device must interpret entries in its Function Arbitration Table as Function Group Numbers rather than Function Numbers. Must be hardwired to 0b by the controller if the MFVC Function Groups Capability bit is 0b. Value After Reset: 0x0 Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:8	ARI_NEXT_FUN_NUM	* Varies	Next Function Number. This field indicates the Function Number of the next higher numbered Function in the Device, or 00h if there are no higher numbered Functions. Function 0 starts this linked list of Functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {(0 +1) != CX_NFUNC) ? (0+1) : 0} Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
7:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
1	ARI_ACS_FUN_GRP_CAP	* Varies	ACS Function Groups Capability. Applicable only for Function 0; must be 0b for all other Functions. If 1b, indicates that the ARI Device supports Function Group level granularity for ACS P2P Egress Control through its ACS Capability structures. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && \$pf==0 && CX_ACS_FUNC_GRP_VALUE && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R <i>Note:</i> This register field is sticky. Value After Reset: {(0 == 0) ? CX_ACS_FUNC_GRP_VALUE : 0} Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
0	ARI_MFVC_FUN_GRP_CAP	R	Multi Functional Virtual Channel (MFVC) Function Groups Capability. Applicable only for Function 0; must be 0b for all other Functions. If 1b, indicates that the ARI Device supports Function Group level arbitration through its Multi-Function Virtual Channel (MFVC) Capability structure. Value After Reset: 0x0 Exists: (CX_ARI_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.15 PF0_SPCIE_CAP Registers

3.15.1 SPCIE_CAP_HEADER_REG

- **Name:** SPCIE Capability Header.
- **Description:** This Register provides Capability Id, Capability Version, and next Offset of SPCIE Structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	EXTENDED_CAP_ID

Table 3-99 Fields for Register: SPCIE_CAP_HEADER_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
15:0	EXTENDED_CAP_ID	* Varies	Secondary PCI Express Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. PCI Express Extended Capability ID for the Secondary PCI Express Extended Capability is 0019h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x19 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

3.15.2 LINK_CONTROL3_REG

- **Name:** Link Control 3 Register.
- **Description:** This Register controls equilization and equilization interrupt.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:2	1	0
RSVDP_2	EQ_REQ_INT_EN	PERFORM_EQ

Table 3-100 Fields for Register: LINK_CONTROL3_REG

Bits	Name	Memory Access	Description
31:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
1	EQ_REQ_INT_EN	* Varies	Link Equalization Request Interrupt Enable. This bit is RW for Downstream Ports and for Upstream Ports when Crosslink Supported is 1b. This bit is not applicable and is RsvdP for Upstream Ports when the Crosslink Supported bit is 0b. If the Port does not support 8.0 GT/s, this bit is permitted to be hardwired to 0b by the controller. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When Set, this bit enables the generation of an interrupt to indicate that the Link Equalization Request 8.0 GT/s bit, the Link Equalization Request 16.0 GT/s bit, or the Link Equalization Request 32.0 GT/s bit has been set. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_CROSSLINK_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_CROSSLINK_ENABLE) ? \"read-only\" : (CX_CROSSLINK_ENABLE) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
0	PERFORM_EQ	* Varies	Perform Equalization. This bit is RW for Downstream Ports and for Upstream Ports when Crosslink Supported is 1b. This bit is not applicable and is RsvdP for Upstream Ports when the Crosslink Supported bit is 0b. If the Port does not support 8.0 GT/s, this bit is permitted to be hardwired to 0b by the controller. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_CROSSLINK_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_CROSSLINK_ENABLE) ? R: (CX_MEMORY_MAP_VIEW==1 && CX_CROSSLINK_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==1 && !CX_CROSSLINK_ENABLE) ? R: (CX_CROSSLINK_ENABLE) ? R/W : R Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When this bit is 1b and a 1b is written to the Retrain Link bit with the Target Link Speed field set to 8.0 GT/s or higher, the Downstream Port must perform Link Equalization. Value After Reset: * Varies Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_CROSSLINK_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_CROSSLINK_ENABLE) ? \"read-only\" : (CX_CROSSLINK_ENABLE) ? \"read-write\" : \"read-only\" "

3.15.3 LANE_ERR_STATUS_REG

- **Name:** Lane Error Status Register.
- **Description:** This Register contains Lane Error Status Bits per Lane.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2))



Table 3-101 Fields for Register: LANE_ERR_STATUS_REG

Bits	Name	Memory Access	Description
x:y	RSVDP_LANE_ERR_STATUS	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Range Variable[x]: 32-(CX_NL) + CX_NL - 1 Range Variable[y]: CX_NL

Bits	Name	Memory Access	Description
x:0	LANE_ERR_STATUS	R/W1C	Lane Error Status Bits per Lane. Each bit indicates if the corresponding Lane detected a Lane-based error. A value of 1b indicates that a Lane based-error was detected on the corresponding Lane Number. For Ports that are narrower than 32 Lanes, the unused upper bits [31: Maximum Link Width] are RsvdZ. For Ports that do not support 8.0 GT/s and do not set these bits based on 8b/10b errors, this field is permitted to be hardwired to 0 by the controller. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true Range Variable[x]: CX_NL - 1

3.15.4 SPCIE_CAP_OFF_iH_REG (for i = 0C; i <= 28)

- **Name:** Lane Equalization Control Register for lanes 1 and 0.
- **Description:** This register provides Transmitter Preset and Receiver Preset Hint for Downstream Port and Upstream Port.
- **Size:** 32 bits
- **Offset:** B+0xc + i*0x4
- **Exists:** (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2))

31	RSVDP_31
30:28	USP_RX_PRESET_HINT1
27:24	USP_TX_PRESET1
23	RSVDP_23
22:20	DSP_RX_PRESET_HINT1
19:16	DSP_TX_PRESET1
15	RSVDP_15
14:12	USP_RX_PRESET_HINT0
11:8	USP_TX_PRESET0
7	RSVDP_7
6:4	DSP_RX_PRESET_HINT0
3:0	DSP_TX_PRESET0

Table 3-102 Fields for Register: SPCIE_CAP_OFF_iH_REG (for i = 0C; i <= 28)

Bits	Name	Memory Access	Description
31	RSVDP_31	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_GEN4_SPEED? 1 : (CX_NL>0))

Bits	Name	Memory Access	Description
30:28	USP_RX_PRESET_HINT1	* Varies	Upstream Port 8.0 GT/s Receiver Preset Hint 1. The write value is gated with the PCIE_CAP_CROSS_LINK_SUPPORT field of LINK_CAPABILITIES2_REG. Field contains the Receiver preset hint 1 value sent or received during 8.0 GT/s Link Equalization. ■ Case A: When Operating port is Downstream port and whether the Crosslink is supported or not, this field representss the value sent on Lane 0 during 8.0 GT/s equalization. ■ Case B: When Operating port is Upstream port and Cross-link Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 0 during Link Equalization. ■ Case C: When Operating port is Upstream port and Cross-link Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is HwInit. For case B, Field is RO. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Value After Reset: {(CX_CROSSLINK_ENABLE== 1) ? "DEFAULT_USP_RX_PRESET_HINT1" : "0x7"} Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && ((CX_GEN4_SPEED && !CX_PL16G_ENABLE)? (CX_NL==1? 1:(CX_NL>1)):(CX_NL>1)) Testable: {(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") } Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27:24	USP_TX_PRESET1	* Varies	Upstream Port 8.0 GT/s Transmitter Preset 1. The write value is gated with the PCIE_CAP_CROSS_LINK_SUPPORT field of LINK_CAPABILITIES2_REG. Field contains the Transmitter preset 1 value sent or received during 8.0 GT/s Link Equalization. ■ Case A: When Operating port is Downstream port and whether the Crosslink is supported or not, this field representss the value sent on Lane 0 during 8.0 GT/s equalization. ■ Case B: When Operating port is Upstream port and Cross-link Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 0 during Link Equalization. ■ Case C: When Operating port is Upstream port and Cross-link Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is HwInit. For case B, Field is RO. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Value After Reset: {(CX_CROSSLINK_ENABLE== 1) ? "DEFAULT_USP_TX_PRESET1" : "DEFAULT_LOCAL_G3_TX_PRESET"} Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && ((CX_GEN4_SPEED && !CX_PL16G_ENABLE)? (CX_NL==1? 1:(CX_NL>1)):(CX_NL>1)) Testable: {(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") } Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\" "
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_GEN4_SPEED? 1 : (CX_NL>0))

Bits	Name	Memory Access	Description
22:20	DSP_RX_PRESET_HINT1	* Varies	Downstream Port 8.0 GT/s Receiver Preset Hint 1. Receiver preset hint 1 value that may be used as a suggested setting for 8.0 GT/s receiver equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP. Otherwise, this field is HwInit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: {(CX_CROSSLINK_ENABLE== 1) ? "DEFAULT_DSP_RX_PRESET_HINT1" : "0x0"} Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && ((CX_GEN4_SPEED && !CX_PL16G_ENABLE)? (CX_NL==1? 1:(CX_NL>1)):(CX_NL>1)) Testable: {(CX_CROSSLINK_ENABLE== 0) ? "unconstrained" : ((CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") } Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
19:16	DSP_TX_PRESET1	* Varies	Downstream Port 8.0 GT/s Transmitter Preset 1. Transmitter preset 1 value used for 8.0 GT/s equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP. Otherwise, this field is Hwlnit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: {(CX_CROSSLINK_ENABLE== 1) ? "DEFAULT_DSP_TX_PRESET1" : "0x0"} Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && ((CX_GEN4_SPEED && !CX_PL16G_ENABLE)? (CX_NL==1? 1:(CX_NL>1)):(CX_NL>1)) Testable: {(CX_CROSSLINK_ENABLE== 0) ? "unconstrained" : ((CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") } Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\" "
15	RSVDP_15	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_GEN4_SPEED? 1 : (CX_NL>0))

Bits	Name	Memory Access	Description
14:12	USP_RX_PRESET_HINT0	* Varies	Upstream Port 8.0 GT/s Receiver Preset Hint 0. The write value is gated with the PCIE_CAP_CROSS_LINK_SUPPORT field of LINK_CAPABILITIES2_REG. Field contains the Receiver preset hint 0 value sent or received during 8.0 GT/s Link Equalization. ■ Case A: When Operating port is Downstream port and whether the Crosslink is supported or not, this field representss the value sent on Lane 0 during 8.0 GT/s equalization. ■ Case B: When Operating port is Upstream port and Cross-link Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 0 during Link Equalization. ■ Case C: When Operating port is Upstream port and Cross-link Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is HwInit. For case B, Field is RO. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Value After Reset: {(CX_CROSSLINK_ENABLE== 1) ? "DEFAULT_USP_RX_PRESET_HINT0" : "0x7"} Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_GEN4_SPEED? 1 : (CX_NL>0)) Testable: {(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") } Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
11:8	USP_TX_PRESET0	* Varies	Upstream Port 8.0 GT/s Transmitter Preset 0. The write value is gated with the PCIE_CAP_CROSS_LINK_SUPPORT field of LINK_CAPABILITIES2_REG. Field contains the Transmitter preset 0 value sent or received during 8.0 GT/s Link Equalization. ■ Case A: When Operating port is Downstream port and whether the Crosslink is supported or not, this field representss the value sent on Lane 0 during 8.0 GT/s equalization. ■ Case B: When Operating port is Upstream port and Cross-link Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 0 during Link Equalization. ■ Case C: When Operating port is Upstream port and Cross-link Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is HwInit. For case B, Field is RO. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Value After Reset: {(CX_CROSSLINK_ENABLE== 1) ? "DEFAULT_USP_TX_PRESET0" : "DEFAULT_LOCAL_G3_TX_PRESET"} Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_GEN4_SPEED? 1 : (CX_NL>0)) Testable: {(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") } Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_GEN4_SPEED? 1 : (CX_NL>0))

Bits	Name	Memory Access	Description
6:4	DSP_RX_PRESET_HINT0	* Varies	Downstream Port 8.0 GT/s Receiver Preset Hint 0. Receiver preset hint 0 value that may be used as a suggested setting for 8.0 GT/s receiver equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP. Otherwise, this field is HwInit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: {(CX_CROSSLINK_ENABLE== 1) ? "DEFAULT_DSP_RX_PRESET_HINT0" : "0x0"} Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_GEN4_SPEED? 1 : (CX_NL>0)) Testable: {(CX_CROSSLINK_ENABLE== 0) ? "unconstrained" : ((CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") } Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
3:0	DSP_TX_PRESET0	* Varies	Downstream Port 8.0 GT/s Transmitter Preset 0. Transmitter preset 0 value used for 8.0 GT/s equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP. Otherwise, this field is Hwlnit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: {(CX_CROSSLINK_ENABLE== 1) ? "DEFAULT_DSP_TX_PRESET0" : "0x0"} Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_GEN4_SPEED? 1 : (CX_NL>0)) Testable: {(CX_CROSSLINK_ENABLE== 0) ? "unconstrained" : ((CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") } Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

3.16 PF0_PL16G_CAP Registers

3.16.1 PL16G_EXT_CAP_HDR_REG

- **Name:** Physical Layer 16.0 GT/s Extended Capability Header.
- **Description:** Physical Layer 16.0 GT/s Extended Capability Header provides information about Capability ID, Version, and next offset.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:20	NEXT_OFFSET
19:16	CAP_VERSION
15:0	EXTENDED_CAP_ID

Table 3-103 Fields for Register: PL16G_EXT_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This bit depends on the version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: wire ? unconstrained : writeAsRead
15:0	EXTENDED_CAP_ID	* Varies	PCI Express Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. The Extended Capability ID for the Physical Layer 16.0 GT/s Capability is 0026h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x26 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: wire ? unconstrained : writeAsRead

3.16.2 PL16G_CAPABILITY_REG

- **Name:** 16.0 GT/s Capabilities Register.
- **Description:** This register is reserved for the future update.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))



Table 3-104 Fields for Register: PL16G_CAPABILITY_REG

Bits	Name	Memory Access	Description
31:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED

3.16.3 PL16G_CONTROL_REG

- **Name:** 16.0 GT/s Control Register.
- **Description:** This register is reserved for the future update.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))



Table 3-105 Fields for Register: PL16G_CONTROL_REG

Bits	Name	Memory Access	Description
31:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED

3.16.4 PL16G_STATUS_REG

- **Name:** 16.0 GT/s Status Register.
- **Description:** 16.0 GT/s Status Register provides status of equalization of 16.0 GT/s speed.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:5	RSVDP_5
4	LINK_EQ_16G_REQ
3	EQ_16G_CPL_P3
2	EQ_16G_CPL_P2
1	EQ_16G_CPL_P1
0	EQ_16G_CPL

Table 3-106 Fields for Register: PL16G_STATUS_REG

Bits	Name	Memory Access	Description
31:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED
4	LINK_EQ_16G_REQ	R/W1C	Link Equalization Request 16.0GT/s. For a Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): This bit is Set by hardware to request the 16.0 GT/s Link equalization process to be performed on the Link. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	EQ_16G_CPL_P3	R	Equalization 16.0GT/s Phase 3 Successful. For a Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): This bit indicates that Phase 3 of the 16.0 GT/s Transmitter Equalization procedure has successfully completed. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true
2	EQ_16G_CPL_P2	R	Equalization 16.0GT/s Phase 2 Successful. For a Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): This bit indicates that Phase 2 of the 16.0 GT/s Transmitter Equalization procedure has successfully completed. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
1	EQ_16G_CPL_P1	R	Equalization 16.0GT/s Phase 1 Successful. For a Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): This bit indicates that Phase 1 of the 16.0 GT/s Transmitter Equalization procedure has successfully completed. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true
0	EQ_16G_CPL	R	Equalization 16.0GT/s Complete. For a Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): This bit indicates that the 16.0 GT/s Transmitter Equalization procedure has completed. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true

3.16.5 PL16G_LC_DPAR_STATUS_REG

- **Name:** 16.0 GT/s Local Data Parity Mismatch Status Register.
- **Description:** The Local Data Parity Mismatch Status register is a 32-bit vector where each bit indicates if the local receiver detected a Data Parity mismatch on the Lane with the corresponding Lane number. This Lane number is the default Lane number which is invariant to Link width and Lane reversal negotiation that occurs during Link training.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

x:y	RSVDP_LC_DPAR_STATUS
x:0	LC_DPAR_STATUS

Table 3-107 Fields for Register: PL16G_LC_DPAR_STATUS_REG

Bits	Name	Memory Access	Description
x:y	RSVDP_LC_DPAR_STATUS	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Range Variable[x]: 32-(CX_NL) + CX_NL - 1 Range Variable[y]: CX_NL

Bits	Name	Memory Access	Description
x:0	LC_DPAR_STATUS	R/W1C	Local Data Parity Mismatch Status. Each bit indicates if the corresponding Lane detected a Data Parity mismatch. A value of 1b indicates that a mismatch was detected on the corresponding Lane Number. The default value of each bit is 0b. For Ports that are narrower than 32 Lanes, the unused upper bits [31: MaximumLink Width] are RsvdZ. Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true Range Variable[x]: CX_NL - 1

3.16.6 PL16G_FIRST_RETIMER_DPAR_STATUS_REG

- **Name:** 16.0 GT/s First Retimer Data Parity Mismatch Status Register.
- **Description:** The First Retimer Data Parity Status register is a 32-bit vector where each bit indicates if the first Retimer of a Path detected a Data Parity mismatch on the Lane with the corresponding Lane number. This Lane number is the default Lane number which is invariant to Link width and Lane reversal negotiation that occurs during Link training.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

x:y	RSVDP_FIRST_RETIMER_DPAR_STATUS
x:0	FIRST_RETIMER_DPAR_STATUS

Table 3-108 Fields for Register: PL16G_FIRST_RETIMER_DPAR_STATUS_REG

Bits	Name	Memory Access	Description
x:y	RSVDP_FIRST_RETIMER_DPAR_STATUS	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Range Variable[x]: 32-(CX_NL) + CX_NL - 1 Range Variable[y]: CX_NL

Bits	Name	Memory Access	Description
x:0	FIRST_RETIMER_DPAR_STATUS	R/W1C	First Retimer Data Parity Mismatch Status. Each bit indicates if the corresponding Lane detected a Data Parity mismatch. A value of 1b indicates that a mismatch was detected on the corresponding Lane Number. The default value of each bit is 0b. The value of this field is undefined when no Retimers are present. For Ports that are narrower than 32 Lanes, the unused upper bits [31: Maximum Link Width] are RsvdZ. Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true Range Variable[x]: CX_NL - 1

3.16.7 PL16G_SECOND_RETIMER_DPAR_STATUS_REG

- **Name:** 16.0 GT/s Second Retimer Data Parity Mismatch Status Register.
- **Description:** The Second Retimer Data Parity Status register is a 32-bit vector where each bit indicates if the second Retimer of a Path detected a Data Parity mismatch on the Lane with the corresponding Lane number. This Lane number is the default Lane number which is invariant to Link width and Lane reversal negotiation that occurs during Link training.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

x:y	RSVDP_SECOND_RETIMER_DPAR_STATUS
x:0	SECOND_RETIMER_DPAR_STATUS

Table 3-109 Fields for Register: PL16G_SECOND_RETIMER_DPAR_STATUS_REG

Bits	Name	Memory Access	Description
x:y	RSVDP_SECOND_RETIMER_DPAR_STATUS	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Range Variable[x]: 32-(CX_NL) + CX_NL - 1 Range Variable[y]: CX_NL

Bits	Name	Memory Access	Description
x:0	SECOND_RETIMER_DPAR_STAT US	R/W1C	Second Retimer Data Parity Mismatch Status. Each bit indicates if the corresponding Lane detected a Data Parity mismatch. A value of 1b indicates that a mismatch was detected on the corresponding Lane Number. The default value of each bit is 0b. The value of this field is undefined when no Retimers are present or only one Retimer is present. For Ports that are narrower than 32 Lanes, the unused upper bits [31: Maximum Link Width] are RsvdZ. Value After Reset: 0x0 Exists: (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true Range Variable[x]: CX_NL - 1

3.16.8 PL16G_CAP_OFF_iH_REG (for i = 20; i <= 2C)

- **Name:** 16.0 GT/s Lane Equalization Control Register for Lane 0-3.
- **Description:** This Equalization Control register consists of control fields required for Lane 0-3 16.0 GT/s equalization.
- **Size:** 32 bits
- **Offset:** B+0x20 + i*0x4
- **Exists:** (CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

USP_16G_TX_PRESET3	31:28
DSP_16G_TX_PRESET3	27:24
USP_16G_TX_PRESET2	23:20
DSP_16G_TX_PRESET2	19:16
USP_16G_TX_PRESET1	15:12
DSP_16G_TX_PRESET1	11:8
USP_16G_TX_PRESET0	7:4
DSP_16G_TX_PRESET0	3:0

Table 3-110 Fields for Register: PL16G_CAP_OFF_iH_REG (for i = 20; i <= 2C)

Bits	Name	Memory Access	Description
31:28	USP_16G_TX_PRESET3	* Varies	Upstream Port 16.0 GT/s Transmitter Preset3. ■ Case A: When Operating port is Downstream port and whether the Crosslink is supported or not, this field represents the value sent on Lane 3 during 16.0 GT/s equalization. ■ Case B: When Operating port is Upstream port and Crosslink Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 3 during Link Equalization. ■ Case C: When Operating port is Upstream port and Crosslink Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is HwInit. For case B, Field is RO. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: <code>{{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_USP_16G_TX_PRESET3" : "DEFAULT_LOCAL_G4_TX_PRESET"}}</code> Exists: <code>(CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED && CX_NL>2</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""</code>

Bits	Name	Memory Access	Description
27:24	DSP_16G_TX_PRESET3	* Varies	Downstream Port 16.0 GT/s Transmitter Preset3. Transmitter Preset of Lane 3 used for 16.0 GT/s equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP. Otherwise, this field is Hwlnit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: <code>{{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_DSP_16G_TX_PRESET3" : "0x0"}}</code> Exists: <code>(CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED && CX_NL>2</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""</code>

Bits	Name	Memory Access	Description
23:20	USP_16G_TX_PRESET2	* Varies	Upstream Port 16.0 GT/s Transmitter Preset2. ■ Case A: When Operating port is Downstream port and whether the Crosslink is supported or not, this field represents the value sent on Lane 2 during 16.0 GT/s equalization. ■ Case B: When Operating port is Upstream port and Crosslink Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 2 during Link Equalization. ■ Case C: When Operating port is Upstream port and Crosslink Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is HwInit. For case B, Field is RO. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: <code>{{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_USP_16G_TX_PRESET2" : "DEFAULT_LOCAL_G4_TX_PRESET"}}</code> Exists: <code>(CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED && CX_NL>2</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""</code>

Bits	Name	Memory Access	Description
19:16	DSP_16G_TX_PRESET2	* Varies	Downstream Port 16.0 GT/s Transmitter Preset2. Transmitter Preset of Lane 2 used for 16.0 GT/s equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP. Otherwise, this field is Hwlnit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: <code>{{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_DSP_16G_TX_PRESET2" : "0x0"}}</code> Exists: <code>(CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED && CX_NL>2</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""</code>

Bits	Name	Memory Access	Description
15:12	USP_16G_TX_PRESET1	* Varies	Upstream Port 16.0 GT/s Transmitter Preset1. ■ Case A: When Operating port is Downstream port and whether the Crosslink is supported or not, this field represents the value sent on Lane 1 during 16.0 GT/s equalization. ■ CaseB: When Operating port is Upstream port and Crosslink Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 1 during Link Equalization. Case C: When Operating port is Upstream port and Crosslink Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is Hwlnit. For case B, Field is RO. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: <code>{{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_USP_16G_TX_PRESET1" : "DEFAULT_LOCAL_G4_TX_PRESET"}}</code> Exists: <code>(CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED && CX_NL>1</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""</code>

Bits	Name	Memory Access	Description
11:8	DSP_16G_TX_PRESET1	* Varies	Downstream Port 16.0 GT/s Transmitter Preset1. Transmitter Preset of Lane 1 used for 16.0 GT/s equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP. Otherwise, this field is Hwlnit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: <code>{{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_DSP_16G_TX_PRESET1" : "0x0"}}</code> Exists: <code>(CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED && CX_NL>1</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""</code>

Bits	Name	Memory Access	Description
7:4	USP_16G_TX_PRESET0	* Varies	Upstream Port 16.0 GT/s Transmitter Preset0. ■ Case A: When Operating port is Downstream port and whether the Crosslink is supported or not, this field represents the value sent on Lane 0 during 16.0 GT/s equalization. ■ Case B: When Operating port is Upstream port and Crosslink Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 0 during Link Equalization. ■ Case C: When Operating port is Upstream port and Crosslink Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is HwInit. For case B, Field is RO. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: <code>{{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_USP_16G_TX_PRESET0" : "DEFAULT_LOCAL_G4_TX_PRESET"}}</code> Exists: <code>(CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""</code>

Bits	Name	Memory Access	Description
3:0	DSP_16G_TX_PRESET0	* Varies	Downstream Port 16.0 GT/s Transmitter Preset0. Transmitter Preset of Lane 0 used for 16.0 GT/s equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP. Otherwise, this field is Hwlnit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: <code>{{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_DSP_16G_TX_PRESET0" : "0x0"}}</code> Exists: <code>(CX_PL16G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""</code>

3.17 PF0_MARGIN_CAP Registers

3.17.1 MARGIN_EXT_CAP_HDR_REG

- **Name:** Margining Extended Capability Header.
- **Description:** This register provides capability ID, capability version and next offset value for Margining Extended Capability.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	EXTENDED_CAP_ID

Table 3-111 Fields for Register: MARGIN_EXT_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field is depends on version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: wire ? unconstrained : writeAsRead
15:0	EXTENDED_CAP_ID	* Varies	PCI Express Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. The Extended Capability ID for the Physical Layer 16.0 GT/s Margining Extended Capability is 0027h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x27 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: wire ? unconstrained : writeAsRead

3.17.2 MARGIN_PORT_CAPABILITIES_STATUS_REG

- **Name:** Margining Port Capabilities and Status Register.
- **Description:** This register indicates the status of the Margining feature.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:18	RSVDP_18
17	MARGINING_SOFTWARE_READY
16	MARGINING_READY
15:1	RSVDP_1
0	MARGINING_USES_DRIVER_SOFTWARE

Table 3-112 Fields for Register: MARGIN_PORT_CAPABILITIES_STATUS_REG

Bits	Name	Memory Access	Description
31:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED

Bits	Name	Memory Access	Description
17	MARGINING_SOFTWARE_READY	R	Margining Software Ready. When Margining uses Driver Software is Set, then this bit, when Set, indicates that the required software has performed the required initialization. The value of this bit is undefined if Margining uses Driver Software is Clear. The default value of this bit is implementation specific. If Margining uses Driver Software is Clear, Margining Ready must be Set no later than 100 ms after the Link trains to 16.0 GT/s. Default value is implementation specific. Value After Reset: * Varies Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Reset Mask: 0x0 Volatile: true
16	MARGINING_READY	R	Margining Ready. Indicates when the Margining feature is ready to accept margining commands. Behavior is undefined if this bit is Clear and, for any Lane, any of the Receiver Number , Margin Type , Usage Model , or Margin Payload fields are written. If Margining uses Driver Software is Set, Margining Ready must be Set no later than 100 ms after the later of Margining Software Ready becoming Set or the link training to 16.0 GT/s. Value After Reset: * Varies Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Reset Mask: 0x0 Volatile: true
15:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED

Bits	Name	Memory Access	Description
0	MARGINING_USES_DRIVER_SOFTWARE	* Varies	Margining uses Driver Software. If Set, indicates that Margining is partially implemented using Device Driver software. Margining Software Ready indicates when this software is initialized. If Clear, Margining does not require device driver software. In this case the value read from Margining Software Ready is undefined. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_MARGINING_USES_DRIVER_SOFTWARE Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: wire ? unconstrained : writeAsRead

3.17.3 MARGIN_LANE_CNTRL_STATUSi_REG (for i = 0; i <= 15)

- **Name:** Margining Lane Control and Status Register for Lane 0.
- **Description:** The Margining Lane Control Register consists of control fields required for per-Lane margining. The number of entries in this register are sized by Maximum Link Width.
- **Size:** 32 bits
- **Offset:** B+0x8 + i*0x4
- **Exists:** (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

MARGIN_PAYLOAD_STATUS	31:24
RSVDP_23	23
USAGE_MODEL_STATUS	22
MARGIN_TYPE_STATUS	21:19
RECEIVER_NUMBER_STATUS	18:16
MARGIN_PAYLOAD	15:8
RSVDP_7	7
USAGE_MODEL	6
MARGIN_TYPE	5:3
RECEIVER_NUMBER	2:0

Table 3-113 Fields for Register: MARGIN_LANE_CNTRL_STATUSi_REG (for i = 0; i <= 15)

Bits	Name	Memory Access	Description
31:24	MARGIN_PAYLOAD_STATUS	R	Margin Payload(Status) for Lane 0. This field is only meaningful, when the Margin Type is a defined encoding other than 'No Command'. This field must be reset to the default value if the Port goes to DL_Down status. Value After Reset: 0x0 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED

Bits	Name	Memory Access	Description
22	USAGE_MODEL_STATUS	R	Usage Model(Status) for Lane 0. This field must be reset to the default value if the Port goes to DL_Down status. Value After Reset: 0x0 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true
21:19	MARGIN_TYPE_STATUS	R	Margin Type(Status) for Lane 0. This field must be reset to the default value if the Port goes to DL_Down status. Value After Reset: 0x0 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true
18:16	RECEIVER_NUMBER_STATUS	R	Receiver Number(Status) for Lane 0. For Downstream Ports, this field must be reset to the default value if the Port goes to DL_Down status. Value After Reset: 0x0 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED Testable: writeAsRead Volatile: true
15:8	MARGIN_PAYLOAD	R/W	Margin Payload for Lane 0. This field must be reset to the default value if the Port goes to DL_Down status. Value After Reset: 0x9c Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED
7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED
6	USAGE_MODEL	R/W	Usage Model for Lane 0. This field must be reset to the default value if the Port goes to DL_Down status. Value After Reset: 0x0 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED

Bits	Name	Memory Access	Description
5:3	MARGIN_TYPE	R/W	Margin Type for Lane 0. The default value is 111b. This field must be reset to the default value if the Port goes to DL_Down status. Value After Reset: 0x7 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED
2:0	RECEIVER_NUMBER	R/W	Receiver Number for Lane 0. This field must be reset to the default value if the Port goes to DL_Down status. Value After Reset: 0x0 Exists: (CX_MARGIN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN4_SPEED

3.18 PF0_PL32G_CAP Registers

3.18.1 PL32G_EXT_CAP_HDR_REG

- **Name:** Physical Layer 32.0 GT/s Extended Capability Header.
- **Description:** This register provides information about Capability ID, Version, and next offset.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:20	NEXT_OFFSET
19:16	CAP_VERSION
15:0	EXTENDED_CAP_ID

Table 3-114 Fields for Register: PL32G_EXT_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This bit depends on the version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: wire ? unconstrained : writeAsRead
15:0	EXTENDED_CAP_ID	* Varies	PCI Express Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. The Extended Capability ID for the Physical Layer 32.0 GT/s Capability is 002Ah. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x2a Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: wire ? unconstrained : writeAsRead

3.18.2 PL32G_CAPABILITY_REG

- **Name:** 32.0 GT/s Capabilities Register.
- **Description:** This register provides Extended Capability of 32.0 GT/s Equalization.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:16	RSVDP_16
15:11	MOD_TS_RSVD_USAGE_MODE
10	MOD_TS_ALT_PROTOCOL_SUPPORT
9	MOD_TS_TRAINING_SET_MSG_SUPPORT
8	MOD_TS_PCIE_SUPPORT
7:2	RSVDP_2
1	NO_EQ_NEEDED_SUPPORT
0	EQ_BYPASS_HIGHEST_RATE_SUPPORT

Table 3-115 Fields for Register: PL32G_CAPABILITY_REG

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT
15:11	MOD_TS_RSVD_USAGE_MODE	R	Modified TS Reserved Usage Modes. Reserved bits for future Usage Modes defined by the PCISIG. Must be 0 0000b. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
10	MOD_TS_ALT_PROTOCOL_SUPPORT	* Varies	Modified TS Usage Mode 2 Supported, Alternate Protocol. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): This field indicates that this Port supports negotiating to use alternate protocols. Value After Reset: DEFAULT_AP_SUPPORT Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_AP_SUPPORT Testable: wire ? unconstrained : writeAsRead
9	MOD_TS_TRAINING_SET_MESSAGE_SUPPORT	R	Modified TS Usage Mode 1 Supported, Training Set Message. Values: ■ 0x1 (SET): Indicates that this Port supports sending and receiving vendor specific Training Set Messages (Modified TS Usage 001b). Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true
8	MOD_TS_PCIE_SUPPORT	* Varies	Modified TS Usage Mode 0 Supported. This bit must be 1b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): Indicates that this Port supports PCI Express (Modified TS Usage 000b). Value After Reset: CX_DEFAULT_MOD_TS_PCIE_SUPPORT Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
7:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT
1	NO_EQ_NEEDED_SUPPORT	* Varies	No Equalization Needed Support. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): When Set, this Port supports controlling whether or not Equalization is needed. Value After Reset: CX_DEFAULT_NO_EQ_NEEDED_SUPPORT Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: wire ? unconstrained : writeAsRead
0	EQ_BYPASS_HIGHEST_RATE_SUPPORT	* Varies	Equalization bypass to highest rate Supported. Must be 1b for Ports that support 32.0 GT/s or higher data rates. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): When Set, this Port supports controlling whether the Port negotiates to skip equalization for speeds other than the highest common supported speed. Value After Reset: CX_DEFAULT_EQ_BYPASS_HIGHEST_RATE_SUPPORT Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: wire ? unconstrained : writeAsRead

3.18.3 PL32G_CONTROL_REG

- **Name:** 32.0 GT/s Control Register.
- **Description:** This register controls the 32.0 GT/s capabilities.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:11	RSVDP_11
10:8	MOD_TS_USAGE_MODE_SELECT
7:2	RSVDP_2
1	NO_EQ_NEEDED_DISABLE
0	EQ_BYPASS_HIGHEST_RATE_DISABLE

Table 3-116 Fields for Register: PL32G_CONTROL_REG

Bits	Name	Memory Access	Description
31:11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT

Bits	Name	Memory Access	Description
10:8	MOD_TS_USAGE_MODE_SELECT	R	Modified TS Usage Mode Selected. This field indicates which Usage Mode is used by this Downstream Port. In Upstream Ports, this field is RsvdP. Default is 000b. All encodings other than the defined encodings are reserved. Unused bits in this field are permitted to be hardwired to 0b. If the only supported usage mode is PCI Express, this field is permitted to be hardwired to 000b. If 001b, PCIe protocol only with vendor defined Training Set Messages. If 010b, Alternate Protocol Negotiation. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: untestable Volatile: true
7:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT
1	NO_EQ_NEEDED_DISABLE	R/W	No Equalization Needed Disable. If No Equalization Needed Supported is Set, this bit is RWS with a default value of 0b. <i>Note:</i> This register field is sticky. Values: 0x0 (CLEAR): When Clear, this Port is permitted to indicate that it does not require equalization. When Set, this Port must always indicate that it requires equalization. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	EQ_BYPASS_HIGHEST_RATE_DISABLE	R/W	Equalization bypass to highest rate Disable. If Equalization bypass to highest rate Supported is Set, this bit is RWS with a default value of 0b. <i>Note:</i> This register field is sticky. Values: 0x0 (CLEAR): When Clear, this Port indicates during Link Training that it wishes to train to the highest common link data rate and skip equalization of intermediate data rates. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

3.18.4 PL32G_STATUS_REG

- **Name:** 32.0 GT/s Status Register.
- **Description:** This register provides information related to 32.0 GT/s equalization.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:11	RSVDP_11
10	NO_EQ_NEEDED_RCVD
9	TX_PRECODE_REQ
8	TX_PRECODING_ON
7:6	RX_ENH_LINK_BEHAVIOR_CTRL
5	MOD_TS_RCVD
4	LINK_EQ_32G_REQ
3	EQ_32G_CPL_P3
2	EQ_32G_CPL_P2
1	EQ_32G_CPL_P1
0	EQ_32G_CPL

Table 3-117 Fields for Register: PL32G_STATUS_REG

Bits	Name	Memory Access	Description
31:11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT
10	NO_EQ_NEEDED_RCVD	R	No Equalization Needed Received. This bit is Set when this Port either received a Modified TS1/TS2 with the No Equalization Needed bit Set or received a non-modified TS1/TS2 was received with the No Equalization Needed encoding. This bit is cleared on DL_Down. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
9	TX_PRECODE_REQ	* Varies	Transmitter Precode Request. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): When Set, this Port will request the transmitter to use Precoding by setting the Precoding Request bit in the TS1s/TS2s it transmits prior to entry to Recovery.Speed. Value After Reset: CX_DEFAULT_TX_PRECODE_REQ Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true
8	TX_PRECODING_ON	R	Transmitter Precoding On. This field indicates whether the Receiver asked this transmitter to enable Precoding. This bit is cleared on DL_Down. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true
7:6	RX_ENH_LINK_BEHAVIOR_CTRL	R	Received Enhanced Link Behavior Control. This field contains the Enhanced Link Behavior Control bits from the most recent TS1 or TS2 received in the Polling or Configuration states. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5	MOD_TS_RCVD	R	Modified TS Received. This bit is Cleared when the Link is Down. This bit is Set when the Modified TS1/TS2 Ordered Set is received. Values: 0x1 (SET): If Set, Received Modified TS Data 1 Register and Received Modified TS Data 2 Register contain meaningful data. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true
4	LINK_EQ_32G_REQ	R/W1C	Link Equalization Request 32.0GT/s. This bit is Set by hardware to request the 32.0 GT/s Link equalization process to be performed on the Link. For a Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true
3	EQ_32G_CPL_P3	R	Equalization 32.0GT/s Phase 3 Successful. For a Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. <i>Note:</i> This register field is sticky. Values: 0x1 (SET): When set to 1b, this bit indicates that Phase 3 of the 32.0 GT/s Transmitter Equalization procedure has successfully completed. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2	EQ_32G_CPL_P2	R	Equalization 32.0GT/s Phase 2 Successful. For a Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. <i>Note:</i> This register field is sticky. Values: 0x1 (SET): When set to 1b, this bit indicates that Phase 2 of the 32.0 GT/s Transmitter Equalization procedure has successfully completed. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true
1	EQ_32G_CPL_P1	R	Equalization 32.0GT/s Phase 1 Successful. For a Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. <i>Note:</i> This register field is sticky. Values: 0x1 (SET): When set to 1b, this bit indicates that Phase 1 of the 32.0 GT/s Transmitter Equalization procedure has successfully completed. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true
0	EQ_32G_CPL	R	Equalization 32.0GT/s Complete. For a Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. <i>Note:</i> This register field is sticky. Values: 0x1 (SET): When Set, this bit indicates that the 32.0 GT/s Transmitter Equalization procedure has completed. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

3.18.5 PL32G_RCVD_MOD_TS_DATA1_REG

- **Name:** Received Modified TS Data 1 Register.
- **Description:** This register contains the values received in the Modified TS1/TS2 Ordered Set. If PCI Express (Usage Mode 0) is the only one supported by a Port, this register is permitted to be hardwired to 0000 0000h by the controller.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

RCVD_MOD_TS_VENDOR_ID	31:16
RCVD_MOD_TS_INFO1	15:3
RCVD_MOD_TS_USAGE_MODE	2:0

Table 3-118 Fields for Register: PL32G_RCVD_MOD_TS_DATA1_REG

Bits	Name	Memory Access	Description
31:16	RCVD_MOD_TS_VENDER_ID	R	Received Modified TS Vendor ID. If Modified TS Received is Set, this field contains the Modified TS Vendor ID field from the Modified TS1/TS2 Ordered Set received. If Modified TS Received is Clear, this field contains 0000h. ■ 15:8 □ Contain the value of Symbol 11. ■ 7:0 □ Contain the value of Symbol 10. If PCI Express (Usage Mode 0) is the only one supported, this field is permitted to be hardwired to 0000h. Values: ■ 0x0 (Ini_value): Zero value. ■ 0xffff (Last_value): Last value. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true
15:3	RCVD_MOD_TS_INFO1	R	Received Modified TS Information 1. If Modified TS Received is Set, this field contains the Modified TS Information 1 field from the Modified TS1/TS2 Ordered Set. If Modified TS Received is Clear, this field contains 0 0000 0000 0000b. ■ 15:8 □ contain the value of Symbol 9 ■ 7:3 □ contain bits 7:3 of Symbol 8 If PCI Express (Usage Mode 0) is the only one supported, this field is permitted to be hardwired to 0 0000 0000 0000b by controller. Values: ■ 0x0 (default): Zero value. ■ 0x1fff (maximum): Zero value. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2:0	RCVD_MOD_TS_USAGE_MODE	R	Received Modified TS Usage Mode. If Modified TS Received is Set, this field contains the Modified TS Usage field from the Modified TS1/TS2 Ordered Set. If Modified TS Received is Clear, this field contains 000b. Unused bits in this field are permitted to be hardwired by controller to 0b. If PCI Express (Usage Mode 0) is the only one supported, this field is permitted to be hardwired to 000b. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

3.18.6 PL32G_RCVD_MOD_TS_DATA2_REG

- **Name:** Received Modified TS Data 2 Register.
- **Description:** This register contains the values received in Symbols 12 through 14 of the Modified TS1/TS2. If Modified TS Usage Mode 1 Supported - Training Set Message and Modified TS Usage Mode 2 Supported - Alternate Protocol are both Clear, this register is permitted to be hardwired to 0000 0000h by the controller.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:26	25:24	23:0
RSVDP_26	RCVD_ALT_PROTOCOL_NEGO_STATUS	RCVD_MOD_TS_INFO2

Table 3-119 Fields for Register: PL32G_RCVD_MOD_TS_DATA2_REG

Bits	Name	Memory Access	Description
31:26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT

Bits	Name	Memory Access	Description
25:24	RCVD_ALT_PROTOCOL_NEGO_STATUS	R	Alternate Protocol Negotiation Status. Indicates the status of the Alternate Protocol Negotiation. If Modified TS Usage Mode 1 Supported - Training Set Message and Modified TS Usage Mode 2 Supported - Alternate Protocol are both Clear, this register is permitted to be hardwired to 0000 0000h by controller. If Modified TS Usage Mode 2 Supported - Alternate Protocol is Clear, this bit is hardwired to 0b. If Modified TS Usage Mode Selected does not equal 2, this bit contains 0b. Values: ■ 0x0 (Not_supported): Alternate Protocol Negotiation not supported - Modified TS Usage Mode 2 Supported - Alternate Protocol is Clear. ■ 0x1 (Disabled): Alternate Protocol Negotiation disabled - Modified TS Usage Mode 2 Supported - Alternate Protocol is Set but Modified TS Usage Mode Selected was not 2 during the appropriate LTSSM State. ■ 0x2 (Failed): Alternate Protocol Negotiation failed - Alternate Protocol Negotiation was attempted and did not locate a protocol that was supported on both ends of the Link. ■ 0x3 (Succeeded): Alternate Protocol Negotiation succeeded - Alternate Protocol Negotiation located one or more protocols that were supported on both ends of the Link and the Downstream Port selected one of those protocols for use. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: untestable

Bits	Name	Memory Access	Description
23:0	RCVD_MOD_TS_INFO2	R	Received Modified TS Information 2. If Modified TS Received is Set, this field contains the Modified TS Information 2 field from the received Modified TS1/TS2 Ordered Set. If Modified TS Received is Clear, this field contains 00 0000h. ■ 23:16 □ Contain the value of Symbol 12. ■ 16:8 □ Contain the value of Symbol 13. ■ 7:0 □ Contain the value of Symbol 14. If PCI Express (Usage Mode 0) is the only one supported, this field is permitted to be hardwired to 00 0000h. Values: ■ 0x0 (Minimum_val): Zero value. ■ 0xfffff (Maximum_val): Maximum value. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

3.18.7 PL32G_TX_MOD_TS_DATA1_REG

- **Name:** Transmitted Modified TS Data 1 Register.
- **Description:** This register contains the values transmitted in the Modified TS1/TS2 Ordered Set. If PCI Express (Usage Mode 0) is the only one supported by a Port, this register is permitted to be hardwired to 0000 0000h by the controller.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:16	TX_MOD_TS_VENDOR_ID
15:3	TX_MOD_TS_INFO1
2:0	TX_MOD_TS_USAGE_MODE

Table 3-120 Fields for Register: PL32G_TX_MOD_TS_DATA1_REG

Bits	Name	Memory Access	Description
31:16	TX_MOD_TS_VENDOR_ID	R	Transmitted Modified TS Vendor ID. If Modified TS Received is Set, this field contains the Modified TS Vendor ID field from the last Modified TS1/TS2 Ordered Set transmitted during the most recent LTSSM State. ■ Bits 15:8 contain the value of Symbol 11. ■ Bits 7:0 contain the value of Symbol 10. If PCI Express (Usage Mode 0) is the only one supported, this field is permitted to be hardwired to 0000h by controller. Values: ■ 0x0 (min_bit_value): Zero value. ■ 0xffff (max_bit_val): max bit value. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:3	TX_MOD_TS_INFO1	R	Transmitted Modified TS Information 1. If Modified TS Received is Set, this field contains the Modified TS Information 1 field from Modified TS2 Ordered Set transmitted during the Configuration.Complete LTSSM State. Bits [15:8] contain the value of Symbol 9 Bits [7:3] contain bits 7:3 of Symbol 8 If PCI Express (Usage Mode 0) is the only one supported, this field is permitted to be hardwired to 0 0000 0000 0000b. Values: ■ 0x0 (init_0): Zero value. ■ 0x1fff (value_13): Zero value. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true
2:0	TX_MOD_TS_USAGE_MODE	R	Transmitted Modified TS Usage Mode. If Modified TS Received is Set, this field contains the Modified TS Usage field from the last Modified TS1/TS2 Ordered Set transmitted during the most recent LTSSM State. Unused bits in this field are permitted to be hardwired to 0b. If PCI Express (Usage Mode 0) is the only one supported, this field is permitted to be hardwired to 000b. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

3.18.8 PL32G_TX_MOD_TS_DATA2_REG

- **Name:** Transmitted Modified TS Data 2 Register.
- **Description:** This register contains the values transmitted in Symbols 12 through 14 of the Modified TS1/TS2. If Modified TS Usage Mode 1 Supported - Training Set Message and Modified TS Usage Mode 2 Supported - Alternate Protocol are both Clear, this register is permitted to be hardwired to 0000 0000h by the controller.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:26	RSVDP_26
25:24	TX_ALT_PROTOCOL_NEGO_STATUS
23:0	TX_MOD_TS_INFO2

Table 3-121 Fields for Register: PL32G_TX_MOD_TS_DATA2_REG

Bits	Name	Memory Access	Description
31:26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT

Bits	Name	Memory Access	Description
25:24	TX_ALT_PROTOCOL_NEGO_STATUS	R	Alternate Protocol Negotiation Status. Indicates the status of the Alternate Protocol Negotiation. Encodings are : If Modified TS Usage Mode 1 Supported - Training Set Message and Modified TS Usage Mode 2 Supported - Alternate Protocol are both Clear, this register is permitted to be hardwired to 0000 0000h by controller. If Modified TS Usage Mode 2 Supported - Alternate Protocol is Clear, this bit is hardwired to 0b. If Modified TS Usage Mode Selected does not equal 2, this bit contains 0b. Values: ■ 0x0 (Not_supported): Alternate Protocol Negotiation not supported - Modified TS Usage Mode 2 Supported - Alternate Protocol is Clear. ■ 0x1 (Disabled): Alternate Protocol Negotiation disabled - Modified TS Usage Mode 2 Supported - Alternate Protocol is Set but Modified TS Usage Mode Selected was not 2 during the appropriate LTSSM State. ■ 0x2 (Failed): Alternate Protocol Negotiation failed - Alternate Protocol Negotiation was attempted and did not locate a protocol that was supported on both ends of the Link. ■ 0x3 (Succeeded): Alternate Protocol Negotiation succeeded - Alternate Protocol Negotiation located one or more protocols that were supported on both ends of the Link and the Downstream Port selected one of those protocols for use. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: untestable
23:0	TX_MOD_TS_INFO2	R	Transmitted Modified TS Information 2. If Modified TS Received is Set, this field contains the Modified TS Information 2 field from the last Modified TS1/TS2 Ordered Set transmitted during the most recent LTSSM State. Bits [23:16] contain the value of Symbol 12. Bits [16:8] contain the value of Symbol 13. Bits [7:0] contain the value of Symbol 14. If PCI Express (Usage Mode 0) is the only one supported, this field is permitted to be hardwired to 00 0000h by the controller. Values: ■ 0x0 (initial_0): Zero value. ■ 0xffff (max_reset): Zero value. Value After Reset: 0x0 Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED CX_AP_SUPPORT Testable: writeAsRead Volatile: true

3.18.9 PL32G_CAP_OFF_iH_REG (for i = 20; i <= 2C)

- **Name:** 32.0 GT/s Lane Equalization Control Register for Lane 0-3.
- **Description:** This register consists of control fields required for Lane 0-3 32.0 GT/s equalization.
- **Size:** 32 bits
- **Offset:** B+0x20 + i*0x4
- **Exists:** (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

USP_32G_TX_PRESET3	31:28
DSP_32G_TX_PRESET3	27:24
USP_32G_TX_PRESET2	23:20
DSP_32G_TX_PRESET2	19:16
USP_32G_TX_PRESET1	15:12
DSP_32G_TX_PRESET1	11:8
USP_32G_TX_PRESET0	7:4
DSP_32G_TX_PRESET0	3:0

Table 3-122 Fields for Register: PL32G_CAP_OFF_iH_REG (for i = 20; i <= 2C)

Bits	Name	Memory Access	Description
31:28	USP_32G_TX_PRESET3	R	Upstream Port 32.0 GT/s Transmitter Preset3. A) When Operating port is Downstream port and whether the Crosslink is supported or not, this field represents the value sent on Lane 3 during 32.0 GT/s equalization. B) When Operating port is Upstream port and Crosslink Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 3 during Link Equalization. C) When Operating port is Upstream port and Crosslink Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is HwInit. For case B, Field is RO. Value After Reset: $\{(CX_GEN5_SPEED \& (CX_MEMORY_MAP_POSITION == 1 CX_CROSSLINK_ENABLE == 1)) ?$ "DEFAULT_USP_32G_TX_PRESET3" : CX_GEN5_SPEED ? "DEFAULT_LOCAL_G5_TX_PRESET" : "0x0" Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE < 2)) && CX_GEN5_SPEED && CX_NL > 2 Testable: $\{(CX_CROSSLINK_ENABLE == 0) ?$ "untestable" : $((CX_DBI_RO_WR_EN == 1 \&\& CX_MEMORY_MAP_VIEW == 0) ?$ "writeAsRead" : "unconstrained") }
27:24	DSP_32G_TX_PRESET3	R	Downstream Port 32.0 GT/s Transmitter Preset3. Transmitter Preset of Lane 3 used for 32.0 GT/s equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP. Otherwise, this field is HwInit. Value After Reset: $\{(CX_GEN5_SPEED \& (CX_MEMORY_MAP_POSITION == 1 CX_CROSSLINK_ENABLE == 1)) ?$ "DEFAULT_DSP_32G_TX_PRESET3" : "0x0" Exists: (CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE < 2)) && CX_GEN5_SPEED && CX_NL > 2 Testable: $\{(CX_CROSSLINK_ENABLE == 0) ?$ "untestable" : $((CX_DBI_RO_WR_EN == 1 \&\& CX_MEMORY_MAP_VIEW == 0) ?$ "writeAsRead" : "unconstrained") }

Bits	Name	Memory Access	Description
23:20	USP_32G_TX_PRESET2	R	Upstream Port 32.0 GT/s Transmitter Preset2. A) When Operating port is Downstream port and whether the Crosslink is supported or not, this field represents the value sent on Lane 2 during 32.0 GT/s equalization. B) When Operating port is Upstream port and Crosslink Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 2 during Link Equalization. C) When Operating port is Upstream port and Crosslink Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is HwInit. For case B, Field is RO. Value After Reset: <code>{(CX_GEN5_SPEED && (CX_MEMORY_MAP_POSITION == 1 CX_CROSSLINK_ENABLE == 1)) ? "DEFAULT_USP_32G_TX_PRESET2" : CX_GEN5_SPEED ? "DEFAULT_LOCAL_G5_TX_PRESET" : "0x0"}</code> Exists: <code>(CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE < 2)) && CX_GEN5_SPEED && CX_NL > 2</code> Testable: <code>{(CX_CROSSLINK_ENABLE == 0) ? "untestable" : ((CX_DBI_RO_WR_EN == 1 && CX_MEMORY_MAP_VIEW == 0) ? "writeAsRead" : "unconstrained") }</code>
19:16	DSP_32G_TX_PRESET2	R	Downstream Port 32.0 GT/s Transmitter Preset2. Transmitter Preset of Lane 2 used for 32.0 GT/s equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP. Otherwise, this field is HwInit. Value After Reset: <code>{(CX_GEN5_SPEED && (CX_MEMORY_MAP_POSITION == 1 CX_CROSSLINK_ENABLE == 1)) ? "DEFAULT_DSP_32G_TX_PRESET2" : "0x0"}</code> Exists: <code>(CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE < 2)) && CX_GEN5_SPEED && CX_NL > 2</code> Testable: <code>{(CX_CROSSLINK_ENABLE == 0) ? "untestable" : ((CX_DBI_RO_WR_EN == 1 && CX_MEMORY_MAP_VIEW == 0) ? "writeAsRead" : "unconstrained") }</code>

Bits	Name	Memory Access	Description
15:12	USP_32G_TX_PRESET1	R	Upstream Port 32.0 GT/s Transmitter Preset1. ■ Case A: When Operating port is Downstream port and whether the Crosslink is supported or not, this field representss the value sent on Lane 1 during 32.0 GT/s equalization. ■ Case B: When Operating port is Upstream port and Cross-link Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 1 during Link Equalization. ■ Case C: When Operating port is Upstream port and Cross-link Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is Hwlnit. For case B, Field is RO. Value After Reset: <code>{{(CX_GEN5_SPEED&&(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1)) ? "DEFAULT_USP_32G_TX_PRESET1" : CX_GEN5_SPEED ? "DEFAULT_LOCAL_G5_TX_PRESET" : "0x0"}}</code> Exists: <code>(CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED && CX_NL>1</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code>
11:8	DSP_32G_TX_PRESET1	R	Downstream Port 32.0 GT/s Transmitter Preset1. Transmitter Preset of Lane 1 used for 32.0 GT/s equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP, otherwise, this field is Hwlnit. Value After Reset: <code>{{(CX_GEN5_SPEED&&(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1)) ? "DEFAULT_DSP_32G_TX_PRESET1" : "0x0"}}</code> Exists: <code>(CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED && CX_NL>1</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code>

Bits	Name	Memory Access	Description
7:4	USP_32G_TX_PRESET0	R	Upstream Port 32.0 GT/s Transmitter Preset0. ■ Case A: When Operating port is Downstream port and whether the Crosslink is supported or not, this field representss the value sent on Lane 0 during 32.0 GT/s equalization. ■ Case B: When Operating port is Upstream port and Cross-link Supported bit is 0, this field is intended for debug and diagnostics. It contains the value captured from Lane 0 during Link Equalization. ■ Case C: When Operating port is Upstream port and Cross-link Supported bit is 1, Field is not used or affected by the current Link Equalization. Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. For case A and C, Field is Hwlnit. For case B, Field is RO. Value After Reset: <code>{{(CX_GEN5_SPEED&&(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1)) ? "DEFAULT_USP_32G_TX_PRESET0" : CX_GEN5_SPEED ? "DEFAULT_LOCAL_G5_TX_PRESET" : "0x0"}}</code> Exists: <code>(CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code>
3:0	DSP_32G_TX_PRESET0	R	Downstream Port 32.0 GT/s Transmitter Preset0. Transmitter Preset of Lane 0 used for 32.0 GT/s equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. For an Upstream Port if Crosslink Supported is 0b, this field is RsvdP, otherwise, this field is Hwlnit. Value After Reset: <code>{{(CX_GEN5_SPEED&&(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1)) ? "DEFAULT_DSP_32G_TX_PRESET0" : "0x0"}}</code> Exists: <code>(CX_PL32G_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) && CX_GEN5_SPEED</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code>

3.19 PF_SRIOV_CAPI Registers

3.19.1 SRIOV_BASE_REG

- **Name:** SR-IOV Capability Header Register.
- **Description:** SR-IOV Extended Capability Header provides information about Capability ID, Version, and Next Offset.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 | CX_MEMORY_MAP_PCIE_TYPE==2))

SRIOV_NEXT_OFFSET	31:20
SRIOV_CAP_VERSION	19:16
SRIOV_PCIE_EXTENDED_CAP_ID	15:0

Table 3-123 Fields for Register: SRIOV_BASE_REG

Bits	Name	Memory Access	Description
31:20	SRIOV_NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xff (MAX_VAL): Max value Value After Reset: Pointer value calculated at configuration time. Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
19:16	SRIOV_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This bit depends on the version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15:0	SRIOV_PCIE_EXTENDED_CAP_ID	* Varies	SRIOV Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. The Extended Capability ID for the SR-IOV Extended Capability is 0010h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x10 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.19.2 CAPABILITIES_REG

- **Name:** SR-IOV Capability Register.
- **Description:** This Register provides Capability present in SR-IOV.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31	RSVDP_31
30:21	SRIOV_VF_MIGRATION_INT_MSG_NUM
20:3	RSVDP_3
2	SRIOV_VF_10BIT_TAG_REQ_CAP
1	SRIOV_ARI_CAP_HIER_RESERVED
0	SRIOV_VF_MIGRATION_CAP

Table 3-124 Fields for Register: CAPABILITIES_REG

Bits	Name	Memory Access	Description
31	RSVDP_31	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
30:21	SRIOV_VF_MIGRATION_INT_MSG_NUM	R	VF Migration Interrupt Message Number. This field indicates the MSI/MSI-X vector used for migration interrupts. The value in this field is undefined if VF Migration Capable is Clear. Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
20:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
2	SRIOV_VF_10BIT_TAG_REQ_CAP	R	VF 10-bit Tag Requester Supported. If the 10-Bit Tag Requester Supported bit in the PF's Device Capabilities 2 Register is Clear, this bit must be Clear. Values: 0x0 (CLEAR): If Clear, VFs must not support 10-Bit Tag Requester capability. 0x1 (SET): If Set, all VFs must support 10-Bit Tag Requester capability. Value After Reset: CX_10BITS_TAG_REQ_VALUE Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
1	SRIOV_ARI_CAP_HIER_PRESERVED	* Varies	ARI Capable Hierarchy Preserved. RCiEP : Not applicable. It is strongly recommended that this bit be hardwired to 0b. <i>Note:</i> The access attributes of this field are as follows: - Wire: R (sticky) - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && \$pf==0) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R <i>Note:</i> This register field is sticky. Values: 0x0 (CLEAR): Clear 0x1 (SET): PCI Express Endpoint: If Set, the ARI Capable Hierarchy bit is preserved across certain power state transitions. Value After Reset: CX_ARI_CAP_HIER_PRSD_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
0	SRIOV_VF_MIGRATION_CAP	R	VF Migration Capable. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): If Set, the PF is Migration Capable and operating under a Migration Capable MR-PCIM. Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.19.3 STATUS_CONTROL_REG

- **Name:** SR-IOV Control and Status Register.
- **Description:** This Register provides status and control of capability.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:6	RSVDP_6
5	SRIOV_VF_10BIT_TAG_REQ_ENABLE
4	SRIOV_ARI_CAPABLE_HIER
3	SRIOV_VF_MSE
2	SRIOV_VF_MIGRATION_INT_ENABLE
1	SRIOV_VF_MIGRATION_ENABLE
0	SRIOV_VF_ENABLE

Table 3-125 Fields for Register: STATUS_CONTROL_REG

Bits	Name	Memory Access	Description
31:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
5	SRIOV_VF_10BIT_TAG_REQ_ENABLE	* Varies	VF 10-bit Tag Requester Enable. If software changes the value of this bit while any VFs have outstanding Non-Posted Requests, the result is undefined. If the VF 10-Bit Tag Requester Supported bit in the SR-IOV Capabilities register is Clear, this bit is permitted to be hardwired by controller to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W but read-value is not always same as write-value ■ Dbi: R/W but read-value is not always same as write-value Values: ■ 0x0 (DISABLE): If Clear, VFs must not use 10-Bit Tags for Non-Posted Requests they generate. ■ 0x1 (ENABLE): If Set, all VFs must use 10-Bit Tags for all Non-Posted Requests they generate. Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead
4	SRIOV_ARI_CAPABLE_HIER	* Varies	ARI Capable Hierarchy (Applies to endpoint only). PCI Express Endpoint: This bit must be RW in the lowest-numbered PF of the Device and hardwired to 0b in all other PFs. If the value of this bit is 1b, the Device is permitted to locate VFs in Function Numbers 8 to 255 of the captured Bus Number. Otherwise, the Device must locate VFs as if it were a non-ARI Device. This bit is not affected by FLR of any PF or VF. RCiEP: Not applicable - this bit must be hardwired to 0b by controller. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W but read-value is not always same as write-value ■ Dbi: R/W but read-value is not always same as write-value Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead

Bits	Name	Memory Access	Description
3	SRIOV_VF_MSE	R/W	VF Memory Space Enable. Memory Space Enable for Virtual Functions. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
2	SRIOV_VF_MIGRATION_INT_ENABLE	R	VF Migration Interrupt Enable. This field used to Enable or Disable F Migration State Change Interrupt. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
1	SRIOV_VF_MIGRATION_ENABLE	R	VF Migration Enable. This field used to Enable or Disable VF Migration Support. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
0	SRIOV_VF_ENABLE	R/W	VF Enable. This field used to Enable or Disable VFs. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

3.19.4 TOTAL_VFS_INITIAL_VFS_REG

- **Name:** TotalVFs InitialVFs Register.
- **Description:** This Register provides value of TotalVFs and InitialVFs.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:16	SRIOV_TOTAL_VFS
15:0	SRIOV_INITIAL_VFS

Table 3-126 Fields for Register: TOTAL_VFS_INITIAL_VFS_REG

Bits	Name	Memory Access	Description
31:16	SRIOV_TOTAL_VFS	R	Total VFs (Max Number of VFs). TotalVFs indicates the maximum number of VFs that could be associated with the PF. The minimum value of TotalVFs is 0. For Devices operating in Single-Root mode, this field is Hwlnit and must contain the same value as InitialVFs. For Devices operating in Multi-Root mode, the value of this field may be changed by MRPCIM. Value After Reset: CX_MAX_VF_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:0	SRIOV_INITIAL_VFS	* Varies	InitialVFs. InitialVFs indicates to SR-PCIM the number of VFs that are initially associated with the PF. The minimum value of InitialVFs is 0. For Devices operating in Single-Root mode, this field is Hwlnit and must contain the same value as TotalVFs. And for Devices operating in Multi-Root mode, the value of this field may be changed by MRPCIM when VF Enable is Clear. If VF Migration Enable is Set and VF Enable is Cleared and then Set, the value of InitialVFs may change. This is necessary since some VFs may have been migrated to other PFs and may no longer be available to this PF. There are two InitialVFs registers; one each for ARI Capable and non-ARI Capable Hierarchies. The "ARI Capable Hierarchy" bit (SRIOV_ARI_CAPABLE_HIER) of the PF0 "SR-IOV Control Register" (STATUS_CONTROL_REG) determines which one is used by the controller when SR-IOV is being used, and which one is accessed by a read request. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_MAX_VF_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.19.5 SRIOV_NUM_VFS

- **Name:** NumVFs and Function Dependency Link Register.
- **Description:** This Register contains value of NumVFs and Functional Dependency Link. There are two of these registers; one each for ARI Capable and non-ARI Capable Hierarchies. The "ARI Capable Hierarchy" bit (SRIOV_ARI_CAPABLE_HIER) of the PF0 "SR-IOV Control Register" (STATUS_CONTROL_REG) determines which one is used by the controller when SR-IOV is being used, and which one is accessed by a read request. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

RSVDP_24	31:24
SRIOV_FDL	23:16
SRIOV_NUM_VFS	15:0

Table 3-127 Fields for Register: SRIOV_NUM_VFS

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
23:16	SRIOV_FDL	R	Functional Dependency Link. The programming model for a Device may have vendor-specific dependencies between sets of Functions. The Function Dependency Link field is used to describe these dependencies. This field describes dependencies between PFs. VF dependencies are the same as the dependencies of their associated PFs. If a PF is independent from other PFs of a Device, this field shall contain its own Function Number. If a PF is dependent on other PFs of a Device, this field shall contain the Function Number of the next PF in the same Function Dependency List. The last PF in a Function Dependency List shall contain the Function Number of the first PF in the Function Dependency List. If PFp and PFq are in the same Function Dependency List, than any SI that is assigned VFp,n shall also be assigned to VFq,n. Value After Reset: {(CX_VF_DEPENDENCY_LINK_SUPP) ? (CX_VF_DEPENDENCY_LINK_0) : 0} Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
15:0	SRIOV_NUM_VFS	* Varies	Number of Visible VFs. NumVFs controls the number of VFs that are visible. SR-PCIM sets NumVFs as part of the process of creating VFs. This number of VFs shall be visible in the PCI Express fabric after both NumVFs is set to a valid value and VF Enable is Set. A visible VF has a Function number reserved for it, but might not exist. If the VF exists, it shall respond to PCI Express transactions targeting them, and shall follow all rules defined by this specification. A VF exists if either: ■ VF Migration Capable is Clear and the VF number is less than or equal to TotalVFs. ■ VF Migration Capable is Set and the associated VF is in either the Active.Available or Dormant.Migrateln state. The results are undefined if NumVFs is set to a value greater than TotalVFs. NumVFs may only be written while VF Enable is Clear. If NumVFs is written when VF Enable is Set, the results are undefined. The initial value of NumVFs is undefined. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: STATUS_CONTROL_REG.SRIOV_VF_ENABLE ? RO : RW ■ Dbi: STATUS_CONTROL_REG.SRIOV_VF_ENABLE ? RO : RW Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead

3.19.6 SRIOV_VF_OFFSET_POSITION

- **Name:** VF Stride and Offset Register.
- **Description:** This Register contains value of VF Stride and First VFs Offset.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 | CX_MEMORY_MAP_PCIE_TYPE==2))

SRIOV_VF_STRIDE	31:16
SRIOV_VF_OFFSET	15:0

Table 3-128 Fields for Register: SRIOV_VF_OFFSET_POSITION

Bits	Name	Memory Access	Description
31:16	SRIOV_VF_STRIDE	* Varies	VF Stride. VF Stride defines the Routing ID offset from one VF to the next one for all VFs associated with the PF that contains this Capability structure. The next VF's 16-bit Routing ID is calculated by adding the contents of this field to the Routing ID of the current VF, ignoring any carry, using unsigned 16-bit arithmetic. This field may change value when the lowest-numbered PF's ARI Capable Hierarchy value changes or when this PF's NumVFs value changes. VF Stride is unused if NumVFs is 0 or 1. If NumVFs is greater than 1, VF Stride must not be zero. There are two VF Stride registers; one each for ARI Capable and non-ARI Capable Hierarchies. The "ARI Capable Hierarchy" bit of the PF0 "SR-IOV Control Register" determines which one is used by the controller when SR-IOV is being used, and which one is accessed by a read request. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
15:0	SRIOV_VF_OFFSET	* Varies	First VF Offset. First VF Offset is a constant and defines the Routing ID offset of the first VF that is associated with the PF that contains this Capability structure. The first VF's 16-bit Routing ID is calculated by adding the contents of this field to the Routing ID of the PF containing this field ignoring any carry, using unsigned, 16-bit arithmetic. This field may change value when the lowest-numbered PF's ARI Capable Hierarchy value changes or when this PF's NumVFs value changes. First VF Offset is unused if NumVFs is 0. If NumVFs is greater than 0, First VF Offset must not be zero. There are two First VF Offset registers at this address location; one each for ARI Capable and non-ARI Capable Hierarchies. The "ARI Capable Hierarchy" bit of the PF0 "SR-IOV Control Register" determines which one is used by the controller when SR-IOV is being used, and which one is accessed by a read request. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true

3.19.7 VF_DEVICE_ID_REG

- **Name:** VF Device ID Register
- **Description:** VF Device ID may be different from the PF Device ID. A VF Device ID must be managed by the vendor. The vendor must ensure that the chosen VF Device ID does not result in the use of an incompatible device driver.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:16	15:0
SRIOV_VF_DEVICE_ID	RSVDP_0

Table 3-129 Fields for Register: VF_DEVICE_ID_REG

Bits	Name	Memory Access	Description
31:16	SRIOV_VF_DEVICE_ID	* Varies	VF Device ID. This field contains the Device ID that should be presented for every VF to the SI. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_VF_DEVICE_ID_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.19.8 SUP_PAGE_SIZES_REG

- **Name:** Supported Page Sizes.
- **Description:** This Register provides the page sizes supported by the PF. This PF supports a page size of $2^{(n+12)}$, if bit n is Set. For example, if bit 0 is Set, the PF supports 4-KB page sizes.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))



Table 3-130 Fields for Register: SUP_PAGE_SIZES_REG

Bits	Name	Memory Access	Description
31:0	SRIOV_SUP_PAGE_SIZE	* Varies	Page Sizes Supported by VF. This field indicates the page sizes supported by the PF. This PF supports a page size of $2^{(n+12)}$ if bit n is Set. For example, if bit 0 is Set, the PF supports 4-KB page sizes. PFs are required to support 4-KB, 8-KB, 64-KB, 256-KB, 1-MB, and 4-MB page sizes. All other page sizes are optional. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_VF_SUPP_PAGE_SIZE_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.19.9 SYSTEM_PAGE_SIZE_REG

- **Name:** System Page Size.
- **Description:** This Register provides the page size the system will use to map the VFs' memory addresses.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

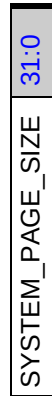


Table 3-131 Fields for Register: SYSTEM_PAGE_SIZE_REG

Bits	Name	Memory Access	Description
31:0	SYSTEM_PAGE_SIZE	R/W	System Page Size. This field defines the page size the system will use to map the VFs' memory addresses. Software must set the value of the System Page Size to one of the page sizes set in the Supported Page Sizes field. As with Supported Page Sizes, if bit n is Set in System Page Size, the VFs associated with this PF are required to support a page size of $2^{(n+12)}$. For example, if bit 1 is Set, the system is using an 8-KB page size. The results are undefined if System Page Size is zero. The results are undefined if more than one bit is set in System Page Size. The results are undefined if a bit is Set in System Page Size that is not Set in Supported Page Sizes. VF Enable must be zero when System Page Size is written. The results are undefined if System Page Size is written when VF Enable is Set. Value After Reset: 0x1 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.19.10 SRIOV_BAR0_REG

- **Name:** VF BAR0.
- **Description:** This register is used to define the BAR0 contents (in VF's Type-1 header) for all VFs in this PF. The actual BARs in the VF's Type-1 header are RO and are derived by the controller from the SRIOV_VF_BAR*_START field in this register in conjunction with SRIOV_VF_OFFSET and SRIOV_VF_STRIDE. The mask for this BAR exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:4	SRIOV_VF_BAR0_START
3	SRIOV_VF_BAR0_PREFETCH
2:1	SRIOV_VF_BAR0_TYPE
0	RSVDP_0

Table 3-132 Fields for Register: SRIOV_BAR0_REG

Bits	Name	Memory Access	Description
31:4	SRIOV_VF_BAR0_START	* Varies	VF BAR0 Base Address. This field provides starting address of BAR0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead

Bits	Name	Memory Access	Description
3	SRIOV_VF_BAR0_PREFETCH	* Varies	VF BAR0 Prefetchable. This field provides whether BAR0 is Prefetchable or not. If this field is set, Data is Prefetchable for this BAR0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR0_ENABLED_\$pf && !SNPS_RSVDPARAM_50) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_PREFETCHABLE0_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
2:1	SRIOV_VF_BAR0_TYPE	* Varies	VF BAR0 32 or 64 bit. This field provides that BAR is 32 bit wide or 64 bit wide. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR0_ENABLED_\$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_BAR0_TYPE_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.19.11 SRIOV_BAR1_REG

- **Name:** VF BAR1.
- **Description:** This register is used to define the BAR1 contents (in VF's Type-1 header) for all VFs in this PF. The actual BARs in the VF's Type-1 header are RO and are derived by the controller from the SRIOV_VF_BAR*_START field in this register in conjunction with SRIOV_VF_OFFSET and SRIOV_VF_STRIDE. The mask for this BAR exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:4	SRIOV_VF_BAR1_START
3	SRIOV_VF_BAR1_PREFETCH
2:1	SRIOV_VF_BAR1_TYPE
0	RSVDP_SRIOV_VF_BAR1

Table 3-133 Fields for Register: SRIOV_BAR1_REG

Bits	Name	Memory Access	Description
31:4	SRIOV_VF_BAR1_START	* Varies	VF BAR1 Base Address. This field provides starting address of BAR1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

Bits	Name	Memory Access	Description
3	SRIOV_VF_BAR1_PREFETCH	* Varies	VF BAR1 Prefetchable. This field provides whether BAR1 is Prefetchable or not. If this field is set, Data is Prefetchable for this BAR1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR1_ENABLED \$pf && !SNPS_RSVDPARAM_50) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_PREFETCHABLE1_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable
2:1	SRIOV_VF_BAR1_TYPE	* Varies	VF BAR1 32 or 64 bit. This field provides that BAR is 32 bit wide or 64 bit wide. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR1_ENABLED \$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_BAR1_TYPE_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
0	RSVDP_SRIOV_VF_BAR1	* Varies	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Memory Access: "(VF_BAR0_TYPE_0 ==2)? \"read-write\": \"read-only\" "

3.19.12 SRIOV_BAR2_REG

- **Name:** VF BAR2.
- **Description:** This register is used to define the BAR2 contents (in VF's Type-1 header) for all VFs in this PF. The actual BARs in the VF's Type-1 header are RO and are derived by the controller from the SRIOV_VF_BAR*_START field in this register in conjunction with SRIOV_VF_OFFSET and SRIOV_VF_STRIDE. The mask for this BAR exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:4	SRIOV_VF_BAR2_START
3	SRIOV_VF_BAR2_PREFETCH
2:1	SRIOV_VF_BAR2_TYPE
0	RSVDP_0

Table 3-134 Fields for Register: SRIOV_BAR2_REG

Bits	Name	Memory Access	Description
31:4	SRIOV_VF_BAR2_START	* Varies	VF BAR2 Base Address. This field provides starting address of BAR2. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead

Bits	Name	Memory Access	Description
3	SRIOV_VF_BAR2_PREFETCH	* Varies	VF BAR2 Prefetchable. This field provides whether BAR2 is Prefetchable or not. If this field is set, Data is Prefetchable for this BAR2. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR2_ENABLED_\$pf && !SNPS_RSVDPARAM_50) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_PREFETCHABLE2_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
2:1	SRIOV_VF_BAR2_TYPE	* Varies	VF BAR2 32 or 64 bit. This field provides that BAR is 32 bit wide or 64 bit wide. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR2_ENABLED_\$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_BAR2_TYPE_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.19.13 SRIOV_BAR3_REG

- **Name:** VF BAR3.
- **Description:** This register is used to define the BAR3 contents (in VF's Type-1 header) for all VFs in this PF. The actual BARs in the VF's Type-1 header are RO and are derived by the controller from the SRIOV_VF_BAR*_START field in this register in conjunction with SRIOV_VF_OFFSET and SRIOV_VF_STRIDE. The mask for this BAR exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:4	SRIOV_VF_BAR3_START
3	SRIOV_VF_BAR3_PREFETCH
2:1	SRIOV_VF_BAR3_TYPE
0	RSVDP_SRIOV_VF_BAR3

Table 3-135 Fields for Register: SRIOV_BAR3_REG

Bits	Name	Memory Access	Description
31:4	SRIOV_VF_BAR3_START	* Varies	VF BAR3 Base Address. This field provides starting address of BAR3. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

Bits	Name	Memory Access	Description
3	SRIOV_VF_BAR3_PREFETCH	* Varies	VF BAR3 Prefetchable. This field provides whether BAR3 is Prefetchable or not. If this field is set, Data is Prefetchable for this BAR3. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR3_ENABLED \$pf && !SNPS_RSVDPARAM_50) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_PREFETCHABLE3_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable
2:1	SRIOV_VF_BAR3_TYPE	* Varies	VF BAR3 32 or 64 bit. This field provides that BAR is 32 bit wide or 64 bit wide. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR3_ENABLED \$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_BAR3_TYPE_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable
0	RSVDP_SRIOV_VF_BAR3	* Varies	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Memory Access: "(VF_BAR2_TYPE_0 ==2)? \"read-write\": \"read-only\" "

3.19.14 SRIOV_BAR4_REG

- **Name:** VF BAR4.
- **Description:** This register is used to define the BAR4 contents (in VF's Type-1 header) for all VFs in this PF. The actual BARs in the VF's Type-1 header are RO and are derived by the controller from the SRIOV_VF_BAR*_START field in this register in conjunction with SRIOV_VF_OFFSET and SRIOV_VF_STRIDE. The mask for this BAR exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x34
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:4	SRIOV_VF_BAR4_START
3	SRIOV_VF_BAR4_PREFETCH
2:1	SRIOV_VF_BAR4_TYPE
0	RSVDP_0

Table 3-136 Fields for Register: SRIOV_BAR4_REG

Bits	Name	Memory Access	Description
31:4	SRIOV_VF_BAR4_START	* Varies	VF BAR4 Base Address. This field provides starting address of BAR4. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead

Bits	Name	Memory Access	Description
3	SRIOV_VF_BAR4_PREFETCH	* Varies	VF BAR4 Prefetchable. This field provides whether BAR4 is Prefetchable or not. If this field is set, Data is Prefetchable for this BAR4. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR4_ENABLED_\$pf && !SNPS_RSVDPARAM_50) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_PREFETCHABLE4_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
2:1	SRIOV_VF_BAR4_TYPE	* Varies	VF BAR4 32 or 64 bit. This field provides that BAR is 32 bit wide or 64 bit wide. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR4_ENABLED_\$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_BAR4_TYPE_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.19.15 SRIOV_BAR5_REG

- **Name:** VF BAR5.
- **Description:** This register is used to define the BAR5 contents (in VF's Type-1 header) for all VFs in this PF. The actual BARs in the VF's Type-1 header are RO and are derived by the controller from the SRIOV_VF_BAR*_START field in this register in conjunction with SRIOV_VF_OFFSET and SRIOV_VF_STRIDE. The mask for this BAR exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x38
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:4	3	2:1	0
SRIOV_VF_BAR5_START	SRIOV_VF_BAR5_PREFETCH	SRIOV_VF_BAR5_TYPE	RSVDP_SRIOV_VF_BAR5

Table 3-137 Fields for Register: SRIOV_BAR5_REG

Bits	Name	Memory Access	Description
31:4	SRIOV_VF_BAR5_START	* Varies	VF BAR5 Base Address. This field provides starting address of BAR5. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

Bits	Name	Memory Access	Description
3	SRIOV_VF_BAR5_PREFETCH	* Varies	VF BAR5 Prefetchable. This field provides whether BAR5 is Prefetchable or not. If this field is set, Data is Prefetchable for this BAR5. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR5_ENABLED \$pf && !SNPS_RSVDPARAM_50) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_PREFETCHABLE5_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable
2:1	SRIOV_VF_BAR5_TYPE	* Varies	VF BAR5 32 or 64 bit. This field provides that BAR is 32 bit wide or 64 bit wide. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: If enabled and (DBI_RO_WR_EN == 1) then R/W else R and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && VF_DEFAULT_BAR5_ENABLED \$pf) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R Value After Reset: VF_BAR5_TYPE_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable
0	RSVDP_SRIOV_VF_BAR5	* Varies	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Memory Access: "(VF_BAR4_TYPE_0 ==2)? \"read-write\": \"read-only\" "

3.19.16 VF_MIGRATION_STATE_ARRAY_REG

- **Name:** VF Migration State Array Offset Register.
- **Description:** If VF Migration Capable is Set and TotalVFs is not zero, this register shall contain a PF BAR relative pointer to the VF Migration State Array. This register is RO Zero if VF Migration Capable is Clear.
- **Size:** 32 bits
- **Offset:** B+0x3c
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:3	SRIOV_VF_MIGRATION_STATE_OFFSET
2:0	SRIOV_VF_MIGRATION_STATE_BIR

Table 3-138 Fields for Register: VF_MIGRATION_STATE_ARRAY_REG

Bits	Name	Memory Access	Description
31:3	SRIOV_VF_MIGRATION_STATE_OFFSET	R	VF Migration State Offset. This field used as an offset from the address contained by one of the Function's Base Address registers to point to the base of the VF Migration State Array. The lower three VF Migration State BIR bits are masked off (set to zero) by software to form a 32-bit QW-aligned offset. This field is undefined if TotalVFs is 0. Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
2:0	SRIOV_VF_MIGRATION_STATE_BIR	R	VF Migration State BIR. This field indicates which one of a Function's Base Address registers , located beginning at 10h in Configuration Space, is used to map the Function's VF Migration State Array into Memory Space. All encodings other than the defined encodings are reserved. For a 64-bit BAR, the VF Migration State BIR indicates the lower DW. This field is undefined if TotalVFs is 0. Values: ■ 0x0 (BAR0): BAR0 10h ■ 0x1 (BAR1): BAR1 14h ■ 0x2 (BAR2): BAR2 18h ■ 0x3 (BAR3): BAR3 1Ch ■ 0x4 (BAR4): BAR4 20h ■ 0x5 (BAR5): BAR5 24h Value After Reset: 0x0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.20 PF_SRIOV_CAP_DBI2i Registers

3.20.1 SHADOW_SRIOV_VF_OFFSET_POSITION

- **Name:** Shadow VF Stride and Offset Register.
- **Description:** This Register contains shadow value of VF Stride and First VFs Offset.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 | CX_MEMORY_MAP_PCIE_TYPE==2))

SHADOW_SRIOV_VF_STRIDE	31:16
SHADOW_SRIOV_VF_OFFSET	15:0

Table 3-139 Fields for Register: SHADOW_SRIOV_VF_OFFSET_POSITION

Bits	Name	Memory Access	Description
31:16	SHADOW_SRIOV_VF_STRIDE	* Varies	Shadow VF Stride. This field is shadow field of VF Stride. Shadow VF Stride defines the Routing ID offset from one VF to the next one for all VFs associated with the PF that contains this Capability structure. The next VF's 16-bit Routing ID is calculated by adding the contents of this field to the Routing ID of the current VF, ignoring any carry, using unsigned 16-bit arithmetic. This field may change value when the lowest-numbered PF's ARI Capable Hierarchy value changes or when this PF's NumVFs value changes. VF Stride is unused if NumVFs is 0 or 1. If NumVFs is greater than 1, VF Stride must not be zero. There are two VF Stride registers; one each for ARI Capable and non-ARI Capable Hierarchies. The "ARI Capable Hierarchy" bit (SRIOV_ARI_CAPABLE_HIER) of the PF0 "SR-IOV Control Register" (STATUS_CONTROL_REG) determines which one is used by the controller when SR-IOV is being used, and which one is accessed by a read request. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W Value After Reset: VF_STRIDE_ARI_CAP_HIER1_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"write-only\""

Bits	Name	Memory Access	Description
15:0	SHADOW_SRIOV_VF_OFFSET	* Varies	Shadow First VF Offset. This field is shadow field of First VF Offset. Shadow First VF Offset is a constant and defines the Routing ID offset of the first VF that is associated with the PF that contains this Capability structure. The first VF's 16-bit Routing ID is calculated by adding the contents of this field to the Routing ID of the PF containing this field ignoring any carry, using unsigned, 16-bit arithmetic. This field may change value when the lowest-numbered PF's ARI Capable Hierarchy value changes or when this PF's NumVFs value changes. First VF Offset is unused if NumVFs is 0. If NumVFs is greater than 0, First VF Offset must not be zero. There are two First VF Offset registers at this address location; one each for ARI Capable and non-ARI Capable Hierarchies. The "ARI Capable Hierarchy" bit (SRIOV_ARI_CAPABLE_HIER) of the PF0 "SR-IOV Control Register" (STATUS_CONTROL_REG) determines which one is used by the controller when SR-IOV is being used, and which one is accessed by a read request. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W Value After Reset: FIRST_VF_OFFSET_ARI_CAP_HIER1_0 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"write-only\""

3.20.2 SRIOV_BAR0_MASK_REG

- **Name:** BAR0 and BAR Mask.
- **Description:** After reading the low order bits to determine if the BAR is a 32-bit BAR or 64-bit BAR pair, determine the size and alignment requirements by writing all 1s to VF BAR_n (or VF BAR_n and 20 VF BAR_n+1 for a 64-bit BAR pair) and reading back the contents of the BAR or BAR pair. Convert the bit mask returned by the read(s) to a size and alignment value as described in the PCI Local Bus Specification. This value is the size and alignment for a single VF. The mask for this BAR0 exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2)) && (VF_DEFAULT_BAR0_ENABLED_n && VF_BAR0_MASK_WRITABLE_n==1)

31:1	PCI_SRIOV_BAR0_MASK
0	PCI_SRIOV_BAR0_ENABLED

Table 3-140 Fields for Register: SRIOV_BAR0_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_SRIOV_BAR0_MASK	* Varies	BAR0 Mask. If this field is set, BAR0 is masked. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_BAR0_MASK_n >> 1 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

Bits	Name	Memory Access	Description
0	PCI_SRIOV_BAR0_ENABLED	* Varies	BAR0 Mask Enabled. If this field is set, Masking of BAR0 is enabled. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_DEFAULT_BAR0_ENABLE_n Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

3.20.3 SRIOV_BAR1_MASK_REG

- **Name:** BAR1 and BAR Mask.
- **Description:** After reading the low order bits to determine if the BAR is a 32-bit BAR or 64-bit BAR pair, determine the size and alignment requirements by writing all 1s to VF BARn (or VF BARn and 20 VF BARn+1 for a 64-bit BAR pair) and reading back the contents of the BAR or BAR pair. Convert the bit mask returned by the read(s) to a size and alignment value as described in the PCI Local Bus Specification. This value is the size and alignment for a single VF. The mask for this BAR1 exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2)) && ((VF_DEFAULT_BAR1_ENABLED_n && VF_BAR1_TYPE_n==0 && VF_BAR1_MASK_WRITABLE_n==1) || (VF_DEFAULT_BAR0_ENABLED_n && VF_BAR0_TYPE_n==2 && VF_BAR0_MASK_WRITABLE_n==1))

31:1	0
PCI_SRIOV_BAR1_MASK	PCI_SRIOV_BAR1_ENABLED

Table 3-141 Fields for Register: SRIOV_BAR1_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_SRIOV_BAR1_MASK	* Varies	BAR1 Mask. If this field is set, BAR1 is masked. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_BAR1_MASK_n >> 1 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable
0	PCI_SRIOV_BAR1_ENABLED	* Varies	BAR1 Mask Enabled. If this field is set, Masking of BAR1 is enabled. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_DEFAULT_BAR1_ENABLED_n Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

3.20.4 SRIOV_BAR2_MASK_REG

- **Name:** BAR2 and BAR Mask.
- **Description:** After reading the low order bits to determine if the BAR is a 32-bit BAR or 64-bit BAR pair, determine the size and alignment requirements by writing all 1s to VF BAR_n (or VF BAR_n and 20 VF BAR_n+1 for a 64-bit BAR pair) and reading back the contents of the BAR or BAR pair. Convert the bit mask returned by the read(s) to a size and alignment value as described in the PCI Local Bus Specification. This value is the size and alignment for a single VF. The mask for this BAR2 exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2)) && (VF_DEFAULT_BAR2_ENABLED_n && VF_BAR2_MASK_WRITABLE_n==1)

31:1	PCI_SRIOV_BAR2_MASK
0	PCI_SRIOV_BAR2_ENABLED

Table 3-142 Fields for Register: SRIOV_BAR2_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_SRIOV_BAR2_MASK	* Varies	BAR2 Mask. If this field is set, BAR2 is masked. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_BAR2_MASK_n >> 1 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

Bits	Name	Memory Access	Description
0	PCI_SRIOV_BAR2_ENABLED	* Varies	BAR2 Mask Enabled. If this field is set, Masking of BAR2 is enabled. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_DEFAULT_BAR2_ENABLE_n Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

3.20.5 SRIOV_BAR3_MASK_REG

- **Name:** BAR3 and BAR Mask.
- **Description:** After reading the low order bits to determine if the BAR is a 32-bit BAR or 64-bit BAR pair, determine the size and alignment requirements by writing all 1s to VF BAR_n (or VF BAR_n and 20 VF BAR_n+1 for a 64-bit BAR pair) and reading back the contents of the BAR or BAR pair. Convert the bit mask returned by the read(s) to a size and alignment value as described in the PCI Local Bus Specification. This value is the size and alignment for a single VF. The mask for this BAR3 exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2)) && ((VF_DEFAULT_BAR3_ENABLED_n && VF_BAR3_TYPE_n==0 && VF_BAR3_MASK_WRITABLE_n==1) || (VF_DEFAULT_BAR2_ENABLED_n && VF_BAR2_TYPE_n==2 && VF_BAR2_MASK_WRITABLE_n==1))

31:1	0
PCI_SRIOV_BAR3_MASK	PCI_SRIOV_BAR3_ENABLED

Table 3-143 Fields for Register: SRIOV_BAR3_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_SRIOV_BAR3_MASK	* Varies	BAR3 Mask. If this fiels is set, BAR3 is masked. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_BAR3_MASK_n >> 1 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable
0	PCI_SRIOV_BAR3_ENABLED	* Varies	BAR3 Mask Enabled. If this field is set, Masking of BAR3 is enabled. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_DEFAULT_BAR3_ENABLED_n Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

3.20.6 SRIOV_BAR4_MASK_REG

- **Name:** BAR4 and BAR Mask.
- **Description:** After reading the low order bits to determine if the BAR is a 32-bit BAR or 64-bit BAR pair, determine the size and alignment requirements by writing all 1s to VF BAR_n (or VF BAR_n and 20 VF BAR_n+1 for a 64-bit BAR pair) and reading back the contents of the BAR or BAR pair. Convert the bit mask returned by the read(s) to a size and alignment value as described in the PCI Local Bus Specification. This value is the size and alignment for a single VF. The mask for this BAR4 exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x34
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2)) && (VF_DEFAULT_BAR4_ENABLED_n && VF_BAR4_MASK_WRITABLE_n==1)

31:1	PCI_SRIOV_BAR4_MASK
0	PCI_SRIOV_BAR4_ENABLED

Table 3-144 Fields for Register: SRIOV_BAR4_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_SRIOV_BAR4_MASK	* Varies	BAR4 Mask. If this field is set, BAR4 is masked. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_BAR4_MASK_n >> 1 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

Bits	Name	Memory Access	Description
0	PCI_SRIOV_BAR4_ENABLED	* Varies	BAR4 Mask Enabled. If this field is set, Masking of BAR4 is enabled. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_DEFAULT_BAR4_ENABLE_n Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

3.20.7 SRIOV_BAR5_MASK_REG

- **Name:** BAR5 and BAR Mask.
- **Description:** After reading the low order bits to determine if the BAR is a 32-bit BAR or 64-bit BAR pair, determine the size and alignment requirements by writing all 1s to VF BAR_n (or VF BAR_n and 20 VF BAR_n+1 for a 64-bit BAR pair) and reading back the contents of the BAR or BAR pair. Convert the bit mask returned by the read(s) to a size and alignment value as described in the PCI Local Bus Specification. This value is the size and alignment for a single VF. The mask for this BAR5 exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.
- **Size:** 32 bits
- **Offset:** B+0x38
- **Exists:** (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2)) && ((VF_DEFAULT_BAR5_ENABLED_n && VF_BAR5_TYPE_n==0 && VF_BAR5_MASK_WRITABLE_n==1) || (VF_DEFAULT_BAR4_ENABLED_n && VF_BAR4_TYPE_n==2 && VF_BAR4_MASK_WRITABLE_n==1))

31:1	0
PCI_SRIOV_BAR5_MASK	PCI_SRIOV_BAR5_ENABLED

Table 3-145 Fields for Register: SRIOV_BAR5_MASK_REG

Bits	Name	Memory Access	Description
31:1	PCI_SRIOV_BAR5_MASK	* Varies	BAR5 Mask. If this fiels is set, BAR5 is masked. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_BAR5_MASK_n >> 1 Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable
0	PCI_SRIOV_BAR5_ENABLED	* Varies	BAR5 Mask Enabled. If this field is set, Masking of BAR5 is enabled. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_DEFAULT_BAR5_ENABLED_n Exists: (CX_SRIOV_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

3.21 PF_TPH_CAPI Registers

3.21.1 TPH_EXT_CAP_HDR_REG

- **Name:** TPH Extended Capability Header.
- **Description:** This register provides Capability Id, Capability Revision, and Next offset of structure Extended Capability Register.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	TPH_REQ_NEXT_PTR
19:16	TPH_REQ_CAP_VER
15:0	PCIE_EXT_CAP_ID

Table 3-146 Fields for Register: TPH_EXT_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	TPH_REQ_NEXT_PTR	* Varies	Next Capability Pointer. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	TPH_REQ_CAP_VER	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	PCIE_EXT_CAP_ID	* Varies	TPH Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. PCI Express Extended Capability ID for the TPH Requester Extended Capability is 0017h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x17 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.21.2 TPH_REQ_CAP_REG_REG

- **Name:** TPH Requester Capability Register.
- **Description:** This register gives information about No ST mode support, Interrupt Vector mode, Device specific mode, Extended TPH requester support, ST table location and size.
SRIOV Note: All VFs in a single PF have the same values for VF_TPH_REQ_CAP_REG_REG. To write this common register, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PF TPH_REQ_CAP_REG_REG register.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_27	31:27
TPH_REQ_CAP_ST_TABLE_SIZE	26:16
RSVDP_11	15:11
TPH_REQ_CAP_ST_TABLE_LOC_1	10
TPH_REQ_CAP_ST_TABLE_LOC_0	9
TPH_REQ_EXTENDED_TPH	8
RSVDP_3	7:3
TPH_REQ_DEVICE_SPEC	2
TPH_REQ_CAP_INT_VEC	1
TPH_REQ_NO_ST_MODE	0

Table 3-147 Fields for Register: TPH_REQ_CAP_REG_REG

Bits	Name	Memory Access	Description
31:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
26:16	TPH_REQ_CAP_ST_TABLE_SIZE	* Varies	ST Table Size. This field indicates the maximum number of ST Table entries the Function may use. Software reads this field to determine the ST Table Size N, which is encoded as N-1. For example, a returned value of 000 0000 0011b indicates a table size of four entries. There is an upper limit of 64 entries when the ST Table is located in the TPH Requester Extended Capability structure. When the ST Table is located in the MSI-X Table, this value is limited by the size of the MSI-X Table. This field is only applicable for Functions that implement an ST Table as indicated by the ST Table Location field. Otherwise, the value in this field is undefined. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: "TPH_ST_TABLE_SIZE_0 - 1" Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
10	TPH_REQ_CAP_ST_TABLE_LOC_1	* Varies	ST Table Location Bit 1. If this field is set, ST Table is Located in the MSI-X Table. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : (CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: "(TPH_ST_TABLE_LOC_0 == 2) ? 1 : 0" Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
9	TPH_REQ_CAP_ST_TABLE_LOC_0	* Varies	ST Table Location Bit 0. If this field is set, ST Table is Located in the TPH Requester Extended Capability structure. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: "(TPH_ST_TABLE_LOC_0 == 1) ? 1 : 0" Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
8	TPH_REQ_EXTENDED_TPH	* Varies	Extended TPH Requester Supported. If Set indicates that the Function is capable of generating Requests with a TPH TLP Prefix . <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_NPRFX>0) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: (CX_DBI_RO_WR_EN && CX_NPRFX>0) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
7:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
2	TPH_REQ_DEVICE_SPEC	* Varies	Device Specific Mode Supported. If set indicates that the Function supports the Device Specific Mode of operation. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: TPH_DS_0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
1	TPH_REQ_CAP_INT_VEC	* Varies	Interrupt Vector Mode Supported. If set indicates that the Function supports the Interrupt Vector Mode of operation. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: TPH_IVEC_0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
0	TPH_REQ_NO_ST_MODE	R	No ST Mode Supported. set indicates that the Function supports the No ST Mode of operation. This mode is required to be supported by all Functions that implement this Capability structure. This bit must have a value of 1b. Value After Reset: 0x1 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.21.3 TPH_REQ_CONTROL_REG_REG

- **Name:** TPH Requester Control Register.
- **Description:** This register provides ST Mode Select and controls TPH Requester Enable.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:10	RSVDP_10
9:8	TPH_REQ_CTRL_REQ_EN
7:3	RSVDP_3
2:0	TPH_REQ_ST_MODE_SELECT

Table 3-148 Fields for Register: TPH_REQ_CONTROL_REG_REG

Bits	Name	Memory Access	Description
31:10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
9:8	TPH_REQ_CTRL_REQ_EN	R/W	TPH Requester Enable Bit. This field controls the ability to issue Request TLPs using either TPH or Extended TPH. Functions that advertise that they do not support Extended TPH are permitted to hardwire bit 9 of this field to 0b. Values: ■ 0x0 (NOT_TPH_AND_NOT_EXT_TPH): Function operating as a Requester is not permitted to issue Requests with TPH or Extended TPH. ■ 0x1 (TPH_AND_NOT_EXT_TPH): Function operating as a Requester is permitted to issue Requests with TPH and is not permitted to issue Requests with Extended TPH. ■ 0x2 (RESERVED): Reserved ■ 0x3 (TPH_AND_EXT_TPH): Function operating as a Requester is permitted to issue Requests with TPH and Extended TPH. Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
2:0	TPH_REQ_ST_MODE_SELECT	* Varies	ST Mode Select. This field selects the ST Mode of operation. All encodings other than the defined encodings are reserved. Functions that support only the No ST Mode of operation must hardwire this field to 000b. Function operation is undefined if software enables a mode of operation that does not correspond to a mode supported by the Function. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (NO_ST_MODE): No ST Mode ■ 0x1 (INT_VECT_MODE): Interrupt Vector Mode ■ 0x2 (DEV_SPECIFIC_MODE): Device Specific Mode Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && TPH_IVEC_0==0 && TPH_DS_0==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && TPH_IVEC_0==0 && TPH_DS_0==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : (TPH_IVEC_0==0 && TPH_DS_0==0) ? \"read-only\" : \"read-write\""

3.21.4 **TPH_ST_TABLE_REG_i (for i = 0; i <= 31)**

- **Name:** TPH ST Table Register 0.
- **Description:** The TPH ST Table 0 must be implemented in the TPH Requester Extended Capability structure if the value of the ST Table Location field is 01b. For all other values, the ST Entry registers must not be implemented. Each implemented ST Entry is 16 bits. The number of ST Entry registers implemented must be equal to the number of ST Table entries supported by the Function, which is the value of the ST Table Size field plus one.
- **Size:** 32 bits
- **Offset:** B+0xc + i*0x4
- **Exists:** (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

TPH_REQ_ST_TABLE_HIGHER_1	31:24
TPH_REQ_ST_TABLE_LOWER_1	23:16
TPH_REQ_ST_TABLE_HIGHER_0	15:8
TPH_REQ_ST_TABLE_LOWER_0	7:0

Table 3-149 Fields for Register: TPH_ST_TABLE_REG_i (for i = 0; i <= 31)

Bits	Name	Memory Access	Description
31:24	TPH_REQ_ST_TABLE_HIGHER_1	* Varies	ST Table 1 Upper Byte. If the Function's Extended TPH Requester Supported bit is Set, then this field contains the upper 8 bits of a Steering Tag (1). Otherwise, this field is RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: this field is RW or Tie to 0 by table size configure ■ Dbi: this field is RW or Tie to 0 by table size configure and more specifically: (CX_NPRFX>0) ? R/W : R Value After Reset: 0x0 Exists: (-1 < (TPH_ST_TABLE_DEPTH_0-1)) && (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
23:16	TPH_REQ_ST_TABLE_LOWER_1	R/W	ST Table 1 Lower Byte. This field contains the lower 8 bits of a Steering Tag1. Value After Reset: 0x0 Exists: (-1 < (TPH_ST_TABLE_DEPTH_0-1)) && (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:8	TPH_REQ_ST_TABLE_HIGHER_0	* Varies	ST Table 0 Upper Byte. If the Function's Extended TPH Requester Supported bit is Set, then this field contains the upper 8 bits of a Steering Tag (0). Otherwise, this field is RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: this field is RW or Tie to 0 by table size configure ■ Dbi: this field is RW or Tie to 0 by table size configure and more specifically: (CX_NPRFX>0) ? R/W : R Value After Reset: 0x0 Exists: (-1 < (TPH_ST_TABLE_DEPTH_0-1)) && (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:0	TPH_REQ_ST_TABLE_LOWER_0	R/W	ST Table 0 Lower Byte. This field contains the lower 8 bits of a Steering Tag0. Value After Reset: 0x0 Exists: (-1 < (TPH_ST_TABLE_DEPTH_0-1)) && (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.22 PF_TPH_CAP_DBI2i Registers

3.22.1 SHADOW_TPH_REQ_CAP_REG_REG

- **Name:** Shadow register TPH Requester Capability Register.
- **Description:** This register is the shadow register of the TPH Requester Capability Register.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_27	31:27
TPH_REQ_CAP_ST_TABLE_SIZE	26:16
RSVDP_11	15:11
TPH_REQ_CAP_ST_TABLE_LOC_1	10
TPH_REQ_CAP_ST_TABLE_LOC_0	9
TPH_REQ_EXTENDED_TPH	8
RSVDP_3	7:3
TPH_REQ_DEVICE_SPEC	2
TPH_REQ_CAP_INT_VEC	1
TPH_REQ_NO_ST_MODE	0

Table 3-150 Fields for Register: SHADOW_TPH_REQ_CAP_REG_REG

Bits	Name	Memory Access	Description
31:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
26:16	TPH_REQ_CAP_ST_TABLE_SIZE	* Varies	ST Table Size in the shadow register. This field indicates the maximum number of ST Table entries the Function may use. Software reads this field to determine the ST Table Size N, which is encoded as N-1. For example, a returned value of 000 0000 0011b indicates a table size of four entries. There is an upper limit of 64 entries when the ST Table is located in the TPH Requester Extended Capability structure. When the ST Table is located in the MSI-X Table, this value is limited by the size of the MSI-X Table. This field is only applicable for Functions that implement an ST Table as indicated by the ST Table Location field. Otherwise, the value in this field is undefined. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: "TPH_ST_TABLE_SIZE_0 - 1" Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
15:11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
10	TPH_REQ_CAP_ST_TABLE_LOC_1	* Varies	ST Table Location Bit 1 in the shadow register. If this field is set, ST Table is Located in the MSI-X Table. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: (CX_MEMORY_MAP_VIEW==2 && CX_DBI_RO_WR_EN && (VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE)) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: "(TPH_ST_TABLE_LOC_0 == 2) ? 1 : 0" Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
9	TPH_REQ_CAP_ST_TABLE_LOC_0	* Varies	ST Table Location Bit 0 in the shadow register. If this field is set, ST Table is Located in the TPH Requester Extended Capability structure. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: "(TPH_ST_TABLE_LOC_0 == 1) ? 1 : 0" Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
8	TPH_REQ_EXTENDED_TPH	* Varies	Extended TPH Requester Supported in the shadow register. If Set indicates that the Function is capable of generating Requests with a TPH TLP Prefix . <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
2	TPH_REQ_DEVICE_SPEC	* Varies	Device Specific Mode Supported in the shadow register. If set indicates that the Function supports the Device Specific Mode of operation. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: TPH_DS_0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
1	TPH_REQ_CAP_INT_VEC	* Varies	Interrupt Vector Mode Supported in the shadow register. If set indicates that the Function supports the Interrupt Vector Mode of operation. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: TPH_IVEC_0 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	TPH_REQ_NO_ST_MODE	R	No ST Mode Supported in the shadow register. set indicates that the Function supports the No ST Mode of operation. This mode is required to be supported by all Functions that implement this Capability structure. This bit must have a value of 1b. Value After Reset: 0x1 Exists: (CX_TPH_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.23 PF_ATS_CAPi Registers

3.23.1 ATS_CAP_HDR_REG

- **Name:** ATS Extended Capability Header.
- **Description:** This register provides value of Capability ID, Capability version, and next offset of Extended capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_ATS_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	CAP_ID

Table 3-151 Fields for Register: ATS_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. The offset to the next PCI Extended Capability structure or 000h if no other items exist in the linked list of capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_ATS_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_ATS_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	CAP_ID	* Varies	ATS Extended Capability ID. This field must return a Capability ID of 000Fh indicating that this is an ATS Extended Capability structure. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xf Exists: (CX_ATS_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.23.2 ATS_CAPABILITIES_CTRL_REG

- **Name:** ATS Requester Capability Register.
- **Description:** This register provides value of Invalidate queue depth, Page alignment request, and Global Invalidate support. It also controls Cache translation and Smallest Translate Unit.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_ATS_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	ENABLE
30:21	RSVDP_21
20:16	STU
15:7	RSVDP_7
6	GLOBAL_INVAL_SPPRTD
5	PAGE_ALIGNED_REQ
4:0	INVALIDATE_Q_DEPTH

Table 3-152 Fields for Register: ATS_CAPABILITIES_CTRL_REG

Bits	Name	Memory Access	Description
31	ENABLE	R/W	Enable (E). When Set, the Function is enabled to cache translations. Behavior is undefined if this bit is Set, and the value of the PASID Enable, Execute Requested Permission Enable, or Privileged Mode Requested Enable bits are changed. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_ATS_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_EXTENSIBLE_VFUNC) ? "untestable" : "unconstrained" }

Bits	Name	Memory Access	Description
30:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_ATS_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20:16	STU	R/W	Smallest Translation Unit (STU). This value indicates to the Function the minimum number of 4096-byte blocks that is indicated in a Translation Completions or Invalidate Requests. This is a power of 2 multiplier and the number of blocks is 2 STU. A value of 0 0000b indicates one block and a value of 1 1111b indicates 2 ³¹ blocks (or 8 TB total). Value After Reset: ATS_STU_0 Exists: (CX_ATS_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_ATS_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
6	GLOBAL_INVALID_SPPRTD	* Varies	Global Invalidate Supported. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_PASID_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: R <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): The Function supports Invalidation Requests that have the Global Invalidate bit Set. ■ 0x0 (CLEAR): The Function ignores the Global Invalidate bit in all Invalidate Requests. This bit is 0b if the Function does not support the PASID TLP Prefix. Value After Reset: DEFAULT_ATS_GLOBAL_INVALID_SPPRTD Exists: (CX_ATS_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
5	PAGE_ALIGNED_REQ	* Varies	Page Aligned Request. If Set, indicates the Untranslated Address is always aligned to a 4096 byte boundary. Setting this field is recommended. This field permits software to distinguish between implementations compatible with this specification and those compatible with an earlier version of this specification in which a Requester was permitted to supply anything in bits [11:2]. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_ATS_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_EXTENSIBLE_VFUNC) ? "untestable" : "unconstrained" }
4:0	INVALIDATE_Q_DEPTH	* Varies	Invalidate Queue Depth. The number of Invalidate Requests that the Function can accept before putting backpressure on the Upstream connection. If 0 0000b, the Function can accept 32 Invalidate Requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: ATS_INV_Q_DPTH_0 Exists: (CX_ATS_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.24 PF_ACS_CAPi Registers

3.24.1 ACS_CAP_HDR_REG

- **Name:** Access Control Services Extended Capability Header Register.
- **Description:** This register provides capability ID, capability version, and next offset value.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_ACS_ENABLE && (((CX_NFUNC==1 || CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) || CX_MEMORY_MAP_PCIE_TYPE!=1))

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	CAP_ID

Table 3-153 Fields for Register: ACS_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Extended Capability structure or 000h, if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field indicates the version of present capability structure. It is a PCI-SIG defined version number. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: wire ? unconstrained : writeAsRead
15:0	CAP_ID	* Varies	ACS Extended Capability ID. This field identifies the type of capability structure that follows this header. It is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. For PCI Express, Extended Capability ID is 000Dh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xd Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: wire ? unconstrained : writeAsRead

3.24.2 ACS_CAPABILITIES_CTRL_REG

- **Name:** Access Control Services Capability Register.
- **Description:** This register is Access Control Services Capability and Control Register.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_ACS_ENABLE && (((CX_NFUNC==1 || CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) || CX_MEMORY_MAP_PCIE_TYPE!=1))

31:23	RSVDP_23
22	ACS_DIRECT_TRANSLATED_P2P_EN
21	ACS_P2P_EGRESS_CONTROL_EN
20	ACS_USP_FORWARDING_EN
19	ACS_P2P_CPL_REDIRCT_EN
18	ACS_P2P_REQ_REDIRCT_EN
17	ACS_AT_BLOCK_EN
16	ACS_SRC_VALID_EN
15:8	ACS_EGRESS_CTRL_SIZE
7	RSVDP_7
6	ACS_DIRECT_TRANSLATED_P2P
5	ACS_P2P_EGRESS_CONTROL
4	ACS_USP_FORWARDING
3	ACS_P2P_CPL_REDIRCT
2	ACS_P2P_REQ_REDIRCT
1	ACS_AT_BLOCK
0	ACS_SRC_VALID

Table 3-154 Fields for Register: ACS_CAPABILITIES_CTRL_REG

Bits	Name	Memory Access	Description
31:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1))

Bits	Name	Memory Access	Description
22	ACS_DIRECT_TRANSLATED_P2P_EN	R/W	ACS Direct Translated Peer-to-Peer Enable. When Set, overrides the ACS_P2P_REQ_REDIRECT and ACS_P2P_EGRESS_CONTROL mechanisms with peer-to-peer Memory Requests whose Address Translation (AT) field indicates a Translated address. This bit is ignored if ACS_AT_BLOCK_EN is 1b. The controller hardwires this bit to 0b if, the ACS Direct Translated P2P functionality is not implemented. Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: writeAsRead Volatile: true
21	ACS_P2P_EGRESS_CONTROL_EN	R/W	ACS Peer-to-Peer Egress Control Enable. In conjunction with the Egress Control Vector plus the ACS_P2P_REQ_REDIRECT and ACS_DIRECT_TRANSLATED_P2P mechanisms, determines when to allow, disallow, or redirect peer-to-peer Requests. The controller hardwires this bit to 0b if, the ACS P2P Egress Control functionality is not implemented. Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: writeAsRead Volatile: true
20	ACS_USP_FORWARDING_EN	R/W	ACS Upstream Forwarding Enable. When Set, the component forwards Upstream any Request or Completion TLPs it receives that were redirected Upstream by a component lower in the hierarchy. This bit only applies to Upstream TLPs arriving at a Downstream Port, and whose normal routing targets the same Downstream Port. The controller hardwires this bit to 0b if, the ACS Upstream Forwarding functionality is not implemented. Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
19	ACS_P2P_CPL_REDIRECT_EN	R/W	ACS Peer-to-Peer Completion Redirect Enable. Determines when the component redirects peer-to-peer Completions Upstream; applicable only to Completions whose Relaxed Ordering Attribute is clear. The controller hardwires this bit to 0b if, the ACS P2P Completion Redirect functionality is not implemented. Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: writeAsRead Volatile: true
18	ACS_P2P_REQ_REDIRECT_EN	R/W	ACS Peer-to-Peer Request Redirect Enable. In conjunction with ACS_P2P_EGRESS_CONTROL and ACS_DIRECT_TRANSLATED_P2P mechanisms, determines when the component redirects peer-to-peer Requests Upstream. With Downstream Ports, this bit only applies to Upstream Requests arriving at the Downstream Port, and whose normal routing targets a different Downstream Port. The controller hardwires this bit to 0b if, the ACS P2P Request Redirect functionality is not implemented. Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: writeAsRead Volatile: true
17	ACS_AT_BLOCK_EN	R/W	ACS Translation Blocking Enable. When Set, the component blocks all Upstream Memory Requests whose Address Translation (AT) field is not set to the default value. The controller hardwires this bit to 0b if, the ACS Translation Blocking functionality is not implemented. Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
16	ACS_SRC_VALID_EN	R/W	ACS Source Validation Enable. When Set, the component validates the Bus Number from the Requester ID of Upstream Requests against the secondary/subordinate Bus Numbers. The controller hardwires this bit to 0b if, the ACS Source Validation functionality is not implemented. Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: writeAsRead Volatile: true
15:8	ACS_EGRESS_CTRL_SIZE	* Varies	ACS Egress control vector size. Encodings 01h-FFh directly indicate the number of applicable bits in the Egress Control Vector; the encoding 00h indicates 256 bits. If the ACS_P2P_EGRESS_CONTROL bit is 0b, the value of the size field is undefined, and the Egress Control Vector register is not required to be present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Value After Reset: CX_ACS_EGRESS_CTRL_SIZE_0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\" "
7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1))

Bits	Name	Memory Access	Description
6	ACS_DIRECT_TRANSLATED_P2P	* Varies	ACS Direct Translated Peer-to-Peer Capable. Required for Root Ports that support Address Translation Services (ATS) and also support peer-to-peer traffic with other Root Ports; required for Switch Downstream Ports; required for Multi-Function Device Functions that support Address Translation Services (ATS) and also support peer-to-peer traffic with other Functions; the controller must be hardwired to 0b otherwise. If 1b, indicates that the component implements ACS Direct Translated P2P. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_ACS_P2P_DIRECT_TRANSL_0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: wire ? unconstrained : writeAsRead
5	ACS_P2P_EGRESS_CONTROL	* Varies	ACS Egress Control. Optional for Root Ports, Switch Downstream Ports, and Multi-Function Device Functions; must be hardwired to 0b otherwise. If 1b, indicates that the component implements ACS P2P Egress Control. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_ACS_P2P_EGRESS_CTRL_0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
4	ACS_USP_FORWARDING	* Varies	ACS Upstream forwarding Capable. Required for Root Ports if the RC supports Redirected Request Validation; required for Switch Downstream Ports; must be hardwired to 0b otherwise. If 1b, indicates that the component implements ACS Upstream Forwarding. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: wire ? unconstrained : writeAsRead
3	ACS_P2P_CPL_REDIRECT	* Varies	ACS Peer-to-peer Completion Redirect Capable. Required for all Functions that support ACS P2P Request Redirect; must be hardwired to 0b otherwise. If 1b, indicates that the component implements ACS P2P Completion Redirect. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_ACS_P2P_COMPL_REDIRECT_0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
2	ACS_P2P_REQ_REDIRECT	* Varies	ACS Peer-to-peer Request Redirect Capable. Required for Root Ports that support peer-to-peer traffic with other Root Ports; required for Switch Downstream Ports; required for Multi-Function Device Functions that support peer-to-peer traffic with other Functions; must be hardwired to 0b otherwise. If 1b, indicates that the component implements ACS P2P Request Redirect. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_ACS_P2P_REQ_REDIRECT_0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: wire ? unconstrained : writeAsRead
1	ACS_AT_BLOCK	* Varies	ACS Translation Blocking Capable. Required for Root Ports and Switch Downstream Ports; Otherwise it is hardwired to 0b. If 1b, indicates that the component implements ACS Translation Blocking. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
0	ACS_SRC_VALID	* Varies	ACS Source Validation Capable. Required for Root Ports and Switch Downstream Ports; Otherwise it is hardwired to 0b. If 1b, indicates that the component implements ACS Source Validation. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: wire ? unconstrained : writeAsRead

3.24.3 ACS_EGRESS_CTRL_VECTOR_REGi (for i = 0; i <= 7)

- **Name:** Access Control Services Egress Control Vector Register #i
- **Description:** This Egress Control Vector register is a read-write register that contains a bit-array. The number of bits in the register is specified by the ACS_EGRESS_CTRL_SIZE field, and the register spans multiple DWORDs, if required.
- **Size:** 32 bits
- **Offset:** B+0x8 + i*0x4
- **Exists:** (CX_ACS_ENABLE && (((CX_NFUNC==1 | | CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) | | CX_MEMORY_MAP_PCIE_TYPE!=1))



Table 3-155 Fields for Register: ACS_EGRESS_CTRL_VECTOR_REGi (for i = 0; i <= 7)

Bits	Name	Memory Access	Description
31:0	ACS_EGRESS_CTRL_VECTOR_REG0	* Varies	Access Control Services Egress Control Vector Register #i. An N-bit bit-array configured by software, where N is given by the value in the ACS_EGRESS_CTRL_SIZE field. When a given bit is set, peer-to-peer Requests targeting the associated Port, Function, or Function Group are blocked or redirected (if enabled). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (ACS_CTRL_VEC_WD_0>0) && CX_ACS_P2P_EGRESS_CTRL_0==1 && (CX_ACS_ENABLE && (((CX_NFUNC==1 CX_SRIOV_ENABLE==0) && CX_MEMORY_MAP_PCIE_TYPE==1) CX_MEMORY_MAP_PCIE_TYPE!=1)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""

3.25 PF_PRS_EXT_CAPi Registers

3.25.1 PRS_EXT_CAP_HDR_REG

- **Name:** Page Request Services Capability Header.
- **Description:** This Register provides Capability ID, Capability version, and Next Offset value of Page Request Service Capability Structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:20	19:16	15:0
PRS_NEXT_OFFSET	PRS_CAP_VERSION	PRS_CAP_ID

Table 3-156 Fields for Register: PRS_EXT_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	PRS_NEXT_OFFSET	* Varies	Page Request Services PCI Express Extended Capability Next Offset. The offset to the next PCI Extended Capability structure or 000h if no other items exist in the linked list of capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: Pointer value calculated at configuration time. Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	PRS_CAP_VERSION	* Varies	Page Request Services PCI Express Extended Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field is depends on the version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
15:0	PRS_CAP_ID	* Varies	Page Request Services PCI Express Extended Capability ID. This field indicates that the associated extended capability structure is a Page Request Extended Capability. This field must return a Capability ID of 0013h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x13 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.25.2 PRS_CONTROL_STATUS_REG

- **Name:** Page Request Services Control and Status Register.
- **Description:** This register provides control and status of different capability such as Reset, Response Failure, ans so on.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 | | CX_MEMORY_MAP_PCIE_TYPE==2))

PRS_PASID	31
RSVDP_25	30:25
PRS_STOPPED	24
RSVDP_18	23:18
PRS_UPRGI	17
PRS_RESP_FAILURE	16
RSVDP_2	15:2
PRS_RESET	1
PRS_ENABLE	0

Table 3-157 Fields for Register: PRS_CONTROL_STATUS_REG

Bits	Name	Memory Access	Description
31	PRS_PASID	* Varies	The PASID TLP Prefix is required. Function behavior is undefined if this bit is Clear and the Function receives a PRG Response Message with a PASID TLP Prefix. Function behavior is undefined if this bit is Set and the Function receives a PRG Response Message with no PASID TLP Prefix when the corresponding Page Requests had a PASID TLP Prefix. This bit is RsvdZ if the Function does not support the PASID TLP Prefix . <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): The Function does not expect PASID TLP Prefixes on any PRG Response Message. ■ 0x1 (SET): The Function expects a PASID TLP Prefix on PRG Response Messages when the corresponding Page Requests had a PASID TLP Prefix. Value After Reset: CX_PRG_PASID_REQUIRED Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
30:25	RSVDP_25	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
24	PRS_STOPPED	R	PRS mechanism stopped requests for additional pages. This field is only meaningful if Enable is Clear. If Enable is Set, this field is undefined. When the Enable field is Cleared, after having been previously Set, the interface transitions to the stopping state and Clears this field. After all page requests currently outstanding at the Function(s) have completed, this field is Set and the interface enters the disabled state. If there were no outstanding page requests, this field may be Set immediately when Enable is Cleared. Resetting the interface will cause an immediate transition to the disabled state. While in the stopping state, receipt of a Response Failure message will result in the immediate transition to the disabled state (Setting this field). For SR-IOV, this field is Set only when all associated Functions (PF and VFs) have stopped issuing page requests. The cfg_prs_stopped output pin reflects the value of this register field. Values: ■ 0x0 (CLEAR): The associated page request interface either has not stopped or has stopped issuing new Page Requests but has outstanding Page Requests. ■ 0x1 (SET): The associated page request interface has stopped issuing additional page requests and that all previously issued Page Requests have completed. Value After Reset: 0x1 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true
23:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
17	PRS_UPRGI	R/W1C	PRS has received a response with Unexpected Page Request Group Index. This field, when Set, indicates that the Function has received a PRG Response Message containing a PRG index that has no matching request. This field is Set by the Function and cleared when a one is written to the field. For SR-IOV, this field is Set in the PF if any associated Function (PF or VF) receives a PRG Response Message that does has no matching request. The <code>cfg_prs_uprgi</code> output pin reflects the value of this register field. Value After Reset: 0x0 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true
16	PRS_RESP_FAILURE	R/W1C	PRS has received a PRG response failure. This field, when Set, indicates that the Function has received a PRG Response Message indicating a Response Failure. The Function expects no further responses from the host (any received are ignored). This field is Set by the Function and Cleared when a one is written to the field. For SR-IOV, this field is Set in the PF if any associated Function (PF or VF) receives a PRG Response Message indicating Response Failure. The <code>cfg_prs_response_failure</code> output pin reflects the value of this register field. Value After Reset: 0x0 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true
15:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
1	PRS_RESET	R	Page Request Services Reset. Reads of this field return 0b. The <code>cfg_prs_reset</code> output pin reflects the value of this register field. Values: ■ 0x0 (CLEAR): No action is initiated if this field is written to 0b, or if this field is written with any value while the Enable field is set. ■ 0x1 (SET): When the Enable field is clear, or is being cleared in the same register update that sets this field, writing a 1b to this field, clears the associated implementation dependent page request credit counter and pending request state for the associated Page Request Interface. Value After Reset: 0x0 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true
0	PRS_ENABLE	R/W	Page Request Services Enabled. If both this field and the Stopped field are Clear, then the Page Request Interface will not issue new page requests, but has outstanding page requests that have been transmitted or are queued for transmission. When the Page Request Interface is transitioned from not-Enabled to Enabled, its status flags (Stopped, Response Failure, and Unexpected Response flags) are cleared. Enabling a Page Request Interface that has not successfully Stopped has indeterminate results. The <code>cfg_prs_enable</code> output pin reflects the value of this register field. Values: ■ 0x0 (CLEAR): The Page Request Interface is not allowed to issue page requests. ■ 0x1 (SET): Indicates that the Page Request Interface is allowed to make page requests. Value After Reset: 0x0 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.25.3 PRS_REQ_CAPACITY_REG

- **Name:** Page Request Services Outstanding Capacity.
- **Description:** This register contains the number of outstanding page request messages the associated Page Request Interface physically supports.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 | | CX_MEMORY_MAP_PCIE_TYPE==2))



Table 3-158 Fields for Register: PRS_REQ_CAPACITY_REG

Bits	Name	Memory Access	Description
31:0	PRS_OUTSTANDING_CAPACITY	* Varies	Page Request Services Outstanding Capacity value. This is the upper limit on the number of pages that can be usefully allocated to the Page Request Interface. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: CX_PRS_OUTSTANDING_CAPACITY_VALUE_0 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.25.4 PRS_REQ_ALLOCATION_REG

- **Name:** Page Request Services Outstanding Allocation.
- **Description:** This register contains the number of outstanding page request messages the associated Page Request Interface is allowed to issue (have outstanding at any given instance).
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

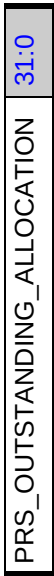


Table 3-159 Fields for Register: PRS_REQ_ALLOCATION_REG

Bits	Name	Memory Access	Description
31:0	PRS_OUTSTANDING_ALLOCATION	R/W	Page Request Services Outstanding Allocation value. This Field contains the number of outstanding page request messages the associated Page Request Interface is allowed to issue. The cfg_prs_outstanding_allocation output pin reflects the value of this register field. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PRS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.26 PF0_LTR_CAP Registers

3.26.1 LTR_CAP_HDR_REG

- **Name:** LTR Extended Capability Header.
- **Description:** This register provides capability ID, capability version and next offset value for LTR(Latency Tolerance Reporting).
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	CAP_ID

Table 3-160 Fields for Register: LTR_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field is depends on version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
15:0	CAP_ID	* Varies	LTR Extended Capacity ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. PCI Express Extended Capability for the LTR Extended Capability is 0018h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x18 Exists: (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.26.2 LTR_LATENCY_REG

- **Name:** LTR Max Snoop and No-Snoop Latency Register.
- **Description:** This register indicates Latency scale and vlaue for Max Snoop and No-Snoop.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:29	RSVDP_29
28:26	MAX_NO_SNOOP_LAT_SCALE
25:16	MAX_NO_SNOOP_LAT
15:13	RSVDP_13
12:10	MAX_SNOOP_LAT_SCALE
9:0	MAX_SNOOP_LAT

Table 3-161 Fields for Register: LTR_LATENCY_REG

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
28:26	MAX_NO_SNOOP_LAT_SCALE	R/W	Max No-Snoop Latency Scale. This register provides a scale for the value contained within the Max No-Snoop LatencyValue field. Encoding is the same as the LatencyScale fields in the LTR Message. It is strongly recommended that any updates to this field are reflected in LTR Message(s) sent by the device within 1 ms. Hardware operation is undefined if software writes a Not Permitted value to this field. Value After Reset: 0x0 Exists: (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
25:16	MAX_NO_SNOOP_LAT	R/W	Max No-Snoop Latency Value. Along with the Max No-Snoop LatencyScale field, this register specifies the maximum no-snoop latency that a device is permitted to request. Software should set this to the platform's maximum supported latency or less. It is strongly recommended that any updates to this field are reflected in LTR Message(s) sent by the device within 1 ms. Value After Reset: 0x0 Exists: (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
15:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
12:10	MAX_SNOOP_LAT_SCALE	R/W	Max Snoop Latency Scale. This register provides a scale for the value contained within the Max Snoop LatencyValue field. Encoding is the same as the LatencyScale fields in the LTR Message. It is strongly recommended that any updates to this field are reflected in LTR Message(s) sent by the device within 1 ms. Hardware operation is undefined if software writes a Not Permitted value to this field. Value After Reset: 0x0 Exists: (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
9:0	MAX_SNOOP_LAT	R/W	Max Snoop Latency Value. Along with the Max Snoop LatencyScale field, this register specifies the maximum snoop latency that a device is permitted to request. Software should set this to the platform's maximum supported latency or less. It is strongly recommended that any updates to this field are reflected in LTR Message(s) sent by the device within 1 ms. Value After Reset: 0x0 Exists: (CX_LTR_M_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.27 PF0_L1SUB_CAP Registers

3.27.1 L1SUB_CAP_HEADER_REG

- **Name:** L1 Substates Extended Capability Header.
- **Description:** L1 Substates Extended Capability Header provides capability ID, capability version and next offset value for L1 Substates.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	EXTENDED_CAP_ID

Table 3-162 Fields for Register: L1SUB_CAP_HEADER_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h, if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. The bottom 2 bits of this offset are Reserved and must be implemented as 00b although software must mask them to allow for future uses of these bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field is depends on version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
15:0	EXTENDED_CAP_ID	* Varies	L1SUB Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. Extended Capability ID for L1 PM Substates is 001Eh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1e Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

3.27.2 L1SUB_CAPABILITY_REG

- **Name:** L1 Substates Capability Register.
- **Description:** This register provides extended capability of L1 Substates.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:24	RSVDP_24
23:19	PWR_ON_VALUE_SUPPORT
18	RSVDP_18
17:16	PWR_ON_SCALE_SUPPORT
15:8	COMM_MODE_SUPPORT
7:5	RSVDP_5
4	L1_PMSUB_SUPPORT
3	L1_1_ASPM_SUPPORT
2	L1_2_ASPM_SUPPORT
1	L1_1_PCIPM_SUPPORT
0	L1_2_PCIPM_SUPPORT

Table 3-163 Fields for Register: L1SUB_CAPABILITY_REG

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
23:19	PWR_ON_VALUE_SUPPORT	* Varies	Port T Power On Value. Along with the Port T_POWER_ON_SCALE field in the L1 PM Substates Capabilities register sets the time (in us) that this Port requires the port on the opposite side of Link to wait in L1.2.Exit after sampling CLKREQ# asserted before actively driving the interface. The value of Port T_POWER_ON is calculated by multiplying the value in this field by the scale value in the Port T_POWER_ON_SCALE field in the L1 PM Substates Capabilities register. Required for all Ports for which either the PCI-PM L1.2 Supported bit is Set, ASPM L1.2 Supported bit is Set, or both are Set, otherwise this field is of type RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: L1SUB_CAPABILITY_REG.L1_2_ASPM_SUPPORT L1SUB_CAPABILITY_REG.L1_2_PCI_PM_SUPPORT ? RWS : RSVDP Value After Reset: DEFAULT_L1SUB_PORT_T_POWER_ON_VALUE Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
17:16	PWR_ON_SCALE_SUPPORT	* Varies	Port T Power On Scale. Specifies the scale used for the Port T_POWER_ON_VALUE field in the L1 PM Substates Capabilities register. Range of values are given below. Required for all Ports for which either the PCI-PM L1.2 Supported bit is Set, ASPM L1.2 Supported bit is Set, or both are Set, otherwise this field is of type RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: L1SUB_CAPABILITY_REG.L1_2_ASPM_SUPPORT L1SUB_CAPABILITY_REG.L1_2_PCI_PM_SUPPORT ? RWS : RSVDP Values: ■ 0x0 (_2us): Scale is 2us ■ 0x1 (_10us): Scale is 10us ■ 0x2 (_100us): Scale is 100us ■ 0x3 (Reserved): Reserved Value After Reset: DEFAULT_L1SUB_PORT_T_POWER_ON_SCALE Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
15:8	COMM_MODE_SUPPORT	* Varies	Port Common Mode Restore Time. Time (in us) required for this Port to re-establish common mode. Required for all Ports for which either the PCI-PM L1.2 Supported bit is Set, ASPM L1.2 Supported bit is Set, or both are Set, otherwise this field is of type RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: L1SUB_CAPABILITY_REG.L1_2_ASPM_SUPPORT L1SUB_CAPABILITY_REG.L1_2_PCI_PM_SUPPORT ? RWS : RSVDP Value After Reset: DEFAULT_L1SUB_PORT_T_COMM_MODE Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
7:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
4	L1_PMSUB_SUPPORT	* Varies	L1 PM Substates ECN Supported. When Set this field indicates that this Port supports L1 PM Substates. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: R/W (sticky) Value After Reset: 0x1 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
3	L1_1_ASPM_SUPPORT	* Varies	ASPM L1.1 Supported. When Set this field indicates that ASPM L1.1 is supported. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: R/W (sticky) Value After Reset: 0x1 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
2	L1_2_ASPM_SUPPORT	* Varies	ASPM L1.2 Supported. When Set this field indicates that ASPM L1.2 is supported. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: R/W (sticky) Value After Reset: CX_LTR_M_ENABLE_VALUE Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
1	L1_1_PCIPM_SUPPORT	* Varies	PCI-PM L11 Supported. When Set this field indicates that PCI-PM L1.1 is supported, and must be Set by all Ports implementing L1 PM Substates. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: R/W (sticky) Value After Reset: 0x1 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
0	L1_2_PCIPM_SUPPORT	* Varies	PCI-PM L12 Supported. When Set this field indicates that PCI-PM L1.2 is supported. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: R/W (sticky) Value After Reset: 0x1 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.27.3 L1SUB_CONTROL1_REG

- **Name:** L1 Substates Control 1 Register.
- **Description:** This register Controls that the individual extended capability is enabled or not.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:29	L1_2_TH_SCA
28:26	RSVDP_26
25:16	L1_2_TH_VAL
15:8	T_COMMON_MODE
7:4	RSVDP_4
3	L1_1_ASPM_EN
2	L1_2_ASPM_EN
1	L1_1_PCIPM_EN
0	L1_2_PCIPM_EN

Table 3-164 Fields for Register: L1SUB_CONTROL1_REG

Bits	Name	Memory Access	Description
31:29	L1_2_TH_SCA	* Varies	LTR L12 Threshold Scale. This field provides a scale for the value contained within the LTR_L1.2_THRESHOLD_Value. Hardware operation is undefined if software writes a Not-Permitted value to this field. Required for all Ports for which the ASPM L12 Supported bit is Set, otherwise this field is of type RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: L1SUB_CAPABILITY_REG.L1_2_ASPM_SUPPORT ? RW : RSVDP ■ Dbi: L1SUB_CAPABILITY_REG.L1_2_ASPM_SUPPORT ? RW : RSVDP Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead
28:26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
25:16	L1_2_TH_VAL	* Varies	LTR L12 Threshold Value. Along with the LTR_L1.2_THRESHOLD_Scale, this field indicates the LTR threshold used to determine if entry into L1 results in L1.1 (if enabled) or L1.2 (if enabled). Required for all Ports for which the ASPM L12 Supported bit is Set, otherwise this field is of type RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: L1SUB_CAPABILITY_REG.L1_2_ASPM_SUPPORT ? RW : RSVDP ■ Dbi: L1SUB_CAPABILITY_REG.L1_2_ASPM_SUPPORT ? RW : RSVDP Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead
15:8	T_COMMON_MODE	R	Common Mode Restore Time. Sets value of TCOMMONMODE (in us), which must be used by the Downstream Ports for timing the re-establishment of common mode. This field is of type RsvdP for Upstream Ports. Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
7:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
3	L1_1_ASPM_EN	R/W	ASPM L11 Enable. When Set this field, enables ASPM L1.1. For Ports for which the ASPM L1.1 Supported bit is clear, this bit is permitted to be hardwired to 0. Required for both Upstream and Downstream Ports. For compatibility with possible future extensions, software must not enable L1 PM Substates unless the L1 PM Substates Supported field in the L1 PM Substates Capabilities Register is Set. Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
2	L1_2_ASPM_EN	R/W	ASPM L12 Enable. When Set this field, enables ASPM L1.2. For Ports for which the ASPM L1.2 Supported bit is clear, this bit is permitted to be hardwired to 0. Required for both Upstream and Downstream Ports. For compatibility with possible future extensions, software must not enable L1 PM Substates unless the L1 PM Substates Supported field in the L1 PM Substates Capabilities Register is Set. Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
1	L1_1_PCIPM_EN	R/W	PCI-PM L11 Enable. When Set this field, enables PCI-PM L1.1. Required for both Upstream and Downstream Ports. For compatibility with possible future extensions, software must not enable L1 PM Substates unless the L1 PM Substates Supported field in the L1 PM Substates Capabilities Register is Set. Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))
0	L1_2_PCIPM_EN	R/W	PCI-PM L12 Enable. When Set this field, enables PCI-PM L1.2. For Ports for which the PCI-PM L1.2 Supported bit is clear, this bit is permitted to be hardwired to 0. Required for both Upstream and Downstream Ports. For compatibility with possible future extensions, software must not enable L1 PM Substates unless the L1 PM Substates Supported field in the L1 PM Substates Capabilities Register is Set. Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

3.27.4 L1SUB_CONTROL2_REG

- **Name:** L1 Substates Control 2 Register.
- **Description:** This register Controls that the individual extended capability is enabled or not.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

RSVDP_8	31:8
T_POWER_ON_VALUE	7:3
RSVDP_2	2
T_POWER_ON_SCALE	1:0

Table 3-165 Fields for Register: L1SUB_CONTROL2_REG

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
7:3	T_POWER_ON_VALUE	* Varies	T Power On Value. Along with the T_POWER_ON_SCALE sets the minimum amount of time (in us) that the Port must wait in L1.2.Exit after sampling CLKREQ# asserted before actively driving the interface. T_POWER_ON is calculated by multiplying the value in this field by the value in the T_POWER_ON_SCALE field. Required for all Ports that support L1.2, otherwise this field is of type RsvdP. This field must only be modified when the ASPM L1.2 Enable and PCI-PM L1.2 Enable bits are both Clear. The Port behavior is undefined if this field is modified when either the ASPM L1.2 Enable and/or PCI-PM L1.2 Enable bit(s) are set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: L1SUB_CAPABILITY_REG.L1_2_ASPM_SUPPORT L1SUB_CAPABILITY_REG.L1_2_PCI_PM_SUPPORT ? RWS : RSVDP ■ Dbi: L1SUB_CAPABILITY_REG.L1_2_ASPM_SUPPORT L1SUB_CAPABILITY_REG.L1_2_PCI_PM_SUPPORT ? RWS : RSVDP Value After Reset: 0x5 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead
2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
1:0	T_POWER_ON_SCALE	* Varies	T Power On Scale. Specifies the scale used for T_POWER_ON_VALUE. Range of values are given below. Required for all Ports that support L1.2, otherwise this field is of type RsvdP. This field must only be modified when the ASPM L1.2 Enable and PCI-PM L1.2 Enable bits are both Clear. The Port behavior is undefined if this field is modified when either the ASPM L1.2 Enable and/or PCI-PM L1.2 Enable bit(s) are set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: L1SUB_CAPABILITY_REG.L1_2_ASPM_SUPPORT L1SUB_CAPABILITY_REG.L1_2_PCI_PM_SUPPORT ? RWS : RSVDP ■ Dbi: L1SUB_CAPABILITY_REG.L1_2_ASPM_SUPPORT L1SUB_CAPABILITY_REG.L1_2_PCI_PM_SUPPORT ? RWS : RSVDP Values: ■ 0x0 (_2us): Scale is 2us ■ 0x1 (_10us): Scale is 10us ■ 0x2 (_100us): Scale is 100us ■ 0x3 (Reserved): Reserved Value After Reset: 0x0 Exists: (CX_L1_SUBSTATES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead

3.28 PF_PASID_CAPI Registers

3.28.1 PASID_EXT_CAP_HDR_REG

- **Name:** PASID Extended Capability Header.
- **Description:** This Register provides Capability ID, Capability Version, and Next offset of capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	CAP_ID

Table 3-166 Fields for Register: PASID_EXT_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. The offset to the next PCI Extended Capability structure or 000h if no other items exist in the linked list of capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	CAP_ID	* Varies	PASID Extended Capability ID. Indicates the PASID Extended Capability structure. This field must return a Capability ID of 001Bh indicating that this is a PASID Extended Capability structure. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1b Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.28.2 PASID_CAP_CNTRL_REG

- **Name:** PASID Capability Control Register.
- **Description:** This register provides value of capability like Execute Permission support, Privileged Mode support, and width of PASID. Also this register controls PASID Enable, Execute Permission Enable, as well as Privileged mode Enable.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:19	RSVDP_19
18	PRIVILEGED_MODE_EN
17	EXECUTE_PERMISSION_EN
16	EN
15:13	RSVDP_13
12:8	MAX_PASID_WIDTH
7:3	RSVDP_3
2	PRIVILEGED_MODE_SUPPORTED
1	EXECUTE_PERMISSION_SUPPORTED
0	RSVDP_0

Table 3-167 Fields for Register: PASID_CAP_CNTRL_REG

Bits	Name	Memory Access	Description
31:19	RSVDP_19	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
18	PRIVILEGED_MODE_EN	* Varies	Privileged Mode Enable (endpoints only). The behavior is undefined if the Endpoint supports ATS and this bit changes value when the Enable bit in the ATS Control register is Set. If Privileged Mode Supported is Clear, this bit is RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PASID_CAP_CNTRL_REG.PRIVILEGED_MODE_SUPPORTED ? RW : RO ■ Dbi: PASID_CAP_CNTRL_REG.PRIVILEGED_MODE_SUPPORTED ? RW : RO Values: ■ 0x1 (SET): The Endpoint is permitted to send Requests that have the Privileged Mode Requested bit Set. ■ 0x0 (CLEAR): The Endpoint is not permitted to send Requests that have the Privileged Mode Requested bit Set. Value After Reset: 0x0 Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
17	EXECUTE_PERMISSION_EN	* Varies	Execute Permission Enable (endpoints only). The behavior is undefined if the Endpoint supports ATS and this bit changes value when the Enable bit in the ATS Control register is Set. If Execute Permission Supported is Clear, this bit is RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PASID_CAP_CNTRL_REG.EXECUTE_PERMISSION_SUPPORTED ? RW : RO ■ Dbi: PASID_CAP_CNTRL_REG.EXECUTE_PERMISSION_SUPPORTED ? RW : RO Values: ■ 0x1 (SET): The Endpoint is permitted to send Requests that have the Execute Requested bit Set. ■ 0x0 (CLEAR): The Endpoint is not permitted to send Requests that have the Execute Requested bit Set. Value After Reset: 0x0 Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
16	EN	R/W	PASID Enable. Behavior is undefined if the Endpoint supports ATS and this bit changes value when the Enable (E) bit in the ATS Control register is Set. Values: 0x1 (SET): The Endpoint is permitted to send and receive TLPs that contain a PASID TLP Prefix. 0x0 (CLEAR): The Endpoint is not permitted to do so. Value After Reset: 0x0 Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
12:8	MAX_PASID_WIDTH	* Varies	Max PASID Width. Indicates the width of the PASID field supported by the Endpoint. The value n indicates support for PASID values 0 through (2ⁿ) - 1 (inclusive). This field must be between '0' and '20' (inclusive). <i>Note:</i> The access attributes of this field are as follows: - Wire: R (sticky) - Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: 0x14 (MAX_VALUE): The value '20' indicates support for all PASID values (20 bits). 0x0 (MIN_VALUE): The value '0' indicates support for a single PASID (0). Value After Reset: DEFAULT_MAX_PASID_WIDTH Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
7:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
2	PRIVILEGED_MODE_SUPPORTED	* Varies	Privileged Mode Supported (endpoints only). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): The Endpoint supports operating in Privileged and Non-Privileged modes, and supports sending requests that have the Privileged Mode Requested bit Set. ■ 0x0 (CLEAR): The Endpoint will never Set the Privileged Mode Requested bit. Value After Reset: DEFAULT_PRVLGD_MODE_SPPRT Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
1	EXECUTE_PERMISSION_SUPPORTED	* Varies	Execute Permission Supported (endpoints only). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): The Endpoint supports sending TLPs that have the Execute Requested bit Set. ■ 0x0 (CLEAR): The Endpoint will never Set the Execute Requested bit. Value After Reset: DEFAULT_EXCT_PRMSN_SPPRT Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PASID_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.29 PF_DPA_CAPi Registers

3.29.1 DPA_EXT_CAP_HDR_REG

- **Name:** DPA Extended Capability Header.
- **Description:** This Register provides Capability ID, Capability Version, and Next offset of capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	EXT_CAP_ID

Table 3-168 Fields for Register: DPA_EXT_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	EXT_CAP_ID	* Varies	DPA Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. PCI Express Extended Capability ID for the DPA Extended Capability is 0016h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: 0x16 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.29.2 DPA_CAP_REG

- **Name:** DPA Capability Register.
- **Description:** This register indicates Maximum Substate Number, Transition Latency Unit, Power Allocation Scale, and Transition Latency.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	XLCY1
23:16	XLCY0
15:14	RSVDP_14
13:12	PAS
11:10	RSVDP_10
9:8	TLUNIT
7:5	RSVDP_5
4:0	SUBSTATE_MAX

Table 3-169 Fields for Register: DPA_CAP_REG

Bits	Name	Memory Access	Description
31:24	XLCY1	* Varies	Transition Latency Value 1 (Xlcy1). This value is multiplied by the Transition Latency Unit to determine the maximum Transition Latency for the substate. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_DPA_XCLY1 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
23:16	XLCY0	* Varies	Transition Latency Value 0 (Xlcy0). This value is multiplied by the Transition Latency Unit to determine the maximum Transition Latency for the substate. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_DPA_XCLY0 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:12	PAS	* Varies	Power Allocation Scale. The encodings provide the scale to determine power allocation per substate in Watts. The value corresponding to the substate in the Substate Power Allocation field is multiplied by this field to determine the power allocation for the substate. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_10.0X): 10.0x ■ 0x1 (_1.0X): 1.0x ■ 0x2 (_0.1X): 0.1x ■ 0x3 (_0.01X): 0.01x Value After Reset: DEFAULT_DPA_PAS Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
11:10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
9:8	TLUNIT	* Varies	Transition Latency Unit (TLUnit). A substate's Transition Latency Value is multiplied by the Transition Latency Unit to determine the maximum Transition Latency for the substate. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_1MS): 1ms ■ 0x1 (_10MS): 10ms ■ 0x2 (_100MS): 100ms ■ 0x3 (RSVD): Reserved Value After Reset: DEFAULT_DPA_TLUNIT Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
7:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:0	SUBSTATE_MAX	* Varies	Maximum Substate Number. Value of this field indicates the maximum substate number, which is the total number of supported substates minus one. A value of 0 0000b indicates support for one substate. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_DPA_SUBSTATE_MAX Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.29.3 DPA_LAT_IND_REG

- **Name:** DPA Latency Indicator Register.
- **Description:** This register provides Transition Latency Indicator Bits.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-170 Fields for Register: DPA_LAT_IND_REG

Bits	Name	Memory Access	Description
x:0	X1_INDICATOR1	* Varies	Transition Latency Indicator Bits. Each bit indicates which Transition Latency Value is associated with the corresponding substate. A value of 0b indicates Transition Latency Value 0; a value of 1b indicates Transition Latency Value 1. Only bits [Substate_Max :0] are defined. Bits above Substate_Max are RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_DPA_TLIB Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Range Variable[x]: (DEFAULT_DPA_SUBSTATE_MAX+1) - 1 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.29.4 DPA_STATUS_CNTRL_REG

- **Name:** DPA Status and Control Register.
- **Description:** This register indicates the current substate of a Function, enables substate control, and controls the substate behaviour.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:21	RSVDP_21
20:16	SUBSTATE_CONTROL
15:9	RSVDP_9
8	SUBSTATE_CONTROL_EN
7:5	RSVDP_5
4:0	SUBSTATE_STATUS

Table 3-171 Fields for Register: DPA_STATUS_CNTRL_REG

Bits	Name	Memory Access	Description
31:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20:16	SUBSTATE_CONTROL	R/W	Substate Control. This field can be used by the software to configure the Function substate. The software can write the substate value in this field to initiate a substate transition. ■ When the Substate Control Enabled bit in the DPA Status Register is Set, this field determines the Function substate. ■ When the Substate Control Enabled bit in the DPA Status Register is Clear, this field has no effect on the Function substate. Value After Reset: 0x0 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
8	SUBSTATE_CONTROL_EN	* Varies	Substate Control Enabled. This field can be used by the software to disable the Substate Control field in the DPA Control Register. Hardware sets this bit following a Conventional Reset or FLR. Software can clear this bit by writing a 1b to it. Software cannot set this bit directly. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RW1C ■ Dbi: R/W Values: ■ 0x1 (SET): The Substate Control field determines the current substate. ■ 0x0 (CLEAR): The Substate Control field has no effect on the current substate. Value After Reset: 0x1 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-write\""
7:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:0	SUBSTATE_STATUS	* Varies	Substate Status. This field indicates current substate for this Function. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.29.5 DPA_PWR_ALLOC_ARRAY0

- **Name:** DSP Substate Power Allocation Register 3..0.
- **Description:** Each Substate Power Allocation register indicates the power allocation value for its associated substate. The number of Substate Power Allocation registers implemented must be equal to the number of substates supported by Function, which is Substate_Max plus one.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	PWR_ALLOC_VAL3
23:16	PWR_ALLOC_VAL2
15:8	PWR_ALLOC_VAL1
7:0	PWR_ALLOC_VAL0

Table 3-172 Fields for Register: DPA_PWR_ALLOC_ARRAY0

Bits	Name	Memory Access	Description
31:24	PWR_ALLOC_VAL3	* Varies	Substate Power Allocation Substate 3. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 3. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1d Exists: DEFAULT_DPA_SUBSTATE_MAX > 2 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
23:16	PWR_ALLOC_VAL2	* Varies	Substate Power Allocation Substate 2. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 2. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1e Exists: DEFAULT_DPA_SUBSTATE_MAX > 1 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
15:8	PWR_ALLOC_VAL1	* Varies	Substate Power Allocation Substate 1. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1f Exists: DEFAULT_DPA_SUBSTATE_MAX > 0 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
7:0	PWR_ALLOC_VAL0	* Varies	Substate Power Allocation Substate 0. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x20 Exists: CX_DPA_ENABLE && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.29.6 DPA_PWR_ALLOC_ARRAY4

- **Name:** DSP Substate Power Allocation Register 7..4.
- **Description:** Each Substate Power Allocation register indicates the power allocation value for its associated substate. The number of Substate Power Allocation registers implemented must be equal to the number of substates supported by Function, which is Substate_Max plus one.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	23:16	15:8	7:0
PWR_ALLOC_VAL7	PWR_ALLOC_VAL6	PWR_ALLOC_VAL5	PWR_ALLOC_VAL4

Table 3-173 Fields for Register: DPA_PWR_ALLOC_ARRAY4

Bits	Name	Memory Access	Description
31:24	PWR_ALLOC_VAL7	* Varies	Substate Power Allocation Substate 7. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 7. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x19 Exists: DEFAULT_DPA_SUBSTATE_MAX > 6 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
23:16	PWR_ALLOC_VAL6	* Varies	Substate Power Allocation Substate 6. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 6. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1a Exists: DEFAULT_DPA_SUBSTATE_MAX > 5 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
15:8	PWR_ALLOC_VAL5	* Varies	Substate Power Allocation Substate 5. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 5. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1b Exists: DEFAULT_DPA_SUBSTATE_MAX > 4 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
7:0	PWR_ALLOC_VAL4	* Varies	Substate Power Allocation Substate 4. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 4. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1c Exists: DEFAULT_DPA_SUBSTATE_MAX > 3 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.29.7 DPA_PWR_ALLOC_ARRAY8

- **Name:** DSP Substate Power Allocation Register 11..8.
- **Description:** Each Substate Power Allocation register indicates the power allocation value for its associated substate. The number of Substate Power Allocation registers implemented must be equal to the number of substates supported by Function, which is Substate_Max plus one.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	23:16	15:8	7:0
PWR_ALLOC_VAL11	PWR_ALLOC_VAL10	PWR_ALLOC_VAL9	PWR_ALLOC_VAL8

Table 3-174 Fields for Register: DPA_PWR_ALLOC_ARRAY8

Bits	Name	Memory Access	Description
31:24	PWR_ALLOC_VAL11	* Varies	Substate Power Allocation Substate 11. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 11. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x15 Exists: DEFAULT_DPA_SUBSTATE_MAX > 10 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
23:16	PWR_ALLOC_VAL10	* Varies	Substate Power Allocation Substate 10. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 10. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x16 Exists: DEFAULT_DPA_SUBSTATE_MAX > 9 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
15:8	PWR_ALLOC_VAL9	* Varies	Substate Power Allocation Substate 9. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 9. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x17 Exists: DEFAULT_DPA_SUBSTATE_MAX > 8 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
7:0	PWR_ALLOC_VAL8	* Varies	Substate Power Allocation Substate 8. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 8. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x18 Exists: DEFAULT_DPA_SUBSTATE_MAX > 7 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.29.8 DPA_PWR_ALLOC_ARRAY12

- **Name:** DSP Substate Power Allocation Register 15..12.
- **Description:** Each Substate Power Allocation register indicates the power allocation value for its associated substate. The number of Substate Power Allocation registers implemented must be equal to the number of substates supported by Function, which is Substate_Max plus one.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	23:16	15:8	7:0
PWR_ALLOC_VAL15	PWR_ALLOC_VAL14	PWR_ALLOC_VAL13	PWR_ALLOC_VAL12

Table 3-175 Fields for Register: DPA_PWR_ALLOC_ARRAY12

Bits	Name	Memory Access	Description
31:24	PWR_ALLOC_VAL15	* Varies	Substate Power Allocation Substate 15. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 15. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x11 Exists: DEFAULT_DPA_SUBSTATE_MAX > 14 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
23:16	PWR_ALLOC_VAL14	* Varies	Substate Power Allocation Substate 14. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 14. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x12 Exists: DEFAULT_DPA_SUBSTATE_MAX > 13 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
15:8	PWR_ALLOC_VAL13	* Varies	Substate Power Allocation Substate 13. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 13. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x13 Exists: DEFAULT_DPA_SUBSTATE_MAX > 12 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
7:0	PWR_ALLOC_VAL12	* Varies	Substate Power Allocation Substate 12. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 12. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x14 Exists: DEFAULT_DPA_SUBSTATE_MAX > 11 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.29.9 DPA_PWR_ALLOC_ARRAY16

- **Name:** DSP Substate Power Allocation Register 19..16.
- **Description:** Each Substate Power Allocation register indicates the power allocation value for its associated substate. The number of Substate Power Allocation registers implemented must be equal to the number of substates supported by Function, which is Substate_Max plus one.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	23:16	15:8	7:0
PWR_ALLOC_VAL19	PWR_ALLOC_VAL18	PWR_ALLOC_VAL17	PWR_ALLOC_VAL16

Table 3-176 Fields for Register: DPA_PWR_ALLOC_ARRAY16

Bits	Name	Memory Access	Description
31:24	PWR_ALLOC_VAL19	* Varies	Substate Power Allocation Substate 19. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 19. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xd Exists: DEFAULT_DPA_SUBSTATE_MAX > 18 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
23:16	PWR_ALLOC_VAL18	* Varies	Substate Power Allocation Substate 18. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 18. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xe Exists: DEFAULT_DPA_SUBSTATE_MAX > 17 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
15:8	PWR_ALLOC_VAL17	* Varies	Substate Power Allocation Substate 17. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 17. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xf Exists: DEFAULT_DPA_SUBSTATE_MAX > 16 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
7:0	PWR_ALLOC_VAL16	* Varies	Substate Power Allocation Substate 16. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 16. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x10 Exists: DEFAULT_DPA_SUBSTATE_MAX > 15 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.29.10 DPA_PWR_ALLOC_ARRAY20

- **Name:** DSP Substate Power Allocation Register 23..20.
- **Description:** Each Substate Power Allocation register indicates the power allocation value for its associated substate. The number of Substate Power Allocation registers implemented must be equal to the number of substates supported by Function, which is Substate_Max plus one.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	23:16	15:8	7:0
PWR_ALLOC_VAL23	PWR_ALLOC_VAL22	PWR_ALLOC_VAL21	PWR_ALLOC_VAL20

Table 3-177 Fields for Register: DPA_PWR_ALLOC_ARRAY20

Bits	Name	Memory Access	Description
31:24	PWR_ALLOC_VAL23	* Varies	Substate Power Allocation Substate 23. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 23. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x9 Exists: DEFAULT_DPA_SUBSTATE_MAX > 22 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
23:16	PWR_ALLOC_VAL22	* Varies	Substate Power Allocation Substate 22. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 22. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xa Exists: DEFAULT_DPA_SUBSTATE_MAX > 21 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
15:8	PWR_ALLOC_VAL21	* Varies	Substate Power Allocation Substate 21. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 21. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xb Exists: DEFAULT_DPA_SUBSTATE_MAX > 20 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
7:0	PWR_ALLOC_VAL20	* Varies	Substate Power Allocation Substate 20. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 20. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xc Exists: DEFAULT_DPA_SUBSTATE_MAX > 19 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.29.11 DPA_PWR_ALLOC_ARRAY24

- **Name:** DSP Substate Power Allocation Register 27..24.
- **Description:** Each Substate Power Allocation register indicates the power allocation value for its associated substate. The number of Substate Power Allocation registers implemented must be equal to the number of substates supported by Function, which is Substate_Max plus one.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	23:16	15:8	7:0
PWR_ALLOC_VAL27	PWR_ALLOC_VAL26	PWR_ALLOC_VAL25	PWR_ALLOC_VAL24

Table 3-178 Fields for Register: DPA_PWR_ALLOC_ARRAY24

Bits	Name	Memory Access	Description
31:24	PWR_ALLOC_VAL27	* Varies	Substate Power Allocation Substate 27. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 27. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x5 Exists: DEFAULT_DPA_SUBSTATE_MAX > 26 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
23:16	PWR_ALLOC_VAL26	* Varies	Substate Power Allocation Substate 26. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 26. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x6 Exists: DEFAULT_DPA_SUBSTATE_MAX > 25 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
15:8	PWR_ALLOC_VAL25	* Varies	Substate Power Allocation Substate 25. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 25. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x7 Exists: DEFAULT_DPA_SUBSTATE_MAX > 24 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
7:0	PWR_ALLOC_VAL24	* Varies	Substate Power Allocation Substate 24. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 24. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x8 Exists: DEFAULT_DPA_SUBSTATE_MAX > 23 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.29.12 DPA_PWR_ALLOC_ARRAY28

- **Name:** DSP Substate Power Allocation Register 31..28.
- **Description:** Each Substate Power Allocation register indicates the power allocation value for its associated substate. The number of Substate Power Allocation registers implemented must be equal to the number of substates supported by Function, which is Substate_Max plus one.
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	23:16	15:8	7:0
PWR_ALLOC_VAL31	PWR_ALLOC_VAL30	PWR_ALLOC_VAL29	PWR_ALLOC_VAL28

Table 3-179 Fields for Register: DPA_PWR_ALLOC_ARRAY28

Bits	Name	Memory Access	Description
31:24	PWR_ALLOC_VAL31	* Varies	Substate Power Allocation Substate 31. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 31. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: DEFAULT_DPA_SUBSTATE_MAX > 30 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
23:16	PWR_ALLOC_VAL30	* Varies	Substate Power Allocation Substate 30. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 30. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x2 Exists: DEFAULT_DPA_SUBSTATE_MAX > 29 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
15:8	PWR_ALLOC_VAL29	* Varies	Substate Power Allocation Substate 29. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 29. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x3 Exists: DEFAULT_DPA_SUBSTATE_MAX > 28 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
7:0	PWR_ALLOC_VAL28	* Varies	Substate Power Allocation Substate 28. The value in this field is multiplied by the Power Allocation Scale to determine power allocation in Watts for the substate 28. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x4 Exists: DEFAULT_DPA_SUBSTATE_MAX > 27 && (CX_DPA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.30 PF_RTR_CAPi Registers

3.30.1 RTR_EXT_CAP_HDR_OFF

- **Name:** Readiness Time Reporting PCI Express Extended Capability Header.
- **Description:** This Register contains detail about Capability ID, Capability Version, and Next Offset of Readiness Time Reporting Extended Capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	RTR_NEXT_OFFSET
19:16	RTR_CAP_VERSION
15:0	RTR_EXT_CAP_ID

Table 3-180 Fields for Register: RTR_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	RTR_NEXT_OFFSET	* Varies	Next Capability Offset. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. Extended Capability ID for the Readiness Time Reporting Extended Capability is 0022h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: Pointer value calculated at configuration time. Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	RTR_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field is version specific. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	RTR_EXT_CAP_ID	* Varies	PCI Express Extended Capability ID. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x22 Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.30.2 READI_TIME_REPORTING1_OFF

- **Name:** Readiness Time Reporting 1.
- **Description:** This Register provides detail about Reset Time, DL_UP_Time, and RTR_Time valid.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	30:24	23:12	11:0
RTR_TIME_VALID	RSVDP_24	DL_UP_TIME	RESET_TIME

Table 3-181 Fields for Register: READI_TIME_REPORTING1_OFF

Bits	Name	Memory Access	Description
31	RTR_TIME_VALID	* Varies	Valid. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Values: ■ 0x0 (CLEAR): If this bit remains Clear and 1 minute has elapsed after all associated device driver(s) have started, software is permitted to assume that this bit will never be set. ■ 0x1 (SET): If Set, indicates that all time values in this capability are valid. If Clear, indicates that the time values in this capability are not yet available. Value After Reset: CX_RN_RTR_VALID_0 Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
30:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:12	DL_UP_TIME	* Varies	DL_UP Time. DL_UP Time is the time the Function requires to become Configuration-Ready after the Downstream Port above the Function reports Data Link Layer Link Active. This field is RsvdP in Functions that are not associated with an Upstream Port. This field is undefined when the Valid bit is Clear. This field must be less than or equal to the encoded value A1Eh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: CX_RN_RTR_DL_UP_VALUE_0 Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
11:0	RESET_TIME	* Varies	Reset Time. Reset Time is the time the Function requires to become Configuration-Ready after the completion of Conventional Reset. This field is RsvdP if the Immediate Readiness bit is Set. This field is undefined when the Valid bit is Clear. This field must be less than or equal to the encoded value A1Eh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: CX_RN_RTR_RESET_VALUE_0 Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.30.3 READI_TIME_REPORTING2_OFF

- **Name:** Readiness Time Reporting 2.
- **Description:** This register gives detail of D3hot to D0 time and FLR time.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	23:12	11:0
RSVDP_24	D3HOT_D0_TIME	FLR_TIME

Table 3-182 Fields for Register: READI_TIME_REPORTING2_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
23:12	D3HOT_D0_TIME	* Varies	D3hot to D0 Time. If Immediate_Readiness_on_Return_to_D0 is Clear, D3hot to D0 Time is the time that the Function requires after it is directed from D3hot to D0 before it is Configuration-Ready and has returned to either D0 uninitialized or D0 active state. This field is RsvdP if the Immediate_Readiness_on_Return_to_D0 bit is Set. This field is undefined when the Valid bit is Clear. This field must be less than or equal to the encoded value 80Ah. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: CX_RN_RTR_D3D0_VALUE_0 Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
11:0	FLR_TIME	* Varies	FLR Time. FLR Time is the time that the Function requires to become Configuration- Ready after it was issued an FLR. This field is RsvdP when the Function Level Reset Capability bit is Clear. This field is undefined when the Valid bit is Clear. This field must be less than or equal to the encoded value A1Eh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: CX_RN_RTR_FLR_VALUE_0 Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.31 PF_RTR_CAP_DBI2i Registers

3.31.1 SHADOW_READI_TIME_REPORTING1_OFF

- **Name:** Shadow Register Readiness Time Reporting 1.
- **Description:** This Register is shadow register of Readiness Time Reporting 1. This register only provides valid signal of RTR_Time and Reset Time.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RTR_TIME_VALID	31
RTR_TIME_REPORTING1_RSVDP	30:12
RESET_TIME	11:0

Table 3-183 Fields for Register: SHADOW_READI_TIME_REPORTING1_OFF

Bits	Name	Memory Access	Description
31	RTR_TIME_VALID	* Varies	Shadow Valid in the shadow register. If this bit remains Clear and 1 minute has elapsed after all associated device driver(s) have started, software is permitted to assume that this bit will never be set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Values: ■ 0x0 (CLEAR): If Clear, indicates that the time values in this capability are not yet available. ■ 0x1 (SET): If Set, indicates that all time values in this capability are valid. Value After Reset: "(CX_EXTENSIBLE_VFUNC) ? EXT_VF_RN_RTR_VALID_0 : VF_RN_RTR_VALID_0 " Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true
30:12	RTR_TIME_REPORTING1_RSVD P	R	Reserved for SHADOW_READI_TIME_REPORTING1_OFF register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable

Bits	Name	Memory Access	Description
11:0	RESET_TIME	* Varies	Shadow Reset Time in the shadow register. Reset Time is the time the Function requires to become Configuration-Ready after the completion of Conventional Reset. This field is RsvdP if the Immediate Readiness bit is Set. This field is undefined when the Valid bit is Clear. This field must be less than or equal to the encoded value A1Eh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: "(CX_EXTENSIBLE_VFUNC) ? EXT_VF_RN_RTR_VALID_0 : VF_RN_RTR_VALID_0 " Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true

3.31.2 SHADOW_READI_TIME_REPORTING2_OFF

- **Name:** Shadow Register Readiness Time Reporting 2.
- **Description:** This Register is shadow register of Readiness Time Reporting 2. This register provides D3hot to D0 time and FLR Time.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	RTR_TIME_REPORTING2_RSVD P
23:12	D3HOT_D0_TIME
11:0	FLR_TIME

Table 3-184 Fields for Register: SHADOW_READI_TIME_REPORTING2_OFF

Bits	Name	Memory Access	Description
31:24	RTR_TIME_REPORTING2_RSVD P	R	Reserved for SHADOW_READI_TIME_REPORTING2_OFF register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable

Bits	Name	Memory Access	Description
23:12	D3HOT_D0_TIME	* Varies	Shadow D3hot to D0 Time in the shadow register. If Immediate_Readiness_on_Return_to_D0 is Clear, D3hot to D0 Time is the time that the Function requires after it is directed from D3hot to D0 before it is Configuration-Ready and has returned to either D0 uninitialized or D0 active state. This field is RsvdP if the Immediate_Readiness_on_Return_to_D0 bit is Set. This field is undefined when the Valid bit is Clear. This field must be less than or equal to the encoded value 80Ah. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: "(CX_EXTENSIBLE_VFUNC) ? EXT_VF_RN_RTR_VALID_0 : VF_RN_RTR_VALID_0 " Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true
11:0	FLR_TIME	* Varies	Shadow FLR Time in the shadow register. FLR Time is the time that the Function requires to become Configuration-Ready after it was issued an FLR. This field is RsvdP when the Function Level Reset Capability bit is Clear. This field is undefined when the Valid bit is Clear. This field must be less than or equal to the encoded value A1Eh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: No access ■ Dbi: No access ■ Dbi2: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: "(CX_EXTENSIBLE_VFUNC) ? EXT_VF_RN_RTR_VALID_0 : VF_RN_RTR_VALID_0 " Exists: (CX_RN_RTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true

3.32 PF_LNR_CAPi Registers

3.32.1 LNR_EXT_CAP_HDR_OFF

- **Name:** LNR Extended Capability Header.
- **Description:** This register provides Capability ID, Capability version, and Next offset of Extended Capability Structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	19:16	15:0
LNR_NEXT_OFFSET	LNR_CAP_VERSION	LNR_EXT_CAP_ID

Table 3-185 Fields for Register: LNR_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	LNR_NEXT_OFFSET	* Varies	Next Capability Offset. This field is a PCI-SIG defined ID number that indicates the nature and format of the extended capability. PCI Express Extended Capability ID for the LNR Extended Capability is 001Ch. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	LNR_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	LNR_EXT_CAP_ID	* Varies	PCI Express Extended Capability ID. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1c Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.32.2 LNR_CAP_CONTROL_OFF

- **Name:** LNR Control Register and Capability Register.
- **Description:** This register provides extended capability for LNR. It also controls the LNR behavior.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_29	31:29
LNR_REGISTRATION_LIMIT	28:24
RSVDP_18	23:18
LNR_CLS	17
LNR_ENABLE	16
RSVDP_13	15:13
LNR_REGISTRATION_MAX	12:8
RSVDP_2	7:2
LNR_128_SUPPORTED	1
LNR_64_SUPPORTED	0

Table 3-186 Fields for Register: LNR_CAP_CONTROL_OFF

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
28:24	LNR_REGISTRATION_LIMIT	R/W	LNR Registration Limit. This field, encoded as a power of 2, imposes a limit on the number of cachelines that this LN Requester is permitted to register concurrently. For example, a value of 00100b indicates that the LN Requester must not register more than 16 cachelines (2^4) concurrently. Value After Reset: 0x1f Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
23:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
17	LNR_CLS	R/W	LNR CLS. This bit controls or indicates the cacheline size used with LN protocol by this Requester. If this LN Requester supports only one cacheline size, this bit is permitted to be hardwired to indicate that size. Otherwise, the default value of this bit is 0b. Values: 0x1 (SET): The Cacheline size is 128 bytes. 0x0 (CLEAR): The cacheline size is 64 bytes. Value After Reset: 0x0 Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
16	LNR_ENABLE	R/W	LNR Enable. When this bit is Set, the Endpoint is enabled to operate as an LN Requester. Software is permitted to Clear this bit at any time. Value After Reset: 0x0 Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
12:8	LNR_REGISTRATION_MAX	* Varies	LNR Registration Max. This field, encoded as a power of 2, indicates the maximum number of cachelines that this LN Requester is capable of registering concurrently. For example, a value of 00101b indicates that the LN Requester might be capable of registering up to 32 cachelines (2^5) concurrently, and is capable of registering greater than 16. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_LN_REG_MAX_VALUE_0 Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
7:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
1	LNR_128_SUPPORTED	* Varies	LNR-128 Supported. This bit must be 1b if the Endpoint supports LN protocol for 128-byte cachelines as a Requester; otherwise, must be 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_LN_128_SUPPORTED_VALUE_0 Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
0	LN_64_SUPPORTED	* Varies	LN-64 Supported. This bit must be 1b if the Endpoint supports LN protocol for 64-byte cachelines as a Requester; otherwise, must be 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_LN_64_SUPPORTED_VALUE_0 Exists: (CX_LN_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.33 PF_RAS_DES_CAPI Registers

3.33.1 RAS_DES_CAP_HEADER_REG

- **Name:** Vendor-Specific Extended Capability Header.
- **Description:** The Vendor-Specific Extended Capability (VSEC Capability) is an optional Extended Capability that is permitted to be implemented by any PCI Express Function or RCRB. This Register contains Capability Id, Capability Version and Next Offset value for Vendor-Specific Extended Capability.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

NEXT_OFFSET	31:20
CAP_VERSION	19:16
EXTENDED_CAP_ID	15:0

Table 3-187 Fields for Register: RAS_DES_CAP_HEADER_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xff (MAX_VAL): Max value Value After Reset: Pointer value calculated at configuration time. Exists: (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. Value of this field is depends on the version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
15:0	EXTENDED_CAP_ID	* Varies	PCI Express Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. Extended Capability ID for the Vendor-Specific Extended Capability is 000Bh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0xb Exists: (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.33.2 VENDOR_SPECIFIC_HEADER_REG

- **Name:** Vendor-Specific Header.
- **Description:** This Register field provides VSEC Length, VSEC ID and VSEC Rev (Version Number). Vendor-specific software must qualify the associated Vendor ID of the PCI Express Function or RCRB before attempting to interpret the values in the VSEC ID or VSEC Rev fields.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	VSEC_LENGTH
19:16	VSEC_REV
15:0	VSEC_ID

Table 3-188 Fields for Register: VENDOR_SPECIFIC_HEADER_REG

Bits	Name	Memory Access	Description
31:20	VSEC_LENGTH	R	VSEC Length. This field indicates the number of bytes in the entire VSEC structure, including the Vendor-Specific Extended Capability Header, the Vendor-Specific Header, and the vendor-specific registers. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: 0x100 Exists: (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
19:16	VSEC_REV	R	VSEC Rev. This field is a vendor-defined version number that indicates the version of the VSEC structure. Software must qualify the Vendor ID and VSEC ID before interpreting this field. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x4 Exists: (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:0	VSEC_ID	R	VSEC ID. This field is a vendor-defined ID number that indicates the nature and format of the VSEC structure. Software must qualify the Vendor ID before interpreting this field. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x2 Exists: (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.33.3 EVENT_COUNTER_CONTROL_REG

- **Name:** Event Counter Control.
- **Description:** This is a viewport control register.
 - Setting the EVENT_COUNTER_EVENT_SELECT and EVENT_COUNTER_LANE_SELECT fields in this register determine the Event Counter data returned by the EVENT_COUNTER_DATA_REG viewport register.
 - Setting the EVENT_COUNTER_ENABLE field in this register enables the Event Counter selected by the EVENT_COUNTER_EVENT_SELECT and EVENT_COUNTER_LANE_SELECT fields in this register.
 - Setting the EVENT_COUNTER_CLEAR field in this register clears the Event Counter selected by the EVENT_COUNTER_EVENT_SELECT and EVENT_COUNTER_LANE_SELECT fields in this register.
 - Reading the EVENT_COUNTER_STATUS field in this register returns the Enable status of the Event Counter selected by the EVENT_COUNTER_EVENT_SELECT and EVENT_COUNTER_LANE_SELECT fields in this register.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_28	31:28
EVENT_COUNTER_EVENT_SELECT	27:16
RSVDP_12	15:12
EVENT_COUNTER_LANE_SELECT	11:8
EVENT_COUNTER_STATUS	7
RSVDP_5	6:5
EVENT_COUNTER_ENABLE	4:2
EVENT_COUNTER_CLEAR	1:0

Table 3-189 Fields for Register: EVENT_COUNTER_CONTROL_REG

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EC_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27:16	EVENT_COUNTER_EVENT_SELECT	R/W	Event Counter Data Select. This field in conjunction with the EVENT_COUNTER_LANE_SELECT field indexes the Event Counter data returned by the EVENT_COUNTER_DATA_REG register. ■ 27-24: Group number(4-bit: 0..0x7) ■ 23-16: Event number(8-bit: 0..0x13) within the Group ■ .. For detailed definitions of Group number and Event number, see the RAS DES chapter in the <i>Databook</i> . <i>Note:</i> This register field is sticky. Values: ■ 0x0 (EBUF_OVERFLOW): Ebuf Overflow ■ 0x1 (EBUF_UNDERRUN): Ebuf Underrun ■ 0x700 (TX_MEM_WRITE): Tx Memory Write ■ 0x713 (RX_MSG_TLP): Rx Message TLP Value After Reset: 0x0 Exists: CX_RAS_DES_EC_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EC_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
11:8	EVENT_COUNTER_LANE_SELECT	R/W	Event Counter Lane Select. This field in conjunction with EVENT_COUNTER_EVENT_SELECT indexes the Event Counter data returned by the EVENT_COUNTER_DATA_REG register. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Lane0 ■ 0xf (MAX_VAL): Lane15 Value After Reset: 0x0 Exists: CX_RAS_DES_EC_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7	EVENT_COUNTER_STATUS	R	Event Counter Status. This register returns the current value of the Event Counter selected by the following fields: ■ EVENT_COUNTER_EVENT_SELECT ■ EVENT_COUNTER_LANE_SELECT <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Set Value After Reset: 0x0 Exists: CX_RAS_DES_EC_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
6:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EC_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:2	EVENT_COUNTER_ENABLE	W	Event Counter Enable. Enables/disables the Event Counter selected by the EVENT_COUNTER_EVENT_SELECT and EVENT_COUNTER_LANE_SELECT fields in this register. By default, all event counters are disabled. You can enable/disable a specific Event Counter by writing the 'per event off' or 'per event on' codes. You can enable/disable all event counters by writing the 'all on' or 'all off' codes. The read value is always '0'. For other values no change. Values: ■ 0x1 (PER_EVENT_OFF): per event off ■ 0x3 (PER_EVENT_ON): per event on ■ 0x5 (ALL_OFF): all off ■ 0x7 (ALL_ON): all on Value After Reset: 0x0 Exists: CX_RAS_DES_EC_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
1:0	EVENT_COUNTER_CLEAR	W	Event Counter Clear. Clears the Event Counter selected by the EVENT_COUNTER_EVENT_SELECT and EVENT_COUNTER_LANE_SELECT fields in this register. You can clear the value of a specific Event Counter by writing the 'per clear' code and you can clear all event counters at once by writing the 'all clear' code. The read value is always '0'. Other values are reserved. Values: ■ 0x0 (NO_CHANGE): no change ■ 0x1 (PER_CLEAR): per clear ■ 0x2 (NO_CHANGE_2): no change ■ 0x3 (ALL_CLEAR): all clear Value After Reset: 0x0 Exists: CX_RAS_DES_EC_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.4 EVENT_COUNTER_DATA_REG

- **Name:** Event Counter Data.
- **Description:** This viewport register returns the data selected by the following fields:
 - EVENT_COUNTER_EVENT_SELECT in EVENT_COUNTER_CONTROL_REG
 - EVENT_COUNTER_LANE_SELECT in EVENT_COUNTER_CONTROL_REGFor more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-190 Fields for Register: **EVENT_COUNTER_DATA_REG**

Bits	Name	Memory Access	Description
31:0	EVENT_COUNTER_DATA	R	Event Counter Data. This register returns the data selected by the following fields: ■ EVENT_COUNTER_EVENT_SELECT in EVENT_COUNTER_CONTROL_REG <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: CX_RAS_DES_EC_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.5 TIME_BASED_ANALYSIS_CONTROL_REG

- **Name:** Time-based Analysis Control.
- **Description:** Used for controlling the measurement of RX/TX data throughput and time spent in each low-power LTSSM state. For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

TIME_BASED_REPORT_SELECT	31:24
RSVDP_16	23:16
TIME_BASED_DURATION_SELECT	15:8
RSVDP_1	7:1
TIMER_START	0

Table 3-191 Fields for Register: TIME_BASED_ANALYSIS_CONTROL_REG

Bits	Name	Memory Access	Description
31:24	TIME_BASED_REPORT_SELECT	R/W	Time-based Report Select. Selects what type of data is measured for the selected duration (TIME_BASED_DURATION_SELECT), and returned in TIME_BASED_ANALYSIS_DATA. Each type of data is measured using one of three types of units: ■ Core_clk Cycles for 2.5GT/s, 5.0GT/s, 8.0GT/s, 16.0GT/s, 32GT/s. Total time in ps is [Value of TIME_BASED_ANALYSIS_DATA returned when TIME_BASED_REPORT_SELECT=0x00] * TIME_BASED_ANALYSIS_DATA. Values 0-4 and 7-8 correspond to Core_clk Cycles for 2.5GT/s, 5.0GT/s, 8.0GT/s, 16.0GT/s, 32GT/s. ■ Aux_clk Cycles. Total time in ps is [Period of platform specific clock] * TIME_BASED_ANALYSIS_DATA. Values 5, 6, and 9 correspond to aux_clk Cycles. ■ Core_clk Cycles for 20GT/s, 25GT/s (CCIX ESM data rate). Total time in ps is [Value of TIME_BASED_ANALYSIS_DATA returned when TIME_BASED_REPORT_SELECT=0x10] * TIME_BASED_ANALYSIS_DATA. Values 10-14 and 17-18 correspond to Core_clk Cycles for 2.5GT/s, 5.0GT/s, 8.0GT/s, 16.0GT/s, 32GT/s. ■ Data Bytes. Actual amount of bytes is 16 * TIME_BASED_ANALYSIS_DATA. Values 20-23 correspond to data bytes. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ONE_CYCLE): Duration of 1 cycle ■ 0x1 (TX_L0S): TxL0s ■ 0x2 (RX_L0S): RxL0s ■ 0x3 (L0): L0 ■ 0x4 (L1): L1 ■ 0x7 (CFG_RCVRY): Configuration/Recovery ■ 0x8 (TX_RX_L0S): TxL0s and RxL0s

Bits	Name	Memory Access	Description
31:24...(cont.)	TIME_BASED_REPORT_SELECT.	R/W	■ 0x5 (L1_1): L1.1 ■ 0x6 (L1_2): L1.2 ■ 0x9 (L1_AUX): L1 aux ■ 0x10 (_1_CYCLE): Duration of 1 cycle ■ 0x11 (TX_L0S_): TxL0s ■ 0x12 (RX_L0S_): RxL0s ■ 0x13 (L0_): L0 ■ 0x14 (L1_): L1 ■ 0x17 (CFG_RCVRY_): Configuration/Recovery ■ 0x18 (TX_RX_L0S_): TxL0s and RxL0s ■ 0x20 (TX_PCIE_TLP): Tx PCIe TLP data payload Bytes ■ 0x21 (RX_PCIE_TLP): Rx PCIe TLP data payload Bytes ■ 0x22 (TX_CCIX_TLP): Tx CCIX TLP data payload Bytes ■ 0x23 (RX_CCIX_TLP): Rx CCIX TLP data payload Bytes Value After Reset: 0x0 Exists: CX_RAS_DES_TBA_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
23:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_TBA_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:8	TIME_BASED_DURATION_SELECT	R/W	Time-based Duration Select. Selects the duration of time-based analysis. When "manual control" is selected and TIMER_START is set to '1', this analysis never stops until TIMER_STOP is set to '0'. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MANUAL): Manual control ■ 0x1 (_1_MS): 1ms ■ 0x2 (_2_MS): 10ms ■ 0x3 (_100_MS): 100ms ■ 0x4 (_1_S): 1s ■ 0x5 (_2_S): 2s ■ 0x6 (_4_S): 4s ■ 0xff (_4_US): 4us (Debug purpose) Value After Reset: 0x1 Exists: CX_RAS_DES_TBA_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_TBA_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	TIMER_START	R/W	Timer Start. This bit will be cleared automatically when the measurement is finished.<i>Note:</i> The app_ras_des_tba_ctrl input also sets the contents of this field and controls the measurement start/stop. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (START_RESTART): Start/Restart ■ 0x0 (STOP): Stop Value After Reset: 0x0 Exists: CX_RAS_DES_TBA_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.6 TIME_BASED_ANALYSIS_DATA_REG

- **Name:** Time-based Analysis Data.
- **Description:** Contains the measurement results of RX/TX data throughput and time spent in each low-power LTSSM state. This viewport register returns the data selected by the TIME_BASED_REPORT_SELECT field in TIME_BASED_ANALYSIS_CONTROL_REG. For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-192 Fields for Register: TIME_BASED_ANALYSIS_DATA_REG

Bits	Name	Memory Access	Description
31:0	TIME_BASED_ANALYSIS_DATA	R	Time Based Analysis Data. This register returns the data selected by the TIME_BASED_REPORT_SELECT field in TIME_BASED_ANALYSIS_CONTROL_REG. The results are cleared when next measurement starts. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: CX_RAS_DES_TBA_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true

3.33.7 TIME_BASED_ANALYSIS_DATA_63_32_REG

- **Name:** Upper 32 bits of Time-based Analysis Data.
- **Description:** This viewport register returns the data selected by the TIME_BASED_REPORT_SELECT field in TIME_BASED_ANALYSIS_CONTROL_REG. For more information, see the "Reliability, Availability, and Serviceability (RAS)" section in the "Controller Operations" chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

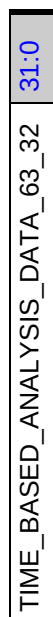


Table 3-193 Fields for Register: TIME_BASED_ANALYSIS_DATA_63_32_REG

Bits	Name	Memory Access	Description
31:0	TIME_BASED_ANALYSIS_DATA_63_32	R	Upper 32 bits of Time Based Analysis Data. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: CX_RAS_DES_TBA_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.8 EINJ_ENABLE_REG

- **Name:** Error Injection Enable.
- **Description:** Each type of error insertion is enabled by the corresponding bit in this register. The specific injection controls for each type of error are defined in the following registers:
 - 0: CRC Error: EINJ0_CRC_REG
 - 1: Sequence Number Error: EINJ1_SEQNUM_REG
 - 2: DLLP Error: EINJ2_DLLP_REG
 - 3: Symbol DataK Mask Error or Sync Header Error: EINJ3_SYMBOL_REG
 - 4: FC Credit Update Error: EINJ4_FC_REG
 - 5: TLP Duplicate/Nullify Error: EINJ5_SP_TLP_REG
 - 6: Specific TLP Error: EINJ6_COMPARE_*_REG/EINJ6_CHANGE_*_REG/EINJ6_TLP_REG

After the errors have been inserted by controller, it will clear each bit here. For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_7	31:7
ERROR_INJECTION6_ENABLE	6
ERROR_INJECTION5_ENABLE	5
ERROR_INJECTION4_ENABLE	4
ERROR_INJECTION3_ENABLE	3
ERROR_INJECTION2_ENABLE	2
ERROR_INJECTION1_ENABLE	1
ERROR_INJECTION0_ENABLE	0

Table 3-194 Fields for Register: EINJ_ENABLE_REG

Bits	Name	Memory Access	Description
31:7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
6	ERROR_INJECTION6_ENABLE	R/W	Error Injection6 Enable (Specific TLP Error). Enables insertion of errors into the packets that you select. You can set this bit to '1' when you have disabled RAS datapath protection (DP) by setting CX_RASDP = CX_RASDP_RAM_PROT =0. You can set this bit to '1' when you have disabled the address translation by setting ADDR_TRANSLATION_SUPPORT_EN=0. For more information, see the EINJ6_COMPARE_*_REG/EINJ6_CHANGE_*_REG/EINJ6_TLP_REG registers. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5	ERROR_INJECTION5_ENABLE	R/W	Error Injection5 Enable (TLP Duplicate/Nullify Error). Enables insertion of duplicate/nullified TLPs. For more information, see the EINJ5_SP_TLP_REG register. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
4	ERROR_INJECTION4_ENABLE	R/W	Error Injection4 Enable (FC Credit Update Error). Enables insertion of errors into UpdateFCs. For more information, see the EINJ4_FC_REG register. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	ERROR_INJECTION3_ENABLE	R/W	Error Injection3 Enable (Symbol DataK Mask Error or Sync Header Error). Enables DataK masking of special symbols or the breaking of the sync header. For more information, see the EINJ3_SYMBOL_REG register. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
2	ERROR_INJECTION2_ENABLE	R/W	Error Injection2 Enable (DLLP Error). Enables insertion of DLLP errors. For more information, see the EINJ2_DLLP_REG register. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
1	ERROR_INJECTION1_ENABLE	R/W	Error Injection1 Enable (Sequence Number Error). Enables insertion of errors into sequence numbers. For more information, see the EINJ1_SEQNUM_REG register. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	ERROR_INJECTION0_ENABLE	R/W	Error Injection0 Enable (CRC Error). Enables insertion of errors into various CRC. For more information, see the EINJ0_CRC_REG register. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.9 EINJ0_CRC_REG

- **Name:** Error Injection Control 0 (CRC Error).
- **Description:** Controls the insertion of errors into the CRC, and parity of ordered sets for the selected type of the packets as follows:
 - **LCRC.** Bad TLP will be detected at the receiver side; receiver responds with NAK DLLP; Data Link Retry starts.
 - **16-bit CRC of ACK/NAK DLLPs.** Bad DLLP occurs at the receiver side; Replay NUM Rollover occurs.
 - **16-bit CRC of UpdateFC DLLPs.** Error insertion continues for the specific time; LTSSM transitions to the Recovery state because of the UpdateFC timeout (if the timeout is implemented at the receiver of the UpdateFCs).
 - **ECRC.** If ECRC check is enabled, ECRC error is detected at the receiver side.
 - **FCRC.** Framing error will be detected, TLP is discarded, and the LTSSM transitions to Recovery state.
 - **Parity of TSOS.** Error insertion continues for the specific time; LTSSM Recovery/Configuration timeout will occur.
 - **Parity of SKPOS.** Lane error will be detected at the receiver side.
- **Size:** 32 bits
- **Offset:** B+0x34
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:12	RSVDP_12
11:8	EINJ0_CRC_TYPE
7:0	EINJ0_COUNT

Table 3-195 Fields for Register: EINJ0_CRC_REG

Bits	Name	Memory Access	Description
31:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
11:8	EINJ0_CRC_TYPE	R/W	Error injection type. Selects the type of CRC error to be inserted. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (TLP_LCRC_ERR): (TX Path) New TLP's LCRC error injection ■ 0x1 (_16b_CRC_ERR_ACK_NAK_DLLP): (TX Path) 16bCRC error injection of ACK/NAK DLLP ■ 0x2 (_16b_CRC_ERR_UPD_FC): (TX Path) 16bCRC error injection of Update-FC DLLP ■ 0x3 (TLP_ECRC_ERR): (TX Path) New TLP's ECRC error injection ■ 0x4 (FCRC_ERR_TLP): (TX Path) TLP's FCRC error injection (128b/130b) ■ 0x5 (PARITY_TSOS_ERR): (TX Path) Parity error of TSOS (128b/130b) ■ 0x6 (PARITY_SKPOS_ERR): (TX Path) Parity error of SKPOS (128b/130b) ■ 0x8 (LCRC_ERR): (RX Path) LCRC error injection ■ 0xb (ECRC_ERR): (RX Path) ECRC error injection Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:0	EINJ0_COUNT	R/W	Error injection count. Indicates the number of errors. This register is decremented when the errors have been inserted. If the counter value is 0x01 and error is inserted, ERROR_INJECTION0_ENABLE in EINJ_ENABLE_REG returns 0b. If the counter value is 0x00 and ERROR_INJECTION0_ENABLE=1, the errors are inserted until ERROR_INJECTION0_ENABLE is set to '0'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.10 EINJ1_SEQNUM_REG

- **Name:** Error Injection Control 1 (Sequence Number Error).
- **Description:** Controls the sequence number of the specific TLPs and ACK/NAK DLLPs. Data Link Protocol Error will be detected at the Rx side of ACK/NAK DLLPs when one of these conditions is true:
 - $((\text{NEXT_TRANSMIT_SEQ} - 1) - \text{AckNak_Seq_Num}) \bmod 4096 > 2048$
 - $(\text{AckNak_Seq_Num} - \text{ACKD_SEQ}) \bmod 4096 \geq 2048$

TLP is treated as Duplicate TLP at the Rx side when all these conditions are true:

 - $\text{Sequence Number} \neq \text{NEXT_RCV_SEQ}$
 - $(\text{NEXT_RCV_SEQ} - \text{Sequence Number}) \bmod 4096 \leq 2048$

TLP is treated as Bad TLP at the Rx side when all these conditions are true:

 - $\text{Sequence Number} \neq \text{NEXT_RCV_SEQ}$ and
 - $(\text{NEXT_RCV_SEQ} - \text{Sequence Number}) \bmod 4096 > 2048$
- **Size:** 32 bits
- **Offset:** B+0x38
- **Exists:** $(\text{CX_RAS_DES_ENABLE} \&\& (\text{CX_MEMORY_MAP_PCIE_TYPE} == 2 \mid \mid \text{CX_MEMORY_MAP_PCIE_TYPE} == 0))$

31:29	28:16	15:9	8	7:0
RSVDP_29	EINJ1_BAD_SEQNUM	RSVDP_9	EINJ1_SEQNUM_TYPE	EINJ1_COUNT

Table 3-196 Fields for Register: EINJ1_SEQNUM_REG

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: $\text{CX_RAS_DES_EINJ_ENABLE} \&\& (\text{CX_RAS_DES_ENABLE} \&\& (\text{CX_MEMORY_MAP_PCIE_TYPE} == 2 \mid \mid \text{CX_MEMORY_MAP_PCIE_TYPE} == 0))$

Bits	Name	Memory Access	Description
28:16	EINJ1_BAD_SEQNUM	R/W	Bad sequence number. Indicates the value to add/subtract from the naturally-assigned sequence numbers. This value is represented by two's complement. For example: ■ Set Type, SEQ# and Count □ EINJ1_SEQNUM_TYPE =0 (Insert errors into new TLPs) □ EINJ1_BAD_SEQNUM =0x1FFD (represents -3) □ EINJ1_COUNT =1 ■ Enable Error Injection □ ERROR_INJECTION1_ENABLE =1 ■ Send a TLP from the controller's Application Interface □ Assume SEQ#5 is given to the TLP. ■ The SEQ# is Changed to #2 by the Error Injection Function in Layer2. □ $5 + (-3) = 2$ ■ The TLP with SEQ#2 is Transmitted to PCIe Link. <i>Note:</i> This register field is sticky. Values: ■ 0xfff (MAX_VAL): +4095 ■ 0x1001 (MIN_VAL): -4095 Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15:9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
8	EINJ1_SEQNUM_TYPE	R/W	Sequence number type. Selects the type of sequence number. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (TLP_ERR): Insertion of New TLP's SEQ# error ■ 0x1 (ACK_NAK_DLLP_ERR): Insertion of ACK/NAK DLLP's SEQ# Error Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:0	EINJ1_COUNT	R/W	Error injection count. Indicates the number of errors. This register is decremented as the errors are being inserted. If the counter value is 0x01 and error is inserted, ERROR_INJECTION1_ENABLE in EINJ_ENABLE_REG returns '0'. If the counter value is 0x00 and ERROR_INJECTION1_ENABLE=1, the errors are inserted until ERROR_INJECTION1_ENABLE is set to '0'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.11 EINJ2_DLLP_REG

- **Name:** Error Injection Control 2 (DLLP Error).
- **Description:** Controls the transmission of DLLPs and inserts the following errors:
 - If "ACK/NAK DLLP's transmission block" is selected, replay timeout error will occur at the transmitter of the TLPs and then Data Link Retry will occur.
 - If "Update FC DLLP's transmission block" is selected, LTSSM will transition to the Recovery state because of the UpdateFC timeout (if the timeout is implemented at the receiver of the UpdateFCs).
 - If "Always Transmission for NAK DLLP" is selected, Data Link Retry will occur at the transmitter of the TLPs. Furthermore, Replay NUM Rollover will occur when the transmitter has been requested four times to send the TLP with the same sequence number.
- **Size:** 32 bits
- **Offset:** B+0x3c
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:10	RSVDP_10
9:8	EINJ2_DLLP_TYPE
7:0	EINJ2_COUNT

Table 3-197 Fields for Register: EINJ2_DLLP_REG

Bits	Name	Memory Access	Description
31:10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
9:8	EINJ2_DLLP_TYPE	R/W	DLLP Type. Selects the type of DLLP errors to be inserted. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ACK_NACK_DLLP): ACK/NAK DLLP's transmission block ■ 0x1 (UPD_FC_DLLP): Update FC DLLP's transmission block ■ 0x2 (NAK_DLLP): Always Transmission for NAK DLLP ■ 0x3 (RSVD): Reserved Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:0	EINJ2_COUNT	R/W	Error injection count. Indicates the number of errors. This register is decremented as the errors are being inserted. If the counter value is 0x01 and the error is inserted, ERROR_INJECTION2_ENABLE in EINJ_ENABLE_REG returns '0'. If the counter value is 0x00 and ERROR_INJECTION2_ENABLE =1, the errors are inserted until ERROR_INJECTION2_ENABLE is set to '0'. This register is affected only when EINJ2_DLLP_TYPE =2'10b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.12 EINJ3_SYMBOL_REG

- **Name:** Error Injection Control 3 (Symbol Error).
- **Description:** When 8b/10b encoding is used, this register controls error insertion into the special (K code) symbols.
 - If TS1/TS2/FTS/E-Idle/SKP is selected, it affects whole of the ordered set. It might cause timeout of the LTSSM.
 - If END/EDB/STP/SDP is selected, TLP/DLLP will be discarded at the receiver side.When 128b/130b encoding is used, this register controls error insertion into the sync-header.
- **Size:** 32 bits
- **Offset:** B+0x40
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:11	10:8	7:0
RSVDP_11	EINJ3_SYMBOL_TYPE	EINJ3_COUNT

Table 3-198 Fields for Register: EINJ3_SYMBOL_REG

Bits	Name	Memory Access	Description
31:11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
10:8	EINJ3_SYMBOL_TYPE	R/W	Error Type. 8b/10b encoding - Mask K symbol. It is not supported to insert errors into the first ordered-set after exiting from TxElectIdle when CX_FREQ_STEP_EN has been enabled. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (RSVD_OR_INVRT_SYNC_HDR): Invert sync header for 128b/130b encoding or this field is reserved for 8b/10b encoding. ■ 0x1 (COM_PAD_TS1): COM/PAD(TS1 Order set) ■ 0x2 (COM_PAD_TS2): COM/PAD(TS2 Order set) ■ 0x3 (COM_FTS): COM/FTS(FTS Order set) ■ 0x4 (COM_IDL): COM/IDL(E-Idle Order set) ■ 0x5 (END_EDB): END/EDB Symbol ■ 0x6 (STP_SDP): STP/SDP Symbol ■ 0x7 (COM_SKP): COM/SKP(SKP Order set) Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:0	EINJ3_COUNT	R/W	Error injection count. Indicates the number of errors. This register is decremented as the errors are being inserted. If the counter value is 0x01 and error is inserted, ERROR_INJECTION3_ENABLE in EINJ_ENABLE_REG returns '0'. If the counter value is 0x00 and ERROR_INJECTION3_ENABLE =1, the errors are inserted until ERROR_INJECTION3_ENABLE is set to '0'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.13 EINJ4_FC_REG

- **Name:** Error Injection Control 4 (FC Credit Error).
- **Description:** Controls error insertion into the credit value in the UpdateFCs. It is possible to insert errors for any of the following types:

- Posted TLP Header credit
- Non-Posted TLP Header credit
- Completion TLP Header credit
- Posted TLP Data credit
- Non-Posted TLP Data credit
- Completion TLP Data credit

These errors are not correctable while error insertion is enabled. Receiver buffer overflow error might occur.

- **Size:** 32 bits
- **Offset:** B+0x44
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:29	RSVDP_29
28:16	EINJ4_BAD_UPDFC_VALUE
15	RSVDP_15
14:12	EINJ4_VC_NUMBER
11	RSVDP_11
10:8	EINJ4_UPDFC_TYPE
7:0	EINJ4_COUNT

Table 3-199 Fields for Register: EINJ4_FC_REG

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
28:16	EINJ4_BAD_UPDFC_VALUE	R/W	Bad update-FC credit value. Indicates the value to add/subtract from the UpdateFC credit. This value is represented by two's complement. <i>Note:</i> This register field is sticky. Values: ■ 0xfff (MAX_VAL): +4095 ■ 0x1001 (MIN_VAL): -4095 Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15	RSVDP_15	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
14:12	EINJ4_VC_NUMBER	R/W	VC Number. Indicates target VC Number. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min Value ■ 0x7 (MAX_VAL): Min Value Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
10:8	EINJ4_UPDFC_TYPE	R/W	Update-FC type. Selects the credit type. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (POSTED_TLP_HDR): Posted TLP Header Credit value control ■ 0x1 (NON_POSTED_TLP_HDR): Non-Posted TLP Header Credit value control ■ 0x2 (CMPL_TLP_HDR): Completion TLP Header Credit value control ■ 0x3 (RSERVED): Reserved ■ 0x4 (POSTED_TLP_DATA): Posted TLP Data Credit value control ■ 0x5 (NON_POSTED_TLP_DATA): Non-Posted TLP Data Credit value control ■ 0x6 (CMPL_TLP_DATA): Completion TLP Data Credit value control ■ 0x7 (RSVD): Reserved Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:0	EINJ4_COUNT	R/W	Error injection count. Indicates the number of errors. This register is decremented as the errors are being inserted. If the counter value is 0x01 and error is inserted, ERROR_INJECTION4_ENABLE in EINJ_ENABLE_REG returns '0'. If the counter value is 0x00 and ERROR_INJECTION4_ENABLE =1, the errors are inserted until ERROR_INJECTION4_ENABLE is set to '0'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.14 EINJ5_SP_TLP_REG

- **Name:** Error Injection Control 5 (Specific TLP Error).
- **Description:** Controls the generation of specified TLPs. Correctable errors will occur which will be fixed by the PCIe protocol.
 - For Duplicate TLP, the controller initiates Data Link Retry by handling ACK DLLP as NAK DLLP. These TLPs will be duplicate TLPs at the receiver side.
 - For Nullified TLP, the TLPs that the controller transmits are changed into nullified TLPs and the original TLPs are stored in the retry buffer. The receiver of these TLPs will detect the lack of seq# and send NAK DLLP at the next TLP. Then the original TLPs are sent from retry buffer and the data controls are recovered. For 128 bit controller or more than 128 bit, the controller inserts errors the number of times of EINJ5_COUNT but does not ensure that the errors are continuously inserted into TLPs.
- **Size:** 32 bits
- **Offset:** B+0x48
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:9	RSVDP_9
8	EINJ5_SPECIFIED_TLP
7:0	EINJ5_COUNT

Table 3-200 Fields for Register: EINJ5_SP_TLP_REG

Bits	Name	Memory Access	Description
31:9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
8	EINJ5_SPECIFIED_TLP	R/W	Specified TLP. Selects the specified TLP to be inserted. <i>Note:</i> This register field is sticky. Values: 0x0 (DUPLICATE_DLLP): Generates duplicate TLPs by handling ACK DLLP as NAK DLLP. 0x1 (NULLIFIED_TLP): Generates Nullified TLP (Original TLP will be stored in retry buffer). Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:0	EINJ5_COUNT	R/W	Error injection count. Indicates the number of errors. This register is decremented as the errors are being inserted. If the counter value is 0x01 and error is inserted, ERROR_INJECTION5_ENABLE in EINJ_ENABLE_REG returns '0'. If the counter value is 0x00 and ERROR_INJECTION5_ENABLE =1, the errors are inserted until ERROR_INJECTION5_ENABLE is set to '0'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.15 EINJ6_COMPARE_POINT_Hi_REG (for $i = 0; i \leq 3$)

- **Name:** Error Injection Control 6 (Compare Point Header DWORD #0).
- **Description:** Program this register for the 1st DWORD of TLP header/prefix. It is necessary to carefully consider the endianness when you program this register.
 Bits [31:0] = TLP_DW0[7:0], TLP_DW0[15:8], TLP_DW0[23:16], TLP_DW0[31:24]
 The Packet Compare Point registers (EINJ6_COMPARE_POINT*) specify which Tx TLP header bits to compare with the corresponding bits in the Packet Compare Value registers (EINJ6_COMPARE_VALUE*). When all specified bits (in the Tx TLP header and EINJ6_COMPARE_VALUE*) match, the controller inserts errors into the TLP. The type and number of errors are specified by the EINJ6_TLP_REG register.
- **Size:** 32 bits
- **Offset:** B+0x4c + i*0x4
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

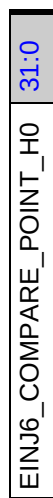


Table 3-201 Fields for Register: EINJ6_COMPARE_POINT_Hi_REG (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
31:0	EINJ6_COMPARE_POINT_H0	R/W	Packet Compare Point: 1st DWORD. Specifies which Tx TLP header DWORD#0 bits to compare with the corresponding bits in the Packet Compare Value registers (EINJ6_COMPARE_VALUE*). When all specified bits (in the Tx TLP header and EINJ6_COMPARE_VALUE*) match, the controller inserts errors into the TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.33.16 EINJ6_COMPARE_VALUE_HI_REG (for i = 0; i <= 3)

- **Name:** Error Injection Control 6 (Compare Value Header DWORD #0).
- **Description:** Program this register for the 1st DWORD of TLP header/prefix. It is necessary to carefully consider the endianness when you program this register.
 Bits [31:0] = TLP_DW0[7:0], TLP_DW0[15:8], TLP_DW0[23:16], TLP_DW0[31:24]
 The Packet Compare Point registers (EINJ6_COMPARE_POINT*) specify which Tx TLP header bits to compare with the corresponding bits in the Packet Compare Value registers (EINJ6_COMPARE_VALUE*). When all specified bits (in the Tx TLP header and EINJ6_COMPARE_VALUE*) match, the controller inserts errors into the TLP. The type and number of errors are specified by the EINJ6_TLP_REG register.
- **Size:** 32 bits
- **Offset:** B+0x5c + i*0x4
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-202 Fields for Register: EINJ6_COMPARE_VALUE_HI_REG (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
31:0	EINJ6_COMPARE_VALUE_HI	R/W	Packet Compare Value: 1st DWORD. Specifies the value to compare against Tx the TLP header DWORD#0 bits specified in the Packet Compare Point registers (EINJ6_COMPARE_POINT*). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.33.17 EINJ6_CHANGE_POINT_Hi_REG (for i = 0; i <= 3)

- **Name:** Error Injection Control 6 (Change Point Header DWORD #0).
- **Description:** Program this register for the 1st DWORD of TLP header/prefix. It is necessary to carefully consider the endianness when you program this register.
Bits [31:0] = TLP_DW0[7:0], TLP_DW0[15:8], TLP_DW0[23:16], TLP_DW0[31:24]
The Packet Change Point registers (EINJ6_CHANGE_POINT*) specify which Tx TLP header bits to replace with the corresponding bits in the Packet Change Value registers (EINJ6_CHANGE_VALUE*). The type and number of errors are specified by the EINJ6_TLP_REG register. Only applies when EINJ6_INVERTED_CONTROL in EINJ6_TLP_REG = 0.
- **Size:** 32 bits
- **Offset:** B+0x6c + i*0x4
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

EINJ6_CHANGE_POINT_H0

31:0

Table 3-203 Fields for Register: EINJ6_CHANGE_POINT_Hi_REG (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
31:0	EINJ6_CHANGE_POINT_H0	R/W	Packet Change Point: 1st DWORD. Specifies which Tx TLP header DWORD#0 bits to replace with the corresponding bits in the Packet Change Value registers (EINJ6_CHANGE_VALUE*). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.33.18 EINJ6_CHANGE_VALUE_Hi_REG (for i = 0; i <= 3)

- **Name:** Error Injection Control 6 (Change Value Header DWORD #0).
- **Description:** Program this register for the 1st DWORD of TLP header/prefix. It is necessary to carefully consider the endianness when you program this register.
 Bits [31:0] = TLP_DW0[7:0], TLP_DW0[15:8], TLP_DW0[23:16], TLP_DW0[31:24]
 The Packet Change Point registers (EINJ6_CHANGE_POINT*) specify which Tx TLP header bits to replace with the corresponding bits in the Packet Change Value registers (EINJ6_CHANGE_VALUE*). The type and number of errors are specified by the EINJ6_TLP_REG register. Only applies when EINJ6_INVERTED_CONTROL in EINJ6_TLP_REG = 0.
- **Size:** 32 bits
- **Offset:** B+0x7c + i*0x4
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

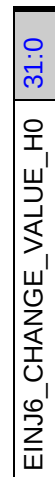


Table 3-204 Fields for Register: EINJ6_CHANGE_VALUE_Hi_REG (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
31:0	EINJ6_CHANGE_VALUE_H0	R/W	Packet Change Value: 1st DWORD. Specifies replacement values for the Tx TLP header DWORD#0 bits defined in the Packet Change Point registers (EINJ6_CHANGE_POINT*). Only applies when the EINJ6_INVERTED_CONTROL field in EINJ6_TLP_REG is '0'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.33.19 EINJ6_TLP_REG

- **Name:** Error Injection Control 6 (Packet Error).
- **Description:**

The Packet Compare Point registers (EINJ6_COMPARE_POINT*) specify which Tx TLP header bits to compare with the corresponding bits in the Packet Compare Value registers (EINJ6_COMPARE_VALUE*). When all specified bits (in the Tx TLP header and EINJ6_COMPARE_VALUE*) match, the controller inserts errors into the TLP. The type and number of errors are specified by the this register.

The Packet Change Point registers (EINJ6_CHANGE_POINT*) specify which Tx TLP header bits to replace with the corresponding bits in the Packet Change Value registers (EINJ6_CHANGE_VALUE*). The type and number of errors are specified by the this register. Only applies when EINJ6_INVERTED_CONTROL in this register =0. The TLP into that errors are injected will not arrive at the transaction layer of the remote device when all of the following conditions are true.

 - Using 128b/130b encoding
 - Injecting errors into TLP Length field / TLP digest bit
- **Size:** 32 bits
- **Offset:** B+0x8c
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_12	31:12
EINJ6_PACKET_TYPE	11:9
EINJ6_INVERTED_CONTROL	8
EINJ6_COUNT	7:0

Table 3-205 Fields for Register: EINJ6_TLP_REG

Bits	Name	Memory Access	Description
31:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:9	EINJ6_PACKET_TYPE	R/W	Packet type. Selects the TLP packets to inject errors into. All encodings other than the specified encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (HEADER): TLP Header ■ 0x1 (TLP_FIRST_4_DW): TLP Prefix 1st 4-DWORDS ■ 0x2 (TLP_SECOND_DW): TLP Prefix 2nd -DWORDS Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
8	EINJ6_INVERTED_CONTROL	R/W	Inverted Error Injection Control. Encoded values given as above. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (REPLACE): EINJ6_CHANGE_VALUE_H[0/1/2/3] is used to replace bits specified by EINJ6_CHANGE_POINT_H[0/1/2/3]. ■ 0x1 (IGNORE): EINJ6_CHANGE_VALUE_H[0/1/2/3] is ignored and inverts bits specified by EINJ6_CHANGE_POINT_H[0/1/2/3]. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
7:0	EINJ6_COUNT	R/W	Error Injection Count. Indicates the number of errors to insert. This counter is decremented while errors are been inserted. If the counter value is 0x01 and error is inserted, ERROR_INJECTION6_ENABLE in EINJ_ENABLE_REG returns '0'. If the counter value is 0x00 and ERROR_INJECTION6_ENABLE=1, errors are inserted until ERROR_INJECTION6_ENABLE is set to '0'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_EINJ_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.20 SD_CONTROL1_REG

- **Name:** Silicon Debug Control 1.
- **Description:** For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xa0
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_24	31:24
LOW_POWER_INTERVAL	23:22
TX_EIOS_NUM	21:20
RSVDP_17	19:17
FORCE_DETECT_LANE_EN	16
FORCE_DETECT_LANE	15:0

Table 3-206 Fields for Register: SD_CONTROL1_REG

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
23:22	LOW_POWER_INTERVAL	R/W	Low Power Entry Interval Time. Interval Time that the controller starts monitoring RXELECIDLE signal after L0s/L1/L2 entry. You should set the value according to the latency from receiving EIOS to, RXELECIDLE assertion at the PHY. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_40NS): 40ns ■ 0x1 (_160NS): 160ns ■ 0x2 (_320NS): 320ns ■ 0x3 (_640NS): 640ns Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
21:20	TX_EIOS_NUM	R/W	Number of Tx EIOS. This register sets the number of transmit EIOS for L0s/L1 entry and Disable/Loopback/Hot-reset exit. The controller selects the greater value between this register and the value defined by the PCI-SIG specification. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (G2_2_EIOS_OTHER_SPEED_1_EIOS): (2.5GT/s, 8.0GT/s or higher) 1 or (5.0GT/s) 2 ■ 0x1 (G2_8_EIOS_OTHER_SPEED_4_EIOS): (2.5GT/s, 8.0GT/s or higher) 4 or (5.0GT/s) 8 ■ 0x2 (G2_16_EIOS_OTHER_SPEED_8_EIOS): (2.5GT/s, 8.0GT/s or higher) 8 or (5.0GT/s) 16 ■ 0x3 (G2_32_EIOS_OTHER_SPEED_16_EIOS): (2.5GT/s, 8.0GT/s or higher) 16 or (5.0GT/s) 32 Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
19:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
16	FORCE_DETECT_LANE_EN	R/W	Force Detect Lane Enable. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When this bit is set, the controller ignores receiver detection from PHY during LTSSM Detect state and uses FORCE_DETECT_LANE. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15:0	FORCE_DETECT_LANE	R/W	Force Detect Lane. This field is a bit vector of lanes to force receiver detection on. When the FORCE_DETECT_LANE_EN field is set, the controller ignores receiver detection from PHY during LTSSM Detect state and uses this value instead. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.33.21 SD_CONTROL2_REG

- **Name:** Silicon Debug Control 2.
- **Description:** For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xa4
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_17	31:17
FRAMING_ERR_RECOVERY_DISABLE	16
RSVDP_11	15:11
DIRECT_LPBKSLV_TO_EXIT	10
DIRECT_POLCOMP_TO_DETECT	9
DIRECT_RECICLE_TO_CONFIG	8
RSVDP_3	7:3
NOACK_FORCE_LINKDOWN	2
RECOVERY_REQUEST	1
HOLD_LTSSM	0

Table 3-207 Fields for Register: SD_CONTROL2_REG

Bits	Name	Memory Access	Description
31:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
16	FRAMING_ERR_RECOVERY_DISABLE	R/W	Framing Error Recovery Disable. This bit disables a transition to Recovery state when a Framing Error is occurred. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15:11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
10	DIRECT_LPBKSLV_TO_EXIT	R/W	Direct Loopback Slave To Exit. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When this bit is set and the LTSSM is in Loopback Slave Active State, the LTSSM transitions to Loopback Slave Exit state. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
9	DIRECT_POLCOMP_TO_DETECT	R/W	Direct Polling.Compliance to Detect. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When this bit is set and the LTSSM is in Polling Compliance State, the LTSSM transitions to Detect state. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
8	DIRECT_RECIDLE_TO_CONFIG	R/W	Direct Recovery.Idle to Configuration. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When this bit is set and the LTSSM is in Recovery Idle State, the LTSSM transitions to Configuration state. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2	NOACK_FORCE_LINKDOWN	R/W	Force LinkDown. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When this bit is set and the controller detects REPLY_NUM rolling over 4 times, the LTSSM transitions to Detect State. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
1	RECOVERY_REQUEST	W	Recovery Request. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When this bit is set to '1' in L0 or L0s, the LTSSM starts transitioning to Recovery State. This request does not cause a speed change or re-equalization. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	HOLD_LTSSM	R/W	Hold and Release LTSSM. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): As long as this field is '1', the controller stays in the current LTSSM. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.33.22 SD_STATUS_L1LANE_REG

- **Name:** Silicon Debug Status(Layer1 Per-lane).
- **Description:** This viewport register returns the data selected by the following field:
 - LANE_SELECT in SD_STATUS_L1LANE_REG
 For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xb0
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	DESKEW_POINTER
23:21	RSVDP_21
20	PIPE_TXELECIDLE
19	PIPE_RXELECIDLE
18	PIPE_RXVALID
17	PIPE_DETECT_LANE
16	PIPE_RXPOLARITY
15:4	RSVDP_4
3:0	LANE_SELECT

Table 3-208 Fields for Register: SD_STATUS_L1LANE_REG

Bits	Name	Memory Access	Description
31:24	DESKEW_POINTER	R	Deskew Pointer. Indicates Deskew pointer of internal Deskew buffer of selected lane number(LANE_SELECT). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20	PIPE_TXELECIDLE	R	PIPE:TxElecIdle. Indicates PIPE TXELECIDLE signal of selected lane number(LANE_SELECT). <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
19	PIPE_RXELECIDLE	R	PIPE:RxElecIdle. Indicates PIPE RXELECIDLE signal of selected lane number(LANE_SELECT). <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
18	PIPE_RXVALID	R	PIPE:RxValid. Indicates PIPE RXVALID signal of selected lane number(LANE_SELECT). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
17	PIPE_DETECT_LANE	R	PIPE:Detect Lane. Indicates whether PHY indicates receiver detection or not on selected lane number(LANE_SELECT). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_CPCIE_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
16	PIPE_RXPOLARITY	R	PIPE:RxPolarity. Indicates PIPE RXPOLARITY signal of selected lane number(LANE_SELECT). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_CPCIE_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
15:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
3:0	LANE_SELECT	R/W	Lane Select. Lane Select register for Silicon Debug Status Register of Layer1-PerLane. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Lane0 ■ 0xf (MAX_VAL): Lane15 Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.33.23 SD_STATUS_L1LTSSM_REG

- **Name:** Silicon Debug Status(Layer1 LTSSM).
- **Description:** For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xb4
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

LTSSM_VARIABLE	31:16
LANE_REVERSAL	15
RSVDP_11	14:11
PIPE_POWER_DOWN	10:8
FRAMING_ERR	7
FRAMING_ERR_PTR	6:0

Table 3-209 Fields for Register: SD_STATUS_L1LTSSM_REG

Bits	Name	Memory Access	Description
31:16	LTSSM_VARIABLE	R	LTSSM Variable. Indicates internal LTSSM variables defined in the PCI Express Base Specification. For other value idle_to_rlock_transitioned. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DIR_SPEED_CHANGE): directed_speed_change ■ 0x1 (CHANGED_SPEED_RCVRY): changed_speed_recovery ■ 0x2 (SUCCESSFUL_SPEED_NEGO): successful_speed_negotiation ■ 0x3 (UPCFG_CAPABLE): upconfigure_capable; Set to '1' if both ports advertised the UpConfigure capability in the last Config.Complete. ■ 0x4 (SEL_DE_EMPHASIS): select_deemphasis ■ 0x5 (START_EQ_W_PRESET): start_equalization_w_preset ■ 0x6 (EQ_DONE_8GT): equalization_done_8GT_data_rate ■ 0x7 (EQ_DONE_16GT): equalization_done_16GT_data_rate Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true
15	LANE_REVERSAL	R	Lane Reversal Operation. Receiver detected lane reversal. This field is only valid in the L0 LTSSM state. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
14:11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
10:8	PIPE_POWER_DOWN	R	PIPE:PowerDown. Indicates PIPE PowerDown signal. <i>Note:</i> This register field is sticky. Value After Reset: 0x2 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7	FRAMING_ERR	R/W1C	Framing Error. Indicates Framing Error detection status. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
6:0	FRAMING_ERR_PTR	R	First Framing Error Pointer. Identifies the first Framing Error using the following encoding. The field contents are only valid value when FRAMING_ERR = 1. ■ Received Unexpected Framing Token (Values 01h to 06h) ■ Received Unexpected STP Token (Values 11h to 13h) ■ Received Unexpected Block (Values 21h to 2Eh) All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (STP_OR_SDP_OR_IDL_RECEIVED_NO_TLP_OR_DLLP): When non- STP/SDP/IDL Token was received and it was not in TLP/DLLP reception ■ 0x2 (CURRENT_NO_VALID_EDB_PREVIOUS_VALID_EDB): When current token was not a valid EDB token and previous token was an EDB. (128/256 bit controller only) ■ 0x3 (SDP_RECEIVED_NOT_EXPECTED): When SDP token was received but not expected. (128 bit & (x8 x16) controller only) ■ 0x4 (STP_RECEIVED_NOT_EXPECTED): When STP token was received but not expected. (128 bit & (x8 x16) controller only) ■ 0x5 (EDS_RECEIVED_NOT_EXPECTED): When EDS token was expected but not received or whenever an EDS token was received but not expected. ■ 0x6 (FRE_ERR_DESKEW_PKT_IN_PROGRESS): When a framing error was detected in the deskew block while a packet has been in progress in token_finder. ■ 0x11 (CRC_STP_NOT_MATCH): When Framing CRC in STP token did not match

Bits	Name	Memory Access	Description
6:0...(cont.)	FRAMING_ERR_PTR.	R	0x12 (PARITY_STP_NOT_MATCH): When Framing Parity in STP token did not match. 0x13 (TLP_LENGTH_SMALLER_THAN_5DW): When Framing TLP Length in STP token was smaller than 5 DWORDs. 0x21 (RECEIVING_OS_AFTER_SDS_IN_DATA_STREAM): When Receiving an OS Block following SDS in Datastream state 0x22 (AFTER_DATA_BLK_OS_BLK_NOT_SKP_EI_EIE): When Data Block followed by OS Block different from SKP, EI, EIE in Datastream state 0x23 (UNDEFINE_BLK_TYPE): When Block with an undefined Block Type in Datastream state 0x24 (DATA_STREAM_WITHOUT_3_CYCLE_DATA_STREAM_S): When Data Stream without data over three cycles in Datastream state 0x25 (OS_BLK_DURING_DATA_STREAM): When OS Block during Data Stream in Datastream state 0x26 (RXSTATUS_ERR_DATA_STREAM_STATE): When RxStatus Error was detected in Datastream state
6:0...(cont.)	FRAMING_ERR_PTR..	R	0x27 (NOT_ALL_LANE_START_RECEIVING_SKP_OS_SAME_T): When Not all active lanes receiving SKP OS starting at same cycle time in SKPOS state 0x28 (2_BLK_TIMEOUT_SKP_OS_SKPOS_STATE): When a 2-Block timeout occurs for SKP OS in SKPOS state 0x29 (CONSECUTIVE_OS_WITHIN_DATA_IN_SKPOS_STATE): When Receiving consecutive OS Blocks within a Data Stream in SKPOS state 0x2a (PHYSTATUS_ERR_IN_SKPOS_STATE): When PHY status error was detected in SKPOS state 0x2b (NOT_ALL_LANE_START_RECEIVING_EIOS_SAME_T): When Not all active lanes receiving EIOS starting at same cycle time in EIOS state 0x2c (AT_LEAST_1_SYM_IS_NOT_EIOS_FROM_4_SYM): When At least one Symbol from the first 4 Symbols is not EIOS Symbol in EIOS state (CX_NB=2 only) 0x2d (NOT_ALL_LANE_START_RECEIVING_EIEOS_SAME_T): When Not all active lanes receiving EIEOS starting at same cycle time in EIEOS state
6:0...(cont.)	FRAMING_ERR_PTR...	R	0x2e (NOT_16_EIEOS_SYM_RECEIVED): When Not full 16 eieos symbols are received in EIEOS state Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.24 SD_STATUS_PM_REG

- **Name:** Silicon Debug Status(PM).
- **Description:** For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xb8
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	RSVDP_24
23:16	LATCHED_NFTS
15:13	L1SUB_STATE
12	PME_RESEND_FLAG
11:8	INTERNAL_PM_SSTATE
7:5	RSVDP_5
4:0	INTERNAL_PM_MSTATE

Table 3-210 Fields for Register: SD_STATUS_PM_REG

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:16	LATCHED_NFTS	R	Latched N_FTS. Indicates the value of N_FTS in the received TS Ordered Sets from the Link Partner. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:13	L1SUB_STATE	R	L1Sub State. Indicates internal state machine of L1Sub state. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (S_L1_U): S_L1_U : idle state ■ 0x1 (S_L1_0): S_L1_0 : wait for aux_clk_active ■ 0x2 (S_L1_0_WAIT4_ACK): S_L1_0_WAIT4_ACK : wait for pclkack ■ 0x3 (S_L1_0_WAIT4_CLKREQ): S_L1_0_WAIT4_CLKREQ : wait for clkreq ■ 0x4 (S_L1_N_ENTRY): S_L1_N_ENTRY : check clkreq_in_n is de-asserted for t_power_off time (only for L1.2, reduces to one cycle for L1.1) ■ 0x5 (S_L1_N): S_L1_N : L1 substate, turn off txcommon-mode circuits (L1.2 only) and rx electrical idle detection circuits ■ 0x6 (S_L1_N_EXIT): S_L1_N_EXIT : locally/remotely initiated exit, assert pclkreq, wait for pclkack ■ 0x7 (S_L1_N_ABORT): S_L1_N_ABORT : wait for pclkack when aborting an attempt to enter L1_N Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_L1_SUBSTATES_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
12	PME_RESEND_FLAG	R/W1C	PME Re-send flag. When the DUT sends a PM_PME message TLP, the DUT sets PME_Status bit. If host software does not clear PME_Status bit for 100ms(+50%/-5%), the DUT resends the PM_PME Message. This bit indicates that a PM_PME was resent. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
11:8	INTERNAL_PM_SSTATE	R	Internal PM State(Slave). Indicates internal state machine of Power Management Slave controller. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (S_IDLE): S_IDLE ■ 0x1 (S_RESPOND_NAK): S_RESPOND_NAK ■ 0x2 (S_BLOCK_TLP): S_BLOCK_TLP ■ 0x3 (S_WAIT_LAST_TLP_ACK): S_WAIT_LAST_TLP_ACK ■ 0x4 (S_WAIT_EIDLE): S_WAIT_EIDLE ■ 0x5 (S_LINK_ENTR_L1): S_LINK_ENTR_L1 ■ 0x6 (S_L1): S_L1 ■ 0x7 (S_L1_EXIT): S_L1_EXIT ■ 0x8 (S_L23RDY): S_L23RDY ■ 0x9 (S_LINK_ENTR_L23): S_LINK_ENTR_L23 ■ 0xa (S_L23RDY_WAIT4ALIVE): S_L23RDY_WAIT4ALIVE ■ 0xb (S_ACK_WAIT4IDLE): S_ACK_WAIT4IDLE ■ 0xc (S_WAIT_LAST_PMDLLP): S_WAIT_LAST_PMDLLP Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:0	INTERNAL_PM_MSTATE	R	Internal PM State(Master). Indicates internal state machine of Power Management Master controller. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (IDLE): IDLE ■ 0x1 (L0): L0 ■ 0x2 (L0S): L0S ■ 0x3 (ENTER_L0S): ENTER_L0S ■ 0x4 (EXIT_L0S): L0S_EXIT ■ 0x5 (WAIT_PMCSR_CPL_SENT): WAIT_PMCSR_CPL_SENT ■ 0x8 (L1): L1 ■ 0x9 (L1_BLOCK_TLP): L1_BLOCK_TLP ■ 0xa (L1_WAIT_LAST_TLP_ACK): L1_WAIT_LAST_TLP_ACK ■ 0xb (L1_WAIT_PMDLLP_ACK): L1_WAIT_PMDLLP_ACK ■ 0xc (L1_LINK_ENTR_L1): L1_LINK_ENTR_L1 ■ 0xd (L1_EXIT): L1_EXIT ■ 0xf (PREP_4L1): PREP_4L1 ■ 0x10 (L23_BLOCK_TLP): L23_BLOCK_TLP ■ 0x11 (L23_WAIT_LAST_TLP_ACK): L23_WAIT_LAST_TLP_ACK ■ 0x12 (L23_WAIT_PMDLLP_ACK): L23_WAIT_PMDLLP_ACK ■ 0x13 (L23_ENTR_L23): L23_ENTR_L23 ■ 0x14 (L23RDY): L23RDY ■ 0x15 (PREP_4L23): PREP_4L23 ■ 0x16 (L23RDY_WAIT4ALIVE): L23RDY_WAIT4ALIVE ■ 0x17 (L0S_BLOCK_TLP): L0S_BLOCK_TLP ■ 0x18 (WAIT_LAST_PMDLLP): WAIT_LAST_PMDLLP ■ 0x19 (WAIT_DSTATE_UPDATE): WAIT_DSTATE_UPDATE Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.25 SD_STATUS_L2_REG

- **Name:** Silicon Debug Status(Layer2).
- **Description:** For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xbc
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RSVDP_28
27	FC_INIT2
26	FC_INIT1
25:24	DLCMSM
23:12	RX_ACK_SEQ_NO
11:0	TX_TLP_SEQ_NO

Table 3-211 Fields for Register: SD_STATUS_L2_REG

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27	FC_INIT2	R	FC_INIT2. Indicates the controller is in FC_INIT2(VC0) state. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
26	FC_INIT1	R	FC_INIT1. Indicates the controller is in FC_INIT1(VC0) state. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
25:24	DLCMSM	R	DLCMSM. Indicates the current DLCMSM. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DL_INACTIVE): DL_INACTIVE ■ 0x1 (DL_FC_INIT): DL_FC_INIT ■ 0x3 (DL_ACTIVE): DL_ACTIVE Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
23:12	RX_ACK_SEQ_NO	R	Tx Ack Sequence Number. Indicates ACKD_SEQ which is updated by receiving ACK/NAK DLLP. <i>Note:</i> This register field is sticky. Value After Reset: 0xfff Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
11:0	TX_TLP_SEQ_NO	R	Tx Tlp Sequence Number. Indicates next transmit sequence number for transmit TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.26 SD_STATUS_L3FC_REG

- **Name:** Silicon Debug Status(Layer3 FC).
 - **Description:** The CREDIT_DATA[0/1] fields in this viewport register return the data for the VC and TLP Type selected by the following fields:
 - ❑ CREDIT_SEL_VC
 - ❑ CREDIT_SEL_CREDIT_TYPE
 - ❑ CREDIT_SEL_TLP_TYPE
 - ❑ CREDIT_SEL_HD
- SD_STATUS_L3FC_REG returns the header credit [7:0] (CREDIT_SEL_HD =0) and the data credit [11:0] (CREDIT_SEL_HD =1) even when the Scaled FC is supported. For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
 - **Offset:** B+0xc0
 - **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

CREDIT_DATA1	31:20
CREDIT_DATA0	19:8
RSVDP_7	7
CREDIT_SEL_HD	6
CREDIT_SEL_TLP_TYPE	5:4
CREDIT_SEL_CREDIT_TYPE	3
CREDIT_SEL_VC	2:0

Table 3-212 Fields for Register: SD_STATUS_L3FC_REG

Bits	Name	Memory Access	Description
31:20	CREDIT_DATA1	R	Credit Data1. Current FC credit data selected by the CREDIT_SEL_VC, CREDIT_SEL_CREDIT_TYPE, CREDIT_SEL_TLP_TYPE, and CREDIT_SEL_HD viewport-select fields. ■ Rx: Credit Allocated Value ■ Tx: Credit Limit Value. This value is valid when DLCMSM=0x3(DL_ACTIVE). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true
19:8	CREDIT_DATA0	R	Credit Data0. Current FC credit data selected by the CREDIT_SEL_VC, CREDIT_SEL_CREDIT_TYPE, CREDIT_SEL_TLP_TYPE, and CREDIT_SEL_HD viewport-select fields. ■ Rx: Credit Received Value ■ Tx: Credit Consumed Value <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
6	CREDIT_SEL_HD	R/W	Credit Select(HeaderData). This field in conjunction with the CREDIT_SEL_VC, CREDIT_SEL_CREDIT_TYPE, and CREDIT_SEL_TLP_TYPE viewport-select fields determines that data that is returned by the CREDIT_DATA0 and CREDIT_DATA1 data fields. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (HEADER_CREDIT): Header Credit ■ 0x1 (DATA_CREDIT): Data Credit Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
5:4	CREDIT_SEL_TLP_TYPE	R/W	Credit Select(TLP Type). This field in conjunction with the CREDIT_SEL_VC, CREDIT_SEL_CREDIT_TYPE, and CREDIT_SEL_HD viewport-select fields determines that data that is returned by the CREDIT_DATA0 and CREDIT_DATA1 data fields. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (POSTED): Posted ■ 0x1 (NON_POSTED): Non-Posted ■ 0x2 (COMPLETION): Completion Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
3	CREDIT_SEL_CREDIT_TYPE	R/W	Credit Select(Credit Type). This field in conjunction with the CREDIT_SEL_VC, CREDIT_SEL_TLP_TYPE, and CREDIT_SEL_HD viewport-select fields determines that data that is returned by the CREDIT_DATA0 and CREDIT_DATA1 data fields. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (RX): Rx ■ 0x1 (TX): Tx Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
2:0	CREDIT_SEL_VC	R/W	Credit Select(VC). This field in conjunction with the CREDIT_SEL_CREDIT_TYPE, CREDIT_SEL_TLP_TYPE, and CREDIT_SEL_HD viewport-select fields determines that data that is returned by the CREDIT_DATA0 and CREDIT_DATA1 data fields. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): VC0 ■ 0x7 (MAX_VAL): VC7 Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.33.27 SD_STATUS_L3_REG

- **Name:** Silicon Debug Status(Layer3).
- **Description:** Silicon Debug Status(Layer3). For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xc4
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:8	7	6:0
RSVDP_8	MFTLP_STATUS	MFTLP_POINTER

Table 3-213 Fields for Register: SD_STATUS_L3_REG

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7	MFTLP_STATUS	R/W1C	Malformed TLP Status. Indicates malformed TLP has occurred. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
6:0	MFTLP_POINTER	R	First Malformed TLP Error Pointer. Indicates the element of the received first malformed TLP. This pointer is validated by MFTLP_STATUS. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ATMC_OP_ALIGN): AtomicOp address alignment ■ 0x2 (ATMC_OP_OPERAND): AtomicOp operand ■ 0x3 (ATMC_OP_BYTE_EN): AtomicOp byte enable ■ 0x4 (TLP_LENGTH_MISMATCH): TLP length miss match ■ 0x5 (MAX_PAYLOAD_SIZE): Max payload size ■ 0x6 (TLP_WITHOUT_TC0): Message TLP without TC0 ■ 0x7 (INVALID_TC): Invalid TC ■ 0x8 (UNXPCTD_ROUTE_BIT_MSG_TLP): Unexpected route bit in Message TLP ■ 0x9 (UNXPCTD_CRS_STATUS_CMPL_TLP): Unexpected CRS status in Completion TLP ■ 0xa (BYTE_ENABLE): Byte enable ■ 0xb (MEM_ADDR_4KB_BOUNDARY): Memory Address 4KB boundary ■ 0xc (TLP_PREFIX_RULES): TLP prefix rules ■ 0xd (TRANSLATION_RULES): Translation request rules ■ 0xe (INVALID_TLP_TYPE): Invalid TLP type ■ 0xf (CMPL_RULES): Completion rules ■ 0x7f (APPLICATION): Application Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.28 SD_EQ_CONTROL1_REG

- **Name:** Silicon Debug EQ Control 1.
- **Description:** This is a viewport control register. Setting the EQ_RATE_SEL and EQ_LANE_SEL fields in this register determine the per-lane Silicon Debug EQ Status data returned by the SD_EQ_STATUS[1/2/3] viewport registers. For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xd0
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	FOM_TARGET
23	FOM_TARGET_ENABLE
22:18	RSVDP_18
17:16	EVAL_INTERVAL_TIME
15:10	RSVDP_10
9:8	EXT_EQ_TIMEOUT
7:6	RSVDP_6
5:4	EQ_RATE_SEL
3:0	EQ_LANE_SEL

Table 3-214 Fields for Register: SD_EQ_CONTROL1_REG

Bits	Name	Memory Access	Description
31:24	FOM_TARGET	R/W	FOM Target. Indicates figure of merit target criteria value of EQ Master(DSP in EQ Phase3/USP in EQ Phase2). This field is only valid when GEN3_EQ_FB_MODE is 0001b(Figure Of Merit). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
23	FOM_TARGET_ENABLE	R/W	FOM Target Enable. Enables the FOM_TARGET fields. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
22:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
17:16	EVAL_INTERVAL_TIME	R/W	Eval Interval Time. Indicates interval time of RxEqEval assertion. This field is used for EQ Master(DSP in EQ Phase3/USP in EQ Phase2). <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_500NS): 500ns ■ 0x1 (_1US): 1us ■ 0x2 (_2US): 2us ■ 0x3 (_4US): 4us Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15:10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
9:8	EXT_EQ_TIMEOUT	R/W	Extends EQ Phase2/3 Timeout. This field is used when the LTSSM is in Recovery.EQ2/3. When this field is set, the value of EQ2/3 timeout is extended. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_24MS_OR_32MS): [EQ Master(DSP in EQ Phase3/USP in EQ Phase2)] 24ms (default) or [EQ Slave(DSP in EQ Phase2/USP in EQ Phase3)] 32ms (default) ■ 0x1 (_48MS_OR_56MS): [EQ Master(DSP in EQ Phase3/USP in EQ Phase2)] 48ms (x2) or [EQ Slave(DSP in EQ Phase2/USP in EQ Phase3)] 56ms (32ms+24ms) ■ 0x2 (_240MS_OR_248MS): [EQ Master(DSP in EQ Phase3/USP in EQ Phase2)] 240ms (x10) or [EQ Slave(DSP in EQ Phase2/USP in EQ Phase3)] 248ms (32ms +9*24ms) ■ 0x3 (NO_TIMEOUT): [EQ Master(DSP in EQ Phase3/USP in EQ Phase2)] No timeout or [EQ Slave(DSP in EQ Phase2/USP in EQ Phase3)] No timeout Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
5:4	EQ_RATE_SEL	R/W	EQ Status Rate Select. Setting this field in conjunction with the EQ_LANE_SEL field determines the per-lane Silicon Debug EQ Status data returned by the SD_EQ_CONTROL[2/3] and SD_EQ_STATUS[1/2/3] viewport registers. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_8GT): 8.0GT/s Speed (include ESM data rate) ■ 0x1 (_16GT): 16.0GT/s Speed (include ESM data rate) ■ 0x2 (_32GT): 32.0GT/s Speed Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
3:0	EQ_LANE_SEL	R/W	EQ Status Lane Select. Setting this field in conjunction with the EQ_RATE_SEL field determines the per-lane Silicon Debug EQ Status data returned by the SD_EQ_CONTROL[2/3] and SD_EQ_STATUS[1/2/3] viewport registers. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Lane0 ■ 0xf (MAX_VAL): Lane15 Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.33.29 SD_EQ_CONTROL2_REG

- **Name:** Silicon Debug EQ Control 2.
- **Description:** This viewport register returns the value for the rate and lane selected by the EQ_RATE_SEL and EQ_LANE_SEL fields in the SD_EQ_CONTROL1_REG register. For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xd4
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	RSVDP_31
30	FORCE_LOCAL_TX_PRESET_ENABLE
29	FORCE_LOCAL_RX_HINT_ENABLE
28	FORCE_LOCAL_TX_COEF_ENABLE
27:24	FORCE_LOCAL_TX_PRESET
23:21	RSVDP_21
20:18	FORCE_LOCAL_RX_HINT
17:12	FORCE_LOCAL_TX_POST_CURSOR
11:6	FORCE_LOCAL_TX_CURSOR
5:0	FORCE_LOCAL_TX_PRE_CURSOR

Table 3-215 Fields for Register: SD_EQ_CONTROL2_REG

Bits	Name	Memory Access	Description
31	RSVDP_31	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
30	FORCE_LOCAL_TX_PRESET_ENABLE	R/W	Force Local Transmitter Preset Enable. Enables the FORCE_LOCAL_TX_PRESET field. If select rate in the EQ_RATE_SEL field is 32.0GT/s Speed, this feature is not available. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
29	FORCE_LOCAL_RX_HINT_ENABLE	R/W	Force Local Receiver Preset Hint Enable. Enables the FORCE_LOCAL_RX_HINT field. If select rate in the EQ_RATE_SEL field is other than 8.0GT/s Speed, this feature is not available. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
28	FORCE_LOCAL_TX_COEF_ENABLE	R/W	Force Local Transmitter Coefficient Enable. Enables the following fields: ■ FORCE_LOCAL_TX_PRE_CURSOR ■ FORCE_LOCAL_TX_CURSOR ■ FORCE_LOCAL_TX_POST_CURSOR <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
27:24	FORCE_LOCAL_TX_PRESET	R/W	Force Local Transmitter Preset. Indicates initial preset value of USP in EQ Slave(EQ Phase2) instead of receiving EQ TS2. If select rate in the EQ_RATE_SEL field is 32.0GT/s Speed, this feature is not available. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
23:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20:18	FORCE_LOCAL_RX_HINT	R/W	Force Local Receiver Preset Hint. Indicates the RxPresetHint value of EQ Slave(DSP in EQ Phase2/USP in EQ Phase3), instead of received or set value. If select rate in the EQ_RATE_SEL field is other than 8.0GT/s Speed, this feature is not available. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
17:12	FORCE_LOCAL_TX_POST_CURSOR	R/W	Force Local Transmitter Post-Cursor. Indicates the coefficient value of EQ Slave(DSP in EQ Phase2/USP in EQ Phase3), instead of the value instructed from link partner. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
11:6	FORCE_LOCAL_TX_CURSOR	R/W	Force Local Transmitter Cursor. Indicates the coefficient value of EQ Slave(DSP in EQ Phase2/USP in EQ Phase3), instead of the value instructed from link partner. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
5:0	FORCE_LOCAL_TX_PRE_CURSOR	R/W	Force Local Transmitter Pre-cursor. Indicates the coefficient value of EQ Slave(DSP in EQ Phase2/USP in EQ Phase3), instead of the value instructed from link partner. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.30 SD_EQ_CONTROL3_REG

- **Name:** Silicon Debug EQ Control 3.
- **Description:** This viewport register returns the value for the rate and lane selected by the EQ_RATE_SEL and EQ_LANE_SEL fields in the SD_EQ_CONTROL1_REG register. For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xd8
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:29	RSVDP_29
28	FORCE_REMOTE_TX_COEF_ENABLE
27:18	RSVDP_18
17:12	FORCE_REMOTE_TX_POST_CURSOR
11:6	FORCE_REMOTE_TX_CURSOR
5:0	FORCE_REMOTE_TX_PRE_CURSOR

Table 3-216 Fields for Register: SD_EQ_CONTROL3_REG

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
28	FORCE_REMOTE_TX_COEF_ENABLE	R/W	Force Remote Transmitter Coefficient Enable. Enables the following fields: ■ FORCE_REMOTE_TX_PRE_CURSOR ■ FORCE_REMOTE_TX_CURSOR ■ FORCE_REMOTE_TX_POST_CURSOR This function can only be used when GEN3_EQ_FB_MODE = 0000b(Direction Change) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
27:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
17:12	FORCE_REMOTE_TX_POST_CURSOR	R/W	Force Remote Transmitter Post-Cursor. Indicates the coefficient value of EQ Master(DSP in EQ Phase3/USP in EQ Phase2), instead of the value instructed from local PHY in dirchange mode. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
11:6	FORCE_REMOTE_TX_CURSOR	R/W	Force Remote Transmitter Cursor. Indicates the coefficient value of EQ Master(DSP in EQ Phase3/USP in EQ Phase2), instead of the value instructed from local PHY in dirchange mode. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
5:0	FORCE_REMOTE_TX_PRE_CURSOR	R/W	Force Remote Transmitter Pre-Cursor. Indicates the coefficient value of EQ Master(DSP in EQ Phase3/USP in EQ Phase2), instead of the value instructed from local PHY in dirchange mode. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.31 SD_EQ_STATUS1_REG

- **Name:** Silicon Debug EQ Status 1.
 - **Description:** This viewport register returns the first of three words of Silicon Debug EQ Status data for the rate and lane selected by the EQ_RATE_SEL and EQ_LANE_SEL fields in the SD_EQ_CONTROL1_REG register. The following fields are available when Equalization finished unsuccessfully (EQ_CONVERGENCE_INFO=2).
 - EQ_RULEA_VIOLATION
 - EQ_RULEB_VIOLATION
 - EQ_RULEC_VIOLATION
 - EQ_REJECT_EVENT
- For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
 - **Offset:** B+0xe0
 - **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:8	7	6	5	4	3	2:1	0
RSVDP_8	EQ_REJECT_EVENT	EQ_RULEC_VIOLATION	EQ_RULEB_VIOLATION	EQ_RULEA_VIOLATION	RSVDP_3	EQ_CONVERGENCE_INFO	EQ_SEQUENCE

Table 3-217 Fields for Register: SD_EQ_STATUS1_REG

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
7	EQ_REJECT_EVENT	R	EQ Reject Event. Indicates that the controller receives two consecutive TS1 OS w/Reject=1b during EQ Master phase(DSP in EQ Phase3/USP in EQ Phase2). This bit is automatically cleared when the controller starts EQ Master phase again. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
6	EQ_RULEC_VIOLATION	R	EQ Rule C Violation. Indicates that coefficients rule C violation is detected in the values provided by PHY using direction change method during EQ Master phase(DSP in EQ Phase3/USP in EQ Phase2). The coefficients rule C correspond to the rules c) from section "Rules for Transmitter Coefficients" in the PCI Express Base Specification. This bit is automatically cleared when the controller starts EQ Master phase again. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5	EQ_RULEB_VIOLATION	R	EQ Rule B Violation. Indicates that coefficients rule B violation is detected in the values provided by PHY using direction change method during EQ Master phase(DSP in EQ Phase3/USP in EQ Phase2). The coefficients rules B correspond to the rules b) from section "Rules for Transmitter Coefficients" in the PCI Express Base Specification. This bit is automatically cleared when the controller starts EQ Master phase again. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
4	EQ_RULEA_VIOLATION	R	EQ Rule A Violation. Indicates that coefficients rule A violation is detected in the values provided by PHY using direction change method during EQ Master phase(DSP in EQ Phase3/USP in EQ Phase2). The coefficients rules A correspond to the rules a) from section "Rules for Transmitter Coefficients" in the PCI Express Base Specification. This bit is automatically cleared when the controller starts EQ Master phase again. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
2:1	EQ_CONVERGENCE_INFO	R	EQ Convergence Info. Indicates equalization convergence information. This bit is automatically cleared when the controller starts EQ Master phase again. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (EQ_NOT_ATTEMPTED): Equalization is not attempted ■ 0x1 (EQ_SUCCESSFUL): Equalization finished successfully ■ 0x2 (EQ_UNSUCCESSFUL): Equalization finished unsuccessfully ■ 0x3 (RSVD): Reserved Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	EQ_SEQUENCE	R	EQ Sequence. Indicates that the controller is starting the equalization sequence. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.32 SD_EQ_STATUS2_REG

- **Name:** Silicon Debug EQ Status 2.
- **Description:** This viewport register returns the second of three words of Silicon Debug EQ Status data for the rate and lane selected by the EQ_RATE_SEL and EQ_LANE_SEL fields in the SD_EQ_CONTROL1_REG register. Each field is available when Equalization finished successfully(EQ_CONVERGENCE_INFO=1). For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xe4
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	EQ_LOCAL_FOM_VALUE
23:21	RSVDP_21
20:18	EQ_LOCAL_RX_HINT
17:12	EQ_LOCAL_POST_CURSOR
11:6	EQ_LOCAL_CURSOR
5:0	EQ_LOCAL_PRE_CURSOR

Table 3-218 Fields for Register: SD_EQ_STATUS2_REG

Bits	Name	Memory Access	Description
31:24	EQ_LOCAL_FOM_VALUE	R	EQ Local Figure of Merit. Indicates Local maximum Figure of Merit value. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20:18	EQ_LOCAL_RX_HINT	R	EQ Local Receiver Preset Hint. Indicates Local Receiver Preset Hint value. If select rate in the EQ_RATE_SEL field is other than 8.0GT/s Speed, this feature is not available. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
17:12	EQ_LOCAL_POST_CURSOR	R	EQ Local Post-Cursor. Indicates Local post cursor coefficient value. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
11:6	EQ_LOCAL_CURSOR	R	EQ Local Cursor. Indicates Local cursor coefficient value. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5:0	EQ_LOCAL_PRE_CURSOR	R	EQ Local Pre-Cursor. Indicates Local pre cursor coefficient value. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.33.33 SD_EQ_STATUS3_REG

- **Name:** Silicon Debug EQ Status 3.
- **Description:** This viewport register returns the third of three words of Silicon Debug EQ Status data for the rate and lane selected by the EQ_RATE_SEL and EQ_LANE_SEL fields in the SD_EQ_CONTROL1_REG register. Each field is available when Equalization finished successfully (EQ_CONVERGENCE_INFO=1). For more information, see the RAS DES section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xe8
- **Exists:** (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:30	RSVDP_30
29:24	EQ_REMOTE_FS
23:18	EQ_REMOTE_LF
17:12	EQ_REMOTE_POST_CURSOR
11:6	EQ_REMOTE_CURSOR
5:0	EQ_REMOTE_PRE_CURSOR

Table 3-219 Fields for Register: SD_EQ_STATUS3_REG

Bits	Name	Memory Access	Description
31:30	RSVDP_30	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
29:24	EQ_REMOTE_FS	R	EQ Remote FS. Indicates Remote FS value. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
23:18	EQ_REMOTE_LF	R	EQ Remote LF. Indicates Remote LF value. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
17:12	EQ_REMOTE_POST_CURSOR	R	EQ Remote Post-Cursor. Indicates Remote post cursor coefficient value. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
11:6	EQ_REMOTE_CURSOR	R	EQ Remote Cursor. Indicates Remote cursor coefficient value. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5:0	EQ_REMOTE_PRE_CURSOR	R	EQ Remote Pre-Cursor. Indicates Remote pre cursor coefficient value. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_SD_ENABLE && CX_GEN3_SPEED && (CX_RAS_DES_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.34 PF_VSECRAS_CAPI Registers

3.34.1 RASDP_EXT_CAP_HDR_OFF

- **Name:** PCIe Extended capability ID, Capability version and Next capability offset.
- **Description:** This Register provides capability ID, Capability version, and Next capability offset for PCIe Extended capability stucture.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

NEXT_OFFSET	31:20
CAP	19:16
ID	15:0

Table 3-220 Fields for Register: RASDP_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. The bottom 2 bits of this offset are Reserved and must be implemented as 00b although software must mask them to allow for future uses of these bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: Pointer value calculated at configuration time. Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. A version of the specification that changes the Extended Capability in a way that is not otherwise identifiable (e.g., through a new Capability field) is permitted to increment this field. All such changes to the Capability structure must be software-compatible. Software must check for Capability Version numbers that are greater than or equal to the highest number defined when the software is written, as Functions reporting any such Capability Version numbers will contain a Capability structure that is compatible with that piece of software. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15:0	ID	* Varies	PCI Express Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0xb Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.34.2 RASDP_VENDOR_SPECIFIC_HDR_OFF

- **Name:** Vendor Specific Header.
- **Description:** This Register provides VSEC Length, VSEC ID and VSEC Rev (Version Number). Vendor-specific software must qualify the associated Vendor ID of the PCI Express Function or RCRB before attempting to interpret the values in the VSEC ID or VSEC Rev fields.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	19:16	15:0
VSEC_LENGTH	VSEC_REV	VSEC_ID

Table 3-221 Fields for Register: RASDP_VENDOR_SPECIFIC_HDR_OFF

Bits	Name	Memory Access	Description
31:20	VSEC_LENGTH	R	VSEC Length. This field indicates the number of bytes in the entire VSEC structure, including the Vendor-Specific Extended Capability Header, the Vendor-Specific Header, and the vendor-specific registers. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: 0x38 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
19:16	VSEC_REV	R	VSEC Rev. This field is a vendor-defined version number that indicates the version of the VSEC structure. Software must qualify the Vendor ID and VSEC ID before interpreting this field. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15:0	VSEC_ID	R	VSEC ID. This field is a vendor-defined ID number that indicates the nature and format of the VSEC structure. Software must qualify the Vendor ID before interpreting this field. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.34.3 RASDP_ERROR_PROT_CTRL_OFF

- **Name:** ECC Error Correction and Control Register.
- **Description:** Allows you to disable ECC error correction for RAMs and datapath.
When the AXI Bridge Module is implemented and the master / slave clocks are asynchronous to the PCIe native controller clock (core_clk), you must not write this register while operations are in progress in the AXI master / slave interface.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_24	31:24
ERROR_PROT_DISABLE_CXS_RX	23
ERROR_PROT_DISABLE_ADM_RX	22
ERROR_PROT_DISABLE_LAYER3_RX	21
ERROR_PROT_DISABLE_LAYER2_RX	20
ERROR_PROT_DISABLE_DMA_READ	19
ERROR_PROT_DISABLE_AXI_BRIDGE_INBOUND_REQUEST	18
ERROR_PROT_DISABLE_AXI_BRIDGE_INBOUND_COMPLETION	17
ERROR_PROT_DISABLE_RX	16
RSVDP_10	15:10
ERROR_PROT_DISABLE_CXL_TX	9
ERROR_PROT_DISABLE_DTIM_TX	8
ERROR_PROT_DISABLE_CXS_TX	7
ERROR_PROT_DISABLE_ADM_TX	6
ERROR_PROT_DISABLE_LAYER3_TX	5
ERROR_PROT_DISABLE_LAYER2_TX	4
ERROR_PROT_DISABLE_DMA_WRITE	3
ERROR_PROT_DISABLE_AXI_BRIDGE_OUTBOUND	2
ERROR_PROT_DISABLE_AXI_BRIDGE_MASTER	1
ERROR_PROT_DISABLE_TX	0

Table 3-222 Fields for Register: RASDP_ERROR_PROT_CTRL_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23	ERROR_PROT_DISABLE_CXS_RX	R/W	Error correction disable for CXS Tx path (PCIe Rx path). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
22	ERROR_PROT_DISABLE_ADM_RX	R/W	Error correction disable for ADM Rx path. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
21	ERROR_PROT_DISABLE_LAYER_3_RX	R/W	Error correction disable for layer 3 Rx path. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
20	ERROR_PROT_DISABLE_LAYER_2_RX	R/W	Error correction disable for layer 2 Rx path. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
19	ERROR_PROT_DISABLE_DMA_READ	R/W	Error correction disable for DMA read engine. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
18	ERROR_PROT_DISABLE_AXI_BRIDGE_INBOUND_REQUEST	R/W	Error correction disable for AXI bridge inbound request path. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
17	ERROR_PROT_DISABLE_AXI_BRIDGE_INBOUND_COMPLETION	R/W	Error correction disable for AXI bridge inbound completion composer. Does not disable the error detection reporting for 1-bit and 2-bit ECC errors. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
16	ERROR_PROT_DISABLE_RX	R/W	Global error correction disable for all Rx layers. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15:10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
9	ERROR_PROT_DISABLE_CXL_TX	R/W	Error correction disable for CXL Transmitter path. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
8	ERROR_PROT_DISABLE_DTIM_TX	R/W	Error correction disable for DTIM Tx path. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7	ERROR_PROT_DISABLE_CXS_TX	R/W	Error correction disable for CXS Rx path (PCIe Tx path). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
6	ERROR_PROT_DISABLE_ADM_TX	R/W	Error correction disable for Adm Tx path. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
5	ERROR_PROT_DISABLE_LAYER3_TX	R/W	Error correction disable for layer 3 Tx path. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
4	ERROR_PROT_DISABLE_LAYER_2_TX	R/W	Error correction disable for layer 2 Tx path. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
3	ERROR_PROT_DISABLE_DMA_WRITE	R/W	Error correction disable for DMA write engine. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
2	ERROR_PROT_DISABLE_AXI_BRIDGE_OUTBOUND	R/W	Error correction disable for AXI bridge outbound request path. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
1	ERROR_PROT_DISABLE_AXI_BRIDGE_MASTER	R/W	Error correction disable for AXI bridge master completion buffer. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
0	ERROR_PROT_DISABLE_TX	R/W	Global error correction disable for all Tx layers. Does not disable the error detection reporting for 1-bit and 2-bit ECC errors. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.34.4 RASDP_CORR_COUNTER_CTRL_OFF

- **Name:** Corrected Error (1-bit ECC) Counter Selection and Control Register.
- **Description:** This is a viewport control register. Setting the CORR_COUNTER_SELECTION_REGION and CORR_COUNTER_SELECTION fields in this register determine the counter data returned by the RASDP_CORR_COUNT_REPORT_OFF viewport data register.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 | | CX_MEMORY_MAP_PCIE_TYPE==0))

CORR_COUNTER_SELECTION	31:24
CORR_COUNTER_SELECTION_REGION	23:20
RSVDP_5	19:5
CORR_EN_COUNTERS	4
RSVDP_1	3:1
CORR_CLEAR_COUNTERS	0

Table 3-223 Fields for Register: RASDP_CORR_COUNTER_CTRL_OFF

Bits	Name	Memory Access	Description
31:24	CORR_COUNTER_SELECTION	R/W	Counter selection. This field selects the counter ID (within the region defined by CORR_COUNTER_SELECTION_REGION) whose contents can be read from the RASDP_CORR_COUNT_REPORT_OFF register. You can cycle this field value from 0 to 255 to access all the counters according to the RASDP_CheckPoints.pdf. For more information on the location of the RASDP_CheckPoints.pdf, see the "RAS Data Protection (RAS DP)" topic of the Databook. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
23:20	CORR_COUNTER_SELECTION_REGION	R/W	Select correctable counter region. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ADM_RX_PATH): Region select for Adm Rx path ■ 0x1 (L3_RX_PATH): Region select for layer 3 Rx path ■ 0x2 (L2_RX_PATH): Region select for layer 2 Rx path ■ 0x3 (DMA_INBOUND_PATH): Region select for DMA inbound path ■ 0x4 (AXI_INBOUND_REQ_PATH): Region select for AXI bridge inbound request path ■ 0x5 (AXI_INBOUND_CMPL_PATH): Region select for AXI bridge inbound completion composer path ■ 0x6 (ADM_TX_PATH): Region select for Adm Tx path ■ 0x7 (L3_TX_PATH): Region select for layer 3 Tx path ■ 0x8 (L2_TX_PATH): Region select for layer 2 Tx path ■ 0x9 (DMA_OUTBOUND_PATH): Region select for DMA outbound path ■ 0xa (AXI_OUTBOUND_REQ_PATH): Region select for AXI bridge outbound request path ■ 0xb (AXI_OUTBOUND_CMPL_PATH): Region select for AXI bridge outbound master completion buffer path ■ 0xc (CXS_TX_RX_PATH): Region select for CXS Tx/Rx path ■ 0xd (DTIM_TX_RX_PATH): Region select for DTIM Tx/Rx path Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
19:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4	CORR_EN_COUNTERS	R/W	Enable correctable errors counters. The counters are enabled by default. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (COUNTER_EN): Counters increment when the controller detects a correctable error ■ 0x0 (COUNTER_FROZEN): Counters are frozen Value After Reset: 0x1 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
3:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	CORR_CLEAR_COUNTERS	W	Clear all correctable error counters. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.34.5 RASDP_CORR_COUNT_REPORT_OFF

- **Name:** Corrected Error (1-bit ECC) Counter Data Register.
- **Description:** This viewport register returns the counter data selected by the CORR_COUNTER_SELECTION_REGION and CORR_COUNTER_SELECTION fields in the RASDP_CORR_COUNTER_CTRL_OFF register.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	CORR_COUNTER_SELECTED
23:20	CORR_COUNTER_SELECTED_REGION
19:8	RSVDP_8
7:0	CORR_COUNTER

Table 3-224 Fields for Register: RASDP_CORR_COUNT_REPORT_OFF

Bits	Name	Memory Access	Description
31:24	CORR_COUNTER_SELECTED	R	Counter selection. Returns the value set in the CORR_COUNTER_SELECTION field of the RASDP_CORR_COUNTER_CTRL_OFF register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:20	CORR_COUNTER_SELECTED_REGION	R	Selected correctable counter region. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ADM_RX_PATH): Region select for Adm Rx path ■ 0x1 (L3_RX_PATH): Region select for layer 3 Rx path ■ 0x2 (L2_RX_PATH): Region select for layer 2 Rx path ■ 0x3 (DMA_INBOUND_PATH): Region select for DMA inbound path ■ 0x4 (AXI_INBOUND_REQ_PATH): Region select for AXI bridge inbound request path ■ 0x5 (AXI_INBOUND_CMPL_PATH): Region select for AXI bridge inbound completion composer path ■ 0x6 (ADM_TX_PATH): Region select for Adm Tx path ■ 0x7 (L3_TX_PATH): Region select for layer 3 Tx path ■ 0x8 (L2_TX_PATH): Region select for layer 2 Tx path ■ 0x9 (DMA_OUTBOUND_PATH): Region select for DMA outbound path ■ 0xa (AXI_OUTBOUND_REQ_PATH): Region select for AXI bridge outbound request path ■ 0xb (AXI_OUTBOUND_CMPL_PATH): Region select for AXI bridge outbound master completion buffer path ■ 0xc (CXS_TX_RX_PATH): Region select for CXS Tx/Rx path ■ 0xd (DTIM_TX_RX_PATH): Region select for DTIM Tx/Rx path Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
19:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
7:0	CORR_COUNTER	R	Current corrected error count for the selected counter. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.34.6 RASDP_UNCORR_COUNTER_CTRL_OFF

- **Name:** Uncorrected Error (2-bit ECC and parity) Counter Selection and Control Register.
- **Description:** This is a viewport control register. Setting the UNCORR_COUNTER_SELECTION_REGION and UNCORR_COUNTER_SELECTION fields in this register determine the counter data returned by the RASDP_UNCORR_COUNT_REPORT_OFF viewport data register.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

UNCORR_COUNTER_SELECTION	31:24
UNCORR_COUNTER_SELECTION_REGION	23:20
RSVDP_5	19:5
UNCORR_EN_COUNTERS	4
RSVDP_1	3:1
UNCORR_CLEAR_COUNTERS	0

Table 3-225 Fields for Register: RASDP_UNCORR_COUNTER_CTRL_OFF

Bits	Name	Memory Access	Description
31:24	UNCORR_COUNTER_SELECTION	R/W	Counter selection. This field selects the counter ID (within the region defined by UNCORR_COUNTER_SELECTION_REGION) whose contents can be read from the RASDP_UNCORR_COUNT_REPORT_OFF register. You can cycle this field value from 0 to 255 to access all the counters according to the RASDP_CheckPoints.pdf. For more information on the location of the RASDP_CheckPointst.pdf, see the "RAS Data Protection (RAS DP)" topic of the Databook. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
23:20	UNCORR_COUNTER_SELECTIO N_REGION	R/W	Select uncorrectable counter region. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ADM_RX_PATH): Region select for Adm Rx path ■ 0x1 (L3_RX_PATH): Region select for layer 3 Rx path ■ 0x2 (L2_RX_PATH): Region select for layer 2 Rx path ■ 0x3 (DMA_INBOUND_PATH): Region select for DMA inbound path ■ 0x4 (AXI_INBOUND_REQ_PATH): Region select for AXI bridge inbound request path ■ 0x5 (AXI_INBOUND_CMPL_PATH): Region select for AXI bridge inbound completion composer path ■ 0x6 (ADM_TX_PATH): Region select for Adm Tx path ■ 0x7 (L3_TX_PATH): Region select for layer 3 Tx path ■ 0x8 (L2_TX_PATH): Region select for layer 2 Tx path ■ 0x9 (DMA_OUTBOUND_PATH): Region select for DMA outbound path ■ 0xa (AXI_OUTBOUND_REQ_PATH): Region select for AXI bridge outbound request path ■ 0xb (AXI_OUTBOUND_CMPL_PATH): Region select for AXI bridge outbound master completion buffer path ■ 0xc (CXS_TX_RX_PATH): Region select for CXS Tx/Rx path ■ 0xd (DTIM_TX_RX_PATH): Region select for DTIM Tx/Rx path Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
19:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4	UNCORR_EN_COUNTERS	R/W	Enable uncorrectable errors counters. The counters are enabled by default. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE_COUNTER): Enables the counters to increment on detected correctable errors ■ 0x0 (COUNTER_FROZEN): Counters are frozen Value After Reset: 0x1 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
3:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	UNCORR_CLEAR_COUNTERS	W	Clear uncorrectable errors counters. When asserted causes all counters tracking the uncorrectable errors to be cleared. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.34.7 RASDP_UNCORR_COUNT_REPORT_OFF

- **Name:** Uncorrected Error (2-bit ECC and parity) Counter Data Register.
- **Description:** This viewport register returns the counter data selected by the UNCORR_COUNTER_SELECTION_REGION and UNCORR_COUNTER_SELECTION fields in the RASDP_UNCORR_COUNTER_CTRL_OFF register.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	UNCORR_COUNTER_SELECTED
23:20	UNCORR_COUNTER_SELECTED_REGION
19:8	RSVDP_8
7:0	UNCORR_COUNTER

Table 3-226 Fields for Register: RASDP_UNCORR_COUNT_REPORT_OFF

Bits	Name	Memory Access	Description
31:24	UNCORR_COUNTER_SELECTED	R	Counter selection. Returns the value set in the UNCORR_COUNTER_SELECTION field of the RASDP_UNCORR_COUNTER_CTRL_OFF register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:20	UNCORR_COUNTER_SELECT D_REGION	R	Selected uncorrectable counter region. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ADM_RX_PATH): Region select for Adm Rx path ■ 0x1 (L3_RX_PATH): Region select for layer 3 Rx path ■ 0x2 (L2_RX_PATH): Region select for layer 2 Rx path ■ 0x3 (DMA_INBOUND_PATH): Region select for DMA inbound path ■ 0x4 (AXI_INBOUND_REQ_PATH): Region select for AXI bridge inbound request path ■ 0x5 (AXI_INBOUND_CMPL_PATH): Region select for AXI bridge inbound completion composer path ■ 0x6 (ADM_TX_PATH): Region select for Adm Tx path ■ 0x7 (L3_TX_PATH): Region select for layer 3 Tx path ■ 0x8 (L2_TX_PATH): Region select for layer 2 Tx path ■ 0x9 (DMA_OUTBOUND_PATH): Region select for DMA outbound path ■ 0xa (AXI_OUTBOUND_REQ_PATH): Region select for AXI bridge outbound request path ■ 0xb (AXI_OUTBOUND_CMPL_PATH): Region select for AXI bridge outbound master completion buffer path ■ 0xc (CXS_TX_RX_PATH): Region select for CXS Tx/Rx path ■ 0xd (DTIM_TX_RX_PATH): Region select for DTIM Tx/Rx path Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
19:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
7:0	UNCORR_COUNTER	R	Current uncorrected error count for the selected counter <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.34.8 RASDP_ERROR_INJ_CTRL_OFF

- **Name:** Error Injection Control Register.
- **Description:** Error injection control for the following features:
 - 1-bit or 2-bit injection
 - Continuous or fixed-number (n) injection modes
 - Global enable/disable
 - Selectable location where injection occurs
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	RSVDP_24
23:16	ERROR_INJ_LOC
15:8	ERROR_INJ_COUNT
7:6	RSVDP_6
5:4	ERROR_INJ_TYPE
3:1	RSVDP_1
0	ERROR_INJ_EN

Table 3-227 Fields for Register: RASDP_ERROR_INJ_CTRL_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
23:16	ERROR_INJ_LOC	R/W	Error injection location. Selects where error injection takes place. You can cycle this field value from 0 to 255 to access all the counters according to the RASDP_CheckPoints.pdf. For more information on the location of the RASDP_CheckPointst.pdf, see the "RAS Data Protection (RAS DP)" topic of the Databook. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15:8	ERROR_INJ_COUNT	R/W	Error injection count. If value is n, n amount of errors injected. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ERR_INJ_EVERY_TLP): Errors are inserted in every TLP until you clear ERROR_INJ_EN. ■ 0x1 (ERR_INJ_1): One errors injected ■ 0x2 (ERR_INJ_2): Two errors injected Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5:4	ERROR_INJ_TYPE	R/W	Error injection type. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NONE): none ■ 0x1 (_1_BIT): 1-bit ■ 0x2 (_2_BIT): 2-bit Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
3:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	ERROR_INJ_EN	R/W	Error injection global enable. When set, enables the error insertion logic. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.34.9 RASDP_CORR_ERROR_LOCATION_OFF

- **Name:** Corrected Errors Location Register.
- **Description:** Corrected errors locations. For more information, see the RAS Data Protection (DP) section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	LOC_LAST_CORR_ERROR
23:20	REG_LAST_CORR_ERROR
19:16	RSVDP_16
15:8	LOC_FIRST_CORR_ERROR
7:4	REG_FIRST_CORR_ERROR
3:0	RSVDP_0

Table 3-228 Fields for Register: RASDP_CORR_ERROR_LOCATION_OFF

Bits	Name	Memory Access	Description
31:24	LOC_LAST_CORR_ERROR	R	Location/ID of the last corrected error within the region defined by REG_LAST_CORR_ERROR. You can cycle this field value from 0 to 255 to access all the counters according to the RASDP_CheckPoints.pdf. For more information on the location of the RASDP_CheckPoints.pdf, see the "RAS Data Protection (RAS DP)" topic of the databook. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:20	REG_LAST_CORR_ERROR	R	Region of the last corrected error. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ADM_RX_PATH): Region select for Adm Rx path ■ 0x1 (L3_RX_PATH): Region select for layer 3 Rx path ■ 0x2 (L2_RX_PATH): Region select for layer 2 Rx path ■ 0x3 (DMA_INBOUND_PATH): Region select for DMA inbound path ■ 0x4 (AXI_INBOUND_REQ_PATH): Region select for AXI bridge inbound request path ■ 0x5 (AXI_INBOUND_CMPL_PATH): Region select for AXI bridge inbound completion composer path ■ 0x6 (ADM_TX_PATH): Region select for Adm Tx path ■ 0x7 (L3_TX_PATH): Region select for layer 3 Tx path ■ 0x8 (L2_TX_PATH): Region select for layer 2 Tx path ■ 0x9 (DMA_OUTBOUND_PATH): Region select for DMA outbound path ■ 0xa (AXI_OUTBOUND_REQ_PATH): Region select for AXI bridge outbound request path ■ 0xb (AXI_OUTBOUND_CMPL_PATH): Region select for AXI bridge outbound master completion buffer path ■ 0xc (CXS_TX_RX_PATH): Region select for CXS Tx/Rx path ■ 0xd (DTIM_TX_RX_PATH): Region select for DTIM Tx/Rx path Value After Reset: 0xf Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
19:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:8	LOC_FIRST_CORR_ERROR	R	Location/ID of the first corrected error within the region defined by REG_FIRST_CORR_ERROR. You can cycle this field value from 0 to 255 to access all the counters according to the RASDP_CheckPoints.pdf. For more information on the location of the RASDP_CheckPoints.pdf, see the "RAS Data Protection (RAS DP)" topic of the Databook. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
7:4	REG_FIRST_CORR_ERROR	R	Region of the first corrected error. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ADM_RX_PATH): Region select for Adm Rx path ■ 0x1 (L3_RX_PATH): Region select for layer 3 Rx path ■ 0x2 (L2_RX_PATH): Region select for layer 2 Rx path ■ 0x3 (DMA_INBOUND_PATH): Region select for DMA inbound path ■ 0x4 (AXI_INBOUND_REQ_PATH): Region select for AXI bridge inbound request path ■ 0x5 (AXI_INBOUND_CMPL_PATH): Region select for AXI bridge inbound completion composer path ■ 0x6 (ADM_TX_PATH): Region select for Adm Tx path ■ 0x7 (L3_TX_PATH): Region select for layer 3 Tx path ■ 0x8 (L2_TX_PATH): Region select for layer 2 Tx path ■ 0x9 (DMA_OUTBOUND_PATH): Region select for DMA outbound path ■ 0xa (AXI_OUTBOUND_REQ_PATH): Region select for AXI bridge outbound request path ■ 0xb (AXI_OUTBOUND_CMPL_PATH): Region select for AXI bridge outbound master completion buffer path ■ 0xc (CXS_TX_RX_PATH): Region select for CXS Tx/Rx path ■ 0xd (DTIM_TX_RX_PATH): Region select for DTIM Tx/Rx path Value After Reset: 0xf Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.34.10 RASDP_UNCORR_ERROR_LOCATION_OFF

- **Name:** Uncorrected Errors Location Register.
- **Description:** Uncorrected errors locations. For more information, see the RAS Data Protection (DP) section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	LOC_LAST_UNCORR_ERROR
23:20	REG_LAST_UNCORR_ERROR
19:16	RSVDP_16
15:8	LOC_FIRST_UNCORR_ERROR
7:4	REG_FIRST_UNCORR_ERROR
3:0	RSVDP_0

Table 3-229 Fields for Register: RASDP_UNCORR_ERROR_LOCATION_OFF

Bits	Name	Memory Access	Description
31:24	LOC_LAST_UNCORR_ERROR	R	Location/ID of the last uncorrected error within the region defined by REG_LAST_UNCORR_ERROR. You can cycle this field value from 0 to 255 to access all the counters according to the RASDP_CheckPoints.pdf. For more information on the location of the RASDP_CheckPoints.pdf, see the "RAS Data Protection (RAS DP)" topic of the databook. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:20	REG_LAST_UNCORR_ERROR	R	Region of the last uncorrected error. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ADM_RX_PATH): Region select for Adm Rx path ■ 0x1 (L3_RX_PATH): Region select for layer 3 Rx path ■ 0x2 (L2_RX_PATH): Region select for layer 2 Rx path ■ 0x3 (DMA_INBOUND_PATH): Region select for DMA inbound path ■ 0x4 (AXI_INBOUND_REQ_PATH): Region select for AXI bridge inbound request path ■ 0x5 (AXI_INBOUND_CMPL_PATH): Region select for AXI bridge inbound completion composer path ■ 0x6 (ADM_TX_PATH): Region select for Adm Tx path ■ 0x7 (L3_TX_PATH): Region select for layer 3 Tx path ■ 0x8 (L2_TX_PATH): Region select for layer 2 Tx path ■ 0x9 (DMA_OUTBOUND_PATH): Region select for DMA outbound path ■ 0xa (AXI_OUTBOUND_REQ_PATH): Region select for AXI bridge outbound request path ■ 0xb (AXI_OUTBOUND_CMPL_PATH): Region select for AXI bridge outbound master completion buffer path ■ 0xc (CXS_TX_RX_PATH): Region select for CXS Tx/Rx path ■ 0xd (DTIM_TX_RX_PATH): Region select for DTIM Tx/Rx path Value After Reset: 0xf Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
19:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:8	LOC_FIRST_UNCORR_ERROR	R	Location/ID of the first uncorrected error within the region defined by REG_FIRST_UNCORR_ERROR. You can cycle this field value from 0 to 255 to access all the counters according to the RASDP_CheckPoints.pdf. For more information on the location of the RASDP_CheckPoints.pdf, see the "RAS Data Protection (RAS DP)" topic of the databook. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
7:4	REG_FIRST_UNCORR_ERROR	R	Region of the first uncorrected error. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ADM_RX_PATH): Region select for Adm Rx path ■ 0x1 (L3_RX_PATH): Region select for layer 3 Rx path ■ 0x2 (L2_RX_PATH): Region select for layer 2 Rx path ■ 0x3 (DMA_INBOUND_PATH): Region select for DMA inbound path ■ 0x4 (AXI_INBOUND_REQ_PATH): Region select for AXI bridge inbound request path ■ 0x5 (AXI_INBOUND_CMPL_PATH): Region select for AXI bridge inbound completion composer path ■ 0x6 (ADM_TX_PATH): Region select for Adm Tx path ■ 0x7 (L3_TX_PATH): Region select for layer 3 Tx path ■ 0x8 (L2_TX_PATH): Region select for layer 2 Tx path ■ 0x9 (DMA_OUTBOUND_PATH): Region select for DMA outbound path ■ 0xa (AXI_OUTBOUND_REQ_PATH): Region select for AXI bridge outbound request path ■ 0xb (AXI_OUTBOUND_CMPL_PATH): Region select for AXI bridge outbound master completion buffer path ■ 0xc (CXS_TX_RX_PATH): Region select for CXS Tx/Rx path ■ 0xd (DTIM_TX_RX_PATH): Region select for DTIM Tx/Rx path Value After Reset: 0xf Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.34.11 RASDP_ERROR_MODE_EN_OFF

- **Name:** RASDP error mode enable. The controller enters RASDP error mode (if ERROR_MODE_EN =1) upon detection of the first uncorrectable error.
- **Description:** RASDP error mode enable. The controller enters RASDP error mode (if ERROR_MODE_EN =1) upon detection of the first uncorrectable error. During this mode:
 - The controller sets cfg_rasdp_error_mode =1 (or mstr_rasdp_error_mode and slv_rasdp_error_mode when AXI bridge is used). When the AXI Bridge Module is implemented and the master / slave clocks are asynchronous to the PCIe native controller clock (core_clk), you must not write this register while operations are in progress in the AXI master / slave interface.
 - Rx TLPs that are forwarded to your application are not guaranteed to be correct; you must discard them.

For more information, see the RAS Data Protection (DP) section in the Controller Operations chapter of the *Databook*.

- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:2	1	0
RSVDP_2	AUTO_LINK_DOWN_EN	ERROR_MODE_EN

Table 3-230 Fields for Register: RASDP_ERROR_MODE_EN_OFF

Bits	Name	Memory Access	Description
31:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
1	AUTO_LINK_DOWN_EN	R/W	Write '1' to enable the controller to bring the link down when the controller enters RASDP error mode. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
0	ERROR_MODE_EN	R/W	Write '1' to enable the controller enter RASDP error mode when it detects an uncorrectable error. <i>Note:</i> This register field is sticky. Value After Reset: RAS_DEFAULT_NULLIFY_EN Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.34.12 RASDP_ERROR_MODE_CLEAR_OFF

- **Name:** RASDP Error Mode Clear Register.
- **Description:** Exit RASDP error mode. For more information, see the RAS Data Protection (DP) section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:1	0
RSVDP_1	ERROR_MODE_CLEAR

Table 3-231 Fields for Register: RASDP_ERROR_MODE_CLEAR_OFF

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	ERROR_MODE_CLEAR	W	Write '1' to take the controller out of RASDP error mode. The controller will then report uncorrectable errors (through AER internal error reporting) and also stop nullifying/discarding TLPs. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.34.13 RASDP_RAM_ADDR_CORR_ERROR_OFF

- **Name:** RAM Address where a corrected error (1-bit ECC) is detected.
- **Description:** For more information, see the RAS Data Protection (DP) section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RAM_INDEX_CORR_ERROR
27	RSVDP_27
26:0	RAM_ADDR_CORR_ERROR

Table 3-232 Fields for Register: RASDP_RAM_ADDR_CORR_ERROR_OFF

Bits	Name	Memory Access	Description
31:28	RAM_INDEX_CORR_ERROR	R	RAM index where a corrected error (1-bit ECC) is detected. You can use this field only for 256-bit datapath configurations, which have two receive header queue RAMs sharing an error checker. The address of the error data is logged and the index indicates which RAM had the error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
26:0	RAM_ADDR_CORR_ERROR	R	RAM Address where a corrected error (1-bit ECC) is detected. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.34.14 RASDP_RAM_ADDR_UNCORR_ERROR_OFF

- **Name:** RAM Address where an uncorrected error (2-bit ECC) is detected.
- **Description:** For more information, see the RAS Data Protection (DP) section in the Controller Operations chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x34
- **Exists:** (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RAM_INDEX_UNCORR_ERROR
27	RSVDP_27
26:0	RAM_ADDR_UNCORR_ERROR

Table 3-233 Fields for Register: RASDP_RAM_ADDR_UNCORR_ERROR_OFF

Bits	Name	Memory Access	Description
31:28	RAM_INDEX_UNCORR_ERROR	R	RAM index where an uncorrected error (2-bit ECC) is detected. You can use this field only for 256-bit datapath configurations, which have two receive header queue RAMs sharing an error checker. The address of the error data is logged and the index indicates which RAM had the error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
26:0	RAM_ADDR_UNCORR_ERROR	R	RAM Address where an uncorrected error (2-bit ECC) is detected. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_RAS_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.35 PF0_DLINK_CAP Registers

3.35.1 DATA_LINK_FEATURE_EXT_HDR_OFF

- **Name:** Data Link Feature Extended Capability Header Register.
- **Description:** This register provides capability ID, capability version, and next offset value.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2))

31:20	19:16	15:0
DLINK_NEXT_OFFSET	DLINK_CAP_VERSION	DLINK_EXT_CAP_ID

Table 3-234 Fields for Register: DATA_LINK_FEATURE_EXT_HDR_OFF

Bits	Name	Memory Access	Description
31:20	DLINK_NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h, if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. The bottom 2 bits of this offset are Reserved and must be implemented as 00b although software must mask them to allow for future uses of these bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
19:16	DLINK_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field depends on the version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
15:0	DLINK_EXT_CAP_ID	* Varies	Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. Extended Capability ID for Data Link Feature is 0025h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x25 Exists: (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

3.35.2 DATA_LINK_FEATURE_CAP_OFF

- **Name:** Data Link Feature Capabilities Register.
- **Description:** This register provides description about extended feature.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2))

31	DL_FEATURE_EXCHANGE_EN
30:23	RSVDP_23
22:1	FUTURE_FEATURE_SUPPORTED
0	SCALED_FLOW_CNTL_SUPPORTED

Table 3-235 Fields for Register: DATA_LINK_FEATURE_CAP_OFF

Bits	Name	Memory Access	Description
31	DL_FEATURE_EXCHANGE_EN	* Varies	Data Link Feature Exchange Enable. If Set, this bit indicates that this Port will enter the DL_Feature negotiation state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: DL_FEATURE_EXCHANGE_ENABLE Exists: (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
30:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
22:1	FUTURE_FEATURE_SUPPORTED	* Varies	Local Future Data Link Feature Supported. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
0	SCALED_FLOW_CNTL_SUPPORTED	* Varies	Local Scaled Flow Control Supported. Bit [0] Local Scaled Flow Control Supported Bits [22:1] are reserved Bits associated with features that this Port is capable of supporting are Hwlnit, defaulting to 1b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: SCALED_FC_SUPPORTED Exists: (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

3.35.3 DATA_LINK_FEATURE_STATUS_OFF

- **Name:** Data Link Feature Status Register.
- **Description:** This register provides status of the capability of data link feature.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2))

31	DATA_LINK_FEATURE_STATUS_VALID
30:23	RSVDP_23
22:0	REMOTE_DATA_LINK_FEATURE_SUPPORTED

Table 3-236 Fields for Register: DATA_LINK_FEATURE_STATUS_OFF

Bits	Name	Memory Access	Description
31	DATA_LINK_FEATURE_STATUS_VALID	R	Remote Data Link Feature Supported Valid. This field indicates that the Port has received a Data Link Feature DLLP in state DL_Feature and that the Remote Data Link Feature Supported and Remote Data Link Feature Ack fields are meaningful. This field is Cleared on entry to state DL_Inactive. Value After Reset: 0x0 Exists: (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
30:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2))
22:0	REMOTE_DATA_LINK_FEATURE_SUPPORTED	R	Remote Data Link Feature Supported. Bit [0] is for Remote Scaled Flow Control Supported. Bits [22:1] are Undefined. Values: ■ 0x0 (min_reset_value): Zero value. ■ 0x7ffff (max_reset): Zero value. Value After Reset: 0x0 Exists: (FC_SCALE_EN && (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true

3.36 PF0_PTM_CAP Registers

3.36.1 PTM_EXT_CAP_HDR_OFF

- **Name:** Precision Time Measurement Capability Header.
- **Description:** This Register provides Capability ID, Capability Version, and Next Offset of Precision Time Measurement Capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	PTM_NEXT_OFFSET
19:16	PTM_CAP_VERSION
15:0	PTM_CAP_ID

Table 3-237 Fields for Register: PTM_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	PTM_NEXT_OFFSET	* Varies	Precision Time Measurement PCI Express Extended Capability Next Offset. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: Pointer value calculated at configuration time. Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	PTM_CAP_VERSION	* Varies	Precision Time Measurement PCI Express Extended Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This bit depends on the version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	PTM_CAP_ID	* Varies	Precision Time Measurement PCI Express Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. PCI Express Extended Capability ID for the Precision Time Measurement Capability is 001Fh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x1f Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.36.2 PTM_CAP_OFF

- **Name:** PTM Capability Register.
- **Description:** This register describes a Function's support for Precision Time Measurement. Not all fields within this register apply to all Functions capable of implementing PTM.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	RSVDP_16
15:8	PTM_CLK_GRAN
7:4	RSVDP_4
3	EPTM_CAPABLE
2	PTM_ROOT_CAPABLE
1	PTM_RES_CAPABLE
0	PTM_REQ_CAPABLE

Table 3-238 Fields for Register: PTM_CAP_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:8	PTM_CLK_GRAN	* Varies	PTM Local Clock Granularity. For other than this value (between b00000001-b11111110) indicates the period of this Time Source's local clock in ns. If the PTM Root Select bit is Set, this local clock is used to provide PTM Master Time. Otherwise, the Time Source uses this local clock to locally track PTM Master Time received from further Upstream within a PTM Hierarchy. This field is reserved for Functions that do not implement the PTM Time Source role. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Values: ■ 0x0 (No_local_clock): Time Source does not implement a local clock. It simply propagates timing information obtained from further Upstream in the PTM Hierarchy when responding to PTM Request messages. ■ 0xff (Greater_than_254ns): Indicates the period of this Time Source's local clock is greater than 254 ns. Value After Reset: {(CX_PTM_ROOT_CAPABLE==1) ? CX_PTM_LOCAL_CLK_GRAN : 0} Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_PTM_ROOT_CAPABLE Testable: writeAsRead Volatile: true
7:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
3	EPTM_CAPABLE	* Varies	ePTM Capable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Indicates Enhanced Precision Time Management (ePTM) is supported. Defaults to 0x1. Value After Reset: CX_EPTM_CAPABLE_VALUE Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
2	PTM_ROOT_CAPABLE	* Varies	PTM Root Capable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Root Ports, RCRBs and Switches are permitted to set this bit to 1b, if they implement a PTM Time Source Role and are capable of serving as the PTM Root. Value After Reset: CX_PTM_ROOT_CAPABLE Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
1	PTM_RES_CAPABLE	* Varies	PTM Responder Capable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Root Ports and RCRB's are permitted to, and Switches supporting PTM must, set this bit to 1b to indicate they implement the PTM Responder role. If PTM Root Capable is Set, this bit must be Set to 1b. Value After Reset: CX_PTMRSPDR_VALUE Exists: (CX_PTMRSPDR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	PTM_REQ_CAPABLE	* Varies	PTM Requester Capable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Endpoints and RCIEPs are permitted to, and Switches supporting PTM must, set this bit to 1b to indicate they implement the PTM Requester role. Value After Reset: CX_PTMRQSTR_VALUE Exists: (CX_PTMRQSTR_EN && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.36.3 PTM_CONTROL_OFF

- **Name:** PTM Control Register.
- **Description:** This register controls a Function's participation in the Precision Time Measurement mechanism. Not all fields within this register apply to all Functions capable of implementing PTM.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	RSVDP_16
15:8	EFF_GRAN
7:2	RSVDP_2
1	ROOT_SELECT
0	PTM_ENABLE

Table 3-239 Fields for Register: PTM_CONTROL_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:8	EFF_GRAN	* Varies	PTM Effective Granularity. For Functions implementing the PTM Requester Role, this field provides information relating to the expected accuracy of the PTM clock, but does not otherwise affect the PTM mechanism. For Endpoints, system software must program this field to the value representing the maximum Local Clock Granularity reported by the PTM Root and all intervening PTM Time Sources. For RCiEPs, system software must set this field to the value reported in the Local Clock Granularity field by the associated PTM Time Source. For other than this value (between b00000001-b111111110) indicates the effective PTM granularity in ns. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: HWINIT Values: ■ 0x0 (No_local_clock): Unknown PTM granularity - one or more Switches between this Function and the PTM Root reported a Local Clock Granularity value of 0000 0000b. ■ 0xff (Greater_than_254ns): Indicates the effective PTM granularity is greater than 254 ns. Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
7:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
1	ROOT_SELECT	* Varies	PTM Root Select. If the value of the PTM Root Capable bit is 0b, this bit is permitted to be hardwired to 0b by the controller. Within each PTM Hierarchy, it is recommended that system software select only the furthest Upstream Time Source to be the PTM Root. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: HWINIT Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When Set, if the PTM Enable bit is also Set, this Time Source is the PTM Root. Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
0	PTM_ENABLE	R/W	PTM Enable. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When Set, this Function is permitted to participate in the PTM mechanism according to its selected role(s). Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.37 PF_PTM_REQ_CAPI Registers

3.37.1 PTM_REQ_CAP_HDR_OFF

- **Name:** Precision Time Measurement Requester Capability Header (VSEC).
- **Description:** This Register provides Capability ID, Capability Version, and Next Offset of Precision Time Measurement Capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

PTM_REQ_EXT_CAP_NEXT_OFFSETS	31:20
PTM_REQ_EXT_CAP_VER	19:16
PTM_REQ_EXT_CAP_ID	15:0

Table 3-240 Fields for Register: PTM_REQ_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	PTM_REQ_EXT_CAP_NEXT_OFFSETS	* Varies	Precision Time Measurement Requester VSEC Next Pointer. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: Pointer value calculated at configuration time. Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	PTM_REQ_EXT_CAP_VER	* Varies	Precision Time Measurement Requester VSEC Version. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: wire ? unconstrained : writeAsRead
15:0	PTM_REQ_EXT_CAP_ID	* Varies	Precision Time Measurement Requester VSEC ID. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0xb Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: wire ? unconstrained : writeAsRead

3.37.2 PTM_REQ_HDR_OFF

- **Name:** Precision Time Measurement Requester Vendor Specific Header.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

31:20	PTM_REQ_VSEC_LENGTH
19:16	PTM_REQ_VSEC_REV
15:0	PTM_REQ_VSEC_ID

Table 3-241 Fields for Register: PTM_REQ_HDR_OFF

Bits	Name	Memory Access	Description
31:20	PTM_REQ_VSEC_LENGTH	* Varies	PTM Requester VSEC Length. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: 0x54 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead

Bits	Name	Memory Access	Description
19:16	PTM_REQ_VSEC_REV	* Varies	PTM Requester VSEC Revision. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead
15:0	PTM_REQ_VSEC_ID	* Varies	PTM Requester VSEC ID. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x3 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead

3.37.3 PTM_REQ_CONTROL_OFF

- **Name:** PTM Requester Vendor Specific Control Register.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

31:17	RSVDP_17
16	PTM_REQ_PDEL_BYTE_REV
15:8	PTM_REQ_LONG_TIMER
7:3	RSVDP_3
2	PTM_REQ_FAST_TIMERS
1	PTM_REQ_START_UPDATE
0	PTM_REQ_AUTO_UPDATE_ENABLED

Table 3-242 Fields for Register: PTM_REQ_CONTROL_OFF

Bits	Name	Memory Access	Description
31:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))
16	PTM_REQ_PDEL_BYTE_REV	R/W	PTM Requester Propagation Delay Byte Reversal. Reverse the order of bytes in the PTM Propagation Delay data word of the PTM ResponseD Message, for compatibility with previous revisions. For more information, see the PTM section in the <i>Databook</i> . <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead

Bits	Name	Memory Access	Description
15:8	PTM_REQ_LONG_TIMER	R/W	PTM Requester Long Timer. Determines the period between each auto update PTM Dialogue in milliseconds. Update period is the register value +1 millisecond. For the Switch product this value must not be set larger than 0x9 for spec compliance. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Value After Reset: 0x9 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead
7:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))
2	PTM_REQ_FAST_TIMERS	R/W	PTM Fast Timers (Debug mode for PTM Timers). The 100us timer output will go high at 30us and the 10ms timer output will go high at 100us (the Long Timer Value is ignored). There is no change to the 1us timer. The requester operation will otherwise remain the same. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead
1	PTM_REQ_START_UPDATE	R/W1C	PTM Requester Start Update. For more information, see the PTM section in the <i>Databook</i>. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When set, the PTM Requester will attempt a PTM Dialogue to update its context; This bit is self clearing. Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	PTM_REQ_AUTO_UPDATE_ENABLED	R/W	PTM Requester Auto Update Enabled. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): When enabled PTM Requester will automatically attempt to update its context every 10ms. Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead

3.37.4 PTM_REQ_STATUS_OFF

- **Name:** PTM Requester Vendor Specific Status Register.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

31:2	RSVDP_2
1	PTM_REQ_MANUAL_UPDATE_ALLOWED
0	PTM_REQ_CONTEXT_VALID

Table 3-243 Fields for Register: PTM_REQ_STATUS_OFF

Bits	Name	Memory Access	Description
31:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))
1	PTM_REQ_MANUAL_UPDATE_ALLOWED	R	PTM Requester Manual Update Allowed. Indicates whether or not a Manual Update can be signalled. For more information, see the PTM section in the <i>Databook</i> . Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	PTM_REQ_CONTEXT_VALID	R	PTM Requester Context Valid. Indicate that the Timing Context is valid. For more information, see the PTM section in the <i>Databook</i>. Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.5 PTM_REQ_LOCAL_LSB_OFF

- **Name:** PTM Requester Local Clock LSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

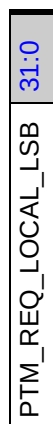


Table 3-244 Fields for Register: PTM_REQ_LOCAL_LSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_LOCAL_LSB	* Varies	PTM Requester Local Clock LSB. Lower 32 bits of local timer value. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: untestable Volatile: true

3.37.6 PTM_REQ_LOCAL_MSB_OFF

- **Name:** PTM Requester Local Clock MSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

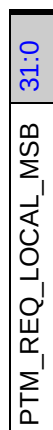


Table 3-245 Fields for Register: PTM_REQ_LOCAL_MSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_LOCAL_MSB	* Varies	PTM Requester Local Clock MSB. Upper 32 bits of local timer value. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.7 PTM_REQ_T1_LSB_OFF

- **Name:** PTM Requester T1 Timestamp LSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

PTM_REQ_T1_LSB 31:0

Table 3-246 Fields for Register: PTM_REQ_T1_LSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_T1_LSB	R	PTM Requester T1 Timestamp LSB. Lower 32 bits of the T1 Timestamp value. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.8 PTM_REQ_T1_MSB_OFF

- **Name:** PTM Requester T1 Timestamp MSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))



Table 3-247 Fields for Register: PTM_REQ_T1_MSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_T1_MSB	R	PTM Requester T1 Timestamp MSB. Upper 32 bits of the T1 Timestamp value. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.9 PTM_REQ_T1P_LSB_OFF

- **Name:** PTM Requester T1 Previous Timestamp LSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

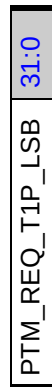


Table 3-248 Fields for Register: PTM_REQ_T1P_LSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_T1P_LSB	R	PTM Requester T1 Previous Timestamp LSB. Lower 32 bits of the previously stored T1 Timestamp value. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.10 PTM_REQ_T1P_MSB_OFF

- **Name:** PTM Requester T1 Previous Timestamp MSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

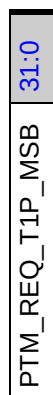


Table 3-249 Fields for Register: PTM_REQ_T1P_MSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_T1P_MSB	R	PTM Requester T1 Previous Timestamp MSB. Upper 32 bits of the previously stored T1 Timestamp value. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.11 PTM_REQ_T4_LSB_OFF

- **Name:** PTM Requester T4 Timestamp LSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

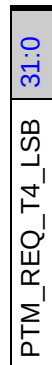


Table 3-250 Fields for Register: PTM_REQ_T4_LSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_T4_LSB	R	PTM Requester T4 Timestamp LSB. Lower 32 bits of the T4 Timestamp value. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.12 PTM_REQ_T4_MSB_OFF

- **Name:** PTM Requester T4 Timestamp MSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

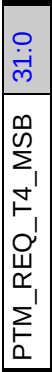


Table 3-251 Fields for Register: PTM_REQ_T4_MSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_T4_MSB	R	PTM Requester T4 Timestamp MSB. Upper 32 bits of the T4 Timestamp value. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.13 PTM_REQ_T4P_LSB_OFF

- **Name:** PTM Requester T4 Previous Timestamp LSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

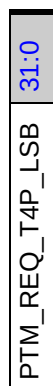
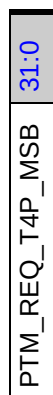


Table 3-252 Fields for Register: PTM_REQ_T4P_LSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_T4P_LSB	R	PTM Requester T4 Previous Timestamp LSB. Lower 32 bits of the previously stored T4 Timestamp value. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.14 PTM_REQ_T4P_MSB_OFF

- **Name:** PTM Requester T4 Previous Timestamp MSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x34
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

**Table 3-253 Fields for Register: PTM_REQ_T4P_MSB_OFF**

Bits	Name	Memory Access	Description
31:0	PTM_REQ_T4P_MSB	R	PTM Requester T4 Previous Timestamp MSB. Upper 32 bits of the previously stored T4 Timestamp value. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.15 PTM_REQ_MASTER_LSB_OFF

- **Name:** >PTM Requester Master Time LSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x38
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

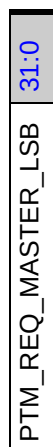
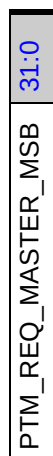


Table 3-254 Fields for Register: PTM_REQ_MASTER_LSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_MASTER_LSB	R	PTM Requester Master Time LSB. Lower 32 bits of Master Time value sent from PTM Responder. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field contains the lower 32 bits of TMU_to_PTM_B parameter extracted from the PTM Responder ResponseD message. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.16 PTM_REQ_MASTER_MSB_OFF

- **Name:** PTM Requester Master Time MSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x3c
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

**Table 3-255 Fields for Register: PTM_REQ_MASTER_MSB_OFF**

Bits	Name	Memory Access	Description
31:0	PTM_REQ_MASTER_MSB	R	PTM Requester Master Time MSB. Upper 32 bits of Master Time value sent from PTM Responder. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field contains the upper 32 bits of TMU_to_PTM_B parameter extracted from the PTM Responder ResponseD message. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.17 PTM_REQ_PROP_DELAY_OFF

- **Name:** PTM Requester Propagation Delay.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x40
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

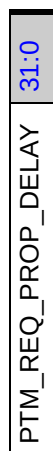


Table 3-256 Fields for Register: PTM_REQ_PROP_DELAY_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_PROP_DELAY	R	PTM Requester Propagation Delay. Propagation Delay sent from the PTM Responder. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field contains the TMU_to_PTM_A parameter extracted from the PTM Responder ResponseD message. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.18 PTM_REQ_MASTERT1_LSB_OFF

- **Name:** PTM Requester Master Time at T1 LSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x44
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))



Table 3-257 Fields for Register: PTM_REQ_MASTERT1_LSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_MASTERT1_LSB	R	PTM Requester Master Time at T1 LSB. Lower 32 bits of the calculated Master Time at T1 prime. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field contains the lower 32 bits of the PTM Master Time reconstructed from TMU_to_PTM_A and TMU_to_PTM_B parameters. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.19 PTM_REQ_MASTERT1_MSB_OFF

- **Name:** PTM Requester Master Time at T1 MSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x48
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

 31:0
PTM_REQ_MASTERT1_MSB

Table 3-258 Fields for Register: PTM_REQ_MASTERT1_MSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_MASTERT1_MSB	R	PTM Requester Master Time at T1 MSB. Upper 32 bits of the calculated Master Time at T1 prime. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field contains the upper 32 bits of the PTM Master Time reconstructed from TMU_to_PTM_A and TMU_to_PTM_B parameters. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.20 PTM_REQ_TX_LATENCY_OFF

- **Name:** PTM Requester TX Latency.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x4c
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

31:12	RSVDP_12
11:0	PTM_REQ_TX_LATENCY

Table 3-259 Fields for Register: PTM_REQ_TX_LATENCY_OFF

Bits	Name	Memory Access	Description
31:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

Bits	Name	Memory Access	Description
11:0	PTM_REQ_TX_LATENCY	R/W	PTM Requester TX Latency. Requester Transmit path latency viewport register. A register is provided for each supported link speed, and the value used for timestamp adjustment is automatically selected based on the current speed. The PTM Requester Latency Register Select register specifies the register to be written or read. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: {CX_PHY_TX_DELAY_MAC + CX_PHY_TX_DELAY_PHY} Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead

3.37.21 PTM_REQ_RX_LATENCY_OFF

- **Name:** PTM Requester RX Latency.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x50
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))



Table 3-260 Fields for Register: PTM_REQ_RX_LATENCY_OFF

Bits	Name	Memory Access	Description
31:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

Bits	Name	Memory Access	Description
11:0	PTM_REQ_RX_LATENCY	R/W	PTM Requester RX Latency - Requester Receive path latency viewport register. A register is provided for each supported link speed, and the value used for timestamp adjustment is automatically selected based on the current speed. The PTM Requester Latency Register Select register specifies the register to be written or read. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: {CX_PHY_RX_DELAY_MAC + CX_PHY_RX_DELAY_PHY} Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead

3.37.22 PTM_REQ_CLOCK_CORR_LSB_OFF

- **Name:** PTM Requester Clock Correction LSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x54
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

PTM_REQ_CLOCK_CORR_LSB

31:0

Table 3-261 Fields for Register: PTM_REQ_CLOCK_CORR_LSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_CLOCK_CORR_LSB	R	PTM Requester Clock Correction LSB. Lower 32 bits of amount by which PTM Requester Local Clock has been corrected. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.23 PTM_REQ_CLOCK_CORR_MSB_OFF

- **Name:** PTM Requester Clock Correction MSB.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x58
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))



Table 3-262 Fields for Register: PTM_REQ_CLOCK_CORR_MSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_CLOCK_CORR_MSB	R	PTM Requester Clock Correction MSB. Upper 32 bits of amount by which PTM Requester Local Clock has been corrected. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.24 PTM_REQ_NOM_CLOCK_T_OFF

- **Name:** PTM Requester Nominal Clock Period.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x5c
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

31:24	RSVDP_24
23:16	PTM_REQ_NOM_CLOCK_T_INT
15:0	PTM_REQ_NOM_CLOCK_T_FRAC

Table 3-263 Fields for Register: PTM_REQ_NOM_CLOCK_T_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

Bits	Name	Memory Access	Description
23:16	PTM_REQ_NOM_CLOCK_T_INT	R	PTM Requester Nominal Clock Period Integral. Integral part of the nominal PTM local clock period. LSB is $1/(2^{16})$ nanoseconds. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: untestable Volatile: true
15:0	PTM_REQ_NOM_CLOCK_T_FRA C	R	PTM Requester Nominal Clock Period Fractional. Fractional part of the nominal PTM local clock period, in nanoseconds. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.25 PTM_REQ_SCALED_CLOCK_T_OFF

- **Name:** PTM Requester Scaled Clock Period.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x60
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

31	PTM_REQ_SCALED_CLOCK_T_EN
30:24	RSVDP_24
23:16	PTM_REQ_SCALED_CLOCK_T_INT
15:0	PTM_REQ_SCALED_CLOCK_T_FRAC

Table 3-264 Fields for Register: PTM_REQ_SCALED_CLOCK_T_OFF

Bits	Name	Memory Access	Description
31	PTM_REQ_SCALED_CLOCK_T_EN	R/W	PTM Requester Scaled Clock Period Enable. This bit is cleared when the core_clk rate starts to change and can only be set when the clock period change is complete. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Use the programmed scaled PTM clock period rather than the nominal values. Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
30:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))
23:16	PTM_REQ_SCALED_CLOCK_T_INT	R/W	PTM Requester Scaled Clock Period Integral. Integral part of the nominal PTM local clock period, in nanoseconds. For more information, see the PTM section in the <i>Databook</i> . <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead
15:0	PTM_REQ_SCALED_CLOCK_T_FRAC	R/W	PTM Requester Scaled Clock Period Fractional. Fractional part of the nominal PTM local clock period. LSB is 1/(2 ¹⁶) nanoseconds. For more information, see the PTM section in the <i>Databook</i> . <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead

3.37.26 PTM_REQ_LATENCY_REG_SEL_OFF

- **Name:** PTM Requester Latency Viewport Register Select.
- **Description:** For more information, see the PTM section in the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x64
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

31:8	RSVDP_8
7	PTM_REQ_ESM_SEL
6:4	RSVDP_4
3:0	PTM_REQ_LATENCY_REG_SEL

Table 3-265 Fields for Register: PTM_REQ_LATENCY_REG_SEL_OFF

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

Bits	Name	Memory Access	Description
7	PTM_REQ_ESM_SEL	R/W	Selection Mode for PTM Requester Tx and Rx Latency Viewport register to be written or read. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: 0x0 (SEL_PCIE_REGS): write or read latency registers for PCIe rates 0x1 (SEL_CCIX_ESM_REGS): write or read latency registers for additional ESM rates Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead
6:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))
3:0	PTM_REQ_LATENCY_REG_SEL	R/W	Selects the PTM Requester Tx and Rx Latency Viewport register to be written or read, together with the PTM_REQ_ESM_SEL field. <i>Note:</i> For USB4, when PTM_REQ_USB4_MODE field of PTM_REQ_USB4_CIO_CTRL register is set, this field is not relevant. For more information, see the PTM section in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VAL): Min value 0xf (MAX_VAL): Max value Value After Reset: 0x0 Exists: (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead

3.37.27 PTM_REQ_USB4_CIO_CTRL_OFF

- **Name:** PTM Requester USB4 Control Register.
- **Description:** You can use this register to control PTM requester operation of the PCIe controller for USB4 configuration.
- **Size:** 32 bits
- **Offset:** B+0x68
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

31	PTM_REQ_USB4_MODE
30:16	RSVDP_16
15:0	PTM_REQ_TMU_REQ_TIMER

Table 3-266 Fields for Register: PTM_REQ_USB4_CIO_CTRL_OFF

Bits	Name	Memory Access	Description
31	PTM_REQ_USB4_MODE	R/W	PTM Requester USB4 mode. <i>Note:</i> If PTM_REQ_USB4_MODE field is set in the PTM responder, it should also be set in the PTM requester to ensure correct operation. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): When set, the controller derives PTM Master Time using TMU_to_PTM_A and TMU_to_PTM_B parameters. Value After Reset: 0x0 Exists: CX_PCIE_OVER_CIO_ENABLE && (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead

Bits	Name	Memory Access	Description
30:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_PCIE_OVER_CIO_ENABLE && (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))
15:0	PTM_REQ_TMU_REQ_TIMER	R/W	PTM Requester TMU Request Timer. The controller instantiates an internal timer with a time period equivalent to this field. Each time the timer expires, the controller ■ Collaborates with USB4 TMU and obtains USB4 TMU timestamp (tmu_t) when USB4 TMU asserts phy_mac_ts_ack ■ Extracts the TMU_to_PTM_A and TMU_to_PTM_B parameters from the most recent ResponseD message ■ Calculates the PTM Master Time using these values You must set this field to a non-zero value greater than 8 for accurate PTM Master Time calculation. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: CX_PCIE_OVER_CIO_ENABLE && (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead

3.37.28 PTM_REQ_TMU_TS_LSB_OFF

- **Name:** PTM Requester Reference USB4 TMU Timestamp Sample LSB Register.
- **Description:** This register holds the lower 32 bits of the reference USB4 TMU timestamp value. For more information, see "Precision Time Management (PTM)" section in the "PCIe for USB4" chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x6c
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

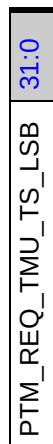


Table 3-267 Fields for Register: PTM_REQ_TMU_TS_LSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_TMU_TS_LSB	R	Lower 32 bits of the reference USB4 TMU timestamp value (tmu_t). <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: CX_PCIE_OVER_CIO_ENABLE && (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.37.29 PTM_REQ_TMU_TS_MSB_OFF

- **Name:** PTM Requester Reference USB4 TMU Timestamp Sample MSB Register.
- **Description:** This register holds the upper 32 bits of the reference USB4 TMU timestamp value. For more information, see "Precision Time Management (PTM)" section in the "PCIe for USB4" chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x70
- **Exists:** (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2))

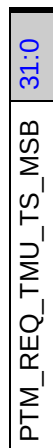


Table 3-268 Fields for Register: PTM_REQ_TMU_TS_MSB_OFF

Bits	Name	Memory Access	Description
31:0	PTM_REQ_TMU_TS_MSB	R	Upper 32 bits of the reference USB4 TMU timestamp value (tmu_t). <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: CX_PCIE_OVER_CIO_ENABLE && (CX_PTM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE<=2)) Testable: writeAsRead Volatile: true

3.38 PF0_CCIX_TP_CAP Registers

3.38.1 CCIX_TP_CAP_HDR_OFF

- **Name:** CCIX Transport DVSEC Capability Header Register.
- **Description:** This register provides, PCIe extended capability ID, version, and next capability offset value.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	CCIX_TP_NEXT_OFFSET
19:16	CCIX_TP_CAP_VERSION
15:0	CCIX_TP_CAP_ID

Table 3-269 Fields for Register: CCIX_TP_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	CCIX_TP_NEXT_OFFSET	* Varies	Next Capability Offset. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CCIX_TP_CAP_VERSION	* Varies	Capability Version. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: wire ? unconstrained : writeAsRead
15:0	CCIX_TP_CAP_ID	* Varies	PCI Express Extended Capability ID. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x23 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: wire ? unconstrained : writeAsRead

3.38.2 CCIX_TP_HDR1_OFF

- **Name:** Designated Vendor Specific Header 1.
- **Description:** This register provides DVSEC vendor ID, capability version, and DVSEC Len0 values.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	CCIX_DVSEC_LEN0
19:16	CCIX_DVSEC_VERSION
15:0	CCIX_DVSEC_VID

Table 3-270 Fields for Register: CCIX_TP_HDR1_OFF

Bits	Name	Memory Access	Description
31:20	CCIX_DVSEC_LEN0	* Varies	DVSEC Len0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x40 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CCIX_DVSEC_VERSION	* Varies	Capability Version. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: wire ? unconstrained : writeAsRead
15:0	CCIX_DVSEC_VID	* Varies	DVSEC Vendor ID (CCIX). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: wire ? unconstrained : writeAsRead

3.38.3 CCIX_TP_CAP_TP_HDR2_OFF

- **Name:** Designated Vendor Specific Header 2 and Transport Capabilities Register.
- **Description:** This register specifies the designated vendor specific header 2 and ESM specific capabilities.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_27	31:27
ESM_QUICK_EQ_TIMEOUT	26:24
RSVDP_23	23
ESM_CALIBRATION_TIME	22:20
ESM_RECALIBRATION	19
ESM_REACH_LENGTH	18:17
ESM_MODE_SUPPORT	16
CCIX_DVSEC_ID0	15:0

Table 3-271 Fields for Register: CCIX_TP_CAP_TP_HDR2_OFF

Bits	Name	Memory Access	Description
31:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

Bits	Name	Memory Access	Description
26:24	ESM_QUICK_EQ_TIMEOUT	* Varies	ESMQuickEqualizationTimeout. Minimum Equalization Phase 2 and Phase 3 to Recovery Speed Timeout supported for Data Rates greater than 16.0 GTs/, when ESMControl.QuickEqualizationTimeoutSelect is programmed to a value greater than 000b. Defined encodings are: All other encodings are Reserved. If supported, devices must advertise ESM Quick Equalization Timeout that is less than or equal to ESMControl.ESMExtendedEqualizationPhase2Timeout on the USP or ESMControl.ESMExtendedEqualizationPhase3Timeout on the DSP. If ESMModeSupported is Clear, this field is RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Values: ■ 0x0 (NOT_SUP): Quick Equalization is not supported by the device ■ 0x1 (_8_TO_16_MS): 8 ms / 16 ms ■ 0x2 (_24_TO_32MS): 24 ms / 32 ms ■ 0x3 (_50_TO_58MS): 50 ms / 58 ms ■ 0x4 (_100_TO_108MS): 100 ms / 108 ms ■ 0x5 (_200_TO_208MS): 200 ms / 208 ms Value After Reset: {DEFAULT_CX_CCIX_ESM_QUICK_EQ_TIME} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

Bits	Name	Memory Access	Description
22:20	ESM_CALIBRATION_TIME	* Varies	ESMCalibrationTime. Indicates the maximum time taken by the Physical Layer to complete the Calibration step. Defined encodings are: If ESMModeSupported is Clear, this field is RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Values: ■ 0x0 (_10_US): 10 us ■ 0x1 (_50_US): 50 us ■ 0x2 (_100_US): 100 us ■ 0x3 (_500_US): 500 us ■ 0x4 (_1_MS): 1 ms ■ 0x5 (_5_MS): 5 ms ■ 0x6 (_10_MS): 10 ms ■ 0x7 (_50_MS): 50 ms Value After Reset: {DEFAULT_CX_CCIX_ESM_CAL_TIME} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	ESM_RECALIBRATION	* Varies	This field is set if the Physical Layer requires re-calibration, if ESMControl.ESMDataRate0 or ESMDataRate1 fields are modified (to change Data Rates), after initial or subsequent calibration and entry into LTSSM state L0 at the Data Rate programmed in ESMDataRate1. This field is clear if the Physical Layer does not require re-calibration, after the initial calibration and if ESMControl.ESMDataRate0 or ESMDataRate1 fields are subsequently modified (to change Data Rates). PHYs that do not need to be re-calibrated are permitted to hardwire this bit to 0b. If ESMModeSupported is Clear, this field is RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: {DEFAULT_CX_CCIX_ESM_RECAL_NEEDED} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
18:17	ESM_REACH_LENGTH	* Varies	ESMPHYReachLengthCapability. Indicates the Reach Length Capability at ESM Data Rate1 only. Defined encodings are: If ESMModeSupported is Clear, this field is RsvdP. At 20.0 GT/s and 25.0 GT/s data rates, Long Reach Capable PHY must also be capable of driving short channels. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Values: ■ 0x0 (SHORT_REACH_CAPA): PHY is Short Reach Capable ■ 0x1 (LONG_REACH_CAPA): PHY is Long Reach Capable ■ 0x2 (SHORT_LONG_REACH_CAPA): PHY is Short Reach and Long Reach Capable ■ 0x3 (RSERVED): Reserved for future use Value After Reset: {DEFAULT_CX_CCIX_ESM_REACH_LENGTH} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	ESM_MODE_SUPPORT	* Varies	ESMModeSupported. This field is set if Physical Layer supports ESM Mode, otherwise it is clear. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: {CCIX_ESM_SUPPORT_VALUE} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:0	CCIX_DVSEC_ID0	* Varies	DVSEC ID0 (TRANSPORT). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: wire ? unconstrained : writeAsRead

3.38.4 ESM_MNDTRY_RATE_CAP_OFF

- **Name:** ESM Mandatory Data Rate Capability Register.
- **Description:** This register advertises the ESM data rates supported.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_15	31:15
CCIX_25G_SUPPORT	14
RSVDP_10	13:10
CCIX_20G_SUPPORT	9
RSVDP_6	8:6
CCIX_16G_SUPPORT	5
RSVDP_3	4:3
CCIX_8G_SUPPORT	2
CCIX_5G_SUPPORT	1
CCIX_2P5G_SUPPORT	0

Table 3-272 Fields for Register: ESM_MNDTRY_RATE_CAP_OFF

Bits	Name	Memory Access	Description
31:15	RSVDP_15	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

Bits	Name	Memory Access	Description
14	CCIX_25G_SUPPORT	* Varies	ESM25GTSupport. Must be Set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: {CCIX_ESM_SUPPORT_VALUE} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
13:10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE
9	CCIX_20G_SUPPORT	* Varies	ESM20GTSupport. Must be Set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: {CCIX_ESM_SUPPORT_VALUE} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
8:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE
5	CCIX_16G_SUPPORT	* Varies	ESM16GTSupport. Must be Set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: 0x1 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\" "
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

Bits	Name	Memory Access	Description
2	CCIX_8G_SUPPORT	* Varies	ESM8GTSupport. Must be Set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: 0x1 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\" "
1	CCIX_5G_SUPPORT	* Varies	ESM5GTSupport. Must be Set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: 0x1 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
0	CCIX_2P5G_SUPPORT	* Varies	ESM2_5GTSupport. Must be Set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: 0x1 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

3.38.5 ESM_STAT_OFF

- **Name:** ESM Status Register.
- **Description:** This register provides information about ESM specific parameters.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:8	RSVDP_8
7	ESM_CALIB_CMPLT
6:0	ESM_CURNT_DATA_RATE

Table 3-273 Fields for Register: ESM_STAT_OFF

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE
7	ESM_CALIB_CMPLT	R	ESMCalibrationComplete. Set after ESM physical layer calibration is complete. Physical layer calibration is initiated by Setting ESMControl.ESMPerformCalibration. Default value of this bit is 0b. Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
6:0	ESM_CURNT_DATA_RATE	R	ESMCurrentDataRate. Hardware indicates the negotiated ESM data rate. Defined encodings are: All other encodings are Reserved. Default value of this field is 00 0000b. Values: ■ 0x0 (ESM_INACTIVE): ESM Inactive ■ 0x1 (_2_5GT): 2.5 GT/s ■ 0x2 (_5GT): 5.0 GT/s ■ 0x3 (_8GT): 8.0 GT/s ■ 0x6 (_16GT): 16.0 GT/s ■ 0xa (_20GT): 20.0 GT/s ■ 0xf (_25GT): 25.0 GT/s Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: writeAsRead Volatile: true

3.38.6 ESM_CNTL_OFF

- **Name:** ESM Control Register.
- **Description:** This register controls ESM specific parameters.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_29	31:29
QUICK_EQ_TIMEOUT	28:26
RSVDP_25	25
LINK_REACH_TARGET	24
RSVDP_23	23
ESM_EXT_EQ3_DSP_TIMEOUT	22:20
ESM_COMPLIANCE	19
ESM_EXT_EQ2_USP_TIMEOUT	18:16
ESM_ENABLE	15
ESM_DATA_RATE1	14:8
ESM_PERFORM_CAL	7
ESM_DATA_RATE0	6:0

Table 3-274 Fields for Register: ESM_CNTL_OFF

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

Bits	Name	Memory Access	Description
28:26	QUICK_EQ_TIMEOUT	R/W	QuickEqualizationTimeoutSelect. Defined encodings are: All other encodings are Reserved. The default value of this field is equal to a value in CCIXLinkTransportCapabilities.ESMQuickEqualizationTimeout. For Data Rates higher than 16.0 GT/s, if CCIXTransportCapabilities.ESMQuickEqualizationTimeout is greater than 000b, setting QuickEqualizationTimeoutSelect to value greater than 000b causes the corresponding QuickEqualizationTimeout value to be used during the Recovery Equalization Phase2/Phase3 to Recovery Speed transition in place of the Timeouts specified in the ESMControl.ESMExtendedEqualizationPhase2Timeout and ESMControl.ESMExtendedEqualizationPhase3Timeout fields. Setting QuickEqualizationTimeoutSelect to 000b disables the Quick Equalization mechanism. If the QuickEqualizationTimeoutSelect field is to be programmed to a non-zero value, it must be programmed to an encoding corresponding to a Timeout equal to or greater than that advertised in CCIXTransportCapabilities.ESMQuickEqualizationTimeout, otherwise the resulting behaviour is undefined. Setting ESMControl.QuickEqualizationTimeoutSelect to a value other than 000b has no effect on behaviour if CCIXTransportCapabilities.ESMQuickEqualizationTimeout is 000b. See Implementation Note "ESM Control Register Quick Equalization Timeout Select field" for usage guidelines. Values: ■ 0x0 (DISABLED): Quick Equalization is Disabled ■ 0x1 (_8_TO_16_MS): 8 ms / 16 ms ■ 0x2 (_24_TO_32MS): 24 ms / 32 ms ■ 0x3 (_50_TO_58MS): 50 ms / 58 ms ■ 0x4 (_100_TO_108MS): 100 ms / 108 ms ■ 0x5 (_200_TO_208MS): 200 ms / 208 ms Value After Reset: {DEFAULT_CX_CCIX_ESM_QUICK_EQ_TIME} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE
25	RSVDP_25	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

Bits	Name	Memory Access	Description
24	LINK_REACH_TARGET	R/W	LinkReachTarget. This bit indicates the Reach Length Target of current link. Defined encodings are: When ESMCompliance is Set, this field is RWS, else it is RW. The default value of this bit is 0b. SSW should configure this bit to indicate the target reach length of the Link before initiating the ESM physical layer calibration, based on the platform channel capability (Compliance to SR or LR) and ESMPHYReachLengthCapability of the DSP and USP. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (SR): Reach Length Target of current link is Short Reach (SR) ■ 0x1 (LR): Reach Length Target of current link is Long Reach (LR) Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: writeAsRead Volatile: true
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

Bits	Name	Memory Access	Description
22:20	ESM_EXT_EQ3_DSP_TIMEOUT	R/W	ESMExtendedEqualizationPhase3Timeout. LTSSM Equalization Phase3 Timeout For Downstream Port / Upstream Port when the current data rate is higher than 16.0 GT/s. Defined encodings are: All other encodings are Reserved. 1. For a DSP, if CCIXTransportCapabilities.ESMPHYReachLengthCapability is a defined value other than 00b, hardware must initialize this field to indicate its Equalization Phase3 Timeout requirement, for a Long Reach Link during ESM Data Rate1 link initialization. 2. For a DSP, if CCIXTransportCapabilities.ESMPHYReachLengthCapability is 00b (PHY is only Short Reach Capable), hardware must initialize this field to 000b 3. For an USP, SSW must program this field taking into account the value of the LinkReachTarget bit and the requested value of ESM Extended Equalization Phase3 Timeout value from the DSP. Values: ■ 0x0 (_24_TO_32MS): 24 ms / 32 ms ■ 0x1 (_50_TO_58MS): 50 ms / 58 ms ■ 0x2 (_100_TO_108MS): 100 ms / 108 ms ■ 0x3 (_200_TO_208MS): 200 ms / 208 ms ■ 0x4 (_400_TO_408MS): 400 ms / 408 ms ■ 0x5 (_600_TO_608MS): 600 ms / 608 ms Value After Reset: DEFAULT_CX_CCIX_ESM_EXT_EQ3_DSP_TIMEOUT Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE
19	ESM_COMPLIANCE	R/W	ESMCompliance. This bit must be Set when performing certain compliance testing. Requirements related to the ESMCompliance bit will be called out by individual compliance procedures. Default value is of this bit is 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

Bits	Name	Memory Access	Description
18:16	ESM_EXT_EQ2_USP_TIMEOUT	R/W	ESMExtendedEqualizationPhase2Timeout. LTSSM Equalization Phase2 Timeout For Upstream Port / Downstream Port when the current data rate is higher than 16.0 GT/s. Defined encodings are: All other encodings are Reserved. 1. For an USP, if CCIXTransportCapabilities.ESMPHYReachLengthCapability is a defined value other than 00b, hardware must initialize this field to indicate its Equalization Phase2 Timeout requirement for a Long Reach Link during ESM Data Rate1 link initialization. 2. For an USP, if CCIXTransportCapabilities.ESMPHYReachLengthCapability is 00b (PHY is only Short Reach Capable) hardware must initialize this field to 000b. 3. For a DSP, SSW must program this field, taking into account the value of the LinkReachTarget bit and the requested value of the ESMExtendedEqualizationPhase2Timeout field from the USP. Values: ■ 0x0 (_24_TO_32MS): 24 ms / 32 ms ■ 0x1 (_50_TO_58MS): 50 ms / 58 ms ■ 0x2 (_100_TO_108MS): 100 ms / 108 ms ■ 0x3 (_200_TO_208MS): 200 ms / 208 ms ■ 0x4 (_400_TO_408MS): 400 ms / 408 ms ■ 0x5 (_600_TO_608MS): 600 ms / 608 ms Value After Reset: {(DEFAULT_CX_CCIX_ESM_REACH_LENGTH == 0) ? 0 : DEFAULT_CX_CCIX_ESM_EXT_EQ2_USP_TIMEOUT } Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

Bits	Name	Memory Access	Description
15	ESM_ENABLE	R/W	ESMEnable. When the bit is 0b, a write of 1b initiates the transition to ESM, when the Link is operating at 2.5 GT/s or 5.0 GT/s, and the Link is operating in PCI Express Mode. When the bit is 0b, writing this bit to 1b when the Data Rate is not 2.5 GT/s, or not operating in PCI Express Mode, produces undefined results. When system software program (set or reset) it from DSP, the software must program it to the remote USP first and then program it to the local DSP. When ESMCompliance is Set, this field is RWS, else it is RW. Default value of this bit is 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: writeAsRead Volatile: true
14:8	ESM_DATA_RATE1	R/W	ESMDataRate1. Sets the ESM Data Rate to be used instead of normal 16.0 GT/s. Programmed by SSW before entering ESM. The programmed data rate must be a supported data rate in either the ESMandatoryDataRateCapability or ESMOptionalDataRateCapability registers, otherwise the result is undefined. Defined encodings are: All other encodings are Reserved. When ESMCompliance is Set, this field is RWS, else it is RW. Default value of this field is b0000000. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NO_SPEED): No Speed ■ 0x6 (_16GT): 16.0 GT/s ■ 0xa (_20GT): 20.0 GT/s ■ 0xf (_25GT): 25.0 GT/s Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
7	ESM_PERFORM_CAL	W	ESMPerformCalibration. A write of 1b initiates the start of physical layer calibration before transition to ESM. Physical layer calibration is performed after the Link has transitioned to a non-D0 state through Programmed Power Management (Link enters the LTSSM L1 state when idle). Hardware sets ESMStatus.ESMCalibrationComplete to indicate that physical layer calibration is complete. When ESMCompliance is Set, this field is RWS, else it is RW. This bit always returns 0b when read. Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: writeAsRead Volatile: true
6:0	ESM_DATA_RATE0	R/W	ESMDataRate0. Sets the ESM Data Rate to be used in place of normal 8.0 GT/s. Programmed by SSW before entering ESM. The programmed data rate must be a supported data rate in either the ESMandatoryDataRateCapability or ESMOptionalDataRateCapability registers, otherwise the result is undefined. Defined encodings are: All other encodings are Reserved. When ESMCompliance is Set, this field is RWS, else it is RW. Default value of this field is 000 0000b. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NO_SPEED): No Speed ■ 0x3 (_8GT): 8.0 GT/s ■ 0x6 (_16GT): 16.0 GT/s Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: writeAsRead Volatile: true

3.38.7 ESM_LN_EQ_CNTL_20G_i_OFF (for i = 0; i <= 12)

- **Name:** ESM Lane Equalization Control Register for 20.0 GT/s for Lanes 0-3.
- **Description:** This register holds information for Lanes 0-3. The entry for each lane contains the value for the Lane, with the corresponding Lane number which is invariant to Link width and Lane reversal negotiation that occurs during Link training.
- **Size:** 32 bits
- **Offset:** B+0x1c + i*0x4
- **Exists:** (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 | | CX_MEMORY_MAP_PCIE_TYPE==0))

ESM_USP_TX_PRESET_20G_3	31:28
ESM_DSP_TX_PRESET_20G_3	27:24
ESM_USP_TX_PRESET_20G_2	23:20
ESM_DSP_TX_PRESET_20G_2	19:16
ESM_USP_TX_PRESET_20G_1	15:12
ESM_DSP_TX_PRESET_20G_1	11:8
ESM_USP_TX_PRESET_20G_0	7:4
ESM_DSP_TX_PRESET_20G_0	3:0

Table 3-275 Fields for Register: ESM_LN_EQ_CNTL_20G_i_OFF (for i = 0; i <= 12)

Bits	Name	Memory Access	Description
31:28	ESM_USP_TX_PRESET_20G_3	* Varies	For 20G Lane3 high, Refer to ESM_USP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: RSVDP Value After Reset: <code>{{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_USP_TX_PRESET_20G3" : "DEFAULT_LOCAL_G4_TX_PRESET"}}</code> Exists: <code>(CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE && CX_NL > 2</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""</code>

Bits	Name	Memory Access	Description
27:24	ESM_DSP_TX_PRESET_20G_3	* Varies	For 20G Lane3 low,Refer to ESM_DSP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: $\{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_DSP_TX_PRESET_20G3" : "0x0"\}$ Exists: $(CX_CCIX_ENABLE \&\& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) \&\& CX_CCIX_ENABLE \&\& CX_NL > 2$ Testable: $\{(CX_CROSSLINK_ENABLE==0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 \&\& CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained")\}$ Memory Access: $"(CX_MEMORY_MAP_VIEW==0 \&\& CX_DBI_RO_WR_EN \&\& CX_CROSSLINK_ENABLE) ? "\text{read-write}" : (CX_MEMORY_MAP_VIEW==0) ? "\text{read-only}" : (CX_MEMORY_MAP_VIEW==1) ? "\text{read-only}" : "\text{read-only}" "$
23:20	ESM_USP_TX_PRESET_20G_2	* Varies	For 20G Lane2 high,Refer to ESM_USP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: RSVDP Value After Reset: $\{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_USP_TX_PRESET_20G2" : "DEFAULT_LOCAL_G4_TX_PRESET"\}$ Exists: $(CX_CCIX_ENABLE \&\& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) \&\& CX_CCIX_ENABLE \&\& CX_NL > 2$ Testable: $\{(CX_CROSSLINK_ENABLE==0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 \&\& CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained")\}$ Memory Access: $"(CX_MEMORY_MAP_VIEW==0 \&\& CX_DBI_RO_WR_EN \&\& CX_CROSSLINK_ENABLE) ? "\text{read-write}" : (CX_MEMORY_MAP_VIEW==0) ? "\text{read-only}" : (CX_MEMORY_MAP_VIEW==1) ? "\text{read-only}" : "\text{read-only}" "$

Bits	Name	Memory Access	Description
19:16	ESM_DSP_TX_PRESET_20G_2	* Varies	For 20G Lane2 low,Refer to ESM_DSP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: $\{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_DSP_TX_PRESET_20G2" : "0x0"\}$ Exists: $(CX_CCIX_ENABLE \&\& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) \&\& CX_CCIX_ENABLE \&\& CX_NL > 2$ Testable: $\{(CX_CROSSLINK_ENABLE==0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 \&\& CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained")\}$ Memory Access: $"(CX_MEMORY_MAP_VIEW==0 \&\& CX_DBI_RO_WR_EN \&\& CX_CROSSLINK_ENABLE) ? \backslash\text{read-write}\backslash" : (CX_MEMORY_MAP_VIEW==0) ? \backslash\text{read-only}\backslash" : (CX_MEMORY_MAP_VIEW==1) ? \backslash\text{read-only}\backslash" : \backslash\text{read-only}\backslash"$
15:12	ESM_USP_TX_PRESET_20G_1	* Varies	For 20G Lane1 high,Refer to ESM_USP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: RSVDP Value After Reset: $\{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_USP_TX_PRESET_20G1" : "DEFAULT_LOCAL_G4_TX_PRESET"\}$ Exists: $(CX_CCIX_ENABLE \&\& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) \&\& CX_CCIX_ENABLE \&\& CX_NL > 1$ Testable: $\{(CX_CROSSLINK_ENABLE==0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 \&\& CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained")\}$ Memory Access: $"(CX_MEMORY_MAP_VIEW==0 \&\& CX_DBI_RO_WR_EN \&\& CX_CROSSLINK_ENABLE) ? \backslash\text{read-write}\backslash" : (CX_MEMORY_MAP_VIEW==0) ? \backslash\text{read-only}\backslash" : (CX_MEMORY_MAP_VIEW==1) ? \backslash\text{read-only}\backslash" : \backslash\text{read-only}\backslash"$

Bits	Name	Memory Access	Description
11:8	ESM_DSP_TX_PRESET_20G_1	* Varies	For 20G Lane1 low,Refer to ESM_DSP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: <code>{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_DSP_TX_PRESET_20G1" : "0x0"}</code> Exists: <code>(CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE && CX_NL > 1</code> Testable: <code>{(CX_CROSSLINK_ENABLE== 0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""</code>

Bits	Name	Memory Access	Description
7:4	ESM_USP_TX_PRESET_20G_0	* Varies	UpstreamPortESMTransmitterPreset. Field contains the Transmit Preset value sent or received during ESM Link Equalization. The field usage varies as follows: ■ Case A: If Operating port direction is Downstream Port, and Crosslink Supported is Any, □ This field contains the value sent on the associated Lane during Link Equalization. □ Field is HwInit. ■ Case B: If Operating port direction is Upstream Port, and Crosslink Supported is 0b, □ Field is intended for debug and diagnostics. It contains the value captured from the associated Link during Link Equalization. □ Field is RO. □ When crosslinks are supported, case C applies and this captured information is not visible to software. □ Vendors are encouraged to provide an alternate mechanism to obtain this information. ■ Case C: If Operating port direction is Upstream Port, and Crosslink Supported is 1b, □ Field is not used or affected by the current Link Equalization. □ Field value will be used if a future crosslink negotiation switches the Operating Port Direction so that case A applies. □ Field is HwInit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: RSVDP Value After Reset: <code>{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_USP_TX_PRESET_20G0" : "DEFAULT_LOCAL_G4_TX_PRESET"}</code> Exists: <code>(CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE</code> Testable:

Bits	Name	Memory Access	Description
7:4...(cont.)	ESM_USP_TX_PRESET_20G_0.	* Varies	{(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") } Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
3:0	ESM_DSP_TX_PRESET_20G_0	* Varies	DownstreamPortESMTransmitterPreset. Transmitter Preset used for ESM equalization by this Port when the Port is operating as a Downstream Port. This field is ignored when the Port is operating as an Upstream Port. See the PCI Express Base Specification, Section 4.2.3.2 for encodings. For an Upstream Port, if Crosslink Supported is 0b, this field is RsvdP. Otherwise, this field is Hwlnit. The default value of this field is 1111b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: {(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_DSP_TX_PRESET_20G0" : "0x0"} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Testable: {(CX_CROSSLINK_ENABLE== 0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") } Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

3.38.8 ESM_LN_EQ_CNTL_25G_i_OFF (for i = 0; i <= 12)

- **Name:** ESM Lane Equalization Control Register for 25.0 GT/s for Lanes 0-3.
- **Description:** This register holds information for Lanes 0-3. The entry for each lane contains the value for the Lane, with the corresponding Lane number which is invariant to Link width and Lane reversal negotiation that occurs during Link training.
- **Size:** 32 bits
- **Offset:** B+0x2c + i*0x4
- **Exists:** (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

ESM_USP_TX_PRESET_25G_3	31:28
ESM_DSP_TX_PRESET_25G_3	27:24
ESM_USP_TX_PRESET_25G_2	23:20
ESM_DSP_TX_PRESET_25G_2	19:16
ESM_USP_TX_PRESET_25G_1	15:12
ESM_DSP_TX_PRESET_25G_1	11:8
ESM_USP_TX_PRESET_25G_0	7:4
ESM_DSP_TX_PRESET_25G_0	3:0

Table 3-276 Fields for Register: ESM_LN_EQ_CNTL_25G_i_OFF (for i = 0; i <= 12)

Bits	Name	Memory Access	Description
31:28	ESM_USP_TX_PRESET_25G_3	* Varies	For 25G Lane3 high, Refer to ESM_USP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: RSVDP Value After Reset: {(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_USP_TX_PRESET_25G3" : "DEFAULT_LOCAL_G4_TX_PRESET"} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE && CX_NL > 2 Testable: {(CX_CROSSLINK_ENABLE== 0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") } Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27:24	ESM_DSP_TX_PRESET_25G_3	* Varies	For 25G Lane3 low,Refer to ESM_DSP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: $\{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_DSP_TX_PRESET_25G3" : "0x0"\}$ Exists: $(CX_CCIX_ENABLE \&\& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) \&\& CX_CCIX_ENABLE \&\& CX_NL > 2$ Testable: $\{(CX_CROSSLINK_ENABLE==0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 \&\& CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained")\}$ Memory Access: $"(CX_MEMORY_MAP_VIEW==0 \&\& CX_DBI_RO_WR_EN \&\& CX_CROSSLINK_ENABLE) ? "\text{read-write}" : (CX_MEMORY_MAP_VIEW==0) ? "\text{read-only}" : (CX_MEMORY_MAP_VIEW==1) ? "\text{read-only}" : "\text{read-only}" "$
23:20	ESM_USP_TX_PRESET_25G_2	* Varies	For 25G Lane2 high,Refer to ESM_USP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: RSVDP Value After Reset: $\{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_USP_TX_PRESET_25G2" : "DEFAULT_LOCAL_G4_TX_PRESET"\}$ Exists: $(CX_CCIX_ENABLE \&\& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) \&\& CX_CCIX_ENABLE \&\& CX_NL > 2$ Testable: $\{(CX_CROSSLINK_ENABLE==0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 \&\& CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained")\}$ Memory Access: $"(CX_MEMORY_MAP_VIEW==0 \&\& CX_DBI_RO_WR_EN \&\& CX_CROSSLINK_ENABLE) ? "\text{read-write}" : (CX_MEMORY_MAP_VIEW==0) ? "\text{read-only}" : (CX_MEMORY_MAP_VIEW==1) ? "\text{read-only}" : "\text{read-only}" "$

Bits	Name	Memory Access	Description
19:16	ESM_DSP_TX_PRESET_25G_2	* Varies	For 25G Lane2 low, Refer to ESM_DSP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: $\{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_DSP_TX_PRESET_25G2" : "0x0"\}$ Exists: $(CX_CCIX_ENABLE \&\& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) \&\& CX_CCIX_ENABLE \&\& CX_NL > 2$ Testable: $\{(CX_CROSSLINK_ENABLE==0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 \&\& CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained")\}$ Memory Access: $"(CX_MEMORY_MAP_VIEW==0 \&\& CX_DBI_RO_WR_EN \&\& CX_CROSSLINK_ENABLE) ? "\text{read-write}" : (CX_MEMORY_MAP_VIEW==0) ? "\text{read-only}" : (CX_MEMORY_MAP_VIEW==1) ? "\text{read-only}" : "\text{read-only}" "$
15:12	ESM_USP_TX_PRESET_25G_1	* Varies	For 25G Lane1 high, Refer to ESM_USP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: RSVDP Value After Reset: $\{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_USP_TX_PRESET_25G1" : "DEFAULT_LOCAL_G4_TX_PRESET"\}$ Exists: $(CX_CCIX_ENABLE \&\& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) \&\& CX_CCIX_ENABLE \&\& CX_NL > 1$ Testable: $\{(CX_CROSSLINK_ENABLE==0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 \&\& CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained")\}$ Memory Access: $"(CX_MEMORY_MAP_VIEW==0 \&\& CX_DBI_RO_WR_EN \&\& CX_CROSSLINK_ENABLE) ? "\text{read-write}" : (CX_MEMORY_MAP_VIEW==0) ? "\text{read-only}" : (CX_MEMORY_MAP_VIEW==1) ? "\text{read-only}" : "\text{read-only}" "$

Bits	Name	Memory Access	Description
11:8	ESM_DSP_TX_PRESET_25G_1	* Varies	For 25G Lane1 low,Refer to ESM_DSP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: $\{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_DSP_TX_PRESET_25G1" : "0x0"\}$ Exists: $(CX_CCIX_ENABLE \&\& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) \&\& CX_CCIX_ENABLE \&\& CX_NL > 1$ Testable: $\{(CX_CROSSLINK_ENABLE==0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 \&\& CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained")\}$ Memory Access: $"(CX_MEMORY_MAP_VIEW==0 \&\& CX_DBI_RO_WR_EN \&\& CX_CROSSLINK_ENABLE) ? \backslash\text{read-write}\backslash" : (CX_MEMORY_MAP_VIEW==0) ? \backslash\text{read-only}\backslash" : (CX_MEMORY_MAP_VIEW==1) ? \backslash\text{read-only}\backslash" : \backslash\text{read-only}\backslash"$
7:4	ESM_USP_TX_PRESET_25G_0	* Varies	For 25G Lane0 high,Refer to ESM_USP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: RSVDP Value After Reset: $\{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_USP_TX_PRESET_25G0" : "DEFAULT_LOCAL_G4_TX_PRESET"\}$ Exists: $(CX_CCIX_ENABLE \&\& (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) \&\& CX_CCIX_ENABLE$ Testable: $\{(CX_CROSSLINK_ENABLE==0) ? "untestable" : ((CX_DBI_RO_WR_EN==1 \&\& CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained")\}$ Memory Access: $"(CX_MEMORY_MAP_VIEW==0 \&\& CX_DBI_RO_WR_EN \&\& CX_CROSSLINK_ENABLE) ? \backslash\text{read-write}\backslash" : (CX_MEMORY_MAP_VIEW==0) ? \backslash\text{read-only}\backslash" : (CX_MEMORY_MAP_VIEW==1) ? \backslash\text{read-only}\backslash" : \backslash\text{read-only}\backslash"$

Bits	Name	Memory Access	Description
3:0	ESM_DSP_TX_PRESET_25G_0	* Varies	For 25G Lane0 low,Refer to ESM_DSP_TX_PRESET_20G_0 field <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RSVDP Value After Reset: <code>{{(CX_MEMORY_MAP_POSITION==1 CX_CROSSLINK_ENABLE==1) ? "DEFAULT_ESM_DSP_TX_PRESET_25G0" : "0x0"}}</code> Exists: <code>(CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE</code> Testable: <code>{{(CX_CROSSLINK_ENABLE== 0) ? "unconstrained" : ((CX_DBI_RO_WR_EN==1 && CX_MEMORY_MAP_VIEW==0) ? "writeAsRead" : "unconstrained") }}</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && CX_CROSSLINK_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""</code>

3.38.9 CCIX_TL_CAP_OFF

- **Name:** Transaction Layer Capabilities Register.
- **Description:** This is the transaction layer capabilities register.
- **Size:** 32 bits
- **Offset:** B+0x3c
- **Exists:** (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:11	10:8	7:1	0
RSVDP_11	CCIX_VC_RESOURCE_CAP	RSVDP_1	CCIX_OPT_TLP_FMT_SUPPORT

Table 3-277 Fields for Register: CCIX_TL_CAP_OFF

Bits	Name	Memory Access	Description
31:11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

Bits	Name	Memory Access	Description
10:8	CCIX_VC_RESOURCE_CAP	* Varies	CCIXVCResourceCapabilityIndex. With reference to the PCIe "Virtual Channel Capability" this is the index used to identify the VC Resource Capability Register corresponding to the VC to be used for CCIX traffic. The valid range is $1 \leq \text{index} \leq n$ (where n is the "Extended VC Count" as defined in the latest PCIe specifications). The VC Resource Offset corresponding to "index" is $10h + \text{index} * 0Ch$ <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: {CCIX_VC_RESOURCE} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
7:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

Bits	Name	Memory Access	Description
0	CCIX_OPT_TLP_FMT_SUPPORT	* Varies	OptimizedTLPFormatSupport. When this field is set, Generation, Reception, and Routing are supported. When this field is clear, Generation, Reception, and Routing are not supported. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: {CCIX_INTERFACE_ENABLE_VALUE} Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

3.38.10 CCIX_TL_CNTL_OFF

- **Name:** Transaction Layer Control Register.
- **Description:** This register controls the transaction layer operation.
- **Size:** 32 bits
- **Offset:** B+0x40
- **Exists:** (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-278 Fields for Register: CCIX_TL_CNTL_OFF

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE
0	EN_CCIX_OPT_TLP	R/W	OptimizedTLPGenerationReceptionRoutingEnable. See Section 4.2.4 of CCIX specification for the behavior controlled by this bit. This bit cannot be modified when VCResourceControl.VCEnable is Set. Functions that do not generate, receive, or forward Optimized TLPs are permitted to hardwire this bit to 0b. Default value of this bit is 0b. Value After Reset: 0x0 Exists: (CX_CCIX_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CCIX_ENABLE

3.39 PF0_CXS_CAP Registers

3.39.1 CXS_CAP_HDR_OFF

- **Name:** CXS Extended Capability Header.
- **Description:** This register provides capability ID, capability version, and next offset value of capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	CXS_NEXT_OFFSET
19:16	CXS_CAP_VERSION
15:0	CXS_CAP_ID

Table 3-279 Fields for Register: CXS_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	CXS_NEXT_OFFSET	* Varies	Next Capability Offset. This field holds the offset to the next PCI Express Capability structure, or 000h, if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CXS_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE Testable: wire ? unconstrained : writeAsRead
15:0	CXS_CAP_ID	* Varies	CXS Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xb Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE Testable: wire ? unconstrained : writeAsRead

3.39.2 CXS_VENDOR_SPECIFIC_HEADER_OFF

- **Name:** CXS VSEC Header Register.
- **Description:** This register provides VSEC ID, VSEC revision, and VSEC length.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	CXS_VSEC_LENGTH
19:16	CXS_VSEC_REV
15:0	CXS_VSEC_ID

Table 3-280 Fields for Register: CXS_VENDOR_SPECIFIC_HEADER_OFF

Bits	Name	Memory Access	Description
31:20	CXS_VSEC_LENGTH	R	CXS VSEC Length. Value After Reset: 0x18 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: restore
19:16	CXS_VSEC_REV	R	CXS VSEC Version. Value After Reset: 0x1 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: restore
15:0	CXS_VSEC_ID	R	CXS VSEC ID. Value After Reset: 0x5 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: restore

3.39.3 CXS_TX_CTRL_OFF

- **Name:** CXS Transmitter Control Register.
- **Description:** This register controls the CXS Transmitter operation.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	15:2	1	0
CXS_TX_DEACT_TIME	RSVDP_2	CXS_TX_PADDING_EN	CXS_TX_STORE_EN

Table 3-281 Fields for Register: CXS_TX_CTRL_OFF

Bits	Name	Memory Access	Description
31:16	CXS_TX_DEACT_TIME	R/W	CXS Transmitter link deactivation timer. Specifies the number of cycles the link remains active before starting the deactivation if no packets are available for the transmission in RADM queue. Value After Reset: {CX_CXS_TX_DEACT_TIMER_VALUE} Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE
15:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE

Bits	Name	Memory Access	Description
1	CXS_TX_PADDING_EN	R/W	CXS Transmitter padding enable. Enables padding for packets with a length smaller than the packet length field. Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE
0	CXS_TX_STORE_EN	R/W	CXS Transmitter store-forward mode enable. When set, the controller uploads complete from RADM queue to CXS Transmitter FIFO before driving it on CXS Interface. Value After Reset: {(CX_CXS_TX_CONTINUOUSDATA==1) ? 1 : 0} Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE

3.39.4 CXS_RX_CTRL_OFF

- **Name:** CXS Receiver Control Register.
- **Description:** This register controls CXS Receiver operation.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:1	RSVDP_1
0	CXS_RX_STORE_EN

Table 3-282 Fields for Register: CXS_RX_CTRL_OFF

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE
0	CXS_RX_STORE_EN	R/W	CXS Receiver store-forward mode enable. When set, the controller buffers the complete packet in CXS Receiver FIFO before driving it on XADM interface. Value After Reset: {(CX_CXS_RX_CONTINUOUSDATA == 1) ? 0 : 1} Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE

3.39.5 CXS_ERROR_CTRL_OFF

- **Name:** Error Control Register.
- **Description:** This register controls the state of CXS error registers.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:1	
RSVDP_1	
0	CXS_ERROR_CLEAR

Table 3-283 Fields for Register: CXS_ERROR_CTRL_OFF

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE
0	CXS_ERROR_CLEAR	W	Resets CXS error status registers. Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE

3.39.6 CXS_ERROR_STAT_OFF

- **Name:** Error Status Register.
- **Description:** This register reports the errors occurred during CXS Transmitter and Receiver operation.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	RSVDP_20
19	CXS_TX_CNTL_ERROR
18	CXS_TX_CNTL_UNCOR
17	CXS_TX_CRDS_OVFLW
16	CXS_TX_FIFO_OVFLW
15:4	RSVDP_4
3	CXS_RX_CNTL_ERROR
2	CXS_RX_CNTL_UNCOR
1	CXS_RX_CRDS_OVFLW
0	CXS_RX_FIFO_OVFLW

Table 3-284 Fields for Register: CXS_ERROR_STAT_OFF

Bits	Name	Memory Access	Description
31:20	RSVDP_20	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE
19	CXS_TX_CNTL_ERROR	R	CXS Transmitter error packet. Indicates detection of a packet marked with error bit in CXS Transmitter Control Bus. Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE

Bits	Name	Memory Access	Description
18	CXS_TX_CNTL_UNCOR	R	CXS Transmitter control bus error. Indicates detection of unrecoverable error on CXS Transmitter Control Bus. Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE
17	CXS_TX_CRDS_OVFLW	R	CXS Transmitter credits overflow. Indicates that pair CXS Receiver device granted more than 15 credits to CXS Transmitter. Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE
16	CXS_TX_FIFO_OVFLW	R	Indicates CXS Transmitter FIFO overflow. Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE
15:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE
3	CXS_RX_CNTL_ERROR	R	CXS Receiver error packet. Indicates detection of a packet marked with error bit in CXS Receiver Control Bus. Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE
2	CXS_RX_CNTL_UNCOR	R	CXS Receiver control bus error. Indicates detection of unrecoverable error on CXS Receiver Control Bus. Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE

Bits	Name	Memory Access	Description
1	CXS_RX_CRDS_OVFLW	R	CXS Receiver credits overflow. Indicates that pair CXS Transmitter device returned more credits than granted by CXS Receiver. Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE
0	CXS_RX_FIFO_OVFLW	R	Indicates CXS Receiver FIFO overflow. Value After Reset: 0x0 Exists: (CX_CXS_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_CXS_ENABLE

3.40 PF_VSECDMA_CAPI Registers

3.40.1 VSECDMA_EXT_CAP_HDR_OFF

- **Name:** PCIe Extended Capability ID, Capability Version, and Next Capability Offset Register.
- **Description:** This register provides capability ID, capability version, and next capability offset for PCIe extended capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	19:16	15:0
NEXT_OFFSET	CAP	ID

Table 3-285 Fields for Register: VSECDMA_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. The bottom 2 bits of this offset are Reserved and must be implemented as 00b although software must mask them to allow for future uses of these bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. A version of the specification that changes the Extended Capability in a way that is not otherwise identifiable (that is, through a new Capability field) is permitted to increment this field. All such changes to the Capability structure must be software-compatible. Software must check for Capability Version numbers that are greater than or equal to the highest number defined when the software is written, as Functions reporting any such Capability Version numbers will contain a Capability structure that is compatible with that piece of software. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15:0	ID	* Varies	PCI Express Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xb Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.40.2 VSECDMA_VENDOR_SPECIFIC_HDR_OFF

- **Name:** Vendor Specific Header Register.
- **Description:** This Register provides VSEC Length, VSEC ID, and VSEC Rev (Version Number). Vendor-specific software must qualify the associated Vendor ID of the PCI Express Function or RCRB before attempting to interpret the values in the VSEC ID or VSEC Rev fields.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	19:16	15:0
VSEC_LENGTH	VSEC_REV	VSEC_ID

Table 3-286 Fields for Register: VSECDMA_VENDOR_SPECIFIC_HDR_OFF

Bits	Name	Memory Access	Description
31:20	VSEC_LENGTH	R	VSEC Length. This field indicates the number of bytes in the entire VSEC structure, including the Vendor-Specific Extended Capability Header, the Vendor-Specific Header, and the vendor-specific registers. Value After Reset: 0x18 Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
19:16	VSEC_REV	R	VSEC Rev. This field is a vendor-defined version number that indicates the version of the VSEC structure. Software must qualify the Vendor ID and VSEC ID before interpreting this field. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:0	VSEC_ID	R	VSEC ID. This field is a vendor-defined ID number that indicates the nature and format of the VSEC structure. Software must qualify the Vendor ID before interpreting this field. Value After Reset: 0x6 Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.40.3 VSECDMA_DEVICE_INFORMATION_OFF

- **Name:** DMA and related AXI Bridge Implementation Information.
- **Description:** This register provides DMA and AXI Bridge implementation-specific information:
 - DMA Architecture: Legacy DMA or Hyper DMA (HDMA).
 - Register Location: Port-logic or Mapped to a Function and BAR.
 - Register Map: Legacy DMA or HDMA.
 - Channel Separation: Address distance between read and write channels.
 - AXI Bridge: Used or Not Used
 - AXI Master Bus Specification: AXI Master Bus Width, Burst Length, and Boundary Pointer Width
 You must use this information along with the IP version registers:
 - PCIE_VERSION_NUMBER_OFF
 - PCIE_VERSION_TYPE_OFF
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_30	31:30
MASTER_PAGE_BOUNDARY_POINTER_WIDTH	29:26
MASTER_BURST_LENGTH	25:23
MASTER_BUS_WIDTH	22:20
AXI	19
CHANNEL_SEPARATION	18:16
PFN	15:11
BARN	10:8
RSVDP_3	7:3
MAP_FORMAT	2:0

Table 3-287 Fields for Register: VSECDMA_DEVICE_INFORMATION_OFF

Bits	Name	Memory Access	Description
31:30	RSVDP_30	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
29:26	MASTER_PAGE_BOUNDARY_POINTER_WIDTH	R	This field provides address page boundary information. It reports the value of CC_MSTR_PAGE_BOUNDARY_PW configuration parameter. Value After Reset: CC_MSTR_PAGE_BOUNDARY_PW Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
25:23	MASTER_BURST_LENGTH	R	Reports the CC_MSTR_BURST_LEN configuration parameter. Values: ■ 0x0 (MSTR_BRSTLEN_0): Reserved ■ 0x1 (MSTR_BRSTLEN_4): 4 beats ■ 0x2 (MSTR_BRSTLEN_8): 8 beats ■ 0x3 (MSTR_BRSTLEN_16): 16 beats ■ 0x4 (MSTR_BRSTLEN_32): 32 beats ■ 0x5 (MSTR_BRSTLEN_64): 64 beats ■ 0x6 (MSTR_BRSTLEN_128): 128 beats ■ 0x7 (MSTR_BRSTLEN_256): 256 beats Value After Reset: CC_MSTR_BURST_LEN_REG Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
22:20	MASTER_BUS_WIDTH	R	This field provides information regarding the AXI master data bus width. It reports the value of MASTER_BUS_DATA_WIDTH configuration parameter. Values: ■ 0x0 (MSTR_BUSWD_32): 32 bits ■ 0x1 (MSTR_BUSWD_64): 64 bits ■ 0x2 (MSTR_BUSWD_128): 128 bits ■ 0x3 (MSTR_BUSWD_256): 256 bits ■ 0x4 (MSTR_BUSWD_512): 512 bits Value After Reset: MASTER_BUS_DATA_WIDTH_REG Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
19	AXI	R	This field provides information about AXI interface usage. It reports the value of AXI_POPULATED configuration parameter. Value After Reset: AXI_POPULATED_VALUE Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
18:16	CHANNEL_SEPARATION	R	If the MAP_FORMAT is set to HDMA_NATIVE, this field specifies the read write channel address separation. Other values are reserved. Values: ■ 0x0 (CHSEP_256): 256 separated ■ 0x1 (CHSEP_512): 512 separated ■ 0x2 (CHSEP_1K): 1k separated ■ 0x3 (CHSEP_2K): 2k separated ■ 0x4 (CHSEP_4K): 4k separated ■ 0x5 (CHSEP_8K): 8k separated ■ 0x6 (CHSEP_16K): 16k separated ■ 0x7 (CHSEP_32K): 32k separated Value After Reset: CX_DMAREG_CHADDR_SPACE Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:11	PFN	R	Physical Function Number. This field provides information regarding the DMA register and physical function mapping. Value After Reset: UNROLL_FUNC_NUM Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
10:8	BARN	R	Bar Number. This field provides information regarding the DMA register and BAR number mapping. Value After Reset: UNROLL_BAR_NUM Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
2:0	MAP_FORMAT	R	Defines the register map format and features to be one of the following values: Other values are reserved. Values: ■ 0x0 (EDMA_LEGACY_PL): Legacy DMA register map accessed by the port-logic registers ■ 0x1 (EDMA_LEGACY_UNROLL): Legacy DMA register map, mapped to a PF/BAR ■ 0x4 (HDMA_COMPATIBILITY_PL): HDMA/DMA compatibility (CC_LEGACY_DMA_MAP =1) register map without access through the Wire ■ 0x5 (HDMA_COMPATIBILITY_MODE): HDMA compatibility mode (CC_LEGACY_DMA_MAP =1) register map, mapped to a PF/BAR ■ 0x6 (HDMA_NATIVE_PL): HDMA native (CC_LEGACY_DMA_MAP =0) register map without access through the Wire ■ 0x7 (HDMA_NATIVE): HDMA native (CC_LEGACY_DMA_MAP =0) register map, mapped to a PF/BAR Value After Reset: CC_VSECDMA_MAP_VERSION Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.40.4 VSECDMA_NUM_CHAN_OFF

- **Name:** Number of Implemented Channels Register.
- **Description:** This register specifies the number of read and write channels implemented.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:26	RSVDP_26
25:16	NUM_DMA_RD_CHAN
15:10	RSVDP_10
9:0	NUM_DMA_WR_CHAN

Table 3-288 Fields for Register: VSECDMA_NUM_CHAN_OFF

Bits	Name	Memory Access	Description
31:26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
25:16	NUM_DMA_RD_CHAN	R	This field provides information regarding the number of implemented read channels. It reports the value of CC_NUM_DMA_RD_CHAN parameter. Value After Reset: CC_NUM_DMA_RD_CHAN Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
9:0	NUM_DMA_WR_CHAN	R	This field provides information regarding the number of implemented write channels. It reports the value of CC_NUM_DMA_WR_CHAN parameter. Value After Reset: CC_NUM_DMA_WR_CHAN Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.40.5 VSECDMA_UNROLL_ADDR_OFFSET_LOW_OFF

- **Name:** DMA Register Map Start Address Offset Low Register.
- **Description:** This register specifies the lower 32 bits of the offset of the start address of the DMA register map. Applicable only if MAP_FORMAT >0, that is, all map formats other than EDMA_LEGACY_PL.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-289 Fields for Register: VSECDMA_UNROLL_ADDR_OFFSET_LOW_OFF

Bits	Name	Memory Access	Description
31:0	UNROLL_ADDR_OFFSET_LOW	R	BAR address offset, 32-bit LSB. Value After Reset: UNROLL_DMA_OFFSET_BAR_LOW Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.40.6 VSECDMA_UNROLL_ADDR_OFFSET_HIGH_OFF

- **Name:** DMA Register Map Start Address Offset High Register.
- **Description:** This register specifies the upper 32 bits of the offset of the start address of the DMA register map. Applicable only if MAP_FORMAT >0, that is, all map formats other than EDMA_LEGACY_PL.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-290 Fields for Register: VSECDMA_UNROLL_ADDR_OFFSET_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	UNROLL_ADDR_OFFSET_HIGH	R	BAR address offset, 32-bit MSB. Value After Reset: UNROLL_DMA_OFFSET_BAR_HIGH Exists: (CC_DMA_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.41 PF0_NPEM_CAP Registers

3.41.1 NPEM_EXT_CAP_HDR_OFF

- **Name:** NPEM Extended Capability Header.
- **Description:** Native PCIe Enclosure Management Extended Capability Header provides the capability ID, capability version, and next offset of NPEM extended capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	19:16	15:0
NPEM_NEXT_OFFSET	NPEM_CAP_VERSION	NPEM_EXT_CAP_ID

Table 3-291 Fields for Register: NPEM_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	NPEM_NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	NPEM_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	NPEM_EXT_CAP_ID	* Varies	PCI Express Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the extended capability. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x29 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.41.2 NPEM_CAP_OFF

- **Name:** NPEM Capability Register.
- **Description:** Native PCIe Enclosure Management Capabilities Register contains an overall NPEM Capable bit and a bit map of states supported in the implementation.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

NPEM_ENCLOSURE_SPECIFIC_CAPABILITIES	31:24
RSVDP_12	23:12
NPEM_DISABLED_CAPABLE	11
NPEM_INVALID_DEVICE_TYPE_CAPABLE	10
NPEM_FAILED_ARRAY_CAPABLE	9
NPEM_CRITICAL_ARRAY_CAPABLE	8
NPEM_HOT_SPARE_CAPABLE	7
NPEM_PFA_CAPABLE	6
NPEM_REBUILD_CAPABLE	5
NPEM_FAIL_CAPABLE	4
NPEM_LOCATE_CAPABLE	3
NPEM_OK_CAPABLE	2
NPEM_RESET_CAPABLE	1
NPEM_CAPABLE	0

Table 3-292 Fields for Register: NPEM_CAP_OFF

Bits	Name	Memory Access	Description
31:24	NPEM_ENCLOSURE_SPECIFIC_CAPABILITIES	* Varies	Enclosure-Specific Capabilities. The definition of enclosure specific bits is outside the scope of this specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENCLOSURE_SPECIFIC_CAPABILITIES_VALUE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
23:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11	NPEM_DISABLED_CAPABLE	* Varies	NPEM Disabled Capable. When Set, this bit indicates that the enclosure has the ability to indicate the NPEM Disabled state. When Clear, this bit indicates that the enclosure has not the ability to indicate the NPEM Disabled state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
10	NPEM_INVALID_DEVICE_TYPE_CAPABLE	* Varies	NPEM Invalid Device Type Capable. When Set, this bit indicates that the enclosure has the ability to indicate the NPEM Invalid Device Type state. When Clear, this bit indicates that the enclosure has not the ability to indicate the NPEM Invalid Device Type state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
9	NPEM_FAILED_ARRAY_CAPABLE	* Varies	NPEM Failed Array Capable. When Set, this bit indicates that the enclosure has the ability to indicate the NPEM Failed Array Capable state. When Clear, this bit indicates that the enclosure has not the ability to indicate the NPEM Failed Array Capable state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
8	NPEM_CRITICAL_ARRAY_CAPABLE	* Varies	NPEM Critical Array Capable. When Set, this bit indicates that the enclosure has the ability to indicate the NPEM Critical Array state. When Clear, this bit indicates that the enclosure has not the ability to indicate the NPEM Critical Array state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
7	NPEM_HOT_SPARE_CAPABLE	* Varies	NPEM Hot Spare Capable. When Set, this bit indicates that the enclosure has the ability to indicate the NPEM Hot Spare state. When Clear, this bit indicates that the enclosure has not the ability to indicate the NPEM Hot Spare state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
6	NPEM_PFA_CAPABLE	* Varies	NPEM PFA Capable. When Set, this bit indicates that the enclosure has the ability to indicate the NPEM PFA state. When Clear, this bit indicates that the enclosure has not the ability to indicate the NPEM PFA state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
5	NPEM_REBUILD_CAPABLE	* Varies	NPEM Rebuild Capable. When Set, this bit indicates that the enclosure has the ability to indicate the NPEM Rebuild state. When Clear, this bit indicates that the enclosure has not the ability to indicate the NPEM Rebuild state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
4	NPEM_FAIL_CAPABLE	* Varies	NPEM Fail Capable. When Set, this bit indicates that the enclosure has the ability to indicate the NPEM Fail state. When Clear, this bit indicates that the enclosure has not the ability to indicate the NPEM Fail state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
3	NPEM_LOCATE_CAPABLE	* Varies	NPEM Locate Capable. When Set, this bit indicates that the enclosure has the ability to indicate the NPEM Locate state. When Clear, this bit indicates that the enclosure has not the ability to indicate the NPEM Locate state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
2	NPEM_OK_CAPABLE	* Varies	NPEM OK Capable. When Set, this bit indicates that the enclosure has the ability to indicate the NPEM OK state. When Clear, this bit indicates that the enclosure has not the ability to indicate the NPEM OK state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
1	NPEM_RESET_CAPABLE	* Varies	NPEM Reset Capable. When Set, this bit indicates support for the NPEM Reset mechanism functionality. When Clear, this bit indicates no support for the NPEM Reset mechanism functionality. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
0	NPEM_CAPABLE	* Varies	NPEM Capable. When Set, this bit indicates that the enclosure has NPEM functionality. When Clear, this bit indicates that the enclosure has not NPEM functionality. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: NPEM_ENABLE Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.41.3 NPEM_CTRL_OFF

- **Name:** NPEM Control Register.
- **Description:** Native PCIe Enclosure Management Control Register contains an overall NPEM Enable bit and a bit map of states that software controls.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

NPEM_ENCLOSURE_SPECIFIC_CONTROL	31:24
RSVDP_12	23:12
NPEM_DISABLED_CONTROL	11
NPEM_INVALID_DEVICE_TYPE_CONTROL	10
NPEM_FAILED_ARRAY_CONTROL	9
NPEM_CRITICAL_ARRAY_CONTROL	8
NPEM_HOT_SPARE_CONTROL	7
NPEM_PFA_CONTROL	6
NPEM_REBUILD_CONTROL	5
NPEM_FAIL_CONTROL	4
NPEM_LOCATE_CONTROL	3
NPEM_OK_CONTROL	2
NPEM_INITIATE_RESET	1
NPEM_ENABLE	0

Table 3-293 Fields for Register: NPEM_CTRL_OFF

Bits	Name	Memory Access	Description
31:24	NPEM_ENCLOSURE_SPECIFIC_CONTROL	* Varies	Enclosure-specific Controls. The definition of enclosure-specific bits is outside the scope of this specification. Enclosure-specific software is permitted to change the value of this field. Other software must preserve the existing value when writing this register. Default value of this field is 00h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && NPEM_ENCLOSURE_SPECIFIC_CAPABILITIES) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !NPEM_ENCLOSURE_SPECIFIC_CAPABILITIES) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && NPEM_ENCLOSURE_SPECIFIC_CAPABILITIES) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !NPEM_ENCLOSURE_SPECIFIC_CAPABILITIES) ? \"read-only\" : \"read-only\" "
23:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
11	NPEM_DISABLED_CONTROL	* Varies	NPEM Disabled Control. Turn on NPEM Disabled indicator. If NPEM Disabled bit in NPEM Capability Register is 0b, this bit is read-only with a value of 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x1 (SET): When Set, this bit specifies that the NPEM Disabled indication be turned ON. ■ 0x0 (CLEAR): When Clear, this bit specifies that the NPEM Disabled indication be turned OFF. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && NPEM_DISABLED_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !NPEM_DISABLED_CAPABLE_VALUE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && NPEM_DISABLED_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !NPEM_RESET_CAPABLE_VALUE) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
10	NPEM_INVALID_DEVICE_TYPE_CONTROL	* Varies	NPEM Invalid Device Type Control. Turn on NPEM Invalid Device Type indicator. If NPEM Invalid Device Type bit in NPEM Capability Register is 0b, this bit is read-only with a value of 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x1 (SET): When Set, this bit specifies that the NPEM Invalid Device Type indication be turned ON. ■ 0x0 (CLEAR): When Clear, this bit specifies that the NPEM Invalid Device Type indication be turned OFF. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && NPEM_INVALID_DEVICE_TYPE_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !NPEM_INVALID_DEVICE_TYPE_CAPABLE_VALUE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && NPEM_INVALID_DEVICE_TYPE_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !NPEM_INVALID_DEVICE_TYPE_CAPABLE_VALUE) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
9	NPEM_FAILED_ARRAY_CONTROL	* Varies	NPEM Failed Array Control. Turn on NPEM Failed Array indicator. If NPEM Failed Array Capable in NPEM Capability Register is 0b, this bit is read-only with a value of 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x1 (SET): When Set, this bit specifies that the NPEM Failed Array indication be turned ON. ■ 0x0 (CLEAR): When Clear, this bit specifies that the NPEM Failed Array indication be turned OFF. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && NPEM_FAILED_ARRAY_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !NPEM_FAILED_ARRAY_CAPABLE_VALUE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && NPEM_FAILED_ARRAY_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !NPEM_FAILED_ARRAY_CAPABLE_VALUE) ? \"read-only\""

Bits	Name	Memory Access	Description
8	NPEM_CRITICAL_ARRAY_CONTROL	* Varies	NPEM Critical Array Control Turn on NPEM Critical Array indicator. If NPEM Critical Array Capable in NPEM Capability Register is 0b, this bit is read-only with a value of 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x1 (SET): When Set, this bit specifies that the NPEM Critical Array indication be turned ON. ■ 0x0 (CLEAR): When Clear, this bit specifies that the NPEM Critical Array indication be turned OFF. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && NPEM_CRITICAL_ARRAY_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !NPEM_CRITICAL_ARRAY_CAPABLE_VALUE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && NPEM_CRITICAL_ARRAY_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !NPEM_CRITICAL_ARRAY_CAPABLE_VALUE) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	NPEM_HOT_SPARE_CONTROL	* Varies	NPEM Hot Spare Control. Turn on NPEM Hot Spare indicator. If NPEM Hot Spare bit Capable in NPEM Capability Register is 0b, this bit is read-only with a value of 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x1 (SET): When Set, this bit specifies that the NPEM Hot Spare indication be turned ON. ■ 0x0 (CLEAR): When Clear, this bit specifies that the NPEM Hot Spare indication be turned OFF. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && NPEM_HOT_SPARE_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !NPEM_HOT_SPARE_CAPABLE_VALUE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && NPEM_HOT_SPARE_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !NPEM_HOT_SPARE_CAPABLE_VALUE) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
6	NPEM_PFA_CONTROL	* Varies	NPEM PFA Control. Turn on NPEM PFA indicator. If NPEM PFA Capable bit in NPEM Capability Register is 0b, this bit is read-only with a value of 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x1 (SET): When Set, this bit specifies that the NPEM PFA indication be turned ON. ■ 0x0 (CLEAR): When Clear, this bit specifies that the NPEM PFA indication be turned OFF. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && NPEM_PFA_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !NPEM_PFA_CAPABLE_VALUE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && NPEM_PFA_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !NPEM_PFA_CAPABLE_VALUE) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	NPEM_REBUILD_CONTROL	* Varies	NPEM Rebuild Control. Turn on NPEM Rebuild indicator. If NPEM Rebuild Capable bit in NPEM Capability Register is 0b, this bit is read-only with a value of 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x1 (SET): When Set, this bit specifies that the NPEM Rebuild indication be turned ON. ■ 0x0 (CLEAR): When Clear, this bit specifies that the NPEM Rebuild indication be turned OFF. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && NPEM_REBUILD_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !NPEM_REBUILD_CAPABLE_VALUE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && NPEM_REBUILD_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !NPEM_REBUILD_CAPABLE_VALUE) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
4	NPEM_FAIL_CONTROL	* Varies	NPEM Fail Control. Turn on NPEM Fail indicator. If NPEM Fail Capable bit in NPEM Capability Register is 0b, this bit is read-only with a value of 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x1 (SET): When Set, this bit specifies that the NPEM Fail indication be turned ON. ■ 0x0 (CLEAR): When Clear, this bit specifies that the NPEM Fail indication be turned OFF. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && NPEM_FAIL_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !NPEM_FAIL_CAPABLE_VALUE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && NPEM_FAIL_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !NPEM_FAIL_CAPABLE_VALUE) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
3	NPEM_LOCATE_CONTROL	* Varies	NPEM Locate Control. Turn on NPEM Locate indicator. If NPEM Locate Capable bit in NPEM Capability Register is 0b, this bit is read-only with a value of 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x1 (SET): When Set, this bit specifies that the NPEM Locate indication be turned ON. ■ 0x0 (CLEAR): When Clear, this bit specifies that the NPEM Locate indication be turned OFF. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && NPEM_LOCATE_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !NPEM_LOCATE_CAPABLE_VALUE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && NPEM_LOCATE_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !NPEM_LOCATE_CAPABLE_VALUE) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
2	NPEM_OK_CONTROL	* Varies	NPEM OK Control. Turn on NPEM OK indicator. If NPEM OK Capable bit in NPEM Capability Register is 0b, this bit is read-only with a value of 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x1 (SET): When Set, this bit specifies that the NPEM OK indication be turned ON. ■ 0x0 (CLEAR): When Clear, this bit specifies that the NPEM OK indication be turned OFF. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && NPEM_OK_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !NPEM_OK_CAPABLE_VALUE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && NPEM_OK_CAPABLE_VALUE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !NPEM_OK_CAPABLE_VALUE) ? \"read-only\" : \"read-only\""
1	NPEM_INITIATE_RESET	R	NPEM Initiate Reset. One-shot pulse that initiates the NPEM Reset. The value read by software from this bit is always 0b. If NPEM Reset Capable bit in NPEM capability Register is 1b, then a write of 1b to this bit initiates NPEM Reset. If NPEM Reset Capable bit in NPEM capability Register is 0b, then this bit is read-only with a value of 0b. Values: ■ 0x1 (SET): NPEM Reset Initiated. ■ 0x0 (CLEAR): Clear. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	NPEM_ENABLE	R/W	NPEM Enable. Values: ■ 0x1 (SET): When Set, this bit enables the NPEM capability. ■ 0x0 (CLEAR): When Clear, this bit disables the NPEM capability. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.41.4 NPEM_STATUS_OFF

- **Name:** NPEM Status Register.
- **Description:** Native PCIe Enclosure Management Status Register.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	NPEM_ENCLOSURE_SPECIFIC_STATUS
23:1	RSVDP_1
0	NPEM_CMD_COMPLETE

Table 3-294 Fields for Register: NPEM_STATUS_OFF

Bits	Name	Memory Access	Description
31:24	NPEM_ENCLOSURE_SPECIFIC_STATUS	R	Enclosure-specific Status. The definition of enclosure specific bits is outside the scope of this specification. Enclosure specific software is permitted to write non-zero values to this field. Other software must write 00h to this field. These bits are read-only. The default value of this field is 00h. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	NPEM_CMD_COMPLETE	R/W1C	NPEM Command completed Status. When set, this bit indicates that an NPEM command has completed, and the NPEM controller is ready to accept a subsequent command. Default value of this bit is 0b. Value After Reset: 0x0 Exists: (NPEM_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.42 PF_DEV3_EXT_CAPi Registers

3.42.1 DEV3_EXT_CAP_HDR_REG

- **Name:** DEV3 Header Register.
- **Description:** Device 3 Extended Capability Header provides the capability ID, capability version, and next offset of DEV3 extended capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:20	DEV3_CAP_NEXT_OFFSET
19:16	DEV3_CAP_VERSION
15:0	DEV3_CAP_ID

Table 3-295 Fields for Register: DEV3_EXT_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	DEV3_CAP_NEXT_OFFSET	* Varies	DEV3 Next Capability Pointer. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	DEV3_CAP_VERSION	* Varies	DEV3 Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	DEV3_CAP_ID	* Varies	DEV3 Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. PCI Express Extended Capability ID for the Device 3 Extended Capability is 002Fh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x2f Exists: (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.42.2 DEVICE_CAPABILITIES3_REG

- **Name:** DEV3 Capabilities Register.
- **Description:** Device 3 Capabilities Register describes the various capabilities of the device for deferrable memory write routing support.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:1	
0	
RSVDP_1	DEV3_CAP_DMWR_ROUTING_SUPP

Table 3-296 Fields for Register: DEVICE_CAPABILITIES3_REG

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
0	DEV3_CAP_DMWR_ROUTING_S UPP	R	Deferrable Memory Write (DMWr) Request Routing Supported. Applicable only to Switch Upstream Ports, Switch Downstream Ports, and Root Ports; must be 0b for other Function types. This bit must be set to 1b if the Port supports this optional capability. For more details, see section 6.30 of <i>PCI Express Base Specification</i>. Value After Reset: 0x0 Exists: (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.42.3 DEVICE_CONTROL3_REG

- **Name:** DEV3 Control Register.
- **Description:** Device 3 Control Register provides various controls the device as a deferrable memory write requester and it's egress blocking feature.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:2	RSVDP_2
1	DEV3_CAP_DMWR_EGRESS_BLK
0	DEV3_CAP_DMWR_REQ_EN

Table 3-297 Fields for Register: DEVICE_CONTROL3_REG

Bits	Name	Memory Access	Description
31:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_DEF_MEM_WR_ENABLE && (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
1	DEV3_CAP_DMWR_EGRESS_BLOCK	R	Deferrable Memory Write (DMWr) Egress Blocking. Applicable and mandatory for Switch Upstream Ports, Switch Downstream Ports, and Root Ports that implement DMWr routing; otherwise the controller hardwires this bit to 0b. When this bit is set, DMWr Requests that target going out this Egress Port must be blocked. For more details, see section 6.30 of <i>PCI Express Base Specification</i>. Reflected on the <code>cfg_dmwr_egress_block</code> output. Value After Reset: 0x0 Exists: <code>CX_DEF_MEM_WR_ENABLE && (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))</code>
0	DEV3_CAP_DMWR_REQ_EN	R/W	Deferrable Memory Write (DMWr) Requester Enable. Applicable only to Endpoints and Root Ports; otherwise the controller hardwires this bit to 0b for other Function types. The Function is allowed to initiate Deferrable Memory Writes only if this bit and the Bus Master Enable bit in the Command register are both Set. This bit is RW if the Endpoint or Root Port is capable of initiating Deferrable Memory Write Requests, but otherwise the controller hardwires to 0b. For more details, see Section 6.30 of <i>PCI Express Base Specification</i>. Value After Reset: 0x0 Exists: <code>(CX_DEF_MEM_WR_REQ_SUP && CX_DEF_MEM_WR_ENABLE && CC_DEVICE_TYPE!=3) && (DEV3_CAP_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))</code>

3.43 PF0_CXL_PORT_CAP Registers

3.43.1 CXL_RCRB_PCIE_EXT_CAP_HDR_OFF

- **Name:** Capability ID, Version ID, Next offset pointer.
- **Description:** This Register Field for Flex Bus Port DVSEC Registers with Capability ID, Version and Next Offset Pointer.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE < 2))

31:20	19:16	15:0
FLEX_BUS_PORT_NXT_OFFSET	FLEX_BUS_PORT_CAP_VERSION	FLEX_BUS_PORT_CAP_ID

Table 3-298 Fields for Register: CXL_RCRB_PCIE_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	FLEX_BUS_PORT_NXT_OFFSET	* Varies	FLEX BUS PORT DVSEC Next Capability Offset. This field contains the offset to the next PCI Express DVSEC PORT Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE < 2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	FLEX_BUS_PORT_CAP_VERSION	* Varies	PCI Express Extended capability Version. This field is a PCIe DVSEC PORT defined version number that indicates the version of the capability structure present. This field is depends on the version of specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
15:0	FLEX_BUS_PORT_CAP_ID	* Varies	PCI Express Extended Capability ID. This PCIe DVSEC PORT defined ID number that indicates the nature and format of the extended capability. And this Device Capability ID field has read-write attribute and sticky reset type for the offset value 00h with [15:0] bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x23 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

3.43.2 CXL_RCRB_DVSEC_PORT_HDR_1_OFF

- **Name:** DVSEC Header 1: Vendor ID, Revision, Length.
- **Description:** Register Fields for Flex Bud Port DVSEC Registers: Designated Vendor Specific Header 1.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2))

31:20	CXL_RCRB_DVSEC_LEN
19:16	CXL_RCRB_DVSEC_REV
15:0	CXL_RCRB_DVSEC_VNDR_ID

Table 3-299 Fields for Register: CXL_RCRB_DVSEC_PORT_HDR_1_OFF

Bits	Name	Memory Access	Description
31:20	CXL_RCRB_DVSEC_LEN	* Varies	FLEX BUS PORT DVSEC HEADER 1 Length. The DVSEC Legth field is set to 16 bytes to accomodate the Flex Bus PORT registers included in the DVSEC. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x14 Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CXL_RCRB_DVSEC_REV	* Varies	FLEX BUS PORT DVSEC HEADER 1 REVISION. the DVSEC revision field is set to 0x0 to show that this is an FLEX Bus PORT DVSEC feature specific header capability structure. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE < 2)) Testable: wire ? unconstrained : writeAsRead
15:0	CXL_RCRB_DVSEC_VNDR_ID	* Varies	PCI Express Extended Capability vendor ID. The Vendor ID register identifies the manufacturer of the Function. Valid vendor identifiers are allocated by the PCI-SIG to ensure uniqueness. It is not permitted to populate this register with a value of FFFFh, which is an invalid value for Vendor ID. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1e98 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE < 2)) Testable: wire ? unconstrained : writeAsRead

3.43.3 CXL_RCRB_DVSEC_HDR_2_FLEXBUS_CAP_OFF

- **Name:** DVSEC HEADER 2 and DVSEC Flex Bus Port Capability.
- **Description:** this register field consists DVSEC Hdr 2 and Flex Bus Port capability.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2))

31:23	RSVDP_23
22	CXL_MUL_LOG_DEV_CAP
21	CXL2P0_CAP
20:19	RSVDP_19
18	MEM_CAP
17	IO_CAP
16	CACHE_CAP
15:0	CXL_RCRB_DVSEC_ID

Table 3-300 Fields for Register: CXL_RCRB_DVSEC_HDR_2_FLEXBUS_CAP_OFF

Bits	Name	Memory Access	Description
31:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
22	CXL_MUL_LOG_DEV_CAP	* Varies	CXL MULTI LOGICAL DEVICE CAPABLE. If set, indicates Multi-Logical Device support available when operating in Flex Bus.CXL mode. This bit is reserved on CXL1.1 components. This bit should be set to 1'b0 on CXL2.0 host Downstream ports. The value must be the same for all functions of an MLD. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {(CX_CXL_MLD_EN) ? "0x1" : "0x0"} Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
21	CXL2P0_CAP	* Varies	CXL 2P0 CAPABLE. If set, indicates CXL Revision 2.0 functionality support available when operating in Flex Bus.CXL mode. This bit is reserved on CXL1.1 components. This must be 1 for all functions of an MLD. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_CXL_2_0_ENABLE Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
20:19	RSVDP_19	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
18	MEM_CAP	* Varies	MEMORY CAPABLE. If set, indicates CXL.mem protocol support when operating in Flex Bus.AL mode. It's at bit 2 at offset 0x0Ah with read-only wire attribute. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: "(CX_CXL_MEM_EN) ? 1 : 0" Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
17	IO_CAP	R	IO CAPABLE. If set, indicates CXL.io protocol support when operating in Flex Bus.AL mode. Must be 1. It's bit value is 1 as read only wire-attribute. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2))
16	CACHE_CAP	* Varies	CACHE CAPABLE. If set, indicates CXL.cache protocol support when operating in Flex Bus.AL mode. It's offset value is 0x0A with 0 bit. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: "(CX_CXL_CACHE_EN) ? 1 : 0" Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15:0	CXL_RCRB_DVSEC_ID	* Varies	FLEX BUS PORT DVSEC ID. DVSEC ID reset value is set to 0x0 to advertise that this is an Flex Bus PORT feature capability structure and it's a unique ID assigned to Flex bus Port. it's offset value is at 04h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x7 Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

3.43.4 CXL_RCRB_FLEXBUS_CNTRL_STATUS_OFF

- **Name:** DVSEC Flex Bus Port Control and Status.
- **Description:** this register offset consists of port control and status fields.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2))

31:29	RSVDP_29
28	CXL_FB_EN_BITS_P2_MISMATCH
27	CXL_RETIMER_PSNT_MISMATCH
26	CXL_UNEXP_PROT_ID_DROP
25	CXL_UNCORR_PROT_ID_FRAMING_ERROR
24	CXL_CORR_PROT_ID_FRAMING_ERROR
23	RSVDP_23
22	CXL_MLD_END
21	CXL2P0_ENABLED
20	DRIFT_BUF_END
19	CXL_SYNC_HDR_BYPASS_END
18	MEM_END
17	IO_END
16	CACHE_END
15:10	RSVDP_10
9	RETIMER_2_PRESENT
8	RETIMER_1_PRESENT
7	DISABLE_CXL_L1P1_TRAINING
6	CXL_MLD_EN
5	CXL2P0_ENABLED
4	DRIFT_BUF_EN
3	CXL_SYNC_HDR_BYPASS_EN
2	MEM_EN
1	IO_EN
0	CACHE_EN

Table 3-301 Fields for Register: CXL_RCRB_FLEXBUS_CNTRL_STATUS_OFF

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
28	CXL_FB_EN_BITS_P2_MISMATCH_H	R	FlexBusEnableBits_Phase2_Mismatch. When set, indicates that the downstream port physical layer detected that the upstream port did not reflect precisely the Flex Bus enable bits located in symbols 12-14 of the modified TS2 during Phase 2 of the negotiation. Please refer to Section 6.3.1.1 for more details. This bit is reserved on upstream ports. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2))
27	CXL_RETIMER_PSNT_MISMATCH_H	R	CXL_Retimers_Present_Mismatched. When set, indicates that the downstream port physical layer detected an inconsistency in the "Retimers Present" or "Two Retimers Present" bits in the received TS2 Ordered Sets during Polling.Config versus Config.Complete LTSSM states. The physical layer will force disable of the sync header bypass optimization when this error condition has been detected. See Section 6.3.1.2.1 for more details. This bit is reserved on upstream ports. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
26	CXL_UNEXP_PROT_ID_DROP	R/W1C	CXL UNEXPECTED PROTOCOL ID DROPPED. When set, indicates that the physical layer dropped a flit with an unexpected protocol ID that is not due to an Uncorrectable Protocol ID Framing Error. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2))
25	CXL_UNCORR_PROT_ID_FRAMING_ERROR	R/W1C	CXL UNCORRECTABLE PROTOCOL ID FRAMING ERROR. An 8-bit encoding with a hamming distance of four is replicated to create the 16-bit encoding for error protection against bit flips. If both 8-bit encoding groups are incorrect, an uncorrectable protocol ID framing error is logged, the flit is dropped, and the physical layer enters into recovery to retrain the link. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
24	CXL_CORR_PROT_ID_FRAMING_ERROR	R/W1C	CXL CORRECTABLE PROTOCOL ID FRAMING ERROR. An 8-bit encoding with a hamming distance of four is replicated to create the 16-bit encoding for error protection against bit flips. A correctable protocol ID framing error is logged but no further error handling action is taken if only one 8-bit encoding group looks incorrect; the correct 8-bit encoding group is used for normal processing. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2))
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2))
22	CXL_MLD_END	R	When set, indicates that CXL Multi-Logical Device operation has been negotiated. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
21	CXL2P0_ENABLED	R	When set, indicates that CXL2.0 protocol operation has been enabled as a result of PCIe alternate protocol negotiation for Flex Bus. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
20	DRIFT_BUF_END	R	DRIFT BUFFER ENABLED. When set, indicates that the physical layer has enabled its drift buffer instead of its elastic buffer. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
19	CXL_SYNC_HDR_BYPASS_END	R	CXL SYNCHRONOUS BYPASS ENABLED. When set, indicates that bypass of the 2-bit sync header by the Flex Bus physical layer has been enabled when operating in Flex Bus.AL mode as a result of PCIe alternate protocol negotiation for Flex Bus. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
18	MEM_END	R	MEMORY ENABLED. When set, indicates that CXL.mem protocol operation has been enabled as a result of PCIe alternate protocol negotiation for Flex Bus. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
17	IO_END	R	IO ENABLED. When set, indicates that CXL.io protocol operation has been enabled as a result of PCIe alternate protocol negotiation for Flex Bus. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
16	CACHE_END	R	CACHE ENABLED. When set, indicates that CXL.cache protocol operation has been enabled as a result of PCIe alternate protocol negotiation for Flex Bus. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
15:10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2))

Bits	Name	Memory Access	Description
9	RETIMER_2_PRESENT	* Varies	RETIMER 2 PRESENT. When set, indicates presence of retimer2. This bit is defined only for a downstream port. This bit is reserved for an upstream port. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_POSITION==1) ? \"read-write\" : \"read-only\" "
8	RETIMER_1_PRESENT	* Varies	RETIMER 1 PRESENT. When set, indicates presence of retimer1. This bit is defined only for a downstream port. This bit is reserved for an upstream port. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_POSITION==1) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
7	DISABLE_CXL_L1P1_TRAINING	* Varies	When set, CXL1.0 training is disabled. Typical usage model is that BIOS will use this bit to disable hot plug of CXL1.1 devices below a CXL2.0 Downstream Port. This bit should default to 1'b0. This bit is reserved on all CXL Upstream ports and on CXL1.1 components. Default value of this field is 0. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1 && MEMORY_MAP_POSITION == 0) then R/W(sticky) else R(sticky) Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
6	CXL_MLD_EN	* Varies	When set, enable Multi-Logical Device operation when in Flex Bus.CXL mode. This bit is reserved on CXL1.1 components. This bit should always be set to 1'b0 for CXL2.0 host Downstream ports. Default value of this field is 0. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1 && MEMORY_MAP_POSITION == 0) then R/W(sticky) else R(sticky) Value After Reset: DEFAULT_CXL_PORT_USP_MLD_ENABLE Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
5	CXL2P0_ENABLE	* Varies	When set, enable CXL2.0 protocol operation when in Flex Bus.CXL mode. This bit is reserved on CXL1.1 components. Default value of this field is 0. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1 && MEMORY_MAP_POSITION == 0) then R/W(sticky) else R(sticky) Value After Reset: DEFAULT_CXL_PORT_USP_CXL2P0_ENABLE Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
4	DRIFT_BUF_EN	* Varies	DRIFT BUFFER ENABLE. When set, enables drift buffer (instead of elastic buffer) if there is a common reference clock. When SerDes Architecture is not enabled (CX_PIPE_SERDES_ARCH = 0), don't set 1 in this field. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1 && MEMORY_MAP_POSITION == 0) then R/W(sticky) else R(sticky) Value After Reset: DEFAULT_CXL_PORT_USP_DRIFT_BUFFER_ENABLE Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	CXL_SYNC_HDR_BYPASS_EN	* Varies	CXL SYNCHRONOUS HEADER BYPASS ENABLE. When set, enables bypass of the 2-bit sync header by the Flex Bus physical layer when operating in Flex Bus.AL mode. This is a performance optimization. When SerDes Architecture is not enabled (CX_PIPE_SERDES_ARCH = 0), don't set 1 in this field. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1 && MEMORY_MAP_POSITION == 0) then R/W(sticky) else R(sticky) Value After Reset: DEFAULT_CXL_PORT_USP_SYNC_HDR_BYPASS_ENABLE Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: writeAsRead Volatile: true
2	MEM_EN	* Varies	MEMORY ENABLE. When set, enables CXL.mem protocol operation when in Flex Bus.AL mode. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1 && MEMORY_MAP_POSITION == 0) then R/W(sticky) else R(sticky) Value After Reset: DEFAULT_CXL_PORT_USP_MEM_ENABLE Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
1	IO_EN	* Varies	IO ENABLE. When set, enables CXL.io protocol operation when in Flex Bus.AL mode. (Must always be set to 1). The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_CXL_IO_ENABLE Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead
0	CACHE_EN	* Varies	CACHE ENABLE. When set, enables CXL.cache protocol operation when in Flex Bus.AL mode. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1 && MEMORY_MAP_POSITION == 0) then R/W(sticky) else R(sticky) Value After Reset: DEFAULT_CXL_PORT_USP_CACHE_ENABLE Exists: (CX_CXL_ENABLE&&(CX_MEMORY_MAP_PCIE_TYPE<2)) Testable: wire ? unconstrained : writeAsRead

3.43.5 CXL_RCRB_FLEXBUS_RCVD_MOD_TS_PH1_OFF

- **Name:** DVSEC flexBus Port Received Modified TS data Phase 1.
- **Description:** this register offset consists of port MODIFIED TS Received register field.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2))

31:24	RSVDP_24
23:0	RCVD_FB_DATA_PH_1

Table 3-302 Fields for Register: CXL_RCRB_FLEXBUS_RCVD_MOD_TS_PH1_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2))
23:0	RCVD_FB_DATA_PH_1	R	This field contains the values received in Symbols 12 through 14 of the Modified TS1 Ordered Set during Phase 1 of CXL alternate protocol negotiation. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE&& (CX_MEMORY_MAP_PCIE_TYPE<2))

3.44 PF0_CXL_DEVICE_CAP Registers

3.44.1 CXL_RCIEP_PCIE_EXT_CAP_HDR_OFF

- **Name:** Device Cap ID, Device Cap Version, Device Next Offset.
- **Description:** This register is the PCI express extended capability header and offset at 00H.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:20	CXL_DEVICE_NEXT_OFFSET
19:16	CXL_DEVICE_CAP_VERSION
15:0	CXL_DEVICE_CAP_ID

Table 3-303 Fields for Register: CXL_RCIEP_PCIE_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	CXL_DEVICE_NEXT_OFFSET	* Varies	CXL DEVICE NEXT OFFSET. This Device next offset field has read-write attribute and sticky reset type for the offset value 00h with [31:20] bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CXL_DEVICE_CAP_VERSION	* Varies	CXL DEVICE CAPABILITY VERSION. This Device Capability version field has read-write attribute and sticky reset type for the offset value 00h with [19:16] bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
15:0	CXL_DEVICE_CAP_ID	* Varies	CXL Device Capability ID. This Device Capability ID field has read-write attribute and sticky reset type for the offset value 00h with [15:0] bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x23 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.44.2 CXL_RCIEP_DVSEC_HDR_1_OFF

- **Name:** DVSEC Vendor ID, DVSEC Revision, DVSEC Length.
- **Description:** This register is for Designated Vendor Specific Header 1 with offset 04H.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:20	DVSEC_LEN
19:16	DVSEC_REV
15:0	DVSEC_VNDR_ID

Table 3-304 Fields for Register: CXL_RCIEP_DVSEC_HDR_1_OFF

Bits	Name	Memory Access	Description
31:20	DVSEC_LEN	* Varies	DVSEC Length. The DVSEC Legth field is set to 16 bytes to accomodate the Flex Bus registers included in the DVSEC. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x38 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	DVSEC_REV	* Varies	DVSEC Revision. the DVSEC revision field is set to 0x0 to show that this is an FLex Bus feature specific header capability structure. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
15:0	DVSEC_VNDR_ID	* Varies	DVSEC vendor ID. This DVSEC Vendor ID has read-write attribute and sticky reset type for the offset value as 04h with [15:0] bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1e98 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.44.3 CXL_RCIEP_DVSEC_HDR_2_FLEXBUS_CAP_OFF

- **Name:** DVSEC ID,cache Capable, IO capable, Memory capable, Memory HwInit Mode, HDM count and Viral capable.
- **Description:** This register is for Designated vendor specific Header 2 with offset 08H.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31	PM_INIT_CMPL_RPT_CAP
30	VIRAL_CAP
29	CXL_LOGICAL_DEV
28	RSVDP_28
27	CXL_RST_MEM_CLR_CAP
26:24	CXL_RST_TIMEOUT
23	CXL_RST_CAP
22	CACHE_WB_INVALID_CAP
21:20	HDM_COUNT
19	MEM_HWINIT_MODE
18	MEM_CAP
17	IO_CAP
16	CACHE_CAP
15:0	DVSEC_ID

Table 3-305 Fields for Register: CXL_RCIEP_DVSEC_HDR_2_FLEXBUS_CAP_OFF

Bits	Name	Memory Access	Description
31	PM_INIT_CMPL_RPT_CAP	* Varies	Power Management complete Report Capable. If set, indicates that the CXL device shall is capable of supporting Power Management Initialization Complete flag. All CXL 2.0 devices shall set this capability bit. CXL 1.1 devices may implement this capability. This capability is not applicable to switches and Root Ports. Switches and Root Ports shall hardwire this bit to 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_PM_INIT_CMPL_RPT_EN} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
30	VIRAL_CAP	* Varies	VIRAL CAPABLE. If set, indicates CXL device supports Viral handling. This value must be the same for all functions of an MLD. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_VIRAL_CAPABLE} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
29	CXL_LOGICAL_DEV	* Varies	Logical Device. If set, indicates this is a Logical Device. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_LOGICAL_DEVICE_EN} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
27	CXL_RST_MEM_CLR_CAP	* Varies	CXL Reset Mem Clr Capable. When set, Device is capable of clearing or randomizing of Volatile HDM Ranges during CLX FLR. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_RESET_MEM_CLR_EN} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
26:24	CXL_RST_TIMEOUT	* Varies	CXL Reset Timeout. If CXL Reset Capable bit is set, this field indicates the maximum time that the device may take to complete the CXL Reset in. If CXL Reset Mem Clr Capable bit is 1, this time also accounts for the time needed for clearing or randomizing of volatile HDM Ranges. If CXL Reset completion status bit in Flex Bus Status2 register is not set after the passage of this time duration, software may assume CXL Reset has failed. This value must be the same for all functions of an MLD. All other encodings are Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (encoding_for_ten_ms): 10 ms ■ 0x1 (encoding_for_hundred_ms): 100 ms ■ 0x2 (encoding_for_one_s): 1 s ■ 0x3 (encoding_for_ten_s): 10 s ■ 0x4 (encoding_for_hundred_s): 100 s Value After Reset: "(CX_CXL_RESET_EN) ? CX_CXL_RESET_TIMEOUT : 0" Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
23	CXL_RST_CAP	* Varies	CXL Reset Capable. If set, indicates the device supports CXL Reset and implements the CXL Reset Timeout field in this register, Initiate CXL Reset bit in Flex Bus Control2 register and CXL Reset Completion Status in Flex Bus Status2 register. This field must report the same value for all functions of an MLD. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_RESET_EN} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
22	CACHE_WB_INVALID_CAP	* Varies	Cache Writeback and Invalidation Cache. If set, indicates the device implements Disable Caching and Initiate Cache Write Back and Invalidation control bits in Flex Bus Control2 register and Cache Invalid status bit in Flex Bus Status2 register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_CACHE_WR_INVLD_EN} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
21:20	HDM_COUNT	* Varies	HDM COUNT. Number of HDM ranges implemented by the CXL device and reported through this function. 00 - Zero ranges. This setting is illegal if CXL.mem Capable=1. 01 - One HDM range. 10 - Two HDM ranges. 11 - Reserved. This field must return 00 if CXL.mem Capable=0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_HDM_COUNT} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
19	MEM_HWINIT_MODE	* Varies	MEMORY HWINIT MODE. If set, indicates this CXL.mem capable device initializes memory with assistance from hardware and firmware located on the device. If clear, indicates memory is initialized by host software such as device driver. This bit should be ignored if CXL.mem Capable=0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_MEM_HWINIT_MODE} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
18	MEM_CAP	* Varies	MEMORY CAPABLE. If set, indicates CXL.mem protocol support when operating in Flex Bus.CXL mode. This must be 1 for all functions of an MLD. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: "(CX_CXL_MEM_EN) ? 1 : 0" Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
17	IO_CAP	R	IO CAPABLE. If set indicates CXL.io protocol support when operating in Flex Bus.CXL mode. Must be 1. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
16	CACHE_CAP	* Varies	CACHE CAPABLE. If set, indicates CXL.cache protocol support when operating in Flex Bus.CXL mode. This must be 0 for all functions of an MLD. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: "(CX_CXL_CACHE_EN) ? 1 : 0" Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15:0	DVSEC_ID	* Varies	DVSEC ID. DVSEC ID is set to 0x0 to advertise that this is an Flex Bus feature capability structure. it's offset value is at 04h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.44.4 CXL_RCIEP_FLEXBUS_CNTRL_STATUS_OFF

- **Name:** Cache Enable, IO Enable, Memory Enable, Cache SF Coverage, Cache SF Granularity, Cache Clean and Viral Enable, Viral Status.
- **Description:** This register controls DVSEC Flex Bus at offset 0CH.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31	RSVDP_31
30	VIRAL_STATUS
29:15	RSVDP_15
14	VIRAL_EN
13:12	RSVDP_12
11	CACHE_CLEAN
10:8	CACHE_SF_GRAN
7:3	CACHE_SF_COV
2	MEM_EN
1	IO_EN
0	CACHE_EN

Table 3-306 Fields for Register: CXL_RCIEP_FLEXBUS_CNTRL_STATUS_OFF

Bits	Name	Memory Access	Description
31	RSVDP_31	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
30	VIRAL_STATUS	R/W1C	VIRAL STATUS. When set, indicates that the CXL device has entered Viral self-isolation mode. Note- Viral condition is cleared by warm reset or cold reset, but Viral_Status bit survives warm reset. Note: Device shall exit Vviral condition upon hot reset, warm reset or cold reset even though this status bit is sticky. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
29:15	RSVDP_15	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
14	VIRAL_EN	* Varies	VIRAL ENABEL. When set, enables Viral handling in the CXL device. Locked by CONFIG_LOCK. Default value of this bit is 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RWL ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-write\""
13:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
11	CACHE_CLEAN	* Varies	CACHE CLEAN EVICTION. Eviction: Performance hint to the device. Locked by CONFIG_LOCK. Default value of this bit is 0 <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RWL ■ Dbi: R/W Values: ■ 0x0 (Clean_eviction_from_DEV): Indicates clean evictions from device caches are needed for best performance ■ 0x1 (Clean_evic_not_required): Indicates clean evictions from device caches are NOT needed for best performance Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-write\""

Bits	Name	Memory Access	Description
10:8	CACHE_SF_GRAN	* Varies	CACHE SF Granularity. Performance hint to the device. Locked by CONFIG_LOCK. Default value of this field is 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RWL ■ Dbi: R/W Values: ■ 0x0 (Granular_Tracking_64B): Indicates 64B granular tracking on the Host ■ 0x1 (Granular_Tracking_128B): Indicates 128B granular tracking on the Host ■ 0x2 (Granular_Tracking_256B): Indicates 256B granular tracking on the Host ■ 0x3 (Granular_Tracking_512B): Indicates 512B granular tracking on the Host ■ 0x4 (Granular_Tracking_1KB): Indicates 1KB granular tracking on the Host ■ 0x5 (Granular_Tracking_2KB): Indicates 2KB granular tracking on the Host ■ 0x6 (Granular_Tracking_4KB): Indicates 4KB granular tracking on the Host ■ 0x7 (Granular_Tracking): Reserved Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \\"read-write\\" : (CX_MEMORY_MAP_VIEW==1) ? \\"read-write\\" : \\"read-write\\""

Bits	Name	Memory Access	Description
7:3	CACHE_SF_COV	* Varies	CACHE SF COVERAGE. Performance hint to the device. Locked by CONFIG_LOCK. For all other values it Indicates Snoop Filter coverage on the Host of $2^{(N+15d)}$ Bytes. For example, if this field contains the value 5, it indicates snoop filter coverage of 1MB. Default value of this field is 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RWL ■ Dbi: R/W Values: ■ 0x0 (Filter_cov): Indicates no Snoop Filter coverage on the Host Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \\"read-write\\" : (CX_MEMORY_MAP_VIEW==1) ? \\"read-write\\" : \\"read-write\\""
2	MEM_EN	* Varies	MEMORY ENABLE. When set, enables CXL.mem protocol operation when in Flex Bus.CXL mode. Locked by CONFIG_LOCK. Default value of this bit is 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RWL ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \\"read-write\\" : (CX_MEMORY_MAP_VIEW==1) ? \\"read-write\\" : \\"read-write\\""

Bits	Name	Memory Access	Description
1	IO_EN	* Varies	IO ENABLE. When set, enables CXL.io protocol operation when in Flex Bus.CXL mode. This bit always returns 1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_CXL_IO_ENABLE Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
0	CACHE_EN	* Varies	CACHE_ENABLE. When set, enables CXL.cache protocol operation when in Flex Bus.CXL mode. Locked by CONFIG_LOCK. Default value of this bit is 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RWL ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-write\""

3.44.5 CXL_RCIEP_FLEXBUS_CNTRL_2_STATUS_2_OFF

- **Name:** Disable Caching, Initiate Cache Write Back and Invalidation, Initiate eFLR, Cache Invalid, eFLR Complete, Power Management Initialization Complete.
- **Description:** This register controls DVSEC Flex Bus at offset 10H.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31	PWR_INIT_CMPL
30:19	RSVDP_19
18	CXL_RST_ERROR
17	CXL_RST_CMPL
16	CACHE_INVALID
15:4	RSVDP_4
3	CXL_RST_MEM_CLR_EN
2	INIT_CXL_RST
1	INIT_CACHE_WR_INVALID
0	DISABLE_CACHE

Table 3-307 Fields for Register: CXL_RCIEP_FLEXBUS_CNTRL_2_STATUS_2_OFF

Bits	Name	Memory Access	Description
31	PWR_INIT_CMPL	R	Power Management Initialization Complete. When set, it indicates that the device has successfully completed Power Management Initialization Flow and is ready to process various Power Management messages.If this bit is not set within 100 ms of link-up, software may conclude that the Power Management initialization has failed and may issue Secondary Bus Reset to force link reinitialization and Power Management re-initialization. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
30:19	RSVDP_19	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
18	CXL_RST_ERROR	R	CXL Reset Error. When set, device has completed CXL Reset with errors. Additional information will be available in error logs. Host software of Fabric Manager may optionally re-issue CXL Reset. Device shall clear this bit upon transition of Initiate CXL Reset bit from 0 to 1, prior to initiating CXL Reset flow. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true
17	CXL_RST_CMPL	R	CXL Reset complete. When set, device has been successfully completed CXL Reset as defined in section Section 9.8. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
16	CACHE_INVALID	R	When set, device guarantees that it does not hold any valid lines and Disable Caching=1. This bit shall read as 0 when Disable Caching=0. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
15:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
3	CXL_RST_MEM_CLR_EN	R/W	CXL Reset Mem Clr Enable. When set, and CXL Reset Mem Clr Capable returns 1, Device shall clear or randomize Volatile HDM ranges as part of the CXL Reset operation. When CXL Reset Mem Clr Capable is clear, this bit is ignored and Volatile HDM ranges may or may not be cleared or randomized during CXL Reset. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
2	INIT_CXL_RST	R/W	Initiate CXL Reset. When set to 1, device shall initiate CXL FLR as defined in Section 9.8. This bit always returns the value of 0 when read by the software. A write of 0 is ignored. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true
1	INIT_CACHE_WR_INVALID	R/W	When set to 1, device shall write back all modified lines in the local cache and invalidate all lines. The device shall send CacheFlushed message to host as required by CXL.Cache protocol to indicate it does not hold any modified lines. If this bit is set when Disable Caching=0, the device behavior is undefined. This bit always returns the value of 0 when read by the software. A write of 0 is ignored. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true
0	DISABLE_CACHE	R/W	When set to 1, device shall no longer cache new modified lines in its local cache. Device shall continue to correctly respond to CXL.cache transactions. Default value of this bit is 0. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true

3.44.6 CXL_RCIEP_FLEX_BUS_LOCK_OFF

- **Name:** Config Lock.
- **Description:** This register is for Flex Bus Lock.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:24	23:20	19:16	15:1	0
CACHE_SIZE	RSVDP_20	CACHE_SIZE_UNIT	RSVDP_1	CONFIG_LOCK

Table 3-308 Fields for Register: CXL_RCIEP_FLEX_BUS_LOCK_OFF

Bits	Name	Memory Access	Description
31:24	CACHE_SIZE	* Varies	Expressed in multiples of Cache Size Unit. If Cache Size=4 and Cache Size Unit=0001b, the device has 256K sized cache. A type 3 CXL device shall return the value of 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_DEVICE_CACHE_SIZE} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
23:20	RSVDP_20	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
19:16	CACHE_SIZE_UNIT	* Varies	Cache Size Unit. All Other are Reserved. A type 3 CXL device shall return the value of 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (No_cache_size): Cache size is not reported. ■ 0x1 (cache_size_64k): 64k ■ 0x2 (cache_size_1M): 1 MB Value After Reset: {CX_CXL_DEVICE_CACHE_SIZE_UNIT} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
15:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
0	CONFIG_LOCK	* Varies	CONFIG LOCK. When set, control register, Memory Base Low and Memory Base High registers become read only. Default value of this bit is 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RW1 ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""

3.44.7 CXL_RCIEP_FLEXBUS_R1_SIZE_HIGH_OFF

- **Name:** Memory size high.
- **Description:** This register is for DVSEC Flex Bus Range 1 size high.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

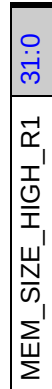


Table 3-309 Fields for Register: CXL_RCIEP_FLEXBUS_R1_SIZE_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	MEM_SIZE_HIGH_R1	R/W	MEMORY SIZE HIGH RANGE 1. Corresponds to bits 63:32 of Flex Bus Range 1 memory size. Value After Reset: {CX_CXL_RANGE_1_MEM_SIZE_HIGH} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true

3.44.8 CXL_RCIEP_FLEXBUS_R1_SIZE_LOW_OFF

- **Name:** Memory Info Valid, Memory Active, Memory Type, Memory Class, Desired Interleave and Memory size low.
- **Description:** This register is for DVSEC Flex Bus Range 1 size LOW.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:28	MEM_SIZE_LOW_R1
27:16	RSVDP_16
15:13	MEMORY_ACTIVE_TIMEOUT_R1
12:8	DESIRED_INTERLEAVE_R1
7:5	MEM_CLASS_R1
4:2	MEDIA_TYPE_R1
1	MEM_ACTIVE_R1
0	MEM_INFO_VALID_R1

Table 3-310 Fields for Register: CXL_RCIEP_FLEXBUS_R1_SIZE_LOW_OFF

Bits	Name	Memory Access	Description
31:28	MEM_SIZE_LOW_R1	* Varies	MEMORY SIZE LOW - RANGE 1. Corresponds to bits 31:20 of Flex Bus Range 1 memory size. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_RANGE_1_MEM_SIZE_LOW} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
27:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
15:13	MEMORY_ACTIVE_TIMEOUT_R1	* Varies	Memory Active Timeout - Range 1. Indicates maximum time that device is permitted to take to set MEM_ACTIVE_R1 field in this register after a conventional reset. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
12:8	DESIRED_INTERLEAVE_R1	* Varies	DESIRED INTERLEAVE - RANGE R1. If a CXL.mem capable CXL 1.1 device is connected to a single CPU via multiple Flex BusCXL links, this field represents the memory interleaving desired by the device. BIOS will configure the CPU to interleave accesses to this HDM range across links at this granularity. or the closest possible value supported by the host. In the case of CXL 2.0 device, this field represents the minimum desired interleave granularity for optimal device performance. Software should program the Interleave Granularity (IG) field in the HDM Decoder 0/n Control Registers (Section 8.2.5.12.7 and Section 8.2.5.12.15) to be an exact match or any larger granularity advertised by the device via CXL HDM Decoder Capability Register (Section 8.2.5.12.1). This field is treated as a hint. The device shall function correctly if the actual programmed value programmed in Interleave Granularity (IG) field in the HDM Decoder 0/n Control Registers is less than what is reported via this field. 00000 - No Interleave 00001 - 256 Byte Granularity 00010 - 4K Interleave The above encodings are backwards compatible with CXL 1.1. The following encodings are new for CXL 2.0 all other settings are reserved. Notes: If a CXL 2.0 Device has different desired interleave values for DPA ranges covered by this CXL Range 1, it should report a value that best fits the requirements for all such ranges (e.g. max of the values). If CXL 2.0 devices in an Interleave Set advertise different values for this field, Software may choose the smallest value that best fits the set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky)
12:8...(cont.)	DESIRED_INTERLEAVE_R1.	* Varies	■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x3 (byte_1_r1): 512 Bytes ■ 0x4 (byte_2_r1): 1024 Bytes ■ 0x5 (byte_3_r1): 2048 Bytes ■ 0x6 (byte_4_r1): 8192 Bytes ■ 0x7 (byte_5_r1): 16384 Bytes Value After Reset: {CX_CXL_RANGE_1_DESIRED_INTERLEAVE} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
7:5	MEM_CLASS_R1	* Varies	MEMORY CLASS - RANGE 1. Indicates the class of memory. All other encodings are reserved. CXL 2.0 and future CXL.mem devices shall set this field to 010. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (mem_class): Memory Class (e.g., normal DRAM). This setting is deprecated starting with CXL 2.0 ■ 0x1 (mme_storage): Storage Class. this setting is deprecated starting with CXL 2.0 ■ 0x2 (mem_cdat): The memory characteristics are communicated via CDAT (Section 8.1.12) and not via this field. Value After Reset: {CX_CXL_RANGE_1_MEM_CLASS} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
4:2	MEDIA_TYPE_R1	* Varies	MEDIA TYPE - RANGE 1. Indicates the memory media characteristics. All other encodings are reserved. CXL 2.0 and future CXL.mem devices shall set this field to 010. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (volatile_mem): Volatile memory. This setting is deprecated starting with CXL 2.0 ■ 0x1 (non-volatile_mem): Non-volatile memory. this setting is deprecated starting with CXL 2.0 ■ 0x2 (mem_char): The memory characteristics are communicated via CDAT (Section 8.1.12) and not via this field. Value After Reset: {CX_CXL_RANGE_1_MEDIA_TYPE} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
1	MEM_ACTIVE_R1	* Varies	MEMORY ACTIVE - RANGE 1. When set, indicates that the Flex Bus Range 1 memory is fully initialized and available for software use. Must be set within 1 second of deassertion of reset to CXL device if CXL.mem HwInit Mode=1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_RANGE_1_MEM_ACTIVE} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
0	MEM_INFO_VALID_R1	* Varies	MEMORY INFO VALID - RANGE1 1. When set, indicates that the Flex Bus Range 1 Size high and Size Low registers are valid. Must be set within 1 second of deassertion of reset to CXL device. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_RANGE_1_MEM_INFO_VALID} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.44.9 CXL_RCIEP_FLEXBUS_R1_BASE_HIGH_OFF

- **Name:** Memory base high.
- **Description:** This register is for DVSEC Flex Bus Range 1 Base HIGH.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))



Table 3-311 Fields for Register: CXL_RCIEP_FLEXBUS_R1_BASE_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	MEM_BASE_HIGH_R1	R/W	MEMORY BASE HIGH - RANGE 1. Corresponds to bits 63:32 of Flex Bus Range 1 base in the host address space. Locked by CONFIG_LOCK. If a device implements CXL HDM Decoder Capability registers and software has enabled HDM Decoder by setting HDM Decoder Enable bit in CXL HDM Decoder Global Control register, the value of this field is not used during address decode. It is recommended that software program this to match CXL HDM Decoder 0 Base High Register for backward compatibility reasons. Default value of this field is 0. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true

3.44.10 CXL_RCIEP_FLEXBUS_R1_BASE_LOW_OFF

- **Name:** Memory base Low.
- **Description:** This register is for DVSEC Flex Bus Range 1 Base LOW.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:28	27:0
MEM_BASE_LOW_R1	RSVDP_0

Table 3-312 Fields for Register: CXL_RCIEP_FLEXBUS_R1_BASE_LOW_OFF

Bits	Name	Memory Access	Description
31:28	MEM_BASE_LOW_R1	R/W	MEMORY BASE LOW - RANGE 1. Corresponds to bits 31:28 of Flex Bus Range 1 base in the host address space. Locked by CONFIG_LOCK. If a device implements CXL HDM Decoder Capability registers and software has enabled HDM Decoder by setting HDM Decoder Enable bit in CXL HDM Decoder Global Control register, the value of this field is not used during address decode. It is recommended that software program this to match CXL HDM Decoder 0 Base Low Register for backward compatibility reasons. Default value of this field is 0 Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
27:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.44.11 CXL_RCIEP_FLEXBUS_R2_SIZE_HIGH_OFF

- **Name:** Memory Size High.
- **Description:** This register is for DVSEC Flex Bus Range 2 size HIGH.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

MEM_SIZE_HIGH_R2 31:0

Table 3-313 Fields for Register: CXL_RCIEP_FLEXBUS_R2_SIZE_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	MEM_SIZE_HIGH_R2	* Varies	MEMORY SIZE HIGH - RANGE 2. Corresponds to bits 63:32 of Flex Bus Range 2 memory size. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_RANGE_2_MEM_SIZE_HIGH} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.44.12 CXL_RCIEP_FLEXBUS_R2_SIZE_LOW_OFF

- **Name:** Memory Info Valid, Memory Active, Memory Type, Memory Class, Desired Interleave and Memory size low.
- **Description:** This register is for DVSEC Flex Bus Range 2 size LOW.
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:28	MEM_SIZE_LOW_R2
27:16	RSVDP_16
15:13	MEMORY_ACTIVE_TIMEOUT_R2
12:8	DESIRED_INTERLEAVE_R2
7:5	MEM_CLASS_R2
4:2	MEDIA_TYPE_R2
1	MEM_ACTIVE_R2
0	MEM_INFO_VALID_R2

Table 3-314 Fields for Register: CXL_RCIEP_FLEXBUS_R2_SIZE_LOW_OFF

Bits	Name	Memory Access	Description
31:28	MEM_SIZE_LOW_R2	* Varies	MEMORY SIZE LOW - RANGE 2. Corresponds to bits 31:20 of Flex Bus Range 2 memory size. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_RANGE_2_MEM_SIZE_LOW} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
27:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
15:13	MEMORY_ACTIVE_TIMEOUT_R2	* Varies	Memory Active Timeout - Range 2. Indicates maximum time that device is permitted to take to set MEM_ACTIVE_R2 field in this register after a conventional reset. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
12:8	DESIRED_INTERLEAVE_R2	* Varies	DESIRED INTERLEAVE - RANGE 2. If a CXL.mem capable CXL 1.1 device is connected to a single CPU via multiple Flex BusCXL links, this field represents the memory interleaving desired by the device. BIOS will configure the CPU to interleave accesses to this HDM range across links at this granularity. or the closest possible value supported by the host. In the case of CXL 2.0 device, this field represents the minimum desired interleave granularity for optimal device performance. Software should program the Interleave Granularity (IG) field in the HDM Decoder 0/n Control Registers (Section 8.2.5.12.7 and Section 8.2.5.12.15) to be an exact match or any larger granularity advertised by the device via CXL HDM Decoder Capability Register (Section 8.2.5.12.1). This field is treated as a hint. The device shall function correctly if the actual programmed value programmed in Interleave Granularity (IG) field in the HDM Decoder 0/n Control Registers is less than what is reported via this field. 00000 - No Interleave 00001 - 256 Byte Granularity 00010 - 4K Interleave The above encodings are backwards compatible with CXL 1.1. The following encodings are new for CXL 2.0 all other settings are reserved. Notes: If a CXL 2.0 Device has different desired interleave values for DPA ranges covered by this CXL Range 1, it should report a value that best fits the requirements for all such ranges (e.g. max of the values). If CXL 2.0 devices in an Interleave Set advertise different values for this field, Software may choose the smallest value that best fits the set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky)
12:8...(cont.)	DESIRED_INTERLEAVE_R2.	* Varies	■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x3 (byte_1_r2): 512 Bytes ■ 0x4 (byte_2_r2): 1024 Bytes ■ 0x5 (byte_3_r2): 2048 Bytes ■ 0x6 (byte_4_r2): 8192 Bytes ■ 0x7 (byte_5_r2): 16384 Bytes Value After Reset: {CX_CXL_RANGE_2_DESIRED_INTERLEAVE} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
7:5	MEM_CLASS_R2	* Varies	MEMORY CLASS - RANGE 2. Indicates the class of memory. All other encodings are reserved. CXL 2.0 and future CXL.mem devices shall set this field to 010. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (mem_class_r2): Memory Class (e.g., normal DRAM). This setting is deprecated starting with CXL 2.0 ■ 0x1 (mme_storage_r2): Storage Class. this setting is deprecated starting with CXL 2.0 ■ 0x2 (mem_cdat_r2): The memory characteristics are communicated via CDAT (Section 8.1.12) and not via this field. Value After Reset: {CX_CXL_RANGE_2_MEM_CLASS} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
4:2	MEDIA_TYPE_R2	* Varies	MEDIA TYPE - RANGE 2. Indicates the memory media characteristics. All other encodings are reserved. CXL 2.0 and future CXL.mem devices shall set this field to 010. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (volatile_mem_r2): Volatile memory. This setting is deprecated starting with CXL 2.0 ■ 0x1 (non-volatile_mem_r2): Non-volatile memory. this setting is deprecated starting with CXL 2.0 ■ 0x2 (mem_char_r2): The memory characteristics are communicated via CDAT (Section 8.1.12) and not via this field. Value After Reset: {CX_CXL_RANGE_2_MEDIA_TYPE} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
1	MEM_ACTIVE_R2	* Varies	MEMORY ACTIVE - RANGE 2. When set, indicates that the Flex Bus Range 2 memory is fully initialized and available for software use. Must be set within 1 second of deassertion of reset to CXL device if CXL.mem HwInit Mode=1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_RANGE_2_MEM_ACTIVE} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
0	MEM_INFO_VALID_R2	* Varies	MEMORY INFO VALID - RANGE 2. When set, indicates that the Flex Bus Range 2 Size high and Size Low registers are valid. Must be set within 1 second of deassertion of reset to CXL device. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: {CX_CXL_RANGE_2_MEM_INFO_VALID} Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.44.13 CXL_RCIEP_FLEXBUS_R2_BASE_HIGH_OFF

- **Name:** Memory base Low.
- **Description:** This register is for DVSEC Flex Bus Range 2 base LOW.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))



Table 3-315 Fields for Register: CXL_RCIEP_FLEXBUS_R2_BASE_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	MEM_BASE_HIGH_R2	R/W	MEMORY BASE HIGH - RANGE 2. Corresponds to bits 63:32 of Flex Bus Range 2 base in the host address space. Configured by system BIOS. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true

3.44.14 CXL_RCIEP_FLEXBUS_R2_BASE_LOW_OFF

- **Name:** Memory base high.
- **Description:** This register is for DVSEC Flex Bus Range 2 base LOW.
- **Size:** 32 bits
- **Offset:** B+0x34
- **Exists:** (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:28	MEM_BASE_LOW_R2
27:0	RSVDP_0

Table 3-316 Fields for Register: CXL_RCIEP_FLEXBUS_R2_BASE_LOW_OFF

Bits	Name	Memory Access	Description
31:28	MEM_BASE_LOW_R2	R/W	MEMORY BASE LOW - RANGE 2. Corresponds to bits 31:20 of Flex Bus Range 2 base in the host address space. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: writeAsRead Volatile: true
27:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.45 PF0_CXL_DEVICE_TEST_CAP_CAP Registers

3.45.1 CXL_TEST_PCIE_EXT_CAP_HDR_OFF

- **Name:** Device Test Capability ID, Device Test Version ID and Device Test Next offset pointer.
- **Description:** This Register Feild for DVSEC Device Test Registers with Capability ID, Version and Next Offset Pointer.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

CXL_DEVICE_TEST_CAP_NEXT_OFFSET	31:20
CXL_DEVICE_TEST_CAP_VER	19:16
CXL_DEVICE_TEST_CAP_ID	15:0

Table 3-317 Fields for Register: CXL_TEST_PCIE_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	CXL_DEVICE_TEST_CAP_NEXT_OFFSET	* Varies	DVSEC DEVICE TEST Next Capability Offset. This field contains the offset to the next PCI Express DVSEC DEVICE TEST Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
19:16	CXL_DEVICE_TEST_CAP_VER	* Varies	PCI Express Extended capability Version for DEVICE TEST. This field is a PCIe DVSEC CXL DEVICE TEST defined version number that indicates the version of the capability structure present. This field is depends on the version of specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15:0	CXL_DEVICE_TEST_CAP_ID	* Varies	PCI Express Extended Capability ID for DVSEC DEVICE TEST. This PCIe DVSEC DEVICE TEST defined ID number that indicates the nature and format of the extended capability. And this Device Test Capability ID field has read-write attribute and sticky reset type for the offset value 00h with [15:0] bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x23 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.45.2 CXL_TEST_DVSEC_HDR_1_OFF

- **Name:** DVSEC Device TEST Header 1: Vendor ID, Revision, Length.
- **Description:** Register Fields for DVSEC DEVICE TEST Registers: Designated Vendor Specific Header 1.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:20	CXL_DEVICE_TEST_CAP_LENGTH
19:16	CXL_DEVICE_TEST_CAP_REV
15:0	CXL_DEVICE_TEST_CAP_VND_ID

Table 3-318 Fields for Register: CXL_TEST_DVSEC_HDR_1_OFF

Bits	Name	Memory Access	Description
31:20	CXL_DEVICE_TEST_CAP_LENGTH	* Varies	DVSEC CXL DEVICE TEST HEADER 1 Length. The DVSEC Legth field is set to 16 bytes to accomodate the DEVIVCE TEST registers included in the DVSEC. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1c Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CXL_DEVICE_TEST_CAP_REV	* Varies	DVSEC CXL DEVICE TEST HEADER 1 REVISION. the DVSEC revision field is set to 0x0 to show that this is an DVSEC DEVICE TEST fetaure specific header capability structure. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
15:0	CXL_DEVICE_TEST_CAP_VND_ID	* Varies	PCI Express Extended Capability vendor ID. The Vendor ID register identifies the manufacturer of the Function. Valid vendor identifiers are allocated by the PCI-SIG to ensure uniqueness. It is not permitted to populate this register with a value of FFFFh, which is an invalid value for Vendor ID. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1e98 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.45.3 CXL_TEST_DVSEC_HDR_2_TEST_LOCK_OFF

- **Name:** DVSEC DEVICE TEST ID and TEST LOCK Capability.
- **Description:** this register field consists DVSEC ID for DEVICE TEST and Test Lock register field.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:17	RSVDP_17
16	TEST_LOCK
15:0	CXL_DEVICE_TEST_CAP_DVSEC_ID

Table 3-319 Fields for Register: CXL_TEST_DVSEC_HDR_2_TEST_LOCK_OFF

Bits	Name	Memory Access	Description
31:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
16	TEST_LOCK	R/W	Software writes 1'b1 to lock the relevant test configuration registers. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: untestable

Bits	Name	Memory Access	Description
15:0	CXL_DEVICE_TEST_CAP_DVSEC_ID	* Varies	DVSEC CXL DEVICE TEST ID. DVSEC ID reset value is set to 0x0 to advertise that this is an DVSEC DEVICE TEST feature capability structure and it's a unique ID assigned to Flex bus Port. it's offset value is at 04h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xa Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.45.4 CXL_TEST_DVSEC_CAPABILITY_OFF

- **Name:** DVSEC CXL TEST CAPABILITY 1.
- **Description:** Defines the CXL Device CTS Capabilities.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

CXL_DEVICE_TEST_CONFIG_SIZE	31:24
RSVDP_21	23:21
CXL_DEVICE_TEST_COMP_TIMEOUT_INJECT	20
CXL_DEVICE_TEST_UNEXPECTED_COMP	19
CXL_DEVICE_TEST_CACHE_FLUSHED	18
CXL_DEVICE_TEST_WR_INV	17
CXL_DEVICE_TEST_WO_WR_INV_F	16
CXL_DEVICE_TEST_WO_WR_INV	15
CXL_DEVICE_TEST_CLEAN_EVC_NO_DATA	14
CXL_DEVICE_TEST_DIRTY_EVICT	13
CXL_DEVICE_TEST_CLEAN_EVICT	12
CXL_DEVICE_TEST_CL_FLUSH	11
CXL_DEVICE_TEST_MEM_WR	10
CXL_DEVICE_TEST_ITOM_WR	9
CXL_DEVICE_TEST_RD_RD_OWN_NO_DATA	8
CXL_DEVICE_TEST_RD_RD_ANY	7
CXL_DEVICE_TEST_RD_RD_SHARED	6
CXL_DEVICE_TEST_RD_RD_OWN	5
CXL_DEVICE_TEST_RD_CURR	4
CXL_DEVICE_TEST_ALGO_2	3
CXL_DEVICE_TEST_ALGO_1B	2
CXL_DEVICE_TEST_ALGO_1A	1
CXL_DEVICE_TEST_SELF_CHECK	0

Table 3-320 Fields for Register: CXL_TEST_DVSEC_CAPABILITY_OFF

Bits	Name	Memory Access	Description
31:24	CXL_DEVICE_TEST_CONFIG_SIZE	* Varies	Size in Bytes of Test configuration control registers. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
23:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
20	CXL_DEVICE_TEST_COMP_TIMEOUT_INJECT	* Varies	Device supports dropping a read in the completion timeout scenario. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
19	CXL_DEVICE_TEST_UNEXPECTED_COMP	* Varies	Device supports sending an unexpected completion on CXL.io. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
18	CXL_DEVICE_TEST_CACHE_FLUSHED	* Varies	Set to 1 if Device supports CXL.cache and CacheFlushed opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
17	CXL_DEVICE_TEST_WR_INV	* Varies	Set to 1 if Device supports CXL.cache and WrInv opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
16	CXL_DEVICE_TEST_WO_WR_INV_F	* Varies	Set to 1 if Device supports CXL.cache and WOWrInvF opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15	CXL_DEVICE_TEST_WO_WR_INV	* Varies	Set to 1 if Device supports CXL.cache and WOWrInv opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
14	CXL_DEVICE_TEST_CLEAN_EV_C_NO_DATA	* Varies	Set to 1 if Device supports CXL.cache and CleanEvictNoData opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
13	CXL_DEVICE_TEST_DIRTY_EVICT	* Varies	Set to 1 if Device supports CXL.cache and DirtyEvict opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
12	CXL_DEVICE_TEST_CLEAN_EVICT	* Varies	Set to 1 if Device supports CXL.cache and CleanEvict opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
11	CXL_DEVICE_TEST_CL_FLUSH	* Varies	Set to 1 if Device supports CXL.cache and CLFlush opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
10	CXL_DEVICE_TEST_MEM_WR	* Varies	Set to 1 if Device supports CXL.cache and MemWr opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
9	CXL_DEVICE_TEST_ITOM_WR	* Varies	Set to 1 if Device supports CXL.cache and ItoMWr opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
8	CXL_DEVICE_TEST_RD_OWN_N O_DATA	* Varies	Set to 1 if Device supports CXL.cache and RdOwnNoData opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
7	CXL_DEVICE_TEST_RD_ANY	* Varies	Set to 1 if Device supports CXL.cache and RdAny opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
6	CXL_DEVICE_TEST_RD_SHARE D	* Varies	Set to 1 if Device supports CXL.cache and RdShared opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
5	CXL_DEVICE_TEST_RD_OWN	* Varies	Set to 1 if Device supports CXL.cache and RdOwn opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
4	CXL_DEVICE_TEST_RD_CURR	* Varies	Set to 1 if Device supports CXL.cache and RdCurr opcodes as requester. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
3	CXL_DEVICE_TEST_ALGO_2	* Varies	Set to 1 if Device supports hardware for test Algorithm 2. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
2	CXL_DEVICE_TEST_ALGO_1B	* Varies	Set to 1 if Device supports hardware for test Algorithm 1b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
1	CXL_DEVICE_TEST_ALGO_1A	* Varies	Set to 1 if Device supports hardware for test Algorithm 1a. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
0	CXL_DEVICE_TEST_SELF_CHECK	* Varies	Set to 1 if Device supports Self Checking. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.45.5 CXL_TEST_DVSEC_CAPABILITY_2_OFF

- **Name:** CACHE SIZE and CACHE SIZE UNITS.
- **Description:** Cache size supported by the device & Units of advertised cache size in CacheSize field.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:16	RSVDP_16
15:14	CXL_DEVICE_TEST_CACHE_SIZE_UNITS
13:0	CXL_DEVICE_TEST_CACHE_SIZE

Table 3-321 Fields for Register: CXL_TEST_DVSEC_CAPABILITY_2_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
15:14	CXL_DEVICE_TEST_CACHE_SIZE_UNITS	* Varies	Units of advertised cache size in CacheSize field. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BYTES): Bytes. ■ 0x1 (KB): KiloBytes. ■ 0x2 (MB): MegaBytes. ■ 0x3 (RSVPD): RESERVED. Value After Reset: CX_CXL_CACHE_SIZE_UNITS Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
13:0	CXL_DEVICE_TEST_CACHE_SIZE	* Varies	Cache size supported by the device. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_CXL_CACHE_SIZE Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead

3.45.6 CXL_TEST_CONFIG_BASE_LOW_OFF

- **Name:** DVSEC CXL TEST CONFIGURATION BASE LOW.
- **Description:** CXL Device CTS Base register type.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))

31:4	CXL_DEVICE_TEST_BASE_LOW
3	RSVDP_3
2:1	CXL_DEVICE_TEST_TYPE
0	CXL_DEVICE_TEST_MEM_SPACE_INDICATOR

Table 3-322 Fields for Register: CXL_TEST_CONFIG_BASE_LOW_OFF

Bits	Name	Memory Access	Description
31:4	CXL_DEVICE_TEST_BASE_LOW	R/W	Bits [31:4] of the base address where the test configuration registers exist. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))
3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

Bits	Name	Memory Access	Description
2:1	CXL_DEVICE_TEST_TYPE	* Varies	CXL Device CTS Base register type. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (base_reg_first_half): Base register is 32 bit wide and can be mapped anywhere in the 32 bit address space. ■ 0x1 (RSVPD_one): Reserved. ■ 0x2 (base_reg_second_half): Base register is 64 bit wide and can be mapped anywhere in the 64 bit address space. ■ 0x3 (RSVPD_two): Reserved. Value After Reset: CX_CXL_CTS_BASE_REG_TYPE Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2)) Testable: wire ? unconstrained : writeAsRead
0	CXL_DEVICE_TEST_MEM_SPAC E_INDICATOR	R	The test configuration registers are in memory space. Device must hardwire this to 1'b0. Value After Reset: 0x1 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.45.7 CXL_TEST_CONFIG_BASE_HIGH_OFF

- **Name:** DVSEC CXL TEST CONFIGURATION BASE REGISTER.
- **Description:** Base High.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==2))



Table 3-323 Fields for Register: CXL_TEST_CONFIG_BASE_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	CXL_DEVICE_TEST_BASE_HIGH	R/W	Bits [63:32] of the base address where the test configuration registers exist. Value After Reset: 0x0 Exists: (CX_CXL_DEVICE_CTS_EN && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==2))

3.46 PF_RESBAR_CAPi Registers

3.46.1 RESBAR_CAP_HDR_REG

- **Name:** Resizable BAR Capability Header.
- **Description:** Resizable Extended Capability Header provides information about Capability ID, Version, and Next Offset.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_NUM_RBARS_\$pf>0)

RESBAR_CAP_NEXT_OFFSET	31:20
RESBAR_CAP_VERSION	19:16
RESBAR_EXT_CAP_ID	15:0

Table 3-324 Fields for Register: RESBAR_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	RESBAR_CAP_NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: Pointer value calculated at configuration time. Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
19:16	RESBAR_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This bit depends on the version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15:0	RESBAR_EXT_CAP_ID	* Varies	Resizable BAR Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. The Extended Capability ID for the Resizable BAR Capability is 0015h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x15 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.46.2 RESBAR_CAP_REG_0_REG

- **Name:** Resizable BAR0 Capability Register.
- **Description:** The Resizable BAR0 Extended Capability indicates that how much BAR size Function can support.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR0_RESIZABLE_\$pf

31	RESBAR_CAP_REG_0_128TB
30	RESBAR_CAP_REG_0_64TB
29	RESBAR_CAP_REG_0_32TB
28	RESBAR_CAP_REG_0_16TB
27	RESBAR_CAP_REG_0_8TB
26	RESBAR_CAP_REG_0_4TB
25	RESBAR_CAP_REG_0_2TB
24	RESBAR_CAP_REG_0_1TB
23	RESBAR_CAP_REG_0_512GB
22	RESBAR_CAP_REG_0_256GB
21	RESBAR_CAP_REG_0_128GB
20	RESBAR_CAP_REG_0_64GB
19	RESBAR_CAP_REG_0_32GB
18	RESBAR_CAP_REG_0_16GB
17	RESBAR_CAP_REG_0_8GB
16	RESBAR_CAP_REG_0_4GB
15	RESBAR_CAP_REG_0_2GB
14	RESBAR_CAP_REG_0_1GB
13	RESBAR_CAP_REG_0_512MB
12	RESBAR_CAP_REG_0_256MB
11	RESBAR_CAP_REG_0_128MB
10	RESBAR_CAP_REG_0_64MB
9	RESBAR_CAP_REG_0_32MB
8	RESBAR_CAP_REG_0_16MB
7	RESBAR_CAP_REG_0_8MB
6	RESBAR_CAP_REG_0_4MB
5	RESBAR_CAP_REG_0_2MB
4	RESBAR_CAP_REG_0_1MB
3:0	RSVDP_0

Table 3-325 Fields for Register: RESBAR_CAP_REG_0_REG

Bits	Name	Memory Access	Description
31	RESBAR_CAP_REG_0_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 TB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CAP_REG_0_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 TB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CAP_REG_0_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 TB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CAP_REG_0_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 TB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CAP_REG_0_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 TB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CAP_REG_0_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 TB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CAP_REG_0_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 TB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	RESBAR_CAP_REG_0_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1TB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	RESBAR_CAP_REG_0_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 GB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CAP_REG_0_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 GB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CAP_REG_0_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 GB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CAP_REG_0_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 GB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CAP_REG_0_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 GB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CAP_REG_0_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 GB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CAP_REG_0_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 GB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CAP_REG_0_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 GB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	RESBAR_CAP_REG_0_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 GB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	RESBAR_CAP_REG_0_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 GB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	RESBAR_CAP_REG_0_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 MB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	RESBAR_CAP_REG_0_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 MB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	RESBAR_CAP_REG_0_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 MB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	RESBAR_CAP_REG_0_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 MB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	RESBAR_CAP_REG_0_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 MB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	RESBAR_CAP_REG_0_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 MB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	RESBAR_CAP_REG_0_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 MB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	RESBAR_CAP_REG_0_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 MB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	RESBAR_CAP_REG_0_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 MB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	RESBAR_CAP_REG_0_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 MB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.46.3 RESBAR_CAP_REG_1_REG

- **Name:** Resizable BAR1 Capability Register.
- **Description:** The Resizable BAR1 Extended Capability indicates that how much BAR size Function can support.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR1_RESIZABLE_\$pf

RESBAR_CAP_REG_1_128TB	31
RESBAR_CAP_REG_1_64TB	30
RESBAR_CAP_REG_1_32TB	29
RESBAR_CAP_REG_1_16TB	28
RESBAR_CAP_REG_1_8TB	27
RESBAR_CAP_REG_1_4TB	26
RESBAR_CAP_REG_1_2TB	25
RESBAR_CAP_REG_1_1TB	24
RESBAR_CAP_REG_1_512GB	23
RESBAR_CAP_REG_1_256GB	22
RESBAR_CAP_REG_1_128GB	21
RESBAR_CAP_REG_1_64GB	20
RESBAR_CAP_REG_1_32GB	19
RESBAR_CAP_REG_1_16GB	18
RESBAR_CAP_REG_1_8GB	17
RESBAR_CAP_REG_1_4GB	16
RESBAR_CAP_REG_1_2GB	15
RESBAR_CAP_REG_1_1GB	14
RESBAR_CAP_REG_1_512MB	13
RESBAR_CAP_REG_1_256MB	12
RESBAR_CAP_REG_1_128MB	11
RESBAR_CAP_REG_1_64MB	10
RESBAR_CAP_REG_1_32MB	9
RESBAR_CAP_REG_1_16MB	8
RESBAR_CAP_REG_1_8MB	7
RESBAR_CAP_REG_1_4MB	6
RESBAR_CAP_REG_1_2MB	5
RESBAR_CAP_REG_1_1MB	4
RSVDP_0	3:0

Table 3-326 Fields for Register: RESBAR_CAP_REG_1_REG

Bits	Name	Memory Access	Description
31	RESBAR_CAP_REG_1_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 TB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CAP_REG_1_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 TB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CAP_REG_1_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 TB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CAP_REG_1_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 TB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CAP_REG_1_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 TB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CAP_REG_1_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 TB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CAP_REG_1_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 TB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	RESBAR_CAP_REG_1_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1TB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	RESBAR_CAP_REG_1_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 GB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CAP_REG_1_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 GB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CAP_REG_1_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 GB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CAP_REG_1_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 GB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CAP_REG_1_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 GB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CAP_REG_1_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 GB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CAP_REG_1_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 GB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CAP_REG_1_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 GB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	RESBAR_CAP_REG_1_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 GB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	RESBAR_CAP_REG_1_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 GB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	RESBAR_CAP_REG_1_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 MB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	RESBAR_CAP_REG_1_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 MB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	RESBAR_CAP_REG_1_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 MB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	RESBAR_CAP_REG_1_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 MB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	RESBAR_CAP_REG_1_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 MB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	RESBAR_CAP_REG_1_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 MB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	RESBAR_CAP_REG_1_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 MB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	RESBAR_CAP_REG_1_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 MB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	RESBAR_CAP_REG_1_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 MB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	RESBAR_CAP_REG_1_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 MB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.46.4 RESBAR_CAP_REG_2_REG

- **Name:** Resizable BAR2 Capability Register.
- **Description:** The Resizable BAR2 Extended Capability indicates that how much BAR size Function can support.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR2_RESIZABLE_\$pf

RESBAR_CAP_REG_2_128TB	31
RESBAR_CAP_REG_2_64TB	30
RESBAR_CAP_REG_2_32TB	29
RESBAR_CAP_REG_2_16TB	28
RESBAR_CAP_REG_2_8TB	27
RESBAR_CAP_REG_2_4TB	26
RESBAR_CAP_REG_2_2TB	25
RESBAR_CAP_REG_2_1TB	24
RESBAR_CAP_REG_2_512GB	23
RESBAR_CAP_REG_2_256GB	22
RESBAR_CAP_REG_2_128GB	21
RESBAR_CAP_REG_2_64GB	20
RESBAR_CAP_REG_2_32GB	19
RESBAR_CAP_REG_2_16GB	18
RESBAR_CAP_REG_2_8GB	17
RESBAR_CAP_REG_2_4GB	16
RESBAR_CAP_REG_2_2GB	15
RESBAR_CAP_REG_2_1GB	14
RESBAR_CAP_REG_2_512MB	13
RESBAR_CAP_REG_2_256MB	12
RESBAR_CAP_REG_2_128MB	11
RESBAR_CAP_REG_2_64MB	10
RESBAR_CAP_REG_2_32MB	9
RESBAR_CAP_REG_2_16MB	8
RESBAR_CAP_REG_2_8MB	7
RESBAR_CAP_REG_2_4MB	6
RESBAR_CAP_REG_2_2MB	5
RESBAR_CAP_REG_2_1MB	4
RSVDP_0	3:0

Table 3-327 Fields for Register: RESBAR_CAP_REG_2_REG

Bits	Name	Memory Access	Description
31	RESBAR_CAP_REG_2_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 TB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CAP_REG_2_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 TB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CAP_REG_2_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 TB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CAP_REG_2_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 TB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CAP_REG_2_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 TB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CAP_REG_2_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 TB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CAP_REG_2_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 TB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	RESBAR_CAP_REG_2_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1TB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	RESBAR_CAP_REG_2_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 GB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CAP_REG_2_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 GB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CAP_REG_2_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 GB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CAP_REG_2_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 GB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CAP_REG_2_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 GB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CAP_REG_2_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 GB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CAP_REG_2_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 GB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CAP_REG_2_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 GB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	RESBAR_CAP_REG_2_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 GB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	RESBAR_CAP_REG_2_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 GB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	RESBAR_CAP_REG_2_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 MB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	RESBAR_CAP_REG_2_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 MB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	RESBAR_CAP_REG_2_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 MB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	RESBAR_CAP_REG_2_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 MB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	RESBAR_CAP_REG_2_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 MB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	RESBAR_CAP_REG_2_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 MB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	RESBAR_CAP_REG_2_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 MB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	RESBAR_CAP_REG_2_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 MB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	RESBAR_CAP_REG_2_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 MB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	RESBAR_CAP_REG_2_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 MB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.46.5 RESBAR_CAP_REG_3_REG

- **Name:** Resizable BAR3 Capability Register.
- **Description:** The Resizable BAR3 Extended Capability indicates that how much BAR size Function can support.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR3_RESIZABLE_\$pf

RESBAR_CAP_REG_3_128TB	31
RESBAR_CAP_REG_3_64TB	30
RESBAR_CAP_REG_3_32TB	29
RESBAR_CAP_REG_3_16TB	28
RESBAR_CAP_REG_3_8TB	27
RESBAR_CAP_REG_3_4TB	26
RESBAR_CAP_REG_3_2TB	25
RESBAR_CAP_REG_3_1TB	24
RESBAR_CAP_REG_3_512GB	23
RESBAR_CAP_REG_3_256GB	22
RESBAR_CAP_REG_3_128GB	21
RESBAR_CAP_REG_3_64GB	20
RESBAR_CAP_REG_3_32GB	19
RESBAR_CAP_REG_3_16GB	18
RESBAR_CAP_REG_3_8GB	17
RESBAR_CAP_REG_3_4GB	16
RESBAR_CAP_REG_3_2GB	15
RESBAR_CAP_REG_3_1GB	14
RESBAR_CAP_REG_3_512MB	13
RESBAR_CAP_REG_3_256MB	12
RESBAR_CAP_REG_3_128MB	11
RESBAR_CAP_REG_3_64MB	10
RESBAR_CAP_REG_3_32MB	9
RESBAR_CAP_REG_3_16MB	8
RESBAR_CAP_REG_3_8MB	7
RESBAR_CAP_REG_3_4MB	6
RESBAR_CAP_REG_3_2MB	5
RESBAR_CAP_REG_3_1MB	4
RSVDP_0	3:0

Table 3-328 Fields for Register: RESBAR_CAP_REG_3_REG

Bits	Name	Memory Access	Description
31	RESBAR_CAP_REG_3_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 TB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CAP_REG_3_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 TB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CAP_REG_3_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 TB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CAP_REG_3_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 TB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CAP_REG_3_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 TB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CAP_REG_3_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 TB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CAP_REG_3_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 TB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	RESBAR_CAP_REG_3_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1TB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	RESBAR_CAP_REG_3_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 GB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CAP_REG_3_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 GB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CAP_REG_3_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 GB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CAP_REG_3_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 GB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CAP_REG_3_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 GB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CAP_REG_3_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 GB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CAP_REG_3_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 GB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CAP_REG_3_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 GB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	RESBAR_CAP_REG_3_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 GB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	RESBAR_CAP_REG_3_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 GB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	RESBAR_CAP_REG_3_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 MB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	RESBAR_CAP_REG_3_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 MB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	RESBAR_CAP_REG_3_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 MB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	RESBAR_CAP_REG_3_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 MB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	RESBAR_CAP_REG_3_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 MB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	RESBAR_CAP_REG_3_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 MB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	RESBAR_CAP_REG_3_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 MB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	RESBAR_CAP_REG_3_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 MB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	RESBAR_CAP_REG_3_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 MB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	RESBAR_CAP_REG_3_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 MB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.46.6 RESBAR_CAP_REG_4_REG

- **Name:** Resizable BAR4 Capability Register.
- **Description:** The Resizable BAR4 Extended Capability indicates that how much BAR size Function can support.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR4_RESIZABLE_\$pf

RESBAR_CAP_REG_4_128TB	31
RESBAR_CAP_REG_4_64TB	30
RESBAR_CAP_REG_4_32TB	29
RESBAR_CAP_REG_4_16TB	28
RESBAR_CAP_REG_4_8TB	27
RESBAR_CAP_REG_4_4TB	26
RESBAR_CAP_REG_4_2TB	25
RESBAR_CAP_REG_4_1TB	24
RESBAR_CAP_REG_4_512GB	23
RESBAR_CAP_REG_4_256GB	22
RESBAR_CAP_REG_4_128GB	21
RESBAR_CAP_REG_4_64GB	20
RESBAR_CAP_REG_4_32GB	19
RESBAR_CAP_REG_4_16GB	18
RESBAR_CAP_REG_4_8GB	17
RESBAR_CAP_REG_4_4GB	16
RESBAR_CAP_REG_4_2GB	15
RESBAR_CAP_REG_4_1GB	14
RESBAR_CAP_REG_4_512MB	13
RESBAR_CAP_REG_4_256MB	12
RESBAR_CAP_REG_4_128MB	11
RESBAR_CAP_REG_4_64MB	10
RESBAR_CAP_REG_4_32MB	9
RESBAR_CAP_REG_4_16MB	8
RESBAR_CAP_REG_4_8MB	7
RESBAR_CAP_REG_4_4MB	6
RESBAR_CAP_REG_4_2MB	5
RESBAR_CAP_REG_4_1MB	4
RSVDP_0	3:0

Table 3-329 Fields for Register: RESBAR_CAP_REG_4_REG

Bits	Name	Memory Access	Description
31	RESBAR_CAP_REG_4_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 TB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CAP_REG_4_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 TB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CAP_REG_4_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 TB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CAP_REG_4_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 TB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CAP_REG_4_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 TB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CAP_REG_4_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 TB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CAP_REG_4_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 TB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	RESBAR_CAP_REG_4_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1TB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	RESBAR_CAP_REG_4_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 GB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CAP_REG_4_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 GB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CAP_REG_4_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 GB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CAP_REG_4_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 GB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CAP_REG_4_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 GB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CAP_REG_4_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 GB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CAP_REG_4_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 GB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CAP_REG_4_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 GB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	RESBAR_CAP_REG_4_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 GB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	RESBAR_CAP_REG_4_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 GB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	RESBAR_CAP_REG_4_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 MB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	RESBAR_CAP_REG_4_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 MB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	RESBAR_CAP_REG_4_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 MB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	RESBAR_CAP_REG_4_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 MB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	RESBAR_CAP_REG_4_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 MB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	RESBAR_CAP_REG_4_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 MB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	RESBAR_CAP_REG_4_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 MB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	RESBAR_CAP_REG_4_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 MB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	RESBAR_CAP_REG_4_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 MB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	RESBAR_CAP_REG_4_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 MB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.46.7 RESBAR_CAP_REG_5_REG

- **Name:** Resizable BAR5 Capability Register.
- **Description:** The Resizable BAR5 Extended Capability indicates that how much BAR size Function can support.
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR5_RESIZABLE_\$pf

RESBAR_CAP_REG_5_128TB	31
RESBAR_CAP_REG_5_64TB	30
RESBAR_CAP_REG_5_32TB	29
RESBAR_CAP_REG_5_16TB	28
RESBAR_CAP_REG_5_8TB	27
RESBAR_CAP_REG_5_4TB	26
RESBAR_CAP_REG_5_2TB	25
RESBAR_CAP_REG_5_1TB	24
RESBAR_CAP_REG_5_512GB	23
RESBAR_CAP_REG_5_256GB	22
RESBAR_CAP_REG_5_128GB	21
RESBAR_CAP_REG_5_64GB	20
RESBAR_CAP_REG_5_32GB	19
RESBAR_CAP_REG_5_16GB	18
RESBAR_CAP_REG_5_8GB	17
RESBAR_CAP_REG_5_4GB	16
RESBAR_CAP_REG_5_2GB	15
RESBAR_CAP_REG_5_1GB	14
RESBAR_CAP_REG_5_512MB	13
RESBAR_CAP_REG_5_256MB	12
RESBAR_CAP_REG_5_128MB	11
RESBAR_CAP_REG_5_64MB	10
RESBAR_CAP_REG_5_32MB	9
RESBAR_CAP_REG_5_16MB	8
RESBAR_CAP_REG_5_8MB	7
RESBAR_CAP_REG_5_4MB	6
RESBAR_CAP_REG_5_2MB	5
RESBAR_CAP_REG_5_1MB	4
RSVDP_0	3:0

Table 3-330 Fields for Register: RESBAR_CAP_REG_5_REG

Bits	Name	Memory Access	Description
31	RESBAR_CAP_REG_5_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 TB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CAP_REG_5_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 TB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CAP_REG_5_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 TB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CAP_REG_5_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 TB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CAP_REG_5_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 TB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CAP_REG_5_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 TB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CAP_REG_5_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 TB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\" "
24	RESBAR_CAP_REG_5_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1TB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
23	RESBAR_CAP_REG_5_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 GB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CAP_REG_5_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 GB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CAP_REG_5_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 GB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CAP_REG_5_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 GB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CAP_REG_5_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 GB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CAP_REG_5_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 GB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CAP_REG_5_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 GB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CAP_REG_5_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 GB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	RESBAR_CAP_REG_5_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 GB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	RESBAR_CAP_REG_5_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 GB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	RESBAR_CAP_REG_5_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 MB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	RESBAR_CAP_REG_5_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 MB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	RESBAR_CAP_REG_5_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 MB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	RESBAR_CAP_REG_5_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 MB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	RESBAR_CAP_REG_5_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 MB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	RESBAR_CAP_REG_5_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 MB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	RESBAR_CAP_REG_5_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 MB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	RESBAR_CAP_REG_5_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 MB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	RESBAR_CAP_REG_5_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 MB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	RESBAR_CAP_REG_5_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 MB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.46.8 RESBAR_CTRL_REG_0_REG

- **Name:** Resizable BAR0 Control Register.
- **Description:** This register provides information about how much BAR0 size supported as well as BAR0 index and Number of BAR0.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR0_RESIZABLE_\$pf

RESBAR_CTRL_REG_0_8EB	31
RESBAR_CTRL_REG_0_4EB	30
RESBAR_CTRL_REG_0_2EB	29
RESBAR_CTRL_REG_0_1EB	28
RESBAR_CTRL_REG_0_512PB	27
RESBAR_CTRL_REG_0_256PB	26
RESBAR_CTRL_REG_0_128PB	25
RESBAR_CTRL_REG_0_64PB	24
RESBAR_CTRL_REG_0_32PB	23
RESBAR_CTRL_REG_0_16PB	22
RESBAR_CTRL_REG_0_8PB	21
RESBAR_CTRL_REG_0_4PB	20
RESBAR_CTRL_REG_0_2PB	19
RESBAR_CTRL_REG_0_1PB	18
RESBAR_CTRL_REG_0_512TB	17
RESBAR_CTRL_REG_0_256TB	16
RSVDP_14	15:14
RESBAR_CTRL_REG_BAR_SIZE	13:8
RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
RESBAR_CTRL_REG_IDX_0	2:0

Table 3-331 Fields for Register: RESBAR_CTRL_REG_0_REG

Bits	Name	Memory Access	Description
31	RESBAR_CTRL_REG_0_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 EB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CTRL_REG_0_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 EB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CTRL_REG_0_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 EB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CTRL_REG_0_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 EB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CTRL_REG_0_512PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 PB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CTRL_REG_0_256PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 PB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CTRL_REG_0_128PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 PB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	RESBAR_CTRL_REG_0_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 PB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	RESBAR_CTRL_REG_0_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 PB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CTRL_REG_0_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 PB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CTRL_REG_0_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 PB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CTRL_REG_0_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 PB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CTRL_REG_0_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 PB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CTRL_REG_0_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 PB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CTRL_REG_0_512TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 TB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CTRL_REG_0_256TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Function supports operating with the BAR0 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 TB. Value After Reset: CX_BAR0_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR0 Size. The default value of this field is equal to the default size of the address space that the BAR0 resource is requesting through the BAR's read-only bits. For backward compatibility with software, the default value must be in the range from 0 to 19. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (MIN_VAL): 1MB(2²⁰ bytes) ■ 0x2b (MAX_VAL): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::resbar_size 0 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""
7:5	RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in Resizable BAR0 Control register (0) (at offset 008h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable BAR for the function ■ 0x2 (BAR2): two resizable BARs for the function. ■ 0x3 (BAR3): three resizable BARs for the function. ■ 0x4 (BAR4): four resizable BARs for the function. ■ 0x5 (BAR5): five resizable BARs for the function. ■ 0x6 (BAR6): six resizable BARs for the function. Value After Reset: [::DWC_pcie_ctl::resbar_nbars 0 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	RESBAR_CTRL_REG_IDX_0	R	BAR Index. This encoded value points to the beginning of the BAR0. For a 64-bit Base Address register, the BAR Index indicates the lower DWORD. This value indicates which BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BAR_10h): BAR located at offset 10h. ■ 0x1 (BAR_14h): BAR located at offset 14h. ■ 0x2 (BAR_18h): BAR located at offset 18h. ■ 0x3 (BAR_1Ch): BAR located at offset 1Ch. ■ 0x4 (BAR_20h): BAR located at offset 20h. ■ 0x5 (BAR_24h): BAR located at offset 24h. Value After Reset: [::DWC_pcie_ctl::resbar_index 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.46.9 RESBAR_CTRL_REG_1_REG

- **Name:** Resizable BAR1 Control Register.
- **Description:** This register provides information about how much BAR1 size supported as well as BAR1 index and Number of BAR1.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR1_RESIZABLE_\$pf

RESBAR_CTRL_REG_1_8EB	31
RESBAR_CTRL_REG_1_4EB	30
RESBAR_CTRL_REG_1_2EB	29
RESBAR_CTRL_REG_1_1EB	28
RESBAR_CTRL_REG_1_512PB	27
RESBAR_CTRL_REG_1_256PB	26
RESBAR_CTRL_REG_1_128PB	25
RESBAR_CTRL_REG_1_64PB	24
RESBAR_CTRL_REG_1_32PB	23
RESBAR_CTRL_REG_1_16PB	22
RESBAR_CTRL_REG_1_8PB	21
RESBAR_CTRL_REG_1_4PB	20
RESBAR_CTRL_REG_1_2PB	19
RESBAR_CTRL_REG_1_1PB	18
RESBAR_CTRL_REG_1_512TB	17
RESBAR_CTRL_REG_1_256TB	16
RSVDP_14	15:14
RESBAR_CTRL_REG_BAR_SIZE	13:8
RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
RESBAR_CTRL_REG_IDX_1	2:0

Table 3-332 Fields for Register: RESBAR_CTRL_REG_1_REG

Bits	Name	Memory Access	Description
31	RESBAR_CTRL_REG_1_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 EB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CTRL_REG_1_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 EB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CTRL_REG_1_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 EB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CTRL_REG_1_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 EB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CTRL_REG_1_512PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 PB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CTRL_REG_1_256PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 PB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CTRL_REG_1_128PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 PB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	RESBAR_CTRL_REG_1_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 PB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	RESBAR_CTRL_REG_1_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 PB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CTRL_REG_1_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 PB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CTRL_REG_1_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 PB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CTRL_REG_1_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 PB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CTRL_REG_1_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 PB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CTRL_REG_1_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 PB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CTRL_REG_1_512TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 TB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CTRL_REG_1_256TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Function supports operating with the BAR1 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 TB. Value After Reset: CX_BAR1_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR1 Size. The default value of this field is equal to the default size of the address space that the BAR1 resource is requesting through the BAR's read-only bits. For backward compatibility with software, the default value must be in the range from 0 to 19. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (MIN_VAL): 1MB(2²⁰ bytes) ■ 0x2b (MAX_VAL): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::resbar_size 1 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""
7:5	RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in Resizable BAR1 Control register (0) (at offset 008h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable BAR for the function ■ 0x2 (BAR2): two resizable BARs for the function. ■ 0x3 (BAR3): three resizable BARs for the function. ■ 0x4 (BAR4): four resizable BARs for the function. ■ 0x5 (BAR5): five resizable BARs for the function. ■ 0x6 (BAR6): six resizable BARs for the function. Value After Reset: [::DWC_pcie_ctl::resbar_nbars 1 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	RESBAR_CTRL_REG_IDX_1	R	BAR Index. This encoded value points to the beginning of the BAR1. For a 64-bit Base Address register, the BAR Index indicates the lower DWORD. This value indicates which BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BAR_10h): BAR located at offset 10h. ■ 0x1 (BAR_14h): BAR located at offset 14h. ■ 0x2 (BAR_18h): BAR located at offset 18h. ■ 0x3 (BAR_1Ch): BAR located at offset 1Ch. ■ 0x4 (BAR_20h): BAR located at offset 20h. ■ 0x5 (BAR_24h): BAR located at offset 24h. Value After Reset: [::DWC_pcie_ctl::resbar_index 1] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.46.10 RESBAR_CTRL_REG_2_REG

- **Name:** Resizable BAR2 Control Register.
- **Description:** This register provides information about how much BAR2 size supported as well as BAR2 index and Number of BAR2.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR2_RESIZABLE_\$pf

RESBAR_CTRL_REG_2_8EB	31
RESBAR_CTRL_REG_2_4EB	30
RESBAR_CTRL_REG_2_2EB	29
RESBAR_CTRL_REG_2_1EB	28
RESBAR_CTRL_REG_2_512PB	27
RESBAR_CTRL_REG_2_256PB	26
RESBAR_CTRL_REG_2_128PB	25
RESBAR_CTRL_REG_2_64PB	24
RESBAR_CTRL_REG_2_32PB	23
RESBAR_CTRL_REG_2_16PB	22
RESBAR_CTRL_REG_2_8PB	21
RESBAR_CTRL_REG_2_4PB	20
RESBAR_CTRL_REG_2_2PB	19
RESBAR_CTRL_REG_2_1PB	18
RESBAR_CTRL_REG_2_512TB	17
RESBAR_CTRL_REG_2_256TB	16
RSVDP_14	15:14
RESBAR_CTRL_REG_BAR_SIZE	13:8
RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
RESBAR_CTRL_REG_IDX_2	2:0

Table 3-333 Fields for Register: RESBAR_CTRL_REG_2_REG

Bits	Name	Memory Access	Description
31	RESBAR_CTRL_REG_2_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 EB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CTRL_REG_2_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 EB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CTRL_REG_2_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 EB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CTRL_REG_2_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 EB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CTRL_REG_2_512PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 PB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CTRL_REG_2_256PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 PB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CTRL_REG_2_128PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 PB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	RESBAR_CTRL_REG_2_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 PB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	RESBAR_CTRL_REG_2_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 PB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CTRL_REG_2_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 PB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CTRL_REG_2_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 PB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CTRL_REG_2_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 PB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CTRL_REG_2_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 PB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CTRL_REG_2_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 PB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CTRL_REG_2_512TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 TB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CTRL_REG_2_256TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Function supports operating with the BAR2 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 TB. Value After Reset: CX_BAR2_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR2 Size. The default value of this field is equal to the default size of the address space that the BAR2 resource is requesting through the BAR's read-only bits. For backward compatibility with software, the default value must be in the range from 0 to 19. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (MIN_VAL): 1MB(2²⁰ bytes) ■ 0x2b (MAX_VAL): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::resbar_size 2 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""
7:5	RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in Resizable BAR2 Control register (0) (at offset 008h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable BAR for the function ■ 0x2 (BAR2): two resizable BARs for the function. ■ 0x3 (BAR3): three resizable BARs for the function. ■ 0x4 (BAR4): four resizable BARs for the function. ■ 0x5 (BAR5): five resizable BARs for the function. ■ 0x6 (BAR6): six resizable BARs for the function. Value After Reset: [::DWC_pcie_ctl::resbar_nbars 2 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	RESBAR_CTRL_REG_IDX_2	R	BAR Index. This encoded value points to the beginning of the BAR2. For a 64-bit Base Address register, the BAR Index indicates the lower DWORD. This value indicates which BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BAR_10h): BAR located at offset 10h. ■ 0x1 (BAR_14h): BAR located at offset 14h. ■ 0x2 (BAR_18h): BAR located at offset 18h. ■ 0x3 (BAR_1Ch): BAR located at offset 1Ch. ■ 0x4 (BAR_20h): BAR located at offset 20h. ■ 0x5 (BAR_24h): BAR located at offset 24h. Value After Reset: [::DWC_pcie_ctl::resbar_index 2] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.46.11 RESBAR_CTRL_REG_3_REG

- **Name:** Resizable BAR3 Control Register.
- **Description:** This register provides information about how much BAR3 size supported as well as BAR3 index and Number of BAR3.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR3_RESIZABLE_\$pf

RESBAR_CTRL_REG_3_8EB	31
RESBAR_CTRL_REG_3_4EB	30
RESBAR_CTRL_REG_3_2EB	29
RESBAR_CTRL_REG_3_1EB	28
RESBAR_CTRL_REG_3_512PB	27
RESBAR_CTRL_REG_3_256PB	26
RESBAR_CTRL_REG_3_128PB	25
RESBAR_CTRL_REG_3_64PB	24
RESBAR_CTRL_REG_3_32PB	23
RESBAR_CTRL_REG_3_16PB	22
RESBAR_CTRL_REG_3_8PB	21
RESBAR_CTRL_REG_3_4PB	20
RESBAR_CTRL_REG_3_2PB	19
RESBAR_CTRL_REG_3_1PB	18
RESBAR_CTRL_REG_3_512TB	17
RESBAR_CTRL_REG_3_256TB	16
RSVDP_14	15:14
RESBAR_CTRL_REG_BAR_SIZE	13:8
RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
RESBAR_CTRL_REG_IDX_3	2:0

Table 3-334 Fields for Register: RESBAR_CTRL_REG_3_REG

Bits	Name	Memory Access	Description
31	RESBAR_CTRL_REG_3_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 EB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CTRL_REG_3_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 EB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CTRL_REG_3_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 EB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CTRL_REG_3_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 EB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CTRL_REG_3_512PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 PB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CTRL_REG_3_256PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 PB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CTRL_REG_3_128PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 PB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	RESBAR_CTRL_REG_3_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 PB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	RESBAR_CTRL_REG_3_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 PB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CTRL_REG_3_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 PB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CTRL_REG_3_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 PB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CTRL_REG_3_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 PB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CTRL_REG_3_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 PB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CTRL_REG_3_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 PB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CTRL_REG_3_512TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 TB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CTRL_REG_3_256TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Function supports operating with the BAR3 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 TB. Value After Reset: CX_BAR3_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR3 Size. The default value of this field is equal to the default size of the address space that the BAR3 resource is requesting through the BAR's read-only bits. For backward compatibility with software, the default value must be in the range from 0 to 19. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (MIN_VAL): 1MB(2²⁰ bytes) ■ 0x2b (MAX_VAL): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::resbar_size 3 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""
7:5	RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in Resizable BAR3 Control register (0) (at offset 008h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable BAR for the function ■ 0x2 (BAR2): two resizable BARs for the function. ■ 0x3 (BAR3): three resizable BARs for the function. ■ 0x4 (BAR4): four resizable BARs for the function. ■ 0x5 (BAR5): five resizable BARs for the function. ■ 0x6 (BAR6): six resizable BARs for the function. Value After Reset: [::DWC_pcie_ctl::resbar_nbars 3 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	RESBAR_CTRL_REG_IDX_3	R	BAR Index. This encoded value points to the beginning of the BAR3. For a 64-bit Base Address register, the BAR Index indicates the lower DWORD. This value indicates which BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BAR_10h): BAR located at offset 10h. ■ 0x1 (BAR_14h): BAR located at offset 14h. ■ 0x2 (BAR_18h): BAR located at offset 18h. ■ 0x3 (BAR_1Ch): BAR located at offset 1Ch. ■ 0x4 (BAR_20h): BAR located at offset 20h. ■ 0x5 (BAR_24h): BAR located at offset 24h. Value After Reset: [::DWC_pcie_ctl::resbar_index 3] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.46.12 RESBAR_CTRL_REG_4_REG

- **Name:** Resizable BAR4 Control Register.
- **Description:** This register provides information about how much BAR4 size supported as well as BAR4 index and Number of BAR4.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR4_RESIZABLE_\$pf

RESBAR_CTRL_REG_4_8EB	31
RESBAR_CTRL_REG_4_4EB	30
RESBAR_CTRL_REG_4_2EB	29
RESBAR_CTRL_REG_4_1EB	28
RESBAR_CTRL_REG_4_512PB	27
RESBAR_CTRL_REG_4_256PB	26
RESBAR_CTRL_REG_4_128PB	25
RESBAR_CTRL_REG_4_64PB	24
RESBAR_CTRL_REG_4_32PB	23
RESBAR_CTRL_REG_4_16PB	22
RESBAR_CTRL_REG_4_8PB	21
RESBAR_CTRL_REG_4_4PB	20
RESBAR_CTRL_REG_4_2PB	19
RESBAR_CTRL_REG_4_1PB	18
RESBAR_CTRL_REG_4_512TB	17
RESBAR_CTRL_REG_4_256TB	16
RSVDP_14	15:14
RESBAR_CTRL_REG_BAR_SIZE	13:8
RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
RESBAR_CTRL_REG_IDX_4	2:0

Table 3-335 Fields for Register: RESBAR_CTRL_REG_4_REG

Bits	Name	Memory Access	Description
31	RESBAR_CTRL_REG_4_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 EB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CTRL_REG_4_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 EB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CTRL_REG_4_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 EB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CTRL_REG_4_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 EB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CTRL_REG_4_512PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 PB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CTRL_REG_4_256PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 PB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CTRL_REG_4_128PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 PB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	RESBAR_CTRL_REG_4_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 PB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	RESBAR_CTRL_REG_4_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 PB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CTRL_REG_4_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 PB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CTRL_REG_4_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 PB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CTRL_REG_4_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 PB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CTRL_REG_4_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 PB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CTRL_REG_4_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 PB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CTRL_REG_4_512TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 TB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CTRL_REG_4_256TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Function supports operating with the BAR4 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 TB. Value After Reset: CX_BAR4_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR4 Size. The default value of this field is equal to the default size of the address space that the BAR4 resource is requesting through the BAR's read-only bits. For backward compatibility with software, the default value must be in the range from 0 to 19. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (MIN_VAL): 1MB(2²⁰ bytes) ■ 0x2b (MAX_VAL): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::resbar_size 4 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""
7:5	RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in Resizable BAR4 Control register (0) (at offset 008h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable BAR for the function ■ 0x2 (BAR2): two resizable BARs for the function. ■ 0x3 (BAR3): three resizable BARs for the function. ■ 0x4 (BAR4): four resizable BARs for the function. ■ 0x5 (BAR5): five resizable BARs for the function. ■ 0x6 (BAR6): six resizable BARs for the function. Value After Reset: [::DWC_pcie_ctl::resbar_nbars 4 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	RESBAR_CTRL_REG_IDX_4	R	BAR Index. This encoded value points to the beginning of the BAR4. For a 64-bit Base Address register, the BAR Index indicates the lower DWORD. This value indicates which BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BAR_10h): BAR located at offset 10h. ■ 0x1 (BAR_14h): BAR located at offset 14h. ■ 0x2 (BAR_18h): BAR located at offset 18h. ■ 0x3 (BAR_1Ch): BAR located at offset 1Ch. ■ 0x4 (BAR_20h): BAR located at offset 20h. ■ 0x5 (BAR_24h): BAR located at offset 24h. Value After Reset: [::DWC_pcie_ctl::resbar_index 4] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.46.13 RESBAR_CTRL_REG_5_REG

- **Name:** Resizable BAR5 Control Register.
- **Description:** This register provides information about how much BAR5 size supported as well as BAR5 index and Number of BAR5.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** ((CX_NUM_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && CX_BAR5_RESIZABLE_\$pf

RESBAR_CTRL_REG_5_8EB	31
RESBAR_CTRL_REG_5_4EB	30
RESBAR_CTRL_REG_5_2EB	29
RESBAR_CTRL_REG_5_1EB	28
RESBAR_CTRL_REG_5_512PB	27
RESBAR_CTRL_REG_5_256PB	26
RESBAR_CTRL_REG_5_128PB	25
RESBAR_CTRL_REG_5_64PB	24
RESBAR_CTRL_REG_5_32PB	23
RESBAR_CTRL_REG_5_16PB	22
RESBAR_CTRL_REG_5_8PB	21
RESBAR_CTRL_REG_5_4PB	20
RESBAR_CTRL_REG_5_2PB	19
RESBAR_CTRL_REG_5_1PB	18
RESBAR_CTRL_REG_5_512TB	17
RESBAR_CTRL_REG_5_256TB	16
RSVDP_14	15:14
RESBAR_CTRL_REG_BAR_SIZE	13:8
RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
RESBAR_CTRL_REG_IDX_5	2:0

Table 3-336 Fields for Register: RESBAR_CTRL_REG_5_REG

Bits	Name	Memory Access	Description
31	RESBAR_CTRL_REG_5_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 EB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	RESBAR_CTRL_REG_5_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 EB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	RESBAR_CTRL_REG_5_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 EB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	RESBAR_CTRL_REG_5_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 EB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	RESBAR_CTRL_REG_5_512PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 PB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	RESBAR_CTRL_REG_5_256PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 PB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	RESBAR_CTRL_REG_5_128PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 128 PB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	RESBAR_CTRL_REG_5_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 64 PB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	RESBAR_CTRL_REG_5_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 32 PB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	RESBAR_CTRL_REG_5_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 16 PB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	RESBAR_CTRL_REG_5_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 8 PB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	RESBAR_CTRL_REG_5_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 4 PB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	RESBAR_CTRL_REG_5_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 2 PB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	RESBAR_CTRL_REG_5_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 1 PB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	RESBAR_CTRL_REG_5_512TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 512 TB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	RESBAR_CTRL_REG_5_256TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Function supports operating with the BAR5 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with BAR sized to 256 TB. Value After Reset: CX_BAR5_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR5 Size. The default value of this field is equal to the default size of the address space that the BAR5 resource is requesting through the BAR's read-only bits. For backward compatibility with software, the default value must be in the range from 0 to 19. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (MIN_VAL): 1MB(2²⁰ bytes) ■ 0x2b (MAX_VAL): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::resbar_size 5 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""
7:5	RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in Resizable BAR5 Control register (0) (at offset 008h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable BAR for the function ■ 0x2 (BAR2): two resizable BARs for the function. ■ 0x3 (BAR3): three resizable BARs for the function. ■ 0x4 (BAR4): four resizable BARs for the function. ■ 0x5 (BAR5): five resizable BARs for the function. ■ 0x6 (BAR6): six resizable BARs for the function. Value After Reset: [::DWC_pcie_ctl::resbar_nbars 5 0] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	RESBAR_CTRL_REG_IDX_5	R	BAR Index. This encoded value points to the beginning of the BAR5. For a 64-bit Base Address register, the BAR Index indicates the lower DWORD. This value indicates which BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BAR_10h): BAR located at offset 10h. ■ 0x1 (BAR_14h): BAR located at offset 14h. ■ 0x2 (BAR_18h): BAR located at offset 18h. ■ 0x3 (BAR_1Ch): BAR located at offset 1Ch. ■ 0x4 (BAR_20h): BAR located at offset 20h. ■ 0x5 (BAR_24h): BAR located at offset 24h. Value After Reset: [::DWC_pcie_ctl::resbar_index 5] Exists: ((CX_NUM_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47 PF_VF_RESBAR_CAPi Registers

3.47.1 VF_RESBAR_CAP_HDR_REG

- **Name:** Resizable BAR Capability Header.
- **Description:** VF Resizable Extended Capability Header provides information about Capability ID, Version, and Next offset.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_NUM_VF_RBARS_\$pf>0)

31:20	VF_RESBAR_CAP_NEXT_OFFSET
19:16	VF_RESBAR_CAP_VERSION
15:0	VF_RESBAR_EXT_CAP_ID

Table 3-337 Fields for Register: VF_RESBAR_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	VF_RESBAR_CAP_NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	VF_RESBAR_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This bit depends on the version of the specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead
15:0	VF_RESBAR_EXT_CAP_ID	* Varies	Resizable BAR Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the extended capability. PCI Express Extended Capability ID for the VF Resizable VF BAR Extended Capability is 0024h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x24 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: wire ? unconstrained : writeAsRead

3.47.2 VF_RESBAR_CAP_REG_0_REG

- **Name:** Resizable BAR0 Capability Register.
- **Description:** The Resizable BAR Extended Capability indicates that how much VF BAR0 size Physical Function(PF) can support.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR0_RESIZABLE_\$pf

VF_RESBAR_CAP_REG_0_128TB	31
VF_RESBAR_CAP_REG_0_64TB	30
VF_RESBAR_CAP_REG_0_32TB	29
VF_RESBAR_CAP_REG_0_16TB	28
VF_RESBAR_CAP_REG_0_8TB	27
VF_RESBAR_CAP_REG_0_4TB	26
VF_RESBAR_CAP_REG_0_2TB	25
VF_RESBAR_CAP_REG_0_1TB	24
VF_RESBAR_CAP_REG_0_512GB	23
VF_RESBAR_CAP_REG_0_256GB	22
VF_RESBAR_CAP_REG_0_128GB	21
VF_RESBAR_CAP_REG_0_64GB	20
VF_RESBAR_CAP_REG_0_32GB	19
VF_RESBAR_CAP_REG_0_16GB	18
VF_RESBAR_CAP_REG_0_8GB	17
VF_RESBAR_CAP_REG_0_4GB	16
VF_RESBAR_CAP_REG_0_2GB	15
VF_RESBAR_CAP_REG_0_1GB	14
VF_RESBAR_CAP_REG_0_512MB	13
VF_RESBAR_CAP_REG_0_256MB	12
VF_RESBAR_CAP_REG_0_128MB	11
VF_RESBAR_CAP_REG_0_64MB	10
VF_RESBAR_CAP_REG_0_32MB	9
VF_RESBAR_CAP_REG_0_16MB	8
VF_RESBAR_CAP_REG_0_8MB	7
VF_RESBAR_CAP_REG_0_4MB	6
VF_RESBAR_CAP_REG_0_2MB	5
VF_RESBAR_CAP_REG_0_1MB	4
RSVDP_0	3:0

Table 3-338 Fields for Register: VF_RESBAR_CAP_REG_0_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CAP_REG_0_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 TB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CAP_REG_0_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 TB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CAP_REG_0_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 TB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CAP_REG_0_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 TB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CAP_REG_0_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 TB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CAP_REG_0_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 TB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CAP_REG_0_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 TB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CAP_REG_0_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 TB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CAP_REG_0_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 GB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CAP_REG_0_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 GB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CAP_REG_0_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 GB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CAP_REG_0_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 GB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CAP_REG_0_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 GB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CAP_REG_0_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 GB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CAP_REG_0_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 GB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CAP_REG_0_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 GB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	VF_RESBAR_CAP_REG_0_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 GB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	VF_RESBAR_CAP_REG_0_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 GB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	VF_RESBAR_CAP_REG_0_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 MB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	VF_RESBAR_CAP_REG_0_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 MB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	VF_RESBAR_CAP_REG_0_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 MB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	VF_RESBAR_CAP_REG_0_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 MB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	VF_RESBAR_CAP_REG_0_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 MB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	VF_RESBAR_CAP_REG_0_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 MB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	VF_RESBAR_CAP_REG_0_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 MB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	VF_RESBAR_CAP_REG_0_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 MB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	VF_RESBAR_CAP_REG_0_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 MB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	VF_RESBAR_CAP_REG_0_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 MB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47.3 VF_RESBAR_CAP_REG_1_REG

- **Name:** Resizable BAR1 Capability Register.
- **Description:** The Resizable BAR Extended Capability indicates that how much VF BAR1 size Physical Function(PF) can support.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR1_RESIZABLE_\$pf

VF_RESBAR_CAP_REG_1_128TB	31
VF_RESBAR_CAP_REG_1_64TB	30
VF_RESBAR_CAP_REG_1_32TB	29
VF_RESBAR_CAP_REG_1_16TB	28
VF_RESBAR_CAP_REG_1_8TB	27
VF_RESBAR_CAP_REG_1_4TB	26
VF_RESBAR_CAP_REG_1_2TB	25
VF_RESBAR_CAP_REG_1_1TB	24
VF_RESBAR_CAP_REG_1_512GB	23
VF_RESBAR_CAP_REG_1_256GB	22
VF_RESBAR_CAP_REG_1_128GB	21
VF_RESBAR_CAP_REG_1_64GB	20
VF_RESBAR_CAP_REG_1_32GB	19
VF_RESBAR_CAP_REG_1_16GB	18
VF_RESBAR_CAP_REG_1_8GB	17
VF_RESBAR_CAP_REG_1_4GB	16
VF_RESBAR_CAP_REG_1_2GB	15
VF_RESBAR_CAP_REG_1_1GB	14
VF_RESBAR_CAP_REG_1_512MB	13
VF_RESBAR_CAP_REG_1_256MB	12
VF_RESBAR_CAP_REG_1_128MB	11
VF_RESBAR_CAP_REG_1_64MB	10
VF_RESBAR_CAP_REG_1_32MB	9
VF_RESBAR_CAP_REG_1_16MB	8
VF_RESBAR_CAP_REG_1_8MB	7
VF_RESBAR_CAP_REG_1_4MB	6
VF_RESBAR_CAP_REG_1_2MB	5
VF_RESBAR_CAP_REG_1_1MB	4
RSVDP_0	3:0

Table 3-339 Fields for Register: VF_RESBAR_CAP_REG_1_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CAP_REG_1_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 TB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CAP_REG_1_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 TB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CAP_REG_1_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 TB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CAP_REG_1_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 TB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CAP_REG_1_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 TB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CAP_REG_1_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 TB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CAP_REG_1_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 TB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CAP_REG_1_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 TB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CAP_REG_1_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 GB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CAP_REG_1_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 GB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CAP_REG_1_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 GB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CAP_REG_1_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 GB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CAP_REG_1_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 GB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CAP_REG_1_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 GB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CAP_REG_1_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 GB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CAP_REG_1_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 GB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	VF_RESBAR_CAP_REG_1_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 GB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	VF_RESBAR_CAP_REG_1_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 GB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	VF_RESBAR_CAP_REG_1_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 MB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	VF_RESBAR_CAP_REG_1_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 MB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	VF_RESBAR_CAP_REG_1_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 MB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	VF_RESBAR_CAP_REG_1_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 MB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	VF_RESBAR_CAP_REG_1_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 MB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	VF_RESBAR_CAP_REG_1_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 MB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	VF_RESBAR_CAP_REG_1_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 MB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	VF_RESBAR_CAP_REG_1_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 MB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	VF_RESBAR_CAP_REG_1_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 MB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	VF_RESBAR_CAP_REG_1_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 MB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47.4 VF_RESBAR_CAP_REG_2_REG

- **Name:** Resizable BAR2 Capability Register.
- **Description:** The Resizable BAR Extended Capability indicates that how much VF BAR2 size Physical Function(PF) can support.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR2_RESIZABLE_\$pf

VF_RESBAR_CAP_REG_2_128TB	31
VF_RESBAR_CAP_REG_2_64TB	30
VF_RESBAR_CAP_REG_2_32TB	29
VF_RESBAR_CAP_REG_2_16TB	28
VF_RESBAR_CAP_REG_2_8TB	27
VF_RESBAR_CAP_REG_2_4TB	26
VF_RESBAR_CAP_REG_2_2TB	25
VF_RESBAR_CAP_REG_2_1TB	24
VF_RESBAR_CAP_REG_2_512GB	23
VF_RESBAR_CAP_REG_2_256GB	22
VF_RESBAR_CAP_REG_2_128GB	21
VF_RESBAR_CAP_REG_2_64GB	20
VF_RESBAR_CAP_REG_2_32GB	19
VF_RESBAR_CAP_REG_2_16GB	18
VF_RESBAR_CAP_REG_2_8GB	17
VF_RESBAR_CAP_REG_2_4GB	16
VF_RESBAR_CAP_REG_2_2GB	15
VF_RESBAR_CAP_REG_2_1GB	14
VF_RESBAR_CAP_REG_2_512MB	13
VF_RESBAR_CAP_REG_2_256MB	12
VF_RESBAR_CAP_REG_2_128MB	11
VF_RESBAR_CAP_REG_2_64MB	10
VF_RESBAR_CAP_REG_2_32MB	9
VF_RESBAR_CAP_REG_2_16MB	8
VF_RESBAR_CAP_REG_2_8MB	7
VF_RESBAR_CAP_REG_2_4MB	6
VF_RESBAR_CAP_REG_2_2MB	5
VF_RESBAR_CAP_REG_2_1MB	4
RSVDP_0	3:0

Table 3-340 Fields for Register: VF_RESBAR_CAP_REG_2_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CAP_REG_2_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 TB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CAP_REG_2_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 TB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CAP_REG_2_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 TB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CAP_REG_2_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 TB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CAP_REG_2_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 TB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CAP_REG_2_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 TB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CAP_REG_2_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 TB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CAP_REG_2_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 TB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CAP_REG_2_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 GB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CAP_REG_2_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 GB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CAP_REG_2_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 GB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CAP_REG_2_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 GB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CAP_REG_2_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 GB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CAP_REG_2_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 GB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CAP_REG_2_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 GB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CAP_REG_2_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 GB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	VF_RESBAR_CAP_REG_2_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 GB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	VF_RESBAR_CAP_REG_2_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 GB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	VF_RESBAR_CAP_REG_2_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 MB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	VF_RESBAR_CAP_REG_2_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 MB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	VF_RESBAR_CAP_REG_2_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 MB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	VF_RESBAR_CAP_REG_2_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 MB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	VF_RESBAR_CAP_REG_2_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 MB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	VF_RESBAR_CAP_REG_2_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 MB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	VF_RESBAR_CAP_REG_2_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 MB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	VF_RESBAR_CAP_REG_2_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 MB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	VF_RESBAR_CAP_REG_2_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 MB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	VF_RESBAR_CAP_REG_2_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 MB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47.5 VF_RESBAR_CAP_REG_3_REG

- **Name:** Resizable BAR3 Capability Register.
- **Description:** The Resizable BAR Extended Capability indicates that how much VF BAR3 size Physical Function(PF) can support.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR3_RESIZABLE_\$pf

VF_RESBAR_CAP_REG_3_128TB	31
VF_RESBAR_CAP_REG_3_64TB	30
VF_RESBAR_CAP_REG_3_32TB	29
VF_RESBAR_CAP_REG_3_16TB	28
VF_RESBAR_CAP_REG_3_8TB	27
VF_RESBAR_CAP_REG_3_4TB	26
VF_RESBAR_CAP_REG_3_2TB	25
VF_RESBAR_CAP_REG_3_1TB	24
VF_RESBAR_CAP_REG_3_512GB	23
VF_RESBAR_CAP_REG_3_256GB	22
VF_RESBAR_CAP_REG_3_128GB	21
VF_RESBAR_CAP_REG_3_64GB	20
VF_RESBAR_CAP_REG_3_32GB	19
VF_RESBAR_CAP_REG_3_16GB	18
VF_RESBAR_CAP_REG_3_8GB	17
VF_RESBAR_CAP_REG_3_4GB	16
VF_RESBAR_CAP_REG_3_2GB	15
VF_RESBAR_CAP_REG_3_1GB	14
VF_RESBAR_CAP_REG_3_512MB	13
VF_RESBAR_CAP_REG_3_256MB	12
VF_RESBAR_CAP_REG_3_128MB	11
VF_RESBAR_CAP_REG_3_64MB	10
VF_RESBAR_CAP_REG_3_32MB	9
VF_RESBAR_CAP_REG_3_16MB	8
VF_RESBAR_CAP_REG_3_8MB	7
VF_RESBAR_CAP_REG_3_4MB	6
VF_RESBAR_CAP_REG_3_2MB	5
VF_RESBAR_CAP_REG_3_1MB	4
RSVDP_0	3:0

Table 3-341 Fields for Register: VF_RESBAR_CAP_REG_3_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CAP_REG_3_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 TB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CAP_REG_3_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 TB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CAP_REG_3_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 TB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CAP_REG_3_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 TB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CAP_REG_3_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 TB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CAP_REG_3_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 TB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CAP_REG_3_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 TB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CAP_REG_3_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 TB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CAP_REG_3_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 GB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CAP_REG_3_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 GB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CAP_REG_3_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 GB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CAP_REG_3_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 GB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CAP_REG_3_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 GB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CAP_REG_3_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 GB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CAP_REG_3_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 GB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CAP_REG_3_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 GB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	VF_RESBAR_CAP_REG_3_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 GB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	VF_RESBAR_CAP_REG_3_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 GB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	VF_RESBAR_CAP_REG_3_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 MB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	VF_RESBAR_CAP_REG_3_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 MB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	VF_RESBAR_CAP_REG_3_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 MB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	VF_RESBAR_CAP_REG_3_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 MB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	VF_RESBAR_CAP_REG_3_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 MB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	VF_RESBAR_CAP_REG_3_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 MB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	VF_RESBAR_CAP_REG_3_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 MB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	VF_RESBAR_CAP_REG_3_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 MB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	VF_RESBAR_CAP_REG_3_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 MB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	VF_RESBAR_CAP_REG_3_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 MB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47.6 VF_RESBAR_CAP_REG_4_REG

- **Name:** Resizable BAR4 Capability Register.
- **Description:** The Resizable BAR Extended Capability indicates that how much VF BAR4 size Physical Function(PF) can support.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR4_RESIZABLE_\$pf

VF_RESBAR_CAP_REG_4_128TB	31
VF_RESBAR_CAP_REG_4_64TB	30
VF_RESBAR_CAP_REG_4_32TB	29
VF_RESBAR_CAP_REG_4_16TB	28
VF_RESBAR_CAP_REG_4_8TB	27
VF_RESBAR_CAP_REG_4_4TB	26
VF_RESBAR_CAP_REG_4_2TB	25
VF_RESBAR_CAP_REG_4_1TB	24
VF_RESBAR_CAP_REG_4_512GB	23
VF_RESBAR_CAP_REG_4_256GB	22
VF_RESBAR_CAP_REG_4_128GB	21
VF_RESBAR_CAP_REG_4_64GB	20
VF_RESBAR_CAP_REG_4_32GB	19
VF_RESBAR_CAP_REG_4_16GB	18
VF_RESBAR_CAP_REG_4_8GB	17
VF_RESBAR_CAP_REG_4_4GB	16
VF_RESBAR_CAP_REG_4_2GB	15
VF_RESBAR_CAP_REG_4_1GB	14
VF_RESBAR_CAP_REG_4_512MB	13
VF_RESBAR_CAP_REG_4_256MB	12
VF_RESBAR_CAP_REG_4_128MB	11
VF_RESBAR_CAP_REG_4_64MB	10
VF_RESBAR_CAP_REG_4_32MB	9
VF_RESBAR_CAP_REG_4_16MB	8
VF_RESBAR_CAP_REG_4_8MB	7
VF_RESBAR_CAP_REG_4_4MB	6
VF_RESBAR_CAP_REG_4_2MB	5
VF_RESBAR_CAP_REG_4_1MB	4
RSVDP_0	3:0

Table 3-342 Fields for Register: VF_RESBAR_CAP_REG_4_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CAP_REG_4_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 TB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CAP_REG_4_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 TB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CAP_REG_4_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 TB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CAP_REG_4_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 TB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CAP_REG_4_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 TB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CAP_REG_4_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 TB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CAP_REG_4_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 TB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CAP_REG_4_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 TB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CAP_REG_4_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 GB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CAP_REG_4_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 GB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CAP_REG_4_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 GB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CAP_REG_4_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 GB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CAP_REG_4_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 GB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CAP_REG_4_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 GB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CAP_REG_4_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 GB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CAP_REG_4_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 GB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	VF_RESBAR_CAP_REG_4_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 GB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	VF_RESBAR_CAP_REG_4_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 GB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	VF_RESBAR_CAP_REG_4_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 MB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	VF_RESBAR_CAP_REG_4_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 MB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	VF_RESBAR_CAP_REG_4_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 MB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	VF_RESBAR_CAP_REG_4_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 MB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	VF_RESBAR_CAP_REG_4_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 MB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	VF_RESBAR_CAP_REG_4_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 MB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	VF_RESBAR_CAP_REG_4_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 MB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	VF_RESBAR_CAP_REG_4_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 MB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	VF_RESBAR_CAP_REG_4_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 MB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	VF_RESBAR_CAP_REG_4_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 MB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47.7 VF_RESBAR_CAP_REG_5_REG

- **Name:** Resizable BAR5 Capability Register.
- **Description:** The Resizable BAR Extended Capability indicates that how much VF BAR5 size Physical Function(PF) can support.
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR5_RESIZABLE_\$pf

VF_RESBAR_CAP_REG_5_128TB	31
VF_RESBAR_CAP_REG_5_64TB	30
VF_RESBAR_CAP_REG_5_32TB	29
VF_RESBAR_CAP_REG_5_16TB	28
VF_RESBAR_CAP_REG_5_8TB	27
VF_RESBAR_CAP_REG_5_4TB	26
VF_RESBAR_CAP_REG_5_2TB	25
VF_RESBAR_CAP_REG_5_1TB	24
VF_RESBAR_CAP_REG_5_512GB	23
VF_RESBAR_CAP_REG_5_256GB	22
VF_RESBAR_CAP_REG_5_128GB	21
VF_RESBAR_CAP_REG_5_64GB	20
VF_RESBAR_CAP_REG_5_32GB	19
VF_RESBAR_CAP_REG_5_16GB	18
VF_RESBAR_CAP_REG_5_8GB	17
VF_RESBAR_CAP_REG_5_4GB	16
VF_RESBAR_CAP_REG_5_2GB	15
VF_RESBAR_CAP_REG_5_1GB	14
VF_RESBAR_CAP_REG_5_512MB	13
VF_RESBAR_CAP_REG_5_256MB	12
VF_RESBAR_CAP_REG_5_128MB	11
VF_RESBAR_CAP_REG_5_64MB	10
VF_RESBAR_CAP_REG_5_32MB	9
VF_RESBAR_CAP_REG_5_16MB	8
VF_RESBAR_CAP_REG_5_8MB	7
VF_RESBAR_CAP_REG_5_4MB	6
VF_RESBAR_CAP_REG_5_2MB	5
VF_RESBAR_CAP_REG_5_1MB	4
RSVDP_0	3:0

Table 3-343 Fields for Register: VF_RESBAR_CAP_REG_5_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CAP_REG_5_128TB	* Varies	Up to 128TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 128 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 TB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[31] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CAP_REG_5_64TB	* Varies	Up to 64TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 64 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 TB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[30] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CAP_REG_5_32TB	* Varies	Up to 32TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 32 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 TB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[29] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CAP_REG_5_16TB	* Varies	Up to 16TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 16 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 TB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[28] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CAP_REG_5_8TB	* Varies	Up to 8TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 8 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 TB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[27] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CAP_REG_5_4TB	* Varies	Up to 4TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 4 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 TB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[26] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CAP_REG_5_2TB	* Varies	Up to 2TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 2 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 TB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[25] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CAP_REG_5_1TB	* Varies	Up to 1TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 1 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 TB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[24] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CAP_REG_5_512GB	* Varies	Up to 512GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 512 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 GB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[23] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CAP_REG_5_256GB	* Varies	Up to 256GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 256 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 GB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[22] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CAP_REG_5_128GB	* Varies	Up to 128GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 128 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 GB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[21] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CAP_REG_5_64GB	* Varies	Up to 64GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 64 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 GB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[20] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CAP_REG_5_32GB	* Varies	Up to 32GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 32 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 GB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[19] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CAP_REG_5_16GB	* Varies	Up to 16GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 16 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 GB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[18] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CAP_REG_5_8GB	* Varies	Up to 8GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 8 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 GB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[17] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CAP_REG_5_4GB	* Varies	Up to 4GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 4 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 GB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[16] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15	VF_RESBAR_CAP_REG_5_2GB	* Varies	Up to 2GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 2 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 GB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[15] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
14	VF_RESBAR_CAP_REG_5_1GB	* Varies	Up to 1GB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 1 GB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 GB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[14] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
13	VF_RESBAR_CAP_REG_5_512MB	* Varies	Up to 512MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 512 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 MB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[13] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12	VF_RESBAR_CAP_REG_5_256MB	* Varies	Up to 256MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 256 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 MB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[12] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11	VF_RESBAR_CAP_REG_5_128MB	* Varies	Up to 128MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 128 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 MB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[11] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
10	VF_RESBAR_CAP_REG_5_64MB	* Varies	Up to 64MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 64 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 MB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[10] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
9	VF_RESBAR_CAP_REG_5_32MB	* Varies	Up to 32MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 32 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 MB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[9] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
8	VF_RESBAR_CAP_REG_5_16MB	* Varies	Up to 16MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 16 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 MB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[8] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
7	VF_RESBAR_CAP_REG_5_8MB	* Varies	Up to 8MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 8 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 MB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[7] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
6	VF_RESBAR_CAP_REG_5_4MB	* Varies	Up to 4MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 4 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 MB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[6] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
5	VF_RESBAR_CAP_REG_5_2MB	* Varies	Up to 2MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 2 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 MB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[5] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
4	VF_RESBAR_CAP_REG_5_1MB	* Varies	Up to 1MB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 1 MB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 MB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[4] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47.8 VF_RESBAR_CTRL_REG_0_REG

- **Name:** Resizable BAR0 Control Register.
- **Description:** This register provides information about how much VF BAR0 size supported as well as VF BAR0 index and Number of VF BAR0.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR0_RESIZABLE_\$pf

VF_RESBAR_CTRL_REG_0_8EB	31
VF_RESBAR_CTRL_REG_0_4EB	30
VF_RESBAR_CTRL_REG_0_2EB	29
VF_RESBAR_CTRL_REG_0_1EB	28
VF_RESBAR_CTRL_REG_0_512PB	27
VF_RESBAR_CTRL_REG_0_256PB	26
VF_RESBAR_CTRL_REG_0_128PB	25
VF_RESBAR_CTRL_REG_0_64PB	24
VF_RESBAR_CTRL_REG_0_32PB	23
VF_RESBAR_CTRL_REG_0_16PB	22
VF_RESBAR_CTRL_REG_0_8PB	21
VF_RESBAR_CTRL_REG_0_4PB	20
VF_RESBAR_CTRL_REG_0_2PB	19
VF_RESBAR_CTRL_REG_0_1PB	18
VF_RESBAR_CTRL_REG_0_512TB	17
VF_RESBAR_CTRL_REG_0_256TB	16
RSVDP_14	15:14
VF_RESBAR_CTRL_REG_BAR_SIZE	13:8
VF_RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
VF_RESBAR_CTRL_REG_IDX_0	2:0

Table 3-344 Fields for Register: VF_RESBAR_CTRL_REG_0_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CTRL_REG_0_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 EB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CTRL_REG_0_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 EB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CTRL_REG_0_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 EB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CTRL_REG_0_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 EB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CTRL_REG_0_512 PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 PB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CTRL_REG_0_256 PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 PB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CTRL_REG_0_128 PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 PB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CTRL_REG_0_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 PB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CTRL_REG_0_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 PB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CTRL_REG_0_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 PB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CTRL_REG_0_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 PB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CTRL_REG_0_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 PB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CTRL_REG_0_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 PB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CTRL_REG_0_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 PB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CTRL_REG_0_512 TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 TB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CTRL_REG_0_256 TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR0 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 TB. Value After Reset: VF_BAR0_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	VF_RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR0 Size. This is an encoded value. ... The default value of this field is equal to the default size of the address space that the VF BAR0 resource is requesting through the VF BAR's read-only bits. When this register field is programmed, the value is immediately reflected in the size of the resource, as encoded in the number of read-only bits in the VF BAR. Software must only write values that correspond to those indicated as supported in the VF Resizable BAR Capability and Control registers. Writing an unsupported value will produce undefined results. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (_1MB): 1MB(2²⁰ bytes) ■ 0x1 (_2MB): 2MB(2²¹ bytes) ■ 0x2 (_4MB): 4MB(2²² bytes) ■ 0x3 (_8MB): 8MB(2²³ bytes) ■ 0x2b (_8EB): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::vf_resbar_size 0 0] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:5	VF_RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable VF BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in VF Resizable BAR0 Control register (0) (at offset 08h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable VF BAR for the function ■ 0x2 (BAR2): two resizable VF BARs for the function. ■ 0x3 (BAR3): three resizable VF BARs for the function. ■ 0x4 (BAR4): four resizable VF BARs for the function. ■ 0x5 (BAR5): five resizable VF BARs for the function. ■ 0x6 (BAR6): six resizable VF BARs for the function. Value After Reset: CX_NUM_VF_RBARS_0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	VF_RESBAR_CTRL_REG_IDX_0	R	BAR Index. This encoded value points to the beginning of the VF BAR0. For a 64-bit Base Address register, the VF BAR Index indicates the lower DWORD. This value indicates which VF BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (VF_BAR_24h): VF BAR located at offset 24h. ■ 0x1 (VF_BAR_28h): VF BAR located at offset 28h. ■ 0x2 (VF_BAR_2Ch): VF BAR located at offset 2Ch. ■ 0x3 (VF_BAR_30h): VF BAR located at offset 30h. ■ 0x4 (VF_BAR_34h): VF BAR located at offset 34h. ■ 0x5 (VF_BAR_38h): VF BAR located at offset 38h. Value After Reset: "0" Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47.9 VF_RESBAR_CTRL_REG_1_REG

- **Name:** Resizable BAR1 Control Register.
- **Description:** This register provides information about how much VF BAR1 size supported as well as VF BAR1 index and Number of VF BAR1.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR1_RESIZABLE_\$pf

VF_RESBAR_CTRL_REG_1_8EB	31
VF_RESBAR_CTRL_REG_1_4EB	30
VF_RESBAR_CTRL_REG_1_2EB	29
VF_RESBAR_CTRL_REG_1_1EB	28
VF_RESBAR_CTRL_REG_1_512PB	27
VF_RESBAR_CTRL_REG_1_256PB	26
VF_RESBAR_CTRL_REG_1_128PB	25
VF_RESBAR_CTRL_REG_1_64PB	24
VF_RESBAR_CTRL_REG_1_32PB	23
VF_RESBAR_CTRL_REG_1_16PB	22
VF_RESBAR_CTRL_REG_1_8PB	21
VF_RESBAR_CTRL_REG_1_4PB	20
VF_RESBAR_CTRL_REG_1_2PB	19
VF_RESBAR_CTRL_REG_1_1PB	18
VF_RESBAR_CTRL_REG_1_512TB	17
VF_RESBAR_CTRL_REG_1_256TB	16
RSVDP_14	15:14
VF_RESBAR_CTRL_REG_BAR_SIZE	13:8
VF_RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
VF_RESBAR_CTRL_REG_IDX_1	2:0

Table 3-345 Fields for Register: VF_RESBAR_CTRL_REG_1_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CTRL_REG_1_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 EB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CTRL_REG_1_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 EB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CTRL_REG_1_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 EB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CTRL_REG_1_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 EB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CTRL_REG_1_512 PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 PB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CTRL_REG_1_256 PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 PB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CTRL_REG_1_128 PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 PB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CTRL_REG_1_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 PB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CTRL_REG_1_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 PB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CTRL_REG_1_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 PB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CTRL_REG_1_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 PB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CTRL_REG_1_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 PB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CTRL_REG_1_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 PB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CTRL_REG_1_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 PB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CTRL_REG_1_512 TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 TB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CTRL_REG_1_256 TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR1 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 TB. Value After Reset: VF_BAR1_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	VF_RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR1 Size. This is an encoded value. ... The default value of this field is equal to the default size of the address space that the VF BAR1 resource is requesting through the VF BAR's read-only bits. When this register field is programmed, the value is immediately reflected in the size of the resource, as encoded in the number of read-only bits in the VF BAR. Software must only write values that correspond to those indicated as supported in the VF Resizable BAR Capability and Control registers. Writing an unsupported value will produce undefined results. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (_1MB): 1MB(2²⁰ bytes) ■ 0x1 (_2MB): 2MB(2²¹ bytes) ■ 0x2 (_4MB): 4MB(2²² bytes) ■ 0x3 (_8MB): 8MB(2²³ bytes) ■ 0x2b (_8EB): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::vf_resbar_size 1 0] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:5	VF_RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable VF BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in VF Resizable BAR1 Control register (0) (at offset 08h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable VF BAR for the function ■ 0x2 (BAR2): two resizable VF BARs for the function. ■ 0x3 (BAR3): three resizable VF BARs for the function. ■ 0x4 (BAR4): four resizable VF BARs for the function. ■ 0x5 (BAR5): five resizable VF BARs for the function. ■ 0x6 (BAR6): six resizable VF BARs for the function. Value After Reset: [::DWC_pcie_ctl::resbar_nbars 1 0] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	VF_RESBAR_CTRL_REG_IDX_1	R	BAR Index. This encoded value points to the beginning of the VF BAR1. For a 64-bit Base Address register, the VF BAR Index indicates the lower DWORD. This value indicates which VF BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (VF_BAR_24h): VF BAR located at offset 24h. ■ 0x1 (VF_BAR_28h): VF BAR located at offset 28h. ■ 0x2 (VF_BAR_2Ch): VF BAR located at offset 2Ch. ■ 0x3 (VF_BAR_30h): VF BAR located at offset 30h. ■ 0x4 (VF_BAR_34h): VF BAR located at offset 34h. ■ 0x5 (VF_BAR_38h): VF BAR located at offset 38h. Value After Reset: "1" Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47.10 VF_RESBAR_CTRL_REG_2_REG

- **Name:** Resizable BAR2 Control Register.
- **Description:** This register provides information about how much VF BAR2 size supported as well as VF BAR2 index and Number of VF BAR2.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR2_RESIZABLE_\$pf

VF_RESBAR_CTRL_REG_2_8EB	31
VF_RESBAR_CTRL_REG_2_4EB	30
VF_RESBAR_CTRL_REG_2_2EB	29
VF_RESBAR_CTRL_REG_2_1EB	28
VF_RESBAR_CTRL_REG_2_512PB	27
VF_RESBAR_CTRL_REG_2_256PB	26
VF_RESBAR_CTRL_REG_2_128PB	25
VF_RESBAR_CTRL_REG_2_64PB	24
VF_RESBAR_CTRL_REG_2_32PB	23
VF_RESBAR_CTRL_REG_2_16PB	22
VF_RESBAR_CTRL_REG_2_8PB	21
VF_RESBAR_CTRL_REG_2_4PB	20
VF_RESBAR_CTRL_REG_2_2PB	19
VF_RESBAR_CTRL_REG_2_1PB	18
VF_RESBAR_CTRL_REG_2_512TB	17
VF_RESBAR_CTRL_REG_2_256TB	16
RSVDP_14	15:14
VF_RESBAR_CTRL_REG_BAR_SIZE	13:8
VF_RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
VF_RESBAR_CTRL_REG_IDX_2	2:0

Table 3-346 Fields for Register: VF_RESBAR_CTRL_REG_2_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CTRL_REG_2_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 EB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CTRL_REG_2_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 EB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CTRL_REG_2_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 EB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CTRL_REG_2_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 EB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CTRL_REG_2_512 PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 PB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CTRL_REG_2_256 PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 PB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CTRL_REG_2_128 PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 PB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CTRL_REG_2_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 PB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CTRL_REG_2_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 PB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CTRL_REG_2_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 PB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CTRL_REG_2_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 PB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CTRL_REG_2_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 PB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CTRL_REG_2_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 PB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CTRL_REG_2_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 PB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CTRL_REG_2_512 TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 TB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CTRL_REG_2_256 TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR2 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 TB. Value After Reset: VF_BAR2_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	VF_RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR2 Size. This is an encoded value. ... The default value of this field is equal to the default size of the address space that the VF BAR2 resource is requesting through the VF BAR's read-only bits. When this register field is programmed, the value is immediately reflected in the size of the resource, as encoded in the number of read-only bits in the VF BAR. Software must only write values that correspond to those indicated as supported in the VF Resizable BAR Capability and Control registers. Writing an unsupported value will produce undefined results. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (_1MB): 1MB(2²⁰ bytes) ■ 0x1 (_2MB): 2MB(2²¹ bytes) ■ 0x2 (_4MB): 4MB(2²² bytes) ■ 0x3 (_8MB): 8MB(2²³ bytes) ■ 0x2b (_8EB): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::vf_resbar_size 2 0] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:5	VF_RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable VF BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in VF Resizable BAR2 Control register (0) (at offset 08h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable VF BAR for the function ■ 0x2 (BAR2): two resizable VF BARs for the function. ■ 0x3 (BAR3): three resizable VF BARs for the function. ■ 0x4 (BAR4): four resizable VF BARs for the function. ■ 0x5 (BAR5): five resizable VF BARs for the function. ■ 0x6 (BAR6): six resizable VF BARs for the function. Value After Reset: [::DWC_pcie_ctl::resbar_nbars 2 0] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	VF_RESBAR_CTRL_REG_IDX_2	R	BAR Index. This encoded value points to the beginning of the VF BAR2. For a 64-bit Base Address register, the VF BAR Index indicates the lower DWORD. This value indicates which VF BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (VF_BAR_24h): VF BAR located at offset 24h. ■ 0x1 (VF_BAR_28h): VF BAR located at offset 28h. ■ 0x2 (VF_BAR_2Ch): VF BAR located at offset 2Ch. ■ 0x3 (VF_BAR_30h): VF BAR located at offset 30h. ■ 0x4 (VF_BAR_34h): VF BAR located at offset 34h. ■ 0x5 (VF_BAR_38h): VF BAR located at offset 38h. Value After Reset: "2" Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47.11 VF_RESBAR_CTRL_REG_3_REG

- **Name:** Resizable BAR3 Control Register.
- **Description:** This register provides information about how much VF BAR3 size supported as well as VF BAR3 index and Number of VF BAR3.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR3_RESIZABLE_\$pf

VF_RESBAR_CTRL_REG_3_8EB	31
VF_RESBAR_CTRL_REG_3_4EB	30
VF_RESBAR_CTRL_REG_3_2EB	29
VF_RESBAR_CTRL_REG_3_1EB	28
VF_RESBAR_CTRL_REG_3_512PB	27
VF_RESBAR_CTRL_REG_3_256PB	26
VF_RESBAR_CTRL_REG_3_128PB	25
VF_RESBAR_CTRL_REG_3_64PB	24
VF_RESBAR_CTRL_REG_3_32PB	23
VF_RESBAR_CTRL_REG_3_16PB	22
VF_RESBAR_CTRL_REG_3_8PB	21
VF_RESBAR_CTRL_REG_3_4PB	20
VF_RESBAR_CTRL_REG_3_2PB	19
VF_RESBAR_CTRL_REG_3_1PB	18
VF_RESBAR_CTRL_REG_3_512TB	17
VF_RESBAR_CTRL_REG_3_256TB	16
RSVDP_14	15:14
VF_RESBAR_CTRL_REG_BAR_SIZE	13:8
VF_RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
VF_RESBAR_CTRL_REG_IDX_3	2:0

Table 3-347 Fields for Register: VF_RESBAR_CTRL_REG_3_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CTRL_REG_3_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 EB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CTRL_REG_3_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 EB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CTRL_REG_3_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 EB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CTRL_REG_3_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 EB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CTRL_REG_3_512 PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 PB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CTRL_REG_3_256 PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 PB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CTRL_REG_3_128 PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 PB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CTRL_REG_3_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 PB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CTRL_REG_3_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 PB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CTRL_REG_3_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 PB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CTRL_REG_3_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 PB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CTRL_REG_3_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 PB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CTRL_REG_3_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 PB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CTRL_REG_3_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 PB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CTRL_REG_3_512 TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 TB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CTRL_REG_3_256 TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR3 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 TB. Value After Reset: VF_BAR3_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	VF_RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR3 Size. This is an encoded value. ... The default value of this field is equal to the default size of the address space that the VF BAR3 resource is requesting through the VF BAR's read-only bits. When this register field is programmed, the value is immediately reflected in the size of the resource, as encoded in the number of read-only bits in the VF BAR. Software must only write values that correspond to those indicated as supported in the VF Resizable BAR Capability and Control registers. Writing an unsupported value will produce undefined results. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (_1MB): 1MB(2²⁰ bytes) ■ 0x1 (_2MB): 2MB(2²¹ bytes) ■ 0x2 (_4MB): 4MB(2²² bytes) ■ 0x3 (_8MB): 8MB(2²³ bytes) ■ 0x2b (_8EB): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::vf_resbar_size 3 0] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:5	VF_RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable VF BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in VF Resizable BAR3 Control register (0) (at offset 08h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable VF BAR for the function ■ 0x2 (BAR2): two resizable VF BARs for the function. ■ 0x3 (BAR3): three resizable VF BARs for the function. ■ 0x4 (BAR4): four resizable VF BARs for the function. ■ 0x5 (BAR5): five resizable VF BARs for the function. ■ 0x6 (BAR6): six resizable VF BARs for the function. Value After Reset: [::DWC_pcie_ctl::resbar_nbars 3 0] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	VF_RESBAR_CTRL_REG_IDX_3	R	BAR Index. This encoded value points to the beginning of the VF BAR3. For a 64-bit Base Address register, the VF BAR Index indicates the lower DWORD. This value indicates which VF BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (VF_BAR_24h): VF BAR located at offset 24h. ■ 0x1 (VF_BAR_28h): VF BAR located at offset 28h. ■ 0x2 (VF_BAR_2Ch): VF BAR located at offset 2Ch. ■ 0x3 (VF_BAR_30h): VF BAR located at offset 30h. ■ 0x4 (VF_BAR_34h): VF BAR located at offset 34h. ■ 0x5 (VF_BAR_38h): VF BAR located at offset 38h. Value After Reset: "3" Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47.12 VF_RESBAR_CTRL_REG_4_REG

- **Name:** Resizable BAR4 Control Register.
- **Description:** This register provides information about how much VF BAR4 size supported as well as VF BAR4 index and Number of VF BAR4.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR4_RESIZABLE_\$pf

VF_RESBAR_CTRL_REG_4_8EB	31
VF_RESBAR_CTRL_REG_4_4EB	30
VF_RESBAR_CTRL_REG_4_2EB	29
VF_RESBAR_CTRL_REG_4_1EB	28
VF_RESBAR_CTRL_REG_4_512PB	27
VF_RESBAR_CTRL_REG_4_256PB	26
VF_RESBAR_CTRL_REG_4_128PB	25
VF_RESBAR_CTRL_REG_4_64PB	24
VF_RESBAR_CTRL_REG_4_32PB	23
VF_RESBAR_CTRL_REG_4_16PB	22
VF_RESBAR_CTRL_REG_4_8PB	21
VF_RESBAR_CTRL_REG_4_4PB	20
VF_RESBAR_CTRL_REG_4_2PB	19
VF_RESBAR_CTRL_REG_4_1PB	18
VF_RESBAR_CTRL_REG_4_512TB	17
VF_RESBAR_CTRL_REG_4_256TB	16
RSVDP_14	15:14
VF_RESBAR_CTRL_REG_BAR_SIZE	13:8
VF_RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
VF_RESBAR_CTRL_REG_IDX_4	2:0

Table 3-348 Fields for Register: VF_RESBAR_CTRL_REG_4_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CTRL_REG_4_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 EB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CTRL_REG_4_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 EB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CTRL_REG_4_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 EB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CTRL_REG_4_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 EB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CTRL_REG_4_512 PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 PB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CTRL_REG_4_256 PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 PB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CTRL_REG_4_128 PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 PB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CTRL_REG_4_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 PB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CTRL_REG_4_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 PB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CTRL_REG_4_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 PB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CTRL_REG_4_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 PB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CTRL_REG_4_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 PB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CTRL_REG_4_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 PB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CTRL_REG_4_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 PB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CTRL_REG_4_512 TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 TB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CTRL_REG_4_256 TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR4 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 TB. Value After Reset: VF_BAR4_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	VF_RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR4 Size. This is an encoded value. ... The default value of this field is equal to the default size of the address space that the VF BAR4 resource is requesting through the VF BAR's read-only bits. When this register field is programmed, the value is immediately reflected in the size of the resource, as encoded in the number of read-only bits in the VF BAR. Software must only write values that correspond to those indicated as supported in the VF Resizable BAR Capability and Control registers. Writing an unsupported value will produce undefined results. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (_1MB): 1MB(2²⁰ bytes) ■ 0x1 (_2MB): 2MB(2²¹ bytes) ■ 0x2 (_4MB): 4MB(2²² bytes) ■ 0x3 (_8MB): 8MB(2²³ bytes) ■ 0x2b (_8EB): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::vf_resbar_size 4 0] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:5	VF_RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable VF BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in VF Resizable BAR4 Control register (0) (at offset 08h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable VF BAR for the function ■ 0x2 (BAR2): two resizable VF BARs for the function. ■ 0x3 (BAR3): three resizable VF BARs for the function. ■ 0x4 (BAR4): four resizable VF BARs for the function. ■ 0x5 (BAR5): five resizable VF BARs for the function. ■ 0x6 (BAR6): six resizable VF BARs for the function. Value After Reset: [::DWC_pcie_ctl::resbar_nbars 4 0] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	VF_RESBAR_CTRL_REG_IDX_4	R	BAR Index. This encoded value points to the beginning of the VF BAR4. For a 64-bit Base Address register, the VF BAR Index indicates the lower DWORD. This value indicates which VF BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (VF_BAR_24h): VF BAR located at offset 24h. ■ 0x1 (VF_BAR_28h): VF BAR located at offset 28h. ■ 0x2 (VF_BAR_2Ch): VF BAR located at offset 2Ch. ■ 0x3 (VF_BAR_30h): VF BAR located at offset 30h. ■ 0x4 (VF_BAR_34h): VF BAR located at offset 34h. ■ 0x5 (VF_BAR_38h): VF BAR located at offset 38h. Value After Reset: "4" Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.47.13 VF_RESBAR_CTRL_REG_5_REG

- **Name:** Resizable BAR5 Control Register.
- **Description:** This register provides information about how much VF BAR5 size supported as well as VF BAR5 index and Number of VF BAR5.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** ((CX_NUM_VF_RBARS_\$pf>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0)) && VF_BAR5_RESIZABLE_\$pf

VF_RESBAR_CTRL_REG_5_8EB	31
VF_RESBAR_CTRL_REG_5_4EB	30
VF_RESBAR_CTRL_REG_5_2EB	29
VF_RESBAR_CTRL_REG_5_1EB	28
VF_RESBAR_CTRL_REG_5_512PB	27
VF_RESBAR_CTRL_REG_5_256PB	26
VF_RESBAR_CTRL_REG_5_128PB	25
VF_RESBAR_CTRL_REG_5_64PB	24
VF_RESBAR_CTRL_REG_5_32PB	23
VF_RESBAR_CTRL_REG_5_16PB	22
VF_RESBAR_CTRL_REG_5_8PB	21
VF_RESBAR_CTRL_REG_5_4PB	20
VF_RESBAR_CTRL_REG_5_2PB	19
VF_RESBAR_CTRL_REG_5_1PB	18
VF_RESBAR_CTRL_REG_5_512TB	17
VF_RESBAR_CTRL_REG_5_256TB	16
RSVDP_14	15:14
VF_RESBAR_CTRL_REG_BAR_SIZE	13:8
VF_RESBAR_CTRL_REG_NUM_BARS	7:5
RSVDP_3	4:3
VF_RESBAR_CTRL_REG_IDX_5	2:0

Table 3-349 Fields for Register: VF_RESBAR_CTRL_REG_5_REG

Bits	Name	Memory Access	Description
31	VF_RESBAR_CTRL_REG_5_8EB	* Varies	Up to 8EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 8 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 EB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[47] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
30	VF_RESBAR_CTRL_REG_5_4EB	* Varies	Up to 4EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 4 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 EB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[46] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
29	VF_RESBAR_CTRL_REG_5_2EB	* Varies	Up to 2EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 2 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 EB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[45] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
28	VF_RESBAR_CTRL_REG_5_1EB	* Varies	Up to 1EB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 1 EB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 EB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[44] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
27	VF_RESBAR_CTRL_REG_5_512 PB	* Varies	Up to 512PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 512 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 PB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[43] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
26	VF_RESBAR_CTRL_REG_5_256 PB	* Varies	Up to 256PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 256 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 PB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[42] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25	VF_RESBAR_CTRL_REG_5_128 PB	* Varies	Up to 128PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 128 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 128 PB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[41] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
24	VF_RESBAR_CTRL_REG_5_64PB	* Varies	Up to 64PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 64 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 64 PB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[40] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23	VF_RESBAR_CTRL_REG_5_32PB	* Varies	Up to 32PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 32 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 32 PB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[39] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
22	VF_RESBAR_CTRL_REG_5_16PB	* Varies	Up to 16PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 16 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 16 PB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[38] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
21	VF_RESBAR_CTRL_REG_5_8PB	* Varies	Up to 8PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 8 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 8 PB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[37] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
20	VF_RESBAR_CTRL_REG_5_4PB	* Varies	Up to 4PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 4 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 4 PB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[36] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
19	VF_RESBAR_CTRL_REG_5_2PB	* Varies	Up to 2PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 2 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 2 PB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[35] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
18	VF_RESBAR_CTRL_REG_5_1PB	* Varies	Up to 1PB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 1 PB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 1 PB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[34] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
17	VF_RESBAR_CTRL_REG_5_512 TB	* Varies	Up to 512TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 512 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 512 TB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[33] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
16	VF_RESBAR_CTRL_REG_5_256 TB	* Varies	Up to 256TB BAR Supported. When Set, indicates that the Physical Function(PF) supports operating with the VF BAR5 sized to 256 TB. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): If Set, function supports operating with VF BAR sized to 256 TB. Value After Reset: VF_BAR5_RESOURCE_AVAIL_n[32] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	VF_RESBAR_CTRL_REG_BAR_SIZE	* Varies	BAR5 Size. This is an encoded value. ... The default value of this field is equal to the default size of the address space that the VF BAR5 resource is requesting through the VF BAR's read-only bits. When this register field is programmed, the value is immediately reflected in the size of the resource, as encoded in the number of read-only bits in the VF BAR. Software must only write values that correspond to those indicated as supported in the VF Resizable BAR Capability and Control registers. Writing an unsupported value will produce undefined results. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (_1MB): 1MB(2²⁰ bytes) ■ 0x1 (_2MB): 2MB(2²¹ bytes) ■ 0x2 (_4MB): 4MB(2²² bytes) ■ 0x3 (_8MB): 8MB(2²³ bytes) ■ 0x2b (_8EB): 8EB(2⁶³ bytes) Value After Reset: [::DWC_pcie_ctl::vf_resbar_size 5 0] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:5	VF_RESBAR_CTRL_REG_NUM_BARS	R	Number of Resizeable BARs. This field indicates the total number of resizable VF BARs in the capability structure for the Function. The value of this field must be in the range of 01h to 06h. The field is valid in VF Resizable BAR5 Control register (0) (at offset 08h), and is RsvdP for all others. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (BAR1): one resizable VF BAR for the function ■ 0x2 (BAR2): two resizable VF BARs for the function. ■ 0x3 (BAR3): three resizable VF BARs for the function. ■ 0x4 (BAR4): four resizable VF BARs for the function. ■ 0x5 (BAR5): five resizable VF BARs for the function. ■ 0x6 (BAR6): six resizable VF BARs for the function. Value After Reset: [::DWC_pcie_ctl::resbar_nbars 5 0] Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	VF_RESBAR_CTRL_REG_IDX_5	R	BAR Index. This encoded value points to the beginning of the VF BAR5. For a 64-bit Base Address register, the VF BAR Index indicates the lower DWORD. This value indicates which VF BAR supports a negotiable size. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (VF_BAR_24h): VF BAR located at offset 24h. ■ 0x1 (VF_BAR_28h): VF BAR located at offset 28h. ■ 0x2 (VF_BAR_2Ch): VF BAR located at offset 2Ch. ■ 0x3 (VF_BAR_30h): VF BAR located at offset 30h. ■ 0x4 (VF_BAR_34h): VF BAR located at offset 34h. ■ 0x5 (VF_BAR_38h): VF BAR located at offset 38h. Value After Reset: "5" Exists: ((CX_NUM_VF_RBARS_0>0) && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48 PF_PORT_LOGIC_i Registers

3.48.1 **ACK_LATENCY_TIMER_OFF**

- **Name:** Ack Latency Timer and Replay Timer Register.
- **Description:** This register holds the ack latency timer limit and replay timer limit values.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	
15:0	
REPLAY_TIME_LIMIT	
ROUND_TRIP_LATENCY_TIME_LIMIT	

Table 3-350 Fields for Register: ACK_LATENCY_TIMER_OFF

Bits	Name	Memory Access	Description
31:16	REPLAY_TIME_LIMIT	R/W	Replay Timer Limit. The replay timer expires when it reaches this limit. The controller initiates a replay upon reception of a NAK or when the replay timer expires. For more information, see "Transmit Replay" in the <i>Databook</i>. - You can modify the effective timer limit through the <code>TIMER_MOD_REPLAY_TIMER</code> field of the <code>TIMER_CTRL_MAX_FUNC_NUM_OFF</code> register. - After reset, the controller updates the default according to the Negotiated Link Width, <code>Max_Payload_Size</code>, and speed. The value is determined from Tables 3-4, 3-5, and 3-6 of the <i>PCI Express Base Specification</i>. - If there is a change in the payload size or link speed, the controller overrides any value that you have written to this register field, and resets the field back to the specification-defined value. The controller does not change the value in the <code>TIMER_MOD_REPLAY_TIMER</code> field of the <code>TIMER_CTRL_MAX_FUNC_NUM_OFF</code> register. Value After Reset: {12429/CX_NB} Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true
15:0	ROUND_TRIP_LATENCY_TIME_LIMIT	R/W	Ack Latency Timer Limit. The Ack latency timer expires when it reaches this limit. For more information, see "ACK/NAK Scheduling" in the <i>Databook</i>. - You can modify the effective timer limit through the <code>TIMER_MOD_ACK_NAK</code> field of the <code>TIMER_CTRL_MAX_FUNC_NUM_OFF</code> register. - After reset, the controller updates the default according to the Negotiated Link Width, <code>Max_Payload_Size</code>, and speed. The value is determined from Tables 3-7, 3-8, and 3-9 of the <i>PCI Express Base Specification</i>. - The limit must reflect the round trip latency from requester to completer. - If there is a change in the payload size or link width, the controller overrides any value that you have written to this register field, and resets the field back to the specification-defined value. The controller does not change the value in the <code>TIMER_MOD_ACK_NAK</code> field of the <code>TIMER_CTRL_MAX_FUNC_NUM_OFF</code> register. Value After Reset: {4143/CX_NB} Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true

3.48.2 VENDOR_SPEC_DLLP_OFF

- **Name:** Vendor Specific DLLP Register.
- **Description:** This register holds the vendor specific DLLP.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

 31:0
VENDOR_SPEC_DLLP

Table 3-351 Fields for Register: VENDOR_SPEC_DLLP_OFF

Bits	Name	Memory Access	Description
31:0	VENDOR_SPEC_DLLP	R/W	Vendor Specific DLLP Register. You can use this register to send a specific PCI Express DLLP. Your application can write 8-bit DLLP Type and 24-bit Payload data into this register, and set the VENDOR_SPECIFIC_DLLP_REQ field of the PORT_LINK_CTRL_OFF, to send the DLLP. ■ Bits[7:0]: DLLP Type ■ Bits[31:8]: Vendor Defined Payload (24 bits) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ZERO): Zero value ■ 0xffffffff (MAX): Max value Value After Reset: 0xffffffff Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.3 PORT_FORCE_OFF

- **Name:** Port Force Link Register.
- **Description:** This register can be used for testing and debuggong the link.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	RSVDP_24
23	DO_DESKEW_FOR_SRIS
22	SUPPORT_PART_LANES_RXEI_EXIT
21:16	LINK_STATE
15	FORCE_EN
14:12	RSVDP_12
11:8	FORCED_LTSSM
7:0	LINK_NUM

Table 3-352 Fields for Register: PORT_FORCE_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
23	DO_DESKEW_FOR_SRIS	R/W	Use the transitions from TS2 to Logical Idle Symbol, SKP OS to Logical Idle Symbol, EIEOS to Logical Idle Symbol, and FTS Sequence to SKP OS to do deskew instead of using received SKP OS or TS1 to TS2 transition if DO_DESKEW_FOR_SRIS is set to '1'. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_DO_DESKEW_FOR_SRIS Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
22	SUPPORT_PART_LANES_RXEI_EXIT	R/W	Support LTSSM transition from Polling.Active to Polling.Config based on Rx 8 TSs on any lanes which are Rx EI exit too from base spec after 24ms timeout. This prevents some lanes detected but not Rx EI exit and LTSSM cannot move to Polling.Config. You must set the parameter CX_AUTO_LANE_FLIP_CTRL_EN true for the auto lanes reversal. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SUPPORT): Any lanes receives 8 consecutive TS OSs, LTSSM moves from Polling.Active to Polling.Config. If all lanes do not receive 8 consecutive TS OSs and any predetermined lanes are still on Rx ElecIdle, LTSSM moves from Polling.Active to Polling.Compliance. ■ 0x0 (UNSUPPORT): Any lanes receives 8 consecutive TS OS and all predetermined lanes have Rx ElecIdle exit, LTSSM moves from Polling.Active to Polling.Config. This is legacy mode from Base Spec. Any predetermined lanes are still on Rx ElecIdle, LTSSM moves from Polling.Active to Polling.Compliance. Value After Reset: DEFAULT_SUPPORT_PART_LANES_RXEI_EXIT_IN_POLL_ACTIVE Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
21:16	LINK_STATE	R/W	Forced LTSSM State. The LTSSM state that the controller is forced to when you set the FORCE_EN bit (Force Link). LTSSM state encoding is defined by the lts_state variable in workspace/src/Layer1/smlh_ltssm.v. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15	FORCE_EN	W	Force Link. The controller supports a testing and debug capability to allow your software to force the LTSSM state machine into a specific state, and to force the controller to transmit a specific Link Command. Asserting this bit triggers the following actions: - Forces the LTSSM to the state specified by the Forced LTSSM State field. - Forces the controller to transmit the command specified by the Forced Link Command field. This is a self-clearing register field. Reading from this register field always returns a '0'. Values: 0x1 (ENABLE): Enable 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
14:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:8	FORCED_LTSSM	R/W	Forced Link Command. The link command that the controller is forced to transmit when you set FORCE_EN bit (Force Link). Link command encoding is defined by the ltssm_cmd variable in workspace/src/Layer1/smlh_ltssm.v. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:0	LINK_NUM	R/W	Link Number. Not used for endpoint. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_LINK_NUM Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.4 ACK_F_ASPM_CTRL_OFF

- **Name:** Ack Frequency and L0-L1 ASPM Control Register.
- **Description:** This register is used to control ack frequency and L0-L1 ASPM behaviour.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	RSVDP_31
30	ENTER_ASPM
29:27	L1_ENTRANCE_LATENCY
26:24	L0S_ENTRANCE_LATENCY
23:16	COMMON_CLK_N_FTS
15:8	ACK_N_FTS
7:0	ACK_FREQ

Table 3-353 Fields for Register: ACK_F_ASPM_CTRL_OFF

Bits	Name	Memory Access	Description
31	RSVDP_31	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
30	ENTER_ASPM	R/W	ASPM L1 Entry Control. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (IDLE_TO_L1): Controller enters ASPM L1 after a period in which it has been idle. ■ 0x0 (L0S_TO_L1): Controller enters ASPM L1 only after idle period during which both receive and transmit are in L0s. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
29:27	L1_ENTRANCE_LATENCY	R/W	L1 Entrance Latency. <i>Note:</i> Programming this timer with a value greater than 32us has no effect unless extended sync is used, or all of the credits are infinite. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_1_US): 1 us ■ 0x1 (_2_US): 2 us ■ 0x2 (_4_US): 4 us ■ 0x3 (_8_US): 8 us ■ 0x4 (_16_US): 16 us ■ 0x5 (_32_US): 32 us ■ 0x6 (_64_US): 64Us ■ 0x7 (_64US_): 64 us Value After Reset: DEFAULT_L1_ENTR_LATENCY Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
26:24	L0S_ENTRANCE_LATENCY	R/W	L0s Entrance Latency. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_1_US): 1 us ■ 0x1 (_2_US): 2 us ■ 0x2 (_3_US): 3 us ■ 0x3 (_4_US): 4 us ■ 0x4 (_5_US): 5 us ■ 0x5 (_6_US): 6 us ■ 0x6 (_7_US): 7 US ■ 0x7 (_7US_): 7 us Value After Reset: DEFAULT_L0S_ENTR_LATENCY Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
23:16	COMMON_CLK_N_FTS	* Varies	Common Clock N_FTS. This is the N_FTS when common clock is used. The number of Fast Training Sequence ordered sets to be transmitted when transitioning from L0s to L0. This field is only writable (sticky) when all of the following configuration parameter equations are true: ■ <code>CX_NFTS != CX_COMM_NFTS</code> ■ <code>DEFAULT_L0S_EXIT_LATENCY != DEFAULT_COMM_L0S_EXIT_LATENCY</code> ■ <code>DEFAULT_L1_EXIT_LATENCY != DEFAULT_COMM_L1_EXIT_LATENCY</code> <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x0 (ZERO): The controller does not support a value of zero; a value of zero can cause the LTSSM to go into the recovery state when exiting from L0s. ■ 0xff (MAX_VAL): The maximum number of FTS ordered-sets that a component can request is 255. Value After Reset: <code>CX_COMM_NFTS</code> Exists: <code>(!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))</code> Memory Access: <code>"(CX_MEMORY_MAP_VIEW==0 && CX_COMM_CLK_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_COMM_CLK_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_COMM_CLK_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_COMM_CLK_EN) ? \"read-only\" : (CX_COMM_CLK_EN) ? \"read-write\" : \"read-only\""</code>

Bits	Name	Memory Access	Description
15:8	ACK_N_FTS	R/W	The number of Fast Training Sequence(N_FTS) ordered sets to be transmitted when transitioning from L0s to L0. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ZERO): The controller does not support a value of zero; a value of zero can cause the LTSSM to go into the recovery state when exiting from L0s. ■ 0xff (MAX_VAL): The maximum number of FTS ordered-sets that a component can request is 255. Value After Reset: CX_NFTS Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:0	ACK_FREQ	R/W	Ack Frequency. The controller accumulates the number of pending ACKs specified here (up to 255) before scheduling an ACK DLLP. ■ 0: Indicates that this Ack Frequency Counter feature is turned off. The controller generates a low-priority ACK request for every TLP that it receives. The controller waits until the ACK Latency Timer expires, then converts the current low-priority ACK request to a high-priority ACK request and schedules the DLLP for transmission to the remote link partner. ■ 1-255: Indicates that the controller will schedule a high-priority ACK after receiving this number of TLPs. It might schedule the ACK before receiving this number of TLPs if the ACK Latency Timer expires, but never later. For a typical system, you do not have to modify the default setting. For more information, see "ACK/NAK Scheduling" in the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): The value '0' indicates that this Ack Frequency Counter feature is turned off. ■ 0xff (MAX_VAL): Any value between 1 and 255 indicates that the controller will schedule a high-priority ACK after receiving the specified number of TLPs. Value After Reset: DEFAULT_ACK_FREQUENCY Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.5 PORT_LINK_CTRL_OFF

- **Name:** Port Link Control Register.
- **Description:** Using this register you can control the port link behaviour.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RSVDP_28
27	TRANSMIT_LANE_REVERSALE_ENABLE
26	EXTENDED_SYNC
25	CORRUPT_LCRC_ENABLE
24	BEACON_ENABLE
23	CROSS_LINK_ACTIVE
22	CROSS_LINK_EN
21:16	LINK_CAPABLE
15:12	RSVDP_12
11:8	LINK_RATE
7	FAST_LINK_MODE
6	LINK_DISABLE
5	DLL_LINK_EN
4	RSVDP_4
3	RESET_ASSERT
2	LOOPBACK_ENABLE
1	SCRAMBLE_DISABLE
0	VENDOR_SPECIFIC_DLLP_REQ

Table 3-354 Fields for Register: PORT_LINK_CTRL_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27	TRANSMIT_LANE_REVERSALE_ENABLE	R/W	TRANSMIT_LANE_REVERSALE_ENABLE is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
26	EXTENDED_SYNCH	R/W	EXTENDED_SYNCH is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
25	CORRUPT_LCRC_ENABLE	R/W	CORRUPT_LCRC_ENABLE is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
24	BEACON_ENABLE	R/W	BEACON_ENABLE is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23	CROSS_LINK_ACTIVE	R	Reserved Values: ■ 0x1 (SET): Indicates a change from upstream to downstream or downstream to upstream. Same as the smlh_crosslink_active output. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: CX_CROSSLINK_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
22	CROSS_LINK_EN	R/W	Reserved <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x1 Exists: CX_CROSSLINK_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
21:16	LINK_CAPABLE	* Varies	Link Mode Enable. Sets the number of lanes in the link that you want to connect to the link partner. When you have unused lanes in your system, then you must change the value in this register to reflect the number of lanes. You must also change the value in the "Predetermined Number of Lanes" field of the "Link Width and Speed Change Control Register". For more information, see "How to Tie Off Unused Lanes". For information on upsizing and downsizing the link width, see "Link Establishment". <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (X1): x1 ■ 0x3 (X2): x2 ■ 0x7 (X4): x4 ■ 0xf (X8): x8 ■ 0x1f (X16): x16 ■ 0x3f (X32): x32 (not supported) Value After Reset: {[:DWC_pcie_ctl::getDefaultLinkCap]} Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(SNPS_RSVDPARAM_25) ? "writeAsRead" : "unconstrained" } Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "
15:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:8	LINK_RATE	R/W	LINK_RATE is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
7	FAST_LINK_MODE	R/W	Fast Link Mode. Sets all internal LTSSM millisecond timers to Fast Mode for speeding up simulation. Forces the LTSSM training (link initialization) to use shorter time-outs and to link up faster. ■ The default scaling factor can be changed using the <code>DEFAULT_FAST_LINK_SCALING_FACTOR</code> parameter or through the <code>FAST_LINK_SCALING_FACTOR</code> field in the <code>TIMER_CTRL_MAX_FUNC_NUM_OFF</code> register. ■ Fast Link Mode can also be activated by setting the <code>diag_ctrl_bus[2]</code> pin to '1'. For more information, see the "Fast Link Simulation Mode" section in the "Integrating the Controller with the PHY or Application RTL or Verification IP" chapter of the <i>User Guide</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (!<code>CX_PL_REG_DISABLE</code> && (<code>CX_MEMORY_MAP_PCIE_TYPE</code>==2 <code>CX_MEMORY_MAP_PCIE_TYPE</code>==0))
6	LINK_DISABLE	R/W	<code>LINK_DISABLE</code> is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!<code>CX_PL_REG_DISABLE</code> && (<code>CX_MEMORY_MAP_PCIE_TYPE</code>==2 <code>CX_MEMORY_MAP_PCIE_TYPE</code>==0))
5	DLL_LINK_EN	R/W	DLL Link Enable. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enables link initialization. ■ 0x0 (DISABLE): The controller does not transmit InitFC DLLPs and does not establish a link. Value After Reset: 0x1 Exists: (!<code>CX_PL_REG_DISABLE</code> && (<code>CX_MEMORY_MAP_PCIE_TYPE</code>==2 <code>CX_MEMORY_MAP_PCIE_TYPE</code>==0))
4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!<code>CX_PL_REG_DISABLE</code> && (<code>CX_MEMORY_MAP_PCIE_TYPE</code>==2 <code>CX_MEMORY_MAP_PCIE_TYPE</code>==0))

Bits	Name	Memory Access	Description
3	RESET_ASSERT	R/W	Reset Assert. Triggers a recovery and forces the LTSSM to the hot reset state (downstream port only). <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): Set ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2	LOOPBACK_ENABLE	* Varies	Loopback Enable. Turns on loopback. For more information, see "Loopback" in the <i>Databook</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \\"read-only\\" : \\"read-write\\" "
1	SCRAMBLE_DISABLE	* Varies	Scramble Disable. Turns off data scrambling. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: {(CX_PCIE_OVER_CIO_ENABLE) ? 1 : 0} Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \\"read-only\\" : \\"read-write\\" "

Bits	Name	Memory Access	Description
0	VENDOR_SPECIFIC_DLLP_REQ	R/W1C	Vendor Specific DLLP Request. Reading from this self-clearing register field always returns a '0'. Values: ■ 0x1 (SET): When software writes a '1' to this bit, the controller transmits the DLLP contained in the VENDOR_SPEC_DLLP field of VENDOR_SPEC_DLLP_OFF ■ 0x0 (CLEAR): This is a self clearing register Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.6 LANE_SKEW_OFF

- **Name:** Lane Skew Register.
- **Description:** This register is used to control the lane skew behaviour.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

DISABLE_LANE_TO_LANE_DESKEW	31
IMPLEMENT_NUM_LANES	30:27
ELASTIC_BUFFER_MODE	26
ACK_NAK_DISABLE	25
FLOW_CTRL_DISABLE	24
INSERT_LANE_SKEW	23:0

Table 3-355 Fields for Register: LANE_SKEW_OFF

Bits	Name	Memory Access	Description
31	DISABLE_LANE_TO_LANE_DESKEW	* Varies	Disable Lane-to-Lane Deskew. Causes the controller to disable the internal Lane-to-Lane deskew logic. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: {(CX_PCIE_OVER_CIO_ENABLE) ? 1 : 0} Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "
30:27	IMPLEMENT_NUM_LANES	R/W	Implementation-specific Number of Lanes. Set the implementation-specific number of lanes. The number of lanes to be used when in Loopback Master. The number of lanes programmed must be equal to or less than the valid number of lanes set in LINK_CAPABLE field. You must configure this field before initiating Loopback by writing in the LOOPBACK_ENABLE field. The controller will transition from Loopback.Entry to Loopback.Active after receiving two consecutive TS1 Ordered Sets with the Loopback bit asserted on the implementation specific number of lanes configured in this field. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_1_LANE): 1 lane ■ 0x1 (_2_LANE): 2 lanes ■ 0x3 (_4_LANE): 4 lanes ■ 0x7 (_8_LANE): 8 lanes ■ 0xf (_16_LANE): 16 lanes Value After Reset: CX_NL_M_1 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
26	ELASTIC_BUFFER_MODE	* Varies	Selects Elasticity Buffer operating mode: <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (HALF_FULL): Nominal Half Full Buffer mode ■ 0x1 (EMPTY): Nominal Empty Buffer Mode Value After Reset: DEFAULT_LANE_SKEW_OFF_26 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "
25	ACK_NAK_DISABLE	R/W	Ack/Nak Disable. Prevents the controller from sending ACK and NAK DLLPs. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
24	FLOW_CTRL_DISABLE	R/W	Flow Control Disable. Prevents the controller from sending FC DLLPs. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
23:0	INSERT_LANE_SKEW	R/W	INSERT_LANE_SKEW is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN): Zero value ■ 0xff (MAX): Max value Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.7 TIMER_CTRL_MAX_FUNC_NUM_OFF

- **Name:** Timer Control and Max Function Number Register.
- **Description:** This register holds the ack frequency, latency, replay, fast link scaling timers, and max function number values.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	RSVDP_31
30:29	FAST_LINK_SCALING_FACTOR
28:24	UPDATE_FREQ_TIMER
23:19	TIMER_MOD_ACK_NAK
18:14	TIMER_MOD_REPLAY_TIMER
13:8	RSVDP_8
7:0	MAX_FUNC_NUM

Table 3-356 Fields for Register: TIMER_CTRL_MAX_FUNC_NUM_OFF

Bits	Name	Memory Access	Description
31	RSVDP_31	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
30:29	FAST_LINK_SCALING_FACTOR	R/W	Fast Link Timer Scaling Factor. Sets the scaling factor of LTSSM timer when FAST_LINK_MODE field in PORT_LINK_CTRL_OFF is set to '1'. Default is set by the hidden configuration parameter DEFAULT_FAST_LINK_SCALING_FACTOR which defaults to '0'. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (SF_1024): Scaling Factor is 1024 (1ms is 1us). When the LTSSM is in Config or L12 Entry State, 1ms timer is 2us, 2ms timer is 4us and 3ms timer is 6us. ■ 0x1 (SF_256): Scaling Factor is 256 (1ms is 4us) ■ 0x2 (SF_64): Scaling Factor is 64 (1ms is 16us) ■ 0x3 (SF_16): Scaling Factor is 16 (1ms is 64us) Value After Reset: DEFAULT_FAST_LINK_SCALING_FACTOR Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
28:24	UPDATE_FREQ_TIMER	R/W	UPDATE_FREQ_TIMER is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:19	TIMER_MOD_ACK_NAK	R/W	Ack Latency Timer Modifier. Increases the timer value for the Ack latency timer in increments of 64 clock cycles. A value of '0' represents no modification to the timer value. For more information, see the ROUND_TRIP_LATENCY_TIME_LIMIT field of the ACK_LATENCY_TIMER_OFF register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
18:14	TIMER_MOD_REPLAY_TIMER	R/W	Replay Timer Limit Modifier. Increases the time-out value for the replay timer in increments of 64 clock cycles at Gen1 or Gen2 speed, and in increments of 256 clock cycles at Gen3 speed. A value of '0' represents no modification to the timer limit. For more information, see the REPLAY_TIME_LIMIT field of the ACK_LATENCY_TIMER_OFF register. At Gen3 speed, the controller automatically changes the value of this field to DEFAULT_GEN3_REPLAY_ADJ. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
13:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:0	MAX_FUNC_NUM	R/W	Maximum function number that can be used in a request. Configuration requests targeted at function numbers above this value are returned with UR (unsupported request). <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN): Zero value ■ 0xff (MAX): Max value Value After Reset: { CX_NFUNC -1 } Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.8 SYMBOL_TIMER_FILTER_1_OFF

- **Name:** Symbol Timer Register and Filter Mask 1 Register.
- **Description:** The Filter Mask 1 Register modifies the RADM filtering and error handling rules. For more information, see the "Receive Filtering" section. In each case, '0' applies the associated filtering rule and '1' masks the associated filtering rule.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 | CX_MEMORY_MAP_PCIE_TYPE==0))

MASK_RADM_1	31:16
DISABLE_FC_WD_TIMER	15
EIDLE_TIMER	14:11
SKP_INT_VAL	10:0

Table 3-357 Fields for Register: SYMBOL_TIMER_FILTER_1_OFF

Bits	Name	Memory Access	Description
31:16	MASK_RADM_1	R/W	Filter Mask 1. The Filter Mask 1 Register modifies the RADM filtering and error handling rules. For more information, see the "Receive Filtering" section. In each case, '0' applies the associated filtering rule and '1' masks the associated filtering rule. 31: CX_FLT_MASK_RC_CFG_DISCARD ■ 0x0: For RADM RC filter to not allow CFG transaction being received ■ 0x1: For RADM RC filter to allow CFG transaction being received 30: CX_FLT_MASK_RC_IO_DISCARD ■ 0x0: For RADM RC filter to not allow IO transaction being received ■ 0x1: For RADM RC filter to allow IO transaction being received 29: CX_FLT_MASK_MSG_DROP ■ 0x0: Drop MSG TLP (except for Vendor MSG). Send decoded message on the SII. ■ 0x1: Do not Drop MSG (except for Vendor MSG). Send message TLPs to your application on TRGT1 and send decoded message on the SII. ■ The default for this bit is the inverse of FLT_DROP_MSG. That is, if FLT_DROP_MSG =1, then the default of this bit is '0' (drop message TLPs). This bit only controls message TLPs other than Vendor MSGs. Vendor MSGs are controlled by Filter Mask Register 2, bits [1:0]. The controller never passes ATS Invalidate messages to the SII interface regardless of this filter rule setting. The controller passes all ATS Invalidate messages to TRGT1 (or AXI bridge master), as they are too big for the SII. 28: CX_FLT_MASK_CPL_ECRC_DISCARD ■ Only used when completion queue is advertised with infinite credits and is in store-and-forward mode. ■ 0x0: Discard completions with ECRC errors

Bits	Name	Memory Access	Description
31:16...(cont.)	MASK_RADM_1.	R/W	■ 0x1: Allow completions with ECRC errors to be passed up ■ Reserved field for SW. 27: CX_FLT_MASK_ECRC_DISCARD ■ 0x0: Discard TLPs with ECRC errors ■ 0x1: Allow TLPs with ECRC errors to be passed up 26: CX_FLT_MASK_CPL_LEN_MATCH ■ 0x0: Enforce length match for completions; a violation results in cpl_abort, and possibly AER of unexp_cpl_err ■ 0x1: Mask length match for completions 25: CX_FLT_MASK_CPL_ATTR_MATCH ■ 0x0: Enforce attribute match for completions; a violation results in a malformed TLP error, and possibly AER of unexp_cpl_err, cpl_rcvd_ur, cpl_rcvd_ca ■ 0x1: Mask attribute match for completions 24: CX_FLT_MASK_CPL_TC_MATCH ■ 0x0: Enforce Traffic Class match for completions; a violation results in a malformed TLP error, and possibly AER of unexp_cpl_err, cpl_rcvd_ur, cpl_rcvd_ca ■ 0x1: Mask Traffic Class match for completions 23: CX_FLT_MASK_CPL_FUNC_MATCH ■ 0x0: Enforce function match for completions; a violation results in cpl_abort, and possibly AER of unexp_cpl_err, cpl_rcvd_ur, cpl_rcvd_ca ■ 0x1: Mask function match for completions 22: CX_FLT_MASK_CPL_REQID_MATCH ■ 0x0: Enforce Req. Id match for completions; a violation result in cpl_abort, and possibly AER of unexp_cpl_err, cpl_rcvd_ur, cpl_rcvd_ca ■ 0x1: Mask Req. Id match for completions 21: CX_FLT_MASK_CPL_TAGERR_MATCH ■ 0x0: Enforce Tag Error Rules for completions; a violation result in cpl_abort, and possibly AER of unexp_cpl_err, cpl_rcvd_ur, cpl_rcvd_ca ■ 0x1: Mask Tag Error Rules for completions 20: CX_FLT_MASK_LOCKED_RD_AS_UR ■ 0x0: Treat locked Read TLPs as UR for EP; Supported for RC ■ 0x1: Treat locked Read TLPs as Supported for EP; UR for RC 19: CX_FLT_MASK_CFG_TYPE1_REQ_AS_UR

Bits	Name	Memory Access	Description
31:16...(cont.)	MASK_RADM_1..	R/W	0x0: Treat CFG type1 TLPs as UR for EP; Supported for RC 0x1: Treat CFG type1 TLPs as Supported for EP; UR for RC - When CX_SRIOV_ENABLE is set then this bit is set to allow the filter to process Type 1 Config requests if the EP consumes more than one bus number. 18: CX_FLT_MASK_UR_OUTSIDE_BAR 0x0: Treat out-of-bar TLPs as UR 0x1: Do not treat out-of-bar TLPs as UR 17: CX_FLT_MASK_UR_POIS 0x0: Treat poisoned request TLPs as UR 0x1: Do not treat poisoned request TLPs as UR - The native controller always passes poisoned completions to your application except when you are using the DMA read channel. 16: CX_FLT_MASK_UR_FUNC_MISMATCH 0x0: Treat Function MisMatched TLPs as UR 0x1: Do not treat Function MisMatched TLPs as UR - CX_FLT_MASK_UR_FUNC_MISMATCH is not allowed to be set unless CONFIG_LIMIT_REG =0 and TARGET_ABOVE_CONFIG_LIMIT_REG =2 in MISC_CONTROL_1_OFF register or CFG_TLP_BY-PASS_EN_REG =1 and TARGET_ABOVE_CONFIG_LIMIT_REG =2 in MISC_CONTROL_1_OFF register. <i>Note:</i> This register field is sticky. Values: 0x0 (Minvalue): Zero value 0xffff (Maxvalue): Max value Value After Reset: DEFAULT_FILTER_MASK_1 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15	DISABLE_FC_WD_TIMER	R/W	Disable FC Watchdog Timer. <i>Note:</i> This register field is sticky. Values: 0x1 (ENABLE): Enable 0x0 (DISABLE): Disable Value After Reset: DEFAULT_FC_WATCH_DOG_DISABLE Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
14:11	EIDLE_TIMER	R/W	EIDLE_TIMER is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
10:0	SKP_INT_VAL	R/W	SKP Interval Value. The number of symbol times to wait between transmitting SKP ordered sets. The controller waits the number of symbol times in this register plus 1, between transmitting SKP ordered sets. Your application must program this register accordingly. For example, if 1536 were programmed into this register (in a 250 MHz controller), then the controller actually transmits SKP ordered sets once every 1537 symbol times. The value programmed to this register is actually clock ticks and not symbol times. In a 125 MHz controller, programming the value programmed to this register should be scaled down by a factor of 2 (because one clock tick =two symbol times in this case). <i>Note:</i> This value is not used at Gen3 speed; the skip interval is hardcoded to 370 blocks. <i>Note:</i> This register field is sticky. Value After Reset: { 1280/CX_PL_FREQ_MULTIPLIER } Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.9 **FILTER_MASK_2_OFF**

- **Name:** Filter Mask 2 Register.
- **Description:** This register modifies the RADM filtering and error handling rules. For more information, see the "Receive Filtering" section. In each case, '0' applies the associated filtering rule and '1' masks the associated filtering rule.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-358 Fields for Register: FILTER_MASK_2_OFF

Bits	Name	Memory Access	Description
31:0	MASK_RADM_2	R/W	Filter Mask 2. This field modifies the RADM filtering and error handling rules. For more information, see the "Receive Filtering" in the <i>Databook</i>. In each case, '0' applies the associated filtering rule and '1' masks the associated filtering rule. 31:12: Reserved 11: CX_FLT_UNMASK_ATOMIC_SPECIFIC_RULES ■ 0x0: Lower Address is checked for Cpls related to AtomicOps Requests. ■ 0x1: Lower Address is not checked for Cpls related to AtomicOps Requests. 10: CX_FLT_UNMASK_ATS_SPECIFIC_RULES ■ 0x0: Cpls for ATS Requests are processed as MemRd-related Cpl. ■ 0x1: Lower Address is not checked for Cpls related to ATS Requests. An ATS-related Cpl completes the request if it has a Byte Count that is equal to four times the Length field. 9: CX_FLT_MASK_CPL_IN_LUT_CHECK ■ 0x0: Disable masking of checking if the tag of CPL is registered in LUT ■ 0x1: Enable masking of checking if the tag of CPL is registered in LUT 8: CX_FLT_MASK_POIS_ERROR_REPORTING ■ 0x0: Disable masking of error reporting for Poisoned TLPs ■ 0x1: Enable masking of error reporting for Poisoned TLPs 7: CX_FLT_MASK_PRS_DROP ■ 0x0: Allow PRS message to pass through ■ 0x1: Drop PRS Messages silently ■ This bit is ignored when the CX_FLT_MASK_MSG_DROP bit in the MASK_RADM_1 field of the SYMBOL_TIMER_FILTER_1_OFF register is set to '1'. 6: CX_FLT_UNMASK_TD ■ 0x0: Disable unmask TD bit if CX_STRIP_ECRC_ENABLE ■ 0x1: Enable unmask TD bit if CX_STRIP_ECRC_ENABLE 5: CX_FLT_UNMASK_UR_POIS_TRGT0

Bits	Name	Memory Access	Description
31:0...(cont.)	MASK_RADM_2.	R/W	■ 0x0: Disable unmask CX_FLT_MASK_UR_POIS with TRGT0 destination ■ 0x1: Enable unmask CX_FLT_MASK_UR_POIS with TRGT0 destination 4: CX_FLT_MASK_LN_VENMSG1_DROP ■ 0x0: Allow LN message to pass through ■ 0x1: Drop LN Messages silently 3: CX_FLT_MASK_HANDLE_FLUSH ■ 0x0: Disable controller Filter to handle flush request ■ 0x1: Enable controller Filter to handle flush request 2: CX_FLT_MASK_DABORT_4UCPL ■ 0x0: Enable DLLP abort for unexpected completion ■ 0x1: Do not enable DLLP abort for unexpected completion 1: CX_FLT_MASK_VENMSG1_DROP ■ 0x0: Vendor MSG Type 1 dropped silently ■ 0x1: Vendor MSG Type 1 not dropped 0: CX_FLT_MASK_VENMSG0_DROP ■ 0x0: Vendor MSG Type 0 dropped with UR error reporting ■ 0x1: Vendor MSG Type 0 not dropped <i>Note:</i> This register field is sticky. Values: ■ 0x0 (Minimumvalue): Zero value ■ 0xffffffff (Maximumvalue): Max value Value After Reset: DEFAULT_FILTER_MASK_2 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.10 PL_DEBUG0_OFF

- **Name:** Debug Register 0.
- **Description:** This register holds cxpl_debug_info[31:0].
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

DEB_REG_0 31:0

Table 3-359 Fields for Register: PL_DEBUG0_OFF

Bits	Name	Memory Access	Description
31:0	DEB_REG_0	R	The value on cxpl_debug_info[31:0]. Values: ■ 0x0 (ZERO): Zero value. ■ 0xffffffff (MAX): Max value. Value After Reset: * Varies Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Reset Mask: 0x0 Volatile: true

3.48.11 PL_DEBUG1_OFF

- **Name:** Debug Register 1.
- **Description:** This register holds cxpl_debug_info[63:32].
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

DEB_REG_1 31:0

Table 3-360 Fields for Register: PL_DEBUG1_OFF

Bits	Name	Memory Access	Description
31:0	DEB_REG_1	R	The value on cxpl_debug_info[63:32]. Values: ■ 0x0 (ZERO): Zero value. ■ 0xffffffff (MAX): Max value. Value After Reset: * Varies Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Reset Mask: 0x0 Volatile: true

3.48.12 TX_P_FC_CREDIT_STATUS_OFF

- **Name:** Transmit Posted FC Credit Status.
- **Description:** This register provides transmit posted FC credit status.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

x:y	RSVDP_TX_P_FC_CREDIT_STATUS
x:y	TX_P_HEADER_FC_CREDIT
x:0	TX_P_DATA_FC_CREDIT

Table 3-361 Fields for Register: TX_P_FC_CREDIT_STATUS_OFF

Bits	Name	Memory Access	Description
x:y	RSVDP_TX_P_FC_CREDIT_STATUS	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Range Variable[x]: {(SCALED_FC_SUPPORTED) ? 4: 12} + {(SCALED_FC_SUPPORTED) ? 28 : 20} - 1 Range Variable[y]: {(SCALED_FC_SUPPORTED) ? 28 : 20}

Bits	Name	Memory Access	Description
x:y	TX_P_HEADER_FC_CREDIT	R	Transmit Posted Header FC Credits. ■ The posted Header credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. ■ Default value depends on the number of advertised credits for header and data ■ Scaled Flow Control: [4'b0, xtlh_xadm_ph_cdts, xtlh_xadm_pd_cdts]; When the number of advertised posted credits (both header and data) are infinite, then the default would be [4'b0, 12'hFFF, 16'hFFFF]. ■ No Scaling: [12'b0, xtlh_xadm_ph_cdts, xtlh_xadm_pd_cdts]; When the number of advertised posted credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF]. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Range Variable[x]: {(SCALED_FC_SUPPORTED) ? 12: 8} + {(SCALED_FC_SUPPORTED) ? 16 : 12} - 1 Range Variable[y]: {(SCALED_FC_SUPPORTED) ? 16 : 12}
x:0	TX_P_DATA_FC_CREDIT	R	Transmit Posted Data FC Credits. ■ The posted Data credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. ■ Default value depends on the number of advertised credits for header and data ■ Scaled Flow Control: [4'b0, xtlh_xadm_ph_cdts, xtlh_xadm_pd_cdts]; When the number of advertised posted credits (both header and data) are infinite, then the default would be [4'b0, 12'hFFF, 16'hFFFF]. ■ No Scaling: [12'b0, xtlh_xadm_ph_cdts, xtlh_xadm_pd_cdts]; When the number of advertised posted credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF]. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Range Variable[x]: {(SCALED_FC_SUPPORTED) ? 16 : 12} - 1

3.48.13 TX_NP_FC_CREDIT_STATUS_OFF

- **Name:** Transmit Non-Posted FC Credit Status.
- **Description:** This register provides the transmit Non-Posted FC credit status.
- **Size:** 32 bits
- **Offset:** B+0x34
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

x:y	RSVDP_TX_NP_FC_CREDIT_STATUS
x:y	TX_NP_HEADER_FC_CREDIT
x:0	TX_NP_DATA_FC_CREDIT

Table 3-362 Fields for Register: TX_NP_FC_CREDIT_STATUS_OFF

Bits	Name	Memory Access	Description
x:y	RSVDP_TX_NP_FC_CREDIT_STATUS	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Range Variable[x]: {(SCALED_FC_SUPPORTED) ? 4: 12} + {(SCALED_FC_SUPPORTED) ? 28 : 20} - 1 Range Variable[y]: {(SCALED_FC_SUPPORTED) ? 28 : 20}

Bits	Name	Memory Access	Description
x:y	TX_NP_HEADER_FC_CREDIT	R	Transmit Non-Posted Header FC Credits. ■ The non-posted Header credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. ■ Default value depends on the number of advertised credits for header and data ■ Scaled Flow Control: [4'b0, xtlh_xadm_nph_cdt, xtlh_xadm_npd_cdt]; When the number of advertised non-posted credits (both header and data) are infinite, then the default would be [4'b0, 12'hFFF, 16'hFFFF]. ■ No Scaling: [12'b0, xtlh_xadm_nph_cdt, xtlh_xadm_npd_cdt]; When the number of advertised non-posted credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF]. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Range Variable[x]: {(SCALED_FC_SUPPORTED) ? 12: 8} + {(SCALED_FC_SUPPORTED) ? 16 : 12} - 1 Range Variable[y]: {(SCALED_FC_SUPPORTED) ? 16 : 12}
x:0	TX_NP_DATA_FC_CREDIT	R	Transmit Non-Posted Data FC Credits. ■ The non-posted Data credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. ■ Default value depends on the number of advertised credits for header and data ■ Scaled Flow Control: [4'b0, xtlh_xadm_nph_cdt, xtlh_xadm_npd_cdt]; When the number of advertised non-posted credits (both header and data) are infinite, then the default would be [4'b0, 12'hFFF, 16'hFFFF]. ■ No Scaling: [12'b0, xtlh_xadm_nph_cdt, xtlh_xadm_npd_cdt]; When the number of advertised non-posted credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF]. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Range Variable[x]: {(SCALED_FC_SUPPORTED) ? 16 : 12} - 1

3.48.14 TX_CPL_FC_CREDIT_STATUS_OFF

- **Name:** Transmit Completion FC Credit Status
- **Description:** This register provides transmit completion FC credit status.
- **Size:** 32 bits
- **Offset:** B+0x38
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

x:y	RSVDP_TX_CPL_FC_CREDIT_STATUS
x:y	TX_CPL_HEADER_FC_CREDIT
x:0	TX_CPL_DATA_FC_CREDIT

Table 3-363 Fields for Register: TX_CPL_FC_CREDIT_STATUS_OFF

Bits	Name	Memory Access	Description
x:y	RSVDP_TX_CPL_FC_CREDIT_STATUS	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Range Variable[x]: {(SCALED_FC_SUPPORTED) ? 4: 12} + {(SCALED_FC_SUPPORTED) ? 28 : 20} - 1 Range Variable[y]: {(SCALED_FC_SUPPORTED) ? 28 : 20}

Bits	Name	Memory Access	Description
x:y	TX_CPL_HEADER_FC_CREDIT	R	Transmit Completion Header FC Credits. ■ The Completion Header credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. ■ Default value depends on the number of advertised credits for header and data ■ Scaled Flow Control: [4'b0, xtlh_xadm_cplh_cdts, xtlh_xadm_cpld_cdts]; When the number of advertised completion credits (both header and data) are infinite, then the default would be [4'b0, 12'hFFF, 16'hFFFF]. ■ No Scaling: [12'b0, xtlh_xadm_cplh_cdts, xtlh_xadm_cpld_cdts]; When the number of advertised completion credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF]. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Range Variable[x]: {(SCALED_FC_SUPPORTED) ? 12: 8} + {(SCALED_FC_SUPPORTED) ? 16 : 12} - 1 Range Variable[y]: {(SCALED_FC_SUPPORTED) ? 16 : 12}
x:0	TX_CPL_DATA_FC_CREDIT	R	Transmit Completion Data FC Credits. ■ The Completion Data credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. ■ Default value depends on the number of advertised credits for header and data ■ Scaled Flow Control: [4'b0, xtlh_xadm_cplh_cdts, xtlh_xadm_cpld_cdts]; When the number of advertised completion credits (both header and data) are infinite, then the default would be [4'b0, 12'hFFF, 16'hFFFF]. ■ No Scaling: [12'b0, xtlh_xadm_cplh_cdts, xtlh_xadm_cpld_cdts]; When the number of advertised completion credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF]. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Range Variable[x]: {(SCALED_FC_SUPPORTED) ? 16 : 12} - 1

3.48.15 QUEUE_STATUS_OFF

- **Name:** Queue Status.
- **Description:** This register provides the queue status.
- **Size:** 32 bits
- **Offset:** B+0x3c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	TIMER_MOD_FLOW_CONTROL_EN
30:29	RSVDP_29
28:16	TIMER_MOD_FLOW_CONTROL
15	RX_SERIALIZATION_Q_READ_ERR
14	RX_SERIALIZATION_Q_WRITE_ERR
13	RX_SERIALIZATION_Q_NON_EMPTY
12:4	RSVDP_4
3	RX_QUEUE_OVERFLOW
2	RX_QUEUE_NON_EMPTY
1	TX_RETRY_BUFFER_NE
0	RX_TLP_FC_CREDIT_NON_RETURN

Table 3-364 Fields for Register: QUEUE_STATUS_OFF

Bits	Name	Memory Access	Description
31	TIMER_MOD_FLOW_CONTROL_EN	R/W	FC Latency Timer Override Enable. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): When this bit is set, the value from the "FC Latency Timer Override Value" field in this register will override the FC latency timer value that the controller calculates according to the PCIe specification. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
30:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
28:16	TIMER_MOD_FLOW_CONTROL	R/W	FC Latency Timer Override Value. When you set the "FC Latency Timer Override Enable" in this register, the value in this field will override the FC latency timer value that the controller calculates according to the PCIe specification. For more information, see "Flow Control" in the <i>Databook</i> . <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15	RX_SERIALIZATION_Q_READ_ERROR	R/W1C	Receive Serialization Read Error. Values: ■ 0x1 (SET): Indicates the serialization queue has attempted to read an incorrectly formatted TLP. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: SNPS_RSVDPARAM_72 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
14	RX_SERIALIZATION_Q_WRITE_ERROR	R/W1C	Receive Serialization Queue Write Error. Values: ■ 0x1 (SET): Indicates insufficient buffer space available to write to the serialization queue. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: SNPS_RSVDPARAM_72 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
13	RX_SERIALIZATION_Q_NON_EMPTY	R	Receive Serialization Queue Not Empty. Values: ■ 0x1 (SET): Indicates there is data in the serialization queue. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
12:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
3	RX_QUEUE_OVERFLOW	R/W1C	Receive Credit Queue Overflow. Values: ■ 0x1 (SET): Indicates insufficient buffer space available to write to the P/NP/CPL credit queue. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
2	RX_QUEUE_NON_EMPTY	R	Receive Credit Queue Not Empty. Values: ■ 0x1 (SET): Indicates there is data in one or more of the receive buffers. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
1	TX_RETRY_BUFFER_NE	R	Transmit Retry Buffer Not Empty. Values: ■ 0x1 (SET): Indicates that there is data in the transmit retry buffer. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	RX_TLP_FC_CREDIT_NON_RETURN	R	Received TLP FC Credits Not Returned. Values: ■ 0x1 (SET): Indicates that the controller has received a TLP but has not yet sent an UpdateFC DLLP indicating that the credits for that TLP have been restored by the receiver at the other end of the link. ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.16 VC_TX_ARBI_1_OFF

- **Name:** VC Transmit Arbitration Register 1.
- **Description:** This register is used for setting the WRR weights for VC0 - VC3.
- **Size:** 32 bits
- **Offset:** B+0x40
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	23:16	15:8	7:0
WRR_WEIGHT_VC_3	WRR_WEIGHT_VC_2	WRR_WEIGHT_VC_1	WRR_WEIGHT_VC_0

Table 3-365 Fields for Register: VC_TX_ARBI_1_OFF

Bits	Name	Memory Access	Description
31:24	WRR_WEIGHT_VC_3	* Varies	WRR Weight for VC3. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: LPVC_WRR_WEIGHT_VC3 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
23:16	WRR_WEIGHT_VC_2	* Varies	WRR Weight for VC2. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: LPVC_WRR_WEIGHT_VC2 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : \"read-write\" : \"read-only\" "
15:8	WRR_WEIGHT_VC_1	* Varies	WRR Weight for VC1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: LPVC_WRR_WEIGHT_VC1 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
7:0	WRR_WEIGHT_VC_0	* Varies	WRR Weight for VC0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: LPVC_WRR_WEIGHT_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : \"read-write\" : \"read-only\" "

3.48.17 VC_TX_ARBI_2_OFF

- **Name:** VC Transmit Arbitration Register 2.
- **Description:** This register is used for setting the WRR weights for VC4 - VC7.
- **Size:** 32 bits
- **Offset:** B+0x44
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	23:16	15:8	7:0
WRR_WEIGHT_VC_7	WRR_WEIGHT_VC_6	WRR_WEIGHT_VC_5	WRR_WEIGHT_VC_4

Table 3-366 Fields for Register: VC_TX_ARBI_2_OFF

Bits	Name	Memory Access	Description
31:24	WRR_WEIGHT_VC_7	* Varies	WRR Weight for VC7. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: LPVC_WRR_WEIGHT_VC7 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : \"read-write\":\"read-only\""

Bits	Name	Memory Access	Description
23:16	WRR_WEIGHT_VC_6	* Varies	WRR Weight for VC6. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: LPVC_WRR_WEIGHT_VC6 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : \"read-write\" : \"read-only\" "
15:8	WRR_WEIGHT_VC_5	* Varies	WRR Weight for VC5. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: LPVC_WRR_WEIGHT_VC5 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
7:0	WRR_WEIGHT_VC_4	* Varies	WRR Weight for VC4. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: LPVC_WRR_WEIGHT_VC4 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_LPVC_WRR_WEIGHT_WRITABLE) ? \"read-only\" : (CX_LPVC_WRR_WEIGHT_WRITABLE)? \"read-write\": \"read-only\" "

3.48.18 VC0_P_RX_Q_CTRL_OFF

- **Name:** Segmented-Buffer VC0 Posted Receive Queue Control.
- **Description:** This register controls segmented-buffer VC0 posted receive queue operation.
- **Size:** 32 bits
- **Offset:** B+0x48
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	VC_ORDERING_RX_Q
30	TLP_TYPE_ORDERING_VC0
29:28	RESERVED5
27:26	VC0_P_DATA_SCALE
25:24	VC0_P_HDR_SCALE
23:21	VC0_P_TLP_Q_MODE
20	RESERVED4
19:12	VC0_P_HEADER_CREDIT
11:0	VC0_P_DATA_CREDIT

Table 3-367 Fields for Register: VC0_P_RX_Q_CTRL_OFF

Bits	Name	Memory Access	Description
31	VC_ORDERING_RX_Q	R/W	VC Ordering for Receive Queues. Determines the VC ordering rule for the receive queues, used only in the segmented-buffer configuration: <i>Note:</i> This register field is sticky. Values: ■ 0x1 (STRICT_ORDERING): Strict ordering, higher numbered VCs have higher priority ■ 0x0 (ROUND_ROBIN): Round robin Value After Reset: CX_RADM_STRICT_VC_PRIORITY Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
30	TLP_TYPE_ORDERING_VC0	R/W	TLP Type Ordering for VC0. Determines the TLP type ordering rule for VC0 receive queues, used only in the segmented-buffer configuration: <i>Note:</i> This register field is sticky. Values: ■ 0x1 (PCIE_ORDERING_RULE): PCIe ordering rules (recommended) ■ 0x0 (STRICT_ORDERING): Strict ordering: posted, completion, then non-posted Value After Reset: CX_RADM_ORDERING_RULES_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
29:28	RESERVED5	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27:26	VC0_P_DATA_SCALE	R/W	VC0 Scale Posted Data Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_PQ_DSCALE_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
25:24	VC0_P_HDR_SCALE	R/W	VC0 Scale Posted Header Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_PQ_HSCALE_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
23:21	VC0_P_TLP_Q_MODE	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: RADM_P_QMODE_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20	RESERVED4	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
19:12	VC0_P_HEADER_CREDIT	* Varies	VC0 Posted Header Credits. The number of initial posted header credits for VC0, used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xff (MAX_VAL): Max value Value After Reset: RADM_PQ_HCRD_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
11:0	VC0_P_DATA_CREDIT	* Varies	VC0 Posted Data Credits. The number of initial posted data credits for VC0, used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xfff (MAX_VAL): Max value Value After Reset: RADM_PQ_DCRD_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_DYNAMIC_FC_CREDIT)? \"read-write\":\"read-only\" "

3.48.19 VC0_NP_RX_Q_CTRL_OFF

- **Name:** Segmented-Buffer VC0 Non-Posted Receive Queue Control.
- **Description:** This register controls the segmented-buffer VC0 non-posted receive queue operation.
- **Size:** 32 bits
- **Offset:** B+0x4c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RESERVED7
27:26	VC0_NP_DATA_SCALE
25:24	VC0_NP_HDR_SCALE
23:21	VC0_NP_TLP_Q_MODE
20	RESERVED6
19:12	VC0_NP_HEADER_CREDIT
11:0	VC0_NP_DATA_CREDIT

Table 3-368 Fields for Register: VC0_NP_RX_Q_CTRL_OFF

Bits	Name	Memory Access	Description
31:28	RESERVED7	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27:26	VC0_NP_DATA_SCALE	R/W	VC0 Scale Non-Posted Data Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_NPQ_DSCALE_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
25:24	VC0_NP_HDR_SCALE	R/W	VC0 Scale Non-Posted Header Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_NPQ_HSCALE_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:21	VC0_NP_TLP_Q_MODE	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: RADM_NP_QMODE_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20	RESERVED6	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
19:12	VC0_NP_HEADER_CREDIT	* Varies	VC0 Non-Posted Header Credits. The number of initial non-posted header credits for VC0, used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xff (MAX_VAL): Max value Value After Reset: RADM_NPQ_HCRD_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_DYNAMIC_FC_CREDIT)? \"read-write\": \"read-only\" "

Bits	Name	Memory Access	Description
11:0	VC0_NP_DATA_CREDIT	* Varies	VC0 Non-Posted Data Credits. The number of initial non-posted data credits for VC0, used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xffff (MAX_VAL): Max value Value After Reset: RADM_NPQ_DCRD_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_DYNAMIC_FC_CREDIT)? \"read-write\":\"read-only\" "

3.48.20 VC0_CPL_RX_Q_CTRL_OFF

- **Name:** Segmented-Buffer VC0 Completion Receive Queue Control.
- **Description:** This register controls the segmented-buffer VC0 completion receive queue operation.
- **Size:** 32 bits
- **Offset:** B+0x50
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RESERVED9
27:26	VC0_CPL_DATA_SCALE
25:24	VC0_CPL_HDR_SCALE
23:21	VC0_CPL_TLP_Q_MODE
20	RESERVED8
19:12	VC0_CPL_HEADER_CREDIT
11:0	VC0_CPL_DATA_CREDIT

Table 3-369 Fields for Register: VC0_CPL_RX_Q_CTRL_OFF

Bits	Name	Memory Access	Description
31:28	RESERVED9	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27:26	VC0_CPL_DATA_SCALE	R/W	VC0 Scale CPL Data Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_CPLQ_DSCALE_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
25:24	VC0_CPL_HDR_SCALE	R/W	VC0 Scale CPL Header Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_CPLQ_HSCALE_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:21	VC0_CPL_TLP_Q_MODE	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: RADM_CPL_QMODE_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20	RESERVED8	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
19:12	VC0_CPL_HEADER_CREDIT	* Varies	VC0 Completion Header Credits. The number of initial Completion header credits for VC0, used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xff (MAX_VAL): Max value Value After Reset: RADM_CPLQ_HCRD_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_DYNAMIC_FC_CREDIT)? \"read-write\": \"read-only\" "

Bits	Name	Memory Access	Description
11:0	VC0_CPL_DATA_CREDIT	* Varies	VC0 Completion Data Credits. The number of initial Completion data credits for VC0, used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xfff (MAX_VAL): Max value Value After Reset: RADM_CPLQ_DCRD_VC0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_DYNAMIC_FC_CREDIT)? \"read-write\": \"read-only\" "

3.48.21 VCI_P_RX_Q_CTRL_OFF (for i = 1; i <= CX_NVC-1)

- **Name:** Segmented-Buffer VC1 Posted Receive Queue Control.
- **Description:** This register controls the segmented-buffer VC1 posted receive queue.
- **Size:** 32 bits
- **Offset:** B+0x54 + i*0x4
- **Exists:** !CX_PL_REG_DISABLE && CX_NVC>i

31	RESERVED2_VC1
30	TLP_TYPE_ORDERING_VC1
29:28	RESERVED1_VC1
27:26	VC1_P_DATA_SCALE
25:24	VC1_P_HDR_SCALE
23:21	VC1_P_TLP_Q_MODE
20	RESERVED0_VC1
19:12	VC1_P_HEADER_CREDIT
11:0	VC1_P_DATA_CREDIT

Table 3-370 Fields for Register: VCI_P_RX_Q_CTRL_OFF (for i = 1; i <= CX_NVC-1)

Bits	Name	Memory Access	Description
31	RESERVED2_VC1	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
30	TLP_TYPE_ORDERING_VC1	R/W	TLP Type Ordering for VC1. Determines the TLP type ordering rule for VC1 receive queues, used only in the segmented-buffer configuration: <i>Note:</i> This register field is sticky. Values: ■ 0x1 (PCIE_ORDERING_RULE): PCIe ordering rules (recommended) ■ 0x0 (STRICT_ORDERING): Strict ordering: posted, completion, then non-posted Value After Reset: CX_RADM_ORDERING_RULES_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
29:28	RESERVED1_VC1	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27:26	VC1_P_DATA_SCALE	R/W	VC1 Scale Posted Data Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_PQ_DSCALE_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
25:24	VC1_P_HDR_SCALE	R/W	VC1 Scale Posted Header Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_PQ_HSCALE_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
23:21	VC1_P_TLP_Q_MODE	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: RADM_P_QMODE_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20	RESERVED0_VC1	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
19:12	VC1_P_HEADER_CREDIT	* Varies	VC1 Posted Header Credits. The number of initial posted header credits for VC <i>#i</i> , used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xff (MAX_VAL): Max value Value After Reset: RADM_PQ_HCRD_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
11:0	VC1_P_DATA_CREDIT	* Varies	VC1 Posted Data Credits. The number of initial posted data credits for VC#i, used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xfff (MAX_VAL): Max value Value After Reset: RADM_PQ_DCRD_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_DYNAMIC_FC_CREDIT)? \"read-write\": \"read-only\" "

3.48.22 VCi_NP_RX_Q_CTRL_OFF (for i = 1; i <= CX_NVC-1)

- **Name:** Segmented-Buffer VC1 Non-Posted Receive Queue Control.
- **Description:** This register controls the segmented-buffer VC1 non-posted receive queue.
- **Size:** 32 bits
- **Offset:** B+0x58 + i*0x4
- **Exists:** !CX_PL_REG_DISABLE && CX_NVC>i

31:28	RESERVED4_VC1
27:26	VC1_NP_DATA_SCALE
25:24	VC1_NP_HDR_SCALE
23:21	VC1_NP_TLP_Q_MODE
20	RESERVED3_VC1
19:12	VC1_NP_HEADER_CREDIT
11:0	VC1_NP_DATA_CREDIT

Table 3-371 Fields for Register: VCi_NP_RX_Q_CTRL_OFF (for i = 1; i <= CX_NVC-1)

Bits	Name	Memory Access	Description
31:28	RESERVED4_VC1	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27:26	VC1_NP_DATA_SCALE	R/W	VC1 Scale Non-Posted Data Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_NPQ_DSCALE_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
25:24	VC1_NP_HDR_SCALE	R/W	VC1 Scale Non-Posted Header Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_NPQ_HSCALE_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:21	VC1_NP_TLP_Q_MODE	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: RADM_NP_QMODE_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20	RESERVED3_VC1	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
19:12	VC1_NP_HEADER_CREDIT	* Varies	VC1 Non-Posted Header Credits. The number of initial non-posted header credits for VC1, used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xff (MAX_VAL): Max value Value After Reset: RADM_NPQ_HCRD_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
11:0	VC1_NP_DATA_CREDIT	* Varies	VC1 Non-Posted Data Credits. The number of initial non-posted data credits for VC1, used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xfff (MAX_VAL): Max value Value After Reset: RADM_NPQ_DCRD_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_DYNAMIC_FC_CREDIT)? \"read-write\": \"read-only\" "

3.48.23 VCI_CPL_RX_Q_CTRL_OFF (for i = 1; i <= CX_NVC-1)

- **Name:** Segmented-Buffer VC1 Completion Receive Queue Control.
- **Description:** This register controls the segmented-buffer VC1 completion receive queue.
- **Size:** 32 bits
- **Offset:** B+0x5c + i*0x4
- **Exists:** !CX_PL_REG_DISABLE && CX_NVC>i

31:28	RESERVED6_VC1
27:26	VC1_CPL_DATA_SCALE
25:24	VC1_CPL_HDR_SCALE
23:21	VC1_CPL_TLP_Q_MODE
20	RESERVED5_VC1
19:12	VC1_CPL_HEADER_CREDIT
11:0	VC1_CPL_DATA_CREDIT

Table 3-372 Fields for Register: VCI_CPL_RX_Q_CTRL_OFF (for i = 1; i <= CX_NVC-1)

Bits	Name	Memory Access	Description
31:28	RESERVED6_VC1	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27:26	VC1_CPL_DATA_SCALE	R/W	VC1 Scale CPL Data Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_CPLQ_DSCALE_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
25:24	VC1_CPL_HDR_SCALE	R/W	VC1 Scale CPL Header Credits. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x3 (MAX_VAL): Max value Value After Reset: RADM_CPLQ_HSCALE_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:21	VC1_CPL_TLP_Q_MODE	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: RADM_CPL_QMODE_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20	RESERVED5_VC1	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
19:12	VC1_CPL_HEADER_CREDIT	* Varies	VC1 Completion Header Credits. The number of initial Completion header credits for VC1, used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xff (MAX_VAL): Max value Value After Reset: RADM_CPLQ_HCRD_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\" : (CX_DYNAMIC_FC_CREDIT)? \"read-write\":\"read-only\" "

Bits	Name	Memory Access	Description
11:0	VC1_CPL_DATA_CREDIT	* Varies	VC1 Completion Data Credits. The number of initial Completion data credits for VC1, used only in the segmented-buffer configuration. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xfff (MAX_VAL): Max value Value After Reset: RADM_CPLQ_DCRD_VC1 Exists: CX_NVC>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_MEMORY_MAP_VIEW==1 && CX_DYNAMIC_FC_CREDIT) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && !CX_DYNAMIC_FC_CREDIT) ? \"read-only\": (CX_DYNAMIC_FC_CREDIT)? \"read-write\": \"read-only\" "

3.48.24 GEN2_CTRL_OFF

- **Name:** Link Width and Speed Change Control Register.
- **Description:** This register controls various functions of the controller related to link training, lane reversal, and equalization.
- **Size:** 32 bits
- **Offset:** B+0x10c
- **Exists:** !CX_PL_REG_DISABLE

SUPPORT_MOD_TS	31
FORCE_LANE_FLIP	30
TX_MOD_CMPL_PATTERN_FOR_LOOPBACK	29
EQ_FOR_LOOPBACK	28
LANE_UNDER_TEST	27:24
SELECTABLE_DEEMPH_BIT_MUX	23
SELECT_DEEMPH_VAR_MUX	22
GEN1_EI_INFERENCE	21
SEL_DEEMPHASIS	20
CONFIG_TX_COMP_RX	19
CONFIG_PHY_TX_CHANGE	18
DIRECT_SPEED_CHANGE	17
AUTO_LANE_FLIP_CTRL_EN	16
PRE_DET_LANE	15:13
NUM_OF_LANES	12:8
FAST_TRAINING_SEQ	7:0

Table 3-373 Fields for Register: GEN2_CTRL_OFF

Bits	Name	Memory Access	Description
31	SUPPORT_MOD_TS	* Varies	Enable the controller to send the Modified TS OS if both sides of the link agree when the SUPPORT_MOD_TS register is set to 1. The default of this field is the CX_MOD_TS_FMT_SUPPORT_VALUE configuration parameter. If the port negotiates alternate protocols or passes a Training Set Message, the bit must be set to 1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: CX_MOD_TS_FMT_SUPPORT_VALUE Exists: CX_MODIFIED_TS_FORMAT_SUPPORT && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
30	FORCE_LANE_FLIP	* Varies	Enable to force the LANE_UNDER_TEST physical lane flips to logical lane 0. All the other physical lanes are turned off. The LINK_CAPABLE register must be set to 1 and only x1 link can be formed if the FORCE_LANE_FLIP register is set to 1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: CX_FORCE_LANE_FLIP_EN && CX_GEN2_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
29	TX_MOD_CMPL_PATTERN_FOR_LOOPBACK	* Varies	Require Loopback slave to transmit Modified Compliance Pattern in Loopback.Active state at Gen5 rate. The default of this field is the CX_DEFAULT_GEN5_TX_MOD_CMPL_PATTERN_FOR_LOOPBACK configuration parameter <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Writing '1' to this field through the DBI instructs the remote Loopback slave to send Modified Compliance Pattern in Gen5 loopback.Active. Value After Reset: CX_DEFAULT_GEN5_TX_MOD_CMPL_PATTERN_FOR_LOOPBACK Exists: CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
28	EQ_FOR_LOOPBACK	* Varies	Perform EQ in Loopback in Gen5 rate. The default of this field is the CX_DEFAULT_GEN5_EQ_FOR_LOOPBACK configuration parameter. Loopback master enters Loopback.Active only because of receiving 2 TS1s with Loopback bit asserted (not the 2ms timeout of Txing Compliance Receive bit) if EQ_FOR_LOOPBACK is set to 1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Writing '1' to this field through the DBI instructs the link to perform Eq in Gen5 loopback if both sides support Gen5 rate. Value After Reset: CX_DEFAULT_GEN5_EQ_FOR_LOOPBACK Exists: CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
27:24	LANE_UNDER_TEST	* Varies	The Lane Under Test is the lane for Forced Lane Flip or for Loopback Eq. Only one lane is configured each time. The default of this field is the CX_DEFAULT_LANE_UNDER_TEST configuration parameter. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero ■ 0x1 (MAX_VAL): CX_NL Value After Reset: CX_DEFAULT_LANE_UNDER_TEST Exists: CX_LANE_UNDER_TEST_EN && CX_GEN2_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
23	SELECTABLE_DEEMPH_BIT_MUX	R/W	The selectable deemphasis bit (Symbol 4 bit 6) of the transmitted TS2 Ordered Sets for DSP in Recovery.RcvrCfg state is muxed between the Selectable De-emphasis field in the Link Control 2 register and the value requested by USP in Recovery.RcvrLock state through Tx TS1s from USP. <i>Note:</i> This register field is sticky. Values: 0x0 (FROM_LINK_CTRL_2_REG): The value from the Selectable De-emphasis field in the Link Control 2 register 0x1 (REQUESTED_BY_USP): The value requested by USP in Recovery.RcvrLock state through Tx TS1s from USP Value After Reset: DEFAULT_SELECTABLE_DEEMPH_BIT_MUX Exists: CX_GEN2_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
22	SELECT_DEEMPH_VAR_MUX	R/W	The select_deemphasis variable for DSP on entry to Recovery.RcvrCfg state is muxed between the Selectable De-emphasis field in the Link Control 2 register and the value requested by the Upstream Port in the eight consecutive TS1 Ordered Sets it received. <i>Note:</i> This register field is sticky. Values: 0x0 (REQUESTED_BY_USP): The value requested by USP in Recovery.RcvrLock state through Tx TS1s from USP 0x1 (FROM_LINK_CTRL_2_REG): The value from the Selectable De-emphasis field in the Link Control 2 register Value After Reset: DEFAULT_SELECT_DEEMPH_VAR_MUX Exists: CX_GEN2_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
21	GEN1_EI_INFERENCE	R/W	Electrical Idle Inference Mode at Gen1 Rate. Programmable mode to determine inferred electrical idle (EI) in Recovery.Speed or Loopback.Active (as slave) state at Gen1 speed by looking for a '1' value on RxElecdle instead of looking for a '0' on RxValid. If the PHY fails to de-assert the RxValid signal in Recovery.Speed or Loopback.Active (because of corrupted EIOS for example), then EI cannot be inferred successfully in the controller by just detecting the condition RxValid=0. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (RXELECIDLE_TO_INFER): Use RxElecdle signal to infer Electrical Idle ■ 0x1 (RXVALID_TO_INFER): Use RxValid signal to infer Electrical Idle Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20	SEL_DEEMPHASIS	* Varies	Used to set the de-emphasis level for upstream ports. This bit selects the level of de-emphasis the link operates at. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (N_6DB): -6 dB ■ 0x1 (N_3_5DB): -3.5 dB Value After Reset: SEL_DE_EMPHASIS Exists: CX_GEN2_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : \"read-write\" "

Bits	Name	Memory Access	Description
19	CONFIG_TX_COMP_RX	* Varies	Config Tx Compliance Receive Bit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): When set to 1, signals LTSSM to transmit TS ordered sets with the compliance receive bit assert (equal to '1'). ■ 0x0 (CLEAR): Clear Value After Reset: 0x0 Exists: CX_GEN2_SPEED CX_PCIE_OVER_CIO_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "
18	CONFIG_PHY_TX_CHANGE	* Varies	Config PHY Tx Swing. Controls the PHY transmitter voltage swing level. The controller drives the mac_phy_txswing output from this register bit field. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (FULL_SWING): Full Swing ■ 0x1 (LOW_SWING): Low Swing Value After Reset: DEFAULT_GEN2_TXSWING Exists: CX_GEN2_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : \"read-write\" "

Bits	Name	Memory Access	Description
17	DIRECT_SPEED_CHANGE	* Varies	Directed Speed Change. When the speed change occurs, the controller will clear the contents of this field; and a read to this field by your software will return a '0'. To manually initiate the speed change: ■ Write to LINK_CONTROL2_LINK_STATUS2_REG.PCIE_CAP_TARGET_LINK_SPEED in the local device ■ De-assert this field ■ Assert this field If you set the default of this field using the DEFAULT_GEN2_SPEED_CHANGE configuration parameter to '1', then the speed change is initiated automatically after link up, and the controller clears the contents of this field. If you want to prevent this automatic speed change, then write a lower speed value to the Target Link Speed field of the Link Control 2 register (LINK_CONTROL2_LINK_STATUS2_OFF.PCIE_CAP_TARGET_LINK_SPEED) through the DBI before link up. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x1 (SET): Writing '1' to this field instructs the LTSSM to initiate a speed change to Gen2 or Gen3 after the link is initialized at Gen1 speed. ■ 0x0 (CLEAR): Clear Value After Reset: DEFAULT_GEN2_SPEED_CHANGE Exists: CX_GEN2_SPEED CX_PCIE_OVER_CIO_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
16	AUTO_LANE_FLIP_CTRL_EN	* Varies	Enable Auto flipping of the lanes. You must set the CX_AUTO_LANE_FLIP_CTRL_EN configuration parameter to include the hardware for this feature in the controller. For more information, see the 'Lane Reversal' appendix in the <i>Databook</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: {(CX_AUTO_LANE_FLIP_CTRL_EN) ? "0x1" : "0x0"} Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(SNPS_RSVDPARAM_25) ? "writeAsRead" : "unconstrained" } Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
15:13	PRE_DET_LANE	* Varies	Predetermined Lane for Auto Flip. This field defines which physical lane is connected to logical Lane0 by the flip operation performed in Detect. This field is used to restrict the receiver detect procedure to a particular lane when the default detect and polling procedure performed on all lanes cannot be successful. A notable example of when it is useful to program this field to a value different from the default, is when a lane is asymmetrically broken, that is, it is detected in Detect LTSSM state but it cannot exit Electrical Idle in Polling LTSSM state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (LANE0): Connect logical Lane0 to physical lane 0 or CX_NL-1 or CX_NL/2-1 or CX_NL/4-1 or CX_NL/8-1, depending on which lane is detected ■ 0x1 (LANE1): Connect logical Lane0 to physical lane 1 ■ 0x2 (LANE3): Connect logical Lane0 to physical lane 3 ■ 0x3 (LANE7): connect logical lane0 to physical lane 7 ■ 0x4 (LANE15): Connect logical Lane0 to physical lane 15 Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(SNPS_RSVDPARAM_25) ? "writeAsRead" : "unconstrained" } Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
12:8	NUM_OF_LANES	* Varies	Predetermined Number of Lanes. Defines the number of lanes which are connected and not bad. Used to limit the effective link width to ignore 'broken' or 'unused' lanes that detect a receiver. Indicates the number of lanes to check for exit from Electrical Idle in Polling.Active and L2.Idle. It is possible that the LTSSM might detect a receiver on a bad or broken lane during the Detect Substate. However, it is also possible that such a lane might also fail to exit Electrical Idle and therefore prevent a valid link from being configured. This value is referred to as the "Predetermined Number of Lanes" in section 4.2.6.2.1 of the <i>PCI Express Base Specification</i>. When you have unused lanes in your system, then you must change the value in this register to reflect the number of lanes. You must also change the value in the "Link Mode Enable" field of PORT_LINK_CTRL_OFF. The value in this register is normally the same as the encoded value in PORT_LINK_CTRL_OFF. If you find that one of your used lanes is bad then you must reduce the value in this register. For more information, see "How to Tie Off Unused Lanes" in the <i>Databook</i>. For information on upsizing and downsizing the link width, see "Link Establishment" in the <i>Databook</i>. Note: The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) Note: This register field is sticky. Values: ■ 0x1 (_1_LANE): 1 lane ■ 0x2 (_2_LANE): 2 lanes ■ 0x3 (_3_LANE): 3 lanes Value After Reset: CX_NL Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(SNPS_RSVDPARAM_25) ? "writeAsRead" : "unconstrained" } Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
7:0	FAST_TRAINING_SEQ	* Varies	Sets the Number of Fast Training Sequences (N_FTS) that the controller advertises as its N_FTS during Gen2 or Gen3 link training. This value is used to inform the link partner about the PHY's ability to recover synchronization after a low power state. The number should be provided by the PHY vendor. Do not set N_FTS to zero; doing so can cause the LTSSM to go into the recovery state when exiting from L0s. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_GEN2_N_FTS Exists: CX_GEN2_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \\"read-write\\" : \\"read-write\\" "

3.48.25 PHY_STATUS_OFF

- **Name:** PHY Status Register.
- **Description:** Memory mapped register from phy_cfg_status GPIO input pins.
- **Size:** 32 bits
- **Offset:** B+0x110
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

PHY_STATUS 31:0

Table 3-374 Fields for Register: PHY_STATUS_OFF

Bits	Name	Memory Access	Description
31:0	PHY_STATUS	R	PHY Status. Data received directly from the phy_cfg_status bus. These is a GPIO register reflecting the values on the static phy_cfg_status input signals. The usage is left completely to the user and does not in any way influence controller functionality. You can use it for any static sideband status signalling requirements that you have for your PHY. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: * Varies Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true

3.48.26 PHY_CONTROL_OFF

- **Name:** PHY Control Register.
- **Description:** Memory mapped register to cfg_phy_control GPIO output pins.
- **Size:** 32 bits
- **Offset:** B+0x114
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-375 Fields for Register: PHY_CONTROL_OFF

Bits	Name	Memory Access	Description
31:0	PHY_CONTROL	R/W	PHY Control. Data sent directly to the cfg_phy_control bus. This is a GPIO register driving the values on the static cfg_phy_control output signals, and does not in any way influence controller functionality. It can be used for any static sideband control signaling requirements that you have for your PHY. Usage of this register and the associated GPIO bus is RESERVED when the controller is connected to a Synopsys PHY. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: DEFAULT_PHY_CONTROL Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.27 TRGT_MAP_CTRL_OFF

- **Name:** Programmable Target Map Control Register.
- **Description:** This register controls the target map.
- **Size:** 32 bits
- **Offset:** B+0x11c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:21	TARGET_MAP_RESERVED_21_31
20:16	TARGET_MAP_INDEX
15:13	TARGET_MAP_RESERVED_13_15
12:7	TARGET_MAP_VF
6	TARGET_MAP_ROM
5:0	TARGET_MAP_PF

Table 3-376 Fields for Register: TRGT_MAP_CTRL_OFF

Bits	Name	Memory Access	Description
31:21	TARGET_MAP_RESERVED_21_31	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: R (sticky) Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
20:16	TARGET_MAP_INDEX	R/W	The number of the PF Function on which the Target Values are set. This register does not respect the Byte Enable setting, any write will affect all register bits. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
15:13	TARGET_MAP_RESERVED_13_15	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: R (sticky) Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
12:7	TARGET_MAP_VF	R/W	Target Values for each BAR on the VF Function selected by the index number. This register does not respect the Byte Enable setting, any write will affect all register bits. Value After Reset: {VF_MEM_FUNC0_BAR0_TARGET_MAP + 2* VF_MEM_FUNC0_BAR1_TARGET_MAP + 4*VF_MEM_FUNC0_BAR2_TARGET_MAP + 8 * VF_MEM_FUNC0_BAR3_TARGET_MAP+16 * VF_MEM_FUNC0_BAR4_TARGET_MAP + 32* VF_MEM_FUNC0_BAR5_TARGET_MAP} Exists: CX_SRIOV_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
6	TARGET_MAP_ROM	R/W	Target Value for the ROM page of the PF Function selected by the index number. This register does not respect the Byte Enable setting, any write will affect all register bits. Value After Reset: {ROM_FUNC0_TARGET_MAP} Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5:0	TARGET_MAP_PF	R/W	Target Values for each BAR on the PF Function selected by the index number. This register does not respect the Byte Enable setting, any write will affect all register bits. Value After Reset: {MEM_FUNC0_BAR0_TARGET_MAP + 2* MEM_FUNC0_BAR1_TARGET_MAP + 4*MEM_FUNC0_BAR2_TARGET_MAP + 8 * MEM_FUNC0_BAR3_TARGET_MAP+16 * MEM_FUNC0_BAR4_TARGET_MAP + 32* MEM_FUNC0_BAR5_TARGET_MAP} Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.28 **CLOCK_GATING_CTRL_OFF**

- **Name:** Clock Gating Control Register.
- **Description:** This register enables you to disable dynamic clock gating. By default dynamic clock gating is on, allowing the controller to autonomously enable and disable its clocks. The clock gating is performed in the clock and reset module, `DWC_pcie_clk_rst.v`, and is initiated by the controllers clock enable signals. The following modules support dynamic clock gating:
 - AXI Bridge
 - RADM
- **Size:** 32 bits
- **Offset:** B+0x18c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:2	1	0
RSVDP_2	AXI_CLK_GATING_EN	RADM_CLK_GATING_EN

Table 3-377 Fields for Register: **CLOCK_GATING_CTRL_OFF**

Bits	Name	Memory Access	Description
31:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
1	AXI_CLK_GATING_EN	R/W	AXI Clock Gating Enable. This register enables the AXI Bridge to autonomously enable and disable the AXI Master clock, the AXI Slave clock and the AXI DBI slave clock. The DWC_pcie_clk_rst.v module provides the gated clock, mstr_axi_aclk_gated, to the AXI Bridge and is enabled when the controllers clock enable signal, mstr_aclk_active, is asserted. For the AXI Slave this module provides the gated clock, slv_axi_aclk_gated, to the AXI Bridge and is enabled when the controllers clock enable signal, slv_aclk_active, is asserted. If the AXI DBI Slave is enabled (DBI_4SLAVE_POPULATED=1) the module provides the gated clock, dbi_axi_aclk_gated, to the AXI Bridge and is enabled when the controllers clock enable signal, dbi_aclk_active, is asserted. The controller de-asserts the clock enable signals when the respective AXI Master/Slave interfaces are idle. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable (default) Value After Reset: 0x1 Exists: (AMBA_INTERFACE>1) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	RADM_CLK_GATING_EN	R/W	RADM Clock Gating Enable. This register, if set, enables the RADM to autonomously enable and disable its clock. The DWC_pcie_clk_rst.v module provides the gated clock, radm_clk_g, to the RADM and is enabled when the controllers clock enable signal, en_radm_clk_g, is asserted. The RADM clock is a gated version of the controller clock, core_clk. The controller de-asserts en_radm_clk_g when there is no Rx traffic, Rx queues and pre/post-queue pipelines are empty, RADM completion LUT is empty, and there are no FLR actions pending. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable (default) Value After Reset: 0x1 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.29 GEN3_RELATED_OFF

- **Name:** Gen3 Control Register.
- **Description:** There is no Gen3-specific N_FTS field. The N_FTS field in the "Link Width and Speed Change Control Register" is used for both Gen2 and Gen3 speed modes. There is no Gen3-specific "Directed Speed Change" field. The "Directed Speed Change" field in the "Link Width and Speed Change Control Register" is used to change to Gen2 or Gen3 speed. A speed change to Gen3 occurs if (1) the "Directed Speed Change" field is set to '1' and (2) the "Target Link Speed" field in the Link Control 2 Register is set to Gen3. Gen3 support is advertised by both sides of the link during link training.
- **Size:** 32 bits
- **Offset:** B+0x190
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:26	RSVDP_26
25:24	RATE_SHADOW_SEL
23	GEN3_EQ_INVREQ_EVAL_DIFF_DISABLE
22	USP_SEND_8GT_EQ_TS2_DISABLE
21	AUTO_EQ_DISABLE
20:19	RSVDP_19
18	GEN3_DC_BALANCE_DISABLE
17	GEN3_DLLP_XMT_DELAY_DISABLE
16	GEN3_EQUALIZATION_DISABLE
15:14	RSVDP_14
13	RXEQ_RGRDLESS_RXTS
12	RXEQ_PH01_EN
11	EQ_REDO
10	EQ_EIEOS_CNT
9	EQ_PHASE_2_3
8	DISABLE_SCRAMBLER_GEN_3
7:2	RSVDP_2
1	NO_SEED_VALUE_CHANGE
0	GEN3_ZRXDC_NONCOMPL

Table 3-378 Fields for Register: GEN3_RELATED_OFF

Bits	Name	Memory Access	Description
31:26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
25:24	RATE_SHADOW_SEL	R/W	Rate Shadow Select. This register value decide the Data Rate of shadow register. The following shadow registers are controlled by this register. ■ GEN3_RELATED_OFF[9] EQ_PHASE_2_3 ■ GEN3_RELATED_OFF[12] RXEQ_PH01_EN ■ GEN3_RELATED_OFF[19] RE_EQ_REQUEST_ENABLE ■ GEN3_RELATED_OFF[21] AUTO_EQ_DISABLE ■ GEN3_RELATED_OFF[22] USP_SEND_8GT_EQ_TS2_DISABLE ■ GEN3_EQ_LOCAL_FS_LF_OFF[5:0] GEN3_EQ_LOCAL_LF ■ GEN3_EQ_LOCAL_FS_LF_OFF[11:6] GEN3_EQ_LOCAL_FS ■ GEN3_EQ_PSET_COEFF_MAP_0[5:0] GEN3_EQ_PRE_CURSOR_PSET ■ GEN3_EQ_PSET_COEFF_MAP_0[11:6] GEN3_EQ_CURSOR_PSET ■ GEN3_EQ_PSET_COEFF_MAP_0[17:12] GEN3_EQ_POSET_CURSOR_PSET ■ GEN3_FCOEF_VEC_DEPTH_OFF[4:0] GEN3_FCOEF_MAX_VEC_DEPTH ■ GEN3_FCOEF_VEC_DEPTH_OFF[11:8] GEN3_FCOEF_ACCESS_VAR ■ GEN3_FCOEF_LANE1_0_OFF[5:0] GEN3_FCOEF_LANE0_PRE_CURSOR ■ GEN3_FCOEF_LANE1_0_OFF[11:6] GEN3_FCOEF_LANE0_POST_CURSOR ■ GEN3_FCOEF_LANE1_0_OFF[21:16] GEN3_FCOEF_LANE1_PRE_CURSOR ■ GEN3_FCOEF_LANE1_0_OFF[27:22] GEN3_FCOEF_LANE1_POST_CURSOR ■ GEN3_FCOEF_LANE3_2_OFF[5:0] GEN3_FCOEF_LANE2_PRE_CURSOR ■ GEN3_FCOEF_LANE3_2_OFF[11:6] GEN3_FCOEF_LANE2_POST_CURSOR ■ GEN3_FCOEF_LANE3_2_OFF[21:16] GEN3_FCOEF_LANE3_PRE_CURSOR ■ GEN3_FCOEF_LANE3_2_OFF[27:22] GEN3_FCOEF_LANE3_POST_CURSOR

Bits	Name	Memory Access	Description
25:24...(cont.)	RATE_SHADOW_SEL.	R/W	■ GEN3_FCOEF_LANE5_4_OFF[5:0] GEN3_FCOEF_LANE4_PRE_CURSOR ■ GEN3_FCOEF_LANE5_4_OFF[11:6] GEN3_FCOEF_LANE4_POST_CURSOR ■ GEN3_FCOEF_LANE5_4_OFF[21:16] GEN3_FCOEF_LANE5_PRE_CURSOR ■ GEN3_FCOEF_LANE5_4_OFF[27:22] GEN3_FCOEF_LANE5_POST_CURSOR ■ GEN3_FCOEF_LANE7_6_OFF[5:0] GEN3_FCOEF_LANE6_PRE_CURSOR ■ GEN3_FCOEF_LANE7_6_OFF[11:6] GEN3_FCOEF_LANE6_POST_CURSOR ■ GEN3_FCOEF_LANE7_6_OFF[21:16] GEN3_FCOEF_LANE7_PRE_CURSOR ■ GEN3_FCOEF_LANE7_6_OFF[27:22] GEN3_FCOEF_LANE7_POST_CURSOR ■ GEN3_FCOEF_LANE9_8_OFF[5:0] GEN3_FCOEF_LANE8_PRE_CURSOR ■ GEN3_FCOEF_LANE9_8_OFF[11:6] GEN3_FCOEF_LANE8_POST_CURSOR ■ GEN3_FCOEF_LANE9_8_OFF[21:16] GEN3_FCOEF_LANE9_PRE_CURSOR ■ GEN3_FCOEF_LANE9_8_OFF[27:22] GEN3_FCOEF_LANE9_POST_CURSOR ■ GEN3_FCOEF_LANE11_10_OFF[5:0] GEN3_FCOEF_LANE10_PRE_CURSOR ■ GEN3_FCOEF_LANE11_10_OFF[11:6] GEN3_FCOEF_LANE10_POST_CURSOR

Bits	Name	Memory Access	Description
25:24...(cont..)	RATE_SHADOW_SEL..	R/W	■ GEN3_FCOEF_LANE11_10_OFF[21:16] GEN3_FCOEF_LANE11_PRE_CURSOR ■ GEN3_FCOEF_LANE11_10_OFF[27:22] GEN3_FCOEF_LANE11_POST_CURSOR ■ GEN3_FCOEF_LANE13_12_OFF[5:0] GEN3_FCOEF_LANE12_PRE_CURSOR ■ GEN3_FCOEF_LANE13_12_OFF[11:6] GEN3_FCOEF_LANE12_POST_CURSOR ■ GEN3_FCOEF_LANE13_12_OFF[21:16] GEN3_FCOEF_LANE13_PRE_CURSOR ■ GEN3_FCOEF_LANE13_12_OFF[27:22] GEN3_FCOEF_LANE13_POST_CURSOR ■ GEN3_FCOEF_LANE15_14_OFF[5:0] GEN3_FCOEF_LANE14_PRE_CURSOR ■ GEN3_FCOEF_LANE15_14_OFF[11:6] GEN3_FCOEF_LANE14_POST_CURSOR ■ GEN3_FCOEF_LANE15_14_OFF[21:16] GEN3_FCOEF_LANE15_PRE_CURSOR ■ GEN3_FCOEF_LANE15_14_OFF[27:22] GEN3_FCOEF_LANE15_POST_CURSOR ■ GEN3_EQ_CONTROL_OFF[3:0] GEN3_EQ_FB_MODE ■ GEN3_EQ_CONTROL_OFF[4] GEN3_EQ_PHASE23_EXIT_MODE ■ GEN3_EQ_CONTROL_OFF[5] GEN3_EQ_EVAL_2MS_DISABLE ■ GEN3_EQ_CONTROL_OFF[23:8] GEN3_EQ_PSET_REQ_VEC

Bits	Name	Memory Access	Description
25:24...(cont...)	RATE_SHADOW_SEL...	R/W	■ GEN3_EQ_CONTROL_OFF[24] GEN3_EQ_FOM_INC_INITIAL_EVAL ■ GEN3_EQ_CONTROL_OFF[25] GEN3_EQ_PSET_REQ_AS_COEF ■ GEN3_EQ_FB_MODE_DIR_CHANGE_OFF[4:0] GEN3_EQ_FMDC_T_MIN_PHASE23 ■ GEN3_EQ_FB_MODE_DIR_CHANGE_OFF[9:5] GEN3_EQ_FMDC_N_EVALS ■ GEN3_EQ_FB_MODE_DIR_CHANGE_OFF[13:10] GEN3_EQ_FMDC_MAX_PRE_CUSROR_DELTA ■ GEN3_EQ_FB_MODE_DIR_CHANGE_OFF[17:14] GEN3_EQ_FMDC_MAX_POST_CUSROR_DELTA <i>Note:</i> This register field is sticky. Values: ■ 0x0 (GEN3): Gen3 Data Rate is selected for shadow register. ■ 0x1 (GEN4): Gen4 Data Rate is selected for shadow register. ■ 0x2 (GEN5): Gen5 Data Rate is selected for shadow register. ■ 0x3 (RSVD): Reserved. Value After Reset: 0x0 Exists: CX_GEN4_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_CCIX_ESM_SUPPORT== 1) ? "writeAsRead" : "unconstrained"}
23	GEN3_EQ_INVREQ_EVAL_DIFF_DISABLE	R/W	Eq InvalidRequest and RxEqEval Different Time Assertion Disable. Disable the assertion of Eq InvalidRequest and RxEqEval at different time. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shared for Gen3 and Gen4/Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_CCIX_ESM_SUPPORT== 1) ? "writeAsRead" : "unconstrained" }

Bits	Name	Memory Access	Description
22	USP_SEND_8GT_EQ_TS2_DISABLE	* Varies	Upstream Port Send 8GT/s or 16GT/s EQ TS2 Disable. The base spec defines that USP can optionally send 8GT or 16GT EQ TS2, which implies that USP can set DSP TxPreset value in Gen4 or Gen5 Data Rate. This applies to upstream ports only; It does not apply to downstream ports. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is RSVD. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. Value after reset in Gen4/Gen5 is DEFAULT_GEN4_USP_SEND_8GT_EQ_TS2_DISABLE/DEFAULT_GEN5_USP_SEND_16GT_EQ_TS2_DISABLE. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: see description ■ Dbi: see description <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): If this register set to 0, USP sends 8GT or 16GT EQ TS2. ■ 0x1 (SET): If this register set to 1, USP does not send 8GT or 16GT EQ TS2. Value After Reset: DEFAULT_GEN4_USP_SEND_8GT_EQ_TS2_DISABLE Exists: CX_GEN4_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true

Bits	Name	Memory Access	Description
21	AUTO_EQ_DISABLE	* Varies	Autonomous Equalization Disable. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is RSVD. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: see description ■ Dbi: see description <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): If the bit is 0, DSP will block DLLPs and direct the link to perform Gen4 EQ in Autonomous Mechanism. ■ 0x1 (SET): When the controller is in L0 state at Gen3 data rate and equalization was completed successfully in Autonomous EQ Mechanism, setting this bit in DSP will not direct the controller to Recovery state to perform Gen4 equalization. Link stays in Gen3 rate and DSP sends DLLPs to USP. Value After Reset: 0x0 Exists: CX_GEN4_SPEED CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
20:19	RSVDP_19	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
18	GEN3_DC_BALANCE_DISABLE	R/W	DC Balance Disable. Disable DC Balance feature. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shared for Gen3 and Gen4/Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
17	GEN3_DLLP_XMT_DELAY_DISABLE	R/W	DLLP Transmission Delay Disable. Disable delay transmission of DLLPs before equalization. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shared for Gen3 and Gen4/Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_CCIX_ESM_SUPPORT== 1) ? "writeAsRead" : "unconstrained" }
16	GEN3_EQUALIZATION_DISABLE	R/W	Equalization Disable. Disable equalization feature. This bit cannot be changed once the LTSSM starts link training. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shared for Gen3 and Gen4/Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_CCIX_ESM_SUPPORT== 1) ? "writeAsRead" : "unconstrained" }
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
13	RXEQ_RGRDLESS_RXTS	* Varies	When set to '1', the controller as Gen3 EQ master asserts RxEqEval to instruct the PHY to do Rx adaptation and evaluation after a 500ns timeout from a new preset request. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: see description ■ Dbi: see description <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ASSERT_1US): mac_phy_rxeqeval asserts after 1us and 2 TS1 received from remote partner. ■ 0x1 (ASSERT_500NS): mac_phy_rxeqeval asserts after 500ns regardless of TS's received or not. Value After Reset: DEFAULT_GEN3_RXEQ_RGRDLESS_RXTS Exists: CX_GEN3_SPEED CX_GEN4_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
12	RXEQ_PH01_EN	* Varies	Rx Equalization Phase 0/Phase 1 Hold Enable. When this bit is set the upstream port holds phase 0 (the downstream port holds phase 1) for 10ms. Holding phase 0 or phase 1 can be used to allow sufficient time for Rx Equalization to be performed by the PHY. This bit is used during Virtex-7 Gen3 equalization. The programmable bits [RXEQ_PH01_EN, EQ_PHASE_2_3] can be used to obtain the following variations of the equalization procedure:<i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: see description ■ Dbi: see description <i>Note:</i> This register field is sticky. Values: ■ 0x0 (TX_EQ_23): Tx equalization only in phase 2/3 ■ 0x1 (NO_TX_EQ_NO_RX_EQ): No Tx equalization Value After Reset: DEFAULT_GEN3_RXEQ_PH01_EN Exists: CX_GEN3_SPEED CX_GEN4_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
11	EQ_REDO	R/W	Equalization Redo Disable. Disable autonomous mechanism for requesting to redo the equalization process. The received presets or coefficients mismatch in Recovery.RcvrLock after Recovery EQ phases causes the EQ redo requests. If the EQ redo is infinite or you do not want eq requests and redo, setting this bit to 1 will stop the EQ requests and EQ redo so that the link can go ahead to L0 state for packet transmissions. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shared for Gen3 and Gen4/Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_GEN3_AUTONOMOUS_EQ_REQ_DISABLE Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_CCIX_ESM_SUPPORT== 1) ? "writeAsRead" : "unconstrained"}

Bits	Name	Memory Access	Description
10	EQ_EIEOS_CNT	R/W	Equalization EIEOS Count Reset Disable. Disable requesting reset of EIEOS count during equalization. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shared for Gen3 and Gen4/Gen5 data rate. <i>Note:</i> This register field is sticky. Values: 0x1 (DISABLE): Disable Value After Reset: DEFAULT_GEN3_REQ_RST_EIEOS_CNT_EQ_DISABLE Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_CCIX_ESM_SUPPORT== 1) ? "writeAsRead" : "unconstrained"}
9	EQ_PHASE_2_3	* Varies	Equalization Phase 2 and Phase 3 Disable. This applies to downstream ports only. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> The access attributes of this field are as follows: - Wire: see description - Dbi: see description <i>Note:</i> This register field is sticky. Values: 0x1 (TX_EQ_23_RX_EQ_PH01): No Rx equalization 0x0 (NO_TX_EQ_RX_EQ_PH01): Rx equalization in phase 0/1 Value After Reset: 0x0 Exists: CX_GEN3_SPEED CX_GEN4_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
8	DISABLE_SCRAMBLER_GEN_3	R/W	Disable Scrambler for Gen3 and Gen4 Data Rate. The Gen3 and Gen4 scrambler/descrambler within the controller needs to be disabled when the scrambling function is implemented outside of the controller (for example within the PHY). <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shared for Gen3 and Gen4/Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
1	NO_SEED_VALUE_CHANGE	R/W	If this bit is set to 1, the seed value of LFSR for scrambler at Gen3 rate does not change after LinkUp = 1. This bit takes effect only when CX_AUTO_LANE_FLIP_CTRL_EN is supported. This feature requires both sides of the link support it. <i>Note:</i> this register is shared for Gen3 and Gen4/Gen5 data rates. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (NO_CHANGE): Not Change Value After Reset: DEFAULT_NO_SEED_VALUE_CHANGE Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
0	GEN3_ZRXDC_NONCOMPL	R/W	Gen3 Receiver Impedance ZRX-DC Not Compliant. Receivers that operate at 8.0 GT/s with an impedance other than the range defined by the ZRX-DC parameter for 2.5 GT/s (40-60 Ohms) must meet additional behavior requirements in the following LTSSM states: Polling, Rx_L0s, L1, L2, and Disabled. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shared for Gen3 and Gen4/Gen5 data rates. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (RCVR_COMPLIES): The receiver complies with the ZRX-DC parameter for 2.5 GT/s when operating at 8 GT/s or higher. ■ 0x1 (RCVR_NOT_COMPLIES): The receiver does not comply with the ZRX-DC parameter for 2.5 GT/s when operating at 8 GT/s or higher. Value After Reset: DEFAULT_GEN3_ZRXDC_NONCOMPL Exists: CX_GEN3_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.30 GEN3_EQ_LOCAL_FS_LF_OFF

- **Name:** Gen3 EQ FS and LF Register.
- **Description:** This register holds the Gen3 EQ FS and LF values.
- **Size:** 32 bits
- **Offset:** B+0x194
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:12	RSVDP_12
11:6	GEN3_EQ_LOCAL_FS
5:0	GEN3_EQ_LOCAL_LF

Table 3-379 Fields for Register: GEN3_EQ_LOCAL_FS_LF_OFF

Bits	Name	Memory Access	Description
31:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
11:6	GEN3_EQ_LOCAL_FS	R/W	Full Swing (FS) Value for Gen3/Gen4 Transmit Equalization. Value Range: 12 through 63 (decimal). Only used when you set the "Gen3 Equalization Presets to Coefficients Mapping Mode" to "Programmable Table", by setting the CX_GEN3_EQ_PSET_COEF_MAP_MODE configuration parameter to '2'. Selects the FS and LF values used by the local PHY. The values are needed in the controller for advertising during Phase1, and for checking during "EQ Master" that the requested coefficients are legal according to rules in section "4.2.3.1 Rules for Transmitter Coefficients" of the <i>PCI Express Base Specification</i>. Note: ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. Note: This register field is sticky. Values: ■ 0xc (MIN_VAL): Min value ■ 0x3f (MAX_VAL): Max value Value After Reset: DEFAULT_GEN3_EQ_LOCAL_FS Exists: CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_GEN4_SPEED== 1) ? "writeAsRead" : "unconstrained" }

Bits	Name	Memory Access	Description
5:0	GEN3_EQ_LOCAL_LF	R/W	Low Frequency (LF) Value for Gen3/Gen4 Transmit Equalization. Only used when you set the "Gen3 Equalization Presets to Coefficients Mapping Mode" to "Programmable Table", by setting the CX_GEN3_EQ_PSET_COEF_MAP_MODE configuration parameter to '2'. Selects the FS and LF values used by the local PHY. The values are needed in the controller for advertising during Phase1, and for checking during "EQ Master" that the requested coefficients are legal according to the rules in section "4.2.3.1 Rules for Transmitter Coefficients" of the <i>PCI Express Base Specification</i>. Note: ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. Note: This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0x3f (MAX_VAL): Max value Value After Reset: DEFAULT_GEN3_EQ_LOCAL_LF Exists: CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_GEN4_SPEED== 1) ? "writeAsRead" : "unconstrained" }

3.48.31 GEN3_EQ_PSET_COEF_MAP__i (for i = 0; i <= 10)

- **Name:** Gen3 EQ Presets to Coefficients Mapping Register.
- **Description:** This register is only used when you set the "Gen3 Equalization Presets to Coefficients Mapping Mod" to "Programmable Table", by setting the CX_GEN3_EQ_PSET_COEF_MAP_MODE configuration parameter to '2'. These registers are programmed through an indirect addressing scheme (using an index register) to reduce the address footprint in the PCI Express extended configuration space. The index register ("Gen3 EQ Preset Index Register") is used to select which Coefficients Mapping Register is being accessed. Therefore, only four bytes of the PCIe extended configuration space address map is used to contain 44 bytes of information. You must write to the index register ("Gen3 EQ Preset Index Register") before you write to this mapping register.
The controller does not automatically update the coefficients table when the "Gen3 EQ FS and LF Register" is updated with new values. Your application must re-program the controller with new coefficient values every time it updates FS and LF. Your application must ensure that the programmed coefficient values satisfy the "Rules for Transmitter Coefficients" defined in Section 4.2.3.1 of the *PCI Express Base Specification*. The controller does not apply the programmed coefficient values to mac_phy_txdeemph if they do not meet those rules.
- **Size:** 32 bits
- **Offset:** B+0x198
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_18	31:18
GEN3_EQ_POST_CURSOR_PSET	17:12
GEN3_EQ_CURSOR_PSET	11:6
GEN3_EQ_PRE_CURSOR_PSET	5:0

Table 3-380 Fields for Register: GEN3_EQ_PSET_COEF_MAP__i (for i = 0; i <= 10)

Bits	Name	Memory Access	Description
31:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
17:12	GEN3_EQ_POST_CURSOR_PSET	R/W	Post-cursor (C+1) Coefficient. The default (DEFAULT_GEN3_EQ_POSTCURSOR_PSETi) is defined in the table in the "Programming Registers" section of the "Gen3 Equalization Details and Example" appendix in the Databook. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: FS_DIV_4 Exists: CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
11:6	GEN3_EQ_CURSOR_PSET	R/W	Cursor (C0) Coefficient. Note: Default is DEFAULT_GEN3_EQ_LOCAL_FS - DEFAULT_GEN3_EQ_PRECURSOR_PSETi - DEFAULT_GEN3_EQ_POSTCURSOR_PSETi that is, $C0 = FS - \text{abs}(C-1) - \text{abs}(C+1)$ <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_CURSOR_P0 Exists: CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
5:0	GEN3_EQ_PRE_CURSOR_PSET	R/W	Pre-cursor (C-1) Coefficient. The default (DEFAULT_GEN3_EQ_PRECURSOR_PSETi) is defined in the table in the "Programming Registers" section of the "Gen3 Equalization Details and Example" appendix in the Databook. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.48.32 GEN3_EQ_PSET_INDEX_OFF

- **Name:** Gen3 EQ Preset Index Register.
- **Description:** The preset registers are programmed through an indirect addressing scheme (using this index register) to reduce the address footprint in the PCI Express extended configuration space. The size of the required port logic space does not depend on the number of regions defined as the index register is used to select which preset is being accessed.
Viewportdepth: 11
- **Size:** 32 bits
- **Offset:** B+0x19c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-381 Fields for Register: GEN3_EQ_PSET_INDEX_OFF

Bits	Name	Memory Access	Description
31:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
3:0	GEN3_EQ_PSET_INDEX	R/W	Preset index i, where i =0, 1,2,...,10. This is the index used for accessing (using register indirect addressing mode) the "Gen3 EQ Presets to Coefficients Mapping Registers". This register is only used when you set the "Gen3 Equalization Presets to Coefficients Mapping Mode" to "Programmable Table", by setting the CX_GEN3_EQ_PSET_COEF_MAP_MODE configuration parameter to '2'. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shared for Gen3 and Gen4/Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.33 GEN3_EQ_COEFF_LEGALITY_STATUS_OFF

- **Name:** Gen3 EQ Status Register.
- **Description:** This register is only used when you set the "Gen3 Equalization Presets to Coefficients Mapping Mode" to "Programmable Table", by setting the CX_GEN3_EQ_PSET_COEF_MAP_MODE configuration parameter to '2'. This is used to validate the FS, LF, and Coefficient values that your application has programmed. Your application should always validate the programmed values. It is set by the controller when you program it with illegal coefficients. For more information on legal coefficient values, see Section "4.2.3.1. Rules for Transmitter Coefficients" of the *PCI Express Base Specification*.
- **Size:** 32 bits
- **Offset:** B+0x1a4
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-382 Fields for Register: GEN3_EQ_COEFF_LEGALITY_STATUS_OFF

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
0	GEN3_EQ_VIOLATE_COEF_RULES	R	Programmable Coefficients Legality Status. This bit is only valid after programming a coefficient into the mapping table. This bit is set if the most recently programmed coefficient ("Gen3 EQ Presets to Coefficients Mapping Registers" violate any of the following rules: ■ $\text{abs}(C-1) \leq (FS/4)$ ■ $\text{abs}(C-1) + C_0 + \text{abs}(C+1) = FS$ ■ $\text{abs}(C_0) - \text{abs}(C-1) - \text{abs}(C+1) \geq LF$ Your application should always check the status after it programs each coefficient, so that it then knows which preset number had the illegal coefficients. <i>Note:</i> You should ignore the value in this field until you have written/updated the programmable coefficients. Although the value of this register field is '0' during reset, it gets set to '1' almost immediately after the reset is removed. This is because the current coefficient defaults to '0', which is always illegal. So if this field is read before you program the coefficient mapping table, then it is '1'. This bit is automatically cleared when you read it. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Set Value After Reset: * Varies Exists: $CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG \&\& (!CX_PL_REG_DISABLE \&\& (CX_MEMORY_MAP_PCIE_TYPE == 2 \parallel CX_MEMORY_MAP_PCIE_TYPE == 0))$ Testable: untestable Reset Mask: 0x0 Volatile: true

3.48.34 GEN3_EQ_CONTROL_OFF

- **Name:** Gen3 EQ Control Register.
- **Description:** This register controls equalization for Phase2 in an upstream port (USP), or Phase3 in a downstream port (DSP).
- **Size:** 32 bits
- **Offset:** B+0x1a8
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	RSVDP_31
30	GEN3_SUPPORT_FFINITE_EQ_REQUEST
29:27	GEN3_EQ_REQ_NUM
26	GEN3_REQ_SEND_CONSEC_EIEOS_FOR_PSET_MAP
25	GEN3_EQ_PSET_REQ_AS_COEF
24	GEN3_EQ_FOM_INC_INITIAL_EVAL
23:8	GEN3_EQ_PSET_REQ_VEC
7	RSVDP_7
6	GEN3_LOWER_RATE_EQ_REDO_ENABLE
5	GEN3_EQ_EVAL_2MS_DISABLE
4	GEN3_EQ_PHASE23_EXIT_MODE
3:0	GEN3_EQ_FB_MODE

Table 3-383 Fields for Register: GEN3_EQ_CONTROL_OFF

Bits	Name	Memory Access	Description
31	RSVDP_31	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_GEN3_SPEED CX_GEN4_SPEED CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
30	GEN3_SUPPORT_FINITE_EQ_REQUEST	R/W	Support finite EQ requests for USP. <i>Note:</i> Gen3, Gen4 and Gen5 share the same register bit and have the same feature. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (UNSUPPORT): Do not support ■ 0x1 (SUPPORT): Support Value After Reset: DEFAULT_SUPPORT_FINITE_EQ_REQUEST Exists: CX_GEN3_SPEED CX_GEN4_SPEED CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
29:27	GEN3_EQ_REQ_NUM	R/W	The number of back-to-back equalization redo requests at a given Gen3, Gen4 and Gen5 data rate for USP. After counting the EQ redo requests which are equal to or larger than the number set, the USP stops any EQ redo requests at the data rate. If changing speed, the USP controller clears the counter to 0. Then re-start to count the requests after changing speed back to the data rate for the USP. <i>Note:</i> Gen3, Gen4 and Gen5 share the same register bits and have the same feature. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0x7 (MAX_VAL): Max value Value After Reset: DEFAULT_EQ_REQ_NUM Exists: CX_GEN3_SPEED CX_GEN4_SPEED CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
26	GEN3_REQ_SEND_CONSEC_EIEOS_FOR_PSET_MAP	R/W	Request controller to send back-to-back EIEOS in Recovery.RcvrLock state until presets to coefficients mapping is complete.<i>Note:</i> Gen3 and Gen4 share the same register bit and have the same feature. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DO_NOT_REQ): Do not request ■ 0x1 (REQ): Request Value After Reset: DEFAULT_GEN3_REQ_SEND_CONSECUTIVE_EIEOS_FOR_PSET_MAP Exists: CX_GEN3_SPEED CX_GEN4_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
25	GEN3_EQ_PSET_REQ_AS_COEF	R/W	GEN3_EQ_PSET_REQ_AS_COEF is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_GEN3_SPEED CX_GEN4_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
24	GEN3_EQ_FOM_INC_INITIAL_EVAL	R/W	Include Initial FOM. Include or not the FOM feedback from the initial preset evaluation performed in the EQ Master, when finding the highest FOM among all preset evaluations.<i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DO_NOT_INCLUDE): Do not include ■ 0x1 (INCLUDE): Include Value After Reset: DEFAULT_GEN3_EQ_FOM_INC_INITIAL_EVAL Exists: CX_GEN3_SPEED CX_GEN4_SPEED CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:8	GEN3_EQ_PSET_REQ_VEC	R/W	Preset Request Vector. Requesting of Presets during the initial part of the EQ Master Phase. Encoding scheme is as follows: Bit [15:0] =0x0: No preset is requested and evaluated in EQ Master Phase. Bit [i] =1: "Preset=i" is requested and evaluated in EQ Master Phase. ■ 0x0000000000000000: No preset be requested and evaluated in EQ Master Phase ■ 0x000000xxxxxxxx1: Preset 0 is requested and evaluated in EQ Master Phase ■ 0x000000xxxxxxxx1x: Preset 1 is requested and evaluated in EQ Master Phase ■ 0x000000xxxxxxxx1xx: Preset 2 is requested and evaluated in EQ Master Phase ■ 0x000000xxxxxxxx1xxx: Preset 3 is requested and evaluated in EQ Master Phase ■ 0x000000xxxxx1xxxx: Preset 4 is requested and evaluated in EQ Master Phase ■ 0x000000xxxx1xxxxx: Preset 5 is requested and evaluated in EQ Master Phase ■ 0x000000xxx1xxxxxx: Preset 6 is requested and evaluated in EQ Master Phase ■ 0x000000xx1xxxxxxx: Preset 7 is requested and evaluated in EQ Master Phase ■ 0x000000x1xxxxxxxx: Preset 8 is requested and evaluated in EQ Master Phase ■ 0x000000x1xxxxxxxxx: Preset 9 is requested and evaluated in EQ Master Phase ■ 0x000001xxxxxxxxxx: Preset 10 is requested and evaluated in EQ Master Phase ■ All other encodings: Reserved <i>Note:</i>

Bits	Name	Memory Access	Description
23:8...(cont.)	GEN3_EQ_PSET_REQ_VEC.	R/W	You must contact your PHY vendor to ensure 24 ms timeout does not occur in presets requests in EQ master phase, that is, you must set a proper value to the GEN3_EQ_PSET_REQ_VEC register so that the EQ tuning for Figure of Merit in the EQ master phase completes before 24 ms timeout.<i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (Min_field_value): Zero value ■ 0xffff (Max_field_value): Max value Value After Reset: DEFAULT_GEN3_EQ_PSET_REQ_VEC Exists: CX_GEN3_SPEED CX_GEN4_SPEED CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_GEN3_SPEED CX_GEN4_SPEED CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
6	GEN3_LOWER_RATE_EQ_REDO_ENABLE	R/W	Support EQ redo and lower rate change. To access this field, RATE_SHADOW_SEL should be set to 0. <i>Note:</i> Gen3 and Gen4 share the same register bit and have the same feature. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NO_SUP): Not supported ■ 0x1 (SUP): Supported Value After Reset: {DEFAULT_GEN3_LOWER_RATE_EQ_REDO_ENABLE} Exists: CX_GEN3_SPEED CX_GEN4_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
5	GEN3_EQ_EVAL_2MS_DISABLE	R/W	Phase2_3 2 ms Timeout Disable. Determine behavior in Phase2 for USP (Phase3 if DSP) when the PHY does not respond within 2ms to the assertion of RxEqEval. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ABORT_CURRENT_EVA): Abort the current evaluation, stop any attempt to modify the remote transmitter settings, Phase2 is terminated by the 24ms timeout ■ 0x1 (IGNORE_2MS_TIMEOUT): Ignore the 2ms timeout and continue as normal. This is used to support PHYs that require more than 2ms to respond to the assertion of RxEqEval. Value After Reset: DEFAULT_GEN3_EQ_EVAL_2MS_DISABLE Exists: CX_GEN3_SPEED CX_GEN4_SPEED CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
4	GEN3_EQ_PHASE23_EXIT_MODE	R/W	Behavior After 24 ms Timeout (when optimal settings are not found). For a USP: Determine next LTSSM state from Phase2 after 24ms Timeout 0: Recovery.Speed 1: Recovery.Equalization.Phase3 When optimal settings are not found then: - Equalization Phase 2 Successful status bit is not set in the "Link Status Register 2" when GEN3_EQ_PHASE23_EXIT_MODE = 0 - Equalization Phase 2 Successful status bit is set in the "Link Status Register 2" when GEN3_EQ_PHASE23_EXIT_MODE = 1 - Equalization Phase 2 Complete status bit is set in the "Link Status Register 2" For a DSP: Determine next LTSSM state from Phase3 after 24ms Timeout 0: Recovery.Speed 1: Recovery.Equalization.RcvrLock When optimal settings are not found then: - Equalization Phase 3 Successful status bit is not set in the "Link Status Register 2" when GEN3_EQ_PHASE23_EXIT_MODE = 0 - Equalization Phase 3 Successful status bit is set in the "Link Status Register 2" when GEN3_EQ_PHASE23_EXIT_MODE = 1 - Equalization Phase 3 Complete status bit is set in the "Link Status Register 2" <i>Note:</i> GEN3_EQ_PHASE23_EXIT_MODE = 1 affects Direction Change feed back mode. EQ requests for Figure Of Merit mode complete before 24ms timeout. For more information, see GEN3_EQ_PSET_REQ_VEC Register. <i>Note:</i> - When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. - If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. - If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate.

Bits	Name	Memory Access	Description
4...(cont.)	GEN3_EQ_PHASE23_EXIT_MODE	R/W	■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (RCVRY_SPEED): Recovery.Speed ■ 0x1 (RCVRY_EQ): USP: Recovery.Equalization.Phase3; DSP: Recovery.Equalization.RcvrLock Value After Reset: DEFAULT_GEN3_EQ_PHASE23_EXIT_MODE Exists: CX_GEN3_SPEED CX_GEN4_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
3:0	GEN3_EQ_FB_MODE	R/W	Feedback Mode. Other values are reserved.<i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is a shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DIR_CHG): Direction Change ■ 0x1 (FOM): Figure Of Merit Value After Reset: DEFAULT_GEN3_EQ_FB_MODE Exists: CX_GEN3_SPEED CX_GEN4_SPEED CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.35 GEN3_EQ_FB_MODE_DIR_CHANGE_OFF

- **Name:** Gen3 EQ Direction Change Feedback Mode Control Register.
- **Description:** Equalization controls to be used in Phase2 (USP) or Phase 3 (DSP), when you set the Feedback Mode in "Gen3 EQ Control Register" to "Direction Change." These fields allow control over the initial starting point for the search of optimal coefficient settings, and allow control over the criteria used to determine when the optimal settings have been achieved. The values are applied to all the lanes.
- **Size:** 32 bits
- **Offset:** B+0x1ac
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:18	RSVDP_18
17:14	GEN3_EQ_FMDC_MAX_POST_CUSROR_DELTA
13:10	GEN3_EQ_FMDC_MAX_PRE_CUSROR_DELTA
9:5	GEN3_EQ_FMDC_N_EVALS
4:0	GEN3_EQ_FMDC_T_MIN_PHASE23

Table 3-384 Fields for Register: GEN3_EQ_FB_MODE_DIR_CHANGE_OFF

Bits	Name	Memory Access	Description
31:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_GEN3_EQ_COEF_CONV_SUPPORTED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
17:14	GEN3_EQ_FMDC_MAX_POST_CUSROR_DELTA	R/W	Convergence Window Aperture for C+1. Post-cursor coefficients maximum delta within the convergence window depth. Allowed range: 0,1,2,...15. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: DEFAULT_GEN3_EQ_FMDC_MAX_POST_CUSROR_DELTA Exists: CX_GEN3_EQ_COEF_CONV_SUPPORTED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_GEN4_SPEED== 1) ? "writeAsRead" : "unconstrained" }

Bits	Name	Memory Access	Description
13:10	GEN3_EQ_FMDC_MAX_PRE_CUSROR_DELTA	R/W	Convergence Window Aperture for C-1. Pre-cursor coefficients maximum delta within the convergence window depth. Allowed range: 0,1,2,...15. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: DEFAULT_GEN3_EQ_FMDC_MAX_PRE_CUSROR_DELTA Exists: CX_GEN3_EQ_COEF_CONV_SUPPORTED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_GEN4_SPEED== 1) ? "writeAsRead" : "unconstrained" }

Bits	Name	Memory Access	Description
9:5	GEN3_EQ_FMDN_N_EVALS	R/W	Convergence Window Depth. Number of consecutive evaluations considered in Phase 2/3 when determining if optimal coefficients have been found. Allowed range: 0,1,2,...16 up to a maximum of CX_GEN3_EQ_COEFQ_DEPTH. When set to 0, EQ Master is performed without sending any requests to the remote partner in Phase 2 for USP and Phase 3 for DSP. Therefore, the remote partner will not change its transmitter coefficients and will move to the next state. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0x1f (MAX_VAL): Maximum of CX_GEN3_EQ_COEFQ_DEPTH Value After Reset: DEFAULT_GEN3_EQ_FMDN_N_EVALS Exists: CX_GEN3_EQ_COEF_CONV_SUPPORTED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_GEN4_SPEED== 1) ? "writeAsRead" : "unconstrained" }

Bits	Name	Memory Access	Description
4:0	GEN3_EQ_FMDC_T_MIN_PHASE 23	R/W	Minimum Time (in ms) To Remain in EQ Master Phase. The LTSSM stays in EQ Master phase for at least this amount of time, before starting to check for convergence of the coefficients. Allowed values 0,1,...,24. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0x18 (MAX_VAL): Max value Value After Reset: DEFAULT_GEN3_EQ_FMDC_T_MIN_PHASE23 Exists: CX_GEN3_EQ_COEF_CONV_SUPPORTED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: {(CX_GEN4_SPEED== 1) ? "writeAsRead" : "unconstrained" }

3.48.36 ORDER_RULE_CTRL_OFF

- **Name:** Order Rule Control Register.
- **Description:** This register controls the order rule.
- **Size:** 32 bits
- **Offset:** B+0x1b4
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	RSVDP_16
15:8	CPL_PASS_P
7:0	NP_PASS_P

Table 3-385 Fields for Register: ORDER_RULE_CTRL_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:8	CPL_PASS_P	R/W	Completion Passing Posted Ordering Rule Control. Determines if CPL can pass halted P queue. Values: ■ 0x0 (CPL_CAN_NOT_PASS): CPL can not pass P (recommended) ■ 0x1 (CPL_CAN_PASS): CPL can pass P Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
7:0	NP_PASS_P	R/W	Non-Posted Passing Posted Ordering Rule Control. Determines if NP can pass halted P queue. Values: ■ 0x0 (NP_CAN_NOT_PASS): NP can not pass P (recommended). ■ 0x1 (NP_CAN_PASS): NP can pass P Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.37 PIPE_LOOPBACK_CONTROL_OFF

- **Name:** PIPE Loopback Control Register.
- **Description:** This register controls the PIPE Loopback.
- **Size:** 32 bits
- **Offset:** B+0x1b8
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	PIPE_LOOPBACK
30:27	RSVDP_27
26:24	RXSTATUS_VALUE
23:22	RSVDP_22
21:16	RXSTATUS_LANE
15:0	LPBK_RXVALID

Table 3-386 Fields for Register: PIPE_LOOPBACK_CONTROL_OFF

Bits	Name	Memory Access	Description
31	PIPE_LOOPBACK	R/W	PIPE Loopback Enable. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_PIPE_LOOPBACK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
30:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_PIPE_LOOPBACK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
26:24	RXSTATUS_VALUE	W	RXSTATUS_VALUE is an internally reserved field. Do not use. Value After Reset: * Varies Exists: CX_PIPE_LOOPBACK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
23:22	RSVDP_22	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_PIPE_LOOPBACK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
21:16	RXSTATUS_LANE	R/W	RXSTATUS_LANE is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_PIPE_LOOPBACK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:0	LPBK_RXVALID	R/W	LPBK_RXVALID is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: { 65535 >> (16 - CX_NL)} Exists: CX_PIPE_LOOPBACK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.38 MISC_CONTROL_1_OFF

- **Name:** DBI Read-Only Write Enable Register.
- **Description:** This is the DBI Read-Only write enable register.
- **Size:** 32 bits
- **Offset:** B+0x1bc
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_25	31:25
RASDES_REG_PFO_ONLY	24
RAS_REG_PFO_ONLY	23
PORT_LOGIC_WR_DISABLE	22
P2P_ERR_RPT_CTRL	21
P2P_TRACK_CPL_TO_REG	20
TARGET_ABOVE_CONFIG_LIMIT_REG	19:18
CONFIG_LIMIT_REG	17:8
CFG_TLP_BYPASS_EN_REG	7
CPLQ_MNG_EN	6
ARI_DEVICE_NUMBER	5
DISABLE_AUTO_LTR_CLR_MSG	4
SIMPLIFIED_REPLAY_TIMER	3
UR_CA_MASK_4_TRGT1	2
DEFAULT_TARGET	1
DBI_RO_WR_EN	0

Table 3-387 Fields for Register: MISC_CONTROL_1_OFF

Bits	Name	Memory Access	Description
31:25	RSVDP_25	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
24	RASDES_REG_PF0_ONLY	* Varies	Allows only Physical Function 0 (PF0) to access the RAS DES Extended Capability registers. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_DES_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\" "
23	RAS_REG_PF0_ONLY	* Varies	Allows only Physical Function 0 (PF0) to access the RAS Extended Capability registers. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_RAS_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\" "
22	PORT_LOGIC_WR_DISABLE	* Varies	Disable port logic register write from wire side. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_PL_WIRE_WR_DISABLE Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
21	P2P_ERR_RPT_CTRL	R/W	Determines whether to enable Peer to Peer (P2P) error reporting. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable P2P error reporting ■ 0x1 (ENABLE): Enable P2P error reporting Value After Reset: CX_P2P_ENABLE_VALUE Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20	P2P_TRACK_CPL_TO_REG	R/W	Determines whether to track completion of transmitted Non-Posted TLPs in P2P mode. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DO_NOT_TRACK): Do not track completion ■ 0x1 (TRACK): Track completion Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
19:18	TARGET_ABOVE_CONFIG_LIMIT_REG	R/W	Configuration requests with an address greater than CONFIG_LIMIT_REG are directed to either ELBI or TRGT1 interface based on the setting of this field. This field can have the following values: <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ELBI): ELBI ■ 0x2 (TRGT1): TRGT1 Value After Reset: TARGET_ABOVE_CONFIG_LIMIT Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
17:8	CONFIG_LIMIT_REG	R/W	Configuration requests are directed either to CDM or ELBI/RTRGT1 based on the value of this field. ■ Configuration requests with an address less CONFIG_LIMIT_REG are directed to the CDM ■ Configuration requests with an address greater than CONFIG_LIMIT_REG are directed to either ELBI or TRGT1 interface based on the setting of TARGET_ABOVE_CONFIG_LIMIT_REG field. Your application must set a proper value for this field based on your extended configuration registers. For more information, see the "CDM/ELBI Register Space Access Through CFG Request" in "Register Module, LBC, and DBI" section in the "Controller Operations" chapter of the <i>Databook</i>. <i>Note:</i> This register field is sticky. Value After Reset: CONFIG_LIMIT Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7	CFG_TLP_BYPASS_EN_REG	R/W	Setting of this field defines how to decide the destination of Configuration requests. <i>Note:</i> When app_req_retry_en is asserted, the setting of this field is ignored. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ONE): Configuration TLPs are routed according to the setting of TARGET_ABOVE_CONFIG_LIMIT_REG, regardless the value of CONFIG_LIMIT_REG. ■ 0x0 (ZERO): Configuration TLPs are routed according to the setting of TARGET_ABOVE_CONFIG_LIMIT_REG, depending on the setting of CONFIG_LIMIT_REG. Refer to the definition of CONFIG_LIMIT_REG for details. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
6	CPLQ_MNG_EN	R/W	This field enables the Completion Queue Management feature. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x1 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
5	ARI_DEVICE_NUMBER	R/W	When ARI is enabled, this field enables use of the device ID. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4	DISABLE_AUTO_LTR_CLR_MSG	R/W	Disable the autonomous generation of LTR clear message in upstream port. This field can have the following values: For more information, see "Latency Tolerance Reporting (LTR) Message Generation [EP Mode]" in "Message Generation" section of the "Controller Operations" chapter of the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ALLOW): Allow the autonomous generation of LTR clear message. ■ 0x1 (DISABLE): Disable the autonomous generation of LTR clear message. Value After Reset: 0x0 Exists: CX_LTR_M_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
3	SIMPLIFIED_REPLAY_TIMER	R/W	Enables Simplified Replay Timer (Gen4). For more information, see "Transmit Replay" in "Transmit TLP Processing" section in the "Controller Operations" chapter of the <i>Databook</i>. Simplified Replay Timer can have the following Values: ■ A value from 24,000 to 31,000 Symbol Times when Extended Synch is 0b. ■ A value from 80,000 to 100,000 Symbol Times when Extended Synch is 1b. The Simplified Replay Timer value must not be changed while the link is in use. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: {(CX_GEN4_MODE == 2) ? 0 : 1 } Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
2	UR_CA_MASK_4_TRGT1	R/W	When this field is set to '1', the controller suppresses error logging, error message generation, and CPL generation for non-posted requests TLPs (with UR filtering status) forwarded to your application (that is, when DEFAULT_TARGET =1). For more information, see "Advanced Error Handling For Received TLPs" chapter of the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Set Value After Reset: CX_MASK_UR_CA_4_TRGT1 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
1	DEFAULT_TARGET	R/W	Default target for an IO or MEM request with UR/CA/CRS received. Based on the value of this field the controller either drops or forwards these requests to your application. For more information, see "ECRC Handling" and "Request TLP Routing Rules" in "Receive Routing" section of the "Controller Operations" chapter of the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DROP_IO_MEM_REQ): The controller drops all incoming I/O or MEM requests (after corresponding error reporting). A completion with UR status is generated for non-posted requests. ■ 0x1 (FWD_IO_MEM_UR_CA_CRS): The controller forwards all incoming I/O or MEM requests with UR/CA/CRS status to your application. Value After Reset: DEFAULT_TARGET Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
0	DBI_RO_WR_EN	R/W	Write to RO Registers Using DBI. For more information, see "Writing to Read-Only Registers" in "Register Module, LBC, and DBI" section in the "Controller Operations" chapter of the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Your application can write to some RO and HwInit register fields through the DBI when you set this field to '1'. Value After Reset: CX_DBI_RO_WR_EN Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.39 MULTI_LANE_CONTROL_OFF

- **Name:** UpConfigure Multi-lane Control Register.
- **Description:** Used when upsizing or downsizing the link width through Configuration state without bringing the link down. For more information, see the "Link Establishment" section in the "Controller Operations" chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x1c0
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:8	RSVDP_8
7	UPCONFIGURE_SUPPORT
6	DIRECT_LINK_WIDTH_CHANGE
5:0	TARGET_LINK_WIDTH

Table 3-388 Fields for Register: MULTI_LANE_CONTROL_OFF

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
7	UPCONFIGURE_SUPPORT	* Varies	Upconfigure Support. The controller sends this value as the Link Upconfigure Capability in TS2 Ordered Sets in Configuration.Complete state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_UPCONFIGURE_SUPPORT Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "
6	DIRECT_LINK_WIDTH_CHANGE	* Varies	Directed Link Width Change. ■ If the <i>upconfigure_capable</i> variable is '1' and the PCIE_CAP_HW_AUTO_WIDTH_DISABLE bit in LINK_CONTROL_LINK_STATUS_REG is '0', the controller starts upconfigure or autonomous width downsizing (to the TARGET_LINK_WIDTH value) in the Configuration state. ■ If TARGET_LINK_WIDTH value is 0x0, the controller does not start upconfigure or autonomous width downsizing in the Configuration state. The controller self-clears this field when the controller accepts this request. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): The controller always moves to Configuration state through Recovery state when this bit is set to '1'. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
5:0	TARGET_LINK_WIDTH	* Varies	Target Link Width. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (NOT_START): Controller does not start upconfigure or autonomous width downsizing in the Configuration state. ■ 0x1 (X1): x1 ■ 0x2 (X2): x2 ■ 0x4 (X4): x4 ■ 0x8 (X8): x8 ■ 0x10 (X16): x16 ■ 0x20 (X32): x32 Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "

3.48.40 PHY_INTEROP_CTRL_OFF

- **Name:** PHY Interoperability Control Register.
- **Description:** This register controls controller PHY dependent behavior.
- **Size:** 32 bits
- **Offset:** B+0x1c4
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 | | CX_MEMORY_MAP_PCIE_TYPE==0))

PIPE_OPTIONAL_PCLKCHANGE_HS	31
PHY_PERST_ON_WARM_RESET	30
PHY_RST_TIMER	29:12
P2NOBEACON_ENABLE	11
L1_CLK_SEL	10
L1_NOWAIT_P1	9
L1SUB_EXIT_MODE	8
RSVDP_7	7
RXSTANDBY_CONTROL	6:0

Table 3-389 Fields for Register: PHY_INTEROP_CTRL_OFF

Bits	Name	Memory Access	Description
31	PIPE_OPTIONAL_PCLKCHANGE_HS	R/W	Enable the optional PIPE5 behavior for the PclkChangeOk/PclkChangeAck handshake. When enabled, the controller performs a PclkChangeOk/PclkChangeAck handshake with the PHY whenever a rate change occurs, even if the new and old rate use the same Pclk frequency. <i>Note:</i> This register field is sticky. Values: 0x0 (DISABLE): Disable the optional PIPE5 behavior for the PclkChangeOk/PclkChangeAck handshake 0x1 (ENABLE): Enable the optional PIPE5 behavior for the PclkChangeOk/PclkChangeAck handshake Value After Reset: CX_PIPE_OPTIONAL_PCLKCHANGE_HS Exists: CX_PIPE_PCLK_MODE_1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
30	PHY_PERST_ON_WARM_RESET	R/W	Control whether the Power Management Controller will drive pm_req_phy_perst during a Warm Reset (PERST# assertion without cycling of power). <i>Note:</i> This register field is sticky. <i>Note:</i> This register field is sticky. Values: 0x0 (DISABLE): DO NOT Drive pm_req_phy_perst during a Warm Reset. 0x1 (ENABLE): Drive pm_req_phy_perst during a Warm Reset. Value After Reset: DEFAULT_PHY_PERST_ON_WARM_RESET Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
29:12	PHY_RST_TIMER	R/W	Control the duration of the PHY reset (PIPE and PMA). Duration is in aux clock cycles (0 to 262,142). <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VAL): Zero value. 0x3ffff (MAX_VAL): Max value. Value After Reset: CX_PHY_RST_TIMER Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
11	P2NOBEACON_ENABLE	R/W	P2.NoBeacon Enable bit.<i>Note:</i>This field is reserved (fixed to '0') if CX_P2NOBEACON_ENABLE is not set. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (NO_BEACON_ENCODING): Controller drives P2.NoBeacon encoding for PHY power down state, when the link goes to L2. ■ 0x0 (ENCODING): Controller drives P2 encoding for PHY power down state, when the link goes to L2. Value After Reset: CX_P2NOBEACON_ENABLE Exists: CX_P2NOBEACON_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
10	L1_CLK_SEL	R/W	L1 Clock control bit. This field is reserved for internal use. You should not write to this field and change the default unless specifically instructed by Synopsys support. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (NO_REQ): Controller does not request aux_clk switch and core_clk gating in L1. ■ 0x0 (CAN_REQ): Controller requests aux_clk switch and core_clk gating in L1. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
9	L1_NOWAIT_P1	* Varies	L1 entry control bit. This field is reserved for internal use. You should not write to this field and change the default unless specifically instructed by Synopsys support. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (NO_WAIT_FOR_ACK): Controller does not wait for PHY to acknowledge transition to P1 before entering L1. ■ 0x0 (WAIT_FOR_ACK): Controller waits for the PHY to acknowledge transition to P1 before entering L1. Value After Reset: {(CX_PIPE43_SUPPORT== 1) ? "0x1" : "0x0"} Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_PIPE43_SUPPORT==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && CX_PIPE43_SUPPORT) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_PIPE43_SUPPORT==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && CX_PIPE43_SUPPORT) ? \"read-only\" : (CX_PIPE43_SUPPORT==0) ? \"read-write\" : \"read-only\""
8	L1SUB_EXIT_MODE	R/W	L1 Exit Control Using phy_mac_pclkack_n. This field is reserved for internal use. You should not write to this field and change the default unless specifically instructed by Synopsys support. If PCLK as PHY input is selected, you should not write to this field and change the default value since PCLK as PHY input does not support any value other than the default. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (NO_WAIT_FOR_PCLKACK): Controller exits L1 without waiting for the PHY to assert phy_mac_pclkack_n. ■ 0x0 (WAIT_FOR_PCLKACK): Controller waits for the PHY to assert phy_mac_pclkack_n before exiting L1. Value After Reset: 0x0 Exists: CX_L1_SUBSTATES_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
6:0	RXSTANDBY_CONTROL	R/W	Rxstandby Control. Bits 0..5 determine if the controller asserts the RxStandby signal (mac_phy_rxstandby) in the indicated condition. Bit 6 enables the controller to perform the RxStandby/RxStandbyStatus handshake. This field is reserved for internal use. You should not write to this field and change the default unless specifically instructed by Synopsys support. ■ [0]: Rx EIOS and subsequent T TX-IDLE-MIN ■ [1]: Rate Change ■ [2]: Inactive lane for upconfigure/downconfigure ■ [3]: PowerDown=P1orP2 ■ [4]: RxL0s.Idle ■ [5]: EI Infer in L0 ■ [6]: Execute RxStandby/RxStandbyStatus Handshake <i>Note:</i> This register field is sticky. Values: ■ 0x0 (Min_bits_value): Zero value ■ 0x7f (Max_bits_value): Max value Value After Reset: CX_RXSTANDBY_CONTROL Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.41 TRGT_CPL_LUT_DELETE_ENTRY_OFF

- **Name:** TRGT_CPL_LUT Delete Entry Control register.
- **Description:** Using this register you can delete one entry in the target completion LUT. You should only use this register when you know that your application will never send the completion because of an FLR or any other reason.
Note:: The target completion LUT (and associated target completion timeout event) is watching for completions (from your application on XALI0/1/2 or AXI master read channel) corresponding to previously received non-posted requests from the PCIe wire.
- **Size:** 32 bits
- **Offset:** B+0x1c8
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	30:0
DELETE_EN	LOOK_UP_ID

Table 3-390 Fields for Register: TRGT_CPL_LUT_DELETE_ENTRY_OFF

Bits	Name	Memory Access	Description
31	DELETE_EN	W	This is a one-shot bit. This is a self-clearing register field. Reading from this register field always returns a '0'. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): A '1' write to this bit triggers the deletion of the target completion LUT entry that is specified in the LOOK_UP_ID field. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
30:0	LOOK_UP_ID	R/W	This number selects one entry to delete of the TRGT_CPL_LUT. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.42 LINK_FLUSH_CONTROL_OFF

- **Name:** Link Reset Request Flush Control Register.
- **Description:** This register controls link reset request flush behaviour.
- **Size:** 32 bits
- **Offset:** B+0x1cc
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:24	RSVD_I_8
23:1	RSVDP_1
0	AUTO_FLUSH_EN

Table 3-391 Fields for Register: LINK_FLUSH_CONTROL_OFF

Bits	Name	Memory Access	Description
31:24	RSVD_I_8	R/W	This is an internally reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0xff Exists: (AMBA_INTERFACE>1) && CC_DEVICE_TYPE!=3 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AMBA_INTERFACE>1) && CC_DEVICE_TYPE!=3 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
0	AUTO_FLUSH_EN	R/W	Enables automatic flushing of pending requests before sending the reset request to the application logic to reset the PCIe controller and the AXI Bridge. The flushing process is initiated if any of the following events occur: ■ Hot reset request. A downstream port (DSP) can "hot reset" an upstream port (USP) by sending two consecutive TS1 ordered sets with the hot reset bit asserted. ■ Warm (Soft) reset request. Generated when exiting from D3 to D0 and <code>cfg_pm_no_soft_rst=0</code>. ■ Link down reset request. A high to low transition on <code>smlh_req_rst_not</code> indicates the link has gone down and the controller is requesting a reset. If you disable automatic flushing, your application is responsible for resetting the PCIe controller and the AXI Bridge. For more information see "Warm and Hot Resets" section in the Architecture chapter of the <i>Databook</i>. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x1 Exists: (AMBA_INTERFACE>1) && CC_DEVICE_TYPE!=3 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.43 AMBA_ERROR_RESPONSE_DEFAULT_OFF

- **Name:** AXI Bridge Slave Error Response Register.
- **Description:** This register holds the AXI bridge slave error responses.
- **Size:** 32 bits
- **Offset:** B+0x1d0
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

AXI_MSTR_CID_EN	31
AXI_SLV_RID_EN	30
RSVDP_16	29:16
AMBA_ERROR_RESPONSE_MAP	15:10
RSVDP_5	9:5
AMBA_ERROR_RESPONSE_CRS	4:3
AMBA_ERROR_RESPONSE_VENDORID	2
RSVDP_1	1
AMBA_ERROR_RESPONSE_GLOBAL	0

Table 3-392 Fields for Register: AMBA_ERROR_RESPONSE_DEFAULT_OFF

Bits	Name	Memory Access	Description
31	AXI_MSTR_CID_EN	* Varies	Enable override of internally generated Completer ID for AXI master completions with value provided in mstr_bmisc_info_cid and mstr_rmisc_info_cid. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Completer ID is generated internally. ■ 0x1 (ENABLE): Completer ID is provided in mstr_bmisc_info_cid and mstr_rmisc_info_cid. Value After Reset: 0x0 Exists: CC_AXI_MSTR_MISC_INFO_CID_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(((CX_MEMORY_MAP_VIEW==0) (CX_MEMORY_MAP_VIEW==2)) && (CC_AXI_MSTR_MISC_INFO_CID_ENABLE==1)) ? \"read-write\" : \"read-only\" "
30	AXI_SLV_RID_EN	* Varies	Enable override of internally generated Requester ID for AXI slave requests with value provided in slv_awmisc_info_rid and slv_armisc_info_rid. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Requester ID is generated internally. ■ 0x1 (ENABLE): Requester ID is provided in slv_awmisc_info_rid and slv_armisc_info_rid. Value After Reset: 0x0 Exists: CC_AXI_SLV_MISC_INFO_RID_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(((CX_MEMORY_MAP_VIEW==0) (CX_MEMORY_MAP_VIEW==2)) && (CC_AXI_SLV_MISC_INFO_RID_ENABLE==1)) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
29:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: AMBA_INTERFACE!=0 && CC_DEVICE_TYPE!=3 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:10	AMBA_ERROR_RESPONSE_MAP	R/W	AXI Slave Response Error Map. Allows you to selectively map the errors received from the PCIe completion (for non-posted requests) to the AXI slave responses, slv_rresp or slv_bresp. The recommended setting is SLVERR. CRS is always mapped to OKAY. ■ 0 □ 0x0: UR (unsupported request) -> DECERR □ 0x1: UR (unsupported request) -> SLVERR ■ 1 □ 0x0: CRS (configuration retry status) -> DECERR □ 0x1: CRS (configuration retry status) -> SLVERR ■ 2 □ 0x0: CA (completer abort) -> DECERR □ 0x1: CA (completer abort) -> SLVERR ■ 3: RESERVED (0x0) ■ 4: RESERVED (0x0) ■ 5 □ 0x0: Completion Timeout -> DECERR □ 0x1: Completion Timeout -> SLVERR. The AXI bridge internally drops (processes internally but not passed to your application) a completion that has been marked by the Rx filter as UC or MLF, and does not pass its status directly down to the slave interface. It waits for a timeout and then signals "Completion Timeout" to the slave interface. The controller sets the AXI slave read databus to 0xFFFF for all error responses. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (Min_value): Zero value ■ 0x3f (Max_value): Max value Value After Reset: 0x27 Exists: (AMBA_INTERFACE>1) && CC_DEVICE_TYPE!=3 && AXI_POPULATED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
9:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: AMBA_INTERFACE!=0 && CC_DEVICE_TYPE!=3 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4:3	AMBA_ERROR_RESPONSE_CR S	R/W	CRS Slave Error Response Mapping. Determines the AXI slave response for CRS completions. For more information see "Error Handling" in the AXI chapter of the <i>Databook</i> . <i>Note:</i> This register field is sticky. Values: 0x0 (OK_): OKAY 0x1 (OK_CRS_CMPL): OKAY with all FFFF_FFFF data for all CRS completions 0x2 (OK_READ_REQ): OKAY with FFFF_0001 data for CRS completions to vendor ID read requests, OKAY with FFFF_FFFF data for all other CRS completions 0x3 (SLVERR_DECERR): SLVERR/DECERR (the AXI_ERROR_RESPONSE_MAP field determines the PCIe-to-AXI Slave error response mapping) Value After Reset: 0x0 Exists: (AMBA_INTERFACE>1) && CC_DEVICE_TYPE!=3 && AXI_POPULATED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2	AMBA_ERROR_RESPONSE_VE NDORID	R/W	Vendor ID Non-existent Slave Error Response Mapping. Determines the AXI slave response for errors on reads to non-existent Vendor ID register. For more information see "Error Handling" in the AXI chapter of the <i>Databook</i> . <i>Note:</i> This register field is sticky. Values: 0x0 (_OK_): OKAY (with FFFF data). 0x1 (_ERR_): SLVERR/DECERR (the AXI_ERROR_RESPONSE_MAP field determines the PCIe-to-AXI Slave error response mapping) Value After Reset: 0x0 Exists: AMBA_INTERFACE!=0 && CC_DEVICE_TYPE!=3 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: AMBA_INTERFACE!=0 && CC_DEVICE_TYPE!=3 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	AMBA_ERROR_RESPONSE_GLOBAL	R/W	Global Slave Error Response Mapping. Determines the AXI slave response for all error scenarios on non-posted requests. For more information see "Error Handling" in the AXI chapter of the <i>Databook</i>. The error response mapping is not applicable to Non-existent Vendor ID register reads. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (OK_): OKAY (with FFFF data for non-posted requests) ■ 0x1 (ERR_): SLVERR/DECERR (the AXI_ERROR_RESPONSE_MAP field determines the PCIe-to-AXI Slave error response mapping) Value After Reset: 0x0 Exists: AMBA_INTERFACE!=0 && CC_DEVICE_TYPE!=3 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.44 AMBA_LINK_TIMEOUT_OFF

- **Name:** Link Down AXI Bridge Slave Timeout Register.
- **Description:** If your application AXI master issues outbound requests to the AXI bridge slave interface before the PCIe link is operational, the controller starts a "flush" timer. The timeout value of the timer is set by this register. If the timer times out before the PCIe link is operational, the bridge TX request queues are flushed. For more information, see the "AXI Bridge Initialization, Clocking and Reset" section in the AXI chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x1d4
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:9	RSVDP_9
8	LINK_TIMEOUT_ENABLE_DEFAULT
7:0	LINK_TIMEOUT_PERIOD_DEFAULT

Table 3-393 Fields for Register: AMBA_LINK_TIMEOUT_OFF

Bits	Name	Memory Access	Description
31:9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AMBA_INTERFACE>1) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
8	LINK_TIMEOUT_ENABLE_DEFAULT	R/W	Disable Flush. <i>Note:</i> This register field is sticky. Values: 0x0 (ENABLE): Enable 0x1 (DISABLE): You can disable the flush feature by setting this field to '1'. Value After Reset: CC_LINK_TIMEOUT_ENABLE_DEFAULT Exists: (AMBA_INTERFACE>1) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:0	LINK_TIMEOUT_PERIOD_DEFAULT	R/W	Timeout Value (ms). The timer will timeout and then flush the bridge TX request queues after this amount of time. The timer counts when there are pending outbound AXI slave interface requests and the PCIe TX link is not transmitting any of these requests. The timer is clocked by core_clk. <i>Note:</i> This register field is sticky. Value After Reset: CC_LINK_TIMEOUT_PERIOD_DEFAULT Exists: (AMBA_INTERFACE>1) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.45 AMBA_ORDERING_CTRL_OFF

- **Name:** AXI Bridge Ordering Control.
- **Description:** This register controls the AXI Bridge Ordering.
- **Size:** 32 bits
- **Offset:** B+0x1d8
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:8	RSVDP_8
7	AX_MSTR_ZEROREAD_FW
6:5	RSVDP_5
4:3	AX_MSTR_ORDR_P_EVENT_SEL
2	RSVDP_2
1	AX_SNP_EN
0	RSVDP_0

Table 3-394 Fields for Register: AMBA_ORDERING_CTRL_OFF

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AMBA_INTERFACE>=2) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
7	AX_MSTR_ZEROLREAD_FW	R/W	AXI Master Zero Length Read Forward to the application. The DW PCIe controller AXI bridge is able to terminate in order with the Posted transactions the zero length read, implementing the PCIe express flush semantics of the Posted transactions. Values: ■ 0x0 (_0_LN_RD_TERMINATE): The zero length Read is terminated at the DW PCIe AXI bridge master ■ 0x1 (_0_LN_RD_FWD): The zero length Read is forward to the application. Value After Reset: 0x0 Exists: (AMBA_INTERFACE>=2) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
6:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AMBA_INTERFACE>=2) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:3	AX_MSTR_ORDR_P_EVENT_SEL	R/W	AXI Master Posted Ordering Event Selector. This field selects how the master interface determines when a P write is completed when enforcing the PCIe ordering rule, "NP must not pass P" at the AXI Master Interface. The AXI protocol does not support ordering between channels. Therefore, NP reads can pass P on your AXI bus fabric. This can result in an ordering violation when the read overtakes a P that is going to the same address. Therefore, the bridge master does not issue any NP requests until all outstanding P writes reach their destination. It does this by waiting for the all of the write responses on the B channel. This can affect the performance of the master read channel. For scenarios where the interconnect serializes the AXI master "AW", "W" and "AR" channels, you can increase the performance by reducing the need to wait until the complete Posted transaction has effectively reached the application slave. Note: This setting will not affect: ■ MSI interrupt catcher and P data ordering. This is always driven by the B'last event. ■ DMA read engine TLP ordering. This is always driven by the B'last event. ■ NP write transactions which are always serialized with P write transactions. Values: ■ 0x0 (B): B'last event: wait for the all of the write responses on the B channel thereby ensuring that the complete Posted transaction has effectively reached the application slave (default). ■ 0x1 (AW): AW'last event: wait until the complete Posted transaction has left the AXI address channel at the bridge master. ■ 0x2 (W): W'last event: wait until the complete Posted transaction has left the AXI data channel at the bridge master. ■ 0x3 (RSVD): Reserved Value After Reset: 0x0 Exists: (AMBA_INTERFACE >= 2) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE == 2 CX_MEMORY_MAP_PCIE_TYPE == 0))
2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AMBA_INTERFACE >= 2) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE == 2 CX_MEMORY_MAP_PCIE_TYPE == 0))

Bits	Name	Memory Access	Description
1	AX_SNP_EN	R/W	AXI Serialize Non-Posted Requests Enable. This field enables the AXI Bridge to serialize same ID Non-Posted Read/Write Requests on the wire. Serialization implies one outstanding same ID NP Read or Write on the wire and used to avoid AXI RAR and WAW hazards at the remote link partner. For more information, see the "Optional Serialization of AXI Slave Non-posted Requests" section in the AXI chapter of the <i>Databook</i>. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: (AMBA_INTERFACE>=2) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (AMBA_INTERFACE>=2) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.46 AMBA_ORDRMGR_WDOG_OFF

- **Name:** AHB Ordering Manager Watchdog Timer.
- **Description:** For more information, see the Ordering section in the AHB chapter of the *Databook*.
- **Size:** 32 bits
- **Offset:** B+0x1dc
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	AMBA_ORDRMGR_WDOG_EN
30:16	RSVDP_16
15:0	AMBA_ORDRMGR_WDOG

Table 3-395 Fields for Register: AMBA_ORDRMGR_WDOG_OFF

Bits	Name	Memory Access	Description
31	AMBA_ORDRMGR_WDOG_EN	R/W	AHB Ordering Manager Watchdog Timer Enable. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DIS): Disable ■ 0x1 (EN): Enable Value After Reset: 0x1 Exists: AMBA_INTERFACE == 1 && TRGT1_POPULATE==1 && CX_RADMQ_MODE==2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
30:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: AMBA_INTERFACE == 1 && TRGT1_POPULATE==1 && CX_RADMQ_MODE==2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:0	AMBA_ORDRMGR_WDOG	R/W	Initial load value for the AHB ordering manager watchdog timer. It has a resolution of 2us. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (_2_US): 2 us ■ 0x2 (_4_US): 4 us ■ 0x3 (_6_US): 6 us ■ 0xffff (MAX_VAL): ~131 ms Value After Reset: 0xffff Exists: AMBA_INTERFACE == 1 && TRGT1_POPULATE==1 && CX_RADMQ_MODE==2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.47 COHERENCY_CONTROL_1_OFF

- **Name:** Cache Coherency Control Register 1.
- **Description:** This register controls the cache coherency operation.
- **Size:** 32 bits
- **Offset:** B+0x1e0
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:2		
CFG_MEMTYPE_BOUNDARY_LOW_ADDR	1	0
RSVDP_1		
CFG_MEMTYPE_VALUE		

Table 3-396 Fields for Register: COHERENCY_CONTROL_1_OFF

Bits	Name	Memory Access	Description
31:2	CFG_MEMTYPE_BOUNDARY_LO W_ADDR	R/W	Boundary Lower Address For Memory Type. Bits [31:0] of dword-aligned address of the boundary for Memory type. The two lower address LSBs are '00'. Addresses up to but not including this value are in the lower address space region; addresses equal or greater than this value are in the upper address space region. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: AMBA_INTERFACE>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: AMBA_INTERFACE>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	CFG_MEMTYPE_VALUE	R/W	Sets the memory type for the lower and upper parts of the address space: <i>Note:</i> This register field is sticky. Values: ■ 0x0 (LOWER_PREPH): lower = Peripheral; upper = Memory ■ 0x1 (LOWER_MEM): lower = Memory type; upper = Peripheral Value After Reset: 0x0 Exists: AMBA_INTERFACE>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.48 COHERENCY_CONTROL_2_OFF

- **Name:** Cache Coherency Control Register 2.
- **Description:** This register controls the cache coherency operation.
- **Size:** 32 bits
- **Offset:** B+0x1e4
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-397 Fields for Register: COHERENCY_CONTROL_2_OFF

Bits	Name	Memory Access	Description
31:0	CFG_MEMTYPE_BOUNDARY_HIGH_ADDR	R/W	Boundary Upper Address For Memory Type. Bits [63:32] of the 64-bit dword-aligned address of the boundary for Memory type. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AMBA_INTERFACE>1) && CX_FLT_Q_ADDR_GT_32 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.49 COHERENCY_CONTROL_3_OFF

- **Name:** Cache Coherency Control Register 3.
- **Description:** This register controls the cache coherency operation.
- **Size:** 32 bits
- **Offset:** B+0x1e8
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_31	31
CFG_MSTR_AWCACHE_VALUE	30:27
RSVDP_26	26
CFG_MSTR_AWDOMAIN_VALUE	25:24
RSVDP_23	23
CFG_MSTR_ARCACHE_VALUE	22:19
RSVDP_18	18
CFG_MSTR_ARDOMAIN_VALUE	17:16
RSVDP_15	15
CFG_MSTR_AWCACHE_MODE	14:11
RSVDP_10	10
CFG_MSTR_AWDOMAIN_MODE	9:8
RSVDP_7	7
CFG_MSTR_ARCACHE_MODE	6:3
RSVDP_2	2
CFG_MSTR_ARDOMAIN_MODE	1:0

Table 3-398 Fields for Register: COHERENCY_CONTROL_3_OFF

Bits	Name	Memory Access	Description
31	RSVDP_31	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
30:27	CFG_MSTR_AWCACHE_VALUE	R/W	Master Write CACHE Signal Value. Value of the individual bits in mstr_awcache when CFG_MSTR_AWCACHE_MODE is '1'. <i>Note:</i> Not applicable to message requests; for message requests the value of mstr_awcache is always '0000'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: AMBA_INTERFACE>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
25:24	CFG_MSTR_AWDOMAIN_VALUE	R/W	Master Write DOMAIN Signal Value. Value of the individual bits in mstr_awdomain when CFG_MSTR_AWDOMAIN_MODE is '1'. <i>Note:</i> Not applicable to message requests; for message requests the value of mstr_awdomain is always '11' <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
22:19	CFG_MSTR_ARCACHE_VALUE	R/W	Master Read CACHE Signal Value. Value of the individual bits in mstr_arcache when CFG_MSTR_ARCACHE_MODE is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: AMBA_INTERFACE>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
17:16	CFG_MSTR_ARDOMAIN_VALUE	R/W	Master Read DOMAIN Signal Value. Value of the individual bits in mstr_ardomain when CFG_MSTR_ARDOMAIN_MODE is '1' <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15	RSVDP_15	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
14:11	CFG_MSTR_AWCACHE_MODE	R/W	Master Write CACHE Signal Behavior. Defines how the individual bits in mstr_awcache are controlled. <i>Note:</i> for message requests the value of mstr_awcache is always "0000" regardless of the value of this bit. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (SET_BY_AXI_MASTER): set automatically by the AXI master ■ 0x1 (SET_BY_FIELD): set by the value of the corresponding bit of the CFG_MSTR_AWCACHE_VALUE field Value After Reset: 0x0 Exists: AMBA_INTERFACE>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
9:8	CFG_MSTR_AWDOMAIN_MODE	R/W	Master Write DOMAIN Signal Behavior. Defines how the individual bits in mstr_awdomain[1:0] are controlled. <i>Note:</i> for message requests the value of mstr_awdomain is always '11' regardless of the value of this bit. <i>Note:</i> This register field is sticky. Values: 0x0 (SET_BY_AXI_MASTER): set automatically by the AXI master 0x1 (SET_BY_FIELD): set by the value of the corresponding bit of the CFG_MSTR_AWDOMAIN_VALUE field Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
6:3	CFG_MSTR_ARCACHE_MODE	R/W	Master Read CACHE Signal Behavior. Defines how the individual bits in mstr_arcache are controlled. <i>Note:</i> This register field is sticky. Values: 0x0 (SET_BY_AXI_MASTER): set automatically by the AXI master 0x1 (SET_BY_FIELD): set by the value of the corresponding bit of the CFG_MSTR_ARCACHE_VALUE field Value After Reset: 0x0 Exists: AMBA_INTERFACE>1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
1:0	CFG_MSTR_ARDOMAIN_MODE	R/W	Master Read DOMAIN Signal Behavior. Defines how the individual bits in mstr_ardomain[1:0] are controlled: <i>Note:</i> This register field is sticky. Values: ■ 0x0 (SET_BY_AXI_MASTER): set automatically by the AXI master ■ 0x1 (SET_BY_FIELD): set the value of the corresponding bit of the CFG_MSTR_ARDOMAIN_VALUE field Value After Reset: 0x0 Exists: CC_ACELITE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.50 AXI_MSTR_MSG_ADDR_LOW_OFF

- **Name:** Lower 32-bits of the Programmable AXI Address.
- **Description:** Lower 20 bits of the programmable AXI address to which Messages coming from wire are mapped. Bits [11:0] of the register are tied to zero for the address to be 4k-aligned. In previous releases, the third and fourth DWORDs of a message (Msg/MsgD) TLP header were delivered through the AXI master address bus (mstr_awaddr). These DWORDS are now supplied through the mstr_awmisc_info_hdr_34dw[63:0] output; and the value on mstr_awaddr is driven to the value you have programmed into the AXI_MSTR_MSG_ADDR_LOW_OFF and AXI_MSTR_MSG_ADDR_HIGH_OFF registers.
- **Size:** 32 bits
- **Offset:** B+0x1f0
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:12	CFG_AXIMSTR_MSG_ADDR_LOW
11:0	CFG_AXIMSTR_MSG_ADDR_LOW_RESERVED

Table 3-399 Fields for Register: AXI_MSTR_MSG_ADDR_LOW_OFF

Bits	Name	Memory Access	Description
31:12	CFG_AXIMSTR_MSG_ADDR_LO W	R/W	Lower 20-bits of the programmable AXI address for Messages. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: AMBA_INTERFACE>=2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
11:0	CFG_AXIMSTR_MSG_ADDR_LO W_RESERVED	R	Reserved for future use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: AMBA_INTERFACE>=2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.51 AXI_MSTR_MSG_ADDR_HIGH_OFF

- **Name:** Upper 32-bits of the Programmable AXI Address.
- **Description:** Upper 32 bits of the programmable AXI address to which Messages coming from wire are mapped.
- **Size:** 32 bits
- **Offset:** B+0x1f4
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

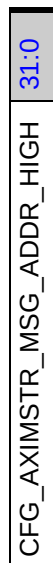


Table 3-400 Fields for Register: AXI_MSTR_MSG_ADDR_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	CFG_AXIMSTR_MSG_ADDR_HIGH	R/W	Upper 32 bits of the programmable AXI address for Messages. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: AMBA_INTERFACE>=2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.52 PCIE_VERSION_NUMBER_OFF

- **Name:** PCIe Controller IIP Release Version Number.
- **Description:** The version number is given in hex format. You should convert each pair of hex characters to ASCII to interpret.
Using 4.70a (GA) as an example:
 - VERSION_NUMBER = 0x3437302a which translates to 470*
 - VERSION_TYPE = 0x67612a2a which translates to ga**
 Using 4.70a-ea01 as an example:
 - VERSION_NUMBER = 0x3437302a which translates to 470*
 - VERSION_TYPE = 0x65613031 which translates to ea01
 GA is a general release available on www.designware.com
 EA is an early release available on a per-customer basis.
- **Size:** 32 bits
- **Offset:** B+0x1f8
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-401 Fields for Register: PCIE_VERSION_NUMBER_OFF

Bits	Name	Memory Access	Description
31:0	VERSION_NUMBER	R	Version Number. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: See IP-XACT file or Register reports in coreConsultant. (32-bit hex). Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.53 PCIE_VERSION_TYPE_OFF

- **Name:** PCIe Controller IIP Release Version Type.
- **Description:** The type is given in hex format. You should convert each pair of hex characters to ASCII to interpret.
Using 4.70a (GA) as an example:
 - VERSION_NUMBER = 0x3437302a which translates to 470*
 - VERSION_TYPE = 0x67612a2a which translates to ga**
 Using 4.70a-ea01 as an example:
 - VERSION_NUMBER = 0x3437302a which translates to 470*
 - VERSION_TYPE = 0x65613031 which translates to ea01
 GA is a general release available on www.designware.com
 EA is an early release available on a per-customer basis.
- **Size:** 32 bits
- **Offset:** B+0x1fc
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

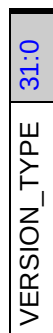


Table 3-402 Fields for Register: PCIE_VERSION_TYPE_OFF

Bits	Name	Memory Access	Description
31:0	VERSION_TYPE	R	Version Type. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: See IP-XACT file or Register reports in coreConsultant. (32-bit hex). Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.54 IATU_VIEWPORT_OFF

- **Name:** iATU Index Register.
- **Description:** The iATU registers are programmed through an indirect addressing scheme (using this index register) to reduce the address footprint in the PCI Express extended configuration space. The size of the required port logic space does not depend on the number of regions defined as the index register is used to select which memory region is being accessed. There are 28 bytes of register space implemented per address region per direction. The number of address regions that are remapped by the iATU is configurable in each direction from '0' (none) to 256. However, only 32 bytes of the PCIe extended configuration space address map is used.
Note: When register does not exist, value is fixed to 32'hFFFF_FFFF
- **Size:** 32 bits
- **Offset:** B+0x200
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	x:y	x:0
REGION_DIR	VP_RSVD	REGION_INDEX

Table 3-403 Fields for Register: IATU_VIEWPORT_OFF

Bits	Name	Memory Access	Description
31	REGION_DIR	R/W	Region Direction. Defines the region being accessed as either outbound or inbound. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (Outbound): Outbound access ■ 0x1 (Inbound): Inbound access Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
x:y	VP_RSVD	R	Reserved. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !ICC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Range Variable[x]: $31 - (\max(1, \lceil \log(\max(CX_ATU_NUM_OUTBOUND_REGIONS, CX_ATU_NUM_INBOUND_REGIONS)) / \log(2) \rceil)) + \max(1, \lceil \log(\max(CX_ATU_NUM_OUTBOUND_REGIONS, CX_ATU_NUM_INBOUND_REGIONS)) / \log(2) \rceil) - 1$ Range Variable[y]: $\max(1, \lceil \log(\max(CX_ATU_NUM_OUTBOUND_REGIONS, CX_ATU_NUM_INBOUND_REGIONS)) / \log(2) \rceil)$
x:0	REGION_INDEX	R/W	Region Index. Defines which region is being accessed when writing to the control, base, limit, and target registers. Must not be set to a number greater than CX_ATU_NUM_OUTBOUND_REGIONS - 1 when an outbound region is being accessed. Must not be set to a value greater than CX_ATU_NUM_INBOUND_REGIONS - 1 when an inbound region is being accessed. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !ICC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Range Variable[x]: $\max(1, \lceil \log(\max(CX_ATU_NUM_OUTBOUND_REGIONS, CX_ATU_NUM_INBOUND_REGIONS)) / \log(2) \rceil) - 1$

3.48.55 IATU_REGION_CTRL_1_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Region Control 1 Register.
- **Description:** This register controls the iATU outbound region access based on the optional iATU outbound features enabled using iATU Region Control 2 Register.
- **Size:** 32 bits
- **Offset:** B+0x204
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_23	31:23
CTRL_1_FUNC_NUM	22:20
PH	19:18
AT	17:16
RSVDP_14	15:14
INCREASE_REGION_SIZE	13
TH	12
IDO	11
ATTR	10:9
TD	8
TC	7:5
TYPE	4:0

Table 3-404 Fields for Register: IATU_REGION_CTRL_1_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
22:20	CTRL_1_FUNC_NUM	R/W	Function Number. When the address of an outbound TLP is matched to this region and the FUNC_BYPASS field in the "iATU Region Control 2 Register" is '0', then the function number used in generating the function part of the requester ID (RID) field of the TLP is taken from this 5-bit register. The value in this register must be 0x0 unless multifunction operation in the controller is enabled (CX_NFUNC > 1). When you are using the AXI Bridge, then this field is swapped before AXI decomposition occurs so that the correct "Max_Read_Request_Size" and "Max_Payload_Size" values are used. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !ICC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
19:18	PH	R/W	When the address of an outbound TLP is matched to this region, then the PH field of the TLP is changed to the value in this register. The setting is independent of the setting of the TH field. Only Valid when the CX_TPH_ENABLE configuration parameter is 1. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !ICC_UNROLL_ENABLE) && CX_TPH_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
17:16	AT	R/W	AT. When the address of an outbound TLP is matched to this region, then the AT field of the TLP is changed to the value in this register. Only valid when the ATS_ENABLE configuration parameter is 1. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !ICC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
13	INCREASE_REGION_SIZE	R/W	Increase the maximum ATU Region size. When set, the maximum ATU Region size is determined by CX_ATU_MAX_REGION_SIZE. When clear, the maximum ATU Region size is 4 GB (default). <i>Note:</i> This register field is sticky. Values: 0x1 (ENABLE): Maximum ATU Region size is determined by CX_ATU_MAX_REGION_SIZE 0x0 (DISABLE): Maximum ATU Region size is 4 GB (default) Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
12	TH	R/W	When the address of an outbound TLP is matched to this region, then the TH field of the TLP is changed to the value in this register. Only Valid when the CX_TPH_ENABLE configuration parameter is 1. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_TPH_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
11	IDO	R/W	When the address of an outbound TLP is matched to this region, then the IDO(ATTR[2]) field of the TLP is changed to the value in this register. Only valid when the CX_IDO_ENABLE configuration parameter is 1. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_IDO_ENABLE && CC_DEVICE_TYPE!=3 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
10:9	ATTR	R/W	When the address of an outbound TLP is matched to this region, then the ATTR field of the TLP is changed to the value in this register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
8	TD	R/W	This is a reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
7:5	TC	R/W	When the address of an outbound TLP is matched to this region, then the TC field of the TLP is changed to the value in this register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
4:0	TYPE	R/W	When the address of an outbound TLP is matched to this region, then the TYPE field of the TLP is changed to the value in this register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !ICC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.48.56 IATU_REGION_CTRL_1_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Region Control 1 Register.
- **Description:** This register controls the iATU inbound region access based on the optional iATU inbound features enabled using iATU Region Control 2 Register.
- **Size:** 32 bits
- **Offset:** +0x904
- **Exists:** !CX_PL_REG_DISABLE && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE)

RSVDP_23	31:23
CTRL_1_FUNC_NUM	22:20
PH	19:18
AT	17:16
RSVDP_14	15:14
INCREASE_REGION_SIZE	13
TH	12
IDO	11
ATTR	10:9
TD	8
TC	7:5
TYPE	4:0

Table 3-405 Fields for Register: IATU_REGION_CTRL_1_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

Bits	Name	Memory Access	Description
22:20	CTRL_1_FUNC_NUM	R/W	Function Number. MEM-I/O: When the Address and BAR matching logic in the controller indicates that a MEM-I/O transaction matches a BAR in the function corresponding to this value, then address translation proceeds. This check is only performed if the "Function Number Match Enable" bit of the "iATU Region Control 2 Register" is set. CFG0/CFG1: When the destination function number as specified in the routing ID of the TLP header matches the function, then address translation proceeds. This check is only performed if the "Function Number Match Enable" bit of the "iATU Region Control 2 Register" is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
19:18	PH	R/W	When the PH field of an inbound TLP is matched to this value, then address translation proceeds. This check is only performed if the "PH Match Enable" bit of the "iATU Region Control 2 Register" is set. The setting is independent of the setting of the TH field. Only Valid when the CX_TPH_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_TPH_ENABLE Testable: writeAsRead

Bits	Name	Memory Access	Description
17:16	AT	R/W	When the AT field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "AT Match Enable" bit of the "iATU Region Control 2 Register" is set. Only valid when the ATS_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
13	INCREASE_REGION_SIZE	R/W	Increase the maximum ATU Region size. When set, the maximum ATU Region size is determined by CX_ATU_MAX_REGION_SIZE. When clear, the maximum ATU Region size is 4 GB (default). <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Maximum ATU Region size is determined by CX_ATU_MAX_REGION_SIZE ■ 0x0 (DISABLE): Maximum ATU Region size is 4 GB (default) Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

Bits	Name	Memory Access	Description
12	TH	R/W	When the TH field of an inbound TLP is matched to this value, then address translation proceeds. This check is only performed if the "TH Match Enable" bit of the "iATU Region Control 2 Register" is set. Only Valid when the CX_TPH_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_TPH_ENABLE Testable: writeAsRead
11	IDO	R/W	When the IDO field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "ATTR Match Enable" bit of the "iATU Region Control 2 Register" is set. Only Valid when the CX_IDO_ENABLE configuration parameter is set to '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_IDO_ENABLE && CC_DEVICE_TYPE!=3 Testable: writeAsRead
10:9	ATTR	R/W	When the ATTR field of an inbound tlp is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). this check is only performed if the "ATTR Match Enable" bit of the "iATU Region Control 2 Register" is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

Bits	Name	Memory Access	Description
8	TD	R/W	When the TD field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TD Match Enable" bit of the "iATU Region Control 2 Register" is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
7:5	TC	R/W	When the TC field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TC Match Enable" bit of the "iATU Region Control 2 Register" is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
4:0	TYPE	R/W	When the TYPE field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

3.48.57 IATU_REGION_CTRL_2_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Region Control 2 Register.
- **Description:** Using this register you can enable/disable the outbound iATU optional features.
- **Size:** 32 bits
- **Offset:** B+0x208
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	REGION_EN
30	RSVDP_30
29	INVERT_MODE
28	CFG_SHIFT_MODE
27	DMA_BYPASS
26:24	RSVDP_24
23	HEADER_SUBSTITUTE_EN
22	INHIBIT_PAYLOAD
21	TLP_HEADER_FIELDS_BYPASS
20	SNP
19	FUNC_BYPASS
18:17	MSB2BITS_TAG
16	TAG_SUBSTITUTE_EN
15:8	TAG
7:0	MSG_CODE

Table 3-406 Fields for Register: IATU_REGION_CTRL_2_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31	REGION_EN	R/W	Region Enable. This bit must be set to '1' for address translation to take place. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
30	RSVDP_30	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
29	INVERT_MODE	R/W	Invert Mode. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
28	CFG_SHIFT_MODE	R/W	CFG Shift Mode. The iATU uses bits [27:12] of the untranslated address (on the XALI0/1/2 interface or AXI slave interface address) to form the BDF number of the outgoing CFG TLP. This supports the Enhanced Configuration Address Mapping (ECAM) mechanism (Section 7.2.2 of the PCI Express Base 3.1 Specification, revision 1.0) by allowing all outgoing I/O and MEM TLPs (that have been translated to CFG) to be mapped from memory space into any 256 MB region of the PCIe configuration space. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
27	DMA_BYPASS	R/W	DMA Bypass Mode. Allows request TLPs which are initiated by the DMA controller to pass through the iATU untranslated. <i>Note:</i> This field is reserved for the SW product. You must set it to '0'. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CC_DMA_ENABLE && CC_DEVICE_TYPE!=3 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
26:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
23	HEADER_SUBSTITUTE_EN	R/W	Header Substitute Enable. When enabled and region address is matched, the iATU fully substitutes bytes 8-11 (for 3 DWORD header) or bytes 12-15 (for 4 DWORD header) of the outbound TLP header with the contents of the LWR_TARGET_RW field in IATU_LWR_TARGET_ADDR_VIEWPORT_OFF_OUTBOUND_i. <i>Note:</i> This register field is sticky. Values: 0x1 (FILL_BYTES): LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_VIEWPORT_OFF_OUTBOUND_i register is used to fill bytes 8-to-11 (for 3 DWORD header) or bytes 12-to-15 (for 4 DWORD header) of the translated TLP header. 0x0 (FORMS_ADDR): LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_VIEWPORT_OFF_OUTBOUND_i register forms the new address of the translated region. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !ICC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
22	INHIBIT_PAYLOAD	R/W	Inhibit TLP Payload Data for TLPs in Matched Region; assign iATU region to be TLP without data. When enabled and region address is matched, the iATU marks all TLPs as having no payload data by forcing the TLP header Fmt[1] bit =0, regardless of the application inputs such as slv_wstrb. Encoding are define as above. <i>Note:</i> This register field is sticky. Values: 0x1 (TLP_NO_DATA): Fmt[1] =0 so that only TLP type without data is sent. For example, a Msg instead of MsgD will be sent. 0x0 (TLP_WITH_OR_WITHOUT_DATA): Fmt[1] =0/1 so that TLPs with or without data can be sent. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !ICC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
21	TLP_HEADER_FIELDS_BYPASS	R/W	TLP Header Fields Translation Bypass. In this mode header fields of the translated TLP is taken from your application transmit interface, or if AMBA is configured, from the AMBA sideband bus (slv_awmisc_info) and not from the corresponding fields of the IATU_REGION_CTRL_1_OFF_OUTBOUND_i or IATU_REGION_CTRL_2_OFF_OUTBOUND_i registers. The header fields are ■ TC ■ PH ■ TH ■ ST ■ AT ■ Attr (IDO, RO and NS) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
20	SNP	R/W	Serialize Non-Posted Requests. In this mode, when the AXI Bridge is populated, same AXI ID Non-Posted Read/Write Requests are transmitted on the wire if there are no other same ID Non-Posted Requests outstanding. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && AXI_POPULATED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
19	FUNC_BYPASS	R/W	Function Number Translation Bypass. In this mode, the function number of the translated TLP is taken from your application transmit interface and not from the CTRL_1_FUNC_NUM field of the "iATU Region Control 1 Register" or the VF_NUMBER field of the "iATU Control 3 Register." <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
18:17	MSB2BITS_TAG	R/W	Not used. Reserved for future use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_10BITS_TAG && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
16	TAG_SUBSTITUTE_EN	R/W	TAG Substitute Enable. When enabled and region address is matched, the iATU substitutes the TAG field of the outbound TLP header with the contents of the TAG field in this register. The expected usage scenario is translation from AXI MWr to Vendor Defined Msg/MsgD. <i>Note (CX_TPH_ENABLE=1):</i> TAG substitution for MWr will not occur because this field (byte 6) in the TLP header is the ST field. ST substitution can still take place using the MSG_CODE field in IATU_REGION_CTRL_2_VIEWPORT_OFF_OUTBOUND_i. <i>Note (CX_10BITS_TAG =1):</i> For 10-bit tags, TAG substitution only applies to the 8 least significant bits of the TAG field (bits 7:0 of Byte 4 of the TLP Header). T9 and T8 of the Header are not substituted. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead
15:8	TAG	R/W	TAG. The substituted TAG field (byte 6) in the outgoing TLP header when TAG_SUBSTITUTE_EN is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
7:0	MSG_CODE	R/W	MSG TLPs (Message Code). When the address of an outbound TLP is matched to this region, and the translated TLP TYPE field is Msg or MsgD; then the message field of the TLP is changed to the value in this register. Memory TLPs: (ST;Steering Tag). When the ST field of an outbound TLP is matched to this region, and the translated TLP TYPE field targets memory space; then the ST field of the TLP is changed to the value in this register. Only Valid when the CX_TPH_ENABLE configuration parameter is 1. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.48.58 IATU_REGION_CTRL_2_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Region Control 2 Register.
- **Description:** Using this register you can enable/disable the inbound iATU optional features.
- **Size:** 32 bits
- **Offset:** +0x908
- **Exists:** !CX_PL_REG_DISABLE && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE)

31	REGION_EN
30	MATCH_MODE
29	INVERT_MODE
28	CFG_SHIFT_MODE
27	FUZZY_TYPE_MATCH_CODE
26	VFBAR_MATCH_MODE_EN
25:24	RESPONSE_CODE
23	SINGLE_ADDR_LOC_TRANS_EN
22	PH_MATCH_EN
21	MSG_CODE_MATCH_EN
20	VF_MATCH_EN
19	FUNC_NUM_MATCH_EN
18	AT_MATCH_EN
17	TH_MATCH_EN
16	ATTR_MATCH_EN
15	TD_MATCH_EN
14	TC_MATCH_EN
13	MSG_TYPE_MATCH_MODE
12:11	RSVDP_11
10:8	BAR_NUM
7:0	MSG_CODE

Table 3-407 Fields for Register: IATU_REGION_CTRL_2_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31	REGION_EN	R/W	Region Enable. This bit must be set to '1' for address translation to take place. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

Bits	Name	Memory Access	Description
30	MATCH_MODE	R/W	Match Mode. Determines Inbound matching mode for TLPs. The mode depends on the type of TLP that is received as follows: For MEM/IO TLPs, this field is interpreted as follows: ■ 0: Address Match Mode. The iATU operates using addresses as in the outbound direction. The Region Base and Limit Registers must be setup. ■ 1: BAR Match Mode. BAR matching is used. The "BAR Number" field is relevant. Not used for RC. For CFG0 TLPs, this field is interpreted as follows: ■ 0: Routing ID Match Mode. The iATU interprets the Routing ID (Bytes 8 to 11 of TLP header) as an address. This corresponds to the upper 16 bits of the address in MEM-I/O transactions. The Routing ID of the TLP must be within the base and limit of the iATU region for matching to proceed. ■ 1: Accept Mode. The iATU accepts all CFG0 transactions as address matches. The routing ID in the CFG0 TLP is ignored. This is useful as all received CFG0 TLPs should be processed regardless of the Bus number. For MSG/MSGD TLPs, this field is interpreted as follows: ■ 0: Address Match Mode. The iATU treats the third dword and fourth dword of the inbound MSG/MSGD TLP as an address and it is matched against the Region Base and Limit Registers.

Bits	Name	Memory Access	Description
30...(cont.)	MATCH_MODE.	R/W	1: Vendor ID Match Mode. This mode is relevant for ID-routed Vendor Defined Messages. The iATU ignores the Routing ID (Bus, Device, Function) in bits [31:16] of the third dword of the TLP header, but matches against the Vendor ID in bits [15:0] of the third dword of the TLP header. Bits [15:0] of the Region Upper Base register should be programmed with the required Vendor ID. The lower Base and Limit Register should be programmed to translate TLPs based on vendor specific information in the fourth dword of the TLP header. - If SINGLE_ADDRESS_LOCATION_TRANSLATE_EN =1 AND MSG_TYPE_MATCH_MODE =1, then Match Mode is ignored. <i>Note:</i> This register field is sticky. Values: 0x0 (ZERO): The interpretation is dependent on TLP type, that is, MEM/IO, CFG0, or MSG/MSGD TLPs. 0x1 (ONE): The interpretation is dependent on TLP type, that is, MEM/IO, CFG0, or MSG/MSGD TLPs. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
29	INVERT_MODE	R/W	Invert Mode Enable. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). When set all regions of that type must use address match mode. <i>Note:</i> This register field is sticky. Values: 0x1 (ENABLE): Invert Mode Enable 0x0 (DISABLE): Invert Mode Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

Bits	Name	Memory Access	Description
28	CFG_SHIFT_MODE	R/W	CFG Shift Enable. This is useful for CFG transactions where the PCIe configuration mechanism maps bits [27:12] of the address to the bus/device and function number. This allows a CFG configuration space to be located in any 256MB window of your application memory space using a 28-bit effective address. Shifts bits [31:16] of the untranslated address to form bits [27:12] of the translated address. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): CFG Shift Enable ■ 0x0 (DISABLE): CFG Shift Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
27	FUZZY_TYPE_MATCH_CODE	R/W	Fuzzy Type Match Enable. When enabled, the iATU relaxes the matching of the TLP TYPE field against the expected TYPE field so that ■ CfgRd0 and CfgRd1 TLPs are seen as identical. Similarly with CfgWr0 and CfgWr1. ■ MWr, MRd and MRdLk TLPs are seen as identical ■ The Routing field of Msg/MsgD TLPs is ignored ■ FetchAdd, Swap and CAS are seen as identical. For example, CFG0 in the TYPE field in the "iATU Region Control 1 Register" matches against an inbound CfgRd0, CfgRd1, CfgWr0, or CfgWr1 TLP. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Fuzzy Type Match Enable ■ 0x0 (DISABLE): Fuzzy Type Match Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

Bits	Name	Memory Access	Description
26	VFBAR_MATCH_MODE_EN	R/W	VF BAR Match Enable. When set, Virtual Function BAR matching is used which allows all VFs in a PF which match a BAR to be matched with a single ATU region. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Virtual Function BAR Match Enable ■ 0x0 (DISABLE): Virtual Function BAR Match Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 Testable: writeAsRead
25:24	RESPONSE_CODE	R/W	Response Code. Defines the type of response to give for accesses matching this region. This overrides the normal RADM filter response. Note that this feature is not available for any region where Single Address Location Translate is enabled. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NORMAL_RADM_FILTER): Normal RADM filter response is used. ■ 0x1 (UNSUPPORTED_REQUEST): Unsupported request (UR) ■ 0x2 (COMPLETER_ABORT): Completer abort (CA) ■ 0x3 (RESERVED): Not used / undefined / reserved. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

Bits	Name	Memory Access	Description
23	SINGLE_ADDR_LOC_TRANS_EN	R/W	Single Address Location Translate Enable. When enabled, Rx TLPs can be translated to a single address location as determined by the target address register of the iATU region. The main usage scenario is translation of Messages (such as Vendor Defined or ATS Messages) to MWr TLPs when the AXI bridge is enabled. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
22	PH_MATCH_EN	R/W	PH Match Enable. Ensures that a successful PH TLP field comparison match (see PH field of the "iATU Region Control 1 Register") occurs for address translation to proceed. Only Valid when the CX_TPH_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): PH Match Enable ■ 0x0 (DISABLE): PH Match Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_TPH_ENABLE Testable: writeAsRead

Bits	Name	Memory Access	Description
21	MSG_CODE_MATCH_EN	R/W	Message Code Match Enable (Msg TLPS). Ensures that a successful message Code TLP field comparison match (see Message Code field of the "iATU Region Control 2 Register") occurs (in MSG transactions) for address translation to proceed. ST Match Enable (Mem TLPS). Ensures that a successful ST TLP field comparison match (see ST field of the "iATU Region Control 2 Register") occurs (in MEM transactions) for address translation to proceed. Only Valid when the CX_TPH_ENABLE configuration parameter is 1 <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Message Code Match Enable (for Msg TLPS) or ST Match Enable (for Mem TLPS) ■ 0x0 (DISABLE): Virtual Function Number Match Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
20	VF_MATCH_EN	R/W	Virtual Function Number Match Enable. Ensures that a successful Virtual Function Number TLP field comparison match (see Virtual Function Number field of the "iATU Control 3 Register") occurs (in MEM-I/O transactions) for address translation to proceed. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Virtual Function Number Match Enable ■ 0x0 (DISABLE): Virtual Function Number Match Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 Testable: writeAsRead

Bits	Name	Memory Access	Description
19	FUNC_NUM_MATCH_EN	R/W	Function Number Match Enable. Ensures that a successful Function Number TLP field comparison match (see Function Number field of the "iATU Region Control 1 Register") occurs (in MEM-I/O and CFG0/CFG1 transactions) for address translation to proceed. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Function Number Match Enable ■ 0x0 (DISABLE): Function Number Match Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
18	AT_MATCH_EN	R/W	AT Match Enable. Ensures that a successful AT TLP field comparison match (see AT field of the "iATU Region Control 1 Register") occurs for address translation to proceed. Only valid when the ATS_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): AT Match Enable ■ 0x0 (DISABLE): AT Match Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE && CC_DEVICE_TYPE!=3) Testable: writeAsRead

Bits	Name	Memory Access	Description
17	TH_MATCH_EN	R/W	TH Match Enable. Ensures that a successful TH TLP field comparison match (see TH field of the "iATU Region Control 1 Register") occurs for address translation to proceed. Only Valid when the CX_TPH_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Values: 0x1 (ENABLE): TH Match Enable 0x0 (DISABLE): TH Match Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_TPH_ENABLE Testable: writeAsRead
16	ATTR_MATCH_EN	R/W	ATTR Match Enable. Ensures that a successful ATTR TLP field comparison match (see ATTR field of the "iATU Region Control 1 Register") occurs for address translation to proceed. <i>Note:</i> This register field is sticky. Values: 0x1 (ENABLE): ATTR Match Enable 0x0 (DISABLE): ATTR Match Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
15	TD_MATCH_EN	R/W	TD Match Enable. Ensures that a successful TD TLP field comparison match (see TD field of the "iATU Region Control 1 Register") occurs for address translation to proceed. <i>Note:</i> This register field is sticky. Values: 0x1 (ENABLE): TD Match Enable 0x0 (DISABLE): TD Match Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

Bits	Name	Memory Access	Description
14	TC_MATCH_EN	R/W	TC Match Enable. Ensures that a successful TC TLP field comparison match (see TC field of the "iATU Region Control 1 Register") occurs for address translation to proceed. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): TC Match Enable ■ 0x0 (DISABLE): TC Match Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
13	MSG_TYPE_MATCH_MODE	R/W	Message Type Match Mode. When enabled, and if single address location translate enable is set, then inbound TLPs of type MSG/MSGd which match the type field of the IATU_REGION_CTRL_1_VIEWPORT_OFF_INBOUND_i register (=>TYPE[4:3]=2'b10) is translated. Message type match mode overrides any value of MATCH_MODE field in this register. Usage scenarios for this are translation of VDM or ATS messages when AXI bridge is configured on client interface. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
12:11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

Bits	Name	Memory Access	Description
10:8	BAR_NUM	R/W	BAR Number. When the BAR number of an inbound MEM or IO TLP "that is matched by the normal internal BAR address matching mechanism" is the same as this field, address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Match Mode" bit of the "iATU Region Control 2 Register" is set. IO translation would require either 00100b or 00101b in the inbound TLP TYPE; the BAR Number set in the range 000b - 101b and that BAR configured as an IO BAR. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BAR0): BAR0 ■ 0x1 (BAR1): BAR1 ■ 0x2 (BAR2): BAR2 ■ 0x3 (BAR3): BAR3 ■ 0x4 (BAR4): BAR4 ■ 0x5 (BAR5): BAR5 ■ 0x6 (ROM): ROM ■ 0x7 (RSVD): Reserved Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead
7:0	MSG_CODE	R/W	MSG TLPs: (Message Code). When the MSG_CODE field of an inbound Msg/MsgD TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Message Code Match Enable" bit of the "iATU Region Control 2 Register" is set and the TYPE field of the inbound TLP is Msg/MsgD. Memory TLPs: (ST:Steering Tag). When the ST field of an inbound TLP is matched to this value, then address translation proceeds. This check is only performed if the "ST Match Enable" bit of the "iATU Control2 Register" is set. The setting is independent of the setting of the TH field. Only Valid when the CX_TPH_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

3.48.59 IATU_LWR_BASE_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Lower Base Address Register.
- **Description:** The CX_ATU_MIN_REGION_SIZE configuration parameter (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ bits are zero.
- **Size:** 32 bits
- **Offset:** B+0x20c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

x:y	LWR_BASE_RW
x:0	LWR_BASE_HW

Table 3-408 Fields for Register: IATU_LWR_BASE_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:y	LWR_BASE_RW	R/W	Forms bits [31:n] of the start address of the address region to be translated. n is $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Range Variable[x]: $32 - (\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)) + \log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2) - 1$ Range Variable[y]: $\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)$

Bits	Name	Memory Access	Description
x:0	LWR_BASE_HW	R	Forms bits [$n-1:0$] of the start address of the address region to be translated. The start address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !ICC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Range Variable[x]: $\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2) - 1$

3.48.60 IATU_LWR_BASE_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Lower Base Address Register.
- **Description:** The CX_ATU_MIN_REGION_SIZE configuration parameter (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB; defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ bits are zero.
- **Size:** 32 bits
- **Offset:** +0x90c
- **Exists:** !CX_PL_REG_DISABLE && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE)

x:y	x:0
LWR_BASE_RW	LWR_BASE_HW

Table 3-409 Fields for Register: IATU_LWR_BASE_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:y	LWR_BASE_RW	R/W	Forms bits [31:n] of the start address of the address region to be translated. n is $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead Range Variable[x]: $32 - (\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)) + \log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2) - 1$ Range Variable[y]: $\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)$

Bits	Name	Memory Access	Description
x:0	LWR_BASE_HW	R	Forms bits $[n-1:0]$ of the start address of the address region to be translated. The start address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead Range Variable[x]: $\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2) - 1$

3.48.61 IATU_UPPER_BASE_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Upper Base Address Register.
- **Description:** This register holds the upper 32-bits of the start (and end) address of the address region to be translated.
- **Size:** 32 bits
- **Offset:** B+0x210
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

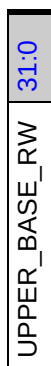


Table 3-410 Fields for Register: IATU_UPPER_BASE_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:0	UPPER_BASE_RW	R/W	Forms bits [63:32] of the start (and end) address of the address region to be translated. In systems with a 32-bit address space, this register is not used and therefore writing to this register has no effect. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.48.62 IATU_UPPER_BASE_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Upper Base Address Register.
- **Description:** This register holds the upper 32 bits of the start (and end) address of the address region to be translated.
- **Size:** 32 bits
- **Offset:** +0x910
- **Exists:** !CX_PL_REG_DISABLE && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE)

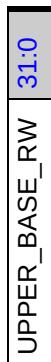


Table 3-411 Fields for Register: IATU_UPPER_BASE_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:0	UPPER_BASE_RW	R/W	Forms bits [63:32] of the start (and end) address of the address region to be translated. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

3.48.63 IATU_LIMIT_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Limit Address Register.
- **Description:** This register holds the end address of the address region to be translated.
- **Size:** 32 bits
- **Offset:** B+0x214
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

x:y	x:4	3:0
LIMIT_ADDR_RW	LIMIT_ADDR_HW	CBUF_INCR

Table 3-412 Fields for Register: IATU_LIMIT_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:y	LIMIT_ADDR_RW	R/W	Forms upper bits of the end address of the address region to be translated. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Range Variable[x]: $32 - (\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)) + \log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2) - 1$ Range Variable[y]: $\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)$

Bits	Name	Memory Access	Description
x:4	LIMIT_ADDR_HW	R	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary, so these bits are always all ones. A write to this location is ignored by the PCIe controller. <i>Note:</i> This register field is sticky. Value After Reset: {(CX_ATU_MIN_REGION_SIZE == 65536) ? "0xfff" : (CX_ATU_MIN_REGION_SIZE == 32768) ? "0x7ff" : (CX_ATU_MIN_REGION_SIZE == 16384) ? "0x3ff" : (CX_ATU_MIN_REGION_SIZE == 8192) ? "0x1ff" : (CX_ATU_MIN_REGION_SIZE == 4096) ? "0xff" : "0xffff"} Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Range Variable[x]: $\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2) - 4 + 3$
3:0	CBUF_INCR	R	Circular Buffer. <i>Note:</i> This register field is sticky. Value After Reset: 0xf Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.48.64 IATU_LIMIT_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Limit Address Register.
- **Description:** This register holds the end address of the address region to be translated.
- **Size:** 32 bits
- **Offset:** +0x914
- **Exists:** !CX_PL_REG_DISABLE && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE)

x:y	x:4	3:0
LIMIT_ADDR_RW	LIMIT_ADDR_HW	CBUF_INCR

Table 3-413 Fields for Register: IATU_LIMIT_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:y	LIMIT_ADDR_RW	R/W	Forms upper bits of the end address of the address region to be translated. When SINGLE_ADDR_LOC_TRANS_EN = 1, MSG_TYPE_MATCH_MODE = 1, and CBUF_INCR > 0 then this field forms the upper bits of the limit address for the circular buffer. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead Range Variable[x]: $32 - (\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)) + \log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2) - 1$ Range Variable[y]: $\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)$

Bits	Name	Memory Access	Description
x:4	LIMIT_ADDR_HW	R	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary, so these bits are always all ones. When SINGLE_ADDR_LOC_TRANS_EN = 1, MSG_TYPE_MATCH_MODE = 1, and CBUF_INCR > 0 then this field forms lower bits of the limit address for the circular buffer. A write to this location is ignored by the PCIe controller. <i>Note:</i> This register field is sticky. Value After Reset: {(CX_ATU_MIN_REGION_SIZE == 65536) ? "0xfff" : (CX_ATU_MIN_REGION_SIZE == 32768) ? "0x7ff" : (CX_ATU_MIN_REGION_SIZE == 16384) ? "0x3ff" : (CX_ATU_MIN_REGION_SIZE == 8192) ? "0x1ff" : (CX_ATU_MIN_REGION_SIZE == 4096) ? "0xff" : "0xffff"} Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead Range Variable[x]: $\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2) - 4 + 3$

Bits	Name	Memory Access	Description
3:0	CBUF_INCR	* Varies	Circular Buffer Increment. When CX_ATU_SLOC_CBUF = 0, then this field is Read-only and forms the lowest bits of the end address of the address region to be translated. When CX_ATU_SLOC_CBUF = 1, then this field is R/W and forms the upper bits of the Circular Buffer Increment size (CBUF_INCR) field for Single Location Address translation. The increment value (in bytes) is decoded as follows: Note: A write to any bit in the CBUF_INCR field resets the circular buffer pointer - that is, the next matched received Message will be translated to the start address of the Circular Buffer. This field must be written to AFTER the target and limit registers have been updated. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (Def_0): 0 (Default; legacy Single Address Location mode) ■ 0x1 (addr_4): 4 ■ 0x2 (addr_8): 8 ■ 0x3 (addr_16): 16 ■ 0x4 (addr_32): 32 ■ 0x5 (addr_64): 64 ■ 0x6 (addr_128): 128 ■ 0x7 (addr_256): 256 ■ 0x8 (addr_512): 512 ■ 0x9 (addr_1024): 1024 ■ 0xa (addr_2048): 2048 ■ 0xb (addr_4096): 4096 ■ 0xc (addr_8192): 8192 ■ 0xd (addr_rvd1): rsvd. ■ 0xe (addr_rvd2): rsvd. ■ 0xf (addr_rvd3): rsvd. Value After Reset: 0x0 Exists:

Bits	Name	Memory Access	Description
3:0...(cont.)	CBUF_INCR.	* Varies	(!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: untestable Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_ATU_SLOC_CBUF) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_ATU_SLOC_CBUF) ? \"read-only\": (CX_MEMORY_MAP_VIEW==1 && CX_ATU_SLOC_CBUF) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && !CX_ATU_SLOC_CBUF) ? \"read-only\": (CX_ATU_SLOC_CBUF)? \"read-write\": \"read-only\" "

3.48.65 IATU_LWR_TARGET_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Lower Target Address Register.
- **Description:** This register holds the Lower Target part of the new address of the translated region, or the outbound TLP header information, depending on the setting of HEADER_SUBSTITUTE_EN field of IATU_REGION_CTRL_2_VIEWPORT_OFF_OUTBOUND_i register.
- **Size:** 32 bits
- **Offset:** B+0x218
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-414 Fields for Register: IATU_LWR_TARGET_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:0	LWR_TARGET_RW_OUTBOUND	R/W	When HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_VIEWPORT_OFF_OUTBOUND_i is '0' (normal operation): ■ LWR_TARGET_RW[31:n] forms MSBs of the Lower Target part of the new address of the translated region; ■ LWR_TARGET_RW[n-1:0] are not used. (The start address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary, so the lower bits of the start address of the new address of the translated region (bits n-1:0) are always '0'). ■ n is $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$. When HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_VIEWPORT_OFF_OUTBOUND_i is '1': ■ LWR_TARGET_RW[31:0] forms bytes 8-11 (for 3 dword header) or bytes 12-15 (for 4 dword header) of the outbound TLP header. Usage scenarios include the transmission of Vendor Defined Messages where the controller determines the content of bytes 12 to 15 of the TLP header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !ICC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.48.66 IATU_LWR_TARGET_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Lower Target Address Register.
- **Description:** This register holds the Lower Target part of the new address of the translated region.
- **Size:** 32 bits
- **Offset:** +0x918
- **Exists:** !CX_PL_REG_DISABLE && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE)

x:y	LWR_TARGET_RW
x:0	LWR_TARGET_HW

Table 3-415 Fields for Register: IATU_LWR_TARGET_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:y	LWR_TARGET_RW	R/W	Forms MSB's of the Lower Target part of the new address of the translated region. These bits are always '0'. ■ Field size depends on $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ in address match mode. ■ Field size depends on $\log_2(\text{BAR_MASK}+1)$ in BAR match mode. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead Range Variable[x]: $32 - (\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)) + \log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2) - 1$ Range Variable[y]: $\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)$

Bits	Name	Memory Access	Description
x:0	LWR_TARGET_HW	R	Forms the LSB's of the Lower Target part of the new address of the translated region. The start address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary (in address match mode); and to the Bar size boundary (in BAR match mode) so that these bits are always '0'. If the BAR is smaller than the iATU region size, then the iATU target address must align to the iATU region size; otherwise it must align to the BAR size. A write to this location is ignored by the PCIe controller. - Field size depends on $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ in address match mode. - Field size depends on $\log_2(\text{BAR_MASK}+1)$ in BAR match mode. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead Range Variable[x]: $\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2) - 1$

3.48.67 IATU_UPPER_TARGET_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Upper Target Address Register.
- **Description:** This register holds the upper 32 bits of the start address (Upper Target part) of the new address of the translated region.
- **Size:** 32 bits
- **Offset:** B+0x21c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

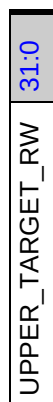


Table 3-416 Fields for Register: IATU_UPPER_TARGET_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:0	UPPER_TARGET_RW	R/W	Forms bits [63:32] of the start address (Upper Target part) of the new address of the translated region. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

3.48.68 IATU_UPPER_TARGET_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Upper Target Address Register.
- **Description:** This register holds the upper 32 bits of the start address (Upper Target part) of the new address of the translated region.
- **Size:** 32 bits
- **Offset:** +0x91c
- **Exists:** !CX_PL_REG_DISABLE && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE)

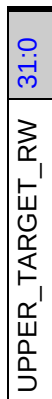


Table 3-417 Fields for Register: IATU_UPPER_TARGET_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:0	UPPER_TARGET_RW	R/W	Forms bits [63:32] of the start address (Upper Target part) of the new address of the translated region. In systems with a 32-bit address space, this register is not used and therefore writing to this register has no effect. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) Testable: writeAsRead

3.48.69 IATU_REGION_CTRL_3_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Region Control 3 Register.
- **Description:** This register holds the virtual function number used for iATU outbound region matching (wherever applicable).
- **Size:** 32 bits
- **Offset:** B+0x220
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	30:y	x:0
VF_ACTIVE	RSVDP_VF_NUMBER	VF_NUMBER

Table 3-418 Fields for Register: IATU_REGION_CTRL_3_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31	VF_ACTIVE	R/W	Virtual Function Active. Indicates that all of the outbound accesses are from virtual functions. For more information, see VF_ACTIVE field. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead

Bits	Name	Memory Access	Description
30:y	RSVDP_VF_NUMBER	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Range Variable[y]: CX_NVFUNC_WD - 1
x:0	VF_NUMBER	R/W	Virtual Function Number. When the address of an outbound TLP is matched to this region and the FUNC_BYPASS field in the "iATU Region Control 2 Register" is '0', then this field used in generating the Function part of the requester ID (RID) field of the TLP. Indicates from which virtual function (VF) the request is mapped. The iATU uses this signal to form the requester ID. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, a value of '0' here refers to the first VF, of the PF that is identified by the CTRL_1_FUNC_NUM field in the "iATU Control 1 Register" For more information, see "RID BDF Number Replacement." The VF_ACTIVE bit must be set to '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Range Variable[x]: CX_NVFUNC_WD - 2

3.48.70 IATU_REGION_CTRL_3_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Region Control 3 Register.
- **Description:** This register holds the virtual function number used for iATU inbound region matching (wherever applicable).
- **Size:** 32 bits
- **Offset:** +0x920
- **Exists:** !CX_PL_REG_DISABLE && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1

31	30:y	x:0
VF_ACTIVE	RSVDP_VF_NUMBER	VF_NUMBER

Table 3-419 Fields for Register: IATU_REGION_CTRL_3_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31	VF_ACTIVE	R	Reserved. Not used. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 Testable: writeAsRead

Bits	Name	Memory Access	Description
30:y	RSVDP_VF_NUMBER	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 Testable: writeAsRead Range Variable[y]: CX_NVFUNC_WD - 1
x:0	VF_NUMBER	R/W	Virtual Function Number. When the Address and BAR matching logic in the controller indicates that a MEM-I/O transaction matches a BAR in the virtual function corresponding to this value, then address translation proceeds. This check is only performed if the "Virtual Function Number Match Enable" bit of the "iATU Region Control 2 Register" is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 Testable: writeAsRead Range Variable[x]: CX_NVFUNC_WD - 2

3.48.71 IATU_UPPR_LIMIT_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Upper Limit Address Register.
- **Description:** The CX_MAX_ATU_REGION_SIZE configuration parameter (Value Range: 0(4GB) to 32(16 EB)) specifies the maximum region size of an address translation region. This register is only used when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_VIEWPORT_OFF_OUTBOUND_i is '1'.
- **Size:** 32 bits
- **Offset:** B+0x224
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:y	x:0
UPPR_LIMIT_ADDR_HW	UPPR_LIMIT_ADDR_RW

Table 3-420 Fields for Register: IATU_UPPR_LIMIT_ADDR_VIEWPORT_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:y	UPPR_LIMIT_ADDR_HW	R	Forms MSBs of the Upper Limit part of the region "end address" to be translated. Only applies to 64-bit systems and when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_VIEWPORT_OFF_OUTBOUND_i is '1'. These bits are always '0'. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_ATU_MAX_REGION_SIZE>0 && CX_ATU_INCR_RGN_SIZE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Range Variable[y]: CX_ATU_MAX_REGION_SIZE

Bits	Name	Memory Access	Description
x:0	UPPR_LIMIT_ADDR_RW	R/W	Forms the LSBs of the Upper Limit part of the region "end address" to be translated. Only applies to 64-bit systems and when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_VIEWPORT_OFF_OUTBOUND_i is '1' <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && !ICC_UNROLL_ENABLE) && CX_ATU_MAX_REGION_SIZE>0 && CX_ATU_INCR_RGN_SIZE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Range Variable[x]: CX_ATU_MAX_REGION_SIZE - 1

3.48.72 IATU_UPPR_LIMIT_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Upper Limit Address Register.
- **Description:** The CX_ATU_MAX_REGION_SIZE configuration parameter (Value Range: 0(4GB) to 32(16 EB)) specifies the maximum region size of an address translation region. This register is only used when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_VIEWPORT_OFF_INBOUND_i is '1'.
- **Size:** 32 bits
- **Offset:** +0x924
- **Exists:** !CX_PL_REG_DISABLE && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_ATU_MAX_REGION_SIZE>0 && CX_ATU_INCR_RGN_SIZE

31:y	UPPR_LIMIT_ADDR_HW
x:0	UPPR_LIMIT_ADDR_RW

Table 3-421 Fields for Register: IATU_UPPR_LIMIT_ADDR_VIEWPORT_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:y	UPPR_LIMIT_ADDR_HW	R	Forms MSB's of the Upper Limit part of the region "end address" to be translated. Only applies to 64-bit systems and when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_VIEWPORT_OFF_INBOUND_i is '1'. These bits are always '0'. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_ATU_MAX_REGION_SIZE>0 && CX_ATU_INCR_RGN_SIZE Testable: writeAsRead Range Variable[y]: CX_ATU_MAX_REGION_SIZE

Bits	Name	Memory Access	Description
x:0	UPPR_LIMIT_ADDR_RW	R/W	Forms the LSB's of the Upper Limit part of the region "end address" to be translated. Only applies to 64-bit systems and when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_VIEWPORT_OFF_INBOUND_i is '1' <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) && (CX_INTERNAL_ATU_ENABLE && !CC_UNROLL_ENABLE) && CX_ATU_MAX_REGION_SIZE>0 && CX_ATU_INCR_RGN_SIZE Testable: writeAsRead Range Variable[x]: CX_ATU_MAX_REGION_SIZE - 1

3.48.73 INTERFACE_TIMER_CONTROL_OFF

- **Name:** Interface Timer Control Register.
- **Description:** This register controls the interface timer.
- **Size:** 32 bits
- **Offset:** B+0x230
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:5	RSVDP_5
4	FORCE_PENDING
3:2	INTERFACE_TIMER_SCALING
1	INTERFACE_TIMER_AER_EN
0	INTERFACE_TIMER_EN

Table 3-422 Fields for Register: INTERFACE_TIMER_CONTROL_OFF

Bits	Name	Memory Access	Description
31:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_INTERFACE_TIMER_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4	FORCE_PENDING	R/W	Writing to this bit forces the value of the pending flags. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_INTERFACE_TIMER_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
3:2	INTERFACE_TIMER_SCALING	R/W	Interface timer scaling. This field can be used to reduce the timer duration for verification purpose. This field should only be programmed when the INTERFACE_TIMER_EN bit is set to 1'b0. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_INTERFACE_TIMER_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
1	INTERFACE_TIMER_AER_EN	R/W	When set to 1 the Interface timer internal uncorrectable error generation is enabled. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): The Interface timer internal uncorrectable error generation is enabled. Value After Reset: 0x0 Exists: CX_INTERFACE_TIMER_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	INTERFACE_TIMER_EN	R/W	Interface timer enable. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_INTERFACE_TIMER_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.74 INTERFACE_TIMER_TARGET_OFF

- **Name:** Interface Timer Target Register.
- **Description:** This is the interface timer target register.
- **Size:** 32 bits
- **Offset:** B+0x234
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	RSVDP_16
15:0	INTERFACE_TIMER_TARGET

Table 3-423 Fields for Register: INTERFACE_TIMER_TARGET_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_INTERFACE_TIMER_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:0	INTERFACE_TIMER_TARGET	R/W	Interface timer target value. This field should only be programmed when the INTERFACE_TIMER_EN bit is set to 1'b0. <i>Note:</i> This register field is sticky. Value After Reset: 0x32 Exists: CX_INTERFACE_TIMER_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.75 INTERFACE_TIMER_STATUS_OFF

- **Name:** Interface Timer Status Register.
- **Description:** This is the interface timer status register.
- **Size:** 32 bits
- **Offset:** B+0x238
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:15	RSVDP_15
14	DBI_SLAVE_RD_ADD_TIMEOUT
13	DBI_SLAVE_WR_DATA_TIMEOUT
12	DBI_SLAVE_WR_ADD_TIMEOUT
11	SLAVE_RD_ADD_TIMEOUT
10	SLAVE_WR_DATA_TIMEOUT
9	SLAVE_WR_ADD_TIMEOUT
8	MASTERH_RD_DATA_TIMEOUT
7	MASTERH_WR_RES_TIMEOUT
6	MASTER_RD_DATA_TIMEOUT
5	MASTER_WR_RES_TIMEOUT
4	CLIENT2_INTERFACE_TIMEOUT
3	CLIENT1_INTERFACE_TIMEOUT
2	CLIENT0_INTERFACE_TIMEOUT
1	CPL_INTERFACE_TIMEOUT
0	MESSAGE_INTERFACE_TIMEOUT

Table 3-424 Fields for Register: INTERFACE_TIMER_STATUS_OFF

Bits	Name	Memory Access	Description
31:15	RSVDP_15	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && DBI_4SLAVE_POPULATED) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
14	DBI_SLAVE_RD_ADD_TIMEOUT	R/W1C	DBI slave read address channel timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && DBI_4SLAVE_POPULATED) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
13	DBI_SLAVE_WR_DATA_TIMEOUT	R/W1C	DBI slave write data channel timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && DBI_4SLAVE_POPULATED) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
12	DBI_SLAVE_WR_ADD_TIMEOUT	R/W1C	DBI slave write address channel timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && DBI_4SLAVE_POPULATED) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
11	SLAVE_RD_ADD_TIMEOUT	R/W1C	Slave read address channel timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && AMBA_POPULATED) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
10	SLAVE_WR_DATA_TIMEOUT	R/W1C	Slave write data channel timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && AMBA_POPULATED) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
9	SLAVE_WR_ADD_TIMEOUT	R/W1C	Slave write address channel timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && AMBA_POPULATED) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
8	MASTERH_RD_DATA_TIMEOUT	R/W1C	High priority Master read data channel timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && MASTER_POPULATED && CC_HP_MASTER) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7	MASTERH_WR_RES_TIMEOUT	R/W1C	High priority Master write response channel timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && MASTER_POPULATED && CC_HP_MASTER) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
6	MASTER_RD_DATA_TIMEOUT	R/W1C	Master read data channel timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && MASTER_POPULATED) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
5	MASTER_WR_RES_TIMEOUT	R/W1C	Master write response channel timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && MASTER_POPULATED) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
4	CLIENT2_INTERFACE_TIMEOUT	R/W1C	Client2 interface timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && CLIENT2_POPULATED) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
3	CLIENT1_INTERFACE_TIMEOUT	R/W1C	Client1 interface timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && CLIENT1_POPULATED) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2	CLIENT0_INTERFACE_TIMEOUT	R/W1C	Client0 interface timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERFACE_TIMER_EN && AMBA_INTERFACE==0) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
1	CPL_INTERFACE_TIMEOUT	R/W1C	CPL interface timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_INTERFACE_TIMER_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	MESSAGE_INTERFACE_TIMEOUT	R/W1C	Message interface timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_INTERFACE_TIMER_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.76 MSIX_ADDRESS_MATCH_LOW_OFF

- **Name:** MSI-X Address Match Low Register.
- **Description:** When you enable the MSI-X Table RAM feature (MSIX_TABLE_EN=1), the controller implements the logic and RAM required to generate MSI-X requests. For more information, see the Interrupts section in the "Controller Operations" chapter of the Databook. This register is only used in AXI configurations. When your local AXI application writes (MWr) to the address defined in this register (and MSIX_ADDRESS_MATCH_HIGH_OFF), the controller will load the MSIX_DOORBELL_OFF register with the contents of the MWr and subsequently create and send MSI-X TLPs
- **Size:** 32 bits
- **Offset:** B+0x240
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:2	1	0
MSIX_ADDRESS_MATCH_LOW	MSIX_ADDRESS_MATCH_RESERVED_1	MSIX_ADDRESS_MATCH_EN

Table 3-425 Fields for Register: MSIX_ADDRESS_MATCH_LOW_OFF

Bits	Name	Memory Access	Description
31:2	MSIX_ADDRESS_MATCH_LOW	R/W	MSI-X Address Match Low Address. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AMBA_INTERFACE>0) && MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
1	MSIX_ADDRESS_MATCH_RESERVED_1	R	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (AMBA_INTERFACE>0) && MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	MSIX_ADDRESS_MATCH_EN	R/W	MSI-X Match Enable. Enable the MSI-X Address Match feature when the AXI bridge is present. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable Value After Reset: 0x0 Exists: (AMBA_INTERFACE>0) && MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.77 MSIX_ADDRESS_MATCH_HIGH_OFF

- **Name:** MSIX Address Match High Register.
- **Description:** MSI-X Address Match High Register. When you enable the MSI-X Table RAM feature (MSIX_TABLE_EN=1), the controller implements the logic and RAM required to generate MSI-X requests. For more information, see the Interrupts section in the "Controller Operations" chapter of the Databook. This register is only used in AXI configurations. When your local AXI application writes (MWr) to the address defined in this register (and MSIX_ADDRESS_MATCH_LOW_OFF), the controller will load the MSIX_DOORBELL_OFF register with the contents of the MWr and subsequently create and send MSI-X TLPs
- **Size:** 32 bits
- **Offset:** B+0x244
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

 31:0
 MSIX_ADDRESS_MATCH_HIGH
Table 3-426 Fields for Register: MSIX_ADDRESS_MATCH_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	MSIX_ADDRESS_MATCH_HIGH	R/W	MSI-X Address Match High Address. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: (AMBA_INTERFACE>0) && MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.78 MSIX_DOORBELL_OFF

- **Name:** MSI-X Doorbell Register.
- **Description:** When you enable the MSI-X Table RAM feature (MSIX_TABLE_EN=1), the controller implements the logic and RAM required to generate MSI-X requests. For more information, see the Interrupts section in the "Controller Operations" chapter of the Databook.
 - For AXI configurations: when your local application writes (MWr) to the address defined in MSIX_ADDRESS_MATCH_LOW_OFF, the controller will load this register with the contents of the MWr and subsequently create and send MSI-X TLPs.
 - For non-AMBA configurations: when your local application writes to this register, the controller will create and send MSI-X TLPs.
- **Size:** 32 bits
- **Offset:** B+0x248
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:29	MSIX_DOORBELL_RESERVED_29_31
28:24	MSIX_DOORBELL_PF
23:16	MSIX_DOORBELL_VF
15	MSIX_DOORBELL_VF_ACTIVE
14:12	MSIX_DOORBELL_TC
11	MSIX_DOORBELL_RESERVED_11
10:0	MSIX_DOORBELL_VECTOR

Table 3-427 Fields for Register: MSIX_DOORBELL_OFF

Bits	Name	Memory Access	Description
31:29	MSIX_DOORBELL_RESERVED_29_31	W	Reserved. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
28:24	MSIX_DOORBELL_PF	W	MSIX Doorbell Physical Function. This register determines the Physical Function for the MSI-X transaction. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:16	MSIX_DOORBELL_VF	W	MSIX Doorbell Virtual Function. This register determines the Virtual Function for the MSI-X transaction. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15	MSIX_DOORBELL_VF_ACTIVE	W	MSIX Doorbell Virtual Function Active. This register determines whether a Virtual Function is used to generate the MSI-X transaction. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
14:12	MSIX_DOORBELL_TC	W	MSIX Doorbell Traffic Class. This register determines which traffic class to generate the MSI-X transaction with. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11	MSIX_DOORBELL_RESERVED_1 1	W	Reserved. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
10:0	MSIX_DOORBELL_VECTOR	W	MSI-X Doorbell Vector. This register determines which vector to generate the MSI-X transaction for. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.79 MSIX_RAM_CTRL_OFF

- **Name:** MSI-X RAM power mode and debug control register.
- **Description:** When you enable the MSI-X Table RAM feature (MSIX_TABLE_EN=1), the controller implements the logic and RAM required to generate MSI-X requests. For more information, see the Interrupts section in the "Controller Operations" chapter of the Databook.
- **Size:** 32 bits
- **Offset:** B+0x24c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

MSIX_RAM_CTRL_RESERVED_26_31	31:26
MSIX_RAM_CTRL_DBG_PBA	25
MSIX_RAM_CTRL_DBG_TABLE	24
MSIX_RAM_CTRL_RESERVED_17_23	23:17
MSIX_RAM_CTRL_BYPASS	16
MSIX_RAM_CTRL_RESERVED_10_15	15:10
MSIX_RAM_CTRL_PBA_SD	9
MSIX_RAM_CTRL_PBA_DS	8
MSIX_RAM_CTRL_RESERVED_2_7	7:2
MSIX_RAM_CTRL_TABLE_SD	1
MSIX_RAM_CTRL_TABLE_DS	0

Table 3-428 Fields for Register: MSIX_RAM_CTRL_OFF

Bits	Name	Memory Access	Description
31:26	MSIX_RAM_CTRL_RESERVED_26_31	R	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
25	MSIX_RAM_CTRL_DBG_PBA	R/W	MSIX PBA RAM Debug Mode. You can also use the dbg_pba input to activate debug mode. Debug mode turns off the PF/VF/Offset-based addressing into the RAM and maps the entire table linearly from the base address of the BAR (indicated by the BIR) in function 0. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable. Use this bit to activate the debug mode and allow direct read/write access to the PBA. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
24	MSIX_RAM_CTRL_DBG_TABLE	R/W	MSIX Table RAM Debug Mode. You can also use the dbg_table input to activate debug mode. Debug mode turns off the PF/VF/Offset-based addressing into the RAM and maps the entire table linearly from the base address of the BAR (indicated by the BIR) in function 0. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (DISABLE): Disable ■ 0x1 (ENABLE): Enable. Use this bit to activate the debug mode and allow direct read/write access to the Table. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:17	MSIX_RAM_CTRL_RESERVED_17_23	R	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
16	MSIX_RAM_CTRL_BYPASS	R/W	MSIX RAM Control Bypass. It is up to the application to ensure the RAMs are in the proper power state before trying to access them. Moreover, the application needs to observe all timing requirements of the RAM low power signals before trying to use the MSIX functionality. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): The bypass field, when set, disables the internal generation of low power signals for both RAMs. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:10	MSIX_RAM_CTRL_RESERVED_10_15	R	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
9	MSIX_RAM_CTRL_PBA_SD	R/W	MSIX PBA RAM Shut Down. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Set this bit to drive the cfg_msix_pba_sd output to signal your external logic to place the MSIX PBA RAM in Shut Down low-power mode. Value After Reset: 0x0 Exists: MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
8	MSIX_RAM_CTRL_PBA_DS	R/W	MSIX PBA RAM Deep Sleep. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Set this bit to drive the <code>cfg_msix_pba_ds</code> output to signal your external logic to place the MSIX PBA RAM in Deep Sleep low-power mode. Value After Reset: 0x0 Exists: <code>MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))</code>
7:2	MSIX_RAM_CTRL_RESERVED_2_7	R	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: <code>MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))</code>
1	MSIX_RAM_CTRL_TABLE_SD	R/W	MSIX Table RAM Shut Down. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Set this bit to drive the <code>cfg_msix_table_sd</code> output to signal your external logic to place the MSIX Table RAM in Shut Down low-power mode. Value After Reset: 0x0 Exists: <code>MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))</code>
0	MSIX_RAM_CTRL_TABLE_DS	R/W	MSIX Table RAM Deep Sleep. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Set this bit to drive the <code>cfg_msix_table_ds</code> output to signal your external logic to place the MSIX Table RAM in Deep Sleep low-power mode. Value After Reset: 0x0 Exists: <code>MSIX_TABLE_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))</code>

3.48.80 SAFETY_MASK_OFF

- **Name:** Functional Safety Interrupt Events Mask Register.
- **Description:** This registers holds the masks for functional safety interrupt events.
- **Size:** 32 bits
- **Offset:** B+0x260
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:6	RSVDP_6
5	SAFETY_INT_MASK_5
4	SAFETY_INT_MASK_4
3	SAFETY_INT_MASK_3
2	SAFETY_INT_MASK_2
1	SAFETY_INT_MASK_1
0	SAFETY_INT_MASK_0

Table 3-429 Fields for Register: SAFETY_MASK_OFF

Bits	Name	Memory Access	Description
31:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5	SAFETY_INT_MASK_5	R/W	Mask for functional safety interrupt event 5 (RASDP correctable). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4	SAFETY_INT_MASK_4	R/W	Mask for functional safety interrupt event 4 (PCIe correctable). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
3	SAFETY_INT_MASK_3	R/W	Mask for functional safety interrupt event 3 (PCIe uncorrectable). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2	SAFETY_INT_MASK_2	R/W	Mask for functional safety interrupt event 2 (CDM register checker). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
1	SAFETY_INT_MASK_1	R/W	Mask for functional safety interrupt event 1 (Interface timers). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	SAFETY_INT_MASK_0	R/W	Mask for functional safety interrupt event 0 (RASDP). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.81 SAFETY_STATUS_OFF

- **Name:** Status for functional safety interrupt events.
- **Description:** This register provides the functional safety interrupt events status.
- **Size:** 32 bits
- **Offset:** B+0x264
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:6	RSVDP_6
5	SAFETY_INT_STATUS_5
4	SAFETY_INT_STATUS_4
3	SAFETY_INT_STATUS_3
2	SAFETY_INT_STATUS_2
1	SAFETY_INT_STATUS_1
0	SAFETY_INT_STATUS_0

Table 3-430 Fields for Register: SAFETY_STATUS_OFF

Bits	Name	Memory Access	Description
31:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5	SAFETY_INT_STATUS_5	R/W1C	Status for functional safety interrupt event 5 (RASDP correctable). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
4	SAFETY_INT_STATUS_4	R/W1C	Status for functional safety interrupt event 4 (PCIe correctable). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
3	SAFETY_INT_STATUS_3	R/W1C	Status for functional safety interrupt event 3 (PCIe uncorrectable). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
2	SAFETY_INT_STATUS_2	R/W1C	Status for functional safety interrupt event 2 (CDM register checker). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
1	SAFETY_INT_STATUS_1	R/W1C	Status for functional safety interrupt event 1 (Interface timers). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	SAFETY_INT_STATUS_0	R/W1C	Status for functional safety interrupt event 0 (RASDP). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_AUTOMOTIVE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.82 PL_APP_BUS_DEV_NUM_STATUS_OFF

- **Name:** Application driven bus and device number register.
- **Description:** This register reflects the application driven bus and device number.
- **Size:** 32 bits
- **Offset:** B+0x410
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	RSVDP_16
15:8	RC_DSW_BUS_NUM
7:3	RC_DSW_DEV_NUM
2:0	RSVDP_0

Table 3-431 Fields for Register: PL_APP_BUS_DEV_NUM_STATUS_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:8	RC_DSW_BUS_NUM	R	This field reflects the value of bus number driven on app_bus_num input signal by your application. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
7:3	RC_DSW_DEV_NUM	R	This field reflects the value of device number driven on app_device_num input signal by your application. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.83 PCIPM_TRAFFIC_CTRL_OFF

- **Name:** TLP Traffic during Non-D0 State Control Register.
- **Description:** This register provides control over TLP Traffic during Non-D0 States.
- **Size:** 32 bits
- **Offset:** B+0x41c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:8	RSVDP_8
7:4	PCIPM_RESERVED_4_7
3	PCIPM_NEW_TLP_CLIENT2_BLOCKED
2	PCIPM_NEW_TLP_CLIENT1_BLOCKED
1	PCIPM_NEW_TLP_CLIENT0_BLOCKED
0	PCIPM_VDM_TRAFFIC_BLOCKED

Table 3-432 Fields for Register: PCIPM_TRAFFIC_CTRL_OFF

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:4	PCIPM_RESERVED_4_7	R	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
3	PCIPM_NEW_TLP_CLIENT2_BLOCKED	R/W	This field indicates that all TLPs transmitted by Client 2 interface are blocked during non-D0 states. <i>Note:</i> This register field is sticky. Value After Reset: CX_CLIENT2_BLOCK_NEW_TLP Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
2	PCIPM_NEW_TLP_CLIENT1_BLOCKED	R/W	This field indicates that all TLPs transmitted by Client 1 interface are blocked during non-D0 states. <i>Note:</i> This register field is sticky. Value After Reset: CX_CLIENT1_BLOCK_NEW_TLP Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
1	PCIPM_NEW_TLP_CLIENT0_BLOCKED	R/W	This field indicates that all TLPs transmitted by Client 0 interface are blocked during non-D0 states. <i>Note:</i> This register field is sticky. Value After Reset: CX_CLIENT0_BLOCK_NEW_TLP Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	PCIPM_VDM_TRAFFIC_BLOCKED	R/W	This field indicates that VDM Message TLPs are blocked during non-D0 states. <i>Note:</i> This register field is sticky. Value After Reset: CX_BLOCK_VDM_TLP Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.84 PL_CHK_REG_CONTROL_STATUS_OFF

- **Name:** CDM Register Checking Control and Status Register.
- **Description:** Controls register checking and displays status of register checking.
- **Size:** 32 bits
- **Offset:** B+0x420
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:19	RSVDP_19
18	CHK_REG_COMPLETE
17	CHK_REG_LOGIC_ERROR
16	CHK_REG_COMPARISON_ERROR
15:2	RSVDP_2
1	CHK_REG_CONTINUOUS
0	CHK_REG_START

Table 3-433 Fields for Register: PL_CHK_REG_CONTROL_STATUS_OFF

Bits	Name	Memory Access	Description
31:19	RSVDP_19	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
18	CHK_REG_COMPLETE	R/W1C	The system has completed a checking cycle. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Indicates the system has completed a checking cycle. Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
17	CHK_REG_LOGIC_ERROR	R/W1C	The system has detected an error in its own checking logic. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Indicates an error in comparison logic Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
16	CHK_REG_COMPARISON_ERROR	R/W1C	The system has detected that there is a bit error in the CDM Register Data. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Indicates comparison error Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
1	CHK_REG_CONTINUOUS	R/W	Set Continuous Checking Sequence. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Set Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	CHK_REG_START	R/W	Begins a checking sequence. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Set Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.85 PL_CHK_REG_START_END_OFF

- **Name:** CDM Register Checking First and Last address to check.
- **Description:** This register holds the first and last address to check.
- **Size:** 32 bits
- **Offset:** B+0x424
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	CHK_REG_END_ADDR
15:0	CHK_REG_START_ADDR

Table 3-434 Fields for Register: PL_CHK_REG_START_END_OFF

Bits	Name	Memory Access	Description
31:16	CHK_REG_END_ADDR	R/W	The last address that is checked by the system. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0xbff Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:0	CHK_REG_START_ADDR	R/W	The first address that is checked by the system. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.86 PL_CHK_REG_ERR_ADDR_OFF

- **Name:** CDM Register Checking Error Address.
- **Description:** This register holds the CDM register checking error address.
- **Size:** 32 bits
- **Offset:** B+0x428
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

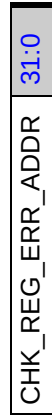


Table 3-435 Fields for Register: PL_CHK_REG_ERR_ADDR_OFF

Bits	Name	Memory Access	Description
31:0	CHK_REG_ERR_ADDR	R	The address at which an error has been detected. Valid only when the CDM Register Checker Comparison Error bit is set in the status register. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0xffffffff (MAX_VAL): Max value Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.87 PL_CHK_REG_ERR_PF_VF_OFF

- **Name:** CDM Register Checking error PF and VF Numbers.
- **Description:** This register holds the CDM Register checking error PF and VF numbers.
- **Size:** 32 bits
- **Offset:** B+0x42c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RSVDP_28
27:16	CHK_REG_VF_ERR_NUMBER
15:5	RSVDP_5
4:0	CHK_REG_PF_ERR_NUMBER

Table 3-436 Fields for Register: PL_CHK_REG_ERR_PF_VF_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27:16	CHK_REG_VF_ERR_NUMBER	R	The VF number at which the error was detected. Valid only when the CDM Register Checker Comparison Error bit is set in the status register. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x7ff (MAX_VAL): Max value Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
15:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4:0	CHK_REG_PF_ERR_NUMBER	R	The PF number at which the error was detected. Valid only when the CDM Register Checker Comparison Error bit is set in the status register. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value ■ 0x1f (MAX_VAL): Max value Value After Reset: 0x0 Exists: CX_CDM_REG_CHK_EN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.88 PL_LTR_LATENCY_OFF

- **Name:** LTR Latency Register.
- **Description:** The function of this register field (and all other fields in this register) differs between an upstream port and a downstream port. For an upstream port, the register fields capture the corresponding fields in the LTR messages that are transmitted by the port. For a downstream port, the register fields capture the corresponding fields in the LTR messages that are received by the port. The full content of the register is reflected on the app_ltr_latency[31:0] output.
- **Size:** 32 bits
- **Offset:** B+0x430
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	NO_SNOOP_LATENCY_REQUIRE
30:29	RSVDP_29
28:26	NO_SNOOP_LATENCY_SCALE
25:16	NO_SNOOP_LATENCY_VALUE
15	SNOOP_LATENCY_REQUIRE
14:13	RSVDP_13
12:10	SNOOP_LATENCY_SCALE
9:0	SNOOP_LATENCY_VALUE

Table 3-437 Fields for Register: PL_LTR_LATENCY_OFF

Bits	Name	Memory Access	Description
31	NO_SNOOP_LATENCY_REQUIRE	* Varies	No Snoop Latency Requirement. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_LTR_M_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
30:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_LTR_M_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
28:26	NO_SNOOP_LATENCY_SCALE	* Varies	No Snoop Latency Scale. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_LTR_M_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
25:16	NO_SNOOP_LATENCY_VALUE	* Varies	No Snoop Latency Value. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_LTR_M_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
15	SNOOP_LATENCY_REQUIRE	* Varies	Snoop Latency Requirement. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_LTR_M_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
14:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_LTR_M_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
12:10	SNOOP_LATENCY_SCALE	* Varies	Snoop Latency Scale. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_LTR_M_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
9:0	SNOOP_LATENCY_VALUE	* Varies	Snoop Latency Value. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_LTR_M_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

3.48.89 AUX_CLK_FREQ_OFF

- **Name:** Auxiliary Clock Frequency Control Register.
- **Description:** This register controls the auxiliary clock frequency.
- **Size:** 32 bits
- **Offset:** B+0x440
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_10	31:10
AUX_CLK_FREQ	9:0

Table 3-438 Fields for Register: AUX_CLK_FREQ_OFF

Bits	Name	Memory Access	Description
31:10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
9:0	AUX_CLK_FREQ	R/W	The aux_clk frequency in MHz. This value is used to provide a 1 us reference for counting time during low-power states with aux_clk when the PHY has removed the pipe_clk. Frequencies lower than 1 MHz are possible but with a loss of accuracy in the time counted. If the actual frequency (f) of aux_clk does not exactly match the programmed frequency (f_prog), then there is an error in the time counted by the controller that can be expressed in percentage as: $err\% = (f_prog/f - 1) * 100$. For example if f=2.5 MHz and f_prog=3 MHz, then $err\% = (3/2.5 - 1) * 100 = 20\%$, meaning that the time counted by the controller on aux_clk will be 20% greater than the time in us programmed in the corresponding time register (for example T_POWER_ON). <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_AUX_CLK_FREQ Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.90 L1_SUBSTATES_OFF

- **Name:** L1 Substates Timing Register.
- **Description:** This register holds L1 substates timing information.
- **Size:** 32 bits
- **Offset:** B+0x444
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:14	RSVDP_14
13:9	L1SUB_T_PCLKACK_HIGH
8	L1SUB_LOW_POWER_CLOCK_SWITCH_MODE
7:6	L1SUB_T_PCLKACK_LOW
5:2	L1SUB_T_L1_2
1:0	L1SUB_T_POWER_OFF

Table 3-439 Fields for Register: L1_SUBSTATES_OFF

Bits	Name	Memory Access	Description
31:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_L1_SUBSTATES_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
13:9	L1SUB_T_PCLKACK_HIGH	R/W	Higher 5-bits of L1SUB_T_PCLKACK. Max delay (in 1us units) between a MAC request to remove the clock on mac_phy_pclkreq_n and a PHY response on phy_mac_pclkack_n. If the PHY does not respond within this time the request is aborted and an internal uncorrectable error is reported. Range is 1..127. Note: When L1SUB_T_PCLKACK==1 or L1SUB_T_PCLKACK_HIGH==0 and L1SUB_T_PCLKACK_LOW==0, the t_pclkack timer is turned off. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0x1f (MAX_VAL): Max value Value After Reset: 0x1f Exists: CX_L1_SUBSTATES_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
8	L1SUB_LOW_POWER_CLOCK_SWITCH_MODE	R/W	If the bit is set to 1'b1 the controller will delay the switching of aux_clk to the slow platform clock until it detects that the link partner has de-asserted CLKREQ#. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): The reference clock may be gated off when CLKREQ# is de-asserted. ■ 0x1 (SET): The reference clock shall be kept running regardless of the CLKREQ# setting. Value After Reset: 0x0 Exists: CX_L1_SUBSTATES_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
7:6	L1SUB_T_PCLKACK_LOW	R/W	Lower 2-bits of L1SUB_T_PCLKACK. Max delay (in 1us units) between a MAC request to remove the clock on mac_phy_pclreq_n and a PHY response on phy_mac_pclack_n. If the PHY does not respond within this time the request is aborted and an internal uncorrectable error is reported. Range is 1..127. Note: When L1SUB_T_PCLKACK==1 or L1SUB_T_PCLKACK_HIGH==0 and L1SUB_T_PCLKACK_LOW==0, the t_pclack timer is turned off. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0x3 (MAX_VAL): Max value Value After Reset: 0x3 Exists: CX_L1_SUBSTATES_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5:2	L1SUB_T_L1_2	R/W	Duration (in 1us units) of L1.2. Range is 1..16. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x4 Exists: CX_L1_SUBSTATES_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
1:0	L1SUB_T_POWER_OFF	R/W	Duration (in 1us units) of L1.2.Entry. Range is 1..4. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0x3 (MAX_VAL): Max value Value After Reset: 0x2 Exists: CX_L1_SUBSTATES_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.91 POWERDOWN_CTRL_STATUS_OFF

- **Name:** Powerdown Control and Status Register.
- **Description:** This is the Powerdown Control and Status register.
- **Size:** 32 bits
- **Offset:** B+0x448
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:12	RSVDP_12
11:8	POWERDOWN_PHY_POWERDOWN
7:4	POWERDOWN_MAC_POWERDOWN
3:2	RSVDP_2
1	POWERDOWN_VMAIN_ACK
0	POWERDOWN_FORCE

Table 3-440 Fields for Register: POWERDOWN_CTRL_STATUS_OFF

Bits	Name	Memory Access	Description
31:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
11:8	POWERDOWN_PHY_POWERDOWN	R	This field represents the Powerdown value that has been acknowledged by the PHY. It is updated with the value of Powerdown driven by the controller, when the PHY has returned the Phystatus acknowledgment for the Powerdown transition. Value After Reset: 0x2 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7:4	POWERDOWN_MAC_POWERDOWN	R	This field represents the Powerdown value driven by the controller to the PHY. Value After Reset: 0x2 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
3:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
1	POWERDOWN_VMAIN_ACK	R/W	Set this bit to 1 if you do not want to perform the handshake with the power-switch after PERST# assertion. By default the controller will perform the handshake with the power-switch if L1 power gating is enabled <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): If you do not want to perform the handshake with the power-switch after PERST# assertion. Value After Reset: CX_VMAIN_ACK_CTRL_RST_VALUE Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
0	POWERDOWN_FORCE	W	This field is a one shot field. This field could be used for debug purposes in event that the P2 Powerdown transition does not complete. It will allow the controller to proceed with the transition to the P1 Powerdown state. This field always reads back as 1'b0. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): Writing a value of 1 to this field causes the controller to complete the P2 Powerdown handshake regardless of whether the PHY has returned Phystatus. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.92 PHY_INTEROP_CTRL_2_OFF

- **Name:** PHY Interoperability Control 2 Register.
- **Description:** This register controls controller PHY dependent behavior.
- **Size:** 32 bits
- **Offset:** B+0x44c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:12	RSVDP_12
11:8	DSP_PCIPM_L1_ENTER_DELAY
7:6	RSVDP_6
5:0	PMA_PIPE_RST_DELAY_TIMER

Table 3-441 Fields for Register: PHY_INTEROP_CTRL_2_OFF

Bits	Name	Memory Access	Description
31:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
11:8	DSP_PCIPM_L1_ENTER_DELAY	R/W	For a DSP component, this register field controls the delay between the reception of PM_Enter_L1 DLLP and the start of PCI-PM L1 entry negotiation in aux clock cycles (1 to 15). When using this feature on legacy DMA configurations, the configured value must be greater than three. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Zero value. ■ 0xf (MAX_VAL): Max value. Value After Reset: CX_DSP_PCIPM_L1_ENTER_DELAY Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5:0	PMA_PIPE_RST_DELAY_TIMER	R/W	Control how long the controller should wait to release a PIPE reset (pm_req_phy_rst) after releasing a PMA reset (pm_req_phy_perst). This register should only be used to ensure a correct sequencing of the resets into the PHY. If the application needs to delay the PIPE reset arbitrarily, for example to configure the PHY firmware, Synopsys recommends using the app_hold_phy_rst control signal. There is no need to modify the value of this register when using Synopsys PHYs. Duration is in aux clock cycles. <i>Note:</i> This register field is sticky. Values: ■ 0xa (MIN_VAL): Min value. ■ 0x3f (MAX_VAL): Max value. Value After Reset: CX_PMA_PIPE_RST_DELAY_TIMER Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.93 PHY_VIEWPORT_CTLSTS_OFF

- **Name:** PHY Register Viewport Control and Status Register.
- **Description:** This register controls the PHY register viewport and provides status information.
- **Size:** 32 bits
- **Offset:** B+0x470
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 | | CX_MEMORY_MAP_PCIE_TYPE==0))

PHY_VIEWPORT_PENDING	31
PHY_VIEWPORT_STATUS	30
RSVDP_23	29:23
PHY_VIEWPORT_TIMEOUT_DISABLE	22
PHY_VIEWPORT_BCWR	21
PHY_VIEWPORT_READ	20
PHY_VIEWPORT_NUM	19:16
PHY_VIEWPORT_ADDR	15:0

Table 3-442 Fields for Register: PHY_VIEWPORT_CTLSTS_OFF

Bits	Name	Memory Access	Description
31	PHY_VIEWPORT_PENDING	R	PHY viewport pending. Indicates the access is pending. This bit is set to 1 when the read/write access has been initiated. This bit is cleared when the access is completed. If phy_reg_clk_g is never activated while this pending bit is 1, this bit will never be cleared. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (PENDING): Pending ■ 0x0 (IDLE): Idle Value After Reset: 0x0 Exists: CX_PHY_VIEWPORT_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
30	PHY_VIEWPORT_STATUS	R	PHY viewport status. Indicates the status of the latest access. If the timeout is detected, this bit is set to 1 and PHY_VIEWPORT_PENDING is cleared. This bit is cleared when the read/write access is initiated. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (TIMEOUT): Timeout ■ 0x0 (NO_ERROR): No errors Value After Reset: 0x0 Exists: CX_PHY_VIEWPORT_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
29:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_PHY_VIEWPORT_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
22	PHY_VIEWPORT_TIMEOUT_DISABLE	R/W	PHY viewport timeout disable. Disables the PHY register access timeout structure. When the timeout is disabled, you must ensure that the PHY register accesses are always acknowledged by cr_para_ack. Otherwise, PHY_VIEWPORT_PENDING bit will never be de-asserted. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_PHY_VIEWPORT_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
21	PHY_VIEWPORT_BCWR	R/W	Broadcast write access to all xN sub-blocks. If you need to broadcast the write accesses into all lanes, both PHY_VIEWPORT_ADDR[15] and this bit should be 1. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_PHY_VIEWPORT_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20	PHY_VIEWPORT_READ	R/W	PHY viewport read. The value is always 0 when this bit is read. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CLEAR): Clear ■ 0x1 (SET): If set, a read access is initiated. Value After Reset: 0x0 Exists: CX_PHY_VIEWPORT_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
19:16	PHY_VIEWPORT_NUM	R/W	PHY viewport xN sub-block number. Selects one of xN sub-blocks. This register is not used when the write access is initiated with PHY_VIEWPORT_BCWR ==1. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_PHY_VIEWPORT_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:0	PHY_VIEWPORT_ADDR	R/W	PHY viewport address. Indicates the PHY's register address. Refer to the PHY databook for the address map definition. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_PHY_VIEWPORT_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.94 PHY_VIEWPORT_DATA_OFF

- **Name:** PHY Register Viewport Data Register.
- **Description:** This register holds the PHY viewport data.
- **Size:** 32 bits
- **Offset:** B+0x474
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	RSVDP_16
15:0	PHY_VIEWPORT_DATA

Table 3-443 Fields for Register: PHY_VIEWPORT_DATA_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_PHY_VIEWPORT_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:0	PHY_VIEWPORT_DATA	R/W	PHY viewport data. A write access to this register initiates the PHY register write access. By a read access to this register, this field returns the latest read data from the PHY register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_PHY_VIEWPORT_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.95 GEN4_LANE_MARGINING_1_OFF

- **Name:** Gen4 Lane Margining 1 Register.
- **Description:** This is the Gen4 Lane Margining 1 Register.
- **Size:** 32 bits
- **Offset:** B+0x480
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_30	31:30
GEN4_MARGINING_MAX_VOLTAGE_OFFSET	29:24
RSVDP_23	23
GEN4_MARGINING_NUM_VOLTAGE_STEPS	22:16
RSVDP_14	15:14
GEN4_MARGINING_MAX_TIMING_OFFSET	13:8
RSVDP_6	7:6
GEN4_MARGINING_NUM_TIMING_STEPS	5:0

Table 3-444 Fields for Register: GEN4_LANE_MARGINING_1_OFF

Bits	Name	Memory Access	Description
31:30	RSVDP_30	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
29:24	GEN4_MARGINING_MAX_VOLTAGE_OFFSET	R/W	MMaxVoltageOffset for 16.0 GT/s for Lane Margining at the Receiver. Offset from default at maximum step value as percentage of one volt. A '0' value might be reported if the vendor chooses not to report the offset when MVoltageSupported is 1b. This value is undefined if MVoltageSupported is 0b. <i>Note:</i> This register field is sticky. Values: ■ 0x5 (MIN_VALUE): Min value. ■ 0x32 (MAX_VALUE): Max value. Value After Reset: DEFAULT_GEN4_RXMARGIN_MAX_VOLTAGE_OFFSET Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
22:16	GEN4_MARGINING_NUM_VOLTAGE_STEPS	R/W	MNumVoltageSteps for 16.0 GT/s for Lane Margining at the Receiver. Number of voltage steps from default (either up or down), minimum range +/-50mV as measured by the reference equalizer. Voltage offset must increase monotonically. The number of steps in both positive and negative direction from the default sample location must be identical. This value is undefined if MVoltageSupported is 0b. <i>Note:</i> This register field is sticky. Values: ■ 0x20 (MIN_VALUE): Min value. ■ 0x7f (MAX_VALUE): Max value. Value After Reset: DEFAULT_GEN4_RXMARGIN_NUM_VOLTAGE_STEPS Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	GEN4_MARGINING_MAX_TIMING_OFFSET	R/W	MMaxTimingOffset for 16.0 GT/s for Lane Margining at the Receiver. Offset from default at maximum step value as percentage of a nominal UI. A '0' value might be reported if the vendor chooses not to report the offset. <i>Note:</i> This register field is sticky. Values: 0x14 (MIN_VALUE): Min value. 0x32 (MAX_VALUE): Max value. Value After Reset: DEFAULT_GEN4_RXMARGIN_MAX_TIMING_OFFSET Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5:0	GEN4_MARGINING_NUM_TIMING_STEPS	R/W	MNumTimingSteps for 16.0 GT/s for Lane Margining at the Receiver. Number of time steps from default (to either left or right), range must be at least +/-0.2 UI . Timing offset must increase monotonically. The number of steps in both positive (toward the end of the unit interval) and negative (toward the beginning of the unit interval) must be identical. <i>Note:</i> This register field is sticky. Values: 0x6 (MIN_VALUE): Min value. 0x3f (MAX_VALUE): Max value. Value After Reset: DEFAULT_GEN4_RXMARGIN_NUM_TIMING_STEPS Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.96 GEN4_LANE_MARGINING_2_OFF

- **Name:** Gen4 Lane Margining 2 Register.
- **Description:** This is the Gen4 Lane Margining 2 Register.
- **Size:** 32 bits
- **Offset:** B+0x484
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

MARGINING_RATE_MODE	31
RSVDP_29	30:29
GEN4_MARGINING_IND_ERROR_SAMPLER	28
GEN4_MARGINING_SAMPLE_REPORTING_METHOD	27
GEN4_MARGINING_IND_LEFT_RIGHT_TIMING	26
GEN4_MARGINING_IND_UP_DOWN_VOLTAGE	25
GEN4_MARGINING_VOLTAGE_SUPPORTED	24
RSVDP_21	23:21
GEN4_MARGINING_MAXLANES	20:16
RSVDP_14	15:14
GEN4_MARGINING_SAMPLE_RATE_TIMING	13:8
RSVDP_6	7:6
GEN4_MARGINING_SAMPLE_RATE_VOLTAGE	5:0

Table 3-445 Fields for Register: GEN4_LANE_MARGINING_2_OFF

Bits	Name	Memory Access	Description
31	MARGINING_RATE_MODE	R/W	Margining Rate Mode. For more information, contact SYNOPSYS support/sales. <i>Note:</i> This register field is sticky. Values: 0x0 (COMPLIANT_MODE): Base Spec compliant mode: Lane Margining at Receiver is available at Gen4 data rate only. 0x1 (PHY_SUP_MODE): Special PHY Support mode: Lane Margining at Receiver is available at Gen4 data rate and at Gen1, Gen2 and Gen3 data rate for debug. Value After Reset: DEFAULT_RXMARGIN_RATE_MODE Exists: CX_MARGIN_ENABLE && CX_RXMARGIN_RATE_MODE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
30:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
28	GEN4_MARGINING_IND_ERROR_SAMPLER	R/W	MIndErrorSampler for 16.0 GT/s for Lane Margining at the Receiver. <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VALUE): Margining might produce errors in the data stream. 0x1 (MAX_VALUE): Margining does not produce errors (change in the error rate) in data stream (that is, error sampler is independent). Value After Reset: DEFAULT_GEN4_RXMARGIN_IND_ERROR_SAMPLER Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27	GEN4_MARGINING_SAMPLE_REPORTING_METHOD	R/W	MSampleReportingMethod for 16.0 GT/s for Lane Margining at the Receiver. Indicates whether sampling rates or sample count is supported. One of the two methods is supported by each device. <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VALUE): Indicates sample count is supported. 0x1 (MAX_VALUE): Indicates sampling rates (MSamplingRateVoltage and MSamplingRateTiming) are supported. Value After Reset: DEFAULT_GEN4_RXMARGIN_SAMPLE_REPORTING_METHOD Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
26	GEN4_MARGINING_IND_LEFT_RIGHT_TIMING	R/W	MIndLeftRightTiming for 16.0 GT/s for Lane Margining at the Receiver. <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VALUE): Indicates independent left/right timing margin is not supported. 0x1 (MAX_VALUE): Indicates independent left/right timing margin is supported. Value After Reset: DEFAULT_GEN4_RXMARGIN_IND_LEFT_RIGHT_TIMING Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
25	GEN4_MARGINING_IND_UP_DOWN_VOLTAGE	R/W	MIndUpDownVoltage for 16.0 GT/s for Lane Margining at the Receiver. <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VALUE): Independent up and down voltage margining is not supported. 0x1 (MAX_VALUE): Independent up and down voltage margining is supported. Value After Reset: DEFAULT_GEN4_RXMARGIN_IND_UP_DOWN_VOLTAGE Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
24	GEN4_MARGINING_VOLTAGE_SUPPORTED	R/W	MVoltageSupported for 16.0 GT/s for Lane Margining at the Receiver. <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VALUE): Voltage margining is not supported. 0x1 (MAX_VALUE): Voltage margining is supported. Value After Reset: DEFAULT_GEN4_RXMARGIN_VOLTAGE_SUPPORTED Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20:16	GEN4_MARGINING_MAXLANES	R/W	MMaxLanes for 16.0 GT/s for Lane Margining at the Receiver. Maximum number of Lanes minus 1 that can be margined at the same time. It is recommended that this value be greater than or equal to the number of Lanes in the Link minus 1. Encoding Behavior is undefined if software attempts to margin more than MMaxLanes+1 at the same time. <i>Note:</i> This value is permitted to exceed the number of Lanes in the Link minus 1. <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VALUE): Min value. 0x1f (MAX_VALUE): Max value. Value After Reset: DEFAULT_GEN4_RXMARGIN_MAXLANES Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
13:8	GEN4_MARGINING_SAMPLE_RATE_TIMING	R/W	MSamplingRateTiming for 16.0 GT/s for Lane Margining at the Receiver. <i>Note:</i> This value is not used when MARGINING_IND_ERROR_SAMPLER is 0b. The MSamplingRateTiming is fixed to 63 internally. The ratio of bits tested to bits received during timing margining. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VALUE): Value of 0 is a ratio of 1:64 (1 bit of every 64 bits received). ■ 0x3f (MAX_VALUE): Value of 63 is a ratio of 64:64 (all bits received). Value After Reset: DEFAULT_GEN4_RXMARGIN_SAMPLE_RATE_TIMING Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5:0	GEN4_MARGINING_SAMPLE_RATE_VOLTAGE	R/W	MSamplingRateVoltage for 16.0 GT/s for Lane Margining at the Receiver. The ratio of bits tested to bits received during voltage margining. <i>Note:</i> This value is not used when MARGINING_IND_ERROR_SAMPLER is 0b. The MSamplingRateVoltage is fixed to 63 internally. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VALUE): Value of 0 is a ratio of 1:64 (1 bit of every 64 bits received). ■ 0x3f (MAX_VALUE): Value of 63 is a ratio of 64:64 (all bits received). Value After Reset: DEFAULT_GEN4_RXMARGIN_SAMPLE_RATE_VOLTAGE Exists: CX_MARGIN_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.97 GEN5_LANE_MARGINING_1_OFF

- **Name:** Gen5 Lane Margining 1 Register.
- **Description:** This is the Gen5 Lane Margining 1 Register.
- **Size:** 32 bits
- **Offset:** B+0x488
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

RSVDP_30	31:30
GEN5_MARGINING_MAX_VOLTAGE_OFFSET	29:24
RSVDP_23	23
GEN5_MARGINING_NUM_VOLTAGE_STEPS	22:16
RSVDP_14	15:14
GEN5_MARGINING_MAX_TIMING_OFFSET	13:8
RSVDP_6	7:6
GEN5_MARGINING_NUM_TIMING_STEPS	5:0

Table 3-446 Fields for Register: GEN5_LANE_MARGINING_1_OFF

Bits	Name	Memory Access	Description
31:30	RSVDP_30	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
29:24	GEN5_MARGINING_MAX_VOLTAGE_OFFSET	R/W	MMaxVoltageOffset for 32.0 GT/s for Lane Margining at the Receiver. Offset from default at maximum step value as percentage of one volt. A '0' value might be reported if the vendor chooses not to report the offset when MVoltageSupported is 1b. This value is undefined if MVoltageSupported is 0b. <i>Note:</i> This register field is sticky. Values: ■ 0x5 (MIN_VALUE): Min value. ■ 0x32 (MAX_VALUE): Max value. Value After Reset: DEFAULT_GEN5_RXMARGIN_MAX_VOLTAGE_OFFSET Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
22:16	GEN5_MARGINING_NUM_VOLTAGE_STEPS	R/W	MNumVoltageSteps for 32.0 GT/s for Lane Margining at the Receiver. Number of voltage steps from default (either up or down), minimum range +/-50mV as measured by the reference equalizer. Voltage offset must increase monotonically. The number of steps in both positive and negative direction from the default sample location must be identical. This value is undefined if MVoltageSupported is 0b. <i>Note:</i> This register field is sticky. Values: ■ 0x20 (MIN_VALUE): Min value. ■ 0x7f (MAX_VALUE): Max value. Value After Reset: DEFAULT_GEN5_RXMARGIN_NUM_VOLTAGE_STEPS Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
13:8	GEN5_MARGINING_MAX_TIMING_OFFSET	R/W	MMaxTimingOffset for 32.0 GT/s for Lane Margining at the Receiver. Offset from default at maximum step value as percentage of a nominal UI. A '0' value might be reported if the vendor chooses not to report the offset. <i>Note:</i> This register field is sticky. Values: ■ 0x14 (MIN_VALUE): Min value. ■ 0x32 (MAX_VALUE): Max value. Value After Reset: DEFAULT_GEN5_RXMARGIN_MAX_TIMING_OFFSET Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
5:0	GEN5_MARGINING_NUM_TIMING_STEPS	R/W	MNumTimingSteps for 32.0 GT/s for Lane Margining at the Receiver. Number of time steps from default (to either left or right), range must be at least +/-0.2 UI . Timing offset must increase monotonically. The number of steps in both positive (toward the end of the unit interval) and negative (toward the beginning of the unit interval) must be identical. <i>Note:</i> This register field is sticky. Values: ■ 0x6 (MIN_VALUE): Min value. ■ 0x3f (MAX_VALUE): Max value. Value After Reset: DEFAULT_GEN5_RXMARGIN_NUM_TIMING_STEPS Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.98 GEN5_LANE_MARGINING_2_OFF

- **Name:** Gen5 Lane Margining 2 Register.
- **Description:** This is the Gen5 Lane Margining 2 Register.
- **Size:** 32 bits
- **Offset:** B+0x48c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:29	RSVDP_29
28	GEN5_MARGINING_IND_ERROR_SAMPLER
27	GEN5_MARGINING_SAMPLE_REPORTING_METHOD
26	GEN5_MARGINING_IND_LEFT_RIGHT_TIMING
25	GEN5_MARGINING_IND_UP_DOWN_VOLTAGE
24	GEN5_MARGINING_VOLTAGE_SUPPORTED
23:21	RSVDP_21
20:16	GEN5_MARGINING_MAXLANES
15:14	RSVDP_14
13:8	GEN5_MARGINING_SAMPLE_RATE_TIMING
7:6	RSVDP_6
5:0	GEN5_MARGINING_SAMPLE_RATE_VOLTAGE

Table 3-447 Fields for Register: GEN5_LANE_MARGINING_2_OFF

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
28	GEN5_MARGINING_IND_ERROR_SAMPLER	R/W	MIndErrorSampler for 32.0 GT/s for Lane Margining at the Receiver. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VALUE): Margining might produce errors in the data stream. ■ 0x1 (MAX_VALUE): Margining does not produce errors (change in the error rate) in data stream (that is, error sampler is independent). Value After Reset: DEFAULT_GEN5_RXMARGIN_IND_ERROR_SAMPLER Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27	GEN5_MARGINING_SAMPLE_REPORTING_METHOD	R/W	MSampleReportingMethod for 32.0 GT/s for Lane Margining at the Receiver. Indicates whether sampling rates or sample count is supported. One of the two methods is supported by each device. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VALUE): Indicates sample count is supported. ■ 0x1 (MAX_VALUE): Indicates sampling rates (MSamplingRateVoltage and MSamplingRateTiming) are supported. Value After Reset: DEFAULT_GEN5_RXMARGIN_SAMPLE_REPORTING_METHOD Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
26	GEN5_MARGINING_IND_LEFT_RIGHT_TIMING	R/W	MIndLeftRightTiming for 32.0 GT/s for Lane Margining at the Receiver. <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VALUE): Indicates independent left/right timing margin is not supported. 0x1 (MAX_VALUE): Indicates independent left/right timing margin is supported. Value After Reset: DEFAULT_GEN5_RXMARGIN_IND_LEFT_RIGHT_TIMING Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
25	GEN5_MARGINING_IND_UP_DOWN_VOLTAGE	R/W	MIndUpDownVoltage for 32.0 GT/s for Lane Margining at the Receiver. <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VALUE): Independent up and down voltage margining is not supported. 0x1 (MAX_VALUE): Independent up and down voltage margining is supported. Value After Reset: DEFAULT_GEN5_RXMARGIN_IND_UP_DOWN_VOLTAGE Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
24	GEN5_MARGINING_VOLTAGE_SUPPORTED	R/W	MVoltageSupported for 32.0 GT/s for Lane Margining at the Receiver. <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VALUE): Voltage margining is not supported. 0x1 (MAX_VALUE): Voltage margining is supported. Value After Reset: DEFAULT_GEN5_RXMARGIN_VOLTAGE_SUPPORTED Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
23:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
20:16	GEN5_MARGINING_MAXLANES	R/W	MMaxLanes for 32.0 GT/s for Lane Margining at the Receiver. Maximum number of Lanes minus 1 that can be margined at the same time. It is recommended that this value be greater than or equal to the number of Lanes in the Link minus 1. Encoding Behavior is undefined if software attempts to margin more than MMaxLanes+1 at the same time. <i>Note:</i> This value is permitted to exceed the number of Lanes in the Link minus 1. <i>Note:</i> This register field is sticky. Values: 0x0 (MIN_VALUE): Min value. 0x1f (MAX_VALUE): Max value. Value After Reset: DEFAULT_GEN5_RXMARGIN_MAXLANES Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
13:8	GEN5_MARGINING_SAMPLE_RATE_TIMING	R/W	MSamplingRateTiming for 32.0 GT/s for Lane Margining at the Receiver. The ratio of bits tested to bits received during timing margining. <i>Note:</i> This value is not used when MARGINING_IND_ERROR_SAMPLER is 0b. The MSamplingRateTiming is fixed to 63 internally. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VALUE): Value of 0 is a ratio of 1:64 (1 bit of every 64 bits received). ■ 0x3f (MAX_VALUE): Value of 63 is a ratio of 64:64 (all bits received). Value After Reset: DEFAULT_GEN5_RXMARGIN_SAMPLE_RATE_TIMING Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
5:0	GEN5_MARGINING_SAMPLE_RATE_VOLTAGE	R/W	MSamplingRateVoltage for 32.0 GT/s for Lane Margining at the Receiver. The ratio of bits tested to bits received during voltage margining. <i>Note:</i> This value is not used when MARGINING_IND_ERROR_SAMPLER is 0b. The MSamplingRateVoltage is fixed to 63 internally. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VALUE): Value of 0 is a ratio of 1:64 (1 bit of every 64 bits received). ■ 0x3f (MAX_VALUE): Value of 63 is a ratio of 64:64 (all bits received). Value After Reset: DEFAULT_GEN5_RXMARGIN_SAMPLE_RATE_VOLTAGE Exists: CX_MARGIN_ENABLE && CX_GEN5_SPEED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.99 PIPE_RELATED_OFF

- **Name:** PIPE Related Register.
- **Description:** This register controls the PIPE's capability, control, and status parameters.
- **Size:** 32 bits
- **Offset:** B+0x490
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:9	RSVDP_9
8	PIPE_GARBAGE_DATA_MODE
7:4	TX_MESSAGE_BUS_MIN_WRITE_BUFFER_DEPTH
3:0	RX_MESSAGE_BUS_WRITE_BUFFER_DEPTH

Table 3-448 Fields for Register: PIPE_RELATED_OFF

Bits	Name	Memory Access	Description
31:9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
8	PIPE_GARBAGE_DATA_MODE	R/W	PIPE Garbage Data Mode. ■ RxValid is de-asserted ■ a valid RxStartBlock is received at 128b/130b encoding ■ a valid COM symbol is received at 8b/10b encoding <i>Note:</i> This register field is sticky. Values: ■ 0x0 (COMPLIANT_MODE): PIPE Spec compliant mode: The MAC discards any symbols received after the electrical idle ordered-set until RxValid is de-asserted. ■ 0x1 (PHY_SUP_MODE): Special PHY Support mode: The MAC discards any symbols received after the electrical idle ordered-set until when any of the following three conditions are true: Value After Reset: DEFAULT_PIPE_GARBAGE_DATA_MODE Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:4	TX_MESSAGE_BUS_MIN_WRITE_BUFFER_DEPTH	R	TXMessageBusMinWriteBufferDepth defined in the <i>PIPE Specification</i>. Indicates the minimum number of write buffer entries that the PHY expects the controller to implement to receive writes from it. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_TX_MESSAGE_BUS_MIN_WRITE_BUFFER_DEPTH Exists: CX_PIPE_MSGBUS_SUPPORT && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
3:0	RX_MESSAGE_BUS_WRITE_BUFFER_DEPTH	R/W	RXMessageBusWriteBufferDepth defined in the <i>PIPE Specification</i>. Indicates the number of write buffer entries that the PHY has implemented to receive writes from the controller. If the value is less than 2 for PIPE 5.1.1 or 1 for PIPE 4.4.1, the controller issues only write_committed commands, never write_uncommitted. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_RX_MESSAGE_BUS_WRITE_BUFFER_DEPTH Exists: CX_PIPE_MSGBUS_SUPPORT && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.100 RX_SERIALIZATION_Q_CTRL_OFF

- **Name:** Receive Serialization Queue Control Register.
- **Description:** This register controls the receive serialization queue.
- **Size:** 32 bits
- **Offset:** B+0x500
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	RX_SERIALIZATION_OVFLW_PRVNTN_EN
30	ALMOST_FULL_THRESHOLD_ADJ_SIGN
29:28	RSVDP_28
27:16	ALMOST_FULL_THRESHOLD_ADJ_VAL
15:0	ALMOST_FULL_THRESHOLD

Table 3-449 Fields for Register: RX_SERIALIZATION_Q_CTRL_OFF

Bits	Name	Memory Access	Description
31	RX_SERIALIZATION_OVFLW_PRVNTN_EN	R/W	Enable Receive Serialization Queue Overflow Prevention. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: RX_SERIALIZATION_OVFLW_PRVNTN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
30	ALMOST_FULL_THRESHOLD_A DJ_SIGN	R/W	Almost full threshold adjustment sign. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: RX_SERIALIZATION_OVFLW_PRVNTN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
29:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: RX_SERIALIZATION_OVFLW_PRVNTN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27:16	ALMOST_FULL_THRESHOLD_A DJ_VAL	R/W	Almost full threshold adjustment value. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: RX_SERIALIZATION_OVFLW_PRVNTN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:0	ALMOST_FULL_THRESHOLD	R	Current almost full threshold. <i>Note:</i> This register field is sticky. Value After Reset: {DEFAULT_RX_SERIALIZATION_Q_ALMOST_FULL_THRES HOLD} Exists: RX_SERIALIZATION_OVFLW_PRVNTN && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true

3.48.101 CCIX_CTRL_OFF

- **Name:** CCIX Control Register.
- **Description:** This is the CCIX Control Register.
- **Size:** 32 bits
- **Offset:** B+0x520
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:16	RSVDP_16
15:0	CCIX_VDM_VID

Table 3-450 Fields for Register: CCIX_CTRL_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CCIX_INTERFACE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:0	CCIX_VDM_VID	R/W	Vendor ID in CCIX Vendor Defined Type1 Message. This value is used to process the CCIX Vendor Defined Type1 Messages at the receiver. <i>Note:</i> This register field is sticky. Value After Reset: CX_CCIX_VDM_VID Exists: CX_CCIX_INTERFACE_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.102 GEN3_FCOEF_VEC_DEPTH_OFF

- **Name:** Gen3 FOM Coefficients Table Depth Register.
- **Description:** This register holds the maximum number of FOM coefficients requests.
- **Size:** 32 bits
- **Offset:** B+0x524
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:12	RSVDP_12
11:8	GEN3_FCOEF_ACCESS_VAR
7:5	RSVDP_5
4:0	GEN3_FCOEF_MAX_VEC_DEPTH

Table 3-451 Fields for Register: GEN3_FCOEF_VEC_DEPTH_OFF

Bits	Name	Memory Access	Description
31:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
11:8	GEN3_FCOEF_ACCESS_VAR	R/W	Set this register field from 0, 1, ... (GEN3_FCOEF_MAX_VEC_DEPTH - 1) to access the different depth pointer of the FOM Coefficients Table Register (GEN3_FCOEF_LANE*_*_OFF). This reduces the address footprint of the PCIe configuration space. <i>Note:</i> When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shared for Gen3 and Gen4/Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
4:0	GEN3_FCOEF_MAX_VEC_DEPTH	R/W	Program this register field to the maximum value of FOM coefficients requests you want. You must contact your PHY vendor to ensure that 24ms timeout does not occur in presets and coefficients requests for FOM mode and DirChange evaluation for Direction Change mode (if applied) in the EQ master phase. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): When programmed to 0, the EQ does not perform the coefficients request in the FOM mode. ■ 0x1 (MAX_VAL): DEFAULT_MAX_COEF_FOM_VECTOR_DEPTH Value After Reset: DEFAULT_MAX_COEF_FOM_VECTOR_DEPTH Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.103 GEN3_FCOEF_LANE1_0_OFF

- **Name:** Gen3 FOM Coefficients Table Register for Lane 1 and 0.
- **Description:** This register array holds the coefficients for FOM mode. You must program the coefficients without breaking the Rules for Transmitter Coefficients in the PCIe Base Specification. The value for the register field GEN3_FCOEF_ACCESS_VAR of GEN3_FCOEF_VEC_DEPTH_OFF must be set to the depth pointer you want to access before programming this register. The Cursor is calculated based on the formula: *Cursor = FS - Post-cursor - Pre-cursor*.
- **Size:** 32 bits
- **Offset:** B+0x528
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RSVDP_28
27:22	GEN3_FCOEF_LANE1_POST_CURSOR
21:16	GEN3_FCOEF_LANE1_PRE_CURSOR
15:12	RSVDP_12
11:6	GEN3_FCOEF_LANE0_POST_CURSOR
5:0	GEN3_FCOEF_LANE0_PRE_CURSOR

Table 3-452 Fields for Register: GEN3_FCOEF_LANE1_0_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27:22	GEN3_FCOEF_LANE1_POST_CURSOR	R/W	Lane 1 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
21:16	GEN3_FCOEF_LANE1_PRE_CURSOR	R/W	Lane 1 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_1 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:6	GEN3_FCOEF_LANE0_POST_CURSOR	R/W	Lane 0 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5:0	GEN3_FCOEF_LANE0_PRE_CU RSOR	R/W	Lane 0 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.104 GEN3_FCOEF_LANE3_2_OFF

- **Name:** Gen3 FOM Coefficients Table Register for Lane 3 and 2.
- **Description:** This register array holds the coefficients for FOM mode. You must program the coefficients without breaking the Rules for Transmitter Coefficients in the PCIe Base Specification. The value for the register field GEN3_FCOEF_ACCESS_VAR of GEN3_FCOEF_VEC_DEPTH_OFF must be set to the depth pointer you want to access before programming this register. The Cursor is calculated based on the formula: *Cursor = FS - Post-cursor - Pre-cursor*.
- **Size:** 32 bits
- **Offset:** B+0x52c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RSVDP_28
27:22	GEN3_FCOEF_LANE3_POST_CURSOR
21:16	GEN3_FCOEF_LANE3_PRE_CURSOR
15:12	RSVDP_12
11:6	GEN3_FCOEF_LANE2_POST_CURSOR
5:0	GEN3_FCOEF_LANE2_PRE_CURSOR

Table 3-453 Fields for Register: GEN3_FCOEF_LANE3_2_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27:22	GEN3_FCOEF_LANE3_POST_CURSOR	R/W	Lane 3 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
21:16	GEN3_FCOEF_LANE3_PRE_CURSOR	R/W	Lane 3 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:6	GEN3_FCOEF_LANE2_POST_CURSOR	R/W	Lane 2 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5:0	GEN3_FCOEF_LANE2_PRE_CU RSOR	R/W	Lane 2 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_2 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.105 GEN3_FCOEF_LANE5_4_OFF

- **Name:** Gen3 FOM Coefficients Table Register for Lane 5 and 4.
- **Description:** This register array holds the coefficients for FOM mode. You must program the coefficients without breaking the Rules for Transmitter Coefficients in the PCIe Base Specification. The value for the register field GEN3_FCOEF_ACCESS_VAR of GEN3_FCOEF_VEC_DEPTH_OFF must be set to the depth pointer you want to access before programming this register. The Cursor is calculated based on the formula: *Cursor = FS - Post-cursor - Pre-cursor*.
- **Size:** 32 bits
- **Offset:** B+0x530
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RSVDP_28
27:22	GEN3_FCOEF_LANE5_POST_CURSOR
21:16	GEN3_FCOEF_LANE5_PRE_CURSOR
15:12	RSVDP_12
11:6	GEN3_FCOEF_LANE4_POST_CURSOR
5:0	GEN3_FCOEF_LANE4_PRE_CURSOR

Table 3-454 Fields for Register: GEN3_FCOEF_LANE5_4_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27:22	GEN3_FCOEF_LANE5_POST_CURSOR	R/W	Lane 5 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
21:16	GEN3_FCOEF_LANE5_PRE_CURSOR	R/W	Lane 5 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:6	GEN3_FCOEF_LANE4_POST_CU RSOR	R/W	Lane 4 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5:0	GEN3_FCOEF_LANE4_PRE_CURSOR	R/W	Lane 4 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.106 GEN3_FCOEF_LANE7_6_OFF

- **Name:** Gen3 FOM Coefficients Table Register for Lane 7 and 6.
- **Description:** This register array holds the coefficients for FOM mode. You must program the coefficients without breaking the Rules for Transmitter Coefficients in the PCIe Base Specification. The value for the register field GEN3_FCOEF_ACCESS_VAR of GEN3_FCOEF_VEC_DEPTH_OFF must be set to the depth pointer you want to access before programming this register. The Cursor is calculated based on the formula: *Cursor = FS - Post-cursor - Pre-cursor*.
- **Size:** 32 bits
- **Offset:** B+0x534
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RSVDP_28
27:22	GEN3_FCOEF_LANE7_POST_CURSOR
21:16	GEN3_FCOEF_LANE7_PRE_CURSOR
15:12	RSVDP_12
11:6	GEN3_FCOEF_LANE6_POST_CURSOR
5:0	GEN3_FCOEF_LANE6_PRE_CURSOR

Table 3-455 Fields for Register: GEN3_FCOEF_LANE7_6_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27:22	GEN3_FCOEF_LANE7_POST_CURSOR	R/W	Lane 7 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
21:16	GEN3_FCOEF_LANE7_PRE_CURSOR	R/W	Lane 7 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:6	GEN3_FCOEF_LANE6_POST_CU RSOR	R/W	Lane 6 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5:0	GEN3_FCOEF_LANE6_PRE_CURSOR	R/W	Lane 6 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_4 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.107 GEN3_FCOEF_LANE9_8_OFF

- **Name:** Gen3 FOM Coefficients Table Register for Lane 9 and 8.
- **Description:** This register array holds the coefficients for FOM mode. You must program the coefficients without breaking the Rules for Transmitter Coefficients in the PCIe Base Specification. The value for the register field GEN3_FCOEF_ACCESS_VAR of GEN3_FCOEF_VEC_DEPTH_OFF must be set to the depth pointer you want to access before programming this register. The Cursor is calculated based on the formula: *Cursor = FS - Post-cursor - Pre-cursor*.
- **Size:** 32 bits
- **Offset:** B+0x538
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RSVDP_28
27:22	GEN3_FCOEF_LANE9_POST_CURSOR
21:16	GEN3_FCOEF_LANE9_PRE_CURSOR
15:12	RSVDP_12
11:6	GEN3_FCOEF_LANE8_POST_CURSOR
5:0	GEN3_FCOEF_LANE8_PRE_CURSOR

Table 3-456 Fields for Register: GEN3_FCOEF_LANE9_8_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27:22	GEN3_FCOEF_LANE9_POST_CURSOR	R/W	Lane 9 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
21:16	GEN3_FCOEF_LANE9_PRE_CURSOR	R/W	Lane 9 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:6	GEN3_FCOEF_LANE8_POST_CURSOR	R/W	Lane 8 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5:0	GEN3_FCOEF_LANE8_PRE_CU RSOR	R/W	Lane 8 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.108 GEN3_FCOEF_LANE11_10_OFF

- **Name:** Gen3 FOM Coefficients Table Register for Lane 11 and 10.
- **Description:** This register array holds the coefficients for FOM mode. You must program the coefficients without breaking the Rules for Transmitter Coefficients in the PCIe Base Specification. The value for the register field GEN3_FCOEF_ACCESS_VAR of GEN3_FCOEF_VEC_DEPTH_OFF must be set to the depth pointer you want to access before programming this register. The Cursor is calculated based on the formula: *Cursor = FS - Post-cursor - Pre-cursor*.
- **Size:** 32 bits
- **Offset:** B+0x53c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RSVDP_28
27:22	GEN3_FCOEF_LANE11_POST_CURSOR
21:16	GEN3_FCOEF_LANE11_PRE_CURSOR
15:12	RSVDP_12
11:6	GEN3_FCOEF_LANE10_POST_CURSOR
5:0	GEN3_FCOEF_LANE10_PRE_CURSOR

Table 3-457 Fields for Register: GEN3_FCOEF_LANE11_10_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27:22	GEN3_FCOEF_LANE11_POST_CURSOR	R/W	Lane 11 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
21:16	GEN3_FCOEF_LANE11_PRE_CURSOR	R/W	Lane 11 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:6	GEN3_FCOEF_LANE10_POST_CURSOR	R/W	Lane 10 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5:0	GEN3_FCOEF_LANE10_PRE_CURSOR	R/W	Lane 10 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.109 GEN3_FCOEF_LANE13_12_OFF

- **Name:** Gen3 FOM Coefficients Table Register for Lane 13 and 12.
- **Description:** This register array holds the coefficients for FOM mode. You must program the coefficients without breaking the Rules for Transmitter Coefficients in the PCIe Base Specification. The value for the register field GEN3_FCOEF_ACCESS_VAR of GEN3_FCOEF_VEC_DEPTH_OFF must be set to the depth pointer you want to access before programming this register. The Cursor is calculated based on the formula: *Cursor = FS - Post-cursor - Pre-cursor*.
- **Size:** 32 bits
- **Offset:** B+0x540
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RSVDP_28
27:22	GEN3_FCOEF_LANE13_POST_CURSOR
21:16	GEN3_FCOEF_LANE13_PRE_CURSOR
15:12	RSVDP_12
11:6	GEN3_FCOEF_LANE12_POST_CURSOR
5:0	GEN3_FCOEF_LANE12_PRE_CURSOR

Table 3-458 Fields for Register: GEN3_FCOEF_LANE13_12_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27:22	GEN3_FCOEF_LANE13_POST_CURSOR	R/W	Lane 13 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
21:16	GEN3_FCOEF_LANE13_PRE_CURSOR	R/W	Lane 13 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:6	GEN3_FCOEF_LANE12_POST_CURSOR	R/W	Lane 12 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5:0	GEN3_FCOEF_LANE12_PRE_CURSOR	R/W	Lane 12 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.110 GEN3_FCOEF_LANE15_14_OFF

- **Name:** Gen3 FOM Coefficients Table Register for Lane 15 and 14.
- **Description:** This register array holds the coefficients for FOM mode. You must program the coefficients without breaking the Rules for Transmitter Coefficients in the PCIe Base Specification. The value for the register field GEN3_FCOEF_ACCESS_VAR of GEN3_FCOEF_VEC_DEPTH_OFF must be set to the depth pointer you want to access before programming this register. The Cursor is calculated based on the formula: *Cursor = FS - Post-cursor - Pre-cursor*.
- **Size:** 32 bits
- **Offset:** B+0x544
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:28	RSVDP_28
27:22	GEN3_FCOEF_LANE15_POST_CURSOR
21:16	GEN3_FCOEF_LANE15_PRE_CURSOR
15:12	RSVDP_12
11:6	GEN3_FCOEF_LANE14_POST_CURSOR
5:0	GEN3_FCOEF_LANE14_PRE_CURSOR

Table 3-459 Fields for Register: GEN3_FCOEF_LANE15_14_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
27:22	GEN3_FCOEF_LANE15_POST_CURSOR	R/W	Lane 15 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
21:16	GEN3_FCOEF_LANE15_PRE_CURSOR	R/W	Lane 15 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
11:6	GEN3_FCOEF_LANE14_POST_CURSOR	R/W	Lane 14 Post-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5:0	GEN3_FCOEF_LANE14_PRE_CURSOR	R/W	Lane 14 Pre-cursor. <i>Note:</i> ■ When CX_GEN4_SPEED/CX_GEN5_SPEED, this register is shadow register for Gen3 and Gen4/Gen5 data rate. ■ If RATE_SHADOW_SEL==00b, this register is for Gen3 data rate. ■ If RATE_SHADOW_SEL==01b, this register is for Gen4 data rate. ■ If RATE_SHADOW_SEL==10b, this register is for Gen5 data rate. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE && CX_NL_GTR_8 && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.111 CXL_VLSM_CR_REG_OFF

- **Name:** CXL VLSM CREDIT REGISTER.
- **Description:** CXL VLSM CREDIT REGISTER at 'h548.
- **Size:** 32 bits
- **Offset:** B+0x548
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:26	RSVDP_26
25	CXL_APN_SW_MODE
24	CXL_APN_BYPASS_ENABLE
23:13	RSVDP_13
12	CXL_VLSM_LP_ALMP_ENABLE
11	CXL_VLSM_TO_ACTIVE
10:8	CXL_VLSM_CR_STATE
7:4	CXL_CACHE_MEM_VLSM_STATE
3:0	CXL_IO_VLSM_STATE

Table 3-460 Fields for Register: CXL_VLSM_CR_REG_OFF

Bits	Name	Memory Access	Description
31:26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
25	CXL_APN_SW_MODE	R/W	Indicates CXL Alternate Protocol Negotiation Switch mode in downstream port with CXL 2.0 capable when receiving CXL 1.1 capable from upstream component. The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> This register field is sticky. Values: 0x0 (ALT_PROT_NEG0): Return the Alternate Protocol Negotiation Status to 01b (not support) according to CXL 2.0 specification (Default) 0x1 (ALT_PROT_NEG1): Return the Alternate Protocol Negotiation Status to 10b (support) for a specific requirement. Value After Reset: DEFAULT_CXL_APN_SW_MODE Exists: (CX_CXL_ENABLE && CX_CXL_2_0_ENABLE && CX_IS_SW) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
24	CXL_APN_BYPASS_ENABLE	R/W	Enable Alternate Protocol Bypass. 1: bypass, 0: no bypass. (default value 0). The Flex Bus physical layer samples the value in this register only during exit from the Detect LTSSM state and ignores any changes to this register in all other LTSSM states. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
23:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
12	CXL_VLSM_LP_ALMP_ENABLE	R/W	Enable ALMP Handshake on entry to Low Power states.(default value 1). <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
11	CXL_VLSM_TO_ACTIVE	R/W	Force VLSM to Active state (default value 0). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
10:8	CXL_VLSM_CR_STATE	R	VLSM State. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
7:4	CXL_CACHE_MEM_VLSM_STATE	R	CXL.cache/me VSLM State. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
3:0	CXL_IO_VLSM_STATE	R	CXL.io VSLM State. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.112 CXL_LRSM_CSR_REG_OFF

- **Name:** CXL LRSM CSR.
- **Description:** CXL LRSM CSR REGISTER at 'h54c.
- **Size:** 32 bits
- **Offset:** B+0x54c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:23	RSVDP_23
22	CXL_LRSM_FATAL_ERROR
21:16	CXL_LRSM_STATE
15:13	RSVDP_13
12:8	CXL_LRSM_MUX_NUM_REINIT
7:5	RSVDP_5
4:0	CXL_LRSM_MUX_NUM_RETRY

Table 3-461 Fields for Register: CXL_LRSM_CSR_REG_OFF

Bits	Name	Memory Access	Description
31:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
22	CXL_LRSM_FATAL_ERROR	R	Asserted when unexpected Retry.Ack is received or LRSM reaches Abort state. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true

Bits	Name	Memory Access	Description
21:16	CXL_LRSM_STATE	R	Reflects the current state of the LRSM. Value After Reset: 0x8 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: untestable Volatile: true
15:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
12:8	CXL_LRSM_MUX_NUM_REINIT	R/W	Maximum number of transitions to recovery before aborting the retry sequence. <i>Note:</i> This register field is sticky. Value After Reset: 0xa Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
7:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
4:0	CXL_LRSM_MUX_NUM_RETRY	R/W	Maximum number of retry attempts before transitioning to Recovery. <i>Note:</i> This register field is sticky. Value After Reset: 0xa Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.113 CXL_LL_CTRL_CR_REG_OFF

- **Name:** CXL Link Layer Control credit.
- **Description:** CXL Link Layer Control credit REGISTER at 'h550.
- **Size:** 32 bits
- **Offset:** B+0x550
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	CXL_AP_PROT_ENA
30:28	CXL_AP_PROT_TYPE
27:26	RSVDP_26
25:16	CXL_CRC_THRESHOLD
15:0	CXL_LRSM_TIMEOUT_THRESHOLD

Table 3-462 Fields for Register: CXL_LL_CTRL_CR_REG_OFF

Bits	Name	Memory Access	Description
31	CXL_AP_PROT_ENA	R/W	CXL_AP_PROT_ENA. Enables Alternate Protocol over CXL Data Link. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_CXL_AP_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
30:28	CXL_AP_PROT_TYPE	R/W	CXL_AP_PROT_TYPE. Selects Alternate Protocol over CXL Data Link: 0 - CCIX 2.0 ; default: N/A. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_CXL_AP_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
27:26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
25:16	CXL_CRC_THRESHOLD	R/W	CRC threshold for Link Layer Timer Credit. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
15:0	CXL_LRSM_TIMEOUT_THRESH OLD	R/W	Sets the threshold for the retry timeout. This indicates the number of flits which can be transmitted while waiting for a Retry.Ack before a retry timeout occurs. <i>Note:</i> This register field is sticky. Value After Reset: 0x1000 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.114 CXL_FATAL_ERR_CR_REG_OFF

- **Name:** CXL FATAL ERROR.
- **Description:** CXL FATAL ERROR credit REGISTER at 'h554.
- **Size:** 32 bits
- **Offset:** B+0x554
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31	RSVDP_31
30	CXL_IDE_RX_PLAIN_IF_ERR
29	CXL_IDE_RX_SECURE_IF_ERR
28	CXL_IDE_TX_PLAIN_IF_ERR
27	CXL_IDE_TX_SECURE_IF_ERR
26	CXL_MDH_ERR
25	CXL_MEM_DATA_VD_ERR
24	CXL_CACHE_DATA_VD_ERR
23	RSVDP_23
22	CXL_H2D_ORD_VIOLATION
21	CXL_AP_VIRAL_REQ
20	CXL_CXS_TX_END_ERR
19	CXL_ULL_CXS_END_ERR
18:14	CXL_LL_RX_FLOW_CTRL_ERR
13	CXL_RETRY_CRC_ERR
12	CXL_RETRY_ESEQ_OVERFLOW
11	CXL_LL_MULTI_INIT_PARAM
10	CXL_LL_INIT_PARAM_ERR
9:5	CXL_LL_TC_FLOW_CTRL_ERR
4	CXL_LRSM_ACK_ERROR
3	CXL_LRSM_ABORT
2	CXL_LL_STATIC_BIT_ERR
1	CXL_RAS_ERROR
0	CXL_FATAL_ERR_RETRY_ACK_VIRAL

Table 3-463 Fields for Register: CXL_FATAL_ERR_CR_REG_OFF

Bits	Name	Memory Access	Description
31	RSVDP_31	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
30	CXL_IDE_RX_PLAIN_IF_ERR	R	IDE RX PlainIF ERROR. IDE RX Data Protection Error on Plain IF. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE<=2 && CX_CXL_IDE_DP_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
29	CXL_IDE_RX_SECURE_IF_ERR	R	IDE RX SecureIF ERROR. IDE RX Data Protection Error on Secure IF. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE<=2 && CX_CXL_IDE_DP_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
28	CXL_IDE_TX_PLAIN_IF_ERR	R	IDE TX PlainIF ERROR. IDE TX Data Protection Error on Plain IF. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE<=2 && CX_CXL_IDE_DP_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
27	CXL_IDE_TX_SECURE_IF_ERR	R	IDE TX SecureIF ERROR. IDE TX Data Protection Error on Secure IF. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE<=2 && CX_CXL_IDE_DP_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
26	CXL_MDH_ERR	R	MDH ERROR. Multi-Data Header is received when MDH is disabled. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE<=2 && CX_CXL_CXS_ENABLE==0) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
25	CXL_MEM_DATA_VD_ERR	R	Mem_Data_Valid_Error. M2S/S2M Data Valid Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_CXL_CHI_DATACHK_EN && CX_CXL_MEM_EN) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
24	CXL_CACHE_DATA_VD_ERR	R	Cache_Data_Valid_Error. H2D/D2H Data Valid Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_CXL_CHI_DATACHK_EN && CX_CXL_CACHE_EN) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
22	CXL_H2D_ORD_VIOLATION	R	H2D_Ordering_Violation. H2D Response/Request ordering violation detected. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_CXL_CACHE_EN) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
21	CXL_AP_VIRAL_REQ	R	App_Viral_Request. Set this bit if the Applications requests to trigger Viral Mode in CXL Controller. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_CXL_CHI_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
20	CXL_CXS_TX_END_ERR	R	If CXS Controller transmits CXS Flit with ENDERROR bit set in the Control Bus then trigger Viral Mode. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_CXL_CXS_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
19	CXL_ULL_CXS_END_ERR	R	ULL Transmits CXS Flit with ENDERROR bit set in the Control Bus to trigger Viral Mode. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_CXL_CXS_ENABLE) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
18:14	CXL_LL_RX_FLOW_CTRL_ERR	R	LL_RX_FlowControl_Error: CXL Receiver Flow Control Error: <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CACHE_REQ_RN): Cache Request Credit Return Overflow. ■ 0x1 (CACHE_RES_RN): Cache Response Credit Return Overflow. ■ 0x2 (CACHE_DATA_RN): Cache Data Credit Return Overflow. ■ 0x3 (MEM_NDR_REQ_RN): Mem NDR/RequestOnly Credit Return Overflow. ■ 0x4 (MEM_DRC_RWD_RN): Mem DRC/RwD Credit Return Overflow. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
13	CXL_RETRY_CRC_ERR	R	Retry_CRC_Error: Retry Buffer CRC error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
12	CXL_RETRY_ESEQ_OVERFLOW	R	Retry_Eseq_Overflow: Retry Buffer Eseq Counter Overflow. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
11	CXL_LL_MULTI_INIT_PARAM	R	LL_Multi_Init_Param: Receipt of an LLCTRL-INIT.Param flit after an LLCTRL-INIT.Param flit has already been received. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
10	CXL_LL_INIT_PARAM_ERR	R	LL_Init_Param_Error: Non-Retry flits received before LLCTRL-INIT.Param flit. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
9:5	CXL_LL_TC_FLOW_CTRL_ERR	R	LL_TX_FlowControl_Error: CXL Transmitter Flow Control Error: <i>Note:</i> This register field is sticky. Values: ■ 0x0 (CACHE_REQ_CR): Cache Request Credit Overflow. ■ 0x1 (CACHE_RES_CR): Cache Response Credit Overflow. ■ 0x2 (CACHE_DATA_CR): Cache Data Credit Overflow. ■ 0x3 (MEM_NDR_REQ_CR): Mem NDR/RequestOnly Credit Overflow. ■ 0x4 (MEM_DRC_RWD_CR): Mem DRC/RwD Credit Overflow. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
4	CXL_LRSM_ACK_ERROR	R	LRSM_Ack_Error: Unexpected Retry.Ack flit is received. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
3	CXL_LRSM_ABORT	R	LRSM_Abort: LRSM FSM reached Abort state. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2	CXL_LL_STATIC_BIT_ERR	R	LL_Static_Bit_Error: Static bit error is detected in the received LL Control Flit. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
1	CXL_RAS_ERROR	R	RAS_Error: RAS uncorrectable error detected. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	CXL_FATAL_ERR_RETRY_ACK_VIRAL	R	Retry_Ack_Viral: Retry.Ack control flit is received with Viral status. Valid if Viral is enabled. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.115 CXL_CACHE_MEM_LL_INJ_OFF

- **Name:** CXL.cache / CXL.mem Link Layer Error Injection Register.
- **Description:** Register 1 for Link Layer Error Injection CXL at address 'h558 in Port Logic.
- **Size:** 32 bits
- **Offset:** B+0x558
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:11	RSVDP_11
10	CACHE_MEM_CRC_INJ_COUNT_BUSY
9:8	CACHE_MEM_CRC_INJ_COUNT
7:6	CACHE_MEM_CRC_INJ
5	IO_POISON_INJ_BUSY
4	IO_POISON_INJ_START
3	MEM_POISON_INJ_BUSY
2	MEM_POISON_INJ_START
1	CACHE_POISON_INJ_BUSY
0	CACHE_POISON_INJ_START

Table 3-464 Fields for Register: CXL_CACHE_MEM_LL_INJ_OFF

Bits	Name	Memory Access	Description
31:11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
10	CACHE_MEM_CRC_INJ_COUNT_BUSY	R	Hardware loads 1'b1 to this bit when the Start bit is written. Hardware must clear this bit to indicate that it has indeed finished CRC injections. Software is permitted to poll on this bit to find out when hardware has finished CRC injection. This bit is required if any of CXL.cache or CXL.mem protocols are supported. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
9:8	CACHE_MEM_CRC_INJ_COUNT	R/W	Software writes to these bits to program the number of CRC injections. This field must be programmed by software before OR at the same time as CacheMemCRCInjection field. The number of flits where CRC bits are flipped is given as follows: This field is required if any of CXL.cache or CXL.mem protocols are supported. Values: ■ 0x0 (DISABLE_CRC): No CRC errors are injected ■ 0x1 (SINGLE_BIT_FLIT): CRC injection is only for 1 flit. CacheMemCRCInjectionBusy bit is cleared after 1 injection ■ 0x2 (TWO_INJ_FLIT): CRC injection is for 2 flits in succession. CacheMemCRCInjectionBusy bit is cleared after 2 injections ■ 0x3 (THREE_INJ_FLIT): CRC injection is for 3 flits in succession. CacheMemCRCInjectionBusy bit is cleared after 3 injections Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
7:6	CACHE_MEM_CRC_INJ	R/W	Software writes to these bits to trigger CRC error injections. The number of CRC bits flipped is given as follows: The specific bit positions that are flipped are implementation specific. This field is required if any of CXL.cache or CXL.mem protocols are supported. Values: ■ 0x0 (DISABLE): No CRC errors are injected ■ 0x1 (SINGLE_FLIT): Single bit flipped in the CRC field for ?n? subsequent Tx flits, where n is the value in CacheMemCRCInjectionCount ■ 0x2 (DOUBLE_FLIT): 2 bits flipped in the CRC field for ?n? subsequent Tx flits, where n is the value in CacheMemCRCInjectionCount ■ 0x3 (TRIPLE_FLIT): 3 bits flipped in the CRC field for ?n? subsequent Tx flits, where n is the value in CacheMemCRCInjectionCount Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
5	IO_POISON_INJ_BUSY	R	Hardware loads 1'b1 to this bit when the Start bit is written. Hardware must clear this bit to indicate that it has indeed finished poisoning a packet. Software is permitted to poll on this bit to find out when hardware has finished poison injection. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
4	IO_POISON_INJ_START	R/W	Software writes 0x1 to this bit to trigger a single poison injection on a CXL.io message in the Tx direction. Hardware must override the poison field in the data header slot of the corresponding message. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	MEM_POISON_INJ_BUSY	R	Hardware loads 1'b1 to this bit when the Start bit is written. Hardware must clear this bit to indicate that it has indeed finished poisoning a packet. Software is permitted to poll on this bit to find out when hardware has finished poison injection. This bit is required only if CXL.mem protocol is supported. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
2	MEM_POISON_INJ_START	R/W	Software writes 0x1 to this bit to trigger a single poison injection on a CXL.mem message in the Tx direction. Hardware must override the poison field in the data header slot of the corresponding. This bit is required only if CXL.mem protocol is supported. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
1	CACHE_POISON_INJ_BUSY	R	Hardware loads 1'b1 to this bit when the Start bit is written. Hardware must clear this bit to indicate that it has indeed finished poisoning a packet. Software is permitted to poll on this bit to find out when hardware has finished poison injection. This bit is required only if CXL.cache protocol is supported. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	CACHE_POISON_INJ_START	R/W	Software writes 0x1 to this bit to trigger a single poison injection on a CXL.cache message in the Tx direction. Hardware must override the poison field in the data header slot of the corresponding message (D2H if device, H2D if Host). This bit is required only if CXL.cache protocol is supported. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.116 CXL_IO_LL_INJ_OFF

- **Name:** CXL.io Link Layer Error Injection Register.
- **Description:** Register 2 for Link Layer Error Injection CXL at address 'h55C in Port Logic.
- **Size:** 32 bits
- **Offset:** B+0x55c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:4	RSVDP_4
3	FLOW_CTRL_ERR_INJ_BUSY
2	FLOW_CTRL_ERR_INJ
1	IO_LL_POISON_INJ_BUSY
0	IO_LL_POISON_INJ_START

Table 3-465 Fields for Register: CXL_IO_LL_INJ_OFF

Bits	Name	Memory Access	Description
31:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
3	FLOW_CTRL_ERR_INJ_BUSY	R	Hardware loads 1'b1 to this bit when the Start bit is written. Hardware must clear this bit to indicate that it has indeed finished Flow Control error injections. Software is permitted to poll on this bit to find out when hardware has finished Flow Control error injection. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2	FLOW_CTRL_ERR_INJ	R/W	Software writes 0x1 to this bit to trigger a Flow Control error on CXL.io only. Hardware must override the Flow Control DLLP. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
1	IO_LL_POISON_INJ_BUSY	R	Hardware loads 1'b1 to this bit when the Start bit is written. Hardware must clear this bit to indicate that it has indeed finished poisoning a packet. Software is permitted to poll on this bit to find out when hardware has finished poison injection. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	IO_LL_POISON_INJ_START	R/W	Software writes 0x1 to this bit to trigger a single poison injection on a CXL.io message in the Tx direction. Hardware must override the poison field in the data header slot of the corresponding message. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.117 FLEXBUS_LOG_PHY_ERR_INJ_OFF

- **Name:** Flex Bus Log Phy Protocol Error Injection Register.
- **Description:** Register 3 for log phy Error Injection at address 'h560 in Port Logic.
- **Size:** 32 bits
- **Offset:** B+0x560
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:4	RSVDP_4
3	FB_PROTO_ID_ERR_INJ_BUSY
2	FB_UNEXP_PROTO_ID_ERR_INJ
1	FB_UNCORR_PROTO_ID_ERR_INJ
0	FB_CORR_PROTO_ID_ERR_INJ

Table 3-466 Fields for Register: FLEXBUS_LOG_PHY_ERR_INJ_OFF

Bits	Name	Memory Access	Description
31:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
3	FB_PROTO_ID_ERR_INJ_BUSY	R	Hardware loads 1'b1 to this bit when the Start bit is written. Hardware must clear this bit to indicate that it has indeed finished Protocol ID error injections. Software is permitted to poll on this bit to find out when hardware has finished Protocol ID error injection. Software should only program one of the bits between correctable, uncorrectable and unexpected protocol ID error injection bits. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
2	FB_UNEXP_PROTO_ID_ERR_INJ	R/W	Software writes 0x1 to this bit to trigger an unexpected protocol ID error on any CXL flit issued by the FlexBus LogPHY. Hardware must override the Protocol ID field in the flit. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
1	FB_UNCORR_PROTO_ID_ERR_INJ	R/W	Software writes 0x1 to this bit to trigger an uncorrectable protocol ID error on any CXL flit issued by the FlexBus LogPHY. Hardware must override the Protocol ID field in the flit. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true
0	FB_CORR_PROTO_ID_ERR_INJ	R/W	Software writes 0x1 to this bit to trigger a correctable protocol ID error on any CXL flit issued by the FlexBus LogPHY. Hardware must override the Protocol ID field in the flit. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.118 CXL_PORT_LOGIC_VIRAL_STS_OFF

- **Name:** CXL Port Logic Viral Status Register.
- **Description:** Viral status and Viral LD-ID reg field at offset 'h560 in Port Logic.
- **Size:** 32 bits
- **Offset:** B+0x564
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:17	RSVDP_17
16:1	CXL_PL_VIRAL_LDID_STS
0	CXL_PL_VIRAL_STATUS

Table 3-467 Fields for Register: CXL_PORT_LOGIC_VIRAL_STS_OFF

Bits	Name	Memory Access	Description
31:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
16:1	CXL_PL_VIRAL_LDID_STS	R/W	CXL Port Logic Viral LDID Status (applicable when MLD is enabled else reserved). Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_CXL_MLD_EN) && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	CXL_PL_VIRAL_STATUS	R/W	CXL Port Logic Viral Status at bit 0. Value After Reset: 0x0 Exists: CX_CXL_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.48.119 ECAM_BASE_ADDR_LWR_OFF

- **Name:** Lower ECAM Base Address Register.
- **Description:** Lower ECAM Base Address register at address 'h570 in Port Logic.
- **Size:** 32 bits
- **Offset:** B+0x570
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))



Table 3-468 Fields for Register: ECAM_BASE_ADDR_LWR_OFF

Bits	Name	Memory Access	Description
31:0	ECAM_BASE_ADDR_LWR	R/W	Lower 32 bits of the 64 bit ECAM Base Address. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_SBSA_ECAM && AXI_POPULATED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.120 ECAM_BASE_ADDR_UPPER_OFF

- **Name:** Upper ECAM Base Address Register.
- **Description:** Upper ECAM Base Address register at address 'h574 in Port Logic.
- **Size:** 32 bits
- **Offset:** B+0x574
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

 31:0
ECAM_BASE_ADDR_UPPER

Table 3-469 Fields for Register: ECAM_BASE_ADDR_UPPER_OFF

Bits	Name	Memory Access	Description
31:0	ECAM_BASE_ADDR_UPPER	R/W	Upper 32 bits of the 64 bit ECAM Base Address. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_SBSA_ECAM && AXI_POPULATED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.121 ECAM_CTRL_OFF

- **Name:** ECAM Control Register.
- **Description:** ECAM Control Register located at address 'h578 in Port Logic.
- **Size:** 32 bits
- **Offset:** B+0x578
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:2	RSVDP_2
1	ECAM_CTRL_TYPE1_TO_TYPE0_DISABLE
0	ECAM_CTRL_DSP_ECAM_EN

Table 3-470 Fields for Register: ECAM_CTRL_OFF

Bits	Name	Memory Access	Description
31:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_SBSA_ECAM && AXI_POPULATED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
1	ECAM_CTRL_TYPE1_TO_TYPE0_DISABLE	R/W	Disable of Type 1 Configuration Request to Type 0 Configuration Request conversion. By default, Type 1 to Type 0 conversion takes place. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_SBSA_ECAM && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))
0	ECAM_CTRL_DSP_ECAM_EN	R	Enable Downstream Port ECAM Base Address Matching. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_SBSA_ECAM && AXI_POPULATED && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.122 DBI_FUNCTION_BANK_CTRL_REG_OFF

- **Name:** DBI Function Bank Control Register.
- **Description:** This register is used to select the PF which is accessible either from lower or upper bank of PFs from DBI.
- **Size:** 32 bits
- **Offset:** B+0x57c
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:1	RSVDP_1
0	DBI_FUNCTION_BANK_CTRL_REG

Table 3-471 Fields for Register: DBI_FUNCTION_BANK_CTRL_REG_OFF

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
0	DBI_FUNCTION_BANK_CTRL_REG	* Varies	DBI Function Bank Select. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Values: ■ 0x0 (BANK0): Functions 0 to 31 are accessible from DBI. ■ 0x1 (BANK1): Functions 32 to 63 are accessible from DBI. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true Memory Access: "(AXI_POPULATED) ? \"read-write\" : \"read-only\" "

3.48.123 UTILITY_OFF

- **Name:** UTILITY register (Reserved).
- **Description:** This is a reserved register.
- **Size:** 32 bits
- **Offset:** B+0x580
- **Exists:** 0

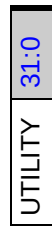


Table 3-472 Fields for Register: UTILITY_OFF

Bits	Name	Memory Access	Description
31:0	UTILITY	R/W	Reserved. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.124 AXIG_CTRL_OFF

- **Name:** AXI-G Control Register.
- **Description:** AXI-G Control Register located at address 'h584 in Port Logic.
- **Size:** 32 bits
- **Offset:** B+0x584
- **Exists:** (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 || CX_MEMORY_MAP_PCIE_TYPE==0))

31:1	
RSVDP_1	AXIG_CTRL_RCHUNK_DISABLE
	0

Table 3-473 Fields for Register: AXIG_CTRL_OFF

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_AXI_RCHUNK_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

Bits	Name	Memory Access	Description
0	AXIG_CTRL_RCHUNK_DISABLE	R/W	Force disable AXI Bridge Master Read Address Channel Chunk Enable (mstr_archunken) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (ENABLE_RCHUNK): Allow Bridge Master transactions with mstr_archunken driven HIGH. ■ 0x1 (DISABLE_RCHUNK): Drive mstr_archunken LOW for all Bridge Master transactions. Value After Reset: 0x0 Exists: CX_AXI_RCHUNK_ENABLE && (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0))

3.48.125 PM_UTILITY_OFF

- **Name:** PM Shadow of UTILITY register (Reserved).
- **Description:** This is the PM shadow copy of the UTILITY register.
- **Size:** 32 bits
- **Offset:** B+0x588
- **Exists:** 0

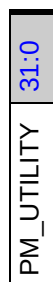


Table 3-474 Fields for Register: PM_UTILITY_OFF

Bits	Name	Memory Access	Description
31:0	PM_UTILITY	R	Reserved. Value After Reset: 0x0 Exists: (!CX_PL_REG_DISABLE && (CX_MEMORY_MAP_PCIE_TYPE==2 CX_MEMORY_MAP_PCIE_TYPE==0)) Testable: writeAsRead Volatile: true

3.49 VF_PF_TYPE0_HDRi Registers

3.49.1 VF_DEVICE_ID_VENDOR_ID_REG

- **Name:** Device ID , RCRB next offset pointer and Vendor ID Register.
- **Description:** This register holds the device ID, next offset pointer for RCRB mode and vendor ID.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1



Table 3-475 Fields for Register: VF_DEVICE_ID_VENDOR_ID_REG

Bits	Name	Memory Access	Description
31:16	PCI_TYPE0_DEVICE_ID	R	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB VERSION[19:16] & RCRB Next Offset Pointer[31:20]. ■ The two CXL based fields will be enabled when target protocol CXL is enabled and the memorymap region is RCRB. ❑ The Version number for the Root Complex Register Block (RCRB). This register space was reserved for RCRB memory mapped region in Type0 and Type1 Configuration Header capability. ❑ The RCRB Next offset pointer register identifies the offset value of the next register field. This register space was reserved for RCRB mode in Type0 and Type1. 0: DEVICE_ID [31:16] ■ The Device ID register identifies the particular Function for PCIe Type0 and Type1 configuration header. This identifier is allocated by the vendor. Databook. Value After Reset: 0xffff Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
15:0	PCI_TYPE0_VENDOR_ID	R	Vendor ID. The Vendor ID register identifies the manufacturer of the Function. Valid vendor identifiers are allocated by the PCI-SIG to ensure uniqueness. It is not permitted to populate this register with a value of FFFFh, which is an invalid value for Vendor ID. Value After Reset: 0xffff Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.49.2 VF_STATUS_COMMAND_REG

- **Name:** Status and Command Register.
- **Description:** This register provides the status and controls the behavior of a function.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

31	DETECTED_PARITY_ERR
30	SIGNALD_SYS_ERR
29	RCVD_MASTER_ABORT
28	RCVD_TARGET_ABORT
27	SIGNALD_TARGET_ABORT
26:25	DEV_SEL_TIMING
24	MASTER_DPE
23	FAST_B2B_CAP
22	RSVDP_22
21	FAST_66MHZ_CAP
20	CAP_LIST
19	INT_STATUS
18:17	RSVDP_17
16	IMM_READINESS
15:11	PCI_TYPE_RESERV
10	PCI_TYPE0_INT_EN
9	RSVDP_9
8	PCI_TYPE0_SERREN
7	PCI_TYPE_IDSEL_STEPPING
6	PCI_TYPE0_PARITY_ERR_EN
5	PCI_TYPE_VGA_PALETTE_SNOOP
4	PCI_TYPE_MWI_ENABLE
3	PCI_TYPE0_SPECIAL_CYCLE_OPERATION
2	PCI_TYPE0_BUS_MASTER_EN
1	PCI_TYPE0_MEM_SPACE_EN
0	PCI_TYPE0_IO_EN

Table 3-476 Fields for Register: VF_STATUS_COMMAND_REG

Bits	Name	Memory Access	Description
31	DETECTED_PARITY_ERR	R/W1C	Detected Parity Error. Values: ■ 0x1 (SET): This bit is set by a Function whenever it receives a Poisoned TLP, regardless of the state the Parity Error Response bit in the Command register. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
30	SIGNALED_SYS_ERR	R/W1C	Signaled System Error. Values: 0x1 (SET): This bit is set when a Function sends an ERR_FATAL or ERR_NONFATAL message, and the SERR# Enable bit in the Command register is 1b. For Functions that do not send ERR_FATAL or ERR_NONFATAL messages, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
29	RCVD_MASTER_ABORT	R/W1C	Received Master Abort. Values: 0x1 (SET): This bit is set when a Requester receives a Completion with Unsupported Request Completion Status. For Functions that do not make Non-Posted Requests on their own behalf, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
28	RCVD_TARGET_ABORT	R/W1C	Received Target Abort. Values: 0x1 (SET): This bit is set when a Requester receives a Completion with Completer Abort Completion Status. For Functions that do not make Non-Posted Requests on their own behalf, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
27	SIGNALED_TARGET_ABORT	R/W1C	Signaled Target Abort. Values: 0x1 (SET): This bit is set when a Function completes a Posted or Non-Posted Request as a Completer Abort error. The controller hardwires this bit to 0b for Functions that do not signal Completer Abort. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
26:25	DEV_SEL_TIMING	R	DEVSEL Timing. This field was originally described in the PCI Local Bus Specification. Its functionality does not apply to PCI Express. The controller hardwires this field to 00b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
24	MASTER_DPE	R/W1C	Master Data Parity Error. This bit is set by a Function if the Parity Error Response bit in the Command register is 1b and either of the following two conditions occurs: ■ Function receives a Poisoned Completion ■ Function transmits a Poisoned Request If the Parity Error Response bit is 0b, this bit is never set. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
23	FAST_B2B_CAP	R	Fast Back to Back Transaction Capable. This bit was originally described in the PCI Local Bus Specification. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
22	RSVDP_22	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
21	FAST_66MHZ_CAP	R	66MHz Capable. This bit was originally described in the PCI Local Bus Specification. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
20	CAP_LIST	R	Capabilities List. Indicates the presence of an Extended Capability list item. Since all PCI Express device Functions are required to implement the PCI Express Capability structure, the controller hardwires this bit to 1b. Value After Reset: 0x1 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
19	INT_STATUS	R	Emulation interrupt pending. When set, indicates that an INTx emulation interrupt is pending internally in the Function. Setting the Interrupt Disable bit has no effect on the state of this bit. For Functions that do not generate INTx interrupts, the controller hardwires this bit to 0b. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true
18:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
16	IMM_READINESS	R	Immediate Readiness. This optional bit, when set, indicates the Function is guaranteed to be ready to successfully complete valid configuration accesses at any time following any reset that the host is capable of issuing Configuration Requests to this Function. When this bit is set, for accesses to this Function, software is exempt from all requirements to delay configuration accesses following any type of reset, including but not limited to the timing requirements defined in "PCI Express Reset - Rules" section of the <i>PCI Express Base Specification</i>. It is permitted that system software/firmware provide mechanisms that supersede the indication provided by this bit, however such software/firmware mechanisms are outside the scope of <i>PCI Express Base Specification</i>. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
15:11	PCI_TYPE_RESERV	R	Reserved. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
10	PCI_TYPE0_INT_EN	R	Interrupt Disable. Controls the ability of a Function to generate INTx emulation interrupts. When set, Functions are prevented from asserting INTx interrupts. Note: Any INTx emulation interrupts already asserted by the Function must be de-asserted when this bit is Set. INTx interrupts use virtual wires that must, if asserted, be de-asserted using the appropriate Deassert_INTx message(s) when this bit is set. Only the INTx virtual wire interrupt(s) associated with the Function(s) for which this bit is set are affected. For functions that generate INTx interrupts, this bit is required. For functions that do not generate INTx interrupts, this bit is optional. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
8	PCI_TYPE0_SERREN	R	SERR# Enable. When set, this bit enables reporting upstream of Non-fatal and Fatal errors detected by the Function. Note: The errors are reported if enabled either through this bit or through the PCI Express specific bits in the Device Control register. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
7	PCI_TYPE_IDSEL_STEPPING	R	IDSEL Stepping/Wait Cycle Control. This bit was originally described in the <i>PCI Local Bus Specification</i>. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
6	PCI_TYPE0_PARITY_ERR_EN	R	Parity Error Response. This bit controls the logging of poisoned TLPs in the Master Data Parity Error bit in the Status register. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
5	PCI_TYPE_VGA_PALETTE_SNOOP	R	VGA Palette Snoop. This bit was originally described in the <i>PCI Local Bus Specification</i> and the <i>PCI-to-PCI Bridge architecture specification</i>. Its functionality does not apply to PCI Express, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
4	PCI_TYPE_MWI_ENABLE	R	Memory Write and Invalidate. This bit was originally described in the <i>PCI Local Bus Specification</i> and the <i>PCI-to-PCI Bridge architecture specification</i>. Its functionality does not apply to PCI Express, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
3	PCI_TYPE0_SPECIAL_CYCLE_OPERATION	R	Special Cycle Enable. This bit was originally described in the <i>PCI Local Bus Specification</i>. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
2	PCI_TYPE0_BUS_MASTER_EN	* Varies	Bus Master Enable. Controls the ability of a Function to issue Memory and I/O Read/Write requests. ■ When this bit is set, the Function is allowed to issue Memory or I/O Requests. ■ When this bit is clear, the Function is not allowed to issue any Memory or I/O Requests. Requests other than Memory or I/O Requests are not controlled by this bit. <i>Note:</i> MSI/MSI-X interrupt Messages are in-band memory writes, setting the Bus Master Enable bit to 0b disables MSI/MSI-X interrupt Messages as well. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && !SNPS_RSVDPARAM_11) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && SNPS_RSVDPARAM_11) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==2 && !SNPS_RSVDPARAM_11) ? \"read-write\" : \"read-only\""
1	PCI_TYPE0_MEM_SPACE_EN	R	Memory Space Enable. Controls a Function's response to Memory Space accesses. ■ When this bit is set, the Function is enabled to decode the address and further process Memory Space accesses. ■ When this bit is clear, all received Memory Space accesses are caused to be handled as Unsupported Requests. For a Function does not support Memory Space accesses, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
0	PCI_TYPE0_IO_EN	R	IO Space Enable. Controls a Function's response to I/O Space accesses. ■ When this bit is set, the Function is enabled to decode the address and further process I/O Space accesses. ■ When this bit is clear, all received I/O accesses are caused to be handled as Unsupported Requests. For a Function that does not support I/O Space accesses, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.49.3 VF_CLASS_CODE_REVISION_ID

- **Name:** Class Code and Revision ID Register.
- **Description:** This register specifies the class code and revision ID of a function.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

31:24	BASE_CLASS_CODE
23:16	SUBCLASS_CODE
15:8	PROGRAM_INTERFACE
7:0	REVISION_ID

Table 3-477 Fields for Register: VF_CLASS_CODE_REVISION_ID

Bits	Name	Memory Access	Description
31:24	BASE_CLASS_CODE	R	Base Class Code. A code that broadly classifies the type of operation the Function performs. Encodings for base class, are provided in the PCI Code and ID Assignment Specification. All unspecified encodings are Reserved. <i>Note:</i> This register field is sticky. Value After Reset: BASE_CLASS_CODE_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:16	SUBCLASS_CODE	R	Sub-Class Code. Specifies a base class sub-class, which identifies more specifically the operation of the Function. Encodings for sub-class are provided in the PCI Code and ID Assignment Specification. All unspecified encodings are Reserved. <i>Note:</i> This register field is sticky. Value After Reset: SUB_CLASS_CODE_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
15:8	PROGRAM_INTERFACE	R	Programming Interface. This field identifies a specific register-level programming interface (if any) so that device independent software can interact with the Function. Encodings for interface are provided in the PCI Code and ID Assignment Specification. All unspecified encodings are Reserved. <i>Note:</i> This register field is sticky. Value After Reset: IF_CODE_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
7:0	REVISION_ID	R	Revision ID. The value in this register specifies a Function specific revision identifier. The value is chosen by the vendor. Zero is an acceptable value. The Revision ID should be viewed as a vendor defined extension to the Device ID. <i>Note:</i> This register field is sticky. Value After Reset: CX_REVISION_ID_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.49.4 VF_BIST_HEADER_TYPE_LATENCY_CACHE_LINE_SIZE_REG

- **Name:** BIST, Header Type, Latency Timer, and Cache Line Size Register.
- **Description:** This register provides the status and controls BIST. It also holds information regarding the header layout, latency timer, and cache line size.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

BIST	31:24
MULTI_FUNC	23
HEADER_TYPE	22:16
LATENCY_MASTER_TIMER	15:8
CACHE_LINE_SIZE	7:0

Table 3-478 Fields for Register: VF_BIST_HEADER_TYPE_LATENCY_CACHE_LINE_SIZE_REG

Bits	Name	Memory Access	Description
31:24	BIST	R	BIST. This register is used for control and status of BIST. For Functions that do not support BIST the controller hardwires the register to 00h. A Function whose BIST is invoked must not prevent normal operation of the PCI Express Link. Bit descriptions: ■ [31]: BIST Capable. When Set, this bit indicates that the Function supports BIST. When Clear, the Function does not support BIST. ■ [30]: Start BIST. If BIST Capable is Set, Set this bit to invoke BIST. The Function resets the bit when BIST is complete. Software is permitted to fail the device if this bit is not Clear (BIST is not complete) 2 seconds after it had been Set. Writing this bit to 0b has no effect. This bit must be hardwired to 0b if BIST Capable is Clear. ■ [29:28]: Reserved. ■ [27:24]: Completion Code. This field encodes the status of the most recent test. A value of 0000b means that the Function has passed its test. Non-zero values mean the Function failed. Function-specific failure codes can be encoded in the non-zero values. This field's value is only meaningful when BIST Capable is Set and Start BIST is Clear. This field must be hardwired to 0000b if BIST Capable is clear. Values: ■ 0x0 (Min_field_value): Zero value ■ 0xff (Max_field_value): Max value Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
23	MULTI_FUNC	R	Multi-Function Device. Except where stated otherwise, it is recommended that this bit be set if there are multiple Functions, and clear if there is only one Function. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): Indicates that the Device may contain multiple Functions, but not necessarily. Software is permitted to probe for Functions other than Function 0 ■ 0x0 (CLEAR): Software must not probe for Functions other than Function 0 unless explicitly indicated by another mechanism, such as an ARI or SR-IOV Capability structure. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
22:16	HEADER_TYPE	R	Header Layout. This field identifies the layout of the second part of the predefined header. The controller uses 000 0000b encoding. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
15:8	LATENCY_MASTER_TIMER	R	Latency Timer. The Latency Timer was originally described in the PCI Local Bus Specification and the PCI-to-PCI Bridge Architecture Specification. Its functionality does not apply to PCI Express. The controller hardwires this register to 00h. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
7:0	CACHE_LINE_SIZE	R	Cache Line Size. The Cache Line Size register is programmed by the system firmware or the operating system to system cache line size. However, legacy conventional PCI software may not always be able to program this register correctly especially in the case of Hot-Plug devices. This read-write register is implemented for legacy compatibility purposes but has no effect on any PCI Express device behavior. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.49.5 VF_BAR0_REG

- **Name:** BAR0 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR0_MASK_REG.PCI_TYPE0_BAR0_ENABLED)

31:4	3	2:1	0
BAR0_START	BAR0_PREFETCH	BAR0_TYPE	BAR0_MEM_IO

Table 3-479 Fields for Register: VF_BAR0_REG

Bits	Name	Memory Access	Description
31:4	BAR0_START	R	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR0[31:4]. ■ The MEMBAR0 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. ■ This is 32bits field, we use this to write MEMBAR0 address into RCRB space. MEMBAR0 registers should be able to access BAR0_REG offset 10h of PCIE memory-mapped space. 0: BAR0_START. BAR0 Base Address. ■ Memory Space: Base Address. ■ IO Space: bits[31:2] are used to map the function into IO space/Base Address. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
3	BAR0_PREFETCH	R	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR0[3]. The MEMBAR0 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. This is 32bits field, we use this to write MEMBAR0 address into RCRB space. MEMBAR0 registers should be able to access BAR0_REG offset 10h of PCIE memory-mapped space. 0: BAR0 Prefetchable. Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must be clear otherwise. IO Space: Not applicable. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2:1	BAR0_TYPE	R	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR0[2:1]. ■ The MEMBAR0 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. ■ This is 32bits field, we use this to write MEMBAR0 address into RCRB space. MEMBAR0 registers should be able to access BAR0_REG offset 10h of PCIe memory-mapped space. 0: BAR0 Type. ■ Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. ■ IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. Values: ■ 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. ■ 0x1 (RSVD_1): Reserved. ■ 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. ■ 0x3 (RSVD_2): Reserved. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR0_MEM_IO	R	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR0[0]. ■ The MEMBAR0 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. ■ This is 32bits field, we use this to write MEMBAR0 address into RCRB space. MEMBAR0 registers should be able to access BAR0_REG offset 10h of PCIe memory-mapped space. 0: BAR0 Memory Space Indicator. ■ This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. ■ Base Address registers that map to I/O Space must return a 1b. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.49.6 VF_BAR1_REG

- **Name:** BAR1 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR1_MASK_REG.PCI_TYPE0_BAR1_ENABLED || BAR0_TYPE_n ==2)

BAR1_START	31:4
BAR1_PREFETCH	3
BAR1_TYPE	2:1
BAR1_MEM_IO	0

Table 3-480 Fields for Register: VF_BAR1_REG

Bits	Name	Memory Access	Description
31:4	BAR1_START	R	- The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR1[31:4]. - The MEMBAR1 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. - This is 32bits field, we use this to write MEMBAR1 address into RCRB space. MEMBAR1 registers should be able to access BAR1_REG offset 14h of PCIe memory-mapped space. 0: BAR1 Base Address. - Memory Space: Base Address. - IO Space: bits[31:2] are used to map the function into IO space/Base Address. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
3	BAR1_PREFETCH	R	The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR1[3]. The MEMBAR1 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. This is 32bits field, we use this to write MEMBAR1 address into RCRB space. MEMBAR1 registers should be able to access BAR1_REG offset 14h of PCIe memory-mapped space. 0: BAR1 Prefetchable. Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must me clear otherwise. IO Space: Not applicable. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2:1	BAR1_TYPE	R	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR1[2:1]. ■ The MEMBAR1 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. ■ This is 32bits field, we use this to write MEMBAR1 address into RCRB space. MEMBAR1 registers should be able to access BAR1_REG offset 14h of PCIE memory-mapped space. 0: BAR1 Type. ■ Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. ■ IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. Values: ■ 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. ■ 0x1 (RSVD_1): Reserved. ■ 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. ■ 0x3 (RSVD_2): Reserved. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR1_MEM_IO	R	■ The name and the function of this field depends on the value of CX_CXL_ENABLE parameter as follows: 1: RCRB MEMBAR1[0]. ■ The MEMBAR1 register field is implemented in Type0 and Type1 Config Header capability when target protocol CXL is enabled and when the memorymap region is RCRB. ■ This is 32bits field, we use this to write MEMBAR1 address into RCRB space. MEMBAR1 registers should be able to access BAR1_REG offset 14h of PCIe memory-mapped space. 0: BAR1 Memory Space Indicator. ■ This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. ■ Base Address registers that map to I/O Space must return a 1b. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.49.7 VF_BAR2_REG

- **Name:** BAR2 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR2_MASK_REG.PCI_TYPE0_BAR2_ENABLED)

31:4	3	2:1	0
BAR2_START	BAR2_PREFETCH	BAR2_TYPE	BAR2_MEM_IO

Table 3-481 Fields for Register: VF_BAR2_REG

Bits	Name	Memory Access	Description
31:4	BAR2_START	R	BAR2 Base Address. ■ Memory Space: Base Address. ■ IO Space: bits[31:2] are used to map the function into IO space/Base. Address. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	BAR2_PREFETCH	R	BAR2 Prefetchable. ■ Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must be clear otherwise. ■ IO Space: Not applicable Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
2:1	BAR2_TYPE	R	BAR2 Type. ■ Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. ■ IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. Values: ■ 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. ■ 0x1 (RSVD_1): Reserved. ■ 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. ■ 0x3 (RSVD_2): Reserved. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR2_MEM_IO	R	BAR2 Memory Space Indicator. This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. Base Address registers that map to I/O Space must return a 1b. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.49.8 VF_BAR3_REG

- **Name:** BAR3 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR3_MASK_REG.PCI_TYPE0_BAR3_ENABLED || BAR2_TYPE_n == 2)

31:4	3	2:1	0
BAR3_START	BAR3_PREFETCH	BAR3_TYPE	BAR3_MEM_IO

Table 3-482 Fields for Register: VF_BAR3_REG

Bits	Name	Memory Access	Description
31:4	BAR3_START	R	BAR3 Base Address. ■ Memory Space: Base Address. ■ IO Space: bits[31:2] are used to map the function into IO space/Base. Address. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	BAR3_PREFETCH	R	BAR3 Prefetchable. ■ Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must be clear otherwise. ■ IO Space: Not applicable Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
2:1	BAR3_TYPE	R	BAR3 Type. ■ Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. ■ IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. Values: ■ 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. ■ 0x1 (RSVD_1): Reserved. ■ 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. ■ 0x3 (RSVD_2): Reserved. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR3_MEM_IO	R	BAR3 Memory Space Indicator. This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. Base Address registers that map to I/O Space must return a 1b. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.49.9 VF_BAR4_REG

- **Name:** BAR4 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR4_MASK_REG.PCI_TYPE0_BAR4_ENABLED)

31:4	BAR4_START
3	BAR4_PREFETCH
2:1	BAR4_TYPE
0	BAR4_MEM_IO

Table 3-483 Fields for Register: VF_BAR4_REG

Bits	Name	Memory Access	Description
31:4	BAR4_START	R	BAR4 Base Address. ■ Memory Space: Base Address. ■ IO Space: bits[31:2] are used to map the function into IO space/Base. Address. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	BAR4_PREFETCH	R	BAR4 Prefetchable. ■ Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must be clear otherwise. ■ IO Space: Not applicable Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
2:1	BAR4_TYPE	R	BAR4 Type. ■ Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. ■ IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. Values: ■ 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. ■ 0x1 (RSVD_1): Reserved. ■ 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. ■ 0x3 (RSVD_2): Reserved. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR4_MEM_IO	R	BAR4 Memory Space Indicator. This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. Base Address registers that map to I/O Space must return a 1b. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.49.10 VF_BAR5_REG

- **Name:** BAR5 Register.
- **Description:** System software must build a consistent address map before booting the machine to an operating system. This means it has to determine how much memory is in the system, and how much address space the Functions in the system require. After determining this information, system software can map the Functions into reasonable locations and proceed with system boot. In order to do this mapping in a device-independent manner, the base registers for this mapping are placed in the predefined header portion of Configuration Space. It is strongly recommended that power-up firmware/software also support the optional Enhanced Configuration Access Mechanism (ECAM).
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** (PFn_TYPE0_HDR_DBI2.BAR5_MASK_REG.PCI_TYPE0_BAR5_ENABLED || BAR4_-TYPE_n ==2)

31:4	3	2:1	0
BAR5_START	BAR5_PREFETCH	BAR5_TYPE	BAR5_MEM_IO

Table 3-484 Fields for Register: VF_BAR5_REG

Bits	Name	Memory Access	Description
31:4	BAR5_START	R	BAR5 Base Address. ■ Memory Space: Base Address. ■ IO Space: bits[31:2] are used to map the function into IO space/Base. Address. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	BAR5_PREFETCH	R	BAR5 Prefetchable. ■ Memory Space: Set to one if data is prefetchable. A Function is permitted to mark a range as prefetchable. If there are no side effects on reads, the function returns all bytes on reads regardless of the byte enables, and host bridges can merge processor writes into this range without causing errors. Bit must be clear otherwise. ■ IO Space: Not applicable Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
2:1	BAR5_TYPE	R	BAR5 Type. ■ Memory Space: Base Address registers that map into Memory Space can be 32 bits or 64 bits wide (to support mapping into a 64-bit address space). The encodings defined in Values: apply. ■ IO Space: Bit 1 is reserved and must return 0b on reads. Bits[31:2] are used to map the function into IO space. The encodings defined in Values: do not apply. Values: ■ 0x0 (BAR_32): Base register is 32 bits wide and can be mapped anywhere in the 32 address bit Memory Space. ■ 0x1 (RSVD_1): Reserved. ■ 0x2 (BAR_64): Base register is 64 bits wide and can be mapped anywhere in the 64 address bit Memory Space. ■ 0x3 (RSVD_2): Reserved. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	BAR5_MEM_IO	R	BAR5 Memory Space Indicator. This bit is used to determine whether the register maps into Memory or I/O Space. Base Address registers that map to Memory Space must return a 0b. Base Address registers that map to I/O Space must return a 1b. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE!=1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.49.11 VF_CARDBUS_CIS_PTR_REG

- **Name:** CardBus CIS Pointer Register.
- **Description:** This register holds the CardBus CIS pointer.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

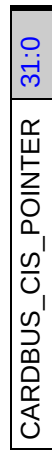


Table 3-485 Fields for Register: VF_CARDBUS_CIS_PTR_REG

Bits	Name	Memory Access	Description
31:0	CARDBUS_CIS_POINTER	R	CardBus CIS Pointer. Its functionality does not apply to PCI Express. It must be hardwired to 0000 0000h. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.49.12 VF_SUBSYSTEM_ID_SUBSYSTEM_VENDOR_ID_REG

- **Name:** Subsystem ID and Subsystem Vendor ID Register.
- **Description:** These registers are used to uniquely identify the add-in card or subsystem where the PCI Express component resides. They provide a mechanism for vendors to distinguish their products from one another even though the assemblies may have the same PCI Express component on them (and, therefore, the same Vendor ID and Device ID).
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1



Table 3-486 Fields for Register: VF_SUBSYSTEM_ID_SUBSYSTEM_VENDOR_ID_REG

Bits	Name	Memory Access	Description
31:16	SUBSYS_DEV_ID	R	Subsystem ID. <i>Note:</i> This register field is sticky. Value After Reset: SUBSYS_DEV_ID_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
15:0	SUBSYS_VENDOR_ID	R	Subsystem Vendor ID. Subsystem Vendor IDs can be obtained from the PCI SIG and are used to identify the vendor of the add-in card or subsystem. Values for the Subsystem ID are vendor-specific. <i>Note:</i> This register field is sticky. Value After Reset: SUBSYS_VENDOR_ID_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.49.13 VF_PCI_CAP_PTR_REG

- **Name:** Capabilities Pointer Register.
- **Description:** This register is used to point to a linked list of capabilities implemented by a Function.
- **Size:** 32 bits
- **Offset:** B+0x34
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1



Table 3-487 Fields for Register: VF_PCI_CAP_PTR_REG

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
7:0	CAP_POINTER	R	Capabilities Pointer. This register points to a valid capability structure. Either this structure is the PCI Express Capability structure, or a subsequent list item points to the PCI Express Capability structure. The bottom two bits are reserved, the controller sets it to 00b. Software must mask these bits off before using this register as a pointer in Configuration Space to the first entry of a linked list of new capabilities. <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.49.14 VF_MAX_LATENCY_MIN_GRANT_INTERRUPT_PIN_INTERRUPT_LINE_REG

- **Name:** Max_Lat, Min_Gnt, Interrupt Pin, and Interrupt Line Register.
- **Description:** The Interrupt Line register communicates interrupt line routing information. The Interrupt Pin register identifies the legacy interrupt Message(s) the Function uses.
- **Size:** 32 bits
- **Offset:** B+0x3c
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

31:16	15:8	7:0
RSVDP_16	INT_PIN	INT_LINE

Table 3-488 Fields for Register: VF_MAX_LATENCY_MIN_GRANT_INTERRUPT_PIN_INTERRUPT_LINE_REG

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
15:8	INT_PIN	R	Interrupt Pin. The Interrupt Pin register identifies the legacy interrupt Message(s) the Function uses. All encodings other than the defined encodings are reserved. PCI Express defines one legacy interrupt Message for a single Function device and up to four legacy interrupt Messages for a multi-Function device. For a single Function device, only INTA may be used. Any Function on a multi-Function device can use any of the INTx Messages. If a device implements a single legacy interrupt Message, it must be INTA; if it implements two legacy interrupt Messages, they must be INTA and INTB; and so forth. For a multi-Function device, all Functions may use the same INTx Message or each may have its own (up to a maximum of four Functions) or any combination thereof. A single Function can never generate an interrupt request on more than one INTx Message. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (INTA): Map to legacy interrupt Messages for INTA ■ 0x2 (INTB): Map to legacy interrupt Messages for INTB ■ 0x3 (INTC): Map to legacy interrupt Messages for INTC ■ 0x4 (INTD): Map to legacy interrupt Messages for INTD ■ 0x0 (NO_INT): Indicates that the Function uses no legacy interrupt Message(s). Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
7:0	INT_LINE	R	Interrupt Line. The Interrupt Line register communicates interrupt line routing information. The register must be implemented by any Function that uses an interrupt pin. Values in this register are programmed by system software and are system architecture specific. The Function itself does not use this value; rather the value in this register is used by device drivers and operating systems. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.50 VF_PF_PM_CAPi Registers

3.50.1 VF_CAP_ID_NXT_PTR_REG

- **Name:** Power Management Capabilities Register.
- **Description:** This register provides information regarding the Power Management Capabilities.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** ((VF_PM_CAP_ENABLE | | EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

PME_SUPPORT	31:27
D2_SUPPORT	26
D1_SUPPORT	25
AUX_CURR	24:22
DSI	21
PME_IMM_READI_RETURN_DO	20
PME_CLK	19
PM_SPEC_VER	18:16
PM_NEXT_POINTER	15:8
PM_CAP_ID	7:0

Table 3-489 Fields for Register: VF_CAP_ID_NXT_PTR_REG

Bits	Name	Memory Access	Description
31:27	PME_SUPPORT	* Varies	PME_Support. This 5-bit field indicates the power states in which the function may generate a PME and/or forward PME messages. A value of 0b for any bit indicates that the function is not capable of asserting PME while in that power state. ■ bit(27) X XXX1b - PME can be generated from D0 ■ bit(28) X XX1Xb - PME can be generated from D1 ■ bit(29) X X1XXb - PME can be generated from D2 ■ bit(30) X 1XXXb - PME can be generated from D3hot ■ bit(31) 1 XXXXb - PME can be generated from D3cold Bit 31 (PME can be asserted from D3cold) represents a special case. Functions that set this bit require some sort of auxiliary power source. Implementation specific mechanisms are recommended to validate that the power source is available before setting this bit. The read value from this field is the write value && (sys_aux_pwr_det, 1'b1, D2_SUPPORT, D1_SUPPORT, 1'b1), where D1_SUPPORT and D2_SUPPORT are fields in this register. The reset value PME_SUPPORT_n && (sys_aux_pwr_det, 1'b1, D2_SUPPORT, D1_SUPPORT, 1'b1), where PME_SUPPORT_n is a configuration parameter. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
26	D2_SUPPORT	R	D2_Support. If this bit is set, this function supports the D2 Power Management state. Functions that do not support D2 must always return a value of 0b for this bit. <i>Note:</i> This register field is sticky. Value After Reset: D2_SUPPORT_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
25	D1_SUPPORT	R	D1_Support. If this bit is set, this function supports the D1 Power Management state. Functions that do not support D1 must always return a value of 0b for this bit. <i>Note:</i> This register field is sticky. Value After Reset: D1_SUPPORT_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
24:22	AUX_CURR	R	Aux_Current. This 3 bit field reports the Vaux auxiliary current requirements for the function. If this function implements the Data Register, the controller hardwires this field to 000b. If PME_Support is 0 xxxb (PME assertion from D3cold is not supported), the controller hardwires this field to 0000b. For functions where PME_Support is 1 xxxb (PME assertion from D3cold is supported), and which do not implement the Data field, the encodings defined in Values: apply: <i>Note:</i> This register field is sticky. Values: ■ 0x7 (_370mA_): 375mA Vaux Max. Current Required ■ 0x6 (_320mA_): 320mA Vaux Max. Current Required ■ 0x5 (_270mA_): 270mA Vaux Max. Current Required ■ 0x4 (_220mA_): 220mA Vaux Max. Current Required ■ 0x3 (_160mA_): 160mA Vaux Max. Current Required ■ 0x2 (_100mA_): 100mA Vaux Max. Current Required ■ 0x1 (_055mA_): 55mA Vaux Max. Current Required ■ 0x0 (SELF_): 0 self powered Value After Reset: AUX_CURRENT_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
21	DSI	R	Device Specific Initialization. The DSI bit indicates whether special initialization of this function is required. When set, indicates that the function requires a device specific initialization sequence following a transition to the D0uninitialized state. <i>Note:</i> This register field is sticky. Value After Reset: DEV_SPEC_INIT_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
20	PME_IMM_READI_RETURN_DO	R	Immediate_Readiness_on_Return_to_D0. If this bit is a '1', this function is guaranteed to be ready to successfully complete valid accesses immediately after being set to D0. These accesses include configuration cycles, and if the function returns to D0initialized, they also include Memory and I/O Cycles. When this bit is '1', for accesses to this function, software is exempt from all requirements to delay accesses following a transition to D0, including but not limited to the 10ms delay; the delays described in Section 5.9 of <i>PCI Express Base Specification</i>. How this guarantee is established is beyond the scope of this document. It is permitted that your system software/firmware provides mechanisms that supersede the indication provided by this bit, however such software/firmware mechanisms are outside the scope of this document. <i>Note:</i> This register field is sticky. Value After Reset: CX_RN_IMM_D0_VALUE_0 Exists: CX_RN_IMM_EN && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: {(VF_IMM_CONVERT_ENABLE) ? "writeAsRead" : "unconstrained" }
19	PME_CLK	R	PME Clock. Does not apply to PCI Express, the controller hardwires it to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
18:16	PM_SPEC_VER	R	Version. This field provides the Power Management specification version. The controller hardwires this field to 011b for functions compliant to <i>PCI Express Base Specification, Revision 4.0, Version 1.0</i>. <i>Note:</i> This register field is sticky. Value After Reset: 0x3 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:8	PM_NEXT_POINTER	R	Next Capability Pointer. This field provides an offset into the function's configuration space pointing to the location of next item in the capabilities list. If there are no additional items in the capabilities list, this field is set to 00h. <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
7:0	PM_CAP_ID	R	Capability ID. This field returns 01h to indicate that this is the PCI Power Management Capability. Each function may have only one item in its capability list with Capability ID set to 01h. Value After Reset: 0x1 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.50.2 VF_CON_STATUS_REG

- **Name:** Power Management Control and Status Register.
- **Description:** This register is used to manage the PCI function's power management state as well as to enable/monitor PMEs.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** ((VF_PM_CAP_ENABLE | | EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:24	DATA_REG_ADD_INFO
23	BUS_PWR_CLK_CON_EN
22	B2_B3_SUPPORT
21:16	RSVDP_16
15	PME_STATUS
14:13	DATA_SCALE
12:9	DATA_SELECT
8	PME_ENABLE
7:4	RSVDP_4
3	NO_SOFT_RST
2	RSVDP_2
1:0	POWER_STATE

Table 3-490 Fields for Register: VF_CON_STATUS_REG

Bits	Name	Memory Access	Description
31:24	DATA_REG_ADD_INFO	R	Data. This field is used to report the state dependent data requested by the Data_Select field. The value of this field is scaled by the value reported by the Data_Scale field. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
23	BUS_PWR_CLK_CON_EN	R	Bus Power/Clock Control Enable. If this field is set, Bus Power/Clock Control is Enable. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
22	B2_B3_SUPPORT	R	B2B3 Support for D3hot. If this field is set, B2B3 support for D3hot is available. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
21:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
15	PME_STATUS	R/W1C	PME_Status. This bit is set when the function normally generates a PME signal. The value of this bit is not affected by the value of the PME_En bit. If PME_Support bit 31 of the Power Management Capabilities register is clear, this bit is permitted to be hardwired to 0b. Functions that consume Aux power must preserve the value of this sticky register when Aux power is available. In such functions, this register value is not modified by Conventional Reset or FLR. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
14:13	DATA_SCALE	R	Data_Scale. This field indicates the scaling factor to be used when interpreting the value of the Data field. The value and meaning of this field varies depending on which data value has been selected by the Data_Select field. For more information, see 7.5.2.3 section of <i>PCI Express Base Specification</i>. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
12:9	DATA_SELECT	R	Data_Select. This 4-bit field is used to select which data is to be reported through the Data and Data_Scale field. If the Data field is not implemented, this field must be hardwired to 0000b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
8	PME_ENABLE	R/W	PME_En. - When set, the function is permitted to generate a PME. - When clear, the function is not permitted to generate a PME. If PME_Support is 1 xxxxb (PME generation from D3cold) or the function consumes Aux power and Aux power is available this bit is RWS and the bit is not modified by Conventional Reset or FLR. If PME_Support is 0 xxxxb, this field is not sticky (RW). If PME_Support is 0 0000b, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
7:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
3	NO_SOFT_RST	R	No_Soft_Reset. This bit indicates the state of the function after writing the PowerState field to transition the function from D3hot to D0. - When set, this transition preserves internal function state. The function is in D0Active and no additional software intervention is required. - When clear, this transition results in undefined internal function state. Regardless of this bit, functions that transition from D3hot to D0 by Fundamental Reset will return to D0Uninitialized with only PME context preserved if PME is supported and enabled. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_NO_SOFT_RESET_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
1:0	POWER_STATE	* Varies	PowerState. This 2-bit field is used both to determine the current power state of a function and to set the function into a new power state. You can write to this register; however, the read-back value is the actual power state, not the write value. If you attempt to write an unsupported, optional state to this field, the write operation completes normally; however, the data is discarded and no state change occurs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (D0): D0 power state ■ 0x1 (D1): D1 power state ■ 0x2 (D2): D2 power state ■ 0x3 (D3hot): D3hot D3hot power state Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PM_CAP_ENABLE EXT_VF_PM_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""

3.51 VF_PF_MSI_CAPI Registers

3.51.1 VF_PCI_MSI_CAP_ID_NEXT_CTRL_REG

- **Name:** MSI Capability Header and Message Control Register.
- **Description:** This register holds MSI Capability Header information and controls the MSI behaviour.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** ((VF_MSI_CAP_ENABLE | | EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

RSVDP_27	31:27
PCI_MSI_EXT_DATA_EN	26
PCI_MSI_EXT_DATA_CAP	25
PCI_PVM_SUPPORT	24
PCI_MSI_64_BIT_ADDR_CAP	23
PCI_MSI_MULTIPLE_MSG_EN	22:20
PCI_MSI_MULTIPLE_MSG_CAP	19:17
PCI_MSI_ENABLE	16
PCI_MSI_CAP_NEXT_OFFSET	15:8
PCI_MSI_CAP_ID	7:0

Table 3-491 Fields for Register: VF_PCI_MSI_CAP_ID_NEXT_CTRL_REG

Bits	Name	Memory Access	Description
31:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
26	PCI_MSI_EXT_DATA_EN	* Varies	Extended Message Data Enable. ■ If set, the function is enabled to provide Extended Message Data. ■ If clear, the function is not enabled to provide Extended Message Data. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PCI_MSI_CAP_ID_NEXT_CTRL_REG.PCI_MSI_EXT_DATA_CAP ? RW : RO ■ Dbi: PCI_MSI_CAP_ID_NEXT_CTRL_REG.PCI_MSI_EXT_DATA_CAP ? RW : RO Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead
25	PCI_MSI_EXT_DATA_CAP	* Varies	Extended Message Data Capable. ■ If set, the function is capable of providing Extended Message Data. ■ If clear, the function does not support providing Extended Message Data. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_EXT_MSI_DATA_CAPABLE Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
24	PCI_PVM_SUPPORT	R	Per-Vector Masking Capable. ■ If set, the function supports MSI Per-Vector Masking. ■ If clear, the function does not support MSI Per-Vector Masking. This bit must be set if the function is a PF or VF within an SR-IOV Device. Value After Reset: MSI_PVM_EN_VALUE Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
23	PCI_MSI_64_BIT_ADDR_CAP	* Varies	64 bit address capable. This bit must be set if the function is a PCI Express Endpoint. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (set_0): If set, the function is capable of sending a 64-bit message address. ■ 0x1 (set_1): If clear, the function is not capable of sending a 64-bit message address. Value After Reset: MSI_64_EN Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
22:20	PCI_MSI_MULTIPLE_MSG_EN	* Varies	Multiple Message Enable. Software writes to this field to indicate the number of allocated vectors (equal to or less than the number of requested vectors). The number of allocated vectors is aligned to a power of two. If a function requests four vectors (indicated by a Multiple Message Capable encoding of 010b), system software can allocate either four, two, or one vector by writing a 010b, 001b, or 000b to this field, respectively. When MSI is enabled, a function will be allocated at least 1 vector. All encodings other than the defined encodings are reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (_1_VECTOR): 1 vector allocated ■ 0x1 (_2_VECTOR): 2 vectors allocated ■ 0x2 (_4_VECTOR): 4 vectors allocated ■ 0x3 (_8_VECTOR): 8 vectors allocated ■ 0x4 (_16_VECTOR): 16 vectors allocated ■ 0x5 (_32_VECTOR): 32 vectors allocated Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && VF_MSI_CONVERT_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !VF_MSI_CONVERT_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && VF_MSI_CONVERT_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !VF_MSI_CONVERT_EN) ? \"read-only\" : \"read-only\" "

Bits	Name	Memory Access	Description
19:17	PCI_MSI_MULTIPLE_MSG_CAP	* Varies	Multiple Message Capable. System software reads this field to determine the number of requested vectors. The number of requested vectors must be aligned to a power of two (if a function requires three vectors, it requests four by initializing this field to 010b). All encodings other than the defined encodings are reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (_1_VECTOR): 1 vector requested ■ 0x1 (_2_VECTOR): 2 vectors requested ■ 0x2 (_4_VECTOR): 4 vectors requested ■ 0x3 (_8_VECTOR): 8 vectors requested ■ 0x4 (_16_VECTOR): 16 vectors requested ■ 0x5 (_32_VECTOR): 32 vectors requested Value After Reset: DEFAULT_MULTI_MSI_CAPABLE_0 Exists: MSI_CAP_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
16	PCI_MSI_ENABLE	* Varies	MSI Enable. ■ If set and the MSI-X Enable bit in the MSI-X Message Control register is clear, the function is permitted to use MSI to request service and is prohibited from using INTx interrupts. System configuration software sets this bit to enable MSI. A device driver is prohibited from writing this bit to mask a function's service request. For more information on control of INTx interrupts, see section 7.5.1.1 of <i>PCI Express Base Specification</i>. ■ If clear, the function is prohibited from using MSI to request service. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && VF_MSI_CONVERT_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !VF_MSI_CONVERT_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && VF_MSI_CONVERT_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !VF_MSI_CONVERT_EN) ? \"read-only\" : \"read-only\" "
15:8	PCI_MSI_CAP_NEXT_OFFSET	R	Next Capability Pointer. This field contains the offset to the next PCI Capability structure or 00h if no other items exist in the linked list of Capabilities. <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
7:0	PCI_MSI_CAP_ID	R	Capability ID. Indicates the MSI Capability structure. This field returns a Capability ID of 05h indicating that this is an MSI Capability structure. Value After Reset: 0x5 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.51.2 VF_MSI_CAP_OFF_04H_REG

- **Name:** Message Address Register for MSI (Offset 04h).
- **Description:** This register holds the system specified message address for an MSI transaction.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** ((VF_MSI_CAP_ENABLE | | EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:2	1:0
PCI_MSI_CAP_OFF_04H	RSVDP_0

Table 3-492 Fields for Register: VF_MSI_CAP_OFF_04H_REG

Bits	Name	Memory Access	Description
31:2	PCI_MSI_CAP_OFF_04H	* Varies	Message Address - System-specified message address. If the Message Enable bit (bit 16 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG register) is set, the contents of this field specify the DWORD-aligned address (Address[31:02]) for the MSI transaction. Address[1:0] are set to 00b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && VF_MSI_CONVERT_EN &&(VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE)) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !VF_MSI_CONVERT_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && VF_MSI_CONVERT_EN && (VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE)) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !VF_MSI_CONVERT_EN) ? \"read-only\" : (VF_MSI_CONVERT_EN && (VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE)) ? \"read-write\" : \"read-only\" "
1:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.51.3 VF_MSI_CAP_OFF_08H_REG

- **Name:** Message Address Register for MSI (Offset 08h).
- **Description:** For a function that supports a 32-bit message address,
 - bits[31:16] of this register represent the Extended Message Data, and
 - bits[15:0] of this register represent the Message Data

For a function that supports a 64-bit message address (bit 23 in PCI_MSI_CAP_ID_NEXT_CTRL_REG register set), this register represents the Message Upper Address Register for MSI (Offset 08h). It specifies the Message Upper Address (System-specified message upper address). This register is required for PCI Express Endpoints and is optional for other function types. If the Message Enable bit (bit 0 of the Message Control register) is set, the contents of this register (if non-zero) specify the upper 32-bits of a 64-bit message address (Address[63:32]). If the contents of this register are zero, the Function uses the 32 bit address specified by the Message Address register.

- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** ((VF_MSI_CAP_ENABLE | | EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:16	PCI_MSI_CAP_OFF_0AH
15:0	PCI_MSI_CAP_OFF_08H

Table 3-493 Fields for Register: VF_MSI_CAP_OFF_08H_REG

Bits	Name	Memory Access	Description
31:16	PCI_MSI_CAP_OFF_0AH	* Varies	For a function that supports a 32-bit message address, this field contains Extended Message Data (System-specified message data). For the MSI Capability structures without per-vector masking, it must be implemented if the Extended Message Data Capable bit is set; otherwise, it is outside the MSI Capability structure and undefined. For the MSI Capability structures with Per-vector Masking, it must be implemented if the Extended Message Data Capable bit is set; otherwise, it is RsvdP. If the Extended Message Data Enable bit (bit 26 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set, the DWORD Memory Write transaction uses Extended Message Data for the upper 16 bits; otherwise, it uses 0000h for the upper 16 bits. For a function that supports a 64-bit message address, it contains upper 16 bits of the Message Upper Address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PCI_MSI_64_BIT_ADDR_CAP `DEFAULT_EXT_MSI_DATA_CAPABLE ? R/W : R</li> <li>■ Dbi: PCI_MSI_64_BIT_ADDR_CAP    `DEFAULT_EXT_MSI_DATA_CAPABLE ? R/W : R and more specifically: (CX_MEMORY_MAP_VIEW==0 &&(MSI_64_EN DEFAULT_EXT_MSI_DATA_CAPABLE)) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R : (CX_MEMORY_MAP_VIEW==1 && (MSI_64_EN DEFAULT_EXT_MSI_DATA_CAPABLE)) ? R/W : (CX_MEMORY_MAP_VIEW==1) ? R : (MSI_64_EN DEFAULT_EXT_MSI_DATA_CAPABLE) ? R/W : R Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead

Bits	Name	Memory Access	Description
15:0	PCI_MSI_CAP_OFF_08H	* Varies	For a function that supports a 32-bit message address, this field contains Message Data (System-specified message data). If the Message Enable bit (bit 16 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set, the function sends a DWORD Memory Write transaction using Message Data for the lower 16 bits. All 4 Byte Enables are set. The Multiple Message Enable field (bits 22:20 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG) defines the number of low order message data bits the function is permitted to modify to generate its system software allocated vectors. For example, a Multiple Message Enable encoding of 010b indicates the function has been allocated four vectors and is permitted to modify message data bits 1 and 0 (a function modifies the lower message data bits to generate the allocated number of vectors). If the Multiple Message Enable field is 000b, the Function is not permitted to modify the message data. For a function that supports a 64-bit message address, it contains lower 16 bits of the Message Upper Address. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && VF_MSI_CONVERT_EN &&(VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE)) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !VF_MSI_CONVERT_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && VF_MSI_CONVERT_EN && (VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE)) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !VF_MSI_CONVERT_EN) ? \"read-only\" : (VF_MSI_CONVERT_EN && (VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE)) ? \"read-write\" : \"read-only\" "

3.51.4 VF_MSI_CAP_OFF_0CH_REG

- **Name:** Message Address Register for MSI (Offset 0Ch).
- **Description:** For a function that supports a 32-bit message address, this register contains the Mask Bits when the Per-Vector Masking Capable bit (bit 24 of PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set. For a function that supports a 64-bit message address, this register contains Message Data.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** ((VF_MSI_CAP_ENABLE | | EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:16	PCI_MSI_CAP_OFF_0EH
15:0	PCI_MSI_CAP_OFF_0CH

Table 3-494 Fields for Register: VF_MSI_CAP_OFF_0CH_REG

Bits	Name	Memory Access	Description
31:16	PCI_MSI_CAP_OFF_0EH	* Varies	For a function that supports a 32-bit message address, this field contains the upper Mask Bits when the Per Vector Masking Capable bit (PCI_MSI_CAP_ID_NEXT_CTRL_REG.PCI_PVM_SUPPORT) is set. For a function that supports a 64-bit message address, this field contains Message Data (System-specified message data). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: (!MSI_64_EN && MSI_PVM_EN_VALUE) ? RW: MSI_64_EN && DEFAULT_EXT_MSI_DATA_CAPABLE ? RW : RO ■ Dbi: (!MSI_64_EN && MSI_PVM_EN_VALUE) ? RW: MSI_64_EN && DEFAULT_EXT_MSI_DATA_CAPABLE ? RW : RO and more specifically: (CX_MEMORY_MAP_VIEW==0 && MSI_64_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1 && MSI_64_EN) ? R/W : (CX_MEMORY_MAP_VIEW==1) ? R: (MSI_64_EN) ? R/W : R Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead

Bits	Name	Memory Access	Description
15:0	PCI_MSI_CAP_OFF_0CH	* Varies	For a function that supports a 32-bit message address, this field contains the lower Mask Bits when the Per Vector Masking Capable bit (PCI_MSI_CAP_ID_NEXT_CTRL_REG.PCI_PVM_SUPPORT) is set. For a function that supports a 64-bit message address, this field contains Message Data (System-specified message data). If the Message Enable bit (bit 16 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set, the function sends a DWORD Memory Write transaction using Message Data for the lower 16 bits. All 4 Byte Enables are set. The Multiple Message Enable field (bits 22:20 of the PCI_MSI_CAP_ID_NEXT_CTRL_REG) defines the number of low order message data bits the function is permitted to modify to generate its system software allocated vectors. For example, a Multiple Message Enable encoding of 010b indicates the function has been allocated four vectors and is permitted to modify message data bits 1 and 0 (a function modifies the lower message data bits to generate the allocated number of vectors). If the Multiple Message Enable field is 000b, the Function is not permitted to modify the message data. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PCI_MSI_64_BIT_ADDR_CAP MSI_PVM_EN ? R/W : R ■ Dbi: PCI_MSI_64_BIT_ADDR_CAP MSI_PVM_EN ? R/W : R and more specifically: (CX_MEMORY_MAP_VIEW==0 && (MSI_64_EN MSI_PVM_EN)) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !MSI_CAP_ENABLE) ? R: (CX_MEMORY_MAP_VIEW==1 && (MSI_64_EN MSI_PVM_EN)) ? R/W : (CX_MEMORY_MAP_VIEW==1 && !MSI_CAP_ENABLE) ? R: (MSI_64_EN MSI_PVM_EN) ? R/W : R Value After Reset: 0x0 Exists: (MSI_64_EN MSI_PVM_EN) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead

3.51.5 VF_MSI_CAP_OFF_10H_REG

- **Name:** Message Address Register for MSI (Offset 10h).
- **Description:** For a function that supports a 32-bit message address, this register contains the Pending Bits when the Per-Vector Masking Capable bit (bit 24 of PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set. For a function that supports a 64-bit message address, this register contains the Mask Bits when the Per-Vector Masking Capable bit (bit 24 of PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** ((VF_MSI_CAP_ENABLE | | EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)



Table 3-495 Fields for Register: VF_MSI_CAP_OFF_10H_REG

Bits	Name	Memory Access	Description
31:0	PCI_MSI_CAP_OFF_10H	* Varies	Used for MSI when the Per Vector Masking Capable bit (PCI_MSI_CAP_ID_NEXT_CTRL_REG.PCI_PVM_SUPPORT) is set. For 32-bit contains Pending Bits. For 64-bit, contains Mask Bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PCI_MSI_64_BIT_ADDR_CAP && MSI_PVM_EN ? R/W : R ■ Dbi: PCI_MSI_64_BIT_ADDR_CAP && MSI_PVM_EN ? R/W : R and more specifically: (CX_MEMORY_MAP_VIEW==0 && VF_MSI_EN && MSI_PVM_EN_VALUE && MSI_64_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !VF_MSI_EN) ? R: (CX_MEMORY_MAP_VIEW==1 && VF_MSI_EN && MSI_PVM_EN_VALUE && MSI_64_EN) ? R/W : (CX_MEMORY_MAP_VIEW==1 && !VF_MSI_EN) ? R: (VF_MSI_EN && MSI_PVM_EN_VALUE && MSI_64_EN) ? R/W : R Value After Reset: 0x0 Exists: MSI_PVM_EN && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead

3.51.6 VF_MSI_CAP_OFF_14H_REG

- **Name:** Message Address Register for MSI (Offset 14h).
- **Description:** Pending Bits Register for MSI. This register is used for a function that supports a 64-bit message address when the Per-Vector Masking Capable bit (bit 24 of PCI_MSI_CAP_ID_NEXT_CTRL_REG) is set.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** ((VF_MSI_CAP_ENABLE | | EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)



Table 3-496 Fields for Register: VF_MSI_CAP_OFF_14H_REG

Bits	Name	Memory Access	Description
31:0	PCI_MSI_CAP_OFF_14H	R	Pending Bits. For each pending bit that is set, the function has a pending associated message. Value After Reset: 0x0 Exists: MSI_64_EN && MSI_PVM_EN && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSI_CAP_ENABLE EXT_VF_MSI_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.52 VF_PF_PCIE_CAPi Registers

3.52.1 VF_PCIE_CAP_ID_PCIE_NEXT_CAP_PTR_PCIE_CAP_REG

- **Name:** PCI Express Capabilities, ID, Next Pointer Register.
- **Description:** This is the PCI Express Capabilities, ID, and Next Pointer Register.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

31	RSVDP_31
30	RSVD
29:25	PCIE_INT_MSG_NUM
24	PCIE_SLOT_IMP
23:20	PCIE_DEV_PORT_TYPE
19:16	PCIE_CAP_REG
15:8	PCIE_CAP_NEXT_PTR
7:0	PCIE_CAP_ID

Table 3-497 Fields for Register: VF_PCIE_CAP_ID_PCIE_NEXT_CAP_PTR_PCIE_CAP_REG

Bits	Name	Memory Access	Description
31	RSVDP_31	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
30	RSVD	R	Reserved. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
29:25	PCIE_INT_MSG_NUM	R	PCIE Interrupt Message Number. Interrupt Message Number. This field indicates which MSI/MSI-X vector is used for the interrupt message generated in association with any of the status bits of this Capability structure. For MSI, the value in this field indicates the offset between the base Message Data and the interrupt message that is generated. Hardware is required to update this field so that it is correct if the number of MSI Messages assigned to the Function changes when software writes to the Multiple Message Enable field in the MSI Message Control register. For MSI-X, the value in this field indicates which MSI-X Table entry is used to generate the interrupt message. The entry must be one of the first 32 entries even if the Function implements more than 32 entries. For a given MSI-X implementation, the entry must remain constant. If both MSI and MSI-X are implemented, they are permitted to use different vectors, though software is permitted to enable only one mechanism at a time. If MSI-X is enabled, the value in this field must indicate the vector for MSI-X. If MSI is enabled or neither is enabled, the value in this field must indicate the vector for MSI. If software enables both MSI and MSI-X at the same time, the value in this field is undefined. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : PCIE_CAP_INT_MSG_NUM_0} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
24	PCIE_SLOT_IMP	R	Slot Implemented. When set, this bit indicates that the Link associated with this Port is connected to a slot (as compared to being connected to a system-integrated device or being disabled). This bit is valid for Downstream Ports. This bit is undefined for Upstream Ports. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : SLOT_IMPLEMENTED_0} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:20	PCIE_DEV_PORT_TYPE	R	Device/Port Type. Indicates the specific type of this PCI Express function. <i>Note:</i> Different functions in a Multi-Function Device can generally be of different types. Defined encodings for functions that implement a Type 00h PCI Configuration Space header are: Defined encodings for functions that implement a Type 01h PCI Configuration Space header are: All other encodings are Reserved. <i>Note:</i> Different Endpoint types have notably different requirements in Section 1.3.2 of <i>PCI Express Base Specification</i> regarding I/O resources, Extended Configuration Space, and other capabilities. Values: ■ 0x0 (PCIE_EP): PCI Express Endpoint ■ 0x1 (PCIE_LEGACY_EP): Legacy PCI Express Endpoint ■ 0x4 (ROOT_PORT_PCIE_RC): Root Port of PCI Express Root Complex ■ 0x5 (USP_PCIE_SWITCH): Upstream Port of PCI Express Switch ■ 0x6 (DSP_PCIE_SWITCH): Downstream Port of PCI Express Switch Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable Reset Mask: 0x0 Volatile: true
19:16	PCIE_CAP_REG	R	Capability Version. Indicates PCI-SIG defined PCI Express Capability structure version number. A version of the specification that changes the PCI Express Capability structure in a way that is not otherwise identifiable (for example, through a new Capability field) is permitted to increment this field. All such changes to the PCI Express Capability structure must be software-compatible. Software must check for Capability Version numbers that are greater than or equal to the highest number defined when the software is written, as functions reporting any such Capability Version numbers will contain a PCI Express Capability structure that is compatible with that piece of software. The controller hardwires this field to 2h for functions compliant to <i>PCI Express Base Specification, Revision 4.0, Version 1.0</i>. <i>Note:</i> This register field is sticky. Value After Reset: 0x2 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
15:8	PCIE_CAP_NEXT_PTR	R	Next Capability Pointer. This field contains the offset to the next PCI Capability structure or 00h if no other items exist in the linked list of Capabilities. <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
7:0	PCIE_CAP_ID	R	Capability ID. Indicates the PCI Express Capability structure. This field must return a Capability ID of 10h indicating that this is a PCI Express Capability structure. Value After Reset: 0x10 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.52.2 VF_DEVICE_CAPABILITIES_REG

- **Name:** Device Capabilities Register.
- **Description:** The Device Capabilities register identifies PCI Express device function specific capabilities.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

RSVDP_29	31:29
PCIE_CAP_FLR_CAP	28
PCIE_CAP_CAP_SLOT_PWR_LMT_SCALE	27:26
PCIE_CAP_CAP_SLOT_PWR_LMT_VALUE	25:18
RSVDP_16	17:16
PCIE_CAP_ROLE_BASED_ERR_REPORT	15
RSVDP_12	14:12
PCIE_CAP_EP_L1_ACCPT_LATENCY	11:9
PCIE_CAP_EP_L0S_ACCPT_LATENCY	8:6
PCIE_CAP_EXT_TAG_SUPP	5
PCIE_CAP_PHANTOM_FUNC_SUPPORT	4:3
PCIE_CAP_MAX_PAYLOAD_SIZE	2:0

Table 3-498 Fields for Register: VF_DEVICE_CAPABILITIES_REG

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
28	PCIE_CAP_FLR_CAP	R	Function Level Reset Capability. A value of 1b indicates the function supports the optional Function Level Reset mechanism described in section 6.6.2 of the <i>PCI Express Base Specification</i>. This bit applies to Endpoints only. For all other function types the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1) ? 0 : CX_FLR_ENABLE_VALUE} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
27:26	PCIE_CAP_CAP_SLOT_PWR_LMT_SCALE	R	Captured Slot Power Limit Scale. Captured Slot Power Limit Scale (Upstream Ports only). Specifies the scale used for the Slot Power Limit Value. This value is set by the Set_Slot_Power_Limit Message or hardwired to 00b (for more details, see section 6.9 of <i>PCI Express Base Specification</i>). Values: ■ 0x0 (_1.0X): 1.0x ■ 0x1 (_0.1X): 0.1x ■ 0x2 (_0.01X): 0.01x ■ 0x3 (_0.001X): 0.001x Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
25:18	PCIE_CAP_CAP_SLOT_PWR_LIMIT_VALUE	R	Captured Slot Power Limit Value. Captured Slot Power Limit Value (Upstream Ports only). In combination with the Captured Slot Power Limit Scale value, specifies the upper limit on power available to the adapter. Power limit (in Watts) is calculated by multiplying the value in this field by the value in the Captured Slot Power Limit Scale field except when the Captured Slot Power Limit Scale field equals 00b (1.0x) and the Captured Slot Power Limit Value exceeds EFh, then alternative encodings are used (for more details, see section 7.5.3.9 of <i>PCI Express Base Specification</i>). This value is set by the Set_Slot_Power_Limit Message or hardwired to 00h (for more details, see section 6.9 of <i>PCI Express Base Specification</i>). Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true
17:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
15	PCIE_CAP_ROLE_BASED_ERR_REPORT	R	Role-Based Error Reporting. When set, this bit indicates that the function implements the functionality originally defined in the Error Reporting ECN for <i>PCI Express Base Specification, Revision 1.0a</i>, and later incorporated into <i>PCI Express Base Specification, Revision 1.1</i>. This bit must be set by all functions conforming to the ECN, <i>PCI Express Base Specification, Revision 1.1</i>, or subsequent <i>PCI Express Base Specification</i> revisions. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? "0x0" : "0x1"} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
14:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
11:9	PCIE_CAP_EP_L1_ACCPT_LATENCY	R	Endpoint L1 Acceptable Latency. This field indicates the acceptable latency that an Endpoint can withstand due to the transition from L1 state to the L0 state. It is essentially an indirect measure of the Endpoint's internal buffering. Power management software uses the reported L1 Acceptable Latency number to compare against the L1 Exit Latencies reported (see below) by all components comprising the data path from this Endpoint to the Root Complex Root Port to determine whether ASPM L1 entry can be used with no loss of performance. For functions other than Endpoints, this field is Reserved and the controller hardwires it to 000b. Values: ■ 0x0 (MAX_1US): Maximum of 1 us ■ 0x1 (MAX_2US): Maximum of 2 us ■ 0x2 (MAX_4US): Maximum of 4 us ■ 0x3 (MAX_8US): Maximum of 8 us ■ 0x4 (MAX_16US): Maximum of 16 us ■ 0x5 (MAX_32US): Maximum of 32 us ■ 0x6 (MAX_64US): Maximum of 64 us ■ 0x7 (NO_LIMIT): No limit Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1) ? 0 : DEFAULT_EP_L1_ACCPT_LATENCY} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
8:6	PCIE_CAP_EP_L0S_ACCPT_LATENCY	R	Endpoint L0s Acceptable Latency. This field indicates the acceptable total latency that an Endpoint can withstand due to the transition from L0s state to the L0 state. It is essentially an indirect measure of the Endpoint's internal buffering. Power management software uses the reported L0s Acceptable Latency number to compare against the L0s exit latencies reported by all components comprising the data path from this Endpoint to the Root Complex Root Port to determine whether ASPM L0s entry can be used with no loss of performance. For functions other than Endpoints, this field is Reserved and the controller hardwires it to 000b. Values: ■ 0x0 (MAX_64NS): Maximum of 64 ns ■ 0x1 (MAX_128NS): Maximum of 128 ns ■ 0x2 (MAX_256NS): Maximum of 256 ns ■ 0x3 (MAX_512NS): Maximum of 512 ns ■ 0x4 (MAX_1US): Maximum of 1 us ■ 0x5 (MAX_2US): Maximum of 2 us ■ 0x6 (MAX_4US): Maximum of 4 us ■ 0x7 (NO_LIMIT): No limit Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1) ? 0 : DEFAULT_EP_L0S_ACCPT_LATENCY} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5	PCIE_CAP_EXT_TAG_SUPP	R/W	Extended Tag Field Supported. This bit, in combination with the 10-Bit Tag Requester Supported bit in the Device Capabilities 2 register, indicates the maximum supported size of the Tag field as a Requester. This bit must be set if the 10-Bit Tag Requester Supported bit is set. <i>Note:</i> 8-bit Tag field generation must be enabled by the Extended Tag Field Enable bit in the Device Control register of the Requester Function before 8-bit Tags can be generated by the Requester. See Section 2.2.6.2 of <i>PCI Express Base Specification</i> for interactions with enabling the use of 10-Bit Tags. Values: 0x0 (_5b_TAG): 5-bit Tag field supported 0x1 (_8b_TAG): 8-bit Tag field supported Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : DEFAULT_EXT_TAG_FIELD_SUPPORTED_0} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
4:3	PCIE_CAP_PHANTOM_FUNC_SUPPORT	R	Phantom Functions Supported. This field indicates the support for use of unclaimed function numbers to extend the number of outstanding transactions allowed by logically combining unclaimed function numbers (called Phantom Functions) with the Tag identifier (see Section 2.2.6.2 of <i>PCI Express Base Specification</i> for a description of Tag Extensions). With every Function in an ARI Device, the Phantom Functions Supported field must be set to 00b. The remainder of this field description applies only to non-ARI Multi-Function Devices. This field indicates the number of most significant bits of the Function Number portion of Requester ID that are logically combined with the Tag identifier. <i>Note:</i> Phantom Function support for the function must be enabled by the Phantom Functions Enable field in the Device Control register before the Function is permitted to use the Function Number field in the Requester ID for Phantom Functions. <i>Note:</i> This register field is sticky. Values: 0x0 (NO_PHNATOM_FUNC): No Function Number bits are used for Phantom Functions. Multi-Function Devices are permitted to implement up to 8 independent functions. 0x1 (MAX_1_PHANTOM_PER_FUNC): The most significant bit of the Function number in Requester ID is used for Phantom Functions; a Multi-Function Device is permitted to implement Functions 0-3. Functions 0, 1, 2, and 3 are permitted to use Function Numbers 4, 5, 6, and 7 respectively as Phantom Functions.

Bits	Name	Memory Access	Description
4:3...(cont.)	PCIE_CAP_PHANTOM_FUNC_SUPPORT.	R	0x2 (MAX_3_PHANTOM_PER_FUNC): The two most significant bits of Function Number in Requester ID are used for Phantom Functions; a Multi-Function Device is permitted to implement Functions 0-1. Function 0 is permitted to use Function Numbers 2, 4, and 6 for Phantom Functions. Function 1 is permitted to use Function Numbers 3, 5, and 7 as Phantom Functions. 0x3 (SINGLE_FUNC_MAX_7_PHANTOM_FUNC): All 3 bits of Function Number in Requester ID used for Phantom Functions. The device must have a single Function 0 that is permitted to use all other Function Numbers as Phantom Functions. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : DEFAULT_PHANTOM_FUNC_SUPPORTED_0} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
2:0	PCIE_CAP_MAX_PAYLOAD_SIZE	R	Max_Payload_Size Supported. This field indicates the maximum payload size that the function can support for TLPs. All encodings other than the defined encodings are reserved. The functions of a Multi-Function Device are permitted to report different values for this field. Values: 0x0 (MAX_128B_SIZE): 128 bytes max payload size 0x1 (MAX_256B_SIZE): 256 bytes max payload size 0x2 (MAX_512B_SIZE): 512 bytes max payload size 0x3 (MAX_1024B_SIZE): 1024 bytes max payload size 0x4 (MAX_2048B_SIZE): 2048 bytes max payload size 0x5 (MAX_4096B_SIZE): 4096 bytes max payload size Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? "0x0" : ((CX_MAX_MTU == 128) ? "0x0" : (CX_MAX_MTU == 256) ? "0x1" : (CX_MAX_MTU == 512) ? "0x2" : (CX_MAX_MTU == 1024) ? "0x3" : (CX_MAX_MTU == 2048) ? "0x4" : (CX_MAX_MTU == 4096) ? "0x5" : "0x0")} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.52.3 VF_DEVICE_CONTROL_DEVICE_STATUS

- **Name:** Device Control and Device Status Register.
- **Description:** This register controls PCI Express device specific parameters and provides information about PCI Express device (function) specific parameters.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

31:23	RSVDP_23
22	PCIE_CAP_EMERGENCY_PWR_RED_DETECTED
21	PCIE_CAP_TRANS_PENDING
20	PCIE_CAP_AUX_POWER_DETECTED
19	PCIE_CAP_UNSUPPORTED_REQ_DETECTED
18	PCIE_CAP_FATAL_ERR_DETECTED
17	PCIE_CAP_NON_FATAL_ERR_DETECTED
16	PCIE_CAP_CORR_ERR_DETECTED
15	PCIE_CAP_INITIATE_FLR
14:12	PCIE_CAP_MAX_READ_REQ_SIZE
11	PCIE_CAP_EN_NO_SNOOP
10	PCIE_CAP_AUX_POWER_PM_EN
9	PCIE_CAP_PHANTOM_FUNC_EN
8	PCIE_CAP_EXT_TAG_EN
7:5	PCIE_CAP_MAX_PAYLOAD_SIZE_CS
4	PCIE_CAP_EN_REL_ORDER
3	PCIE_CAP_UNSUPPORTED_REQ_REP_EN
2	PCIE_CAP_FATAL_ERR_REPORT_EN
1	PCIE_CAP_NON_FATAL_ERR_REPORT_EN
0	PCIE_CAP_CORR_ERR_REPORT_EN

Table 3-499 Fields for Register: VF_DEVICE_CONTROL_DEVICE_STATUS

Bits	Name	Memory Access	Description
31:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
22	PCIE_CAP_EMERGENCY_PWR_RED_DETECTED	R	Emergency Power Reduction Detected. When set, this bit indicates that the function is in the Emergency Power Reduction State. Values: ■ 0x0 (CLEAR): Function is not in the Emergency Power Reduction State. ■ 0x1 (SET): Function is in the Emergency Power Reduction State. Value After Reset: 0x0 Exists: EMERGENCY_POWER_REDUCTION_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable Volatile: true
21	PCIE_CAP_TRANS_PENDING	R	Transactions Pending. Endpoints: When set, this bit indicates that the function has issued Non-Posted Requests that have not been completed. A Function reports this bit cleared only when all outstanding Non-Posted Requests have completed or have been terminated by the Completion Timeout mechanism. This bit must also be cleared upon the completion of an FLR. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true
20	PCIE_CAP_AUX_POWER_DETECTED	R	AUX Power Detected. Functions that require Aux power report this bit as set if Aux power is detected by the function. This bit is derived by sampling the sys_aux_pwr_det input. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
19	PCIE_CAP_UNSUPPORTED_REQUEST_DETECTED	R/W1C	Unsupported Request Detected. This bit indicates that the function received an Unsupported Request. Errors are logged in this register regardless of whether error reporting is enabled or not in the Device Control register. For a Multi-Function Device, each function indicates status of errors as perceived by the respective function. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
18	PCIE_CAP_FATAL_ERROR_DETECTED	R/W1C	Fatal Error Detected. This bit indicates status of Fatal errors detected. Errors are logged in this register regardless of whether error reporting is enabled or not in the Device Control register. For a Multi-Function device, each function indicates status of errors as perceived by the respective Function. For Functions supporting Advanced Error Handling, errors are logged in this register regardless of the settings of the Uncorrectable Error Mask register. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
17	PCIE_CAP_NON_FATAL_ERROR_DETECTED	R/W1C	Non-Fatal Error Detected. This bit indicates status of Non-fatal errors detected. Errors are logged in this register regardless of whether error reporting is enabled or not in the Device Control register. For a Multi-Function device, each function indicates status of errors as perceived by the respective Function. For functions supporting Advanced Error Handling, errors are logged in this register regardless of the settings of the Uncorrectable Error Mask register. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
16	PCIE_CAP_CORR_ERR_DETECT ED	R/W1C	Correctable Error Detected. This bit indicates status of correctable errors detected. Errors are logged in this register regardless of whether error reporting is enabled or not in the Device Control register. For a Multi-Function device, each function indicates status of errors as perceived by the respective function. For functions supporting Advanced Error Handling, errors are logged in this register regardless of the settings of the Correctable Error Mask register. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
15	PCIE_CAP_INITIATE_FLR	W	Initiate Function Level Reset (applicable to Endpoints with Function Level Reset Capability set to 1b). A write of 1b initiates Function Level Reset to the Function. The value read by software from this bit is always 0b. Value After Reset: 0x0 Exists: (CX_FLR_ENABLE && !CX_PCIE_BRIDGE) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: {(CX_EXTENSIBLE_VFUNC) ? "untestable" : "unconstrained" }

Bits	Name	Memory Access	Description
14:12	PCIE_CAP_MAX_READ_REQ_SIZE	R	Max_Read_Request_Size. This field sets the maximum Read Request size for the function as a Requester. The function must not generate Read Requests with a size exceeding the set value. For functions that do not generate Read Requests larger than 128 bytes and functions that do not generate Read Requests on their own behalf, the controller implements this field as Read Only (RO) with a value of 000b. Values: ■ 0x0 (MAX_128B_SIZE): 128 bytes maximum Read Request size ■ 0x1 (MAX_256B_SIZE): 256 bytes maximum Read Request size ■ 0x2 (MAX_512B_SIZE): 512 bytes maximum Read Request size ■ 0x3 (MAX_1024B_SIZE): 1024 bytes maximum Read Request size ■ 0x4 (MAX_2048B_SIZE): 2048 bytes maximum Read Request size ■ 0x5 (MAX_4096B_SIZE): 4096 bytes maximum Read Request size ■ 0x6 (RESERVED1): RESERVED ■ 0x7 (RESERVED2): RESERVED Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
11	PCIE_CAP_EN_NO_SNOOP	R	Enable No Snoop. If this bit is set, the function is permitted to Set the No Snoop bit in the Requester Attributes of transactions it initiates that do not require hardware enforced cache coherency (see section 2.2.6.5 in <i>PCI Express Base Specification</i>). <i>Note:</i> Setting this bit to 1b should not cause a function to set the No Snoop attribute on all transactions that it initiates. Even when this bit is set, a function is only permitted to set the No Snoop attribute on a transaction when it can guarantee that the address of the transaction is not stored in any cache in the system. The controller hardwires this bit 0b if a function would never set the No Snoop attribute in transactions it initiates. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
10	PCIE_CAP_AUX_POWER_PM_EN	R	Aux Power PM Enable. This bit is derived by sampling the sys_aux_pwr_det input. When set this bit, enables a function to draw Aux power independent of PME Aux power. Functions that require Aux power on legacy operating systems should continue to indicate PME Aux power requirements. Aux power is allocated as requested in the Aux_Current field of the Power Management Capabilities register (PMC), independent of the PME_En bit in the Power Management Control/Status register (PMCSR). For Multi-Function devices, a component is allowed to draw Aux power if at least one of the functions has this bit set. <i>Note:</i> Functions that consume Aux power must preserve the value of this sticky register when Aux power is available. In such functions, this bit is not modified by Conventional Reset. For functions that do not implement this capability, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
9	PCIE_CAP_PHANTOM_FUNC_EN	R	Phantom Functions Enable. This bit, in combination with the 10-Bit Tag Requester Enable bit in the Device Control 2 register, determines how many Tag field bits a Requester is permitted to use. When the 10-Bit Tag Requester Enable bit is clear, ■ If this bit is set, it enables a function to use unclaimed functions as Phantom functions to extend the number of outstanding transaction identifiers ■ If this bit is clear, the function is not allowed to use Phantom functions For more information, see section 2.2.6.2 of <i>PCI Express Base Specification</i>. Software should not change the value of this bit while the function has outstanding Non-Posted Requests; otherwise, the result is undefined. For functions that do not implement this capability, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
8	PCIE_CAP_EXT_TAG_EN	R	Extended Tag Field Enable. This bit, in combination with the 10-Bit Tag Requester Enable bit in the Device Control 2 register, determines how many Tag field bits a Requester is permitted to use. When the 10-Bit Tag Requester Enable bit is clear, ■ If the Extended Tag Field Enable bit is set, the function is permitted to use an 8-bit Tag field as a Requester ■ If the Extended Tag Field Enable bit is clear, the Function is restricted to a 5-bit Tag field See section 2.2.6.2 of <i>PCI Express Base Specification</i> for required behavior when the 10-Bit Tag Requester Enable bit is set. If software changes the value of the Extended Tag Field Enable bit while the function has outstanding Non-Posted Requests, the result is undefined. For functions that do not implement this capability, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
7:5	PCIE_CAP_MAX_PAYLOAD_SIZE_CS	R	Max_Payload_Size. This field sets maximum TLP payload size for the Function. As a Receiver, the Function must handle TLPs as large as the set value. As a Transmitter, the Function must not generate TLPs exceeding the set value. Permissible values that can be programmed are indicated by the Max_Payload_Size Supported field (PCIE_CAP_MAX_PAYLOAD_SIZE) in the Device Capabilities (DEVICE_CAPABILITIES_REG) register (for more details, see section 7.5.3.3 of <i>PCI Express Base Specification</i>). This field sets the maximum Read Request size for the function as a Requester. The function must not generate Read Requests with a size exceeding the set value. For Functions that support only the 128-byte max payload size, the controller hardwires this field to 000b. System software is not required to program the same value for this field for all the Functions of a Multi-Function device (for more details, see section 2.2.2 of <i>PCI Express Base Specification</i>). For ARI Devices, Max_Payload_Size is determined solely by the setting in Function0. The settings in the other Functions always return whatever value software programmed for each, but otherwise are ignored by the component. Values: ■ 0x0 (MAX_128B_SIZE): 128 bytes maximum Read Request size ■ 0x1 (MAX_256B_SIZE): 256 bytes maximum Read Request size ■ 0x2 (MAX_512B_SIZE): 512 bytes maximum Read Request size ■ 0x3 (MAX_1024B_SIZE): 1024 bytes maximum Read Request size ■ 0x4 (MAX_2048B_SIZE): 2048 bytes maximum Read Request size ■ 0x5 (MAX_4096B_SIZE): 4096 bytes maximum Read Request size ■ 0x6 (RESERVED1): RESERVED ■ 0x7 (RESERVED2): RESERVED Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
4	PCIE_CAP_EN_REL_ORDER	R	Enable Relaxed Ordering. If this bit is set, the function is permitted to set the Relaxed Ordering bit in the Attributes field of transactions it initiates that do not require strong write ordering (for more details, see section 2.2.6.4 and section 2.4 of <i>PCI Express Base Specification</i>). For a function that never sets the Relaxed Ordering attribute in transactions it initiates as a Requester, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
3	PCIE_CAP_UNSUPPORT_REQ_REP_EN	R	Unsupported Request Reporting Enable. This bit, in conjunction with other bits, controls the signaling of Unsupported Request Errors by sending error Messages (for more details, see section 6.2.5 and section 6.2.6 of <i>PCI Express Base Specification</i>). For a Multi-Function Device, this bit controls error reporting for each Function from point-of-view of the respective Function. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
2	PCIE_CAP_FATAL_ERR_REPORT_EN	R	Fatal Error Reporting Enable. This bit, in conjunction with other bits, controls sending ERR_FATAL Messages (for more details, see section 6.2.5 and section 6.2.6 of <i>PCI Express Base Specification</i>). For a Multi-Function device, this bit controls error reporting for each function from point-of-view of the respective function. For a Root Port, the reporting of Fatal errors is internal to the root. No external ERR_FATAL Message is generated. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
1	PCIE_CAP_NON_FATAL_ERR_REPORT_EN	R	Non-Fatal Error Reporting Enable. This bit, in conjunction with other bits, controls sending ERR_NONFATAL Messages (for more details, see section 6.2.5 and Section 6.2.6 of <i>PCI Express Base Specification</i>). For a Multi-Function Device, this bit controls error reporting for each function from point-of-view of the respective Function. For a Root Port, the reporting of Non-fatal errors is internal to the root. No external ERR_NONFATAL Message is generated. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
0	PCIE_CAP_CORR_ERR_REPOR T_EN	R	Correctable Error Reporting Enable. This bit, in conjunction with other bits, controls sending ERR_COR Messages (for more details, see section 6.2.5, section 6.2.6, and section 6.2.10.2 of <i>PCI Express Base Specification</i>). For a Multi-Function device, this bit controls error reporting for each function from point-of-view of the respective function. For a Root Port, the reporting of correctable errors is internal to the root. No external ERR_COR Message is generated. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.52.4 VF_LINK_CAPABILITIES_REG

- **Name:** Link Capabilities Register.
- **Description:** The Link Capabilities register identifies PCI Express Link specific capabilities and CXL-RCRB link specific capabilities.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

31:24	PCIE_CAP_PORT_NUM
23	RSVDP_23
22	PCIE_CAP_ASPM_OPT_COMPLIANCE
21	PCIE_CAP_LINK_BW_NOT_CAP
20	PCIE_CAP_DLL_ACTIVE_REP_CAP
19	PCIE_CAP_SURPRISE_DOWN_ERR_REP_CAP
18	PCIE_CAP_CLOCK_POWER_MAN
17:15	PCIE_CAP_L1_EXIT_LATENCY
14:12	PCIE_CAP_L0S_EXIT_LATENCY
11:10	PCIE_CAP_ACTIVE_STATE_LINK_PM_SUPPORT
9:4	PCIE_CAP_MAX_LINK_WIDTH
3:0	PCIE_CAP_MAX_LINK_SPEED

Table 3-500 Fields for Register: VF_LINK_CAPABILITIES_REG

Bits	Name	Memory Access	Description
31:24	PCIE_CAP_PORT_NUM	R	Port Number. This field indicates the PCI Express Port number for the given PCI Express Link. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all functions. Value After Reset: PORT_NUM Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable Volatile: true

Bits	Name	Memory Access	Description
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
22	PCIE_CAP_ASPM_OPT_COMPLIANCE	R	ASPM Optionality Compliance. This field must be set to 1b in all functions. Components implemented against certain earlier versions of this specification will have this bit set to 0b. Software is permitted to use the value of this bit to help determine whether to enable ASPM or whether to run ASPM compliance tests. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: untestable Volatile: true
21	PCIE_CAP_LINK_BW_NOT_CAP	R	Link Bandwidth Notification Capability. A value of 1b indicates support for the Link Bandwidth Notification status and interrupt mechanisms. This capability is required for all Root Ports and Switch Downstream Ports supporting Links wider than x1 and/or multiple Link speeds. This field is not applicable and is Reserved for Endpoints, PCI Express to PCI/PCI-X bridges, and Upstream Ports of Switches. For functions that do not implement the Link Bandwidth Notification Capability the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
20	PCIE_CAP_DLL_ACTIVE_REP_CAP	R	Data Link Layer Link Active Reporting Capable. For a Downstream Port, the controller hardwires this bit to 1b if the component supports the optional capability of reporting the DL_Active state of the Data Link Control and Management State Machine. For a hot-plug capable Downstream Port (as indicated by the Hot-Plug Capable bit of the Slot Capabilities register) or a Downstream Port that supports Link speeds greater than 5.0 GT/s, the controller hardwires this bit to 1b. For Upstream Ports and components that do not support this optional capability, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
19	PCIE_CAP_SURPRISE_DOWN_ERR_REP_CAP	R	Surprise Down Error Reporting Capable. For a Downstream Port, this bit must be set if the component supports the optional capability of detecting and reporting a Surprise Down error condition. For Upstream Ports and components that do not support this optional capability, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
18	PCIE_CAP_CLOCK_POWER_MAN	R	Clock Power Management. For Upstream Ports, a value of 1b in this bit indicates that the component tolerates the removal of any reference clock(s) through the "clock request" (CLKREQ#) mechanism when the Link is in the L1 and L2/L3 Ready Link states. A value of 0b indicates the component does not have this capability and that reference clock(s) must not be removed in these Link states. L1 PM Substates defines other semantics for the CLKREQ# signal, which are managed independently of Clock Power Management. This Capability is applicable only in form factors that support "clock request" (CLKREQ#) capability. For a Multi-Function device associated with an Upstream Port, each Function indicates its capability independently. Power Management configuration software must only permit reference clock removal if all functions of the Multi-Function device indicate a 1b in this bit. For ARI Devices, all Functions must indicate the same value in this bit. For Downstream Ports, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_CLK_PM_CAP Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
17:15	PCIE_CAP_L1_EXIT_LATENCY	R	L1 Exit Latency. This field indicates the L1 exit latency for the given PCI Express Link. The value reported indicates the length of time this Port requires to complete transition from ASPM L1 to L0. If ASPM L1 is not supported, the value is undefined. <i>Note:</i> Exit latencies may be influenced by PCI Express reference clock configuration depending upon whether a component uses a common or separate reference clock. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all functions. There are two each of these register fields, this one and a shadow one at the same address. The Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG) determines which one is used by the controller and which one is accessed by a read request. Common Clock operation is supported (possible) in the controller when one or more of the following expressions is true: ■ CX_NFTS !=CX_COMM_NFTS ■ DEFAULT_L0S_EXIT_LATENCY !=DEFAULT_COMM_L0S_EXIT_LATENCY ■ DEFAULT_L1_EXIT_LATENCY !=DEFAULT_COMM_L1_EXIT_LATENCY Common Clock operation is enabled in the controller when you set the Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG). The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the shadow field at this location. Values: ■ 0x0 (LESS_THAN_1US): Less than 1us ■ 0x1 (_1US_TO_2US): 1 us to less than 2 us ■ 0x2 (_2US_TO_4US): 2 us to less than 4 us
17:15...(cont.)	PCIE_CAP_L1_EXIT_LATENCY.	R	■ 0x3 (_4US_TO_8US): 4 us to less than 8 us ■ 0x4 (_8US_TO_16US): 8 us to less than 16 us ■ 0x5 (_16US_TO_32US): 16 us to less than 32 us ■ 0x6 (_32US_TO_64US): 32 us to 64 us ■ 0x7 (GREATER_THAN_64US): More than 64 us Value After Reset: DEFAULT_L1_EXIT_LATENCY Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
14:12	PCIE_CAP_L0S_EXIT_LATENCY	R	L0s Exit Latency. This field indicates the L0s exit latency for the given PCI Express Link. The value reported indicates the length of time this Port requires to complete transition from L0s to L0. If L0s is not supported, the value is undefined; however, see the Implementation Note "Potential Issues With Legacy Software When L0s is Not Supported" in section 5.4.1.1 of <i>PCI Express Base Specification</i> for the recommended value. <i>Note:</i> Exit latencies may be influenced by PCI Express reference clock configuration depending upon whether a component uses a common or separate reference clock. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all functions. There are two each of these register fields, this one and a shadow one at the same address. The Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG) determines which one is used by the controller and which one is accessed by a read request. Common Clock operation is supported (possible) in the controller when one or more of the following expressions is true: ■ CX_NFTS !=CX_COMM_NFTS ■ DEFAULT_L0S_EXIT_LATENCY !=DEFAULT_COMM_L0S_EXIT_LATENCY ■ DEFAULT_L1_EXIT_LATENCY !=DEFAULT_COMM_L1_EXIT_LATENCY Common Clock operation is enabled in the controller when you set the Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG). The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the shadow field at this location. Values:

Bits	Name	Memory Access	Description
14:12...(cont.)	PCIE_CAP_L0S_EXIT_LATENCY.	R	■ 0x0 (LESS_THAN_64NS): Less than 64 ns ■ 0x1 (_64NS_TO_128NS): 64 ns to less than 128 ns ■ 0x2 (_128NS_TO_256NS): 128 ns to less than 256 ns ■ 0x3 (_256NS_TO_512NS): 256 ns to less than 512 ns ■ 0x4 (_512NS_TO_1US): 512 ns to less than 1 us ■ 0x5 (_1US_TO_2US): 1 us to less than 2 us ■ 0x6 (_2US_TO_4US): 2 us to 4 us ■ 0x7 (GREATER_THAN_4US): More than 4 us Value After Reset: DEFAULT_L0S_EXIT_LATENCY Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
11:10	PCIE_CAP_ACTIVE_STATE_LINK_PM_SUPPORT	R	Active State Power Management (ASPM) Support. This field indicates the level of ASPM supported on the given PCI Express Link. For more information on ASPM support requirements, see section 5.4.1 of <i>PCI Express Base Specification</i>. Multi-Function devices associated with an Upstream Port must report the same value in this field for all functions. Values: ■ 0x0 (NO_ASPM_SUP): No ASPM Support ■ 0x1 (L0S_SUP): L0s Supported ■ 0x2 (L1_SUP): L1 Supported ■ 0x3 (L0S_L1_SUP): L0s and L1 Supported Value After Reset: AS_LINK_PM_SUPT Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
9:4	PCIE_CAP_MAX_LINK_WIDTH	R	Maximum Link Width. This field indicates the maximum Link width (xN - corresponding to N Lanes) implemented by the component. This value is permitted to exceed the number of Lanes routed to the slot (Downstream Port), adapter connector (Upstream Port), or in the case of component-to-component connections, the actual wired connection width. All encodings other than the defined encodings are reserved. Multi-Function devices associated with an Upstream Port must report the same value in this field for all functions. Values: ■ 0x1 (X1): x1 ■ 0x2 (X2): x2 ■ 0x4 (X4): x4 ■ 0x8 (X8): x8 ■ 0xc (X12): x12 ■ 0x10 (X16): x16 ■ 0x20 (X32): x32 Value After Reset: CX_NL Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3:0	PCIE_CAP_MAX_LINK_SPEED	R	Max Link Speed. This field indicates the maximum Link speed of the associated Port. The encoded value specifies a bit location in the Supported Link Speeds Vector (in the Link Capabilities 2 register) that corresponds to the maximum Link speed. All encodings other than the defined encodings are reserved. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all functions. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SUP_LINK_SPEED_FIELD_BIT_0): Supported Link Speeds Vector field bit 0 ■ 0x2 (SUP_LINK_SPEED_FIELD_BIT_1): Supported Link Speeds Vector field bit 1 ■ 0x3 (SUP_LINK_SPEED_FIELD_BIT_2): Supported Link Speeds Vector field bit 2 ■ 0x4 (SUP_LINK_SPEED_FIELD_BIT_3): Supported Link Speeds Vector field bit 3 ■ 0x5 (SUP_LINK_SPEED_FIELD_BIT_4): Supported Link Speeds Vector field bit 4 ■ 0x6 (SUP_LINK_SPEED_FIELD_BIT_5): Supported Link Speeds Vector field bit 5 ■ 0x7 (SUP_LINK_SPEED_FIELD_BIT_6): Supported Link Speeds Vector field bit 6 Value After Reset: MAX_LINK_SP Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

3.52.5 VF_LINK_CONTROL_LINK_STATUS_REG

- **Name:** Link Control and Link Status Register.
- **Description:** This register controls and provides information about PCI Express Link specific parameters as well as RCRB link associated parameters.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

PCIE_CAP_LINK_AUTO_BW_STATUS	31
PCIE_CAP_LINK_BW_MAN_STATUS	30
PCIE_CAP_DLL_ACTIVE	29
PCIE_CAP_SLOT_CLK_CONFIG	28
PCIE_CAP_LINK_TRAINING	27
RSVDP_26	26
PCIE_CAP_NEGO_LINK_WIDTH	25:20
PCIE_CAP_LINK_SPEED	19:16
PCIE_CAP_DRS_SIGNALING_CONTROL	15:14
RSVDP_12	13:12
PCIE_CAP_LINK_AUTO_BW_INT_EN	11
PCIE_CAP_LINK_BW_MAN_INT_EN	10
PCIE_CAP_HW_AUTO_WIDTH_DISABLE	9
PCIE_CAP_EN_CLK_POWER_MAN	8
PCIE_CAP_EXTENDED_SYNC	7
PCIE_CAP_COMMON_CLK_CONFIG	6
PCIE_CAP_RETRAIN_LINK	5
PCIE_CAP_LINK_DISABLE	4
PCIE_CAP_RCB	3
RSVDP_2	2
PCIE_CAP_ACTIVE_STATE_LINK_PM_CONTROL	1:0

Table 3-501 Fields for Register: VF_LINK_CONTROL_LINK_STATUS_REG

Bits	Name	Memory Access	Description
31	PCIE_CAP_LINK_AUTO_BW_STATUS	* Varies	Link Autonomous Bandwidth Status. This bit is set by hardware to indicate that hardware has autonomously changed Link speed or width, without the Port transitioning through DL_Down status, for reasons other than to attempt to correct unreliable Link operation. This bit must be set if the Physical Layer reports a speed or width change was initiated by the Downstream component that was indicated as an autonomous change. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG. This bit is not applicable and is Reserved for Endpoints, PCI Express-to-PCI/PCI-X bridges, and Upstream Ports of Switches. For functions that do not implement the Link Bandwidth Notification Capability, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: R Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
30	PCIE_CAP_LINK_BW_MAN_STAT US	* Varies	Link Bandwidth Management Status. This bit is set by hardware to indicate that either of the following has occurred without the Port transitioning through DL_Down status: - A Link retraining has completed following a write of 1b to the Retrain Link bit. <i>Note:</i> This bit is set following any write of 1b to the Retrain Link bit, including when the Link is in the process of retraining for some other reason. Hardware has changed Link speed or width to attempt to correct unreliable Link operation, either through an LTSSM timeout or a higher level process. This bit must be set if the Physical Layer reports a speed or width change was initiated by the Downstream component that was not indicated as an autonomous change. This bit is not applicable and is Reserved for Endpoints, PCI Express-to-PCI/PCI-X bridges, and Upstream Ports of Switches. For functions that do not implement the Link Bandwidth Notification Capability, the controller hardwires this bit to 0b. The default value of this bit is 0b. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: R Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
29	PCIE_CAP_DLL_ACTIVE	R	Data Link Layer Link Active. This bit indicates the status of the Data Link Control and Management State Machine. It returns a 1b to indicate the DL_Active state, 0b otherwise. This bit must be implemented if the Data Link Layer Link Active Reporting Capable bit is 1b. Otherwise, the controller hardwires it to 0b. Value After Reset: 0x0 Exists: (CX_MEMORY_MAP_PCIE_TYPE<2) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
28	PCIE_CAP_SLOT_CLK_CONFIG	* Varies	Slot Clock Configuration. This bit indicates that the component uses the same physical reference clock that the platform provides on the connector. If the device uses an independent clock irrespective of the presence of a reference clock on the connector, this bit must be clear. For a Multi-Function Device, each Function must report the same value for this bit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true
27	PCIE_CAP_LINK_TRAINING	* Varies	Link Training. This read-only bit indicates that the Physical Layer LTSSM is in the Configuration or Recovery state, or that 1b was written to the Retrain Link bit but Link training has not yet begun. Hardware clears this bit when the LTSSM exits the Configuration/Recovery state. This bit is not applicable and Reserved for Endpoints, PCI Express to PCI/PCI-X bridges, and Upstream Ports of Switches, and the controller hardwires it to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: R Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
25:20	PCIE_CAP_NEGO_LINK_WIDTH	R	Negotiated Link Width. This field indicates the negotiated width of the given PCI Express Link. All encodings other than the defined encodings are reserved. The value in this field is undefined when the Link is not up. Values: ■ 0x1 (X1): x1 ■ 0x2 (X2): x2 ■ 0x4 (X4): x4 ■ 0x8 (X8): x8 ■ 0xc (X12): x12 ■ 0x10 (X16): x16 ■ 0x20 (X32): x32 Value After Reset: * Varies Exists: (CX_MEMORY_MAP_PCIE_TYPE<2) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
19:16	PCIE_CAP_LINK_SPEED	R	Current Link Speed. This field indicates the negotiated Link speed of the given PCI Express Link. The encoded value specifies a bit location in the Supported Link Speeds Vector (in the Link Capabilities 2 register) that corresponds to the current Link speed. All encodings other than the defined encodings are reserved. The value in this field is undefined when the Link is not up. Values: ■ 0x1 (SUP_LINK_SPEED_FIELD_BIT_0): Supported Link Speeds Vector field bit 0 ■ 0x2 (SUP_LINK_SPEED_FIELD_BIT_1): Supported Link Speeds Vector field bit 1 ■ 0x3 (SUP_LINK_SPEED_FIELD_BIT_2): Supported Link Speeds Vector field bit 2 ■ 0x4 (SUP_LINK_SPEED_FIELD_BIT_3): Supported Link Speeds Vector field bit 3 ■ 0x5 (SUP_LINK_SPEED_FIELD_BIT_4): Supported Link Speeds Vector field bit 4 ■ 0x6 (SUP_LINK_SPEED_FIELD_BIT_5): Supported Link Speeds Vector field bit 5 ■ 0x7 (SUP_LINK_SPEED_FIELD_BIT_6): Supported Link Speeds Vector field bit 6 Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
15:14	PCIE_CAP_DRS_SIGNALING_CONTROL	R	DRS Signaling Control. Indicates the mechanism used to report reception of a DRS message. Must be implemented for Downstream Ports with the DRS Supported bit Set in the Link Capabilities 2 Register. Encodings are: If DRS Supported is set, receiving a DRS Message will set DRS Message Received in the Link Status 2 Register but will otherwise have no effect If the DRS Message Received bit in the Link Status 2 Register transitions from 0 to 1, and either MSI or MSI-X is enabled, an MSI or MSI-X interrupt is generated using the vector in Interrupt Message Number (section 7.5.3.2) If the DRS Message Received bit in the Link Status 2 Register transitions from 0 to 1, the Port must send an FRS Message Upstream with the FRS Reason field set to DRS Message Received. Behavior is undefined if this field is set to 10b and the FRS Supported bit in the Device Capabilities 2 Register is Clear. Behavior is undefined if this field is set to 11b. For Downstream Ports with the DRS Supported bit clear in the Link Capabilities 2 register, the controller hardwires this field to 00b. This field is Reserved for Upstream Ports. Values: ■ 0x0 (DRS_NOT_REPORTED): DRS not Reported ■ 0x1 (DRS_INTERRUPT_EN): DRS Interrupt Enabled ■ 0x2 (FRS_EN): DRS to FRS Signaling Enabled Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
13:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
11	PCIE_CAP_LINK_AUTO_BW_INT_EN	* Varies	Link Autonomous Bandwidth Management Interrupt Enable. When set, this bit enables the generation of an interrupt to indicate that the Link Autonomous Bandwidth Status bit has been set. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG. This bit is not applicable and is Reserved for Endpoints, PCI Express-to-PCI/PCI-X bridges, and Upstream Ports of Switches. For functions that do not implement the Link Bandwidth Notification Capability, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: LINK_CAPABILITIES_REG.PCIE_CAP_LINK_BW_NOT_CAP ? RW : RO ■ Dbi: LINK_CAPABILITIES_REG.PCIE_CAP_LINK_BW_NOT_CAP ? RW : RO Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true
10	PCIE_CAP_LINK_BW_MAN_INT_EN	* Varies	Link Bandwidth Management Interrupt Enable. When set, this bit enables the generation of an interrupt to indicate that the Link Bandwidth Management Status bit has been set. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG. This bit is not applicable and is Reserved for Endpoints, PCI Express-to-PCI/PCI-X bridges, and Upstream Ports of Switches. For functions that do not implement the Link Bandwidth Notification Capability, the controller hardwires this bit to 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: LINK_CAPABILITIES_REG.PCIE_CAP_LINK_BW_NOT_CAP ? RW : RO ■ Dbi: LINK_CAPABILITIES_REG.PCIE_CAP_LINK_BW_NOT_CAP ? RW : RO Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
9	PCIE_CAP_HW_AUTO_WIDTH_DISABLE	R	Hardware Autonomous Width Disable. When set, this bit disables hardware from changing the Link width for reasons other than attempting to correct unreliable Link operation by reducing Link width. For a Multi-Function Device associated with an Upstream Port, the bit in Function 0 is of type RW, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is of type RsvdP. For components that do not implement the ability autonomously to change Link width, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
8	PCIE_CAP_EN_CLK_POWER_MAN	* Varies	Enable Clock Power Management. Applicable only for Upstream Ports and with form factors that support a "Clock Request" (CLKREQ#) mechanism, this bit operates as follows: For a non-ARI Multi-Function Device, power-management-configuration software must only Set this bit if all Functions of the Multi-Function Device indicate a 1b in the Clock Power Management bit of the Link Capabilities register. The component is permitted to use the CLKREQ# signal to power manage Link clock only if this bit is Set for all Functions. For ARI Devices, Clock Power Management is enabled solely by the setting in Function 0. The settings in the other Functions always return whatever value software programmed for each, but otherwise are ignored by the component. The CLKREQ# signal may also be controlled through the L1 PM Substates mechanism. Such control is not affected by the setting of this bit. For Downstream Ports and components that do not support Clock Power Management (as indicated by a 0b value in the Clock Power Management bit of the Link Capabilities register), the controller hardwires this bit to 0b. The write value is gated with the PCIE_CAP_CLOCK_POWER_MAN field in LINK_CAPABILITIES_REG. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: LINK_CAPABILITIES_REG.PCIE_CAP_CLOCK_POWER_MAN ? RWS : ROS ■ Dbi: LINK_CAPABILITIES_REG.PCIE_CAP_CLOCK_POWER_MAN ? RWS : ROS <i>Note:</i> This register field is sticky. Values: ■ 0x0 (PM_DISABLE_LOW_CLKREQ): Clock power management is disabled and device must hold CLKREQ# signal low.
8...(cont.)	PCIE_CAP_EN_CLK_POWER_MAN.	* Varies	■ 0x1 (CLKREQ_IN_USE): When this bit is set, the device is permitted to use CLKREQ# signal to power manage Link clock according to protocol defined in appropriate form factor specification. Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
7	PCIE_CAP_EXTENDED_SYNCH	R/W	Extended Synch. When set, this bit forces the transmission of additional Ordered Sets when exiting the L0s state (see section 4.2.4.5 of <i>PCI Express Base Specification</i>) and when in the Recovery state (see section 4.2.6.4.1 of <i>PCI Express Base Specification</i>). This mode provides external devices (for example, logic analyzers) monitoring the Link time to achieve bit and Symbol lock before the Link enters the L0 state and resumes communication. For Multi-Function devices if any function has this bit set, then the component must transmit the additional Ordered Sets when exiting L0s or when in Recovery. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
6	PCIE_CAP_COMMON_CLK_CONFIG	R	Common Clock Configuration. When set, this bit indicates that this component and the component at the opposite end of this Link are operating with a distributed common reference clock. A value of 0b indicates that this component and the component at the opposite end of this Link are operating with asynchronous reference clock. For non-ARI Multi-Function Devices, software must program the same value for this bit in all Functions. If not all Functions are Set, then the component must as a whole assume that its reference clock is not common with the Upstream component. For ARI Devices, Common Clock Configuration is determined solely by the setting in Function 0. The settings in the other Functions always return whatever value software programmed for each, but otherwise are ignored by the component. Components utilize this common clock configuration information to report the correct L0s and L1 Exit Latencies. After changing the value in this bit in both components on a Link, software must trigger the Link to retrain by writing a 1b to the Retrain Link bit of the Downstream Port. Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
5	PCIE_CAP_RETRAIN_LINK	* Varies	Retrain Link. A write of 1b to this bit initiates Link retraining by directing the Physical Layer LTSSM to the Recovery state. If the LTSSM is already in Recovery or Configuration, re-entering Recovery is permitted but not required. If the Port is in DPC when a write of 1b to this bit occurs, the result is undefined. Reads of this bit always return 0b. It is permitted to write 1b to this bit while simultaneously writing modified values to other fields in this register. If the LTSSM is not already in Recovery or Configuration, the resulting Link training must use the modified values. If the LTSSM is already in Recovery or Configuration, the modified values are not required to affect the Link training that's already in progress. This bit is not applicable and is Reserved for Endpoints, PCI Express to PCI/PCI-X bridges, and Upstream Ports of Switches. This bit always returns 0b when read. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: see description ■ Dbi: see description Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
4	PCIE_CAP_LINK_DISABLE	* Varies	Link Disable. This bit disables the Link by directing the LTSSM to the Disabled state when set; this bit is Reserved on Endpoints, PCI Express to PCI/PCI-X bridges, and Upstream Ports of Switches. Writes to this bit are immediately reflected in the value read from the bit, regardless of actual Link state. After clearing this bit, software must honor timing requirements defined in Section 6.6.1 with respect to the first Configuration Read following a Conventional Reset. In a DSP that supports crosslink, the controller gates the write value with the CROSS_LINK_EN field in PORT_LINK_CTRL_OFF. Note: The access attributes of this field are as follows: ■ Wire: CX_CROSSLINK_ENABLE=1 && PORT_LINK_CTRL_OFF.CROSS_LINK_EN=1 CX_CROSSLINK_ENABLE=0 && dsp=1 ? RW : RO ■ Dbi: CX_CROSSLINK_ENABLE=1 && PORT_LINK_CTRL_OFF.CROSS_LINK_EN=1 CX_CROSSLINK_ENABLE=0 && dsp=1? RW : RO Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true
3	PCIE_CAP_RCB	R	Read Completion Boundary (RCB). <i>Endpoints and Bridges:</i> Optionally set by configuration software to indicate the RCB value of the Root Port Upstream from the Endpoint or Bridge. Refer to Section 2.3.1.1 of <i>PCI Express Base Specification</i> for the definition of the parameter RCB is same as Root Port. Configuration software must only set this bit if the Root Port Upstream from the Endpoint or Bridge reports an RCB value of 128 bytes (a value of 1b in the Read Completion Boundary bit). For functions that do not implement this feature, the controller hardwires this bit to 0b. Values: ■ 0x0 (_64B): 64 byte ■ 0x1 (_128B): 128 byte Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
1:0	PCIE_CAP_ACTIVE_STATE_LINK_PM_CONTROL	R	Active State Power Management (ASPM) Control. This field controls the level of ASPM enabled on the given PCI Express Link. See section 5.4.1.3 of <i>PCI Express Base Specification</i> for requirements on when and how to enable ASPM. <i>Note:</i> "L0s Entry Enabled" enables the Transmitter to enter L0s. If L0s is supported, the Receiver must be capable of entering L0s even when the Transmitter is disabled from entering L0s (00b or 10b). ASPM L1 must be enabled by software in the Upstream component on a Link prior to enabling ASPM L1 in the Downstream component on that Link. When disabling ASPM L1, software must disable ASPM L1 in the Downstream component on a Link prior to disabling ASPM L1 in the Upstream component on that Link. ASPM L1 must only be enabled on the Downstream component if both components on a Link support ASPM L1. For Multi-Function Devices (including ARI Devices), it is recommended that software program the same value for this field in all Functions. For non-ARI Multi-Function Devices, only capabilities enabled in all Functions are enabled for the component as a whole. For ARI Devices, ASPM Control is determined solely by the setting in Function0, regardless of Function 0's D-state. The settings in the other Functions always return whatever value software programmed for each, but otherwise are ignored by the component. Software must not enable L0s in either direction on a given Link unless components on both sides of the Link each support L0s; otherwise, the result is undefined. Values: ■ 0x0 (DISABLED): Disabled ■ 0x1 (L0S_ENTRY_EN): L0s Entry Enabled ■ 0x2 (L1_ENTRY_EN): L1 Entry Enabled ■ 0x3 (L0S_L1_ENTRY_EN): L0s and L1 Entry Enabled Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.52.6 VF_DEVICE_CAPABILITIES2_REG

- **Name:** Device Capabilities 2 Register.
- **Description:** This register identifies PCI Express device specific capabilities; in addition to the Device Capabilities Register.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

PCIE_CAP_FRS_SUPPORTED	31
PCIE_CAP_DMWR_LEN_SUPP	30:29
PCIE_CAP_DMWR_CPL_SUPP	28
RSVDP_27	27
PCIE_CAP2_EMERGENCY_PWR_RED_INIT_REQUIRED	26
PCIE_CAP2_EMERGENCY_PWR_RED_SUPPORT	25:24
PCIE_CAP2_CFG_MAX_END2END_TLP_PRFXS	23:22
PCIE_CAP2_CFG_END2END_TLP_PRFX_SUPPORT	21
PCIE_CAP2_CFG_EXTND_FMT_SUPPORT	20
PCIE_CAP_OBFF_SUPPORT	19:18
PCIE_CAP2_10_BIT_TAG_REQ_SUPPORT	17
PCIE_CAP2_10_BIT_TAG_COMP_SUPPORT	16
PCIE_CAP2_LN_SYS_CLS	15:14
PCIE_CAP_TPH_CMPLT_SUPPORT_1	13
PCIE_CAP_TPH_CMPLT_SUPPORT_0	12
PCIE_CAP_LTR_SUPP	11
PCIE_CAP_NO_RO_EN_PR2PR_PAR	10
PCIE_CAP_128_CAS_CPL_SUPP	9
PCIE_CAP_64_ATOMIC_CPL_SUPP	8
PCIE_CAP_32_ATOMIC_CPL_SUPP	7
PCIE_CAP_ATOMIC_ROUTING_SUPP	6
PCIE_CAP_ARI_FORWARD_SUPPORT	5
PCIE_CAP_CPL_TIMEOUT_DISABLE_SUPPORT	4
PCIE_CAP_CPL_TIMEOUT_RANGE	3:0

Table 3-502 Fields for Register: VF_DEVICE_CAPABILITIES2_REG

Bits	Name	Memory Access	Description
31	PCIE_CAP_FRS_SUPPORTED	* Varies	FRS Supported. When set, indicates support for the optional Function Readiness Status (FRS) capability. Must be set for all Functions that support generation or reception capabilities of FRS Messages. Must not be set by Switch Functions that do not generate FRS Messages on their own behalf. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) and more specifically: (CX_MEMORY_MAP_PCIE_TYPE==1 && 1) ? R : (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : R Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_RN_FRS_VALUE_0} Exists: CX_RN_FRS_SUPPORTED && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
30:29	PCIE_CAP_DMWR_LEN_SUPP	R	Deferrable Memory Write (DMWr) Lengths Supported. Applicable to Functions with either the DMWr Routing Supported bit Set or the DMWr Completer Supported bit Set (or both). This field indicates the largest DMWr TLP that this Function can receive. When applicable, all Functions in a Multi-Function Device associated with an Upstream Port must report the same value in this field. This field is RsvdP if both DMWr Completer Supported and DMWr Routing Supported are Clear. For more information, see Section 6.30 of <i>PCI Express Base Specification</i>. Values: ■ 0x0 (up_to_64_bytes): 0 up to 64 bytes ■ 0x1 (up_to_128_bytes): 1 up to 128 bytes Value After Reset: {((CX_DEF_MEM_WR_CPL_EN==1) (CX_DEF_MEM_WR_ROUTING_EN==1)) ? CX_DEF_MEM_WR_LEN_SUPP : 0} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
28	PCIE_CAP_DMWR_CPL_SUPP	R	Deferrable Memory Write (DMWr) Completer Supported. Applicable to Functions with Memory Space BARs as well as all Root Ports; This bit must be Set if the Function can serve as a DMWr Completer. For more information, see Section 6.30 of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_DEF_MEM_WR_CPL_EN} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
26	PCIE_CAP2_EMERGENCY_PWR_RED_INIT_REQUIRED	* Varies	Emergency Power Reduction Initialization Required. Indicates if Software initialization upon exit from the Emergency Power Reduction State is required. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(Sticky) else R(Sticky) Values: ■ 0x0 (CLEAR): The Function requires no software intervention to return to normal operation upon exit from the Emergency Power Reduction State. ■ 0x1 (SET): The Function requires complete or partial initialization upon exit from the Emergency Power Reduction State. Value After Reset: 0x0 Exists: EMERGENCY_POWER_REDUCTION_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
25:24	PCIE_CAP2_EMERGENCY_PWR_RED_SUPPORT	* Varies	Emergency Power Reduction Supported. Indicates support level of the optional Emergency Power Reduction State feature. For Multi-Function Devices associated with an Upstream Port, all Functions that report a non-zero value for this field, must report the same non-zero value for this field. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(Sticky) else R(Sticky) Values: ■ 0x0 (NO_SUP): Emergency Power Reduction State not supported. ■ 0x1 (SUP_01b): Emergency Power Reduction State is supported and is triggered by Device Specific mechanism(s). ■ 0x2 (SUP_10b): Emergency Power Reduction State is supported and is triggered either by the mechanism defined in the corresponding Form Factor specification or by Device Specific mechanism(s). ■ 0x3 (RSVD_4): Reserved Value After Reset: 0x0 Exists: EMERGENCY_POWER_REDUCTION_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
23:22	PCIE_CAP2_CFG_MAX_END2END_TLP_PRFXS	R	Max End-End TLP Prefixes. Indicates the maximum number of End-End TLP Prefixes supported by this Function. For more information, see Section 2.2.10.2 of <i>PCI Express Base Specification</i>. If End-End TLP Prefix Supported is clear, this field is RsvdP. Different Root Ports that have the End-End TLP Prefix Supported bit set are permitted to report different values for this field. For Switches where End-End TLP Prefix Supported is set, this field must be 00b indicating support for up to four End-End TLP Prefixes. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ONE_END_END_TLP_PREFIX): 1 End-End TLP Prefix ■ 0x2 (TWO_END_END_TLP_PREFIX): 2 End-End TLP Prefixes ■ 0x3 (THREE_END_END_TLP_PREFIX): 3 End-End TLP Prefixes ■ 0x0 (FOUR_END_END_TLP_PREFIX): 4 End-End TLP Prefixes Value After Reset: 0x0 Exists: CX_TLP_PREFIX_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
21	PCIE_CAP2_CFG_END2END_TLP_PRFX_SUPPORT	R	End-End TLP Prefix Supported. Indicates whether End-End TLP Prefix support is offered by a Function. Values are: All Ports of a Switch must have the same value for this bit. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NO_SUP): No Support ■ 0x1 (SUP): Support is provided to receive TLPs containing End-End TLP Prefixes. Value After Reset: 0x0 Exists: CX_TLP_PREFIX_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
20	PCIE_CAP2_CFG_EXTND_FMT_SUPPORT	R	Extended Fmt Field Supported. If set, the Function supports the 3-bit definition of the Fmt field. If clear, the Function supports a 2-bit definition of the Fmt field. For more information, see section 2.2 of <i>PCI Express Base Specification</i>. Must be set for Functions that support End-End TLP Prefixes. All Functions in an Upstream Port must have the same value for this bit. Each Downstream Port of a component may have a different value for this bit. It is strongly recommended that Functions support the 3-bit definition of the Fmt field. <i>Note:</i> This register field is sticky. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? "0x0" : "0x1"} Exists: CX_TLP_PREFIX_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
19:18	PCIE_CAP_OBFF_SUPPORT	R	OBFF Supported. This field indicates if OBFF is supported and, if so, what signaling mechanism is used. The value reported in this field must indicate support for WAKE# signaling only if: ■ for a Downstream Port, driving the WAKE# signal for OBFF is supported and the connector or component connected Downstream is known to receive that same WAKE# signal ■ for an Upstream Port, receiving the WAKE# signal for OBFF is supported and, if the component is on an add-in-card, that the component is connected to the WAKE# signal on the connector. Root Ports, Switch Ports, and Endpoints are permitted to implement this capability. For a Multi-Function Device associated with an Upstream Port, each Function must report the same value for this field. For Bridges and Ports that do not implement this capability, the controller hardwires this field to 00b. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NOT_SUP): OBFF Not Supported ■ 0x1 (SUP_USING_MSG): OBFF supported using Message signaling only ■ 0x2 (SUP_USING_WAKE): OBFF supported using WAKE# signaling only ■ 0x3 (SUP_USING_MSG_AND_WAKE): OBFF supported using WAKE# and Message signaling Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_OBFF_SUPPORT} Exists: CX_OBFF_SUPPORT && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
17	PCIE_CAP2_10_BIT_TAG_REQ_SUPPORT	R	10-Bit Tag Requester Supported. If this bit is set, the Function supports 10-Bit Tag Requester capability; otherwise, the Function does not. This bit must not be set if the 10-Bit Tag Completer Supported bit is clear. <i>Note:</i> 10-Bit Tag field generation must be enabled by the 10-Bit Tag Requester Enable bit in the Device Control 2 register of the Requester Function before 10-Bit Tags can be generated by the Requester. For more information, see section 2.2.6.2. of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_10BITS_TAG_REQ_VALUE} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
16	PCIE_CAP2_10_BIT_TAG_COMP_SUPPORT	R	10-Bit Tag Completer Supported. If this bit is set, the Function supports 10-Bit Tag Completer capability; otherwise, the Function does not. For more information, see section 2.2.6.2. of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_10BITS_TAG_VALUE} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
15:14	PCIE_CAP2_LN_SYS_CLS	R	LN System CLS. Applicable only to Root Ports and RCRBs; must be 00b for all other Function types. This field indicates if the Root Port or RCRB supports LN protocol as an LN Completer, and if so, what cacheline size is in effect. All encodings other than the defined encodings are reserved. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (LN_COMPLETER_NO_SUP): LN Completer either not supported or not in effect ■ 0x1 (LN_COMPLETER_64B_CACHE): LN Completer with 64-byte cachelines in effect ■ 0x2 (LN_COMPLETER_128B_CACHE): LN Completer with 128-byte cachelines in effect Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
13	PCIE_CAP_TPH_CMPLT_SUPP_ORT_1	R	TPH Completer Supported Bit 1. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
12	PCIE_CAP_TPH_CMPLT_SUPP0RT_0	R	TPH Completer Supported Bit 0. Value of this bit along with TPH Completer Supported Bit 1 indicates Completer support for TPH or Extended TPH. Applicable only to Root Ports and Endpoints. For all other Functions, this field is Reserved. For more information, see section 6.17 of <i>PCI Express Base Specification</i>. Values: 0x0 (TPH_EXT_TPH_CMPL_NOT_SUP_OR_RSVD): TPH and Extended TPH Completer not supported (if TPH Completer Supported Bit is 0) or Reserved (if TPH Completer Supported Bit is 1). 0x1 (TPH_SUP_EXT_TPH_NOT_SUP_OR_BOTH_SUPP): TPH Completer supported; Extended TPH Completer not supported (if TPH Completer Supported Bit is 0) or Both TPH and Extended TPH Completer supported (if TPH Completer Supported Bit is 1). Value After Reset: $\{((VF_TPH_ENABLE==1) (EXT_VF_TPH_ENABLE==1)) ? CX_TPH_ENABLE_VALUE : 0\}$ Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
11	PCIE_CAP_LTR_SUPP	R	LTR Mechanism Supported. A value of 1b indicates support for the optional Latency Tolerance Reporting (LTR) mechanism. Root Ports, Switches and Endpoints are permitted to implement this capability. For a Multi-Function Device associated with an Upstream Port, each Function must report the same value for this bit. For Bridges and other Functions that do not implement this capability, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: $\{(CX_MEMORY_MAP_PCIE_TYPE==1 \&\& !CX_MEMORY_MAP_POSITION) ? 0 : CX_LTR_M_ENABLE_VALUE\}$ Exists: CX_LTR_M_ENABLE_VALUE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
10	PCIE_CAP_NO_RO_EN_PR2PR_PAR	R	No RO-enabled PR-PR Passing. If this bit is set, the routing element never carries out the passing permitted by Table 2-39 of <i>PCI Express Base Specification</i> entry A2b that is associated with the Relaxed Ordering Attribute field being Set. This bit applies only for Switches and RCs that support peer-to-peer traffic between Root Ports. This bit applies only to Posted Requests being forwarded through the Switch or RC and does not apply to traffic originating or terminating within the Switch or RC itself. All Ports on a Switch or RC must report the same value for this bit. For all other functions, this bit must be 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
9	PCIE_CAP_128_CAS_CPL_SUPP	R	128-bit CAS Completer Supported. Applicable to Functions with Memory Space BARs as well as all Root Ports; must be 0b otherwise. This bit must be set to 1b if the Function supports this optional capability. For more information, see section 6.15 of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_ATOMIC_128_CAS_EN} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
8	PCIE_CAP_64_ATOMIC_CPL_SU PP	R	64-bit AtomicOp Completer Supported. Applicable to Functions with Memory Space BARs as well as all Root Ports; must be 0b otherwise. Includes FetchAdd, Swap, and CAS AtomicOps. This bit must be set to 1b if the Function supports this optional capability. For more information on additional RC requirements, see section 6.15.3.1 of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_ATOMIC_64_CPL_EN} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
7	PCIE_CAP_32_ATOMIC_CPL_SUPP	R	32-bit AtomicOp Completer Supported. Applicable to Functions with Memory Space BARs as well as all Root Ports; must be 0b otherwise. Includes FetchAdd, Swap, and CAS AtomicOps. This bit must be set to 1b if the Function supports this optional capability. For more information on additional RC requirements, see section 6.15.3.1 of <i>PCI Express Base Specification</i>. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : CX_ATOMIC_32_CPL_EN} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
6	PCIE_CAP_ATOMIC_ROUTING_SUPP	R	AtomicOp Routing Supported. Applicable only to Switch Upstream Ports, Switch Downstream Ports, and Root Ports; must be 0b for other Function types. This bit must be set to 1b if the Port supports this optional capability. For more information, see section 6.15 of <i>PCI Express Base Specification</i>. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
5	PCIE_CAP_ARI_FORWARD_SUPPORT	R	ARI Forwarding Supported. Applicable only to Switch Downstream Ports and Root Ports; must be 0b for other Function types. This bit must be set to 1b if a Switch Downstream Port or Root Port supports this optional capability. For more information, see section 6.13 of <i>PCI Express Base Specification</i>. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
4	PCIE_CAP_CPL_TIMEOUT_DISABLE_SUPPORT	R	Completion Timeout Disable Supported. A value of 1b indicates support for the Completion Timeout Disable mechanism. The Completion Timeout Disable mechanism is required for Endpoints that issue Requests on their own behalf and PCI Express to PCI/PCI-X Bridges that take ownership of Requests issued on PCI Express. This mechanism is optional for Root Ports. For all other Functions this field is Reserved and the controller hardwires this bit to 0b. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? "0x0" : "0x1"} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
3:0	PCIE_CAP_CPL_TIMEOUT_RANGE	R	Completion Timeout Ranges Supported. This field indicates device Function support for the optional Completion Timeout programmability mechanism. This mechanism allows system software to modify the Completion Timeout value. This field is applicable only to Root Ports, Endpoints that issue Requests on their own behalf, and PCI Express to PCI/PCI-X Bridges that take ownership of Requests issued on PCI Express. For all other Functions this field is Reserved and must be hardwired to 0000b. Four time value ranges are defined: ■ Range A: 50 us to 10 ms ■ Range B: 10 ms to 250 ms ■ Range C: 250 ms to 4 s ■ Range D: 4 s to 64 s Bits are set according to the list below to show timeout value ranges supported. All encodings other than the defined encodings are reserved. It is strongly recommended that the Completion Timeout mechanism not expire in less than 10 ms. Values: ■ 0x0 (NOT_SUP): Completion Timeout programming not supported, the Function must implement a timeout value in the range 50 us to 50 ms. ■ 0x1 (RANGE_A): Range A ■ 0x2 (RANGE_B): Range B ■ 0x3 (RANGE_A_B): Ranges A and B ■ 0x6 (RANGE_B_C): Ranges B and C ■ 0x7 (RANGE_A_B_C): Ranges A, B, and C ■ 0xe (RANGE_B_C_D): Ranges B, C, and D ■ 0xf (RANGE_A_B_C_D): Ranges A, B, C, and D Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? "0x0" : ((CX_CPL_TO_RANGES_ENABLE==1) ? 0xf : 0)} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.52.7 VF_DEVICE_CONTROL2_DEVICE_STATUS2_REG

- **Name:** Device Control 2 and Status 2 Register.
- **Description:** This register controls PCI Express device specific parameters and provides information about PCI Express device (function) specific parameters; in addition to the Device Control and Device Status Register.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

31:16	RSVDP_16
15	PCIE_CTRL2_CFG_END2END_TLP_PFX_BLK
14:13	PCIE_CAP_OBFF_EN
12	PCIE_CAP_10BITS_TAG_REQ_EN
11	PCIE_CAP_EMERGENCY_PWR_RED_REQUEST
10	PCIE_CAP_LTR_EN
9	PCIE_CAP_IDO_CPL_EN
8	PCIE_CAP_IDO_REQ_EN
7	PCIE_CAP_ATOMIC_EGRESS_BLK
6	PCIE_CAP_ATOMIC_REQ_EN
5	PCIE_CAP_ARI_FORWARD_SUPPORT_CS
4	PCIE_CAP_CPL_TIMEOUT_DISABLE
3:0	PCIE_CAP_CPL_TIMEOUT_VALUE

Table 3-503 Fields for Register: VF_DEVICE_CONTROL2_DEVICE_STATUS2_REG

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_TLP_PREFIX_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
15	PCIE_CTRL2_CFG_END2END_TLP_PFX_BLK	R	End-End TLP Prefix Blocking. Controls whether the routing function is permitted to forward TLPs containing an End-End TLP Prefix. This bit affects TLPs that exit the Switch/Root Complex using the associated Port. It does not affect TLPs forwarded internally within the Switch/Root Complex. It does not affect TLPs that enter through the associated Port, that originate in the associated Port or originate in a Root Complex Integrated Device integrated with the associated Port. Blocked TLPs are reported by the TLP Prefix Blocked Error. This bit is hardwired to 1b in Root Ports that support End-End TLP Prefixes but do not support forwarding of End-End TLP Prefixes. This bit is applicable to Root Ports and Switch Ports where the End-End TLP Prefix Supported bit is set. This bit is not applicable and is RsvdP in all other cases. Values: ■ 0x0 (FORWARDING_EN): Forwarding Enabled, Function is permitted to send TLPs with End-End TLP Prefixes. ■ 0x1 (FORWARDING_BLK): Forwarding Blocked, Function is not permitted to send TLPs with End-End TLP Prefixes. Value After Reset: 0x0 Exists: CX_TLP_PREFIX_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
14:13	PCIE_CAP_OBFF_EN	R	OBFF Enable. This field enables the OBFF mechanism and selects the signaling method. For more information on these encodings, see section 6.19 of <i>PCI Express Base Specification</i>. This field is required for all Ports that support the OBFF Capability. For a Multi-Function Device associated with an Upstream Port of a Device that implements OBFF, the field in Function 0 is of type RW, and only Function 0 controls the Component's behavior. In all other Functions of that Device, this field is of type RsvdP. Ports that do not implement OBFF are permitted to hardwire this field to 00b Note: RW for function #0 and RsvdP for all other functions. Values: ■ 0x0 (DISABLED): Disabled ■ 0x1 (EN_USING_MSG_SIG_A): Enabled using Message signaling [Variation A] ■ 0x2 (EN_USING_MSG_SIG_B): Enabled using Message signaling [Variation B] ■ 0x3 (EN_USING_WAKE_SIG): Enabled using WAKE# signaling Value After Reset: 0x0 Exists: CX_OBFF_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
12	PCIE_CAP_10BITS_TAG_REQ_EN	R	10-Bit Tag Requester Enable. 10-Bit Tag Requester Enable. This bit, in combination with the Extended Tag Field Enable bit in the Device Control register, determines how many Tag field bits a Requester is permitted to use. When the 10-Bit Tag Requester Enable bit is Set, the Requester is permitted to use 10-Bit Tags. For more information, see section 2.2.6.2 of <i>PCI Express Base Specification</i>. If software changes the value of this bit while the Function has outstanding Non-Posted Requests, the result is undefined. Functions that do not implement 10-Bit Tag Requester capability are permitted to hardwire this bit to 0b. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : 0} Exists: CX_10BITS_TAG_REQ && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
11	PCIE_CAP_EMERGENCY_PWR_RED_REQUEST	* Varies	Emergency Power Reduction Request. This bit is implemented in the lowest numbered Function associated with an Upstream Port that has a non-zero value in the Emergency Power Reduction Supported field. This bit is RsvdP in all other Functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDZ ■ Dbi: RSVDZ Values: ■ 0x0 (CLEAR): If Clear these Functions must exit the Emergency Power Reduction State if no other reasons exist to preclude exiting this state. ■ 0x1 (SET): If Set, all Functions in the component that support Emergency Power Reduction State must enter the Emergency Power Reduction State. Value After Reset: 0x0 Exists: EMERGENCY_POWER_REDUCTION_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Memory Access: "(CX_MEMORY_MAP_VIEW==0 && 0==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && 0==0) ? \"read-only\": \"read-only\" "

Bits	Name	Memory Access	Description
10	PCIE_CAP_LTR_EN	* Varies	LTR Mechanism Enable. When set to 1b, this bit enables Upstream Ports to send LTR messages and Downstream Ports to process LTR Messages. For a Multi-Function Device associated with an Upstream Port of a device that implements LTR, the bit in Function 0 is RW, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is RsvdP. Functions that do not implement the LTR mechanism are permitted to hardwire this bit to 0b. For Downstream Ports, this bit must be reset to the default value if the Port goes to DL_Down status. The write value is gated with the PCIE_CAP_LTR_SUPP field of DEVICE_CAPABILITIES2_REG. <i>Note:</i> RW for function #0 and RsvdP for all other functions. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: if (pf=0 && DEVICE_CAPABILITIES2_REG.PCIE_CAP_LTR_SUPP) then R/W else R ■ Dbi: if (pf=0 && DEVICE_CAPABILITIES2_REG.PCIE_CAP_LTR_SUPP) then R/W else R Value After Reset: 0x0 Exists: CX_LTR_M_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead
9	PCIE_CAP_IDO_CPL_EN	R	IDO Completion Enable. If this bit is set, the Function is permitted to set the ID-Based Ordering (IDO) bit (Attr[2]) of Completions it returns (for more details, see section 2.2.6.3 and Section 2.4 of <i>PCI Express Base Specification</i>). Endpoints and Root Ports are permitted to implement this capability. A Function is permitted to hardwire this bit to 0b if it never sets the IDO attribute in Completions. Value After Reset: 0x0 Exists: CX_IDO_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
8	PCIE_CAP_IDO_REQ_EN	R	IDO Request Enable. If this bit is set, the Function is permitted to set the ID-Based Ordering (IDO) bit (Attr[2]) of Requests it initiates (for more details, see section 2.2.6.3 and section 2.4 of <i>PCI Express Base Specification</i>). Endpoints and Root Ports are permitted to implement this capability. A Function is permitted to hardwire this bit to 0b if it never sets the IDO attribute in Requests. Value After Reset: 0x0 Exists: (CX_IDO_ENABLE && CC_DEVICE_TYPE!=3) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
7	PCIE_CAP_ATOMIC_EGRESS_BLOCK	R	AtomicOp Egress Blocking. AtomicOp Egress Blocking. Applicable and mandatory for Switch Upstream Ports, Switch Downstream Ports, and Root Ports that implement AtomicOp routing capability; otherwise the controller hardwires this bit to 0b. When this bit is set, AtomicOp Requests that target going out this Egress Port must be blocked. For more information, see section 6.15.2 of <i>PCI Express Base Specification</i>. Reflected on the cfg_atomic_egress_block output. Value After Reset: 0x0 Exists: CX_ATOMIC_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
6	PCIE_CAP_ATOMIC_REQ_EN	R	AtomicOp Requester Enable. Applicable only to Endpoints and Root Ports; the controller hardwires this bit to 0b for other Function types. The Function is allowed to initiate AtomicOp Requests only if this bit and the Bus Master Enable bit in the Command register are both set. This bit is required to be RW if the Endpoint or Root Port is capable of initiating AtomicOp Requests, but otherwise is permitted to be hardwired to 0b. This bit does not serve as a capability bit. This bit is permitted to be RW even if no AtomicOp Requester capabilities are supported by the Endpoint or Root Port. Reflected on the cfg_atomic_req_en output. Value After Reset: 0x0 Exists: CX_ATOMIC_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
5	PCIE_CAP_ARI_FORWARD_SUP PORT_CS	R	ARI Forwarding Enable. When set, the Downstream Port disables its traditional Device Number field being 0 enforcement when turning a Type 1 Configuration Request into a Type 0 Configuration Request, permitting access to Extended Functions in an ARI Device immediately below the Port. For more information, see Section 6.13 of <i>PCI Express Base Specification</i>. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
4	PCIE_CAP_CPL_TIMEOUT_DISABLE	R	Completion Timeout Disable. When set, this bit disables the Completion Timeout mechanism. This bit is required for all Functions that support the Completion Timeout Disable Capability. Functions that do not support this optional capability are permitted to hardwire this bit to 0b Software is permitted to set or clear this bit at any time. When set, the Completion Timeout detection mechanism is disabled. If there are outstanding Requests when the bit is cleared, it is permitted but not required for hardware to apply the completion timeout mechanism to the outstanding Requests. If this is done, it is permitted to base the start time for each Request on either the time this bit was cleared or the time each Request was issued. Value After Reset: {(CX_MEMORY_MAP_PCIE_TYPE==1 && !CX_MEMORY_MAP_POSITION) ? 0 : 0} Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
3:0	PCIE_CAP_CPL_TIMEOUT_VALUE	R	Completion Timeout Value. In device Functions that support Completion Timeout programmability, this field allows system software to modify the Completion Timeout value. This field is applicable to Root Ports, Endpoints that issue Requests on their own behalf, and PCI Express to PCI/PCI-X Bridges that take ownership of Requests issued on PCI Express. For all other Functions this field is Reserved and controller hardwires it to 0000b. A Function that does not support this optional capability must hardwire this field to 0000b and is required to implement a timeout value in the range 50 us to 50 ms. Functions that support Completion Timeout programmability must support the values given below corresponding to the programmability ranges indicated in the Completion Timeout Ranges Supported field. All encodings other than the defined encodings are reserved. It is strongly recommended that the Completion Timeout mechanism not expire in less than 10 ms. Values available if Range A (50 us to 10 ms) programmability range is supported: Values available if Range B (10 ms to 250 ms) programmability range is supported: Values available if Range C (250 ms to 4 s) programmability range is supported: Values available if the Range D (4 s to 64 s) programmability range is supported: Software is permitted to change the value in this field at any time. For Requests already pending when the Completion Timeout Value is changed, hardware is permitted to use either the new or the old value for the outstanding Requests, and is permitted to base the start time for each Request either on when this value was changed or on when each request was issued. Values:
3:0...(cont.)	PCIE_CAP_CPL_TIMEOUT_VALUE.	R	■ 0x0 (DEFAULT): Default range: 50 us to 50 ms ■ 0x1 (_50_TO_100_US): 50 us to 100 us ■ 0x2 (_1_TO_10_MS): 1 ms to 10 ms ■ 0x5 (_16_TO_55_MS): 16 ms to 55 ms ■ 0x6 (_65_TO_210_MS): 65 ms to 210 ms ■ 0x9 (_260_TO_900_MS): 260 ms to 900 ms ■ 0xa (_1_TO_3_5_S): 1 s to 3.5 s ■ 0xd (_4_TO_13_S): 4 s to 13 s ■ 0xe (_17_TO_64_S): 17 s to 64 s Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.52.8 VF_LINK_CAPABILITIES2_REG

- **Name:** Link Capabilities 2 Register.
- **Description:** This register identifies PCI Express Link specific capabilities and CXL-RCRB link specific capabilities; in addition to the Link Capabilities Register.
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

DRS_SUPPORTED	31
RSVDP_25	30:25
PCIE_CAP_TWO_RETIMERS_PRE_DET_SUPPORT	24
PCIE_CAP_RETIMER_PRE_DET_SUPPORT	23
RSVDP_9	22:9
PCIE_CAP_CROSS_LINK_SUPPORT	8
PCIE_CAP_SUPPORT_LINK_SPEED_VECTOR	7:1
RSVDP_0	0

Table 3-504 Fields for Register: VF_LINK_CAPABILITIES2_REG

Bits	Name	Memory Access	Description
31	DRS_SUPPORTED	R	DRS Supported. When set, indicates support for the optional Device Readiness Status (DRS) capability. Must be Set in Downstream Ports that support DRS. Must be Set in Downstream Ports that support FRS. For Upstream Ports that support DRS, it is strongly recommended that this bit be Set in Function 0. For all other Functions associated with an Upstream Port, this bit must be Clear.¹²⁷ Must be Clear in Functions that are not associated with a Port. RsvdP in all other Functions. Value After Reset: 0x0 Exists: CX_RN_DRS_SUPPORTED && (CX_MEMORY_MAP_PCIE_TYPE<2) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
30:25	RSVDP_25	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
24	PCIE_CAP_TWO_RETIMERS_PRE_DET_SUPPORT	R	Two Retimers Presence Detect Supported. When set to 1b, this bit indicates that the associated Port supports detection and reporting of two Retimers presence. This bit must be set to 1b in a Port when the Supported Link Speeds Vector of the Link Capabilities 2 register indicates support for a Link speed of 16.0 GT/s or higher. It is permitted to be set to 1b regardless of the supported Link speeds if the Retimer Presence Detect Supported bit is also set to 1b. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all Functions. Value After Reset: 0x1 Exists: CX_GEN4_SPEED && CX_PL16G_ENABLE && CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23	PCIE_CAP_RETIMER_PRE_DET_SUPPORT	R	Retimer Presence Detect Supported. When set to 1b, this bit indicates that the associated Port supports detection and reporting of Retimer presence. This bit must be set to 1b in a Port when the Supported Link Speeds Vector of the Link Capabilities 2 register indicates support for a Link speed of 16.0 GT/s or higher. It is permitted to be set to 1b regardless of the supported Link speeds. Multi-Function Devices associated with an Upstream Port must report the same value in this field for all Functions. Value After Reset: 0x1 Exists: CX_GEN4_SPEED && CX_PL16G_ENABLE && CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
22:9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
8	PCIE_CAP_CROSS_LINK_SUPPORT	R	Crosslink Supported. When set to 1b, this bit indicates that the associated Port supports crosslinks (for more details, see section 4.2.6.3.1 of <i>PCI Express Base Specification</i>). When set to 0b on a Port that supports Link speeds of 8.0 GT/s or higher, this bit indicates that the associated Port does not support crosslinks. When set to 0b on a Port that only supports Link speeds of 2.5 GT/s or 5.0 GT/s, this bit provides no information regarding the Port's level of crosslink support. It is recommended that this bit be Set in any Port that supports crosslinks even though doing so is only required for Ports that also support operating at 8.0 GT/s or higher Link speeds. Note: Software should use this bit when referencing fields whose definition depends on whether or not the Port supports crosslinks (for more details, see section 7.7.3.4 of <i>PCI Express Base Specification</i>). Multi-Function Devices associated with an Upstream Port must report the same value in this field for all Functions. Value After Reset: CX_CROSSLINK_ENABLE Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
7:1	PCIE_CAP_SUPPORT_LINK_SPEED_VECTOR	R	Supported Link Speeds Vector. This field indicates the supported Link speed(s) of the associated Port. For each bit, a value of 1b indicates that the corresponding Link speed is supported; otherwise, the Link speed is not supported. For more information, see section 8.2.1 of <i>PCI Express Base Specification</i>. Bit definitions within this field are: ■ 0: 2.5 GT/s ■ 1: 5.0 GT/s ■ 2: 8.0 GT/s ■ 3: 16.0 GT/s ■ 4: 32.0 GT/s ■ 6:5 RsvdP Multi-Function Devices associated with an Upstream Port must report the same value in this field for all Functions. This field has a default of (PCIE_CAP_MAX_LINK_SPEED == 0101) ? 0011111 : (PCIE_CAP_MAX_LINK_SPEED == 0100) ? 0001111 : (PCIE_CAP_MAX_LINK_SPEED == 0011) ? 0000111 : (PCIE_CAP_MAX_LINK_SPEED == 0010) ? 0000011 : 0000001 where PCIE_CAP_MAX_LINK_SPEED is a field in the LINK_CAPABILITIES_REG register. Values: ■ 0x0 (ini_val): Zero value. ■ 0x7f (def_last_val): last default value. Value After Reset: * Varies Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Reset Mask: 0x0 Volatile: true
0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_MEMORY_MAP_PCIE_TYPE<2 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.52.9 VF_LINK_CONTROL2_LINK_STATUS2_REG

- **Name:** Link Control 2 and Status 2 Register.
- **Description:** This register controls and provides information about PCI Express Link specific parameters as well as RCRB link associated parameters; in addition to the Link Control and Link Status Register.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

31	DRS_MESSAGE_RECEIVED
30:28	DOWNSTREAM_COMPO_PRESENCE
27:26	RSVDP_26
25:24	PCIE_CAP_CROSSLINK_RESOLUTION
23	PCIE_CAP_TWO_RETIMERS_PRE_DET
22	PCIE_CAP_RETIMER_PRE_DET
21	PCIE_CAP_LINK_EQ_REQ
20	PCIE_CAP_EQ_CPL_P3
19	PCIE_CAP_EQ_CPL_P2
18	PCIE_CAP_EQ_CPL_P1
17	PCIE_CAP_EQ_CPL
16	PCIE_CAP_CURR_DEEMPHASIS
15:12	PCIE_CAP_COMPLIANCE_PRESET
11	PCIE_CAP_COMPLIANCE_SOS
10	PCIE_CAP_ENTER_MODIFIED_COMPLIANCE
9:7	PCIE_CAP_TX_MARGIN
6	PCIE_CAP_SEL_DEEMPHASIS
5	PCIE_CAP_HW_AUTO_SPEED_DISABLE
4	PCIE_CAP_ENTER_COMPLIANCE
3:0	PCIE_CAP_TARGET_LINK_SPEED

Table 3-505 Fields for Register: VF_LINK_CONTROL2_LINK_STATUS2_REG

Bits	Name	Memory Access	Description
31	DRS_MESSAGE_RECEIVED	R	DRS Message Received. This bit must be set whenever the Port receives a DRS Message. This bit must be cleared in DL_Down. This bit must be implemented in any Downstream Port where the DRS Supported bit is Set in the Link Capabilities 2 register. This bit is RsvdZ for all other Functions. Value After Reset: 0x0 Exists: (CX_RN_DRS_SUPPORTED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
30:28	DOWNSTREAM_COMPO_PRESENCE	R	Downstream Component Presence. This field indicates the presence and DRS status for the Downstream Component, if any, connected to the Link. All encodings other than the defined encodings are reserved. Component Presence state must be determined by the logical "OR" of the Physical Layer in-band presence detect mechanism and, if present, any out-of-band presence detect mechanism implemented for the Link. If no out-of-band presence detect mechanism is implemented, then Component Presence state must be determined solely by the Physical Layer in-band presence detect mechanism. This field must be implemented in any Downstream Port where the DRS Supported bit is Set in the Link Capabilities 2 register. This field is RsvdZ for all other Functions. Values: ■ 0x0 (LINK_DN_NOT_DETERMINED): Link Down - Presence Not Determined ■ 0x1 (LINK_DN_DSP_NOT_PRESENT): Link Down - Component Not Present indicates the Downstream Port (DP) has determined that a Downstream Component is not present ■ 0x2 (LINK_DN_DLL_NOT_ACTIVE): Link Down - Component Present indicates the DP has determined that a Downstream Component is present, but the Data Link Layer is not active ■ 0x4 (LINK_UP_DRS_NOT_RECEIVED): Link Up - Component Present indicates the DP has determined that a Downstream Component is present, but no DRS Message has been received since the Data Link Layer became active ■ 0x5 (LINK_UP_DRS_RECEIVED): Link Up - Component Present and DRS Received indicates the DP has received a DRS Message since the Data Link Layer became active Value After Reset: 0x0 Exists: (CX_RN_DRS_SUPPORTED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
27:26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
25:24	PCIE_CAP_CROSSLINK_RESOLUTION	R	Crosslink Resolution. This field indicates the state of the Crosslink negotiation. It must be implemented if Crosslink Supported is Set and the Port supports 16.0 GT/s or higher data rate. It is permitted to be implemented in all other Ports. If Crosslink Supported is clear, the controller hardwires this field to 01b or 10b. Once a value of 01b or 10b is returned in this field, that value must continue to be returned while the Link is Up. Values: ■ 0x0 (CROSSLINK_RESOLUTION_NOT_SUP): Crosslink Resolution is not supported. No information is provided regarding the status of the Crosslink negotiation. ■ 0x1 (CROSSLINK_NEGOTIATION_RESOLVED_USP): Crosslink negotiation resolved as an Upstream Port. ■ 0x2 (CROSSLINK_NEGOTIATION_RESOLVED_DSP): Crosslink negotiation resolved as a Downstream Port. ■ 0x3 (CROSSLINK_NEGOTIATION_NOT_CMPL): Crosslink negotiation is not completed. Value After Reset: {(CX_CROSSLINK_ENABLE) ? 0x3 : 0x1} Exists: (CX_GEN4_SPEED && CX_MEMORY_MAP_PCIE_TYPE<2) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
23	PCIE_CAP_TWO_RETIMERS_PRE_DET	R	Two Retimers Presence Detected. When set to 1b, this bit indicates that two Retimers were present during the most recent Link negotiation. For more information, see section 4.2.6.3.5.1 of <i>PCI Express Base Specification</i>. This bit is required for Ports that have the Two Retimers Presence Detect Supported bit of the Link Capabilities 2 register set to 1b. Ports that have the Two Retimers Presence Detect Supported bit set to 0b are permitted to hardwire this bit to 0b. For Multi-Function Devices associated with an Upstream Port, this bit must be implemented in Function 0 and RsvdZ in all other Functions. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN4_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
22	PCIE_CAP_RETIMER_PRE_DET	R	Retimer Presence Detected. When set to 1b, this bit indicates that a Retimer was present during the most recent Link negotiation. For more information, see section 4.2.6.3.5.1 of <i>PCI Express Base Specification</i>. This bit is required for Ports that have the Retimer Presence Detect Supported bit of the Link Capabilities 2 register set to 1b. For Ports that have the Retimer Presence Detect Supported bit set to 0b, the controller hardwires this bit to 0b. For Multi-Function Devices associated with an Upstream Port, this bit must be implemented in Function 0 and is RsvdZ in all other Functions. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN4_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
21	PCIE_CAP_LINK_EQ_REQ	R	Link Equalization Request 8.0 GT/s. This bit is set by hardware to request the 8.0 GT/s Link equalization process to be performed on the Link. For more information, see sections 4.2.3 and 4.2.6.4.2 of <i>PCI Express Base Specification</i>. For Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. For components that only support speeds below 8.0 GT/s, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
20	PCIE_CAP_EQ_CPL_P3	R	EEqualization 8.0 GT/s Phase 3 Successful. When set to 1b, this bit indicates that Phase 3 of the 8.0 GT/s Transmitter Equalization procedure has successfully completed. Details of the Transmitter Equalization process and when this bit needs to be set to 1b is provided in section 4.2.6.4.2 of <i>PCI Express Base Specification</i>. For Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. For components that only support speeds below 8.0 GT/s, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
19	PCIE_CAP_EQ_CPL_P2	R	Equalization 8.0 GT/s Phase 2 Successful. When set to 1b, this bit indicates that Phase 2 of the 8.0 GT/s Transmitter Equalization procedure has successfully completed. Details of the Transmitter Equalization process and when this bit needs to be set to 1b is provided in section 4.2.6.4.2 of <i>PCI Express Base Specification</i>. For Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. For components that only support speeds below 8.0 GT/s, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
18	PCIE_CAP_EQ_CPL_P1	R	Equalization 8.0 GT/s Phase 1 Successful. When set to 1b, this bit indicates that Phase 1 of the 8.0 GT/s Transmitter Equalization procedure has successfully completed. Details of the Transmitter Equalization process and when this bit needs to be set to 1b is provided in section 4.2.6.4.2 of <i>PCI Express Base Specification</i>. For Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. For components that only support speeds below 8.0 GT/s, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
17	PCIE_CAP_EQ_CPL	R	Equalization 8.0 GT/s Complete. When set to 1b, this bit indicates that the Transmitter Equalization procedure at the 8.0 GT/s data rate has completed. Details of the Transmitter Equalization process and when this bit needs to be set to 1b is provided in section 4.2.6.4.2 of <i>PCI Express Base Specification</i>. For Multi-Function Upstream Port, this bit must be implemented in Function 0 and RsvdZ in other Functions. For components that only support speeds below 8.0 GT/s, the controller hardwires this bit to 0b. Value After Reset: 0x0 Exists: (CX_GEN3_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
16	PCIE_CAP_CURR_DEEMPHASIS	R	Current De-emphasis Level. When the Link is operating at 5.0 GT/s speed, this bit reflects the level of de-emphasis. The value in this bit is undefined when the Link is not operating at 5.0 GT/s speed. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 0b. For components that support speeds greater than 2.5 GT/s, Multi-Function Devices associated with an Upstream Port must report the same value in this field for all Functions of the Port. In C-PCIe mode, its contents are derived by sampling the PIPE. Values: ■ 0x1 (_3_5DB): -3.5 dB ■ 0x0 (_6DB): -6 dB Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && CX_MEMORY_MAP_PCIE_TYPE<2) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:12	PCIE_CAP_COMPLIANCE_PRES ET	R	Compliance Preset/De-emphasis. ■ For 8.0 GT/s and higher Data Rate: This field sets the Transmitter Preset in Polling.Compliance state if the entry occurred due to the Enter Compliance bit being 1b. The encodings are defined in section 4.2.3.2 of <i>PCI Express Base Specification</i> . Results are undefined if a reserved preset encoding is used when entering Polling.Compliance in this way. ■ For 5.0 GT/s Data Rate: This field sets the de-emphasis level in Polling.Compliance state if the entry occurred due to the Enter Compliance bit being 1b. ■ When the Link is operating at 2.5 GT/s, the setting of this field has no effect. Components that support only 2.5 GT/s speed are permitted to hardwire this field to 0000b. ■ For a Multi-Function Device associated with an Upstream Port, the field in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this field is of type RsvdP. ■ This field is intended for debug and compliance testing purposes. System firmware and software is allowed to modify this field only during debug or compliance testing. In all other cases, the system must ensure that this field is set to the default value. Note: This register field is sticky. Values: ■ 0x1 (_3_5_DB): -3.5 dB (for 5.0 GT/s Data Rate) ■ 0x0 (_6_DB): -6 dB (for 5.0 GT/s Data Rate) Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
11	PCIE_CAP_COMPLIANCE_SOS	R	Compliance SOS. When set to 1b, the LTSSM is required to send SKP Ordered Sets between sequences when sending the Compliance Pattern or Modified Compliance Pattern. For a Multi-Function Device associated with an Upstream Port, the bit in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is of type RsvdP. This bit is applicable when the Link is operating at 2.5 GT/s or 5.0 GT/s data rates only. For components that support only the 2.5 GT/s speed, the controller hardwires this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true
10	PCIE_CAP_ENTER_MODIFIED_COMPLIANCE	R	Enter Modified Compliance. When this bit is set to 1b, the device transmits Modified Compliance Pattern if the LTSSM enters Polling.Compliance substate. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 0b. For a Multi-Function Device associated with an Upstream Port, the bit in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is of type RsvdP. This bit is intended for debug, compliance testing purposes only. System firmware and software is allowed to modify this bit only during debug or compliance testing. In all other cases, the system must ensure that this bit is set to the default value. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
9:7	PCIE_CAP_TX_MARGIN	R/W	Transmit Margin, This field controls the value of the non-deemphasized voltage level at the Transmitter pins. This field is reset to 000b on entry to the LTSSM Polling.Configuration substate (see Chapter 4 of <i>PCI Express Base Specification</i> for details of how the Transmitter voltage level is determined in various states). 001b-111b: As defined in Section 8.3.4 not all encodings are required to be implemented. For a Multi-Function Device associated with an Upstream Port, the field in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this field is of type RsvdP. For components that support only the 2.5 GT/s speed, the controller hardwires this bit to 000b. This field is intended for debug, compliance testing purposes only. System firmware and software is allowed to modify this field only during debug or compliance testing. In all other cases, the system must ensure that this field is set to the default value. <i>Note:</i> This register field is sticky. Values: 0x0 (NORMAL_RANGE): Normal operating range Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
6	PCIE_CAP_SEL_DEEMPHASIS	R	Selectable De-emphasis. When the Link is operating at 5.0 GT/s speed, this bit is used to control the transmit de-emphasis of the link in specific situations. For more information, see section 4.2.6 of <i>PCI Express Base Specification</i>. When the Link is not operating at 5.0 GT/s speed, the setting of this bit has no effect. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 0b. This bit is not applicable and Reserved for Endpoints, PCI Express to PCI/PCI-X bridges, and Upstream Ports of Switches. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (_3_5_DB): -3.5 dB ■ 0x0 (_6_DB): -6 dB Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
5	PCIE_CAP_HW_AUTO_SPEED_DISABLE	R	Hardware Autonomous Speed Disable. When set, this bit disables hardware from changing the Link speed for device-specific reasons other than attempting to correct unreliable Link operation by reducing Link speed. Initial transition to the highest supported common link speed is not blocked by this bit. For a Multi-Function Device associated with an Upstream Port, the bit in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is of type RsvdP. Functions that do not implement the associated mechanism are permitted to hardwire this bit to 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
4	PCIE_CAP_ENTER_COMPLIANCE	* Varies	Enter Compliance. Software is permitted to force a Link to enter Compliance mode (at the speed indicated in the Target Link Speed field and the de-emphasis/preset level indicated by the Compliance Preset/De-emphasis field) by setting this bit to 1b in both components on a Link and then initiating a hot reset on the Link. Default value of this bit following Fundamental Reset is 0b. For a Multi-Function Device associated with an Upstream Port, the bit in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this bit is of type RsvdP. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 0b. This bit is intended for debug, compliance testing purposes only. System firmware and software is allowed to modify this bit only during debug or compliance testing. In all other cases, the system must ensure that this bit is set to the default value. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_GEN2_SPEED && (CX_MEMORY_MAP_PCIE_TYPE<2)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead Volatile: true Memory Access: "(CX_PCIE_OVER_CIO_ENABLE) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
3:0	PCIE_CAP_TARGET_LINK_SPEED	R	Target Link Speed. For Downstream Ports, this field sets an upper limit on Link operational speed by restricting the values advertised by the Upstream component in its training sequences. The encoded value specifies a bit location in the Supported Link Speeds Vector (in the Link Capabilities 2 register) that corresponds to the desired target Link speed. All encodings other than the defined encodings are reserved. If a value is written to this field that does not correspond to a supported speed (as indicated by the Supported Link Speeds Vector), the result is undefined. If either of the Enter Compliance or Enter Modified Compliance bits are implemented, then this field must also be implemented. The default value of this field is the highest Link speed supported by the component (as reported in the Max Link Speed field of the Link Capabilities register) unless the corresponding platform/form factor requires a different default value. For both Upstream and Downstream Ports, this field is used to set the target compliance mode speed when software is using the Enter Compliance bit to force a Link into compliance mode. For Upstream Ports, if the Enter Compliance bit is Clear, this field is permitted to have no effect. For a Multi-Function Device associated with an Upstream Port, the field in Function 0 is of type RWS, and only Function 0 controls the component's Link behavior. In all other Functions of that device, this field is of type RsvdP. Components that support only the 2.5 GT/s speed are permitted to hardwire this field to 0000b. <i>Note:</i> This register field is sticky. Values:
3:0...(cont.)	PCIE_CAP_TARGET_LINK_SPEED.	R	■ 0x1 (SUP_LINK_SPEED_VECTOR_BIT_0): Supported Link Speeds Vector field bit 0 ■ 0x2 (SUP_LINK_SPEED_VECTOR_BIT_1): Supported Link Speeds Vector field bit 1 ■ 0x3 (SUP_LINK_SPEED_VECTOR_BIT_2): Supported Link Speeds Vector field bit 2 ■ 0x4 (SUP_LINK_SPEED_VECTOR_BIT_3): Supported Link Speeds Vector field bit 3 ■ 0x5 (SUP_LINK_SPEED_VECTOR_BIT_4): Supported Link Speeds Vector field bit 4 ■ 0x6 (SUP_LINK_SPEED_VECTOR_BIT_5): Supported Link Speeds Vector field bit 5 ■ 0x7 (SUP_LINK_SPEED_VECTOR_BIT_6): Supported Link Speeds Vector field bit 6 Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.53 VF_PF_MSIX_CAPi Registers

3.53.1 VF_PCI_MSIX_CAP_ID_NEXT_CTRL_REG

- **Name:** MSI-X Capability ID, Next Pointer, Control Register.
- **Description:** This Register holds MSI-X Capability ID, Next Capability pointer. It also controls the MSI-X behaviour.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** ((VF_MSIX_CAP_ENABLE || EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31	PCI_MSIX_ENABLE
30	PCI_MSIX_FUNCTION_MASK
29:27	RSVDP_27
26:16	PCI_MSIX_TABLE_SIZE
15:8	PCI_MSIX_CAP_NEXT_OFFSET
7:0	PCI_MSIX_CAP_ID

Table 3-506 Fields for Register: VF_PCI_MSIX_CAP_ID_NEXT_CTRL_REG

Bits	Name	Memory Access	Description
31	PCI_MSIX_ENABLE	R	MSI-X Enable. If Set and the MSI Enable bit in the MSI Message Control Register for MSI is Clear, the Function is permitted to use MSI-X to request service and is prohibited from using INTx interrupts (if implemented). System configuration software Sets this bit to enable MSI-X. A device driver is prohibited from writing this bit to mask a Function's service request. If Clear, the Function is prohibited from using MSI-X to request service. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
30	PCI_MSIX_FUNCTION_MASK	* Varies	Function Mask. If Set, all of the vectors associated with the Function are masked, regardless of their per-vector Mask bit values. If Clear, each vector's Mask bit determines whether the vector is masked or not. Setting or Clearing the MSI-X Function Mask bit has no effect on the value of the per-vector Mask bits. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && VF_MSIX_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !VF_MSIX_CAP_ENABLE) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && VF_MSIX_CAP_ENABLE) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !VF_MSIX_CAP_ENABLE) ? \"read-only\" : \"read-only\""
29:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
26:16	PCI_MSIX_TABLE_SIZE	* Varies	MSI-X Table Size. System software reads this field to determine the MSI-X Table Size N, which is encoded as N-1. For example, a returned value of 000 0000 0011b indicates a table size of 4. SRIOV Note: All VFs in a single PF have the same value for "MSI-X Table Size" (PCI_MSIX_TABLE_SIZE field in SHADOW_PCI_MSIX_CAP_ID_NEXT_CTRL_REG). To write this common value, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PCI_MSIX_TABLE_SIZE field in the PF PCI_MSIX_CAP_ID_NEXT_CTRL_REG register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R ■ Dbi2: R (sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R : (CX_MEMORY_MAP_VIEW==1) ? R : (MSIX_CAP_ENABLE && CX_DBI_RO_WR_EN) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
15:8	PCI_MSIX_CAP_NEXT_OFFSET	R	MSI-X Next Capability Pointer. This field contains the offset to the next PCI Capability structure or 00h if no other items exist in the linked list of Capabilities. <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
7:0	PCI_MSIX_CAP_ID	R	MSI-X Capability ID. This field indicates the MSI-X Capability structure. This field must return a Capability ID of 11h indicating that this is an MSI-X Capability structure. Value After Reset: 0x11 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.53.2 VF_MSIX_TABLE_OFFSET_REG

- **Name:** MSI-X Table Offset and BIR Register.
- **Description:** This register provides Table BIR and MSI-x Table offset select.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** ((VF_MSIX_CAP_ENABLE | |EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_P-CIE_TYPE!=1)

31:3	2:0
PCI_MSIX_TABLE_OFFSET	PCI_MSIX_BIR

Table 3-507 Fields for Register: VF_MSIX_TABLE_OFFSET_REG

Bits	Name	Memory Access	Description
31:3	PCI_MSIX_TABLE_OFFSET	* Varies	MSI-X Table Offset. Used as an offset from the address contained by one of the Function's Base Address Registers to point to the base of the MSI-X Table. The lower 3 Table BIR bits are masked off (set to zero) by software to form a 32-bit QWORD-aligned offset. SRIOV Note: All VFs in a single PF have the same value for "MSI-X Table Offset" (PCI_MSIX_TABLE_OFFSET field in SHADOW_MSIX_TABLE_OFFSET_REG). To write this common value, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PCI_MSIX_TABLE_OFFSET field in the PF MSIX_TABLE_OFFSET_REG register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R ■ Dbi2: R (sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: (MSIX_CAP_ENABLE && CX_DBI_RO_WR_EN) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: VF_MSIX_TABLE_OFFSET_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2:0	PCI_MSIX_BIR	* Varies	MSI-X Table BAR Indicator Register Field. This field indicates which one of a Function's Base Address Registers , located beginning at 10h in Configuration Space, or entry in the Enhanced Allocation capability with a matching BEI , is used to map the Function's MSI-X Table into Memory Space. All encodings other than the defined encodings are reserved. For a 64-bit Base Address Register , the Table BIR indicates the lower DWORD. For Functions with Type 1 Configuration Space headers, BIR values 2 through 5 are also Reserved. SRIOV Note: All VFs in a single PF have the same value for "MSI-X Table BAR Indicator Register" (PCI_MSIX_BIR field in SHADOW_MSIX_TABLE_OFFSET_REG). To write this common value, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PCI_MSIX_BIR field in the PF MSIX_TABLE_OFFSET_REG register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R ■ Dbi2: R (sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: (MSIX_CAP_ENABLE && CX_DBI_RO_WR_EN) ? R/W : R <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BAR_10): Base Address Register 10h ■ 0x1 (BAR_14): Base Address Register 14h ■ 0x2 (BAR_18): Base Address Register 18h ■ 0x3 (BAR_1C): Base Address Register 1Ch ■ 0x4 (BAR_20): Base Address Register 20h ■ 0x5 (BAR_24): Base Address Register 24h Value After Reset: VF_MSIX_TABLE_BIR_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

3.53.3 VF_MSIX_PBA_OFFSET_REG

- **Name:** MSI-X PBA Offset and BIR Register.
- **Description:** This register provides PBA Offset and PBA BIR value.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** ((VF_MSIX_CAP_ENABLE | |EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_P-CIE_TYPE!=1)

31:3	2:0
PCI_MSIX_PBA_OFFSET	PCI_MSIX_PBA_BIR

Table 3-508 Fields for Register: VF_MSIX_PBA_OFFSET_REG

Bits	Name	Memory Access	Description
31:3	PCI_MSIX_PBA_OFFSET	* Varies	MSI-X PBA Offset. Used as an offset from the address contained by one of the Function's Base Address Registers to point to the base of the MSI-X PBA. The lower 3 PBA BIR bits are masked off (set to zero) by software to form a 32-bit QWORD-aligned offset. SRIOV Note: All VFs in a single PF have the same value for "MSI-X PBA Offset" (PCI_MSIX_PBA_OFFSET field in SHADOW_MSIX_PBA_OFFSET_REG). To write this common value, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PCI_MSIX_PBA_OFFSET field in the PF MSIX_PBA_OFFSET_REG register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R ■ Dbi2: R (sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: (MSIX_CAP_ENABLE && CX_DBI_RO_WR_EN) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: VF_MSIX_PBA_OFFSET_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: untestable

Bits	Name	Memory Access	Description
2:0	PCI_MSIX_PBA_BIR	* Varies	MSI-X PBA BIR. This field indicates which one of a Function's Base Address Registers , located beginning at 10h in Configuration Space, or entry in the Enhanced Allocation capability with a matching BEI, is used to map the Function's MSI-X PBA into Memory Space. The PBA BIR value definitions are identical to those for the Table BIR . SRIOV Note: All VFs in a single PF have the same value for "MSI-X PBA BIR" (PCI_MSIX_PBA_BIR field in SHADOW_MSIX_PBA_OFFSET_REG). To write this common value, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PCI_MSIX_PBA_BIR field in the PF MSIX_PBA_OFFSET_REG register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R ■ Dbi2: R (sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN && MSIX_CAP_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0) ? R: (CX_MEMORY_MAP_VIEW==1) ? R: (MSIX_CAP_ENABLE && CX_DBI_RO_WR_EN) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: VF_MSIX_PBA_BIR_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_MSIX_CAP_ENABLE EXT_VF_MSIX_CAP_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: untestable

3.54 VF_PF_AER_CAPi Registers

3.54.1 VF_AER_EXT_CAP_HDR_OFF

- **Name:** Advanced Error Reporting Extended Capability Header.
- **Description:** Advanced Error Reporting Extended Capability Header provides information about Capability ID, Version, and next offset.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	CAP_ID

Table 3-509 Fields for Register: VF_AER_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field must be 2h if the End-End TLP Prefix Supported bit is set and must be 1h or 2h otherwise. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x2 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead
15:0	CAP_ID	* Varies	AER Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. The Extended Capability ID for the Advanced Error Reporting Capability is 0001h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

3.54.2 VF_UNCORR_ERR_STATUS_OFF

- **Name:** Uncorrectable Error Status Register.
- **Description:** The Uncorrectable Error Status Register (Offset 04h) indicates error detection status of individual errors on a PCI Express device Function. An individual error status bit that is Set indicates that a particular error was detected; software may clear an error status by writing a 1b to the respective bit. Register bits not implemented by the Function are hardwired to 0b.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

RSVDP_28	31:28
DMWR_EGRESS_BLOCKED_ERR_STATUS	27
PSND_TLP_EGRSS_BLOCKED_ERR_STATUS	26
TLP_PREFX_BLOCKED_ERR_STATUS	25
ATOMIC_EGRESS_BLOCKED_ERR_STATUS	24
RSVDP_23	23
INTERNAL_ERR_STATUS	22
ACS_VIOLATION_STATUS	21
UNSUPPORTED_REQ_ERR_STATUS	20
ECRC_ERR_STATUS	19
MALF_TLP_ERR_STATUS	18
REC_OVERFLOW_ERR_STATUS	17
UNEXP_CMPLT_ERR_STATUS	16
CMPLT_ABORT_ERR_STATUS	15
CMPLT_TIMEOUT_ERR_STATUS	14
FC_PROTOCOL_ERR_STATUS	13
POIS_TLP_ERR_STATUS	12
RSVDP_6	11:6
SURPRISE_DOWN_ERR_STATUS	5
DL_PROTOCOL_ERR_STATUS	4
RSVDP_0	3:0

Table 3-510 Fields for Register: VF_UNCORR_ERR_STATUS_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
27	DMWR_EGRESS_BLOCKED_ERR_STATUS	R/W1C	Deferrable Memory Write Egress Blocked Status. Status bit for Deferrable Memory Write Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: CX_DEF_MEM_WR_ROUTING_SUP && CC_DEVICE_TYPE==3 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
26	PSND_TLP_EGRSS_BLOCKED_ERR_STATUS	R/W1C	Poisoned TLP Egress Blocked Status. This field provides status of Poisoned TLP Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: AER_ENABLE && CX_DPC_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
25	TLP_PRFX_BLOCKED_ERR_STATUS	R/W1C	TLP Prefix Blocked Error Status. Status bit for TLP Prefix Blocked Error. <i>Note:</i> Not supported. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_TLP_PREFIX_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
24	ATOMIC_EGRESS_BLOCKED_ERROR_STATUS	R/W1C	AtomicOp Egress Blocked Status. Status bit for AtomicOp Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: ATOMIC_ROUTING_SUP && CC_DEVICE_TYPE==3 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
22	INTERNAL_ERR_STATUS	R/W1C	Uncorrectable Internal Error Status. This field gives status of the Uncorrectable Internal Error. The controller sets this bit when your application asserts app_err_bus[9]. It does not set this bit when it detects internal uncorrectable internal errors such as parity and ECC failures. You should use the outputs from these errors to drive the app_err_bus[9] input. For more information, see the "Data Integrity (Wire, Datapath, and RAM Protection)" section in the <i>Databook</i>. This register field is reserved for VFs. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_INTERNAL_ERR_REPORTING && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
21	ACS_VIOLATION_STATUS	R/W1C	ACS Violation Status. ACS Violations may occur due to either hardware or software defects/failures. The ACS Violation Status bit provides positive identification of the error. The controller sets this bit when your application asserts app_err_bus[12]. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_ACS_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
20	UNSUPPORTED_REQ_ERR_STATUS	R/W1C	Unsupported Request Error Status. This field represents status of Unsupported Request Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
19	ECRC_ERR_STATUS	R/W1C	ECRC Error Status. This field represents status of ECRC Error. <i>Note:</i> If CX_ECRC_ENABLE=0 the register field always reads 0. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
18	MALF_TLP_ERR_STATUS	R	Malformed TLP Status. This field represents status of Malformed TLP. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
17	REC_OVERFLOW_ERR_STATUS	R/W1C	Receiver Overflow Status. Status bit for Receiver Overflow. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
16	UNEXP_CMPLT_ERR_STATUS	R/W1C	Unexpected Completion Status. Status bit for Unexpected Completion. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
15	CMPLT_ABORT_ERR_STATUS	R/W1C	Completer Abort Status. Status bit for Completer Abort. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
14	CMPLT_TIMEOUT_ERR_STATUS	R/W1C	Completion Timeout Status. Status for Completion Timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
13	FC_PROTOCOL_ERR_STATUS	R/W1C	Flow Control Protocol Error Status. Status bit for Flow Control Protocol Error. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
12	POIS_TLP_ERR_STATUS	R/W1C	Poisoned TLP Status. Status bit for Poisoned TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
11:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
5	SURPRISE_DOWN_ERR_STATUS	R/W1C	Surprise Down Error Status (Optional). Status bit for Surprise Down Error. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
4	DL_PROTOCOL_ERR_STATUS	R/W1C	Data Link Protocol Error Status. Status bit for Data Link Protocol Error. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.54.3 VF_UNCORR_ERR_MASK_OFF

- **Name:** Uncorrectable Error Mask Register.
- **Description:** The Uncorrectable Error Mask Register controls reporting of individual errors by the device Function to the PCI Express Root Complex through a PCI Express error Message. A masked error (respective bit Set in the mask register) is not recorded or reported in the Header Log, TLP Prefix Log, or First Error Pointer, and is not reported to the PCI Express Root Complex by this Function. There is a mask bit per error bit of the Uncorrectable Error Status register. Register fields for bits not implemented by the Function are hardwired to 0b.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

RSVDP_28	31:28
DMWR_EGRESS_BLOCKED_ERR_MASK	27
RSVDP_26	26
TLP_PREFIX_BLOCKED_ERR_MASK	25
ATOMIC_EGRESS_BLOCKED_ERR_MASK	24
RSVDP_23	23
INTERNAL_ERR_MASK	22
ACS_VIOLATION_MASK	21
UNSUPPORTED_REQ_ERR_MASK	20
ECRC_ERR_MASK	19
MALF_TLP_ERR_MASK	18
REC_OVERFLOW_ERR_MASK	17
UNEXP_CMPLT_ERR_MASK	16
CMPLT_ABORT_ERR_MASK	15
CMPLT_TIMEOUT_ERR_MASK	14
FC_PROTOCOL_ERR_MASK	13
POIS_TLP_ERR_MASK	12
RSVDP_6	11:6
SURPRISE_DOWN_ERR_MASK	5
DL_PROTOCOL_ERR_MASK	4
RSVDP_0	3:0

Table 3-511 Fields for Register: VF_UNCORR_ERR_MASK_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
27	DMWR_EGRESS_BLOCKED_ERR_MASK	R	Deferrable Memory Write Egress Block Mask (Optional). Mask bit for Deferrable Memory Write Egress Block Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
25	TLP_PRFX_BLOCKED_ERR_MASK	R	TLP Prefix Blocked Error Mask. Mask bit for TLP Prefix Blocked Error. <i>Note:</i> Not supported. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
24	ATOMIC_EGRESS_BLOCKED_ERR_MASK	R	AtomicOp Egress Block Mask (Optional). Mask bit for AtomicOp Egress Block Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
22	INTERNAL_ERR_MASK	R	Uncorrectable Internal Error Mask (Optional). Mask bit for Uncorrectable Internal Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_INTERNAL_ERR_REPORTING && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
21	ACS_VIOLATION_MASK	R	ACS Violation Mask. Setting the ACS Violation Mask bit disables error logging and signaling for ACS Violation errors. The bit is Read-Only Zero for upstream ports, when ACS P2P Egress Control Enable is not set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_ACS_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
20	UNSUPPORTED_REQ_ERR_MASK	R	Unsupported Request Error Mask. Mask bit for Unsupported Request Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
19	ECRC_ERR_MASK	R	ECRC Error Mask (Optional). Mask bit for ECRC Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_ECRC_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
18	MALF_TLP_ERR_MASK	R	Malformed TLP Mask. Mask bit for Malformed TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
17	REC_OVERFLOW_ERR_MASK	R	Receiver Overflow Mask (Optional). This field represents Receiver Overflow Mask. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
16	UNEXP_CMPLT_ERR_MASK	R	Unexpected Completion Mask. Mask bit for Unexpected Completion Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
15	CMPLT_ABORT_ERR_MASK	R	Completer Abort Error Mask (Optional). Mask bit for Completer Abort Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
14	CMPLT_TIMEOUT_ERR_MASK	R	Completion Timeout Error Mask. Mask bit for Completion Timeout Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
13	FC_PROTOCOL_ERR_MASK	R	Flow Control Protocol Error Mask. Mask bit for Flow Control Protocol Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
12	POIS_TLP_ERR_MASK	R	Poisoned TLP Error Mask. Mask bit for Poisoned TLP Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
11:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
5	SURPRISE_DOWN_ERR_MASK	R	Surprise Down Error Mask. Mask bit for Surprise Down Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
4	DL_PROTOCOL_ERR_MASK	R	Data Link Protocol Error Mask. This field informs whether Data Link Protocol Error is masked or not. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.54.4 VF_UNCORR_ERR_SEV_OFF

- **Name:** Uncorrectable Error Severity Register.
- **Description:** The Uncorrectable Error Severity Register controls whether an individual error is reported as a Non-fatal or Fatal error. An error is reported as fatal when the corresponding error bit in the severity register is Set. If the bit is Clear, the corresponding error is considered non-fatal.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

RSVDP_28	31:28
DMWR_EGRESS_BLOCKED_ERR_SEVERITY	27
PSND_TLP_EGRSS_BLOCKED_ERR_SEVERITY	26
TLP_PRFX_BLOCKED_ERR_SEVERITY	25
ATOMIC_EGRESS_BLOCKED_ERR_SEVERITY	24
RSVDP_23	23
INTERNAL_ERR_SEVERITY	22
ACS_VIOLATION_SEVERITY	21
UNSUPPORTED_REQ_ERR_SEVERITY	20
ECRC_ERR_SEVERITY	19
MALF_TLP_ERR_SEVERITY	18
REC_OVERFLOW_ERR_SEVERITY	17
UNEXP_CMPLT_ERR_SEVERITY	16
CMPLT_ABORT_ERR_SEVERITY	15
CMPLT_TIMEOUT_ERR_SEVERITY	14
FC_PROTOCOL_ERR_SEVERITY	13
POIS_TLP_ERR_SEVERITY	12
RSVDP_6	11:6
SURPRISE_DOWN_ERR_SVRITY	5
DL_PROTOCOL_ERR_SEVERITY	4
RSVDP_0	3:0

Table 3-512 Fields for Register: VF_UNCORR_ERR_SEV_OFF

Bits	Name	Memory Access	Description
31:28	RSVDP_28	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
27	DMWR_EGRESS_BLOCKED_ERR_SEVERITY	R	Deferrable Memory Write Egress Blocked Severity (Optional). Severity bit for Deferrable Memory Write Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
26	PSND_TLP_EGRSS_BLOCKED_ERR_SEVERITY	R/W	Poisoned TLP Egress Blocked Severity. Severity bit for Poisoned TLP Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: AER_ENABLE && CX_DPC_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
25	TLP_PRFX_BLOCKED_ERR_SEVERITY	R	TLP Prefix Blocked Error Severity (Optional). Severity bit for TLP Prefix Blocked Error. <i>Note:</i> Not supported. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
24	ATOMIC_EGRESS_BLOCKED_ERR_SEVERITY	R	AtomicOp Egress Blocked Severity (Optional). Severity bit for AtomicOp Egress Blocked Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: ATOMIC_ROUTING_SUP && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
22	INTERNAL_ERR_SEVERITY	R	Uncorrectable Internal Error Severity (Optional). Severity bit for Uncorrectable Internal Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
21	ACS_VIOLATION_SEVERITY	* Varies	ACS Violation Severity. Setting the ACS Violation Severity bit sets ACS Violation error severity to Fatal. The bit is Read-Only Zero for upstream ports, when ACS P2P Egress Control Enable is not set. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: if (acs_viol_svrity_mask_wr_en == 1) then R/W (Sticky) else R(Sticky) ■ Dbi: if (acs_viol_svrity_mask_wr_en == 1) then R/W (Sticky) else R(Sticky) and more specifically: (CX_MEMORY_MAP_VIEW==0 && CX_ACS_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !CX_ACS_ENABLE) ? R : (CX_MEMORY_MAP_VIEW==1 && CX_ACS_ENABLE) ? R/W : (CX_MEMORY_MAP_VIEW==1 && !CX_ACS_ENABLE) ? R : (CX_ACS_ENABLE) ? R/W : R <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_ACS_ENABLE && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
20	UNSUPPORTED_REQ_ERR_SEVERITY	R	Unsupported Request Error Severity. Severity bit for Unsupported Request Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
19	ECRC_ERR_SEVERITY	R	ECRC Error Severity (Optional). Severity bit for ECRC Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
18	MALF_TLP_ERR_SEVERITY	R	Malformed TLP Severity. Severity bit for Malformed TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
17	REC_OVERFLOW_ERR_SEVERITY	R	Receiver Overflow Error Severity (Optional). Severity bit for Receiver Overflow Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
16	UNEXP_CMPLT_ERR_SEVERITY	R	Unexpected Completion Error Severity. Severity bit for Unexpected Completion Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
15	CMPLT_ABORT_ERR_SEVERITY	R	Completer Abort Error Severity (Optional). Severity bit for Completer Abort Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
14	CMPLT_TIMEOUT_ERR_SEVERITY	R	Completion Timeout Error Severity. Severity bit for Completion Timeout Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
13	FC_PROTOCOL_ERR_SEVERITY	R	Flow Control Protocol Error Severity (Optional). Severity bit for Flow Control Protocol Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
12	POIS_TLP_ERR_SEVERITY	R	Poisoned TLP Severity. Severity bit for Poisoned TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
11:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
5	SURPRISE_DOWN_ERR_SVRITY	R	Surprise Down Error Severity (Optional). Severity bit for Surprise Down Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
4	DL_PROTOCOL_ERR_SEVERITY	R	Data Link Protocol Error Severity. Severity bit for Data Link Protocol Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.54.5 VF_CORR_ERR_STATUS_OFF

- **Name:** Correctable Error Status Register.
- **Description:** The Correctable Error Status register reports error status of individual correctable error sources on a PCI Express device Function. When an individual error status bit is Set, it indicates that a particular error occurred; software may clear an error status by writing a 1b to the respective bit. Register bits not implemented by the Function are hardwired to 0b by the controller.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:16	RSVDP_16
15	HEADER_LOG_OVERFLOW_STATUS
14	CORRECTED_INT_ERR_STATUS
13	ADVISORY_NON_FATAL_ERR_STATUS
12	RPL_TIMER_TIMEOUT_STATUS
11:9	RSVDP_9
8	REPLAY_NO_ROLEOVER_STATUS
7	BAD_DLLP_STATUS
6	BAD_TLP_STATUS
5:1	RSVDP_1
0	RX_ERR_STATUS

Table 3-513 Fields for Register: VF_CORR_ERR_STATUS_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
15	HEADER_LOG_OVERFLOW_STATUS	R/W1C	Header Log Overflow Error Status (Optional). This field provides status of Header Log Overflow Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: !VF_AER_ENABLE !VF_HDR_LOG_SHARED && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
14	CORRECTED_INT_ERR_STATUS	R/W1C	Corrected Internal Error Status (Optional). This field provides status of Corrected Internal Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_INTERNAL_ERR_REPORTING && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
13	ADVISORY_NON_FATAL_ERR_STATUS	R/W1C	Advisory Non-Fatal Error Status. Status bit for Advisory Non-Fatal Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
12	RPL_TIMER_TIMEOUT_STATUS	R/W1C	Replay Timer Timeout Status. Status bit for Replay Timer Timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
11:9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
8	REPLAY_NO_ROLEOVER_STATUS	R/W1C	REPLAY_NUM Rollover Status. Status bit for REPLAY_NUM Rollover. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
7	BAD_DLLP_STATUS	R/W1C	Bad DLLP Status. Status bit for Bad DLLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
6	BAD_TLP_STATUS	R/W1C	Bad TLP Status. Status bit for Bad TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
5:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
0	RX_ERR_STATUS	R/W1C	Receiver Error Status (Optional). This field provides status of Receiver Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

3.54.6 VF_CORR_ERR_MASK_OFF

- **Name:** Correctable Error Mask Register.
- **Description:** The Correctable Error Mask Register controls reporting of individual correctable errors by this Function to the PCI Express Root Complex through a PCI Express error Message. A masked error (respective bit Set in the mask register) is not reported to the PCI Express Root Complex by this Function. There is a mask bit per error bit in the Correctable Error Status register. Register fields for bits not implemented by the Function are hardwired to 0b by the controller.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:16	RSVDP_16
15	HEADER_LOG_OVERFLOW_MASK
14	CORRECTED_INT_ERR_MASK
13	ADVISORY_NON_FATAL_ERR_MASK
12	RPL_TIMER_TIMEOUT_MASK
11:9	RSVDP_9
8	REPLAY_NO_ROLEOVER_MASK
7	BAD_DLLP_MASK
6	BAD_TLP_MASK
5:1	RSVDP_1
0	RX_ERR_MASK

Table 3-514 Fields for Register: VF_CORR_ERR_MASK_OFF

Bits	Name	Memory Access	Description
31:16	RSVDP_16	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
15	HEADER_LOG_OVERFLOW_MASK	R/W	Header Log Overflow Error Mask (Optional). Masking bit for Header Log Overflow Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: !VF_HDR_LOG_SHARED && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
14	CORRECTED_INT_ERR_MASK	R	Corrected Internal Error Mask (Optional). Masking bit for Corrected Internal Error Mask. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
13	ADVISORY_NON_FATAL_ERR_MASK	R	Advisory Non-Fatal Error Mask. Masking bit for Advisory Non-Fatal Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
12	RPL_TIMER_TIMEOUT_MASK	R	Replay Timer Timeout Mask. Masking bit for Replay Timer Timeout. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
11:9	RSVDP_9	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
8	REPLAY_NO_ROLEOVER_MASK	R	REPLAY_NUM Rollover Mask. Masking bit for REPLAY_NUM Rollover. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
7	BAD_DLLP_MASK	R	Bad DLLP Mask. Masking bit for Bad DLLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
6	BAD_TLP_MASK	R	Bad TLP Mask. Masking bit for Bad TLP. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
5:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
0	RX_ERR_MASK	R	Receiver Error Mask (Optional). Masking bit for Receiver Error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.54.7 VF_ADV_ERR_CAP_CTRL_OFF

- **Name:** Advanced Error Capabilities and Control Register.
- **Description:** Advanced Error Capabilities and Control Register provides information whether the individual capability is supported or not. If the capability is supported then it is enabled or not.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:13	RSVDP_13
12	CTO_PRFX_HDR_LOG_CAP
11	TLP_PRFX_LOG_PRESENT
10	MULTIPLE_HEADER_EN
9	MULTIPLE_HEADER_CAP
8	ECRC_CHECK_EN
7	ECRC_CHECK_CAP
6	ECRC_GEN_EN
5	ECRC_GEN_CAP
4:0	FIRST_ERR_POINTER

Table 3-515 Fields for Register: VF_ADV_ERR_CAP_CTRL_OFF

Bits	Name	Memory Access	Description
31:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
12	CTO_PRFX_HDR_LOG_CAP	R	Completion Timeout Prefix/Header Log Capable. If Set, this bit indicates that the Function records the prefix/header of Request TLPs that experience a Completion Timeout error. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
11	TLP_PRFX_LOG_PRESENT	R	TLP Prefix Log Present. If Set and the First Error Pointer is valid, indicates that the TLP Prefix Log register contains valid information. If Clear or if First Error Pointer is invalid, the TLP Prefix Log register is undefined. Default value of this bit is 0. This bit is RsvdP if the End-End TLP Prefix Supported bit is Clear. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
10	MULTIPLE_HEADER_EN	* Varies	Multiple Header Recording Enable. When Set, this bit enables the Function to record more than one error header. Functions that do not implement the associated mechanism are permitted to hardwire this bit to 0b. Default value of this bit is 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && (VF_AER_ENABLE==0 VF_HDR_LOG_SHARED==0) && (VF_HDR_LOG_DEPTH >=2)) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && (VF_AER_ENABLE==0 VF_HDR_LOG_SHARED==0) && (VF_HDR_LOG_DEPTH >=2)) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : ((VF_AER_ENABLE==0 VF_HDR_LOG_SHARED==0) && VF_HDR_LOG_DEPTH >=2) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
9	MULTIPLE_HEADER_CAP	R	Multiple Header Recording Capable. If Set, this bit indicates that the Function is capable of recording more than one error header. <i>Note:</i> This register field is sticky. Value After Reset: "(VF_HDR_LOG_DEPTH > 1) ? 1 : 0" Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
8	ECRC_CHECK_EN	R	ECRC Check Enable. When Set, ECRC checking is enabled. Functions that do not implement the associated mechanism are permitted to hardwire this bit to 0b. Default value of this bit is 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
7	ECRC_CHECK_CAP	R	ECRC Check Capable. If Set, this bit indicates that the Function is capable of checking ECRC. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_ECRC_CHK_CAP_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
6	ECRC_GEN_EN	R	ECRC Generation Enable. When Set, ECRC generation is enabled. Functions that do not implement the associated mechanism are permitted to hardwire this bit to 0b. Default value of this bit is 0b. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
5	ECRC_GEN_CAP	R	ECRC Generation Capable. If Set, this bit indicates that the Function is capable of generating ECRC. <i>Note:</i> This register field is sticky. Value After Reset: DEFAULT_ECRC_GEN_CAP_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
4:0	FIRST_ERR_POINTER	R	First Error Pointer. The First Error Pointer is a field that identifies the bit position of the first error reported in the Uncorrectable Error Status register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

3.54.8 VF_HDR_LOG_0_OFF

- **Name:** Header Log Register 0.
- **Description:** The Header Log Register 0 contains the header for the TLP corresponding to a detected error; The header is captured such that, when read using DW accesses, the fields of the header are laid out in the same way the headers are presented in the specification. Therefore, byte 0 of the header is located in byte 3 of the Header Log Register 0, byte 1 of the header is in byte 2 of the Header Log Register 0, and so forth. For 12-byte headers, only bytes 0 through 11 of the Header Log Register are used and values in bytes 12 through 15 are undefined. In certain cases where a Malformed TLP is reported, the Header Log Register may contain TLP Prefix information.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:24	FIRST_DWORD_FOURTH_BYTE
23:16	FIRST_DWORD_THIRD_BYTE
15:8	FIRST_DWORD_SECOND_BYTE
7:0	FIRST_DWORD_FIRST_BYTE

Table 3-516 Fields for Register: VF_HDR_LOG_0_OFF

Bits	Name	Memory Access	Description
31:24	FIRST_DWORD_FOURTH_BYTE	R	Byte 3 of Header log register of First 32-bit Data Word. This field represents fourth byte of First DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:16	FIRST_DWORD_THIRD_BYTE	R	Byte 2 of Header log register of First 32-bit Data Word. This field represents third byte of First DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
15:8	FIRST_DWORD_SECOND_BYTE	R	Byte 1 of Header log register of First 32-bit Data Word. This field represents second byte of First DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
7:0	FIRST_DWORD_FIRST_BYTE	R	Byte 0 of Header log register of First 32-bit Data Word. This field represents first byte of First DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

3.54.9 VF_HDR_LOG_1_OFF

- **Name:** Header Log Register 1.
- **Description:** The Header Log Register 1 contains the header for the TLP corresponding to a detected error; The header is captured such that, when read using DW accesses, the fields of the header are laid out in the same way the headers are presented in the specification. Therefore, byte 0 of the header is located in byte 3 of the Header Log Register 1, byte 1 of the header is in byte 2 of the Header Log Register 1 and so forth. For 12-byte headers, only bytes 0 through 11 of the Header Log Register are used and values in bytes 12 through 15 are undefined. In certain cases where a Malformed TLP is reported, the Header Log Register may contain TLP Prefix information.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:24	SECOND_DWORD_FOURTH_BYTE
23:16	SECOND_DWORD_THIRD_BYTE
15:8	SECOND_DWORD_SECOND_BYTE
7:0	SECOND_DWORD_FIRST_BYTE

Table 3-517 Fields for Register: VF_HDR_LOG_1_OFF

Bits	Name	Memory Access	Description
31:24	SECOND_DWORD_FOURTH_BYTE	R	Byte 3 of Header log register of Second 32-bit Data Word. This field represents fourth byte of Second DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:16	SECOND_DWORD_THIRD_BYTE	R	Byte 2 of Header log register of Second 32-bit Data Word. This field represents third byte of Second DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
15:8	SECOND_DWORD_SECOND_BYTE	R	Byte 1 of Header log register of Second 32-bit Data Word. This field represents second byte of Second DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
7:0	SECOND_DWORD_FIRST_BYTE	R	Byte 0 of Header log register of Second 32-bit Data Word. This field represents first byte of Second DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

3.54.10 VF_HDR_LOG_2_OFF

- **Name:** Header Log Register 2.
- **Description:** The Header Log Register 2 contains the header for the TLP corresponding to a detected error; The header is captured such that, when read using DW accesses, the fields of the header are laid out in the same way the headers are presented in the specification. Therefore, byte 0 of the header is located in byte 3 of the Header Log Register 2, byte 1 of the header is in byte 2 of the Header Log Register 2 and so forth. For 12-byte headers, only bytes 0 through 11 of the Header Log Register are used and values in bytes 12 through 15 are undefined. In certain cases where a Malformed TLP is reported, the Header Log Register may contain TLP Prefix information.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:24	THIRD_DWORD_FOURTH_BYTE
23:16	THIRD_DWORD_THIRD_BYTE
15:8	THIRD_DWORD_SECOND_BYTE
7:0	THIRD_DWORD_FIRST_BYTE

Table 3-518 Fields for Register: VF_HDR_LOG_2_OFF

Bits	Name	Memory Access	Description
31:24	THIRD_DWORD_FOURTH_BYTE	R	Byte 3 of Header log register of Third 32-bit Data Word. This field represents fourth byte of Third DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:16	THIRD_DWORD_THIRD_BYTE	R	Byte 2 of Header log register of Third 32-bit Data Word. This field represents third byte of Third DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
15:8	THIRD_DWORD_SECOND_BYTE	R	Byte 1 of Header log register of Third 32-bit Data Word. This field represents second byte of Third DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
7:0	THIRD_DWORD_FIRST_BYTE	R	Byte 0 of Header log register of Third 32-bit Data Word. This field represents first byte of Third DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

3.54.11 VF_HDR_LOG_3_OFF

- **Name:** Header Log Register 3.
- **Description:** The Header Log Register 3 contains the header for the TLP corresponding to a detected error; The header is captured such that, when read using DW accesses, the fields of the header are laid out in the same way the headers are presented in the specification. Therefore, byte 0 of the header is located in byte 3 of the Header Log Register 3, byte 1 of the header is in byte 2 of the Header Log Register 3 and so forth. For 12-byte headers, only bytes 0 through 11 of the Header Log Register are used and values in bytes 12 through 15 are undefined. In certain cases where a Malformed TLP is reported, the Header Log Register may contain TLP Prefix information.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:24	FOURTH_DWORD_FOURTH_BYTE
23:16	FOURTH_DWORD_THIRD_BYTE
15:8	FOURTH_DWORD_SECOND_BYTE
7:0	FOURTH_DWORD_FIRST_BYTE

Table 3-519 Fields for Register: VF_HDR_LOG_3_OFF

Bits	Name	Memory Access	Description
31:24	FOURTH_DWORD_FOURTH_BYTE	R	Byte 3 of Header log register of Fourth 32-bit Data Word. This field represents fourth byte of Fourth DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
23:16	FOURTH_DWORD_THIRD_BYTE	R	Byte 2 of Header log register of Fourth 32-bit Data Word. This field represents third byte of Fourth DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
15:8	FOURTH_DWORD_SECOND_BYTE	R	Byte 1 of Header log register of Fourth 32-bit Data Word. This field represents second byte of Fourth DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
7:0	FOURTH_DWORD_FIRST_BYTE	R	Byte 0 of Header log register of Fourth 32-bit Data Word. This field represents first byte of Fourth DW of Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

3.54.12 VF_TLP_PREFIX_LOG_i_OFF (for i = 1; i <= 4)

- **Name:** TLP Prefix Log Register 1.
- **Description:** The First TLP Prefix Log Register contains the first End-End TLP Prefix from the TLP corresponding to the detected error. The TLP Prefix Log Register is only meaningful when the TLP Prefix Log Present bit is Set. The TLP Prefixes are captured such that, when read using DW accesses, the fields of the TLP Prefix are laid out in the same way the fields of the TLP Prefix are described. Therefore, byte 0 of a TLP Prefix is located in byte 3 of the associated TLP Prefix Log Register; byte 1 of a TLP Prefix is located in byte 2; and so forth. The TLP Prefix Log Registers beyond the number supported by the Function are hardwired to zero by controller. If the End-End TLP Prefix Supported bit is Clear, the TLP Prefix Log Register is not required to be implemented.
- **Size:** 32 bits
- **Offset:** B+0x38 + i*0x4
- **Exists:** ((VF_AER_ENABLE | | EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

CFG_TLP_PFX_LOG_1_FOURTH_BYTE	31:24
CFG_TLP_PFX_LOG_1_THIRD_BYTE	23:16
CFG_TLP_PFX_LOG_1_SECOND_BYTE	15:8
CFG_TLP_PFX_LOG_1_FIRST_BYTE	7:0

Table 3-520 Fields for Register: VF_TLP_PREFIX_LOG_i_OFF (for i = 1; i <= 4)

Bits	Name	Memory Access	Description
31:24	CFG_TLP_PFX_LOG_1_FOURTH_BYTE	R	Byte 3 of Error TLP Prefix Log 1. This field contains fourth byte of First DW of TLP Prefix. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
23:16	CFG_TLP_PFX_LOG_1_THIRD_BYTE	R	Byte 2 of Error TLP Prefix Log 1. This field contains third byte of First DW of TLP Prefix. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
15:8	CFG_TLP_PFX_LOG_1_SECOND_BYTE	R	Byte 1 of Error TLP Prefix Log 1. This field contains second byte of First DW of TLP Prefix. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
7:0	CFG_TLP_PFX_LOG_1_FIRST_BYTE	R	Byte 0 of Error TLP Prefix Log 1. This field contains first byte of First DW of TLP Prefix. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_AER_ENABLE EXT_VF_AER_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

3.55 VF_PF_ARI_CAPI Registers

3.55.1 VF_ARI_BASE

- **Name:** ARI Capability Header.
- **Description:** This register provides Capability Id, Capability version, and Next offset of Capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

31:20	ARI_NEXT_OFFSET
19:16	ARI_CAP_VERSION
15:0	ARI_PCIE_EXTENDED_CAP_ID

Table 3-521 Fields for Register: VF_ARI_BASE

Bits	Name	Memory Access	Description
31:20	ARI_NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	ARI_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the capability structure present. This field is depends on the version of specification. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: wire ? unconstrained : writeAsRead
15:0	ARI_PCIE_EXTENDED_CAP_ID	* Varies	ARI Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the extended capability. PCI Express Extended Capability ID for the ARI Capability is 000Eh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xe Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: wire ? unconstrained : writeAsRead

3.55.2 VF_CAP_REG

- **Name:** ARI Capability and Control Register.
- **Description:** This Register provides information regarding supported ARI Capability, and controls that capability.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** CX_MEMORY_MAP_PCIE_TYPE!=1

31:23	RSVDP_23
22:20	ARI_FUN_GRP
19:18	RSVDP_18
17	ARI_ACS_FUN_GRP_EN
16	ARI_MFVC_FUN_GRP_EN
15:8	ARI_NEXT_FUN_NUM
7:2	RSVDP_2
1	ARI_ACS_FUN_GRP_CAP
0	ARI_MFVC_FUN_GRP_CAP

Table 3-522 Fields for Register: VF_CAP_REG

Bits	Name	Memory Access	Description
31:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
22:20	ARI_FUN_GRP	* Varies	Function Group. Assigns a Function Group Number to this Function. Must be hardwired to 000b by the controller if in Function 0, the MFVC Function Groups Capability bit and ACS Function Groups Capability bit are both 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: CAP_REG.ARI_ACS_FUN_GRP_CAP ? RW : RO ■ Dbi: CAP_REG.ARI_ACS_FUN_GRP_CAP ? RW : RO and more specifically: (CX_MEMORY_MAP_VIEW==0 && VF_ACS_FUNC_GRP && CX_ACS_FUNC_GRP_VALUE) ? R/W : (CX_MEMORY_MAP_VIEW==0 && !VF_ACS_FUNC_GRP) ? R: (CX_MEMORY_MAP_VIEW==1 && VF_ACS_FUNC_GRP && CX_ACS_FUNC_GRP_VALUE) ? R/W: (CX_MEMORY_MAP_VIEW==1 && !VF_ACS_FUNC_GRP) ? R: (VF_ACS_FUNC_GRP && CX_ACS_FUNC_GRP_VALUE) ? R/W: R Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1 Testable: writeAsRead
19:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
17	ARI_ACS_FUN_GRP_EN	R	ACS Function Groups Enable. Applicable only for Function 0; must be hardwired to 0b for all other Functions. When set, each Function in the ARI Device must associate bits within its Egress Control Vector with Function Group Numbers rather than Function Numbers. Must be hardwired to 0b if the ACS Function Groups Capability bit is 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
16	ARI_MFVC_FUN_GRP_EN	R	MFVC Function Groups Enable. Applicable only for Function 0; must be hardwired to 0b for all other Functions. When set, the ARI Device must interpret entries in its Function Arbitration Table as Function Group Numbers rather than Function Numbers. Must be hardwired to 0b by the controller if the MFVC Function Groups Capability bit is 0b. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

Bits	Name	Memory Access	Description
15:8	ARI_NEXT_FUN_NUM	R	Next Function Number. This field indicates the Function Number of the next higher numbered Function in the Device, or 00h if there are no higher numbered Functions. Function 0 starts this linked list of Functions. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
7:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
1	ARI_ACS_FUN_GRP_CAP	R	ACS Function Groups Capability. Applicable only for Function 0; must be 0b for all other Functions. If 1b, indicates that the ARI Device supports Function Group level granularity for ACS P2P Egress Control through its ACS Capability structures. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1
0	ARI_MFVC_FUN_GRP_CAP	R	Multi Functional Virtual Channel (MFVC) Function Groups Capability. Applicable only for Function 0; must be 0b for all other Functions. If 1b, indicates that the ARI Device supports Function Group level arbitration through its Multi-Function Virtual Channel (MFVC) Capability structure. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1

3.56 VF_PF_ACS_CAPi Registers

3.56.1 VF_ACS_CAP_HDR_REG

- **Name:** Access Control Services Extended Capability Header Register.
- **Description:** This register provides capability ID, capability version, and next offset value.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	CAP_ID

Table 3-523 Fields for Register: VF_ACS_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. This field contains the offset to the next PCI Express Extended Capability structure or 000h, if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field indicates the version of present capability structure. It is a PCI-SIG defined version number. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead
15:0	CAP_ID	* Varies	ACS Extended Capability ID. This field identifies the type of capability structure that follows this header. It is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. For PCI Express, Extended Capability ID is 000Dh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0xd Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

3.56.2 VF_ACS_CAPABILITIES_CTRL_REG

- **Name:** Access Control Services Capability Register.
- **Description:** This register is Access Control Services Capability and Control Register.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1)

RSVDP_23	31:23
ACS_DIRECT_TRANSLATED_P2P_EN	22
ACS_P2P_EGRESS_CONTROL_EN	21
ACS_USP_FORWARDING_EN	20
ACS_P2P_CPL_REDIRCT_EN	19
ACS_P2P_REQ_REDIRCT_EN	18
ACS_AT_BLOCK_EN	17
ACS_SRC_VALID_EN	16
ACS_EGRESS_CTRL_SIZE	15:8
RSVDP_7	7
ACS_DIRECT_TRANSLATED_P2P	6
ACS_P2P_EGRESS_CONTROL	5
ACS_USP_FORWARDING	4
ACS_P2P_CPL_REDIRCT	3
ACS_P2P_REQ_REDIRCT	2
ACS_AT_BLOCK	1
ACS_SRC_VALID	0

Table 3-524 Fields for Register: VF_ACS_CAPABILITIES_CTRL_REG

Bits	Name	Memory Access	Description
31:23	RSVDP_23	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
22	ACS_DIRECT_TRANSLATED_P2P_EN	R/W	ACS Direct Translated Peer-to-Peer Enable. When Set, overrides the ACS_P2P_REQ_REDIRECT and ACS_P2P_EGRESS_CONTROL mechanisms with peer-to-peer Memory Requests whose Address Translation (AT) field indicates a Translated address. This bit is ignored if ACS_AT_BLOCK_EN is 1b. The controller hardwires this bit to 0b if, the ACS Direct Translated P2P functionality is not implemented. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
21	ACS_P2P_EGRESS_CONTROL_EN	R/W	ACS Peer-to-Peer Egress Control Enable. In conjunction with the Egress Control Vector plus the ACS_P2P_REQ_REDIRECT and ACS_DIRECT_TRANSLATED_P2P mechanisms, determines when to allow, disallow, or redirect peer-to-peer Requests. The controller hardwires this bit to 0b if, the ACS P2P Egress Control functionality is not implemented. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
20	ACS_USP_FORWARDING_EN	R/W	ACS Upstream Forwarding Enable. When Set, the component forwards Upstream any Request or Completion TLPs it receives that were redirected Upstream by a component lower in the hierarchy. This bit only applies to Upstream TLPs arriving at a Downstream Port, and whose normal routing targets the same Downstream Port. The controller hardwires this bit to 0b if, the ACS Upstream Forwarding functionality is not implemented. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
19	ACS_P2P_CPL_REDIRECT_EN	R/W	ACS Peer-to-Peer Completion Redirect Enable. Determines when the component redirects peer-to-peer Completions Upstream; applicable only to Completions whose Relaxed Ordering Attribute is clear. The controller hardwires this bit to 0b if, the ACS P2P Completion Redirect functionality is not implemented. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
18	ACS_P2P_REQ_REDIRECT_EN	R/W	ACS Peer-to-Peer Request Redirect Enable. In conjunction with ACS_P2P_EGRESS_CONTROL and ACS_DIRECT_TRANSLATED_P2P mechanisms, determines when the component redirects peer-to-peer Requests Upstream. With Downstream Ports, this bit only applies to Upstream Requests arriving at the Downstream Port, and whose normal routing targets a different Downstream Port. The controller hardwires this bit to 0b if, the ACS P2P Request Redirect functionality is not implemented. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
17	ACS_AT_BLOCK_EN	R/W	ACS Translation Blocking Enable. When Set, the component blocks all Upstream Memory Requests whose Address Translation (AT) field is not set to the default value. The controller hardwires this bit to 0b if, the ACS Translation Blocking functionality is not implemented. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
16	ACS_SRC_VALID_EN	R/W	ACS Source Validation Enable. When Set, the component validates the Bus Number from the Requester ID of Upstream Requests against the secondary/subordinate Bus Numbers. The controller hardwires this bit to 0b if, the ACS Source Validation functionality is not implemented. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:8	ACS_EGRESS_CTRL_SIZE	* Varies	ACS Egress control vector size. Encodings 01h-FFh directly indicate the number of applicable bits in the Egress Control Vector; the encoding 00h indicates 256 bits. If the ACS_P2P_EGRESS_CONTROL bit is 0b, the value of the size field is undefined, and the Egress Control Vector register is not required to be present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Value After Reset: CX_ACS_EGRESS_CTRL_SIZE_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_DBI_RO_WR_EN) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_DBI_RO_WR_EN) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1)
6	ACS_DIRECT_TRANSLATED_P2P	* Varies	ACS Direct Translated Peer-to-Peer Capable. Required for Root Ports that support Address Translation Services (ATS) and also support peer-to-peer traffic with other Root Ports; required for Switch Downstream Ports; required for Multi-Function Device Functions that support Address Translation Services (ATS) and also support peer-to-peer traffic with other Functions; the controller must be hardwired to 0b otherwise. If 1b, indicates that the component implements ACS Direct Translated P2P. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: CX_ACS_P2P_DIRECT_TRANSL_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
5	ACS_P2P_EGRESS_CONTROL	* Varies	ACS Egress Control. Optional for Root Ports, Switch Downstream Ports, and Multi-Function Device Functions; must be hardwired to 0b otherwise. If 1b, indicates that the component implements ACS P2P Egress Control. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: CX_ACS_P2P_EGRESS_CTRL_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
4	ACS_USP_FORWARDING	* Varies	ACS Upstream forwarding Capable. Required for Root Ports if the RC supports Redirected Request Validation; required for Switch Downstream Ports; must be hardwired to 0b otherwise. If 1b, indicates that the component implements ACS Upstream Forwarding. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead
3	ACS_P2P_CPL_REDIRECT	* Varies	ACS Peer-to-peer Completion Redirect Capable. Required for all Functions that support ACS P2P Request Redirect; must be hardwired to 0b otherwise. If 1b, indicates that the component implements ACS P2P Completion Redirect. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: CX_ACS_P2P_COMPL_REDIRECT_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2	ACS_P2P_REQ_REDIRECT	* Varies	ACS Peer-to-peer Request Redirect Capable. Required for Root Ports that support peer-to-peer traffic with other Root Ports; required for Switch Downstream Ports; required for Multi-Function Device Functions that support peer-to-peer traffic with other Functions; must be hardwired to 0b otherwise. If 1b, indicates that the component implements ACS P2P Request Redirect. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Value After Reset: CX_ACS_P2P_REQ_REDIRECT_0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
1	ACS_AT_BLOCK	* Varies	ACS Translation Blocking Capable. Required for Root Ports and Switch Downstream Ports; Otherwise it is hardwired to 0b. If 1b, indicates that the component implements ACS Translation Blocking. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead
0	ACS_SRC_VALID	* Varies	ACS Source Validation Capable. Required for Root Ports and Switch Downstream Ports; Otherwise it is hardwired to 0b. If 1b, indicates that the component implements ACS Source Validation. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

3.56.3 VF_ACS_EGRESS_CTRL_VECTOR_REGi (for i = 0; i <= 7)

- **Name:** Access Control Services Egress Control Vector Register #i
- **Description:** This Egress Control Vector register is a read-write register that contains a bit-array. The number of bits in the register is specified by the ACS_EGRESS_CTRL_SIZE field, and the register spans multiple DWORDs, if required.
- **Size:** 32 bits
- **Offset:** B+0x8 + i*0x4
- **Exists:** (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1)



Table 3-525 Fields for Register: VF_ACS_EGRESS_CTRL_VECTOR_REGi (for i = 0; i <= 7)

Bits	Name	Memory Access	Description
31:0	ACS_EGRESS_CTRL_VECTOR_REG0	* Varies	Access Control Services Egress Control Vector Register #i. An N-bit bit-array configured by software, where N is given by the value in the ACS_EGRESS_CTRL_SIZE field. When a given bit is set, peer-to-peer Requests targeting the associated Port, Function, or Function Group are blocked or redirected (if enabled). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (ACS_CTRL_VEC_WD_0>0) && CX_ACS_P2P_EGRESS_CTRL_0==1 && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && (VF_ACS_ENABLE && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : \"read-only\""

3.57 VF_PF_ATS_CAPi Registers

3.57.1 VF_ATS_CAP_HDR_REG

- **Name:** ATS Extended Capability Header.
- **Description:** This register provides value of Capability ID, Capability version, and next offset of Extended capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** ((VF_ATS_ENABLE | | EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	CAP_ID

Table 3-526 Fields for Register: VF_ATS_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	R	Next Capability Offset. The offset to the next PCI Extended Capability structure or 000h if no other items exist in the linked list of capabilities. <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_ATS_ENABLE EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
19:16	CAP_VERSION	R	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_ATS_ENABLE EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
15:0	CAP_ID	R	ATS Extended Capability ID. This field must return a Capability ID of 000Fh indicating that this is an ATS Extended Capability structure. <i>Note:</i> This register field is sticky. Value After Reset: 0xf Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_ATS_ENABLE EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.57.2 VF_ATS_CAPABILITIES_CTRL_REG

- **Name:** ATS Requester Capability Register.
- **Description:** This register provides value of Invalidate queue depth, Page alignment request, and Global Invalidate support. It also controls Cache translation and Smallest Translate Unit.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** ((VF_ATS_ENABLE | | EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31	ENABLE
30:21	RSVDP_21
20:16	STU
15:7	RSVDP_7
6	GLOBAL_INVAL_SPPRTD
5	PAGE_ALIGNED_REQ
4:0	INVALIDATE_Q_DEPTH

Table 3-527 Fields for Register: VF_ATS_CAPABILITIES_CTRL_REG

Bits	Name	Memory Access	Description
31	ENABLE	R/W	Enable (E). When Set, the Function is enabled to cache translations. Behavior is undefined if this bit is Set, and the value of the PASID Enable, Execute Requested Permission Enable, or Privileged Mode Requested Enable bits are changed. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_ATS_ENABLE EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: {(CX_EXTENSIBLE_VFUNC) ? "untestable" : "unconstrained" }

Bits	Name	Memory Access	Description
30:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_ATS_ENABLE EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
20:16	STU	R	Smallest Translation Unit (STU). This value indicates to the Function the minimum number of 4096-byte blocks that is indicated in a Translation Completions or Invalidate Requests. This is a power of 2 multiplier and the number of blocks is 2 STU. A value of 0 0000b indicates one block and a value of 1 1111b indicates 2 ³¹ blocks (or 8 TB total). Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_ATS_ENABLE EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
15:7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_ATS_ENABLE EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
6	GLOBAL_INVALID_SPPRTD	* Varies	Global Invalidate Supported. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): The Function supports Invalidation Requests that have the Global Invalidate bit Set. ■ 0x0 (CLEAR): The Function ignores the Global Invalidate bit in all Invalidate Requests. This bit is 0b if the Function does not support the PASID TLP Prefix. Value After Reset: DEFAULT_ATS_GLOBAL_INVALID_SPPRTD Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_ATS_ENABLE EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
5	PAGE_ALIGNED_REQ	* Varies	Page Aligned Request. If Set, indicates the Untranslated Address is always aligned to a 4096 byte boundary. Setting this field is recommended. This field permits software to distinguish between implementations compatible with this specification and those compatible with an earlier version of this specification in which a Requester was permitted to supply anything in bits [11:2]. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_ATS_ENABLE EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: {(CX_EXTENSIBLE_VFUNC) ? "untestable" : "unconstrained" }
4:0	INVALIDATE_Q_DEPTH	R	Invalidate Queue Depth. The number of Invalidate Requests that the Function can accept before putting backpressure on the Upstream connection. If 0 0000b, the Function can accept 32 Invalidate Requests. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_ATS_ENABLE EXT_VF_ATS_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.58 VF_PF_LNR_CAPi Registers

3.58.1 VF_LNR_EXT_CAP_HDR_OFF

- **Name:** LNR Extended Capability Header.
- **Description:** This register provides Capability ID, Capability version, and Next offset of Extended Capability Structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** ((VF_LN_ENABLE | | EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:20	19:16	15:0
LNR_NEXT_OFFSET	LNR_CAP_VERSION	LNR_EXT_CAP_ID

Table 3-528 Fields for Register: VF_LNR_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	LNR_NEXT_OFFSET	* Varies	Next Capability Offset. This field is a PCI-SIG defined ID number that indicates the nature and format of the extended capability. PCI Express Extended Capability ID for the LNR Extended Capability is 001Ch. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	LNR_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead
15:0	LNR_EXT_CAP_ID	* Varies	PCI Express Extended Capability ID. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1c Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

3.58.2 VF_LNR_CAP_CONTROL_OFF

- **Name:** LNR Control Register and Capability Register.
- **Description:** This register provides extended capability for LNR. It also controls the LNR behavior.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** ((VF_LN_ENABLE | | EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:29	RSVDP_29
28:24	LNR_REGISTRATION_LIMIT
23:18	RSVDP_18
17	LNR_CLS
16	LNR_ENABLE
15:13	RSVDP_13
12:8	LNR_REGISTRATION_MAX
7:2	RSVDP_2
1	LNR_128_SUPPORTED
0	LNR_64_SUPPORTED

Table 3-529 Fields for Register: VF_LNR_CAP_CONTROL_OFF

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
28:24	LNR_REGISTRATION_LIMIT	R/W	LNR Registration Limit. This field, encoded as a power of 2, imposes a limit on the number of cachelines that this LN Requester is permitted to register concurrently. For example, a value of 00100b indicates that the LN Requester must not register more than 16 cachelines (2^4) concurrently. Value After Reset: 0x1f Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
23:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
17	LNR_CLS	R/W	LNR CLS. This bit controls or indicates the cacheline size used with LN protocol by this Requester. If this LN Requester supports only one cacheline size, this bit is permitted to be hardwired to indicate that size. Otherwise, the default value of this bit is 0b. Values: 0x1 (SET): The Cacheline size is 128 bytes. 0x0 (CLEAR): The cacheline size is 64 bytes. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
16	LNR_ENABLE	R/W	LNR Enable. When this bit is Set, the Endpoint is enabled to operate as an LN Requester. Software is permitted to Clear this bit at any time. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
15:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
12:8	LNR_REGISTRATION_MAX	* Varies	LNR Registration Max. This field, encoded as a power of 2, indicates the maximum number of cachelines that this LN Requester is capable of registering concurrently. For example, a value of 00101b indicates that the LN Requester might be capable of registering up to 32 cachelines (2^5) concurrently, and is capable of registering greater than 16. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_LN_REG_MAX_VALUE Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead
7:2	RSVDP_2	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
1	LNR_128_SUPPORTED	* Varies	LNR-128 Supported. This bit must be 1b if the Endpoint supports LN protocol for 128-byte cachelines as a Requester; otherwise, must be 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_LN_128_SUPPORTED_VALUE Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
0	LN_64_SUPPORTED	* Varies	LN-64 Supported. This bit must be 1b if the Endpoint supports LN protocol for 64-byte cachelines as a Requester; otherwise, must be 0b. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: VF_LN_64_SUPPORTED_VALUE Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_LN_ENABLE EXT_VF_LN_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

3.59 VF_PF_PASID_CAPI Registers

3.59.1 VF_PASID_EXT_CAP_HDR_REG

- **Name:** PASID Extended Capability Header.
- **Description:** This Register provides Capability ID, Capability Version, and Next offset of capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** ((VF_PASID_ENABLE || EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:20	19:16	15:0
NEXT_OFFSET	CAP_VERSION	CAP_ID

Table 3-530 Fields for Register: VF_PASID_EXT_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	NEXT_OFFSET	* Varies	Next Capability Offset. The offset to the next PCI Extended Capability structure or 000h if no other items exist in the linked list of capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead
15:0	CAP_ID	* Varies	PASID Extended Capability ID. Indicates the PASID Extended Capability structure. This field must return a Capability ID of 001Bh indicating that this is a PASID Extended Capability structure. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1b Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

3.59.2 VF_PASID_CAP_CNTRL_REG

- **Name:** PASID Capability Control Register.
- **Description:** This register provides value of capability like Execute Permission support, Privileged Mode support, and width of PASID. Also this register controls PASID Enable, Execute Permission Enable, as well as Privileged mode Enable.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** ((VF_PASID_ENABLE | | EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

RSVDP_19	31:19
PRIVILEGED_MODE_EN	18
EXECUTE_PERMISSION_EN	17
EN	16
RSVDP_13	15:13
MAX_PASID_WIDTH	12:8
RSVDP_3	7:3
PRIVILEGED_MODE_SUPPORTED	2
EXECUTE_PERMISSION_SUPPORTED	1
RSVDP_0	0

Table 3-531 Fields for Register: VF_PASID_CAP_CNTRL_REG

Bits	Name	Memory Access	Description
31:19	RSVDP_19	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
18	PRIVILEGED_MODE_EN	* Varies	Privileged Mode Enable (endpoints only). The behavior is undefined if the Endpoint supports ATS and this bit changes value when the Enable bit in the ATS Control register is Set. If Privileged Mode Supported is Clear, this bit is RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PASID_CAP_CNTRL_REG.PRIVILEGED_MODE_SUPPORTED ? RW : RO ■ Dbi: PASID_CAP_CNTRL_REG.PRIVILEGED_MODE_SUPPORTED ? RW : RO Values: ■ 0x1 (SET): The Endpoint is permitted to send Requests that have the Privileged Mode Requested bit Set. ■ 0x0 (CLEAR): The Endpoint is not permitted to send Requests that have the Privileged Mode Requested bit Set. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
17	EXECUTE_PERMISSION_EN	* Varies	Execute Permission Enable (endpoints only). The behavior is undefined if the Endpoint supports ATS and this bit changes value when the Enable bit in the ATS Control register is Set. If Execute Permission Supported is Clear, this bit is RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: PASID_CAP_CNTRL_REG.EXECUTE_PERMISSION_SUPPORTED ? RW : RO ■ Dbi: PASID_CAP_CNTRL_REG.EXECUTE_PERMISSION_SUPPORTED ? RW : RO Values: ■ 0x1 (SET): The Endpoint is permitted to send Requests that have the Execute Requested bit Set. ■ 0x0 (CLEAR): The Endpoint is not permitted to send Requests that have the Execute Requested bit Set. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
16	EN	R/W	PASID Enable. Behavior is undefined if the Endpoint supports ATS and this bit changes value when the Enable (E) bit in the ATS Control register is Set. Values: 0x1 (SET): The Endpoint is permitted to send and receive TLPs that contain a PASID TLP Prefix. 0x0 (CLEAR): The Endpoint is not permitted to do so. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
15:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
12:8	MAX_PASID_WIDTH	* Varies	Max PASID Width. Indicates the width of the PASID field supported by the Endpoint. The value n indicates support for PASID values 0 through (2^n) - 1 (inclusive). This field must be between '0' and '20' (inclusive). <i>Note:</i> The access attributes of this field are as follows: - Wire: R (sticky) - Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: 0x14 (MAX_VALUE): The value '20' indicates support for all PASID values (20 bits). 0x0 (MIN_VALUE): The value '0' indicates support for a single PASID (0). Value After Reset: DEFAULT_MAX_PASID_WIDTH Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
7:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
2	PRIVILEGED_MODE_SUPPORTED	* Varies	Privileged Mode Supported (endpoints only). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): The Endpoint supports operating in Privileged and Non-Privileged modes, and supports sending requests that have the Privileged Mode Requested bit Set. ■ 0x0 (CLEAR): The Endpoint will never Set the Privileged Mode Requested bit. Value After Reset: DEFAULT_PRVLGD_MODE_SPPRT Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
1	EXECUTE_PERMISSION_SUPPORTED	* Varies	Execute Permission Supported (endpoints only). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x1 (SET): The Endpoint supports sending TLPs that have the Execute Requested bit Set. ■ 0x0 (CLEAR): The Endpoint will never Set the Execute Requested bit. Value After Reset: DEFAULT_EXCT_PRMSSN_SPPRT Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-write\""
0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_PASID_ENABLE EXT_VF_PASID_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.60 VF_PF_RTR_CAPI Registers

3.60.1 VF_RTR_EXT_CAP_HDR_OFF

- **Name:** Readiness Time Reporting PCI Express Extended Capability Header.
- **Description:** This Register contains detail about Capability ID, Capability Version, and Next Offset of Readiness Time Reporting Extended Capability structure.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** ((VF_RTR_ENABLE | | EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:20	RTR_NEXT_OFFSET
19:16	RTR_CAP_VERSION
15:0	RTR_EXT_CAP_ID

Table 3-532 Fields for Register: VF_RTR_EXT_CAP_HDR_OFF

Bits	Name	Memory Access	Description
31:20	RTR_NEXT_OFFSET	* Varies	Next Capability Offset. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. Extended Capability ID for the Readiness Time Reporting Extended Capability is 0022h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xfff (MAX_VAL): Max value Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_RTR_ENABLE EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	RTR_CAP_VERSION	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. This field is version specific. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xf (MAX_VAL): Max value Value After Reset: 0x1 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_RTR_ENABLE EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead
15:0	RTR_EXT_CAP_ID	* Varies	PCI Express Extended Capability ID. This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of Capabilities. For Extended Capabilities implemented in Configuration Space, this offset is relative to the beginning of PCI-compatible Configuration Space and thus must always be either 000h (for terminating list of Capabilities) or greater than 0FFh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (MIN_VAL): Min value ■ 0xffff (MAX_VAL): Max value Value After Reset: 0x22 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_RTR_ENABLE EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

3.60.2 VF_READI_TIME_REPORTING1_OFF

- **Name:** Readiness Time Reporting 1.
- **Description:** This Register provides detail about Reset Time, DL_UP_Time, and RTR_Time valid.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** ((VF_RTR_ENABLE | | EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31	RTR_TIME_VALID
30:24	RSVDP_24
23:12	DL_UP_TIME
11:0	RESET_TIME

Table 3-533 Fields for Register: VF_READI_TIME_REPORTING1_OFF

Bits	Name	Memory Access	Description
31	RTR_TIME_VALID	* Varies	Valid. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: RO. Written to using CS2 on parent PF Values: ■ 0x0 (CLEAR): If this bit remains Clear and 1 minute has elapsed after all associated device driver(s) have started, software is permitted to assume that this bit will never be set. ■ 0x1 (SET): If Set, indicates that all time values in this capability are valid. If Clear, indicates that the time values in this capability are not yet available. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_RTR_ENABLE EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
30:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_RTR_ENABLE EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
23:12	DL_UP_TIME	* Varies	DL_UP Time. DL_UP Time is the time the Function requires to become Configuration-Ready after the Downstream Port above the Function reports Data Link Layer Link Active. This field is RsvdP in Functions that are not associated with an Upstream Port. This field is undefined when the Valid bit is Clear. This field must be less than or equal to the encoded value A1Eh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RO. Written to using CS2 on parent PF Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_RTR_ENABLE EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

Bits	Name	Memory Access	Description
11:0	RESET_TIME	* Varies	Reset Time. Reset Time is the time the Function requires to become Configuration-Ready after the completion of Conventional Reset. This field is RsvdP if the Immediate Readiness bit is Set. This field is undefined when the Valid bit is Clear. This field must be less than or equal to the encoded value A1Eh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: RO. Written to using CS2 on parent PF Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_RTR_ENABLE EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

3.60.3 VF_READI_TIME_REPORTING2_OFF

- **Name:** Readiness Time Reporting 2.
- **Description:** This register gives detail of D3hot to D0 time and FLR time.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** ((VF_RTR_ENABLE | | EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:24	23:12	11:0
RSVDP_24	D3HOT_D0_TIME	FLR_TIME

Table 3-534 Fields for Register: VF_READI_TIME_REPORTING2_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_RTR_ENABLE EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
23:12	D3HOT_D0_TIME	* Varies	D3hot to D0 Time. If Immediate_Readiness_on_Return_to_D0 is Clear, D3hot to D0 Time is the time that the Function requires after it is directed from D3hot to D0 before it is Configuration-Ready and has returned to either D0 uninitialized or D0 active state. This field is RsvdP if the Immediate_Readiness_on_Return_to_D0 bit is Set. This field is undefined when the Valid bit is Clear. This field must be less than or equal to the encoded value 80Ah. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: HWINIT ■ Dbi: RO. Written to using CS2 on parent PF Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_RTR_ENABLE EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""
11:0	FLR_TIME	* Varies	FLR Time. FLR Time is the time that the Function requires to become Configuration- Ready after it was issued an FLR. This field is RsvdP when the Function Level Reset Capability bit is Clear. This field is undefined when the Valid bit is Clear. This field must be less than or equal to the encoded value A1Eh. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: RSVDP ■ Dbi: RO. Written to using CS2 on parent PF Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_RTR_ENABLE EXT_VF_RTR_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-only\" : \"read-only\""

3.61 VF_PF_TPH_CAPi Registers

3.61.1 VF_TPH_EXT_CAP_HDR_REG

- **Name:** TPH Extended Capability Header.
- **Description:** This register provides Capability Id, Capability Revision, and Next offset of structure Extended Capability Register.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** ((VF_TPH_ENABLE | | EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:20	TPH_REQ_NEXT_PTR
19:16	TPH_REQ_CAP_VER
15:0	PCIE_EXT_CAP_ID

Table 3-535 Fields for Register: VF_TPH_EXT_CAP_HDR_REG

Bits	Name	Memory Access	Description
31:20	TPH_REQ_NEXT_PTR	* Varies	Next Capability Pointer. This field contains the offset to the next PCI Express Extended Capability structure or 000h if no other items exist in the linked list of Capabilities. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: Pointer value calculated at configuration time. Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
19:16	TPH_REQ_CAP_VER	* Varies	Capability Version. This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead
15:0	PCIE_EXT_CAP_ID	* Varies	TPH Extended Capability ID. This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. PCI Express Extended Capability ID for the TPH Requester Extended Capability is 0017h. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x17 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: wire ? unconstrained : writeAsRead

3.61.2 VF_TPH_REQ_CAP_REG_REG

- **Name:** TPH Requester Capability Register.
- **Description:** This register gives information about No ST mode support, Interrupt Vector mode, Device specific mode, Extended TPH requester support, ST table location and size.
SRIOV Note: All VFs in a single PF have the same values for VF_TPH_REQ_CAP_REG_REG. To write this common register, you must perform a DBI_CS2 write (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) while accessing the PF TPH_REQ_CAP_REG_REG register.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** ((VF_TPH_ENABLE | |EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:27	RSVDP_27
26:16	TPH_REQ_CAP_ST_TABLE_SIZE
15:11	RSVDP_11
10	TPH_REQ_CAP_ST_TABLE_LOC_1
9	TPH_REQ_CAP_ST_TABLE_LOC_0
8	TPH_REQ_EXTENDED_TPH
7:3	RSVDP_3
2	TPH_REQ_DEVICE_SPEC
1	TPH_REQ_CAP_INT_VEC
0	TPH_REQ_NO_ST_MODE

Table 3-536 Fields for Register: VF_TPH_REQ_CAP_REG_REG

Bits	Name	Memory Access	Description
31:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
26:16	TPH_REQ_CAP_ST_TABLE_SIZE	R	ST Table Size. This field indicates the maximum number of ST Table entries the Function may use. Software reads this field to determine the ST Table Size N, which is encoded as N-1. For example, a returned value of 000 0000 0011b indicates a table size of four entries. There is an upper limit of 64 entries when the ST Table is located in the TPH Requester Extended Capability structure. When the ST Table is located in the MSI-X Table, this value is limited by the size of the MSI-X Table. This field is only applicable for Functions that implement an ST Table as indicated by the ST Table Location field. Otherwise, the value in this field is undefined. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
15:11	RSVDP_11	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
10	TPH_REQ_CAP_ST_TABLE_LOC_1	R	ST Table Location Bit 1. If this field is set, ST Table is Located in the MSI-X Table. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
9	TPH_REQ_CAP_ST_TABLE_LOC_0	R	ST Table Location Bit 0. If this field is set, ST Table is Located in the TPH Requester Extended Capability structure. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
8	TPH_REQ_EXTENDED_TPH	R	Extended TPH Requester Supported. If Set indicates that the Function is capable of generating Requests with a TPH TLP Prefix . <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
7:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
2	TPH_REQ_DEVICE_SPEC	R	Device Specific Mode Supported. If set indicates that the Function supports the Device Specific Mode of operation. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true

Bits	Name	Memory Access	Description
1	TPH_REQ_CAP_INT_VEC	R	Interrupt Vector Mode Supported. If set indicates that the Function supports the Interrupt Vector Mode of operation. <i>Note:</i> This register field is sticky. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true
0	TPH_REQ_NO_ST_MODE	R	No ST Mode Supported. set indicates that the Function supports the No ST Mode of operation. This mode is required to be supported by all Functions that implement this Capability structure. This bit must have a value of 1b. Value After Reset: * Varies Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Reset Mask: 0x0 Volatile: true

3.61.3 VF_TPH_REQ_CONTROL_REG_REG

- **Name:** TPH Requester Control Register.
- **Description:** This register provides ST Mode Select and controls TPH Requester Enable.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** ((VF_TPH_ENABLE | | EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

31:10	RSVDP_10
9:8	TPH_REQ_CTRL_REQ_EN
7:3	RSVDP_3
2:0	TPH_REQ_ST_MODE_SELECT

Table 3-537 Fields for Register: VF_TPH_REQ_CONTROL_REG_REG

Bits	Name	Memory Access	Description
31:10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
9:8	TPH_REQ_CTRL_REQ_EN	R/W	TPH Requester Enable Bit. This field controls the ability to issue Request TLPs using either TPH or Extended TPH. Functions that advertise that they do not support Extended TPH are permitted to hardwire bit 9 of this field to 0b. Values: ■ 0x0 (NOT_TPH_AND_NOT_EXT_TPH): Function operating as a Requester is not permitted to issue Requests with TPH or Extended TPH. ■ 0x1 (TPH_AND_NOT_EXT_TPH): Function operating as a Requester is permitted to issue Requests with TPH and is not permitted to issue Requests with Extended TPH. ■ 0x2 (RESERVED): Reserved ■ 0x3 (TPH_AND_EXT_TPH): Function operating as a Requester is permitted to issue Requests with TPH and Extended TPH. Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true
7:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

Bits	Name	Memory Access	Description
2:0	TPH_REQ_ST_MODE_SELECT	* Varies	ST Mode Select. This field selects the ST Mode of operation. All encodings other than the defined encodings are reserved. Functions that support only the No ST Mode of operation must hardwire this field to 000b. Function operation is undefined if software enables a mode of operation that does not correspond to a mode supported by the Function. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (NO_ST_MODE): No ST Mode ■ 0x1 (INT_VECT_MODE): Interrupt Vector Mode ■ 0x2 (DEV_SPECIFIC_MODE): Device Specific Mode Value After Reset: 0x0 Exists: (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0 && TPH_IVEC_0==0 && TPH_DS_0==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && TPH_IVEC_0==0 && TPH_DS_0==0) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1) ? \"read-write\" : (TPH_IVEC_0==0 && TPH_DS_0==0) ? \"read-only\" : \"read-write\""

3.61.4 VF_TPH_ST_TABLE_REG_i (for i = 0; i <= 31)

- **Name:** TPH ST Table Register 0.
- **Description:** The TPH ST Table 0 must be implemented in the TPH Requester Extended Capability structure if the value of the ST Table Location field is 01b. For all other values, the ST Entry registers must not be implemented. Each implemented ST Entry is 16 bits. The number of ST Entry registers implemented must be equal to the number of ST Table entries supported by the Function, which is the value of the ST Table Size field plus one.
- **Size:** 32 bits
- **Offset:** B+0xc + i*0x4
- **Exists:** ((VF_TPH_ENABLE | | EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

TPH_REQ_ST_TABLE_HIGHER_1	31:24
TPH_REQ_ST_TABLE_LOWER_1	23:16
TPH_REQ_ST_TABLE_HIGHER_0	15:8
TPH_REQ_ST_TABLE_LOWER_0	7:0

Table 3-538 Fields for Register: VF_TPH_ST_TABLE_REG_i (for i = 0; i <= 31)

Bits	Name	Memory Access	Description
31:24	TPH_REQ_ST_TABLE_HIGHER_1	* Varies	ST Table 1 Upper Byte. If the Function's Extended TPH Requester Supported bit is Set, then this field contains the upper 8 bits of a Steering Tag (1). Otherwise, this field is RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: this field is RW or Tie to 0 by table size configure ■ Dbi: this field is RW or Tie to 0 by table size configure and more specifically: (CX_NPRFX>0) ? R/W : R Value After Reset: 0x0 Exists: (-1 < (TPH_ST_TABLE_DEPTH_0-1)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead
23:16	TPH_REQ_ST_TABLE_LOWER_1	R/W	ST Table 1 Lower Byte. This field contains the lower 8 bits of a Steering Tag1. Value After Reset: 0x0 Exists: (-1 < (TPH_ST_TABLE_DEPTH_0-1)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)
15:8	TPH_REQ_ST_TABLE_HIGHER_0	* Varies	ST Table 0 Upper Byte. If the Function's Extended TPH Requester Supported bit is Set, then this field contains the upper 8 bits of a Steering Tag (0). Otherwise, this field is RsvdP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: this field is RW or Tie to 0 by table size configure ■ Dbi: this field is RW or Tie to 0 by table size configure and more specifically: (CX_NPRFX>0) ? R/W : R Value After Reset: 0x0 Exists: (-1 < (TPH_ST_TABLE_DEPTH_0-1)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1) Testable: writeAsRead
7:0	TPH_REQ_ST_TABLE_LOWER_0	R/W	ST Table 0 Lower Byte. This field contains the lower 8 bits of a Steering Tag0. Value After Reset: 0x0 Exists: (-1 < (TPH_ST_TABLE_DEPTH_0-1)) && (CX_MAX_VF_0>0) && CX_SRIOV_ENABLE && ((VF_TPH_ENABLE EXT_VF_TPH_ENABLE) && CX_MEMORY_MAP_PCIE_TYPE!=1)

3.62 PF0_MEMBAR0_HEADER_CAP Registers

3.62.1 MEMBAR0_HEADER_CAP_REG_OFF

- **Name:** CXL Capability Header Register of CXL.cache and CXL.mem Registers.
- **Description:** This register field has Capability Id, Capability version, Cache-Mem Version and the array Size.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_CXL_ENABLE&& CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_CACHE_MEM_SIZE	31:24
CXL_RCRB_MEMBAR0_CACHE_MEM_VER	23:20
CXL_RCRB_MEMBAR0_CAP_VER	19:16
CXL_RCRB_MEMBAR0_CAP_ID	15:0

Table 3-539 Fields for Register: MEMBAR0_HEADER_CAP_REG_OFF

Bits	Name	Memory Access	Description
31:24	CXL_RCRB_MEMBAR0_CACHE_MEM_SIZE	* Varies	This defines the number of elements present in the CXL_Capability array, not including the CXL_Capability_Header element. Each element is 1 DWORD in size and is located contiguous with previous elements. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: "(CX_CXL_MBAR0_CAP_ARRAY_SIZE)" Exists: (CX_CXL_ENABLE&&CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead
23:20	CXL_RCRB_MEMBAR0_CACHE_MEM_VER	* Varies	This defines the version of the CXL Cache Mem Protocol supported. For the first generation, this field must be 0x1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE&&CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead
19:16	CXL_RCRB_MEMBAR0_CAP_VE R	* Varies	This defines the version number of the CXL_Capability structure present. For the first generation, this field must be 0x1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE&&CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

Bits	Name	Memory Access	Description
15:0	CXL_RCRB_MEMBAR0_CAP_ID	* Varies	This defines the nature and format of the CXL_Capability register. For the CXL_Capability_Header register, this field must be 0x1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE&&CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.62.2 MEMBAR0_RAS_CAP_REG_OFF

- **Name:** CXL RAS Capability Header.
- **Description:** This register field has cap ID for RAS capability, it's Version and the next capability pointer for RAS capability.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_CXL_ENABLE&& CX_MEMORY_MAP_PCIE_TYPE==1)

31:20	CXL_RCRB_MEMBAR0_RAS_CAP_PTR
19:16	CXL_RCRB_MEMBAR0_RAS_CAP_VER
15:0	CXL_RCRB_MEMBAR0_RAS_CAP_ID

Table 3-540 Fields for Register: MEMBAR0_RAS_CAP_REG_OFF

Bits	Name	Memory Access	Description
31:20	CXL_RCRB_MEMBAR0_RAS_CAP_PTR	R	This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register. Value After Reset: 0x20 Exists: (CX_CXL_ENABLE&& CX_MEMORY_MAP_PCIE_TYPE==1) Testable: untestable Volatile: true

Bits	Name	Memory Access	Description
19:16	CXL_RCRB_MEMBAR0_RAS_CAP_VER	* Varies	This defines the version number of the CXL_Capability structure present. For the first generation, this field must be 0x1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x2 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE == 1) Testable: wire ? unconstrained : writeAsRead
15:0	CXL_RCRB_MEMBAR0_RAS_CAP_ID	* Varies	This defines the nature and format of the CXL_Capability register. For the CXL_RAS_Capability_Pointer register, this field should be 0x2. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x2 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE == 1) Testable: wire ? unconstrained : writeAsRead

3.62.3 MEMBAR0_SECURITY_CAP_REG_OFF

- **Name:** CXL Security Capability Header.
- **Description:** This register field has cap ID for Security Capability, it's version and the next capability pointer for Security Capability.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:20	CXL_RCRB_MEMBAR0_SEC_CAP_PTR
19:16	CXL_RCRB_MEMBAR0_SEC_CAP_VER
15:0	CXL_RCRB_MEMBAR0_SEC_CAP_ID

Table 3-541 Fields for Register: MEMBAR0_SECURITY_CAP_REG_OFF

Bits	Name	Memory Access	Description
31:20	CXL_RCRB_MEMBAR0_SEC_CAP_PTR	R	This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register. Value After Reset: 0x100 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_CXL_ENABLE && CX_MEMORY_MAP_POSITION==1) Testable: untestable Volatile: true

Bits	Name	Memory Access	Description
19:16	CXL_RCRB_MEMBAR0_SEC_CAP_VER	* Varies	This defines the version number of the CXL_Capability structure present. For the first generation, this field must be 0x1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_CXL_ENABLE && CX_MEMORY_MAP_POSITION==1) Testable: wire ? unconstrained : writeAsRead
15:0	CXL_RCRB_MEMBAR0_SEC_CAP_ID	* Varies	This defines the nature and format of the CXL_Capability register. For the CXL_Security_Capability_Pointer register, this field should be 0x3. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x3 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_CXL_ENABLE && CX_MEMORY_MAP_POSITION==1) Testable: wire ? unconstrained : writeAsRead

3.62.4 MEMBAR0_LINK_CAP_REG_OFF

- **Name:** CXL Link Capability Header.
- **Description:** This Register field has cap ID of Link Capability, it's version and the next capability pointer for Link Capability.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_CXL_ENABLE&& CX_MEMORY_MAP_PCIE_TYPE==1)

31:20	CXL_RCRB_MEMBAR0_LINK_CAP_PTR
19:16	CXL_RCRB_MEMBAR0_LINK_CAP_VER
15:0	CXL_RCRB_MEMBAR0_LINK_CAP_ID

Table 3-542 Fields for Register: MEMBAR0_LINK_CAP_REG_OFF

Bits	Name	Memory Access	Description
31:20	CXL_RCRB_MEMBAR0_LINK_CAP_PTR	R	This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register. Value After Reset: 0x110 Exists: (CX_CXL_ENABLE&& CX_MEMORY_MAP_PCIE_TYPE==1) Testable: untestable Volatile: true

Bits	Name	Memory Access	Description
19:16	CXL_RCRB_MEMBAR0_LINK_CAP_VER	* Varies	This defines the version number of the CXL_Capability structure present. For the first generation, this field must be 0x1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x2 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE == 1) Testable: wire ? unconstrained : writeAsRead
15:0	CXL_RCRB_MEMBAR0_LINK_CAP_ID	* Varies	This defines the nature and format of the CXL_Capability register. For the CXL_Link_Capability_Pointer register, this field should be 0x4. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x4 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE == 1) Testable: wire ? unconstrained : writeAsRead

3.63 PF0_MEMBAR0_RAS_CAP Registers

3.63.1 MEMBAR0_RAS_UNCOR_ERROR_STATUS_REG_OFF

- **Name:** CXL RCRB MEMBAR0 Uncorrectable Error Status Register.
- **Description:** Uncorrectable Error Status Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

RSVDP_17	31:17
CXL_RCRB_MEMBAR0_CXL_IDE_RX_ER	16
CXL_RCRB_MEMBAR0_CXL_IDE_TX_ER	15
CXL_RCRB_MEMBAR0_CXL_INTERNAL_ERR	14
RSVDP_12	13:12
CXL_RCRB_MEMBAR0_RCVR_OVERFLOW	11
CXL_RCRB_MEMBAR0_POISON_RVD	10
CXL_RCRB_MEMBAR0_ENCODED_VIOLATION	9
CXL_RCRB_MEMBAR0_REINIT_THRESHOLD	8
CXL_RCRB_MEMBAR0_MEM_DATA_ECC	7
CXL_RCRB_MEMBAR0_MEM_BE_PAR	6
CXL_RCRB_MEMBAR0_MEM_ADD_PAR	5
CXL_RCRB_MEMBAR0_MEM_DATA_PAR	4
CXL_RCRB_MEMBAR0_CACHE_DATA_ECC	3
CXL_RCRB_MEMBAR0_CACHE_BE_PAR	2
CXL_RCRB_MEMBAR0_CACHE_ADD_PAR	1
CXL_RCRB_MEMBAR0_CACHE_DATA_PAR	0

Table 3-543 Fields for Register: MEMBAR0_RAS_UNCOR_ERROR_STATUS_REG_OFF

Bits	Name	Memory Access	Description
31:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
16	CXL_RCRB_MEMBAR0_CXL_IDE_RX_ER	R/W1C	See Section 8.2.5.14.6 for the next level details for CXL IDE RX ERROR. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && CX_CXL_2_0_ENABLE Testable: writeAsRead Volatile: true
15	CXL_RCRB_MEMBAR0_CXL_IDE_TX_ER	R/W1C	See Section 8.2.5.14.6 for the next level details fro CXL IDE TX ERROR. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && CX_CXL_2_0_ENABLE Testable: writeAsRead Volatile: true
14	CXL_RCRB_MEMBAR0_CXL_INTERNAL_ERR	R/W1C	Internal_Error: Component specific error. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && CX_CXL_2_0_ENABLE Testable: writeAsRead Volatile: true
13:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
11	CXL_RCRB_MEMBAR0_RCVR_O VERFLOW	R/W1C	First 3b of the Header Log status are relevant as they also shows which buffer had the overflow and should be interpreted as the following. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
10	CXL_RCRB_MEMBAR0_POISON _RVD	R/W1C	Received Poison status from the peer. Header Log contains the entire flit received. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
9	CXL_RCRB_MEMBAR0_ENCOD VIOLATION	R/W1C	Received unrecognized encoding status. Header Log contains the entire flit received. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
8	CXL_RCRB_MEMBAR0_REINIT_ THRESHOLD	R/W1C	REINIT Threshold Hit status. Header Log not applicable. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
7	CXL_RCRB_MEMBAR0_MEM_DATA_ECC	R/W1C	Internal Data ECC error status on CXL.mem. Header Log contains M2S RxD Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
6	CXL_RCRB_MEMBAR0_MEM_BE_PAR	R/W1C	Internal Byte Enable Parity error status on CXL.mem. Header Log contains M2S RxD Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
5	CXL_RCRB_MEMBAR0_MEM_ADDR_PAR	R/W1C	Internal Address Parity error status on CXL.mem. If Bit 0 of Header Log is `0, rest of Header Log contains M2S Req. If Bit 0 of Header Log is `1, rest of Header Log contains M2S RxD Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
4	CXL_RCRB_MEMBAR0_MEM_DATA_PAR	R/W1C	Internal Data Parity error status on CXL.mem. Header Log contains M2S RxD Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
3	CXL_RCRB_MEMBAR0_CACHE_DATA_ECC	R/W1C	Internal Data ECC error status on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
2	CXL_RCRB_MEMBAR0_CACHE_BE_PAR	R/W1C	Internal Byte Enable Parity error status on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
1	CXL_RCRB_MEMBAR0_CACHE_ADD_PAR	R/W1C	Internal Address Parity error status on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
0	CXL_RCRB_MEMBAR0_CACHE_DATA_PAR	R/W1C	Internal Data Parity error status on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true

3.63.2 MEMBAR0_RAS_UNCOR_ERROR_MASK_REG_OFF

- **Name:** CXL RCRB MEMBAR0 Uncorrectable Error Mask Register.
- **Description:** Uncorrectable Error Mask Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

RSVDP_17	31:17
CXL_RCRB_MEMBAR0_CXL_IDE_RX_MASK	16
CXL_RCRB_MEMBAR0_CXL_IDE_TX_MASK	15
CXL_RCRB_MEMBAR0_CXL_INTERNAL_ER_MASK	14
RSVDP_12	13:12
CXL_RCRB_MEMBAR0_RCVR_OVERFLOW_MASK	11
CXL_RCRB_MEMBAR0_POISON_RVD_MASK	10
CXL_RCRB_MEMBAR0_ENCODED_VIOLATION_MASK	9
CXL_RCRB_MEMBAR0_REINIT_THRESHOLD_MASK	8
CXL_RCRB_MEMBAR0_MEM_DATA_ECC_MASK	7
CXL_RCRB_MEMBAR0_MEM_BE_PAR_MASK	6
CXL_RCRB_MEMBAR0_MEM_ADD_PAR_MASK	5
CXL_RCRB_MEMBAR0_MEM_DATA_PAR_MASK	4
CXL_RCRB_MEMBAR0_CACHE_DATA_ECC_MASK	3
CXL_RCRB_MEMBAR0_CACHE_BE_PAR_MASK	2
CXL_RCRB_MEMBAR0_CACHE_ADD_PAR_MASK	1
CXL_RCRB_MEMBAR0_CACHE_DATA_PAR_MASK	0

Table 3-544 Fields for Register: MEMBAR0_RAS_UNCOR_ERROR_MASK_REG_OFF

Bits	Name	Memory Access	Description
31:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
16	CXL_RCRB_MEMBAR0_CXL_IDE_RX_MASK	R/W	Cold reset Default is 1 for CXL IDE RX MASK. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
15	CXL_RCRB_MEMBAR0_CXL_IDE_TX_MASK	R/W	Cold reset Default is 1 for CXL IDE TX MASK. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
14	CXL_RCRB_MEMBAR0_CXL_INTERNAL_ER_MASK	R/W	Internal_Error Mask: Default value for this field is 1. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
13:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
11	CXL_RCRB_MEMBAR0_RCVR_OVERFLOW_MASK	R/W	First 3b of the Header Log mask are relevant as they also shows which buffer had the overflow and should be interpreted as the following. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
10	CXL_RCRB_MEMBAR0_POISON_RVD_MASK	R/W	Received Poison mask from the peer. Header Log contains the entire flit received. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
9	CXL_RCRB_MEMBAR0_ENCODE_VIOLATION_MASK	R/W	Received unrecognized encoding mask. Header Log contains the entire flit received. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
8	CXL_RCRB_MEMBAR0_REINIT_THRESHOLD_MASK	R/W	REINIT Threshold Hit mask. Header Log not applicable. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
7	CXL_RCRB_MEMBAR0_MEM_DATA_ECC_MASK	R/W	Internal Data ECC error mask on CXL.mem. Header Log contains M2S Rwd Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
6	CXL_RCRB_MEMBAR0_MEM_BYTE_ENABLE_PAR_MASK	R/W	Internal Byte Enable Parity error mask on CXL.mem. Header Log contains M2S Rwd Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
5	CXL_RCRB_MEMBAR0_MEM_ADDRESS_PAR_MASK	R/W	Internal Address Parity error mask on CXL.mem. If Bit 0 of Header Log is `0, rest of Header Log contains M2S Req. If Bit 0 of Header Log is `1, rest of Header Log contains M2S Rwd Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
4	CXL_RCRB_MEMBAR0_MEM_DATA_PAR_MASK	R/W	Internal Data Parity error mask on CXL.mem. Header Log contains M2S Rwd Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
3	CXL_RCRB_MEMBAR0_CACHE_DATA_ECC_MASK	R/W	Internal Data ECC error mask on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
2	CXL_RCRB_MEMBAR0_CACHE_BE_PAR_MASK	R/W	Internal Byte Enable Parity error mask on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
1	CXL_RCRB_MEMBAR0_CACHE_ADD_PAR_MASK	R/W	Internal Address Parity error mask on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
0	CXL_RCRB_MEMBAR0_CACHE_DATA_PAR_MASK	R/W	Internal Data Parity error mask on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

3.63.3 MEMBAR0_RAS_UNCOR_ERROR_SEVERITY_REG_OFF

- **Name:** CXL RCRB MEMBAR0 Uncorrectable Error Severity Register.
- **Description:** Uncorrectable Error Severity Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

RSVDP_17	31:17
CXL_RCRB_MEMBAR0_CXL_IDE_RX_SEV	16
CXL_RCRB_MEMBAR0_CXL_IDE_TX_SEV	15
CXL_RCRB_MEMBAR0_CXL_INTERNAL_ER_SEV	14
RSVDP_12	13:12
CXL_RCRB_MEMBAR0_RCVR_OVERFLOW_SEV	11
CXL_RCRB_MEMBAR0_POISON_RVD_SEV	10
CXL_RCRB_MEMBAR0_ENCODED_VIOLATION_SEV	9
CXL_RCRB_MEMBAR0_REINIT_THRESHOLD_SEV	8
CXL_RCRB_MEMBAR0_MEM_DATA_ECC_SEV	7
CXL_RCRB_MEMBAR0_MEM_BE_PAR_SEV	6
CXL_RCRB_MEMBAR0_MEM_ADD_PAR_SEV	5
CXL_RCRB_MEMBAR0_MEM_DATA_PAR_SEV	4
CXL_RCRB_MEMBAR0_CACHE_DATA_ECC_SEV	3
CXL_RCRB_MEMBAR0_CACHE_BE_PAR_SEV	2
CXL_RCRB_MEMBAR0_CACHE_ADD_PAR_SEV	1
CXL_RCRB_MEMBAR0_CACHE_DATA_PAR_SEV	0

Table 3-545 Fields for Register: MEMBAR0_RAS_UNCOR_ERROR_SEVERITY_REG_OFF

Bits	Name	Memory Access	Description
31:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
16	CXL_RCRB_MEMBAR0_CXL_IDE_RX_SEV	R/W	Cold reset Default is 1 for CXL IDE RX SEVERITY. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
15	CXL_RCRB_MEMBAR0_CXL_IDE_TX_SEV	R/W	Cold reset Default is 1 for CXL IDE TX SEVERITY. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
14	CXL_RCRB_MEMBAR0_CXL_INTERNAL_ER_SEV	R/W	Internal_Error Severity: Default value for this field is 1. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
13:12	RSVDP_12	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
11	CXL_RCRB_MEMBAR0_RCVR_OVERFLOW_SEV	R/W	First 3b of the Header Log severity are relevant as they also shows which buffer had the overflow and should be interpreted as the following. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
10	CXL_RCRB_MEMBAR0_POISON_RVD_SEV	R/W	Received Poison severity from the peer. Header Log contains the entire flit received. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
9	CXL_RCRB_MEMBAR0_ENCODING_VIOLATION_SEV	R/W	Received unrecognized encoding severity. Header Log contains the entire flit received. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
8	CXL_RCRB_MEMBAR0_REINIT_THRESHOLD_SEV	R/W	REINIT Threshold Hit severity. Header Log not applicable. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
7	CXL_RCRB_MEMBAR0_MEM_DATA_ECC_SEV	R/W	Internal Data ECC error severity on CXL.mem. Header Log contains M2S RxD Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
6	CXL_RCRB_MEMBAR0_MEM_BYTE_PAR_SEV	R/W	Internal Byte Enable Parity error severity on CXL.mem. Header Log contains M2S RxD Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
5	CXL_RCRB_MEMBAR0_MEM_ADDRESS_PAR_SEV	R/W	Internal Address Parity error severity on CXL.mem. If Bit 0 of Header Log is '0, rest of Header Log contains M2S Req. If Bit 0 of Header Log is 1, rest of Header Log contains M2S RxD Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
4	CXL_RCRB_MEMBAR0_MEM_DATA_PAR_SEV	R/W	Internal Data Parity error severity on CXL.mem. Header Log contains M2S RxD Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
3	CXL_RCRB_MEMBAR0_CACHE_DATA_ECC_SEV	R/W	Internal Data ECC error severity on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
2	CXL_RCRB_MEMBAR0_CACHE_BE_PAR_SEV	R/W	Internal Byte Enable Parity error severity on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
1	CXL_RCRB_MEMBAR0_CACHE_ADD_PAR_SEV	R/W	Internal Address Parity error severity on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
0	CXL_RCRB_MEMBAR0_CACHE_DATA_PAR_SEV	R/W	Internal Data Parity error severity on CXL.cache. Header Log contains H2D Data Header. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

3.63.4 MEMBAR0_RAS_CORR_ERROR_STATUS_REG_OFF

- **Name:** CXL RCRB MEMBAR0 Correctable Error Status Register.
- **Description:** Correctable Error Status Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:7	RSVDP_7
6	CXL_RCRB_MEMBAR0_PHY_LAYER_ERR
5	CXL_RCRB_MEMBAR0_MEM_POISON_RVD
4	CXL_RCRB_MEMBAR0_CACHE_POISON_RVD
3	CXL_RCRB_MEMBAR0_CORR_RETRY_THRESHOLD
2	CXL_RCRB_MEMBAR0_CORR_CRC_THRESHOLD
1	CXL_RCRB_MEMBAR0_CORR_MEM_DATA_ECC
0	CXL_RCRB_MEMBAR0_CORR_CACHE_DATA_ECC

Table 3-546 Fields for Register: MEMBAR0_RAS_CORR_ERROR_STATUS_REG_OFF

Bits	Name	Memory Access	Description
31:7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
6	CXL_RCRB_MEMBAR0_PHY_LAYER_ERR	R/W1C	Received error status indication from Physical Layer. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
5	CXL_RCRB_MEMBAR0_MEM_POISON_RVD	R/W1C	Received Poison status from the peer on CXL.mem. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
4	CXL_RCRB_MEMBAR0_CACHE_POISON_RVD	R/W1C	Received Poison status from the peer on CXL.cache. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
3	CXL_RCRB_MEMBAR0_CORR_RETRY_THRESHOLD	R/W1C	Retry Threshold Hit status. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
2	CXL_RCRB_MEMBAR0_CORR_CRC_THRESHOLD	R/W1C	CRC Threshold Hit status. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
1	CXL_RCRB_MEMBAR0_CORR_MEM_DATA_ECC	R/W1C	Internal Data ECC error status on CXL.mem. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
0	CXL_RCRB_MEMBAR0_CORR_CACHE_DATA_ECC	R/W1C	Internal Data ECC error status on CXL.cache. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true

3.63.5 MEMBAR0_RAS_CORR_ERROR_MASK_REG_OFF

- **Name:** CXL RCRB MEMBAR0 Correctable Error Mask Register.
- **Description:** Correctable Error Mask Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:7	RSVDP_7
6	CXL_RCRB_MEMBAR0_PHY_LAYER_ERR_MASK
5	CXL_RCRB_MEMBAR0_MEM_POISON_RVD_MASK
4	CXL_RCRB_MEMBAR0_CACHE_POISON_RVD_MASK
3	CXL_RCRB_MEMBAR0_CORR_RETRY_THRESHOLD_MASK
2	CXL_RCRB_MEMBAR0_CORR_CRC_THRESHOLD_MASK
1	CXL_RCRB_MEMBAR0_CORR_MEM_DATA_ECC_MASK
0	CXL_RCRB_MEMBAR0_CORR_CACHE_DATA_ECC_MASK

Table 3-547 Fields for Register: MEMBAR0_RAS_CORR_ERROR_MASK_REG_OFF

Bits	Name	Memory Access	Description
31:7	RSVDP_7	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
6	CXL_RCRB_MEMBAR0_PHY_LAYER_ERR_MASK	R/W	Received error mask indication from Physical Layer. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
5	CXL_RCRB_MEMBAR0_MEM_POISON_RVD_MASK	R/W	Received Poison mask from the peer on CXL.mem. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
4	CXL_RCRB_MEMBAR0_CACHE_POISON_RVD_MASK	R/W	Received Poison mask from the peer on CXL.cache. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
3	CXL_RCRB_MEMBAR0_CORR_RETRY_THRESHOLD_MASK	R/W	Retry Threshold Hit mask. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
2	CXL_RCRB_MEMBAR0_CORR_CRC_THRESHOLD_MASK	R/W	CRC Threshold Hit mask. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
1	CXL_RCRB_MEMBAR0_CORR_MEM_DATA_ECC_MASK	R/W	Internal Data ECC error mask on CXL.mem. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
0	CXL_RCRB_MEMBAR0_CORR_CACHE_DATA_ECC_MASK	R/W	Internal Data ECC error mask on CXL.cache. <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

3.63.6 MEMBAR0_RAS_ERR_CAP_CNTRL_REG_OFF

- **Name:** CXL RCRB MEMBAR0 ERROR CAPABILITIES AND CONTROL STATUS.
- **Description:** Error Capabilities and Control Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:14	RSVDP_14
13	CXL_RCRB_MEMBAR0_POISON_END
12:10	RSVDP_10
9	CXL_RCRB_MEMBAR0_MUL_HDR_RECORDING_CAP
8:6	RSVDP_6
5:0	CXL_RCRB_MEMBAR0_ERR_PTR

Table 3-548 Fields for Register: MEMBAR0_RAS_ERR_CAP_CNTRL_REG_OFF

Bits	Name	Memory Access	Description
31:14	RSVDP_14	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
13	CXL_RCRB_MEMBAR0_POISON_END	* Varies	If this bit is 0, CXL 1.1 Upstream Ports, CXL 1.1 Downstream Ports and CXL 2.0 Root Port shall treat poison received on .cache or .mem as uncorrectable error and log the error in Uncorrectable Error Status Register. If this bit is 1, these ports shall treat poison received on .cache or .mem as correctable error and log the error in Correctable Error Status Register. This bit defaults to 1. This bit is hardwired to 1 in CXL 2.0 Upstream Switch Port, CXL 2.0 Downstream Switch Port and CXL 2.0 device. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W (sticky) ■ Dbi: R/W (sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x1 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Memory Access: "(CX_MEMORY_MAP_POSITION==0 && CX_CXL_2_0_ENABLE) ? \"read-only\" : \"read-write\" "
12:10	RSVDP_10	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
9	CXL_RCRB_MEMBAR0_MUL_HD_R_RECORDING_CAP	R	This indicates if recording more than one error header is supported. For the first generation, this will be set to 'b0'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
8:6	RSVDP_6	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
5:0	CXL_RCRB_MEMBAR0_ERR_PT R	* Varies	This identifies the bit position of the first error reported in the Uncorrectable Error Status register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: untestable

3.63.7 MEMBAR0_RAS_HEADER_LOG_REG_00_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 18H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 00.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)



Table 3-549 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_00_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_00	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 00. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.8 MEMBAR0_RAS_HEADER_LOG_REG_01_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 1CH.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 01.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_HDR_LOG_01

31:0

Table 3-550 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_01_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_01	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 01. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.9 MEMBAR0_RAS_HEADER_LOG_REG_02_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 20H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 02.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)



Table 3-551 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_02_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_02	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 02. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.10 MEMBAR0_RAS_HEADER_LOG_REG_03_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 24H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 03.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_HDR_LOG_03

31:0

Table 3-552 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_03_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_03	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 03. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.11 MEMBAR0_RAS_HEADER_LOG_REG_04_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 28H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 04.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_HDR_LOG_04

31:0

Table 3-553 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_04_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_04	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 04. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.12 MEMBAR0_RAS_HEADER_LOG_REG_05_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 2CH.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 05.
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_HDR_LOG_05

31:0

Table 3-554 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_05_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_05	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 05. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.13 MEMBAR0_RAS_HEADER_LOG_REG_06_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 30H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 06.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_HDR_LOG_06

31:0

Table 3-555 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_06_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_06	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 06. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.14 MEMBAR0_RAS_HEADER_LOG_REG_07_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 34H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 07.
- **Size:** 32 bits
- **Offset:** B+0x34
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_HDR_LOG_07

31:0

Table 3-556 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_07_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_07	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 07. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.15 MEMBAR0_RAS_HEADER_LOG_REG_08_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 38H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 08.
- **Size:** 32 bits
- **Offset:** B+0x38
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_HDR_LOG_08

31:0

Table 3-557 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_08_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_08	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 08. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.16 MEMBAR0_RAS_HEADER_LOG_REG_09_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 3CH.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 09.
- **Size:** 32 bits
- **Offset:** B+0x3c
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)



Table 3-558 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_09_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_09	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 09. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.17 MEMBAR0_RAS_HEADER_LOG_REG_10_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 40H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 10.
- **Size:** 32 bits
- **Offset:** B+0x40
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

**Table 3-559 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_10_OFF**

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_10	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 10. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.18 MEMBAR0_RAS_HEADER_LOG_REG_11_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 44H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 11.
- **Size:** 32 bits
- **Offset:** B+0x44
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:0
CXL_RCRB_MEMBAR0_HDR_LOG_11

Table 3-560 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_11_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_11	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 11. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.19 MEMBAR0_RAS_HEADER_LOG_REG_12_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 48H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 12.
- **Size:** 32 bits
- **Offset:** B+0x48
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_HDR_LOG_12

31:0

Table 3-561 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_12_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_12	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 12. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.20 MEMBAR0_RAS_HEADER_LOG_REG_13_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 4CH.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 13.
- **Size:** 32 bits
- **Offset:** B+0x4c
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_HDR_LOG_13

31:0

Table 3-562 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_13_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_13	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 13. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.21 MEMBAR0_RAS_HEADER_LOG_REG_14_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 50H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 14.
- **Size:** 32 bits
- **Offset:** B+0x50
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

**Table 3-563 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_14_OFF**

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_14	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 14. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.63.22 MEMBAR0_RAS_HEADER_LOG_REG_15_OFF

- **Name:** CXL RCRB MEMBAR0 HEADER LOG REGISTER 54H.
- **Description:** Header Log for recorded Uncorrectable Error Status bit 15.
- **Size:** 32 bits
- **Offset:** B+0x54
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_HDR_LOG_15

31:0

Table 3-564 Fields for Register: MEMBAR0_RAS_HEADER_LOG_REG_15_OFF

Bits	Name	Memory Access	Description
31:0	CXL_RCRB_MEMBAR0_HDR_LOG_15	* Varies	The information logged here depends on the type of Uncorrectable Error Status bit 15. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead

3.64 PF0_MEMBAR0_LINK_CAP Registers

3.64.1 MEMBAR0_LINK_LAYER_CAP_REG_OFF

- **Name:** CXL LINK LAYER CAPABILITY REG.
- **Description:** This register contains the lower 32-bits of the 64-bit CXL Link Layer Capability Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:29	CXL_RCRB_MEMBAR0_NUM_PHY_REINIT_RCVD
28:24	CXL_RCRB_MEMBAR0_NUM_RETRY_RCVD
23:16	CXL_RCRB_MEMBAR0_LLR_VAL_RCVD
15:8	CXL_RCRB_MEMBAR0_LLR_VAL_SUPP
7:4	CXL_RCRB_MEMBAR0_LNK_VER_RCVD
3:0	CXL_RCRB_MEMBAR0_LNK_VER_SUPP

Table 3-565 Fields for Register: MEMBAR0_LINK_LAYER_CAP_REG_OFF

Bits	Name	Memory Access	Description
31:29	CXL_RCRB_MEMBAR0_NUM_PHY_REINIT_RCVD	R	Num_Phy_Reinit value reflected in the last Retry for bit [29:31].Req message received. Used for debug. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
28:24	CXL_RCRB_MEMBAR0_NUM_RETRY_RCVD	R	Num_Retry value reflected in the last Retry.Req message received. Used for debug. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
23:16	CXL_RCRB_MEMBAR0_LLRR_VAL_RCVD	R	LLR Wrap value received from INIT.Param flit. Used for debug. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
15:8	CXL_RCRB_MEMBAR0_LLRR_VAL_SUPP	R/W	LLR Wrap value supported by this entity. Used for debug. <i>Note:</i> This register field is sticky. Value After Reset: 0xff Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
7:4	CXL_RCRB_MEMBAR0_LNK_VERSION_RCVD	R	Version of AL received from INIT.Param flit. Used for debug. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
3:0	CXL_RCRB_MEMBAR0_LNK_VERSION_SUPP	R/W	Version of AL the port is compliant with. For CXL 1.0, this should be 'b001. <i>Note:</i> This register field is sticky. Value After Reset: "(CX_CXL_2_0_ENABLE) ? 2 : 1" Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

3.64.2 MEMBAR0_LINK_LAYER_CAP_2_REG_OFF

- **Name:** CXL LINK LAYER CAPABILITY Second REG.
- **Description:** This register contains the upper 32-bits of the 64-bit CXL Link Layer Capability Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x4
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:27	RSVDP_27
26	CXL_RCRB_MEMBAR0_NO_LL_RESET_SUPPORT
25:18	CXL_RCRB_MEMBAR0_NUM_FREE_BUF_RCVD
17:10	CXL_RCRB_MEMBAR0_ECHO_ESEQ_RCVD
9:2	CXL_RCRB_MEMBAR0_WR_PTR_RCVD
1:0	CXL_RCRB_MEMBAR0_NUM_PHY_REINIT_RCVD_2

Table 3-566 Fields for Register: MEMBAR0_LINK_LAYER_CAP_2_REG_OFF

Bits	Name	Memory Access	Description
31:27	RSVDP_27	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
26	CXL_RCRB_MEMBAR0_NO_LL_RESET_SUPPORT	* Varies	No_LL_Reset_Support: If set, indicates that the LL_Reset configuration bit is not supported. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: wire ? unconstrained : writeAsRead
25:18	CXL_RCRB_MEMBAR0_NUM_FREE_BUF_RCVD	R	Num_Free_Buf value reflected in the last Retry.Ack message received. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
17:10	CXL_RCRB_MEMBAR0_ECHO_ESEQ_RCVD	R	Echo_Eseq value reflected in the last Retry.Ack message received. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
9:2	CXL_RCRB_MEMBAR0_WR_PTR_RCVD	R	Wr_Ptr value reflected in the last Retry.Ack message received. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
1:0	CXL_RCRB_MEMBAR0_NUM_PHY_REINIT_RCVD_2	R	Num_Phy_Reinit value reflected in the last Retry for bit [0:1].Req message received. Used for debug. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true

3.64.3 MEMBAR0_LINK_CNTRL_STATUS_REG_OFF

- **Name:** CXL LINK LAYER CONTROL AND STATUS REG.
- **Description:** CXL Link Layer Control and Status Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:13	RSVDP_13
12:5	CXL_RCRB_MEMBAR0_LINK_LL_RETRY_BUF_CONSUME
4:3	CXL_RCRB_MEMBAR0_LINK_INIT_STATE
2	CXL_RCRB_MEMBAR0_LINK_LL_CRD_STALL
1	CXL_RCRB_MEMBAR0_LINK_LL_INIT_STALL
0	CXL_RCRB_MEMBAR0_LINK_LL_RST

Table 3-567 Fields for Register: MEMBAR0_LINK_CNTRL_STATUS_REG_OFF

Bits	Name	Memory Access	Description
31:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
12:5	CXL_RCRB_MEMBAR0_LINK_LL_RETRY_BUF_CONSUME	R	Snapshot of link layer retry buffer consumed. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
4:3	CXL_RCRB_MEMBAR0_LINK_INIT_STATE	R	This field reflects the current initialization status of the Link Layer, including any stall conditions controlled by bits 2:1. Values: 0x0 (NOT_RDY_FOR_INIT): LLCTRLINIT Param flit not sent. 0x1 (PARAM_EX): LCTRL-INIT Param sent and waiting to receive it. 0x2 (CRD_RETURN_STALL): RETURNSTAL Parameter exchanged successfully and Credit return is stalled. 0x3 (INIT_DONE): LCTRLINIT Param flit sent and received, and initial credit refund not stalled. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true
2	CXL_RCRB_MEMBAR0_LINK_LL_CRD_STALL	R/W	If set, link layer stalls credit initialization until this bit is cleared. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
1	CXL_RCRB_MEMBAR0_LINK_LL_INIT_STALL	R/W	If set, link layer stalls the transmission of the LLCTRL-INIT.Param flit until this bit is cleared. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
0	CXL_RCRB_MEMBAR0_LINK_LL_RST	R/W	Re-initialize without resetting values in sticky registers. Write 1 to reset link - this is a destructive reset all link layer state. When link layer reset completes, HW will clear the bit to '0. Entity triggering soft reset should ensure that link is quiesced. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) Testable: writeAsRead Volatile: true

3.64.4 MEMBAR0_LINK_RX_CR_CNTRL_REG_OFF

- **Name:** CXL LINK RX CREDIT CONTROL REG.
- **Description:** This register contains the lower 32-bits of the 64-bit CXL Link Layer Rx Credit Control Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

CXL_RCRB_MEMBAR0_LINK_CTL_MEM_REQ_RSP_CRD	31:30
CXL_RCRB_MEMBAR0_LINK_CTL_CACHE_DATA_CRD	29:20
CXL_RCRB_MEMBAR0_LINK_CTL_CACHE_RSP_CRD	19:10
CXL_RCRB_MEMBAR0_LINK_CTL_CACHE_REQ_CRD	9:0

Table 3-568 Fields for Register: MEMBAR0_LINK_RX_CR_CNTRL_REG_OFF

Bits	Name	Memory Access	Description
31:30	CXL_RCRB_MEMBAR0_LINK_CTL_MEM_REQ_RSP_CRD	R/W	Credits to advertise for Mem Request or Response channel at init for bits [30:31]. <i>Note:</i> This register field is sticky. Value After Reset: { (CX_CXL_CHI_ENABLE && CX_CXL_CXS_ENABLE==0) ? (CX_MEMORY_MAP_VIEW==0 ? ((CX_CXL_MEM_RSP_DP == 4) ? 3 : 0) : ((CX_CXL_MEM_REQ_DP == 4) ? 3 : 0)) : "0x0" } Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CHI_ENABLE)
29:20	CXL_RCRB_MEMBAR0_LINK_CTL_CACHE_DATA_CRD	R/W	Credits to advertise for Cache Data channel at init. <i>Note:</i> This register field is sticky. Value After Reset: { (CX_CXL_CHI_ENABLE && CX_CXL_CXS_ENABLE==0) ? ((CX_CXL_CACHE_DQ_DP == 4) ? 1023 : 2**(6+ CX_CXL_CACHE_DQ_DP)) : "0x0" } Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CHI_ENABLE)
19:10	CXL_RCRB_MEMBAR0_LINK_CTL_CACHE_RSP_CRD	R/W	Credits to advertise for Cache Response channel at init. <i>Note:</i> This register field is sticky. Value After Reset: { (CX_CXL_CHI_ENABLE && CX_CXL_CXS_ENABLE==0) ? ((CX_CXL_CACHE_RSP_DP == 4) ? 1023 : 2**(6+ CX_CXL_CACHE_RSP_DP)) : "0x0" } Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CHI_ENABLE)
9:0	CXL_RCRB_MEMBAR0_LINK_CTL_CACHE_REQ_CRD	R/W	Credits to advertise for Cache Request channel at init. <i>Note:</i> This register field is sticky. Value After Reset: { (CX_CXL_CXS_ENABLE==0 && CX_CXL_CHI_ENABLE) ? (CX_MEMORY_MAP_VIEW==0 ? ((CX_CXL_CACHE_REQ_DP == 4) ? 1023 : 2**(6+ CX_CXL_CACHE_REQ_DP)) : ((CX_CXL_CACHE_REQ_DP == 4) ? 1023 : 2**(6+ CX_CXL_CACHE_REQ_DP))) : "0x0" } Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CHI_ENABLE)

3.64.5 MEMBAR0_LINK_RX_CR_CNTRL_2_REG_OFF

- **Name:** CXL LINK RX CREDIT CONTROL Second REG.
- **Description:** This register contains the upper 32-bits of the 64-bit CXL Link Layer Rx Credit Control Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:18	RSVDP_18
17:8	CXL_RCRB_MEMBAR0_LINK_CTL_MEM_DATA_CRD
7:0	CXL_RCRB_MEMBAR0_LINK_CTL_MEM_REQ_RSP_CRD_2

Table 3-569 Fields for Register: MEMBAR0_LINK_RX_CR_CNTRL_2_REG_OFF

Bits	Name	Memory Access	Description
31:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CHI_ENABLE)

Bits	Name	Memory Access	Description
17:8	CXL_RCRB_MEMBAR0_LINK_CTL_MEM_DATA_CRD	R/W	Credits to advertise for Mem Data channel at init. <i>Note:</i> This register field is sticky. Value After Reset: { (CX_CXL_CHI_ENABLE && CX_CXL_CXS_ENABLE==0) ? ((CX_CXL_MEM_DQ_DP == 4) ? 1023 : 2**(6 + CX_CXL_MEM_DQ_DP)) : "0x0" } Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CHI_ENABLE)
7:0	CXL_RCRB_MEMBAR0_LINK_CTL_MEM_REQ_RSP_CRD_2	R/W	Credits to advertise for Mem Request or Response channel at init for bits [7:0]. <i>Note:</i> This register field is sticky. Value After Reset: { (CX_CXL_CHI_ENABLE && CX_CXL_CXS_ENABLE==0) ? (CX_MEMORY_MAP_VIEW==0 ? ((CX_CXL_MEM_RSP_DP == 4) ? 255 : 2**(4+ CX_CXL_MEM_RSP_DP)) : ((CX_CXL_MEM_REQ_DP == 4) ? 255 : 2**(4+ CX_CXL_MEM_REQ_DP))) : "0x0" } Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CHI_ENABLE)

3.64.6 MEMBAR0_LINK_RX_CR_RTN_STATUS_REG_OFF

- **Name:** CXL LINK RX CREDIT STATUS REG.
- **Description:** This register contains the lower 32-bits of the 64-bit CXL Link Layer Rx Credit Return Status Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:30	CXL_RCRB_MEMBAR0_LINK_ST_MEM_REQ_RSP_CRD
29:20	CXL_RCRB_MEMBAR0_LINK_ST_CACHE_DATA_CRD
19:10	CXL_RCRB_MEMBAR0_LINK_ST_CACHE_RSP_CRD
9:0	CXL_RCRB_MEMBAR0_LINK_ST_CACHE_REQ_CRD

Table 3-570 Fields for Register: MEMBAR0_LINK_RX_CR_RTN_STATUS_REG_OFF

Bits	Name	Memory Access	Description
31:30	CXL_RCRB_MEMBAR0_LINK_ST_MEM_REQ_RSP_CRD	R	Running snapshot of accumulated Mem Request or Response credits to be returned for bits [31:30]. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
29:20	CXL_RCRB_MEMBAR0_LINK_ST_CACHE_DATA_CRD	R	Running snapshot of accumulated Cache Data credits to be returned. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true
19:10	CXL_RCRB_MEMBAR0_LINK_ST_CACHE_RSP_CRD	R	Running snapshot of accumulated Cache Response credits to be returned. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true
9:0	CXL_RCRB_MEMBAR0_LINK_ST_CACHE_REQ_CRD	* Varies	Running snapshot of accumulated Cache Request credits to be returned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R ■ Dbi: R Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true Memory Access: "(CX_CXL_CXS_ENABLE==0 && CX_CXL_CHI_ENABLE) ? \"read-only\" : \"read-only\" "

3.64.7 MEMBAR0_LINK_RX_CR_RTN_STATUS_2_REG_OFF

- **Name:** CXL LINK RX CREDIT STATUS Second REG.
- **Description:** This register contains the upper 32-bits of the 64-bit CXL Link Layer Rx Credit Return Status Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:18	RSVDP_18
17:8	CXL_RCRB_MEMBAR0_LINK_ST_MEM_DATA_CRD
7:0	CXL_RCRB_MEMBAR0_LINK_ST_MEM_REQ_RSP_CRD_2

Table 3-571 Fields for Register: MEMBAR0_LINK_RX_CR_RTN_STATUS_2_REG_OFF

Bits	Name	Memory Access	Description
31:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0)

Bits	Name	Memory Access	Description
17:8	CXL_RCRB_MEMBAR0_LINK_ST_MEM_DATA_CRD	R	Running snapshot of accumulated Mem Data credits to be returned. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true
7:0	CXL_RCRB_MEMBAR0_LINK_ST_MEM_REQ_RSP_CRD_2	R	Running snapshot of accumulated Mem Request or Response credits to be returned for bits [7:0]. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true

3.64.8 MEMBAR0_LINK_TX_CR_STATUS_REG_OFF

- **Name:** CXL LAYER TX CREDIT STATUS.
- **Description:** This register contains the lower 32-bits of the 64-bit CXL Link Layer Tx Credit Status Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x20
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:30	CXL_RCRB_MEMBAR0_LINK_TX_MEM_REQ_RSP_CRD
29:20	CXL_RCRB_MEMBAR0_LINK_TX_CACHE_DATA_CRD
19:10	CXL_RCRB_MEMBAR0_LINK_TX_CACHE_RSP_CRD
9:0	CXL_RCRB_MEMBAR0_LINK_TX_CACHE_REQ_CRD

Table 3-572 Fields for Register: MEMBAR0_LINK_TX_CR_STATUS_REG_OFF

Bits	Name	Memory Access	Description
31:30	CXL_RCRB_MEMBAR0_LINK_TX_MEM_REQ_RSP_CRD	R	Running snapshot of Mem Req or Response credits for Tx for bits [31:30]. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
29:20	CXL_RCRB_MEMBAR0_LINK_TX_CACHE_DATA_CRD	R	Running snapshot of Cache Data credits for Tx. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true
19:10	CXL_RCRB_MEMBAR0_LINK_TX_CACHE_RSP_CRD	R	Running snapshot of Cache Response credits for Tx. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true
9:0	CXL_RCRB_MEMBAR0_LINK_TX_CACHE_REQ_CRD	R	Running snapshot of Cache Request credits for Tx. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true

3.64.9 MEMBAR0_LINK_TX_CR_STATUS_2_REG_OFF

- **Name:** CXL LAYER TX CREDIT STATUS Second REG.
- **Description:** This register contains the upper 32-bits of the 64-bit CXL Link Layer Tx Credit Status Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x24
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:18	RSVDP_18
17:8	CXL_RCRB_MEMBAR0_LINK_TX_MEM_DATA_CRD
7:0	CXL_RCRB_MEMBAR0_LINK_TX_MEM_REQ_RSP_CRD_2

Table 3-573 Fields for Register: MEMBAR0_LINK_TX_CR_STATUS_2_REG_OFF

Bits	Name	Memory Access	Description
31:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0)

Bits	Name	Memory Access	Description
17:8	CXL_RCRB_MEMBAR0_LINK_TX_MEM_DATA_CRD	R	Running snapshot of Mem Data credits for Tx. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true
7:0	CXL_RCRB_MEMBAR0_LINK_TX_MEM_REQ_RSP_CRD_2	R	Running snapshot of Mem Req or Response credits for Tx for bits [7:0]. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CXS_ENABLE==0) Testable: writeAsRead Volatile: true

3.64.10 MEMBAR0_LINK_ACK_TMR_CNTRL_REG_OFF

- **Name:** CXL LINK ACK TIMER CONTROL REG.
- **Description:** CXL Link Layer Ack Timer Control Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x28
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:18	RSVDP_18
17:8	CXL_RCRB_MEMBAR0_LINK_ACK_FLUSH_RETIMER
7:0	CXL_RCRB_MEMBAR0_LINK_ACK_FRC_THRESHOLD

Table 3-574 Fields for Register: MEMBAR0_LINK_ACK_TMR_CNTRL_REG_OFF

Bits	Name	Memory Access	Description
31:18	RSVDP_18	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
17:8	CXL_RCRB_MEMBAR0_LINK_ACK_FLUSH_RETIMER	R/W	This specifies how many link layer clock cycles the entity should wait in case of idle, before flushing accumulated Acks using a LLCRD. This applies for any case where accumulated Acks is greater than 1 or accumulated CRD for any channel is greater than 0. The recommended default/minimum value is 0x20. <i>Note:</i> This register field is sticky. Value After Reset: 0x20 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
7:0	CXL_RCRB_MEMBAR0_LINK_ACK_FRC_THRESHOLD	R/W	This specifies how many Flit Acks the Link Layer should accumulate before injecting a LLCRD. <i>Note:</i> This register field is sticky. Value After Reset: 0x10 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

3.64.11 MEMBAR0_LINK_DEFEATURE_REG_OFF

- **Name:** CXL LINK DEFEATURE REG.
- **Description:** CXL Link Layer Defeature Register defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:1	
0	
RSVDP_1	CXL_RCRB_MEMBAR0_LINK_MDH_DISABLE

Table 3-575 Fields for Register: MEMBAR0_LINK_DEFEATURE_REG_OFF

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CHI_ENABLE)

Bits	Name	Memory Access	Description
0	CXL_RCRB_MEMBAR0_LINK_MDH_DISABLE	R/W	Write '1 to disable MDH. Software needs to ensure it programs this value consistently on the UP & DP. After programming, a warm reset is required for the disable to take effect. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1) && (CX_MEMORY_MAP_PCIE_TYPE==1 && CX_CXL_CHI_ENABLE)

3.65 PF0_MEMBAR0_ARB_MUX_CAP Registers

3.65.1 MEMBAR0_ARB_MUX_CXL_IO_REG_OFF

- **Name:** ARB/MUX Arbitration Control Register for CXL.io.
- **Description:** This register contains the lower 32-bits of the 64-bit ARB/MUX Arbitration Control Register for CXL.io defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x0
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:8	7:4	3:0
RSVDP_8	CXL_RCRB_MEMBAR0_ARB_MUX_CXL_IO_RND_ROBIN	RSVDP_0

Table 3-576 Fields for Register: MEMBAR0_ARB_MUX_CXL_IO_REG_OFF

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
7:4	CXL_RCRB_MEMBAR0_ARB_MU X_CXL_IO_RND_ROBIN	R/W	CXL.io Weighted Round Robin Arbitration Weight. This is the weight assigned to CXL.io in the weighted round robin arbitration between CXL protocols. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

3.65.2 MEMBAR0_ARB_MUX_CXL_CACHE_MEM_REG_OFF

- **Name:** ARB/MUX Arbitration Control Register for CXL.cache and CXL.mem.
- **Description:** This register contains the lower 32-bits of the 64-bit ARB/MUX Arbitration Control Register for CXL.cache and CXL.mem defined in the CXL Specification.
- **Size:** 32 bits
- **Offset:** B+0x40
- **Exists:** (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

31:8	RSVDP_8
7:4	CXL_RCRB_MEMBAR0_ARB_MUX_CXL_CACHE_MEM_RND_ROBIN
3:0	RSVDP_0

Table 3-577 Fields for Register: MEMBAR0_ARB_MUX_CXL_CACHE_MEM_REG_OFF

Bits	Name	Memory Access	Description
31:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

Bits	Name	Memory Access	Description
7:4	CXL_RCRB_MEMBAR0_ARB_MU X_CXL_CACHE_MEM_RND_ROB IN	R/W	CXL.cache and CXL.mem Weighted Round Robin Arbitration Weight. This is the weight assigned to CXL.cache and CXL.mem in the weighted round robin arbitration between CXL protocols. Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)
3:0	RSVDP_0	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CX_CXL_ENABLE && CX_MEMORY_MAP_PCIE_TYPE==1)

3.66 PF0_ATU_CAP Registers

3.66.1 IATU_REGION_CTRL_1_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Region Control 1 Register.
- **Description:** This register controls the iATU outbound region access based on the optional iATU outbound features enabled using iATU Region Control 2 Register.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x0
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

CTRL_1_FUNC_NUM	x:20
PH	19:18
AT	17:16
Rsvd	15:14
INCREASE_REGION_SIZE	13
TH	12
IDO	11
ATTR	10:9
TD	8
TC	7:5
TYPE	4:0

Table 3-578 Fields for Register: IATU_REGION_CTRL_1_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:20	CTRL_1_FUNC_NUM	R/W	Function Number. When the address of an outbound TLP is matched to this region and the FUNC_BYPASS field in the "iATU Region Control 2 Register" is '0', then the function number used in generating the function part of the requester ID (RID) field of the TLP is taken from this 5-bit register. The value in this register must be 0x0 unless multifunction operation in the controller is enabled (CX_NFUNC > 1). When you are using the AXI Bridge, then this field is swapped before AXI decomposition occurs so that the correct "Max_Read_Request_Size" and "Max_Payload_Size" values are used. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0) Range Variable[x]: (CX_NFUNC_WD) + 19
19:18	PH	R/W	When the address of an outbound TLP is matched to this region, then the PH field of the TLP is changed to the value in this register. The setting is independent of the setting of the TH field. Only Valid when the CX_TPH_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_TPH_ENABLE && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
17:16	AT	R/W	AT. When the address of an outbound TLP is matched to this region, then the AT field of the TLP is changed to the value in this register. Only valid when the ATS_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && ATS_TX_ENABLE && CC_UNROLL_ENABLE) && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
15:14			Reserved Field: Yes
13	INCREASE_REGION_SIZE	R/W	Increase the maximum ATU Region size. When set, the maximum ATU Region size is determined by CX_ATU_MAX_REGION_SIZE. When clear, the maximum ATU Region size is 4 GB (default). <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Maximum ATU Region size is determined by CX_ATU_MAX_REGION_SIZE ■ 0x0 (DISABLE): Maximum ATU Region size is 4 GB (default) Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
12	TH	R/W	When the address of an outbound TLP is matched to this region, then the TH field of the TLP is changed to the value in this register. Only Valid when the CX_TPH_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_TPH_ENABLE && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
11	IDO	R/W	When the address of an outbound TLP is matched to this region, then the IDO(ATTR[2]) field of the TLP is changed to the value in this register. Only valid when the CX_IDO_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_IDO_ENABLE && CC_DEVICE_TYPE!=3 && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
10:9	ATTR	R/W	When the address of an outbound TLP is matched to this region, then the ATTR field of the TLP is changed to the value in this register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
8	TD	R/W	This is a reserved field. Do not use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
7:5	TC	R/W	When the address of an outbound TLP is matched to this region, then the TC field of the TLP is changed to the value in this register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
4:0	TYPE	R/W	When the address of an outbound TLP is matched to this region, then the TYPE field of the TLP is changed to the value in this register. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

3.66.2 IATU_REGION_CTRL_2_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Region Control 2 Register.
- **Description:** Using this register you can enable/disable the outbound iATU optional features.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x4
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

31	REGION_EN
30	Rsvd
29	INVERT_MODE
28	CFG_SHIFT_MODE
27	DMA_BYPASS
26:24	Rsvd
23	HEADER_SUBSTITUTE_EN
22	INHIBIT_PAYLOAD
21	TLP_HEADER_FIELDS_BYPASS
20	SNP
19	FUNC_BYPASS
18:17	MSB2BITS_TAG
16	TAG_SUBSTITUTE_EN
15:8	TAG
7:0	MSG_CODE

Table 3-579 Fields for Register: IATU_REGION_CTRL_2_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31	REGION_EN	R/W	Region Enable. This bit must be set to '1' for address translation to take place. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
30			Reserved Field: Yes

Bits	Name	Memory Access	Description
29	INVERT_MODE	R/W	Invert Mode. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
28	CFG_SHIFT_MODE	R/W	CFG Shift Mode. The iATU uses bits [27:12] of the untranslated address (on the XALI0/1/2 interface or AXI slave interface address) to form the BDF number of the outgoing CFG TLP. This supports the Enhanced Configuration Address Mapping (ECAM) mechanism (Section 7.2.2 of the PCI Express Base 3.1 Specification, revision 1.0) by allowing all outgoing I/O and MEM TLPs (that have been translated to CFG) to be mapped from memory space into any 256 MB region of the PCIe configuration space. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
27	DMA_BYPASS	R/W	DMA Bypass Mode. Allows request TLPs which are initiated by the DMA controller to pass through the iATU untranslated. <i>Note:</i> This field is reserved for the SW product. You must set it to '0'. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_DEVICE_TYPE!=3 && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
26:24			Reserved Field: Yes
23	HEADER_SUBSTITUTE_EN	R/W	Header Substitute Enable. When enabled and region address is matched, the iATU fully substitutes bytes 8-11 (for 3 DWORD header) or bytes 12-15 (for 4 DWORD header) of the outbound TLP header with the contents of the LWR_TARGET_RW field in IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i. Encodings are as above. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (FILL_BYTES): LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i register is used to fill bytes 8-to-11 (for 3 DWORD header) or bytes 12-to-15 (for 4 DWORD header) of the translated TLP header. ■ 0x0 (FORM_ADDR): LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i register forms the new address of the translated region. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
22	INHIBIT_PAYLOAD	R/W	Inhibit TLP Payload Data for TLP's in Matched Region; assign iATU region to be TLP without data. When enabled and region address is matched, the iATU marks all TLPs as having no payload data by forcing the TLP header Fmt[1] bit =0, regardless of the application inputs such as slv_wstrb. Encoding are define as above. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (NO_SUPPORT_FOR_DATA_TLP): Fmt[1] =0 so that only TLP type without data is sent. For example, a MsgD instead of MsgD will be sent. ■ 0x0 (SUPPORT_DATA_TLP): Fmt[1] =0/1 so that TLPs with or without data can be sent. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
21	TLP_HEADER_FIELDS_BYPASS	R/W	TLP Header Fields Translation Bypass. In this mode header fields of the translated TLP is taken from your application transmit interface or, if AMBA is configured, from the AMBA sideband bus (slv_awmisc_info) and not from the corresponding fields of the IATU_REGION_CTRL_1_OFF_OUTBOUND_i or IATU_REGION_CTRL_2_OFF_OUTBOUND_i registers. The header fields are ■ TC ■ PH ■ TH ■ ST ■ AT ■ Attr (IDO, RO and NS). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
20	SNP	R/W	Serialize Non-Posted Requests. In this mode, when the AXI Bridge is populated, same AXI ID Non-Posted Read/Write Requests are transmitted on the wire if there are no other same ID Non-Posted Requests outstanding. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: AXI_POPULATED && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
19	FUNC_BYPASS	R/W	Function Number Translation Bypass. In this mode, the function number of the translated TLP is taken from your application transmit interface and not from the CTRL_1_FUNC_NUM field of the "iATU Region Control 1 Register" or the VF_NUMBER field of the "iATU Control 3 Register." <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
18:17	MSB2BITS_TAG	R/W	Not used. Reserved for future use. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_10BITS_TAG && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
16	TAG_SUBSTITUTE_EN	R/W	TAG Substitute Enable. When enabled and region address is matched, the iATU substitutes the TAG field of the outbound TLP header with the contents of the TAG field in this register. The expected usage scenario is translation from AXI MWr to Vendor Defined Msg/MsgD. <i>Note (CX_TPH_ENABLE=1):</i> TAG substitution for MWr will not occur because this field (byte 6) in the TLP header is the ST field. ST substitution can still take place using the MSG_CODE field in IATU_REGION_CTRL_2_OFF_OUTBOUND_i. <i>Note (CX_10BIT_TAG =1):</i> For 10-bit tags, TAG substitution only applies to the 8 least significant bits of the TAG field (bits 7:0 of Byte 4 of the TLP Header). T9 and T8 of the Header are not substituted. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)
15:8	TAG	R/W	TAG. The substituted TAG field (byte 6) in the outgoing TLP header when TAG_SUBSTITUTE_EN is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
7:0	MSG_CODE	R/W	MSG TLPs (Message Code). When the address of an outbound TLP is matched to this region, and the translated TLP TYPE field is Msg or MsgD; then the message field of the TLP is changed to the value in this register. Memory TLPs: (ST: Steering Tag). When the ST field of an outbound TLP is matched to this region, and the translated TLP TYPE field targets memory space; then the ST field of the TLP is changed to the value in this register. Only Valid when the CX_TPH_ENABLE configuration parameter is 1. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

3.66.3 IATU_LWR_BASE_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Lower Base Address Register.
- **Description:** The CX_ATU_MIN_REGION_SIZE configuration parameter (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ bits are zero.
- **Size:** 32 bits
- **Offset:** $B+i*2*0x100+0x8$
- **Exists:** $(\text{CX_INTERNAL_ATU_ENABLE} \&\& \text{CC_UNROLL_ENABLE} \&\& (\text{CX_MEMORY_MAP_PCIE_TYPE}==0 \mid \mid \text{CX_MEMORY_MAP_PCIE_TYPE}==1)) \&\& (\text{CX_ATU_NUM_OUTBOUND_REGIONS}>0)$

x:y	LWR_BASE_RW
x:0	LWR_BASE_HW

Table 3-580 Fields for Register: IATU_LWR_BASE_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:y	LWR_BASE_RW	R/W	Forms bits [31:n] of the start address of the address region to be translated. n is $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: $(\text{CX_INTERNAL_ATU_ENABLE} \&\& \text{CC_UNROLL_ENABLE} \&\& (\text{CX_MEMORY_MAP_PCIE_TYPE}==0 \mid \mid \text{CX_MEMORY_MAP_PCIE_TYPE}==1)) \&\& (\text{CX_ATU_NUM_OUTBOUND_REGIONS}>0)$ Range Variable[x]: $32 - (\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2)) + \log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2) - 1$ Range Variable[y]: $\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2)$

Bits	Name	Memory Access	Description
x:0	LWR_BASE_HW	R	Forms bits [$n-1:0$] of the start address of the address region to be translated. The start address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0) Range Variable[x]: $\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2)$ - 1

3.66.4 IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Upper Base Address Register.
- **Description:** This register holds the upper 32-bits of the start (and end) address of the address region to be translated.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0xc
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

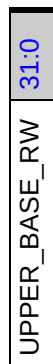


Table 3-581 Fields for Register: IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:0	UPPER_BASE_RW	R/W	Forms bits [63:32] of the start (and end) address of the address region to be translated. In systems with a 32-bit address space, this register is not used and therefore writing to this register has no effect. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

3.66.5 IATU_LIMIT_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Limit Address Register.
- **Description:** This register holds the end address of the address region to be translated.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x10
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

x:y	x:4	3:0
LIMIT_ADDR_RW	LIMIT_ADDR_HW	CBUF_INCR

Table 3-582 Fields for Register: IATU_LIMIT_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:y	LIMIT_ADDR_RW	R/W	Forms upper bits of the end address of the address region to be translated. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0) Range Variable[x]: $32 - (\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)) + \log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2) - 1$ Range Variable[y]: $\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)$

Bits	Name	Memory Access	Description
x:4	LIMIT_ADDR_HW	R	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary, so these bits are always all ones. A write to this location is ignored by the PCIe controller. <i>Note:</i> This register field is sticky. Value After Reset: {(CX_ATU_MIN_REGION_SIZE == 65536) ? "0xff" : (CX_ATU_MIN_REGION_SIZE == 32768) ? "0x7ff" : (CX_ATU_MIN_REGION_SIZE == 16384) ? "0x3ff" : (CX_ATU_MIN_REGION_SIZE == 8192) ? "0x1ff" : (CX_ATU_MIN_REGION_SIZE == 4096) ? "0xff" : "0xffff"} Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0) Range Variable[x]: $\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2)-4 + 3$
3:0	CBUF_INCR	R	Circular Buffer. <i>Note:</i> This register field is sticky. Value After Reset: 0xf Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

3.66.6 IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Lower Target Address Register.
- **Description:** This register holds the Lower Target part of the new address of the translated region, or the outbound TLP header information, depending on the setting of HEADER_SUBSTITUTE_EN field of IATU_REGION_CTRL_2_VIEWPORT_OFF_OUTBOUND_i register.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x14
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)



Table 3-583 Fields for Register: IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:0	LWR_TARGET_RW_OUTBOUND	R/W	When HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_OFF_OUTBOUND_ is '0' (normal operation): ■ LWR_TARGET_RW[31:n] forms MSB's of the Lower Target part of the new address of the translated region; ■ LWR_TARGET_RW[n-1:0] are not used. (The start address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary, so the lower bits of the start address of the new address of the translated region (bits n-1:0) are always '0'). ■ n is log2(CX_ATU_MIN_REGION_SIZE). When HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_OFF_OUTBOUND_i is '1': ■ LWR_TARGET_RW[31:0] forms bytes 8-11 (for 3 dword header) or bytes 12-15 (for 4 dword header) of the outbound TLP header. Usage scenarios include the transmission of Vendor Defined Messages where the controller determines the content of bytes 12 to 15 of the TLP header. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

3.66.7 IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Upper Target Address Register.
- **Description:** This register holds the upper 32 bits of the start address (Upper Target part) of the new address of the translated region.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x18
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

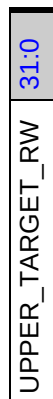


Table 3-584 Fields for Register: IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:0	UPPER_TARGET_RW	R/W	Forms bits [63:32] of the start address (Upper Target part) of the new address of the translated region. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

3.66.8 IATU_REGION_CTRL_3_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Region Control 3 Register.
- **Description:** This register holds the virtual function number used for iATU outbound region matching (wherever applicable).
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x1c
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

31	VF_ACTIVE
30:y	RSVDP_VF_NUMBER
x:0	VF_NUMBER

Table 3-585 Fields for Register: IATU_REGION_CTRL_3_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31	VF_ACTIVE	R/W	Virtual Function Active. Indicates that all of the outbound accesses are from virtual functions. For more information, see VF_ACTIVE field. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
30:y	RSVDP_VF_NUMBER	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0) Range Variable[y]: CX_NVFUNC_WD - 1
x:0	VF_NUMBER	R/W	Virtual Function Number. When the address of an outbound TLP is matched to this region and the FUNC_BYPASS field in the "iATU Region Control 2 Register" is '0', then this field used in generating the Function part of the requester ID (RID) field of the TLP. Indicates from which virtual function (VF) the request is mapped. The iATU uses this signal to form the requester ID. The PCIe SR-IOV specification starts numbering VFs at 1. To ease timing during synthesis, the PCIe controller starts numbering VFs at 0. For example, a value of '0' here refers to the first VF, of the PF that is identified by the CTRL_1_FUNC_NUM field in the "iATU Control 1 Register" For more information, see "RID BDF Number Replacement." The VF_ACTIVE bit must be set to '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0) Range Variable[x]: CX_NVFUNC_WD - 2

3.66.9 IATU_UPPR_LIMIT_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

- **Name:** iATU Upper Limit Address Register.
- **Description:** The CX_MAX_ATU_REGION_SIZE configuration parameter (Value Range: 0->32, 8=1TB) specifies the maximum region size of an address translation region. This register is only used when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_OFF_OUTBOUND_i is '1'.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x20
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0)

31:y	UPPR_LIMIT_ADDR_HW
x:0	UPPR_LIMIT_ADDR_RW

Table 3-586 Fields for Register: IATU_UPPR_LIMIT_ADDR_OFF_OUTBOUND_i (for i = 0; i <= CX_ATU_NUM_OUTBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:y	UPPR_LIMIT_ADDR_HW	R	Forms MSB's of the Upper Limit part of the region "end address" to be translated. Only applies to 64-bit systems and when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_OFF_OUTBOUND_i is '1'. These bits are always '0'. Value After Reset: 0x0 Exists: CX_ATU_MAX_REGION_SIZE>0 && CX_ATU_INCR_RGN_SIZE && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0) Range Variable[y]: CX_ATU_MAX_REGION_SIZE

Bits	Name	Memory Access	Description
x:0	UPPR_LIMIT_ADDR_RW	R/W	Forms the LSB's of the Upper Limit part of the region "end address" to be translated. Only applies to 64-bit systems and when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_OFF_OUTBOUND_i is '1' <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_ATU_MAX_REGION_SIZE>0 && CX_ATU_INCR_RGN_SIZE && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_OUTBOUND_REGIONS>0) Range Variable[x]: CX_ATU_MAX_REGION_SIZE - 1

3.66.10 IATU_REGION_CTRL_1_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Region Control 1 Register.
- **Description:** This register controls the iATU inbound region access based on the optional iATU inbound features enabled using iATU Region Control 2 Register.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x0+0x100
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

CTRL_1_FUNC_NUM	x:20
PH	19:18
AT	17:16
Rsvd	15:14
INCREASE_REGION_SIZE	13
TH	12
IDO	11
ATTR	10:9
TD	8
TC	7:5
TYPE	4:0

Table 3-587 Fields for Register: IATU_REGION_CTRL_1_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:20	CTRL_1_FUNC_NUM	R/W	Function Number. MEM-I/O: When the Address and BAR matching logic in the controller indicate that a MEM-I/O transaction matches a BAR in the function corresponding to this value, then address translation proceeds. This check is only performed if the "Function Number Match Enable" bit of the "iATU Region Control 2 Register" is set. CFG0/CFG1: When the destination function number as specified in the routing ID of the TLP header matches the function, then address translation proceeds. This check is only performed if the "Function Number Match Enable" bit of the "iATU Region Control 2 Register" is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0) Range Variable[x]: (CX_NFUNC_WD) + 19
19:18	PH	R/W	When the PH field of an inbound TLP is matched to this value, then address translation proceeds. This check is only performed if the "PH Match Enable" bit of the "iATU Region Control 2 Register" is set. The setting is independent of the setting of the TH field. Only Valid when the CX_TPH_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_TPH_ENABLE && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
17:16	AT	R/W	When the AT field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "AT Match Enable" bit of the "iATU Region Control 2 Register" is set. Only valid when the ATS_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && ATS_RX_ENABLE && CC_UNROLL_ENABLE) && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
15:14			Reserved Field: Yes
13	INCREASE_REGION_SIZE	R/W	Increase the maximum ATU Region size. When set, the maximum ATU Region size is determined by CX_ATU_MAX_REGION_SIZE. When clear, the maximum ATU Region size is 4 GB (default). <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Maximum ATU Region size is determined by CX_ATU_MAX_REGION_SIZE ■ 0x0 (DISABLE): Maximum ATU Region size is 4 GB (default) Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
12	TH	R/W	When the TH field of an inbound TLP is matched to this value, then address translation proceeds. This check is only performed if the "TH Match Enable" bit of the "iATU Region Control 2 Register" is set. Only Valid when the CX_TPH_ENABLE configuration parameter is 1. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_TPH_ENABLE && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
11	IDO	R/W	When the IDO field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "ATTR Match Enable" bit of the "iATU Region Control 2 Register" is set. Only Valid when the CX_IDO_ENABLE configuration parameter is set to '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_IDO_ENABLE && CC_DEVICE_TYPE!=3 && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
10:9	ATTR	R/W	When the ATTR field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "ATTR Match Enable" bit of the "iATU Region Control 2 Register" is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
8	TD	R/W	When the TD field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TD Match Enable" bit of the "iATU Region Control 2 Register" is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
7:5	TC	R/W	When the TC field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TC Match Enable" bit of the "iATU Region Control 2 Register" is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
4:0	TYPE	R/W	When the TYPE field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

3.66.11 IATU_REGION_CTRL_2_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Region Control 2 Register.
- **Description:** Using this register you can enable/disable the inbound iATU optional features.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x4+0x100
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

31	REGION_EN
30	MATCH_MODE
29	INVERT_MODE
28	CFG_SHIFT_MODE
27	FUZZY_TYPE_MATCH_CODE
26	VFBAR_MATCH_MODE_EN
25:24	RESPONSE_CODE
23	SINGLE_ADDR_LOC_TRANS_EN
22	PH_MATCH_EN
21	MSG_CODE_MATCH_EN
20	VF_MATCH_EN
19	FUNC_NUM_MATCH_EN
18	AT_MATCH_EN
17	TH_MATCH_EN
16	ATTR_MATCH_EN
15	TD_MATCH_EN
14	TC_MATCH_EN
13	MSG_TYPE_MATCH_MODE
12:11	Rsvd
10:8	BAR_NUM
7:0	MSG_CODE

Table 3-588 Fields for Register: IATU_REGION_CTRL_2_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31	REGION_EN	R/W	Region Enable. This bit must be set to '1' for address translation to take place. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
30	MATCH_MODE	R/W	Match Mode. Determines Inbound matching mode for TLPs. The mode depends on the type of TLP that is received as follows: For MEM I/O TLPs, this field is interpreted as follows: ■ 0: Address Match Mode. The iATU operates using addresses as in the outbound direction. The Region Base and Limit Registers must be setup. ■ 1: BAR Match Mode. BAR matching is used. The "BAR Number" field is relevant. Not used for RC. For CFG0 TLPs, this field is interpreted as follows: ■ 0: Routing ID Match Mode. The iATU interprets the Routing ID (Bytes 8 to 11 of TLP header) as an address. This corresponds to the upper 16 bits of the address in MEM-I/O transactions. The Routing ID of the TLP must be within the base and limit of the iATU region for matching to proceed. ■ 1: Accept Mode. The iATU accepts all CFG0 transactions as address matches. The routing ID in the CFG0 TLP is ignored. This is useful as all received CFG0 TLPs should be processed regardless of the Bus number. For MSG/MSGD TLPs, this field is interpreted as follows: ■ 0: Address Match Mode. The iATU treats the third dword and fourth dword of the inbound MSG/MSGD TLP as an address and it is matched against the Region Base and Limit Registers.

Bits	Name	Memory Access	Description
30...(cont.)	MATCH_MODE.	R/W	1: Vendor ID Match Mode. This mode is relevant for ID-routed Vendor Defined Messages. The iATU ignores the Routing ID (Bus, Device, Function) in bits [31:16] of the third dword of the TLP header, but matches against the Vendor ID in bits [15:0] of the third dword of the TLP header. Bits [15:0] of the Region Upper Base register should be programmed with the required Vendor ID. The lower Base and Limit Register should be programmed to translate TLPs based on vendor specific information in the fourth dword of the TLP header. - If SINGLE_ADDRESS_LOCATION_TRANSLATE_EN = 1 AND MSG_TYPE_MATCH_MODE = 1, then Match Mode is ignored. <i>Note:</i> This register field is sticky. Values: 0x0 (ZERO): The interpretation is dependent on TLP type, that is, MEM/IO, CFG0, or MSG/MSGD TLPs. 0x1 (ONE): The interpretation is dependent on TLP type, that is, MEM/IO, CFG0, or MSG/MSGD TLPs. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
29	INVERT_MODE	R/W	Invert Mode Enable. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). When set all regions of that type must use address match mode. <i>Note:</i> This register field is sticky. Values: 0x1 (ENABLE): Invert Mode Enable 0x0 (DISABLE): Invert Mode Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
28	CFG_SHIFT_MODE	R/W	CFG Shift Enable. This is useful for CFG transactions where the PCIe configuration mechanism maps bits [27:12] of the address to the bus/device and function number. This allows a CFG configuration space to be located in any 256MB window of your application memory space using a 28-bit effective address. Shifts bits [31:16] of the untranslated address to form bits [27:12] of the translated address. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): CFG Shift Enable ■ 0x0 (DISABLE): CFG Shift Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
27	FUZZY_TYPE_MATCH_CODE	R/W	Fuzzy Type Match Enable. When enabled, the iATU relaxes the matching of the TLP TYPE field against the expected TYPE field so that ■ CfgRd0 and CfgRd1 TLPs are seen as identical. Similarly with CfgWr0 and CfgWr1. ■ MWr, MRd, and MRdLk TLPs are seen as identical ■ The Routing field of Msg/MsgD TLPs is ignored ■ FetchAdd, Swap, and CAS are seen as identical. For example, CFG0 in the TYPE field in the "iATU Region Control 1 Register" matches against an inbound CfgRd0, CfgRd1, CfgWr0, or CfgWr1 TLP. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Fuzzy Type Match Enable ■ 0x0 (DISABLE): Fuzzy Type Match Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
26	VFBAR_MATCH_MODE_EN	R/W	VF BAR Match Enable. When set, Virtual Function BAR matching is used which allows all VFs in a PF which match a BAR to be matched with a single ATU region. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Virtual Function BAR Match Enable ■ 0x0 (DISABLE): Virtual Function BAR Match Disable Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
25:24	RESPONSE_CODE	R/W	Response Code. Defines the type of response to give for accesses matching this region. This overrides the normal RADM filter response. Note that this feature is not available for any region where Single Address Location Translate is enabled. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (NORMAL_RADM): Normal RADM filter response is used. ■ 0x1 (UNSUP_REQ): Unsupported request (UR) ■ 0x2 (COMPL_ABORT): Completer abort (CA) ■ 0x3 (NOT_USED): Not used / undefined / reserved Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
23	SINGLE_ADDR_LOC_TRANS_EN	R/W	Single Address Location Translate Enable. When enabled, Rx TLPs can be translated to a single address location as determined by the target address register of the iATU region. The main usage scenario is translation of Messages (such as Vendor Defined or ATS Messages) to MWr TLPs when the AXI bridge is enabled. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
22	PH_MATCH_EN	R/W	PH Match Enable. Ensures that a successful PH TLP field comparison match (see PH field of the "iATU Region Control 1 Register") occurs for address translation to proceed. Only Valid when the CX_TPH_ENABLE configuration parameter is 1. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): PH Match Enable ■ 0x0 (DISABLE): PH Match Disable Value After Reset: 0x0 Exists: CX_TPH_ENABLE && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
21	MSG_CODE_MATCH_EN	R/W	Message Code Match Enable (Msg TLPs). Ensures that a successful message Code TLP field comparison match (see Message Code field of the "iATU Region Control 2 Register") occurs (in MSG transactions) for address translation to proceed. ST Match Enable (Mem TLPs). Ensures that a successful ST TLP field comparison match (see ST field of the "iATU Region Control 2 Register") occurs (in MEM transactions) for address translation to proceed. Only Valid when the CX_TPH_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Message Code Match Enable (for Msg TLPs) or ST Match Enable (for Mem TLPs) ■ 0x0 (DISABLE): Virtual Function Number Match Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
20	VF_MATCH_EN	R/W	Virtual Function Number Match Enable. Ensures that a successful Virtual Function Number TLP field comparison match (see Virtual Function Number field of the "iATU Control 3 Register") occurs (in MEM-I/O transactions) for address translation to proceed. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Virtual Function Number Match Enable ■ 0x0 (DISABLE): Virtual Function Number Match Disable Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
19	FUNC_NUM_MATCH_EN	R/W	Function Number Match Enable. Ensures that a successful Function Number TLP field comparison match (see Function Number field of the "iATU Region Control 1 Register") occurs (in MEM-I/O and CFG0/CFG1 transactions) for address translation to proceed. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Function Number Match Enable ■ 0x0 (DISABLE): Function Number Match Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
18	AT_MATCH_EN	R/W	AT Match Enable. Ensures that a successful AT TLP field comparison match (see AT field of the "iATU Region Control 1 Register") occurs for address translation to proceed. Only valid when the ATS_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): AT Match Enable ■ 0x0 (DISABLE): AT Match Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && ATS_RX_ENABLE && CC_UNROLL_ENABLE && CC_DEVICE_TYPE!=3) && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
17	TH_MATCH_EN	R/W	TH Match Enable. Ensures that a successful TH TLP field comparison match (see TH field of the "iATU Region Control 1 Register") occurs for address translation to proceed. Only Valid when the CX_TPH_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): TH Match Enable ■ 0x0 (DISABLE): TH Match Disable Value After Reset: 0x0 Exists: CX_TPH_ENABLE && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
16	ATTR_MATCH_EN	R/W	ATTR Match Enable. Ensures that a successful ATTR TLP field comparison match (see ATTR field of the "iATU Region Control 1 Register") occurs for address translation to proceed. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): ATTR Match Enable ■ 0x0 (DISABLE): ATTR Match Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
15	TD_MATCH_EN	R/W	TD Match Enable. Ensures that a successful TD TLP field comparison match (see TD field of the "iATU Region Control 1 Register") occurs for address translation to proceed. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): TD Match Enable ■ 0x0 (DISABLE): TD Match Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

Bits	Name	Memory Access	Description
14	TC_MATCH_EN	R/W	TC Match Enable. Ensures that a successful TC TLP field comparison match (see TC field of the "iATU Region Control 1 Register") occurs for address translation to proceed. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): TC Match Enable ■ 0x0 (DISABLE): TC Match Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
13	MSG_TYPE_MATCH_MODE	R/W	Message Type Match Mode. When enabled, and if single address location translate enable is set, then inbound TLPs of type MSG/MSGd which match the type field of the IATU_REGION_CTRL_1_VIEWPORT_OFF_INBOUND_i register (TYPE[4:3]=2'b10) will be translated. Message type match mode overrides any value of MATCH_MODE field in this register. Usage scenarios for this are translation of VDM or ATS messages when AXI bridge is configured on client interface. <i>Note:</i> This register field is sticky. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
12:11			Reserved Field: Yes

Bits	Name	Memory Access	Description
10:8	BAR_NUM	R/W	BAR Number. When the BAR number of an inbound MEM or IO TLP " that is matched by the normal internal BAR address matching mechanism " is the same as this field, address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Match Mode" bit of the "iATU Region Control 2 Register" is set. IO translation would require either 00100b or 00101b in the inbound TLP TYPE; the BAR Number set in the range 000b - 101b and that BAR configured as an IO BAR. <i>Note:</i> This register field is sticky. Values: ■ 0x0 (BAR0): BAR0 ■ 0x1 (BAR1): BAR1 ■ 0x2 (BAR2): BAR2 ■ 0x3 (BAR3): BAR3 ■ 0x4 (BAR4): BAR4 ■ 0x5 (BAR5): BAR5 ■ 0x6 (ROM): ROM ■ 0x7 (RSVD): reserved Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
7:0	MSG_CODE	R/W	MSG TLPs: (Message Code). When the MSG_CODE field of an inbound Msg/MsgD TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Message Code Match Enable" bit of the "iATU Region Control 2 Register" is set and the TYPE field of the inbound TLP is Msg/MsgD. Memory TLPs: (ST: Steering Tag). When the ST field of an inbound TLP is matched to this value, then address translation proceeds. This check is only performed if the "ST Match Enable" bit of the "iATU Region Control 2 Register" is set. The setting is independent of the setting of the TH field. Only Valid when the CX_TPH_ENABLE configuration parameter is '1'. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

3.66.12 IATU_LWR_BASE_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Lower Base Address Register.
- **Description:** The CX_ATU_MIN_REGION_SIZE configuration parameter (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ bits are zero.
- **Size:** 32 bits
- **Offset:** $B+i*2*0x100+0x8+0x100$
- **Exists:** $(\text{CX_INTERNAL_ATU_ENABLE} \&\& \text{CC_UNROLL_ENABLE} \&\& (\text{CX_MEMORY_MAP_PCIE_TYPE}==0 \mid \mid \text{CX_MEMORY_MAP_PCIE_TYPE}==1)) \&\& (\text{CX_ATU_NUM_INBOUND_REGIONS}>0)$

x:y	x:0
LWR_BASE_RW	LWR_BASE_HW

Table 3-589 Fields for Register: IATU_LWR_BASE_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:y	LWR_BASE_RW	R/W	Forms bits [31:n] of the start address of the address region to be translated. n is $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: $(\text{CX_INTERNAL_ATU_ENABLE} \&\& \text{CC_UNROLL_ENABLE} \&\& (\text{CX_MEMORY_MAP_PCIE_TYPE}==0 \mid \mid \text{CX_MEMORY_MAP_PCIE_TYPE}==1)) \&\& (\text{CX_ATU_NUM_INBOUND_REGIONS}>0)$ Range Variable[x]: $32 - (\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2)) + \log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2) - 1$ Range Variable[y]: $\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2)$

Bits	Name	Memory Access	Description
x:0	LWR_BASE_HW	R	Forms bits [$n-1:0$] of the start address of the address region to be translated. The start address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0) Range Variable[x]: $\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2)$ - 1

3.66.13 IATU_UPPER_BASE_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Upper Base Address Register.
- **Description:** This register holds the upper 32 bits of the start (and end) address of the address region to be translated.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0xc+0x100
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

UPPER_BASE_RW 31:0

Table 3-590 Fields for Register: IATU_UPPER_BASE_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:0	UPPER_BASE_RW	R/W	Forms bits [63:32] of the start (and end) address of the address region to be translated. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

3.66.14 IATU_LIMIT_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Limit Address Register.
- **Description:** This register holds the end address of the address region to be translated.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x10+0x100
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

x:y	x:4	3:0
LIMIT_ADDR_RW	LIMIT_ADDR_HW	CBUF_INCR

Table 3-591 Fields for Register: IATU_LIMIT_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:y	LIMIT_ADDR_RW	R/W	Forms upper bits of the end address of the address region to be translated. When SINGLE_ADDR_LOC_TRANS_EN = 1, MSG_TYPE_MATCH_MODE = 1, and CBUF_INCR > 0 then this field forms the upper bits of the limit address for the circular buffer. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0) Range Variable[x]: $32 - (\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)) + \log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2) - 1$ Range Variable[y]: $\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)$

Bits	Name	Memory Access	Description
x:4	LIMIT_ADDR_HW	R	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary, so these bits are always all ones. When SINGLE_ADDR_LOC_TRANS_EN = 1, MSG_TYPE_MATCH_MODE = 1, and CBUF_INCR > 0 then this field forms lower bits of the limit address for the circular buffer. A write to this location is ignored by the PCIe controller. <i>Note:</i> This register field is sticky. Value After Reset: {(CX_ATU_MIN_REGION_SIZE == 65536) ? "0xfff" : (CX_ATU_MIN_REGION_SIZE == 32768) ? "0x7ff" : (CX_ATU_MIN_REGION_SIZE == 16384) ? "0x3ff" : (CX_ATU_MIN_REGION_SIZE == 8192) ? "0x1ff" : (CX_ATU_MIN_REGION_SIZE == 4096) ? "0xff" : "0xffff"} Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0) Range Variable[x]: $\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2)-4 + 3$

Bits	Name	Memory Access	Description
3:0	CBUF_INCR	* Varies	Circular Buffer Increment. When CX_ATU_SLOC_CBUF = 0, then this field is Read-only and forms the lowest bits of the end address of the address region to be translated. When CX_ATU_SLOC_CBUF = 1, then this field is R/W and forms the upper bits of the Circular Buffer Increment size (CBUF_INCR) field for Single Location Address translation. The increment value (in bytes) is decoded as follows: Note: A write to any bit in the CBUF_INCR field resets the circular buffer pointer - that is, the next matched received Message will be translated to the start address of the Circular Buffer. This field must be written to AFTER the target and limit registers have been updated. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R (sticky) ■ Dbi: R (sticky) <i>Note:</i> This register field is sticky. Values: ■ 0x0 (Def_0): 0 (Default; legacy Single Address Location mode) ■ 0x1 (addr_4): 4 ■ 0x2 (addr_8): 8 ■ 0x3 (addr_16): 16 ■ 0x4 (addr_32): 32 ■ 0x5 (addr_64): 64 ■ 0x6 (addr_128): 128 ■ 0x7 (addr_256): 256 ■ 0x8 (addr_512): 512 ■ 0x9 (addr_1024): 1024 ■ 0xa (addr_2048): 2048 ■ 0xb (addr_4096): 4096 ■ 0xc (addr_8192): 8192 ■ 0xd (addr_rvd1): rsvd. ■ 0xe (addr_rvd2): rsvd. ■ 0xf (addr_rvd3): rsvd. Value After Reset: 0x0 Exists:

Bits	Name	Memory Access	Description
3:0...(cont.)	CBUF_INCR.	* Varies	(CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0 && CX_ATU_SLOC_CBUF) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==0 && !CX_ATU_SLOC_CBUF) ? \"read-only\" : (CX_MEMORY_MAP_VIEW==1 && CX_ATU_SLOC_CBUF) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !CX_ATU_SLOC_CBUF) ? \"read-only\" : (CX_ATU_SLOC_CBUF) ? \"read-write\" : \"read-only\""

3.66.15 IATU_LWR_TARGET_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Lower Target Address Register.
- **Description:** This register holds the Lower Target part of the new address of the translated region.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x14+0x100
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

x:y	LWR_TARGET_RW
x:0	LWR_TARGET_HW

Table 3-592 Fields for Register: IATU_LWR_TARGET_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
x:y	LWR_TARGET_RW	R/W	Forms MSB's of the Lower Target part of the new address of the translated region. These bits are always '0'. ■ Field size depends on $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ in address match mode. ■ Field size depends on $\log_2(\text{BAR_MASK}+1)$ in BAR match mode. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0) Range Variable[x]: $32 - (\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)) + \log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2) - 1$ Range Variable[y]: $\log(\text{CX_ATU_MIN_REGION_SIZE}) / \log(2)$

Bits	Name	Memory Access	Description
x:0	LWR_TARGET_HW	R	Forms the LSB's of the Lower Target part of the new address of the translated region. The start address must be aligned to a CX_ATU_MIN_REGION_SIZE kB boundary (in address match mode); and to the Bar size boundary (in BAR match mode) so that these bits are always '0'. If the BAR is smaller than the iATU region size, then the iATU target address must align to the iATU region size; otherwise it must align to the BAR size. A write to this location is ignored by the PCIe controller. - Field size depends on $\log_2(\text{CX_ATU_MIN_REGION_SIZE})$ in address match mode. - Field size depends on $\log_2(\text{BAR_MASK}+1)$ in BAR match mode. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0) Range Variable[x]: $\log(\text{CX_ATU_MIN_REGION_SIZE})/\log(2) - 1$

3.66.16 IATU_UPPER_TARGET_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Upper Target Address Register.
- **Description:** This register holds the upper 32 bits of the start address (Upper Target part) of the new address of the translated region.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x18+0x100
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

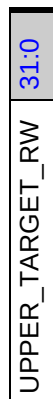


Table 3-593 Fields for Register: IATU_UPPER_TARGET_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:0	UPPER_TARGET_RW	R/W	Forms bits [63:32] of the start address (Upper Target part) of the new address of the translated region. In systems with a 32-bit address space, this register is not used and therefore writing to this register has no effect. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: (CX_INTERNAL_ATU_ENABLE && CC_IN64 && CC_UNROLL_ENABLE) && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

3.66.17 IATU_REGION_CTRL_3_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Region Control 3 Register.
- **Description:** This register holds the virtual function number used for iATU inbound region matching (wherever applicable).
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x1c+0x100
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

31	30:y	x:0
VF_ACTIVE	RSVDP_VF_NUMBER	VF_NUMBER

Table 3-594 Fields for Register: IATU_REGION_CTRL_3_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31	VF_ACTIVE	R	Reserved. Not used. Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)
30:y	RSVDP_VF_NUMBER	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0) Range Variable[y]: CX_NVFUNC_WD - 1

Bits	Name	Memory Access	Description
x:0	VF_NUMBER	R/W	Virtual Function Number. When the Address and BAR matching logic in the controller indicates that a MEM-I/O transaction matches a BAR in the virtual function corresponding to this value, then address translation proceeds. This check is only performed if the "Virtual Function Number Match Enable" bit of the "iATU Region Control 2 Register" is set. <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DEVICE_TYPE!=3 && CC_DEVICE_TYPE!=1 && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0) Range Variable[x]: CX_NVFUNC_WD - 2

3.66.18 IATU_UPPR_LIMIT_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

- **Name:** iATU Upper Limit Address Register.
- **Description:** The CX_MAX_ATU_REGION_SIZE configuration parameter (Value Range: 0(4GB) to 32(16 EB)) specifies the maximum region size of an address translation region. This register is only used when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_OFF_INBOUND_i is '1'.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x20+0x100
- **Exists:** (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0)

31:y	UPPR_LIMIT_ADDR_HW
x:0	UPPR_LIMIT_ADDR_RW

Table 3-595 Fields for Register: IATU_UPPR_LIMIT_ADDR_OFF_INBOUND_i (for i = 0; i <= CX_ATU_NUM_INBOUND_REGIONS-1)

Bits	Name	Memory Access	Description
31:y	UPPR_LIMIT_ADDR_HW	R	Forms MSB's of the Upper Limit part of the region "end address" to be translated. Only applies to 64-bit systems and when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_OFF_INBOUND_i is '1'. These bits are always '0'. Value After Reset: 0x0 Exists: CX_ATU_MAX_REGION_SIZE>0 && CX_ATU_INCR_RGN_SIZE && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0) Range Variable[y]: CX_ATU_MAX_REGION_SIZE

Bits	Name	Memory Access	Description
x:0	UPPR_LIMIT_ADDR_RW	R/W	Forms the LSB's of the Upper Limit part of the region "end address" to be translated. Only applies to 64-bit systems and when the INCREASE_REGION_SIZE field in IATU_REGION_CTRL_1_OFF_INBOUND_i is '1' <i>Note:</i> This register field is sticky. Value After Reset: 0x0 Exists: CX_ATU_MAX_REGION_SIZE>0 && CX_ATU_INCR_RGN_SIZE && (CX_INTERNAL_ATU_ENABLE && CC_UNROLL_ENABLE && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CX_ATU_NUM_INBOUND_REGIONS>0) Range Variable[x]: CX_ATU_MAX_REGION_SIZE - 1

3.67 PF0_DMA_CAP Registers

3.67.1 DMA_CTRL_DATA_ARB_PRIOR_OFF

- **Name:** DMA Arbitration Scheme for TRGT1 Interface.
 - **Description:** This register is used to control traffic priorities among various sources that are delivered to your application through TRGT1 where 0x0 represents the highest priority.
 - Non-DMA Rx Requests
 - DMA Write Channel MRd Requests (DMA data requests and LL element/descriptor access)
 - DMA Read Channel MRd Requests (LL element/descriptor access)
 - DMA Read Channel MWr Requests
- Concurrent traffic from channels with same priority are sorted according to Round-Robin arbitration rules. The arbitration priority defaults to Non-DMA requests (highest), Write Channel MRd, Read Channel MRd, Read Channel MWr. For more information, see the "Internal Architecture" section in the DMA chapter of the *Databook*.
- **Size:** 32 bits
 - **Offset:** B+0x0
 - **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:17	16:12	11:9	8:6	5:3	2:0
RSVDP_17	VERSION	RDBUFF_TRGT_WEIGHT	RD_CTRL_TRGT_WEIGHT	WR_CTRL_TRGT_WEIGHT	RTRGT1_WEIGHT

Table 3-596 Fields for Register: DMA_CTRL_DATA_ARB_PRIOR_OFF

Bits	Name	Memory Access	Description
31:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && ICX_DMA_PF_ENABLE

Bits	Name	Memory Access	Description
16:12	VERSION	R	Reports the version of Register Map of eDMA. Value After Reset: {(CC_LEGACY_DMA_MAP == 1) ? 0 : 1} Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE
11:9	RDBUFF_TRGT_WEIGHT	* Varies	DMA Read Channel MWr Requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x3 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
8:6	RD_CTRL_TRGT_WEIGHT	* Varies	DMA Read Channel MRd Requests (for LL element/descriptor access). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x2 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
5:3	WR_CTRL_TRGT_WEIGHT	* Varies	DMA Write Channel MRd Requests (for DMA data requests and LL element/descriptor access). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
2:0	RTRGT1_WEIGHT	* Varies	Non-DMA Rx Requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.2 DMA_CTRL_OFF

- **Name:** DMA Number of Channels Register.
- **Description:** This register provides information regarding the number of configured DMA read and write channels.
Note: When this register does not exist, value is fixed to 32'hFFFF_FFFF.
- **Size:** 32 bits
- **Offset:** B+0x8
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:26	RSVDP_26
25	DIS_C2W_CACHE_RD
24	DIS_C2W_CACHE_WR
23:20	RSVDP_20
19:16	NUM_DMA_RD_CHAN
15:4	RSVDP_4
3:0	NUM_DMA_WR_CHAN

Table 3-597 Fields for Register: DMA_CTRL_OFF

Bits	Name	Memory Access	Description
31:26	RSVDP_26	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && ICX_DMA_PF_ENABLE

Bits	Name	Memory Access	Description
25	DIS_C2W_CACHE_RD	* Varies	Disable DMA Read Channel's "completion to memory write" context cache pre-fetch function. <i>Note:</i> For internal debugging only. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
24	DIS_C2W_CACHE_WR	* Varies	Disable DMA Write Channel's "completion to memory write" context cache pre-fetch function. <i>Note:</i> For internal debugging only. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
23:20	RSVDP_20	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE
19:16	NUM_DMA_RD_CHAN	R	Number of Read Channels. You can read this register to determine the number of read channels configured. Value After Reset: CC_NUM_DMA_RD_CHAN Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE
15:4	RSVDP_4	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE
3:0	NUM_DMA_WR_CHAN	R	Number of Write Channels. You can read this register to determine the number of write channels configured. Value After Reset: CC_NUM_DMA_WR_CHAN Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

3.67.3 DMA_WRITE_ENGINE_EN_OFF

- **Name:** DMA Write Engine Enable Register.
- **Description:** This register indicates the status of DMA write engine and the status of DMA write engine handshake feature (per-channel).
- **Size:** 32 bits
- **Offset:** B+0xc
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	RSVDP_24
23	DMA_WRITE_ENGINE_EN_HSHAKE_CH7
22	DMA_WRITE_ENGINE_EN_HSHAKE_CH6
21	DMA_WRITE_ENGINE_EN_HSHAKE_CH5
20	DMA_WRITE_ENGINE_EN_HSHAKE_CH4
19	DMA_WRITE_ENGINE_EN_HSHAKE_CH3
18	DMA_WRITE_ENGINE_EN_HSHAKE_CH2
17	DMA_WRITE_ENGINE_EN_HSHAKE_CH1
16	DMA_WRITE_ENGINE_EN_HSHAKE_CH0
15:1	RSVDP_1
0	DMA_WRITE_ENGINE

Table 3-598 Fields for Register: DMA_WRITE_ENGINE_EN_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
23	DMA_WRITE_ENGINE_EN_HSHAKE_CH7	* Varies	Enable Handshake for DMA Write Engine Channel 7. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
22	DMA_WRITE_ENGINE_EN_HSHAKE_CH6	* Varies	Enable Handshake for DMA Write Engine Channel 6. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
21	DMA_WRITE_ENGINE_EN_HSHAKE_CH5	* Varies	Enable Handshake for DMA Write Engine Channel 5. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
20	DMA_WRITE_ENGINE_EN_HSHAKE_CH4	* Varies	Enable Handshake for DMA Write Engine Channel 4. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
19	DMA_WRITE_ENGINE_EN_HSHAKE_CH3	* Varies	Enable Handshake for DMA Write Engine Channel 3. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
18	DMA_WRITE_ENGINE_EN_HSHAKE_CH2	* Varies	Enable Handshake for DMA Write Engine Channel 2. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
17	DMA_WRITE_ENGINE_EN_HSHAKE_CH1	* Varies	Enable Handshake for DMA Write Engine Channel 1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
16	DMA_WRITE_ENGINE_EN_HSHAKE_CH0	* Varies	Enable Handshake for DMA Write Engine Channel 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
15:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
0	DMA_WRITE_ENGINE	* Varies	DMA Write Engine Enable. For normal operation, you must initially set this bit to '1', before any other software setup action. You do not need to toggle or rewrite to this bit during normal operation. You should set this bit to '0' when you want to "Soft Reset" the DMA controller write logic. There are three possible reasons for resetting the DMA controller write logic: ■ The "Abort Interrupt Status" bit is set in the DMA Write Interrupt Status Register (DMA_WRITE_INT_STATUS_OFF), and any of the bits in the DMA Write Error Status Register (DMA_WRITE_ERR_STATUS_OFF) are set. Resetting the DMA controller write logic re-initializes the control logic, ensuring that the next DMA write transfer is executed successfully. ■ You have executed the procedure outlined in "Stop Bit", after which, the "Abort Interrupt Status" bit is set and the Channel Status(CS) field of the DMA Write Channel Control 1 Register (DMA_CH_CTRL1_OFF_WRCH_i) is set to "Stopped". Resetting the DMA controller write logic re-initializes the control logic ensuring that the next DMA write transfer is executed successfully. ■ During software development, when you incorrectly program the DMA write engine. To "Soft Reset" the DMA controller write logic, you must: ■ De-assert the DMA write engine enable bit. ■ Wait for the DMA to complete any in-progress TLP transfer, by waiting until a read on the DMA write engine enable bit returns a '0'.

Bits	Name	Memory Access	Description
0...(cont .)	DMA_WRITE_ENGINE.	* Varies	■ Assert the DMA write engine enable bit. This "Soft Reset" does not clear the DMA configuration registers. The DMA write transfer does not start until you write to the DMA Write Doorbell Register (DMA_WRITE_DOORBELL_OFF). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable (Soft Reset) Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: untestable Volatile: true Memory Access:
0...(cont ..)	DMA_WRITE_ENGINE..	* Varies	"(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.4 DMA_WRITE_DOORBELL_OFF

- **Name:** DMA Write Doorbell Register.
- **Description:** This register controls the DMA write channel doorbell.
- **Size:** 32 bits
- **Offset:** B+0x10
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31	WR_STOP
30:3	RSVDP_3
2:0	WR_DOORBELL_NUM

Table 3-599 Fields for Register: DMA_WRITE_DOORBELL_OFF

Bits	Name	Memory Access	Description
31	WR_STOP	* Varies	Stop. Set in conjunction with the Doorbell Number field. The DMA write channel stops issuing requests, sets the channel status to "Stopped", and asserts the "Abort" interrupt if it is enabled. Before setting the Stop bit, you must read the channel Status field (CS) of the DMA Write Channel Control 1 Register (DMA_CH_CONTROL1_OFF_WRCH_0) to ensure that the write channel is "Running" (transferring data). For more information, see "Stopping the DMA Transfer (Software Stop)". <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
30:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
2:0	WR_DOORBELL_NUM	* Varies	Doorbell Number. You must write the channel number to this register to start the DMA write transfer for that channel. The DMA detects a write to this register field even if the value of this field does not change. You do not need to toggle or write any other value to this register to start a new transfer. The range of this field is 0x0 to 0x7, where 0x0 corresponds to channel 0. <i>Note:</i> A write to this field triggers the controller to exit L1 substates. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.5 DMA_WRITE_ENGINE_PF_OFF

- **Name:** DMA Write Engine Pre-Fetch Register.
- **Description:** This is the DMA write engine pre-fetch register.
- **Size:** 32 bits
- **Offset:** B+0x14
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:20	19:10	9:0
RSVDP_20	RSVDP_WRITE_ENGINE_PF	DMA_WRITE_ENGINE_MAX_PRE_FETCH

Table 3-600 Fields for Register: DMA_WRITE_ENGINE_PF_OFF

Bits	Name	Memory Access	Description
31:20	RSVDP_20	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_PF_ENABLE

Bits	Name	Memory Access	Description
19:10	RSVDP_WRITE_ENGINE_PF	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_PF_ENABLE
9:0	DMA_WRITE_ENGINE_MAX_PREFETCH	R/W	This field controls the linked list prefetch allocation space for each of the HDMA write engine channels. Each channel will be able to store as many linked list elements as specified here. Therefore this field must be within the following range: [0 : floor(CX_DMA_WR_LLQ * CC_NUM_DMA_WR_CHAN / Number_of_active_active_channels) - 1] <i>Note:</i> If this field is setup with a value outside the specified range data corruption and channel crosstalk may happen. If changing the default value of this register, limit the doorbelled channels to the range: [0 : Number_of_active_active_channels - 1]. The actual prefetch allocation space is the value of this field plus one. So, for example, setting it to zero means that it has allocated space for one descriptor. Value After Reset: {CC_HDMA_REG_WR_PF_DEPTH} Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_PF_ENABLE

3.67.6 DMA_WRITE_CHANNEL_ARB_WEIGHT_LOW_OFF

- **Name:** DMA Write Engine Channel Arbitration Weight Low Register.
- **Description:** The 5-bit channel weight (for write channels 0-3) specifies the maximum number of TLP requests that the DMA can issue for that channel before it must return to the arbitration routine. When the channel weight count is reached or DMA channel request transfer size reaches zero, the WWR arbiter selects the next channel to be processed. Your software must initialize this register before ringing the doorbell. For more information, see "Multichannel Arbitration". Value range is (0-0x1F) corresponding to (1-32) transaction requests.
- **Size:** 32 bits
- **Offset:** B+0x18
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:20	RSVDP_20
19:15	WRITE_CHANNEL3_WEIGHT
14:10	WRITE_CHANNEL2_WEIGHT
9:5	WRITE_CHANNEL1_WEIGHT
4:0	WRITE_CHANNEL0_WEIGHT

Table 3-601 Fields for Register: DMA_WRITE_CHANNEL_ARB_WEIGHT_LOW_OFF

Bits	Name	Memory Access	Description
31:20	RSVDP_20	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && ICX_DMA_PF_ENABLE

Bits	Name	Memory Access	Description
19:15	WRITE_CHANNEL3_WEIGHT	* Varies	Channel 3 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
14:10	WRITE_CHANNEL2_WEIGHT	* Varies	Channel 2 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
9:5	WRITE_CHANNEL1_WEIGHT	* Varies	Channel 1 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
4:0	WRITE_CHANNEL0_WEIGHT	* Varies	Channel 0 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.7 DMA_WRITE_CHANNEL_ARB_WEIGHT_HIGH_OFF

- **Name:** DMA Write Engine Channel Arbitration Weight High Register.
- **Description:** The 5-bit channel weight (for write channels 4-7) specifies the maximum number of TLP requests that the DMA can issue for that channel before it must return to the arbitration routine. When the channel weight count is reached or DMA channel request transfer size reaches zero, the WWR arbiter selects the next channel to be processed. Your software must initialize this register before ringing the doorbell. For more information, see "Multichannel Arbitration". Value range is (0-0x1F) corresponding to (1-32) transaction requests.
- **Size:** 32 bits
- **Offset:** B+0x1c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:20	RSVDP_20
19:15	WRITE_CHANNEL7_WEIGHT
14:10	WRITE_CHANNEL6_WEIGHT
9:5	WRITE_CHANNEL5_WEIGHT
4:0	WRITE_CHANNEL4_WEIGHT

Table 3-602 Fields for Register: DMA_WRITE_CHANNEL_ARB_WEIGHT_HIGH_OFF

Bits	Name	Memory Access	Description
31:20	RSVDP_20	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE

Bits	Name	Memory Access	Description
19:15	WRITE_CHANNEL7_WEIGHT	* Varies	Channel 7 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
14:10	WRITE_CHANNEL6_WEIGHT	* Varies	Channel 6 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
9:5	WRITE_CHANNEL5_WEIGHT	* Varies	Channel 5 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
4:0	WRITE_CHANNEL4_WEIGHT	* Varies	Channel 4 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.8 DMA_READ_ENGINE_EN_OFF

- **Name:** DMA Read Engine Enable Register.
- **Description:** This register indicates the status of DMA read engine and the status of DMA read engine handshake feature (per-channel).
- **Size:** 32 bits
- **Offset:** B+0x2c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	RSVDP_24
23	DMA_READ_ENGINE_EN_HSHAKE_CH7
22	DMA_READ_ENGINE_EN_HSHAKE_CH6
21	DMA_READ_ENGINE_EN_HSHAKE_CH5
20	DMA_READ_ENGINE_EN_HSHAKE_CH4
19	DMA_READ_ENGINE_EN_HSHAKE_CH3
18	DMA_READ_ENGINE_EN_HSHAKE_CH2
17	DMA_READ_ENGINE_EN_HSHAKE_CH1
16	DMA_READ_ENGINE_EN_HSHAKE_CH0
15:1	RSVDP_1
0	DMA_READ_ENGINE

Table 3-603 Fields for Register: DMA_READ_ENGINE_EN_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
23	DMA_READ_ENGINE_EN_HSHAKE_CH7	* Varies	Enable Handshake for DMA Read Engine Channel 7. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
22	DMA_READ_ENGINE_EN_HSHAKE_CH6	* Varies	Enable Handshake for DMA Read Engine Channel 6. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
21	DMA_READ_ENGINE_EN_HSHA KE_CH5	* Varies	Enable Handshake for DMA Read Engine Channel 5. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
20	DMA_READ_ENGINE_EN_HSHA KE_CH4	* Varies	Enable Handshake for DMA Read Engine Channel 4. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
19	DMA_READ_ENGINE_EN_HSHA KE_CH3	* Varies	Enable Handshake for DMA Read Engine Channel 3. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
18	DMA_READ_ENGINE_EN_HSHA KE_CH2	* Varies	Enable Handshake for DMA Read Engine Channel 2. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
17	DMA_READ_ENGINE_EN_HSHAKE_CH1	* Varies	Enable Handshake for DMA Read Engine Channel 1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
16	DMA_READ_ENGINE_EN_HSHAKE_CH0	* Varies	Enable Handshake for DMA Read Engine Channel 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
15:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
0	DMA_READ_ENGINE	* Varies	DMA Read Engine Enable. For normal operation, you must initially set this bit to '1', before any other software setup action. You do not need to toggle or rewrite to this bit during normal operation. You should set this field to '0' when you want to "Soft Reset" the DMA controller read logic. There are three possible reasons for resetting the DMA controller read logic: ■ The "Abort Interrupt Status" bit is set in the DMA Read Interrupt Status Register (DMA_READ_INT_STATUS_OFF), and any of the bits in the DMA Read Error Status Low Register (DMA_READ_ERR_STATUS_LOW_OFF) is set. Resetting the DMA controller read logic re-initializes the control logic, ensuring that the next DMA read transfer is executed successfully. ■ You have executed the procedure outlined in "Stop Bit", after which, the "Abort Interrupt Status" bit is set and the channel Status field (CS) of the DMA Read Channel Control 1 Register (DMA_CH_CONTROL1_OFF_WRCH_i) is set to "Stopped". Resetting the DMA controller read logic re-initializes the control logic ensuring that the next DMA read transfer is executed successfully. ■ During software development, when you incorrectly program the DMA read engine. To "Soft Reset" the DMA controller read logic, you must: ■ De-assert the DMA read engine enable bit. ■ Wait for the DMA to complete any in-progress TLP transfer, by waiting until a read on the DMA read engine enable bit returns a '0'.

Bits	Name	Memory Access	Description
0...(cont .)	DMA_READ_ENGINE.	* Varies	■ Assert the DMA read engine enable bit. This "Soft Reset" does not clear the DMA configuration registers. The DMA read transfer does not start until you write to the DMA Read Doorbell Register (DMA_READ_DOORBELL_OFF). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable (Soft Reset) Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: untestable Volatile: true Memory Access:
0...(cont ..)	DMA_READ_ENGINE..	* Varies	"(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.9 DMA_READ_DOORBELL_OFF

- **Name:** DMA Read Doorbell Register.
- **Description:** This register controls the DMA read channel doorbell.
- **Size:** 32 bits
- **Offset:** B+0x30
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

RD_STOP	31
RSVDP_3	30:3
RD_DOORBELL_NUM	2:0

Table 3-604 Fields for Register: DMA_READ_DOORBELL_OFF

Bits	Name	Memory Access	Description
31	RD_STOP	* Varies	Stop. Set in conjunction with the Doorbell Number field. The DMA read channel stops issuing requests, sets the channel status to "Stopped", and asserts the "Abort" interrupt if it is enabled. Before setting the Stop bit, you must read the channel Status field (CS) of the DMA Channel Control 1 Register (DMA_CH_CONTROL1_OFF_RDCH_i) to ensure that the read channel is "Running" (transferring data). For more information, see "Stopping the DMA Transfer (Software Stop)". <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
30:3	RSVDP_3	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
2:0	RD_DOORBELL_NUM	* Varies	Doorbell Number. You must write 0x0 to this register to start the DMA read transfer for that channel. The DMA detects a write to this register field even if the value of this field does not change. The range of this field is 0x0 to 0x7, where 0x0 corresponds to channel 0. <i>Note:</i> A write to this field triggers the controller to exit L1 substates. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.10 DMA_READ_ENGINE_PF_OFF

- **Name:** DMA Write Engine Pre-Fetch Register.
- **Description:** This register controls the DMA Write Engine Pre-Fetch.
- **Size:** 32 bits
- **Offset:** B+0x34
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:20	RSVDP_20
19:10	RSVDP_READ_ENGINE_PF
9:0	DMA_READ_ENGINE_MAX_PRE_FETCH

Table 3-605 Fields for Register: DMA_READ_ENGINE_PF_OFF

Bits	Name	Memory Access	Description
31:20	RSVDP_20	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_PF_ENABLE

Bits	Name	Memory Access	Description
19:10	RSVDP_READ_ENGINE_PF	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_PF_ENABLE
9:0	DMA_READ_ENGINE_MAX_PREFETCH	R/W	This field controls the linked list prefetch allocation space for each of the HDMA read engine channels. Each channel will be able to store as many linked list elements as specified here. Therefore this field must be within the following range: $[0 : \text{floor}(\text{CX_DMA_RD_LLQ} * \text{CC_NUM_DMA_RD_CHAN} / \text{Number_of_active_active_channels}) - 1]$. <i>Note:</i> If this field is setup with a value outside the specified range data corruption and channel crosstalk may happen. If changing the default value of this register, limit the doorbelled channels to the range: $[0 : \text{Number_of_active_active_channels} - 1]$. The actual prefetch allocation space is the value of this field plus one. So, for example, setting it to zero means that it has allocated space for one descriptor. Value After Reset: {CC_HDMA_REG_RD_PF_DEPTH} Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_PF_ENABLE

3.67.11 DMA_READ_CHANNEL_ARB_WEIGHT_LOW_OFF

- **Name:** DMA Read Engine Channel Arbitration Weight Low Register.
- **Description:** The 5-bit channel weight (for read channels 0-3) specifies the maximum number of TLP requests that the DMA can issue for that channel before it must return to the arbitration routine. When the channel weight count is reached or DMA channel request transfer size reaches zero, the WWR arbiter selects the next channel to be processed. Your software must initialize this register before ringing the doorbell. For more information, see "Multichannel Arbitration". Value range is (0-0x1F) corresponding to (1-32) transaction requests.
- **Size:** 32 bits
- **Offset:** B+0x38
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:20	19:15	14:10	9:5	4:0
RSVDP_20	READ_CHANNEL3_WEIGHT	READ_CHANNEL2_WEIGHT	READ_CHANNEL1_WEIGHT	READ_CHANNEL0_WEIGHT

Table 3-606 Fields for Register: DMA_READ_CHANNEL_ARB_WEIGHT_LOW_OFF

Bits	Name	Memory Access	Description
31:20	RSVDP_20	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE

Bits	Name	Memory Access	Description
19:15	READ_CHANNEL3_WEIGHT	* Varies	Channel 3 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
14:10	READ_CHANNEL2_WEIGHT	* Varies	Channel 2 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
9:5	READ_CHANNEL1_WEIGHT	* Varies	Channel 1 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
4:0	READ_CHANNEL0_WEIGHT	* Varies	Channel 0 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.12 DMA_READ_CHANNEL_ARB_WEIGHT_HIGH_OFF

- **Name:** DMA Read Engine Channel Arbitration Weight High Register.
- **Description:** The 5-bit channel weight (for read channels 4-7) specifies the maximum number of TLP requests that the DMA can issue for that channel before it must return to the arbitration routine. When the channel weight count is reached or DMA channel request transfer size reaches zero, the WWR arbiter selects the next channel to be processed. Your software must initialize this register before ringing the doorbell. For more information, see "Multichannel Arbitration". Value range is (0-0x1F) corresponding to (1-32) transaction requests.
- **Size:** 32 bits
- **Offset:** B+0x3c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:20	RSVDP_20
19:15	READ_CHANNEL7_WEIGHT
14:10	READ_CHANNEL6_WEIGHT
9:5	READ_CHANNEL5_WEIGHT
4:0	READ_CHANNEL4_WEIGHT

Table 3-607 Fields for Register: DMA_READ_CHANNEL_ARB_WEIGHT_HIGH_OFF

Bits	Name	Memory Access	Description
31:20	RSVDP_20	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE

Bits	Name	Memory Access	Description
19:15	READ_CHANNEL7_WEIGHT	* Varies	Channel 7 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
14:10	READ_CHANNEL6_WEIGHT	* Varies	Channel 6 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
9:5	READ_CHANNEL5_WEIGHT	* Varies	Channel 5 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
4:0	READ_CHANNEL4_WEIGHT	* Varies	Channel 4 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x1 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.13 DMA_WRITE_INT_STATUS_OFF

- **Name:** DMA Write Interrupt Status Register.
- **Description:** This register indicates the status of Abort and Done interrupts for the DMA write channels.
- **Size:** 32 bits
- **Offset:** B+0x4c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	RSVDP_24
23:16	WR_ABORT_INT_STATUS
15:8	RSVDP_8
7:0	WR_DONE_INT_STATUS

Table 3-608 Fields for Register: DMA_WRITE_INT_STATUS_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
23:16	WR_ABORT_INT_STATUS	* Varies	Abort Interrupt Status. The DMA write channel has detected an error, or you manually stopped the transfer as described in "Error Handling Assistance by Remote Software". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. ■ Enabling: For more information, see "Interrupts and Error Handling". ■ Masking: The DMA write interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the DMA write interrupt Clear register to clear this interrupt bit. Note: You can write to this register to emulate interrupt generation, during software or hardware testing. A write to the address triggers an interrupt, but the DMA does not set the Done or Abort bits in this register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
7:0	WR_DONE_INT_STATUS	* Varies	Done Interrupt Status. The DMA write channel has successfully completed the DMA transfer. For more information, see "Interrupts and Error Handling". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. ■ Enabling: For more information, see "Interrupts and Error Handling". ■ Masking: The DMA write interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the DMA write interrupt Clear register to clear this interrupt bit. Note: You can write to this register to emulate interrupt generation, during software or hardware testing. A write to the address triggers an interrupt, but the DMA does not set the Done or Abort bits in this register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.14 DMA_WRITE_INT_MASK_OFF

- **Name:** DMA Write Interrupt Mask Register.
- **Description:** This register holds the Abort and Done DMA write interrupt mask.
- **Size:** 32 bits
- **Offset:** B+0x54
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	RSVDP_24
x:16	WR_ABORT_INT_MASK
15:8	RSVDP_8
x:0	WR_DONE_INT_MASK

Table 3-609 Fields for Register: DMA_WRITE_INT_MASK_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
x:16	WR_ABORT_INT_MASK	* Varies	Abort Interrupt Mask. Prevents the Abort interrupt status field in the DMA write interrupt status register from asserting the edma_int output. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: {(CC_NUM_DMA_WR_CHAN == 0) ? "0x0" : (CC_NUM_DMA_WR_CHAN == 1) ? "0x1" : (CC_NUM_DMA_WR_CHAN == 2) ? "0x3" : (CC_NUM_DMA_WR_CHAN == 3) ? "0x7" : (CC_NUM_DMA_WR_CHAN == 4) ? "0xf" : (CC_NUM_DMA_WR_CHAN == 5) ? "0x1f" : (CC_NUM_DMA_WR_CHAN == 6) ? "0x3f" : (CC_NUM_DMA_WR_CHAN == 7) ? "0x7f" : "0xff"} Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Range Variable[x]: (CC_NUM_DMA_WR_CHAN) + 15 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
x:0	WR_DONE_INT_MASK	* Varies	Done Interrupt Mask. Prevents the Done interrupt status field in the DMA write interrupt status register from asserting the edma_int output. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: {(CC_NUM_DMA_WR_CHAN == 0) ? "0x0" : (CC_NUM_DMA_WR_CHAN == 1) ? "0x1" : (CC_NUM_DMA_WR_CHAN == 2) ? "0x3" : (CC_NUM_DMA_WR_CHAN == 3) ? "0x7" : (CC_NUM_DMA_WR_CHAN == 4) ? "0xf" : (CC_NUM_DMA_WR_CHAN == 5) ? "0x1f" : (CC_NUM_DMA_WR_CHAN == 6) ? "0x3f" : (CC_NUM_DMA_WR_CHAN == 7) ? "0x7f" : "0xff"} Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Range Variable[x]: (CC_NUM_DMA_WR_CHAN) - 1 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.15 DMA_WRITE_INT_CLEAR_OFF

- **Name:** DMA Write Interrupt Clear Register.
- **Description:** You can write to this register to clear the Abort and Done write interrupts.
- **Size:** 32 bits
- **Offset:** B+0x58
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	x:16	15:8	x:0
RSVDP_24	WR_ABORT_INT_CLEAR	RSVDP_8	WR_DONE_INT_CLEAR

Table 3-610 Fields for Register: DMA_WRITE_INT_CLEAR_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
x:16	WR_ABORT_INT_CLEAR	W	Abort Interrupt Clear. You must write a 1'b1 to clear the corresponding bit in the Abort interrupt status field of the DMA write interrupt status register. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: Reading from this self-clearing register field always returns a '0'. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true Range Variable[x]: CC_NUM_DMA_WR_CHAN + 15
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE
x:0	WR_DONE_INT_CLEAR	W	Done Interrupt Clear. You must write a 1'b1 to clear the corresponding bit in the Done interrupt status field of the DMA write interrupt status register. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: Reading from this self-clearing register field always returns a '0'. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true Range Variable[x]: CC_NUM_DMA_WR_CHAN - 1

3.67.16 DMA_WRITE_ERR_STATUS_OFF

- **Name:** DMA Write Error Status Register.
- **Description:** This register provides information pertaining to errors incurred during write channel operation.
- **Size:** 32 bits
- **Offset:** B+0x5c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	RSVDP_24
23:16	LINKLIST_ELEMENT_FETCH_ERR_DETECT
15:8	RSVDP_8
7:0	APP_READ_ERR_DETECT

Table 3-611 Fields for Register: DMA_WRITE_ERR_STATUS_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
23:16	LINKLIST_ELEMENT_FETCH_ERROR_DETECT	R	Linked List Element Fetch Error Detected. The DMA write channel has received an error response from the AXI bus (or TRGT1 interface when the AXI Bridge is not used) while reading a linked list element from local memory. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. ■ Enabling: For more information, see "Interrupts and Error Handling". ■ Masking: The DMA write interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the DMA Write Interrupt Clear Register (DMA_WRITE_INT_CLEAR_OFF) to clear this error bit. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
7:0	APP_READ_ERR_DETECT	R	Application Read Error Detected. The DMA write channel has received an error response from the AXI bus (or TRGT1 interface when the AXI Bridge is not used) while reading data from it. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. ■ Enabling: For more information, see "Interrupts and Error Handling". ■ Masking: The DMA write interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the DMA Write Interrupt Clear Register (DMA_WRITE_INT_ - CLEAR_OFF) to clear this error bit. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true

3.67.17 DMA_WRITE_DONE_IMWR_LOW_OFF

- **Name:** DMA Write Done IMWr Address Low Register.
- **Description:** This register holds the lower 32 bits of the Done IMWr TLP address.
- **Size:** 32 bits
- **Offset:** B+0x60
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

DMA_WRITE_DONE_LOW_REG 31:0

Table 3-612 Fields for Register: DMA_WRITE_DONE_IMWR_LOW_OFF

Bits	Name	Memory Access	Description
31:0	DMA_WRITE_DONE_LOW_REG	* Varies	The DMA uses this field to generate bits [31:0] of the address field for the Done IMWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.18 DMA_WRITE_DONE_IMWR_HIGH_OFF

- **Name:** DMA Write Done IMWr Interrupt Address High Register.
- **Description:** This register holds the higher 32 bits of the Done IMWr TLP address.
- **Size:** 32 bits
- **Offset:** B+0x64
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

DMA_WRITE_DONE_HIGH_REG 31:0

Table 3-613 Fields for Register: DMA_WRITE_DONE_IMWR_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	DMA_WRITE_DONE_HIGH_REG	* Varies	The DMA uses this field to generate bits [63:32] of the address field for the Done IMWr TLP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.19 DMA_WRITE_ABORT_IMWR_LOW_OFF

- **Name:** DMA Write Abort IMWr Address Low Register.
- **Description:** This register holds the lower 32 bits of Abort IMWr TLP.
- **Size:** 32 bits
- **Offset:** B+0x68
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

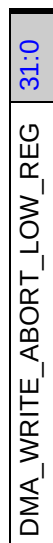


Table 3-614 Fields for Register: DMA_WRITE_ABORT_IMWR_LOW_OFF

Bits	Name	Memory Access	Description
31:0	DMA_WRITE_ABORT_LOW_REG	* Varies	The DMA uses this field to generate bits [31:0] of the address field for the Abort IMWr TLP it generates. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.20 DMA_WRITE_ABORT_IMWR_HIGH_OFF

- **Name:** DMA Write Abort IMWr Address High Register.
- **Description:** This register holds the higher 32 bits of Abort IMWr TLP.
- **Size:** 32 bits
- **Offset:** B+0x6c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

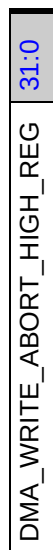


Table 3-615 Fields for Register: DMA_WRITE_ABORT_IMWR_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	DMA_WRITE_ABORT_HIGH_REG	* Varies	The DMA uses this field to generate bits [63:32] of the address field for the Abort IMWr TLP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.21 DMA_WRITE_CH01_IMWR_DATA_OFF

- **Name:** DMA Write Channel 1 and 0 IMWr Data Register.
- **Description:** This register holds the Channel 1 and 0 IMWr Done or Abort TLP Data.
- **Size:** 32 bits
- **Offset:** B+0x70
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

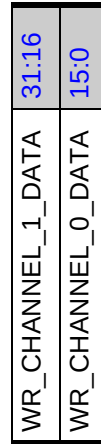


Table 3-616 Fields for Register: DMA_WRITE_CH01_IMWR_DATA_OFF

Bits	Name	Memory Access	Description
31:16	WR_CHANNEL_1_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
15:0	WR_CHANNEL_0_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.22 DMA_WRITE_CH23_IMWR_DATA_OFF

- **Name:** DMA Write Channel 3 and 2 IMWr Data Register.
- **Description:** This register holds the Channel 3 and 2 IMWr Done or Abort TLP Data.
- **Size:** 32 bits
- **Offset:** B+0x74
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:16	WR_CHANNEL_3_DATA
15:0	WR_CHANNEL_2_DATA

Table 3-617 Fields for Register: DMA_WRITE_CH23_IMWR_DATA_OFF

Bits	Name	Memory Access	Description
31:16	WR_CHANNEL_3_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 3. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
15:0	WR_CHANNEL_2_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 2. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.23 DMA_WRITE_CH45_IMWR_DATA_OFF

- **Name:** DMA Write Channel 5 and 4 IMWr Data Register.
- **Description:** This register holds the Channel 5 and 4 IMWr Done or Abort TLP Data.
- **Size:** 32 bits
- **Offset:** B+0x78
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:16	WR_CHANNEL_5_DATA
15:0	WR_CHANNEL_4_DATA

Table 3-618 Fields for Register: DMA_WRITE_CH45_IMWR_DATA_OFF

Bits	Name	Memory Access	Description
31:16	WR_CHANNEL_5_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 5. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
15:0	WR_CHANNEL_4_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 4. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.24 DMA_WRITE_CH67_IMWR_DATA_OFF

- **Name:** DMA Write Channel 7 and 6 IMWr Data Register.
- **Description:** This register holds the Channel 7 and 6 IMWr Done or Abort TLP Data.
- **Size:** 32 bits
- **Offset:** B+0x7c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:16	WR_CHANNEL_7_DATA
15:0	WR_CHANNEL_6_DATA

Table 3-619 Fields for Register: DMA_WRITE_CH67_IMWR_DATA_OFF

Bits	Name	Memory Access	Description
31:16	WR_CHANNEL_7_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 7. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
15:0	WR_CHANNEL_6_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 6. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.25 DMA_WRITE_LINKED_LIST_ERR_EN_OFF

- **Name:** DMA Write Linked List Error Enable Register.
- **Description:** The LIE and RIE bits in the LL element enable the channel "done" interrupts (local and remote). The LLLAIE and LLRAIE bits enable the channel "abort" interrupts (local and remote).
- **Size:** 32 bits
- **Offset:** B+0x90
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	RSVDP_24
x:16	WR_CHANNEL_LLLAIE
15:8	RSVDP_8
x:0	WR_CHANNEL_LLRAIE

Table 3-620 Fields for Register: DMA_WRITE_LINKED_LIST_ERR_EN_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
x:16	WR_CHANNEL_LLLAIE	* Varies	Write Channel LL Local Abort Interrupt Enable (LLLAIE). You enable the write channel local abort interrupt through this bit. The LIE and RIE bits in the LL element enable the write channel done interrupts. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Used in linked list mode only. For more information, see "Interrupt Handling". <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Range Variable[x]: CC_NUM_DMA_WR_CHAN + 15 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
x:0	WR_CHANNEL_LLRAIE	* Varies	Write Channel LL Remote Abort Interrupt Enable (LLRAIE). You enable the write channel remote abort interrupt through this bit. The LIE and RIE bits in the LL element enable the write channel done interrupts. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Used in linked list mode only. For more information, see "Interrupt Handling". <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Range Variable[x]: CC_NUM_DMA_WR_CHAN - 1 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.26 DMA_READ_INT_STATUS_OFF

- **Name:** DMA Read Interrupt Status Register.
- **Description:** This register indicates the status of Abort and Done interrupts for the DMA read channels.
- **Size:** 32 bits
- **Offset:** B+0xa0
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	RSVDP_24
23:16	RD_ABORT_INT_STATUS
15:8	RSVDP_8
7:0	RD_DONE_INT_STATUS

Table 3-621 Fields for Register: DMA_READ_INT_STATUS_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
23:16	RD_ABORT_INT_STATUS	* Varies	Abort Interrupt Status. The DMA read channel has detected an error, or you manually stopped the transfer as described in "Stopping the DMA Transfer (Software Stop)". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. You can read the DMA Read Error Status Low Register (DMA_READ_ERR_STATUS_LOW_OFF) and DMA Read Error Status High Register (DMA_READ_ERR_STATUS_HIGH_OFF) to determine the source of the error. ■ Enabling: For more information, see "Interrupts and Error Handling". ■ Masking: The DMA read interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the DMA read interrupt Clear register to clear this interrupt bit. Note: You can write to this register to emulate interrupt generation, during software or hardware testing. A write to the address triggers an interrupt, but the DMA does not set the Done or Abort bits in this register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
7:0	RD_DONE_INT_STATUS	* Varies	Done Interrupt Status. The DMA read channel has successfully completed the DMA read transfer. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. ■ Enabling: For more information, see "Interrupts and Error Handling". ■ Masking: The DMA read interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the DMA read interrupt Clear register to clear this interrupt bit. Note: You can write to this register to emulate interrupt generation, during software or hardware testing. A write to the address triggers an interrupt, but the DMA does not set the Done or Abort bits in this register. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.27 DMA_READ_INT_MASK_OFF

- **Name:** DMA Read Interrupt Mask Register.
- **Description:** This register holds the Abort and Done DMA read interrupt mask.
- **Size:** 32 bits
- **Offset:** B+0xa8
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	RSVDP_24
x:16	RD_ABORT_INT_MASK
15:8	RSVDP_8
x:0	RD_DONE_INT_MASK

Table 3-622 Fields for Register: DMA_READ_INT_MASK_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
x:16	RD_ABORT_INT_MASK	* Varies	Abort Interrupt Mask. Prevents the Abort interrupt status field in the DMA read interrupt status register from asserting the edma_int output. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: {(CC_NUM_DMA_RD_CHAN == 0) ? "0x0" : (CC_NUM_DMA_RD_CHAN == 1) ? "0x1" : (CC_NUM_DMA_RD_CHAN == 2) ? "0x3" : (CC_NUM_DMA_RD_CHAN == 3) ? "0x7" : (CC_NUM_DMA_RD_CHAN == 4) ? "0xf" : (CC_NUM_DMA_RD_CHAN == 5) ? "0x1f" : (CC_NUM_DMA_RD_CHAN == 6) ? "0x3f" : (CC_NUM_DMA_RD_CHAN == 7) ? "0x7f" : "0xff"} Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Range Variable[x]: CC_NUM_DMA_RD_CHAN + 15 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
x:0	RD_DONE_INT_MASK	* Varies	Done Interrupt Mask. Prevents the Done interrupt status field in the DMA read interrupt status register from asserting the edma_int output. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: {(CC_NUM_DMA_RD_CHAN == 0) ? "0x0" : (CC_NUM_DMA_RD_CHAN == 1) ? "0x1" : (CC_NUM_DMA_RD_CHAN == 2) ? "0x3" : (CC_NUM_DMA_RD_CHAN == 3) ? "0x7" : (CC_NUM_DMA_RD_CHAN == 4) ? "0xf" : (CC_NUM_DMA_RD_CHAN == 5) ? "0x1f" : (CC_NUM_DMA_RD_CHAN == 6) ? "0x3f" : (CC_NUM_DMA_RD_CHAN == 7) ? "0x7f" : "0xff"} Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Range Variable[x]: CC_NUM_DMA_RD_CHAN - 1 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.28 DMA_READ_INT_CLEAR_OFF

- **Name:** DMA Read Interrupt Clear Register.
- **Description:** You can write to this register to clear the Abort and Done read interrupts.
- **Size:** 32 bits
- **Offset:** B+0xac
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

RSVDP_24	31:24
RD_ABORT_INT_CLEAR	23:16
RSVDP_8	15:8
RD_DONE_INT_CLEAR	7:0

Table 3-623 Fields for Register: DMA_READ_INT_CLEAR_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
23:16	RD_ABORT_INT_CLEAR	W	Abort Interrupt Clear. You must write a 1'b1 to clear the corresponding bit in the Abort interrupt status field of the DMA read interrupt status register. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: Reading from this self-clearing register field always returns a '0'. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE
7:0	RD_DONE_INT_CLEAR	W	Done Interrupt Clear. You must write a 1'b1 to clear the corresponding bit in the Done interrupt status field of the DMA read interrupt status register. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: Reading from this self-clearing register field always returns a '0'. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true

3.67.29 DMA_READ_ERR_STATUS_LOW_OFF

- **Name:** DMA Read Error Status Low Register.
- **Description:** This register provides information pertaining to linked list element fetch error and application write error.
- **Size:** 32 bits
- **Offset:** B+0xb4
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	RSVDP_24
23:16	LINK_LIST_ELEMENT_FETCH_ERR_DETECT
15:8	RSVDP_8
7:0	APP_WR_ERR_DETECT

Table 3-624 Fields for Register: DMA_READ_ERR_STATUS_LOW_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
23:16	LINK_LIST_ELEMENT_FETCH_ERROR_DETECT	R	Linked List Element Fetch Error Detected. ■ The DMA read channel has received an error response from the AXI bus while reading a linked list element from local memory. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. ■ Enabling: For details, see "Interrupts and Error Handling". ■ Masking: The DMA read interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the DMA Read Interrupt Clear Register (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this clears all bits in this register, and also the DMA Read Error Status High register (DMA_READ_ERR_STATUS_HIGH_OFF). Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
7:0	APP_WR_ERR_DETECT	R	Application Write Error Detected. The DMA read channel has received an error response from the AXI bus (or TRGT1 interface when the AXI Bridge is not used) while writing data to it. This error is fatal. You must restart the transfer from the beginning, as the channel context is corrupted, and the transfer is not rolled back. For more information, see "Linked List Mode". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. ■ Enabling: For more information, see "Interrupts and Error Handling". ■ Masking: The DMA read interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the "DMA Read Interrupt Clear Register" (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this clears all bits in this register, and also the DMA Read Error Status High register (DMA_READ_ERR_STATUS_HIGH_OFF). Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true

3.67.30 DMA_READ_ERR_STATUS_HIGH_OFF

- **Name:** DMA Read Error Status High Register.
- **Description:** This register provides information pertaining to completion errors.
- **Size:** 32 bits
- **Offset:** B+0xb8
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	DATA_POISONING
23:16	CPL_TIMEOUT
15:8	CPL_ABORT
7:0	UNSUPPORTED_REQ

Table 3-625 Fields for Register: DMA_READ_ERR_STATUS_HIGH_OFF

Bits	Name	Memory Access	Description
31:24	DATA_POISONING	R	Data Poisoning. The DMA read channel has detected data poisoning in the completion from the remote device (in response to the MRd request). The DMA read channel will drop the completion and then be halted. The CX_FLT_MASK_UR_POIS filter rule does not affect this behavior. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. ■ Enabling: For more information, see "Interrupts and Error Handling". ■ Masking: The DMA read interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the DMA Read Interrupt Clear Register (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this also clears the other error bits for the same channel in this register and in the DMA Read Error Status Low register. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true
23:16	CPL_TIMEOUT	R	Completion Time Out. The DMA read channel has timed-out while waiting for the remote device to respond to the MRd request, or a malformed CplD has been received. For more information, see "Linked List Mode". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. ■ Enabling: For details, see "Interrupts and Error Handling". ■ Masking: The DMA read interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the DMA Read Interrupt Clear Register (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this also clears the other error bits for the same channel in this register and in the DMA Read Error Status Low register. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
15:8	CPL_ABORT	R	Completer Abort. The DMA read channel has received a PCIe completer abort completion status from the remote device in response to the MRd request. For more information, see "Linked List Mode". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. ■ Enabling: For more information, see "Interrupts and Error Handling". ■ Masking: The DMA read interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the DMA Read Interrupt Clear Register (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this also clears the other error bits for the same channel in this register and in the DMA Read Error Status Low register. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true
7:0	UNSUPPORTED_REQ	R	Unsupported Request. The DMA read channel has received a PCIe unsupported request completion status from the remote device in response to the MRd request. For more information, see "Linked List Mode". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. ■ Enabling: For more information, see "Interrupts and Error Handling". ■ Masking: The DMA read interrupt Mask register has no effect on this register. ■ Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the DMA Read Interrupt Clear Register (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this also clears the other error bits for the same channel in this register and in the DMA Read Error Status Low register. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Testable: writeAsRead Volatile: true

3.67.31 DMA_READ_LINKED_LIST_ERR_EN_OFF

- **Name:** DMA Read Linked List Error Enable Register.
- **Description:** The LIE and RIE bits in the LL element enable the channel "done" interrupts (local and remote). The LLLAIE and LLRAIE bits enable the channel "abort" interrupts (local and remote).
- **Size:** 32 bits
- **Offset:** B+0xc4
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:24	RSVDP_24
x:16	RD_CHANNEL_LLLAIE
15:8	RSVDP_8
x:0	RD_CHANNEL_LLRAIE

Table 3-626 Fields for Register: DMA_READ_LINKED_LIST_ERR_EN_OFF

Bits	Name	Memory Access	Description
31:24	RSVDP_24	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
x:16	RD_CHANNEL_LLLAIE	* Varies	Read Channel LL Local Abort Interrupt Enable (LLLAIE). You enable the read channel Local Abort interrupt through this bit. The LIE and RIE bits in the LL element enable the read channel done interrupts. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Used in linked list mode only. For more information, see "Interrupt Handling". <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Range Variable[x]: CC_NUM_DMA_RD_CHAN + 15 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
15:8	RSVDP_8	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE

Bits	Name	Memory Access	Description
x:0	RD_CHANNEL_LLRAIE	* Varies	Read Channel LL Remote Abort Interrupt Enable (LLRAIE). You enable the read channel Remote Abort interrupt through this bit. The LIE and RIE bits in the LL element enable the read channel done interrupts. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Used in linked list mode only. For more information, see "Interrupt Handling". <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Range Variable[x]: CC_NUM_DMA_RD_CHAN - 1 Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.32 DMA_READ_DONE_IMWR_LOW_OFF

- **Name:** DMA Read Done IMWr Address Low Register.
- **Description:** This register holds the lower 32 bits of the Done IMWr TLP address.
- **Size:** 32 bits
- **Offset:** B+0xcc
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))



Table 3-627 Fields for Register: DMA_READ_DONE_IMWR_LOW_OFF

Bits	Name	Memory Access	Description
31:0	DMA_READ_DONE_LOW_REG	* Varies	The DMA uses this field to generate bits [31:0] of the address field for the Done IMWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.33 DMA_READ_DONE_IMWR_HIGH_OFF

- **Name:** DMA Read Done IMWr Address High Register.
- **Description:** This register holds the higher 32 bits of the Done IMWr TLP address.
- **Size:** 32 bits
- **Offset:** B+0xd0
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

DMA_READ_DONE_HIGH_REG 31:0

Table 3-628 Fields for Register: DMA_READ_DONE_IMWR_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	DMA_READ_DONE_HIGH_REG	* Varies	The DMA uses this field to generate bits [63:32] of the address field for the Done IMWr TLP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.34 DMA_READ_ABORT_IMWR_LOW_OFF

- **Name:** DMA Read Abort IMWr Address Low Register.
- **Description:** This register holds the lower 32 bits of the Abort IMWr TLP address.
- **Size:** 32 bits
- **Offset:** B+0xd4
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

DMA_READ_ABORT_LOW_REG 31:0

Table 3-629 Fields for Register: DMA_READ_ABORT_IMWR_LOW_OFF

Bits	Name	Memory Access	Description
31:0	DMA_READ_ABORT_LOW_REG	* Varies	The DMA uses this field to generate bits [31:0] of the address field for the Abort IMWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.35 DMA_READ_ABORT_IMWR_HIGH_OFF

- **Name:** DMA Read Abort IMWr Address High Register.
- **Description:** This register holds the higher 32 bits of the Abort IMWr TLP address.
- **Size:** 32 bits
- **Offset:** B+0xd8
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

DMA_READ_ABORT_HIGH_REG 31:0

Table 3-630 Fields for Register: DMA_READ_ABORT_IMWR_HIGH_OFF

Bits	Name	Memory Access	Description
31:0	DMA_READ_ABORT_HIGH_REG	* Varies	The DMA uses this field to generate bits [63:32] of the address field for the Abort IMWr TLP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.36 DMA_READ_CH01_IMWR_DATA_OFF

- **Name:** DMA Read Channel 1 and 0 IMWr Data Register.
- **Description:** This register holds the Channel 1 and 0 IMWr Done or Abort TLP Data.
- **Size:** 32 bits
- **Offset:** B+0xdc
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:16	RD_CHANNEL_1_DATA
15:0	RD_CHANNEL_0_DATA

Table 3-631 Fields for Register: DMA_READ_CH01_IMWR_DATA_OFF

Bits	Name	Memory Access	Description
31:16	RD_CHANNEL_1_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
15:0	RD_CHANNEL_0_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 0. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.37 DMA_READ_CH23_IMWR_DATA_OFF

- **Name:** DMA Read Channel 3 and 2 IMWr Data Register.
- **Description:** This register holds the Channel 3 and 2 IMWr Done or Abort TLP Data.
- **Size:** 32 bits
- **Offset:** B+0xe0
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:16	RD_CHANNEL_3_DATA
15:0	RD_CHANNEL_2_DATA

Table 3-632 Fields for Register: DMA_READ_CH23_IMWR_DATA_OFF

Bits	Name	Memory Access	Description
31:16	RD_CHANNEL_3_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 3. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
15:0	RD_CHANNEL_2_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 2. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.38 DMA_READ_CH45_IMWR_DATA_OFF

- **Name:** DMA Read Channel 5 and 4 IMWr Data Register.
- **Description:** This register holds the Channel 5 and 4 IMWr Done or Abort TLP Data.
- **Size:** 32 bits
- **Offset:** B+0xe4
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:16	RD_CHANNEL_5_DATA
15:0	RD_CHANNEL_4_DATA

Table 3-633 Fields for Register: DMA_READ_CH45_IMWR_DATA_OFF

Bits	Name	Memory Access	Description
31:16	RD_CHANNEL_5_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 5. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
15:0	RD_CHANNEL_4_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 4. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.39 DMA_READ_CH67_IMWR_DATA_OFF

- **Name:** DMA Read Channel 7 and 6 IMWr Data Register.
- **Description:** This register holds the Channel 7 and 6 IMWr Done or Abort TLP Data.
- **Size:** 32 bits
- **Offset:** B+0xe8
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:16	RD_CHANNEL_7_DATA
15:0	RD_CHANNEL_6_DATA

Table 3-634 Fields for Register: DMA_READ_CH67_IMWR_DATA_OFF

Bits	Name	Memory Access	Description
31:16	RD_CHANNEL_7_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 7. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
15:0	RD_CHANNEL_6_DATA	* Varies	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 6. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.40 DMA_WRITE_ENGINE_CHGROUP_OFF

- **Name:** DMA Write Engine Channel Grouping Control Register.
- **Description:** This register controls the DMA Write Engine Channel Grouping.
- **Size:** 32 bits
- **Offset:** B+0x100
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:17	RSVDP_17
16	DMA_WRITE_ENGINE_CHGROUP_ENABLE
15:8	DMA_WRITE_ENGINE_CHGROUP_END
7:0	DMA_WRITE_ENGINE_CHGROUP_START

Table 3-635 Fields for Register: DMA_WRITE_ENGINE_CHGROUP_OFF

Bits	Name	Memory Access	Description
31:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_CHANNEL_GROUP_ENABLE

Bits	Name	Memory Access	Description
16	DMA_WRITE_ENGINE_CHGROU P_ENABLE	R/W	Channel Grouping Enable. To enable channel grouping, set this field to '1'. When channel grouping is enabled, a doorbell to channel CHGRP_START activates channels CHGRP_START to CHGRP_END. This bit must be set to '0', if any of the channels in the range, CHGRP_START to CHGRP_END, are to be used individually. For more information refer to the Channel Grouping section in DMA. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_CHANNEL_GROUP_ENABLE
15:8	DMA_WRITE_ENGINE_CHGROU P_END	R/W	Channel Group End Channel Number. This field indicates the highest channel number in the channel group. The value can range from 0 to CC_NUM_DMA_WR_CHAN. CHGRP_END is greater than CHGRP_START. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_CHANNEL_GROUP_ENABLE
7:0	DMA_WRITE_ENGINE_CHGROU P_START	R/W	Channel Group Start Channel Number. This field indicates the lowest channel number in the channel group. The value can range from 0 to CC_NUM_DMA_WR_CHAN. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_CHANNEL_GROUP_ENABLE

3.67.41 DMA_READ_ENGINE_CHGROUP_OFF

- **Name:** DMA Read Engine Channel Grouping Control Register.
- **Description:** This register controls the DMA read engine channel grouping.
- **Size:** 32 bits
- **Offset:** B+0x104
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:17	RSVDP_17
16	DMA_READ_ENGINE_CHGROUP_ENABLE
15:8	DMA_READ_ENGINE_CHGROUP_END
7:0	DMA_READ_ENGINE_CHGROUP_START

Table 3-636 Fields for Register: DMA_READ_ENGINE_CHGROUP_OFF

Bits	Name	Memory Access	Description
31:17	RSVDP_17	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_CHANNEL_GROUP_ENABLE

Bits	Name	Memory Access	Description
16	DMA_READ_ENGINE_CHGROUP_ENABLE	R/W	Channel Grouping Enable. To enable read channel grouping, set this field to '1'. When channel grouping is enabled, a doorbell to channel CHGRP_START activates channels CHGRP_START to CHGRP_END. This bit must be set to '0', if any of the channels in the range CHGRP_START to CHGRP_END are to be used individually. For more information, refer to the Channel Grouping section in DMA. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_CHANNEL_GROUP_ENABLE
15:8	DMA_READ_ENGINE_CHGROUP_END	R/W	Channel Group End Channel Number. This field indicates the highest channel number in the channel group. The value can range from 0 to CC_NUM_DMA_RD_CHAN. CHGRP_END is greater than CHGRP_START. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_CHANNEL_GROUP_ENABLE
7:0	DMA_READ_ENGINE_CHGROUP_START	R/W	Channel Group Start Channel Number. This field indicates the lowest channel number in the channel group. The value can range from 0 to CC_NUM_DMA_RD_CHAN. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CX_DMA_CHANNEL_GROUP_ENABLE

3.67.42 DMA_WRITE_ENGINE_HSHAKE_CNT_LOW_OFF

- **Name:** DMA Write Engine Handshake Counter Channel 0/1/2/3 Register.
- **Description:** This register holds the handshake counters write channels 0 - 3.
- **Size:** 32 bits
- **Offset:** B+0x108
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:29	RSVDP_29
28:24	DMA_WRITE_ENGINE_HSHAKE_CNT_CH3
23:21	RSVDP_21
20:16	DMA_WRITE_ENGINE_HSHAKE_CNT_CH2
15:13	RSVDP_13
12:8	DMA_WRITE_ENGINE_HSHAKE_CNT_CH1
7:5	RSVDP_5
4:0	DMA_WRITE_ENGINE_HSHAKE_CNT_CH0

Table 3-637 Fields for Register: DMA_WRITE_ENGINE_HSHAKE_CNT_LOW_OFF

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

Bits	Name	Memory Access	Description
28:24	DMA_WRITE_ENGINE_HSHAKE_CNT_CH3	R	DMA handshake counter for DMA Write Engine Channel 3. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
23:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
20:16	DMA_WRITE_ENGINE_HSHAKE_CNT_CH2	R	DMA handshake counter for DMA Write Engine Channel 2. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
15:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
12:8	DMA_WRITE_ENGINE_HSHAKE_CNT_CH1	R	DMA handshake counter for DMA Write Engine Channel 1. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

Bits	Name	Memory Access	Description
7:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
4:0	DMA_WRITE_ENGINE_HSHAKE_CNT_CH0	R	DMA handshake counter for DMA Write Engine Channel 0. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

3.67.43 DMA_WRITE_ENGINE_HSHAKE_CNT_HIGH_OFF

- **Name:** DMA Write Engine Handshake Counter Channel 4/5/6/7 Register.
- **Description:** This register holds the handshake counters write channels 4 - 7.
- **Size:** 32 bits
- **Offset:** B+0x10c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:29	RSVDP_29
28:24	DMA_WRITE_ENGINE_HSHAKE_CNT_CH7
23:21	RSVDP_21
20:16	DMA_WRITE_ENGINE_HSHAKE_CNT_CH6
15:13	RSVDP_13
12:8	DMA_WRITE_ENGINE_HSHAKE_CNT_CH5
7:5	RSVDP_5
4:0	DMA_WRITE_ENGINE_HSHAKE_CNT_CH4

Table 3-638 Fields for Register: DMA_WRITE_ENGINE_HSHAKE_CNT_HIGH_OFF

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

Bits	Name	Memory Access	Description
28:24	DMA_WRITE_ENGINE_HSHAKE_CNT_CH7	R	DMA handshake counter for DMA Write Engine Channel 7. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
23:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
20:16	DMA_WRITE_ENGINE_HSHAKE_CNT_CH6	R	DMA handshake counter for DMA Write Engine Channel 6. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
15:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
12:8	DMA_WRITE_ENGINE_HSHAKE_CNT_CH5	R	DMA handshake counter for DMA Write Engine Channel 5. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

Bits	Name	Memory Access	Description
7:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
4:0	DMA_WRITE_ENGINE_HSHAKE_CNT_CH4	R	DMA handshake counter for DMA Write Engine Channel 4. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

3.67.44 DMA_READ_ENGINE_HSHAKE_CNT_LOW_OFF

- **Name:** DMA Read Engine Handshake Counter Channel 0/1/2/3 Register.
- **Description:** This register holds the handshake counters read channels 0 - 3.
- **Size:** 32 bits
- **Offset:** B+0x118
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

RSVDP_29	31:29
DMA_READ_ENGINE_HSHAKE_CNT_CH3	28:24
RSVDP_21	23:21
DMA_READ_ENGINE_HSHAKE_CNT_CH2	20:16
RSVDP_13	15:13
DMA_READ_ENGINE_HSHAKE_CNT_CH1	12:8
RSVDP_5	7:5
DMA_READ_ENGINE_HSHAKE_CNT_CH0	4:0

Table 3-639 Fields for Register: DMA_READ_ENGINE_HSHAKE_CNT_LOW_OFF

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

Bits	Name	Memory Access	Description
28:24	DMA_READ_ENGINE_HSHAKE_CNT_CH3	R	DMA handshake counter for DMA Read Engine Channel 3. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
23:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
20:16	DMA_READ_ENGINE_HSHAKE_CNT_CH2	R	DMA handshake counter for DMA Read Engine Channel 2. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
15:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
12:8	DMA_READ_ENGINE_HSHAKE_CNT_CH1	R	DMA handshake counter for DMA Read Engine Channel 1. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

Bits	Name	Memory Access	Description
7:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
4:0	DMA_READ_ENGINE_HSHAKE_CNT_CH0	R	DMA handshake counter for DMA Read Engine Channel 0. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

3.67.45 DMA_READ_ENGINE_HSHAKE_CNT_HIGH_OFF

- **Name:** DMA Read Engine Handshake Counter Channel 4/5/6/7 Register.
- **Description:** This register holds the handshake counters read channels 4 - 7.
- **Size:** 32 bits
- **Offset:** B+0x11c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

RSVDP_29	31:29
DMA_READ_ENGINE_HSHAKE_CNT_CH7	28:24
RSVDP_21	23:21
DMA_READ_ENGINE_HSHAKE_CNT_CH6	20:16
RSVDP_13	15:13
DMA_READ_ENGINE_HSHAKE_CNT_CH5	12:8
RSVDP_5	7:5
DMA_READ_ENGINE_HSHAKE_CNT_CH4	4:0

Table 3-640 Fields for Register: DMA_READ_ENGINE_HSHAKE_CNT_HIGH_OFF

Bits	Name	Memory Access	Description
31:29	RSVDP_29	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

Bits	Name	Memory Access	Description
28:24	DMA_READ_ENGINE_HSHAKE_CNT_CH7	R	DMA handshake counter for DMA Read Engine Channel 7. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
23:21	RSVDP_21	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
20:16	DMA_READ_ENGINE_HSHAKE_CNT_CH6	R	DMA handshake counter for DMA Read Engine Channel 6. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
15:13	RSVDP_13	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
12:8	DMA_READ_ENGINE_HSHAKE_CNT_CH5	R	DMA handshake counter for DMA Read Engine Channel 5. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

Bits	Name	Memory Access	Description
7:5	RSVDP_5	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE
4:0	DMA_READ_ENGINE_HSHAKE_CNT_CH4	R	DMA handshake counter for DMA Read Engine Channel 4. If CC_DMA_HSHAKE =1, the data transfer in Linked List mode starts only when the counter is non-zero. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && CC_DMA_HSHAKE

3.67.46 DMA_WRITE_CHi_PWR_EN_OFF (for i = 0; i <= 7)

- **Name:** DMA Write Channel 0 Power Enable Register.
- **Description:** This register controls power enable/disable for write channel 0.
- **Size:** 32 bits
- **Offset:** B+0x128 + i*0x4
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:1	0
RSVDP_1	DMA_WRITE_CH0_PWR_EN

Table 3-641 Fields for Register: DMA_WRITE_CHi_PWR_EN_OFF (for i = 0; i <= 7)

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && SNPS_RSVDPARAM_60 && (!CX_DMA_PF_ENABLE && CC_NUM_DMA_WR_CHAN>0)

Bits	Name	Memory Access	Description
0	DMA_WRITE_CH0_PWR_EN	R/W	DMA Write Channel 0 Power enable/disable. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && SNPS_RSVDPARAM_60 && (!CX_DMA_PF_ENABLE && CC_NUM_DMA_WR_CHAN>0)

3.67.47 DMA_READ_CHi_PWR_EN_OFF (for i = 0; i <= 7)

- **Name:** DMA Read Channel 0 Power Enable Register.
- **Description:** This register controls power enable/disable for read channel 0.
- **Size:** 32 bits
- **Offset:** B+0x168 + i*0x4
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1))

31:1	
	0
RSVDP_1	DMA_READ_CH0_PWR_EN

Table 3-642 Fields for Register: DMA_READ_CHi_PWR_EN_OFF (for i = 0; i <= 7)

Bits	Name	Memory Access	Description
31:1	RSVDP_1	R	Reserved for future use. Reserved Field: Yes Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && SNPS_RSVDPARAM_60 && (!CX_DMA_PF_ENABLE && CC_NUM_DMA_RD_CHAN>0)

Bits	Name	Memory Access	Description
0	DMA_READ_CH0_PWR_EN	R/W	DMA Read Channel 0 Power enable/disable. Values: ■ 0x1 (ENABLE): Enable ■ 0x0 (DISABLE): Disable Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && SNPS_RSVDPARAM_60 && (!CX_DMA_PF_ENABLE && CC_NUM_DMA_RD_CHAN>0)

3.67.48 DMA_CH_CONTROL1_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** DMA Write Channel Control 1 Register.
- **Description:** This register controls the DMA write channel operation.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x200
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

DMA_AT	31:30
DMA_TC	29:27
DMA_RESERVED5	26
DMA_RO	25
DMA_NS_SRC	24
DMA_NS_DST	23
DMA_MEM_TYPE	22
DMA_ELEMENT_PREFETCH	21:17
DMA_FUNC_NUM	16:12
DMA_RESERVED1	11:10
LLE	9
CCS	8
DMA_RESERVED0	7
CS	6:5
RIE	4
LIE	3
LLP	2
TCB	1
CB	0

Table 3-643 Fields for Register: DMA_CH_CONTROL1_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:30	DMA_AT	* Varies	Address Translation TLP Header Bit (AT). The DMA uses this TLP header field when generating MRd/MWr (not IMWr) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
29:27	DMA_TC	* Varies	Traffic Class TLP Header Bit (TC). The DMA uses this TLP header field when generating MRd/MWr (not IMWr) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
26	DMA_RESERVED5	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
25	DMA_RO	* Varies	Relaxed Ordering TLP Header Bit (RO). DMA uses this TLP header field when generating MRd/MWr (not IMWr) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
24	DMA_NS_SRC	* Varies	Source No Snoop TLP Header Bit. The DMA uses this TLP header field when generating MRd (SAR addressing space) (not IMWr) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
23	DMA_NS_DST	* Varies	Destination No Snoop TLP Header Bit. The DMA uses this TLP header field when generating MWr (DAR addressing space) (not IMWr) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
22	DMA_MEM_TYPE	* Varies	Master AXI ACE-Lite Cache Coherency Control. This field sets the DMA channel memory type of the address space of the data transfer as follows: For more information, see "ACE-Lite Features and Limitations" section of the <i>Databook</i>. Irrespective of the value of this field, Linked list descriptor requests have this attribute set to 0x1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (PERIPHERAL): peripheral type ■ 0x1 (MEM_TYPE): memory type Value After Reset: 0x0 Exists: CC_DMA_ENABLE && AXI_POPULATED && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
21:17	DMA_ELEMENT_PREFETCH	* Varies	This field controls how many elements the HDMA Write Channel should prefetch and its updated by the linked list element pointer loaded by HDMA. The valid value range is [0 : DMA_WRITE_ENGINE_PF_OFF.DMA_WRITE_ENGINE_MAX_PREFETCH]. <i>Note:</i> The actual prefetch amount is the value of this register plus one. For example, if set with 0, the prefetch is 1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_DMA_PF_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
16:12	DMA_FUNC_NUM	* Varies	DMA Channel Physical Function Number. This field indicates the physical function associated with the DMA channel. The controller uses this field to form the requester ID of the generated MRd/MWr DMA TLPs. <i>Note:</i> ■ When SR-IOV is enabled, you must consider DMA_VF and DMA_VF_EN fields of DMA_CH_CONTROL2_OFF_* as well to determine the requester ID for the requests. ■ The behavior is undefined if any value outside of the valid range is set for this field. ■ The DMA drives the value of this field in the XADM/RADM signals d_client[0]1_tlp_func_num/d_radm_trgt1_func_num. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Physical Channel Function Number starts at 0. ■ 0x1f (Max_Val): The maximum number of Physical Functions is given by CX_NFUNC, so the maximum value of this field is CX_NFUNC-1. Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
11:10	DMA_RESERVED1	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
9	LLE	* Varies	Linked List Enable (LLE). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x1 (ENABLE): Enable linked list operation ■ 0x0 (DISABLE): Disable linked list operation Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
8	CCS	* Varies	Consumer Cycle State (CCS). Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more information, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". You must initialize this bit. The DMA updates this bit during linked list operation. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
7	DMA_RESERVED0	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
6:5	CS	R/W	Channel Status (CS). The channel status bits identify the current operational state of the DMA channel. The operation state encoding for each DMA channel is as follows: Values: ■ 0x0 (RESERVED): Reserved ■ 0x1 (RUNNING): This channel is active and transferring data ■ 0x2 (HALTED): An error condition has been detected, and the DMA has stopped this channel ■ 0x3 (STOPPED): The DMA has transferred all data for this channel, or you have prematurely stopped this channel by writing to the Stop field of the DMA Write Doorbell Register (DMA_WRITE_DOORBELL_OFF) or DMA Read Doorbell Register (DMA_READ_DOORBELL_OFF) Value After Reset: {(CX_DMA_PF_ENABLE==0) ? 0 : 3} Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
4	RIE	* Varies	Remote Interrupt Enable (RIE). You must set this bit to enable the generation of the Done or Abort Remote interrupts. For more information, see "Interrupts and Error Handling". In LL mode, the DMA overwrites this with the RIE of the LL element. The RIE bit in a LL element only enables the Done interrupt. In non-LL mode, the RIE bit enables the Done and Abort interrupts. This field is not defined in a <i>link</i> LL element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
3	LIE	* Varies	Local Interrupt Enable (LIE). You must set this bit to enable the generation of the Done or Abort Local interrupts. For more information, see "Interrupts and Error Handling". In LL mode, the DMA overwrites this with the LIE of the LL element. The LIE bit in a LL element only enables the Done interrupt. In non-LL mode, the LIE bit enables the Done and Abort interrupts. This field is not defined in a <i>link</i> LL element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
2	LLP	* Varies	Load Link Pointer (LLP). Used in linked list mode only. Indicates that this linked list element is a link element, and its LL element pointer dwords are pointing to the next (non-contiguous) element. The DMA loads this field with the LLP of the linked list element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
1	TCB	* Varies	Toggle Cycle Bit (TCB). Indicates to the DMA to toggle its interpretation of the CB. Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more information, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". The DMA loads this field with the TCB of the linked list element. this field is not defined in a data LL element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\": \"read-write\" "

Bits	Name	Memory Access	Description
0	CB	* Varies	Cycle Bit (CB). Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more information, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". The DMA loads this field with the CB of the linked list element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.49 DMA_CH_CONTROL2_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** DMA Write Channel Control 2 Register.
- **Description:** This register controls the DMA write channel operation. This register provides control features in addition to DMA Write Channel Control 1 Register.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x204
- **Exists:** !CX_PL_REG_DISABLE && CC_DMA_ENABLE && (CX_SRIOV_ENABLE || CX_TPH_ENABLE)

DMA_TH	31
DMA_PH	30:29
DMA_RESERVED3	28:25
DMA_ST	24:17
DMA_VF_RESERVED	16:9
DMA_VF	8:1
DMA_VF_EN	0

Table 3-644 Fields for Register: DMA_CH_CONTROL2_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31	DMA_TH	* Varies	TLP Processing Hints Present (TH). This field is mapped onto the corresponding TLP field for wire DMA MWr requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && CC_DMA_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
30:29	DMA_PH	* Varies	Processing Hints (PH). This field is mapped onto the corresponding TLP field for wire DMA MWr requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && CC_DMA_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
28:25	DMA_RESERVED3	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && CC_DMA_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
24:17	DMA_ST	* Varies	Steering Tag (ST). This field is mapped onto the corresponding TLP field for wire DMA MWr requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && CC_DMA_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
16:9	DMA_VF_RESERVED	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DMA_ENABLE && CC_DEVICE_TYPE != 1 && CC_DEVICE_TYPE != 3 && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
8:1	DMA_VF	* Varies	DMA Channel Virtual Function Number. When the DMA_VF_EN field is '1', the controller uses both the DMA_FUNC_NUM field in DMA Read Channel Control 1 Register (DMA_CH_CONTROL2_OFF_*) and this field to form the requester ID of the generated MRd/MWr DMA TLPs. <i>Note:</i> ■ When the DMA_VF_EN field is '0', the controller uses the DMA_FUNC_NUM field in DMA Read Channel Control 1 Register (DMA_CH_CONTROL1_OFF_RDCH_i) for generating the requester ID of the generated MRd/MWr DMA TLPs. ■ The behavior is undefined if any value outside of the valid range is set for this field. ■ The DMA drives the value of this field in the XADM/RADM signals d_client[0]1_tlp_vfunc_num/d_radm_trgt1_vfunc_num. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Virtual Channel Function Number starts at 0. ■ 0xff (Max_Val): The maximum number of internal virtual functions varies from 2 to CX_NVFUNC, so the maximum value of this field varies from 1 to CX_NVFUNC-1. Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DMA_ENABLE && CC_DEVICE_TYPE != 1 && CC_DEVICE_TYPE != 3 && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
0	DMA_VF_EN	* Varies	DMA Channel Virtual Function Enable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): When this field is '1', the controller uses both the DMA_FUNC_NUM field in DMA Read Channel Control 1 Register (DMA_CH_CONTROL1_OFF_RDCH_i) and the DMA_VF field of this registers to form the requester ID of the generated MRd/MWr DMA TLPs. Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DMA_ENABLE && CC_DEVICE_TYPE != 1 && CC_DEVICE_TYPE != 3 && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.50 DMA_TRANSFER_SIZE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** DMA Write Transfer Size Register.
- **Description:** This register holds the DMA write transfer size.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x208
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

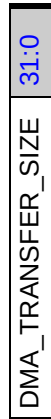


Table 3-645 Fields for Register: DMA_TRANSFER_SIZE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	DMA_TRANSFER_SIZE	* Varies	DMA Transfer Size. You program this register with the size of the DMA transfer. The maximum DMA transfer size is 4Gbytes. The minimum transfer size is one byte (0x1). This field is automatically decremented by the DMA as the DMA write channel transfer progresses. This field indicates the number bytes remaining to be transferred. When all bytes are successfully transferred the current transfer size is zero. In LL mode, the DMA overwrites this register with the corresponding dword of the LL element. You can read this register to monitor the transfer progress, however in some scenarios this register is updated after a delay. For example, when less than 3 channels are doorbelled, this register is updated only after a descriptor finishes(linked list mode), or the transfer ends (non-linked list mode). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.51 DMA_SAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** DMA Write SAR Low Register.
- **Description:** This register holds the lower 32 bits of the Source Address Register (SAR).
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x20c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

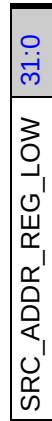


Table 3-646 Fields for Register: DMA_SAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	SRC_ADDR_REG_LOW	* Varies	Source Address Register (lower 32 bits). Indicates the next address to be read from. The DMA increments the SAR as the DMA transfer progresses. In LL mode, the DMA overwrites this with the corresponding dword of the LL element. ■ DMA Read: The SAR is the address of the remote memory. ■ DMA Write: The SAR is the address of the local memory. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.52 DMA_SAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** DMA Write SAR High Register.
- **Description:** This register holds the higher 32 bits of the Source Address Register (SAR).
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x210
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

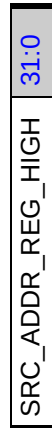


Table 3-647 Fields for Register: DMA_SAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	SRC_ADDR_REG_HIGH	* Varies	Source Address Register (higher 32 bits). In LL mode, the DMA overwrites this with the corresponding dword of the LL element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.53 DMA_DAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** DMA Write DAR Low Register.
- **Description:** This register holds the lower 32 bits of the Destination Address Register (DAR).
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x214
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)



Table 3-648 Fields for Register: DMA_DAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	DST_ADDR_REG_LOW	* Varies	Destination Address Register (lower 32 bits). Indicates the next address to be written to. The DMA increments the DAR as the DMA transfer progresses. In LL mode, the DMA overwrites this with the corresponding dword of the LL element. ■ DMA Read: The DAR is the address of the local memory. ■ DMA Write: The DAR is the address of the remote memory. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.54 DMA_DAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** DMA Write DAR High Register.
- **Description:** This register holds the higher 32 bits of the Destination Address Register (DAR).
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x218
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

31:0
DST_ADDR_REG_HIGH

Table 3-649 Fields for Register: DMA_DAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	DST_ADDR_REG_HIGH	* Varies	Destination Address Register (Higher 32 bits). In LL mode, the DMA overwrites this with the corresponding dword of the LL element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.55 DMA_LL_P_LOW_OFF_WRCN_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** DMA Write Linked List Pointer Low Register.
- **Description:** This register holds the lower 32 bits of the DMA write linked list pointer.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x21c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

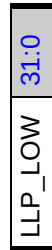


Table 3-650 Fields for Register: DMA_LL_P_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	LLP_LOW	* Varies	Lower bits of the address of the linked list transfer list in local memory. Used in linked list mode only. When using legacy DMA to point to the next element in the transfer list after the previous element is consumed. ■ When the current element is a data element; this field is incremented by 6 DWORDs. ■ When the current element is a link element; this field is overwritten by the LL Element Pointer of the element. When using HDMA with CC_LEGACY_DMA_MAP =1, it fetches descriptors from local memory respecting the following patterns: ■ When the current element is a data element; HDMA increments this field by 6 DWORDs times the prefetch depth. For more information, see registers HDMA_ELEM_P_F_OFF_WRCH* and HDMA_QOS_OFF_WRCH* ■ When the current element is a link element; HDMA overwrites this field with the LL Element Pointer of the next LL element structure. The current fetched memory address pointer is not directly visible in this register, instead HDMA updates this register on the following occurrences: ■ Watermark interrupt event ■ Channel Status ABORT event ■ Channel Status STOP event On a watermark interrupt event this register points to the descriptor's address that triggered it. A channel status STOP or ABORT events are end of transfer events that makes this register point to the next data to transfer after the event. <i>Note:</i> The access attributes of this field are as follows:
31:0...(cont.)	LLP_LOW.	* Varies	■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.56 DMA_LL_P_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** DMA Write Linked List Pointer High Register.
- **Description:** This register holds the higher 32 bits of the DMA write linked list pointer.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x220
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

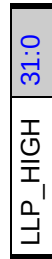


Table 3-651 Fields for Register: DMA_LL_P_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	LLP_HIGH	* Varies	Higher 32 bits of the address of the linked list transfer list in local memory. Used in linked list mode only. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.57 DMA_CH_CONTROL1_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** DMA Read Channel Control 1 Register.
- **Description:** This register controls the DMA read channel operation.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x200+0x100
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)

DMA_AT	31:30
DMA_TC	29:27
DMA_RESERVED5	26
DMA_RO	25
DMA_NS_SRC	24
DMA_NS_DST	23
DMA_MEM_TYPE	22
DMA_ELEMENT_PREFETCH	21:17
DMA_FUNC_NUM	16:12
DMA_RESERVED1	11:10
LLE	9
CCS	8
DMA_RESERVED0	7
CS	6:5
RIE	4
LIE	3
LLP	2
TCB	1
CB	0

Table 3-652 Fields for Register: DMA_CH_CONTROL1_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:30	DMA_AT	* Varies	Address Translation TLP Header Bit (AT). The DMA uses this TLP header field when generating MRd/MWr (not IMWr) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
29:27	DMA_TC	* Varies	Traffic Class TLP Header Bit (TC). The DMA uses this TLP header field when generating MRd/MWr (not IMWr) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
26	DMA_RESERVED5	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
25	DMA_RO	* Varies	Relaxed Ordering TLP Header Bit (RO) The DMA uses this TLP header field when generating MRd/MWrr (not IMWrr) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
24	DMA_NS_SRC	* Varies	Source No Snoop TLP Header Bit. The DMA uses this TLP header field when generating MRd (SAR addressing space) (not IMWr) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
23	DMA_NS_DST	* Varies	Destination No Snoop TLP Header Bit. The DMA uses this TLP header field when generating MWr (DAR addressing space) (not IMWr) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
22	DMA_MEM_TYPE	* Varies	Master AXI ACE-Lite Cache Coherency Control. This field sets the DMA channel memory type of the address space of the data transfer as follows: For more information, see "ACE-Lite Features and Limitations" section of the <i>Databook</i>. Irrespective of the value of this field, Linked list descriptor requests have this attribute set to 0x1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (PERIPHERAL): peripheral ■ 0x1 (MEM_TYPE): main memory Value After Reset: 0x0 Exists: CC_DMA_ENABLE && AXI_POPULATED && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
21:17	DMA_ELEMENT_PREFETCH	* Varies	This field controls how many elements the HDMA Read Channel should prefetch and its updated by the linked list element pointer loaded by HDMA. The valid value range is [0 : DMA_READ_ENGINE_PF_OFF.DMA_READ_ENGINE_MAX_PRE_FETCH]. <i>Note:</i> The actual prefetch amount is the value of this register plus one. For example, if set with 0, the prefetch is 1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_DMA_PF_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
16:12	DMA_FUNC_NUM	* Varies	DMA Channel Physical Function Number. This field indicates the physical function associated with the DMA channel. The controller uses this field to form the requester ID of the generated MRd/MWr DMA TLPs. <i>Note:</i> ■ When SR-IOV is enabled, you must consider DMA_VF and DMA_VF_EN fields of DMA_CH_CONTROL2_OFF_* as well to determine the requester ID for the requests. ■ The behavior is undefined if any value outside of the valid range is set for this field. ■ The DMA drives the value of this field in the XADM/RADM signals d_client[0]1_tlp_func_num/d_radm_trgt1_func_num. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Physical Channel Function Number starts at 0. ■ 0x1f (Max_Val): The maximum number of Physical Functions is given by CX_NFUNC, so the maximum value of this field is CX_NFUNC-1. Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
11:10	DMA_RESERVED1	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
9	LLE	* Varies	Linked List Enable (LLE). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x1 (ENABLE): Enable linked list operation ■ 0x0 (DISABLE): Disable linked list operation Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
8	CCS	* Varies	Consumer Cycle State (CCS). Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more information, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". You must initialize this bit. The DMA updates this bit during linked list operation. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
7	DMA_RESERVED0	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
6:5	CS	R/W	Channel Status (CS). The channel status bits identify the current operational state of the DMA channel. The operation state encoding for each DMA channel is as follows: Values: ■ 0x0 (RSVD): Reserved ■ 0x1 (RUN): Running. This channel is active and transferring data. ■ 0x2 (HALT): Halted. An error condition has been detected, and the DMA has stopped this channel. ■ 0x3 (STOP): Stopped. The DMA has transferred all data for this channel, or you have prematurely stopped this channel by writing to the Stop field of the DMA Read Doorbell Register (DMA_WRITE_DOORBELL_OFF) or DMA Read Doorbell Register (DMA_READ_DOORBELL_OFF). Value After Reset: {(CX_DMA_PF_ENABLE==0) ? 0 : 3} Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
4	RIE	* Varies	Remote Interrupt Enable (RIE). You must set this bit to enable the generation of the Done or Abort Remote interrupts. For more information, see "Interrupts and Error Handling". In LL mode, the DMA overwrites this with the RIE of the LL element. The RIE bit in a LL element only enables the Done interrupt. In non-LL mode, the RIE bit enables the Done and Abort interrupts. This field is not defined in a <i>link</i> LL element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
3	LIE	* Varies	Local Interrupt Enable (LIE). You must set this bit to enable the generation of the Done or Abort Local interrupts. For more information, see "Interrupts and Error Handling". In LL mode, the DMA overwrites this with the LIE of the LL element. The LIE bit in a LL element only enables the Done interrupt. In non-LL mode, the LIE bit enables the Done and Abort interrupts. This field is not defined in a <i>link</i> LL element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
2	LLP	* Varies	Load Link Pointer (LLP). Used in linked list mode only. Indicates that this linked list element is a link element, and its LL element pointer dwords are pointing to the next (non-contiguous) element. The DMA loads this field with the LLP of the linked list element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CX_DMA_PF_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
1	TCB	* Varies	Toggle Cycle Bit (TCB). Indicates to the DMA to toggle its interpretation of the CB. Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more information, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". The DMA loads this field with the TCB of the linked list element. This field is not defined in a data LL element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
0	CB	* Varies	Cycle Bit (CB). Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more information, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". The DMA loads this field with the CB of the linked list element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.58 DMA_CH_CONTROL2_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** DMA Read Channel Control 2 Register.
- **Description:** This register controls the DMA read channel operation. This register provides control features in addition to DMA Read Channel Control 1 Register.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x204+0x100
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)

DMA_TH	31
DMA_PH	30:29
DMA_RESERVED3	28:25
DMA_ST	24:17
DMA_VF_RESERVED	16:9
DMA_VF	8:1
DMA_VF_EN	0

Table 3-653 Fields for Register: DMA_CH_CONTROL2_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31	DMA_TH	* Varies	TLP Processing Hints Present (TH). This field is mapped onto the corresponding TLP field for application DMA MWr requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && CC_DMA_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
30:29	DMA_PH	* Varies	Processing Hints (PH). This field is mapped onto the corresponding TLP field for application DMA MWr requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && CC_DMA_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
28:25	DMA_RESERVED3	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && CC_DMA_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\": \"read-write\" "
24:17	DMA_ST	* Varies	Steering Tag (ST). This field is mapped onto the corresponding TLP field for application DMA MWr requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && CC_DMA_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\": \"read-write\" "

Bits	Name	Memory Access	Description
16:9	DMA_VF_RESERVED	* Varies	Reserved. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DMA_ENABLE && CC_DEVICE_TYPE != 1 && CC_DEVICE_TYPE != 3 && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
8:1	DMA_VF	* Varies	DMA Channel Virtual Function Number. When the DMA_VF_EN field is '1', the controller uses both the DMA_FUNC_NUM field in DMA Read Channel Control 1 Register (DMA_CH_CONTROL2_OFF_*) and this field to form the requester ID of the generated MRd/MWr DMA TLPs. <i>Note:</i> ■ When the DMA_VF_EN field is '0', the controller uses the DMA_FUNC_NUM field in DMA Read Channel Control 1 Register (DMA_CH_CONTROL1_OFF_RDCH_i) for generating the requester ID of the generated MRd/MWr DMA TLPs. ■ The behavior is undefined if any value outside of the valid range is set for this field. ■ The DMA drives the value of this field in the XADM/RADM signals d_client[0]1_tlp_vfunc_num/d_radm_trgt1_vfunc_num. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Virtual Channel Function Number starts at 0. ■ 0xff (Max_Val): The maximum number of internal virtual functions varies from 2 to CX_NVFUNC, so the maximum value of this field varies from 1 to CX_NVFUNC-1. Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DMA_ENABLE && CC_DEVICE_TYPE != 1 && CC_DEVICE_TYPE != 3 && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
0	DMA_VF_EN	* Varies	DMA Channel Virtual Function Enable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): When this field is '1', the controller uses both the DMA_FUNC_NUM field in DMA Read Channel Control 1 Register (DMA_CH_CONTROL1_OFF_RDCH_i) and the DMA_VF field of this registers to form the requester ID of the generated MRd/MWr DMA TLPs. Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && CC_DMA_ENABLE && CC_DEVICE_TYPE != 1 && CC_DEVICE_TYPE != 3 && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\": \"read-write\" "

3.67.59 DMA_TRANSFER_SIZE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** DMA Read Transfer Size Register.
- **Description:** This register holds the DMA read transfer size.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x208+0x100
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)

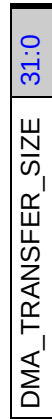


Table 3-654 Fields for Register: DMA_TRANSFER_SIZE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	DMA_TRANSFER_SIZE	* Varies	DMA Transfer Size. You program this register with the size of the DMA transfer. The maximum DMA transfer size is 4Gbytes. The minimum transfer size is one byte (0x1). This field is automatically decremented by the DMA as the DMA read channel transfer progresses. This field indicates the number bytes remaining to be transferred. When all bytes are successfully transferred the current transfer size is zero. In LL mode, the DMA overwrites this register with the corresponding dword of the LL element. You can read this register to monitor the transfer progress, however in some scenarios this register is updated after a delay. For example, when less than 3 channels are doorbelled, this register is updated only after a descriptor finishes(linked list mode), or the transfer ends (non-linked list mode). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.60 DMA_SAR_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** DMA Read SAR Low Register.
- **Description:** This register holds the lower 32 bits of the Source Address Register (SAR).
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x20c+0x100
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)

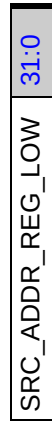


Table 3-655 Fields for Register: DMA_SAR_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	SRC_ADDR_REG_LOW	* Varies	Source Address Register (Lower 32 bits). Indicates the next address to be read from. The DMA increments the SAR as the DMA transfer progresses. In LL mode, the DMA overwrites this with the corresponding dword of the LL element. ■ DMA Read: The SAR is the address of the remote memory. ■ DMA Read: The SAR is the address of the local memory. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.61 DMA_SAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** DMA Read SAR High Register.
- **Description:** This register holds the higher 32 bits of the Source Address Register (SAR).
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x210+0x100
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)

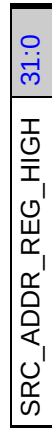


Table 3-656 Fields for Register: DMA_SAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	SRC_ADDR_REG_HIGH	* Varies	Source Address Register (higher 32 bits). In LL mode, the DMA overwrites this with the corresponding dword of the LL element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.62 DMA_DAR_LOW_OFF_RDCH_i (for $i = 0; i \leq CC_NUM_DMA_RD_CHAN-1$)

- **Name:** DMA Read DAR Low Register.
- **Description:** This register holds the lower 32 bits of the Destination Address Register (DAR).
- **Size:** 32 bits
- **Offset:** $B+i*2*0x100+0x214+0x100$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ || \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$

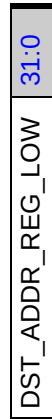


Table 3-657 Fields for Register: DMA_DAR_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	DST_ADDR_REG_LOW	* Varies	Destination Address Register (Lower 32 bits). Indicates the next address to be written to. The DMA increments the DAR as the DMA transfer progresses. In LL mode, the DMA overwrites this with the corresponding dword of the LL element. ■ DMA Read: The DAR is the address of the local memory. ■ DMA Read: The DAR is the address of the remote memory. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.63 DMA_DAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** DMA Read DAR High Register.
- **Description:** This register holds the higher 32 bits of the Destination Address Register (DAR).
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x218+0x100
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)



Table 3-658 Fields for Register: DMA_DAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	DST_ADDR_REG_HIGH	* Varies	Destination Address Register (higher 32 bits). In LL mode, the DMA overwrites this with the corresponding dword of the LL element. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.67.64 DMA_LL_P_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** DMA Read Linked List Pointer Low Register.
- **Description:** This register holds the lower 32 bits of the DMA read linked list pointer.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x21c+0x100
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)

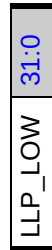


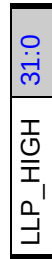
Table 3-659 Fields for Register: DMA_LL_P_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	LLP_LOW	* Varies	Lower bits of the address of the linked list transfer list in local memory. Used in linked list mode only. Used in linked list mode only. When using legacy DMA to point to the next element in the transfer list after the previous element is consumed. ■ When the current element is a data element; this field is incremented by 6 DWORDs. ■ When the current element is a link element; this field is overwritten by the LL Element Pointer of the element. When using HDMA with CC_LEGACY_DMA_MAP =1, it fetches descriptors from local memory respecting the following patterns: ■ When the current element is a data element; HDMA increments this field by 6 DWORDs times the prefetch depth. For more information, see registers HDMA_ELEM_P_F_OFF_RDCH* and HDMA_QOS_OFF_RDCH*. ■ When the current element is a link element; HDMA overwrites this field with the LL Element Pointer of the next LL element structure. The current fetched memory address pointer is not directly visible in this register, instead HDMA updates this register on the following occurrences: ■ Watermark interrupt event ■ Channel Status ABORT event ■ Channel Status STOP event On a watermark interrupt event this register points to the descriptor's address that triggered it. A channel status STOP or ABORT events are end of transfer events that makes this register point to the next data to transfer after the event. <i>Note</i>

Bits	Name	Memory Access	Description
31:0...(cont.)	LLP_LOW.	* Varies	: The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.67.65 DMA_LLP_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** DMA Read Linked List Pointer High Register.
- **Description:** This register holds the higher 32 bits of the DMA read linked list pointer.
- **Size:** 32 bits
- **Offset:** B+i*2*0x100+0x220+0x100
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)

**Table 3-660 Fields for Register: DMA_LLP_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)**

Bits	Name	Memory Access	Description
31:0	LLP_HIGH	* Varies	Higher 32 bits of the address of the linked list transfer list in local memory. Used in linked list mode only. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CC_DMA_ENABLE && CC_UNROLL_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68 PF0_HDMA_CAP Registers

3.68.1 HDMA_EN_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Enable Register.
- **Description:** This register enables an HDMA write channel.
- **Size:** 32 bits
- **Offset:** $B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x0$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_WR_CHAN>0)$

31:1	0
Rsvd	ENABLE

Table 3-661 Fields for Register: HDMA_EN_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	ENABLE	* Varies	HDMA Write Channel Enable. The controller checks this field for power management purposes. If this field is enabled for any one of the read, or write channel, the controller exits low power state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): Enable Value After Reset: 0x0 Exists: $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_WR_CHAN>0)$ Memory Access: $"(CX_MEMORY_MAP_VIEW==0) ? \backslash\text{"read-write"} : (CX_MEMORY_MAP_VIEW==1 \&\& !SNPS_RSVDPARAM_38) ? \backslash\text{"read-write"} : (CX_MEMORY_MAP_VIEW==1 \&\& SNPS_RSVDPARAM_38) ? \backslash\text{"read-only"} : \backslash\text{"read-write"}"$

3.68.2 HDMA_DOORBELL_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Doorbell Register.
- **Description:** This register controls the Doorbell state of an HDMA write channel.
- **Size:** 32 bits
- **Offset:** $B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x4}$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_WR_CHAN>0)$

31:2	1	0
Rsvd	DB_STOP	DB_START

Table 3-662 Fields for Register: HDMA_DOORBELL_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	DB_STOP	W	HDMA Write Channel Doorbell Stop. You must set this field to stop the write transfer for this channel. HDMA stops the write transfer for this channel at the earliest when this field is set. <i>Note:</i> Only after the HDMA_STATUS_OFF_WRCH_i.STATUS = 0x03, you can consider this channel to be in stop state. Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Set Value After Reset: 0x0 Exists: $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_WR_CHAN>0)$ Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	DB_START	W	HDMA Write Channel Doorbell Start. You must set this field to start the write transfer for this channel. Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Set Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true

3.68.3 HDMA_ELEM_PF_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Prefetch Register.
- **Description:** This register holds information regarding descriptor prefetch of an HDMA write channel.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x8
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)



Table 3-663 Fields for Register: HDMA_ELEM_PF_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes

Bits	Name	Memory Access	Description
6:0	ELEMENT_PREFETCH	* Varies	This field controls the number of linked list elements (descriptors) this HDMA write channel prefetches. The linked list element pointer loaded by HDMA updates this field. <i>Note:</i> The actual prefetch value is ELEMENT_PREFETCH plus '1'. For example, if ELEMENT_PREFETCH =0, one descriptor is prefetched by this channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0x7f (Max_Val): The maximum value of ELEMENT_PREFETCH must be $\leq \text{HDMA_QOS_OFF_WRCH_i.PF_DEPTH}$ Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.4 HDMA_HANDSHAKE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Handshake Register.
- **Description:** This register controls the handshake operation of an HDMA write channel.
- **Size:** 32 bits
- **Offset:** $B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0xc}$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_WR_CHAN>0)$

31:7	6:1	0
Rsvd	HANDSHAKE_COUNT	HS_EN

Table 3-664 Fields for Register: HDMA_HANDSHAKE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes
6:1	HANDSHAKE_COUNT	R	HDMA Write Channel Handshake Counter. Values: ■ 0x0 (Min_Val): Min Val ■ 0x3f (Max_Val): Max Val Value After Reset: 0x0 Exists: $CC_DMA_HSHAKE \&\& (CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_WR_CHAN>0)$ Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	HS_EN	* Varies	Enable Handshake for an HDMA write channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): Enable Value After Reset: 0x0 Exists: CC_DMA_HSHAKE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.5 HDMA_LLP_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Linked List Pointer Low Register.
- **Description:** This register holds the lower 32 bits of the address of a write channel transfer list. It is used in linked list mode only.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))} + 0x10$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE == 0 \ || \ CX_MEMORY_MAP_PCIE_TYPE == 1)) \ \&\& \ (CC_NUM_DMA_WR_CHAN > 0)$

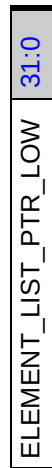


Table 3-665 Fields for Register: HDMA_LL_P_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	ELEMENT_LIST_PTR_LOW	* Varies	Lower 32 bits of the address of the transfer list in the local memory. Used in linked list mode only. HDMA fetches descriptors from local memory respecting the following patterns: ■ When the current element is a data element; HDMA increments this field by 6 DWORDs times the prefetch depth. For more information, see registers HDMA_ELEM_P-F_OFF_WRCH* and HDMA_QOS_OFF_WRCH* ■ When the current element is a link element; HDMA overwrites this field with the LL Element Pointer of the next LL element structure. The current fetched memory address pointer is not directly visible in this register, instead HDMA updates this register on the following occurrences: ■ Watermark interrupt event ■ Channel Status ABORT event ■ Channel Status STOP event On a watermark interrupt event this register points to the descriptor's address that triggered it. A channel status STOP or ABORT events are end of transfer events that makes this register point to the next data to transfer after the event. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.6 HDMA_LL_P_HIGH_OFF_WRCN_i (for i = 0; i <= CC_NUM_DMA_WRCN-1)

- **Name:** HDMA Write Channel Linked List Pointer High Register.
- **Description:** This register holds the higher 32 bits of the address of a write channel transfer list. It is used in linked list mode only.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x14
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WRCN>0)



Table 3-666 Fields for Register: HDMA_LL_P_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	ELEMENT_LIST_PTR_HIGH	* Varies	Higher 32 bits of the address of the transfer list in the local memory. HDMA fetches descriptors from local memory respecting the following patterns: ■ When the current element is a data element; HDMA increments this field by 6 DWORDs times the prefetch depth. For more information, see registers HDMA_ELEM_P_F_OFF_WRCH* and HDMA_QOS_OFF_WRCH* ■ When the current element is a link element; HDMA overwrites this field with the LL Element Pointer of the next LL element structure. The current fetched memory address pointer is not directly visible in this register, instead HDMA updates this register on the following occurrences: ■ Watermark interrupt event ■ Channel Status ABORT event ■ Channel Status STOP event On a watermark interrupt event this register points to the descriptor's address that triggered it. A channel status STOP or ABORT events are end of transfer events that makes this register point to the next data to transfer after the event. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.7 HDMA_CYCLE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Producer-Consumer Cycle Synchronization Register.
- **Description:** This register is used to synchronize the producer (software) and the consumer (HDMA) (used in linked list mode only). For more information, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization" in the *Databook*.
Note: The HDMA updates this register while operating on a linked list.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x18
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

31:2	1	0
Rsvd	CYCLE_STATE	CYCLE_BIT

Table 3-667 Fields for Register: HDMA_CYCLE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	CYCLE_STATE	* Varies	Consumer Cycle State. You must initialize this field. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
0	CYCLE_BIT	* Varies	Consumer Cycle Bit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\": \"read-write\" "

3.68.8 HDMA_XFERSIZE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Transfer Size Register.
- **Description:** This register holds the transfer size of an HDMA channel.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))} + 0x1c$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE == 0 \ || \ CX_MEMORY_MAP_PCIE_TYPE == 1)) \ \&\& \ (CC_NUM_DMA_WR_CHAN > 0)$

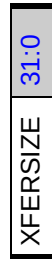


Table 3-668 Fields for Register: HDMA_XFERSIZE_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	XFERSIZE	* Varies	HDMA Write Channel Transfer Size. You program this register with the size of the HDMA transfer. The maximum HDMA transfer size is 4Gbytes. The minimum transfer size is one byte (0x1). HDMA automatically decrements the value of this field as the write channel transfer progresses. This field indicates the number of bytes remaining to be requested. When all bytes are requested the current transfer size is zero. In LL mode, the HDMA overwrites this register with the corresponding dword of the LL element. You can read this register to monitor the transfer progress, but it is not reliable for that specific usage, as it is updated after read requests and not after write requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.9 HDMA_SAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel SAR Low Register.
- **Description:** This register holds the lower 32 bits of the Source Address Register (SAR) of an HDMA write channel.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x20
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)



Table 3-669 Fields for Register: HDMA_SAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	SAR_PTR_LOW	* Varies	Source Address Register (lower 32 bits). Indicates the address of the local memory from which HDMA reads. The HDMA increments the SAR as the HDMA transfer progresses. In LL mode, the HDMA overwrites this field with the corresponding dword of the LL element. The SAR is the address of the local memory. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.10 HDMA_SAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel SAR High Register.
- **Description:** This register holds the higher 32 bits of the Source Address Register (SAR) of an HDMA write channel.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x24
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)



Table 3-670 Fields for Register: HDMA_SAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	SAR_PTR_HIGH	* Varies	Source Address Register (higher 32 bits). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.11 HDMA_DAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel DAR Low Register.
- **Description:** This register holds the lower 32 bits of the Destination Address Register (DAR) of an HDMA write channel.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x28
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)



Table 3-671 Fields for Register: HDMA_DAR_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	DAR_PTR_LOW	* Varies	Destination Address Register (lower 32 bits). Indicates the address to which HDMA writes. The HDMA increments the DAR as the HDMA transfer progresses. In LL mode, the HDMA overwrites this field with the corresponding dword of the LL element. The DAR is the address of the remote memory. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.12 HDMA_DAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel DAR High Register.
- **Description:** This register holds the higher 32 bits of the Destination Address Register (DAR) of an HDMA write channel.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x2c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)



Table 3-672 Fields for Register: HDMA_DAR_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	DAR_PTR_HIGH	* Varies	Destination Address Register (higher 32 bits). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.13 HDMA_WATERMARK_EN_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Linked-list Watermark Enable Register.
- **Description:** This register controls the watermark interrupts generated after processing a water-marked LL element. HDMA updates this field as LL elements are processed.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x30
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

31:2	1	0
Rsvd	LWIE	RWIE

Table 3-673 Fields for Register: HDMA_WATERMARK_EN_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	LWIE	* Varies	Enables Local Interrupts at watermark events (end of linked list element) in an HDMA write channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): Enable Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
0	RWIE	* Varies	Enables Remote Interrupts at watermark events (end of linked list element) in an HDMA write channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): Enable Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.14 HDMA_CONTROL1_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Control Settings 1 Register.
- **Description:** This register controls the operation of the HDMA write channel.
- **Size:** 32 bits
- **Offset:** $B+i*2*2^{(\text{CX_DMAREG_CHADDR_SPACE}+8)}+0x34$
- **Exists:** $(\text{CC_DMA_ENABLE} \&\& \text{CC_UNROLL_ENABLE} \&\& !\text{CC_LEGACY_DMA_MAP} \&\& (\text{CX_MEMORY_MAP_PCIE_TYPE}==0 \mid \mid \text{CX_MEMORY_MAP_PCIE_TYPE}==1)) \&\& (\text{CC_NUM_DMA_WR_CHAN}>0)$

31:27	26:19	18:17	16	15:7	6:5	4	3	2	1	0
Rsvd	ST	PH	TH	Rsvd	AT	RO	DST_SNOOP	SRC_SNOOP	MEM_TYPE	LLEN

Table 3-674 Fields for Register: HDMA_CONTROL1_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:27			Reserved Field: Yes
26:19	ST	* Varies	Steering Tag TLP Header Bit (ST). The HDMA write channel uses this TLP header field when generating MWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: $\text{CX_TPH_ENABLE} \&\& (\text{CC_DMA_ENABLE} \&\& \text{CC_UNROLL_ENABLE} \&\& !\text{CC_LEGACY_DMA_MAP} \&\& (\text{CX_MEMORY_MAP_PCIE_TYPE}==0 \mid \mid \text{CX_MEMORY_MAP_PCIE_TYPE}==1)) \&\& (\text{CC_NUM_DMA_WR_CHAN}>0)$ Memory Access: $(\text{CX_MEMORY_MAP_VIEW}==0) ? \text{"read-write"} : (\text{CX_MEMORY_MAP_VIEW}==1 \&\& !\text{SNPS_RSVDPARAM_38}) ? \text{"read-write"} : (\text{CX_MEMORY_MAP_VIEW}==1 \&\& \text{SNPS_RSVDPARAM_38}) ? \text{"read-only"} : \text{"read-write"}$

Bits	Name	Memory Access	Description
18:17	PH	* Varies	Processing Hints TLP Header Bit (PH). The HDMA write channel uses this TLP header field when generating MWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
16	TH	* Varies	TLP Processing Hints Present Header Bit (TH). The HDMA write channel uses this TLP header field when generating MWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
15:7			Reserved Field: Yes

Bits	Name	Memory Access	Description
6:5	AT	* Varies	Address Translation Services TLP Header Bit (AT). The HDMA write channel uses this TLP header field when generating MRd/MWr/IMWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
4	RO	* Varies	Relaxed Ordering TLP Header Bit. HDMA write channel uses this TLP header field when generating MRd/MWr/IMWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
3	DST_SNOOP	* Varies	Destination No Snoop TLP Header Bit. The HDMA write channel uses this TLP header field when generating MWr/IMWr (DAR addressing space) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
2	SRC_SNOOP	* Varies	Source No Snoop TLP Header Bit. The HDMA write channel uses this TLP header field when generating MRd (SAR addressing space) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
1	MEM_TYPE	* Varies	Master AXI ACE-Lite Cache Coherency Control. This field sets the HDMA write channel memory type of the address space of the data transfer as follows: For more information, see "ACE-Lite Features and Limitations" section of the <i>Databook</i>. Irrespective of the value of this field, Linked list descriptor requests have this attribute set to 0x1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (PERIPHERAL): peripheral type ■ 0x1 (MEM_TYPE): memory type Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
0	LLEN	* Varies	This field enables the HDMA write channel Linked List mode. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): Enable Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.15 HDMA_FUNC_NUM_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Function Number Register.
- **Description:** This register controls HDMA write channel to Physical/Virtual function mapping.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x38
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

31:25	24:17	16	15:5	4:0
Rsvd	VF	VF_EN	Rsvd	PF

Table 3-675 Fields for Register: HDMA_FUNC_NUM_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:25			Reserved Field: Yes

Bits	Name	Memory Access	Description
24:17	VF	* Varies	DMA Channel Virtual Function Number. -When the VF_EN field of this register '1', the controller uses both the PF and this field for generating the requester ID of the generated MRd/MWr DMA TLPs. The PCIe controller supports 256 virtual functions and starts numbering VFs at 0, that maps to virtual function 1. The behavior is undefined if any value outside of the valid range is set to this field. -When the DMA_VF_EN field is '0', the controller uses the PF field of this register for generating the requester ID of the generated MRd/MWr DMA TLPs. At the signal interfaces, DMA drives the XADM / RADM signals d_client[0]1_tlp_vfunc_num / d_radm_trgt1_vfunc_num. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
16	VF_EN	* Varies	DMA Channel Virtual Function Enable. At the signal interfaces, DMA drives the XADM / RADM signals d_client[0]1_tlp_vfunc_active / d_radm_trgt1_vfunc_active. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): When this field is '1', the controller uses both the PF field and the VF field of this registers for generating the requester ID of the generated MRd/MWr DMA TLPs. The PCIe controller supports 256 virtual functions. Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
15:5			Reserved Field: Yes

Bits	Name	Memory Access	Description
4:0	PF	* Varies	DMA Channel Physical Function Number. DMA channel wide physical function of the generated TLP. The controller uses this field to form the requester ID for requests. Function numbering starts at '0'. When you have enabled SR-IOV, refer to VF and VF_EN fields of this register as well. The behavior is undefined if any value outside of the valid range is set to this field. At the signal interfaces, DMA drives the XADM / RADM signals d_client[0]1_tlp_func_num / d_radm_trgt1_func_num. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.68.16 HDMA_QOS_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel QoS Settings Register.
- **Description:** This register controls the HDMA write channel QoS features.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x3c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

31:26	25:16	15:8	7:3	2:0
Rsvd	PF_DEPTH	Rsvd	WEIGHT	TC

Table 3-676 Fields for Register: HDMA_QOS_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:26			Reserved Field: Yes

Bits	Name	Memory Access	Description
25:16	PF_DEPTH	* Varies	This field controls the linked list prefetch allocation space for an HDMA write channel. Each write channel stores PF_DEPTH plus '1' linked list elements. For example, if PF_DEPTH =0, the write channel stores one linked list element. This field must be within the range: [0 : floor(CX_DMA_WR_LLQ * CC_NUM_DMA_WR_CHAN / number_of_active_channels) - 1]. <i>Note:</i> If this field is set to a value outside the specified range, data corruption or channel crosstalk may happen. If PF_DEPTH is changed at run time, only channels in the range: [0 : number_of_active_channels - 1] can be doorbelled. This field cannot be changed while any write engine channel is in doorbell state. Doing so might yield undefined results. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: CC_HDMA_REG_WR_PF_DEPTH Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
15:8			Reserved Field: Yes

Bits	Name	Memory Access	Description
7:3	WEIGHT	* Varies	HDMA Write Channel Weight. Specifies the maximum number of TLP MRd requests that the HDMA can issue for a channel, before it returns to the arbitration routine. When a channel reaches its weight count or HDMA channel request transfer size reaches zero, the weighted round robin (WRR) arbiter selects the next channel to be processed. Your software must initialize this field before ringing the doorbell. For more information, see "Multichannel Arbitration" in the <i>Databook</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: CC_HDMA_REG_WEIGHT Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
2:0	TC	* Varies	Traffic Class TLP Header Field (TC). The HDMA write channel uses this TLP header field when generating MRd/MWr/IMWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.17 HDMA_STATUS_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Status Register.
- **Description:** This register specifies the current operational state of a channel.
- **Size:** 32 bits
- **Offset:** $B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x80}$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ || \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_WR_CHAN>0)$

31:3	2:0
Rsvd	STATUS

Table 3-677 Fields for Register: HDMA_STATUS_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	STATUS	R	HDMA Write Channel Status. Values: ■ 0x0 (RESERVED): Reserved ■ 0x1 (RUNNING): This channel is active and transferring data. ■ 0x2 (ABORTED): An error condition is detected, and the HDMA has stopped this channel. ERROR field of the HDMA_INT_STATUS_OFF_WRCH_0 register specifies the error. ■ 0x3 (STOPPED): The HDMA has transferred all data for this channel. Value After Reset: CC_HDMA_REG_STATUS Exists: $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_WR_CHAN>0)$ Testable: writeAsRead Volatile: true

3.68.18 HDMA_INT_STATUS_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Interrupt Status Register.
- **Description:** This register provides information regarding the HDMA write channel interrupt status.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0x84$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_WR_CHAN>0)$

31:7	6:3	2	1	0
Rsvd	ERROR	ABORT	WATERMARK	STOP

Table 3-678 Fields for Register: HDMA_INT_STATUS_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes

Bits	Name	Memory Access	Description
6:3	ERROR	R	HDMA Write Channel Interrupt Error Status. Specifies the error that caused the HDMA to change HDMA_STATUS_OFF_WRCH_i.STATUS to ABORTED state. Values: ■ 0x0 (Clear): No error to report. This field is cleared when HDMA_INT_CLEAR_OFF_WRCH_i.ABORT_CLEAR =1 ■ 0x1 (LL_CPL_UR): Unsupported request CPL while reading a linked list element. ■ 0x2 (LL_CPL_CA): Completer abort CPL while reading a linked list element. ■ 0x3 (LL_CPL_EP): Poisoned data CPL while reading a linked list element. ■ 0x8 (DATA_CPL_TIMEOUT): CPL timeout while reading data from the source. ■ 0x9 (DATA_CPL_UR): Unsupported request CPL while reading data from the source. ■ 0xa (DATA_CPL_CA): Completer abort CPL while reading data from the source. ■ 0xb (DATA_CPL_EP): Poisoned data CPL while reading data from the source. Only valid if CC_DMA_ABORT_ON_POISONED is set. ■ 0xc (DATA_MWR): Error response received while writing data to the destination. Only valid for Write Channels with AXI bridge (AMBA_INTERFACE >2). Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2	ABORT	R	HDMA Write Channel Abort Interrupt Status. ABORT and STOP register fields are mutually exclusive. Values: ■ 0x0 (Clear): This field is cleared when HDMA_INT_CLEAR_OFF_WRCH_i.ABORT_CLEAR =1 ■ 0x1 (Set): If HDMA_INT_SETUP_OFF_WRCH_i.LAIE =1 and HDMA_INT_SETUP_OFF_WRCH_i.ABORT_MASK =0, and a local interrupt is generated, this field indicates that an error is detected for this channel. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true
1	WATERMARK	R	HDMA Write Channel Watermark Interrupt Status. Values: ■ 0x0 (Clear): This field is cleared when HDMA_INT_CLEAR_OFF_WRCH_i.WATERMARK_CLEAR =1 ■ 0x1 (Set): If HDMA_WATERMARK_EN_OFF_WRCH_i.LWIE =1 and HDMA_INT_SETUP_OFF_WRCH_i.WATERMARK_MASK =0, and a local interrupt is generated, this field indicates that HDMA has transferred all the data for an element with the LWIE bit set. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	STOP	R	HDMA Write Channel Stop Interrupt Status. STOP and ABORT register fields are mutually exclusive. Values: ■ 0x0 (Clear): This field is cleared when HDMA_INT_CLEAR_OFF_WRCH_i.STOP_CLEAR =1 ■ 0x1 (Set): If HDMA_INT_SETUP_OFF_WRCH_i.LSIE =1 and HDMA_INT_SETUP_OFF_WRCH_i.STOP_MASK =0, and a local interrupt is generated, this field indicates that HDMA has transferred all the data for this channel. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true

3.68.19 HDMA_INT_SETUP_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Interrupt Setup Register.
- **Description:** This register is used to configure interrupts for a write channel.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x88
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)

31:7	6	5	4	3	2	1	0
Rsvd	LAIE	RAIE	LSIE	RSIE	ABORT_MASK	WATERMARK_MASK	STOP_MASK

Table 3-679 Fields for Register: HDMA_INT_SETUP_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes

Bits	Name	Memory Access	Description
6	LAIE	* Varies	HDMA Write Channel Local Abort Interrupt Enable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field enables local interrupts in abort conditions. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
5	RAIE	* Varies	HDMA Write Channel Remote Abort Interrupt Enable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field enables remote interrupts in abort conditions. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
4	LSIE	* Varies	HDMA Write Channel Local Stop Interrupt Enable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field enables local interrupts in stop conditions. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
3	RSIE	* Varies	HDMA Write Channel Remote Stop Interrupt Enable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field enables remote interrupts in stop conditions. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
2	ABORT_MASK	* Varies	HDMA Write Channel Abort Interrupt Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field masks the local interrupt pin for abort interrupts. Value After Reset: CC_HDMA_REG_INT_MASK Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
1	WATERMARK_MASK	* Varies	HDMA Write Channel Watermark Interrupt Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field masks the local interrupt pin for watermark interrupts. Value After Reset: CC_HDMA_REG_INT_MASK Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
0	STOP_MASK	* Varies	HDMA Write Channel Stop Interrupt Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field masks the local interrupt pin for stop interrupts. Value After Reset: CC_HDMA_REG_INT_MASK Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.20 HDMA_INT_CLEAR_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Interrupt Clear Register.
- **Description:** This register indicates interrupt clear status of a write channel. It is a self-clearing register. Reads to this register returns '0'.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0x8c$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ || \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_WR_CHAN>0)$

31:3	2	1	0
Rsvd	ABORT_CLEAR	WATERMARK_CLEAR	STOP_CLEAR

Table 3-680 Fields for Register: HDMA_INT_CLEAR_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2	ABORT_CLEAR	W	HDMA Abort Channel Abort Interrupt Clear. Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field clears the ABORT and ERROR fields of the HDMA_INT_STATUS_OFF_WRCH_i register. Value After Reset: 0x0 Exists: $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_WR_CHAN>0)$ Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
1	WATERMARK_CLEAR	W	HDMA Write Channel Watermark Interrupt Clear. Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field clears the WATERMARK field of the HDMA_INT_STATUS_OFF_WRCH_i register. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true
0	STOP_CLEAR	W	HDMA Write Channel Stop Interrupt Clear. Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field clears the STOP field of the HDMA_INT_STATUS_OFF_WRCH_i register. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Testable: writeAsRead Volatile: true

3.68.21 HDMA_MSI_STOP_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Read Stop Remote Interrupt Address Low Register.
- **Description:** This register holds the lower 32 bits of the Stop Interrupt MWr TLP address.
- **Size:** 32 bits
- **Offset:** $B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x90}$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_WR_CHAN>0)$

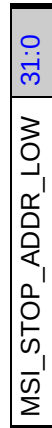


Table 3-681 Fields for Register: HDMA_MSI_STOP_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_STOP_ADDR_LOW	* Varies	The HDMA uses this field to generate bits [31:0] of the address field for the Stop Interrupt MWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.22 HDMA_MSI_STOP_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Read Stop Remote Interrupt Address High Register.
- **Description:** This register holds the higher 32 bits of the Stop Interrupt MWr TLP address.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))} + 0x94$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE == 0 \ || \ CX_MEMORY_MAP_PCIE_TYPE == 1)) \ \&\& \ (CC_NUM_DMA_WR_CHAN > 0)$

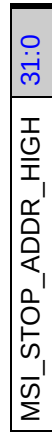


Table 3-682 Fields for Register: HDMA_MSI_STOP_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_STOP_ADDR_HIGH	* Varies	The HDMA uses this field to generate bits [63:32] of the address field for the Stop Interrupt MWr TLP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.23 HDMA_MSI_WATERMARK_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Read Watermark Remote Interrupt Address Low Register.
- **Description:** This register holds the lower 32 bits of the Watermark Interrupt MWr TLP address.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x98
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)



Table 3-683 Fields for Register: HDMA_MSI_WATERMARK_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_WATERMARK_ADDR_LOW	* Varies	The HDMA uses this field to generate bits [31:0] of the address field for the Watermark Interrupt MWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.24 HDMA_MSI_WATERMARK_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Read Watermark Remote Interrupt Address High Register.
- **Description:** This register holds the lower 32 bits of the Watermark Interrupt MWr TLP address.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x9c
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)



Table 3-684 Fields for Register: HDMA_MSI_WATERMARK_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_WATERMARK_ADDR_HIGH	* Varies	The HDMA uses this field to generate bits [31:0] of the address field for the Watermark Interrupt MWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.25 HDMA_MSI_ABORT_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Read Abort Remote Interrupt Address Low Register.
- **Description:** This register holds the lower 32 bits of the Abort Interrupt MWr TLP address.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (\text{CX_DMAREG_CHADDR_SPACE} + 8))} + 0\text{xa}0$
- **Exists:** $(\text{CC_DMA_ENABLE} \ \&\& \ \text{CC_UNROLL_ENABLE} \ \&\& \ !\text{CC_LEGACY_DMA_MAP} \ \&\& \ (\text{CX_MEMORY_MAP_PCIE_TYPE} == 0 \ || \ \text{CX_MEMORY_MAP_PCIE_TYPE} == 1)) \ \&\& \ (\text{CC_NUM_DMA_WR_CHAN} > 0)$



Table 3-685 Fields for Register: HDMA_MSI_ABORT_LOW_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_ABORT_ADDR_LOW	* Varies	The HDMA uses this field to generate bits [31:0] of the address field for the Abort Interrupt MWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.26 HDMA_MSI_ABORT_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Read Abort Remote Interrupt Address High Register.
- **Description:** This register holds the lower 32 bits of the Abort Interrupt MWr TLP address.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))} + 0xa4$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE == 0 \ || \ CX_MEMORY_MAP_PCIE_TYPE == 1)) \ \&\& \ (CC_NUM_DMA_WR_CHAN > 0)$



Table 3-686 Fields for Register: HDMA_MSI_ABORT_HIGH_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_ABORT_ADDR_HIGH	* Varies	The HDMA uses this field to generate bits [31:0] of the address field for the Abort Interrupt MWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.27 HDMA_MSI_MSGD_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

- **Name:** HDMA Write Channel Remote Interrupt Data Register.
- **Description:** This register holds the IMWr TLP Data of an HDMA write channel.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0xa8
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0)



Table 3-687 Fields for Register: HDMA_MSI_MSGD_OFF_WRCH_i (for i = 0; i <= CC_NUM_DMA_WR_CHAN-1)

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	MSI_MESSAGE	* Varies	The HDMA write channel uses this field to generate the data field for every IMWr TLPs it generates. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_WR_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.28 HDMA_EN_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Enable.
- **Description:** This register enables an HDMA read channel.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0x0+\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ || \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$

31:1	0
Rsvd	ENABLE

Table 3-688 Fields for Register: HDMA_EN_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	ENABLE	* Varies	HDMA Read Channel Enable. The controller checks this field for power management purposes. If this field is enabled for any one of the read, or write channel, the controller exits low power state. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): Enable Value After Reset: 0x0 Exists: $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$ Memory Access: $(CX_MEMORY_MAP_VIEW==0) ? \backslash\text{"read-write"} : (CX_MEMORY_MAP_VIEW==1 \ \&\& \ !SNPS_RSVDPARAM_38) ? \backslash\text{"read-write"} : (CX_MEMORY_MAP_VIEW==1 \ \&\& \ SNPS_RSVDPARAM_38) ? \backslash\text{"read-only"} : \backslash\text{"read-write"}$

3.68.29 HDMA_DOORBELL_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Doorbell Register.
- **Description:** This register controls the Doorbell state of an HDMA read channel.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0x4+\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_RD_CHAN>0)$

31:2	1	0
Rsvd	DB_STOP	DB_START

Table 3-689 Fields for Register: HDMA_DOORBELL_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	DB_STOP	W	HDMA Read Channel Doorbell Stop. You must set this field to stop the read transfer for this channel. HDMA stops the read transfer for this channel at the earliest when this field is set. <i>Note:</i> Only after the HDMA_STATUS_OFF_RDCH_i.STATUS = 0x03, you can consider this channel to be in stop state. Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Set Value After Reset: 0x0 Exists: $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_RD_CHAN>0)$ Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	DB_START	W	HDMA Read Channel Doorbell Start. You must set this field to start the read transfer for this channel. Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Set Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true

3.68.30 HDMA_ELEM_PF_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Prefetch Register.
- **Description:** This register holds information regarding descriptor prefetch of an HDMA read channel.
- **Size:** 32 bits
- **Offset:** $B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x8+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ || \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$



Table 3-690 Fields for Register: HDMA_ELEM_PF_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes

Bits	Name	Memory Access	Description
6:0	ELEMENT_PREFETCH	* Varies	This field controls the number of linked list elements (descriptors) this HDMA read channel prefetches. The linked list element pointer loaded by HDMA updates this field. <i>Note:</i> The actual prefetch value is ELEMENT_PREFETCH plus '1'. For example, if ELEMENT_PREFETCH =0, one descriptor is prefetched by this channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0x7f (Max_Val): The maximum value of ELEMENT_PREFETCH must be $\leq \text{HDMA_QOS_OFF_RDCH_i.PF_DEPTH}$ Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.31 HDMA_HANDSHAKE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Handshake Register.
- **Description:** This register controls the handshake operation of an HDMA read channel.
- **Size:** 32 bits
- **Offset:** $B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0xc+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_RD_CHAN>0)$

31:7	6:1	0
Rsvd	HANDSHAKE_COUNT	HS_EN

Table 3-691 Fields for Register: HDMA_HANDSHAKE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes
6:1	HANDSHAKE_COUNT	R	HDMA Read Channel Handshake Counter. Values: ■ 0x0 (Min_Val): Min Val ■ 0x3f (Max_Val): Max Val Value After Reset: 0x0 Exists: $CC_DMA_HSHAKE \&\& (CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_RD_CHAN>0)$ Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	HS_EN	* Varies	Enable Handshake for an HDMA read channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): Enable Value After Reset: 0x0 Exists: CC_DMA_HSHAKE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.32 HDMA_LL_P_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Linked List Pointer Low Register.
- **Description:** This register holds the lower 32 bits of the address of a read channel transfer list. It is used in linked list mode only.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8)) + 0x10 + \text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))}$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE == 0 \ || \ CX_MEMORY_MAP_PCIE_TYPE == 1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN > 0)$

31:0
ELEMENT_LIST_PTR_LOW

Table 3-692 Fields for Register: HDMA_LL_P_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	ELEMENT_LIST_PTR_LOW	* Varies	Lower 32 bits of the address of the transfer list in the local memory. Used in linked list mode only. HDMA fetches descriptors from local memory respecting the following patterns: ■ When the current element is a data element; HDMA increments this field by 6 DWORDs times the prefetch depth. For more information, see registers HDMA_ELEM_P_F_OFF_RDCH* and HDMA_QOS_OFF_RDCH* ■ When the current element is a link element; HDMA overwrites this field with the LL Element Pointer of the next LL element structure. The current fetched memory address pointer is not directly visible in this register, instead HDMA updates this register on the following occurrences: ■ Watermark interrupt event ■ Channel Status ABORT event ■ Channel Status STOP event On a watermark interrupt event this register points to the descriptor's address that triggered it. A channel status STOP or ABORT events are end of transfer events that makes this register point to the next data to transfer after the event <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.33 HDMA_LL_P_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Linked List Pointer High Register.
- **Description:** This register holds the higher 32 bits of the address of a read channel transfer list. It is used in linked list mode only.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8)) + 0x14 + \text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))}$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE == 0 \ || \ CX_MEMORY_MAP_PCIE_TYPE == 1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN > 0)$

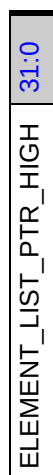


Table 3-693 Fields for Register: HDMA_LL_P_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	ELEMENT_LIST_PTR_HIGH	* Varies	Higher 32 bits of the address of the transfer list in the local memory. HDMA fetches descriptors from local memory respecting the following patterns: ■ When the current element is a data element; HDMA increments this field by 6 DWORDs times the prefetch depth. For more information, see registers HDMA_ELEM_P_F_OFF_RDCH* and HDMA_QOS_OFF_RDCH* ■ When the current element is a link element; HDMA overwrites this field with the LL Element Pointer of the next LL element structure. The current fetched memory address pointer is not directly visible in this register, instead HDMA updates this register on the following occurrences: ■ Watermark interrupt event ■ Channel Status ABORT event ■ Channel Status STOP event On a watermark interrupt event this register points to the descriptor's address that triggered it. A channel status STOP or ABORT events are end of transfer events that makes this register point to the next data to transfer after the event. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.34 HDMA_CYCLE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Producer-Consumer Cycle Synchronization Register.
- **Description:** This register is used to synchronize the producer (software) and the consumer (HDMA) (used in linked list mode only). For more information, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization" in the *Databook*.
Note: The HDMA updates this register while operating on a linked list.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0x18+\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ || \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$

31:2	1	0
Rsvd	CYCLE_STATE	CYCLE_BIT

Table 3-694 Fields for Register: HDMA_CYCLE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	CYCLE_STATE	* Varies	Consumer Cycle State. You must initialize this field. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$ Testable: writeAsRead Volatile: true Memory Access: $(CX_MEMORY_MAP_VIEW==0) ? \backslash \text{"read-write"} : (CX_MEMORY_MAP_VIEW==1 \ \&\& \ !SNPS_RSVDPARAM_38) ? \backslash \text{"read-write"} : (CX_MEMORY_MAP_VIEW==1 \ \&\& \ SNPS_RSVDPARAM_38) ? \backslash \text{"read-only"} : \backslash \text{"read-write"}$

Bits	Name	Memory Access	Description
0	CYCLE_BIT	* Varies	Consumer Cycle Bit. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\": (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\": \"read-write\" "

3.68.35 HDMA_XFERSIZE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Transfer Size Register.
- **Description:** This register holds the transfer size of an HDMA channel.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))} + 0x1c + \text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE == 0 \ || \ CX_MEMORY_MAP_PCIE_TYPE == 1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN > 0)$

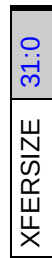


Table 3-695 Fields for Register: HDMA_XFERSIZE_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	XFERSIZE	* Varies	HDMA Read Channel Transfer Size. You program this register with the size of the HDMA transfer. The maximum HDMA transfer size is 4Gbytes. The minimum transfer size is one byte (0x1). HDMA automatically decrements the value of this field as the read channel transfer progresses. This field indicates the number of bytes remaining to be requested. When all bytes are requested the current transfer size is zero. In LL mode, the HDMA overwrites this register with the corresponding dword of the LL element. You can read this register to monitor the transfer progress, but it is not reliable for that specific usage, as it is updated after read requests and not after write requests. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.36 HDMA_SAR_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel SAR Low Register.
- **Description:** This register holds the lower 32 bits of the Source Address Register (SAR) of an HDMA read channel.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))} + 0x20 + \text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE == 0 \ || \ CX_MEMORY_MAP_PCIE_TYPE == 1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN > 0)$



Table 3-696 Fields for Register: HDMA_SAR_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	SAR_PTR_LOW	* Varies	Source Address Register (lower 32 bits). Indicates the address of the local memory from which HDMA reads. The HDMA increments the SAR as the HDMA transfer progresses. In LL mode, the HDMA overwrites this field with the corresponding dword of the LL element. The SAR is the address of the remote memory. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.37 HDMA_SAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel SAR High Register.
- **Description:** This register holds the higher 32 bits of the Source Address Register (SAR) of an HDMA read channel.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))} + 0x24 + \text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE == 0 \ || \ CX_MEMORY_MAP_PCIE_TYPE == 1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN > 0)$

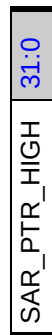


Table 3-697 Fields for Register: HDMA_SAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	SAR_PTR_HIGH	* Varies	Source Address Register (higher 32 bits). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.38 HDMA_DAR_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel DAR Low Register.
- **Description:** This register holds the lower 32 bits of the Destination Address Register (DAR) of an HDMA read channel.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))} + 0x28 + \text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE == 0 \ || \ CX_MEMORY_MAP_PCIE_TYPE == 1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN > 0)$



Table 3-698 Fields for Register: HDMA_DAR_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	DAR_PTR_LOW	* Varies	Destination Address Register (lower 32 bits). Indicates the address to which HDMA writes. The HDMA increments the DAR as the HDMA transfer progresses. In LL mode, the HDMA overwrites this field with the corresponding dword of the LL element. The DAR is the address of the local memory. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.39 HDMA_DAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel DAR High Register.
- **Description:** This register holds the higher 32 bits of the Destination Address Register (DAR) of an HDMA read channel.
- **Size:** 32 bits
- **Offset:** $B + i * 2^{\text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))} + 0x2c + \text{pow}(2, (CX_DMAREG_CHADDR_SPACE + 8))$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE == 0 \ || \ CX_MEMORY_MAP_PCIE_TYPE == 1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN > 0)$



Table 3-699 Fields for Register: HDMA_DAR_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	DAR_PTR_HIGH	* Varies	Destination Address Register (higher 32 bits). <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.40 HDMA_WATERMARK_EN_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Linked-list Watermark Enable Register.
- **Description:** This register controls the watermark interrupts generated after processing a water-marked LL element. HDMA updates this field as LL elements are processed.
- **Size:** 32 bits
- **Offset:** $B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x30+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ || \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$

31:2	1	0
Rsvd	LWIE	RWIE

Table 3-700 Fields for Register: HDMA_WATERMARK_EN_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes

Bits	Name	Memory Access	Description
1	LWIE	* Varies	Enables Local Interrupts at watermark events (end of linked list element) in an HDMA read channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): Enable Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
0	RWIE	* Varies	Enables Remote Interrupts at watermark events (end of linked list element) in an HDMA read channel. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): Enable Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.41 HDMA_CONTROL1_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Control Settings 1 Register.
- **Description:** This register controls the operation of the HDMA read channel.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0x34+\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ || \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$

31:27	26:19	18:17	16	15:7	6:5	4	3	2	1	0
Rsvd	ST	PH	TH	Rsvd	AT	RO	DST_SNOOP	SRC_SNOOP	MEM_TYPE	LLEN

Table 3-701 Fields for Register: HDMA_CONTROL1_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:27			Reserved Field: Yes
26:19	ST	* Varies	Steering Tag TLP Header Bit (ST). The HDMA read channel uses this TLP header field when generating MWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: $CX_TPH_ENABLE \ \&\& \ (CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$ Memory Access: $(CX_MEMORY_MAP_VIEW==0) ? \backslash\text{"read-write"} : (CX_MEMORY_MAP_VIEW==1 \ \&\& \ !SNPS_RSVDPARAM_38) ? \backslash\text{"read-write"} : (CX_MEMORY_MAP_VIEW==1 \ \&\& \ SNPS_RSVDPARAM_38) ? \backslash\text{"read-only"} : \backslash\text{"read-write"}$

Bits	Name	Memory Access	Description
18:17	PH	* Varies	Processing Hints TLP Header Bit (PH). The HDMA read channel uses this TLP header field when generating MWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
16	TH	* Varies	TLP Processing Hints Present Header Bit (TH). The HDMA read channel uses this TLP header field when generating MWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_TPH_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
15:7			Reserved Field: Yes

Bits	Name	Memory Access	Description
6:5	AT	* Varies	Address Translation Services TLP Header Bit (AT). The HDMA read channel uses this TLP header field when generating MRd/MWr/IMWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
4	RO	* Varies	Relaxed Ordering TLP Header Bit. HDMA read channel uses this TLP header field when generating MRd/MWr/IMWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
3	DST_SNOOP	* Varies	Destination No Snoop TLP Header Bit. The HDMA read channel uses this TLP header field when generating MWr/IMWr (DAR addressing space) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
2	SRC_SNOOP	* Varies	Source No Snoop TLP Header Bit. The HDMA read channel uses this TLP header field when generating MRd (SAR addressing space) TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
1	MEM_TYPE	* Varies	Master AXI ACE-Lite Cache Coherency Control. This field sets the HDMA read channel memory type of the address space of the data transfer as follows: For more information, see "ACE-Lite Features and Limitations" section of the <i>Databook</i>. Irrespective of the value of this field, Linked list descriptor requests have this attribute set to 0x1. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (PERIPHERAL): peripheral type ■ 0x1 (MEM_TYPE): memory type Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
0	LLEN	* Varies	This field enables the HDMA read channel Linked List mode. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): Enable Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.42 HDMA_FUNC_NUM_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Function Number Register.
- **Description:** This register controls HDMA read channel to Physical/Virtual function mapping.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0x38+\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ || \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$

31:25	24:17	16	15:5	4:0
Rsvd	VF	VF_EN	Rsvd	PF

Table 3-702 Fields for Register: HDMA_FUNC_NUM_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:25			Reserved Field: Yes

Bits	Name	Memory Access	Description
24:17	VF	* Varies	DMA Channel Virtual Function Number. -When the VF_EN field of this register '1', the controller uses both the PF and this field for generating the requester ID of the generated MRd/MWr DMA TLPs. The PCIe controller supports 256 virtual functions and starts numbering VFs at 0, that maps to virtual function 1. The behavior is undefined if any value outside of the valid range is set to this field. -When the DMA_VF_EN field is '0', the controller uses the PF field of this register for generating the requester ID of the generated MRd/MWr DMA TLPs. At the signal interfaces, DMA drives the XADM / RADM signals d_client[0]1_tlp_vfunc_num / d_radm_trgt1_vfunc_num. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
16	VF_EN	* Varies	DMA Channel Virtual Function Enable. At the signal interfaces, DMA drives the XADM / RADM signals d_client[0]1_tlp_vfunc_active / d_radm_trgt1_vfunc_active. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Disable): Disable ■ 0x1 (Enable): When this field is '1', the controller uses both the PF field and the VF field of this registers for generating the requester ID of the generated MRd/MWr DMA TLPs. The PCIe controller supports 256 virtual functions. Value After Reset: 0x0 Exists: CX_SRIOV_ENABLE && (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
15:5			Reserved Field: Yes

Bits	Name	Memory Access	Description
4:0	PF	* Varies	DMA Channel Physical Function Number. DMA channel wide physical function of the generated TLP. The controller uses this field to form the requester ID for requests. Function numbering starts at '0'. When you have enabled SR-IOV, refer to VF and VF_EN fields of this register as well. The behavior is undefined if any value outside of the valid range is set to this field. At the signal interfaces, DMA drives the XADM / RADM signals d_client[0]1_tlp_func_num / d_radm_trgt1_func_num. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.68.43 HDMA_QOS_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel QoS Settings Register.
- **Description:** This register controls the HDMA read channel QoS features.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0x3c+\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_RD_CHAN>0)$

31:26	25:16	15:8	7:3	2:0
Rsvd	PF_DEPTH	Rsvd	WEIGHT	TC

Table 3-703 Fields for Register: HDMA_QOS_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:26			Reserved Field: Yes

Bits	Name	Memory Access	Description
25:16	PF_DEPTH	* Varies	This field controls the linked list prefetch allocation space for an HDMA read channel. Each read channel stores PF_DEPTH plus '1' linked list elements. For example, if PF_DEPTH =0, the read channel stores one linked list element. This field must be within the range: [0 : floor(CX_DMA_RD_LLQ * CC_NUM_DMA_RD_CHAN / number_of_active_channels) - 1]. <i>Note:</i> ■ If this field is set to a value outside the specified range, data corruption or channel crosstalk may happen. ■ If PF_DEPTH is changed at run time, only channels in the range: [0 : number_of_active_channels - 1] can be doorbelled. ■ This field cannot be changed while any read engine channel is in doorbell state. Doing so might yield undefined results. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Field_value): Zero value ■ 0x3ff (Max_Field_value): Max value Value After Reset: CC_HDMA_REG_RD_PF_DEPTH Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
15:8			Reserved Field: Yes

Bits	Name	Memory Access	Description
7:3	WEIGHT	* Varies	HDMA Read Channel Weight. Specifies the maximum number of TLP MRd requests that the HDMA can issue for a channel, before it returns to the arbitration routine. When a channel reaches its weight count or HDMA channel request transfer size reaches zero, the weighted round robin (WRR) arbiter selects the next channel to be processed. Your software must initialize this field before ringing the doorbell. For more information, see "Multichannel Arbitration" in the <i>Databook</i>. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: CC_HDMA_REG_WEIGHT Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
2:0	TC	* Varies	Traffic Class TLP Header Field (TC). The HDMA read channel uses this TLP header field when generating MRd/MWr/IMWr TLPs. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.44 HDMA_STATUS_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Status Register.
- **Description:** This register specifies the current operational state of a channel.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0x80+\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ || \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$

31:3	2:0
Rsvd	STATUS

Table 3-704 Fields for Register: HDMA_STATUS_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	STATUS	R	HDMA Read Channel Status. Values: ■ 0x0 (RESERVED): Reserved ■ 0x1 (RUNNING): This channel is active and transferring data. ■ 0x2 (ABORTED): An error condition is detected, and the HDMA has stopped this channel. ERROR field of the HDMA_INT_STATUS_OFF_RDCH_0 register specifies the error. ■ 0x3 (STOPPED): The HDMA has transferred all data for this channel. Value After Reset: CC_HDMA_REG_STATUS Exists: $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$ Testable: writeAsRead Volatile: true

3.68.45 HDMA_INT_STATUS_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Interrupt Status Register.
- **Description:** This register provides information regarding the HDMA read channel interrupt status.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0x84+\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_RD_CHAN>0)$

31:7	6:3	2	1	0
Rsvd	ERROR	ABORT	WATERMARK	STOP

Table 3-705 Fields for Register: HDMA_INT_STATUS_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes

Bits	Name	Memory Access	Description
6:3	ERROR	R	HDMA Read Channel Error Interrupt Status. Specifies the error that caused the HDMA to change HDMA_STATUS_OFF_RDCH_i.STATUS to ABORTED state. Values: ■ 0x0 (NONE): No error to report. This field is cleared when HDMA_INT_CLEAR_OFF_RDCH_i.ABORT_CLEAR =1 ■ 0x1 (LL_CPL_UR): Unsuported request CPL while reading a linked list element. ■ 0x2 (LL_CPL_CA): Completer abort CPL while reading a linked list element. ■ 0x3 (LL_CPL_EP): Poisoned data CPL while reading a linked list element. ■ 0x8 (DATA_CPL_TIMEOUT): CPL timeout while reading data from the source. Only valid for Read Channels. ■ 0x9 (DATA_CPL_UR): Unsuported request CPL while reading data from the source. ■ 0xa (DATA_CPL_CA): Completer abort CPL while reading data from the source. ■ 0xb (DATA_CPL_EP): Poisoned data CPL while reading data from the source. Only valid if CC_DMA_ABORT_ON_POISONED is set. ■ 0xc (DATA_MWR): Error response received while writing data to the destination. Only valid for Read Channels with AXI bridge (AMBA_INTERFACE >2). Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
2	ABORT	R	HDMA Read Channel Abort Interrupt Status. ABORT and STOP register fields are mutually exclusive. Values: ■ 0x0 (Clear): This field is cleared when HDMA_INT_CLEAR_OFF_RDCH_i.ABORT_CLEAR =1 ■ 0x1 (Set): If HDMA_INT_SETUP_OFF_RDCH_i.LAIE =1 and HDMA_INT_SETUP_OFF_RDCH_i.ABORT_MASK =0, and a local interrupt is generated, this field indicates that an error is detected for this channel. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true
1	WATERMARK	R	HDMA Read Channel Watermark Interrupt Status. Values: ■ 0x0 (Clear): This field is cleared when HDMA_INT_CLEAR_OFF_RDCH_i.WATERMARK_CLEAR =1 ■ 0x1 (Set): If HDMA_WATERMARK_EN_OFF_RDCH_i.LWIE =1 and HDMA_INT_SETUP_OFF_RDCH_i.WATERMARK_MASK =0, and a local interrupt is generated, this field indicates that HDMA has transferred all the data for an element with the LWIE bit set. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
0	STOP	R	HDMA Read Channel Stop Interrupt Status. STOP and ABORT register fields are mutually exclusive. Values: ■ 0x0 (Clear): This field is cleared when HDMA_INT_CLEAR_OFF_RDCH_i.STOP_CLEAR =1 ■ 0x1 (Set): If HDMA_INT_SETUP_OFF_RDCH_i.LSIE =1 and HDMA_INT_SETUP_OFF_RDCH_i.STOP_MASK =0, and a local interrupt is generated, this field indicates that HDMA has transferred all the data for this channel. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true

3.68.46 HDMA_INT_SETUP_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Interrupt Setup Register.
- **Description:** This register is used to configure interrupts for a read channel.
- **Size:** 32 bits
- **Offset:** $B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x88+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_RD_CHAN>0)$

31:7	6	5	4	3	2	1	0
Rsvd	LAIE	RAIE	LSIE	RSIE	ABORT_MASK	WATERMARK_MASK	STOP_MASK

Table 3-706 Fields for Register: HDMA_INT_SETUP_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes

Bits	Name	Memory Access	Description
6	LAIE	* Varies	HDMA Read Channel Local Abort Interrupt Enable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field enables local interrupts in abort conditions. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
5	RAIE	* Varies	HDMA Read Channel Remote Abort Interrupt Enable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field enables remote interrupts in abort conditions. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
4	LSIE	* Varies	HDMA Read Channel Local Stop Interrupt Enable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field enables local interrupts in stop conditions. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""
3	RSIE	* Varies	HDMA Read Channel Remote Stop Interrupt Enable. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field enables remote interrupts in stop conditions. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
2	ABORT_MASK	* Varies	HDMA Read Channel Abort Interrupt Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field masks the local interrupt pin for abort interrupts. Value After Reset: CC_HDMA_REG_INT_MASK Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "
1	WATERMARK_MASK	* Varies	HDMA Read Channel Watermark Interrupt Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field masks the local interrupt pin for watermark interrupts. Value After Reset: CC_HDMA_REG_INT_MASK Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
0	STOP_MASK	* Varies	HDMA Read Channel Stop Interrupt Mask. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field masks the local interrupt pin for stop interrupts. Value After Reset: CC_HDMA_REG_INT_MASK Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\" "

3.68.47 HDMA_INT_CLEAR_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Interrupt Clear Register.
- **Description:** This register indicates interrupt clear status of a read channel. It is a self-clearing register. Reads to this register returns '0'.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0x8c+\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))$
- **Exists:** $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ || \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$

31:3	2	1	0
Rsvd	ABORT_CLEAR	WATERMARK_CLEAR	STOP_CLEAR

Table 3-707 Fields for Register: HDMA_INT_CLEAR_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2	ABORT_CLEAR	W	HDMA Abort Channel Abort Interrupt Clear. Setting this field clears the ABORT and ERROR fields. Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field clears the ABORT and ERROR fields of the HDMA_INT_STATUS_OFF_RDCH_i register. Value After Reset: 0x0 Exists: $(CC_DMA_ENABLE \ \&\& \ CC_UNROLL_ENABLE \ \&\& \ !CC_LEGACY_DMA_MAP \ \&\& \ (CX_MEMORY_MAP_PCIE_TYPE==0 \ \ CX_MEMORY_MAP_PCIE_TYPE==1)) \ \&\& \ (CC_NUM_DMA_RD_CHAN>0)$ Testable: writeAsRead Volatile: true

Bits	Name	Memory Access	Description
1	WATERMARK_CLEAR	W	HDMA Read Channel Watermark Interrupt Clear. Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field clears the WATERMARK field of the HDMA_INT_STATUS_OFF_RDCH_i register. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true
0	STOP_CLEAR	W	HDMA Read Channel Stop Interrupt Clear. Values: ■ 0x0 (Clear): Clear ■ 0x1 (Set): Setting this field clears the STOP field of the HDMA_INT_STATUS_OFF_RDCH_i register. Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Testable: writeAsRead Volatile: true

- 3.68.48 HDMA_MSI_STOP_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)
- **Name:** HDMA Read Stop Remote Interrupt Address Low Register.
 - **Description:** This register holds the lower 32 bits of the Stop Interrupt MWr TLP address.
 - **Size:** 32 bits
 - **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x90*pow(2,(CX_DMAREG_CHADDR_SPACE+8))
 - **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)



Table 3-708 Fields for Register: HDMA_MSI_STOP_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_STOP_ADDR_LOW	* Varies	The HDMA uses this field to generate bits [31:0] of the address field for the Stop Interrupt MWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.49 HDMA_MSI_STOP_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Stop Remote Interrupt Address High Register.
- **Description:** This register holds the higher 32 bits of the Stop Interrupt MWr TLP address.
- **Size:** 32 bits
- **Offset:** $B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x94+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_RD_CHAN>0)$



Table 3-709 Fields for Register: HDMA_MSI_STOP_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_STOP_ADDR_HIGH	* Varies	The HDMA uses this field to generate bits [63:32] of the address field for the Stop Interrupt MWr TLP. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.50 HDMA_MSI_WATERMARK_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Watermark Remote Interrupt Address Low Register.
- **Description:** This register holds the lower 32 bits of the Watermark Interrupt MWr TLP address.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x98*pow(2,(CX_DMAREG_CHADDR_SPACE+8))
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)



Table 3-710 Fields for Register: HDMA_MSI_WATERMARK_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_WATERMARK_ADDR_LOW	* Varies	The HDMA uses this field to generate bits [31:0] of the address field for the Watermark Interrupt MWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.51 HDMA_MSI_WATERMARK_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Watermark Remote Interrupt Address High Register.
- **Description:** This register holds the lower 32 bits of the Watermark Interrupt MWr TLP address.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0x9c*pow(2,(CX_DMAREG_CHADDR_SPACE+8))
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)



Table 3-711 Fields for Register: HDMA_MSI_WATERMARK_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_WATERMARK_ADDR_HIGH	* Varies	The HDMA uses this field to generate bits [31:0] of the address field for the Watermark Interrupt MWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.52 HDMA_MSI_ABORT_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Abort Remote Interrupt Address Low Register.
- **Description:** This register holds the lower 32 bits of the Abort Interrupt MWr TLP address.
- **Size:** 32 bits
- **Offset:** B+i*2*pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0xa0*pow(2,(CX_DMAREG_CHADDR_SPACE+8))
- **Exists:** (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 || CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0)



Table 3-712 Fields for Register: HDMA_MSI_ABORT_LOW_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_ABORT_ADDR_LOW	* Varies	The HDMA uses this field to generate bits [31:0] of the address field for the Abort Interrupt MWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffffffff (Max_Val): Max Va Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

3.68.53 HDMA_MSI_ABORT_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Abort Remote Interrupt Address High.
- **Description:** This register holds the lower 32 bits of the Abort Interrupt mwr TLP address.
- **Size:** 32 bits
- **Offset:** $B+i*2*\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))+0xa4+\text{pow}(2,(CX_DMAREG_CHADDR_SPACE+8))$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_RD_CHAN>0)$



Table 3-713 Fields for Register: HDMA_MSI_ABORT_HIGH_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:0	MSI_ABORT_ADDR_HIGH	* Varies	The HDMA uses this field to generate bits [31:0] of the address field for the Abort Interrupt MWr TLP. Bits [1:0] must be '00' as this address must be dword aligned. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Value After Reset: 0x0 Exists: $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_RD_CHAN>0)$ Memory Access: $(CX_MEMORY_MAP_VIEW==0) ? \text{"read-write"} : (CX_MEMORY_MAP_VIEW==1 \&\& !SNPS_RSVDPARAM_38) ? \text{"read-write"} : (CX_MEMORY_MAP_VIEW==1 \&\& SNPS_RSVDPARAM_38) ? \text{"read-only"} : \text{"read-write"}$

3.68.54 HDMA_MSI_MSGD_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

- **Name:** HDMA Read Channel Remote Interrupt Data Register.
- **Description:** This register holds the IMWr TLP Data of an HDMA read channel.
- **Size:** 32 bits
- **Offset:** $B+i*2^{pow(2,(CX_DMAREG_CHADDR_SPACE+8))+0xa8+pow(2,(CX_DMAREG_CHADDR_SPACE+8))}$
- **Exists:** $(CC_DMA_ENABLE \&\& CC_UNROLL_ENABLE \&\& !CC_LEGACY_DMA_MAP \&\& (CX_MEMORY_MAP_PCIE_TYPE==0 \mid \mid CX_MEMORY_MAP_PCIE_TYPE==1)) \&\& (CC_NUM_DMA_RD_CHAN>0)$

31:16	15:0
Rsvd	MSI_MESSAGE

Table 3-714 Fields for Register: HDMA_MSI_MSGD_OFF_RDCH_i (for i = 0; i <= CC_NUM_DMA_RD_CHAN-1)

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes

Bits	Name	Memory Access	Description
15:0	MSI_MESSAGE	* Varies	The HDMA read channel uses this field to generate the data field for every IMWr TLPs it generates. <i>Note:</i> The access attributes of this field are as follows: ■ Wire: R/W ■ Dbi: R/W Values: ■ 0x0 (Min_Val): Min Val ■ 0xffff (Max_Val): Max Val Value After Reset: 0x0 Exists: (CC_DMA_ENABLE && CC_UNROLL_ENABLE && !CC_LEGACY_DMA_MAP && (CX_MEMORY_MAP_PCIE_TYPE==0 CX_MEMORY_MAP_PCIE_TYPE==1)) && (CC_NUM_DMA_RD_CHAN>0) Memory Access: "(CX_MEMORY_MAP_VIEW==0) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && !SNPS_RSVDPARAM_38) ? \"read-write\" : (CX_MEMORY_MAP_VIEW==1 && SNPS_RSVDPARAM_38) ? \"read-only\" : \"read-write\""

A

Internal Parameter Descriptions

Provides a description of the internal parameters that might be indirectly referenced in expressions in the Signals, Parameters, or Registers chapters. These parameters are not visible in the coreConsultant GUI and most of them are derived automatically from visible parameters. **You must not set any of these parameters directly.**

Some expressions might refer to TCL functions or procedures (sometimes identified as **function_of**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the core in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

Table A-1 Internal Parameters

Parameter Name	Equals To
1	DT_RATE_GEN1
2	DT_RATE_GEN2
3	DT_RATE_GEN3
4	DT_RATE_GEN4
5	DT_RATE_GEN5
ACS_CTRL_VEC_WD	[function_of: ACS_CTRL_VEC_WD]
ACS_CTRL_VEC_WD_0	((CX_ACS_P2P_EGRESS_CTRL_0) * ((CX_ACS_EGRESS_CTRL_SIZE_0 == 0) ? 256 : CX_ACS_EGRESS_CTRL_SIZE_0))
ACTIVE	0x1
ADDR_TRANSLATION_SUPPORT_EN	(CX_INTERNAL_ATU_ENABLE)
ADDR_WIDTH	FLT_Q_ADDR_WIDTH
AHB_CLK_DIFF_ENABLE	AHB_POPULATED
AHB_ENABLED	AHB_POPULATED

Parameter Name	Equals To
AHB_POPULATED	AMBA_INTERFACE==1
AMBA_DECOMPOSER_DEF_DEPTH_FACTOR	2
AMBA_POPULATED	AMBA_INTERFACE!=0
APP_CRD_WD	(CX_NW_GTR_4_VALUE?((((NW/4)+1)>32768)?16:(((NW/4)+1)>16384)?15:((((NW/4)+1)>8192)?14:((((NW/4)+1)>4096)?13:((((NW/4)+1)>2048)?12:((((NW/4)+1)>1024)?11:((((NW/4)+1)>512)?10:((((NW/4)+1)>256)?9:((((NW/4)+1)>128)?8:((((NW/4)+1)>64)?7:((((NW/4)+1)>32)?6:((((NW/4)+1)>16)?5:((((NW/4)+1)>8)?4:((((NW/4)+1)>4)?3:((((NW/4)+1)>2)?2:((((NW/4)+1)>1)?1:1)))))))))))*CX_NVC):CX_NVC)
APP_PAR_ERR_OUT_EN	(CX_RAS_EN)
ARC_256_P2	CX_NW==8
ARC_WIDTH	[function_of: CX_DL_NB CX_NL]
ATOMIC_ROUTING_SUP	CX_ATOMIC_ROUTING_EN
ATS_RX_ENABLE	CX_ATS_ENABLE && (CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_DM AMBA_POPULATED)
ATS_RX_ENABLE_VALUE	ATS_RX_ENABLE
ATS_TX_ENABLE	CX_ATS_ENABLE && (CC_DEVICE_TYPE==CC_EP CC_DEVICE_TYPE==CC_DM AMBA_POPULATED)
ATTR_WD	FLT_Q_ATTR_WIDTH
ATU_IN_MIN1	((ATU_IN_REGIONS>0)?ATU_IN_REGIONS:1)
ATU_IN_REGIONS	CX_ATU_NUM_INBOUND_REGIONS
ATU_IN_SINGLE_TRGT_ADDR_ENABLE	CX_INTERNAL_ATU_ENABLE
AXI_ENABLED	AXI_POPULATED
AXI_IF_PAR_ECC_ENABLED	CX_AXI_IF_ECC_PAR_PROT_EN_VALUE
AXI_MULTIFUNCTION_ENABLED	((AXI_POPULATED == 1) && (CX_NFUNC != 1))
AXI_POPULATED	((AMBA_INTERFACE==2) (AMBA_INTERFACE==3) (AMBA_INTERFACE==4))
AXI_POPULATED_VALUE	AXI_POPULATED
AXI_RADM_SEG_BUF_ENABLE	AXI_POPULATED && RADM_SEG_BUF
BUSNUM_WD	CX_BUSNUM_WD
CAS	0x31

Parameter Name	Equals To
CC_AHB_SNUM_MASTERS	(AHB_POPULATED && SLAVE_POPULATED && SPLIT_SUPPORT ? CC_SLV_NUM_MASTERS : 1)
CC_AMBA_ATS_NW	CX_ATS_ENABLE
CC_AMBA_TLP_SEQ_MAX_COUNT	(CC_WR_CH_P_REQ_QUEUE_DP + CC_IB_WR_REQ_CDC_RAM_DP + CC_IB_WREQ_PTRK_EN * CC_IB_WR_REQ_PTRK_HDR_RAM_DP+12+2)
CC_AMBA_TLP_SEQ_WD	[function_of: [function_of: CC_AMBA_TLP_SEQ_MAX_COUNT]] + 2
CC_AXI3_ENABLE	(AMBA_INTERFACE==2)
CC_AXI4_ENABLE	((AMBA_INTERFACE==3) (AMBA_INTERFACE==4))
CC_AXI_DEF_MEM_WR_CPL_EN	CX_DEF_MEM_WR_CPL_EN
CC_AXI_DEF_MEM_WR_REQ_EN	CX_DEF_MEM_WR_REQ_EN
CC_AXI_LOCK_WIDTH	CC_AXI4_ENABLE ? 1 : 2
CC_AXI_MSTR_MISC_INFO_CID_ENABLE	0
CC_AXI_RCHUNKNUM_WD	[function_of: [function_of: 32768/MASTER_BUS_DATA_WIDTH - 1]]
CC_AXI_RCHUNKSTRB_WD	MASTER_BUS_DATA_WIDTH/128
CC_AXI_SLV_MISC_INFO RID_ENABLE	0
CC_COMPOSER_MTUSIZE_AVAIL	((((CC_DMA_ENABLE ? CC_DMA_RD_TAG_LOW - 1 : CX_MAX_TAG)*(128 < CX_MAX_MTU ? 128 : CX_MAX_MTU)) >= CC_SLV_RD_REQ_SIZE) ? 1 : 0)
CC_COMPOSER_SIZE_DEFAULT	(CC_COMPOSER_MTUSIZE_AVAIL == 1 && (CX_MAX_MTU < CC_SLV_RD_REQ_SIZE)) ? CX_MAX_MTU : CC_SLV_RD_REQ_SIZE
CC_CORE_ADDR_BUS_WD	(CX_RAS_EN==1) ? ((FLT_Q_ADDR_WIDTH/CX_RAS_PROT_RANGE) + FLT_Q_ADDR_WIDTH) : FLT_Q_ADDR_WIDTH
CC_DBI_BURST_LEN	(SHARED_DBI_ENABLED ? CC_SLV_BURST_LEN : 16)
CC_DBI_BUS_ADDR_PROT_WIDTH	((CC_DBI_SLV_BUS_ADDR_WIDTH - 1) / CX_RAS_PROT_RANGE) + 1

Parameter Name	Equals To
CC_DBI_BUS_DATA_PROT_WIDTH	$(CX_AXI_IF_TYPE_OF_PROT < 2) ?$ $(CC_DBI_SLV_BUS_DATA_WIDTH / CX_RAS_PROT_RANGE) : (CC_DBI_SLV_BUS_DATA_WIDTH == 32 ?$ $(CX_AXI_IF_ECC_GRAN == 0 ? 7 :$ $(CX_AXI_IF_ECC_GRAN == 1 ? 7 : 7)) :$ $CC_DBI_SLV_BUS_DATA_WIDTH == 64 ?$ $(CX_AXI_IF_ECC_GRAN == 0 ? 8 :$ $(CX_AXI_IF_ECC_GRAN == 1 ? 14 : 8)) :$ $CC_DBI_SLV_BUS_DATA_WIDTH == 128 ?$ $(CX_AXI_IF_ECC_GRAN == 0 ? 9 :$ $(CX_AXI_IF_ECC_GRAN == 1 ? 28 : 16)) :$ $CC_DBI_SLV_BUS_DATA_WIDTH == 256 ?$ $(CX_AXI_IF_ECC_GRAN == 0 ? 10 :$ $(CX_AXI_IF_ECC_GRAN == 1 ? 56 : 32)) :$ $CC_DBI_SLV_BUS_DATA_WIDTH == 512 ?$ $(CX_AXI_IF_ECC_GRAN == 0 ? 11 :$ $(CX_AXI_IF_ECC_GRAN == 1 ? 112 : 64)) : 0)$
CC_DBI_SLV_AR_EXTRA_PYLD_PAR_WIDTH	$((CC_DBI_SLV_AR_EXTRA_PYLD_WIDTH - 1) / CX_RAS_PROT_RANGE) + 1$
CC_DBI_SLV_AR_EXTRA_PYLD_WIDTH	$CC_ACELITE_ENABLE * 8$
CC_DBI_SLV_AR_USER_WIDTH	$CX_AXI_IF_ECC_PAR_PROT_EN ? (4 +$ $CC_DBI_BUS_ADDR_PROT_WIDTH +$ $CC_DBI_SLV_AR_EXTRA_PYLD_PAR_WIDTH +$ $CC_AXI4_ENABLE) : 0$
CC_DBI_SLV_AW_EXTRA_PYLD_PAR_WIDTH	$((CC_DBI_SLV_AW_EXTRA_PYLD_WIDTH - 1) / CX_RAS_PROT_RANGE) + 1$
CC_DBI_SLV_AW_EXTRA_PYLD_WIDTH	$CC_ACELITE_ENABLE * 7$
CC_DBI_SLV_AW_USER_WIDTH	$CX_AXI_IF_ECC_PAR_PROT_EN ? (4 +$ $CC_DBI_BUS_ADDR_PROT_WIDTH +$ $CC_DBI_SLV_AW_EXTRA_PYLD_PAR_WIDTH +$ $CC_AXI4_ENABLE) : 0$
CC_DBISLV_BURST_LEN_CALC_PW	[function_of: CC_DBI_BURST_LEN]
CC_DBISLV_BURST_LEN_MIN_PW	$(AMBA_INTERFACE \geq 3 ? 8 : 4)$
CC_DBISLV_BURST_LEN_PW	$(CC_DBISLV_BURST_LEN_CALC_PW >$ $CC_DBISLV_BURST_LEN_MIN_PW ?$ $CC_DBISLV_BURST_LEN_CALC_PW :$ $CC_DBISLV_BURST_LEN_MIN_PW)$
CC_DBI_SLV_B_USER_WIDTH	$CX_AXI_IF_ECC_PAR_PROT_EN ? 2 : 0$
CC_DBI_SLV_ID_WD	$((AHB_ENABLED == 1) ? CC_MAX_SLV_TAG_PW : 4)$
CC_DBI_SLV_R_USER_WIDTH	$CX_AXI_IF_ECC_PAR_PROT_EN ? (2 +$ $CC_DBI_BUS_DATA_PROT_WIDTH) : 0$

Parameter Name	Equals To
CC_DBI_SLV_W_USER_WIDTH	$CX_AXI_IF_ECC_PAR_PROT_EN ? (2 + CC_AXI3_ENABLE + CC_DBI_BUS_DATA_PROT_WIDTH) : 0$
CC_DM	2
CC_DMA_ABORT_ON_POISONED	CC_DMA_ENABLE
CC_DMA_CTX_RAM_ADDRP_WIDTH	$(CX_RAS_EN == 1 ? [function_of: CC_DMA_CTX_RAM_ADDR_WIDTH, CX_RAS_PROT_RANGE] : 1)$
CC_DMA_CTX_RAM_ADDR_WIDTH	$[function_of: CC_DMA_CTX_RAM_DEPTH]$
CC_DMA_CTX_RAM_DEPTH	$(2 + CC_NUM_DMA_RD_CHAN + CC_NUM_DMA_WR_CHAN)$
CC_DMA_ENABLE_VALUE	CC_DMA_ENABLE
CC_DMA_RD_TAG_LOW	$((CX_MAX_TAG > 767) ? (768 - CC_NUM_DMA_RD_TAG) : (CX_MAX_TAG + 1 - CC_NUM_DMA_RD_TAG))$
CC_DMA_SEG_BUF_DATA_WECC_WIDTH	$(CX_RAS_EN == 1 ? [function_of: CC_DMA_SEG_BUF_DATA_WIDTH, CX_RAS_RAM_PROT_RANGE] + CC_DMA_SEG_BUF_DATA_WIDTH : CC_DMA_SEG_BUF_DATA_WIDTH)$
CC_DMA_SEG_BUF_DATA_WIDTH	$(CX_NW * 32)$
CC_DMA_SEG_BUF_DEPTH	$(CC_NUM_DMA_RD_TAG * CC_DMA_SEG_DEPTH)$
CC_DMA_SEG_BUF_NW_ADDRP_WIDTH	$(CX_RAS_EN == 1 ? [function_of: CC_DMA_SEG_BUF_NW_ADDR_WIDTH, CX_RAS_PROT_RANGE] : 1)$
CC_DMA_SEG_BUF_NW_ADDR_WIDTH	$[function_of: CC_DMA_SEG_BUF_DEPTH]$
CC_DMA_SEG_DEPTH	$CX_NW_GTR_1 ? ((CX_MAX_MTU / (CX_NW * 4)) + 2) : (CX_MAX_MTU / (CX_NW * 4))$
CC_DMATAGS	$(CX_REMOTE_MAX_TAGS > CC_NUM_DMA_RD_TAG) ? CX_REMOTE_MAX_TAGS : CC_NUM_DMA_RD_TAG$
CC_EDMA_SINGLE_CHANNEL	$((CC_NUM_DMA_WR_CHAN == 1) \&\& (CC_NUM_DMA_RD_CHAN == 1))$
CC_EP	0
CC_GENERIC_PHY	8
CC_GRANT_TRGT1_LATENCY	8
CC_HDMA_REG_INT_MASK	1

Parameter Name	Equals To
CC_HDMA_REG_RD_PF_DEPTH	CX_DMA_RD_LLQ - 1
CC_HDMA_REG_STATUS	3
CC_HDMA_REG_WEIGHT	1
CC_HDMA_REG_WR_PF_DEPTH	CX_DMA_WR_LLQ - 1
CC_HP_MASTER	0
CC_IB_MCPL_CDC_RAM_ADDRP_WD	(CX_RAS_EN==1 ? [function_of: CC_IB_MCPL_CDC_RAM_ADDR_WD CX_RAS_PROT_RANGE] : 1)
CC_IB_MCPL_CDC_RAM_ADDR_WD	[function_of: CC_IB_MCPL_CDC_RAM_DP]
CC_IB_MCPL_CDC_RAM_DATA_WD	(CX_RAS_EN==1 ? [function_of: CC_MSTR_RSP_INTF_WD_NP CX_RAS_RAM_PROT_RANGE] + CC_MSTR_RSP_INTF_WD_NP : CC_MSTR_RSP_INTF_WD)
CC_IB_MCPL_CDC_RAM_DP	AHB_POPULATED ? (4096/(4*CC_MSTR_NW)) : M_C_CDC_FIFO_DEPTH
CC_IB_MCPL_SEG_BUF_RAM_ADDRP_WD	(CX_RAS_EN==1 ? [function_of: CC_IB_MCPL_SEG_BUF_RAM_ADDR_WD CX_RAS_PROT_RANGE] : 1)
CC_IB_MCPL_SEG_BUF_RAM_ADDR_WD	CC_MSTR_CPL_SEG_BUF_ADDR_WD
CC_IB_RD_REQ_CDC_RAM_ADDRP_WD	(CX_RAS_EN==1 ? [function_of: CC_IB_RD_REQ_CDC_RAM_ADDR_WD CX_RAS_PROT_RANGE] : 1)
CC_IB_RD_REQ_CDC_RAM_ADDR_WD	[function_of: CC_IB_RD_REQ_CDC_RAM_DP]
CC_IB_RD_REQ_CDC_RAM_DATA_WD	(CX_RAS_EN==1 ? [function_of: CC_IB_RD_REQ_CDC_RAM_DATA_WD_NOECC CX_RAS_RAM_PROT_RANGE] + CC_IB_RD_REQ_CDC_RAM_DATA_WD_NOECC : CC_IB_RD_REQ_CDC_RAM_DATA_WD_NOECC)
CC_IB_RD_REQ_CDC_RAM_DATA_WD_NOECC	DC_INBD_HEADER_PNP_WD_NP_RAM
CC_IB_RD_REQ_CDC_RAM_DP	C_M_CDC_FIFO_DEPTH
CC_IB_RD_REQ_ORDR_RAM_ADDRP_WD	(CX_RAS_EN==1 ? [function_of: CC_IB_RD_REQ_ORDR_RAM_ADDR_WD CX_RAS_PROT_RANGE] : 1)
CC_IB_RD_REQ_ORDR_RAM_ADDR_WD	CX_REMOTE_LUT_PTR_WIDTH

Parameter Name	Equals To
CC_IB_RD_REQ_ORDR_RAM_DATA_WD	(CX_RAS_EN==1 ? [function_of: DC_INBD_HEADER_PNP_1STG_WD_NP_RAM CX_RAS_RAM_PROT_RANGE] + DC_INBD_HEADER_PNP_1STG_WD_NP_RAM : DC_INBD_HEADER_PNP_1STG_WD_RAM)
CC_IB_WR_REQ_CDC_RAM_ADDRP_WD	(CX_RAS_EN==1 ? [function_of: CC_IB_WR_REQ_CDC_RAM_ADDR_WD CX_RAS_PROT_RANGE] : 1)
CC_IB_WR_REQ_CDC_RAM_ADDR_WD	[function_of: CC_IB_WR_REQ_CDC_RAM_DP]
CC_IB_WR_REQ_CDC_RAM_DATA_WD	(CX_RAS_EN==1 ? [function_of: CC_IB_WR_REQ_CDC_RAM_DATA_WD_NOECC CX_RAS_RAM_PROT_RANGE] + CC_IB_WR_REQ_CDC_RAM_DATA_WD_NOECC : CC_IB_WR_REQ_CDC_RAM_DATA_WD_NOECC)
CC_IB_WR_REQ_CDC_RAM_DATA_WD_NOECC	(MASTER_BUS_DATA_WIDTH > PCIE_CORE_DATA_BUS_WD) ? MASTER_BUS_DATA_WIDTH + DC_INBD_HEADER_PNP_WD_NP : PCIE_CORE_DATA_BUS_WD + DC_INBD_HEADER_PNP_WD_NP
CC_IB_WR_REQ_CDC_RAM_DP	C_M_CDC_FIFO_DEPTH
CC_IB_WR_REQ_MAX_NP_DATA_REQ_BYTES	(CC_AXI_DEF_MEM_WR_CPL_EN ? ((CX_DEF_MEM_WR_LEN_SUPP+1)*64):32)
CC_IB_WR_REQ_MAX_NP_DATA_REQ_SIZE	((CC_IB_WR_REQ_MAX_NP_DATA_REQ_BYTES <= (4 * C X_NW)) ? 1 : (CC_IB_WR_REQ_MAX_NP_DATA_REQ_BYTES / (4 * CX NW)))
CC_IB_WR_REQ_PTRK_DATA_RAM_ADDRP_WD	(CX_RAS_EN==1 ? [function_of: CC_IB_WR_REQ_PTRK_DATA_RAM_ADDR_WD CX_RAS_PROT_RANGE] : 1)
CC_IB_WR_REQ_PTRK_DATA_RAM_ADDR_WD	[function_of: CC_IB_WR_REQ_PTRK_DATA_RAM_DP]
CC_IB_WR_REQ_PTRK_DATA_RAM_DATA_WD	(CX_RAS_EN==1 ? [function_of: CC_IB_WR_REQ_PTRK_DATA_RAM_DATA_WD_NOECC CX_RAS_RAM_PROT_RANGE] + CC_IB_WR_REQ_PTRK_DATA_RAM_DATA_WD_NOECC : CC_IB_WR_REQ_PTRK_DATA_RAM_DATA_WD_NOECC)
CC_IB_WR_REQ_PTRK_DATA_RAM_DATA_WD_NOECC	PCIE_CORE_DATA_BUS_WD

Parameter Name	Equals To
CC_IB_WR_REQ_PTRK_DATA_RAM_DP	$(CC_IB_WR_REQ_PTRK_P_DATA_RAM_DP + CC_MSTR_ATOMIC_OR_DMWR_EN * CC_IB_WR_REQ_PTRK_NP_DATA_RAM_DP)$
CC_IB_WR_REQ_PTRK_DATA_RAM_MIN_DP	16
CC_IB_WR_REQ_PTRK_HDR_RAM_ADDRP_WD	$(CX_RAS_EN == 1 ? [function_of: CC_IB_WR_REQ_PTRK_HDR_RAM_ADDR_WD, CX_RAS_PROT_RANGE] : 1)$
CC_IB_WR_REQ_PTRK_HDR_RAM_ADDR_WD	$[function_of: CC_IB_WR_REQ_PTRK_HDR_RAM_DP]$
CC_IB_WR_REQ_PTRK_HDR_RAM_DATA_WD	$(CX_RAS_EN == 1 ? [function_of: CC_IB_WR_REQ_PTRK_HDR_RAM_DATA_WD_NOECC, CX_RAS_RAM_PROT_RANGE] + CC_IB_WR_REQ_PTRK_HDR_RAM_DATA_WD_NOECC : CC_IB_WR_REQ_PTRK_HDR_RAM_DATA_WD_NOECC)$
CC_IB_WR_REQ_PTRK_HDR_RAM_DATA_WD_NOECC	$(CX_INTERNAL_ATU_ENABLE \&\& RADM_SEG_BUF_VALUE) ? (DC_INBD_HEADER_PNP_1STG_WD_NP - 3 + 64) : (DC_INBD_HEADER_PNP_1STG_WD_NP - 3)$
CC_IB_WR_REQ_PTRK_HDR_RAM_DP	$(CC_IB_WR_REQ_PTRK_P_HDR_RAM_DP + CC_MSTR_ATOMIC_OR_DMWR_EN * CC_IB_WR_REQ_PTRK_NP_HDR_RAM_DP)$
CC_IB_WR_REQ_PTRK_HDR_RAM_MIN_DP	10
CC_IB_WR_REQ_PTRK_NP_DATA_RAM_DP	$(CC_IB_WR_REQ_MAX_NP_DATA_REQ_SIZE * (CC_IB_WR_REQ_PTRK_NP_DATA_REQ_NUM + CC_GRANT_TRGT1_LATENCY + 3 + 3 * CC_DMA_ENABLE_VALUE))$
CC_IB_WR_REQ_PTRK_NP_DATA_REQ_NUM	2
CC_IB_WR_REQ_PTRK_NP_HDR_RAM_DP	$(CC_IB_WR_REQ_PTRK_NP_HDR_REQ_SIZE + CC_GRANT_TRGT1_LATENCY + (3 + 3 * CC_DMA_ENABLE_VALUE))$
CC_IB_WR_REQ_PTRK_NP_HDR_REQ_NUM	4
CC_IB_WR_REQ_PTRK_NP_HDR_REQ_SIZE	$(CC_IB_WR_REQ_PTRK_NP_HDR_REQ_NUM)$
CC_IB_WR_REQ_PTRK_P_DATA_RAM_CALC_DP	$((((CX_MAX_MTU) / (4 * CX_NW)) > (3 + 3 * CC_DMA_ENABLE_VALUE)) ? ((2 * CX_MAX_MTU) / (4 * CX_NW)) : ((3 + 3 * CC_DMA_ENABLE_VALUE) + (CX_MAX_MTU) / (4 * CX_NW)))$

Parameter Name	Equals To
CC_IB_WR_REQ_PTRK_P_DATA_RAM_DP	$((CC_IB_WR_REQ_PTRK_P_DATA_RAM_CALC_DP > CC_IB_WR_REQ_PTRK_DATA_RAM_MIN_DP) ? CC_IB_WR_REQ_PTRK_P_DATA_RAM_CALC_DP : CC_IB_WR_REQ_PTRK_DATA_RAM_MIN_DP)$
CC_IB_WR_REQ_PTRK_P_HDR_RAM_CALC_DP	$(((((CX_MAX_MTU)/(4 * CX_NW) > (3 + 3 * CC_DMA_ENABLE_VALUE))) ? ((CX_MAX_MTU)/(4 * CX_NW)) : ((3 + 3 * CC_DMA_ENABLE_VALUE) + 1)))$
CC_IB_WR_REQ_PTRK_P_HDR_RAM_DP	$((CC_IB_WR_REQ_PTRK_P_HDR_RAM_CALC_DP > CC_IB_WR_REQ_PTRK_HDR_RAM_MIN_DP) ? CC_IB_WR_REQ_PTRK_P_HDR_RAM_CALC_DP : CC_IB_WR_REQ_PTRK_HDR_RAM_MIN_DP)$
CC_IF_RDCTRL_LL_RAM_BASE_WD	$1 + 1 + 1 + 64 + 64 + 1 + [function_of: 2 * [function_of: [function_of: (CC_NUM_DMA_RD_CHAN * CX_DMA_RD_LLQ) + 1]]]$
CC_IF_RDCTRL_LL_RAM_ECC_WD	$CX_RAS_EN ? [function_of: CC_IF_RDCTRL_LL_RAM_BASE_WD CX_RAS_RAM_PROT_RANGE] : 0$
CC_IF_RDCTRL_LL_RAM_PW	$(CC_NUM_DMA_RD_CHAN_WIDTH)$
CC_IF_RDCTRL_LL_RAM_PW_PAR	$([function_of: CC_IF_RDCTRL_LL_RAM_PW CX_RAS_PROT_RANGE])$
CC_IF_RDCTRL_LL_RAM_WD	$CC_IF_RDCTRL_LL_RAM_BASE_WD + CC_IF_RDCTRL_LL_RAM_ECC_WD$
CC_IF_RD_CTXC2W_RAM_BASE_WD	$241 + (CC_NUM_DMA_RD_CHAN_WIDTH)$
CC_IF_RD_CTXC2W_RAM_ECC_WD	$CX_RAS_EN ? [function_of: CC_IF_RD_CTXC2W_RAM_BASE_WD CX_RAS_RAM_PROT_RANGE] : 0$
CC_IF_RD_CTXC2W_RAM_PW	$[function_of: CC_DMATAGS]$
CC_IF_RD_CTXC2W_RAM_PW_PAR	$([function_of: CC_IF_RD_CTXC2W_RAM_PW CX_RAS_PROT_RANGE])$
CC_IF_RD_CTXC2W_RAM_WD	$CC_IF_RD_CTXC2W_RAM_BASE_WD + CC_IF_RD_CTXC2W_RAM_ECC_WD$
CC_IF_RDCTX_LLQ_OVERLAY_RAM_BASE_WD	236
CC_IF_RDCTX_LLQ_OVERLAY_RAM_ECC_WD	$CX_RAS_EN ? [function_of: CC_IF_RDCTX_LLQ_OVERLAY_RAM_BASE_WD CX_RAS_RAM_PROT_RANGE] : 0$
CC_IF_RDCTX_LLQ_OVERLAY_RAM_PW	$[function_of: [function_of: CX_DMA_RD_LLQ * CC_NUM_DMA_RD_CHAN]]$

Parameter Name	Equals To
CC_IF_RDCTX_LLQ_OVERLAY_RAM_PW_PAR	([function_of: CC_IF_RDCTX_LLQ_OVERLAY_RAM_PW CX_RAS_PROT_RANGE])
CC_IF_RDCTX_LLQ_OVERLAY_RAM_WD	CC_IF_RDCTX_LLQ_OVERLAY_RAM_BASE_WD + CC_IF_RDCTX_LLQ_OVERLAY_RAM_ECC_WD
CC_IF_RD_CTXSTSH_RAM_BASE_WD	241 + (CC_NUM_DMA_RD_CHAN_WIDTH)
CC_IF_RD_CTXSTSH_RAM_ECC_WD	CX_RAS_EN ? [function_of: CC_IF_RD_CTXSTSH_RAM_BASE_WD CX_RAS_RAM_PROT_RANGE] : 0
CC_IF_RD_CTXSTSH_RAM_PW	[function_of: CC_DMATAGS]
CC_IF_RD_CTXSTSH_RAM_PW_PAR	([function_of: CC_IF_RD_CTXSTSH_RAM_PW CX_RAS_PROT_RANGE])
CC_IF_RD_CTXSTSH_RAM_WD	CC_IF_RD_CTXSTSH_RAM_BASE_WD + CC_IF_RD_CTXSTSH_RAM_ECC_WD
CC_IF_RD_MSI_RAM_BASE_WD	208
CC_IF_RD_MSI_RAM_DP	(CC_NUM_DMA_RD_CHAN)
CC_IF_RD_MSI_RAM_ECC_WD	CX_RAS_EN ? [function_of: CC_IF_RD_MSI_RAM_BASE_WD CX_RAS_RAM_PROT_RANGE] : 0
CC_IF_RD_MSI_RAM_PW	[function_of: CC_IF_RD_MSI_RAM_DP]
CC_IF_RD_MSI_RAM_PW_PAR	([function_of: CC_IF_RD_MSI_RAM_PW CX_RAS_PROT_RANGE])
CC_IF_RD_MSI_RAM_WD	CC_IF_RD_MSI_RAM_BASE_WD + CC_IF_RD_MSI_RAM_ECC_WD
CC_IF_WRCTRL_LL_RAM_BASE_WD	1+1+1+64+64+1+ [function_of: 2 * [function_of: [function_of: (CC_NUM_DMA_WR_CHAN * CX_DMA_WR_LLQ) + 1]]]
CC_IF_WRCTRL_LL_RAM_ECC_WD	CX_RAS_EN ? [function_of: CC_IF_WRCTRL_LL_RAM_BASE_WD CX_RAS_RAM_PROT_RANGE] : 0
CC_IF_WRCTRL_LL_RAM_PW	(CC_NUM_DMA_WR_CHAN_WIDTH)
CC_IF_WRCTRL_LL_RAM_PW_PAR	([function_of: CC_IF_WRCTRL_LL_RAM_PW CX_RAS_PROT_RANGE])
CC_IF_WRCTRL_LL_RAM_WD	CC_IF_WRCTRL_LL_RAM_BASE_WD + CC_IF_WRCTRL_LL_RAM_ECC_WD
CC_IF_WR_CTXC2W_RAM_BASE_WD	241 + (CC_NUM_DMA_WR_CHAN_WIDTH)

Parameter Name	Equals To
CC_IF_WR_CTXC2W_RAM_ECC_WD	CX_RAS_EN ? [function_of: CC_IF_WR_CTXC2W_RAM_BASE_WD CX_RAS_RAM_PROT_RANGE] : 0
CC_IF_WR_CTXC2W_RAM_PW	[function_of: CC_DMATAGS]
CC_IF_WR_CTXC2W_RAM_PW_PAR	([function_of: CC_IF_WR_CTXC2W_RAM_PW CX_RAS_PROT_RANGE])
CC_IF_WR_CTXC2W_RAM_WD	CC_IF_WR_CTXC2W_RAM_BASE_WD + CC_IF_WR_CTXC2W_RAM_ECC_WD
CC_IF_WRCTX_LLQ_OVERLAY_RAM_BASE_WD	236
CC_IF_WRCTX_LLQ_OVERLAY_RAM_ECC_WD	CX_RAS_EN ? [function_of: CC_IF_WRCTX_LLQ_OVERLAY_RAM_BASE_WD CX_RAS_RAM_PROT_RANGE] : 0
CC_IF_WRCTX_LLQ_OVERLAY_RAM_PW	[function_of: [function_of: CX_DMA_WR_LLQ * CC_NUM_DMA_WR_CHAN]]
CC_IF_WRCTX_LLQ_OVERLAY_RAM_PW_PAR	([function_of: CC_IF_WRCTX_LLQ_OVERLAY_RAM_PW CX_RAS_PROT_RANGE])
CC_IF_WRCTX_LLQ_OVERLAY_RAM_WD	CC_IF_WRCTX_LLQ_OVERLAY_RAM_BASE_WD + CC_IF_WRCTX_LLQ_OVERLAY_RAM_ECC_WD
CC_IF_WR_CTXSTSH_RAM_BASE_WD	241 + (CC_NUM_DMA_WR_CHAN_WIDTH)
CC_IF_WR_CTXSTSH_RAM_ECC_WD	CX_RAS_EN ? [function_of: CC_IF_WR_CTXSTSH_RAM_BASE_WD CX_RAS_RAM_PROT_RANGE] : 0
CC_IF_WR_CTXSTSH_RAM_PW	[function_of: CC_DMATAGS]
CC_IF_WR_CTXSTSH_RAM_PW_PAR	([function_of: CC_IF_WR_CTXSTSH_RAM_PW CX_RAS_PROT_RANGE])
CC_IF_WR_CTXSTSH_RAM_WD	CC_IF_WR_CTXSTSH_RAM_BASE_WD + CC_IF_WR_CTXSTSH_RAM_ECC_WD
CC_IF_WR_MSI_RAM_BASE_WD	208
CC_IF_WR_MSI_RAM_DP	(CC_NUM_DMA_WR_CHAN)
CC_IF_WR_MSI_RAM_ECC_WD	CX_RAS_EN ? [function_of: CC_IF_WR_MSI_RAM_BASE_WD CX_RAS_RAM_PROT_RANGE] : 0
CC_IF_WR_MSI_RAM_PW	[function_of: CC_IF_WR_MSI_RAM_DP]
CC_IF_WR_MSI_RAM_PW_PAR	([function_of: CC_IF_WR_MSI_RAM_PW CX_RAS_PROT_RANGE])

Parameter Name	Equals To
CC_IF_WR_MSI_RAM_WD	CC_IF_WR_MSI_RAM_BASE_WD + CC_IF_WR_MSI_RAM_ECC_WD
CC_IN64	(AMBA_INTERFACE != 0) ? (MASTER_BUS_ADDR_WIDTH > 32) : (FLT_Q_ADDR_WIDTH > 32)
CCIX_ESM_SUPPORT_VALUE	CX_CCIX_ESM_SUPPORT
CCIX_INTERFACE_ENABLE_VALUE	CX_CCIX_INTERFACE_ENABLE
CCIX_VC_RESOURCE	(CX_NVC-1)
CC_LEGACY_DMA_MAP_VALUE	CC_LEGACY_DMA_MAP
CC_LINK_TIMEOUT_ENABLE_DEFAULT	(CC_DEVICE_TYPE==CC_EP) ? 1 : 0
CC_LINK_TIMEOUT_PERIOD_DEFAULT	0x32
CC_MAX_MSTRL_TAGS_AXI	(CC_HP_MASTER ? (CC_MAX_MSTR_TAGS_AXI - CC_MAX_MSTRH_TAGS_AXI) : CC_MAX_MSTR_TAGS_AXI)
CC_MAX_MSTR_TAG_PW	[function_of: CC_MAX_MSTR_TAGS]
CC_MAX_MSTR_TAGS	(AXI_POPULATED ? CC_MAX_MSTR_TAGS_AXI : CC_MAX_MSTR_TAGS_AHB)
CC_MAX_MSTR_TAGS_AHB	1
CC_MAX_NON_DMA_10BIT_TAG_HIGH	((CC_DMA_ENABLE) ? ((CX_MAX_TAG > 767) ? (767 - CC_NUM_DMA_RD_TAG) : (CX_MAX_TAG - CC_NUM_DMA_RD_TAG)) : 2)
CC_MAX_NON_DMA_5BIT_TAG_HIGH	((CC_DMA_ENABLE) ? ((CX_MAX_TAG > 31) ? (31 - CC_NUM_DMA_RD_5BIT_TAG) : (CX_MAX_TAG - CC_NUM_DMA_RD_5BIT_TAG)) : 2)
CC_MAX_NON_DMA_8BIT_TAG_HIGH	((CC_DMA_ENABLE) ? ((CX_MAX_TAG > 255) ? (255 - CC_NUM_DMA_RD_TAG) : (CX_MAX_TAG - CC_NUM_DMA_RD_TAG)) : 2)
CC_MAX_NON_DMA_TAG_HIGH	((CX_10BITS_TAG_REQ) ? (CC_MAX_NON_DMA_10BIT_TAG_HIGH) : (CC_MAX_NON_DMA_8BIT_TAG_HIGH))
CC_MAX_NON_DMA_TAG_HIGH_WIDTH	[function_of: CC_MAX_NON_DMA_TAG_HIGH]
CC_MAX_SLV_TAG_PW	[function_of: CC_MAX_SLV_TAG]
CC_MSTR_AR_EXTRA_PYLD_PAR_WIDTH	((CC_MSTR_AR_EXTRA_PYLD_WIDTH - 1) / CX_RAS_PROT_RANGE) + 1

Parameter Name	Equals To
CC_MSTR_AR_EXTRA_PYLD_WIDTH	$(\text{MSTR_MISC_INFO_PW} + \text{CC_MSTR_AR_MISC_INFO_WIDTH} + \text{CC_ACELITE_ENABLE} * 8)$
CC_MSTR_AR_MISC_INFO_WIDTH	$(1 + 1 + \text{ATS_RX_ENABLE} * \text{CC_AMBA_ATS_NW} + \text{ATS_RX_ENABLE} * 2 + \text{CX_AMBA_ARI_OR_IOV} * \text{CX_NFUNC_WD} + \text{CX_SRIOV_ENABLE} * \text{CX_NVFUNC_WD} + \text{CC_DMA_ENABLE} * (\text{CC_NUM_DMA_MAX_CHAN_WIDTH} + \text{DC_DMA_TYPE_WIDTH}) + \text{CX_TLP_PREFIX_ENABLE} * (32 * \text{CX_NPRFX}) + \text{CX_LN_ENABLE} + \text{CX_TPH_ENABLE} * 11 + \text{CX_IDO_ENABLE})$
CC_MSTR_AR_USER_WIDTH	$\text{CX_RASDP} == 0 ? 0 : \text{CX_AXI_IF_ECC_PAR_PROT_EN} ? (4 + \text{CC_MSTR_BUS_ADDR_PROT_WIDTH} + \text{CC_MSTR_AR_EXTRA_PYLD_PAR_WIDTH} + \text{CC_AXI4_ENABLE}) : \text{CC_MSTR_BUS_ADDR_PROT_WIDTH}$
CC_MSTR_ATOMIC_OR_DMWR_EN	$(\text{CX_ATOMIC_ENABLE} \text{CC_AXI_DEF_MEM_WR_CPL_EN})$
CC_MSTR_AW_EXTRA_PYLD_PAR_WIDTH	$((\text{CC_MSTR_AW_EXTRA_PYLD_WIDTH} - 1) / \text{CX_RAS_PROT_RANGE}) + 1$
CC_MSTR_AW_EXTRA_PYLD_WIDTH	$(\text{MSTR_MISC_INFO_PW} + \text{CC_MSTR_AW_MISC_INFO_WIDTH} + \text{CX_ATOMIC_ENABLE} * 6 + \text{CC_ACELITE_ENABLE} * 7)$
CC_MSTR_AW_MISC_INFO_WIDTH	$(1 + 1 + 64 + \text{ATS_RX_ENABLE} * \text{CC_AMBA_ATS_NW} + \text{ATS_RX_ENABLE} * 2 + \text{CX_AMBA_ARI_OR_IOV} * \text{CX_NFUNC_WD} + \text{CX_SRIOV_ENABLE} * \text{CX_NVFUNC_WD} + \text{CC_DMA_ENABLE} * (\text{CC_NUM_DMA_RD_CHAN_WIDTH} + \text{DC_DMA_TYPE_WIDTH}) + \text{CX_TLP_PREFIX_ENABLE} * (32 * \text{CX_NPRFX}) + \text{CX_LN_ENABLE} + \text{CX_TPH_ENABLE} * 11 + \text{CX_IDO_ENABLE})$
CC_MSTR_AW_USER_WIDTH	$\text{CX_RASDP} == 0 ? 0 : \text{CX_AXI_IF_ECC_PAR_PROT_EN} ? (4 + \text{CC_MSTR_BUS_ADDR_PROT_WIDTH} + \text{CC_MSTR_AW_EXTRA_PYLD_PAR_WIDTH} + \text{CC_AXI4_ENABLE}) : \text{CC_MSTR_BUS_ADDR_PROT_WIDTH}$
CC_MSTR_BE_CTRL_WIDTH	$(\text{AMBA_POPULATED} ? 8 : 0)$
CC_MSTR_B_EXTRA_PYLD_PAR_WIDTH	$((\text{CC_MSTR_B_EXTRA_PYLD_WIDTH} - 1) / \text{CX_RAS_PROT_RANGE}) + 1$
CC_MSTR_B_EXTRA_PYLD_WIDTH	$(\text{CC_MSTR_B_MISC_INFO_WIDTH})$

Parameter Name	Equals To
CC_MSTR_B_MISC_INFO_WIDTH	$(3 + CC_MSTR_RESP_TLP_PREFIX_ENABLE * (32 * CX_NPRFX) + CX_LN_ENABLE + (CC_AXI_MSTR_MISC_INFO_CID_ENABLE) * (16))$
CC_MSTR_BURST_LEN_CALC_PW	[function_of: CC_MSTR_BURST_LEN]
CC_MSTR_BURST_LEN_MIN_PW	$(AMBA_INTERFACE \geq 3 ? 8 : 4)$
CC_MSTR_BURST_LEN_PW	$(CC_MSTR_BURST_LEN_CALC_PW > CC_MSTR_BURST_LEN_MIN_PW ? CC_MSTR_BURST_LEN_CALC_PW : CC_MSTR_BURST_LEN_MIN_PW)$
CC_MSTR_BURST_LEN_REG	CC_MSTR_BURST_LEN_CALC_PW-1
CC_MSTR_BURST_TYPE	1
CC_MSTR_BUS_ADDR_PROT_WIDTH	MASTER_BUS_ADDR_WIDTH/CX_RAS_PROT_RANGE
CC_MSTR_BUS_DATA_PROT_WIDTH	$(CX_AXI_IF_TYPE_OF_PROT < 2 ? (MASTER_BUS_DATA_WIDTH/CX_RAS_PROT_RANGE) : (MASTER_BUS_DATA_WIDTH == 32 ? (CX_AXI_IF_ECC_GRAN == 0 ? 7 : (CX_AXI_IF_ECC_GRAN == 1 ? 7 : 7)) : MASTER_BUS_DATA_WIDTH == 64 ? (CX_AXI_IF_ECC_GRAN == 0 ? 8 : (CX_AXI_IF_ECC_GRAN == 1 ? 14 : 8)) : MASTER_BUS_DATA_WIDTH == 128 ? (CX_AXI_IF_ECC_GRAN == 0 ? 9 : (CX_AXI_IF_ECC_GRAN == 1 ? 28 : 16)) : MASTER_BUS_DATA_WIDTH == 256 ? (CX_AXI_IF_ECC_GRAN == 0 ? 10 : (CX_AXI_IF_ECC_GRAN == 1 ? 56 : 32)) : MASTER_BUS_DATA_WIDTH == 512 ? (CX_AXI_IF_ECC_GRAN == 0 ? 11 : (CX_AXI_IF_ECC_GRAN == 1 ? 112 : 64)) : 0)$
CC_MSTR_B_USER_WIDTH	$CX_AXI_IF_ECC_PAR_PROT_EN ? (2 + CC_MSTR_B_EXTRA_PYLD_PAR_WIDTH) : 0$
CC_MSTR_BUS_ID_WIDTH	CC_MAX_MSTR_TAG_PW + 1
CC_MSTR_BUS_WSTRB_WIDTH	$(MASTER_BUS_DATA_WIDTH / 8)$
CC_MSTR_CHNK_CPL_DATA_SEG_BYTE_LEN	$CX_MAX_MTU < CC_MSTR_MTU ? CX_MAX_MTU : CC_MSTR_MTU$
CC_MSTR_CHNK_CPL_DATA_SEG_DP	$CC_MSTR_DATA_WD_LT_CORE_DATA_WD ? ((CC_MSTR_CHNK_CPL_DATA_SEG_BYTE_LEN / (4 * CX_MSTR_NW)) + 2) : ((CC_MSTR_CHNK_CPL_DATA_SEG_BYTE_LEN / (4 * CC_MSTR_NW)) + 2)$

Parameter Name	Equals To
CC_MSTR_CPL_DATA_SEG_DP	AXI_POPULATED ? (CX_AXI_RCHUNK_ENABLE ? CC_MSTR_CHNK_CPL_DATA_SEG_DP : CC_MSTR_MIN_CPL_DATA_SEG_DP) : (MSTR_CLK_DIFF_ENABLE ? CC_MSTR_MIN_CPL_DATA_SEG_DP : CC_MSTR_MAX_CPL_DATA_SEG_DP)
CC_MSTR_CPL_DATA_WD	CX_RAS_EN ? PCIE_CORE_DATA_BUS_WD + CC_PCIE_CORE_DATA_BUS_PAR_BITS : PCIE_CORE_DATA_BUS_WD
CC_MSTR_CPL_SEG_BUF_ADDR_WD	[function_of: CC_MSTR_CPL_SEG_BUF_DP]
CC_MSTR_CPL_SEG_BUF_DATA_WD	(CX_RAS_EN==1 ? [function_of: CC_MSTR_CPL_SEG_BUF_DATA_WD_NOECC CX_RAS_RAM_PROT_RANGE] + CC_MSTR_CPL_SEG_BUF_DATA_WD_NOECC : CC_MSTR_CPL_SEG_BUF_DATA_WD_NOECC)
CC_MSTR_CPL_SEG_BUF_DATA_WD_NOECC	CC_MSTR_DATA_WD_LT_CORE_DATA_WD ? (PCIE_CORE_DATA_BUS_WD + CX_NW + 1) : (MASTER_BUS_DATA_WIDTH + CC_MSTR_NW + 1)
CC_MSTR_CPL_SEG_BUF_DP	AXI_POPULATED ? (CC_MSTR_CPL_DATA_SEG_DP*(CC_MAX_MSTRL_T AGS_AXI)) : (CC_MSTR_CPL_DATA_SEG_DP*(CX_REMOTE_MAX_T AGS))
CC_MSTR_CPL_SEG_BUF_SLICE_WD	CX_AXI_RCHUNK_ENABLE ? (CX_RAS_EN==1 ? [function_of: 132 CX_RAS_RAM_PROT_RANGE] + 132 : 132) : CC_MSTR_CPL_SEG_BUF_DATA_WD
CC_MSTR_CPL_SEG_DATA_NUM_BYTES	(CC_MSTR_CPL_SEG_RAW_DATA_WD / 8)
CC_MSTR_CPL_SEG_DATA_WD	CC_MSTR_DATA_WD_LT_CORE_DATA_WD ? CC_MSTR_CPL_DATA_WD : CC_MSTR_RSP_DATA_WD
CC_MSTR_CPL_SEG_DATA_WD_NP	CC_MSTR_DATA_WD_LT_CORE_DATA_WD ? PCIE_CORE_DATA_BUS_WD : MASTER_BUS_DATA_WIDTH
CC_MSTR_CPL_SEG_N_SLICES	CX_AXI_RCHUNK_ENABLE ? (CC_MSTR_CPL_SEG_RAW_DATA_WD/128) : 1
CC_MSTR_CPL_SEG_RAW_DATA_WD	CC_MSTR_DATA_WD_LT_CORE_DATA_WD ? PCIE_CORE_DATA_BUS_WD : MASTER_BUS_DATA_WIDTH
CC_MSTR_DATA_BUS_PAR_BITS	CX_RAS_EN ? (MASTER_BUS_DATA_WIDTH/CX_RAS_PROT_RANGE): 0

Parameter Name	Equals To
CC_MSTR_DATA_WD_LT_CORE_DATA_WD	MASTER_BUS_DATA_WIDTH < PCIE_CORE_DATA_BUS_WD
CC_MSTR_MAX_CPL_DATA_SEG_DP	$CC_MSTR_DATA_WD_LT_CORE_DATA_WD ? ((4096/(4 * CX_NW)) + 2) : ((4096/(4 * CC_MSTR_NW)) + 2)$
CC_MSTR_MIN_CPL_DATA_SEG_DP	$CC_MSTR_DATA_WD_LT_CORE_DATA_WD ? ((CX_MAX_MTU/(4 * CX_NW)) + 2) : ((CX_MAX_MTU/(4 * CC_MSTR_NW)) + 2)$
CC_MSTR_MISC_INFO_REQID	1
CC_MSTR_MISC_INFO_TAG	1
CC_MSTR_MTU_PW	[function_of: [function_of: CC_MSTR_MTU+1]]
CC_MSTR_NW	(MASTER_BUS_DATA_WIDTH/32)
CC_MSTR_RD_REQ_SIZE	$(CC_MSTR_BURST_LEN * MASTER_BUS_DATA_WIDTH / 8)$
CC_MSTR_RESP_TLP_PREFIX_ENABLE	0
CC_MSTR_R_EXTRA_PYLD_PAR_WIDTH	$((CC_MSTR_R_EXTRA_PYLD_WIDTH - 1) / CX_RAS_PROT_RANGE) + 1$
CC_MSTR_R_EXTRA_PYLD_WIDTH	(MSTR_RESP_MISC_INFO_PW + CC_MSTR_R_MISC_INFO_WIDTH)
CC_MSTR_R_MISC_INFO_WIDTH	$(3 + CC_MSTR_RESP_TLP_PREFIX_ENABLE * (32 * CX_NPRFX) + CX_LN_ENABLE + (CC_AXI_MSTR_MISC_INFO_CID_ENABLE) * (16))$
CC_MSTR_RSP_DATA_WD	$CX_RAS_EN ? MASTER_BUS_DATA_WIDTH + CC_MSTR_DATA_BUS_PAR_BITS : MASTER_BUS_DATA_WIDTH$
CC_MSTR_RSP_INTF_WD	$1 + CC_MAX_MSTR_TAG_PW + CC_MSTR_CPL_SEG_DATA_NUM_BYTES + CC_MSTR_CPL_SEG_DATA_WD + 3 * (AXI_ENABLED) + 2 + 4 + CC_MSTR_RESP_TLP_PREFIX_ENABLE * ((32 * CX_NPRFX) + CX_PRFX_PAR_WD) + 1 + CC_MAX_MSTR_TAGS + CX_LN_ENABLE + 2 * (CX_ATOMIC_ENABLE) + (CC_AXI_MSTR_MISC_INFO_CID_ENABLE) * (16) + (CX_AXI_RCHUNK_ENABLE ? CC_MAX_MSTR_TAG_PW + 1 + CC_AXI_RCHUNKNUM_WD + CC_AXI_RCHUNKSTRB_WD : 0)$

Parameter Name	Equals To
CC_MSTR_RSP_INTF_WD_NP	$1 + \text{CC_MAX_MSTR_TAG_PW} + \text{CC_MSTR_CPL_SEG_DATA_NUM_BYTES} +$ $(\text{MSTR_CLK_DIFF_ENABLE} ? \text{CC_MSTR_CPL_SEG_DATA_WD_NP} :$ $\text{CC_MSTR_CPL_SEG_DATA_WD}) + 3 * (\text{AXI_ENABLED}) +$ $2 + 4 + \text{CC_MSTR_RESP_TLP_PREFIX_ENABLE} * ((32 * \text{CX_NPR}$ $\text{FX}) + \text{CX_PRFX_PAR_WD}) + 1 + \text{CC_MAX_MSTR_TAGS} +$ $\text{CX_LN_ENABLE} + 2 * (\text{CX_ATOMIC_ENABLE}) +$ $(\text{CC_AXI_MSTR_MISC_INFO_CID_ENABLE}) * (16) +$ $(\text{CX_AXI_RCHUNK_ENABLE} ? \text{CC_MAX_MSTR_TAG_PW} + 1 +$ $\text{CC_AXI_RCHUNKNUM_WD} + \text{CC_AXI_RCHUNKSTRB_WD} : 0)$
CC_MSTR_R_USER_WIDTH	$\text{CX_RASDP} == 0 ? 0 : \text{CX_AXI_IF_ECC_PAR_PROT_EN}$ $? (2 + \text{CC_MSTR_BUS_DATA_PROT_WIDTH} +$ $\text{CC_MSTR_R_EXTRA_PYLD_PAR_WIDTH}) :$ $\text{CC_MSTR_BUS_DATA_PROT_WIDTH}$
CC_MSTR_W_USER_WIDTH	$\text{CX_RASDP} == 0 ? 0 : \text{CX_AXI_IF_ECC_PAR_PROT_EN}$ $? (2 + \text{CC_AXI3_ENABLE} +$ $\text{CC_MSTR_BUS_DATA_PROT_WIDTH}) :$ $\text{CC_MSTR_BUS_DATA_PROT_WIDTH}$
CC_NUM_DMA_MAX_CHAN	$\text{CC_NUM_DMA_RD_CHAN_GTR_WR} ?$ $\text{CC_NUM_DMA_RD_CHAN} : \text{CC_NUM_DMA_WR_CHAN}$
CC_NUM_DMA_MAX_CHAN_WIDTH	[function_of: CC_NUM_DMA_MAX_CHAN]
CC_NUM_DMA_RD_5BIT_TAG	$((\text{CC_DMA_ENABLE}) ? ((\text{CX_MAX_TAG} \leq 31) ?$ $\text{CC_NUM_DMA_RD_TAG} : (((\text{CX_MAX_TAG} + 1) / ($ $\text{CC_NUM_DMA_RD_TAG})) \leq 8) ? (($ $\text{CC_NUM_DMA_RD_TAG} * 32) / (\text{CX_MAX_TAG} + 1)) : 2$ $)) : 2)$
CC_NUM_DMA_RD_CHAN_GTR_WR	$(\text{CC_NUM_DMA_RD_CHAN} >$ $\text{CC_NUM_DMA_WR_CHAN}) ? 1 : 0$
CC_NUM_DMA_RD_CHAN_WIDTH	[function_of: CC_NUM_DMA_RD_CHAN]
CC_NUM_DMA_WR_CHAN_WIDTH	[function_of: CC_NUM_DMA_WR_CHAN]
CC_OB_CCMP_DATA_RAM_ADDRP_WD	$(\text{CX_RAS_EN} == 1 ? [\text{function_of:}$ $\text{CC_OB_CCMP_DATA_RAM_ADDR_WD}$ $\text{CX_RAS_PROT_RANGE}] : 1)$
CC_OB_CCMP_DATA_RAM_ADDR_WD	[function_of: CC_OB_CCMP_DATA_RAM_DP]

Parameter Name	Equals To
CC_OB_CCMP_DATA_RAM_DATA_WD	(CX_RAS_EN==1 ? [function_of: CC_OB_CCMP_DATA_RAM_DATA_WD_NOECC CX_RAS_RAM_PROT_RANGE] + CC_OB_CCMP_DATA_RAM_DATA_WD_NOECC : CC_OB_CCMP_DATA_RAM_DATA_WD_NOECC)
CC_OB_CCMP_DATA_RAM_DATA_WD_NOECC	(SLAVE_BUS_DATA_WIDTH <= PCIE_CORE_DATA_BUS_WD) ? (PCIE_CORE_DATA_BUS_WD + CX_NW + 1) : (SLAVE_BUS_DATA_WIDTH + CC_SLV_NW + 1)
CC_OB_CCMP_DATA_RAM_DP	(CC_RADMX_COMPOSER_SEG_DP * (CC_DMA_ENABLE ? CC_DMA_RD_TAG_LOW : CX_MAX_TAG + 1))
CC_OB_NPDCMP_RAM_ADDRP_WD	(CX_RAS_EN==1 ? [function_of: CC_OB_NPDCMP_RAM_ADDR_WD CX_RAS_PROT_RANGE] : 1)
CC_OB_NPDCMP_RAM_ADDR_WD	[function_of: CC_OB_NPDCMP_RAM_DP]
CC_OB_NPDCMP_RAM_DATA_WD	(CX_RAS_EN==1 ? [function_of: CC_OB_NPDCMP_RAM_DATA_WD_NOECC CX_RAS_RAM_PROT_RANGE] + CC_OB_NPDCMP_RAM_DATA_WD_NOECC : CC_OB_NPDCMP_RAM_DATA_WD_NOECC)
CC_OB_NPDCMP_RAM_DATA_WD_NOECC	CX_LN_ENABLE+CX_TLP_PREFIX_ENABLE*(32*CX_NP RFX+CX_PRFX_PAR_WD)+(CC_AXI_SLV_MISC_INFO_R ID_ENABLE*16)+11*CX_TPH_ENABLE+CX_ATU_CTRL_ EN+ATS_TX_ENABLE*3+CX_SRIOV_ENABLE*(CX_NVF UNC_WD-1+1)+AXI_MULTIFUNCTION_ENABLED*CX_N FUNC_WD+1+1+1+3+CC_SLV_MTU_PW+CC_MAX_SLV _TAG_PW+CC_SLV_BUS_ID_WIDTH+1+((CC_SLV_MTU _PW < CC_SLV_RD_REQ_SIZE_PW) ? CC_SLV_MTU_PW : CC_SLV_RD_REQ_SIZE_PW)+1+1+2+CX_IDO_ENABLE +8+2+5+SLAVE_BUS_ADDR_WIDTH+CC_AXI_DEF_ME M_WR_REQ_EN*CC_SLV_WRAP_ENABLE*2+(CC_AXI_ DEF_MEM_WR_REQ_EN ? (CX_DEF_MEM_WR_LEN_SUPP+1)*64*8 : CX_ATOMIC_ENABLE ? 256 : 32)+((CC_DMA_ENABLE_VALUE==0) ? ((CX_10BITS_TAG_REQ) ? 10 : CX_LUT_PTR_WIDTH) : CC_MAX_NON_DMA_TAG_HIGH_WIDTH)
CC_OB_NPDCMP_RAM_DP	CC_MAX_SLV_TAG
CC_OB_PDCMP_DATA_RAM_ADDRP_WD	(CX_RAS_EN==1 ? [function_of: CC_OB_PDCMP_DATA_RAM_ADDR_WD CX_RAS_PROT_RANGE] : 1)

Parameter Name	Equals To
CC_OB_PDCMP_DATA_RAM_ADDR_WD	[function_of: CC_OB_PDCMP_DATA_RAM_DP]
CC_OB_PDCMP_DATA_RAM_DATA_WD	(CX_RAS_EN==1 ? [function_of: CC_OB_PDCMP_DATA_RAM_DATA_WD_NOECC CX_RAS_RAM_PROT_RANGE] + CC_OB_PDCMP_DATA_RAM_DATA_WD_NOECC : CC_OB_PDCMP_DATA_RAM_DATA_WD_NOECC)
CC_OB_PDCMP_DATA_RAM_DATA_WD_NOECC	CC_PRF_TLP_DATA_WIDTH_NP
CC_OB_PDCMP_DATA_RAM_DP	((SLAVE_BUS_DATA_WIDTH <= PCIE_CORE_DATA_BUS_WD) ? ((S_C_CDC_FIFO_DEPTH <= ((AMBA_DECOMPOSER_DEF_DEPTH_FACTOR*CC_SLV_MTU)/(CX_NW*4)+4+2)) ? ((AMBA_DECOMPOSER_DEF_DEPTH_FACTOR*CC_SLV_MTU)/(CX_NW*4)+4+2) : S_C_CDC_FIFO_DEPTH) : ((S_C_CDC_FIFO_DEPTH <= ((AMBA_DECOMPOSER_DEF_DEPTH_FACTOR*CC_SLV_MTU)/(CX_NW*4)+4+2)) ? ((AMBA_DECOMPOSER_DEF_DEPTH_FACTOR*CC_SLV_MTU)/(CX_NW*4)+4+2) : S_C_CDC_FIFO_DEPTH))
CC_OB_PDCMP_HDR_RAM_ADDRP_WD	(CX_RAS_EN==1 ? [function_of: CC_OB_PDCMP_HDR_RAM_ADDR_WD CX_RAS_PROT_RANGE] : 1)
CC_OB_PDCMP_HDR_RAM_ADDR_WD	[function_of: CC_OB_PDCMP_HDR_RAM_DP]
CC_OB_PDCMP_HDR_RAM_DATA_WD	(CX_RAS_EN==1 ? [function_of: CC_OB_PDCMP_HDR_RAM_DATA_WD_NOECC CX_RAS_RAM_PROT_RANGE] + CC_OB_PDCMP_HDR_RAM_DATA_WD_NOECC : CC_OB_PDCMP_HDR_RAM_DATA_WD_NOECC)
CC_OB_PDCMP_HDR_RAM_DATA_WD_NOECC	(CX_TAG_SIZE+CX_LN_ENABLE+CX_TLP_PREFIX_ENABLE*((32*CX_NPRFX)+ CX_PRFX_PAR_WD)+(CC_AXI_SLV_MISC_INFO_RID_ENABLE*16)+CX_TPH_ENABLE*(1+2+8)+CX_ATU_CTRL_EN+ATS_TX_ENABLE*(2+1)+CX_SRIOV_ENABLE*(CX_NVFUNC_WD-1+1)+AXI_MULTIFUNCTION_ENABLED*CX_NFUNC_WD+1+1+CC_SLV_BUS_ID_WIDTH+1+(CC_SLV_MTU_PW < (CX_MAX_MTU_PW+1) ? CC_SLV_MTU_PW : CX_MAX_MTU_PW+1)+3+1+1+2+CX_IDO_ENABLE+8+2+5+SLAVE_BUS_ADDR_WIDTH)
CC_OB_PDCMP_HDR_RAM_DP	S_C_CDC_FIFO_DEPTH

Parameter Name	Equals To
CC_PCIE_CORE_DATA_BUS_PAR_BITS	$CX_RAS_EN ?$ $(PCIE_CORE_DATA_BUS_WD / CX_RAS_PROT_RANGE) :$ 0
CC_PRF_TLP_DATA_WIDTH_NP	$CC_SLV_DATA_WD_LT_CORE_DATA_WD ?$ $PCIE_CORE_DATA_BUS_WD :$ $SLAVE_BUS_DATA_WIDTH$
CC_RADMX_COMPOSER_SEG_DP	$(SLAVE_BUS_DATA_WIDTH \leq$ $PCIE_CORE_DATA_BUS_WD) ?$ $((CC_RADMX_COMPOSER_MAX_SEG_BYTE_LEN / (4 * C$ $X_NW)) + 2) :$ $((CC_RADMX_COMPOSER_MAX_SEG_BYTE_LEN / (4 * C$ $C_SLV_NW)) + 2)$
CC_RC	1
CC_RESPONSE_MODE	0
CC_SHARED_DBI_ECAM_EN	$CX_SBSA_ECAM ? 1 : 0$
CC_SLV_AR_EXTRA_PYLD_PAR_WIDTH	$((CC_SLV_AR_EXTRA_PYLD_WIDTH - 1) /$ $CX_RAS_PROT_RANGE) + 1$
CC_SLV_AR_EXTRA_PYLD_WIDTH	$(SLV_MISC_INFO_PW +$ $CC_SLV_AR_MISC_INFO_WIDTH +$ $CC_ACELITE_ENABLE * 8)$
CC_SLV_AR_MISC_INFO_WIDTH	$(ATS_RX_ENABLE * 3 + CX_AMBA_ARI_OR_IOV *$ $CX_NFUNC_WD + CX_SRIOV_ENABLE *$ $CX_NVFUNC_WD + CX_TLP_PREFIX_ENABLE * (32 * CX_NPRFX)$ $+ (CC_AXI_SLV_MISC_INFO_RID_ENABLE * 16) +$ $CX_LN_ENABLE + CX_TPH_ENABLE * 11 +$ $CX_ATU_CTRL_EN + CX_IDO_ENABLE)$
CC_SLV_AR_USER_WIDTH	$CX_RASDP == 0 ? 0 : CX_AXI_IF_ECC_PAR_PROT_EN$ $? (4 + CC_SLV_BUS_ADDR_PROT_WIDTH +$ $CC_SLV_AR_EXTRA_PYLD_PAR_WIDTH +$ $CC_AXI4_ENABLE) :$ $CC_SLV_BUS_ADDR_PROT_WIDTH$
CC_SLV_ATOMIC_OR_DMWR_EN	$(CX_ATOMIC_ENABLE $ $CC_AXI_DEF_MEM_WR_REQ_EN)$
CC_SLV_AW_EXTRA_PYLD_PAR_WIDTH	$((CC_SLV_AW_EXTRA_PYLD_WIDTH - 1) /$ $CX_RAS_PROT_RANGE) + 1$
CC_SLV_AW_EXTRA_PYLD_WIDTH	$(SLV_MISC_INFO_PW +$ $CC_SLV_AW_MISC_INFO_WIDTH +$ $CX_ATOMIC_ENABLE * 6 + CC_ACELITE_ENABLE * 7)$

Parameter Name	Equals To
CC_SLV_AW_MISC_INFO_WIDTH	$(64 + CX_TAG_SIZE + ATS_RX_ENABLE * 3 + CX_AMBA_ARI_OR_IOV * CX_NFUNC_WD + CX_SRIOV_ENABLE * CX_NVFUNC_WD + CX_TLP_PREFIX_ENABLE * (32 * CX_NPRFX) + (CC_AXI_SLV_MISC_INFO_RID_ENABLE * 16) + CX_LN_ENABLE + CX_TPH_ENABLE * 11 + CX_ATU_CTRL_EN + CX_IDO_ENABLE)$
CC_SLV_AW_USER_WIDTH	$CX_RASDP == 0 ? 0 : CX_AXI_IF_ECC_PAR_PROT_EN ? (4 + CC_SLV_BUS_ADDR_PROT_WIDTH + CC_SLV_AW_EXTRA_PYLD_PAR_WIDTH + CC_AXI4_ENABLE) : CC_SLV_BUS_ADDR_PROT_WIDTH$
CC_SLV_B_EXTRA_PYLD_PAR_WIDTH	$((CC_SLV_B_EXTRA_PYLD_WIDTH - 1) / CX_RAS_PROT_RANGE) + 1$
CC_SLV_B_EXTRA_PYLD_WIDTH	$(SLV_RESP_MISC_INFO_PW + CC_SLV_B_MISC_INFO_WIDTH)$
CC_SLV_B_MISC_INFO_WIDTH	$(CC_SLV_RESP_TLP_PREFIX_ENABLE * (32 * CX_NPRFX) + CX_LN_ENABLE)$
CC_SLV_BURST_LEN_5BIT_INT	$((CC_SLV_RD_REQ_SIZE_5BIT * 8) / SLAVE_BUS_DATA_WIDTH)$
CC_SLV_BURST_LEN_CALC_PW	[function_of: CC_SLV_BURST_LEN]
CC_SLV_BURST_LEN_MIN_PW	$(AMBA_INTERFACE \geq 3 ? 8 : 4)$
CC_SLV_BURST_LEN_PW	$(CC_SLV_BURST_LEN_CALC_PW > CC_SLV_BURST_LEN_MIN_PW ? CC_SLV_BURST_LEN_CALC_PW : CC_SLV_BURST_LEN_MIN_PW)$
CC_SLV_BUS_ADDR_PROT_WIDTH	$SLAVE_BUS_ADDR_WIDTH / CX_RAS_PROT_RANGE$

Parameter Name	Equals To
CC_SLV_BUS_DATA_PROT_WIDTH	$(CX_AXI_IF_TYPE_OF_PROT < 2) ?$ $(SLAVE_BUS_DATA_WIDTH/CX_RAS_PROT_RANGE) : ($ $SLAVE_BUS_DATA_WIDTH == 32 ?$ $(CX_AXI_IF_ECC_GRAN == 0 ? 7 :$ $(CX_AXI_IF_ECC_GRAN == 1 ? 7 : 7)) :$ $SLAVE_BUS_DATA_WIDTH == 64 ?$ $(CX_AXI_IF_ECC_GRAN == 0 ? 8 :$ $(CX_AXI_IF_ECC_GRAN == 1 ? 14 : 8)) :$ $SLAVE_BUS_DATA_WIDTH == 128 ?$ $(CX_AXI_IF_ECC_GRAN == 0 ? 9 :$ $(CX_AXI_IF_ECC_GRAN == 1 ? 28 : 16)) :$ $SLAVE_BUS_DATA_WIDTH == 256 ?$ $(CX_AXI_IF_ECC_GRAN == 0 ? 10 :$ $(CX_AXI_IF_ECC_GRAN == 1 ? 56 : 32)) :$ $SLAVE_BUS_DATA_WIDTH == 512 ?$ $(CX_AXI_IF_ECC_GRAN == 0 ? 11 :$ $(CX_AXI_IF_ECC_GRAN == 1 ? 112 : 64)) : 0)$
CC_SLV_BUS_DATA_WIDTH	$(CX_RAS_EN == 1) ?$ $((SLAVE_BUS_DATA_WIDTH/CX_RAS_PROT_RANGE) +$ $SLAVE_BUS_DATA_WIDTH) : SLAVE_BUS_DATA_WIDTH$
CC_SLV_B_USER_WIDTH	$CX_AXI_IF_ECC_PAR_PROT_EN ? (2 +$ $CC_SLV_B_EXTRA_PYLD_PAR_WIDTH) : 0$
CC_SLV_BUS_NPDATA_NBYTES	$(CC_AXI_DEF_MEM_WR_REQ_EN ?$ $(CX_DEF_MEM_WR_LEN_SUPP + 1) * 64 :$ $CX_ATOMIC_ENABLE ? 32 : 4)$
CC_SLV_BUS_NPWSTRB_WIDTH	$(CC_AXI_DEF_MEM_WR_REQ_EN ?$ $CC_SLV_BUS_NPDATA_NBYTES : 4)$
CC_SLV_BUS_WSTRB_WIDTH	$(SLAVE_BUS_DATA_WIDTH / 8)$
CC_SLV_DATA_WD_LT_CORE_DATA_WD	$SLAVE_BUS_DATA_WIDTH <$ $PCIE_CORE_DATA_BUS_WD$
CC_SLV_INT_BURST_LEN_PW	$(CC_SLV_BURST_LEN_CALC_PW)$
CC_SLV_MTU_PW	$[function_of: [function_of: CC_SLV_MTU + 1]]$
CC_SLV_NPW_SAB_RAM_ADDRP_WD	$(CX_RAS_EN == 1 ? [function_of:$ $CC_SLV_NPW_SAB_RAM_ADDR_WD$ $CX_RAS_PROT_RANGE] : 1)$
CC_SLV_NPW_SAB_RAM_ADDR_WD	$[function_of: CC_SLV_NPW_SAB_RAM_DP]$
CC_SLV_NPW_SAB_RAM_DATA_WD	$(CX_RAS_EN == 1 ? [function_of:$ $CC_SLV_NPW_SAB_RAM_DATA_WD_NOECC$ $CX_RAS_RAM_PROT_RANGE] +$ $CC_SLV_NPW_SAB_RAM_DATA_WD_NOECC :$ $CC_SLV_NPW_SAB_RAM_DATA_WD_NOECC)$

Parameter Name	Equals To
CC_SLV_NPW_SAB_RAM_DATA_WD_NOECC	$ \begin{aligned} & (CX_LN_ENABLE + CC_SLV_BUS_ID_WIDTH + \\ & SLAVE_BUS_ADDR_WIDTH + \\ & CC_SLV_INT_BURST_LEN_PW + 8 + \\ & (CC_AXI_DEF_MEM_WR_REQ_EN ? \\ & (CX_DEF_MEM_WR_LEN_SUPP + 1) * 64 * 8 : \\ & CX_ATOMIC_ENABLE ? 256 : 32) + \\ & (CC_AXI_DEF_MEM_WR_REQ_EN ? \\ & CC_SLV_BUS_NPWSTRB_WIDTH : 4) + 7 + \\ & SHARED_DBI_ENABLED * 2 + \\ & CC_SHARED_DBI_ECAM_EN + CX_NFUNC_WD + \\ & CX_SRIOV_ENABLE * CX_NVFUNC_WD + \\ & CX_TLP_PREFIX_ENABLE * ((32 * CX_NPRFX) + CX_PRFX \\ & _PAR_WD) + \\ & (CC_AXI_SLV_MISC_INFO_RID_ENABLE * 16) + \\ & CX_ATU_CTRL_EN + \\ & CC_SLV_ATOMIC_OR_DMWR_EN * 2 + \\ & CC_SLV_ATOMIC_OR_DMWR_EN * (ATS_TX_ENABLE * 3) \\ & + \\ & CC_AXI_DEF_MEM_WR_REQ_EN * CC_SLV_WRAP_EN \\ & ABLE * 2 + \\ & CC_AXI_DEF_MEM_WR_REQ_EN * CX_IDO_ENABLE + \\ & CC_SLV_ATOMIC_OR_DMWR_EN * (CX_TPH_ENABLE * 1 \\ & 1) + CX_AXI_UIDI_ENABLE) \end{aligned} $
CC_SLV_NPW_SAB_RAM_DP	$ ((CX_IS_EP \&\& !CX_ATOMIC_ENABLE \&\& !CC_AXI_DEF_MEM_WR_REQ_EN) ? 2 : CC_SLV_NUM_OUTSTND_CPU_WR_REQ) $
CC_SLV_NUM_MASTERS	4
CC_SLV_NW	(SLAVE_BUS_DATA_WIDTH/32)
CC_SLV_RD_REQ_SIZE	$ (((CC_SLV_BURST_LEN * SLAVE_BUS_DATA_WIDTH / 8) > 4096) ? 4096 : (CC_SLV_BURST_LEN * SLAVE_BUS_DATA_WIDTH / 8)) $
CC_SLV_RD_REQ_SIZE_5BIT	$ \begin{aligned} & (CC_SLV_RD_REQ_SIZE_5BIT_INT \geq 4096) ? 4096 : \\ & (CC_SLV_RD_REQ_SIZE_5BIT_INT \geq 2048) ? 2048 : \\ & (CC_SLV_RD_REQ_SIZE_5BIT_INT \geq 1024) ? 1024 : \\ & (CC_SLV_RD_REQ_SIZE_5BIT_INT \geq 512) ? 512 : \\ & (CC_SLV_RD_REQ_SIZE_5BIT_INT \geq 256) ? 256 : \\ & (CC_SLV_RD_REQ_SIZE_5BIT_INT \geq 128) ? 128 : \\ & (CC_SLV_RD_REQ_SIZE_5BIT_INT \geq 64) ? 64 : \\ & (CC_SLV_RD_REQ_SIZE_5BIT_INT \geq 32) ? 32 : \\ & (CC_SLV_RD_REQ_SIZE_5BIT_INT \geq 16) ? 16 : \\ & (CC_SLV_RD_REQ_SIZE_5BIT_INT \geq 8) ? 8 : 4 \end{aligned} $

Parameter Name	Equals To
CC_SLV_RD_REQ_SIZE_5BIT_INT	$((1 + CX_MAX_TAG) - ((EXT_TAG_SUPPORTED \parallel CC_DMA_ENABLE) ? (1 + CX_MAX_TAG) - (CC_DMA_ENABLE ? (CC_MAX_NON_DMA_5BIT_TAG_HIGH + 1) : 32) + 1 : 0)) * (128 < CC_RADMX_COMPOSER_MAX_SEG_BYTE_LEN ? 128 : CC_RADMX_COMPOSER_MAX_SEG_BYTE_LEN))$
CC_SLV_RD_REQ_SIZE_PW	[function_of: [function_of: CC_SLV_RD_REQ_SIZE + 1]]
CC_SLV_RESP_TLP_PREFIX_ENABLE	0
CC_SLV_R_EXTRA_PYLD_PAR_WIDTH	$((CC_SLV_R_EXTRA_PYLD_WIDTH - 1) / CX_RAS_PROT_RANGE) + 1$
CC_SLV_R_EXTRA_PYLD_WIDTH	$(SLV_RESP_MISC_INFO_PW + CC_SLV_R_MISC_INFO_WIDTH)$
CC_SLV_R_MISC_INFO_WIDTH	$(CC_SLV_RESP_TLP_PREFIX_ENABLE * (32 * CX_NPRFX) + CX_LN_ENABLE + CX_IDO_ENABLE)$
CC_SLV_R_USER_WIDTH	$CX_RASDP == 0 ? 0 : CX_AXI_IF_ECC_PAR_PROT_EN ? (2 + CC_SLV_BUS_DATA_PROT_WIDTH + CC_SLV_R_EXTRA_PYLD_PAR_WIDTH) : CC_SLV_BUS_DATA_PROT_WIDTH$
CC_SLV_W_EXTRA_PYLD_PAR_WIDTH	$((CC_SLV_W_EXTRA_PYLD_WIDTH - 1) / CX_RAS_PROT_RANGE) + 1$
CC_SLV_W_EXTRA_PYLD_WIDTH	CC_SLV_W_MISC_INFO_WIDTH
CC_SLV_W_MISC_INFO_WIDTH	2
CC_SLV_W_USER_WIDTH	$CX_RASDP == 0 ? 0 : CX_AXI_IF_ECC_PAR_PROT_EN ? (2 + CC_AXI3_ENABLE + CC_SLV_BUS_DATA_PROT_WIDTH + CC_SLV_W_EXTRA_PYLD_PAR_WIDTH) : CC_SLV_BUS_DATA_PROT_WIDTH$
CC_SW	3
CC_UNROLL_EN	$(CC_DMA_ENABLE \parallel CX_INTERNAL_ATU_ENABLE) \&\& !AHB_POPULATED$
CC_UNROLL_ENABLE	$((CC_UNROLL_EN == 0) ? 0 : 1)$
CC_VSECDMA_MAP_VERSION	$((CX_HDMA_ENABLE_VALUE << 2) \mid (!CC_LEGACY_DMA_MAP_VALUE << 1) \mid (ENABLE_MEM_MAP_UNROLL_DMA_REG << 0))$
CC_WR_CH_P_REQ_QUEUE_DP	$(CC_HP_MASTER == 1) ? (RADM_MAX_OUTSTD_P_REQ / 2) : RADM_MAX_OUTSTD_P_REQ$

Parameter Name	Equals To
CLIENT1_POPULATED	1
CLIENT_HDR_PROT_WD	(CX_RASDP_EN==1)? [function_of: CLIENT_HDR_WD CX_RAS_PROT_RANGE RAS_BASE_PROT_WD]:0
CLIENT_HDR_WD	132 + CX_TAG_SIZE + CX_IDO_ENABLE_VALUE + CX_NFUNC_WD + CX_LN_VALUE + ((CX_NVFUNC_EWD+1)*CX_SRIOV_ENABLE_VALUE) + (CX_TPH_ENABLE_VALUE*11)
C_M_CDC_FIFO_DEPTH	14
CONTROL	1
CPL_LUT_DEPTH	(CX_MAX_TAG+1)
CPL_LUT_PTR_WD	CX_LUT_PTR_WIDTH
CPL_LUT_VFINDEX_REGOUT	VFINDEX_CALC_REGOUT
CUT_THROUGH_INVOLVED	(RADM_P_QMODE_VC0==2 RADM_NP_QMODE_VC0==2 RADM_CPL_QMODE_VC0==2 RADM_P_QMODE_VC1==2 RADM_NP_QMODE_VC1==2 RADM_CPL_QMODE_VC1==2 RADM_P_QMODE_VC2==2 RADM_NP_QMODE_VC2==2 RADM_CPL_QMODE_VC2==2 RADM_P_QMODE_VC3==2 RADM_NP_QMODE_VC3==2 RADM_CPL_QMODE_VC3==2 RADM_P_QMODE_VC4==2 RADM_NP_QMODE_VC4==2 RADM_CPL_QMODE_VC4==2 RADM_P_QMODE_VC5==2 RADM_NP_QMODE_VC5==2 RADM_CPL_QMODE_VC5==2 RADM_P_QMODE_VC6==2 RADM_NP_QMODE_VC6==2 RADM_CPL_QMODE_VC6==2 RADM_P_QMODE_VC7==2 RADM_NP_QMODE_VC7==2 RADM_CPL_QMODE_VC7==2)
CX_10BITS_TAG_REQ	(CX_10BITS_TAG== 1 && CX_MAX_TAG > 255)? 1: 0
CX_10BITS_TAG_REQ_VALUE	(CX_10BITS_TAG_REQ == 1)? 1: 0
CX_10BITS_TAG_VALUE	(CX_10BITS_TAG == 1)? 1: 0
CX_2ND_SPEED	(CX_GEN2_SPEED)
CX_5GTS_SPEED	(CX_CPCIE_ENABLE && CX_GEN2_SPEED)

Parameter Name	Equals To
CX_ACS_FUNC_GRP_VALUE	CX_ACS_FUNC_GRP
CX_ADM_ADAPTOR_ENABLE	(CX_PL_MODE == 1)
CX_ADM_ADAPTOR_RADM_PIPE_STAGES	16
CX_ADM_RFC_WD	CX_RX_NDLLP + (32*CX_RX_NDLLP) + CX_NVC
CX_ADM_TLH_WD	10 + CX_TX_HW_W_PAR + CX_DW_W_PAR + CX_NW + ((CX_TLP_PREFIX_ENABLE == 1) ? (32*CX_NPRFX + 15) : 0) + ((CX_P2P_ENABLE_VALUE == 1) ? 1 : 0)
CX_AMBA_ARI_OR_IOV	(AMBA_INTERFACE != 0 && (CX_SRIOV_ENABLE CX_ARI_ENABLE))
CX_APP_L1SUB_CONTROL	(CX_L1_SUBSTATES_ENABLE == 1) ? 1 : 0
CX_APP_NPD_CONTROL	0
CX_ATU_CTRL_EN	CX_INTERNAL_ATU_ENABLE
CX_ATU_INCR_RGN_SIZE	((AMBA_INTERFACE != 0) ? ((CX_ATU_MAX_REGION_SIZE > 0) && AMBA_INTERFACE != 1 && SLAVE_BUS_ADDR_WIDTH > 32) : CX_ATU_MAX_REGION _SIZE > 0)
CX_ATU_SLOC_CBUF	CX_ATU_SLOC_CBUF_ENABLE
CX_ATU_SLOC_CBUF_ENABLE	ATU_IN_SINGLE_TRGT_ADDR_ENABLE
CX_AXI_IF_ECC_PAR_PROT_EN	((CX_AXI_IF_TYPE_OF_PROT != 0) && CX_RASDP_EN)
CX_AXI_IF_ECC_PAR_PROT_EN_VALUE	CX_AXI_IF_ECC_PAR_PROT_EN
CX_BAR0_RESIZABLE_0	(BAR0_SIZING_SCHEME_0 == 2) ? 1 : 0
CX_BAR1_RESIZABLE_0	(BAR1_SIZING_SCHEME_0 == 2) ? 1 : 0
CX_BAR2_RESIZABLE_0	(BAR2_SIZING_SCHEME_0 == 2) ? 1 : 0
CX_BAR3_RESIZABLE_0	(BAR3_SIZING_SCHEME_0 == 2) ? 1 : 0
CX_BAR4_RESIZABLE_0	(BAR4_SIZING_SCHEME_0 == 2) ? 1 : 0
CX_BAR5_RESIZABLE_0	(BAR5_SIZING_SCHEME_0 == 2) ? 1 : 0
CX_BUSNUM_WD	(MULTI_DEVICE_AND_BUS_PER_FUNC_EN == 0) ? 8 : (CX_NFUNC*8)

Parameter Name	Equals To
CX_CALC_DDEPTH	(CX_ADM_ADAPTOR_ENABLE==1) ? [function_of: CX_NW CX_MAX_MTU CX_ADM_ADAPTOR_RADM_PIPE_STAGES] : ([function_of: CX_DL_NB CX_NL CX_MAX_MTU !CX_ECRC_STRIP_ENABLE CX_INTERNAL_DELAY CX_GEN5_SPEED CX_GEN4_SPEED CX_GEN3_SPEED CX_2ND_SPEED FC_SCALE_EN CX_RETIMER_LATENCY])
CX_CALC_HDEPTH	(CX_ADM_ADAPTOR_ENABLE==1) ? [function_of: CX_NW CX_ADM_ADAPTOR_RADM_PIPE_STAGES] : ([function_of: CX_DL_NB CX_NL CX_MAX_MTU !CX_ECRC_STRIP_ENABLE CX_INTERNAL_DELAY CX_RADM_MAXPKT CX_GEN5_SPEED CX_GEN4_SPEED CX_GEN3_SPEED CX_2ND_SPEED FC_SCALE_EN CX_RETIMER_LATENCY])
CX_CCIX_DATA_WD	DW_W_PAR
CX_CCIX_HDR_WD	TX_HW_W_PAR
CX_CCIX_RX_WD	(CX_CCIX_INTERFACE_ENABLE) ? 1 : 0
CX_CCIX_VDM_VID	7724
CX_CLIENT_PAR_MODE	(CX_RAS_EN==1) ? 8 : 0
CX_COMM_CLK_EN	((CX_NFTS==CX_COMM_NFTS) && (DEFAULT_L0S_EXIT_LATENCY==DEFAULT_COMM_L0S_EXIT_LATENCY) && (DEFAULT_L1_EXIT_LATENCY==DEFAULT_COMM_L1_EXIT_LATENCY)) ? 0 : 1
CX_CPCIE_ENABLE	(CX_S_CPCIE_MODE)
CX_CPCIE_INTERNAL_DELAY	CX_PHY_TX_DELAY_MAC + CX_PHY_TX_DELAY_PHY + CX_PHY_TX_DELAY_SERDES_PIPE + CX_PHY_RX_DELAY_MAC + CX_PHY_RX_DELAY_PHY + CX_PHY_RX_DELAY_SERDES_PIPE
CX_CXL_2_0_ENABLE	(CX_CXL_ENABLE && (CX_CXL_VERSION==2))
CX_CXL_AP_ENABLE	(CX_CXL_CXS_ENABLE && CX_CXL_CCIX_ENABLE)
CX_CXL_CACHE_D2H_DATA_CHK_SIZE	(int(((CX_CXL_CACHE_D2H_DATA_SIZE - 1)*1.0/64 + 1) * 8)
CX_CXL_CACHE_D2H_DATA_SIZE	(CX_CXL_CACHE_D2H_DHDR_SIZE + CX_CXL_CACHE_D2H_DDATA_SIZE + CX_CXL_CACHE_D2H_DBE_SIZE)
CX_CXL_CACHE_D2H_DBE_SIZE	64

Parameter Name	Equals To
CX_CXL_CACHE_D2H_DDATA_CHK_SIZE	$(\text{int}(((\text{CX_CXL_CACHE_D2H_DDATA_SIZE} - 1) * 1.0 / 64 + 1) * 8))$
CX_CXL_CACHE_D2H_DDATA_SIZE	512
CX_CXL_CACHE_D2H_DHDR_SIZE	17
CX_CXL_CACHE_D2H_REQ_CHK_SIZE	$(\text{int}(((\text{CX_CXL_CACHE_D2H_REQ_SIZE} - 1) * 1.0 / 64 + 1) * 8))$
CX_CXL_CACHE_D2H_REQ_SIZE	79
CX_CXL_CACHE_D2H_RESP_CHK_SIZE	$(\text{int}(((\text{CX_CXL_CACHE_D2H_RESP_SIZE} - 1) * 1.0 / 64 + 1) * 8))$
CX_CXL_CACHE_D2H_RESP_SIZE	20
CX_CXL_CACHE_H2D_DATA_CHK_SIZE	$(\text{int}(((\text{CX_CXL_CACHE_H2D_DATA_SIZE} - 1) * 1.0 / 64 + 1) * 8))$
CX_CXL_CACHE_H2D_DATA_SIZE	$(\text{CX_CXL_CACHE_H2D_DHDR_SIZE} + \text{CX_CXL_CACHE_H2D_DDATA_SIZE})$
CX_CXL_CACHE_H2D_DDATA_CHK_SIZE	$(\text{int}(((\text{CX_CXL_CACHE_H2D_DDATA_SIZE} - 1) * 1.0 / 64 + 1) * 8))$
CX_CXL_CACHE_H2D_DDATA_SIZE	512
CX_CXL_CACHE_H2D_DHDR_SIZE	24
CX_CXL_CACHE_H2D_REQ_CHK_SIZE	$(\text{int}(((\text{CX_CXL_CACHE_H2D_REQ_SIZE} - 1) * 1.0 / 64 + 1) * 8))$
CX_CXL_CACHE_H2D_REQ_SIZE	64
CX_CXL_CACHE_H2D_RESP_CHK_SIZE	$(\text{int}(((\text{CX_CXL_CACHE_H2D_RESP_SIZE} - 1) * 1.0 / 64 + 1) * 8))$
CX_CXL_CACHE_H2D_RESP_PTR_DP_SIZE	1
CX_CXL_CACHE_H2D_RESP_PTR_SIZE	$((6 + \text{CX_CXL_CACHE_RSP_DP}) + 1)$
CX_CXL_CACHE_H2D_RESP_SIZE	32
CX_CXL_CACHE_MAX_DATA_NUM	$(\text{CX_CXL_IS_RC} ? 4 : 4)$
CX_CXL_CHI_DATACHK_EN	$(\text{CX_CXL_CHI_DATACHK} > 0) ? \text{CX_CXL_ENABLE} : 0$
CX_CXL_CHI_ENABLE	$(\text{CX_CXL_ENABLE} \&\& !\text{CX_CXL_CXS_ENABLE})$
CX_CXL_IDE_DP_ENABLE	0
CX_CXL_IDE_ENABLE	0
CX_CXL_IS_EP	$(\text{CX_IS_EP} \parallel \text{CX_IS_DM} \parallel \text{CX_IS_SW})$

Parameter Name	Equals To
CX_CXL_IS_RC	(CX_IS_RC CX_IS_DM CX_IS_SW)
CX_CXL_MEM_M2S_DATA_CHK_SIZE	$(\text{int}((\text{CX_CXL_MEM_M2S_DATA_SIZE} - 1) * 1.0 / 64 + 1) * 8)$
CX_CXL_MEM_M2S_DATA_SIZE	$(\text{CX_CXL_MEM_M2S_DHDR_SIZE} + \text{CX_CXL_MEM_M2S_DDATA_SIZE} + \text{CX_CXL_MEM_M2S_DBE_SIZE})$
CX_CXL_MEM_M2S_DBE_SIZE	64
CX_CXL_MEM_M2S_DDATA_CHK_SIZE	$(\text{int}((\text{CX_CXL_MEM_M2S_DDATA_SIZE} - 1) * 1.0 / 64 + 1) * 8)$
CX_CXL_MEM_M2S_DDATA_SIZE	512
CX_CXL_MEM_M2S_DHDR_SIZE	87
CX_CXL_MEM_M2S_NDR_CHK_SIZE	$(\text{int}((\text{CX_CXL_MEM_M2S_NDR_SIZE} - 1) * 1.0 / 64 + 1) * 8)$
CX_CXL_MEM_M2S_NDR_SIZE	87
CX_CXL_MEM_MAX_DATA_NUM	$(\text{CX_CXL_IS_EP} ? 4 : 1)$
CX_CXL_MEM_REQ_DP	CX_CXL_MEM_NDR_DP
CX_CXL_MEM_RSP_DP	CX_CXL_MEM_NDR_DP
CX_CXL_MEM_S2M_DATA_CHK_SIZE	$(\text{int}((\text{CX_CXL_MEM_S2M_DATA_SIZE} - 1) * 1.0 / 64 + 1) * 8)$
CX_CXL_MEM_S2M_DATA_SIZE	$(\text{CX_CXL_MEM_S2M_DHDR_SIZE} + \text{CX_CXL_MEM_S2M_DDATA_SIZE})$
CX_CXL_MEM_S2M_DDATA_CHK_SIZE	$(\text{int}((\text{CX_CXL_MEM_S2M_DDATA_SIZE} - 1) * 1.0 / 64 + 1) * 8)$
CX_CXL_MEM_S2M_DDATA_SIZE	512
CX_CXL_MEM_S2M_DHDR_SIZE	40
CX_CXL_MEM_S2M_NDR_CHK_SIZE	$(\text{int}((\text{CX_CXL_MEM_S2M_NDR_SIZE} - 1) * 1.0 / 64 + 1) * 8)$
CX_CXL_MEM_S2M_NDR_SIZE	$\text{CX_CXL_2_0_ENABLE} ? 30 : 28$
CX_CXL_MST_D2H_DATA_RAM_ADDRW	$(\text{CX_CXL_IS_RC} ? (6 + \text{CX_CXL_CACHE_DQ_DP}) : 1)$
CX_CXL_MST_D2H_DATA_RAM_DATAW	$(\text{CX_CXL_IS_RC} ? (((\text{CX_CXL_CHI_DATACHK_EN}) ? (\text{CX_CXL_CACHE_D2H_DATA_CHK_SIZE}) : 0) + \text{CX_CXL_CACHE_D2H_DATA_SIZE}) : 1)$
CX_CXL_MST_D2H_REQ_RAM_ADDRW	$(\text{CX_CXL_IS_RC} ? ((6 + \text{CX_CXL_CACHE_REQ_DP}) - 2) : 1)$
CX_CXL_MST_D2H_REQ_RAM_DATAW	$(\text{CX_CXL_IS_RC} ? (((\text{CX_CXL_CHI_DATACHK_EN}) ? (\text{CX_CXL_CACHE_D2H_REQ_CHK_SIZE}) : 0) + \text{CX_CXL_CACHE_D2H_REQ_SIZE}) : 1)$

Parameter Name	Equals To
CX_CXL_MST_D2H_RESP_RAM_ADDRW	$(CX_CXL_IS_RC ? ((6 + CX_CXL_CACHE_RSP_DP) - 1) : 1)$
CX_CXL_MST_D2H_RESP_RAM_DATAW	$(CX_CXL_IS_RC ? (((CX_CXL_CHI_DATACHK_EN) ? (CX_CXL_CACHE_D2H_RESP_CHK_SIZE) : 0) + CX_CXL_CACHE_D2H_RESP_SIZE) : 1)$
CX_CXL_MST_H2D_DATA_RAM_ADDRW	$(CX_CXL_IS_EP ? (6 + CX_CXL_CACHE_DQ_DP) : 1)$
CX_CXL_MST_H2D_DATA_RAM_DATAW	$(CX_CXL_IS_EP ? (((CX_CXL_CHI_DATACHK_EN) ? (CX_CXL_CACHE_H2D_DATA_CHK_SIZE) : 0) + CX_CXL_CACHE_H2D_DATA_SIZE) : 1)$
CX_CXL_MST_H2D_REQ_RAM_ADDRW	$(CX_CXL_IS_EP ? ((6 + CX_CXL_CACHE_REQ_DP) - 1) : 1)$
CX_CXL_MST_H2D_REQ_RAM_DATAW	$(CX_CXL_IS_EP ? (((CX_CXL_CHI_DATACHK_EN) ? (CX_CXL_CACHE_H2D_REQ_CHK_SIZE) : 0) + (CX_CXL_CACHE_H2D_RESP_PTR_DP_SIZE + CX_CXL_CACHE_H2D_RESP_PTR_SIZE + CX_CXL_CACHE_H2D_REQ_SIZE)) : 1)$
CX_CXL_MST_H2D_RESP_RAM_ADDRW	$(CX_CXL_IS_EP ? ((6 + CX_CXL_CACHE_RSP_DP) - 2) : 1)$
CX_CXL_MST_H2D_RESP_RAM_DATAW	$(CX_CXL_IS_EP ? (((CX_CXL_CHI_DATACHK_EN) ? (CX_CXL_CACHE_H2D_RESP_CHK_SIZE) : 0) + CX_CXL_CACHE_H2D_RESP_SIZE) : 1)$
CX_CXL_MST_HD_DATA_RAM_ADDRW	$((CX_CXL_MST_H2D_DATA_RAM_ADDRW > CX_CXL_MST_D2H_DATA_RAM_ADDRW) ? CX_CXL_MST_H2D_DATA_RAM_ADDRW : CX_CXL_MST_D2H_DATA_RAM_ADDRW)$
CX_CXL_MST_HD_DATA_RAM_DATAW	$((CX_CXL_MST_H2D_DATA_RAM_DATAW > CX_CXL_MST_D2H_DATA_RAM_DATAW) ? CX_CXL_MST_H2D_DATA_RAM_DATAW : CX_CXL_MST_D2H_DATA_RAM_DATAW)$
CX_CXL_MST_HREQ_DRESP_RAM_ADDRW	$((CX_CXL_MST_H2D_REQ_RAM_ADDRW > CX_CXL_MST_D2H_RESP_RAM_ADDRW) ? CX_CXL_MST_H2D_REQ_RAM_ADDRW : CX_CXL_MST_D2H_RESP_RAM_ADDRW)$
CX_CXL_MST_HREQ_DRESP_RAM_DATAW	$((CX_CXL_MST_H2D_REQ_RAM_DATAW > CX_CXL_MST_D2H_RESP_RAM_DATAW) ? CX_CXL_MST_H2D_REQ_RAM_DATAW : CX_CXL_MST_D2H_RESP_RAM_DATAW)$
CX_CXL_MST_HRESP_DREQ_RAM_ADDRW	$((CX_CXL_MST_H2D_RESP_RAM_ADDRW > CX_CXL_MST_D2H_REQ_RAM_ADDRW) ? CX_CXL_MST_H2D_RESP_RAM_ADDRW : CX_CXL_MST_D2H_REQ_RAM_ADDRW)$

Parameter Name	Equals To
CX_CXL_MST_HRESP_DREQ_RAM_DATAW	$((CX_CXL_MST_H2D_RESP_RAM_DATAW > CX_CXL_MST_D2H_REQ_RAM_DATAW) ? CX_CXL_MST_H2D_RESP_RAM_DATAW : CX_CXL_MST_D2H_REQ_RAM_DATAW)$
CX_CXL_MST_M2S_DATA_RAM_ADDRW	$(CX_CXL_IS_EP ? (6 + CX_CXL_MEM_DQ_DP) : 1)$
CX_CXL_MST_M2S_DATA_RAM_DATAW	$(CX_CXL_IS_EP ? (((CX_CXL_CHI_DATACHK_EN) ? (CX_CXL_MEM_M2S_DATA_CHK_SIZE) : 0) + (CX_CXL_MEM_M2S_DATA_SIZE)) : 1)$
CX_CXL_MST_M2S_REQ_RAM_ADDRW	$(CX_CXL_IS_EP ? ((6 + CX_CXL_MEM_REQ_DP) - 1) : 1)$
CX_CXL_MST_M2S_REQ_RAM_DATAW	$(CX_CXL_IS_EP ? (((CX_CXL_CHI_DATACHK_EN) ? (CX_CXL_MEM_M2S_NDR_CHK_SIZE) : 0) + (CX_CXL_MEM_M2S_NDR_SIZE)) : 1)$
CX_CXL_MST_MS_DATA_RAM_ADDRW	$((CX_CXL_MST_M2S_DATA_RAM_ADDRW > CX_CXL_MST_S2M_DATA_RAM_ADDRW) ? CX_CXL_MST_M2S_DATA_RAM_ADDRW : CX_CXL_MST_S2M_DATA_RAM_ADDRW)$
CX_CXL_MST_MS_DATA_RAM_DATAW	$((CX_CXL_MST_M2S_DATA_RAM_DATAW > CX_CXL_MST_S2M_DATA_RAM_DATAW) ? CX_CXL_MST_M2S_DATA_RAM_DATAW : CX_CXL_MST_S2M_DATA_RAM_DATAW)$
CX_CXL_MST_MS_NDR_RAM_ADDRW	$((CX_CXL_MST_M2S_REQ_RAM_ADDRW > CX_CXL_MST_S2M_RSP_RAM_ADDRW) ? CX_CXL_MST_M2S_REQ_RAM_ADDRW : CX_CXL_MST_S2M_RSP_RAM_ADDRW)$
CX_CXL_MST_MS_NDR_RAM_DATAW	$((CX_CXL_MST_M2S_REQ_RAM_DATAW > CX_CXL_MST_S2M_RSP_RAM_DATAW) ? CX_CXL_MST_M2S_REQ_RAM_DATAW : CX_CXL_MST_S2M_RSP_RAM_DATAW)$
CX_CXL_MST_S2M_DATA_RAM_ADDRW	$(CX_CXL_IS_RC ? (6 + CX_CXL_MEM_DQ_DP) : 1)$
CX_CXL_MST_S2M_DATA_RAM_DATAW	$(CX_CXL_IS_RC ? (((CX_CXL_CHI_DATACHK_EN) ? (CX_CXL_MEM_S2M_DATA_CHK_SIZE) : 0) + CX_CXL_MEM_S2M_DATA_SIZE) : 1)$
CX_CXL_MST_S2M_RSP_RAM_ADDRW	$(CX_CXL_IS_RC ? ((6 + CX_CXL_MEM_RSP_DP) - 1) : 1)$
CX_CXL_MST_S2M_RSP_RAM_DATAW	$(CX_CXL_IS_RC ? (((CX_CXL_CHI_DATACHK_EN) ? (CX_CXL_MEM_S2M_NDR_CHK_SIZE) : 0) + CX_CXL_MEM_S2M_NDR_SIZE) : 1)$
CX_CXL_PM_INIT_CMPL_RPT_EN	$(CX_CXL_2_0_ENABLE)$

Parameter Name	Equals To
CX_CXL_RAS_HDR_LOG_SIZE	$(CX_CXL_MEM_EN ? (CX_CXL_IS_EP ? (CX_CXL_MEM_M2S_DHDR_SIZE + 1) : CX_CXL_MEM_S2M_DHDR_SIZE) : (CX_CXL_IS_EP ? CX_CXL_CACHE_H2D_DHDR_SIZE : CX_CXL_CACHE_D2H_DHDR_SIZE))$
CX_CXL_RBUF_RAM_ADDRW	$(CX_CXL_RBUF_WD)$
CX_CXL_RBUF_RAM_DATAW	528
CX_CXL_RBUF_WD	$([function_of: CX_CXL_RBUF_DEPTH])$
CX_CXL_SLV_HD_DATA_FIFO_DEPTH	$(CX_CXL_CHI_MAXCREDITS + 2 * CX_CXL_CACHE_MAX_DATA_NUM)$
CX_CXL_SLV_HD_DATA_FIFO_DEPTH_1	$(CX_CXL_SLV_HD_DATA_FIFO_DEPTH - 1)$
CX_CXL_SLV_HD_DATA_RAM_ADDRW	$([function_of: CX_CXL_SLV_HD_DATA_FIFO_DEPTH_1])$
CX_CXL_SLV_HD_DATA_RAM_DATAW	$(CX_CXL_IS_RC ? (CX_CXL_CACHE_H2D_DDATA_SIZE + (CX_CXL_CHI_DATACHK_EN ? CX_CXL_CACHE_H2D_DDATA_CHK_SIZE : 0)) : (CX_CXL_CACHE_D2H_DDATA_SIZE + (CX_CXL_CHI_DATACHK_EN ? CX_CXL_CACHE_D2H_DDATA_CHK_SIZE : 0)))$
CX_CXL_SLV_MS_DATA_FIFO_DEPTH	$(CX_CXL_CHI_MAXCREDITS + 2 * CX_CXL_MEM_MAX_DATA_NUM)$
CX_CXL_SLV_MS_DATA_FIFO_DEPTH_1	$(CX_CXL_SLV_MS_DATA_FIFO_DEPTH - 1)$
CX_CXL_SLV_MS_DATA_RAM_ADDRW	$([function_of: CX_CXL_SLV_MS_DATA_FIFO_DEPTH_1])$
CX_CXL_SLV_MS_DATA_RAM_DATAW	$(CX_CXL_IS_RC ? (CX_CXL_MEM_M2S_DDATA_SIZE + (CX_CXL_CHI_DATACHK_EN ? CX_CXL_MEM_M2S_DDATA_CHK_SIZE : 0)) : (CX_CXL_MEM_S2M_DDATA_SIZE + (CX_CXL_CHI_DATACHK_EN ? CX_CXL_MEM_S2M_DDATA_CHK_SIZE : 0)))$
CX_CXL_VIRAL_CAPABLE	1
CX_CXPL_EN	$(CX_PL_MODE == 0)$
CX_CXS_CCIXDATAWIDTH	$((CX_NW == 16) ? 512 : (CX_NW == 8) ? 256 : 128)$
CX_CXS_CNTLCHKWIDTH	$((CX_CXS_RX_DATACHECK == 2) ? 8 : ((CX_CXS_RX_DATACHECK == 1) ? 1 : ((CX_CXS_TX_DATACHECK == 2) ? 8 : 1)))$
CX_CXS_CNTLWIDTH	$(CX_CXS_MAXPKTPERFLIT * (3 + ([function_of: CX_CXS_DATAFLITWIDTH] - 5) + ([function_of: CX_CXS_DATAFLITWIDTH] - 7)))$
CX_CXS_DATACHKWIDTH	$(CX_CXS_DATAFLITWIDTH / 8)$

Parameter Name	Equals To
CX_CXS_LAST_ENABLE	(CX_CXL_CXS_ENABLE)
CX_CXS_PRCL_ENABLE	(CX_CXS_ENABLE && CX_CCIX_INTERFACE_ENABLE && CX_CXL_CXS_ENABLE)
CX_CXS_RX_BUFF_ADDRW	([function_of: CX_CXS_RX_BUFF_DEPTH])
CX_CXS_RX_BUFF_DATAW	(CX_CXS_LAST_ENABLE + 3*CX_CXS_PRCL_ENABLE + (CXS_FLIT_LOWER_DATA_WIDTH ? (CX_CXS_CCIXDATAWIDTH + CX_CXS_RX_CCIXCNTLWIDTH + (CX_RAS_EN ? (CX_CXS_CCIXDATAWIDTH/8 + 8) : 0)) : (CX_CXS_DATAFLITWIDTH + CX_CXS_CNTLWIDTH + (CX_RAS_EN ? (CX_CXS_DATACHKWIDTH + 8) : 0)))
CX_CXS_RX_BUFF_DEPTH	(CX_CXS_RX_BUFF_FIFO_DEPTH + CX_CXS_RX_MEM_DEPTH_ADJ)
CX_CXS_RX_BUFF_FIFO_DEPTH	(CX_CXS_RX_CREDITS + 1 + (((CX_CXS_RX_CONTINUOUSDATA && (CX_S_FLIT_EQUALS_DATA_WIDTH==1)) ? 0 : (4096/(CX_S_FLIT_LOWER_DATA_WIDTH ? CX_CXS_CCIXDATAWIDTH : CX_CXS_DATAFLITWIDTH))))
CX_CXS_RX_CCIXCNTLWIDTH	((CX_CXS_CCIXDATAWIDTH/128)*(3+([function_of: CX_CXS_CCIXDATAWIDTH]-5)+([function_of: CX_CXS_CCIXDATAWIDTH]-7)))
CX_CXS_RX_MEM_DEPTH_ADJ	([function_of: CX_CXS_RX_BUFF_FIFO_DEPTH] ? 0 : (CX_CXS_RX_BUFF_FIFO_DEPTH % 2) ? 1 : 2
CX_CXS_TX_BUFF_ADDRW	([function_of: CX_CXS_TX_BUFF_DEPTH])
CX_CXS_TX_BUFF_DATAW	(CX_CXS_LAST_ENABLE + 3*CX_CXS_PRCL_ENABLE + (CX_S_FLIT_LOWER_DATA_WIDTH ? (CX_CXS_CCIXDATAWIDTH + CX_CXS_TX_CCIXCNTLWIDTH + (CX_RAS_EN ? (CX_CXS_CCIXDATAWIDTH/8 + 8) : 0)) : (CX_CXS_DATAFLITWIDTH + CX_CXS_CNTLWIDTH + (CX_RAS_EN ? (CX_CXS_DATACHKWIDTH + 8) : 0)))
CX_CXS_TX_BUFF_DEPTH	(CX_CCIX_INTERFACE_ENABLE ? (4*(4096/(CX_S_FLIT_LOWER_DATA_WIDTH ? CX_CXS_CCIXDATAWIDTH : CX_CXS_DATAFLITWIDTH))) : 16)
CX_CXS_TX_CCIXCNTLWIDTH	((CX_S_FLIT_EQUALS_DATA_WIDTH ? CX_CXS_MAXPKTPERFLIT : 1)*(3+([function_of: CX_CXS_CCIXDATAWIDTH]-5)+([function_of: CX_CXS_CCIXDATAWIDTH]-7)))

Parameter Name	Equals To
CX_DBI_ISO_EN	(DBISLV_CLK_DIFF_ENABLE==1 && CX_ENHANCED_PM_EN==1)
CX_DEFAULT_EQ_BYPASS_HIGHEST_RATE_SUPPORT	CX_SKIP_LOWER_RATE_EQ
CX_DEFAULT_GEN5_TX_MOD_CMPL_PATTERN_FOR_LOOPBACK	CX_DEFAULT_GEN5_EQ_FOR_LOOPBACK
CX_DEFAULT_LANE_UNDER_TEST	0x0
CX_DEFAULT_MOD_TS_PCIE_SUPPORT	0x1
CX_DEFAULT_NO_EQ_NEEDED_SUPPORT	CX_NO_EQ_NEEDED
CX_DEF_MEM_WR_REQ_SUP	CX_DEF_MEM_WR_REQ_EN
CX_DEF_MEM_WR_ROUTING_SUP	CX_DEF_MEM_WR_ROUTING_EN
CX_DEVNUM_WD	(MULTI_DEVICE_AND_BUS_PER_FUNC_EN == 0) ? 5 : (CX_NFUNC*5)
CX_DL_NB	(CX_FREQ_STEP_DL_EN==1 && CX_FREQ_STEP_DL_RATIO==2) ? (CX_NB * 2) : (CX_FREQ_STEP_DL_EN==1 && CX_FREQ_STEP_DL_RATIO==4) ? (CX_NB * 4) : CX_NB
CX_DMA_CHANNEL_GROUP_ENABLE	0
CX_DMA_NATIVE	(AMBA_POPULATED==0 && CC_DMA_ENABLE)
CX_DSCALE	CX_CALC_DDEPTH <= 2047 ? 1 : CX_CALC_DDEPTH <= 8188 ? 4 : 16
CX_DSP_PCIPM_L1_ENTER_DELAY	0x0
CX_DW	[function_of: CX_DL_NB CX_NL]
CX_DW_W_PAR	32*CX_NW+TRGT_DATA_PROT_WD
CX_ECC_PIPE_EN	0
CX_EPTM_CAPABLE_VALUE	CX_PTM_ENABLE
CX_EQ_MASTER_COEF_REQ_FOR_FOM_ENABLE	0
CX_ERR_BUS_WD	15
CX_EXPORT_REMOTE_TX_COEF_PORT	0
CX_EXTDPC_ENABLE	0
CX_EXTENSIBLE_CONVERT_IN	0
CX_FLR_ENABLE_VALUE	CX_FLR_ENABLE

Parameter Name	Equals To
CX_FLT_Q_ADDR_GT_32	(FLT_Q_ADDR_WIDTH > 32) ? 1 : 0
CX_FREQ	(CX_MAC_SMODE_GEN1==1) ? 1 : (CX_MAC_SMODE_GEN1==2) ? 2 : (CX_MAC_SMODE_GEN1==4) ? 3 : 3
CX_FREQ_STEP_DL_EN	(CX_FREQ_STEP_DL > 0)
CX_FREQ_STEP_DL_RATIO	((CX_FREQ_STEP_DL > 0) ? CX_FREQ_STEP_DL : 1)
CX_FX_TLP	((CX_NW==16) ? 1 : (CX_NW==8) ? 2 : 1)
CX_GEN2_DYNAMIC_FREQ	(CX_GEN2_MODE == GEN2_DF)
CX_GEN2_DYNAMIC_WIDTH	(CX_GEN2_MODE == GEN2_DW)
CX_GEN2_MODE	(CX_MAX_PCIE_SPEED<2) ? 2 : (CX_MAC_SMODE_GEN1==1) && (CX_MAC_SMODE_GEN2==2) ? 1 : (CX_MAC_SMODE_GEN1==2) && (CX_MAC_SMODE_GEN2==4) ? 1 : (CX_MAC_SMODE_GEN1==1) && (CX_MAC_SMODE_GEN2==1) ? 0 : (CX_MAC_SMODE_GEN1==2) && (CX_MAC_SMODE_GEN2==2) ? 0 : (CX_MAC_SMODE_GEN1==4) && (CX_MAC_SMODE_GEN2==4) ? 0 : 2
CX_GEN2_SPEED	CX_GEN2_MODE != GEN2_DISABLED
CX_GEN3_DYNAMIC_FREQ	(CX_GEN3_MODE == GEN3_DF)
CX_GEN3_DYNAMIC_WIDTH	(CX_GEN3_MODE == GEN3_DW)
CX_GEN3_EQ_PSET_COEF_MAP_MODE_PROG	(CX_GEN3_EQ_PSET_COEF_MAP_MODE==2)
CX_GEN3_MODE	(CX_MAX_PCIE_SPEED<3) ? 2 : (CX_MAC_SMODE_GEN2==2) && (CX_MAC_SMODE_GEN3==4) ? 1 : (CX_MAC_SMODE_GEN2==4) && (CX_MAC_SMODE_GEN3==8) ? 1 : (CX_MAC_SMODE_GEN2==1) && (CX_MAC_SMODE_GEN3==4) ? 4 : (CX_MAC_SMODE_GEN2==2) && (CX_MAC_SMODE_GEN3==8) ? 4 : (CX_MAC_SMODE_GEN2==4) && (CX_MAC_SMODE_GEN3==16) ? 4 : (CX_MAC_SMODE_GEN2==1) && (CX_MAC_SMODE_GEN3==1) ? 0 : (CX_MAC_SMODE_GEN2==2) && (CX_MAC_SMODE_GEN3==2) ? 0 : (CX_MAC_SMODE_GEN2==4) && (CX_MAC_SMODE_GEN3==4) ? 0 : 2

Parameter Name	Equals To
CX_GEN3_SPEED	CX_GEN3_MODE != GEN3_DISABLED
CX_GEN3_SPEED_VALUE	CX_GEN3_SPEED
CX_GEN4_DYNAMIC_FREQ	CX_GEN4_SPEED ? (CX_GEN4_MODE == GEN4_DF) : 0
CX_GEN4_DYNAMIC_WIDTH	CX_GEN4_SPEED ? (CX_GEN4_MODE == GEN4_DW) : 0
CX_GEN4_MODE	(CX_MAX_PCIE_SPEED < 4) ? 2 : (CX_MAC_SMODE_GEN3 == 2) && (CX_MAC_SMODE_GEN4 == 4) ? 1 : (CX_MAC_SMODE_GEN3 == 4) && (CX_MAC_SMODE_GEN4 == 8) ? 1 : (CX_MAC_SMODE_GEN3 == 8) && (CX_MAC_SMODE_GEN4 == 16) ? 1 : (CX_MAC_SMODE_GEN3 == 1) && (CX_MAC_SMODE_GEN4 == 1) ? 0 : (CX_MAC_SMODE_GEN3 == 2) && (CX_MAC_SMODE_GEN4 == 2) ? 0 : (CX_MAC_SMODE_GEN3 == 4) && (CX_MAC_SMODE_GEN4 == 4) ? 0 : (CX_MAC_SMODE_GEN3 == 8) && (CX_MAC_SMODE_GEN4 == 8) ? 0 : (CX_MAC_SMODE_GEN3 == 16) && (CX_MAC_SMODE_GEN4 == 16) ? 0 : 2
CX_GEN4_SPEED	CX_GEN4_MODE != GEN4_DISABLED
CX_GEN4_SPEED_VALUE	CX_GEN4_SPEED
CX_GEN5_SPEED	CX_MAX_PCIE_SPEED >= 5
CX_HDMA_ENABLE_VALUE	CX_DMA_PF_ENABLE
CX_HEADER_RSVD_ENABLE	CX_P2P_ENABLE == 1 CC_DEVICE_TYPE == CC_SW
CX_HSCALE	CX_CALC_HDEPTH <= 127 ? 1 : CX_CALC_HDEPTH <= 508 ? 4 : 16
CX_IDO_ENABLE_VALUE	CX_IDO_ENABLE
CX_INTERFACE_TIMER_EN	(CX_AUTOMOTIVE_ENABLE == 1) ? 1 : 0
CX_INTERNAL_ERR_REPORTING	CX_INTERNAL_ERR_REPORTING_EN
CX_INTERNAL_ERR_REPORTING_EN	0x1
CX_IS_DM	(CC_DEVICE_TYPE == CC_DM) ? 1 : 0
CX_IS_EP	(CC_DEVICE_TYPE == CC_EP) ? 1 : 0
CX_IS_RC	(CC_DEVICE_TYPE == CC_RC) ? 1 : 0

Parameter Name	Equals To
CX_IS_SW	(CC_DEVICE_TYPE==CC_SW) ? 1 : 0
CX_L1_PG_ENABLE	CX_L12_PG_EN
CX_LANE_REVERSE	(CX_PCIE_MODE == PCIE_OVER_CIO) ? 0 : 1
CX_LANE_UNDER_TEST_EN	(CX_GEN5_SPEED CX_FORCE_LANE_FLIP_EN) && (CX_NL > 1)
CXL_LRSM_ABORT	0x10
CX_LN_VALUE	CX_LN_ENABLE
CX_LTR_M_ENABLE_VALUE	CX_LTR_M_ENABLE
CX_LTSSM_EMU_WD	6+5
CX_LUT_PTR_WIDTH	[function_of: CX_MAX_TAG]
CX_MARGIN_ENABLE	(CX_GEN4_SPEED)
CX_MAX_APP_RD_REQ_SIZE_AMBA_DMA	((CC_SLV_MTU < CX_MAX_MTU) ? CX_MAX_MTU : CC_SLV_MTU)

Parameter Name	Equals To
CX_MAX_CORECLK_FREQ	<pre> ((CX_MAX_PCIE_SPEED==1) && (CX_MAC_SMODE_GEN1 == 1) && (CX_CCIX_ESM_SUPPORT == 0)) ? 250 : ((CX_MAX_PCIE_SPEED==1) && (CX_MAC_SMODE_GEN1 == 2) && (CX_CCIX_ESM_SUPPORT == 0)) ? 125 : ((CX_MAX_PCIE_SPEED==1) && (CX_MAC_SMODE_GEN1 == 4) && (CX_CCIX_ESM_SUPPORT == 0)) ? 63 : ((CX_MAX_PCIE_SPEED==2) && (CX_MAC_SMODE_GEN2 == 1) && (CX_CCIX_ESM_SUPPORT == 0)) ? 500 : ((CX_MAX_PCIE_SPEED==2) && (CX_MAC_SMODE_GEN2 == 2) && (CX_CCIX_ESM_SUPPORT == 0)) ? 250 : ((CX_MAX_PCIE_SPEED==2) && (CX_MAC_SMODE_GEN2 == 4) && (CX_CCIX_ESM_SUPPORT == 0)) ? 125 : ((CX_MAX_PCIE_SPEED==3) && (CX_MAC_SMODE_GEN3 == 1) && (CX_CCIX_ESM_SUPPORT == 0)) ? 1000 : ((CX_MAX_PCIE_SPEED==3) && (CX_MAC_SMODE_GEN3 == 2) && (CX_CCIX_ESM_SUPPORT == 0)) ? 500 : ((CX_MAX_PCIE_SPEED==3) && (CX_MAC_SMODE_GEN3 == 4) && (CX_CCIX_ESM_SUPPORT == 0)) ? 250 : ((CX_MAX_PCIE_SPEED==3) && (CX_MAC_SMODE_GEN3 == 8) && (CX_CCIX_ESM_SUPPORT == 0)) ? 125 : ((CX_MAX_PCIE_SPEED==3) && (CX_MAC_SMODE_GEN3 == 16) && (CX_CCIX_ESM_SUPPORT == 0)) ? 63 : ((CX_MAX_PCIE_SPEED==4) && (CX_MAC_SMODE_GEN4 == 1) && (CX_CCIX_ESM_SUPPORT == 0)) ? 2000 : ((CX_MAX_PCIE_SPEED==4) && </pre>

Parameter Name	Equals To
CX_MAX_CORECLK_FREQ ...(cont.)	(CX_MAC_SMODE_GEN4 == 2) && (CX_CCIX_ESM_SUPPORT == 0) ? 1000 : ((CX_MAX_PCIE_SPEED == 4) && (CX_MAC_SMODE_GEN4 == 4) && (CX_CCIX_ESM_SUPPORT == 0) ? 500 : ((CX_MAX_PCIE_SPEED == 4) && (CX_MAC_SMODE_GEN4 == 8) && (CX_CCIX_ESM_SUPPORT == 0) ? 250 : ((CX_MAX_PCIE_SPEED == 4) && (CX_MAC_SMODE_GEN4 == 16) && (CX_CCIX_ESM_SUPPORT == 0) ? 125 : ((CX_MAX_PCIE_SPEED == 4) && (CX_MAC_SMODE_GEN4 == 2) && (CX_CCIX_ESM_SUPPORT == 1) ? 1562 : ((CX_MAX_PCIE_SPEED == 4) && (CX_MAC_SMODE_GEN4 == 4) && (CX_CCIX_ESM_SUPPORT == 1) ? 781 : ((CX_MAX_PCIE_SPEED == 4) && (CX_MAC_SMODE_GEN4 == 8) && (CX_CCIX_ESM_SUPPORT == 1) ? 390 : ((CX_MAX_PCIE_SPEED == 4) && (CX_MAC_SMODE_GEN4 == 16) && (CX_CCIX_ESM_SUPPORT == 1) ? 195 : ((CX_MAX_PCIE_SPEED == 5) && (CX_MAC_SMODE_GEN5 == 1) ? 4000 : ((CX_MAX_PCIE_SPEED == 5) && (CX_MAC_SMODE_GEN5 == 2) ? 2000 : ((CX_MAX_PCIE_SPEED == 5) && (CX_MAC_SMODE_GEN5 == 4) ? 1000 : ((CX_MAX_PCIE_SPEED == 5) && (CX_MAC_SMODE_GEN5 == 8) ? 500 : ((CX_MAX_PCIE_SPEED == 5) && (CX_MAC_SMODE_GEN5 == 16) ? 250 : 125
CX_MAX_DELAY_LAYER2_LAYER3	16
CX_MAX_END2END_TLP_PRFXS	(CX_NPRFX < 4) ? CX_NPRFX : 0
CX_MAX_END2END_TLP_PRFXS_VALUE	(CX_NPRFX < 4) ? CX_NPRFX : 4
CX_MAX_L0S_LTIME	CX_MAX_NFTS
CX_MAX_MTU_PW	[function_of: CX_MAX_MTU]
CX_MAX_NFTS	((CX_GEN2_SPEED == 1) ? ((DEFAULT_GEN2_N_FTS > CX_MAX_NFTS_TMP) ? DEFAULT_GEN2_N_FTS : CX_MAX_NFTS_TMP) : CX_MAX_NFTS_TMP)
CX_MAX_NFTS_TMP	((CX_NFTS > CX_COMM_NFTS) ? CX_NFTS : CX_COMM_NFTS)
CX_MAX_TAG_GTR_255	(CX_MAX_TAG > 255) ? 1 : 0

Parameter Name	Equals To
CX_MEMORY_MAP_PCIE_TYPE	0
CX_MOD_TS_FMT_SUPPORT_VALUE	DEFAULT_AP_ENABLE
CX_MPCIE_ENABLE	(CX_S_MPCIE_MODE CX_SEL_PHY_MODE)
CX_MSI_CTRL_EN	CX_MSI_CTRL_ENABLE
CX_MSI_CTRL_EN_VALUE	CX_MSI_CTRL_ENABLE
CX_MSIX_PBA_PW	[function_of: CX_MSIX_PBA_RAM_DEPTH]
CX_MSIX_PBA_RAM_DEPTH	[function_of: pba]
CX_MSIX_TABLE_PW	[function_of: CX_MSIX_TABLE_RAM_DEPTH]
CX_MSIX_TABLE_RAM_DEPTH	[function_of: table]
CX_MSTR_ISO_EN	(MSTR_CLK_DIFF_ENABLE==1 && CX_ENHANCED_PM_EN==1)
CX_NB	(CX_NB_GEN5)
CX_NB_GEN1	(CX_MAC_SMODE_GEN1)
CX_NB_GEN2	(CX_GEN2_SPEED ? CX_MAC_SMODE_GEN2 : CX_NB_GEN1)
CX_NB_GEN3	(CX_GEN3_SPEED ? CX_MAC_SMODE_GEN3 : CX_NB_GEN2)
CX_NB_GEN4	(CX_GEN4_SPEED ? CX_MAC_SMODE_GEN4 : CX_NB_GEN3)
CX_NB_GEN5	(CX_GEN5_SPEED ? CX_MAC_SMODE_GEN5 : CX_NB_GEN4)
CX_NDQ	(CX_NW==16) ? 4 : (CX_NW==8) ? 2 : 1
CX_NFUNC_WD	(CX_ARI_ENABLE) ? [function_of: CX_NFUNC] : 3
CX_NFUNC_WD_OPT	([function_of: CX_NFUNC])
CX_NHQ	(CX_NW==8) ? 2 : 1
CX_NL_GTR_1	(CX_NL > 1) ? 1 : 0
CX_NL_GTR_2	(CX_NL > 2) ? 1 : 0
CX_NL_GTR_4	(CX_NL > 4) ? 1 : 0
CX_NL_GTR_8	(CX_NL > 8) ? 1 : 0
CX_NL_M_1	CX_NL-1

Parameter Name	Equals To
CX_NP_CALC_DDEPTH	(CX_ADM_ADAPTOR_ENABLE==1) ? [function_of: CX_NW CX_ADM_ADAPTOR_RADM_PIPE_STAGES] : (CX_ATOMIC_ENABLE ? (RADM_NPQ_HCRD_VC0*CX_HSCALE)/2 : (RADM_NPQ_HCRD_VC0*CX_HSCALE)/4)
CX_NP_DSCALE	CX_NP_CALC_DDEPTH <= 2047 ? 1 : CX_NP_CALC_DDEPTH <= 8188 ? 4 : 16
CX_NUM_RBARS_0	[function_of: CX_BAR0_RESIZABLE_0 CX_BAR1_RESIZABLE_0 CX_BAR2_RESIZABLE_0 CX_BAR3_RESIZABLE_0 CX_BAR4_RESIZABLE_0 CX_BAR5_RESIZABLE_0]
CX_NUM_VF_RBARS_0	[function_of: VF_BAR0_RESIZABLE_0 VF_BAR1_RESIZABLE_0 VF_BAR2_RESIZABLE_0 VF_BAR3_RESIZABLE_0 VF_BAR4_RESIZABLE_0 VF_BAR5_RESIZABLE_0]
CX_NVC_LOG2	[function_of: CX_NVC]
CX_NVC_XALI_EXPANSION	CX_MULTISTREAM ? CX_NVC : 1
CX_NVFUNC_EWD	[function_of: CX_NVFUNC]
CX_NVFUNC_NUM_WD	CX_NVFUNC_WD
CX_NVFUNC_WD	[function_of: CX_NVFUNC] + 1
CX_NW_GTR_1	(CX_NW > 1) ? 1 : 0
CX_NW_GTR_4	(CX_NW > 4) ? 1 : 0
CX_NW_GTR_4_VALUE	CX_NW_GTR_4
CX_OBFF_ENABLE	(CX_OBFF_SUPPORT > 0) ? 1 : 0
CX_P2P_ENABLE_VALUE	CX_P2P_ENABLE
CX_PCIE_BRIDGE	CX_PCIE_BRIDGE_ENABLE
CX_PCIE_BRIDGE_ENABLE	0x0
CX_PCIE_OVER_CIO_ENABLE	(CX_PCIE_MODE == PCIE_OVER_CIO)

Parameter Name	Equals To
CX_PHY_FREQ	(CX_PHY_SMODE_GEN1==1) ? 1 : (CX_PHY_SMODE_GEN1==2) ? 2 : (CX_PHY_SMODE_GEN1==4) ? 3 : (CX_PHY_SMODE_GEN1==102) ? 0 : (CX_PHY_SMODE_GEN1==104) ? 4 : (CX_PHY_SMODE_GEN1==202) ? 1 : (CX_PHY_SMODE_GEN1==204) ? 0 : (CX_PHY_SMODE_GEN1==208) ? 4 : (CX_PHY_SMODE_GEN1==402) ? 2 : (CX_PHY_SMODE_GEN1==404) ? 1 : (CX_PHY_SMODE_GEN1==408) ? 0 : (CX_PHY_SMODE_GEN1==4016) ? 4 : 3
CX_PHY_GEN1_DT_RATE	(CX_PHY_SMODE_GEN1==102) ? 2 : (CX_PHY_SMODE_GEN1==104) ? 4 : (CX_PHY_SMODE_GEN1==202) ? 2 : (CX_PHY_SMODE_GEN1==204) ? 4 : (CX_PHY_SMODE_GEN1==208) ? 8 : (CX_PHY_SMODE_GEN1==402) ? 2 : (CX_PHY_SMODE_GEN1==404) ? 4 : (CX_PHY_SMODE_GEN1==408) ? 8 : (CX_PHY_SMODE_GEN1==4016) ? 16 : 1
CX_PHY_GEN2_DT_RATE	(CX_PHY_SMODE_GEN2==102) ? 2 : (CX_PHY_SMODE_GEN2==202) ? 2 : (CX_PHY_SMODE_GEN2==204) ? 4 : (CX_PHY_SMODE_GEN2==402) ? 2 : (CX_PHY_SMODE_GEN2==404) ? 4 : (CX_PHY_SMODE_GEN2==408) ? 8 : 1
CX_PHY_GEN2_MODE	(CX_MAX_PCIE_SPEED<2) ? 2 : (CX_PHY_SMODE_GEN1==1) && (CX_PHY_SMODE_GEN2==2) ? 1 : (CX_PHY_SMODE_GEN1==2) && (CX_PHY_SMODE_GEN2==4) ? 1 : (CX_PHY_SMODE_GEN1==1) && (CX_PHY_SMODE_GEN2==1) ? 0 : (CX_PHY_SMODE_GEN1==2) && (CX_PHY_SMODE_GEN2==2) ? 0 : (CX_PHY_SMODE_GEN1==4) && (CX_PHY_SMODE_GEN2==4) ? 0 : (CX_PHY_SMODE_GEN1==202) && (CX_PHY_SMODE_GEN2==202) ? 0 : (CX_PHY_SMODE_GEN1==102) && (CX_PHY_SMODE_GEN2==1) ? 1 : 2

Parameter Name	Equals To
CX_PHY_GEN3_DT_RATE	(CX_PHY_SMODE_GEN3==202) ? 2 : (CX_PHY_SMODE_GEN3==402) ? 2 : (CX_PHY_SMODE_GEN3==404) ? 4 : (CX_PHY_SMODE_GEN3==802) ? 2 : (CX_PHY_SMODE_GEN3==804) ? 4 : (CX_PHY_SMODE_GEN3==808) ? 8 : (CX_PHY_SMODE_GEN3==162) ? 2 : (CX_PHY_SMODE_GEN3==164) ? 4 : (CX_PHY_SMODE_GEN3==168) ? 8 : (CX_PHY_SMODE_GEN3==1616) ? 16 : 1
CX_PHY_GEN3_MODE	(CX_MAX_PCIE_SPEED<3) ? 2 : (CX_PHY_SMODE_GEN2==1) && (CX_PHY_SMODE_GEN3==2) ? 1 : (CX_PHY_SMODE_GEN2==2) && (CX_PHY_SMODE_GEN3==4) ? 1 : (CX_PHY_SMODE_GEN2==4) && (CX_PHY_SMODE_GEN3==8) ? 1 : (CX_PHY_SMODE_GEN2==1) && (CX_PHY_SMODE_GEN3==4) ? 4 : (CX_PHY_SMODE_GEN2==4) && (CX_PHY_SMODE_GEN3==16) ? 4 : (CX_PHY_SMODE_GEN2==1) && (CX_PHY_SMODE_GEN3==1) ? 0 : (CX_PHY_SMODE_GEN2==2) && (CX_PHY_SMODE_GEN3==2) ? 0 : (CX_PHY_SMODE_GEN2==4) && (CX_PHY_SMODE_GEN3==4) ? 0 : (CX_PHY_SMODE_GEN2==202) && (CX_PHY_SMODE_GEN3==2) ? 3 : 2
CX_PHY_GEN4_DT_RATE	(CX_PHY_SMODE_GEN4==402) ? 2 : (CX_PHY_SMODE_GEN4==802) ? 2 : (CX_PHY_SMODE_GEN4==804) ? 4 : (CX_PHY_SMODE_GEN4==162) ? 2 : (CX_PHY_SMODE_GEN4==164) ? 4 : (CX_PHY_SMODE_GEN4==168) ? 8 : 1

Parameter Name	Equals To
CX_PHY_GEN4_MODE	$(CX_MAX_PCIE_SPEED < 4) ? 2 :$ $(CX_PHY_SMODE_GEN3 < CX_PHY_SMODE_GEN4) ? 1$ $: (CX_PHY_SMODE_GEN3 == 1) \&\&$ $(CX_PHY_SMODE_GEN4 == 2) ? 1 :$ $(CX_PHY_SMODE_GEN3 == 2) \&\&$ $(CX_PHY_SMODE_GEN4 == 4) ? 1 :$ $(CX_PHY_SMODE_GEN3 == 4) \&\&$ $(CX_PHY_SMODE_GEN4 == 8) ? 1 :$ $(CX_PHY_SMODE_GEN3 == 8) \&\&$ $(CX_PHY_SMODE_GEN4 == 1) ? 0 :$ $(CX_PHY_SMODE_GEN3 == 2) \&\&$ $(CX_PHY_SMODE_GEN4 == 2) ? 0 :$ $(CX_PHY_SMODE_GEN3 == 4) \&\&$ $(CX_PHY_SMODE_GEN4 == 4) ? 0 :$ $(CX_PHY_SMODE_GEN3 == 8) \&\&$ $(CX_PHY_SMODE_GEN4 == 8) ? 0 :$ $(CX_PHY_SMODE_GEN3 == 16) \&\&$ $(CX_PHY_SMODE_GEN4 == 16) ? 0 : 2$
CX_PHY_GEN5_DT_RATE	1
CX_PHY_INTERFACE	$(CX_GEN3_SPEED_VALUE == 1)$
CX_PHY_NB	$(CX_MAX_PCIE_SPEED == 1) ? CX_PHY_NB_GEN1 :$ $(CX_MAX_PCIE_SPEED == 2) ? CX_PHY_NB_GEN2 :$ $(CX_MAX_PCIE_SPEED == 3) ? CX_PHY_NB_GEN3 :$ $(CX_MAX_PCIE_SPEED == 4) ? CX_PHY_NB_GEN4 :$ $(CX_MAX_PCIE_SPEED == 5) ? CX_PHY_NB_GEN5 :$ $CX_PHY_NB_GEN5$
CX_PHY_NB_GEN1	$(CX_PHY_SMODE_GEN1 == 1) ? 1 :$ $(CX_PHY_SMODE_GEN1 == 2) ? 2 :$ $(CX_PHY_SMODE_GEN1 == 4) ? 4 :$ $(CX_PHY_SMODE_GEN1 == 102) ? 1 :$ $(CX_PHY_SMODE_GEN1 == 104) ? 1 :$ $(CX_PHY_SMODE_GEN1 == 202) ? 2 :$ $(CX_PHY_SMODE_GEN1 == 204) ? 2 :$ $(CX_PHY_SMODE_GEN1 == 208) ? 2 :$ $(CX_PHY_SMODE_GEN1 == 402) ? 4 :$ $(CX_PHY_SMODE_GEN1 == 404) ? 4 :$ $(CX_PHY_SMODE_GEN1 == 408) ? 4 :$ $(CX_PHY_SMODE_GEN1 == 4016) ? 4 : 2$
CX_PHY_NB_GEN2	$(CX_PHY_SMODE_GEN2 == 1) ? 1 :$ $(CX_PHY_SMODE_GEN2 == 2) ? 2 :$ $(CX_PHY_SMODE_GEN2 == 4) ? 4 :$ $(CX_PHY_SMODE_GEN2 == 102) ? 1 :$ $(CX_PHY_SMODE_GEN2 == 202) ? 2 :$ $(CX_PHY_SMODE_GEN2 == 204) ? 2 :$ $(CX_PHY_SMODE_GEN2 == 402) ? 4 :$ $(CX_PHY_SMODE_GEN2 == 404) ? 4 :$ $(CX_PHY_SMODE_GEN2 == 408) ? 4 : 2$

Parameter Name	Equals To
CX_PHY_NB_GEN3	(CX_PHY_SMODE_GEN3==1) ? 1 : (CX_PHY_SMODE_GEN3==2) ? 2 : (CX_PHY_SMODE_GEN3==4) ? 4 : (CX_PHY_SMODE_GEN3==8) ? 8 : (CX_PHY_SMODE_GEN3==16) ? 16 : (CX_PHY_SMODE_GEN3==202) ? 2 : (CX_PHY_SMODE_GEN3==402) ? 4 : (CX_PHY_SMODE_GEN3==404) ? 4 : (CX_PHY_SMODE_GEN3==802) ? 8 : (CX_PHY_SMODE_GEN3==804) ? 8 : (CX_PHY_SMODE_GEN3==808) ? 8 : (CX_PHY_SMODE_GEN3==162) ? 16 : (CX_PHY_SMODE_GEN3==164) ? 16 : (CX_PHY_SMODE_GEN3==168) ? 16 : (CX_PHY_SMODE_GEN3==1616) ? 16 : 2
CX_PHY_NB_GEN4	(CX_PHY_SMODE_GEN4==1) ? 1 : (CX_PHY_SMODE_GEN4==2) ? 2 : (CX_PHY_SMODE_GEN4==4) ? 4 : (CX_PHY_SMODE_GEN4==8) ? 8 : (CX_PHY_SMODE_GEN4==16) ? 16 : (CX_PHY_SMODE_GEN4==402) ? 4 : (CX_PHY_SMODE_GEN4==802) ? 8 : (CX_PHY_SMODE_GEN4==804) ? 8 : (CX_PHY_SMODE_GEN4==162) ? 16 : (CX_PHY_SMODE_GEN4==164) ? 16 : (CX_PHY_SMODE_GEN4==168) ? 16 : 2
CX_PHY_NB_GEN5	(CX_PHY_SMODE_GEN5==1) ? 1 : (CX_PHY_SMODE_GEN5==2) ? 2 : (CX_PHY_SMODE_GEN5==4) ? 4 : (CX_PHY_SMODE_GEN5==8) ? 8 : (CX_PHY_SMODE_GEN5==16) ? 16 : 2
CX_PHY_NUM_MACROS	[function_of:]
CX_PHY_PDOWN_WD	4
CX_PHY_RATE_WD	(CX_GEN5_SPEED) ? 3 : (CX_GEN3_SPEED==1) ? 2 : 1
CX_PHY_RST_TIMER	10*DEFAULT_AUX_CLK_FREQ
CX_PHY_RX_DELAY_MAC	4
CX_PHY_RX_DELAY_SERDES_PIPE	(CX_PIPE_SERDES_ARCH==0) ? 0 : (CX_FREQ_STEP_EN) ? (CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL) : (CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY)

Parameter Name	Equals To
CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY	(!CX_PIPE_SERDES_ARCH) ? 0 : ([function_of: CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY_GEN1 CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY_GEN2 CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY_GEN3 CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY_GEN4 CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY_GEN5])
CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY_GEN1	([function_of: CX_SRIS_SUPPORT CX_PHY_NB_GEN1 CX_SERDES_ARCH_SYNC_TYPE CX_SERDES_ARCH_RX_REGIN CX_SERDES_ARCH_FPGA])
CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY_GEN2	(!CX_GEN2_SPEED) ? 0 : ([function_of: CX_SRIS_SUPPORT CX_PHY_NB_GEN2 CX_SERDES_ARCH_SYNC_TYPE CX_SERDES_ARCH_RX_REGIN CX_SERDES_ARCH_FPGA])
CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY_GEN3	(!CX_GEN3_SPEED) ? 0 : ([function_of: CX_SRIS_SUPPORT CX_PHY_NB_GEN3 CX_SERDES_ARCH_SYNC_TYPE CX_SERDES_ARCH_RX_REGIN CX_SERDES_ARCH_FPGA])
CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY_GEN4	(!CX_GEN4_SPEED) ? 0 : ([function_of: CX_SRIS_SUPPORT CX_PHY_NB_GEN4 CX_SERDES_ARCH_SYNC_TYPE CX_SERDES_ARCH_RX_REGIN CX_SERDES_ARCH_FPGA])
CX_PHY_RX_DELAY_SERDES_PIPE_EMPTY_GEN5	(!CX_GEN5_SPEED) ? 0 : ([function_of: CX_SRIS_SUPPORT CX_PHY_NB_GEN5 CX_SERDES_ARCH_SYNC_TYPE CX_SERDES_ARCH_RX_REGIN CX_SERDES_ARCH_FPGA])
CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL	(!CX_PIPE_SERDES_ARCH) ? 0 : ([function_of: CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL_GEN1 CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL_GEN2 CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL_GEN3 CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL_GEN4 CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL_GEN5])
CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL_GEN1	([function_of: CX_SRIS_SUPPORT CX_PHY_NB_GEN1 CX_SERDES_ARCH_SYNC_TYPE CX_SERDES_ARCH_RX_REGIN CX_SERDES_ARCH_FPGA])

Parameter Name	Equals To
CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL_GEN2	(!CX_GEN2_SPEED) ? 0 : ((function_of: CX_SRIS_SUPPORT CX_PHY_NB_GEN2 CX_SERDES_ARCH_SYNC_TYPE CX_SERDES_ARCH_RX_REGIN CX_SERDES_ARCH_FPGA))
CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL_GEN3	(!CX_GEN3_SPEED) ? 0 : ((function_of: CX_SRIS_SUPPORT CX_PHY_NB_GEN3 CX_SERDES_ARCH_SYNC_TYPE CX_SERDES_ARCH_RX_REGIN CX_SERDES_ARCH_FPGA))
CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL_GEN4	(!CX_GEN4_SPEED) ? 0 : ((function_of: CX_SRIS_SUPPORT CX_PHY_NB_GEN4 CX_SERDES_ARCH_SYNC_TYPE CX_SERDES_ARCH_RX_REGIN CX_SERDES_ARCH_FPGA))
CX_PHY_RX_DELAY_SERDES_PIPE_HALFFULL_GEN5	(!CX_GEN5_SPEED) ? 0 : ((function_of: CX_SRIS_SUPPORT CX_PHY_NB_GEN5 CX_SERDES_ARCH_SYNC_TYPE CX_SERDES_ARCH_RX_REGIN CX_SERDES_ARCH_FPGA))
CX_PHY_RXSB_WD	(CX_PIPERX_MULTI_BLOCK==1) ? (CX_PHY_NB / 4) : (SNPS_RSVDPARAM_71) ? 2 : 1
CX_PHY_RXSH_WD	((CX_PIPERX_MULTI_BLOCK==1) && (SNPS_RSVDPARAM_71==0)) ? (CX_PHY_NB / 4) * 2 : 2
CX_PHY_TXDEEMPH_WD	(PHY_IS_PIPE==1 && CX_GEN3_SPEED==1) ? (CX_NL*18) : 2
CX_PHY_TX_DELAY_MAC	4
CX_PHY_TX_DELAY_SERDES_PIPE	((CX_PIPE_SERDES_ARCH) && (CX_SERDES_ARCH_TX_REGOUT)) ? 1 : 0
CX_PHY_TXEI_WD	(CX_PIPE5_SUPPORT) ? ((CX_PHY_NB>8) ? (CX_PHY_NB/2) : 4) : 1
CX_PHY_VIEWPORT_DATA	16
CX_PHY_WIDTH_WD	((CX_GEN3_SPEED==1) && (CX_PHY_NB==16)) ? 3 : 2
CX_PIPE44_SUPPORT	(CX_PIPE_VER>=2)
CX_PIPE5_SUPPORT	(CX_PIPE_VER>=3)
CX_PIPE_LEGACY_MSGBUS_SUPPORT	CX_GEN4_SPEED && (!CX_PIPE44_SUPPORT && !CX_PIPE5_SUPPORT)
CX_PIPE_LOOPBACK_EN	1

Parameter Name	Equals To
CX_PIPE_LOW_PIN_COUNT	(CX_PIPE5_SUPPORT)
CX_PIPE_MSGBUS_SUPPORT	(CX_GEN4_SPEED && CX_PIPE44_SUPPORT) CX_PIPE5_SUPPORT
CX_PIPE_PCLK_AS_PHY_INPUT	CX_PIPE_PCLK_MODE > 0
CX_PIPE_PCLK_MODE_1	(CX_PIPE_PCLK_MODE==1)
CX_PIPE_REGIF_L1_PG_ALWAYS_ON	(CX_L12_PG_EN && CX_PIPE_REGIF_SUPPORT) && (CX_PIPE_LOW_PIN_COUNT CX_CCIX_ESM_SUPPORT)
CX_PIPE_REGIF_SUPPORT	CX_PIPE_MSGBUS_SUPPORT CX_PIPE_LEGACY_MSGBUS_SUPPORT
CX_PIPE_SERDES_ARCH_VALUE	CX_PIPE_SERDES_ARCH
CX_PIPE_WIDTH_	(8 * (CX_NB))
CX_PL16G_ENABLE	(CX_GEN4_SPEED)
CX_PL32G_ENABLE	(CX_GEN5_SPEED CX_AP_SUPPORT)
CX_PL_FREQ_MULTIPLIER	(CX_PL_FREQ_VALUE == 0) ? 1 : (CX_PL_FREQ_VALUE == 1) ? 2 : 4
CX_PL_FREQ_VALUE	(CX_FREQ - 1)
CX_PL_MODE	0
CX_PL_REG_DISABLE	0
CX_PMA_PIPE_RST_DELAY_TIMER	0x10
CX_PRFX_PAR_WD	(CX_RASDP_EN) ? [function_of: CX_NW CX_NPRFX CX_RAS_PROT_RANGE CX_RAS_PROT_TYPE] : 0
CX_PTM_REQUESTER_CAPABLE	(CX_PTM_ENABLE==1 && CC_DEVICE_TYPE!=CC_RC)
CX_PTM_REQUESTER_VALUE	CX_PTM_REQUESTER_CAPABLE
CX_PTM_RESPONDER_CAPABLE	(CX_PTM_ROOT_CAPABLE (CX_PTM_ENABLE && CC_DEVICE_TYPE!=CC_EP))
CX_PTM_RESPONDER_VALUE	CX_PTM_RESPONDER_CAPABLE
CX_PTM_ROOT_CAPABLE	(CX_PTM_ENABLE && CC_DEVICE_TYPE!=CC_EP)
CX_RADM_CPL_LUT_PIPE_EN	((CX_MAX_CORECLK_FREQ > 500) ((CX_MAX_CORECLK_FREQ==500)&& (CX_NW==8)&& (CX_MAX_TAG>127)))

Parameter Name	Equals To
CX_RADM_DATAQ_DEPTH	$(RADM_PQ_DDP_VC0 + RADM_NPQ_DDP_VC0 + RADM_CPLQ_DDP_VC0) + (CX_NVC \leq 1 ? 0 : (RADM_PQ_DDP_VC1 + RADM_NPQ_DDP_VC1 + RADM_CPLQ_DDP_VC1)) + (CX_NVC \leq 2 ? 0 : (RADM_PQ_DDP_VC2 + RADM_NPQ_DDP_VC2 + RADM_CPLQ_DDP_VC2)) + (CX_NVC \leq 3 ? 0 : (RADM_PQ_DDP_VC3 + RADM_NPQ_DDP_VC3 + RADM_CPLQ_DDP_VC3)) + (CX_NVC \leq 4 ? 0 : (RADM_PQ_DDP_VC4 + RADM_NPQ_DDP_VC4 + RADM_CPLQ_DDP_VC4)) + (CX_NVC \leq 5 ? 0 : (RADM_PQ_DDP_VC5 + RADM_NPQ_DDP_VC5 + RADM_CPLQ_DDP_VC5)) + (CX_NVC \leq 6 ? 0 : (RADM_PQ_DDP_VC6 + RADM_NPQ_DDP_VC6 + RADM_CPLQ_DDP_VC6)) + (CX_NVC \leq 7 ? 0 : (RADM_PQ_DDP_VC7 + RADM_NPQ_DDP_VC7 + RADM_CPLQ_DDP_VC7)) + (CX_NVC * 3)$
CX_RADM_DATASIZE	$(CX_NW == 16) ? CX_MAX_MTU / (4 * CX_NW) + (CX_ECRC_STRIP_ENABLE == 0) : CX_NHQ * CX_RADM_MAXPKT / (4 * CX_NW)$
CX_RADM_FORMQ_CCIX_DS	$(CX_CCIX_INTERFACE_ENABLE) ? CX_CCIX_RX_WD : 0$
CX_RADM_FORMQ_DATA_DS	$((CX_DW + TRGT_DATA_PROT_WD) / (CX_NW < 4 ? 1 : CX_NW / 4))$
CX_RADM_FORMQ_DEBUG_W_ALIGNED_RAM_WD	0
CX_RADM_FORMQ_HDR_DS	128 + CX_RAS_PCIE_HDR_PROT_WD
CX_RADM_FORMQ_NQW	CX_RADM_SERQ_NUM_RAM
CX_RADM_FORMQ_QDW_CTRL_WD	$(10 + ((CX_TLP_PREFIX_ENABLE) ? 1 : 0) + CX_RADM_FORMQ_RASDES_DS + CX_RADM_FORMQ_CCIX_DS)$
CX_RADM_FORMQ_QDW_DS	$((CX_RADM_FORMQ_HDR_DS > CX_RADM_FORMQ_DATA_DS) ? CX_RADM_FORMQ_HDR_DS : CX_RADM_FORMQ_DATA_DS)$
CX_RADM_FORMQ_QDW_FF_WD	$(CX_TLP_PREFIX_ENABLE) ? 4 : 3$
CX_RADM_FORMQ_QDW_RAM_PW	CX_RADM_SERQ_QDW_RAM_PW
CX_RADM_FORMQ_QDW_RAM_WD	$CX_RADM_FORMQ_QDW_DS + CX_RADM_FORMQ_QDW_CTRL_WD + 6 * (CX_RADM_FORMQ_DEBUG_W_ALIGNED_RAM_WD) - CX_RADM_FORMQ_QDW_FF_WD + CX_RASDP_FORMQ_RAM_ECC_WD - (((CX_RASDP > 0) \&\& (CX_RASDP < 3)) ? 16 : 0)$

Parameter Name	Equals To
CX_RADM_FORMQ_RASDES_DS	$(CX_RAS_DES_ENABLE) ? 8 : 0$
CX_RADM_HDRQ_DEPTH	$(RADM_PQ_HDP_VC0 + RADM_NPQ_HDP_VC0 +$ $RADM_CPLQ_HDP_VC0) + (CX_NVC \leq 1 ? 0 :$ $(RADM_PQ_HDP_VC1 + RADM_NPQ_HDP_VC1 +$ $RADM_CPLQ_HDP_VC1)) + (CX_NVC \leq 2 ? 0 :$ $(RADM_PQ_HDP_VC2 + RADM_NPQ_HDP_VC2 +$ $RADM_CPLQ_HDP_VC2)) + (CX_NVC \leq 3 ? 0 :$ $(RADM_PQ_HDP_VC3 + RADM_NPQ_HDP_VC3 +$ $RADM_CPLQ_HDP_VC3)) + (CX_NVC \leq 4 ? 0 :$ $(RADM_PQ_HDP_VC4 + RADM_NPQ_HDP_VC4 +$ $RADM_CPLQ_HDP_VC4)) + (CX_NVC \leq 5 ? 0 :$ $(RADM_PQ_HDP_VC5 + RADM_NPQ_HDP_VC5 +$ $RADM_CPLQ_HDP_VC5)) + (CX_NVC \leq 6 ? 0 :$ $(RADM_PQ_HDP_VC6 + RADM_NPQ_HDP_VC6 +$ $RADM_CPLQ_HDP_VC6)) + (CX_NVC \leq 7 ? 0 :$ $(RADM_PQ_HDP_VC7 + RADM_NPQ_HDP_VC7 +$ $RADM_CPLQ_HDP_VC7)) + (CX_NVC * 3)$
CX_RADM_MAXPKT	$(CX_MAX_MTU + 28)$
CX_RADM_ORDERING_RULES_VC0	CX_RADM_ORDERING_RULES
CX_RADM_ORDERING_RULES_VC1	CX_RADM_ORDERING_RULES
CX_RADM_PQ_D_ADDRBITS	$CX_NDQ * (CX_RADMQ_MODE == 2 ?$ $CX_RADM_SBUF_DATAQ_PW : (RADM_PQ_DPW *$ $CX_NVC))$
CX_RADM_PQ_H_ADDRBITS	$CX_NHQ * (CX_RADMQ_MODE == 2 ?$ $CX_RADM_SBUF_HDRQ_PW : (RADM_PQ_HPW *$ $CX_NVC))$
CX_RADM_PQ_H_DATABITS	$CX_NHQ * (CX_RADMQ_MODE == 2 ?$ $CX_RADM_SBUF_HDRQ_WD : (CX_RADM_PQ_HWD *$ $CX_NVC))$
CX_RADM_PQ_H_DATABITS_OUT	$(!CX_ECC_PIPE_EN) ? CX_RADM_PQ_H_DATABITS :$ $(CX_RADMQ_MODE == 2) ?$ $(CX_RADM_SBUF_HDRQ_WD +$ $2 * (RADM_ECRC_ERR_WD + RADM_DLLP_ABORT_WD$ $+ RADM_TLP_ABORT_WD) + P_HDRQ_NECC_BITS) *$ $CX_NHQ : (CX_RADM_PQ_HWD + 16 +$ $P_HDRQ_NECC_BITS) * CX_NVC * CX_NHQ$
CX_RADM_PQ_HWD	[function_of: RADM_P_HWD 6]
CX_RADM_Q_DATABITS	$(CX_RADMQ_MODE == 2) ?$ $CX_RADM_SBUF_DATAQ_RAM_WD * CX_NDQ :$ $(CX_NVC * ((32 * CX_NW) + CX_NW + 5 * CX_NDQ))$

Parameter Name	Equals To
CX_RADM_Q_DATABITS_OUT	$(!CX_ECC_PIPE_EN) ? CX_RADM_Q_DATABITS : (CX_RADMQ_MODE == 2) ? CX_NDQ * (CX_RADM_SBUF_DATAQ_RAM_WD + 2 * (CX_RADM_SBUF_DATAQ_CTRL_WD - CX_RADM_SBUF_DATAQ_CTRLQ_WD) + P_DATAQ_NECC_BITS) : (CX_NW * 32 + 3 * (CX_NW + 5) + P_DATAQ_NECC_BITS) * CX_NVC$
CX_RADM_Q_D_CTRLBITS	$CX_NDQ * (CX_RADMQ_MODE == 2 ? 1 : CX_NVC)$
CX_RADM_Q_H_CTRLBITS	$CX_NHQ * (CX_RADMQ_MODE == 2 ? 1 : CX_NVC)$
CX_RADMQ_MODE	$(CC_DEVICE_TYPE == CC_SW CX_P2P_ENABLE == 1 CX_NW == 16) ? 2 : 0 == 1 ? 0 : 2$
CX_RADM_RXQ_NUM_QDWS	$CX_RADM_HDRQ_DEPTH + CX_RADM_DATAQ_DEPTH * (CX_NW / 4)$
CX_RADM_RXQ_NUM_QDWS_PER_SERQ_RAM	$CX_RADM_HDRQ_DEPTH < CX_RADM_SERQ_NUM_RAMS ? 1 : ([function_of: CX_RADM_RXQ_NUM_QDWS CX_RADM_SERQ_NUM_RAMS])$
CX_RADM_SBUF_DATAQ_CTRLQ_WD	RADM_EOT_WD
CX_RADM_SBUF_DATAQ_CTRL_WD	$RADM_DLLP_ABORT_WD + RADM_ECRC_ERR_WD + RADM_TLP_ABORT_WD + RADM_EOT_WD$
CX_RADM_SBUF_DATAQ_DATA_WD	$(32 * CX_NW) / CX_NDQ$
CX_RADM_SBUF_DATAQ_PROT_WD	$(CX_RAS_EN == 1) ? [function_of: CX_RADM_SBUF_DATAQ_DATA_WD CX_RADM_SBUF_DATAQ_CTRL_WD CX_RASDP_EN CX_RAS_PROT_TYPE CX_RAS_PROT_RANGE CX_RAS_RAM_PROT_RANGE RAS_BASE_PROT_WD RAS_RAM_BASE_PROT_WD] : 0$
CX_RADM_SBUF_DATAQ_PW	[function_of: CX_RADM_DATAQ_DEPTH]
CX_RADM_SBUF_DATAQ_RAM_WD	$CX_RADM_SBUF_DATAQ_WD - CX_RADM_SBUF_DATAQ_CTRLQ_WD$
CX_RADM_SBUF_DATAQ_WD	$CX_RADM_SBUF_DATAQ_DATA_WD + CX_RADM_SBUF_DATAQ_CTRL_WD + CX_RADM_SBUF_DATAQ_PROT_WD$
CX_RADM_SBUF_HDRQ_CTRL_WD	$RADM_DLLP_ABORT_WD + RADM_ECRC_ERR_WD + RADM_TLP_ABORT_WD$

Parameter Name	Equals To
CX_RADM_SBUF_HDRQ_PROT_WD	(CX_RAS_EN==1)? [function_of: CX_RAS_RADM_P_HWD_TOPROT CX_RADM_SBUF_HDRQ_CTRL_WD CX_RASDP_EN CX_RAS_PROT_TYPE CX_RAS_PROT_RANGE CX_RAS_RAM_PROT_RANGE RAS_BASE_PROT_WD RAS_RAM_BASE_PROT_WD]:0
CX_RADM_SBUF_HDRQ_PW	[function_of: CX_RADM_HDRQ_DEPTH]
CX_RADM_SBUF_HDRQ_WD	CX_RAS_RADM_P_HWD + RADM_DLLP_ABORT_WD + RADM_ECRC_ERR_WD + RADM_TLP_ABORT_WD + CX_RADM_SBUF_HDRQ_PROT_WD
CX_RADM_SERQ_AF_NUM_QDWS	CX_RADM_SERQ_AF_NUM_QDWS_PER_RAM * CX_RADM_SERQ_NUM_RAM
CX_RADM_SERQ_AF_NUM_QDWS_PER_RAM	[function_of: [function_of: CX_RADM_RXQ_NUM_QDWS_PER_SERQ_RAM * CX_RADM_SERQ_TO_RXQ_SIZE_RATIO] 100]
CX_RADM_SERQ_MARGIN_NUM_QDWS	[function_of: CX_MAX_MTU CX_MAX_DELAY_LAYER2_LAYER3 CX_NPRFX]
CX_RADM_SERQ_MARGIN_NUM_QDWS_PER_RAM	[function_of: CX_RADM_SERQ_MARGIN_NUM_QDWS CX_RADM_SERQ_NUM_RAM]
CX_RADM_SERQ_NUM_RAM	(CX_NW == 16) ? ((CX_TLP_PREFIX_ENABLE)? 7 : 6) : (CX_NW == 8) ? 3 : 2
CX_RADM_SERQ_OPT_AF_NUM_QDWS	[function_of: CX_PHY_TX_DELAY_PHY CX_PHY_RX_DELAY_PHY CX_NB CX_PHY_NB CX_RETIMER_LATENCY CX_NPRFX]
CX_RADM_SERQ_OPT_AF_NUM_QDWS_PER_RAM	[function_of: CX_RADM_SERQ_OPT_AF_NUM_QDWS CX_RADM_SERQ_NUM_RAM]
CX_RADM_SERQ_QDW_RAM_DP	CX_RADM_SERQ_AF_NUM_QDWS_PER_RAM + CX_RADM_SERQ_MARGIN_NUM_QDWS_PER_RAM
CX_RADM_SERQ_QDW_RAM_PW	[function_of: CX_RADM_SERQ_QDW_RAM_DP]
CX_RAM_PROTECTION_MODE	0
CX_RAS_DES_EC_G0_BW	4
CX_RAS_DES_EC_G1_BW	8
CX_RAS_DES_EC_G2_BW	8
CX_RAS_DES_EC_G3_BW	8
CX_RAS_DES_EC_G4_BW	4
CX_RAS_DES_EC_G5_BW	32

Parameter Name	Equals To
CX_RAS_DES_EC_G6_BW	32
CX_RAS_DES_EC_G7_BW	32
CX_RAS_DES_EC_INFO_CMN_BW	177
CX_RAS_DES_EC_INFO_PLN_BW	13
CX_RAS_DES_EC_RAM_ADDR_WIDTH	[function_of: CX_RAS_DES_EC_RAM_DEPTH]
CX_RAS_DES_EC_RAM_DATA_WIDTH	64
CX_RAS_DES_EC_RAM_DEPTH	[function_of: CX_RAS_DES_EC_G0_BW CX_RAS_DES_EC_G1_BW CX_RAS_DES_EC_G2_BW CX_RAS_DES_EC_G3_BW CX_RAS_DES_EC_G4_BW CX_RAS_DES_EC_G5_BW CX_RAS_DES_EC_G6_BW CX_RAS_DES_EC_G7_BW CX_RAS_DES_EC_ENABLE CX_NL CX_GEN2_SPEED CX_GEN3_SPEED CX_GEN4_SPEED CX_L1_SUBSTATES_ENABLE SNPS_RSVDPARAM_7 CX_CCIX_INTERFACE_ENABLE CX_DEF_MEM_WR_ENABLE]
CX_RAS_DES_SD_INFO_CMN_BW	79
CX_RAS_DES_SD_INFO_PLN_BW	205
CX_RAS_DES_SD_INFO_PVC_BW	240
CX_RAS_DES_TBA_DATA_WIDTH	[function_of: CX_MAX_CORECLK_FREQ]
CX_RAS_DES_TBA_INFO_CMN_BW	7
CX_RAS_DES_TBA_RAM_ADDR_WIDTH	[function_of: CX_RAS_DES_TBA_RAM_DEPTH]
CX_RAS_DES_TBA_RAM_DATA_WIDTH	[function_of: CX_RAS_DES_TBA_DATA_WIDTH]
CX_RAS_DES_TBA_RAM_DEPTH	[function_of: CX_RAS_DES_TBA_ENABLE CX_L1_SUBSTATES_ENABLE CX_CCIX_ESM_SUPPORT]
CX_RASDP_BYPASS_HDR_PROT_WD	(CX_RASDP_EN==1)? [function_of: CX_RASDP_BYPASS_HDR_WDCX_RAS_PROT_RANGE RAS_BASE_PROT_WD]: 0
CX_RASDP_BYPASS_HDR_WD	(CX_RASDP_EN)? ((CX_IS_SW)? 50: 55) + CX_TAG_SIZE + (ATS_RX_ENABLE_VALUE*2) + FLT_Q_ADDR_WIDTH + (CX_TPH_ENABLE_VALUE*11) + ((CX_NVFUNC_EWD+1)*CX_SRIOV_ENABLE_VALUE) + CX_NVFUNC_WD + FLT_Q_ATTR_WIDTH +CX_LN_VALUE: 0
CX_RASDP_EN	((CX_RASDP==0)? 0: 1)

Parameter Name	Equals To
CX_RASDP_FORMQ_ERR_SYND_WD	$((CX_RAS_EN \&\& CX_NW==16)? ((CX_RASDP>2)? CX_RASDP_FORMQ_RAM_ECC_WD + TRGT_DATA_PROT_WD/(CX_NW/4): CX_RASDP_FORMQ_RAM_ECC_WD) : 0)$
CX_RASDP_FORMQ_RAM_ECC_WD	$(CX_RAS_EN ? ((CX_RASDP >2 \&\& CX_RASDP_EN==1)? 6 : 22) : 0)$
CX_RASDP_HDRQ_ERR_SYND_WD	$(CX_RASDP_EN \&\& CX_TLP_PREFIX_ENABLE \&\& CX_RAS_PROT_TYPE==0) ? CX_RADM_SBUF_HDRQ_PROT_WD + CX_RX_PRFX_PAR_WD : CX_RADM_SBUF_HDRQ_PROT_WD$
CX_RASDP_TRGT1_HDR_PROT_WD	$(CX_RASDP_EN==1)? [function_of: CX_RASDP_TRGT1_HDR_WD CX_RAS_PROT_RANGE RAS_BASE_PROT_WD] : 0$
CX_RASDP_TRGT1_HDR_WD	$CX_RASDP_EN ? (CX_IS_SW ? CX_ATU_CTRL_EN \&\& (CX_RADMQ_MODE==2) ? 51 + FLT_Q_ADDR_WIDTH : 50 + FLT_Q_ADDR_WIDTH : CX_IS_RC ? CX_ATU_CTRL_EN \&\& (CX_RADMQ_MODE==2) ? 52 + FLT_Q_ADDR_WIDTH : 51 + FLT_Q_ADDR_WIDTH : CX_ATU_CTRL_EN \&\& (CX_RADMQ_MODE==2) ? 56 + FLT_Q_ADDR_WIDTH : 55 + FLT_Q_ADDR_WIDTH) + CX_TAG_SIZE + (ATS_RX_ENABLE_VALUE*2) + FLT_Q_ADDR_WIDTH + (CX_TPH_ENABLE_VALUE*11) + ((CX_NVFUNC_EWD+1)*CX_SRIOV_ENABLE_VALUE) + CX_NVFUNC_WD + FLT_Q_ATTR_WIDTH + CX_LN_VALUE : 0$
CX_RAS_EN	$((CX_RASDP==0)\&\& (CX_RASDP_RAM_PROT==0))? 0 : 1$
CX_RAS_EXT_IF	0
CX_RAS_PCIE_HDR_PROT_WD	$((CX_RASDP_EN==1)? [function_of: CX_RAS_PROT_RANGE RAS_BASE_PROT_WD] : 0)$
CX_RAS_PROT_RANGE	$(CX_RAS_PROT_TYPE==0)? ((CX_NW>1)? 64 : 32) : 8$
CX_RAS_PROT_TYPE	$((CX_RASDP ==1) (CX_RASDP ==2))$
CX_RAS_RADM_P_HWD	$(CX_RASDP_EN \&\& CX_TLP_PREFIX_ENABLE)? ((CX_RAS_PROT_TYPE==0) ? RADM_P_HWD : RADM_P_HWD-CX_RX_PRFX_PAR_WD) : RADM_P_HWD$
CX_RAS_RADM_P_HWD_TOPROT	$(CX_RASDP_EN \&\& CX_TLP_PREFIX_ENABLE)? ((CX_RAS_PROT_TYPE==0) ? RADM_P_HWD-FLT_Q_PRFX_WIDTH : RADM_P_HWD-CX_RX_PRFX_PAR_WD) : RADM_P_HWD$

Parameter Name	Equals To
CX_RAS_RAM_PROT_RANGE	((CX_NW>1)? 64: 32)
CX_RAS_SOTBUF_PROT_WD	(CX_RAS_EN==1)? ([function_of: CX_RBUF_DEPTH] > 26)? 7: 6): 0
CX_RBARS_INCLUDED	[function_of:]
CX_RBUF_CTRL_WD	(CX_TLP_PREFIX_ENABLE && CX_GEN3_SPEED==1)? (11+CX_NW+CX_RBUF_SOT_WD): (CX_NW+CX_RBUF_SOT_WD)
CX_RBUF_SOT_WD	(CX_NW>2)? ((CX_NW==16)? 5 : (CX_NW==8)? CX_NW/2: CX_NW-1): CX_NW
CX_RCB	(CC_DEVICE_TYPE == CC_RC) ? 128 : 64
CX_REMOTE_LOOKUPID_WD	(CC_DMA_ENABLE) ? 8 : (CX_10BITS_TAG) ? 10 : 8
CX_REMOTE_LUT_PTR_WIDTH	[function_of: CX_REMOTE_MAX_TAG]
CX_REMOTE_MAX_TAGS	CX_REMOTE_MAX_TAG+1
CX_REMOTE_MAX_TAGS_WD	[function_of: CX_REMOTE_MAX_TAGS]
CX_RET_CORE_CLK_UG_EN	(CX_PIPE_REGIF_L1_PG_ALWAYS_ON && (!CX_FREQ_STEP_EN && !CX_FREQ_STEP_DL_EN))
CX_RET_CORE_PL_CLK_UG_EN	(CX_PIPE_REGIF_L1_PG_ALWAYS_ON && (!CX_FREQ_STEP_EN && CX_FREQ_STEP_DL_EN))
CX_RET_PIPE_CLK_EN	(CX_PIPE_REGIF_L1_PG_ALWAYS_ON && CX_FREQ_STEP_EN)
CX_RP_PIO_IMPSPEC	0
CX_RTLH_SIMPLE_EXTRACT	CX_RTLH_SIMPLE_EXTRACT_ENABLE
CX_RTLH_SIMPLE_EXTRACT_ENABLE	CX_NW==16
CX_RX_HEADER_RSVD_ENABLE	CX_HEADER_RSVD_ENABLE==1 CX_CCIX_INTERFACE_ENABLE==1
CX_RXMARGIN_RATE_MODE_ENABLE	0
CX_RX_NDLLP	(CX_NW >> 1 == 0) ? 1 : (CX_NW >> 1)
CX_RX_PRFX_PAR_WD	(CX_RASDP_EN) ? [function_of: CX_NW CX_NPRFX CX_RAS_PROT_RANGE CX_RAS_PROT_TYPE] : 0
CX_RX_TLP	(CX_NW==16) ? 4 : (CX_NW==8) ? 2 : 1
CXSCNTLCHKWIDTH	CX_CXS_CNTLCHKWIDTH
CXSCNTLWIDTH	CX_CXS_CNTLWIDTH

Parameter Name	Equals To
CX_S_CPCIE_MODE	((CX_PCIE_MODE == SINGLE_CPCIE) (CX_PCIE_MODE == PCIE_OVER_CIO))
CXSDATACHKWIDTH	CX_CXS_DATACHKWIDTH
CXSDATAFLITWIDTH	CX_CXS_DATAFLITWIDTH
CX_SEL_PHY_MODE	CX_PCIE_MODE == DUAL_CMPCIE
CX_SERDES_ARCH_FPGA	(DWC_PCIE_FPGA && CX_PIPE_SERDES_ARCH)
CX_SERDES_ARCH_RX_REGIN	1
CX_SERDES_ARCH_SYNC_TYPE	3
CX_SERDES_ARCH_TX_REGOUT	1
CXS_FLIT_EQUALS_DATA_WIDTH	(CX_CXS_DATAFLITWIDTH==CX_CXS_CCIXDATAWIDTH)
CXS_FLIT_LOWER_DATA_WIDTH	(CX_CXS_DATAFLITWIDTH<CX_CXS_CCIXDATAWIDTH)
CX_SINGLE_FUNC	(CX_NFUNC == 1)
CX_SKEW_MAC_IMPL_CCIX20G_CORECLK	2
CX_SKEW_MAC_IMPL_CCIX25G_CORECLK	2
CX_SKEW_MAC_IMPL_GEN1_CORECLK	(CX_NB_GEN1 == 4 CX_NB_GEN1 == 2) ? 3 : 2
CX_SKEW_MAC_IMPL_GEN2_CORECLK	(CX_NB_GEN2 == 4 CX_NB_GEN2 == 2) ? 3 : 2
CX_SKEW_MAC_IMPL_GEN3_CORECLK	2
CX_SKEW_MAC_IMPL_GEN4_CORECLK	2
CX_SKEW_MAC_IMPL_GEN5_CORECLK	2
CX_SKEW_PIPEIF_CCIX20G_CORECLK	(CX_DESKEW_DEPTH_CPCIE - CX_SKEW_MAC_IMPL_CCIX20G_CORECLK - CX_SKEW_MAC_MARGIN_CORECLK)
CX_SKEW_PIPEIF_CCIX20G_PCLK	(CX_SKEW_PIPEORGIF_CCIX20G_PCLK - CX_SKEW_SERDESPIPE_IMPL_CCIX20G_PCLK)
CX_SKEW_PIPEIF_CCIX25G_CORECLK	(CX_DESKEW_DEPTH_CPCIE - CX_SKEW_MAC_IMPL_CCIX25G_CORECLK - CX_SKEW_MAC_MARGIN_CORECLK)
CX_SKEW_PIPEIF_CCIX25G_PCLK	(CX_SKEW_PIPEORGIF_CCIX25G_PCLK - CX_SKEW_SERDESPIPE_IMPL_CCIX25G_PCLK)
CX_SKEW_PIPEIF_GEN1_CORECLK	(CX_DESKEW_DEPTH_CPCIE - CX_SKEW_MAC_IMPL_GEN1_CORECLK - CX_SKEW_MAC_MARGIN_CORECLK)

Parameter Name	Equals To
CX_SKEW_PIPEIF_GEN1_PCLK	(CX_SKEW_PIPEORGIF_GEN1_PCLK - CX_SKEW_SERDESPIPE_IMPL_E_GEN1_PCLK)
CX_SKEW_PIPEIF_GEN2_CORECLK	(CX_DESKEW_DEPTH_CPCIE - CX_SKEW_MAC_IMPL_E_GEN2_CORECLK - CX_SKEW_MAC_MARGIN_CORECLK)
CX_SKEW_PIPEIF_GEN2_PCLK	(CX_SKEW_PIPEORGIF_GEN2_PCLK - CX_SKEW_SERDESPIPE_IMPL_E_GEN2_PCLK)
CX_SKEW_PIPEIF_GEN3_CORECLK	(CX_DESKEW_DEPTH_CPCIE - CX_SKEW_MAC_IMPL_E_GEN3_CORECLK - CX_SKEW_MAC_MARGIN_CORECLK)
CX_SKEW_PIPEIF_GEN3_PCLK	(CX_SKEW_PIPEORGIF_GEN3_PCLK - CX_SKEW_SERDESPIPE_IMPL_E_GEN3_PCLK)
CX_SKEW_PIPEIF_GEN4_CORECLK	(CX_DESKEW_DEPTH_CPCIE - CX_SKEW_MAC_IMPL_E_GEN4_CORECLK - CX_SKEW_MAC_MARGIN_CORECLK)
CX_SKEW_PIPEIF_GEN4_PCLK	(CX_SKEW_PIPEORGIF_GEN4_PCLK - CX_SKEW_SERDESPIPE_IMPL_E_GEN4_PCLK)
CX_SKEW_PIPEIF_GEN5_CORECLK	(CX_DESKEW_DEPTH_CPCIE - CX_SKEW_MAC_IMPL_E_GEN5_CORECLK - CX_SKEW_MAC_MARGIN_CORECLK)
CX_SKEW_PIPEIF_GEN5_PCLK	(CX_SKEW_PIPEORGIF_GEN5_PCLK - CX_SKEW_SERDESPIPE_IMPL_E_GEN5_PCLK)
CX_SKEW_PIPEORGIF_CCIX20G_PCLK	[function_of: CX_SKEW_PIPEIF_CCIX20G_CORECLK CX_PHY_NB_GEN4 CX_NB_GEN4 1 CX_PIPE_SERDES_ARCH]
CX_SKEW_PIPEORGIF_CCIX25G_PCLK	[function_of: CX_SKEW_PIPEIF_CCIX25G_CORECLK CX_PHY_NB_GEN4 CX_NB_GEN4 1 CX_PIPE_SERDES_ARCH]
CX_SKEW_PIPEORGIF_GEN1_PCLK	[function_of: CX_SKEW_PIPEIF_GEN1_CORECLK CX_PHY_NB_GEN1 CX_NB_GEN1 CX_PHY_GEN1_DT_RATE CX_PIPE_SERDES_ARCH]
CX_SKEW_PIPEORGIF_GEN2_PCLK	[function_of: CX_SKEW_PIPEIF_GEN2_CORECLK CX_PHY_NB_GEN2 CX_NB_GEN2 CX_PHY_GEN2_DT_RATE CX_PIPE_SERDES_ARCH]
CX_SKEW_PIPEORGIF_GEN3_PCLK	[function_of: CX_SKEW_PIPEIF_GEN3_CORECLK CX_PHY_NB_GEN3 CX_NB_GEN3 CX_PHY_GEN3_DT_RATE CX_PIPE_SERDES_ARCH]
CX_SKEW_PIPEORGIF_GEN4_PCLK	[function_of: CX_SKEW_PIPEIF_GEN4_CORECLK CX_PHY_NB_GEN4 CX_NB_GEN4 CX_PHY_GEN4_DT_RATE CX_PIPE_SERDES_ARCH]

Parameter Name	Equals To
CX_SKEW_PIPEORGIF_GEN5_PCLK	[function_of: CX_SKEW_PIPEIF_GEN5_CORECLK CX_PHY_NB_GEN5 CX_NB_GEN5 CX_PHY_GEN5_DT_RATE CX_PIPE_SERDES_ARCH]
CX_SKEW_SERDESPIPE_IMPLER_CCIX20G_PCLK	(CX_PIPE_SERDES_ARCH==0) ? 0 : (CX_PHY_NB_GEN4 == 1) ? 9 : (CX_PHY_NB_GEN4 == 2) ? 6 : (CX_PHY_NB_GEN4 == 4) ? 5 : (CX_PHY_NB_GEN4 == 8) ? 5 : 5
CX_SKEW_SERDESPIPE_IMPLER_CCIX25G_PCLK	(CX_PIPE_SERDES_ARCH==0) ? 0 : (CX_PHY_NB_GEN4 == 1) ? 9 : (CX_PHY_NB_GEN4 == 2) ? 6 : (CX_PHY_NB_GEN4 == 4) ? 5 : (CX_PHY_NB_GEN4 == 8) ? 5 : 5
CX_SKEW_SERDESPIPE_IMPLER_GEN1_PCLK	(CX_PIPE_SERDES_ARCH==0) ? 0 : (CX_PHY_NB_GEN1 == 1) ? 5 : (CX_PHY_NB_GEN1 == 2) ? 5 : 5
CX_SKEW_SERDESPIPE_IMPLER_GEN2_PCLK	(CX_PIPE_SERDES_ARCH==0) ? 0 : (CX_PHY_NB_GEN2 == 1) ? 5 : (CX_PHY_NB_GEN2 == 2) ? 5 : 5
CX_SKEW_SERDESPIPE_IMPLER_GEN3_PCLK	(CX_PIPE_SERDES_ARCH==0) ? 0 : (CX_PHY_NB_GEN3 == 1) ? 9 : (CX_PHY_NB_GEN3 == 2) ? 6 : (CX_PHY_NB_GEN3 == 4) ? 5 : (CX_PHY_NB_GEN3 == 8) ? 5 : 5
CX_SKEW_SERDESPIPE_IMPLER_GEN4_PCLK	(CX_PIPE_SERDES_ARCH==0) ? 0 : (CX_PHY_NB_GEN4 == 1) ? 9 : (CX_PHY_NB_GEN4 == 2) ? 6 : (CX_PHY_NB_GEN4 == 4) ? 5 : (CX_PHY_NB_GEN4 == 8) ? 5 : 5
CX_SKEW_SERDESPIPE_IMPLER_GEN5_PCLK	(CX_PIPE_SERDES_ARCH==0) ? 0 : (CX_PHY_NB_GEN5 == 1) ? 9 : (CX_PHY_NB_GEN5 == 2) ? 6 : (CX_PHY_NB_GEN5 == 4) ? 5 : (CX_PHY_NB_GEN5 == 8) ? 5 : 5
CX_SLV_ISO_EN	(SLV_CLK_DIFF_ENABLE==1 && CX_ENHANCED_PM_EN==1)
CXSMAXPKTPERFLIT	2
CX_S_MPCIE_MODE	CX_PCIE_MODE == SINGLE_MPCIE
CX_SRIOV_ENABLE_VALUE	CX_SRIOV_ENABLE
CXS_RX_BUFF_ADDRW	CX_CXS_RX_BUFF_ADDRW
CXS_RX_BUFF_DATAW	CX_CXS_RX_BUFF_DATAW
CXSRXREPWIDTH	CX_CXS_RX_REPLICATION
CXS_TX_BUFF_ADDRW	CX_CXS_TX_BUFF_ADDRW

Parameter Name	Equals To
CXS_TX_BUFF_DATAW	CX_CXS_TX_BUFF_DATAW
CXSTXREPWIDTH	CX_CXS_TX_REPLICATION
CX_SYS_INT_GRANT_EN	(CX_IS_RC!=1)
CX_TAG_SIZE	(CX_10BITS_TAG== 1)? 10: 8
CX_TLP_PREFIX_ENABLE	(CX_NPRFX == 0) ? 0 : 1
CX_TLP_PREFIX_ENABLE_VALUE	CX_TLP_PREFIX_ENABLE
CX_TPH_ENABLE_VALUE	CX_TPH_ENABLE
CX_TX_HW_W_PAR	128+CX_RAS_PCIE_HDR_PROT_WD
CX_VF_RBARS_INCLUDED	[function_of:]
CX_WIRE_SKEW_CCIX20G_PCLK	[function_of: CX_WIRE_SKEW_CCIX20G_NS CX_PHY_NB_GEN4 1 50000 8 CX_PIPE_SERDES_ARCH]
CX_WIRE_SKEW_CCIX25G_PCLK	[function_of: CX_WIRE_SKEW_CCIX25G_NS CX_PHY_NB_GEN4 1 40000 8 CX_PIPE_SERDES_ARCH]
CX_WIRE_SKEW_GEN1_PCLK	[function_of: CX_WIRE_SKEW_GEN1_NS CX_PHY_NB_GEN1 CX_PHY_GEN1_DT_RATE 400000 10 CX_PIPE_SERDES_ARCH]
CX_WIRE_SKEW_GEN2_PCLK	[function_of: CX_WIRE_SKEW_GEN2_NS CX_PHY_NB_GEN2 CX_PHY_GEN2_DT_RATE 200000 10 CX_PIPE_SERDES_ARCH]
CX_WIRE_SKEW_GEN3_PCLK	[function_of: CX_WIRE_SKEW_GEN3_NS CX_PHY_NB_GEN3 CX_PHY_GEN3_DT_RATE 125000 8 CX_PIPE_SERDES_ARCH]
CX_WIRE_SKEW_GEN4_PCLK	[function_of: CX_WIRE_SKEW_GEN4_NS CX_PHY_NB_GEN4 CX_PHY_GEN4_DT_RATE 62500 8 CX_PIPE_SERDES_ARCH]
CX_WIRE_SKEW_GEN5_PCLK	[function_of: CX_WIRE_SKEW_GEN5_NS CX_PHY_NB_GEN5 CX_PHY_GEN5_DT_RATE 31250 8 CX_PIPE_SERDES_ARCH]
CX_XADM_CLI_INTF_DEDICATED	CX_MULTISTREAM
CX_XADM_VC_ARB_WRR	CX_MULTISTREAM
DATA_PAR_WD	TRGT_DATA_PROT_WD
DBI_SLV_ADDR_WD	CC_DBI_SLV_BUS_ADDR_WIDTH
DBI_SLV_AR_USER_WIDTH	CC_DBI_SLV_AR_USER_WIDTH

Parameter Name	Equals To
DBI_SLV_AW_USER_WIDTH	CC_DBI_SLV_AW_USER_WIDTH
DBISLV_BURST_LEN_PW	CC_DBISLV_BURST_LEN_PW
DBI_SLV_B_USER_WIDTH	CC_DBI_SLV_B_USER_WIDTH
DBISLV_CLK_DIFF_ENABLE	((AXI_POPULATED AHB_CLK_DIFF_ENABLE) && DBI_4SLAVE_POPULATED)
DBI_SLV_DATA_WD	CC_DBI_SLV_BUS_DATA_WIDTH
DBI_SLV_ID_WD	CC_DBI_SLV_ID_WD
DBI_SLV_LOCK_WD	CC_AXI_LOCK_WIDTH
DBI_SLV_R_USER_WIDTH	CC_DBI_SLV_R_USER_WIDTH
DBI_SLV_W_USER_WIDTH	CC_DBI_SLV_W_USER_WIDTH
DC_DMA_CTX_RAM_DATA_WECC_WIDTH	(CX_RAS_EN==1 ? [function_of: DC_DMA_CTX_RAM_DATA_WIDTH CX_RAS_RAM_PROT_RANGE] + DC_DMA_CTX_RAM_DATA_WIDTH : DC_DMA_CTX_RAM_DATA_WIDTH)
DC_DMA_CTX_RAM_DATA_WIDTH	((CX_SRIOV_ENABLE==1) (CX_TPH_ENABLE==1)) ? 288 : 256)
DC_DMA_TYPE_WIDTH	3
DC_INBD_HEADER_PNP_1STG_WD	CX_LN_ENABLE + 3*ATS_RX_ENABLE_VALUE + CC_AMBA_TLP_SEQ_WD + CC_CORE_ADDR_BUS_WD + CC_DMA_ENABLE_VALUE*DC_DMA_TYPE_WIDTH + CC_DMA_ENABLE_VALUE*CC_NUM_DMA_MAX_CHAN _WIDTH + CX_MSI_CTRL_EN_VALUE + CX_NFUNC_WD + CX_NW + CX_SRIOV_ENABLE_VALUE*(CX_NVFUNC_EWD - 1) + 2*CX_SRIOV_ENABLE_VALUE + CX_TLP_PREFIX_ENABLE_VALUE*(FLT_Q_PRFX_WIDT H - 1) + CX_TLP_PREFIX_ENABLE_VALUE + 11*CX_TPH_ENABLE_VALUE + RADM_SEG_BUF_VALUE*DC_MULTI_VC_VALUE*(CX_N VC_LOG2 - 1) + RADM_SEG_BUF_VALUE*DC_MULTI_VC_VALUE + FLT_Q_ATTR_WIDTH + RADM_SEG_BUF_VALUE + CX_TAG_SIZE + 59 + (CX_INTERNAL_ATU_ENABLE*RADM_SEG_BUF_VALU E) + CX_REMOTE_MAX_TAGS_WD

Parameter Name	Equals To
DC_INBD_HEADER_PNP_1STG_WD_NP	$ \begin{aligned} &CX_LN_ENABLE + 3*ATS_RX_ENABLE_VALUE + \\ &CC_AMBA_TLP_SEQ_WD + FLT_Q_ADDR_WIDTH + \\ &CC_DMA_ENABLE_VALUE*DC_DMA_TYPE_WIDTH + \\ &CC_DMA_ENABLE_VALUE*CC_NUM_DMA_MAX_CHAN_ \\ &_WIDTH + CX_MSI_CTRL_EN_VALUE + CX_NFUNC_WD \\ &+ CX_NW + \\ &CX_SRIOV_ENABLE_VALUE*(CX_NVFUNC_EWD - 1) + \\ &2*CX_SRIOV_ENABLE_VALUE + \\ &CX_TLP_PREFIX_ENABLE_VALUE*(FLT_Q_PRFX_WIDT \\ &H - 1) + CX_TLP_PREFIX_ENABLE_VALUE + \\ &11*CX_TPH_ENABLE_VALUE + \\ &RADM_SEG_BUF_VALUE*DC_MULTI_VC_VALUE*(CX_N \\ &VC_LOG2 - 1) + \\ &RADM_SEG_BUF_VALUE*DC_MULTI_VC_VALUE + \\ &FLT_Q_ATTR_WIDTH + RADM_SEG_BUF_VALUE + \\ &CX_TAG_SIZE + 59 + \\ &(CX_INTERNAL_ATU_ENABLE*RADM_SEG_BUF_VALU \\ &E)+ CX_REMOTE_MAX_TAGS_WD \end{aligned} $
DC_INBD_HEADER_PNP_1STG_WD_NP_RAM	$ \begin{aligned} &DC_INBD_HEADER_PNP_1STG_WD_NP - (13 + \\ &!CX_FLT_Q_ADDR_GT_32 + CX_NW + RADM_SEG_BUF \\ &+ CX_MSI_CTRL_EN + \\ &CC_EDMA_SINGLE_CHANNEL*CC_DMA_ENABLE + \\ &CX_ATU_CTRL_EN*RADM_SEG_BUF + \\ &(CX_SINGLE_FUNC ? CX_NFUNC_WD : \\ &CX_NFUNC_WD-CX_NFUNC_WD_OPT) + \\ &(ATS_RX_ENABLE*(!CC_AMBA_ATS_NW))) \end{aligned} $
DC_INBD_HEADER_PNP_1STG_WD_RAM	$ \begin{aligned} &DC_INBD_HEADER_PNP_1STG_WD - (13 + \\ &!CX_FLT_Q_ADDR_GT_32 + CX_NW + RADM_SEG_BUF \\ &+ CX_MSI_CTRL_EN + \\ &CC_EDMA_SINGLE_CHANNEL*CC_DMA_ENABLE + \\ &CX_ATU_CTRL_EN*RADM_SEG_BUF + \\ &(CX_SINGLE_FUNC ? CX_NFUNC_WD : \\ &CX_NFUNC_WD-CX_NFUNC_WD_OPT) + \\ &(ATS_RX_ENABLE*(!CC_AMBA_ATS_NW))) \end{aligned} $

Parameter Name	Equals To
DC_INBD_HEADER_PNP_WD_NP	$ \begin{aligned} &CX_LN_ENABLE + 3*ATS_RX_ENABLE_VALUE + \\ &CC_AMBA_TLP_SEQ_WD + FLT_Q_ADDR_WIDTH + \\ &CC_DMA_ENABLE_VALUE*DC_DMA_TYPE_WIDTH + \\ &CC_DMA_ENABLE_VALUE*CC_NUM_DMA_MAX_CHAN_ \\ &WIDTH + CC_MSTR_BE_CTRL_WIDTH + \\ &CX_MSI_CTRL_EN_VALUE + CX_NFUNC_WD + CX_NW \\ &+ CX_SRIOV_ENABLE_VALUE*(CX_NVFUNC_EWD - 1) + \\ &2*CX_SRIOV_ENABLE_VALUE + \\ &CX_TLP_PREFIX_ENABLE_VALUE*(FLT_Q_PRFX_WIDT \\ &H - 1) + CX_TLP_PREFIX_ENABLE_VALUE + \\ &11*CX_TPH_ENABLE_VALUE + \\ &RADM_SEG_BUF_VALUE*DC_MULTI_VC*(CX_NVC_LO \\ &G2 - 1) + RADM_SEG_BUF_VALUE*DC_MULTI_VC + \\ &FLT_Q_ATTR_WIDTH + RADM_SEG_BUF_VALUE + \\ &CX_TAG_SIZE + 73 + \\ &(CX_INTERNAL_ATU_ENABLE*RADM_SEG_BUF_VALU \\ &E) + CX_REMOTE_MAX_TAGS_WD \end{aligned} $
DC_INBD_HEADER_PNP_WD_NP_RAM	$ \begin{aligned} &DC_INBD_HEADER_PNP_WD_NP - (13 + \\ &!CX_FLT_Q_ADDR_GT_32 + CX_NW + RADM_SEG_BUF \\ &+ CX_MSI_CTRL_EN + \\ &CC_EDMA_SINGLE_CHANNEL*CC_DMA_ENABLE + \\ &CX_ATU_CTRL_EN*RADM_SEG_BUF + \\ &(CX_SINGLE_FUNC ? CX_NFUNC_WD : \\ &CX_NFUNC_WD - CX_NFUNC_WD_OPT) + \\ &(13 - CC_MSTR_MTU_PW) + \\ &CC_AXI_NCBE_NODECOMP*(CC_MSTR_BE_CTRL_WI \\ &DTH + 1) + (ATS_RX_ENABLE*(!CC_AMBA_ATS_NW)) + \\ &CC_AXI_NCBE_NODECOMP) \end{aligned} $
DC_MULTI_VC	$(CX_NVC > 1)$
DC_MULTI_VC_VALUE	DC_MULTI_VC
DCRD_WD	$(SCALED_FC_SUPPORTED ? 16 : 12)$
DEFAULT_ACK_FREQUENCY	0x0
DEFAULT_AP_ENABLE	CX_AP_SUPPORT
DEFAULT_AP_SUPPORT	CX_AP_SUPPORT
DEFAULT_CURSOR_P0	$(DEFAULT_GEN3_EQ_LOCAL_FS - FS_DIV_4)$
DEFAULT_CXL_IO_ENABLE	$(CX_CXL_CACHE_EN \parallel CX_CXL_MEM_EN)$
DEFAULT_CXL_PORT_USP_DRIFT_BUFFER_ENABLE	0
DEFAULT_CXL_PORT_USP_SYNC_HDR_BYPASS_ENABLE	0
DEFAULT_DO_DESKEW_FOR_SRIS	CX_SRIS_SUPPORT

Parameter Name	Equals To
DEFAULT_DPA_PAS	0x1
DEFAULT_DPA_TLIB	0x0
DEFAULT_DPA_TLUNIT	0x1
DEFAULT_DPA_XCLY0	0x80
DEFAULT_DPA_XCLY1	0xff
DEFAULT_DSP_16G_TX_PRESET1	(DEFAULT_DSP_16G_TX_PRESET0)
DEFAULT_DSP_16G_TX_PRESET2	(DEFAULT_DSP_16G_TX_PRESET0)
DEFAULT_DSP_16G_TX_PRESET3	(DEFAULT_DSP_16G_TX_PRESET0)
DEFAULT_DSP_32G_TX_PRESET1	(DEFAULT_DSP_32G_TX_PRESET0)
DEFAULT_DSP_32G_TX_PRESET2	(DEFAULT_DSP_32G_TX_PRESET0)
DEFAULT_DSP_32G_TX_PRESET3	(DEFAULT_DSP_32G_TX_PRESET0)
DEFAULT_DSP_RX_PRESET_HINT1	(DEFAULT_DSP_RX_PRESET_HINT0)
DEFAULT_DSP_TX_PRESET1	(DEFAULT_DSP_TX_PRESET0)
DEFAULT_EQ_REQ_NUM	1
DEFAULT_ESM_DSP_TX_PRESET_20G1	(DEFAULT_ESM_DSP_TX_PRESET_20G0)
DEFAULT_ESM_DSP_TX_PRESET_20G2	(DEFAULT_ESM_DSP_TX_PRESET_20G0)
DEFAULT_ESM_DSP_TX_PRESET_20G3	(DEFAULT_ESM_DSP_TX_PRESET_20G0)
DEFAULT_ESM_DSP_TX_PRESET_25G1	(DEFAULT_ESM_DSP_TX_PRESET_25G0)
DEFAULT_ESM_DSP_TX_PRESET_25G2	(DEFAULT_ESM_DSP_TX_PRESET_25G0)
DEFAULT_ESM_DSP_TX_PRESET_25G3	(DEFAULT_ESM_DSP_TX_PRESET_25G0)
DEFAULT_ESM_USP_TX_PRESET_20G1	(DEFAULT_ESM_USP_TX_PRESET_20G0)
DEFAULT_ESM_USP_TX_PRESET_20G2	(DEFAULT_ESM_USP_TX_PRESET_20G0)
DEFAULT_ESM_USP_TX_PRESET_20G3	(DEFAULT_ESM_USP_TX_PRESET_20G0)
DEFAULT_ESM_USP_TX_PRESET_25G1	(DEFAULT_ESM_USP_TX_PRESET_25G0)
DEFAULT_ESM_USP_TX_PRESET_25G2	(DEFAULT_ESM_USP_TX_PRESET_25G0)
DEFAULT_ESM_USP_TX_PRESET_25G3	(DEFAULT_ESM_USP_TX_PRESET_25G0)
DEFAULT_EXT_TAG_FIELD_SUPPORTED_0	(CX_MAX_TAG>31) ? 1 : 0
DEFAULT_EXT_TAG_FIELD_SUPPORTED_1	DEFAULT_EXT_TAG_FIELD_SUPPORTED_0
DEFAULT_EXT_TAG_FIELD_SUPPORTED_2	DEFAULT_EXT_TAG_FIELD_SUPPORTED_0

Parameter Name	Equals To
DEFAULT_EXT_TAG_FIELD_SUPPORTED_3	DEFAULT_EXT_TAG_FIELD_SUPPORTED_0
DEFAULT_EXT_TAG_FIELD_SUPPORTED_4	DEFAULT_EXT_TAG_FIELD_SUPPORTED_0
DEFAULT_EXT_TAG_FIELD_SUPPORTED_5	DEFAULT_EXT_TAG_FIELD_SUPPORTED_0
DEFAULT_EXT_TAG_FIELD_SUPPORTED_6	DEFAULT_EXT_TAG_FIELD_SUPPORTED_0
DEFAULT_EXT_TAG_FIELD_SUPPORTED_7	DEFAULT_EXT_TAG_FIELD_SUPPORTED_0
DEFAULT_EXT_TAG_FIELD_SUPPORTED_8	DEFAULT_EXT_TAG_FIELD_SUPPORTED_0
DEFAULT_EXT_TAG_FIELD_SUPPORTED_9	DEFAULT_EXT_TAG_FIELD_SUPPORTED_0
DEFAULT_FAST_LINK_SCALING_FACTOR	(CX_PCIE_MODE == PCIE_OVER_CIO) ? 2 : 0
DEFAULT_GEN2_TXSWING	0
DEFAULT_GEN3_AUTONOMOUS_EQ_REQ_DISABLE	0
DEFAULT_GEN3_EQ_EVAL_2MS_DISABLE	1
DEFAULT_GEN3_LOWER_RATE_EQ_REDO_ENABLE	CX_GEN3_SPEED
DEFAULT_GEN3_REQ_RST_EIEOS_CNT_EQ_DISABLE	0
DEFAULT_GEN3_REQ_SEND_CONSECUTIVE_EIEOS_F OR_PSET_MAP	1
DEFAULT_GEN3_RXEQ_PH01_EN	0
DEFAULT_GEN3_ZRXDC_NONCOMPL	0x1
DEFAULT_GEN4_USP_SEND_8GT_EQ_TS2_DISABLE	1
DEFAULT_GEN5_USP_SEND_16GT_EQ_TS2_DISABLE	0
DEFAULT_LANE_SKEW_OFF_26	CX_PIPE_HYBRID_MODE
DEFAULT_LOCAL_G3_TX_PRESET	0x4
DEFAULT_LOCAL_G4_TX_PRESET	0x4
DEFAULT_LOCAL_G5_TX_PRESET	0x4
DEFAULT_MAX_COEF_FOM_VECTOR_DEPTH	0x8
DEFAULT_NO_SEED_VALUE_CHANGE	0x0
DEFAULT_PHANTOM_FUNC_SUPPORTED_0	0x0
DEFAULT_PHY_CONTROL	[function_of:] ? 0x1 : 0x0
DEFAULT_PIPE_GARBAGE_DATA_MODE	(RIO_POPULATED==1) ? 1 : 0
DEFAULT_ROM_BAR_ENABLED_0	ROM_BAR_ENABLED_0

Parameter Name	Equals To
DEFAULT_RXMARGIN_RATE_MODE	CX_RXMARGIN_RATE_MODE_ENABLE
DEFAULT_RX_SERIALIZATION_Q_ALMOST_FULL_THRESHOLD	CX_RADM_SERQ_AF_NUM_QDWS
DEFAULT_SELECTABLE_DEEMPH_BIT_MUX	0x0
DEFAULT_SELECT_DEEMPH_VAR_MUX	0x0
DEFAULT_SUPPORT_FINITE_EQ_REQUEST	0
DEFAULT_SUPPORT_PART_LANES_RXEI_EXIT_IN_POLL_ACTIVE	0x0
DEFAULT_USP_16G_TX_PRESET1	(DEFAULT_USP_16G_TX_PRESET0)
DEFAULT_USP_16G_TX_PRESET2	(DEFAULT_USP_16G_TX_PRESET0)
DEFAULT_USP_16G_TX_PRESET3	(DEFAULT_USP_16G_TX_PRESET0)
DEFAULT_USP_32G_TX_PRESET1	(DEFAULT_USP_32G_TX_PRESET0)
DEFAULT_USP_32G_TX_PRESET2	(DEFAULT_USP_32G_TX_PRESET0)
DEFAULT_USP_32G_TX_PRESET3	(DEFAULT_USP_32G_TX_PRESET0)
DEFAULT_USP_RX_PRESET_HINT1	(DEFAULT_USP_RX_PRESET_HINT0)
DEFAULT_USP_TX_PRESET1	(DEFAULT_USP_TX_PRESET0)
DEFAULT_VC_ARB_TABLE_OFFSET	CX_XADM_VC_ARB_WRR ? (0x70 >> 4) : 0
DEMUX_EXCLUSIVE_WIDTH	[function_of: FLT_Q_DW_LENGTH_WIDTH FLT_Q_IN_MEMBAR_RANGE_WIDTH FLT_Q_ROM_IN_RANGE_WIDTH FLT_Q_IO_REQ_IN_RANGE_WIDTH FLT_Q_BYTE_CNT_WIDTH FLT_Q_CMPLTR_ID_WIDTH FLT_Q_BCM_WIDTH FLT_Q_CPL_LOWER_ADDR_WIDTH FLT_Q_TD_WIDTH FLT_Q_EP_WIDTH FLT_Q_CPL_LAST_WIDTH]
DEV3_CAP_ENABLE	CX_DEF_MEM_WR_ENABLE
DEVNUM_WD	CX_DEVNUM_WD
DIRFEEDBACK_WD	6
DMA_CTX_RAM_ADDRP_WIDTH	CC_DMA_CTX_RAM_ADDRP_WIDTH
DMA_CTX_RAM_ADDR_WIDTH	CC_DMA_CTX_RAM_ADDR_WIDTH
DMA_CTX_RAM_DATA_WIDTH	DC_DMA_CTX_RAM_DATA_WECC_WIDTH
DMA_SEG_BUF_DATA_WIDTH	CC_DMA_SEG_BUF_DATA_WECC_WIDTH

Parameter Name	Equals To
DMA_SEG_BUF_NW_ADDRP_WIDTH	CC_DMA_SEG_BUF_NW_ADDRP_WIDTH
DMA_SEG_BUF_NW_ADDR_WIDTH	CC_DMA_SEG_BUF_NW_ADDR_WIDTH
DPC_PRFX_LOG_SIZE	CX_MAX_END2END_TLP_PRFXS_VALUE
DPC_PRFX_LOG_WD	(32*DPC_PRFX_LOG_SIZE)
DT_RATE_GEN1	0x1
DT_RATE_GEN2	0x1
DT_RATE_GEN3	0x1
DT_RATE_GEN4	0x1
DT_RATE_GEN5	0x1
DUAL_CMPCIE	2
DW	((CC_DEVICE_TYPE==CC_EP)?(RADM_PARBITS_OUT_VALUE?((32*NW)+DATA_PAR_WD):(32*NW)):(32*NW))
DW_W_PAR	((32*NW)+DATA_PAR_WD)
ECC_PROTECTION_EN	((CX_RASDP_EN==1) && (CX_RAS_PROT_TYPE==0))
EP_INTERFACE_ACTIVATED	((CC_DEVICE_TYPE==CC_DM) (CC_DEVICE_TYPE==CC_EP))
ERR_BUS_WD	CX_ERR_BUS_WD

Parameter Name	Equals To
EXT_TAG_SUPPORTED	DEFAULT_EXT_TAG_FIELD_SUPPORTED_0 DEFAULT_EXT_TAG_FIELD_SUPPORTED_1 DEFAULT_EXT_TAG_FIELD_SUPPORTED_2 DEFAULT_EXT_TAG_FIELD_SUPPORTED_3 DEFAULT_EXT_TAG_FIELD_SUPPORTED_4 DEFAULT_EXT_TAG_FIELD_SUPPORTED_5 DEFAULT_EXT_TAG_FIELD_SUPPORTED_6 DEFAULT_EXT_TAG_FIELD_SUPPORTED_7 DEFAULT_EXT_TAG_FIELD_SUPPORTED_8 DEFAULT_EXT_TAG_FIELD_SUPPORTED_9 DEFAULT_EXT_TAG_FIELD_SUPPORTED_10 DEFAULT_EXT_TAG_FIELD_SUPPORTED_11 DEFAULT_EXT_TAG_FIELD_SUPPORTED_12 DEFAULT_EXT_TAG_FIELD_SUPPORTED_13 DEFAULT_EXT_TAG_FIELD_SUPPORTED_14 DEFAULT_EXT_TAG_FIELD_SUPPORTED_15 DEFAULT_EXT_TAG_FIELD_SUPPORTED_16 DEFAULT_EXT_TAG_FIELD_SUPPORTED_17 DEFAULT_EXT_TAG_FIELD_SUPPORTED_18 DEFAULT_EXT_TAG_FIELD_SUPPORTED_19 DEFAULT_EXT_TAG_FIELD_SUPPORTED_20 DEFAULT_EXT_TAG_FIELD_SUPPORTED_21 DEFAULT_EXT_TAG_FIELD_SUPPORTED_22 DEFAULT_EXT_TAG_FIELD_SUPPORTED_23 DEFAULT_EXT_TAG_FIELD_SUPPORTED_24 DEFAULT_EXT_TAG_FIELD_SUPPORTED_25 DEFAULT_EXT_TAG_FIELD_SUPPORTED_26 DEFAULT_EXT_TAG_FIELD_SUPPORTED_27 DEFAULT_EXT_TAG_FIELD_SUPPORTED_28 DEFAULT_EXT_TAG_FIELD_SUPPORTED_29 DEFAULT_EXT_TAG_FIELD_SUPPORTED_30 DEFAULT_EXT_TAG_FIELD_SUPPORTED_31
EXT_VF_MSI_EN	EXT_VF_MSI_CAP_ENABLE
EXT_VF_PASID_ENABLE	0
EXT_VF_PM_CAP_ENABLE	0x0
EXT_VF_TPH_ENABLE	EXT_VF_TPH_ENABLE_VALUE
FLT_Q_ATTR_WIDTH	CX_IDO_ENABLE ? 3 : 2
FLT_Q_AT_WIDTH	ATS_RX_ENABLE ? 2 : 0
FLT_Q_BCM_WIDTH	1
FLT_Q_BYTE_CNT_WIDTH	12
FLT_Q_CCIX_WIDTH	CX_CCIX_INTERFACE_ENABLE ? CX_CCIX_RX_WD : 0
FLT_Q_CMPLTR_ID_WIDTH	16

Parameter Name	Equals To
FLT_Q_COMMON_WIDTH	[function_of: FLT_Q_DESTINATION_WIDTH FLT_Q_FMT_WIDTH FLT_Q_TYPE_WIDTH FLT_Q_TC_WIDTH FLT_Q_ATTR_WIDTH FLT_Q_LN_WIDTH FLT_Q_REQID_WIDTH FLT_Q_TAG_WIDTH FLT_Q_FUNC_NMBR_WIDTH FLT_Q_CPL_STATUS_WIDTH FLT_Q_VF_WIDTH FLT_Q_PRFX_WIDTH]
FLT_Q_CPL_LAST_WIDTH	1
FLT_Q_CPL_LOWER_ADDR_WIDTH	7
FLT_Q_CPL_STATUS_WIDTH	3
FLT_Q_DESTINATION_WIDTH	2
FLT_Q_DWADDR_WIDTH	[function_of: FLT_Q_ADDR_WIDTH]
FLT_Q_DW_LENGTH_WIDTH	10
FLT_Q_EP_WIDTH	1
FLT_Q_FMT_WIDTH	2
FLT_Q_FRSTDW_BE_WIDTH	4
FLT_Q_FUNC_NMBR_WIDTH	CX_ARI_ENABLE ? CX_NFUNC_WD : 3
FLT_Q_HDR_RSVD_DW0_WIDTH	CX_RX_HEADER_RSVD_ENABLE*8
FLT_Q_HDR_RSVD_DW2_WIDTH	CX_RX_HEADER_RSVD_ENABLE
FLT_Q_IN_MEMBAR_RANGE_WIDTH	(CC_DEVICE_TYPE==CC_RC CC_DEVICE_TYPE==CC_SW) ? 0 : [function_of: CC_DEVICE_TYPE CC_DM CC_EP]
FLT_Q_IO_REQ_IN_RANGE_WIDTH	1
FLT_Q_LN_WIDTH	CX_LN_ENABLE ? 1 : 0
FLT_Q_LSTDW_BE_WIDTH	4
FLT_Q_PRFX_WIDTH	CX_TLP_PREFIX_ENABLE ? CX_NPRFX*32 + CX_RX_PRFX_PAR_WD : 0
FLT_Q_REQID_WIDTH	16
FLT_Q_ROM_IN_RANGE_WIDTH	EP_INTERFACE_ACTIVATED==1
FLT_Q_SATA_WIDTH	SATA_CAP_ENABLE ? 1 : 0
FLT_Q_TAG_WIDTH	CX_TAG_SIZE
FLT_Q_TC_WIDTH	3
FLT_Q_TD_WIDTH	1

Parameter Name	Equals To
FLT_Q_TH_WIDTH	CX_TPH_ENABLE ? 1 : 0
FLT_Q_TYPE_WIDTH	5
FLT_Q_VF_WIDTH	CX_SRIOV_ENABLE ? CX_NVFUNC_NUM_WD : 0
FLUSH_CNTRL_ENABLE	(AXI_RADM_SEG_BUF_ENABLE CX_DPC_ENABLE)
FOMFEEDBACK_WD	8
FREQ_125	2
FREQ_250	1
FREQ_500	0
FREQ_62_5	3
FS_DIV_4	[function_of: DEFAULT_GEN3_EQ_LOCAL_FS]
FULL_SWING	0
FX_TLP	CX_FX_TLP
GEN2_DF	0
GEN2_DISABLED	2
GEN2_DW	1
GEN3_DF	0
GEN3_DISABLED	2
GEN3_DW	1
GEN4_DF	0
GEN4_DISABLED	2
GEN4_DW	1
HCRD_WD	(SCALED_FC_SUPPORTED?12:8)
HD_DATA_RAM_ADDRW	CX_CXL_SLV_HD_DATA_RAM_ADDRW
HD_DATA_RAM_DATAW	CX_CXL_SLV_HD_DATA_RAM_DATAW
HD_DATA_RX_RAM_ADDRW	CX_CXL_MST_HD_DATA_RAM_ADDRW
HD_DATA_RX_RAM_DATAW	CX_CXL_MST_HD_DATA_RAM_DATAW
HREQ_DRSP_RX_RAM_ADDRW	CX_CXL_MST_HREQ_DRESP_RAM_ADDRW
HREQ_DRSP_RX_RAM_DATAW	CX_CXL_MST_HREQ_DRESP_RAM_DATAW
HRSP_DREQ_RX_RAM_ADDRW	CX_CXL_MST_HRESP_DREQ_RAM_ADDRW

Parameter Name	Equals To
HRSP_DREQ_RX_RAM_DATAW	CX_CXL_MST_HRESP_DREQ_RAM_DATAW
IDLE	0x0
INIT	0xc
INTERNAL_VF_ENABLE	CX_SRIOV_ENABLE && (CX_INTERNAL_NVFUNC != 0)
INT_NVF	CX_INTERNAL_NVFUNC
L2	0x8
LBC_EXT_AW	CX_LBC_EXT_AW
LLCRD	0x0
LOOKUPID_WD	CX_REMOTE_LOOKUPID_WD
LTSSM_EMU_IN_WD	CX_LTSSM_EMU_WD
LTSSM_EMU_OUT_WD	CX_LTSSM_EMU_WD
MASTER_BUS_DATA_WIDTH_REG	[function_of: MASTER_BUS_DATA_WIDTH] - 5
MASTER_POPULATED	(AMBA_POPULATED)
MAX_LINK_SP	(CX_GEN5_SPEED ? 5 : CX_GEN4_SPEED ? 4 : CX_GEN3_SPEED ? 3 : CX_GEN2_SPEED ? 2 : 1)
M_C_CDC_FIFO_DEPTH	14
MS_DATA_RAM_ADDRW	CX_CXL_SLV_MS_DATA_RAM_ADDRW
MS_DATA_RAM_DATAW	CX_CXL_SLV_MS_DATA_RAM_DATAW
MS_DATA_RX_RAM_ADDRW	CX_CXL_MST_MS_DATA_RAM_ADDRW
MS_DATA_RX_RAM_DATAW	CX_CXL_MST_MS_DATA_RAM_DATAW
MSI_64_EN	MSI_CAP_ENABLE==1
MSI_PVM_EN_VALUE	MSI_PVM_EN
MSIX_EN	MSIX_CAP_ENABLE
MSIX_PBA_OFFSET_0_BASELINE	MSIX_TABLE_BIR_0 == MSIX_PBA_BIR_0 ? (MSIX_TABLE_OFFSET_0 << 3) + MSIX_TABLE_SIZE_0_BYTES : 0
MSIX_PBA_PW	CX_MSIX_PBA_PW

Parameter Name	Equals To
MSIX_TABLE_OFFSET_0_BASELINE	(!ENABLE_MEM_MAP_UNROLL_DMA_REG UNROLL_DMA_OFFSET_BAR != 0) && (!ENABLE_MEM_MAP_UNROLL_ATU_REG UNROLL_ATU_OFFSET_BAR != 0) ? 0 : (UNROLL_FUNC_NUM == 0 && UNROLL_BAR_NUM == MSIX_TABLE_BIR_0 ? (UNROLL_DMA_OFFSET_BAR > UNROLL_ATU_OFFSET_BAR ? UNROLL_DMA_OFFSET_BAR + UNROLL_DMA_SIZE : UNROLL_ATU_OFFSET_BAR + UNROLL_ATU_SIZE) : 0)
MSIX_TABLE_PW	CX_MSIX_TABLE_PW
MSIX_TABLE_SIZE_0_BYTES	(MSIX_TABLE_SIZE_0+1)*16
MS_NDR_RX_RAM_ADDRW	CX_CXL_MST_MS_NDR_RAM_ADDRW
MS_NDR_RX_RAM_DATAW	CX_CXL_MST_MS_NDR_RAM_DATAW
MSTR_ADDR_WD	MASTER_BUS_ADDR_WIDTH
MSTR_AR_USER_WIDTH	CC_MSTR_AR_USER_WIDTH
MSTR_AW_USER_WIDTH	CC_MSTR_AW_USER_WIDTH
MSTR_BURST_LEN_PW	CC_MSTR_BURST_LEN_PW
MSTR_B_USER_WIDTH	CC_MSTR_B_USER_WIDTH
MSTR_DATA_WD	MASTER_BUS_DATA_WIDTH
MSTR_ID_WD	CC_MSTR_BUS_ID_WIDTH
MSTR_LOCK_WD	CC_AXI_LOCK_WIDTH
MSTR_MISC_INFO_PW	CX_TAG_SIZE + 16 + CC_MSTR_MISC_INFO_REQID*16 + CC_MSTR_MISC_INFO_TAG*8
MSTR_RESP_MISC_INFO_PW	13
MSTR_R_USER_WIDTH	CC_MSTR_R_USER_WIDTH
MSTR_WSTRB_WD	CC_MSTR_BUS_WSTRB_WIDTH
MSTR_W_USER_WIDTH	CC_MSTR_W_USER_WIDTH
NDQ	CX_NDQ
NF	CX_NFUNC
NFUNC_GTR_THAN_32	(CX_NFUNC>32)
NHQ	CX_NHQ
NL	CX_NL
NONE	0x0

Parameter Name	Equals To
NPEM_CRITICAL_ARRAY_CAPABLE_VALUE	NPEM_ENABLE
NPEM_DISABLED_CAPABLE_VALUE	NPEM_ENABLE
NPEM_ENCLOSURE_SPECIFIC_CAPABILITIES_VALUE	NPEM_ENCLOSURE_SPECIFIC_CAPABILITIES
NPEM_FAIL_CAPABLE_VALUE	NPEM_ENABLE
NPEM_FAILED_ARRAY_CAPABLE_VALUE	NPEM_ENABLE
NPEM_HOT_SPARE_CAPABLE_VALUE	NPEM_ENABLE
NPEM_INVALID_DEVICE_TYPE_CAPABLE_VALUE	NPEM_ENABLE
NPEM_LOCATE_CAPABLE_VALUE	NPEM_ENABLE
NPEM_OK_CAPABLE_VALUE	NPEM_ENABLE
NPEM_PFA_CAPABLE_VALUE	NPEM_ENABLE
NPEM_REBUILD_CAPABLE_VALUE	NPEM_ENABLE
NPEM_RESET_CAPABLE_VALUE	NPEM_ENABLE
NQW	CX_RADM_FORMQ_NQW
NVC	CX_NVC
NVC_XALI_EXP	CX_NVC_XALI_EXPANSION
NVF	CX_NVFUNC
NW	CX_NW
NW_PRFX	CX_NPRFX
OB_CCMP_DATA_RAM_ADDRP_WD	CC_OB_CCMP_DATA_RAM_ADDRP_WD
OB_CCMP_DATA_RAM_ADDR_WD	CC_OB_CCMP_DATA_RAM_ADDR_WD
OB_CCMP_DATA_RAM_DATA_WD	CC_OB_CCMP_DATA_RAM_DATA_WD
OB_NPDCMP_RAM_ADDRP_WD	CC_OB_NPDCMP_RAM_ADDRP_WD
OB_NPDCMP_RAM_ADDR_WD	CC_OB_NPDCMP_RAM_ADDR_WD
OB_NPDCMP_RAM_DATA_WD	CC_OB_NPDCMP_RAM_DATA_WD
OB_PDCMP_DATA_RAM_ADDRP_WD	CC_OB_PDCMP_DATA_RAM_ADDRP_WD
OB_PDCMP_DATA_RAM_ADDR_WD	CC_OB_PDCMP_DATA_RAM_ADDR_WD
OB_PDCMP_DATA_RAM_DATA_WD	CC_OB_PDCMP_DATA_RAM_DATA_WD
OB_PDCMP_HDR_RAM_ADDRP_WD	CC_OB_PDCMP_HDR_RAM_ADDRP_WD
OB_PDCMP_HDR_RAM_ADDR_WD	CC_OB_PDCMP_HDR_RAM_ADDR_WD

Parameter Name	Equals To
OB_PDCMP_HDR_RAM_DATA_WD	CC_OB_PDCMP_HDR_RAM_DATA_WD
ORIG_DATA_WD	(PHY_NB*8)
P0	0x0
P1	0x2
P2	0x3
PCIE_CORE_DATA_BUS_WD	(CX_NW * 32)
PCIE_OVER_CIO	3
PCLK_RATE_WD	(CX_CCIX_ESM_SUPPORT) ? 4 : 3
P_DATAQ_NECC_BITS	(CX_RADMQ_MODE == 2) ? [function_of: CX_RADM_SBUF_DATAQ_RAM_WD] : [function_of: RADM_PQ_DWD]
PDWN_WIDTH	CX_PHY_PDOWN_WD
PF_WD	CX_NFUNC_WD
P_HDRQ_NECC_BITS	(CX_RADM_SBUF_HDRQ_WD > 247) ? 10 : 9
PHY_IS_PIPE	(CX_CPCIE_ENABLE && (CX_PHY_INTERFACE == PIPE_INTERFACE))
PHY_IS_V7	(CX_PHY_INTERFACE == V7_INTERFACE)
PHY_NB	CX_PHY_NB
PHY_RXSB_WD	CX_PHY_RXSB_WD
PHY_RXSH_WD	CX_PHY_RXSH_WD
PHY_TXEI_WD	CX_PHY_TXEI_WD
PHY_VPT_DATA	CX_PHY_VIEWPORT_DATA
PHY_VPT_NUM	CX_PHY_NUM_MACROS
PIPE_DATA_WD	(CX_PIPE_SERDES_ARCH_VALUE?(NL*SERDES_DATA _WD):(NL*ORIG_DATA_WD))
PIPE_INTERFACE	1
PM_MST_WD	5
PM_SLV_WD	5

Parameter Name	Equals To
PRFX_DW	$((CC_DEVICE_TYPE == CC_SW) ? ((32 * NW_PRFX) + PRFX_PAR_WD) : (CX_TLP_PREFIX_ENABLE_VALUE ? ((32 * NW_PRFX) + PRFX_PAR_WD) : 32))$
PRFX_PAR_WD	CX_PRFX_PAR_WD
PRFX_WIDTH	$(AXI_IF_PAR_ECC_ENABLED ? (32 * CX_NPRFX) : ((32 * CX_NPRFX) + CX_PRFX_PAR_WD))$
PRFX_W_PAR	$(CX_TLP_PREFIX_ENABLE_VALUE ? FLT_Q_PRFX_WIDTH : 32)$
PROTOCOL	0
P_R_WD	PCLK_RATE_WD
PSET_ID_WD	$(CX_PIPE5_SUPPORT == 1) ? 6 : (CX_PIPE44_SUPPORT == 1) ? 5 : 4$
RADM_DATAQ_WD	$CX_NW + 5 * CX_NDQ + CX_DW$
RADM_DEPTH_DECOUPLE_VC0	0
RADM_DEPTH_DECOUPLE_VC1	$CX_NVC \leq 1 \ \&\& \ RADM_DEPTH_DECOUPLE_VC0$
RADM_DLLP_ABORT_WD	CUT_THROUGH_INVOLVED
RADM_ECRC_ERR_WD	ECRC_ERR_PASS_THROUGH
RADM_EOT_WD	1
RADM_MAX_OUTSTD_P_REQ	$(CC_HP_MASTER == 1) ? ((PCIE_CORE_DATA_BUS_WD == 32) ? 64 : (PCIE_CORE_DATA_BUS_WD == 64) ? 128 : (PCIE_CORE_DATA_BUS_WD == 128) ? 128 : 256) : ((PCIE_CORE_DATA_BUS_WD == 32) ? 32 : (PCIE_CORE_DATA_BUS_WD == 64) ? 64 : (PCIE_CORE_DATA_BUS_WD == 128) ? 64 : 128)$
RADM_P_HWD	[function_of: CX_RADMQ_MODE RADM_CPL_QMODE_VC0 TRGT1_POPULATE FLT_Q_COMMON_WIDTH TRGT1_TRGT0_INCLUSIVE_WIDTH TRGT0_EXCLUSIVE_WIDTH DEMUX_EXCLUSIVE_WIDTH]
RADM_PQ_D_ADDRBITS	CX_RADM_PQ_D_ADDRBITS
RADM_PQ_DPW	[function_of: RADM_PQ_DDP_VC0]
RADM_PQ_DWD	$RADM_DATAQ_WD / CX_NDQ$

Parameter Name	Equals To
RADM_PQ_H_ADDRBITS	CX_RADM_PQ_H_ADDRBITS
RADM_PQ_H_DATABITS	CX_RADM_PQ_H_DATABITS
RADM_PQ_H_DATABITS_O	CX_RADM_PQ_H_DATABITS_OUT
RADM_PQ_HP_W	[function_of: RADM_PQ_HDP_VC0]
RADM_Q_DATABITS	CX_RADM_Q_DATABITS
RADM_Q_DATABITS_O	CX_RADM_Q_DATABITS_OUT
RADM_Q_D_CTRLBITS	CX_RADM_Q_D_CTRLBITS
RADM_Q_H_CTRLBITS	CX_RADM_Q_H_CTRLBITS
RADM_RFC_OUT_WD	CX_ADM_RFC_WD
RADM_SBUF_HDRQ_ERR_ADDR_WD	(CX_RADM_SBUF_HDRQ_PW+((NHQ>32768)?16:((NHQ>16384)?15:((NHQ>8192)?14:((NHQ>4096)?13:((NHQ>2048)?12:((NHQ>1024)?11:((NHQ>512)?10:((NHQ>256)?9:((NHQ>128)?8:((NHQ>64)?7:((NHQ>32)?6:((NHQ>16)?5:((NHQ>8)?4:((NHQ>4)?3:((NHQ>2)?2:((NHQ>1)?1:1))))))))))))))
RADM_SEG_BUF	(CX_RADMQ_MODE==2)
RADM_SEG_BUF_CT_DPT_ADJ	(CX_RADMQ_MODE==2 && CUT_THROUGH_INVOLVED) ? CX_RADM_DATASIZE*2/CX_NHQ + 4: 0
RADM_SEG_BUF_MIN_DPT	3
RADM_SEG_BUF_VALUE	RADM_SEG_BUF
RADM_TAG_PORT_EN	0
RADM_TLP_ABORT_WD	1
RADM_VFINDEX_REGOUT	(VFINDEX_CALC_REGOUT)
RAM_PW	CX_RADM_FORMQ_QDW_RAM_PW
RAM_WD	CX_RADM_FORMQ_QDW_RAM_WD
RAS_BASE_PROT_WD	(CX_RAS_PROT_TYPE==1)? 1: (CX_RAS_PROT_RANGE<=16)? 6: (CX_RAS_PROT_RANGE==32)? 7: (CX_RAS_PROT_RANGE==64)? 8: 9
RAS_DEFAULT_NULLIFY_EN	1
RASDES_EC_INFO_CM	CX_RAS_DES_EC_INFO_CMN_BW
RASDES_EC_INFO_PL	CX_RAS_DES_EC_INFO_PLN_BW

Parameter Name	Equals To
RASDES_EC_RAM_ADDR_WIDTH	CX_RAS_DES_EC_RAM_ADDR_WIDTH
RASDES_EC_RAM_DATA_WIDTH	CX_RAS_DES_EC_RAM_DATA_WIDTH
RASDES_SD_INFO_CM	CX_RAS_DES_SD_INFO_CMN_BW
RASDES_SD_INFO_PL	CX_RAS_DES_SD_INFO_PLN_BW
RASDES_SD_INFO_PV	CX_RAS_DES_SD_INFO_PVC_BW
RASDES_TBA_INFO_CM	CX_RAS_DES_TBA_INFO_CMN_BW
RASDES_TBA_RAM_ADDR_WIDTH	CX_RAS_DES_TBA_RAM_ADDR_WIDTH
RASDES_TBA_RAM_DATA_WIDTH	CX_RAS_DES_TBA_RAM_DATA_WIDTH
RASDP_BYPASS_HDR_PROT_WD	CX_RASDP_BYPASS_HDR_PROT_WD
RASDP_DATAQ_ERR_SYND_WD	CX_RADM_SBUF_DATAQ_PROT_WD
RASDP_FORMQ_ERR_SYND_WD	CX_RASDP_FORMQ_ERR_SYND_WD
RASDP_HDRQ_ERR_SYND_WD	CX_RASDP_HDRQ_ERR_SYND_WD
RASDP_RBUF_ERR_SYND_WD	RBUF_PROT_WD
RASDP_SOTBUF_ERR_SYND_WD	CX_RAS_SOTBUF_PROT_WD
RASDP_TRGT1_HDR_PROT_WD	CX_RASDP_TRGT1_HDR_PROT_WD
RAS_PCIE_HDR_PROT_WD	CX_RAS_PCIE_HDR_PROT_WD
RAS_RAM_BASE_PROT_WD	(CX_RAS_RAM_PROT_RANGE==32)? 7: 8
RATE_WIDTH	CX_PHY_RATE_WD
RBUF_PROT_WD	(CX_RAS_EN==1)? [function_of: (32*CX_NW) CX_RBUF_CTRL_WD CX_RASDP_EN CX_RAS_PROT_TYPE CX_RAS_PROT_RANGE CX_RAS_RAM_PROT_RANGE RAS_BASE_PROT_WD RAS_RAM_BASE_PROT_WD]: 0
RBUF_PW	[function_of: CX_RBUF_DEPTH]
RESERVED	0x2
RIO_POPULATED	0
RTLH_RAMD_IN_WD	CX_ADM_TLH_WD
RX_PSET_WD	3
RX_SERIALIZATION_OVFLW_PRVNTN	CX_NW==16
RX_TLP	CX_RX_TLP
SATA_CAP_ENABLE	0x0

Parameter Name	Equals To
SCALED_FC_SUPPORTED	FC_SCALE_EN
S_C_CDC_FIFO_DEPTH	14
SERDES_DATA_WD	(PHY_NB*10)
SINGLE_CPCIE	0
SINGLE_MPCIE	1
S_L1_0	0x1
S_L1_0_WAIT4_ACK	0x2
S_L1_0_WAIT4_CLKREQ	0x3
S_L1_N	0x5
S_L1_N_ABORT	0x7
S_L1_N_ENTRY	0x4
S_L1_N_EXIT	0x6
S_L1_U	0x0
SLAVE_BUS_ADDR_WIDTH_IS_64	(SLAVE_BUS_ADDR_WIDTH==64)
SLAVE_POPULATED	(AMBA_POPULATED)
SLV_ADDR_WD	SLAVE_BUS_ADDR_WIDTH
SLV_AR_USER_WIDTH	CC_SLV_AR_USER_WIDTH
SLV_AW_USER_WIDTH	CC_SLV_AW_USER_WIDTH
SLV_BURST_LEN_PW	CC_SLV_BURST_LEN_PW
SLV_B_USER_WIDTH	CC_SLV_B_USER_WIDTH
SLV_DATA_WD	SLAVE_BUS_DATA_WIDTH
SLV_ID_WD	CC_SLV_BUS_ID_WIDTH
SLV_LOCK_WD	CC_AXI_LOCK_WIDTH
SLV_MISC_INFO_PW	((AMBA_INTERFACE == 1) ? ((CX_NFUNC != 1) ? 29 : 26) : ((CX_NFUNC != 1) ? 25 : 22))
SLV_NPW_SAB_RAM_ADDRP_WD	CC_SLV_NPW_SAB_RAM_ADDRP_WD
SLV_NPW_SAB_RAM_ADDR_WD	CC_SLV_NPW_SAB_RAM_ADDR_WD
SLV_NPW_SAB_RAM_DATA_WD	CC_SLV_NPW_SAB_RAM_DATA_WD
SLV_RESP_MISC_INFO_PW	((AMBA_POPULATED && (CX_NFUNC != 1)) ? 14 : 11)

Parameter Name	Equals To
SLV_R_USER_WIDTH	CC_SLV_R_USER_WIDTH
SLV_WSTRB_WD	CC_SLV_BUS_WSTRB_WIDTH
SLV_W_USER_WIDTH	CC_SLV_W_USER_WIDTH
SNPS_RSVDPARAM_11	0
SNPS_RSVDPARAM_18	0
SNPS_RSVDPARAM_19	SNPS_RSVDPARAM_69 ? 4096 : 256
SNPS_RSVDPARAM_25	0
SNPS_RSVDPARAM_28	0
SNPS_RSVDPARAM_35	0
SNPS_RSVDPARAM_36	(SNPS_RSVDPARAM_64==1 && CX_GEN4_SPEED==1) ? 1 : 0
SNPS_RSVDPARAM_37	(SNPS_RSVDPARAM_64==1 && SNPS_RSVDPARAM_36 && (CX_GEN5_SPEED==1 CX_CCIX_ENABLE==1)) ? 1 : 0
SNPS_RSVDPARAM_38	0
SNPS_RSVDPARAM_49	0
SNPS_RSVDPARAM_50	(SNPS_RSVDPARAM_25 && !SNPS_RSVDPARAM_18)
SNPS_RSVDPARAM_54	(SNPS_RSVDPARAM_28) ? 8 : ((CX_MAX_MTU / (4 * CX_NW)) + 8)
SNPS_RSVDPARAM_60	(CC_DMA_ENABLE && CC_UNROLL_ENABLE) ? 1 : 0
SNPS_RSVDPARAM_61	CX_EXTENSIBLE_VFUNC ? 0xFFFF : SNPS_RSVDPARAM_69 ? 4096 : 256
SNPS_RSVDPARAM_64	((CX_MAC_SMODE_GEN3==16) (CX_MAC_SMODE_GEN4==16) (CX_MAC_SMODE_GEN5==16))
SNPS_RSVDPARAM_65	0x0
SNPS_RSVDPARAM_69	0
SNPS_RSVDPARAM_7	0
SNPS_RSVDPARAM_71	((CX_PHY_FREQ == FREQ_125) && (CX_PHY_GEN2_MODE == GEN2_DF) && (CX_PHY_GEN3_MODE == GEN3_DW) && (CX_PHY_GEN4_MODE == GEN4_DW) && (CX_PHY_INTERFACE == V7_INTERFACE)) ? 1 : 0
SNPS_RSVDPARAM_72	(CX_NW == 16) ? 1 : 0

Parameter Name	Equals To
SOTBUF_L2DEPTH	[function_of: CX_DL_NB CX_NL CX_MAX_MTU CX_ECRC_ENABLE CX_TLP_PREFIX_ENABLE_VALUE CX_NPRFX CX_INTERNAL_DELAY CX_MAX_L0S_LTIME CX_RBUF_AUTOSIZE CX_SOTBUF_DEPTH CX_GEN5_SPEED CX_GEN4_SPEED CX_GEN3_SPEED CX_5GTS_SPEED CX_RETIMER_LATENCY]
SOTBUF_PW	SOTBUF_L2DEPTH
SOTBUF_WD	SOTBUF_WIDTH
SPLIT_SUPPORT	(CC_RESPONSE_MODE==1)
SRIOV_OR_DBI_MFBAR_EN	(CX_SRIOV_ENABLE DBI_MULTI_FUNC_BAR_EN CX_ARI_ENABLE)
STATUS	0
TAG_SIZE	CX_TAG_SIZE
TPH_ST_TABLE_DEPTH_0	(TPH_ST_TABLE_LOC_0==1) ? (((function_of: TPH_ST_TABLE_SIZE_0 % 2] == 0) ? (TPH_ST_TABLE_SIZE_0/2) : ((TPH_ST_TABLE_SIZE_0 + 1)/2)) : 0
TRGT0_EXCLUSIVE_WIDTH	[function_of: FLT_Q_IN_MEMBAR_RANGE_WIDTH FLT_Q_ROM_IN_RANGE_WIDTH FLT_Q_IO_REQ_IN_RANGE_WIDTH FLT_Q_FRSTDW_BE_WIDTH FLT_Q_DWADDR_WIDTH FLT_Q_LSTDW_BE_WIDTH FLT_Q_DW_LENGTH_WIDTH]
TRGT1_POPULATE	1
TRGT1_TRGT0_INCLUSIVE_WIDTH	[function_of: FLT_Q_DW_LENGTH_WIDTH FLT_Q_IN_MEMBAR_RANGE_WIDTH FLT_Q_ROM_IN_RANGE_WIDTH FLT_Q_IO_REQ_IN_RANGE_WIDTH FLT_Q_FRSTDW_BE_WIDTH FLT_Q_ADDR_WIDTH FLT_Q_TD_WIDTH FLT_Q_EP_WIDTH FLT_Q_CPL_LAST_WIDTH FLT_Q_LSTDW_BE_WIDTH FLT_Q_SATA_WIDTH FLT_Q_AT_WIDTH FLT_Q_TH_WIDTH FLT_Q_HDR_RSVD_DW0_WIDTH FLT_Q_HDR_RSVD_DW2_WIDTH FLT_Q_CCIX_WIDTH]
TRGT_DATA_PROT_WD	(CX_RASDP_EN==1)? [function_of: (32*CX_NW) CX_RAS_PROT_RANGE RAS_BASE_PROT_WD]:0
TRGT_DATA_WD	(CX_RASDP_EN)? (32*CX_NW) + TRGT_DATA_PROT_WD : (32*CX_NW)
TX_COEF_WD	18
TX_CRCSR_WD	6

Parameter Name	Equals To
TX_DEEMPH_WD	CX_PHY_TXDEEMPH_WD
TX_FS_WD	6
TX_HW_W_PAR	(128+RAS_PCIE_HDR_PROT_WD)
TX_PSET_WD	4
UNROLL_ATU_SIZE	ENABLE_MEM_MAP_UNROLL_ATU_REG ? [function_of: CX_ATU_NUM_INBOUND_REGIONS CX_ATU_NUM_OUTBOUND_REGIONS] : 0
UNROLL_DMA_OFFSET_BAR_HIGH	(UNROLL_DMA_OFFSET_BAR >> 32)
UNROLL_DMA_OFFSET_BAR_LOW	(UNROLL_DMA_OFFSET_BAR & 0x00000000ffffff)
UNROLL_DMA_SIZE	ENABLE_MEM_MAP_UNROLL_DMA_REG ? (2**CX_DMAREG_CHADDR_SPACE)*[function_of: CC_NUM_DMA_RD_CHAN CC_NUM_DMA_WR_CHAN] + (CC_LEGACY_DMA_MAP_VALUE ? 0x200 : 0) : 0
V7_INTERFACE	2
VF_AER_EN	VF_AER_ENABLE
VF_BAR0_RESIZABLE_0	(VF_BAR0_SIZING_SCHEME_0==2) ? 1 : 0
VF_BAR1_RESIZABLE_0	(VF_BAR1_SIZING_SCHEME_0==2) ? 1 : 0
VF_BAR2_RESIZABLE_0	(VF_BAR2_SIZING_SCHEME_0==2) ? 1 : 0
VF_BAR3_RESIZABLE_0	(VF_BAR3_SIZING_SCHEME_0==2) ? 1 : 0
VF_BAR4_RESIZABLE_0	(VF_BAR4_SIZING_SCHEME_0==2) ? 1 : 0
VF_BAR5_RESIZABLE_0	(VF_BAR5_SIZING_SCHEME_0==2) ? 1 : 0
VF_IMM_CONVERT_ENABLE	(VF_IMM_ENABLE EXT_VF_IMM_ENABLE)
VFI_WD	((NVF>32768)?16:((NVF>16384)?15:((NVF>8192)?14:((NVF>4096)?13:((NVF>2048)?12:((NVF>1024)?11:((NVF>512)?10:((NVF>256)?9:((NVF>128)?8:((NVF>64)?7:((NVF>32)?6:((NVF>16)?5:((NVF>8)?4:((NVF>4)?3:((NVF>2)?2:((NVF>1)?1:1))))))))))))))))))
VF_LN_128_SUPPORTED_VALUE	0x0
VF_LN_64_SUPPORTED_VALUE	0x0
VF_LN_REG_MAX_VALUE	0x0
VF_MSI_CONVERT_EN	(VF_MSI_EN EXT_VF_MSI_EN)
VF_MSI_EN	VF_MSI_CAP_ENABLE

Parameter Name	Equals To
VF_MSIX_PBA_OFFSET_0_BASELINE	VF_MSIX_TABLE_BIR_0 == VF_MSIX_PBA_BIR_0 ? (VF_MSIX_TABLE_OFFSET_0 << 3) + MSIX_TABLE_SIZE_0_BYTES : 0
VF_PASID_ENABLE	0
VF_TPH_ENABLE	VF_TPH_ENABLE_VALUE
VF_WD	(((NVF>32768)?16:(((NVF>16384)?15:(((NVF>8192)?14:((N VF>4096)?13:(((NVF>2048)?12:(((NVF>1024)?11:(((NVF>51 2)?10:(((NVF>256)?9:(((NVF>128)?8:(((NVF>64)?7:(((NVF>3 2)?6:(((NVF>16)?5:(((NVF>8)?4:(((NVF>4)?3:(((NVF>2)?2:((N VF>1)?1:1)))))))))))))))))))+1)
WIDTH_WIDTH	CX_PHY_WIDTH_WD
XADM_RFC_IN_WD	CX_ADM_RFC_WD
XADM_XTLH_OUT_WD	CX_ADM_TLH_WD